

Modern Fractal Geometry

Volume II: Entropy, Inverse Theorems, and Arithmetic Rigidity

Reader's edition — 7 October 2026

Reader's guide

This volume develops the entropy and additive-combinatorial methods behind modern rigidity results for self-similar measures. It begins with finite-scale entropy and components, proves inverse theorems on the line and in higher dimensions, and then applies them to dimension drop, projections, Bernoulli convolutions, absolute continuity, L^q spectra, exact overlaps, and recent analytic and holomorphic extensions.

The expected background is the material in Knapp's *Basic Real Analysis* and *Advanced Real Analysis*, together with the designated working books on harmonic analysis, additive combinatorics, and foundational dynamical systems. Every other technical term is defined before use. Earlier results from Volume I are restated at the interfaces where their exact normalization matters.

Definitions, motivation, theorem statements, proofs, and consequences are interleaved in the order in which the mathematics needs them. Green and blue boxes carry definitions and proved results; amber boxes identify guided or cited results whose proofs are not reproduced in full; red boxes identify a conditional statement or an explicit research boundary. A citation records provenance and is never meant to hide an unstated hypothesis.

No exercises or solutions are included in this edition. They are deferred until proof writing for the full series is complete.

Proof-status convention

A result is *proved in full* when the argument printed here is complete relative to the declared background corpus, earlier fully proved results in the series, and any precisely stated classical tools cited at the point of use. Routine probability and algebraic-number lemmas may be used in this way; a research theorem is not silently treated as background.

A *guided result* includes a substantial derivation but leaves a named research input external. A *cited result* is stated precisely, with an exact primary reference, but its proof is not reproduced. An *open or conditional result* is never used as an unconditional premise. The color and heading of each theorem box distinguish these cases, so the reader can tell the logical status of a result without interrupting the mathematical narrative. A separate theorem-by-theorem index records every result not proved in full.

The October 2026 additions have a more specific distinction. The unrestricted entropy-rate formula uses the earlier proved Feng–Hu theorem, and the quartic-Salem and degree-31 singularity proofs are independent of the arithmetic classification. The latter’s binary function-field character bound is proved here, including its imprimitive characters and trivial factors, by an explicit cover and an auxiliary-function count. Neither a general Weil theorem nor a general Riemann–Roch theorem is an omitted input. Its arithmetic consequences therefore have full internal proofs relative to the declared background. The degree-31 proof is computer-assisted: a retained exact-rational certificate is combined with the written analytic error bounds.

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Chapter II.1

Entropy at a Geometric Scale

Geometric intuition

Entropy at a geometric scale is the book's quantitative microscope. This chapter fixes the dyadic and smoothed conventions used throughout the volume, proves the comparison and stability estimates that make them interchangeable, and explains how normalized entropy records dimension without pretending that a single finite scale is an asymptotic invariant.

II.1.1 Entropy of a partition

At a fixed scale, a measure may be replaced by the probability vector recording its mass in each cell. Entropy measures how evenly that mass is distributed. For example, the uniform law on N cells has entropy $\log N$, while a point mass has entropy zero. This observation will make entropy a scale-dependent version of a cell count.

All logarithms are natural. Thus entropy is measured in *nats*; division by $\log 2$ converts nats to bits.

Definition II.1.1.1 (II.1.01: Partitions and Shannon entropy). Let $(X, \mathcal{B}, \mu)$ be a probability space. A **countable measurable partition** $\mathcal{P}$ of X is a finite or countably infinite family of pairwise disjoint, nonempty sets in $\mathcal{B}$ whose union is X . Its members are called its **atoms**. Null atoms may be retained, but they make no contribution to entropy.

For $0 \leq t \leq 1$, put

$$\eta(t) = \begin{cases} -t \log t, & t > 0, \\ 0, & t = 0. \end{cases}$$

The **Shannon entropy of μ with respect to $\mathcal{P}$** is

$$H_\mu(\mathcal{P}) = \sum_{P \in \mathcal{P}} \eta(\mu(P)).$$

For a probability vector $p = (p_i)_{i \in I}$ on a finite or countable index set, write $H(p) = \sum_i \eta(p_i)$. The convention $0 \log 0 = 0$ is always in force.

Every countable sum of nonnegative terms is interpreted by monotone convergence over finite subsums. The first estimates concern probability vectors alone. They supply the elementary convexity input for all the partition identities that follow.

Definition II.1.1.2 (II.1.03: Relative entropy). For probability vectors $p = (p_i)$ and $q = (q_i)$ on the same finite or countable set, their **relative entropy** is

$$D(p\|q) = \sum_{i: p_i > 0} p_i \log \frac{p_i}{q_i} \in \mathbb{R} \cup \{+\infty\},$$

where the value is $+\infty$ if some $p_i > 0 = q_i$. The series is well defined because its negative part is finite: indeed,

$$\sum_{i: p_i < q_i} p_i \log \frac{q_i}{p_i} \leq \sum_{i: p_i < q_i} (q_i - p_i) \leq 1.$$

II.1.1.1 Scalar information inequalities

Lemma II.1.1.3 (L-II.1.01 — Log-sum and Gibbs inequalities). *Let I be finite or countable and let $(a_i), (b_i)$ be nonnegative families with*

$$0 < A = \sum_i a_i < \infty, \quad 0 < B = \sum_i b_i < \infty.$$

Then

$$\sum_i a_i \log \frac{a_i}{b_i} \geq A \log \frac{A}{B}, \quad (1.1)$$

where a positive a_i over $b_i = 0$ gives $+\infty$. Equality holds if and only if

$$\frac{a_i}{A} = \frac{b_i}{B} \quad (i \in I). \quad (1.2)$$

Consequently, $D(p\|q) \geq 0$ for countable probability vectors, with equality if and only if $p = q$.

Proof

Normalize

$$p_i = \frac{a_i}{A}, \quad q_i = \frac{b_i}{B}.$$

If $p_i > 0 = q_i$ for some i , the conclusion is immediate. Otherwise put $S = \{i : p_i > 0\}$. The inequality

$$\log x \leq x - 1 \quad (x > 0) \quad (1.3)$$

follows because $x - 1 - \log x$ has derivative $1 - 1/x$, its unique minimum is at $x = 1$, and its value there is zero. It gives

$$p_i \log \frac{q_i}{p_i} \leq q_i - p_i \quad (i \in S). \quad (1.4)$$

The positive part of the series on the left is bounded termwise by $(q_i - p_i)_+$, whose sum is at most 1. Thus the series is well defined in $[-\infty, \infty)$, and summing (1.4) yields

$$\sum_{i \in S} p_i \log \frac{q_i}{p_i} \leq \sum_{i \in S} (q_i - p_i) = q(S) - 1 \leq 0. \quad (1.5)$$

Therefore $D(p\|q) \geq 0$. Since

$$\sum_i a_i \log \frac{a_i}{b_i} = AD(p\|q) + A \log \frac{A}{B}, \quad (1.6)$$

(1.1) follows.

Equality in (1.5) requires both $q(S) = 1$ and equality in (1.3) for every $i \in S$, hence $q_i/p_i = 1$ on S ; it also forces $q_i = 0$ off S . Thus $p = q$. The converse is immediate, proving (1.2) and the Gibbs equality criterion. $\square$

Lemma II.1.1.4 (L-II.1.02 – Entropy extrema and maximal atoms). *Let $p = (p_i)$ be a finite or countable probability vector and let $p^* = \sup_i p_i$. Then*

$$H(p) \geq 0, \quad H(p) \geq -\log p^*. \quad (1.7)$$

The first entropy is zero if and only if one coordinate equals 1. If at most $N < \infty$ coordinates are positive, then

$$H(p) \leq \log N, \quad (1.8)$$

and equality holds if and only if precisely N coordinates are positive and all equal $1/N$.

Proof

Each $\eta(p_i)$ is nonnegative, so $H(p) \geq 0$. If the sum is zero, then every positive coordinate satisfies $\eta(p_i) = 0$, hence $p_i = 1$; because the coordinates sum to one, there is exactly one such coordinate. The converse is direct.

For every positive coordinate, $p_i \leq p^*$, so

$$H(p) = \sum_{p_i > 0} p_i \log \frac{1}{p_i} \geq \sum_{p_i > 0} p_i \log \frac{1}{p^*} = -\log p^*. \quad (1.9)$$

If the positive support has cardinality $M \leq N$, apply L-II.1.01 to its law p and the uniform law $u_i = 1/M$:

$$0 \leq D(p\|u) = \sum_i p_i \log p_i + \log M = \log M - H(p). \quad (1.10)$$

Thus $H(p) \leq \log M \leq \log N$. Equality with $\log N$ forces $M = N$ and equality in Gibbs' inequality, so $p = u$. A uniform law on N points gives equality. $\square$

II.1.2 Chain rules and information

Refining a partition is a two-stage experiment: first identify the coarse atom, and then identify the finer atom inside it. The chain rule says that the total information is exactly the coarse information plus the average information still unresolved. The direct definition below remains meaningful even when the corresponding unconditional entropies are infinite.

Definition II.1.2.1 (II.1.02: Joins, refinement, and conditional entropy). For countable measurable partitions $\mathcal{P}, \mathcal{Q}$, their **join** is

$$\mathcal{P} \vee \mathcal{Q} = \{P \cap Q : P \in \mathcal{P}, Q \in \mathcal{Q}, P \cap Q \neq \emptyset\}.$$

The partition $\mathcal{Q}$ **refines** $\mathcal{P}$, written $\mathcal{P} \leq \mathcal{Q}$, if every atom of $\mathcal{Q}$ is contained in an atom of $\mathcal{P}$. Equivalently, $\mathcal{P}$ is a **coarsening** of $\mathcal{Q}$.

For $Q \in \mathcal{Q}$ with $\mu(Q) > 0$, the **conditional component on Q** is

$$\mu_Q(A) = \frac{\mu(A \cap Q)}{\mu(Q)}, \quad A \in \mathcal{B}.$$

The **conditional entropy of $\mathcal{P}$ given $\mathcal{Q}$** is defined by the nonnegative sum

$$H_\mu(\mathcal{P} \mid \mathcal{Q}) = \sum_{\substack{Q \in \mathcal{Q} \\ \mu(Q) > 0}} \mu(Q) H_{\mu_Q}(\mathcal{P}) \in [0, \infty].$$

Null conditioning atoms contribute zero.

Theorem II.1.2.2 (T-II.1.03 – Chain rules and refinement). For countable measurable partitions $\mathcal{P}, \mathcal{Q}, \mathcal{R}$,

$$H_\mu(\mathcal{P} \vee \mathcal{Q}) = H_\mu(\mathcal{Q}) + H_\mu(\mathcal{P} \mid \mathcal{Q}), \quad (2.1)$$

and

$$H_\mu(\mathcal{P} \vee \mathcal{Q} \mid \mathcal{R}) = H_\mu(\mathcal{Q} \mid \mathcal{R}) + H_\mu(\mathcal{P} \mid \mathcal{Q} \vee \mathcal{R}). \quad (2.2)$$

Both are identities in $[0, \infty]$. If $\mathcal{P} \leq \mathcal{Q}$, then

$$H_\mu(\mathcal{Q}) = H_\mu(\mathcal{P}) + H_\mu(\mathcal{Q} \mid \mathcal{P}) \geq H_\mu(\mathcal{P}). \quad (2.3)$$

Proof

Write $p_{P,Q} = \mu(P \cap Q)$ and $q_Q = \mu(Q)$. If $q_Q > 0$, then

$$\begin{aligned} \sum_{P \in \mathcal{P}} \eta(p_{P,Q}) &= - \sum_P p_{P,Q} \log p_{P,Q} \\ &= - \sum_P p_{P,Q} \log q_Q - q_Q \sum_P \frac{p_{P,Q}}{q_Q} \log \frac{p_{P,Q}}{q_Q} \\ &= \eta(q_Q) + q_Q H_{\mu_Q}(\mathcal{P}). \end{aligned} \quad (2.4)$$

When $q_Q = 0$, both sides of (2.4) are zero. Summing over Q is legitimate by nonnegativity and gives (2.1).

For every R with $\mu(R) > 0$, apply (2.1) to the probability μ_R and then multiply by $\mu(R)$. Summing over R gives

$$\sum_R \mu(R) H_{\mu_R}(\mathcal{P} \vee \mathcal{Q}) = \sum_R \mu(R) H_{\mu_R}(\mathcal{Q}) + \sum_{R,Q} \mu(R \cap Q) H_{\mu_{R \cap Q}}(\mathcal{P}), \quad (2.5)$$

where null intersections contribute zero. The three expressions are exactly the three terms in (2.2).

If $\mathcal{P} \preceq \mathcal{Q}$, deleting empty atoms gives $\mathcal{P} \vee \mathcal{Q} = \mathcal{Q}$. Substitute this in (2.1) with the roles of $\mathcal{P}, \mathcal{Q}$ arranged as displayed in (2.3). Conditional entropy is nonnegative, so the inequality follows. $\square$

Definition II.1.2.3 (Mutual information). Two partitions $\mathcal{P}, \mathcal{Q}$ are **independent under μ** if

$$\mu(P \cap Q) = \mu(P)\mu(Q) \quad (P \in \mathcal{P}, Q \in \mathcal{Q}).$$

Their **mutual information** is the relative entropy of the joint atom law from the product of its marginals:

$$I_\mu(\mathcal{P}; \mathcal{Q}) = \sum_{P,Q: \mu(P \cap Q) > 0} \mu(P \cap Q) \log \frac{\mu(P \cap Q)}{\mu(P)\mu(Q)}.$$

Mutual information measures the part of two observations that is shared. The product of the marginal laws is the joint law one would see under independence, so Gibbs' inequality makes mutual information nonnegative and detects independence exactly. Coarsening either observation deliberately forgets distinctions; the data-processing inequality asserts that this cannot create shared information.

Theorem II.1.2.4 (T-II.1.04 – Mutual information and data processing). *For countable partitions $\mathcal{P}, \mathcal{Q}$,*

$$I_\mu(\mathcal{P}; \mathcal{Q}) = I_\mu(\mathcal{Q}; \mathcal{P}) \geq 0, \quad (2.6)$$

and equality holds if and only if $\mathcal{P}, \mathcal{Q}$ are independent. If $H_\mu(\mathcal{P}) < \infty$, then

$$I_\mu(\mathcal{P}; \mathcal{Q}) = H_\mu(\mathcal{P}) - H_\mu(\mathcal{P} \mid \mathcal{Q}), \quad (2.7)$$

so

$$0 \leq H_\mu(\mathcal{P} \mid \mathcal{Q}) \leq H_\mu(\mathcal{P}). \quad (2.8)$$

If $\mathcal{P}' \preceq \mathcal{P}$ and $\mathcal{Q}' \preceq \mathcal{Q}$, then

$$I_\mu(\mathcal{P}'; \mathcal{Q}') \leq I_\mu(\mathcal{P}; \mathcal{Q}). \quad (2.9)$$

Proof

Let

$$r_{P,Q} = \mu(P \cap Q), \quad s_{P,Q} = \mu(P)\mu(Q).$$

These are probability vectors on $\mathcal{P} \times \mathcal{Q}$, and the definition gives

$$I_\mu(\mathcal{P}; \mathcal{Q}) = D(r \parallel s). \quad (2.10)$$

Gibbs' inequality proves nonnegativity and says equality holds exactly when $r = s$, which is precisely independence. Symmetry follows by interchanging the coordinates.

We next prove (2.8) without subtracting infinite quantities. The function η is concave on $[0, 1]$: it is continuous at zero and $\eta''(t) = -1/t < 0$ for $t > 0$. Put

$$q_Q = \mu(Q), \quad r_{P|Q} = \frac{\mu(P \cap Q)}{q_Q}$$

when $q_Q > 0$. For each fixed P , concavity gives

$$\sum_{Q: q_Q > 0} q_Q \eta(r_{P|Q}) \leq \eta\left(\sum_Q q_Q r_{P|Q}\right) = \eta(\mu(P)). \quad (2.11)$$

For a countable partition $\mathcal{Q}$, retain finitely many weights and aggregate the remaining weight q_* at its weighted mean r_* . Finite Jensen gives the inequality with the nonnegative extra term $q_* \eta(r_*)$. Dropping that term shows that every finite partial sum of the left side of (2.11) is bounded by its right side. Monotone convergence proves (2.11) for the countable family. Summing (2.11) over P and using nonnegativity gives

$$H_\mu(\mathcal{P} \mid \mathcal{Q}) \leq H_\mu(\mathcal{P}). \quad (2.12)$$

Under the hypothesis of (2.7), both sides of (2.12) are finite. Hence the two nonnegative double series below are finite, and termwise subtraction is legitimate:

$$\begin{aligned}
H_\mu(\mathcal{P}) - H_\mu(\mathcal{P} \mid \mathcal{Q}) &= \sum_{P,Q} r_{P,Q} \log \frac{1}{\mu(P)} - \sum_{P,Q} r_{P,Q} \log \frac{\mu(Q)}{r_{P,Q}} \\
&= \sum_{P,Q} r_{P,Q} \log \frac{r_{P,Q}}{\mu(P)\mu(Q)} \\
&= I_\mu(\mathcal{P}; \mathcal{Q}).
\end{aligned} \tag{2.13}$$

This proves (2.7)–(2.8); symmetry gives the analogous formula with Q .

For data processing, each atom $P' \in \mathcal{P}'$ is a disjoint union of a subfamily $A(P') \subset \mathcal{P}$. For each $Q' \in \mathcal{Q}'$, let $B(Q') \subset \mathcal{Q}$ be the subfamily whose disjoint union is Q' . Apply the log-sum inequality to each rectangle $A(P') \times B(Q')$:

$$\sum_{\substack{P \subset P' \\ Q \subset Q'}} r_{P,Q} \log \frac{r_{P,Q}}{s_{P,Q}} \geq r'_{P',Q'} \log \frac{r'_{P',Q'}}{s'_{P',Q'}}, \tag{2.14}$$

where

$$r'_{P',Q'} = \mu(P' \cap Q'), \quad s'_{P',Q'} = \mu(P')\mu(Q'). \tag{2.15}$$

The rectangles partition $\mathcal{P} \times \mathcal{Q}$. Summing (2.14), with the extended-value convention from L-II.1.01, gives (2.9). This regrouping is legitimate because the negative part of every relative-entropy series is finite; equivalently, the nonnegative blockwise log-sum deficits may be summed by monotone convergence. $\square$

The next result explains how entropy behaves when a probability measure is assembled from finitely many component measures. The extra term $H(\alpha)$ is exactly the information carried by the component label.

Definition II.1.2.5 (II.1.04: Finite mixtures). Let J be a nonempty finite set, let $\alpha = (\alpha_j)_{j \in J}$ be a probability vector, and let μ_j be probability measures on the same measurable space. Their **mixture with mixing law** α is

$$\mu = \sum_{j \in J} \alpha_j \mu_j.$$

The **mixing entropy** is $H(\alpha)$. Indices with $\alpha_j = 0$ may be deleted and contribute zero to all displayed expressions.

Theorem II.1.2.6 (T-II.1.05 – Concavity and entropy of mixtures). *Let J be finite and*

$$\mu = \sum_{j \in J} \alpha_j \mu_j.$$

For countable partitions $\mathcal{P}, \mathcal{Q}$,

$$\sum_j \alpha_j H_{\mu_j}(\mathcal{P}) \leq H_\mu(\mathcal{P}) \leq H(\alpha) + \sum_j \alpha_j H_{\mu_j}(\mathcal{P}), \tag{2.16}$$

and

$$\sum_j \alpha_j H_{\mu_j}(\mathcal{P} \mid \mathcal{Q}) \leq H_\mu(\mathcal{P} \mid \mathcal{Q}) \leq H(\alpha) + \sum_j \alpha_j H_{\mu_j}(\mathcal{P} \mid \mathcal{Q}). \quad (2.17)$$

Proof

For every P , concavity of η gives

$$\eta\left(\sum_j \alpha_j \mu_j(P)\right) \geq \sum_j \alpha_j \eta(\mu_j(P)). \quad (2.18)$$

Summing over P proves the lower bound in (2.16).

For the upper bound, put a probability $\tilde{\mu}$ on $J \times X$ by

$$\tilde{\mu}(\{j\} \times A) = \alpha_j \mu_j(A). \quad (2.19)$$

Let $\mathcal{J} = \{\{j\} \times X : j \in J\}$ and lift $\mathcal{P}$ to $J \times X$ by $\tilde{\mathcal{P}} = \{J \times P : P \in \mathcal{P}\}$. The chain rule gives

$$H_{\tilde{\mu}}(\mathcal{J} \vee \tilde{\mathcal{P}}) = H(\alpha) + \sum_j \alpha_j H_{\mu_j}(\mathcal{P}), \quad (2.20)$$

$$H_{\tilde{\mu}}(\mathcal{J} \vee \tilde{\mathcal{P}}) = H_\mu(\mathcal{P}) + H_{\tilde{\mu}}(\mathcal{J} \mid \tilde{\mathcal{P}}) \geq H_\mu(\mathcal{P}).$$

This proves the upper bound in (2.16) without subtraction.

For conditional concavity, fix Q with $\mu(Q) > 0$ and set

$$\beta_{j,Q} = \frac{\alpha_j \mu_j(Q)}{\mu(Q)} \quad (2.21)$$

when the numerator is positive. Then

$$\mu_Q = \sum_j \beta_{j,Q} (\mu_j)_Q. \quad (2.22)$$

Apply the already proved lower bound in (2.16) to (2.22), multiply by $\mu(Q)$, and sum over Q . The result is the lower bound in (2.17).

For the upper bound, lift $\mathcal{Q}$ to $\tilde{\mathcal{Q}}$ on $J \times X$. Conditional refinement and the conditional chain rule give

$$\begin{aligned} H_\mu(\mathcal{P} \mid \mathcal{Q}) &\leq H_{\tilde{\mu}}(\mathcal{J} \vee \tilde{\mathcal{P}} \mid \tilde{\mathcal{Q}}) \\ &= H_{\tilde{\mu}}(\mathcal{J} \mid \tilde{\mathcal{Q}}) + H_{\tilde{\mu}}(\tilde{\mathcal{P}} \mid \mathcal{J} \vee \tilde{\mathcal{Q}}) \\ &\leq H(\alpha) + \sum_j \alpha_j H_{\mu_j}(\mathcal{P} \mid \mathcal{Q}). \end{aligned} \quad (2.23)$$

The last inequality uses T-II.1.04 for $H_{\tilde{\mu}}(\mathcal{J} \mid \tilde{\mathcal{Q}}) \leq H_{\tilde{\mu}}(\mathcal{J}) = H(\alpha)$, and the last conditional entropy is the displayed weighted sum by (2.19). This proves (2.17). $\square$

II.1.3 Comparing partitions and scales

A translated dyadic grid is not the same partition, and a measure charging a grid boundary can change its finite-scale entropy discontinuously under an arbitrarily small translation. What remains uniformly controlled is the number of atoms of one grid that can meet a fixed atom of the other. This bounded-overlap principle is the robust substitute for continuity.

Definition II.1.3.1 (II.1.05: Geometric and dyadic scale entropy). For $\delta > 0$, define the half-open cubical partition

$$\mathcal{Q}_\delta = \{\delta(k + [0, 1)^d) : k \in \mathbb{Z}^d\}$$

of $\mathbb{R}^d$. The **entropy of μ at geometric scale δ** is $H(\mu; \delta) = H_\mu(\mathcal{Q}_\delta)$.

For $n \in \mathbb{Z}$, the **level- n dyadic partition** and **unnormalized dyadic entropy** are

$$\mathcal{D}_n = \mathcal{Q}_{2^{-n}}, \quad H_n(\mu) = H_\mu(\mathcal{D}_n).$$

For $n \geq 1$, the **normalized dyadic entropy** is

$$h_n(\mu) = \frac{H_n(\mu)}{n \log 2}.$$

The lower and upper **entropy dimensions** are

$$\underline{\dim}_e \mu = \liminf_{n \rightarrow \infty} h_n(\mu), \quad \overline{\dim}_e \mu = \limsup_{n \rightarrow \infty} h_n(\mu).$$

For $a \in \mathbb{R}^d$, the **translated dyadic grid** is $\mathcal{D}_n + a = \{Q + a : Q \in \mathcal{D}_n\}$. The notation H_n always denotes unnormalized entropy.

The normalization has geometric dimension units: a roughly uniform distribution over 2^{dn} level- n cubes has entropy about $dn \log 2$ and normalized entropy about d .

Definition II.1.3.2 (II.1.06: Partition overlap). For countable partitions $\mathcal{P}, \mathcal{Q}$ of the same set, their directed **intersection multiplicity** is

$$\kappa(\mathcal{P}, \mathcal{Q}) = \sup_{P \in \mathcal{P}} \#\{Q \in \mathcal{Q} : P \cap Q \neq \emptyset\} \in \mathbb{N} \cup \{\infty\}.$$

Lemma II.1.3.3 (L-II.1.06 – Bounded-intersection comparison). *Suppose*

$$\kappa(\mathcal{P}, \mathcal{Q}) \leq k, \quad \kappa(\mathcal{Q}, \mathcal{P}) \leq l, \tag{3.1}$$

where k, l are positive integers. Then

$$H_\mu(\mathcal{Q} \mid \mathcal{P}) \leq \log k, \quad H_\mu(\mathcal{P} \mid \mathcal{Q}) \leq \log l. \tag{3.2}$$

If either unconditional entropy is finite, both are finite and

$$H_\mu(\mathcal{P}) - \log l \leq H_\mu(\mathcal{Q}) \leq H_\mu(\mathcal{P}) + \log k. \quad (3.3)$$

Proof

For every positive-mass P , the μ_P -law of $\mathcal{Q}$ has at most k positive coordinates. L-II.1.02 gives

$$H_{\mu_P}(\mathcal{Q}) \leq \log k.$$

Average with weights $\mu(P)$ to obtain the first inequality in (3.2). Reverse the partitions for the second.

The two chain-rule expansions

$$H_\mu(\mathcal{P} \vee \mathcal{Q}) = H_\mu(\mathcal{P}) + H_\mu(\mathcal{Q} \mid \mathcal{P}) = H_\mu(\mathcal{Q}) + H_\mu(\mathcal{P} \mid \mathcal{Q}) \quad (3.4)$$

show first that finiteness of either unconditional entropy implies finiteness of the other. Subtracting only after finiteness is established and using (3.2) yields (3.3). $\square$

Theorem II.1.3.4 (T-II.1.07 – Dyadic scale calculus). *Let $m \geq n$ be integers and suppose $H_n(\mu) < \infty$. Then*

$$H_m(\mu) = H_n(\mu) + H_\mu(\mathcal{D}_m \mid \mathcal{D}_n), \quad (3.5)$$

and

$$0 \leq H_m(\mu) - H_n(\mu) \leq d(m - n) \log 2. \quad (3.6)$$

If $\mu([0, 1)^d) = 1$, then for $n \geq 1$,

$$0 \leq H_n(\mu) \leq dn \log 2, \quad 0 \leq h_n(\mu) \leq d. \quad (3.7)$$

Proof

The partition $\mathcal{D}_m$ refines $\mathcal{D}_n$. Each level- n cube is the disjoint union of exactly

$$2^{d(m-n)} \quad (3.8)$$

level- m cubes. The refinement identity T-II.1.03 gives (3.5), and L-II.1.02 applied inside each positive level- n component gives

$$0 \leq H_\mu(\mathcal{D}_m \mid \mathcal{D}_n) \leq \log 2^{d(m-n)} = d(m - n) \log 2. \quad (3.9)$$

This also proves $H_m(\mu) < \infty$. If μ is carried by $[0, 1)^d$, then one atom of $\mathcal{D}_0$ has mass one, so $H_0(\mu) = 0$. Apply (3.6) with coarse level 0 and fine level n , then divide by $n \log 2$, proving (3.7). $\square$

Corollary II.1.3.5 (C-II.1.08 – Translation stability). For $T_a(x) = x + a$, every $a \in \mathbb{R}^d$, and every integer n with $H_n(\mu) < \infty$,

$$|H_n((T_a)_*\mu) - H_n(\mu)| \leq d \log 2. \quad (3.10)$$

For $n \geq 1$, the normalized difference is at most d/n .

Proof

Pull-back of the partition gives

$$H_n((T_a)_*\mu) = H_\mu(\mathcal{D}_n - a). \quad (3.11)$$

In one coordinate, an interval of length 2^{-n} in either of two translated grids meets at most two intervals of the other grid. Taking Cartesian products,

$$\kappa(\mathcal{D}_n, \mathcal{D}_n - a) \leq 2^d, \quad \kappa(\mathcal{D}_n - a, \mathcal{D}_n) \leq 2^d. \quad (3.12)$$

L-II.1.06 gives (3.10). Division by $n \log 2$ gives d/n . $\square$

Definition II.1.3.6 (Bounded perturbations at a dyadic scale). Let $(X, \mathcal{B}, \mu)$ be a probability space and let $f, g : X \rightarrow \mathbb{R}^d$ be Borel. They are a **C-bounded perturbation at dyadic level n** if

$$\|f(x) - g(x)\|_\infty \leq C2^{-n} \quad \text{for } \mu\text{-almost every } x,$$

where $\|x\|_\infty = \max_{1 \leq j \leq d} |x_j|$.

Proposition II.1.3.7 (P-II.1.09 – Bounded-perturbation stability). Let $f, g : X \rightarrow \mathbb{R}^d$ be Borel and suppose

$$\|f(x) - g(x)\|_\infty \leq C2^{-n} \quad (3.13)$$

for μ -almost every x , where $C \geq 0$. If either entropy is finite, then

$$|H_n(f_*\mu) - H_n(g_*\mu)| \leq d \log(2[C] + 1). \quad (3.14)$$

Proof

The set on which (3.13) fails is Borel and null. Redefine $g = f$ on that set. This changes neither push-forward and makes (3.13) hold everywhere.

Put $c = [C]$. Suppose for some x that $f(x)$ lies in the dyadic cube with index $k \in \mathbb{Z}^d$ and $g(x)$ lies in the cube with index j . Coordinate by coordinate, (3.13) implies

$$k_r - c \leq j_r \leq k_r + c. \quad (3.15)$$

Indeed, $j_r \geq k_r + c + 1$ or $j_r \leq k_r - c - 1$ would force the coordinate distance to exceed $C2^{-n}$; the half-open endpoint convention rules out equality at the forbidden outer endpoint. Thus a fixed atom of $f^{-1}\mathcal{D}_n$ meets at most

$$(2c + 1)^d \quad (3.16)$$

atoms of $g^{-1}\mathcal{D}_n$. The same argument with f, g reversed gives the reverse intersection bound. Empty pull-back atoms are discarded. Finally,

$$H_\mu(f^{-1}\mathcal{D}_n) = H_n(f_*\mu), \quad H_\mu(g^{-1}\mathcal{D}_n) = H_n(g_*\mu), \quad (3.17)$$

so L-II.1.06 with $k = l = (2c + 1)^d$ proves (3.14). $\square$

II.1.4 Entropy and covering

Covering numbers describe where a measure is allowed to live; entropy describes how the mass is distributed among the available cells. A set that meets N cells can support at most $\log N$ entropy, but support alone gives no reverse inequality. A lower entropy bound requires a quantitative mass-spread hypothesis, for which the maximal cell mass is the simplest choice.

Definition II.1.4.1 (II.1.07: Occupancy, maximal mass, and covering). For a nonempty bounded set $E \subset \mathbb{R}^d$, its **dyadic occupancy number at level n** is

$$N_n^{\mathcal{D}}(E) = \#\{Q \in \mathcal{D}_n : Q \cap E \neq \emptyset\}.$$

For a probability μ with finite $H_n(\mu)$, define its **maximal level- n cell mass**, **positive-mass cell count**, and **effective level- n cell count** by

$$p_n^*(\mu) = \sup_{Q \in \mathcal{D}_n} \mu(Q), \quad N_n^+(\mu) = \#\{Q \in \mathcal{D}_n : \mu(Q) > 0\}, \quad N_{n,\text{eff}}(\mu) = \exp H_n(\mu).$$

For $r > 0$, $N_r(E)$ denotes the least number of closed Euclidean radius- r balls whose union contains E , as in Volume I. A probability μ is **carried by E** if some Borel set $F \subset E$ satisfies $\mu(F) = 1$. This is meaningful even when E itself is not measurable; for Borel E it is equivalent to $\mu(\mathbb{R}^d \setminus E) = 0$.

Theorem II.1.4.2 (T-II.1.10 – Cell occupancy and covering comparison). *Let $E \subset \mathbb{R}^d$ be nonempty and bounded, put $\delta = 2^{-n}$, and let μ be carried by E . Then*

$$N_{\sqrt{d}\delta}(E) \leq N_n^{\mathcal{D}}(E) \leq 4^d N_\delta(E). \quad (4.1)$$

Moreover,

$$\frac{1}{p_n^*(\mu)} \leq \exp H_n(\mu) \leq N_n^+(\mu) \leq N_n^{\mathcal{D}}(E), \quad (4.2)$$

and hence

$$-\log p_n^*(\mu) \leq H_n(\mu) \leq \log N_n^{\mathcal{D}}(E) \leq \log N_\delta(E) + d \log 4. \quad (4.3)$$

If $p_n^*(\mu) \leq C/N_n^{\mathcal{D}}(E)$ for $C \geq 1$, then

$$H_n(\mu) \geq \log N_n^{\mathcal{D}}(E) - \log C. \quad (4.4)$$

Proof

For every occupied dyadic cube Q , choose $x_Q \in E \cap Q$. If $y \in E \cap Q$, then

$$\|y - x_Q\|_2 \leq \sqrt{d} \delta. \quad (4.5)$$

Thus the balls $B(x_Q, \sqrt{d}\delta)$ cover E , proving the first inequality in (4.1).

Conversely, fix a closed Euclidean ball $B(x, \delta)$. Its projection onto each coordinate axis is contained in an interval of length 2δ , which meets at most four half-open intervals of the grid of mesh δ . Hence the ball meets at most 4^d dyadic cubes. A cover of E by $N_\delta(E)$ such balls therefore meets every occupied cube among at most $4^d N_\delta(E)$ cubes. This proves the second inequality in (4.1).

Because E is bounded, $N_n^{\mathcal{D}}(E)$ is finite. Every positive-mass dyadic cell meets E , since $\mu(\mathbb{R}^d \setminus E) = 0$, so

$$N_n^+(\mu) \leq N_n^{\mathcal{D}}(E). \quad (4.6)$$

Apply L-II.1.02 to the finite probability vector $(\mu(Q))_{\mu(Q)>0}$. Its support-size upper bound and maximal-coordinate lower bound give

$$-\log p_n^*(\mu) \leq H_n(\mu) \leq \log N_n^+(\mu). \quad (4.7)$$

Exponentiating and adjoining (4.6) proves (4.2); combining with (4.1) proves (4.3). Substitution of the hypothesis in the left side of (4.3) gives (4.4). $\square$

Corollary II.1.4.3 (C-II.1.11 – Entropy dimension versus box dimension). *If μ is carried by a nonempty bounded set E , then*

$$\underline{\dim}_e \mu \leq \underline{\dim}_B E, \quad \overline{\dim}_e \mu \leq \overline{\dim}_B E. \quad (4.8)$$

Proof

Set $\delta = 2^{-n}$ in (4.3) and divide by $n \log 2$:

$$h_n(\mu) \leq \frac{\log N_{2^{-n}}(E)}{n \log 2} + \frac{d \log 4}{n \log 2}. \quad (4.9)$$

By L-I.2.02, the liminf and limsup defining the box dimensions may be taken along the bounded-ratio scale sequence 2^{-n} . The last term in (4.9) tends to zero. Taking liminf and limsup gives the two inequalities in (4.8). $\square$

For a uniform measure on a finite set, the entropy loss from $\log |A|$ is the average logarithm of the number of points that collide in one dyadic cell. A separated set has no collisions and retains all of its discrete entropy. The exact formula below is the basic dictionary between finite-set cardinality and entropy.

Proposition II.1.4.4 (P-II.1.12 – Finite-set occupancy dictionary). *Let $A \subset \mathbb{R}^d$ be finite and nonempty, put*

$$\mu_A = \frac{1}{|A|} \sum_{a \in A} \delta_a, \quad c_Q = |A \cap Q|, \quad M_n(A) = \max_Q c_Q. \quad (4.10)$$

Then

$$H_n(\mu_A) = \log |A| - \frac{1}{|A|} \sum_Q c_Q \log c_Q, \quad (4.11)$$

and

$$\log \frac{|A|}{M_n(A)} \leq H_n(\mu_A) \leq \log \min\{|A|, N_n^{\otimes}(A)\}. \quad (4.12)$$

If A is 2^{-n} -separated in ℓ^∞ , then

$$H_n(\mu_A) = \log |A|. \quad (4.13)$$

Proof

Since $\mu_A(Q) = c_Q/|A|$, direct expansion gives

$$\begin{aligned} H_n(\mu_A) &= - \sum_{Q: c_Q > 0} \frac{c_Q}{|A|} \log \frac{c_Q}{|A|} \\ &= \log |A| - \frac{1}{|A|} \sum_Q c_Q \log c_Q, \end{aligned} \quad (4.14)$$

proving (4.11). Since $c_Q \leq M_n(A)$ on occupied cells and $\sum_Q c_Q = |A|$,

$$\frac{1}{|A|} \sum_Q c_Q \log c_Q \leq \log M_n(A), \quad (4.15)$$

which proves the lower bound in (4.12). The positive cell-mass vector has at most $N_n^{\otimes}(A)$ coordinates, and at most $|A|$ because every occupied cell contains a point. L-II.1.02 proves the upper bound.

Two distinct points in one half-open dyadic cube have ℓ^∞ distance strictly less than 2^{-n} . Under the separation hypothesis every c_Q is zero or one, so the second term in (4.11) vanishes. This proves (4.13). $\square$

II.1.5 Examples and boundary cases

II.1.5.1 E-II.1.13 – Infinite entropy makes subtraction illegal

On $\mathbb{N}_{\geq 3} = \{3, 4, \dots\}$, let

$$p_k = \frac{c}{k(\log k)^2}, \quad c^{-1} = \sum_{k=3}^{\infty} \frac{1}{k(\log k)^2}.$$

The normalizing constant is finite and positive because

$$\int_3^{\infty} \frac{dx}{x(\log x)^2} = \frac{1}{\log 3} < \infty,$$

and comparison on each interval $[k, k+1]$ gives convergence of the defining series. Thus $\sum p_k = 1$. For all sufficiently large k , $\log(1/p_k) \geq (1/2) \log k$; hence the integral test gives

$$H(p) \geq \frac{c}{2} \sum_{k \geq k_0} \frac{\log k}{k(\log k)^2} = \frac{c}{2} \sum_{k \geq k_0} \frac{1}{k \log k} = \infty. \quad (5.1)$$

Indeed, the corresponding integral equals $\log(\log R) - \log(\log k_0)$ and tends to infinity. Take $P = Q$ to be the singleton partition. Then

$$H(P) = H(Q) = H(P \vee Q) = \infty, \quad H(P | Q) = 0. \quad (5.2)$$

Thus the chain rule is meaningful as $\infty = \infty + 0$, whereas defining conditional entropy by $H_{\mu}(\mathcal{P} \vee \mathcal{Q}) - H_{\mu}(\mathcal{Q})$ would produce the undefined symbol $\infty - \infty$.

II.1.5.2 E-II.1.14 – Translation stability is not translation continuity

Fix $0 < \varepsilon < 1/2$ on the line and let

$$\mu = \frac{1}{2}(\delta_{-\varepsilon} + \delta_0).$$

At level zero, the two atoms lie in different half-open cells, so $H_0(\mu) = \log 2$. After translation by ε , both lie in $[0, 1)$, so

$$H_0((T_{\varepsilon})_*\mu) = 0. \quad (5.3)$$

Hence the error $\log 2$ in C-II.1.08 is attained in dimension one. Product examples attain $d \log 2$. In particular, fixed-grid entropy need not vary continuously with the translation parameter at measures charging grid boundaries.

II.1.5.3 E-II.1.15 – Occupancy alone gives no entropy lower bound

For every $N \geq 2$, choose a bounded set E meeting N level- n cubes and put a Dirac probability at one point of E . Then

$$N_n^{\mathcal{D}}(E) = N, \quad H_n(\mu) = 0. \quad (5.4)$$

Thus the upper bound in T-II.1.10 has no converse without a quantitative mass-spread hypothesis such as $p_n^*(\mu) \leq C/N$.

II.1.5.4 E-II.1.16 — Uniform cell mass realizes covering entropy

If E meets exactly N level- n cubes and μ assigns mass $1/N$ to each of those cubes, then

$$H_n(\mu) = \log N, \quad N_{n,\text{eff}}(\mu) = N. \quad (5.5)$$

Equality holds throughout the entropy/support-size part of (4.2).

II.1.6 What the chapter provides

The chain rule turns entropy into a multiscale budget, while bounded intersection multiplicity makes that budget stable under changes of grid and bounded perturbations. The covering comparison and finite-set formula then connect the probabilistic quantity to geometric covering and additive combinatorics. Chapter II.2 applies these principles to conditional measures seen under repeated magnification.

Chapter II.2

Components, Rescaling, and Local Entropy Averages

Geometric intuition

Global entropy identities become useful only after they are localized. We introduce components, rescaled components, and local entropy averages, then track the boundary errors that arise when Euclidean balls, dyadic cells, and linear projections are compared. These tools are the common language of the inverse theorems later in the volume.

II.2.1 Components as conditional microscopes

The entropy conventions of Chapter II.1 remain in force. In particular,

$$H_n(\mu) = H_\mu(\mathcal{D}_n), \quad h_n(\mu) = \frac{H_n(\mu)}{n \log 2} \quad (n \geq 1). \quad (0.1)$$

Thus H_n is unnormalized. A component records what the measure looks like after one learns the dyadic cell containing a sampled point; rescaling then enlarges that cell back to the unit cube.

Definition II.2.1.1 (II.2.01: Raw and rescaled dyadic components). Let μ be a Borel probability on $\mathbb{R}^d$, let $n \in \mathbb{Z}$, and let $k \in \mathbb{Z}^d$. For

$$Q = 2^{-n}(k + [0, 1)^d) \in \mathcal{D}_n^d, \quad \mu(Q) > 0, \quad (0.2)$$

put

$$\mu_Q = \frac{\mu|_Q}{\mu(Q)}, \quad T_Q(x) = 2^n x - k, \quad \mu^Q = (T_Q)_* \mu_Q. \quad (0.3)$$

These are respectively the **raw Q -component**, the canonical orientation-preserving rescaling map, and the **rescaled Q -component**. If $\mathcal{D}_n(x)$ has positive mass, write

$$\mu_{x,n} = \mu_{\mathcal{D}_n(x)}, \quad \mu^{x,n} = \mu^{\mathcal{D}_n(x)}. \quad (0.4)$$

At each fixed level the union of the zero-mass cells is null. Consequently (0.4) is simultaneously defined at every nonnegative level on a set of full μ -measure.

Definition II.2.1.2 (II.2.02: Random components and component expectations). A **random level- n component** is obtained by sampling x with law μ and returning $\mu_{x,n}$ or $\mu^{x,n}$. Equivalently, the component at $Q \in \mathcal{D}_n$ is selected with probability $\mu(Q)$. If a scale is also sampled uniformly from a nonempty finite $I \subset \mathbb{Z}$, independently of x , the corresponding expectation is

$$\mathbb{E}_{i \in I, x \sim \mu} F(\mu^{x,i}) = \frac{1}{|I|} \sum_{i \in I} \int F(\mu^{x,i}) d\mu(x), \quad (0.5)$$

whenever the cellwise-constant integrand is nonnegative or integrable.

This mass-weighted sampling convention is precisely what turns the ordinary conditional entropy formula into the geometric identities below.

II.2.2 Components and the global entropy average

Lemma II.2.2.1 (L-II.2.01 — Component disintegration and conditional entropy). *Let μ be a Borel probability on $\mathbb{R}^d$ and let $n \in \mathbb{Z}$. For every function F on the positive-mass rescaled level- n components such that the resulting cellwise-constant function is nonnegative or integrable,*

$$\int F(\mu^{x,n}) d\mu(x) = \sum_{Q \in \mathcal{D}_n: \mu(Q) > 0} \mu(Q) F(\mu^Q). \quad (1.1)$$

For every $m \geq 1$,

$$\int H_m(\mu^{x,n}) d\mu(x) = H_\mu(\mathcal{D}_{n+m} \mid \mathcal{D}_n). \quad (1.2)$$

Moreover,

$$\mu = \sum_{Q \in \mathcal{D}_n} \mu(Q) \mu_Q \quad (1.3)$$

as Borel probability measures, with null summands omitted.

Proof

The function in (1.1) is constant with value $F(\mu^Q)$ on each positive-mass cell Q . Countable additivity, applied separately to positive and negative parts in the integrable case, gives (1.1).

The map T_Q sends the level- $n + m$ subcells of Q bijectively onto the level- m dyadic cells of $[0, 1]^d$. Hence

$$H_m(\mu^Q) = H_{\mu_Q}(\mathcal{D}_{n+m}). \quad (1.4)$$

Insert (1.4) into (1.1) and use the definition of conditional partition entropy from II.1:

$$\sum_{Q \in \mathcal{D}_n} \mu(Q) H_{\mu_Q}(\mathcal{D}_{n+m}) = H_\mu(\mathcal{D}_{n+m} \mid \mathcal{D}_n). \quad (1.5)$$

For a Borel set A , the right side of (1.3), evaluated on A , is

$$\sum_{Q: \mu(Q) > 0} \mu(Q) \frac{\mu(A \cap Q)}{\mu(Q)} = \sum_Q \mu(A \cap Q) = \mu(A). \quad (1.6)$$

This proves all three claims. $\square$

The entropy of a length- m component is the information gained between two levels separated by m . Summing over all starting positions counts each interior one-step increment exactly m times; only the first and last $m - 1$ levels have smaller multiplicity. This is the source of the boundary error in the next theorem.

Theorem II.2.2.2 (T-II.2.02 – Global local entropy averages). *If $\mu([0, 1]^d) = 1$ and $N, m \geq 1$, then*

$$\left| \frac{H_N(\mu)}{N \log 2} - \frac{1}{N} \sum_{i=0}^{N-1} \int \frac{H_m(\mu^{x,i})}{m \log 2} d\mu(x) \right| \leq \frac{d(m-1)}{N}. \quad (1.7)$$

For every fixed m , the liminf and limsup of the averaged expression are therefore $\underline{\dim}_e \mu$ and $\overline{\dim}_e \mu$, respectively.

Proof

Put $H_j = H_j(\mu)$ and $H_0 = 0$. The chain rule in T-II.1.03 and (1.2) give

$$\int H_m(\mu^{x,i}) d\mu(x) = H_{i+m} - H_i. \quad (1.8)$$

Consequently

$$\sum_{i=0}^{N-1} (H_{i+m} - H_i) = \sum_{r=0}^{m-1} H_{N+r} - \sum_{j=0}^{m-1} H_j. \quad (1.9)$$

At one dyadic refinement each atom has at most 2^d children. The entropy increment estimate T-II.1.07 therefore yields, for $v \geq u \geq 0$,

$$0 \leq H_v - H_u \leq d(v - u) \log 2. \quad (1.10)$$

Subtract mH_N from (1.9), divide by m , and use (1.10):

$$\left| \frac{1}{m} \sum_{i=0}^{N-1} (H_{i+m} - H_i) - H_N \right| \leq \frac{1}{m} \sum_{r=0}^{m-1} (H_{N+r} - H_N) + \frac{1}{m} \sum_{j=0}^{m-1} H_j \quad (1.11)$$

$$\leq d(m-1) \log 2.$$

Dividing by $N \log 2$ proves (1.7). For fixed m its right side tends to zero. The definitions of lower and upper entropy dimension then give the last assertion. $\square$

II.2.3 Dyadic cells and Euclidean local dimension

A point can approach dyadic boundaries at infinitely many scales, so there is no uniform deterministic comparison between its dyadic cell and a centered ball. The remedy is measure-theoretic: a cell is bad only when it has exponentially less mass than its bounded family of neighbors, and the total mass of bad cells is summable over the scales.

Definition II.2.3.1 (II.2.05: Neighbor cells and dyadic local dimensions). For $Q \in \mathcal{D}_n^d$, let Q^* be the union of all level- n cubes whose closures meet $\bar{Q}$. There are at most 3^d such cubes, and

$$\bar{B}(x, 2^{-n}) \subset Q^* \quad \text{whenever } x \in Q. \quad (0.11)$$

The lower and upper **dyadic local dimensions** are

$$\underline{d}_{\mu, \mathcal{D}}(x) = \liminf_{n \rightarrow \infty} \frac{-\log \mu(\mathcal{D}_n(x))}{n \log 2}, \quad \bar{d}_{\mu, \mathcal{D}}(x) = \limsup_{n \rightarrow \infty} \frac{-\log \mu(\mathcal{D}_n(x))}{n \log 2}. \quad (0.12)$$

They are not definitionally identified with the Euclidean local dimensions; the equality is the content of [T-II.2.04](#).

Lemma II.2.3.2 (L-II.2.03 — Neighbor-mass Borel–Cantelli estimate). *For every Borel probability μ on $\mathbb{R}^d$ and every $\varepsilon > 0$,*

$$\sum_{n=1}^{\infty} \mu \{x : \mu(\mathcal{D}_n(x)) < e^{-\varepsilon n} \mu(\mathcal{D}_n(x)^*)\} < \infty. \quad (2.1)$$

Proof

At a fixed level n , call a positive-mass cell Q bad when

$$\mu(Q) < e^{-\varepsilon n} \mu(Q^*). \quad (2.2)$$

Every level- n cell occurs in R^* for at most 3^d choices of R . Therefore

$$\begin{aligned}
\mu\{x : \mathcal{D}_n(x) \text{ is bad}\} &= \sum_{Q \text{ bad}} \mu(Q) \\
&\leq e^{-\varepsilon n} \sum_{Q \text{ bad}} \mu(Q^*) \\
&\leq 3^d e^{-\varepsilon n} \sum_{R \in \mathcal{D}_n} \mu(R) = 3^d e^{-\varepsilon n}.
\end{aligned} \tag{2.3}$$

The geometric series is finite, proving (2.1). Borel–Cantelli says that almost every point belongs to only finitely many bad cells. $\square$

Theorem II.2.3.3 (T-II.2.04 – Dyadic cells recover Euclidean local dimensions). *For every Borel probability μ on $\mathbb{R}^d$, for μ -almost every x ,*

$$\underline{d}_{\mu, \mathcal{D}}(x) = \underline{d}_{\mu}(x), \quad \bar{d}_{\mu, \mathcal{D}}(x) = \bar{d}_{\mu}(x). \tag{2.4}$$

Proof

Apply L-II.2.03 simultaneously to every positive rational ε . On the resulting full-measure set, for each $\varepsilon > 0$ and all sufficiently large n , with $Q_n = \mathcal{D}_n(x)$,

$$\mu(Q_n) \geq e^{-\varepsilon n} \mu(Q_n^*). \tag{2.5}$$

The elementary geometric inclusions are

$$Q_n \subset \bar{B}(x, \sqrt{d} 2^{-n}), \quad \bar{B}(x, 2^{-n}) \subset Q_n^*. \tag{2.6}$$

Thus (2.5)–(2.6) imply

$$\frac{-\log \mu(\bar{B}(x, \sqrt{d} 2^{-n}))}{n \log 2} \leq \frac{-\log \mu(Q_n)}{n \log 2} \leq \frac{-\log \mu(\bar{B}(x, 2^{-n}))}{n \log 2} + \frac{\varepsilon}{\log 2}. \tag{2.7}$$

The geometric-scale sampling lemma proved in I.4 says that the liminf and limsup of the ball quotients along 2^{-n} equal the corresponding full limits as $r \downarrow 0$. Replacing 2^{-n} by $\sqrt{d} 2^{-n}$ changes the denominator by the fixed constant $-(1/2) \log d$ and hence changes neither limit. Taking liminf and limsup in (2.7), then letting rational $\varepsilon \downarrow 0$, proves both identities in (2.4). Notice that no distance from x to the dyadic boundary was assumed. $\square$

II.2.4 Martingale control of information

Along the dyadic address of a random point, realized information differs from its conditional mean by martingale noise. We first isolate the elementary probabilistic estimate that will make this noise sublinear.

Definition II.2.4.1 (II.2.06: Martingale differences and maximal partial sums). On a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_n)_{n \geq 0})$, a sequence $(Z_n)_{n \geq 1}$ is a **square-integrable martingale-difference sequence** if

$$Z_n \in L^2(\mathbb{P}), \quad Z_n \text{ is } \mathcal{F}_n\text{-measurable}, \quad \mathbb{E}(Z_n \mid \mathcal{F}_{n-1}) = 0. \quad (0.13)$$

Its partial sums and maximal partial sum are $S_N = \sum_{n=1}^N Z_n$ and $S_N^* = \max_{1 \leq k \leq N} |S_k|$.

Lemma II.2.4.2 (L-II.2.05 — Martingale maximal inequality and strong law). *Let (Z_n) be a real square-integrable martingale-difference sequence and put $S_k = \sum_{n=1}^k Z_n$. For every $N \geq 1$ and $\lambda > 0$,*

$$\mathbb{P} \left(\max_{1 \leq k \leq N} |S_k| \geq \lambda \right) \leq \frac{4 \mathbb{E}|S_N|^2}{\lambda^2}. \quad (3.1)$$

If $\sup_n \mathbb{E}|Z_n|^2 \leq C < \infty$, then

$$\frac{S_N}{N} \longrightarrow 0 \quad \text{almost surely.} \quad (3.2)$$

Proof

If $i < j$, then Z_i is $\mathcal{F}_{j-1}$ -measurable, and therefore

$$\mathbb{E}(Z_i Z_j) = \mathbb{E}\{Z_i \mathbb{E}(Z_j \mid \mathcal{F}_{j-1})\} = 0. \quad (3.3)$$

It follows that (S_k) is a square-integrable martingale and

$$\mathbb{E}|S_N|^2 = \sum_{n=1}^N \mathbb{E}|Z_n|^2. \quad (3.4)$$

We prove the needed maximal estimate rather than cite it. Put $M_N = \max_{k \leq N} |S_k|$. For $\lambda > 0$, let τ be the first $k \leq N$ for which $|S_k| \geq \lambda$, on the event $A_\lambda = \{M_N \geq \lambda\}$. Conditional Jensen and the martingale property give

$$\mathbb{E}(|S_N| \mid \mathcal{F}_k) \geq |\mathbb{E}(S_N \mid \mathcal{F}_k)| = |S_k|. \quad (3.5)$$

Since $\{\tau = k\}$ belongs to $\mathcal{F}_k$, summing (3.5) over k yields

$$\lambda \mathbb{P}(A_\lambda) \leq \mathbb{E}(|S_N| \mathbf{1}_{A_\lambda}). \quad (3.6)$$

The layer-cake identity, Tonelli's theorem, and (3.6) imply

$$\begin{aligned} \mathbb{E}M_N^2 &= \int_0^\infty 2\lambda \mathbb{P}(M_N \geq \lambda) d\lambda \\ &\leq 2 \int_0^\infty \mathbb{E}(|S_N| \mathbf{1}_{\{M_N \geq \lambda\}}) d\lambda \\ &= 2\mathbb{E}(|S_N| M_N) \\ &\leq 2(\mathbb{E}|S_N|^2)^{1/2} (\mathbb{E}M_N^2)^{1/2}. \end{aligned} \quad (3.7)$$

If $\mathbb{E}M_N^2 = 0$ the conclusion is immediate; otherwise divide by its square root and square to obtain

$$\mathbb{E}M_N^2 \leq 4\mathbb{E}|S_N|^2. \quad (3.8)$$

Markov's inequality proves (3.1).

Under the uniform second-moment hypothesis, (3.4) gives $\mathbb{E}|S_N|^2 \leq CN$. At $N = 2^j$, (3.1) gives

$$\mathbb{P}\{M_{2^j} \geq \varepsilon 2^j\} \leq \frac{4C}{\varepsilon^2 2^j}. \quad (3.9)$$

The series over j converges. Borel–Cantelli, first for positive rational ε and then by intersection, gives

$$\frac{M_{2^j}}{2^j} \rightarrow 0 \quad \text{almost surely.} \quad (3.10)$$

If $2^{j-1} < N \leq 2^j$, then

$$\frac{|S_N|}{N} \leq 2 \frac{M_{2^j}}{2^j}. \quad (3.11)$$

Equations (3.10)–(3.11) prove (3.2). $\square$

We now return to dyadic components. At one step the realized information is $-\log p$, where p is the conditional probability of the chosen child. Its conditional mean is the entropy of the child distribution. Finite branching is decisive: even for very small p , the weighted second moment $p \log^2(1/p)$ remains uniformly bounded.

Definition II.2.4.3 (II.2.03: Dyadic filtration and information increments). Assume that $\mu([0, 1]^d) = 1$. Let $\mathcal{F}_n$ be the finite sigma-algebra generated on $[0, 1]^d$ by $\mathcal{D}_n^d$, with $\mathcal{F}_0$ trivial. For $n \geq 1$, the **one-step information increment** and its conditional mean are

$$I_n(x) = -\log \frac{\mu(\mathcal{D}_n(x))}{\mu(\mathcal{D}_{n-1}(x))}, \quad a_n(x) = \mathbb{E}_\mu(I_n \mid \mathcal{F}_{n-1})(x). \quad (0.6)$$

The quotient is positive almost everywhere. Its conditional distribution has at most 2^d outcomes, and direct evaluation on each parent cell gives

$$a_n(x) = H_1(\mu^{x,n-1}). \quad (0.7)$$

Lemma II.2.4.4 (L-II.2.06 – Uniform square integrability of information). Let $p = (p_1, \dots, p_M)$ be a probability vector and let J have law p . Then

$$\mathbb{E} \left(\log^2 \frac{1}{p_J} \right) = \sum_{j: p_j > 0} p_j \log^2 \frac{1}{p_j} \leq \frac{4M}{e^2}. \quad (3.12)$$

Consequently, in the d -dimensional dyadic filtration,

$$\sup_n \mathbb{E}|I_n - a_n|^2 \leq \frac{4 \cdot 2^d}{e^2}. \quad (3.13)$$

Proof

For $0 < t \leq 1$, put $\phi(t) = t \log^2(1/t)$. With $u = \log(1/t) \geq 0$, $\phi(t) = e^{-u} u^2$, whose maximum is $4/e^2$ at $u = 2$. Summing this bound over at most M positive coordinates proves (3.12).

Condition on a level- $n - 1$ dyadic cell. The next digit has a probability vector with at most 2^d entries, and its realized information is $\log(1/p_J)$. Hence (3.12) gives

$$\mathbb{E}(I_n^2 \mid \mathcal{F}_{n-1}) \leq \frac{4 \cdot 2^d}{e^2} \quad \text{almost surely.} \quad (3.14)$$

Because conditional centering subtracts the conditional mean,

$$\mathbb{E}((I_n - a_n)^2 \mid \mathcal{F}_{n-1}) = \mathbb{E}(I_n^2 \mid \mathcal{F}_{n-1}) - a_n^2. \quad (3.15)$$

Integrating (3.15) and using (3.14) proves (3.13). $\square$

Theorem II.2.4.5 (T-II.2.07 – One-step pointwise information law). *If $\mu([0, 1)^d) = 1$, then for μ -almost every x ,*

$$\frac{-\log \mu(\mathcal{D}_N(x))}{N \log 2} - \frac{1}{N} \sum_{i=0}^{N-1} \frac{H_1(\mu^{x,i})}{\log 2} \longrightarrow 0. \quad (3.16)$$

Proof

Put $Z_n = I_n - a_n$. Equations (0.6)–(0.7) show that (Z_n) is a martingale-difference sequence, and L-II.2.06 gives its uniform square-integrability. L-II.2.05 therefore yields

$$\frac{1}{N} \sum_{n=1}^N (I_n - a_n) \longrightarrow 0 \quad \text{almost surely.} \quad (3.17)$$

The information increments telescope:

$$\sum_{n=1}^N I_n(x) = -\log \mu(\mathcal{D}_N(x)), \quad (3.18)$$

because $\mu(\mathcal{D}_0(x)) = 1$. Substitute $a_n(x) = H_1(\mu^{x,n-1})$ in (3.17), divide by $\log 2$, and reindex. This is (3.16). $\square$

Definition II.2.4.6 (II.2.04: Block information and local entropy averages). For $i \geq 0$ and $m \geq 1$, the **length- m block information starting at level i** and its conditional mean are

$$I_{i,m}(x) = -\log \frac{\mu(\mathcal{D}_{i+m}(x))}{\mu(\mathcal{D}_i(x))} = \sum_{j=i+1}^{i+m} I_j(x), \quad (0.8)$$

$$A_{i,m}(x) = \mathbb{E}_\mu(I_{i,m} \mid \mathcal{F}_i)(x) = H_m(\mu^{x,i}). \quad (0.9)$$

The **normalized empirical local entropy average** is

$$\mathcal{A}_{N,m}(\mu, x) = \frac{1}{N} \sum_{i=0}^{N-1} \frac{H_m(\mu^{x,i})}{m \log 2}. \quad (0.10)$$

Unless a theorem states otherwise, m is fixed before N tends to infinity.

Length- m blocks beginning at consecutive levels overlap, so their centered errors are not a martingale-difference sequence in the naive order. Splitting the starting levels into residue classes modulo m makes the blocks in each subsequence disjoint. After the noise is controlled on each class, an exact multiplicity count compares the overlapping blocks with m times the total one-step information.

Theorem II.2.4.7 (T-II.2.08 — Fixed-block pointwise local entropy averages). *If $\mu([0, 1]^d) = 1$, then for every fixed $m \geq 1$ and for μ -almost every x ,*

$$\frac{-\log \mu(\mathcal{D}_N(x))}{N \log 2} - \mathcal{A}_{N,m}(\mu, x) \longrightarrow 0. \quad (3.19)$$

There is one full-measure set on which (3.19) holds for every fixed positive integer m .

Proof

Fix m and define

$$Z_i^{(m)} = I_{i,m} - A_{i,m} \quad (i \geq 0). \quad (3.20)$$

For a residue $r \in \{0, \dots, m-1\}$, the variables

$$Z_r^{(m)}, Z_{r+m}^{(m)}, Z_{r+2m}^{(m)}, \dots \quad (3.21)$$

are martingale differences relative to $\mathcal{F}_r, \mathcal{F}_{r+m}, \mathcal{F}_{r+2m}, \dots$: the variable beginning at i is $\mathcal{F}_{i+m}$ -measurable and has conditional mean zero given $\mathcal{F}_i$. Conditionally on $\mathcal{F}_i$, the block information has a law with at most 2^{dm} outcomes. The scalar estimate (3.12), now with $M = 2^{dm}$, gives a second-moment bound depending only on d, m , uniformly in i . Applying L-II.2.05 to each of the finitely many residue classes and recombining them gives

$$\frac{1}{N} \sum_{i=0}^{N-1} (I_{i,m} - A_{i,m}) \longrightarrow 0 \quad \text{almost surely.} \quad (3.22)$$

It remains to compare overlapping realized blocks with the total information. Equation (3.14) and Markov's inequality imply, for every $\varepsilon > 0$,

$$\sum_{j=1}^{\infty} \mu\{I_j > \varepsilon j\} \leq \frac{4 \cdot 2^d}{e^2 \varepsilon^2} \sum_{j=1}^{\infty} \frac{1}{j^2} < \infty. \quad (3.23)$$

Hence

$$\frac{I_j(x)}{j} \longrightarrow 0 \quad \text{for almost every } x. \quad (3.24)$$

For $N \geq m$, counting how often each I_j occurs in the overlapping blocks gives the exact identity

$$\begin{aligned} \sum_{i=0}^{N-1} I_{i,m} - m \sum_{j=1}^N I_j &= - \sum_{j=1}^{m-1} (m-j) I_j \\ &\quad + \sum_{r=1}^{m-1} (m-r) I_{N+r}. \end{aligned} \quad (3.25)$$

After division by Nm , the first sum tends to zero because it is fixed, and the second tends to zero by (3.24). Therefore

$$\frac{1}{Nm} \sum_{i=0}^{N-1} I_{i,m} - \frac{1}{N} \sum_{j=1}^N I_j \longrightarrow 0. \quad (3.26)$$

Combine (3.22), (3.26), (0.9), and the telescoping identity (3.18), then divide by $\log 2$. This proves (3.19) for the fixed m . Intersecting the full-measure sets over the countable set of positive integers proves the simultaneous assertion. $\square$

Corollary II.2.4.8 (C-II.2.09 – Euclidean dimension recovery). *Let $d \geq 1$, let $\mu([0, 1]^d) = 1$, and fix $m \geq 1$. For μ -almost every x ,*

$$\underline{d}_\mu(x) = \liminf_{N \rightarrow \infty} \mathcal{A}_{N,m}(\mu, x), \quad \bar{d}_\mu(x) = \limsup_{N \rightarrow \infty} \mathcal{A}_{N,m}(\mu, x). \quad (3.27)$$

Consequently:

- (1) if $\liminf_N \mathcal{A}_{N,m}(\mu, x) \geq s$ almost everywhere, then $\underline{\dim}_H \mu \geq s$;
- (2) if $\limsup_N \mathcal{A}_{N,m}(\mu, x) \leq t$ almost everywhere, then $\underline{\dim}_P \mu \leq t$; and
- (3) if $\mathcal{A}_{N,m}(\mu, x) \rightarrow \alpha$ almost everywhere, then μ is exact dimensional of dimension α .

Proof

T-II.2.08 identifies both limits in (3.27) with the corresponding dyadic local dimensions. T-II.2.04 identifies those dyadic dimensions with the Euclidean local dimensions, proving (3.27).

Under the first hypothesis, $\underline{d}_\mu(x) \geq s$ almost everywhere. The Hausdorff local-to-global dictionary T-I.4.04 gives

$$\underline{\dim}_H \mu = \operatorname{ess\,inf}_\mu \underline{d}_\mu \geq s. \quad (3.28)$$

Under the second hypothesis, $\bar{d}_\mu(x) \leq t$ almost everywhere. The packing local-to-global dictionary T-I.4.07 gives

$$\overline{\dim}_p \mu = \operatorname{ess\,sup}_\mu \bar{d}_\mu \leq t. \quad (3.29)$$

Under the third hypothesis, (3.27) says that both local dimensions equal α almost everywhere, which is exactly D-I.4.04. $\square$

II.2.5 Finite-alphabet trees

Dyadic addresses form a finite-branching tree, and in the associated ultrametric a cylinder is literally a closed ball. This removes Euclidean boundary effects and lets local entropy be converted directly into mass decay.

Definition II.2.5.1 (II.2.07: Finite-alphabet metric trees). Let Λ be a nonempty finite alphabet. A **finite-alphabet tree** is a nonempty closed set $X \subset \Lambda^\mathbb{N}$. For a word $u \in \Lambda^n$, its **cylinder** is

$$[u]_X = \{x \in X : x_1 \cdots x_n = u\}; \quad (0.14)$$

empty cylinders are discarded. Given $0 < \rho < 1$, the associated ρ -**tree metric** is

$$d_\rho(x, y) = \begin{cases} 0, & x = y, \\ \rho^{\min\{j \geq 0 : x_{j+1} \neq y_{j+1}\}}, & x \neq y. \end{cases} \quad (0.15)$$

In particular,

$$\bar{B}_{d_\rho}(x, \rho^n) = [x_1 \cdots x_n]_X. \quad (0.16)$$

For a probability σ on X and a positive-mass cylinder $[u]_X$, its **tree component** is $\sigma_{[u]} = \sigma|_{[u]_X} / \sigma([u]_X)$. Its lower and upper **tree local dimensions** are

$$\underline{d}_\sigma(x) = \liminf_{r \downarrow 0} \frac{\log \sigma(\bar{B}_{d_\rho}(x, r))}{\log r}, \quad \bar{d}_\sigma(x) = \limsup_{r \downarrow 0} \frac{\log \sigma(\bar{B}_{d_\rho}(x, r))}{\log r}.$$

The lower Hausdorff dimension of σ is

$$\underline{\dim}_H \sigma = \inf \{ \dim_H^{d_\rho} A : A \subset X \text{ Borel, } \sigma(A) > 0 \}. \quad (0.17)$$

The probability σ is **tree exact dimensional of dimension α** when both tree local dimensions equal α at σ -almost every point.

Theorem II.2.5.2 (T-II.2.10 — Tree mass decay from local entropy). *Let $X \subset \Lambda^\mathbb{N}$ be a finite-alphabet ρ -tree and let μ be a Borel probability on X . For a positive-mass prefix $u = x_1 \cdots x_{n-1}$, put*

$$p_u(a) = \frac{\mu([ua]_X)}{\mu([u]_X)}, \quad h_n(x) = - \sum_{a: p_u(a) > 0} p_u(a) \log p_u(a). \quad (4.1)$$

Then for μ -almost every x ,

$$-\frac{1}{N} \log \mu([x_1 \cdots x_N]_X) - \frac{1}{N} \sum_{n=1}^N h_n(x) \longrightarrow 0. \quad (4.2)$$

If

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N h_n(x) \geq \alpha \quad \text{for } \mu\text{-almost every } x, \quad (4.3)$$

then

$$\underline{\dim}_H \mu \geq \frac{\alpha}{\log(1/\rho)}. \quad (4.4)$$

Proof

Let $\mathcal{G}_n$ be the sigma-algebra generated by level- n cylinders and define

$$J_n(x) = -\log p_{x_1 \cdots x_{n-1}}(x_n). \quad (4.5)$$

Conditionally on $\mathcal{G}_{n-1}$, this is the information of a probability vector with at most $|\Lambda|$ entries, and

$$\mathbb{E}(J_n \mid \mathcal{G}_{n-1}) = h_n. \quad (4.6)$$

[L-II.2.06](#), with $M = |\Lambda|$, uniformly bounds the second moments of the centered variables $J_n - h_n$. They are martingale differences, so [L-II.2.05](#) gives

$$\frac{1}{N} \sum_{n=1}^N (J_n - h_n) \longrightarrow 0 \quad \text{almost surely.} \quad (4.7)$$

The product of the conditional probabilities in (4.5) is the cylinder mass:

$$\sum_{n=1}^N J_n(x) = -\log \mu([x_1 \cdots x_N]_X). \quad (4.8)$$

Equations (4.7)–(4.8) prove (4.2).

We record the metric-space step in full. Suppose a probability σ on a finite-alphabet ρ -tree satisfies

$$\underline{d}_\sigma(x) \geq s \quad \text{for } \sigma\text{-almost every } x. \quad (4.9)$$

Fix $0 < q < s$ and a Borel set A with $\sigma(A) > 0$. Since tree balls at radii ρ^n are cylinders, the geometric sampling argument gives, for almost every x ,

$$\sigma([x_1 \cdots x_n]) \leq \rho^{nq} \quad \text{for every sufficiently large } n. \quad (4.10)$$

Let G_N be the set on which (4.10) holds for all $n \geq N$. The union of the sets G_N has full measure, so $B = A \cap G_N$ has positive measure for some N . Normalize the restriction of σ to B and call it τ .

If $U \subset X$ has sufficiently small positive diameter, choose $n \geq N$ so that

$$\rho^{n+1} < \text{diam } U \leq \rho^n. \quad (4.11)$$

Any two points of U share their first n symbols, so U lies in one level- n cylinder. If this cylinder meets B , choose x in the intersection and use (4.10); if it does not, $\tau^*(U) = 0$, where τ^* is the outer measure induced by τ . The containing cylinder is Borel, so in either case

$$\tau^*(U) \leq \frac{\rho^{nq}}{\sigma(B)} \leq \frac{\rho^{-q}}{\sigma(B)} (\text{diam } U)^q. \quad (4.12)$$

For every sufficiently fine countable cover (U_j) of B , countable subadditivity and (4.12) give the following inequality. A zero-diameter member is a singleton; (4.10) makes its τ -mass zero, so it contributes nothing. We use outer measure because cover members need not be Borel.

$$1 = \tau(B) \leq \sum_j \tau^*(U_j) \leq \frac{\rho^{-q}}{\sigma(B)} \sum_j (\text{diam } U_j)^q. \quad (4.13)$$

Hence $\mathcal{H}^q(B) > 0$, so $\dim_H A \geq \dim_H B \geq q$. Letting $q \uparrow s$ shows that every positive- σ Borel set has Hausdorff dimension at least s ; therefore

$$\underline{\dim}_H \sigma \geq s. \quad (4.14)$$

The case $s = 0$ is automatic, and endpoint values follow by approximation from below. Return to μ . Equations (4.2)–(4.3) and the cylinder-ball identity (0.16) give

$$\underline{d}_\mu(x) \geq \frac{\alpha}{\log(1/\rho)} \quad \text{almost everywhere.} \quad (4.15)$$

Apply (4.14) to obtain (4.4). $\square$

A tree morphism may collapse several source children into the same output child. The relevant local quantity is therefore the entropy of the induced output-symbol law, rather than the entropy of the source children themselves.

Definition II.2.5.3 (II.2.08: Tree morphisms and conditional output entropy). Let $X \subset \Lambda_X^{\mathbb{N}}$ and $Y \subset \Lambda_Y^{\mathbb{N}}$ be finite-alphabet trees. A Borel map $f : X \rightarrow Y$ is a **tree morphism** if every level- n source cylinder is mapped into a level- n target cylinder. The target cylinder is unique. If $[u]_X$ maps into $[v]_Y$, define the **output fiber**

$$F_{u,b} = \{a \in \Lambda_X : [ua]_X \neq \emptyset, f([ua]_X) \subset [vb]_Y\}. \quad (0.18)$$

For $\mu([u]_X) > 0$, its conditional output law and **conditional output entropy** are

$$q_u(b) = \sum_{a \in F_{u,b}} \frac{\mu([ua]_X)}{\mu([u]_X)}, \quad e_u^f(\mu) = - \sum_b q_u(b) \log q_u(b). \quad (0.19)$$

For $x \in [u]_X$, $|u| = n$, write $e_n^f(\mu, x) = e_u^f(\mu)$.

The proof uses random pruning. At each node, source children are grouped by their output symbol; one child from each group is retained at random, in proportion to its original mass, and the whole group mass is transferred to it. The expected pruned measure remains the original measure, while the map becomes an isometry on every pruned support and the retained transition entropies are precisely the output entropies above.

Theorem II.2.5.4 (T-II.2.11 – Tree-morphism local entropy theorem). *Let X, Y be finite-alphabet ρ -trees, let $f : X \rightarrow Y$ be a tree morphism, and let μ be a Borel probability on X . If $\alpha \geq 0$ and*

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} e_n^f(\mu, x) \geq \alpha \quad \text{for } \mu\text{-almost every } x, \quad (4.16)$$

then

$$\underline{\dim}_H(f_*\mu) \geq \frac{\alpha}{\log(1/\rho)}. \quad (4.17)$$

Proof

We give the complete random fiber-pruning argument. For every source node u of positive μ -mass, set

$$p_u(a) = \frac{\mu([ua]_X)}{\mu([u]_X)}, \quad q_u(b) = \sum_{a \in F_{u,b}} p_u(a). \quad (4.18)$$

For each pair (u, b) with $q_u(b) > 0$, choose one symbol $A_{u,b} \in F_{u,b}$ randomly according to

$$\mathbb{P}\{A_{u,b} = a\} = \frac{p_u(a)}{q_u(b)}. \quad (4.19)$$

Make all these countably many choices independent. Define random transition probabilities at u by

$$r_u(a) = \begin{cases} q_u(b), & a = A_{u,b} \text{ for some } b, \\ 0, & \text{otherwise.} \end{cases} \quad (4.20)$$

The fibers in (0.18) partition the valid children, and hence $\sum_a r_u(a) = \sum_b q_u(b) = 1$. Starting with $v([\emptyset]) = 1$, define recursively

$$v([ua]_X) = v([u]_X) r_u(a). \quad (4.21)$$

The consistency identity $\sum_a \nu([ua]) = \nu([u])$ defines a probability premeasure on the algebra of finite unions of cylinders. If a countable disjoint union in this algebra again belongs to the algebra, compactness of the clopen cylinders reduces the union to a finite one; hence the premeasure is countably additive. The A-RA measure-extension theorem gives a unique Borel probability ν on the compact tree X . Nodes of zero μ -mass may be assigned arbitrary transitions; the induction below shows that they have zero ν -mass almost surely.

The choice made at u is independent of the choices at its strict ancestors, which determine $\nu([u])$. Moreover, if $a \in F_{u,b}$, then

$$\mathbb{E}r_u(a) = q_u(b) \frac{p_u(a)}{q_u(b)} = p_u(a), \quad (4.22)$$

with both sides zero when $p_u(a) = 0$. Induction on word length using (4.21)–(4.22) proves

$$\mathbb{E}\nu([u]_X) = \mu([u]_X) \quad \text{for every cylinder.} \quad (4.23)$$

For completeness, let C be the class of Borel sets A for which $\omega \mapsto \nu_\omega(A)$ is measurable. It contains the cylinder algebra; it is closed under complements in X and under countable disjoint unions by pointwise limits. The pi-lambda theorem therefore gives $C = B(X)$. Tonelli's theorem then shows that

$$A \longmapsto \mathbb{E}\nu(A)$$

is a finite Borel measure. The cylinders are a generating pi-system, so a second application of the pi-lambda theorem extends (4.23) to

$$\mathbb{E}\nu(A) = \mu(A) \quad \text{for every Borel } A \subset X. \quad (4.24)$$

We next identify the geometry and entropy of each realization. At a retained node u , (4.20) retains at most one child from each output fiber. If two points $x, y \in \text{supp}(\nu)$ first differ immediately after u , their two children must therefore belong to distinct output fibers. The morphism property says that $f(x), f(y)$ share the target prefix corresponding to u and then differ in the next target symbol. Hence

$$d_\rho(f(x), f(y)) = d_\rho(x, y) \quad (x, y \in \text{supp } \nu). \quad (4.25)$$

Thus f is an isometry on $\text{supp}(\nu)$. Also, the nonzero transition masses of ν at a retained node u are exactly $(q_u(b))_b$. Their Shannon entropy is therefore

$$h_{|u|+1}^\nu(x) = e_u^f(\mu) \quad \text{for every } x \in [u]_X \cap \text{supp } \nu. \quad (4.26)$$

Let G be the Borel set of points satisfying (4.16). It has full μ -measure. Equation (4.24) gives

$$\mathbb{E}\nu(G) = \mu(G) = 1. \quad (4.27)$$

Since $0 \leq \nu(G) \leq 1$, it follows that $\nu(G) = 1$ almost surely. For every such realization, (4.26), (4.16), and T-II.2.10 give

$$\underline{\dim}_H \nu \geq \frac{\alpha}{\log(1/\rho)}. \quad (4.28)$$

The isometry (4.25) preserves Hausdorff dimension on $\text{supp}(\nu)$ and is injective there. Taking positive-mass Borel subsets in the definition (0.17) therefore gives

$$\underline{\dim}_H(f_*\nu) = \underline{\dim}_H \nu \geq \frac{\alpha}{\log(1/\rho)} \quad \text{almost surely.} \quad (4.29)$$

It remains to pass from the random pruned measures to the original image. Pushing (4.24) forward gives

$$(f_*\mu)(A) = E(f_*\nu)(A) \quad \text{for every Borel } A \subset Y. \quad (4.30)$$

We use the following elementary mixture fact. If a random probability θ satisfies $\underline{\dim}_H \theta \geq s$ almost surely and $\eta(A) = E\theta(A)$, then $\underline{\dim}_H \eta \geq s$. Indeed, if $\eta(A) > 0$, then $E\theta(A) > 0$, so $\theta(A) > 0$ for a positive-probability set of realizations. At least one of these is also a realization on which the dimension bound holds, and consequently $\underline{\dim}_H A \geq s$. Since this is true for every positive- η Borel set, the claim follows.

Apply the mixture fact to $\theta = f_*\nu$, $\eta = f_*\mu$, and $s = \alpha / \log(1/\rho)$. Equations (4.29)–(4.30) prove (4.17). $\square$

II.2.6 Dyadic-level-preserving Euclidean images

The symbolic theorem returns to Euclidean space through canonical dyadic addresses. No inverse-Lipschitz property of the coding is needed: a Euclidean set of diameter comparable to 2^{-n} meets only boundedly many level- n dyadic cubes, which is enough to compare Hausdorff dimensions.

Definition II.2.6.1 (Dyadic-level-preserving maps). A Borel map $g : [0, 1]^d \rightarrow [0, 1]^k$ is **dyadic-level preserving on E** if, for every $n \geq 0$ and every $Q \in \mathcal{D}_n^d$, there is $R \in \mathcal{D}_n^k$ such that

$$g(E \cap Q) \subset R. \quad (0.20)$$

No continuity, linearity, or differentiability is included in this term.

Corollary II.2.6.2 (C-II.2.12 — Euclidean image theorem). Let $\mu([0, 1]^d) = 1$, let $E \subset [0, 1]^d$ be Borel with $\mu(E) = 1$, and let $g : [0, 1]^d \rightarrow [0, 1]^k$ be Borel and dyadic-level preserving on E . For a positive-mass $Q = \mathcal{D}_n(x)$, set

$$e_n^g(\mu, x) = H_{g_*\mu_{x,n}}(\mathcal{D}_{n+1}^k). \quad (5.1)$$

The measure in (5.1) is carried by one level- n target cube, so $0 \leq e_n^g(\mu, x) \leq k \log 2$. If $s \geq 0$ and

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} e_n^g(\mu, x) \geq s \log 2 \quad \text{for } \mu\text{-almost every } x, \quad (5.2)$$

then

$$\underline{\dim}_H(g_*\mu) \geq s. \quad (5.3)$$

Proof

Let $\Lambda_d = \{0, 1\}^d$. Every $x \in [0, 1)^d$ has a unique **canonical dyadic address**

$$c_d(x) = (a_1, a_2, \dots) \in \Lambda_d^{\mathbb{N}} \quad (5.4)$$

defined by requiring x to lie in the half-open dyadic cube indexed by $a_1 \dots a_n$ for every n . The half-open convention chooses the terminating binary address at dyadic rationals. The coding map is Borel and injective, and its inverse on its image is

$$\pi_d((a_j)_{j \geq 1}) = \sum_{j=1}^{\infty} 2^{-j} a_j, \quad (5.5)$$

with the sum taken coordinatewise. The canonical image is Borel: it is the set of sequences for which no coordinate is eventually constantly 1.

We first prove the exact dimension comparison that will be used at the end: for every $A \subset [0, 1)^k$,

$$\dim_H^{| \cdot |} A = \dim_H^{d_{1/2}} c_k(A). \quad (5.6)$$

Indeed, if two addresses share n symbols, their Euclidean images are at distance at most $\sqrt{k}2^{-n}$. Thus π_k is $\sqrt{k}$ -Lipschitz and

$$\dim_H A \leq \dim_H^{d_{1/2}} c_k(A). \quad (5.7)$$

For the reverse inequality, let $U \subset [0, 1)^k$ have positive diameter r and choose n with

$$2^{-(n+1)} < r \leq 2^{-n}. \quad (5.8)$$

The projection of U on each coordinate axis lies in an interval of length at most r , so U meets at most 2^k level- n dyadic cubes; the weaker bound 3^k also suffices. Their address cylinders have tree diameter at most

$$2^{-n} < 2r. \quad (5.9)$$

Applying this replacement to every member of a sufficiently fine Hausdorff cover shows, for every $q \geq 0$,

$$\mathcal{H}_{d_{1/2}}^q(c_k(A)) \leq 3^k 2^q \mathcal{H}_{| \cdot |}^q(A) \quad (5.10)$$

at the level of limiting outer measures. Zero-diameter cover elements can be replaced by arbitrarily small balls. Equations (5.7)–(5.10) prove (5.6).

Now set

$$\tilde{\mu} = (c_d)_* \mu, \quad X = \text{supp } \tilde{\mu} \subset \Lambda_d^{\mathbb{N}}, \quad Y = \Lambda_k^{\mathbb{N}}, \quad (5.11)$$

both with the $1/2$ -tree metric. If a word u occurs in X , then its cylinder has positive $\tilde{\mu}$ -mass, so the corresponding half-open source cube Q_u has positive μ -mass. Since E is a full-measure carrier, $E \cap Q_u$ is nonempty. The hypothesis (0.20) places $g(E \cap Q_u)$ in a unique target cube $R_{v(u)}$ of level $|u|$.

If u' extends u , then $E \cap Q_{u'}$ is nonempty and its image lies in both $R_{v(u')}$ and $R_{v(u)}$. Hence $v(u')$ extends $v(u)$. For $\omega \in X$, the nested words

$$v(\omega|_1), v(\omega|_2), \dots \quad (5.12)$$

therefore define a unique $F(\omega) \in Y$. The map $F : X \rightarrow Y$ is a tree morphism. For μ -almost every $x \in E$, all source prefixes occur in X , and the construction gives

$$F(c_d(x)) = c_k(g(x)). \quad (5.13)$$

Consequently

$$F_* \tilde{\mu} = (c_k)_* (g_* \mu). \quad (5.14)$$

Let u have length n , let Q_u be its source cube, and let $R_{v(u)b}$ range over the children of its target cube. The output transition probability of F is

$$\begin{aligned} q_u(b) &= \frac{\mu\{x \in Q_u : c_k(g(x)) \in [v(u)b]\}}{\mu(Q_u)} \\ &= (g_* \mu_{Q_u})(R_{v(u)b}). \end{aligned} \quad (5.15)$$

Null subsets outside E do not affect either side. Since $g_* \mu_{Q_u}$ is carried by $R_{v(u)}$, (5.15) gives the exact entropy identity

$$e_n^F(\tilde{\mu}, c_d(x)) = H_{g_* \mu_{x,n}}(\mathcal{D}_{n+1}^k) = e_n^g(\mu, x) \quad (5.16)$$

for almost every $x \in Q_u$. Thus (5.2), (5.16), and T-II.2.11 with $\rho = 1/2$ imply

$$\underline{\dim}_{\mathbb{H}}^{d_{1/2}}(F_* \tilde{\mu}) \geq s. \quad (5.17)$$

Let $\eta = g_* \mu$ and let $A \subset [0, 1)^k$ be Borel with $\eta(A) > 0$. The canonical coding is a Borel isomorphism onto its Borel image, so $c_k(A)$ is Borel. Equations (5.14) and (5.17) show that

$$\underline{\dim}_{\mathbb{H}}^{d_{1/2}} c_k(A) \geq s. \quad (5.18)$$

By (5.6), $\dim_{\mathbb{H}} A \geq s$. This holds for every positive- η Borel set, which is precisely (5.3).

□

II.2.7 Where the method stops

The Euclidean image theorem applies only when the map respects every dyadic level. An arbitrary orthogonal projection can send one dyadic cube across several target-grid boundaries, and a nonlinear map introduces an additional scale-dependent linearization error. Later chapters develop the lifting and bounded-error arguments needed for those settings. The present chapter has isolated the part that is purely entropic: components, martingale control, tree mass decay, and the passage from symbolic to Euclidean dimension.

Chapter II.3

Entropy Growth Under Convolution

Geometric intuition

Convolution cannot usually preserve a genuinely nonuniform measure at every scale. The chapter turns that heuristic into quantitative entropy growth, separating the elementary identities from the structural alternatives that will drive the line and higher-dimensional inverse theorems.

All logarithms are natural, and the entropy and component conventions of Chapters II.1–II.2 remain in force. In particular,

$$H_n(\mu) = H_\mu(\mathcal{D}_n), \quad h_n(\mu) = \frac{H_n(\mu)}{n \log 2} \quad (n \geq 1). \quad (0.1)$$

The central question is simple to state: when can addition fail to create entropy? We begin with the algebra of convolution and the one unavoidable dyadic complication, namely binary carries. We then pass from global entropy growth to the growth of typical component pairs. That localization is the input for the inverse theorem of Chapter II.4.

II.3.1 Convolution as addition

If independent variables are observed separately, their joint entropy is the sum of their entropies. Observing only their sum forgets which pair produced it, so different pairs may collide. Nevertheless, in a discrete group the entropy of the sum cannot fall below that of either summand: revealing one summand recovers a translate of the other. At dyadic resolution there is one additional issue. The cell indices of two points add only up to one binary carry in each coordinate, accounting for the $d \log 2$ errors below.

Definition II.3.1.1 (II.3.01: Convolution and convolution powers). For Borel probabilities μ, ν on $\mathbb{R}^d$, define

$$\mu * \nu = A_*(\mu \times \nu), \quad A(x, y) = x + y. \quad (0.2)$$

Thus $\mu * \nu$ is the law of $X + Y$ when $X \sim \mu$, $Y \sim \nu$ are independent. For $z \in \mathbb{R}^d$, $\tau_z(x) = x + z$, so $\delta_z * \mu = (\tau_z)_*\mu$. Put $\mu^{*1} = \mu$ and $\mu^{*(k+1)} = \mu^{*k} * \mu$.

II.3.2 Algebra and entropy at one scale

The first lemma records the algebraic identities that allow us to pass freely between a convolution and its representation as a mixture of translates.

Lemma II.3.2.1 (L-II.3.01 — Convolution algebra and mixture identities). *For Borel probabilities on $\mathbb{R}^d$, convolution is commutative and associative, δ_0 is its identity, and*

$$\mu * \nu = \int (\tau_y)_*\mu d\nu(y) = \int (\tau_x)_*\nu d\mu(x) \quad (1.1)$$

as measure identities. It is the law of the sum of independent variables with laws μ, ν .

Proof

For every Borel $E \subset \mathbb{R}^d$, the definition (0.2) and Tonelli's theorem give

$$\begin{aligned} (\mu * \nu)(E) &= \iint \mathbf{1}_E(x + y) d\mu(x) d\nu(y) \\ &= \int (\tau_y)_*\mu(E) d\nu(y). \end{aligned} \quad (1.2)$$

Interchanging x, y gives the second identity in (1.1) and commutativity. For three probabilities, Tonelli and $(x + y) + z = x + (y + z)$ show that both parenthesizations assign to E the integral

$$\iiint \mathbf{1}_E(x + y + z) d\mu(x) d\nu(y) d\lambda(z), \quad (1.3)$$

so convolution is associative. Taking one variable identically zero proves the identity law. Equation (1.2) is also exactly the distribution formula for an independent sum. $\square$

We now encode dyadic cells by lattice points. The following definition makes the carry term precise.

Definition II.3.2.2 (II.3.02: Dyadic rounding and carry). For $n \in \mathbb{Z}$ and $x \in \mathbb{R}^d$, define coordinatewise

$$q_n(x) = \lfloor 2^n x \rfloor \in \mathbb{Z}^d, \quad r_n(x) = 2^n x - q_n(x) \in [0, 1)^d. \quad (0.3)$$

The rounded law is $[\mu]_n = (q_n)_*\mu$, a probability on $\mathbb{Z}^d$, and

$$H([\mu]_n) = H_n(\mu). \quad (0.4)$$

For $x, y \in \mathbb{R}^d$, the *carry vector* is

$$c_n(x, y) = \lfloor r_n(x) + r_n(y) \rfloor \in \{0, 1\}^d, \quad (0.5)$$

and direct substitution gives

$$q_n(x + y) = q_n(x) + q_n(y) + c_n(x, y). \quad (0.6)$$

Lemma II.3.2.3 (L-II.3.02 — Shannon entropy of independent sums). *Let X, Y be independent random variables in a countable abelian group, with finite entropies. Then*

$$\max\{H(X), H(Y)\} \leq H(X + Y) \leq H(X) + H(Y). \quad (1.4)$$

Proof

Conditioning reduces entropy by T-II.1.04. Given $Y = y$, the map $X \mapsto X + y$ is a bijection, so

$$H(X + Y) \geq H(X + Y \mid Y) = H(X). \quad (1.5)$$

Interchanging the variables gives the other lower bound. The sum is a function of the pair, hence its point partition is coarser than the pair partition. Refinement monotonicity and independence give

$$H(X + Y) \leq H(X, Y) = H(X) + H(Y). \quad (1.6)$$

This proves (1.4). $\square$

For later localization it is useful to absorb the maximal dyadic carry loss into the definition of local growth. This makes the corrected quantity nonnegative and will allow a direct application of Markov's inequality.

Definition II.3.2.4 (II.3.03: Convolution entropy growth). Whenever the displayed entropies are finite, the *level- n convolution entropy growth* is

$$\Delta_n(\mu; \nu) = h_n(\mu * \nu) - h_n(\mu). \quad (0.7)$$

For probabilities σ, τ on $[0, 1]^d$ and $m \geq 1$, their *carry-corrected local growth* is

$$\gamma_m(\sigma; \tau) = h_m(\sigma * \tau) - h_m(\sigma) + \frac{d}{m}. \quad (0.8)$$

The correction is part of the definition. Its nonnegativity is the final assertion of the next theorem.

Theorem II.3.2.5 (T-II.3.03 — Dyadic convolution bounds with carry). *Let μ, ν be Borel probabilities on $\mathbb{R}^d$ with finite level- n entropy. Then*

$$\max\{H_n(\mu), H_n(\nu)\} - d \log 2 \leq H_n(\mu * \nu) \leq H_n(\mu) + H_n(\nu) + d \log 2. \quad (1.7)$$

For $n \geq 1$, division by $n \log 2$ gives

$$\max\{h_n(\mu), h_n(\nu)\} - \frac{d}{n} \leq h_n(\mu * \nu) \leq h_n(\mu) + h_n(\nu) + \frac{d}{n}. \quad (1.8)$$

In particular, $\gamma_m(\sigma; \tau) \geq 0$ for unit-cube probabilities.

Proof

Let $X \sim \mu, Y \sim \nu$ be independent and put

$$U = q_n(X), \quad V = q_n(Y), \quad K = c_n(X, Y), \quad W = q_n(X + Y). \quad (1.9)$$

By (0.6), $W = U + V + K$. The variable W is a function of (U, V, K) , while K has at most 2^d values. The chain rule and independence of U, V give

$$\begin{aligned} H_n(\mu * \nu) &= H(W) \leq H(U, V, K) \\ &= H(U, V) + H(K | U, V) \\ &\leq H_n(\mu) + H_n(\nu) + d \log 2. \end{aligned} \quad (1.10)$$

For the lower bound, use the first mixture representation in (1.1). If $p_D(y) = (\tau_y)_* \mu(D)$ for $D \in \mathcal{D}_n$, then

$$(\mu * \nu)(D) = \int p_D(y) d\nu(y). \quad (1.11)$$

The function $\eta(t) = -t \log t$, with $\eta(0) = 0$, is concave. Jensen's inequality in (1.11), followed by Tonelli over the nonnegative summands, gives

$$H_n(\mu * \nu) \geq \int H_n((\tau_y)_* \mu) d\nu(y). \quad (1.12)$$

The translated-grid estimate C-II.1.08 says

$$H_n((\tau_y)_* \mu) \geq H_n(\mu) - d \log 2 \quad (1.13)$$

for every y . Equations (1.12)–(1.13) prove the lower bound based on μ ; commutativity gives the one based on ν . Formula (0.8) and the first inequality in (1.8) now give $\gamma_m \geq 0$. $\square$

II.3.3 Saturation, localization, and convolution powers

Two elementary mechanisms can make convolution growth small. A measure that is already close to maximal entropy has little room to gain, while a factor supported in one tiny neighborhood acts almost as a translation. These are the model cases for the uniform and atomic alternatives in Chapter II.4.

Definition II.3.3.1 (II.3.05: Saturation, supported localization, and near-invariance).

A probability σ on $[0, 1)^d$ is (ε, m) *entropy-saturated* if

$$h_m(\sigma) \geq d - \varepsilon. \quad (0.10)$$

A probability τ on $\mathbb{R}^d$ is (C, m, z) *supported-localized* if

$$\tau\{y : \|y - z\|_\infty \leq C2^{-m}\} = 1. \quad (0.11)$$

This is a support condition, not merely a high-probability concentration condition. A pair (σ, τ) of probabilities on $[0, 1)^d$ is (ε, m) *entropy-near-invariant* if $\gamma_m(\sigma; \tau) \leq \varepsilon$.

Corollary II.3.3.2 (C-II.3.04 – Saturation and supported localization). *Let $m \geq 1$.*

(1) *If σ, τ are probabilities on $[0, 1)^d$ and $h_m(\sigma) \geq d - \varepsilon$, then*

$$h_m(\sigma * \tau) \geq d - \varepsilon - \frac{d}{m}. \quad (2.1)$$

(2) *If σ is a probability on $\mathbb{R}^d$ with finite H_m and τ is (C, m, z) supported-localized, then*

$$|H_m(\sigma * \tau) - H_m(\sigma)| \leq d \log(2[C] + 1) + d \log 2. \quad (2.2)$$

If $z \in 2^{-m}\mathbb{Z}^d$, the last term is absent.

Proof

The first inequality follows immediately from the lower half of (1.8):

$$h_m(\sigma * \tau) \geq h_m(\sigma) - \frac{d}{m} \geq d - \varepsilon - \frac{d}{m}. \quad (2.3)$$

For the second assertion, work on the product probability space with law $\sigma \times \tau$ and compare

$$f(x, y) = x + y, \quad g(x, y) = x + z. \quad (2.4)$$

The support condition (0.11) gives

$$\|f(x, y) - g(x, y)\|_\infty \leq C2^{-m} \quad (2.5)$$

almost everywhere. Their push-forwards are $\sigma * \tau$ and $(\tau_z)_* \sigma$. Bounded-perturbation stability P-II.1.09 therefore gives

$$|H_m(\sigma * \tau) - H_m((\tau_z)_* \sigma)| \leq d \log(2[C] + 1). \quad (2.6)$$

Translated-grid stability C-II.1.08 gives

$$|H_m((\tau_z)_* \sigma) - H_m(\sigma)| \leq d \log 2. \quad (2.7)$$

The triangle inequality proves (2.2). If $z \in 2^{-m}\mathbb{Z}^d$, translation by z permutes the cells of $\mathcal{D}_m$, so the left side of (2.7) is zero. $\square$

The same one-scale inequalities also control the entropy profile formed by adding further independent copies of a fixed measure.

Definition II.3.3.3 (II.3.06: Repeated-convolution entropy profile). For a probability μ , a fixed resolution n , and integers $k \geq 1$, the *repeated-convolution entropy profile* is

$$k \mapsto h_n(\mu^{*k}).$$

No monotonicity is included in the definition; the next corollary proves only monotonicity up to the explicit carry loss d/n .

Corollary II.3.3.4 (C-II.3.05 — Repeated-convolution entropy profile). *Let μ be a probability on $\mathbb{R}^d$, let $n \geq 1$, and assume $H_n(\mu) < \infty$. For every $k \geq 1$,*

$$h_n(\mu) - \frac{d}{n} \leq h_n(\mu^{*k}) \leq kh_n(\mu) + (k-1)\frac{d}{n}, \quad (2.8)$$

and

$$-\frac{d}{n} \leq h_n(\mu^{*(k+1)}) - h_n(\mu^{*k}) \leq h_n(\mu) + \frac{d}{n}. \quad (2.9)$$

Proof

Apply the upper half of (1.8) recursively:

$$\begin{aligned} h_n(\mu^{*(k+1)}) &\leq h_n(\mu^{*k}) + h_n(\mu) + \frac{d}{n} \\ &\leq (k+1)h_n(\mu) + k\frac{d}{n}. \end{aligned} \quad (2.10)$$

The base case is equality at $k = 1$, so induction proves the upper bound and also shows all entropies are finite. For the lower bound, $k = 1$ is immediate. When $k \geq 2$, write $\mu^{*k} = \mu * \mu^{*(k-1)}$ and retain the μ term in the lower half of (1.8):

$$h_n(\mu^{*k}) \geq h_n(\mu) - \frac{d}{n}. \quad (2.11)$$

Finally apply both halves of (1.8) to $\mu^{*(k+1)} = \mu^{*k} * \mu$, retaining μ^{*k} in the lower bound. After subtracting $h_n(\mu^{*k})$, the two resulting inequalities are exactly (2.9). $\square$

II.3.4 Components of a convolution

The global convolution is a mixture of convolutions of raw components. After rescaling, each component pair lives in a common bounded box, so the local entropy can be compared

at one fixed resolution. Concavity of conditional entropy is what lets us pass from these local models back to the whole convolution.

Definition II.3.4.1 (II.3.04: Independent component pairs). For probabilities μ, ν and a level i , an *independent same-level component pair* is obtained by sampling $x \sim \mu$ and $y \sim \nu$ independently and returning

$$(\mu^{x,i}, \nu^{y,i}), \quad x \sim \mu, \quad y \sim \nu. \quad (0.9)$$

The raw pair $(\mu_{x,i}, \nu_{y,i})$ is defined analogously. Their *component convolution* is respectively $\mu^{x,i} * \nu^{y,i}$ or $\mu_{x,i} * \nu_{y,i}$. Every probability statement over x, y uses the product law $\mu \times \nu$.

Lemma II.3.4.2 (L-II.3.06 – Random-measure concavity and component disintegration). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let Θ_ω be Borel probabilities on a common metric space X such that $\omega \mapsto \Theta_\omega(A)$ is measurable for every Borel $A \subset X$. Define the barycenter by

$$\bar{\Theta}(A) = \mathbb{E}\Theta_\omega(A). \quad (3.1)$$

For finite measurable partitions $\mathcal{P}, \mathcal{Q}$,

$$H_{\bar{\Theta}}(\mathcal{P}) \geq \mathbb{E}H_{\Theta_\omega}(\mathcal{P}), \quad H_{\bar{\Theta}}(\mathcal{P} \mid \mathcal{Q}) \geq \mathbb{E}H_{\Theta_\omega}(\mathcal{P} \mid \mathcal{Q}). \quad (3.2)$$

For Borel probabilities μ, ν on $\mathbb{R}^d$ and every level i ,

$$\mu * \nu = \mathbb{E}_{x \sim \mu, y \sim \nu}(\mu_{x,i} * \nu_{y,i}) \quad (3.3)$$

as Borel measures.

Proof

Monotone convergence shows that (3.1) is countably additive, and its total mass is one. For an atom $P \in \mathcal{P}$, concavity of $\eta(t) = -t \log t$ gives

$$\eta(\bar{\Theta}(P)) = \eta(\mathbb{E}\Theta_\omega(P)) \geq \mathbb{E}\eta(\Theta_\omega(P)). \quad (3.4)$$

Summing the finitely many inequalities proves the first part of (3.2).

For the conditional claim, define on nonnegative vectors $v = (v_1, \dots, v_r)$

$$\Phi(v) = \begin{cases} -\sum_{j=1}^r v_j \log \frac{v_j}{s(v)}, & s(v) = \sum_j v_j > 0, \\ 0, & s(v) = 0. \end{cases} \quad (3.5)$$

This is the perspective of Shannon entropy and is concave. To verify that fact rather than invoke it, let $0 \leq t \leq 1$ and let v, w have masses s_v, s_w . If $s = ts_v + (1-t)s_w > 0$, the normalized vector of $tv + (1-t)w$ is

$$\frac{ts_v}{s} \frac{v}{s_v} + \frac{(1-t)s_w}{s} \frac{w}{s_w}, \quad (3.6)$$

with zero-mass terms omitted. Entropy concavity from [T-II.1.05](#), multiplied by s , gives

$$\Phi(tv + (1-t)w) \geq t\Phi(v) + (1-t)\Phi(w). \quad (3.7)$$

The zero-mass cases follow from $\Phi(0) = 0$. The function is continuous on the compact simplex of subprobability vectors, so Jensen's inequality applies to random vectors as well.

For each $Q \in \mathcal{Q}$, take

$$V_Q(\Theta_\omega) = (\Theta_\omega(P \cap Q))_{P \in \mathcal{P}}. \quad (3.8)$$

The contribution of Q to conditional entropy is exactly $\Phi(V_Q)$. Jensen for (3.7), followed by the finite sum over Q , gives the second inequality in (3.2).

To prove (3.3), use the raw component identities of [L-II.2.01](#) independently for μ and ν . For a Borel $E \subset \mathbb{R}^d$, Fubini–Tonelli gives

$$\begin{aligned} \mathbb{E}_{x,y}(\mu_{x,i} * \nu_{y,i})(E) &= \iint \iint \mathbf{1}_E(u+v) d\mu_{x,i}(u) d\nu_{y,i}(v) d\mu(x) d\nu(y) \\ &= \iint \mathbf{1}_E(u+v) d\mu(u) d\nu(v) = (\mu * \nu)(E). \end{aligned} \quad (3.9)$$

This proves the measure identity. $\square$

Theorem II.3.4.3 (T-II.3.07 – One-window component-convolution lower bound).

Let μ, ν be probabilities on $[0, 1]^d$. For $i \geq 0$ and $m \geq 1$,

$$H_{\mu * \nu}(\mathcal{D}_{i+m} \mid \mathcal{D}_i) \geq \mathbb{E}_{x,y} H_m(\mu^{x,i} * \nu^{y,i}) - d \log 2. \quad (3.10)$$

Proof

Proof strategy. Disintegrate the global convolution into raw component convolutions, use concavity of conditional entropy, and rescale one fixed pair of parent cells. The only loss is the entropy of its level-zero carry cell.

The convolution $\mu * \nu$ is carried by $[0, 2)^d$, so only finitely many atoms of the displayed partitions have positive mass. Apply the conditional concavity in [L-II.3.06](#) to the random-measure decomposition (3.3):

$$H_{\mu * \nu}(\mathcal{D}_{i+m} \mid \mathcal{D}_i) \geq \mathbb{E}_{x,y} H_{\mu_{x,i} * \nu_{y,i}}(\mathcal{D}_{i+m} \mid \mathcal{D}_i). \quad (3.11)$$

Fix positive component cells

$$Q = 2^{-i}(k + [0, 1)^d), \quad R = 2^{-i}(\ell + [0, 1)^d), \quad (3.12)$$

and put $S(z) = 2^i z - (k + \ell)$. Then

$$S_*(\mu_Q * \nu_R) = \mu^Q * \nu^R. \quad (3.13)$$

The map S sends the level- $i + m$ grid to the level- m grid and the level- i grid to the level-zero grid, up to a permutation of cells. Hence

$$H_{\mu_Q * \nu_R}(\mathcal{D}_{i+m} \mid \mathcal{D}_i) = H_{\mu^Q * \nu^R}(\mathcal{D}_m \mid \mathcal{D}_0). \quad (3.14)$$

The measure on the right is carried by $[0, 2)^d$, which meets exactly 2^d level-zero cells. Since $\mathcal{D}_m$ refines $\mathcal{D}_0$, the chain rule and entropy extremum give

$$\begin{aligned} H_{\mu^Q * \nu^R}(\mathcal{D}_m \mid \mathcal{D}_0) &= H_m(\mu^Q * \nu^R) - H_0(\mu^Q * \nu^R) \\ &\geq H_m(\mu^Q * \nu^R) - d \log 2. \end{aligned} \quad (3.15)$$

Insert (3.14)–(3.15) into (3.11) to obtain (3.10). $\square$

The next step averages these one-window estimates over their starting scales. Each interior entropy increment is counted (m) times. For the lower bound, the initial boundary block has the favorable sign, so only the final ($m-1$) increments need to be discarded. This is why the endpoint error in the theorem is one-sided.

Theorem II.3.4.4 (T-II.3.08 – Global entropy from local convolution entropy). *Let μ, ν be probabilities on $[0, 1)^d$, and let $N, m \geq 1$. Then*

$$h_N(\mu * \nu) \geq \frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_{x,y} h_m(\mu^{x,i} * \nu^{y,i}) - \frac{d}{m} - \frac{d(m-1)}{2N}. \quad (3.16)$$

Proof

Proof strategy. Write each m -step conditional entropy as $H_{i+m} - H_i$, sum over the starting scale, and inspect the two endpoint blocks. The initial block has the favorable sign; the terminal block is bounded by the entropy created in ($m-1$) binary refinements.

Put $\lambda = \mu * \nu$, $H_j = H_j(\lambda)$, and

$$C_i = H_\lambda(\mathcal{D}_{i+m} \mid \mathcal{D}_i) = H_{i+m} - H_i. \quad (3.17)$$

The equality is the finite-entropy chain rule. Cancellation gives

$$\frac{1}{m} \sum_{i=0}^{N-1} C_i = H_N + A - B, \quad (3.18)$$

where

$$A = \frac{1}{m} \sum_{r=0}^{m-1} (H_{N+r} - H_N), \quad B = \frac{1}{m} \sum_{j=0}^{m-1} H_j. \quad (3.19)$$

Both are nonnegative. A refinement by r dyadic levels has at most 2^{dr} descendants per cell, so T-II.1.07 gives

$$0 \leq A \leq \frac{d \log 2}{m} \sum_{r=0}^{m-1} r = \frac{d(m-1) \log 2}{2}. \quad (3.20)$$

The term B has a favorable sign in the required lower bound. Equations (3.18)–(3.20) therefore yield

$$H_N \geq \frac{1}{m} \sum_{i=0}^{N-1} C_i - \frac{d(m-1) \log 2}{2}. \quad (3.21)$$

By T-II.3.07,

$$C_i \geq \mathbb{E}_{x,y} H_m(\mu^{x,i} * \nu^{y,i}) - d \log 2. \quad (3.22)$$

Sum (3.22), insert it into (3.21), and divide by $N \log 2$. The result is (3.16). $\square$

II.3.5 Quantitative localization of small growth

If the global entropy gain is small, the average corrected growth of component pairs is small. Since corrected growth is nonnegative, two successive applications of Markov's inequality show first that most sampled triples (i, x, y) have small growth, and then that at most scales most component pairs have small growth. We introduce notation for these two conclusions before proving them.

Definition II.3.5.1 (II.3.07: Bad-pair frequency and good scales). Let μ, ν be probabilities on $[0, 1]^d$, and let $N, m \geq 1$ and $\eta > 0$. The *bad-pair frequency* is

$$B_{N,m,\eta}(\mu, \nu) = \mathbb{P}_{\substack{0 \leq i < N \\ x \sim \mu, y \sim \nu}} \{Y_m(\mu^{x,i}; \nu^{y,i}) > \eta\}, \quad (0.12)$$

where i is uniform and independent of x, y . For $0 < \theta \leq 1$, the corresponding *good-scale set* is

$$G_{N,m,\eta,\theta}(\mu, \nu) = \{0 \leq i < N : \mathbb{P}_{x,y} \{Y_m(\mu^{x,i}; \nu^{y,i}) \leq \eta\} \geq 1 - \theta\}. \quad (0.13)$$

These quantities diagnose local near-invariance; they do not yet classify components as atomic or uniform.

Theorem II.3.5.2 (T-II.3.09 – Global-to-local corrected growth). Let μ, ν be probabilities on $[0, 1]^d$, and let $N, m \geq 1$. Then

$$\frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_{x,y} Y_m(\mu^{x,i}; \nu^{y,i}) \leq \Delta_N(\mu; \nu) + \frac{2d}{m} + \frac{3d(m-1)}{2N}. \quad (4.1)$$

In particular, the right side is nonnegative.

Proof

Proof strategy. Compare the lower bound for the local convolution entropies with the upper local-entropy average for the first factor. Their difference is precisely corrected component growth, up to the displayed carry and endpoint errors.

T-II.2.02, applied to μ , gives the upper estimate

$$h_N(\mu) \leq \frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_x h_m(\mu^{x,i}) + \frac{d(m-1)}{N}. \quad (4.2)$$

Subtract (4.2) from the lower bound (3.16). The additional expectation over y does not change a term depending only on x , so

$$\begin{aligned} \Delta_N(\mu; \nu) &\geq \frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_{x,y} (h_m(\mu^{x,i} * \nu^{y,i}) - h_m(\mu^{x,i})) \\ &\quad - \frac{d}{m} - \frac{3d(m-1)}{2N}. \end{aligned} \quad (4.3)$$

By (0.8), the difference inside the average is

$$\gamma_m(\mu^{x,i}, \nu^{y,i}) - \frac{d}{m}. \quad (4.4)$$

Insert (4.4) into (4.3) and rearrange to obtain (4.1). Each γ_m is nonnegative by T-II.3.03, so its average is nonnegative; (4.1) forces the right side to be nonnegative as well. $\square$

Corollary II.3.5.3 (C-II.3.10 – Few bad component pairs under small growth). *Let $d, N, m \geq 1$, let μ, ν be probabilities on $[0, 1)^d$, and let $\eta > 0$. Define*

$$R_{N,m}(\mu, \nu) = \Delta_N(\mu; \nu) + \frac{2d}{m} + \frac{3d(m-1)}{2N}. \quad (4.5)$$

For every $\eta > 0$,

$$B_{N,m,\eta}(\mu, \nu) \leq \min \left\{ 1, \frac{R_{N,m}(\mu, \nu)}{\eta} \right\}. \quad (4.6)$$

Proof

Let

$$Z(i, x, y) = \gamma_m(\mu^{x,i}, \nu^{y,i}).$$

It is nonnegative, and (4.1) says $\mathbb{E}Z \leq R_{N,m}$ under the product sampling in (0.12). Markov's inequality gives

$$\mathbb{P}\{Z > \eta\} \leq \frac{\mathbb{E}Z}{\eta} \leq \frac{R_{N,m}}{\eta}.$$

A probability is also at most one, proving (4.6). $\square$

Lemma II.3.5.4 (L-II.3.11 — Extraction of good scales). *Let $d, N, m \geq 1$, let μ, ν be probabilities on $[0, 1]^d$, and let $\eta > 0$. With $R_{N,m}$ from C-II.3.10, for $0 < \theta \leq 1$,*

$$\frac{|G_{N,m,\eta,\theta}(\mu, \nu)|}{N} \geq 1 - \min \left\{ 1, \frac{R_{N,m}(\mu, \nu)}{\eta\theta} \right\}. \quad (4.7)$$

If $R_{N,m}(\mu, \nu) \leq \eta\theta^2$, then at least $(1 - \theta)N$ scales belong to $G_{N,m,\eta,\theta}$.

Proof

For $0 \leq i < N$, put

$$b_i = \mathbb{P}_{x,y} \{ \gamma_m(\mu^{x,i}; \nu^{y,i}) > \eta \}. \quad (4.8)$$

Then

$$\frac{1}{N} \sum_{i=0}^{N-1} b_i = B_{N,m,\eta}(\mu, \nu) \leq \min \left\{ 1, \frac{R_{N,m}}{\eta} \right\}. \quad (4.9)$$

A scale fails to belong to the good-scale set exactly when $b_i > \theta$. Markov's inequality on the finite uniform probability space of scales gives

$$\frac{1}{N} |\{i : b_i > \theta\}| \leq \min \left\{ 1, \frac{R_{N,m}}{\eta\theta} \right\}. \quad (4.10)$$

Taking the complementary set proves (4.7). Under the final hypothesis, the second term in (4.10) is at most θ , proving the stated specialization. $\square$

II.3.6 Finite-set dictionary

For uniform measures on finite subsets of a dyadic lattice, entropy before addition is exactly logarithmic cardinality, while the entropy of the sum is bounded above by the logarithm of the sumset size. This turns a small-doubling estimate into a small-entropy-growth estimate. The converse is false without additional hypotheses: a distribution may put very small mass on many points and therefore have a large support but little entropy.

Proposition II.3.6.1 (P-II.3.12 — Dyadic finite sets and sumsets). *Let $n \in \mathbb{Z}$, let A, B be nonempty finite subsets of $2^{-n}\mathbb{Z}^d$, and let μ_A, μ_B be their uniform probabilities. If X, Y are independent and uniform on A, B , then*

$$H_n(\mu_A) = \log |A|, \quad H_n(\mu_A * \mu_B) = H(X + Y), \quad (5.1)$$

and

$$\max\{\log |A|, \log |B|\} \leq H(X + Y) \leq \log |A + B|. \quad (5.2)$$

Consequently $|A + B| \geq \max\{|A|, |B|\}$. If $|A + B| \leq K|A|$, then

$$H_n(\mu_A * \mu_B) - H_n(\mu_A) \leq \log K. \quad (5.3)$$

Proof

The map q_n is injective on $2^{-n}\mathbb{Z}^d$, and it is also injective on the sumset, which lies in the same lattice. Thus the level- n cell index of X is uniform on $|A|$ points, while the cell index of $X + Y$ has exactly the same probability vector as the discrete sum. This proves (5.1).

The lower bound in (5.2) is L-II.3.02. The sum variable is carried by $A + B$, so the entropy-support bound L-II.1.02 gives

$$H(X + Y) \leq \log |A + B|.$$

Combining the two sides proves the cardinality inequality. Under $|A + B| \leq K|A|$, equations (5.1)–(5.2) give

$$H_n(\mu_A * \mu_B) - H_n(\mu_A) \leq \log |A + B| - \log |A| \leq \log K,$$

which is (5.3). $\square$

The chapter has therefore reduced small global entropy growth to local near-invariance of component pairs. Chapter II.4 supplies the genuinely structural step: on the line, such pairs must exhibit an atomic–uniform dichotomy at most scales.

Chapter II.4

Hochman's Inverse Theorem on the Line

Geometric intuition

Hochman's inverse theorem on the line explains near-failure of entropy growth: at many scales one factor must look nearly atomic or the other nearly uniform. The proof is developed with explicit components, variance control, and a quantitative Berry–Esseen estimate, leading to a precise finite-scale statement for later applications.

All logarithms are natural, and the entropy, component, and convolution conventions of the preceding chapters remain in force. On the line, the two local structures relevant to inverse theory are high entropy, which models uniformity, and low entropy, which models atomicity.

Definition II.4.0.1 (II.4.01: Uniformity and atomicity on the line). For a probability θ on $[0, 1)$ and $m \geq 1$, its *normalized level- m component entropy* is

$$e_m(\theta) = \frac{H_m(\theta)}{m \log 2} \in [0, 1]. \quad (0.1)$$

For $0 < \varepsilon < 1$, the measure θ is

- (ε, m) *entropy-uniform* if $e_m(\theta) \geq 1 - \varepsilon$;
- (ε, m) *entropy-atomic* if $e_m(\theta) \leq \varepsilon$; and
- *geometrically ε -atomic* if some interval $I \subset \mathbb{R}$ of length at most ε satisfies $\theta(I) \geq 1 - \varepsilon$.

The two atomicity notions are not synonymous; Lemma L-II.4.05 compares them quantitatively. The weak endpoint inequalities ensure that complements have no unclassified equality case.

Repeated convolution creates the uniform alternative. If a component has definite

variance, the variance of a sum of independent copies grows linearly. After square-root rescaling, the sum approaches a Gaussian, whose density is nearly constant within a sufficiently fine dyadic cell. Its still finer children consequently have nearly equal masses. We first introduce the moment and discrepancy notation needed to make this argument precise.

Definition II.4.0.2 (II.4.02: Moments, Gaussian laws, and interval discrepancy). For a probability θ on $\mathbb{R}$ with the indicated finite moments, define

$$a(\theta) = \int x d\theta(x), \quad \text{Var}(\theta) = \int (x - a(\theta))^2 d\theta(x),$$

and its *centered third absolute moment*

$$\rho_3(\theta) = \int |x - a(\theta)|^3 d\theta(x).$$

For $a \in \mathbb{R}$ and $v > 0$, let $G_{a,v}$ be the Gaussian probability with mean a , variance v , and density

$$g_{a,v}(x) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x-a)^2}{2v}\right).$$

For a real random variable X with law θ , the notations $a(X)$, $\text{Var}(X)$, and $\rho_3(X)$ denote the corresponding quantities of θ . The *interval discrepancy* between probabilities λ, τ on $\mathbb{R}$ is

$$d_1(\lambda, \tau) = \sup\{|\lambda(I) - \tau(I)| : I \subset \mathbb{R} \text{ is an interval}\}.$$

II.4.1 From normal approximation to component entropy

Lemma II.4.1.1 (L-II.4.01 – Finite-alphabet entropy continuity). *Let p, q be probability vectors on an alphabet of cardinality $M \geq 2$, and put $t = \|p - q\|_TV$. Then*

$$|H(p) - H(q)| \leq b(t) + t \log(M - 1), \quad (1.1)$$

where $b(t) = -t \log t - (1 - t) \log(1 - t)$. Consequently, if P, Q are laws on $A \times B$, where A is finite, $|B| = M \geq 2$, and $\|P - Q\|_TV = t$, then

$$|H_P(B | A) - H_Q(B | A)| \leq 2b(t) + t \log M. \quad (1.2)$$

Proof

Write $r_i = \min\{p_i, q_i\}$. Since

$$\sum_i r_i = 1 - t, \quad (1.3)$$

there is a coupling (X, Y) with laws p, q which puts mass r_i on (i, i) and couples the two residual laws off the diagonal. The residual supports are disjoint, so

$$\mathbb{P}(X \neq Y) = t. \quad (1.4)$$

Let $E = 1_{X \neq Y}$. Given Y and $E = 1$, the variable X has at most $M - 1$ possible values. The entropy chain rule and conditioning reduction give

$$\begin{aligned} H(X) &\leq H(X, Y) \\ &= H(Y) + H(E | Y) + H(X | Y, E) \\ &\leq H(Y) + b(t) + t \log(M - 1). \end{aligned} \quad (1.5)$$

Interchanging X, Y proves (1.1).

For (1.2), take a maximal coupling $(A, B), (A', B')$ of the two joint laws, and let E indicate that the coupled pairs differ. Then $\mathbb{P}(E = 1) = t$. When $t < 1$, put

$$h_0 = H(B | A, E = 0) = H(B' | A', E = 0), \quad (1.6)$$

where equality holds because the pairs agree on $E = 0$; put $h_0 = 0$ when $t = 1$. The chain rule gives, for either joint law,

$$(1 - t)h_0 - b(t) \leq H(B | A) \leq b(t) + (1 - t)h_0 + t \log M. \quad (1.6a)$$

Indeed, the upper bound follows by revealing E , and the lower bound from $H(B | A, E) \leq H(B | A) + H(E)$. Apply opposite sides of (1.6a) to the two laws and then interchange them. This proves (1.2). $\square$

Lemma II.4.1.2 (L-II.4.02 – Uniform Gaussian component entropy). *Fix $0 < \sigma \leq 1$ and $m \geq 1$. Uniformly for $a \in \mathbb{R}$ and $\sigma \leq \nu \leq 1$,*

$$H_{G_{a,\nu}}(\mathcal{D}_{p+m} | \mathcal{D}_p) \longrightarrow m \log 2 \quad (p \rightarrow \infty). \quad (1.7)$$

Proof

Put $h = 2^{-p}$ and fix $R > 1$. Let U be the union of all level- p dyadic cells meeting $[a - R, a + R]$. Chebyshev's inequality gives

$$G_{a,\nu}(U^c) \leq G_{a,\nu}\{|x - a| > R\} \leq \frac{\nu}{R^2} \leq R^{-2}. \quad (1.8)$$

On any level- p cell $Q \subset U$, every point has distance at most $R + h$ from a . Since

$$\frac{d}{dx} \log g_{a,\nu}(x) = -\frac{x - a}{\nu}, \quad (1.9)$$

the mean-value theorem implies

$$\frac{\sup_Q g_{a,\nu}}{\inf_Q g_{a,\nu}} \leq \exp\left(\frac{(R + h)h}{\sigma}\right). \quad (1.10)$$

Let $q_1, \dots, q_{2^m}$ be the conditional masses of the level- $p + m$ children of Q . Every child has length $2^{-m}|Q|$; hence (1.10) yields

$$2^m q_j \leq \exp\left(\frac{(R+h)h}{\sigma}\right). \quad (1.11)$$

The entropy deficit from the uniform vector is its relative entropy to that vector, so

$$m \log 2 - H(q_1, \dots, q_{2^m}) = \sum_j q_j \log(2^m q_j) \leq \frac{(R+h)h}{\sigma}. \quad (1.12)$$

On cells outside U the deficit is at most $m \log 2$. Averaging (1.12) and using (1.8),

$$0 \leq m \log 2 - H_{G_{a,v}}(\mathcal{D}_{p+m} \mid \mathcal{D}_p) \leq \frac{(R+h)h}{\sigma} + \frac{m \log 2}{R^2}. \quad (1.13)$$

First choose R large and then p large. The right side tends to zero uniformly in a, v , proving (1.7). $\square$

The only external premise in this chapter is the following classical normal approximation estimate. Everything after the estimate—tail truncation, entropy continuity, Gaussian cell analysis, and component extraction—is proved here. The following interval form follows from [Hoc14, Theorem 4.1] by applying its raw-third-moment estimate to the centered variables $X_j - a_j$ and translating the intervals back by $\sum_j a_j$.

Cited result — X-BE-II.4 — One-dimensional Berry–Esseen estimate

We use the centered interval-discrepancy form deduced immediately above from [Hoc14, Theorem 4.1]. There is a universal constant C_{BE} such that the following holds. Let $X_1, \dots, X_k$ be independent real random variables with means a_j , variances v_j , and centered third absolute moments ρ_j . Put

$$S = \sum_{j=1}^k X_j, \quad V = \sum_{j=1}^k v_j > 0,$$

and let G be the Gaussian law with the same mean and variance as S . Then

$$\sup_{I \text{ interval}} |\mathbb{P}(S \in I) - G(I)| \leq C_{\text{BE}} \frac{\sum_{j=1}^k \rho_j}{V^{3/2}}.$$

Proposition II.4.1.3 (P-II.4.03 — Bounded-convolution entropy smoothing). *Fix $\sigma > 0$, $0 < \varepsilon < 1$, and $m \geq 1$. There are integers $p \geq 1$ and K such that, whenever $k \geq K$ and every θ_j is carried by an interval of length at most one,*

$$\eta = \theta_1 * \dots * \theta_k, \quad \text{Var}(\eta) \geq \sigma k, \quad (1.14)$$

and

$$r = \lfloor \log_2 \sqrt{k} \rfloor, \quad q = p - r, \quad (1.15)$$

one has

$$\mathbb{P}_{x \sim \eta} \{e_m(\eta^{x,q}) \geq 1 - \varepsilon\} \geq 1 - \varepsilon. \quad (1.16)$$

Proof

Proof strategy. Normalize the variance of the sum into a fixed compact range, apply the Berry–Esseen input, and compare the resulting finite collections of dyadic cell masses by entropy continuity. A final component disintegration and Markov estimate convert high conditional entropy into uniformity for most components.

Let X_j be independent with laws θ_j , put $S = \sum_j X_j$, and let $V = \text{Var}(S)$. Since a random variable in an interval of length at most one has variance at most $1/4$,

$$\sigma k \leq V \leq k/4. \quad (1.17)$$

Set $c_k = 2^{-r}$ and let λ be the law of $c_k S$. From $2^r \leq \sqrt{k} < 2^{r+1}$,

$$\sigma \leq c_k^2 V < 1. \quad (1.18)$$

Let G be the Gaussian law with the same mean and variance as λ . The centered third absolute moment of each X_j is at most one. The scale-invariance of $X - BE - II.4$ therefore gives

$$d_1(\lambda, G) \leq C_{\text{BE}} \frac{k}{(\sigma k)^{3/2}} = C_{\text{BE}} \sigma^{-3/2} k^{-1/2} =: \delta_k. \quad (1.19)$$

We now justify the entropy consequence of (1.19). Let a be the common mean, choose $R > 1$, and retain the level- p cells meeting $[a - R, a + R]$. Their number is at most

$$L_R = 2R 2^p + 3. \quad (1.20)$$

Both λ and G have variance at most one, so each gives mass at most R^{-2} to the complement of their retained union. For either measure, let C be the level- p cell label and $F \in \{0, \dots, 2^m - 1\}$ the level- $p + m$ child label. Collapse all discarded cells to one symbol and set $F = 0$ there. The total variation distance of the two collapsed joint laws is at most

$$t_{R,k} = \min \left\{ 1, \frac{1}{2} (L_R 2^m \delta_k + 2R^{-2}) \right\}. \quad (1.21)$$

Indeed, each retained fine-cell atom is an interval and contributes at most δ_k , while the two collapsed tail masses have total absolute difference at most $2R^{-2}$.

Before collapse, the discarded coarse cells contribute at most $R^{-2} m \log 2$ to either conditional entropy. Applying L-II.4.01 to the collapsed laws, on alphabets of sizes at most $L_R + 1$ and 2^m , therefore shows that

$$\begin{aligned} & |H_\lambda(\mathcal{D}_{p+m} \mid \mathcal{D}_p) - H_G(\mathcal{D}_{p+m} \mid \mathcal{D}_p)| \\ & \leq 2R^{-2} m \log 2 + 2b(t_{R,k}) + t_{R,k} m \log 2. \end{aligned} \quad (1.22)$$

Given any $\zeta > 0$, choose R so the tail terms are small, then use L-II.4.02 to choose p so the Gaussian deficit is small, and finally choose K so the remaining terms in (1.22) are small for $k \geq K$. With $\zeta = \varepsilon^2 m \log 2$, these choices give

$$m \log 2 - H_\lambda(\mathcal{D}_{p+m} \mid \mathcal{D}_p) \leq \varepsilon^2 m \log 2. \quad (1.23)$$

By component disintegration,

$$H_\lambda(\mathcal{D}_{p+m} \mid \mathcal{D}_p) = \int H_m(\lambda^{z,p}) d\lambda(z). \quad (1.24)$$

The nonnegative deficit $m \log 2 - H_m(\lambda^{z,p})$ is therefore larger than $\varepsilon m \log 2$ on a set of λ -mass at most ε , by Markov's inequality. Thus at least $1 - \varepsilon$ of the level- p components of λ are (ε, m) entropy-uniform.

Finally, the dilation $x \mapsto 2^{-r}x$ sends level- q cells exactly to level- p cells because $p = q + r$, and it preserves the rescaled component laws. Hence the last assertion for λ is exactly (1.16) for η . $\square$

II.4.2 Variance, atomicity, and inheritance

The complementary mechanism is concentration. Small variance places most of a measure near its mean, hence in very few cells at a suitable diagnostic scale. This bounds the entropy available in a longer window, and nonnegativity of entropy increments then forces most subcomponents to have small entropy. We keep entropy atomicity and geometric atomicity distinct; the latter will be deduced quantitatively rather than built into the terminology.

Definition II.4.2.1 (II.4.03: Component variance profile). For a probability μ on $[0, 1]$ and an integer $i \geq 0$, define

$$\lambda_i(\mu) = \int \text{Var}(\mu^{x,i}) d\mu(x) \in [0, 1/4].$$

For $a > 0$, $0 < \varepsilon < 1$, and $m \geq 1$, the *variance-atomic class* is

$$\mathcal{V}(a; \varepsilon, m) = \{\theta \in \mathcal{P}([0, 1]) : \text{Var}(\theta) < a \text{ and } e_m(\theta) \leq \varepsilon\}.$$

The *variance-atomic event* for a random component is the inverse image of this class under the component map.

Lemma II.4.2.2 (L-II.4.04 — Variance algebra and a uniform bounded weak law). *If X, Y are independent with finite second moments, then*

$$a(X + Y) = a(X) + a(Y), \quad \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y). \quad (2.1)$$

If $W_1, \dots, W_k$ are independent copies of a random variable in $[0, 1/4]$ with mean $\lambda \geq \sigma > 0$, then

$$\mathbb{P}\left\{k^{-1} \sum_{j=1}^k W_j < \sigma/2\right\} \leq \frac{1}{4k\sigma^2}. \quad (2.2)$$

Proof

Linearity gives the mean identity. Expanding the centered square gives

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E}[(X - a(X))^2] + \mathbb{E}[(Y - a(Y))^2] \\ &\quad + 2\mathbb{E}[(X - a(X))(Y - a(Y))]. \end{aligned} \quad (2.3)$$

The last term is zero by independence, proving (2.1).

Since $0 \leq W_j \leq 1/4$, $\text{Var}(W_j) \leq \mathbb{E}(W_j - \lambda)^2 \leq 1/16$. If the event in (2.2) occurs, then the sample mean differs from λ by at least $\sigma/2$. Chebyshev and independence give

$$\mathbb{P}\left\{\left|k^{-1} \sum_j W_j - \lambda\right| \geq \sigma/2\right\} \leq \frac{k/16}{k^2\sigma^2/4} = \frac{1}{4k\sigma^2}. \quad (2.4)$$

For a random level- i component, the last assertion is the definition of $\lambda_i(\mu)$. $\square$

Lemma II.4.2.3 (L-II.4.05 – Entropy, variance, and geometric atomicity). *Let $\theta \in \mathcal{P}([0, 1])$, $s \geq 1$, and*

$$t = 2^{2s+4} \text{Var}(\theta) \leq 1/2.$$

Then

$$H_s(\theta) \leq b(t) + (1 - t) \log 2 + t s \log 2. \quad (2.5)$$

For every such θ , without a smallness assumption,

$$\text{Var}(\theta) \leq \frac{1}{2} \left(2^{-2s} + 1 - e^{-2H_s(\theta)}\right). \quad (2.6)$$

Proof

Let $a = a(\theta)$, put $r = 2^{-s-2}$, and let $E = \{|x - a| \leq r\}$. Chebyshev gives

$$p := \theta(E^c) \leq \frac{\text{Var}(\theta)}{r^2} = t. \quad (2.7)$$

The interval $[a - r, a + r]$ has length 2^{-s-1} and hence meets at most two level- s dyadic cells. On E^c at most 2^s cells of $[0, 1)$ occur. Writing C for the cell label and first revealing 1_E ,

$$H_s(\theta) = H(C) \leq b(p) + (1 - p) \log 2 + p s \log 2. \quad (2.8)$$

The right side is nondecreasing for $0 \leq p \leq 1/2$: its derivative is

$$\log \frac{1-p}{p} + (s-1) \log 2 \geq 0. \quad (2.9)$$

Thus (2.7) and $t \leq 1/2$ prove (2.5).

For the converse, let p_* be the largest level- s cell mass. If C is the random cell label, then

$$H_s(\theta) = \mathbb{E} \log \frac{1}{\mathbb{P}(C)} \geq \log \frac{1}{p_*}, \quad p_* \geq e^{-H_s(\theta)}. \quad (2.10)$$

Let X, X' be independent with law θ . On the event that both occupy a cell of mass p_* , $|X - X'| \leq 2^{-s}$; always $|X - X'| \leq 1$. Hence

$$\begin{aligned} 2 \operatorname{Var}(\theta) &= \mathbb{E}(X - X')^2 \\ &\leq p_*^2 2^{-2s} + (1 - p_*^2) \leq 2^{-2s} + 1 - e^{-2H_s(\theta)}, \end{aligned} \quad (2.11)$$

which is (2.6).

If $e_s(\theta) \leq a$, then (2.10) gives a level- s cell of mass at least 2^{-as} . When $2^{-s} \leq \xi$ and $2^{-as} \geq 1 - \xi$, that cell itself is an interval witnessing geometric ξ -atomicity. $\square$

Lemma II.4.2.4 (L-II.4.06 — Heredity of entropy atomicity and uniformity). *Let $\theta \in \mathcal{P}([0, 1])$ and set*

$$A_j = H_j(\theta) - H_{j-1}(\theta), \quad j \geq 1. \quad (2.12)$$

Then $0 \leq A_j \leq \log 2$ and

$$\int H_m(\theta^{x,i}) d\theta(x) = A_{i+1} + \cdots + A_{i+m}. \quad (2.13)$$

If $e_N(\theta) \leq \alpha$, then

$$\mathbb{P}_{0 \leq i < N, x} \{e_m(\theta^{x,i}) > \beta\} \leq \frac{\alpha + m/N}{\beta}. \quad (2.14)$$

If $m \leq N$ and $e_N(\theta) \geq 1 - \alpha$, then

$$\mathbb{P}_{0 \leq i \leq N-m, x} \{e_m(\theta^{x,i}) < 1 - \beta\} \leq \frac{\alpha N}{\beta(N - m + 1)}. \quad (2.15)$$

Proof

Each A_j is a conditional entropy for one binary refinement, proving its bounds and (2.13). Summing (2.13) for $0 \leq i < N$, each A_j occurs at most m times. Therefore

$$\sum_{i=0}^{N-1} \int H_m(\theta^{x,i}) d\theta(x) \leq m H_{N+m}(\theta) \leq m(H_N(\theta) + m \log 2). \quad (2.16)$$

After division by $Nm \log 2$, the mean of e_m is at most $\alpha + m/N$; Markov proves (2.14).

For (2.15), set $B_j = \log 2 - A_j \geq 0$. The component entropy deficit is

$$m \log 2 - \int H_m(\theta^{x,i}) d\theta(x) = B_{i+1} + \dots + B_{i+m}. \quad (2.17)$$

Summing for $0 \leq i \leq N - m$, each B_j , $1 \leq j \leq N$, occurs at most m times, so the mean normalized deficit is at most

$$\frac{m(N \log 2 - H_N(\theta))}{(N - m + 1)m \log 2} \leq \frac{\alpha N}{N - m + 1}. \quad (2.18)$$

Markov applied to the deficit proves (2.15). $\square$

Corollary II.4.2.5 (C-II.4.07 – Finite atomic variance cascade). *For every $a > 0$, $0 < \varepsilon, \eta < 1$, $m \geq 1$, and prescribed $L_0 \geq 1$, there exist $L \geq L_0$ and $\rho > 0$ such that*

$$\text{Var}(\theta) < \rho \implies \mathbb{P}_{0 \leq i < L, x} \{ \text{Var}(\theta^{x,i}) < a, e_m(\theta^{x,i}) \leq \varepsilon \} \geq 1 - \eta. \quad (2.19)$$

Proof

Choose $s \geq 1$ so that $2^{-2s}/2 < a$. By continuity at zero, choose $c > 0$ so small that

$$\frac{1}{2}(2^{-2s} + 1 - e^{-2c}) < a. \quad (2.20)$$

Choose $L \geq L_0$ large enough that

$$\frac{2}{L\varepsilon} + \frac{2s \log 2}{Lc} < \eta. \quad (2.21)$$

Put $M = L + \max\{m, s\}$. The right side of (2.5), at scale M , tends to $\log 2$ as $\text{Var}(\theta)$ tends to zero. We may therefore choose $\rho > 0$ so that

$$\text{Var}(\theta) < \rho \implies H_M(\theta) < 2 \log 2. \quad (2.22)$$

For $u = m$ and $u = s$, the sliding-block count used in (2.16), now over only $0 \leq i < L$, gives

$$\sum_{i=0}^{L-1} \int H_u(\theta^{x,i}) d\theta(x) \leq u H_{L+u}(\theta) \leq 2u \log 2. \quad (2.23)$$

Thus the probability that $e_m(\theta^{x,i}) > \varepsilon$ is at most $2/(L\varepsilon)$. The probability that $H_s(\theta^{x,i}) \geq c$ is at most $2s \log 2/(Lc)$. Outside their union, (2.6) and (2.20) imply $\text{Var}(\theta^{x,i}) < a$. Equation (2.21) now proves (2.19). $\square$

II.4.3 Repeated convolutions

The repeated-convolution argument must coordinate a variance gap, a convolution power, the scale shift induced by that power, and an atomic window longer than the shift. Choosing these quantities only after inspecting the final measure would make the dependence of

the theorem circular. We instead freeze a finite menu of choices and use a pigeonhole argument to select one. The following notation keeps the selection argument explicit.

Definition II.4.3.1 (II.4.04: Scale windows, shifts, and convolution-power menus). For an integer q and $L \geq 1$, the *length- L scale window* is

$$W_L(q) = \{q, q + 1, \dots, q + L - 1\}.$$

For $A \subset \mathbb{Z}$ and an integer ℓ , put $A - \ell = \{a - \ell : a \in A\}$. The *relative density* of a finite set A in a nonempty finite set S is $|A \cap S|/|S|$.

A *finite convolution-power menu* is a nonempty finite set $K \subset \mathbb{N}$. When a theorem selects a power from K , the menu depends only on the displayed theorem parameters, whereas the selected $k \in K$ may depend on the measure and on the terminal scale.

Lemma II.4.3.2 (L-II.4.08 – Dense windows with a prescribed shift). *Let B, U be finite subsets of $\mathbb{Z}$, let $L \geq 1$, $0 \leq \ell < L$, and suppose*

$$|U \cap W_L(q)| \geq (1 - \alpha)L \quad (q \in B). \quad (3.1)$$

Then there is $B_0 \subset B$ whose windows are pairwise disjoint and cover B . If

$$U' = U \cap \bigcup_{q \in B_0} W_L(q), \quad J = U' \cap (U' - \ell), \quad (3.2)$$

then

$$|J| \geq \max \left\{ 0, 1 - 2\alpha - \frac{\ell}{L} \right\} |B|. \quad (3.3)$$

Proof

Select the least element of B , delete every point of B in its window, and repeat. The selected starting points form B_0 ; their windows are pairwise disjoint and their union contains B . In particular,

$$L|B_0| = \left| \bigcup_{q \in B_0} W_L(q) \right| \geq |B|. \quad (3.4)$$

Fix $q \in B_0$. Among the $L - \ell$ integers $t = q, \dots, q + L - 1 - \ell$, at most αL fail to lie in U , and at most αL have $t + \ell \notin U$, by (3.1). Hence at least

$$L - \ell - 2\alpha L \quad (3.5)$$

of them belong to U' and have their ℓ -translate in U' . They therefore lie in J . The truncated windows for distinct $q \in B_0$ are disjoint. Summing (3.5), using (3.4), and replacing a negative lower bound by zero proves (3.3). $\square$

Proposition II.4.3.3 (P-II.4.09 – Local repeated-convolution smoothing). *Fix $\sigma > 0$, $0 < \varepsilon, \eta < 1$, and $m \geq 1$. There are integers $k \geq 1$ and $\ell \geq 0$ such that, for every $\mu \in \mathcal{P}([0, 1])$, every $i \geq \ell$, and $\nu = \mu^{*k}$,*

$$\lambda_i(\mu) \geq \sigma \implies \mathbb{P}_{y \sim \nu} \{e_m(\nu^{y, i-\ell}) \geq 1 - \varepsilon\} \geq 1 - \eta. \quad (3.6)$$

Proof

Choose $0 < \zeta < \min\{1/2, \varepsilon\eta/4\}$. Apply P-II.4.03 with variance threshold $\sigma/2$, entropy tolerance ζ , and block length m , obtaining p and a lower bound on k . Increase k , if necessary, so that

$$\frac{1}{4k\sigma^2} \leq \zeta, \quad r := \lfloor \log_2 \sqrt{k} \rfloor \geq p, \quad (3.7)$$

and put $\ell = r - p$.

Fix $i \geq \ell$. Sample independently $x_1, \dots, x_k$ with law μ . The level- i raw component of the product measure is

$$\mu_{x_1, i} \times \cdots \times \mu_{x_k, i}. \quad (3.8)$$

Write $\theta_j = \mu^{x_j, i}$ and $W_j = \text{Var}(\theta_j)$. These are independent, lie in $[0, 1/4]$, and have common mean $\lambda_i(\mu)$. If that mean is at least σ , L-II.4.04 and (3.7) give

$$\mathbb{P} \left\{ \sum_{j=1}^k W_j \geq \frac{\sigma k}{2} \right\} \geq 1 - \zeta. \quad (3.9)$$

For every tuple in this event, translate each θ_j by the integer address of its raw parent cell. Each translated law is still carried by an interval of length one, and P-II.4.03 shows that at least $1 - \zeta$ of the level- $p - r = -\ell$ components of their convolution have entropy at least $(1 - \zeta)m \log 2$. The common dilation by 2^{-i} sends the canonical level- $-\ell$ grid exactly to the canonical absolute level $i - \ell$ grid. Thus, if $\eta_\omega = \mu_{x_1, i} * \cdots * \mu_{x_k, i}$ and

$$S(\eta_\omega) = H_{\eta_\omega}(\mathcal{D}_{i-\ell+m} \mid \mathcal{D}_{i-\ell}), \quad (3.10)$$

then on the event (3.9),

$$S(\eta_\omega) \geq (1 - \zeta)^2 m \log 2. \quad (3.11)$$

Iterating the component-convolution disintegration L-II.3.06 gives

$$\nu = \mathbb{E} \eta_\omega. \quad (3.12)$$

Conditional entropy is concave for arbitrary barycenters, also by L-II.3.06. Therefore (3.9)–(3.12) imply

$$H_\nu(\mathcal{D}_{i-\ell+m} \mid \mathcal{D}_{i-\ell}) \geq (1 - \zeta)^3 m \log 2. \quad (3.13)$$

The mean component deficit is consequently at most $[1 - (1 - \zeta)^3]m \log 2 \leq 3\zeta m \log 2$. Markov shows that the proportion of components with deficit greater than $\varepsilon m \log 2$ is at most

$$\frac{3\zeta}{\varepsilon} < \eta. \quad (3.14)$$

This is (3.6). $\square$

Theorem II.4.3.4 (T-II.4.10 — Finite-menu repeated-convolution dichotomy). *Fix $0 < \kappa, \varepsilon_u, \varepsilon_a < 1$ and $m_u, m_a \geq 1$. There are a finite set $K \subset \mathbb{N}$ and N_0 such that every $\mu \in \mathcal{P}([0, 1])$ and $N \geq N_0$ admit $k \in K$ and disjoint $I, J \subset \{0, \dots, N-1\}$ with*

$$|I \cup J| \geq (1 - \kappa)N, \quad (3.15)$$

for which, with $\nu = \mu^{*k}$,

$$\begin{aligned} \mathbb{P}_{x \sim \nu} \{e_{m_u}(\nu^{x,i}) \geq 1 - \varepsilon_u\} &\geq 1 - \kappa \quad (i \in I), \\ \mathbb{P}_{x \sim \mu} \{e_{m_a}(\mu^{x,i}) \leq \varepsilon_a\} &\geq 1 - \kappa \quad (i \in J). \end{aligned} \quad (3.16)$$

Proof

Proof strategy. Freeze a descending finite list of variance thresholds. For each adjacent gap, freeze both a smoothing power and a sufficiently long atomic window. After the measure is given, select a sparsely occupied gap; high-variance scales yield uniform components, low-variance scales yield atomic components, and a common shift makes the two scale sets disjoint.

Replacing κ by a smaller number strengthens the conclusion, so assume $\kappa < 1/2$. Set

$$R = \left\lceil \frac{8}{\kappa} \right\rceil, \quad \sigma_0 = \frac{1}{4}. \quad (3.17)$$

We recursively construct positive numbers $\sigma_0 > \sigma_1 > \dots > \sigma_R$ and, for each $1 \leq s \leq R$, integers k_s, ℓ_s, L_s .

Suppose σ_{s-1} is fixed. Apply P-II.4.09 with parameters

$$(\sigma, \varepsilon, \eta) = (\sigma_{s-1}, \varepsilon_u, \kappa/4) \quad (3.18)$$

to obtain k_s, ℓ_s . Put

$$a_s = \min\{\kappa/4, \sigma_{s-1}\}, \quad \zeta_s = \frac{a_s \kappa}{256}. \quad (3.19)$$

Apply C-II.4.07 with variance target $\sigma_{s-1}/2$, atomic parameters ε_a, m_a , error ζ_s , and the prescribed lower bound $32(\ell_s + 1)/\kappa$ on its window length. This gives L_s, ρ_s with

$$\frac{\ell_s}{L_s} \leq \frac{\kappa}{32}. \quad (3.20)$$

Finally choose

$$0 < \sigma_s < \min\{\sigma_{s-1}/2, \rho_s \zeta_s\}. \quad (3.21)$$

This finishes the finite recursion. Set

$$K = \{k_1, \dots, k_R\}. \quad (3.22)$$

Now fix μ, N and abbreviate $\lambda_i = \lambda_i(\mu)$. The sets

$$G_s = \{0 \leq i < N : \sigma_s \leq \lambda_i < \sigma_{s-1}\} \quad (1 \leq s \leq R) \quad (3.23)$$

are disjoint. Choose s so that

$$|G_s| \leq N/R \leq \kappa N/8. \quad (3.24)$$

For this s , write $\ell = \ell_s$, $L = L_s$, $k = k_s$, and define

$$A = \{i : \lambda_i \geq \sigma_{s-1}\}, \quad B = \{i : \lambda_i < \sigma_s\}, \quad (3.25)$$

with $0 \leq i < N$. Then A, B are disjoint and

$$|A| + |B| \geq (1 - \kappa/8)N. \quad (3.26)$$

Let

$$A_0 = A \cap \{\ell, \dots, N-1\}, \quad I = A_0 - \ell. \quad (3.27)$$

By [P-II.4.09](#), every $i \in I$ satisfies the first line of [\(3.16\)](#), with the stronger exceptional bound $\kappa/4$.

We next treat B . Define, for every nonnegative scale t ,

$$f(t) = \mathbb{P}_{x \sim \mu} \{\text{Var}(\mu^{x,t}) < \sigma_{s-1}/2, e_{m_a}(\mu^{x,t}) \leq \varepsilon_a\}, \quad (3.28)$$

and

$$U = \{t : f(t) > 1 - a_s\}. \quad (3.29)$$

Fix $q \in B$ with $q + L - 1 < N$. Markov and [\(3.21\)](#) show that a random level- q component $\theta = \mu^{x,q}$ satisfies

$$\mathbb{P}\{\text{Var}(\theta) \geq \rho_s\} \leq \frac{\lambda_q}{\rho_s} < \zeta_s. \quad (3.30)$$

For every remaining parent component, [C-II.4.07](#) gives the desired event on at least $1 - \zeta_s$ of the joint relative-scale/component choices in its length- L window. The tower property for components therefore gives

$$\frac{1}{L} \sum_{t=q}^{q+L-1} f(t) > (1 - \zeta_s)^2 \geq 1 - 2\zeta_s. \quad (3.31)$$

At a scale outside U , $1 - f(t) \geq a_s$. Hence [\(3.31\)](#) implies

$$|U \cap W_L(q)| \geq \left(1 - \frac{2\zeta_s}{a_s}\right)L = \left(1 - \frac{\kappa}{128}\right)L. \quad (3.32)$$

Apply L-II.4.08 to

$$B_0 = B \cap \{0, \dots, N - L\}, \quad U_N = U \cap \{0, \dots, N - 1\}, \quad L, \quad \ell, \quad \alpha = \kappa/128, \quad (3.33)$$

and call the resulting shifted intersection J . Then

$$|J| \geq \left(1 - \frac{\kappa}{64} - \frac{\ell}{L}\right)|B_0| \geq (1 - 3\kappa/64)|B_0|. \quad (3.34)$$

Every $j \in J$ belongs to U , so the second line of (3.16) holds because $a_s \leq \kappa/4$.

It remains to prove disjointness. If $j \in I$, then

$$\lambda_{j+\ell} \geq \sigma_{s-1}. \quad (3.35)$$

If $j \in J$, then $j + \ell \in U$. Since component variances never exceed $1/4$, (3.28)–(3.29) give

$$\lambda_{j+\ell} \leq \frac{\sigma_{s-1}}{2} + \frac{a_s}{4} \leq \frac{3\sigma_{s-1}}{4} < \sigma_{s-1}, \quad (3.36)$$

contradicting (3.35). Thus $I \cap J$ is empty.

Finally, (3.26), (3.27), and (3.34) imply

$$\begin{aligned} |I| + |J| &\geq |A| + |B| - (\ell + L) - \frac{3\kappa}{64}|B| \\ &\geq \left(1 - \frac{\kappa}{8} - \frac{3\kappa}{64}\right)N - (\ell + L). \end{aligned} \quad (3.37)$$

The construction contains only finitely many pairs (ℓ_s, L_s) . Choose N_0 so large that $(\ell_s + L_s)/N \leq \kappa/8$ for all s and $N \geq N_0$. Then the right side of (3.37) is greater than $(1 - \kappa)N$, proving (3.15). $\square$

II.4.4 Diminishing entropy increments

The structural dichotomy above is obtained for a repeated convolution v^{*k} , but the final inverse theorem concerns the original factor v . Kaimanovich–Vershik diminishing increments provide the bridge: the entropy gain after k additions is at most k times the first gain. Dyadic rounding does not commute exactly with addition, so we isolate and bound the accumulated rounding displacement before transferring the group inequality back to dyadic entropy.

Theorem II.4.4.1 (T-II.4.11 – Kaimanovich–Vershik diminishing increments). *Let Γ be a countable abelian group and let $\mu, \nu \in \mathcal{P}(\Gamma)$ have finite entropy. Put*

$$a_k = H(\mu * \nu^{*k}), \quad d_k = a_{k+1} - a_k. \quad (4.1)$$

Then $d_{k+1} \leq d_k$ and

$$a_k \leq a_0 + k(a_1 - a_0). \quad (4.2)$$

Proof

Let X_0 have law μ ; let $Z_1, Z_2, \dots$ be independent with common law ν , also independent of X_0 ; and put

$$X_n = X_0 + Z_1 + \dots + Z_n. \quad (4.3)$$

All entropies below are finite because $H(X_n) \leq H(X_0, Z_1, \dots, Z_n)$. Conditioned on $Z_1 = g$, the law of X_n is a translate of the law of X_{n-1} . Hence, for $n \geq 1$,

$$\begin{aligned} H(Z_1 | X_n) &= H(Z_1) + H(X_n | Z_1) - H(X_n) \\ &= H(\nu) + a_{n-1} - a_n. \end{aligned} \quad (4.4)$$

Given X_n , the next state is $X_{n+1} = X_n + Z_{n+1}$, and Z_{n+1} is independent of (Z_1, X_n) . Therefore Z_1 and X_{n+1} are conditionally independent given X_n , so

$$H(Z_1 | X_n) = H(Z_1 | X_n, X_{n+1}) \leq H(Z_1 | X_{n+1}). \quad (4.5)$$

Insert (4.4), first with n and then with $n + 1$, into (4.5):

$$a_n - a_{n-1} \geq a_{n+1} - a_n. \quad (4.6)$$

Thus the increments decrease. Summing $a_{j+1} - a_j \leq a_1 - a_0$ for $0 \leq j < k$ proves (4.2). Notice that the second term in (4.2) is $a_1 - a_0 = H(\mu * \nu) - H(\mu)$; replacing the final $H(\mu)$ by $H(\nu)$ would not follow from the increment statement. $\square$

Definition II.4.4.2 (II.4.05: Dyadic discretization). For $n \in \mathbb{Z}$, define the dyadic lattice and left-endpoint rounding map by

$$M_n = 2^{-n}\mathbb{Z}, \quad \sigma_n(x) = 2^{-n}\lfloor 2^n x \rfloor.$$

The *level- n discretization* of a probability μ is $(\sigma_n)_*\mu$. A probability is *level- n discrete* if it is carried by M_n . Each cell of $\mathcal{D}_n$ contains exactly one point of M_n , namely its left endpoint.

Lemma II.4.4.3 (L-II.4.12 — Exact dyadic discretization error). For $n \in \mathbb{Z}$, let

$$\sigma_n(x) = 2^{-n}\lfloor 2^n x \rfloor. \quad (4.7)$$

If $\mu_1, \dots, \mu_\ell$ are probabilities on $\mathbb{R}$ and $H_n(\mu_j) < \infty$ for every j , then

$$|H_n(\mu_1 * \cdots * \mu_\ell) - H_n((\sigma_n)_* \mu_1 * \cdots * (\sigma_n)_* \mu_\ell)| \leq \log(2\ell + 1). \quad (4.8)$$

Also $H((\sigma_n)_* \mu) = H_n(\mu)$ whenever either side is finite.

Proof

The dyadic convolution upper bound T-II.3.03, iterated $\ell - 1$ times, shows that the first entropy in (4.8) is finite. The rounded variables have finite atom entropies $H_n(\mu_j)$, so their sum also has finite entropy.

On the product probability space, define

$$F(x_1, \dots, x_\ell) = \sum_{j=1}^{\ell} x_j, \quad G(x_1, \dots, x_\ell) = \sum_{j=1}^{\ell} \sigma_n(x_j). \quad (4.9)$$

Since $0 \leq x - \sigma_n(x) < 2^{-n}$,

$$|F - G| < \ell 2^{-n}. \quad (4.10)$$

The bounded-perturbation estimate P-II.1.09, with $C = \ell$ in dimension one, gives (4.8). The points of $2^{-n}\mathbb{Z}$ are in one-to-one correspondence with the half-open cells of $\mathcal{D}_n$, so the atom entropy of $(\sigma_n)_* \mu$ is exactly $H_n(\mu)$. $\square$

Proposition II.4.4.4 (P-II.4.13 – Dyadic Kaimanovich–Vershik inequality). *For $\mu, \nu \in \mathcal{P}(\mathbb{R})$, $n \geq 1$, and $k \geq 1$, assume $H_n(\mu), H_n(\nu) < \infty$. Then*

$$\begin{aligned} H_n(\mu * \nu^{*k}) &\leq H_n(\mu) + k(H_n(\mu * \nu) - H_n(\mu)) \\ &\quad + k \log 5 + \log(2k + 3). \end{aligned} \quad (4.11)$$

Proof

Put $\tilde{\mu} = (\sigma_n)_* \mu$ and $\tilde{\nu} = (\sigma_n)_* \nu$. These are probabilities on the countable abelian group $2^{-n}\mathbb{Z}$. T-II.4.11 gives

$$H(\tilde{\mu} * \tilde{\nu}^{*k}) \leq H(\tilde{\mu}) + k(H(\tilde{\mu} * \tilde{\nu}) - H(\tilde{\mu})). \quad (4.12)$$

By L-II.4.12,

$$\begin{aligned} H(\tilde{\mu}) &= H_n(\mu), \\ H(\tilde{\mu} * \tilde{\nu}) &\leq H_n(\mu * \nu) + \log 5, \\ H_n(\mu * \nu^{*k}) &\leq H(\tilde{\mu} * \tilde{\nu}^{*k}) + \log(2k + 3). \end{aligned} \quad (4.13)$$

The last convolution has $k + 1$ summands, which explains $2k + 3$. Substitution of (4.13) into (4.12) proves (4.11). $\square$

Lemma II.4.4.5 (L-II.4.14 – Support-free corrected local growth). *Let $\mu \in \mathcal{P}([0, 1])$, let τ be a compactly supported probability on $\mathbb{R}$, and let $N, m \geq 1$. Then*

$$\frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_{x,y} \gamma_m(\mu^{x,i}; \tau^{y,i}) \leq h_N(\mu * \tau) - h_N(\mu) + \frac{2}{m} + \frac{3(m-1)}{2N}. \quad (4.14)$$

Proof

The component disintegration applies to arbitrary Borel probabilities; see L-II.3.06. Conditional-entropy concavity therefore gives, at every $i \geq 0$,

$$H_{\mu * \tau}(\mathcal{D}_{i+m} \mid \mathcal{D}_i) \geq \mathbb{E}_{x,y} H_{\mu_{x,i} * \tau_{y,i}}(\mathcal{D}_{i+m} \mid \mathcal{D}_i). \quad (4.15)$$

After translating and rescaling the two parent cells, the measure inside the expectation becomes $\mu^{x,i} * \tau^{y,i}$, supported on $[0, 2)$. Its level-zero cell label has entropy at most $\log 2$. Hence

$$H_{\mu * \tau}(\mathcal{D}_{i+m} \mid \mathcal{D}_i) \geq \mathbb{E}_{x,y} H_m(\mu^{x,i} * \tau^{y,i}) - \log 2. \quad (4.16)$$

Put $K_j = H_j(\mu * \tau)$. Summing the left side of (4.16) gives

$$\sum_{i=0}^{N-1} (K_{i+m} - K_i) = \sum_{r=0}^{m-1} K_{N+r} - \sum_{r=0}^{m-1} K_r. \quad (4.17)$$

Because one dyadic refinement has at most two children, $K_{N+r} \leq K_N + r \log 2$; dropping the nonnegative second sum in (4.17) yields

$$K_N \geq \frac{1}{m} \sum_{i=0}^{N-1} (K_{i+m} - K_i) - \frac{m-1}{2} \log 2. \quad (4.18)$$

Combine (4.16)–(4.18) and divide by $N \log 2$:

$$h_N(\mu * \tau) \geq \frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E} e_m(\mu^{x,i} * \tau^{y,i}) - \frac{1}{m} - \frac{m-1}{2N}. \quad (4.19)$$

T-II.2.02 for the unit-interval measure μ gives

$$h_N(\mu) \leq \frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E} e_m(\mu^{x,i}) + \frac{m-1}{N}. \quad (4.20)$$

Subtract (4.20) from (4.19), and use

$$\gamma_m(\sigma; \theta) = e_m(\sigma * \theta) - e_m(\sigma) + \frac{1}{m}. \quad (4.21)$$

Rearrangement is exactly (4.14). Compact support guarantees finiteness of all entropies; no bound on its diameter entered the constants. $\square$

II.4.5 The inverse theorem

At a scale where a repeated copy of the second factor is nearly uniform but the first factor is not, the exact carry inequality forces positive corrected local growth. If such scales have positive density, the support-free local-to-global estimate gives positive global growth. The Kaimanovich–Vershik inequality then says that this growth would already have been visible after adding the original second factor. Thus small global growth leaves only the uniform alternative for the first factor or the atomic alternative for the second.

The proof first works with one large master block. We later pass to the requested shorter windows by heredity and choose one common shift for the two structural scale sets; the common shift preserves their disjointness.

Definition II.4.5.1 (II.4.06: Inverse parameters and finite-scale entropy porosity). An *inverse-theorem parameter card* is

$$\mathfrak{p} = (\eta, \varepsilon_u, m_u, \varepsilon_a, m_a),$$

where $0 < \eta, \varepsilon_u, \varepsilon_a < 1$ and $m_u, m_a \in \mathbb{N}$. Here η controls both the exceptional proportion of scales and the exceptional component mass; ε_u, m_u control the uniform alternative; and ε_a, m_a control the entropy-atomic alternative.

Let $0 < \alpha \leq 1$, $m, N \geq 1$, and $0 \leq \rho < 1$. A probability μ on $[0, 1)$ is (α, m, ρ) *entropy porous through level N* if

$$\frac{1}{N} \sum_{i=0}^{N-1} \mu\{x : e_m(\mu^{x,i}) \leq 1 - \alpha\} \geq 1 - \rho.$$

Thus α is the entropy deficit, m the observation window, and ρ the exceptional proportion in the joint scale–component sampling.

The large-block core

We first prove the following internal claim.

Large-block claim. For every $0 < a < 1/4$ there is $M_*(a)$ such that, for every $M \geq M_*(a)$, there are $\delta_*(a, M) > 0$ and $N_*(a, M)$ with the following property. If $N \geq N_*$, $\mu, \nu \in \mathcal{P}([0, 1))$, and

$$h_N(\mu * \nu) < h_N(\mu) + \delta_*, \quad (5.1)$$

then there are disjoint $I, J \subset \{0, \dots, N-1\}$, covering at least $(1-a)N$ scales, such that

$$\begin{aligned} \mathbb{P}_x\{e_M(\mu^{x,i}) \geq 1-a\} &> 1-a \quad (i \in I), \\ \mathbb{P}_y\{e_M(\nu^{y,i}) \leq a\} &> 1-a \quad (i \in J). \end{aligned} \quad (5.2)$$

Proof of the large-block claim

Proof strategy. Apply the finite-menu dichotomy to a suitable power $\tau = \nu^{*k}$. If too many of its uniform scales fail to be uniform for μ , the exact carry inequality forces positive

average local growth. The local-to-global estimate and the dyadic Kaimanovich–Vershik inequality then contradict the assumed one-step small growth.

Put $b = a/4$. Apply T-II.4.10 to ν , with coverage/reliability b , uniform tolerance b , atomic tolerance a , and both block lengths equal to M . It supplies a finite menu K such that, for some selected $k \in K$ and $\tau = \nu^{*k}$, there are disjoint sets A, B satisfying

$$|A \cup B| \geq (1 - b)N, \quad (5.3)$$

$$\begin{aligned} \mathbb{P}_y\{e_M(\tau^{y,i}) \geq 1 - b\} &\geq 1 - b \quad (i \in A), \\ \mathbb{P}_y\{e_M(\nu^{y,i}) \leq a\} &\geq 1 - b \quad (i \in B). \end{aligned} \quad (5.4)$$

Let

$$I = \{i \in A : \mathbb{P}_x(e_M(\mu^{x,i}) \geq 1 - a) > 1 - a\}, \quad J = B. \quad (5.5)$$

If $|I \cup J| \geq (1 - a)N$, (5.2) follows because $b < a$. Suppose instead that this inequality fails, and put $A_1 = A \setminus I$. From (5.3),

$$|A_1| > (a - b)N. \quad (5.6)$$

For $i \in A_1$, the probability that $\mu^{x,i}$ is not (a, M) uniform is at least a ; by (5.4), the probability that the independent component $\tau^{y,i}$ is (b, M) uniform is at least $1 - b$. Their intersection has probability at least $a - b$. On this intersection,

$$e_M(\tau^{y,i}) - e_M(\mu^{x,i}) > a - b. \quad (5.7)$$

The exact carry lower bound T-II.3.03 gives

$$\begin{aligned} \gamma_M(\mu^{x,i}; \tau^{y,i}) &= e_M(\mu^{x,i} * \tau^{y,i}) - e_M(\mu^{x,i}) + M^{-1} \\ &\geq e_M(\tau^{y,i}) - e_M(\mu^{x,i}). \end{aligned} \quad (5.8)$$

Corrected growth is nonnegative off the event in (5.7). Equations (5.6)–(5.8) therefore yield

$$\frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_{x,y} \gamma_M(\mu^{x,i}; \tau^{y,i}) > (a - b)^3 =: c_a. \quad (5.9)$$

By L-II.4.14,

$$h_N(\mu * \tau) - h_N(\mu) > c_a - \frac{2}{M} - \frac{3(M-1)}{2N}. \quad (5.10)$$

Choose $M_*(a)$ so that $2/M \leq c_a/8$ for $M \geq M_*$, and then take N large enough that the last term in (5.10) is at most $c_a/8$. Thus

$$h_N(\mu * \nu^{*k}) - h_N(\mu) > \frac{3c_a}{4}. \quad (5.11)$$

On the other hand, P-II.4.13 gives

$$h_N(\mu * v^{*k}) - h_N(\mu) \leq k(h_N(\mu * v) - h_N(\mu)) + \frac{k \log 5 + \log(2k + 3)}{N \log 2}. \quad (5.12)$$

Let $K_{\max} = \max K$, choose

$$\delta_* \leq \frac{c_a}{8K_{\max}}, \quad (5.13)$$

and enlarge N_* so that the last term of (5.12) is at most $c_a/8$ for every $k \in K$. Under (5.1), (5.12) is then less than $c_a/4$, contradicting (5.11). The supposition following (5.5) is impossible, and the large-block claim follows. $\square$

Theorem II.4.5.2 (T-II.4.15 – Full normalized inverse theorem on the line). *Fix an inverse-parameter card*

$$\mathfrak{p} = (\eta, \varepsilon_u, m_u, \varepsilon_a, m_a). \quad (5.14)$$

There are $\delta > 0$ and N_0 with the following property. If $N \geq N_0$, $\mu, \nu \in \mathcal{P}([0, 1])$, and

$$h_N(\mu * \nu) < h_N(\mu) + \delta,$$

then there are disjoint sets $I, J \subset \{0, \dots, N-1\}$ with $|I \cup J| \geq (1 - \eta)N$ such that

$$\mathbb{P}_{x \sim \mu} \{e_{m_u}(\mu^{x,i}) \geq 1 - \varepsilon_u\} > 1 - \eta \quad (i \in I),$$

$$\mathbb{P}_{y \sim \nu} \{e_{m_a}(\nu^{y,i}) \leq \varepsilon_a\} > 1 - \eta \quad (i \in J).$$

Proof

Proof strategy. Choose a large master block for the large-block claim. Heredity passes uniformity and atomicity to most shorter subblocks. Averaging the two counts together selects one common shift, preserving disjointness while meeting the prescribed parameter card.

Replacing η by a smaller value strengthens the conclusion, so assume $\eta < 1/2$. Put

$$\varepsilon_0 = \min\{\varepsilon_u, \varepsilon_a\}. \quad (5.15)$$

Choose $0 < a < \min\{\eta/4, 1/4\}$ so small that

$$\beta_0 := \frac{a + 3a/\varepsilon_0}{\eta} < \frac{\eta}{4}. \quad (5.16)$$

Choose a master block length

$$M \geq \max\{M_*(a), 2m_u + 2, 2m_a + 2\}, \quad (5.17)$$

and set $L = \lfloor M/2 \rfloor$. Apply the large-block claim with parameters a, M . Under its small-growth hypothesis, it gives disjoint U_1, U_2 covering at least $(1 - a)N$ scales such that a proportion greater than $1 - a$ of the level- i parent components are (a, M) uniform for $i \in U_1$, and (a, M) entropy-atomic for $i \in U_2$.

Consider $i \in U_1$ and a good parent $\theta = \mu^{x,i}$. For $0 \leq t < L$, the proof of L-II.4.06, restricted to these L starting scales, gives

$$\frac{1}{L} \sum_{t=0}^{L-1} \mathbb{E}_y[1 - e_{m_u}(\theta^{y,t})] \leq \frac{aM}{L} \leq 3a. \quad (5.18)$$

The inequality $L + m_u - 1 \leq M$ follows from (5.17). Markov shows that, among the joint relative-scale/child choices of this good parent, the failure of (ε_u, m_u) uniformity has probability at most $3a/\varepsilon_u$. Allowing the bad parent components, the total failure probability is at most

$$\zeta := a + 3a/\varepsilon_0. \quad (5.19)$$

For $i \in U_2$, the corresponding restricted proof of the atomic half of L-II.4.06 gives

$$\frac{1}{L} \sum_{t=0}^{L-1} \mathbb{E}_y e_{m_a}(\theta^{y,t}) \leq \frac{aM}{L} \leq 3a \quad (5.20)$$

for every good (a, M) atomic parent. Hence the joint failure probability for (ε_a, m_a) atomicity is also at most ζ .

Define absolute good-scale sets

$$\begin{aligned} V_1 &= \{j : \mathbb{P}_x(e_{m_u}(\mu^{x,j}) \geq 1 - \varepsilon_u) > 1 - \eta\}, \\ V_2 &= \{j : \mathbb{P}_y(e_{m_a}(\nu^{y,j}) \leq \varepsilon_a) > 1 - \eta\}. \end{aligned} \quad (5.21)$$

The tower property for components converts (5.19)–(5.20) into

$$|V_r \cap W_L(i)| \geq (1 - \zeta/\eta)L = (1 - \beta_0)L \quad (i \in U_r, r = 1, 2). \quad (5.22)$$

Remove from U_1, U_2 the starting scales larger than $N - L$; in total this removes at most L integers. Averaging over $t = 0, \dots, L - 1$, (5.22) gives

$$\begin{aligned} \frac{1}{L} \sum_{t=0}^{L-1} (|\{i \in U_1 : i + t \in V_1\}| + |\{i \in U_2 : i + t \in V_2\}|) \\ \geq (1 - \beta_0)(|U_1| + |U_2| - L). \end{aligned} \quad (5.23)$$

Choose one t_0 attaining at least the right side and set

$$I = (U_1 + t_0) \cap V_1, \quad J = (U_2 + t_0) \cap V_2, \quad (5.24)$$

using the trimmed sets. Translation is injective and U_1, U_2 are disjoint, so I, J are disjoint. By (5.16),

$$|I| + |J| \geq (1 - \beta_0)((1 - a)N - L). \quad (5.25)$$

For N large enough that $L/N < \eta/4$, the right side exceeds $(1 - \eta)N$. Membership in V_1, V_2 gives the required component probabilities. Taking $\delta = \delta_*(a, M)$ and enlarging N_0 to cover all preceding lower bounds proves the theorem. $\square$

II.4.5.1 Geometric-atomic and self-convolution forms

Fix $0 < \xi < 1$. Choose m_a and ε_a so that

$$2^{-m_a} \leq \xi, \quad 2^{-\varepsilon_a m_a} \geq 1 - \xi. \quad (5.26)$$

L-II.4.05 then converts every (ε_a, m_a) entropy-atomic component in T-II.4.15 into a geometrically ξ -atomic component. This is the geometric form of Hochman's inverse theorem [Hoc14, Theorem 2.7]. Taking $\nu = \mu$ gives the self-convolution form [Hoc14, Theorem 2.9]; no new proof input is needed.

Entropy porosity rules out the uniform alternative for the first measure on most scale-component pairs. If the second measure has entropy density above the porosity exceptional proportion by a positive margin, it cannot be atomic on all remaining scales. The inverse theorem therefore forces a definite entropy gain.

Theorem II.4.5.3 (T-II.4.16 – Entropy porosity forces growth above a margin). *Fix $0 < \alpha \leq 1$, $m \geq 1$, and $0 \leq \rho < \beta \leq 1$. Then there are $\delta > 0, N_0$ such that, for $N \geq N_0$ and $\mu, \nu \in \mathcal{P}([0, 1])$,*

$$\frac{1}{N} \sum_{i=0}^{N-1} \mathbb{P}_x \{e_m(\mu^{x,i}) \leq 1 - \alpha\} \geq 1 - \rho, \quad h_N(\nu) \geq \beta \quad (5.27)$$

imply

$$h_N(\mu * \nu) \geq h_N(\mu) + \delta. \quad (5.28)$$

Proof

Choose $\eta_0 > 0$ and $\varepsilon_a > 0$ so small that

$$\frac{\rho}{1 - \eta_0} + 2\eta_0 + \varepsilon_a < \beta. \quad (5.29)$$

This is possible because $\rho < \beta$. Apply T-II.4.15 with card

$$(\eta_0, \alpha/2, m, \varepsilon_a, 1), \quad (5.30)$$

and let δ, N_0 be its constants. Suppose (5.28) fails. Then the inverse theorem gives disjoint I, J , covering at least $(1 - \eta_0)N$ scales.

The event $e_m \geq 1 - \alpha/2$ is disjoint from $e_m \leq 1 - \alpha$. By (5.27), its mean probability over all scales is at most ρ . At every $i \in I$ its probability is greater than $1 - \eta_0$; hence

$$\frac{|I|}{N} \leq \frac{\rho}{1 - \eta_0}. \quad (5.31)$$

Since I, J are disjoint and their union misses at most $\eta_0 N$ scales,

$$\frac{|J^c|}{N} \leq \eta_0 + \frac{\rho}{1 - \eta_0}. \quad (5.32)$$

For $i \in J$, the expected value of $e_1(v^{y,i})$ is at most $\varepsilon_a + \eta_0$; outside J it is at most one. The exact block-one local entropy identity T-II.2.02 now gives

$$\begin{aligned} h_N(v) &= \frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_y e_1(v^{y,i}) \\ &\leq \varepsilon_a + \eta_0 + \frac{|J^c|}{N} \\ &\leq \varepsilon_a + 2\eta_0 + \frac{\rho}{1 - \eta_0} < \beta. \end{aligned} \quad (5.33)$$

contradicting (5.27). Thus (5.28) holds. $\square$

II.4.6 Cylinder laws and overlap information

For a homogeneous dyadic iterated function system, concatenating words turns exactly into convolution of their cylinder-location laws. The uncertainty of a word after its dyadic cluster has been revealed measures the address information lost at that resolution. This makes the inverse theorem applicable to a prefix law and a rescaled suffix law without any additional approximation.

Definition II.4.6.1 (II.4.07: Homogeneous dyadic cylinders and clusters). Fix an integer $s \geq 1$, a finite alphabet Λ , probabilities $p_i > 0$ summing to one, and maps

$$f_i(x) = 2^{-s}x + t_i$$

that send $[0, 1)$ into itself. This is a *homogeneous dyadic IFS*. For $u = u_1 \cdots u_q \in \Lambda^q$, define $f_u = f_{u_1} \circ \cdots \circ f_{u_q}$ and put

$$p_u = \prod_{j=1}^q p_{u_j}, \quad t_u = f_u(0) = t_{u_1} + 2^{-s}t_{u_2} + \cdots + 2^{-s(q-1)}t_{u_q}.$$

The *level- q cylinder-location law* is

$$\kappa_q = \sum_{u \in \Lambda^q} p_u \delta_{t_u};$$

equal locations are aggregated as one atom. For $N \geq 1$, an *N -dyadic cylinder cluster* is a nonempty fiber of

$$u \mapsto \lfloor 2^N t_u \rfloor.$$

It records words whose cylinder base points lie in the same half-open level- N dyadic cell; it does not assert equality of the cylinder maps.

Proposition II.4.6.2 (P-II.4.17 — Cylinder clustering and the cylinder inverse theorem).

Let the homogeneous dyadic IFS and its location laws κ_q be as in D-II.4.07. For $q, r \geq 1$,

$$\kappa_{q+r} = \kappa_q * (S_{2^{-sq}})_* \kappa_r, \quad S_c(x) = cx. \quad (6.1)$$

If U_q is a random word with probabilities (p_u) and $C_N = \lfloor 2^N t_{U_q} \rfloor$, then

$$H_N(\kappa_q) = H(C_N), \quad qH(p) - H_N(\kappa_q) = H(U_q | C_N). \quad (6.2)$$

If μ is the invariant measure of the IFS, then also

$$\mu = \kappa_q * (S_{2^{-sq}})_* \mu. \quad (6.3)$$

Fix an inverse-parameter card $\mathfrak{p} = (\eta, \varepsilon_u, m_u, \varepsilon_a, m_a)$, and let δ, N_0 be the constants from T-II.4.15. If $N \geq N_0$ and

$$h_N(\kappa_{q+r}) < h_N(\kappa_q) + \delta, \quad (6.6)$$

then there are disjoint $I, J \subset \{0, \dots, N-1\}$, covering at least $(1-\eta)N$ scales, such that

$$\mathbb{P}_{x \sim \kappa_q} \{e_{m_u}(\kappa_q^{x,i}) \geq 1 - \varepsilon_u\} > 1 - \eta \quad (i \in I),$$

$$\mathbb{P}_{y \sim (S_{2^{-sq}})_* \kappa_r} \{e_{m_a}(((S_{2^{-sq}})_* \kappa_r)^{y,i}) \leq \varepsilon_a\} > 1 - \eta \quad (i \in J).$$

At every $i \geq sq$, a rescaled level- i component of the scaled suffix law agrees with the corresponding rescaled level- $i - sq$ component of κ_r .

Proof

Proof strategy. First compute the convolution and entropy identities from word concatenation. Then apply the inverse theorem to the prefix and scaled suffix laws and identify the scaled components by the exact dyadic change of scale.

For words $u \in \Lambda^q, v \in \Lambda^r$, direct composition gives

$$t_{uv} = f_{uv}(0) = t_u + 2^{-sq} t_v, \quad p_{uv} = p_u p_v. \quad (6.4)$$

Summing the point masses in (6.4) proves (6.1). The law of C_N is the dyadic rounding of κ_q , hence its entropy is $H_N(\kappa_q)$. The letters of U_q are independent with law p , so $H(U_q) = qH(p)$. Since C_N is a function of U_q , the finite entropy chain rule gives

$$qH(p) = H(U_q) = H(C_N) + H(U_q | C_N), \quad (6.5)$$

which is (6.2). Iterating the invariance identity $\mu = \sum_i p_i(f_i)_* \mu$ and using $f_u(x) = t_u + 2^{-sq}x$ proves (6.3).

Inverse-theorem step.

Apply T-II.4.15 to the two factors in (6.1). It yields disjoint sets I, J , covering at least $(1-\eta)N$ scales, on which the prefix-law components are (ε_u, m_u) uniform and the scaled suffix-

law components are (ε_a, m_a) entropy-atomic, respectively, with component exceptional mass below η .

For $i \geq sq$,

$$\lfloor 2^i(2^{-sq}x) \rfloor = \lfloor 2^{i-sq}x \rfloor. \quad (6.7)$$

The canonical rescaling maps also agree after this identification. Thus a random rescaled level- i component of $(S_{2^{-sq}})_* \kappa_r$ is exactly a random rescaled level- $i - sq$ component of κ_r . This proves the printed cylinder inverse conclusion.

Equation (6.2) supplies the promised overlap interpretation: its right side is the average uncertainty of the word after its N -dyadic cylinder cluster is revealed. It is zero precisely when every positive-probability cluster contains one word, and is positive exactly when resolution 2^{-N} fails to distinguish some positive-probability cylinder addresses. $\square$

The identity (6.2) is the bridge to overlap theory: it measures how much symbolic address information is invisible at resolution 2^{-N} . Turning persistent loss of address information into dimension drop or super-exponential concentration requires the separation and dimension arguments developed in later chapters.

Chapter II.5

Higher-Dimensional Inverse Entropy

Geometric intuition

In higher dimension, “uniform” and “atomic” are replaced by saturation and concentration along subspaces. This chapter builds the subspace calculus, proves the convolution and action inverse mechanisms used later, and states the precise research inputs needed to connect these mechanisms to the deepest applications.

The central question of this chapter is geometric: what structure can make convolution, a group action, or a smooth two-input map create very little new entropy? In several dimensions the answer cannot be encoded by a binary atomic–uniform dichotomy. A measure may be spread along some directions and thin across the others, so the relevant object is a subspace of available entropy.

The argument has three layers. The first develops the Euclidean inverse theorem for convolution. The second localizes an isometry action and recovers its group parameters from spatial images. The third combines finite-window linearization with the Euclidean theorem to treat nondegenerate smooth maps and matrix operations. Definitions are introduced where these layers first need them; theorem statements are placed immediately before their proofs.

All logarithms are natural. Put $r = \dim V$ and $W = V^\perp$ whenever a subspace V is fixed. Constants are enlarged only when their displayed subscripts allow it.

II.5.1 Product-coordinate entropy

A rotated product grid almost never agrees with the ambient dyadic grid. What matters is that their cells meet with uniformly bounded multiplicity. Consequently saturation is conditional entropy along V , up to a bounded unnormalized error. For a measure on a horizontal line in $\mathbb{R}^2$, the transverse projection is atomic and this conditional entropy is simply the entropy in the horizontal coordinate.

Definition II.5.1.1 (II.5.01: Subspace neighborhoods and intrinsic grids). For a nonempty set $A \subset \mathbb{R}^d$ and $\varepsilon > 0$, write

$$A^{(\varepsilon)} = \{x \in \mathbb{R}^d : \text{dist}(x, A) < \varepsilon\}.$$

For a linear subspace $V \leq \mathbb{R}^d$, π_V is the orthogonal projection onto V , $V^\perp$ is its orthogonal complement, and B_1 is the closed Euclidean unit ball. For sets $A, B \subset \mathbb{R}^d$, write

$$A \subseteq B \iff A \cap B_1 \subseteq B.$$

For each subspace V , fix an orthonormal coordinate isometry $J_V : \mathbb{R}^{\dim V} \rightarrow V$. Its intrinsic level- n dyadic partition is

$$\mathcal{D}_n^V = \{J_V(2^{-n}(k + [0, 1)^{\dim V})) : k \in \mathbb{Z}^{\dim V}\}, \quad n \in \mathbb{Z}.$$

For $V = \{0\}$ this is the one-atom partition. A coordinate partition on V used for a measure on $\mathbb{R}^d$ is pulled back by π_V , and may be written $\mathcal{P}_n^V = \pi_V^{-1}\mathcal{D}_n^V$. These intrinsic partitions are distinct from the restriction of the ambient standard grid to V ; L-II.5.01 proves their bounded entropy comparison.

Definition II.5.1.2 (II.5.02: Concentration on one or finitely many affine translates). Let $V \leq \mathbb{R}^d$, $\varepsilon > 0$, and let μ be a Borel probability on $\mathbb{R}^d$. It is (V, ε) -**concentrated** if some $a \in \mathbb{R}^d$ satisfies

$$\mu((a + V)^{(\varepsilon)}) \geq 1 - \varepsilon.$$

For $M \in \mathbb{N}$, it is $(V, \varepsilon)_M$ -**concentrated** if there are $a_1, \dots, a_M \in \mathbb{R}^d$ for which

$$\mu\left(\bigcup_{j=1}^M (a_j + V)^{(\varepsilon)}\right) \geq 1 - \varepsilon.$$

Repeated translates may be deleted, so “at most M ” and the displayed definition are equivalent. Concentration is always near affine translates; it means concentration near V itself only when the translate is zero. For $\varepsilon \geq 1$ the mass requirement is automatic. Exact support on an affine subspace is stated separately rather than encoded by the parameter zero.

Definition II.5.1.3 (II.5.03: Uniformity and saturation on a subspace). Let $V \leq \mathbb{R}^d$, $\varepsilon \geq 0$, and $m \geq 1$. A compactly supported probability μ is (V, ε, m) -**uniform** if it is $(V, 2^{-m})$ -concentrated and

$$h_m(\mu) \geq \dim V - \varepsilon.$$

It is (V, ε, m) -**saturated** if

$$h_m(\mu) \geq h_m((\pi_{V^\perp})_*\mu) + \dim V - \varepsilon. \quad (\text{II.5.01})$$

Uniformity combines a geometric tube condition with high total entropy; saturation asserts nearly maximal conditional entropy along V . Neither is defined to imply the other.

For a compactly supported probability θ , its **saturation deficit** at V and window m is

$$\text{def}_{V,m}(\theta) = \dim V - h_m(\theta) + h_m((\pi_{V^\perp})_*\theta).$$

Thus (V, ε, m) -saturation is exactly $\text{def}_{V,m}(\theta) \leq \varepsilon$. The deficit may be negative; its positive part is

$$(\text{def}_{V,m}(\theta))_+ = \max\{\text{def}_{V,m}(\theta), 0\}.$$

The next lemma gives the product-coordinate meaning of saturation and shows that it is stable under similarities. Here a **similarity** of $\mathbb{R}^d$ means a map $g(x) = rUx + a$, where $r > 0$, $U \in O(d) = \{U : U^T U = I_d\}$, and $a \in \mathbb{R}^d$.

Lemma II.5.1.4 (L-II.5.01 – Product-coordinate characterization and covariance). *There is $C_d \geq 1$ with the following properties. For every compactly supported probability μ on $\mathbb{R}^d$, every subspace V , and every $m \geq 1$,*

$$\left| h_m(\mu) - h_m((\pi_{V^\perp})_*\mu) - \frac{H_\mu(\mathcal{D}_m^V | \mathcal{D}_m^{V^\perp})}{m \log 2} \right| \leq \frac{C_d}{m}. \quad (\text{II.5.02})$$

Here the two factor partitions are pulled back to $\mathbb{R}^d$. In particular, (V, ε, m) -saturation implies

$$\frac{H_\mu(\mathcal{D}_m^V | \mathcal{D}_m^{V^\perp})}{m \log 2} \geq \dim V - \varepsilon - \frac{C_d}{m}.$$

Conversely, this conditional-entropy lower bound implies $(V, \varepsilon + 2C_d/m, m)$ -saturation. A lower bound without the C_d/m subtraction implies $(V, \varepsilon + C_d/m, m)$ -saturation.

If $g(x) = 2^t Ux + a$ is a similarity and μ is (V, ε, m) -saturated, then g_μ is*

$$\left(UV, \varepsilon + \frac{C_d(1 + |t|)}{m}, m \right)\text{-saturated}. \quad (\text{II.5.03})$$

The proof first establishes the bounded grid comparison and then applies the entropy chain rule. Similarity covariance follows by tracking the bounded change of observing level.

Fix orthonormal coordinates on V and W . Let $\mathcal{P}_m^V = \pi_V^{-1} \mathcal{D}_m^V$, $\mathcal{P}_m^W = \pi_W^{-1} \mathcal{D}_m^W$, and

$$\mathcal{R}_m(V) = \mathcal{P}_m^V \vee \mathcal{P}_m^W.$$

Grid comparison lemma

There is $K_d < \infty$ such that every atom of $\mathcal{D}_m$ intersects at most K_d atoms of $\mathcal{R}_m(V)$, and conversely, uniformly in m and V . The analogous assertion holds between the dyadic

partition induced on W from $\mathbb{R}^d$ and $\mathcal{D}_m^W$.

Proof

After dilation by 2^m , each atom has diameter at most $\sqrt{d}$. Its extent along every coordinate axis of the other orthonormal grid is therefore at most $\sqrt{d}$, so it meets at most $(\lceil \sqrt{d} \rceil + 2)^d$ grid cubes. On W , every induced ambient cell still has diameter at most $\sqrt{d}$; conversely, every intrinsic cell on W has ambient coordinate extents at most $\sqrt{d}$. The same count applies in both directions. No positive inradius is assumed for a sliced cell. By L-II.1.06, mutually K_d -intersecting partitions have entropies differing by at most $\log K_d$. $\square$

Proof. The chain rule gives the exact identity

$$H_\mu(\mathcal{R}_m(V)) = H_\mu(\mathcal{P}_m^W) + H_\mu(\mathcal{P}_m^V \mid \mathcal{P}_m^W). \quad (1.1)$$

Moreover,

$$H_\mu(\mathcal{P}_m^W) = H_{(\pi_W)_*\mu}(\mathcal{D}_m^W).$$

Apply the grid comparison lemma once to the first term of (1.1) and once to the left side. For a constant C_d ,

$$\left| H_m(\mu) - H_m((\pi_W)_*\mu) - H_\mu(\mathcal{P}_m^V \mid \mathcal{P}_m^W) \right| \leq C_d \log 2. \quad (1.2)$$

Division by $m \log 2$ proves (II.5.02) and both implications in the stated characterization of saturation.

Now let $g(x) = 2^t Ux + a$. Pulling the product-coordinate partitions for $UV + (UV)^\perp$ back by g gives translated, rotated product grids on $V + W$ with mesh $2^{-(m+t)}$. Put $q = \lfloor m + t \rfloor$. These pulled-back grids have uniformly bounded intersection multiplicity with the level- q product grids. For $q \geq m$, write each conditional entropy as joint entropy minus W -entropy. The joint entropy increment lies in $[0, d(q - m) \log 2]$ and the W -entropy increment in $[0, (d - r)(q - m) \log 2]$. Consequently,

$$\left| H_\mu(\mathcal{P}_q^V \mid \mathcal{P}_q^W) - H_\mu(\mathcal{P}_m^V \mid \mathcal{P}_m^W) \right| \leq d|q - m| \log 2. \quad (1.3)$$

For $q < m$, exchange m, q in this absolute-value estimate. Since $|q - m| \leq |t| + 1$, (1.2), (1.3), and the saturation hypothesis yield

$$\frac{H_{g_*\mu}(\mathcal{P}_m^{UV} \mid \mathcal{P}_m^{(UV)^\perp})}{m \log 2} \geq r - \varepsilon - \frac{C_d(1 + |t|)}{m}.$$

One last use of (1.2) proves (II.5.03). $\square$

II.5.2 Uniform, mixed, and perturbed measures

Saturation is useful only if it survives the elementary operations used later: replacing a uniform tube law by conditional entropy, mixing components, perturbing a law, and estimating a measure concentrated near a plane.

Lemma II.5.2.1 (L-II.5.02 – Uniformity, mixtures, perturbations, and tube entropy). *For every $R \geq 1$ there is $C_{d,R}$ such that, for $m \geq 1$, subspaces V , and probabilities supported on $[-R, R]^d$, the following hold.*

1. (V, ε, m) -uniformity implies $(V, C_{d,R}(\varepsilon + 1/m), m)$ -saturation.
2. A finite convex combination of (V, ε, m) -saturated probabilities is $(V, \varepsilon + C_d/m, m)$ -saturated.
3. If μ is (V, ε, m) -saturated and $\|\mu - \nu\|_{\text{TV}} \leq \delta \leq 1$, then ν is

$$\left(V, \varepsilon + C_{d,R} \left(\delta + \frac{b(\delta)}{m \log 2} + \frac{1}{m} \right), m\right)\text{-saturated,} \quad (\text{II.5.04})$$

where $b(\delta) = -\delta \log \delta - (1 - \delta) \log(1 - \delta)$.

4. If μ is $(V, 2^{-m})$ -concentrated, then

$$h_m(\mu) \leq \dim V + C_{d,R} \frac{\log(m+1)}{m}. \quad (\text{II.5.05})$$

Proof of part (1)

Proof. Assume $\mu \in \mathcal{P}([-R, R]^d)$ is (V, ε, m) -uniform. There is a translate $a + V$ such that, with $s \leq 2^{-m}$,

$$\mu = (1 - s)\mu_0 + s\mu_1, \quad \mu_0((a + V)^{(2^{-m})}) = 1. \quad (2.1)$$

The projection $(\pi_W)_*\mu_0$ is carried by a radius- 2^{-m} ball and hence by at most C_d level- m cells of W . The projection of μ_1 is carried by a fixed d, R -dependent ball and hence by at most $C_{d,R}2^{(d-r)m}$ such cells. The mixture upper bound T-II.1.05 gives

$$\begin{aligned} H_m((\pi_W)_*\mu) &\leq b(s) + (1 - s) \log C_d + s((d - r)m \log 2 + \log C_{d,R}) \\ &\leq C_{d,R}. \end{aligned} \quad (2.2)$$

The last inequality uses $s \leq 2^{-m}$. Since $h_m(\mu) \geq r - \varepsilon$, (2.2) proves

$$h_m(\mu) - h_m((\pi_W)_*\mu) \geq r - \varepsilon - C_{d,R}/m.$$

This is the asserted saturation estimate. □

Proof of part (2)

Proof. Let $\mu = \sum_{j=1}^J a_j \mu_j$, where every μ_j is (V, ε, m) -saturated. By L-II.5.01,

$$H_{\mu_j}(\mathcal{P}_m^V \mid \mathcal{P}_m^W) \geq (r - \varepsilon)m \log 2 - C_d \log 2. \quad (2.3)$$

Conditional entropy is concave in the measure, so

$$H_\mu(\mathcal{P}_m^V \mid \mathcal{P}_m^W) \geq \sum_{j=1}^J a_j H_{\mu_j}(\mathcal{P}_m^V \mid \mathcal{P}_m^W).$$

Insert (2.3) and apply L-II.5.01 again. After enlarging C_d , μ is $(V, \varepsilon + C_d/m, m)$ -saturated. $\square$

Proof of part (3)

Proof. The joint atom laws of $\mathcal{P}_m^W \vee \mathcal{P}_m^V$ under μ, ν have total-variation distance at most δ . On the fixed support box, the V -coordinate partition has at most

$$M \leq C_{d,R} 2^{rm} \quad (2.4)$$

non-null values. The all-range conditional-entropy continuity theorem L-II.4.01, applied with the actual total-variation distance $t \leq \delta$, therefore gives the following convenient upper-bound form:

$$\left| H_\mu(\mathcal{P}_m^V \mid \mathcal{P}_m^W) - H_\nu(\mathcal{P}_m^V \mid \mathcal{P}_m^W) \right| \leq 2b(\delta) + 2\delta \log M. \quad (2.5)$$

For $\delta \leq 1/2$ this follows from monotonicity of b on $[0, 1/2]$. For $\delta > 1/2$, both conditional entropies lie in $[0, \log M]$, so their difference is at most $\log M \leq 2\delta \log M$. If the coordinate partition has only one value, both conditional entropies are zero.

Combine (2.5), (2.4), the product-coordinate form of the saturation of μ , and L-II.5.01. Dividing by $m \log 2$ gives exactly (II.5.04), after increasing $C_{d,R}$. This proof remains valid at $\delta = 0, 1$ by the adopted endpoint values of b . $\square$

Proof of part (4)

Proof. Suppose $\mu((a+V)^{(2^{-m})}) \geq 1 - 2^{-m}$ and decompose it as in (2.1). The intersection of the tube with $[-R, R]^d$ meets at most

$$C_{d,R} 2^{rm} \quad (2.6)$$

ambient level- m cubes: project their centers to V , use a bounded-overlap 2^{-m} net in the bounded projection of the box, and note that each net point has only C_d possible transverse cubes. The full box meets at most $C_{d,R} 2^{dm}$ cubes. By the entropy maximum and mixture bounds,

$$\begin{aligned} H_m(\mu) &\leq b(s) + (1-s)(rm \log 2 + \log C_{d,R}) \\ &\quad + s(dm \log 2 + \log C_{d,R}) \\ &\leq rm \log 2 + C_{d,R}. \end{aligned}$$

This is stronger than (II.5.05), since $1 \leq \log(m+1)/\log 2$. $\square$

II.5.3 Components and retuning

Global saturation bounds the average positive local deficit. Markov's inequality produces good components, concentration passes to descendants by conditioning, and disjoint windows retune a long-scale statement to a prescribed shorter window.

Proposition II.5.3.1 (P-II.5.03 – Component inheritance and retuning). *There is $C_d \geq 1$ such that the following hold.*

1. *If $\mu \in \mathcal{P}([0, 1]^d)$ is (V, ε, N) -saturated and $1 \leq m < N$, then*

$$\mathbb{P}_{\substack{0 \leq i < N \\ x \sim \mu}} \left\{ \mu^{x,i} \text{ is } \left(V, C_d \sqrt{\varepsilon} + \frac{C_d}{\sqrt{m}} + C_d \sqrt{\frac{m}{N}}, m \right) \text{-saturated} \right\} \geq 1 - C_d \sqrt{\varepsilon} - \frac{C_d}{\sqrt{m}} - C_d \sqrt{\frac{m}{N}}. \quad (\text{II.5.06})$$

2. *If μ is (V, ε) -concentrated and $1 \leq j \leq \log_2(1/\varepsilon)$, then*

$$\mathbb{P}_{x \sim \mu} \left\{ \mu^{x,j} \text{ is } (V, \sqrt{2^j \varepsilon})\text{-concentrated} \right\} \geq 1 - \sqrt{2^{-j} \varepsilon}. \quad (\text{II.5.07})$$

3. *For every $0 < \eta < 1$, there is $M = M(d, \eta)$ such that for every $m \geq M$ there is $K = K(d, \eta, m)$ such that, for every $k \geq K$, there are $\delta_0 = \delta_0(d, \eta, m, k) > 0$ and $N_0 = N_0(d, \eta, m, k, \delta_0)$ such that whenever μ, ν are Borel probabilities on $\mathbb{R}^d$, $0 < \delta \leq \delta_0$, $N \geq N_0$, and level-only subspaces $V_0, \dots, V_N$ satisfy*

$$\mathbb{P}_{\substack{0 \leq i \leq N \\ x \sim \mu, y \sim \nu}} \left\{ \mu^{x,i} \text{ is } (V_i, \delta, k)\text{-saturated, } \nu^{y,i} \text{ is } (V_i, \delta)\text{-concentrated} \right\} \geq 1 - \delta,$$

then there are level-only $V'_0, \dots, V'_N$ for which the same probability is at least $1 - \eta$ with (δ, k) replaced by (η, m) and δ -concentration replaced by η -concentration. If every $V_i = V$, one may take every $V'_i = V$.

Proof strategy

We first express the mean component deficit by conditional entropy and deduce saturation inheritance by Markov's inequality. Concentration inheritance follows by conditioning the mass outside a fixed tube. Finally, we tile a long scale interval by disjoint windows and combine the two inheritance estimates, keeping the endpoint loss explicit.

Saturation inheritance

Let $X \sim \mu$. Write $A_j = \mathcal{D}_j(X)$ for its ambient dyadic cell and $B_j = \mathcal{D}_j^W(\pi_W X)$ for the cell of its projection in the fixed dyadic grid on W . Uniform bounded intersection of an ambient cube with projected tubes gives

$$H(B_j \mid A_j) \leq C_d \quad (3.1)$$

for every j . Since canonical rescaling maps the descendants of A_i to the standard grid, component disintegration gives

$$\mathbb{E}_x H_m(\mu^{x,i}) = H(A_{i+m} \mid A_i). \quad (3.2)$$

On the atom $A_i = Q$, the projected rescaled vector is $\pi_W(T_Q X) = 2^i \pi_W X - \pi_W k_Q$. Translation changes its level- m entropy by at most C_d , uniformly in Q . Therefore

$$|\mathbb{E}_x H_m((\pi_W)_* \mu^{x,i}) - H(B_{i+m} \mid A_i)| \leq C_d. \quad (3.3)$$

Here the entropy on the left is computed with the ambient standard grid, whereas B_{i+m} uses the fixed grid on W ; the final $O_d(1)$ is precisely the bounded-intersection comparison between these two induced grids from the grid comparison lemma.

The chain rule, (3.1), and the fact that B_N refines B_{i+m} show

$$\begin{aligned} & H(A_{i+m} \mid A_i) - H(B_{i+m} \mid A_i) \\ &= H(A_{i+m} \mid A_i, B_{i+m}) - H(B_{i+m} \mid A_i, A_{i+m}) \\ &\geq H(A_{i+m} \mid A_i, B_N) - C_d. \end{aligned} \quad (3.4)$$

For $1 \leq j \leq N$, put $c_j = H(A_j \mid A_{j-1}, B_N)$. Because A_0 is constant on $[0, 1)^d$,

$$\sum_{j=1}^N c_j = H(A_N \mid B_N) \geq H(A_N) - H(B_N) \geq (r - \varepsilon)N \log 2 - C_d. \quad (3.5)$$

The last inequality is the saturation hypothesis and terminal grid comparison. Also,

$$H(A_{i+m} \mid A_i, B_N) = \sum_{j=i+1}^{i+m} c_j. \quad (3.6)$$

Sum (3.6) over $0 \leq i \leq N - m$. Every c_j is counted at most m times, and only the first and last $m - 1$ indices are counted fewer than m times. We may assume $N \geq 2m$: otherwise the asserted exceptional bound in (II.5.06) is at least one after enlarging C_d . Since $0 \leq c_j \leq d \log 2$, (3.5) yields

$$\frac{1}{N - m + 1} \sum_{i=0}^{N-m} \frac{H(A_{i+m} \mid A_i, B_N)}{m \log 2} \geq r - \varepsilon - C_d \frac{m}{N} - \frac{C_d}{N}. \quad (3.7)$$

Combining (3.2)–(3.4) with (3.7), the mean positive saturation deficit of the components is at most

$$C_d \left(\varepsilon + \frac{1}{m} + \frac{m}{N} \right). \quad (3.8)$$

Indeed the component entropy difference is at most $r + C_d/m$, by the same fiber-cell count as in L-II.5.01; hence its positive deficit below r is bounded in mean by r minus its mean, plus C_d/m .

Markov's inequality at threshold

$$a = C_d \left(\sqrt{\varepsilon} + \frac{1}{\sqrt{m}} + \sqrt{\frac{m}{N}} \right)$$

shows that, apart from a proportion at most a , the component entropy difference is at least $r - a$. Restoring the last $m - 1$ starting scales costs at most $m/N \leq \sqrt{m/N}$. Enlarging C_d proves (II.5.06).

Concentration inheritance

Let $L = a + V$ satisfy $\mu(L^{(\varepsilon)}) \geq 1 - \varepsilon$, and fix $1 \leq j \leq \log_2(1/\varepsilon)$. For $Q \in \mathcal{D}_j$, the canonical rescaling sends

$$Q \cap L^{(\varepsilon)} \quad \text{into} \quad (T_Q L)^{(2^j \varepsilon)}.$$

Put $\rho = \sqrt{2^j \varepsilon} \leq 1$, and let E be the family of level- j cells for which

$$\mu_Q(Q \setminus L^{(\varepsilon)}) > \rho.$$

Then

$$\varepsilon \geq \sum_{Q \in E} \mu(Q) \mu_Q(Q \setminus L^{(\varepsilon)}) > \rho \mu\left(\bigcup E\right).$$

Thus $\mu(\bigcup E) < \varepsilon/\rho = \sqrt{2^{-j} \varepsilon}$. On every other cell, μ^Q gives mass at least $1 - \rho$ to a $2^j \varepsilon$ -tube, which is contained in the ρ -tube because $2^j \varepsilon = \rho^2 \leq \rho$. This proves (II.5.07).

Scale-window retuning

Let $C = C_d$ be the constant in P-II.5.03(1). Choose M , then K , so that every $m \geq M$ and $k \geq K(d, \eta, m)$ satisfy

$$k > m, \quad C/\sqrt{m} \leq \eta/12, \quad C\sqrt{m/k} \leq \eta/12.$$

Next choose $\delta_0 > 0$ such that

$$(C + 3)\sqrt{\delta_0} \leq \eta/12, \quad \sqrt{2^k \delta_0} \leq \eta/4, \quad \delta_0 \leq 2^{-k}. \quad (3.9)$$

Finally choose N_0 so that $k/(N + 1) \leq \eta/4$ for every $N \geq N_0$. All subsequent estimates hold uniformly for $0 < \delta \leq \delta_0$. This is the exact order of quantifiers printed in P-II.5.03(3).

Let I consist of those levels i for which the conditional probability, over the independent components $\mu^{x,i}, \nu^{y,i}$, of the hypothesis event is at least $1 - \sqrt{\delta}$. Markov gives

$$|I| \geq (1 - \sqrt{\delta})(N + 1). \quad (3.10)$$

Choose the least remaining member of I as a starting point and delete all remaining members through its $k - 1$ successor. Repetition gives $I_0 \subset I$ whose integer intervals $[i, i + k - 1]$ are disjoint and cover I . On such an interval define $V'_{i+j} = V_i$ for $0 \leq j < k$. Apply the two inheritance estimates to every component pair in the hypothesis event. For the concentration estimate, (3.9) permits every $1 \leq j < k$; at $j = 0$ the original δ -concentration already implies η -concentration. Thus the concentration tolerance is at most $\sqrt{2^k \delta} \leq \eta/4$, and its failure probability at every inner scale is at most $\sqrt{\delta}$. The saturation estimate P-II.5.03(1), with outer window k and target window m , has tolerance and exceptional probability at most

$$a = C\sqrt{\delta} + C/\sqrt{m} + C\sqrt{m/k} \leq \eta/4.$$

For completeness, if Q is a level- i cube and $Q' \subset Q$ a level- $i + j$ descendant, then $T_{T_Q Q'} \circ T_Q = T_{Q'}$ and $\mu(Q)\mu_Q(Q') = \mu(Q')$. Thus conditioning once and then sampling a descendant produces precisely the mass-weighted global component at level $i + j$. This is [L-II.2.01](#) applied twice, and the same calculation holds independently for v . The initial bad pairs at any selected level have probability at most $\sqrt{\delta}$. Union bounds and averaging therefore give

$$\mathbb{P}_{\substack{q \in U \\ z \sim \mu, w \sim \nu}} \{ \mu^{z,q} \text{ is } (V'_q, \eta, m)\text{-saturated, } \nu^{w,q} \text{ is } (V'_q, \eta)\text{-concentrated} \} \geq 1 - a - 2\sqrt{\delta}, \quad (3.11)$$

where $U = \cup_{i \in I_0} [i, i + k - 1]$ and q is uniform on U . Only the final interval can extend beyond N . Write $T = \{0, \dots, N\}$, $a_0 = |T \setminus U|$, $b_0 = |U \setminus T| < k$, and $p = a + 2\sqrt{\delta}$. We have $a_0 \leq \sqrt{\delta}(N + 1)$ by [\(3.10\)](#). The total expected number of failed levels in T is at most $p|U| + a_0$; dividing by $N + 1$ bounds the failure probability by

$$p + (1 - p) \frac{a_0}{N + 1} + p \frac{b_0}{N + 1} \leq a + 3\sqrt{\delta} + \frac{k}{N + 1} \leq \eta/2. \quad (3.12)$$

Define V'_q arbitrarily on $T \setminus U$. The last estimate proves [P-II.5.03\(3\)](#), with room in the required error η . If every $V_i = V$, the construction assigns $V'_q = V$ on U ; assigning the same value off U proves the final assertion. $\square$

II.5.4 Changing the subspace

The inverse theorem initially produces planes that depend on samples and scales. We therefore need quantitative stability when a plane is enlarged, slightly rotated, or combined with a transverse plane.

Definition II.5.4.1 (II.5.11: Grassmannian distance and transverse angle). For linear subspaces $V, W \leq \mathbb{R}^d$, their **gap metric** is

$$d_G(V, W) = \|\pi_V - \pi_W\|_{\text{op}}.$$

The equality with the infimum of the two approximate-containment tolerances is a theorem rather than a second definition; it is proved in [L-II.5.04](#) and displayed in [\(4.1\)](#). The **Grassmannian of r -planes** is $Gr(d, r) = \{V \leq \mathbb{R}^d : \dim V = r\}$, equipped with d_G . The space of all linear subspaces is the disjoint union of $Gr(d, r)$, $0 \leq r \leq d$. Put $E = V \cap W$. The **transverse angle** between V and W is zero when one contains the other; otherwise it is

$$\angle(V, W) = \inf\{\|v - w\| : v \in V \cap E^\perp, w \in W \cap E^\perp, \|v\| = \|w\| = 1\}.$$

This is the symmetric gap-type angle used in [\[Hoc20\]](#), rather than the smallest-principal-angle convention.

Lemma II.5.4.2 (L-II.5.04 – Grassmannian and subspace-stability calculus). *Let $V, W \leq \mathbb{R}^d$. In assertions 3–5 assume in addition that $\mu \in \mathcal{P}([0, 1]^d)$.*

1. $0 \leq d_G(V, W) \leq 1$. If $0 < \varepsilon < 1$ and $V \sqsubseteq W^{(\varepsilon)}$, then $\dim V \leq \dim W$; equality implies $d_G(V, W) \leq \varepsilon$. 2. $\angle(V, W) \leq \sqrt{2}d_G(V, W)$. 3. If $\mu \in \mathcal{P}([0, 1]^d)$ is (V, ε) -concentrated and $d_G(V, W) \leq \delta$, then it is $(W, \varepsilon + C_d\delta)$ -concentrated. 4. If μ is (V, ε, m) -saturated and $W \leq V$, then it is $(W, \varepsilon + C_d/m, m)$ -saturated. If instead $d_G(V, W) \leq 2^{-m}$, it is $(W, \varepsilon + C_d/m, m)$ -saturated. 5. If μ is saturated on both V_1, V_2 with parameters (ε, m) and $\angle(V_1, V_2) \geq \delta > 0$, then it is

$$\left(V_1 + V_2, 2\varepsilon + C_d \frac{1 + \log(1/\delta)}{m}, m \right)\text{-saturated.} \quad (\text{II.5.08})$$

Proof

The provenance for L-II.5.04 is [Hoc20, Lemmas 3.20–3.21, printed pp. 34–36]. All parts are proved below.

Gap metric and near inclusion

Write $P = \pi_V, Q = \pi_W$ and define the one-sided gap

$$\alpha(V, W) = \|(I - Q)P\| = \sup_{v \in V \cap B_1} \text{dist}(v, W).$$

Since $P(I - Q)x$ and $(I - P)Qx$ are orthogonal,

$$\begin{aligned} (P - Q)x &= P(I - Q)x - (I - P)Qx, \\ \|(P - Q)x\|^2 &\leq \alpha(V, W)^2 \|(I - Q)x\|^2 + \alpha(W, V)^2 \|Qx\|^2. \end{aligned}$$

Testing $(P - Q)$ on V and W supplies the reverse inequality, so

$$\|P - Q\| = \max\{\alpha(V, W), \alpha(W, V)\} \leq 1. \quad (4.1)$$

This also proves the infimum characterization of d_G in D-II.5.11. If $\alpha(V, W) < 1$, then $Q|_V$ is injective, because a unit vector in its kernel would have distance one from W . Hence $\dim V \leq \dim W$. When the dimensions agree and are positive, choose orthonormal bases. The matrices of $Q|_V$ and $P|_W$ are transposes, with the same smallest singular value s . Thus both one-sided gaps equal $\sqrt{1 - s^2}$. Dimension zero is immediate. This proves L-II.5.04(1), including its equal-dimension conclusion. It also proves the useful dual identity

$$\alpha(V, W) = \alpha(W^\perp, V^\perp), \quad (4.2)$$

because $(I - Q)P$ and $P(I - Q)$ are transposes.

The transverse angle

Put $E = V \cap W$. If one subspace contains the other, the defined angle is zero. Otherwise choose a unit $v \in V \cap E^\perp$. If $Qv \neq 0$, put $w = Qv/\|Qv\|$. Then $w \in W \cap E^\perp$ and

$$\|v - w\|^2 = 2(1 - \|Qv\|) \leq 2(1 - \|Qv\|^2) = 2 \operatorname{dist}(v, W)^2 \leq 2d_G(V, W)^2.$$

If $Qv = 0$, then $d_G(V, W) = 1$ and any unit $w \in W \cap E^\perp$ gives $\|v - w\| = \sqrt{2}$. Taking the infimum proves L-II.5.04(2).

Concentration under a change of direction

Assume first $0 < \varepsilon < 1$. Represent the affine concentration plane as $a + V$ with $a \in V^\perp$. Its ε -tube has positive μ -mass inside the unit cube; hence $\|a\| < \sqrt{d} + \varepsilon$. For every point x of that tube in the cube,

$$\begin{aligned} \operatorname{dist}(x, a + W) &= \|(I - Q)(x - a)\| \\ &\leq \|(I - P)(x - a)\| + \|(P - Q)(x - a)\| \\ &< \varepsilon + (2\sqrt{d} + 1)\delta. \end{aligned}$$

The same set has mass at least $1 - \varepsilon$, which proves L-II.5.04(3) with $C_d = 2\sqrt{d} + 1$. For $\varepsilon \geq 1$ the mass requirement is automatic.

Saturation on a smaller or nearby subspace

For $W \leq V$, put $K = V \cap W^\perp$ and $T = V^\perp$, so $\mathbb{R}^d = W \oplus K \oplus T$ is an orthogonal decomposition. Use the dyadic coordinate partitions on these three factors. The grid comparison from Section 1 works for this decomposition as well. Saturation on V gives

$$H_\mu(\mathcal{P}_m^W \vee \mathcal{P}_m^K \mid \mathcal{P}_m^T) \geq (\dim V - \varepsilon)m \log 2 - C_d.$$

Since the projection of the unit cube onto K is contained in a ball of radius $\sqrt{d}$, its grid meets at most $C_d 2^m \dim K$ cells. Therefore the chain rule yields

$$\begin{aligned} H_\mu(\mathcal{P}_m^W \mid \mathcal{P}_m^K \vee \mathcal{P}_m^T) &\geq (\dim V - \varepsilon)m \log 2 - C_d - H_\mu(\mathcal{P}_m^K \mid \mathcal{P}_m^T) \\ &\geq (\dim W - \varepsilon)m \log 2 - C_d. \end{aligned}$$

Product-coordinate comparison now gives the asserted $(W, \varepsilon + C_d/m, m)$ -saturation.

If instead $d_G(V, W) \leq 2^{-m} < 1$, then $\dim V = \dim W$ by (4.1) and the dimension assertion. For $X \sim \mu$,

$$\|\pi_{V^\perp} X - \pi_{W^\perp} X\| \leq \sqrt{d} 2^{-m}.$$

Each of the two dyadic cell labels, conditional on the other, consequently has at most $(2\sqrt{d} + 3)^d$ values. Their entropies differ by at most the logarithm of this number. The total entropy of μ does not change, so its saturation inequality transfers with error C_d/m . This proves L-II.5.04(4).

Adding transverse saturated directions

First suppose $V_1 \cap V_2 = \{0\}$, put $U = V_1 + V_2$, $T = U^\perp$, and suppose $\angle(V_1, V_2) \geq \delta > 0$. Necessarily $\delta \leq \sqrt{2}$. For unit $a \in V_1$, $b \in V_2$, applying the angle condition to both b and $-b$ gives

$$|\langle a, b \rangle| \leq 1 - \delta^2/2.$$

Thus the sum map $S(a, b, c) = a + b + c$ on $V_1 \times V_2 \times T$ satisfies

$$\frac{\delta^2}{2}(\|a\|^2 + \|b\|^2) + \|c\|^2 \leq \|S(a, b, c)\|^2 \leq 2(\|a\|^2 + \|b\|^2) + \|c\|^2. \quad (4.3)$$

For $X \sim \mu$ write $X = A + B + C$ in these algebraic coordinates and let A_m, B_m, C_m be their intrinsic dyadic cell labels. A linear change of coordinates whose norm and inverse norm are at most $K \geq 1$ changes a grid atom into a set with coordinate extents at most $K\sqrt{d}2^{-m}$. The grid intersection count from Section 1 is therefore at most $(\lceil K\sqrt{d} \rceil + 2)^d$ in both directions. For the maps below its logarithm is bounded by $C_d \log(2/\delta)$.

In particular, (4.3) compares the joint label (A_m, B_m, C_m) with the ambient level- m label of X . Also the map

$$(b, c) \mapsto \pi_{V_1^\perp} b + c$$

is an isomorphism onto $V_1^\perp$ with norm at most one and inverse norm at most $\sqrt{2}/\delta$. Indeed $\|\pi_{V_1^\perp} b\| \geq \delta\|b\|/\sqrt{2}$ follows from the inner-product bound above. Its two summands are orthogonal. It follows that (B_m, C_m) has entropy within $C_d \log(2/\delta)$ of the level- m entropy of $\pi_{V_1^\perp} X$. Subtraction and saturation on V_1 give

$$H(A_m | B_m, C_m) \geq (\dim V_1 - \varepsilon)m \log 2 - C_d \log(2/\delta).$$

Interchanging V_1, V_2 gives the analogous bound for $H(B_m | A_m, C_m)$. By the chain rule and reduction of entropy under conditioning,

$$\begin{aligned} H(A_m, B_m | C_m) &= H(A_m | B_m, C_m) + H(B_m | C_m) \\ &\geq H(A_m | B_m, C_m) + H(B_m | A_m, C_m) \\ &\geq (\dim U - 2\varepsilon)m \log 2 - C_d \log(2/\delta). \end{aligned} \quad (4.4)$$

Comparison of the joint grid with the ambient grid and of the C -grid with the ambient grid on T converts (4.4) into saturation on U .

In general put $E = V_1 \cap V_2$ and replace V_2 by $V_2' = V_2 \cap E^\perp$. Part (4) transfers saturation to V_2' with loss C_d/m . The sum is unchanged, the new intersection is zero, and the transverse angle is unchanged: maximizing the absolute inner product with a unit vector of V_2' only uses the $E^\perp$ component of V_1 . Apply the preceding argument. Absorb the extra C_d/m and use $\log(2/\delta) \leq C(1 + \log(1/\delta))$ for $0 < \delta \leq \sqrt{2}$. This proves (II.5.08), completing L-II.5.04.

□

II.5.5 Finite constructions of envelopes and cores

Plane stability permits finite geometric compression. An approximate envelope records all directions occurring in a family, while an approximate core records their common directions. The explicit moduli below avoid a nonquantitative compactness argument.

Definition II.5.5.1 (II.5.12: Approximate envelopes, cores, and moduli). Let $\mathcal{W}$ be a nonempty family of subspaces and let $t, \rho > 0$. A **t -to- ρ approximate envelope** is a subspace E such that

$$W \subseteq E^{(\rho)} \quad (W \in \mathcal{W}), \quad [\forall W \in \mathcal{W}, W \subseteq U^{(t)}] \implies E \subseteq U^{(\rho)} \quad (U \leq \mathbb{R}^d).$$

A **t -to- ρ approximate core** is a subspace C such that

$$C \subseteq W^{(\rho)} \quad (W \in \mathcal{W}), \quad [\forall W \in \mathcal{W}, U \subseteq W^{(t)}] \implies U \subseteq C^{(\rho)} \quad (U \leq \mathbb{R}^d).$$

A **modulus** is a function $\omega : (0, 1) \rightarrow (0, \infty)$ with $\lim_{t \downarrow 0} \omega(t) = 0$; no monotonicity is implied unless it is stated separately. Existence of envelopes and cores with the tolerances and finite witnesses used below is the content of L-II.5.05.

Lemma II.5.5.2 (L-II.5.05 – Thickened spans, envelopes, and cores). For $d \geq 1$ and $0 < t < 1$, put

$$c_d = 2^{d+1}d, \quad \omega_d^+(t) = \omega_d^-(t) = c_d t^{1/(d+1)}.$$

These explicit moduli have the following properties.

1. If $v_1, \dots, v_r \in B_1$, $\text{dist}(v_j, \text{span}(v_1, \dots, v_{j-1})) \geq \delta > 0$, and $v = \sum_j a_j v_j$, then

$$\|(a_1, \dots, a_r)\|_2 \leq \sqrt{r} 2^r \delta^{-r} \|v\|. \quad (\text{II.5.09})$$

2. For every nonempty family $\mathcal{W}$ of subspaces and $0 < t < 1$, there is an **approximate envelope** V^+ such that

$$W \subseteq (V^+)^{(\omega_d^+(t))} \quad (W \in \mathcal{W}),$$

and whenever every $W \in \mathcal{W}$ satisfies $W \subseteq U^{(t)}$, then $V^+ \subseteq U^{(\omega_d^+(t))}$. 3. There is an **approximate core** V^- such that

$$V^- \subseteq W^{(\omega_d^-(t))} \quad (W \in \mathcal{W}),$$

and whenever $U \subseteq W^{(t)}$ for every $W \in \mathcal{W}$, then $U \subseteq (V^-)^{(\omega_d^-(t))}$. More precisely, there are $W_1, \dots, W_s \in \mathcal{W}$ with $0 \leq s \leq d - \dim V^-$ such that

$$B_1 \cap \bigcap_{j=1}^s W_j^{(t)} \subseteq (V^-)^{(\omega_d^-(t))}.$$

The empty intersection means $\mathbb{R}^d$. For an increasing finite or countable sequence of nonempty families $\mathcal{W}_1 \subseteq \mathcal{W}_2 \subseteq \dots$, the cores can be chosen nested: $V_{i+1}^- \leq V_i^-$, with the same modulus and the same witness bound for each family.

The proof constructs explicit power moduli; no compactness-only existence is used in downstream quantitative statements.

Proof

The coefficient estimate below is [Hoc20, Lemma 3.22, printed pp. 36–37]. The envelope/core conclusions correspond to Propositions 3.27–3.28, printed pp. 38–40. The following direct construction supplies its own explicit moduli and the finite witness statement.

A coefficient estimate

Suppose $v_1, \dots, v_r$ satisfy L-II.5.05(1), and write $u = \sum_{j=1}^r a_j v_j$. Necessarily $0 < \delta \leq 1$. Projection onto the orthogonal complement of $\text{span}(v_1, \dots, v_{r-1})$ gives

$$|a_r| \delta \leq \|u\|.$$

The remainder $u' = u - a_r v_r$ therefore satisfies $\|u'\| \leq (1 + 1/\delta)\|u\|$. Induction on r , starting at $r = 1$, now gives, for every j ,

$$|a_j| \leq (1 + 1/\delta)^r \|u\| \leq 2^r \delta^{-r} \|u\|.$$

Taking the Euclidean norm proves (II.5.09). In particular,

$$\sum_{j=1}^r |a_j| \leq r 2^r \delta^{-r} \|u\|. \quad (5.1)$$

Greedy envelope

Fix $0 < t < 1$ and put $\tau = t^{1/(d+1)}$. Start with $E_0 = \{0\}$. If some $W \in \mathcal{W}$ has a vector $v \in W \cap B_1$ with $\text{dist}(v, E_j) \geq \tau$, append it to the list and set $E_{j+1} = E_j + \mathbb{R}v$. Each step increases dimension by one. Hence the construction stops after $r \leq d$ steps, with $E = \text{span}(v_1, \dots, v_r)$ and

$$W \cap B_1 \subseteq E^{(\tau)} \quad (W \in \mathcal{W}). \quad (5.2)$$

If a subspace U satisfies $W \subseteq U^{(t)}$ for all members of the family, then every selected v_j has distance less than t from U . A unit $u = \sum a_j v_j \in E$ consequently satisfies

$$\text{dist}(u, U) \leq \sum_j |a_j| \text{dist}(v_j, U) \leq d 2^d t \tau^{-d} = d 2^d t^{1/(d+1)} < \omega_d^+(t). \quad (5.3)$$

For $r = 0$ this conclusion is immediate. Equations (5.2)–(5.3) prove L-II.5.05(2) with $V^+ = E$ and the declared modulus $\omega_d^+(t) = 2^{d+1} d t^{1/(d+1)}$.

Dual core and its witnesses

Apply the same construction to $\{W^\perp : W \in \mathcal{W}\}$. Keep the selected vectors $v_j \in W_j^\perp \cap B_1$ and put $V^- = E^\perp$. Equation (5.2) and the one-sided dual identity (4.2) imply $\alpha(V^-, W) \leq \tau$ for every W . The strict enlargement $\omega_d^-(t) > \tau$ proves the required approximate inclusions.

Now let $x \in B_1 \cap \bigcap_{j=1}^r W_j^{(t)}$. For every unit $e = \sum a_j v_j \in E$, (5.1) gives

$$|\langle x, e \rangle| \leq \sum_j |a_j| |\langle x, v_j \rangle| \leq d 2^d t \tau^{-d} < \omega_d^-(t),$$

because $|\langle x, v_j \rangle| \leq \text{dist}(x, W_j) < t$. Taking the supremum over unit $e \in E$ shows

$$\text{dist}(x, V^-) = \|\pi_E x\| < \omega_d^-(t). \quad (5.4)$$

The bound is uniform with a factor of two to spare, so the supremum preserves the strict final inequality. There are $r = \dim E = d - \dim V^-$ witnesses; deleting repetitions can only reduce their number. If $r = 0$, then $V^- = \mathbb{R}^d$ and the empty-intersection assertion holds. If $U \subseteq W^{(t)}$ for every member, apply (5.4) to each $x \in U \cap B_1$. This proves L-II.5.05(3).

For increasing families, retain all vectors already chosen and continue the greedy procedure when the next family is introduced. All earlier vectors still belong to the new union, the separation condition at the time of their choice is unchanged, and at most d vectors are ever selected. The resulting envelopes increase by inclusion, so their orthogonal complements decrease by inclusion. The same proof of (5.4) uses witnesses from the current family at every stage. This proves the nested-core assertion for finite and countable increasing sequences. $\square$

II.5.6 Canonical choices and slow variation

Envelopes and cores now produce canonical planes. This turns sample-dependent geometry into level-dependent flags and bounds the number of scales at which those flags can change substantially.

Definition II.5.6.1 (II.5.08: Flags and canonical subspaces). A **flag** here is a finite or countable family of linear subspaces nested by inclusion; repetitions are allowed. An increasing flag satisfies $V_i \subseteq V_{i+1}$, a decreasing flag the reverse. The phrase **slowly varying** refers only to the explicitly quantified gap-distance counts in the relevant theorem, not to a separate unquantified regularity condition. A **canonical subspace** in P-II.5.06 means any choice satisfying its displayed approximate minimality or maximality properties; uniqueness is not asserted.

Proposition II.5.6.2 (P-II.5.06 — Canonical subspaces and slowly varying flags).

There are dimension-dependent moduli $\omega_d(t) \rightarrow 0$ as $t \downarrow 0$ such that:

1. *For every $\eta \in \mathcal{P}([0, 1]^d)$ and sufficiently small ε , there is $V_c(\eta, \varepsilon)$ for which η is $(V_c, \omega_d(\varepsilon))$ -concentrated, and every W on which η is (W, ε) -concentrated satisfies $V_c \subseteq W^{(\omega_d(\varepsilon))}$.*
2. *For every $\theta \in \mathcal{P}([0, 1]^d)$ and $m \geq 2$, there is $V_s(\theta, m)$ for which θ is*

$(V_s, C_d \log(m+1)/m, m)$ -saturated, and every W on which θ is $(W, 1/m, m)$ -saturated satisfies $W \subseteq V_s^{(C_d \log(m+1)/m)}$. 3. Fix $\ell \geq 1$, let $\eta \in \mathcal{P}(\mathbb{R}^d)$, suppose $0 < \varepsilon < 2^{-2\ell}$ and $N \geq \ell$, and put

$$L = \left\lfloor \frac{1}{2} \log_2(1/\varepsilon) \right\rfloor + 1, \quad B = \left\lceil \frac{N+1}{L} \right\rceil.$$

Suppose for every $0 \leq q \leq N$ a subspace W_q satisfies

$$\mathbb{P}_{x \sim \eta} \{ \eta^{x,q} \text{ is } (W_q, \varepsilon)\text{-concentrated} \} \geq 1 - \varepsilon.$$

Then there are $V_q \leq W_q$ such that for every q

$$\mathbb{P}_{x \sim \eta} \{ \eta^{x,q} \text{ is } (V_q, \omega_d(\varepsilon))\text{-concentrated} \} \geq 1 - \omega_d(\varepsilon),$$

and

$$\frac{1}{N+1} \# \{ \ell \leq q \leq N : d_G(V_q, V_{q-\ell}) \leq \omega_d(\varepsilon) \} \geq 1 - \frac{(d+1)\ell B}{N+1} \geq 1 - (d+1)\ell \left(\frac{1}{L} + \frac{1}{N+1} \right). \quad (\text{II.5.10})$$

This retains the initial-scale and final-block losses. In particular, the simpler error $2(d+1)\ell / \log_2(1/\varepsilon)$ requires a boundary correction and is not asserted for finite N .

Proof

Here “canonical” refers to the stated approximate minimality or maximality, not to uniqueness. [P-II.5.06](#) asserts existence for each measure. A measurable choice for a random measure is an additional obligation of the random-subspace proof [P-II.5.11](#); it is not inferred merely from existence. The source counterparts are [[Hoc20](#), Propositions 3.29–3.31 and Corollary 3.32, printed pp. 40–43].

A common explicit modulus

Let $c_d = 2^{d+1}d$ be the constant in Section 5, and set

$$e_d = \frac{1}{4(d+1)}, \quad b_d = 2^{1/(d+1)}c_d, \quad s_d = 2\sqrt{d} + 1, \\ K_d = 10(d+1)(1+s_d)b_d, \quad \varepsilon_d = (2K_d)^{-4(d+1)}.$$

For parts (1) and (3) take

$$\omega_d(\varepsilon) = \begin{cases} K_d \varepsilon^{e_d}, & 0 < \varepsilon < \varepsilon_d, \\ 2, & \varepsilon_d \leq \varepsilon < 1. \end{cases} \quad (6.1)$$

It tends to zero with ε . In the small-parameter branch, if $a = \varepsilon^{1/4}$ and $\Omega = b_d \varepsilon^{e_d}$, then

$$2a < 1, \quad da < \frac{1}{2}, \quad \Omega < \frac{1}{4}, \quad \sqrt{d}\Omega + da + s_d\Omega < \omega_d(\varepsilon), \quad 2d\sqrt{\varepsilon} < \omega_d(\varepsilon). \quad (6.2)$$

These follow from $\varepsilon^{e_d} < 1/(2K_d)$, $a \leq \varepsilon^{e_d}$, and the displayed definition of K_d .

A minimally concentrated subspace

Let $\mathcal{W}$ be the family of all subspaces on which η is ε -concentrated. It is nonempty because it contains $\mathbb{R}^d$. Assume $\varepsilon < \varepsilon_d$ and apply L-II.5.05(3) with $t = 2\varepsilon$. It gives a core V_c , at most d witnesses $W_1, \dots, W_r$, and modulus

$$\Omega_c = c_d(2\varepsilon)^{1/(d+1)} \leq b_d \varepsilon^{e_d}.$$

For each witness choose an affine tube $(a_j + W_j)^{(\varepsilon)}$ with mass at least $1 - \varepsilon$. Their intersection inside the unit cube has mass at least $1 - r\varepsilon > 0$. Choose a point x_0 in this intersection. Every other point x of it satisfies

$$\|(x - x_0)/\sqrt{d}\| \leq 1, \quad \text{dist}((x - x_0)/\sqrt{d}, W_j) < 2\varepsilon.$$

The witness conclusion (5.4) therefore puts x in the $\sqrt{d}\Omega_c$ -tube about $x_0 + V_c$. Its mass is at least $1 - r\varepsilon$; (6.2) proves $(V_c, \omega_d(\varepsilon))$ -concentration. If there are no witnesses, $V_c = \mathbb{R}^d$ and the same conclusion is immediate. The core's other property gives $V_c \subseteq W^{(\Omega_c)} \subseteq W^{(\omega_d(\varepsilon))}$ for every $W \in \mathcal{W}$. This proves P-II.5.06(1).

A maximally saturated subspace

Let C_0 be a constant valid for the downward-saturation bound in L-II.5.04(4), and C_1 a constant such that the transverse-sum proof gives error $C_1 \log(2/\delta)/m$. Choose $A \geq \max(1 + C_0, 2C_1, 1)$ and put

$$q_m = \frac{\log(m+1)}{m}, \quad \delta_j = A3^j q_m \quad (0 \leq j \leq d).$$

If $2\delta_d \geq 1$, take $V_s = \{0\}$. It is saturated with zero error, and every subspace is approximately contained in it with tolerance $4\delta_d \geq 2$. Hence the desired conclusion holds with $C_d = 4A3^d$ in this case.

Suppose now $2\delta_d < 1$. Choose V_s of maximal dimension among subspaces V for which θ is $(V, \delta_{\dim V}, m)$ -saturated. This family contains $\{0\}$, and its set of dimensions is a nonempty finite subset of $\{0, \dots, d\}$, so such a choice exists without a compactness argument.

Write $j = \dim V_s$. Suppose θ is $(W, 1/m, m)$ -saturated but W is not approximately contained in V_s with tolerance $2\delta_d$. There is then a unit $w \in W$ with $\text{dist}(w, V_s) \geq 2\delta_d$. Put $L = \mathbb{R}w$. Downward saturation gives $(L, (1 + C_0)/m, m)$ -saturation, hence (L, δ_j, m) -saturation. If $j = 0$, this already gives a one-dimensional admissible subspace with error at most δ_1 , a contradiction. For $j > 0$ the line L is not contained in V_s and

$$\angle(V_s, L) \geq \text{dist}(w, V_s) \geq 2\delta_d > \delta_d.$$

L-II.5.04(5) therefore makes θ saturated on $V_s + L$ with error at most

$$2\delta_j + \frac{C_1 \log(2/\delta_d)}{m} \leq 2\delta_j + \frac{2C_1 \log(m+1)}{m} \leq 3\delta_j = \delta_{j+1}. \quad (6.3)$$

Here $\delta_d \geq 1/m$ and $\log(2m) \leq 2 \log(m+1)$ were used. Since $\dim(V_s + L) = j+1$, (6.3) contradicts maximality. Thus every such W is contained approximately with tolerance $2\delta_d$, while θ is saturated on V_s with error $\delta_j \leq \delta_d$. Enlarging both to $C_d q_m$ proves P-II.5.06(2). $\square$

Nested cores on one scale block

We prove the concentration assertion in P-II.5.06(3) and keep every finite-scale loss. In the branch $\omega_d(\varepsilon) = 2$ take $V_q = W_q$: the concentration requirement is automatic and every gap is at most one. The final counting bound follows because at most the first ℓ indices are excluded. It remains to treat $\varepsilon < \varepsilon_d$.

Let $L = \text{floor}((1/2) \log_2(1/\varepsilon)) + 1$. Fix a block $J = \{s, \dots, s+h-1\}$ of length $h \leq L$. For each $q \in J$ apply the nested-core construction to the increasing families

$$\mathcal{W}_q = \{W_s, W_{s+1}, \dots, W_q\}, \quad t = 2\varepsilon^{1/4}.$$

Write their cores as U_q . They are nested by exact inclusion and satisfy $\alpha(U_q, W_q) \leq \Omega$, where $\Omega = c_d t^{1/(d+1)} = b_d \varepsilon^{ed}$. At level q , retain witnesses $W_{k_1}, \dots, W_{k_r}$, where $s \leq k_j \leq q$ and $r \leq d$.

For each witness, the hypothesis at level k_j and concentration inheritance (II.5.07) imply

$$\mathbb{P}_{x \sim \eta} \{ \eta^{x,q} \text{ is } (W_{k_j}, \varepsilon^{1/4})\text{-concentrated} \} \geq 1 - \varepsilon - \sqrt{\varepsilon} \geq 1 - 2\sqrt{\varepsilon}. \quad (6.4)$$

Indeed $q - k_j \leq L - 1 \leq (1/2) \log_2(1/\varepsilon)$, so $\sqrt{2^{q-k_j} \varepsilon} \leq \varepsilon^{1/4}$. At $q = k_j$ use the original concentration condition directly. The two-stage sampling calculation in Section 3 identifies the descendant with the level- q component in (6.4).

With probability at least $1 - 2d\sqrt{\varepsilon}$ all witness events hold. Fix a component θ in this event. Its witness tubes have error $a = \varepsilon^{1/4}$; their intersection has mass at least $1 - ra > 0$. Choosing an anchor point in that intersection and using the same difference-and-rescaling argument as for V_c shows that θ gives mass at least $1 - ra$ to a $\sqrt{d}\Omega$ -tube about a translate of U_q . It is therefore $(U_q, \sqrt{d}\Omega + da)$ -concentrated.

To enforce the required exact inclusion in W_q , set

$$V_q = \pi_{W_q} U_q.$$

The restriction $\pi_{W_q}|_{U_q}$ is injective because $\Omega < 1$. For every unit $u \in U_q$ its image belongs to V_q and differs from u by at most Ω . Equal dimensions and (4.1) imply

$$V_q \leq W_q, \quad \dim V_q = \dim U_q, \quad d_G(V_q, U_q) \leq \Omega. \quad (6.5)$$

Apply concentration stability with the constant $s_d = 2\sqrt{(d)+1}$. By (6.2), every component in the preceding good event is $(V_q, \omega_d(\varepsilon))$ -concentrated. Its probability is at least $1 - 2d\sqrt{\varepsilon} \geq 1 - \omega_d(\varepsilon)$. This proves the pointwise in q concentration conclusion throughout the block.

Counting the exceptional comparisons

Partition $\{0, \dots, N\}$ into consecutive blocks of length L , except possibly the final shorter block. Their number is $B = \text{ceil}((N+1)/L)$. Make the preceding construction in each

block. Within any one block the dimensions of U_q decrease at most d times. If q and $q - \ell$ lie in that block and no dimension drop occurs between them, nestedness forces $U_q = U_{q-\ell}$. Equation (6.5) then gives

$$d_G(V_q, V_{q-\ell}) \leq 2\Omega \leq \omega_d(\varepsilon).$$

At most ℓ indices per block have their predecessor outside the block or below zero, and each of at most d dimension drops affects at most ℓ comparisons. Thus at most $(d + 1)\ell$ indices per block fail, including all initially excluded indices. Summing over the B blocks gives

$$\#\{\ell \leq q \leq N : d_G(V_q, V_{q-\ell}) \leq \omega_d(\varepsilon)\} \geq N + 1 - (d + 1)\ell B. \quad (6.6)$$

Finally $B \leq (N + 1)/L + 1$. Divide (6.6) by $N + 1$ to obtain (II.5.10), completing P-II.5.06.

□

The proof of part (3) holds for every Borel probability η on $\mathbb{R}^d$, as stated above. All diameter and entropy bounds concern its rescaled components, which are unit-cube probabilities; the only cross-level step is the general resampling identity. No bound on the location, diameter or entropy of the original measure was used. Parts (1)–(2) retain their unit-cube hypotheses.

II.5.7 Covariance and spectral concentration

Covariance identifies candidate directions. Small transverse covariance places most mass near a spectral plane, while independent sums add covariances exactly. Covariance does not prove the inverse theorem by itself; it supplies the planes later converted into entropy saturation.

Definition II.5.7.1 (II.5.04: Covariance, spectral subspaces, Gaussian laws, and discrepancy). For a probability μ on $\mathbb{R}^d$ with finite second moment, define

$$m(\mu) = \int x \, d\mu(x), \quad \Sigma(\mu) = \int (x - m(\mu))(x - m(\mu))^T \, d\mu(x).$$

For a positive semidefinite matrix Σ , list its eigenvalues with multiplicity as

$$\lambda_1(\Sigma) \geq \dots \geq \lambda_d(\Sigma) \geq 0, \quad \lambda_0(\Sigma) = d, \quad \lambda_{d+1}(\Sigma) = 0.$$

The indices $0, d + 1$ are auxiliary conventions, not extra eigenvalues. In particular $\lambda_0 = d$ is not an upper bound for a general covariance matrix; the ordered list above runs only from 1 to d .

When $\lambda_r(\Sigma) > \lambda_{r+1}(\Sigma)$, the **top spectral r -plane** $E_r(\Sigma)$ is the direct sum of the eigenspaces for $\lambda_1, \dots, \lambda_r$; set $E_0(\Sigma) = \{0\}$ and $E_d(\Sigma) = \mathbb{R}^d$.

Let $Z = (Z_1, \dots, Z_d)$ have independent coordinates with the scalar standard normal density $(2\pi)^{-1/2} \exp(-x^2/2)$. The Gaussian probability $N(a, \Sigma)$ is the law of $a + BZ$, where B is a real $d \times d$ matrix satisfying $BB^T = \Sigma$. Section 7 proves that the law does not depend on the chosen factor B . In particular, $N(0, I_d)$ denotes the law of Z . For

probabilities α, β , their **convex-set discrepancy** is

$$d_{\text{cvx}}(\alpha, \beta) = \sup\{|\alpha(C) - \beta(C)| : C \subset \mathbb{R}^d \text{ is convex and Borel}\}.$$

For a probability μ on $\mathbb{R}^d$ and an integer $i \geq 0$, its **mean rescaled-component covariance** is

$$\bar{\Sigma}_i(\mu) = \int \Sigma(\mu^{x,i}) d\mu(x).$$

This integral is entrywise. Each rescaled component is carried by the unit cube, so its entries are bounded Borel functions and the integral exists even if μ has no global second moment.

For a real matrix $M = (M_{ab})$, its **Frobenius norm** is $\|M\|_F = (\sum_{a,b} |M_{ab}|^2)^{1/2}$.

Lemma II.5.7.2 (L-II.5.07 – Covariance and spectral concentration). *Let μ have finite second moment.*

1. *For every subspace V ,*

$$m((\pi_V)_*\mu) = \pi_V m(\mu), \quad \Sigma((\pi_V)_*\mu) = \pi_V \Sigma(\mu) \pi_V.$$

The covariance forms agree on $V \times V$; as an ambient matrix the right side is a compression, not a restriction to an invariant subspace. More generally, for a linear map T and vector a , $\Sigma((x \mapsto Tx + a)_\mu) = T\Sigma(\mu)T^T$. Covariances and means add under convolution of probabilities with finite second moments. 2. Fix $0 \leq r \leq d$ and suppose either $r \in \{0, d\}$ or $\lambda_r(\Sigma(\mu)) > \lambda_{r+1}(\Sigma(\mu))$. If $V = E_r(\Sigma(\mu))$, then*

$$\mu \text{ is } (V, C_d \lambda_{r+1}(\Sigma(\mu))^{1/3})\text{-concentrated.} \quad (\text{II.5.11})$$

This display is for $\lambda_{r+1} > 0$. If $\lambda_{r+1} = 0$, the conclusion is the exact statement $\mu(m(\mu) + V) = 1$.

3. *If $\mu \in \mathcal{P}([0, 1]^d)$ is (W, ε) -concentrated and $\dim W = r$, then $\lambda_{r+1}(\Sigma(\mu)) \leq C_d \varepsilon$, and μ is $(E_r(\Sigma(\mu)), C_d \varepsilon^{1/3})$ -concentrated whenever the displayed top spectral plane is uniquely defined. 4. Eigenvalues are continuous functions of a positive semidefinite matrix. On every compact set with $\lambda_r \geq \tau > \sigma \geq \lambda_{r+1}$, the map $\Sigma \mapsto E_r(\Sigma)$ is uniformly continuous in d_G .*

The random-covariance form used later is included: if Θ is a random probability on $[0, 1]^d$, $0 < \varepsilon \leq 1$, $A = \mathbb{E}\Sigma(\Theta)$, $\lambda_{r+1}(A) \leq \varepsilon$, and either $r \in \{0, d\}$ or $\lambda_r(A) > \lambda_{r+1}(A)$, then for $V = E_r(A)$ there is a measurable event E such that

$$\mathbb{P}(E) \geq 1 - C_d \sqrt{\varepsilon}, \quad \omega \in E \implies \Theta(\omega) \text{ is } (V, C_d \varepsilon^{1/6})\text{-concentrated.} \quad (\text{II.5.12})$$

One may take $E = \{\text{tr}(\pi_{V^\perp} \Sigma(\Theta) \pi_{V^\perp}) \leq \sqrt{\varepsilon}\}$. This explicit event avoids an unstated measurability assertion about a union over arbitrary affine translates.

Proof

The source counterparts for L-II.5.07 are [Hoc20, Lemmas 4.4–4.7, printed pp. 45–48]. The proof here uses traces and finite-dimensional spectral estimates to keep every exceptional case explicit.

Affine maps, independent sums, and the Gaussian definition

Let $X \sim \mu$, $m = \mathbb{E}X$ and $\Sigma = \mathbb{E}[(X - m)(X - m)^\top]$. For a linear map T and vector a , integrability and linearity give

$$\begin{aligned}\mathbb{E}(TX + a) &= Tm + a, \\ \mathbb{E}[(TX - Tm)(TX - Tm)^\top] &= T\Sigma T^\top.\end{aligned}\tag{7.1}$$

Taking $T = \pi_V$ proves the projection/compression assertion. For independent X, Y of finite second moment the cross terms in the covariance of $X + Y$ vanish:

$$\mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)^\top] = \mathbb{E}(X - \mathbb{E}X) \mathbb{E}(Y - \mathbb{E}Y)^\top = 0.$$

The mean and covariance identities for two convolands, then for any finite number by induction, follow. This proves L-II.5.07(1).

The Gaussian definition is consistent as follows. The product density of Z in D-II.5.04 is $(2\pi)^{-d/2} \exp(-\|z\|^2/2)$. The change-of-variables formula shows that QZ has the same law for every orthogonal matrix Q . Existence of a factor follows by writing $\Sigma = U \operatorname{diag}(\lambda_1, \dots, \lambda_d) U^\top$ and taking $B = U \operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_d})$. Suppose $BB^\top = CC^\top$. The rows of B and C have the same Gram matrix. The map taking the i th row of B to the i th row of C is therefore a well-defined isometry between their row spans: every linear combination has the same squared norm on both sides. Extend orthonormal bases of those spans to orthonormal bases of $\mathbb{R}^d$. The isometry extends to an orthogonal matrix, giving $C = BQ$ for some orthogonal Q . Thus $CZ = BQZ$ and BZ have the same law. The scalar normal moments from D-II.4.02 and independence give $\mathbb{E}Z = 0$ and $\Sigma(Z) = I_d$, so (7.1) also verifies the mean and covariance of $N(a, \Sigma)$. Singular covariance matrices are included.

Small transverse covariance implies concentration

For any subspace V put $P = \pi_{V^\perp}$. Directly from the definition of covariance,

$$\mathbb{E} \operatorname{dist}(X, m + V)^2 = \mathbb{E} \|P(X - m)\|^2 = \operatorname{tr}(P\Sigma P).\tag{7.2}$$

If $V = E_r(\Sigma)$ is defined and $\lambda = \lambda_{r+1}(\Sigma) > 0$, this trace is at most $d\lambda$. For $\rho = (2d\lambda)^{1/3}$, Markov gives

$$\mu\{x : \operatorname{dist}(x, m + V) \geq \rho\} \leq \frac{d\lambda}{\rho^2} = \frac{\rho}{2}.$$

Thus the open ρ -tube has mass at least $1 - \rho/2 \geq 1 - \rho$, proving (II.5.11) with $C_d = (2d)^{1/3}$. If $\lambda = 0$, the nonnegative random variable in (7.2) has expectation zero and vanishes almost surely. Hence $\mu(m + V) = 1$ exactly. This includes $r = d$; $r = 0$ works with $V = \{0\}$.

Concentration controls the spectral tail

Suppose $\mu([0, 1]^d) = 1$ and it is (W, ε) -concentrated, with $\dim W = r < d$ and $0 < \varepsilon < 1$. Write a witnessing affine subspace as $a + W$ with $a \in W^\perp$. As in Section 4, $\|a\| < \sqrt{d} + 1$. The projected mean minimizes mean-square distance to a constant, so

$$\begin{aligned} \operatorname{tr}(\pi_{W^\perp} \Sigma \pi_{W^\perp}) &= \mathbb{E} \|\pi_{W^\perp} X - \pi_{W^\perp} m\|^2 \\ &\leq \mathbb{E} \|\pi_{W^\perp} X - a\|^2 \\ &\leq \varepsilon^2 + (2\sqrt{d} + 1)^2 \varepsilon \leq C_d \varepsilon. \end{aligned} \tag{7.3}$$

Choose orthonormal eigenvectors of Σ in decreasing eigenvalue order. The span of the first $r + 1$ eigenvectors intersects $W^\perp$ in a nonzero vector. For a unit vector u in this intersection,

$$\lambda_{r+1}(\Sigma) \leq u^\top \Sigma u \leq \operatorname{tr}(\pi_{W^\perp} \Sigma \pi_{W^\perp}) \leq C_d \varepsilon.$$

The middle inequality follows by extending u to an orthonormal basis of $W^\perp$ and using nonnegativity of every diagonal entry of the compressed positive semidefinite matrix. Combine this bound with (7.2) and the preceding concentration argument. If the tail eigenvalue is zero, exact support implies concentration for every positive tolerance. This proves L-II.5.07(3). For $r = d$ it is immediate; for $\varepsilon \geq 1$ the concentration conclusion is automatic and the bounded covariance gives the eigenvalue estimate after enlarging C_d .

Random covariance

Let Θ and A be as in (II.5.12), and put $V = E_r(A)$, $P = \pi_{V^\perp}$. The nonnegative random variable

$$Z = \operatorname{tr}(P \Sigma(\Theta) P)$$

satisfies $\mathbb{E} Z = \operatorname{tr}(P A P) \leq d \varepsilon$. Its measurability follows by expressing covariance entries as first and second moments: on the unit cube these coordinate functions have bounded continuous extensions to $\mathbb{R}^d$, so their integrals are Borel functions of the random probability. Markov gives

$$\mathbb{P}\{Z > \sqrt{\varepsilon}\} \leq d \sqrt{\varepsilon}. \tag{7.4}$$

For any realization with $Z \leq \sqrt{\varepsilon}$, apply (7.2) to Θ and the fixed V . At radius $\rho = 2\varepsilon^{1/6}$ the mass outside the open tube about $m(\Theta) + V$ is at most

$$\frac{\sqrt{\varepsilon}}{4\varepsilon^{1/3}} = \frac{1}{4}\varepsilon^{1/6} \leq \rho.$$

Taking $E = \{Z \leq \sqrt{\varepsilon}\}$ proves (II.5.12) on an explicitly measurable event. No eigenplane of the random matrix $\Sigma(\Theta)$ is selected: the deterministic gap is in A alone.

Continuity of eigenvalues and spectral planes

For a symmetric matrix A the eigenbasis argument gives

$$\lambda_j(A) = \max_{\dim U=j} \min_{\substack{u \in U \\ \|u\|=1}} u^\top A u. \quad (7.5)$$

Indeed the top j -eigenvector span achieves $\lambda_j(A)$. Every other j -plane meets the span of eigenvectors numbered $j, \dots, d$ nontrivially, so its minimum is no larger. Rayleigh quotients for A and B differ by at most $\|A - B\|$ on the unit sphere. Applying (7.5) in both directions gives

$$|\lambda_j(A) - \lambda_j(B)| \leq \|A - B\|. \quad (7.6)$$

Now suppose both matrices satisfy $\lambda_r \geq \tau > \sigma \geq \lambda_{r+1}$, with $1 \leq r < d$. Let $u_{r+1}, \dots, u_d$ be orthonormal lower eigenvectors of A with eigenvalues $a_i \leq \sigma$, and let v_j be a unit top eigenvector of B with eigenvalue $b_j \geq \tau$. Symmetry gives

$$(b_j - a_i) \langle u_i, v_j \rangle = \langle u_i, (B - A)v_j \rangle.$$

Squaring, summing in i , and using $b_j - a_i \geq \tau - \sigma$ yield

$$\|\pi_{E_r(A)^\perp} v_j\| \leq \frac{\|A - B\|}{\tau - \sigma}.$$

Expand an arbitrary unit vector of $E_r(B)$ in its r orthonormal top eigenvectors and use Cauchy–Schwarz on the coefficients. Then

$$\alpha(E_r(B), E_r(A)) \leq \frac{\sqrt{r} \|A - B\|}{\tau - \sigma}.$$

The planes have equal dimension, so (4.1) gives the same bound for their gap metric. This proves uniform continuity even without compactness. The cases $r = 0, d$ have constant spectral planes. Equations (7.6) and the gap estimate prove L-II.5.07(4), completing the lemma. $\square$

II.5.8 Convolution preserves saturated directions

Lemma II.5.8.1 (L-II.5.08 – Convolution stability on subspaces). *There is C_d such that for compactly supported probabilities μ, ν , subspaces V , and $m \geq 1$:*

$$h_m(\mu * \nu) \geq h_m(\mu) - \frac{C_d}{m}. \quad (\text{II.5.13})$$

*If μ is (V, ε, m) -saturated, then $\mu * \nu$ is $(V, \varepsilon + C_d/m, m)$ -saturated. Hence, if μ is not $(V, 2\varepsilon, m)$ -saturated while ν is (V, ε, m) -saturated, then*

$$h_m(\mu * \nu) - h_m(\mu) \geq \varepsilon - \frac{C_d}{m}. \quad (\text{II.5.14})$$

If $0 < \varepsilon < 1$ and $\mu((a + V)^{(\varepsilon)}) \geq 1 - \varepsilon$, then for every $k \geq 1$,

$$\mu^{*k}((ka + V)^{(k\varepsilon)}) \geq (1 - \varepsilon)^k. \quad (\text{II.5.15})$$

Proof

The source counterparts for L-II.5.08 are [Hoc20, Lemma 4.1 and Corollary 4.2, printed pp. 43–44]. Its final tube estimate is the direct independent-event statement underlying Lemma 4.3 on p. 44.

For a vector y let $\tau_y(x) = x + y$. Translated standard dyadic grids have bounded intersection multiplicity independent of y . Thus

$$H_m((\tau_y)_*\mu) \geq H_m(\mu) - C_d.$$

For every dyadic cube Q , $(\mu * \nu)(Q) = \int (\tau_y)_*\mu(Q) d\nu(y)$. Apply concavity of $-t \log t$ to this identity and sum over the finitely many relevant cubes. This gives

$$H_m(\mu * \nu) \geq \int H_m((\tau_y)_*\mu) d\nu(y) \geq H_m(\mu) - C_d,$$

which proves (II.5.13).

We also need conditional entropy concavity for this integral mixture. Let A, B denote the $V, V^\perp$ coordinate cell labels. Write their joint law under $(\tau_y)_*\mu$ as $p_y(a, b)$, with $p_y(b) = \sum_a p_y(a, b)$ and $p(b) = \int p_y(b) d\nu(y)$. For $p(b) > 0$ the conditional law of A given $B = b$ in the mixture is

$$p(a | b) = \int p_y(a | b) \frac{p_y(b)}{p(b)} d\nu(y).$$

The scalar entropy concavity inequality, applied to this probability mixture and summed in a , gives

$$p(b)H(p(\cdot | b)) \geq \int p_y(b)H(p_y(\cdot | b)) d\nu(y).$$

The zero-mass cases contribute zero. Sum in b to obtain

$$H_{\mu*\nu}(A | B) \geq \int H_{(\tau_y)_*\mu}(A | B) d\nu(y). \quad (8.1)$$

There are only finitely many relevant labels on the fixed compact sum of the supports, so all expressions are finite. By L-II.5.01 with scale shift zero, each translate of an (V, ε, m) -saturated measure is $(V, \varepsilon + C_d/m, m)$ -saturated. Product-coordinate comparison, (8.1), and a final product-coordinate comparison imply $(V, \varepsilon + C_d/m, m)$ -saturation of $\mu * \nu$, after enlarging the dimension constant.

For the growth assertion assume ν is (V, ε, m) -saturated and μ is not $(V, 2\varepsilon, m)$ -saturated. Put $W = V^\perp$. Projection commutes with convolution because it is linear; hence

$$\begin{aligned} h_m(\mu * \nu) &\geq h_m((\pi_W)_*\mu * (\pi_W)_*\nu) + \dim V - \varepsilon - C_d/m \\ &\geq h_m((\pi_W)_*\mu) + \dim V - \varepsilon - C_d/m \\ &> h_m(\mu) + \varepsilon - C_d/m. \end{aligned} \quad (8.2)$$

This proves (II.5.14), with its weak inequality as stated.

Finally let $X_1, \dots, X_k$ be independent with law μ and suppose $0 < \varepsilon < 1$ and $\mu((a+V)^{(\varepsilon)}) \geq 1 - \varepsilon$. With probability at least $(1 - \varepsilon)^k$ every X_j belongs to this open tube. On that event write $X_j = a + v_j + e_j$ with $v_j \in V$ and $\|e_j\| < \varepsilon$. Then

$$\sum_{j=1}^k X_j = ka + \sum_{j=1}^k v_j + \sum_{j=1}^k e_j \in (ka + V)^{(k\varepsilon)}.$$

This proves the exact mass bound (II.5.15), completing L-II.5.08. $\square$

II.5.9 Multidimensional dyadic Kaimanovich–Vershik transfer

Lemma II.5.9.1 (L-II.5.09 — Exact multidimensional dyadic KV transfer). *Let μ, ν be probabilities on $\mathbb{R}^d$, let $n, k \geq 1$, and assume $H_n(\mu), H_n(\nu) < \infty$. Then*

$$\begin{aligned} H_n(\mu * \nu^{*k}) &\leq H_n(\mu) + k(H_n(\mu * \nu) - H_n(\mu)) \\ &\quad + kd \log 5 + d \log(2k + 3). \end{aligned} \tag{II.5.16}$$

The last line divided by $n \log 2$ is the normalized terminal error. The subtracted term is $H_n(\mu)$, not $H_n(\nu)$.

Proof

We use the fully proved countable-group theorem T-II.4.11. The $\mathbb{R}^d$ discretization is proved here; it incurs no new external input.

For $n \in \mathbb{Z}$ use the coordinatewise $q_n(x) = \lfloor 2^n x \rfloor$ from D-II.3.02 and define $\sigma_n(x) = 2^{-n} q_n(x)$. This is the coordinatewise extension of D-II.4.05. Let $X_1, \dots, X_\ell$ be independent $\mathbb{R}^d$ -valued variables with laws $\mu_j = \text{Law}(X_j)$ satisfying $H_n(\mu_j) < \infty$. Put

$$A = \sum_{j=1}^{\ell} q_n(X_j), \quad Q = q_n\left(\sum_{j=1}^{\ell} X_j\right).$$

Every coordinate satisfies the exact carry identity

$$Q = A + K, \quad K \in \{0, 1, \dots, \ell - 1\}^d. \tag{9.1}$$

Indeed K is the coordinatewise integer part of the sum of the fractional parts of $2^n X_j$. Independence and the finite entropy of the individual quantizations imply $H(A) < \infty$. Since $H(K) \leq d \log \ell$, the chain rule gives

$$H(Q) \leq H(A) + d \log \ell, \quad H(A) \leq H(Q) + d \log \ell. \tag{9.2}$$

The first inequality also establishes finiteness of $H(Q)$, so there is no subtraction of infinite entropies. The normalized lattice sum $2^{-n} A$ has Shannon entropy $H(A)$. Thus (9.2) proves

$$|H_n(\mu_1 * \cdots * \mu_\ell) - H((\sigma_n)_* \mu_1 * \cdots * (\sigma_n)_* \mu_\ell)| \leq d \log \ell. \quad (9.3)$$

In particular $H((\sigma_n)_* \mu) = H_n(\mu)$. For the hypotheses of L-II.5.09 put $\mu_n = (\sigma_n)_* \mu$ and $\nu_n = (\sigma_n)_* \nu$. These are finite-entropy probabilities on the countable abelian group $2^{-n} \mathbb{Z}^d$. Apply (9.3) with $\ell = k + 1$, then T-II.4.11, then (9.3) with $\ell = 2$:

$$\begin{aligned} H_n(\mu * \nu^{*k}) &\leq H(\mu_n * \nu_n^{*k}) + d \log(k + 1) \\ &\leq H(\mu_n) + k(H(\mu_n * \nu_n) - H(\mu_n)) + d \log(k + 1) \\ &\leq H_n(\mu) + k(H_n(\mu * \nu) - H_n(\mu)) + kd \log 2 + d \log(k + 1). \end{aligned} \quad (9.4)$$

This is stronger than (II.5.16), because $\log 2 \leq \log 5$ and $\log(k + 1) \leq \log(2k + 3)$. Division by $n \log 2$ for $n \geq 1$ gives the stated normalized error. In every step the subtracted entropy is that of μ , as required by the telescope in T-II.4.11. This completes L-II.5.09. $\square$

II.5.10 Entropy localization for isometry actions

An action has two levels of conditioning. First choose a raw group component, then choose a spatial component:

$$(g, x) \sim \nu \times \mu \longrightarrow (\nu_{g,i}, \mu^{x,i}) \longrightarrow (h, y) \mapsto hy.$$

Keeping these choices separate is essential when a signed entropy increment is converted into a high-probability statement.

Definition II.5.10.1 (II.5.05: Similarities, Euclidean isometries, and raw group components). The similarity group is

$$G(d) = \{x \mapsto rUx + a : r > 0, U \in O(d), a \in \mathbb{R}^d\}.$$

Its Euclidean isometry subgroup is

$$G_0(d) = \{g = (U_g, a_g) : x \mapsto U_g x + a_g, U_g \in O(d), a_g \in \mathbb{R}^d\}.$$

Composition is $(U, a)(V, b) = (UV, Ub + a)$, and we use the metric

$$d_0((U, a), (V, b)) = \|U - V\|_{\text{op}} + \|a - b\|.$$

Embed $G_0(d)$ in $\mathbb{R}^{d^2+d}$ by its matrix entries and translation coordinates. Its **level- n dyadic partition** is

$$\mathcal{D}_n^{G_0} = \{G_0(d) \cap Q : Q \in \mathcal{D}_n^{d^2+d}, G_0(d) \cap Q \neq \emptyset\}.$$

For a probability ν on $G_0(d)$, a positive-mass atom D , and $g \in D$, its **raw group components** are

$$\nu_D = \frac{\nu|_D}{\nu(D)}, \quad \nu_{g,n} = \nu_{\mathcal{D}_n^{G_0}(g)}.$$

There is no canonical rescaled group component in this chapter.

Definition II.5.10.2 (II.5.06: Action measures, component actions, and window entropy). For probabilities $\nu \in \mathcal{P}(G_0(d))$ and $\mu \in \mathcal{P}(\mathbb{R}^d)$, define the **action measure**

$$\nu \cdot \mu = A_*(\nu \times \mu), \quad A(g, x) = gx.$$

For $x \in \mathbb{R}^d$, $\nu \cdot x = (g \mapsto gx)_* \nu$. For a raw component $\nu_{g,i}$, its **rescaled image component at x** is

$$\mathcal{B}_i(\nu_{g,i}, x) = (S_i)_*(\nu_{g,i} \cdot x), \quad S_i(z) = 2^i z.$$

Its rotation-normalized **component action at x** is

$$\mathcal{A}_i(\nu_{g,i}, x) = (S_i \circ U_g^{-1})_*(\nu_{g,i} \cdot x) = (U_g^{-1})_* \mathcal{B}_i(\nu_{g,i}, x).$$

The two measures use different coordinate frames. Concentration of $\mathcal{B}_i$ near a subspace V is equivalent to concentration of $\mathcal{A}_i$ near $U_g^{-1}V$, with the same error.

We also use fixed image coordinates without changing the group grid. For an integer $b \geq 0$, an orthogonal map Q , a point $x_0 \in \mathbb{R}^d$ and a vector $t \in \mathbb{R}^d$, put

$$F(h) = 2^b Q(hx_0 - t), \quad \mathcal{F}_j = \mathcal{D}_{b+j}^{G_0}, \quad \mathcal{B}_j^F(\nu_D) = (S_j)_* F_* \nu_D \quad (D \in \mathcal{F}_j, \nu(D) > 0).$$

Here $\nu_D = \nu|_D / \nu(D)$. These are image laws of atoms of the original, nested group partitions. They are not defined by repartitioning a transformed group measure.

For a probability θ on $\mathbb{R}^d$ and integers $i \in \mathbb{Z}$, $m \geq 1$, define the dimension-normalized **window entropy**

$$h_{i,m}(\theta) = \frac{H_{i+m}(\theta)}{m \log 2}.$$

For ν on $G_0(d)$, put

$$h_{i,m}^{G_0}(\nu) = \frac{H_\nu(\mathcal{D}_{i+m}^{G_0})}{m \log 2}.$$

For $N \geq 1$, its normalized level entropy is

$$h_N^{G_0}(\nu) = \frac{H_\nu(\mathcal{D}_N^{G_0})}{N \log 2}.$$

These window entropies are used only when the measure is supported in boundedly many level- i atoms; then subtracting the level- i entropy changes the value by $O(1/m)$ with the stated support dependence. They are not, by definition, conditional entropies. Whenever a conditional window is needed, its numerator is written explicitly as $H_{i+m} - H_i$. For a raw group component supported on one level- i atom, $H_i = 0$, so the two quantities agree exactly.

Lemma II.5.10.3 (L-II.5.10 — Local entropy decomposition for actions). *Fix $R \geq 1$. There is $C_{d,R}$ such that, when $v \in \mathcal{P}(G_0(d))$ and $\mu \in \mathcal{P}(\mathbb{R}^d)$ have supports of d_0 /Euclidean diameter at most R , and $1 \leq m < n$,*

$$h_n(v \cdot \mu) \geq \mathbb{E}_{\substack{0 \leq i < n \\ g \sim v}} \frac{H_{i+m}(v_{g,i} \cdot \mu) - H_i(v_{g,i} \cdot \mu)}{m \log 2} - C_{d,R} \left(\frac{1}{m} + \frac{m}{n} \right). \quad (\text{II.5.17})$$

For every fixed $k \geq 0$,

$$\mathbb{E}_{\substack{g \sim v \\ x \sim \mu}} (h_n(v_{g,k} \cdot \mu) - h_n(\mu_{x,k})) \leq h_n(v \cdot \mu) - h_n(\mu) + \frac{C_{d,R,k}}{n}. \quad (\text{II.5.18})$$

In fact, (II.5.17) holds with the smaller error $d(m-1)/n$, and the error in (II.5.18) can be taken to be $H_{\min(k,n)}(\mu)/(n \log 2) \leq d \log(2^k R + 2)/(n \log 2)$. Raw spatial components are used in (II.5.18). Their canonical rescaling increases the subtracted entropy by a number in $[0, dk/n]$, so the same upper bound also holds after that replacement.

The joint-component version is also included. For $N \geq k$, set $\Delta_N(v, \mu) = h_N(v \cdot \mu) - h_N(\mu)$ and $Z_k(g, x) = \Delta_N(v_{g,k}, \mu_{x,k})$. A dimension constant $c_{\text{iso},d}$ independent of the supports and the levels satisfies

$$\mathbb{E} Z_k \leq \Delta_N(v, \mu) + \frac{H_k(\mu)}{N \log 2}, \quad Z_k \geq -\frac{c_{\text{iso},d}}{N \log 2}.$$

Consequently, for $\tau > 0$,

$$\mathbb{P}\{Z_k > \tau\} \leq \frac{\Delta_N(v, \mu) + (H_k(\mu) + c_{\text{iso},d})/(N \log 2)}{\tau + c_{\text{iso},d}/(N \log 2)}.$$

These inequalities are proved in Section 10, (10.6)–(10.8). The conditioning of both factors is essential for the later local action theorem, and the lower bound is essential for the stated Markov step.

Proof

The provenance is [Hoc20, Lemmas 3.7–3.8, printed pp. 27–29]. We retain conditional entropy throughout the local-average argument. A raw group component acting on all of μ need not have small spatial diameter: even a point mass at the identity leaves μ unchanged.

Conditional local averages

Put $\theta = v \cdot \mu$. The bounded-diameter hypotheses imply bounded supports, although their locations may be arbitrary. The action image is therefore bounded and every dyadic entropy below is finite. For every integer $i \geq 0$, group-component disintegration gives the probability-mixture identity

$$\theta = \int (v_{g,i} \cdot \mu) dv(g).$$

Conditional entropy concavity, proved for integral mixtures in Section 8, implies

$$\int H_{v_{g,i}\mu}(\mathcal{D}_{i+m} \mid \mathcal{D}_i) dv(g) \leq H_\theta(\mathcal{D}_{i+m} \mid \mathcal{D}_i). \quad (10.1)$$

For $j \geq 1$ put $c_j = H_\theta(\mathcal{D}_j \mid \mathcal{D}_{j-1})$. Then $0 \leq c_j \leq d \log 2$, since each dyadic cell has 2^d children. The chain rule and counting how often each increment occurs give

$$\begin{aligned} \sum_{i=0}^{n-1} H_\theta(\mathcal{D}_{i+m} \mid \mathcal{D}_i) &= \sum_{i=0}^{n-1} \sum_{j=i+1}^{i+m} c_j \\ &\leq m \sum_{j=1}^{n+m-1} c_j \\ &= m(H_{n+m-1}(\theta) - H_0(\theta)) \\ &\leq m(H_n(\theta) + d(m-1) \log 2). \end{aligned} \quad (10.2)$$

Divide (10.1)–(10.2) by $nm \log 2$ and use $H(\mathcal{D}_{i+m} \mid \mathcal{D}_i) = H_{i+m} - H_i$. This proves (II.5.17) with error $d(m-1)/n$, hence with the larger declared error as well. No support-diameter bound for $v_{g,i} \cdot \mu$ was used.

Localization at one fixed coarse level

Unconditional entropy concavity gives the exact inequality

$$\int H_n(v_{g,k} \cdot \mu) dv(g) \leq H_n(v \cdot \mu). \quad (10.3)$$

For the spatial components, disintegration gives

$$\int H_n(\mu_{x,k}) d\mu(x) = H_\mu(\mathcal{D}_n \mid \mathcal{D}_k) = H_n(\mu) - H_{\min(n,k)}(\mu). \quad (10.4)$$

The last equality uses nesting: when $n < k$ the conditional entropy is zero, and when $n \geq k$ the chain rule subtracts $H_k(\mu)$. Subtract (10.4) from (10.3) and divide by $n \log 2$. This proves (II.5.18) with exact error $H_{\min(n,k)}(\mu)/(n \log 2)$.

A set of Euclidean diameter at most R has extent at most R in each coordinate, and therefore intersects at most $(2^k R + 2)^d$ level- k dyadic cells. Thus

$$H_{\min(n,k)}(\mu) \leq H_k(\mu) \leq d \log(2^k R + 2), \quad (10.5)$$

which gives the required $C_{d,R,k}/n$ uniformly in the location of the support. Finally, on a level- k cube canonical rescaling changes $H_n(\mu_{x,k})$ to $H_{n+k}(\mu_{x,k})$. Their difference lies in $[0, dk \log 2]$. This proves the precise rescaling assertion after (II.5.18), completing L-II.5.10.

□

Joint coarse-component localization and a signed Markov bound

The pair used in a small-support action theorem is $(v_{g,k}, \mu_{x,k})$, not $(v_{g,k}, \mu)$. For $N \geq k$ put

$$\Delta_N(v, \mu) = h_N(v \cdot \mu) - h_N(\mu), \quad Z_k(g, x) = \Delta_N(v_{g,k}, \mu_{x,k}).$$

Joint component disintegration and entropy concavity give

$$\mathbb{E}_{g,x} H_N(v_{g,k} \cdot \mu_{x,k}) \leq H_N(v \cdot \mu).$$

Subtract (10.4), now with $n = N \geq k$. The exact result is

$$\mathbb{E} Z_k \leq \Delta_N(v, \mu) + \frac{H_k(\mu)}{N \log 2}. \quad (10.6)$$

There is also a uniform lower bound. A rotated and translated dyadic cube has coordinate extents at most $\sqrt{d}$ times its side length. It therefore meets at most J_d cells of the same-scale ambient grid, where we fix

$$J_d = (\lceil \sqrt{d} \rceil + 2)^d, \quad c_{\text{iso},d} = \log J_d.$$

The same bound holds in the reverse direction. The grid-comparison lemma therefore gives $|H_N(h_*\theta) - H_N(\theta)| \leq \log J_d$ for every isometry h and every bounded probability θ . Entropy concavity for the mixture over h implies

$$Z_k(g, x) \geq -\frac{c_{\text{iso},d}}{N \log 2} \quad (10.7)$$

for every positive-mass component pair. This lower bound has no dependence on the location of the support, on k , or on the particular components.

For $\tau > 0$, apply Markov to the nonnegative random variable $Z_k + c_{\text{iso},d}/(N \log 2)$. Equations (10.6)–(10.7) imply

$$\mathbb{P}\{Z_k > \tau\} \leq \frac{\Delta_N(v, \mu) + (H_k(\mu) + c_{\text{iso},d})/(N \log 2)}{\tau + c_{\text{iso},d}/(N \log 2)}. \quad (10.8)$$

In particular, if $0 < \eta < 1$,

$$\Delta_N(v, \mu) \leq \eta\tau/4, \quad \frac{H_k(\mu) + c_{\text{iso},d}}{N \log 2} \leq \eta\tau/4,$$

then $\mathbb{P}\{Z_k > \tau\} \leq \eta/2$. The coarse entropy is bounded by (10.5), with R replaced by the support diameter if necessary. These choices can consequently be made before the measures are given.

This proves the joint-component addition to L-II.5.10. One cannot apply Markov directly to a signed entropy increment, nor apply a theorem about two small supports while leaving the spatial factor unconditioned. $\square$

II.5.11 Removing the random subspace

A good subspace may initially depend on both sampled components. The decoupling argument discretizes the possible planes, uses independence to find a popular common plane, replaces the surviving family by an envelope or core, and finally resamples to obtain a level-only field.

Definition II.5.11.1 (II.5.13: Random probabilities, kernels, and subspace fields). For measurable spaces $(\Omega, \mathcal{F})$ and $(K, \mathcal{B})$, a **probability kernel** is a function $\kappa : \Omega \times \mathcal{B} \rightarrow [0, 1]$ such that $\kappa(\omega, \cdot)$ is a probability measure for every ω , and $\omega \mapsto \kappa(\omega, A)$ is measurable for every $A \in \mathcal{B}$. A **measurable certifying event** for a property $\mathcal{P}(\omega)$ is an event $E \in \mathcal{F}$ on which $\mathcal{P}(\omega)$ holds. A probability lower bound for such an event does not assert that the full set on which $\mathcal{P}$ holds is Borel.

For a Borel subset K of Euclidean space, a **random probability on K** is a measurable map into $\mathcal{P}(K)$ with its weak Borel structure. The **weak topology** is the coarsest topology making $\theta \mapsto \int \phi d\theta$ continuous for every bounded continuous real function ϕ on K ; the **weak Borel structure** is its Borel sigma-algebra. This convention does not require K to be compact.

Conditional sampling $x \sim \Theta$ uses the kernel $\omega \mapsto \Theta(\omega)$. Independent samples from two random probabilities use the product of their kernels conditionally on the two measures. The measure-valued random variables themselves are independent only when this is stated separately.

The space of all subspaces carries the Borel structure induced by d_G . A **random subspace field** over a random object Z is a Borel map $V = V(Z)$ into that space. A field $V(i, x, y)$ over random components may depend on every displayed argument unless a theorem declares it deterministic or level-only.

When asymptotic notation such as

$$0 < \delta \ll_{d,R,\varepsilon,m} 1, \quad k \gg_{d,R,\varepsilon,m,\delta} 1, \quad n \gg_{d,R,\varepsilon,m,\delta,k} 1$$

appears in explanatory prose, it records the displayed order of parameter choice. Every formal theorem replaces it by quantified functions such as $\delta_0(d, R, \varepsilon, m)$, $k_0(d, R, \varepsilon, m, \delta)$, and $n_0(d, R, \varepsilon, m, \delta, k)$, in that order.

Proposition II.5.11.2 (P-II.5.11 – Random-subspace decoupling). *For every $0 < \varepsilon < 1$, there is $M = M(d, \varepsilon)$ such that for every $m \geq M$ there are $\varepsilon' = \varepsilon'(d, \varepsilon, m) > 0$ and $m' = m'(d, \varepsilon, m) > m$ with the following property. Let Θ, Ξ be independent random probabilities on $[0, 1]^d$, and let $V = V(\Theta, \Xi)$ be a random subspace. Suppose there is a measurable event E_0 such that*

$$\mathbb{P}(E_0) \geq 1 - \varepsilon', \quad E_0 \subseteq \{\Theta \text{ is } (V, \varepsilon', m')\text{-saturated and } \Xi \text{ is } (V, \varepsilon')\text{-concentrated}\}. \quad (\text{II.5.19})$$

Then there is a deterministic subspace V_ and, after independently sampling i uniformly from $\{0, \dots, m' - 1\}$ and $x \sim \Theta, y \sim \Xi$, a measurable event E_1 such that*

$$\mathbb{P}(E_1) \geq 1 - \varepsilon, \quad E_1 \subseteq \{\Theta^{x,i} \text{ is } (V_*, \varepsilon, m)\text{-saturated and } \Xi^{y,i} \text{ is } (V_*, \varepsilon)\text{-concentrated}\}. \quad (\text{II.5.20})$$

The explicit measurable events specify the meaning of the assertions for general random probabilities; no measurability of an uncountable union over translates is assumed.

The scale-indexed corollary is included in this node. For the same ε, m range there are $\varepsilon'' > 0, m'' > m$, and N_0 such that, for $N \geq N_0, \mu, \nu \in \mathcal{P}([0, 1]^d)$, and a measurable field $V(i, x, y)$, the hypothesis

$$\mathbb{P}_{\substack{0 \leq i \leq N \\ x \sim \mu, y \sim \nu}} \left\{ \begin{array}{l} \mu^{x,i} \text{ is } (V(i, x, y), \varepsilon'', m'')\text{-saturated,} \\ \nu^{y,i} \text{ is } (V(i, x, y), \varepsilon'')\text{-concentrated} \end{array} \right\} \geq 1 - \varepsilon''$$

implies that level-only subspaces $V'_0, \dots, V'_N$ exist with

$$\mathbb{P}_{\substack{0 \leq i \leq N \\ x \sim \mu, y \sim \nu}} \left\{ \begin{array}{l} \mu^{x,i} \text{ is } (V'_i, \varepsilon, m)\text{-saturated,} \\ \nu^{y,i} \text{ is } (V'_i, \varepsilon)\text{-concentrated} \end{array} \right\} \geq 1 - \varepsilon.$$

The hypothesis is again certified on a measurable event. Once a subspace is assigned to each positive-mass component pair, all component events here are countable unions of measurable cells.

Proof

The provenance is [Hoc20, Proposition 5.1 and Corollary 5.2, printed pp. 58–62]. We prove the measurable choices explicitly by finite menus before using independence.

Measurable finite menus

On $\mathcal{P}([0, 1]^d)$, the map $\theta \mapsto \theta(B)$ is Borel for every Borel set B . To see this first for an open set O , use the increasing continuous functions $f_j(x) = \min(1, j \operatorname{dist}(x, O^c))$ (and $f_j = 1$ if O^c is empty). Their integrals increase to $\theta(O)$. Complements and countable disjoint unions then extend the claim to all Borel sets by the monotone-class theorem. The same conclusion holds on the subspace of probabilities carried by $[0, 1]^d$.

It follows that entropy in any fixed finite collection of dyadic or projected-dyadic cells is Borel in θ . For a positive-mass dyadic cell Q and any Borel B ,

$$\theta^Q(B) = \frac{\theta(Q \cap T_Q^{-1}B)}{\theta(Q)}. \quad (11.1)$$

Thus component evaluation is Borel as well. Give zero-mass cells the arbitrary value δ_0 when a total map is needed. Their sampling weight is zero.

We also need quantitative finite nets of subspaces. A projection matrix has entries in $[-1, 1]$. Partition that matrix box into cubes of side $\rho/(2d)$ and choose one projection matrix from each occupied cube. Since $\|M\|_{\text{op}} \leq d \max_{i,j} |M_{ij}|$, the corresponding subspaces form a finite ρ -net in d_G , of size at most

$$N_d(\rho) \leq C_d^{\text{net}} \rho^{-d^2}, \quad 0 < \rho < 1. \quad (11.2)$$

Here one may take $C_d^{\text{net}} = (4d + 2)^{d^2}$. A nearest net element has the same dimension as the original subspace, because two different dimensions have gap one. Fix an ordering of every net used below.

A measurable concentrated choice

For all sufficiently small $t > 0$ we construct finite-valued Borel maps $\xi \mapsto V_c(\xi)$ and $\xi \mapsto a_c(\xi)$ with

$$\begin{aligned} \xi((a_c(\xi) + V_c(\xi))^{(\gamma_c)}) &\geq 1 - \gamma_c, \\ \xi \text{ is } (W, t)\text{-concentrated} &\implies \alpha(V_c(\xi), W) \leq \gamma_c, \quad \gamma_c = C_d^c t^{1/(d+1)}. \end{aligned} \quad (11.3)$$

Use a subspace net of mesh $h = t/(4s_d)$, where $s_d = 2\sqrt{d} + 1$ is the concentration-stability constant from Section 4. Also fix a finite grid of possible affine offsets which approximates every vector of norm at most $\sqrt{d} + 1$ within $t/4$. For a measure ξ retain exactly those net planes W for which one of the offset vectors a satisfies

$$\xi((a + W)^{(4t)}) > 1 - 4t. \quad (11.4)$$

This is a finite family determined by finitely many Borel tests; it contains $\mathbb{R}^d$. Choose the first successful offset for each retained plane.

If ξ is (W_0, t) -concentrated for an arbitrary plane W_0 , its witness can be written with perpendicular offset of norm less than $\sqrt{d} + 1$. A net plane within h and a grid offset within $t/4$ give a tube of radius at most $t + s_d h + t/4 < 4t$ containing the original good set. Its mass is at least $1 - t > 1 - 4t$, so this net plane is retained.

Apply L-II.5.05(3) to the retained family, with parameter $8t$. For each possible retained family, fix once and for all its core and its at most d witnesses. Because there are finitely many families, these choices yield finite-valued Borel functions of ξ . The witness tubes chosen in (11.4) have intersection of mass greater than $1 - 4dt > 0$ for small t . For each possible family/offset configuration choose an anchor in that intersection with the unit cube; the geometric intersection is fixed by the configuration. There are only finitely many such configurations, so the anchor can also be fixed in advance and chosen Borel measurably. Configurations with empty intersection cannot occur.

The difference-and-rescaling argument in Section 6 gives concentration on the core with radius $\sqrt{d}c_d(8t)^{1/(d+1)}$ and exceptional mass at most $4dt$. Its near inclusion in any original W_0 costs at most $c_d(8t)^{1/(d+1)} + h$. Taking a dimension constant C_d^c larger than these coefficients proves (11.3). The witness affine offset is precisely the chosen anchor a_c . This proof does not apply a measurable-selection theorem to an uncountable family.

A measurable saturated choice

For every sufficiently large integer p there is a finite-valued Borel map $\theta \mapsto V_s(\theta)$ such that

$$\begin{aligned} \theta &\text{ is } (V_s(\theta), \gamma_s, p)\text{-saturated,} \\ \theta &\text{ is } (W, 1/p, p)\text{-saturated} \implies \alpha(W, V_s(\theta)) \leq \gamma_s, \quad \gamma_s = C_d^s \frac{\log(p+1)}{p}. \end{aligned} \quad (11.5)$$

Here is the finite-menu modification of Section 6's maximality argument. Fix a subspace net of mesh less than 2^{-p} , containing $\{0\}$ and $\mathbb{R}^d$. Let C_0 control the downward and nearby-subspace errors in L-II.5.04, and C_1 control its $\log(2/\delta)/p$ transverse-sum error. Choose $A \geq \max(1 + 2C_0, 2C_1, 1)$ and set

$$\delta_j = A4^j \frac{\log(p+1)}{p} \quad (0 \leq j \leq d).$$

Choose a net plane of largest dimension satisfying $(V, \delta_{\dim V}, p)$ -saturation, breaking ties by the fixed order. The zero plane is always admissible. Every test is Borel by (11.1) and the finite-label entropy observation above.

Assume $2\delta_d < 1$, as holds for all sufficiently large p . If θ is $(W, 1/p, p)$ -saturated but W escapes the chosen V_s by distance $2\delta_d$, choose a line in W in such a direction. When $j = \dim V_s > 0$, the transverse-sum argument of Section 6 makes $V_s + L$ saturated with error at most $3\delta_j$. A net plane of the same dimension within 2^{-p} is saturated with error at most $3\delta_j + C_0/p \leq 4\delta_j = \delta_{j+1}$. This contradicts maximality. When $j = 0$, downward saturation and net approximation of the line cost at most $(1 + 2C_0)/p$, also less than δ_1 , with the same contradiction. Thus (11.5) holds, for example with $C_d^s = 4A4^d$.

A one-sided change of concentration plane

We shall use the following consequence of the proof in Section 4: for a unit-cube probability ζ , if it is (U, a) -concentrated with $0 < a < 1$ and $\alpha(U, W) \leq b$, then it is $(W, a + s_d b)$ -concentrated. Indeed, for a perpendicular offset z of a witnessing translate of U , $\|z\| < \sqrt{d} + 1$, and

$$\begin{aligned} \text{dist}(x, z + W) &\leq \|(I - \pi_W)\pi_U(x - z)\| + \|(I - \pi_U)(x - z)\| \\ &< b(2\sqrt{d} + 1) + a \end{aligned}$$

on the original good set. Equal dimensions are not needed. We also repeatedly use

$$\alpha(U, W) \leq \alpha(U, Z) + \alpha(Z, W), \quad (11.6)$$

obtained by projecting a unit vector of U onto Z , whose projection still has norm at most one.

Parameter order

Let C_I be the saturation-inheritance constant in P-II.5.03(1), and let C_S control each saturation change in L-II.5.04(4). For the principal assertion choose m sufficiently large that

$$\frac{C_I}{\sqrt{m}} + \frac{2C_S}{m} < \frac{\varepsilon}{12}.$$

Next choose $p = m' > m$ so large that, with γ_s from (11.5),

$$\begin{aligned} C_I\sqrt{\gamma_s} + C_I\sqrt{m/p} &< \varepsilon/12, \quad 4\gamma_s < 1, \\ \Gamma = c_d(4\gamma_s)^{1/(d+1)} &< \min\{2^{-m}, \varepsilon/(24s_d)\}. \end{aligned} \quad (11.7)$$

These conditions are possible since $\gamma_s \rightarrow 0$ as $p \rightarrow \infty$. Finally put $D = d^2$, $b = 1/(4(D+1))$ and choose $t = \varepsilon' > 0$ sufficiently small that, with $\delta_c = t^b$,

$$\begin{aligned}
t \leq 1/p, \quad C_d^c t^{1/(d+1)} \leq \delta_c \leq \gamma_s, \quad \sqrt{2^p \delta_c} < \varepsilon/12, \\
\delta_c + \sqrt{\delta_c} + \sqrt{t} < \varepsilon/6, \quad \frac{\delta_c^{D+1}}{C_d^{\text{net}}} > \sqrt{t}.
\end{aligned} \tag{11.8}$$

Also require t to be in the range of (11.3). The last condition is possible because $\delta_c^{D+1} = t^{1/4}$, whereas $\sqrt{t} = t^{1/2}$. Since $b < 1/(d+1)$, all other requirements are also met by taking t small after p is fixed. This proves the required parameter order without a circular scale choice.

Independence produces a deterministic intermediate plane

Apply (11.3) to Ξ and (11.5) to Θ . Write the resulting independent finite-valued subspaces as V_c, V_s . On E_0 , (11.3), (11.5), $t \leq 1/p$ and (11.6) give

$$\alpha(V_c, V_s) \leq \delta_c + \gamma_s =: \delta.$$

Thus this Borel relation holds with probability at least $1 - t$. Take a δ_c -net $\mathcal{W}$ with $N \leq C_d^{\text{net}} \delta_c^{-D}$ elements, and retain its heavy members

$$\mathcal{W}_0 = \{W \in \mathcal{W} : \mathbb{P}\{d_G(V_c, W) < \delta_c\} > \delta_c/N\}.$$

The balls around the whole net cover every subspace. The union of the balls around light members has probability at most δ_c . Hence $\mathcal{W}_0$ is nonempty and

$$\mathbb{P}\{V_c \text{ lies within } \delta_c \text{ of some } W \in \mathcal{W}_0\} \geq 1 - \delta_c. \tag{11.9}$$

Apply L-II.5.05(2) to $\mathcal{W}_0$ with parameter $2\delta < 1$, and call its envelope V_* . Put $\delta' = c_d(2\delta)^{1/(d+1)} \leq \Gamma$. Then every heavy W is approximately contained in V_* with tolerance δ' . In particular (11.9) gives

$$\mathbb{P}\{\alpha(V_c, V_*) \leq 2\delta'\} \geq 1 - \delta_c, \tag{11.10}$$

where $\delta_c \leq \delta'$ follows from $\delta_c \leq \delta < 1$.

For each possible value S of V_s , let $q(S) = \mathbb{P}_{\Xi}\{\alpha(V_c, S) > \delta\}$. Independence and the preceding joint relation imply $\mathbb{E}_{\Theta} q(V_s) \leq t$. Thus, with probability at least $1 - \sqrt{t}$ over Θ ,

$$q(V_s) \leq \sqrt{t}. \tag{11.11}$$

Fix any S satisfying (11.11). For every heavy W , the event $d_G(V_c, W) < \delta_c$ has probability greater than $\delta_c/N \geq \delta_c^{D+1}/C_d^{\text{net}} > \sqrt{t}$. It must meet the event $\alpha(V_c, S) \leq \delta$. Consequently $\alpha(W, S) < \delta_c + \delta \leq 2\delta$. The envelope's minimality therefore gives $\alpha(V_*, S) \leq \delta'$. We have proved

$$\mathbb{P}_{\Theta}\{\alpha(V_*, V_s) \leq \delta'\} \geq 1 - \sqrt{t}. \tag{11.12}$$

All choices of V_* use the laws of the random measures, not their realizations. It is the required deterministic plane.

Passing to subcomponents

For a realization of Ξ in (11.10), its certified concentration on V_c has error at most δ_c . Concentration inheritance, including the direct case $i = 0$, shows that for every $0 \leq i < p$, outside conditional probability at most $\sqrt{\delta_c}$, its component is concentrated on V_c with error at most

$$\rho = \sqrt{2^p \delta_c} < \varepsilon/12.$$

The one-sided change-of-plane estimate and $\alpha(V_c, V_*) \leq 2\delta'$ give concentration on V_* with error at most $\rho + 2s_d\delta' < \varepsilon/6$. The union of the component failure and the exceptional realizations of Ξ has probability at most $\sqrt{\delta_c} + \delta_c$. The affine centers in (11.3), followed by the explicit tube-mass tests in the inheritance proof, certify these events measurably using (11.1).

Every realization of Θ is saturated on V_s with error γ_s at window p . Saturation inheritance gives a good component event of probability at least $1 - a_s$, where

$$a_s = C_I\sqrt{\gamma_s} + C_I/\sqrt{m} + C_I\sqrt{m/p}.$$

For a realization in (11.12), set $W = \pi_{V_s}V_*$. Its dimension equals that of V_* and $d_G(W, V_*) \leq \delta' < 2^{-m}$, by the projection argument (6.5). Downward saturation from V_s to W , then nearby-plane stability from W to V_* , give saturation with error at most

$$a_s + 2C_S/m < \varepsilon/6 \tag{11.13}$$

by the choices of m, p . This fails only on probability at most $a_s + \sqrt{t}$. All entropy tests concern fixed finite-menu planes and countably many component cells, so these are Borel events.

Take E_1 to be the intersection of the two explicitly certified good events just constructed. By (11.8), (11.13), and a union bound,

$$\mathbb{P}(E_1) \geq 1 - \delta_c - \sqrt{\delta_c} - \sqrt{t} - a_s > 1 - \varepsilon/3 \geq 1 - \varepsilon.$$

On it both conditions of (II.5.20) hold, proving the principal assertion.

The scale-indexed corollary

Let M_0 denote the lower bound used by the principal proof, and take the combined theorem's bound to be $M(d, \varepsilon) = M_0(d, \varepsilon/4)$. Apply the principal proof at $(\varepsilon/4, m)$ to obtain parameters t_0, p_0 . Decreasing its input tolerance if needed is harmless, so assume $t_0 \leq \varepsilon/4$. Set

$$m'' = p_0, \quad \varepsilon'' = \min(t_0^2, t_0, \varepsilon/4), \quad N_0 + 1 \geq 4p_0/\varepsilon.$$

For each level i let p_i be the conditional probability of the hypothesis event over the independent spatial components. Markov gives a set I of levels with $p_i \geq 1 - t_0$ and

$$|I| \geq (1 - t_0)(N + 1). \tag{11.14}$$

At a fixed level there are only countably many distinct pairs of rescaled component measures. Whenever the hypothesis succeeds for any point pair producing a given measure

pair, choose one witnessing subspace for that measure pair; assign an arbitrary plane to other pairs. This replacement depends only on the component measures, is cellwise measurable, and can only increase the success probability. The two random component measures are independent, so at every $i \in I$ the principal assertion applies. It gives a deterministic plane W_i whose descendant pairs over the length- p_0 window starting at i satisfy the target properties with error and failure probability at most $\varepsilon/4$.

Use the greedy disjoint length- p_0 intervals covering I from Section 3. Assign $V'_q = W_i$ on the interval beginning at i . The exact truncation count used in (3.12) bounds the global failure probability by

$$\varepsilon/4 + t_0 + p_0/(N+1) \leq 3\varepsilon/4.$$

Outside the chosen intervals define V'_q arbitrarily. The component-of-a-component identity carries each local assertion to the corresponding global level. The resulting level-only family satisfies the displayed conclusion of the corollary. This completes P-II.5.11. $\square$

II.5.12 From spatial image components to group components

Image-space structure does not immediately give group-space structure: an isometry has both a rotation and a translation. We first localize the group law, merge finitely many parallel image tubes, and then transfer the resulting structure back to raw group components.

Proposition II.5.12.1 (P-II.5.12 – From image components to action components).

For every $R \geq 1$ and $0 < \varepsilon < 1$, there is $M = M(d, \varepsilon)$ such that for every $m \geq M$ there are $\varepsilon' > 0$, $m' \geq m$, and N_0 such that the following holds for $N \geq N_0$. Let $v \in \mathcal{P}(G_0(d))$, $\mu \in \mathcal{P}(\mathbb{R}^d)$, $x_0 \in [-R, R]^d$, put $\eta = v \cdot x_0$, and suppose level-only subspaces $V_0, \dots, V_N$ satisfy

$$\mathbb{P}_{x \sim \mu, y \sim \eta} \{ \mu^{x,i} \text{ is } (V_i, \varepsilon', m')\text{-saturated and } \eta^{y,i} \text{ is } (V_i, \varepsilon')\text{-concentrated} \} \geq 1 - \varepsilon'. \quad (\text{II.5.21})$$

Then there are level-only $V'_0, \dots, V'_N$ such that

$$\mathbb{P}_{x \sim \mu, g \sim v} \{ \mu^{x,i} \text{ is } (V'_i, \varepsilon, m)\text{-saturated and } \mathcal{B}_i(v_{g,i}, x_0) \text{ is } (V'_i, \varepsilon)\text{-concentrated} \} \geq 1 - \varepsilon. \quad (\text{II.5.22})$$

The proof contains the finite-union-of-translates intermediate lemma; it is not an undeclared premise. This is an assertion in the ambient image coordinates. In rotation-normalized coordinates its concentration plane is $U_g^{-1}V'_i$, not generally V'_i .

Separated-accuracy clause. *For every $R \geq 1$, $0 < a < 1$ there is $M = M(d, a)$ such that for every $m \geq M$ and $0 < \zeta < 1$ there are $s > 0$, $L_0 > m$ with the following property. For every $L \geq L_0$ there are $z > 0$, N_0 such that joint saturation (V_i, s, L) and spatial-image concentration (V_i, z) with probability at least $1 - s$ imply joint saturation (W_i, a, m) and*

group-image concentration (W_i, ζ) with probability at least $1 - a$, for $N \geq N_0$. The exact probability statements are (12A.3)–(12A.4).

The clause permits the image map $F(h) = 2^b Q(hx_0 - t)$ and the original group partitions $\mathcal{D}_{b+j}^{G_0}$, for any integer $b \geq 0$, orthogonal Q , $x_0 \in [-R, R]^d$ and $t \in \mathbb{R}^d$. Its level- j group-image law is $(S_j)_* F_* \nu_{\mathcal{D}_{b+j}^{G_0}(h)}$. Parameters are uniform in these choices. In particular, s precedes the freely enlargable window L , and the concentration input z follows L . The proof below also establishes the fixed post-rotation variant.

Proof

The source counterparts are [Hoc20, Lemma 5.4 and Propositions 5.5–5.6, printed pp. 62–65]. Throughout this section the image components are $\mathcal{B}_i = (S_i)_*(\nu_{g,i} \cdot x_0)$. No rotation of the image coordinate frame is performed.

Diameter of an image of a group cell

If g, h belong to the same level- j group cell, then their matrix entries and translation coordinates differ by at most 2^{-j} . For $x_0 \in [-R, R]^d$,

$$\|gx_0 - hx_0\| \leq d2^{-j}\|x_0\| + \sqrt{d}2^{-j} \leq C2^{-j}, \quad C = d\sqrt{d}R + \sqrt{d}. \quad (12.1)$$

Consequently the image of a level- j group cell meets at most $M = ([C] + 2)^d$ ambient level- j spatial cells. The constants are independent of the translations and the cell's location.

Merging finitely many parallel tubes

We first prove a precise auxiliary statement. Let Z be a random vector, and let $F_0, \dots, F_{L-1}$ be countable measurable partitions of its underlying probability space. Assume the conditional values of Z on each positive-mass F_j -atom have diameter at most $C2^{-j}$. Suppose Z belongs, with probability at least $1 - \lambda$, to the union of at most M open λ -tubes about parallel affine planes $a_u + V$, where $0 < \lambda \leq 2^{-L}$. Fix $0 < \rho < 1$.

For each j , conditional Markov bounds the probability of atoms whose mass outside that union exceeds $\sqrt{\lambda}$ by $\sqrt{\lambda}$. Write $d_{uv} = \|\pi_{V^\perp}(a_u - a_v)\|$ and delete repeated parallel planes. A single conditional support meeting both tubes has

$$d_{uv} \leq C2^{-j} + 2\lambda \leq (C + 2)2^{-j}. \quad (12.2)$$

Call a scale j exceptional if for some pair

$$\rho 2^{-j} < d_{uv} \leq (C + 2)2^{-j}.$$

For each pair this restricts j to an interval of length $\log_2((C + 2)/\rho)$. Thus the number of exceptional integer scales is at most

$$T = M^2 \left(2 + \log_2 \frac{C + 2}{\rho} \right). \quad (12.3)$$

At a nonexceptional scale, all tubes meeting any one conditional support have their parallel planes at pairwise distance at most $\rho 2^{-j}$, by (12.2). If the atom also has union mass at least $1 - \sqrt{\lambda}$, choose one of its meeting planes. The part of the union within that support lies in its $(\lambda + \rho 2^{-j})$ -tube. After dilation by 2^j , the conditional measure is therefore (V, σ) -concentrated, with

$$\sigma = \max\{\sqrt{\lambda}, \rho + 2^L \lambda\}.$$

Averaging over j uniformly in $\{0, \dots, L-1\}$ proves the bound

$$\mathbb{P}\{\text{dilated conditional law is } (V, \sigma)\text{-concentrated}\} \geq 1 - \sqrt{\lambda} - T/L. \quad (12.4)$$

There are finitely many planes and countably many partition atoms, so these concentration claims are certified by measurable tube-mass events. The auxiliary statement does not require a choice of a rescaled group element.

Transferring one spatial scale

Fix a level i and suppose, with $\eta = \nu \cdot x_0$, that

$$\mathbb{P}_{y \sim \eta}\{\eta^{y,i} \text{ is } (V, \delta)\text{-concentrated}\} \geq 1 - \sqrt{\delta}, \quad 0 < \delta < 1. \quad (12.5)$$

For each good positive-mass spatial cell Q choose a witnessing affine tube in the original coordinates, of width $2^{-i}\delta$. Let T_i be the union of its intersection with Q over all good cells. There are countably many such cells, so T_i is Borel, and

$$\eta(T_i) \geq (1 - \sqrt{\delta})(1 - \delta) \geq 1 - 2\sqrt{\delta}. \quad (12.6)$$

For each positive-mass level- i group cell D put $\eta_D = (g \mapsto gx_0)_* \nu_D$. Group disintegration and (12.6) give $\sum_D \nu(D) \eta_D(T_i^c) \leq 2\sqrt{\delta}$. With

$$\lambda = \sqrt{2\sqrt{\delta}},$$

Markov shows that, except for ν -mass at most λ , $\eta_D(T_i) \geq 1 - \lambda$. For such a cell, (12.1) implies that T_i on its image uses at most M spatial cells and hence at most M parallel tubes. Dilation by 2^i makes their width $\delta \leq \lambda$. Thus $\mathcal{B}_i(\nu_D, x_0)$ is $(V, \lambda)_M$ -concentrated.

Now work conditionally in one such group cell D , put $Z(g) = 2^i g x_0$ and use its level- $i+j$ descendant partition for F_j . By (12.1) the conditional values have diameter at most $C2^{-j}$. The dilated conditional law appearing in (12.4) is exactly

$$(S_j)_*(S_i)_*(\nu_{g,i+j} \cdot x_0) = \mathcal{B}_{i+j}(\nu_{g,i+j}, x_0).$$

Therefore, whenever $\lambda \leq 2^{-L}$, the combination of the outer Markov estimate and (12.4) gives

$$\mathbb{P}_{0 \leq j < L, \substack{g \sim \nu}}\{\mathcal{B}_{i+j}(\nu_{g,i+j}, x_0) \text{ is } (V, \sigma)\text{-concentrated}\} \geq 1 - \lambda - \sqrt{\lambda} - T/L. \quad (12.7)$$

This proves the one-scale transfer with its full parameter dependence.

Choosing parameters and combining scales

Let C_I be the constant in saturation inheritance. Choose $M(d, \varepsilon)$ so that for $m \geq M(d, \varepsilon)$,

$$C_I/\sqrt{m} \leq \varepsilon/10.$$

Put $\rho = \varepsilon/10$, and let C, M, T be as in (12.1)–(12.3). Choose $L > m$ large enough that

$$C_I\sqrt{m/L} \leq \varepsilon/10, \quad T/L \leq \varepsilon/10.$$

Set $m' = L$. Next choose $\delta = \varepsilon' > 0$ so small that

$$\begin{aligned} (C_I + 2)\sqrt{\delta} &\leq \varepsilon/10, & \lambda + \sqrt{\lambda} &\leq \varepsilon/10, & 2^L\lambda &\leq \varepsilon/10, \\ \lambda &= \sqrt{2\sqrt{\delta}} < 1. \end{aligned} \tag{12.8}$$

In particular $\lambda \leq 2^{-L}$. Finally choose N_0 so that $L/(N+1) \leq \varepsilon/10$ for every $N \geq N_0$. This is the parameter order of P-II.5.12: m is fixed before m', ε' before N_0 , and every dependence on R enters through the displayed image-diameter constants.

Under hypothesis (II.5.21), let I be the set of levels at which the joint success probability is at least $1 - \sqrt{\delta}$. Markov gives $|I| \geq (1 - \sqrt{\delta})(N + 1)$. At any $i \in I$ the spatial-image marginal satisfies (12.5), so (12.7) applies with $V = V_i$. Its concentration tolerance is at most $\sigma \leq \varepsilon/5$ by (12.8).

The other marginal at level i has saturation success probability at least $1 - \sqrt{\delta}$. On its successful components, P-II.5.03(1) with outer window L gives target saturation tolerance and exceptional probability at most

$$a = C_I\sqrt{\delta} + C_I/\sqrt{m} + C_I\sqrt{m/L} \leq 3\varepsilon/10.$$

The component-of-a-component identity transfers this to global levels $i, \dots, i + L - 1$. Combining the two marginals by a union bound, on this interval the average joint failure probability is at most

$$p = \sqrt{\delta} + a + \lambda + \sqrt{\lambda} + T/L.$$

Choose greedily disjoint length- L integer intervals starting in I and covering I , as in Section 3. Define $V'_q = V_i$ on the interval starting at i . The exact boundary count from (3.12) bounds the global failure probability by

$$\begin{aligned} p + \sqrt{\delta} + \frac{L}{N+1} &= \frac{C_I}{\sqrt{m}} + C_I\sqrt{\frac{m}{L}} + (C_I + 2)\sqrt{\delta} + \lambda + \sqrt{\lambda} + \frac{T}{L} + \frac{L}{N+1} \\ &\leq 6\varepsilon/10 < \varepsilon. \end{aligned} \tag{12.9}$$

Assign arbitrary planes at any remaining levels. Both the saturation and concentration tolerances are less than ε . This proves (II.5.22) and completes P-II.5.12. $\square$

II.5.13 Image transfer with separately chosen concentration accuracy

The single-tolerance statement P-II.5.12 does not allow the concentration accuracy to be chosen independently of the saturation accuracy. The following refinement supplies precisely that freedom and keeps the original group partition throughout the image transfer.

Proposition II.5.13.1 (Image transfer with separately chosen concentration accuracy).

Fix $R \geq 1$. For every $0 < a < 1$ there is $M = M(d, a)$ such that, for every $m \geq M$ and $0 < \zeta < 1$, there are $s > 0$ and $L_0 > m$ with the following property: for every $L \geq L_0$ there are $z > 0$ and N_0 for which the conclusion below holds. Crucially, s and L_0 are fixed **before** L , whereas z, N_0 may depend on L .

Let $b \geq 0$ be an integer, Q an orthogonal map, $x_0 \in [-R, R]^d$, and $t \in \mathbb{R}^d$. On the group space put

$$F(h) = 2^b Q(hx_0 - t), \quad \mathcal{F}_j = \mathcal{D}_{b+j}^{G_0} \quad (j \geq 0), \quad \eta = F_* v. \quad (12A.1)$$

Here v is a Borel probability on $G_0(d)$. For a positive-mass atom D of $\mathcal{F}_j$ write $v_D = v|_D / v(D)$, and define

$$\mathcal{B}_j^F(v_D) = (S_j)_* F_* v_D, \quad S_j(y) = 2^j y. \quad (12A.2)$$

Thus these are image laws of the **original** group partition, not components of a rotated group measure. Let μ be a Borel probability on $\mathbb{R}^d$, $N \geq N_0$, and let $V_0, \dots, V_N$ be level-only planes. Suppose

$$\mathbb{P}_{\substack{0 \leq i \leq N \\ x \sim \mu, y \sim \eta}} \left\{ \begin{array}{l} \mu^{x,i} \text{ is } (V_i, s, L)\text{-saturated,} \\ \eta^{y,i} \text{ is } (V_i, z)\text{-concentrated} \end{array} \right\} \geq 1 - s. \quad (12A.3)$$

Then there are level-only planes $W_0, \dots, W_N$ such that

$$\mathbb{P}_{\substack{0 \leq i \leq N \\ x \sim \mu, h \sim v}} \left\{ \begin{array}{l} \mu^{x,i} \text{ is } (W_i, a, m)\text{-saturated,} \\ \mathcal{B}_i^F(v_{\mathcal{F}_i(h)}) \text{ is } (W_i, \zeta)\text{-concentrated} \end{array} \right\} \geq 1 - a. \quad (12A.4)$$

The parameters are uniform in b, Q, x_0, t, μ, v, N subject to the stated bounds. All assertions are certified by measurable events. The case $b = 0, Q = I, t = 0$ uses the ordinary unrotated image component.

Proof. For points in one $\mathcal{F}_j$ atom, the calculation (12.1) and orthogonality of Q give

$$\text{diam } F(D) \leq C 2^{-j}, \quad C = d\sqrt{d} R + \sqrt{d}. \quad (12A.5)$$

Consequently at most $M_* = ([C] + 2)^d$ level- j spatial cells meet $F(D)$. Translations and the initial level b have disappeared from this bound.

We first separate the two errors in the tube-merging argument. Suppose a random vector lies, with probability at least $1 - u$, in at most M_* parallel tubes of radius z , and its conditional supports at level j have diameter at most $C 2^{-j}$. Assume $z \leq 2^{-L}$. The proof of (12.2)–(12.4), with the mass error and radius kept distinct, gives, for every $0 < \rho < 1$,

$$\mathbb{P}_{\substack{0 \leq j < L \\ \text{conditional atom}}} \left\{ \begin{array}{l} \text{the dilated conditional law is} \\ (V, \max\{\sqrt{u}, \rho + 2^L z\})\text{-concentrated} \end{array} \right\} \geq 1 - \sqrt{u} - T/L, \quad (12A.6)$$

$$T = M_*^2 \left(2 + \log_2 \frac{C+2}{\rho} \right).$$

To verify this directly, conditional Markov leaves mass at least $1 - \sqrt{u}$ in the union except for conditional-atom probability $\sqrt{u}$. At all but T scales, any two tubes meeting the same conditional support have parallel-plane distance at most $\rho 2^{-j}$: a larger distance would lie between $\rho 2^{-j}$ and $(C+2)2^{-j}$, one of the counted exceptional intervals. The union within the support therefore lies in one tube of radius $z + \rho 2^{-j}$. Dilation proves (12A.6). The mass error $\sqrt{u}$ is not multiplied by 2^L .

Choose the parameters as follows. Take $m \geq M(d, a)$ so that

$$C_I/\sqrt{m} \leq a/20, \quad \rho = \zeta/4, \quad 0 < u \leq \min\{\zeta^2/16, a^2/400\}. \quad (12A.7)$$

Here C_I is the saturation-inheritance constant. Choose $0 < s < 1$ so small that

$$C_I\sqrt{s} \leq a/20, \quad 2\sqrt{s} \leq a/20, \quad \sqrt{s}/u \leq a/20. \quad (12A.8)$$

Next choose $L_0 > m$ so that, for every $L \geq L_0$,

$$C_I\sqrt{m/L} \leq a/20, \quad T/L \leq a/20. \quad (12A.9)$$

For the selected $L \geq L_0$ choose

$$0 < z \leq \min\{2^{-L}\zeta/4, au/20\}, \quad N_0 + 1 \geq 20L/a. \quad (12A.10)$$

These inequalities have exactly the order stated in the theorem.

Let I be the levels at which the conditional failure probability in (12A.3) is at most $\sqrt{s}$. Markov gives

$$|I| \geq (1 - \sqrt{s})(N + 1).$$

Fix $i \in I$. The good η -component cells have total η -mass at least $1 - \sqrt{s}$. Choose their witnessing affine tubes of original radius $2^{-i}z$ and intersect each tube with its cell; let T_i be their union. Then

$$\eta(T_i^c) \leq \sqrt{s} + z. \quad (12A.11)$$

Disintegrate ν over $\mathcal{F}_i$. Except for group-cell probability at most $(\sqrt{s} + z)/u$, the law $F_*\nu_D$ puts mass at least $1 - u$ in T_i . By (12A.5), on such an image T_i uses at most M_* parallel tubes. After dilation by 2^i their radii are z .

Condition within this group cell and use the nested partitions $\mathcal{F}_{i+j}$. Their dilated image diameters are $C2^{-j}$, so (12A.6) applies. Its laws are exactly $\mathcal{B}_{i+j}^F(\nu_{\mathcal{F}_{i+j}(h)})$. By (12A.7), (12A.10), their concentration tolerance is at most

$$\max\{\sqrt{u}, \rho + 2^L z\} \leq \zeta/2. \quad (12A.12)$$

The average group-image failure on this length- L window is at most $(\sqrt{s}+z)/u + \sqrt{u} + T/L$.

The spatial marginal at level i has saturation success at least $1 - \sqrt{s}$. On its successful components, P-II.5.03 gives target saturation tolerance and additional failure at most

$$a_s = C_I \sqrt{s} + C_I / \sqrt{m} + C_I \sqrt{m/L} \leq 3a/20.$$

Resampling identifies these descendants with the original spatial components. Thus, on the length- L window starting at i , joint failure is at most

$$\sqrt{s} + a_s + \frac{\sqrt{s} + z}{u} + \sqrt{u} + T/L.$$

Choose the greedy disjoint windows starting in I and assign $W_j = V_i$ throughout the window starting at i . The final truncation costs at most $L/(N+1)$; uncovered levels cost at most $\sqrt{s}$. Total failure is at most

$$2\sqrt{s} + a_s + \frac{\sqrt{s} + z}{u} + \sqrt{u} + T/L + \frac{L}{N+1} \leq 9a/20 < a. \quad (12A.13)$$

Saturation tolerance is $a_s < a$ and concentration tolerance is at most $\zeta/2 < \zeta$. Countable cell choices and explicit tube-mass tests certify all events. This proves (12A.4). $\square$

The new clause allows L to be enlarged after s is fixed, for example to meet the Euclidean inverse theorem's threshold $L \geq M(d, s)$. Its concentration input z is then chosen after L , which that inverse theorem explicitly permits. No condition of the form $L \geq M(d, z)$ is introduced.

II.5.14 Recovering group entropy from an affine frame

Definition II.5.14.1 (II.5.07: Non-affinity and independent affine frames). Let $0 < \varepsilon < 1$ and $\sigma > 0$. A probability μ on $\mathbb{R}^d$ is (ε, σ) -**non-affine** if

$$\mu(L^{(\sigma)}) < \varepsilon$$

for every proper affine subspace L of $\mathbb{R}^d$.

Points $x_0, \dots, x_d \in \mathbb{R}^d$ form a σ -**independent affine frame** if

$$\text{dist}(x_j, \text{aff}\{x_\ell : 0 \leq \ell \leq d, \ell \neq j\}) \geq \sigma \quad (0 \leq j \leq d),$$

where the affine hull on the right is formed from the other d points.

The images of an independent affine frame determine an isometry quantitatively. Non-affinity supplies such frames with positive probability, allowing entropy of spatial images to recover entropy of the group component.

Lemma II.5.14.2 (L-II.5.13 — Affine-frame recovery of group entropy). *Fix $R \geq 1$, $\sigma > 0$, and an integer k . There is $C_{d,R,\sigma}$, independent of k , such that if $x_0, \dots, x_d \in [-R, R]^d$ are a σ -independent affine frame and v is supported on one atom of $\mathcal{D}_k^{G_0}$, then, for every $m \geq 1$,*

$$h_{k,m}^{G_0}(v) \leq \sum_{j=0}^d h_{k,m}(v \cdot x_j) + \frac{C_{d,R,\sigma}}{m}. \quad (\text{II.5.23})$$

If $\mu \in \mathcal{P}([-R, R]^d)$ is (ε, σ) -non-affine, then

$$\mu \left\{ x : h_{k,m}(v \cdot x) > \frac{1}{d+1} h_{k,m}^{G_0}(v) - \frac{C_{d,R,\sigma}}{m} \right\} \geq 1 - ((d+1)\varepsilon)^{1/(d+1)}. \quad (\text{II.5.24})$$

Proof

The provenance is [Hoc20, Definition 5.7, Proposition 5.8, Lemma 5.9, and Corollary 5.10, printed pp. 66–67].

Quantitative recovery of an isometry

Let $x_0, \dots, x_d$ be a σ -independent affine frame in the fixed box. These points are affinely independent: an affine dependence would place one of them in the affine span of the others. Consequently the matrix $L = [x_1 - x_0, \dots, x_d - x_0]$ is invertible.

The j th row of L^{-1} , viewed as a linear functional, vanishes on all columns except $x_j - x_0$ and takes value one there. Its norm is the reciprocal of the distance from $x_j - x_0$ to the span of the other columns. That distance is at least σ by the frame condition. Hence

$$\|L^{-1}\|_{\text{op}} \leq \sqrt{d}/\sigma. \quad (13.1)$$

For isometries $g = (U, a)$ and $h = (V, b)$, put $y_j = gx_j - hx_j$ and $\eta = \max_j \|y_j\|$. Then $(U - V)L$ has columns $y_j - y_0$ and operator norm at most $2\sqrt{d}\eta$. Equation (13.1) gives

$$\begin{aligned} \|U - V\|_{\text{op}} &\leq (2d/\sigma)\eta, \\ \|a - b\| &= \|y_0 - (U - V)x_0\| \leq (1 + 2d\sqrt{d}R/\sigma)\eta, \\ d_0(g, h) &\leq (1 + 2d(1 + \sqrt{d}R)/\sigma)\eta. \end{aligned} \quad (13.2)$$

The reverse Lipschitz bound $\max_j \|gx_j - hx_j\| \leq \max(\sqrt{d}R, 1)d_0(g, h)$ follows immediately from the action formula. Thus the map $F(g) = (gx_0, \dots, gx_d)$ is injective and has the stated quantitative inverse on its image.

Let $n = k + m$. If two isometries have F -images in the same level- n product dyadic cell, then $\eta \leq \sqrt{d}2^{-n}$. By (13.2), the preimage of that output cell meets at most $K_{d,R,\sigma}$ level- n group cells: each matrix entry and translation coordinate ranges over an interval of length at most $C_{d,R,\sigma}2^{-n}$. Conditional entropy comparison and subadditivity therefore give

$$\begin{aligned}
H_\nu(\mathcal{D}_n^{G_0}) &\leq H_{F_\nu}(\mathcal{D}_n^{d(d+1)}) + \log K_{d,R,\sigma} \\
&\leq \sum_{j=0}^d H_n(\nu \cdot x_j) + \log K_{d,R,\sigma}.
\end{aligned} \tag{13.3}$$

All these entropies are finite because ν is carried by a single group cell. Divide by $m \log 2$ to prove (II.5.23). In particular, the proof works for every integer k with a constant independent of that level.

Finding an affine frame in a large set

Suppose μ is (ε, σ) -non-affine, and let A be Borel with $\mu(A) > ((d+1)\varepsilon)^{1/(d+1)}$. Sample $X_0, \dots, X_d$ independently from μ . For each j the affine span L_j of the other d points is proper, and X_j is independent of those points. Non-affinity gives

$$\mathbb{P}\{X_j \in L_j^{(\sigma)}\} < \varepsilon.$$

These events are measurable. For fixed other points, distance to their affine span is the infimum over rational coefficients of distances to their affine combinations; this is a countable infimum of continuous functions of the entire tuple. The displayed bound can therefore be integrated over the other points by Fubini.

A union bound and independence of the X_j show

$$\begin{aligned}
&\mathbb{P}\{X_j \in A \text{ and } \text{dist}(X_j, L_j) \geq \sigma \text{ for every } j\} \\
&\geq \mu(A)^{d+1} - (d+1)\varepsilon > 0.
\end{aligned} \tag{13.4}$$

Any tuple in this event is the required σ -independent affine frame in A .

Most points detect group entropy

Let C_0 be the error constant in (II.5.23), and enlarge the common constant $C_{d,R,\sigma}$ to at least $C_0 + 1$. Define the bad set

$$A = \left\{ x \in [-R, R]^d : h_{k,m}(\nu \cdot x) \leq \frac{h_{k,m}^{G_0}(\nu)}{d+1} - \frac{C_{d,R,\sigma}}{m} \right\}.$$

It is Borel: for a dyadic cell Q the function $x \mapsto \int 1_Q(gx) d\nu(g)$ is Borel by parameter integration of a Borel function, and only finitely many cells are relevant on the fixed spatial box and group cell.

If $\mu(A) > ((d+1)\varepsilon)^{1/(d+1)}$, (13.4) supplies an affine frame $x_0, \dots, x_d$ in A . Applying (II.5.23) to that frame gives

$$h_{k,m}^{G_0}(\nu) \leq h_{k,m}^{G_0}(\nu) - \frac{(d+1)C_{d,R,\sigma} - C_0}{m},$$

which is impossible. Thus $\mu(A)$ is at most the stated threshold. Its complement is the strict-superlevel set in (II.5.24), proving that inequality and completing L-II.5.13. $\square$

II.5.15 Linearizing the isometry action

Localization freezes the base point and rotation. The next lemma measures the quadratic error and records how the linearized image changes when the base point is moved.

Lemma II.5.15.1 (L-II.5.14 — Linearization of the isometry action). *For $g = (U, a)$, $g_0 = (U_0, a_0)$, and $x, x_0 \in \mathbb{R}^d$,*

$$gx = gx_0 + U_0(x - x_0) + (U - U_0)(x - x_0). \quad (\text{II.5.25})$$

Consequently, for every $R \geq 1$ and $m \geq 1$ there is $k_0 = k_0(d, R, m)$ such that, for $k \geq k_0$, if v and μ are supported in single level- k group/spatial dyadic cells, $g_0 \in \text{supp } v$, and $x_0 \in \text{supp } \mu$, then

$$|h_{k,m}(v \cdot \mu) - h_{k,m}((v \cdot x_0) * (U_0)_* \mu)| \leq \frac{C_{d,R}}{m}. \quad (\text{II.5.26})$$

Translations may be inserted to center the two factors. Rotation comparison also gives

$$|h_{k,m}(v \cdot \mu) - h_{k,m}(((U_0^{-1})_*(v \cdot x_0)) * \mu)| \leq C_{d,R}/m.$$

Also, for every $\varepsilon > 0$, all sufficiently large k satisfy

$$\mathcal{A}_k(v_{g,k}, x) (V, \varepsilon)\text{-concentrated} \implies \mathcal{A}_k(v_{g,k}, y) (V, 2\varepsilon)\text{-concentrated} \quad (\text{II.5.27})$$

whenever x, y lie in one level- k spatial cell and the displayed supports stay in the fixed radius- R ball.

Proof

The provenance is [Hoc20, identities (63)–(65) and Lemmas 5.11–5.12, printed pp. 65–69]. We use direct couplings and grid counts.

The quadratic remainder

Expand $g = (U, a)$ and $g_0 = (U_0, a_0)$. For every x, x_0 ,

$$gx = gx_0 + U_0(x - x_0) + (U - U_0)(x - x_0),$$

which is (II.5.25). If the group and spatial inputs are in their respective level- k cells, then

$$\|U - U_0\| \leq d2^{-k}, \quad \|x - x_0\| \leq \sqrt{d} 2^{-k},$$

including when the chosen support points are in the closures of those cells. Hence the last term has norm at most $J_d 2^{-2k}$, where $J_d = d^{3/2}$. Choose

$$k_0 \geq m + \lceil \log_2 J_d \rceil.$$

For independent $g \sim v$ and $x \sim \mu$, the coupled vectors $Y = gx$ and $Y' = gx_0 + U_0(x - x_0)$ then differ by at most $2^{-(k+m)}$. Each coordinate of their level- $k + m$ cell indices differs by a

bounded integer, so each cell index conditional on the other has at most 5^d possible values. The entropy chain rule gives

$$|H_{k+m}(\text{Law}(Y)) - H_{k+m}(\text{Law}(Y'))| \leq d \log 5. \quad (14.1)$$

The law of Y' is

$$(\nu \cdot x_0) * ((U_0 \circ \tau_{-x_0})_* \mu).$$

It is a translate by $-U_0 x_0$ of $(\nu \cdot x_0) * (U_0)_* \mu$. Translation changes dyadic entropy by at most $d \log 2$. Divide (14.1), including this translation error, by $m \log 2$ to obtain (II.5.26). Applying U_0^{-1} to the entire convolution changes entropy by a dimension-dependent constant, by the orthonormal grid comparison in Section 1. Its image is $((U_0^{-1})_*(\nu \cdot x_0)) * \mu$. This proves the rotated convolution variant too.

Changing the base point

Fix a positive-mass group cell containing g and let $h \sim \nu_{g,k}$. For spatial points x, y in one level- k cell, the two random vectors in the normalized component actions satisfy

$$2^k U_g^{-1} h y - 2^k U_g^{-1} h x = 2^k (y - x) + 2^k U_g^{-1} (U_h - U_g)(y - x). \quad (14.2)$$

The second term has norm at most $J_d 2^{-k}$, whereas the first term is a fixed translation. If k is so large that $J_d 2^{-k} \leq \varepsilon$, a coupling maps every point in an open ε -tube about $a + V$ for the first law into the open 2ε -tube about $a + 2^k(y - x) + V$ for the second law. The mass of that tube is at least $1 - \varepsilon \geq 1 - 2\varepsilon$. This proves (II.5.27). The same proof without the fixed inverse rotation applies to the unrotated $\mathcal{B}_k$ actions as well. It completes L-II.5.14. $\square$

II.5.16 Localized action-dependent decoupling

Proposition II.5.16.1 (P-II.5.11-Loc – Localized action-dependent decoupling). *For every $d \geq 1$, $R \geq 1$ and $0 < \varepsilon < 1$ there is $M = M(d, R, \varepsilon)$ such that for every $m \geq M$ there are $\varepsilon_0 > 0$, integers $p > m$, k_0 and N_0 with the following property. Let $k \geq k_0$, $N \geq N_0$, $\mu \in \mathcal{P}([-R, R]^d)$ and $\nu \in \mathcal{P}(G_0(d))$. Require that each measure is carried by a single level- k dyadic cell in its respective spatial/group space.*

Suppose a measurable subspace field $V(i, x, g)$ and measurable event E_0 certify, for independent uniform $0 \leq i \leq N$, $x \sim \mu$ and $g \sim \nu$, joint probability at least $1 - \varepsilon_0$ of

$$\mu^{x,i} \text{ is } (V(i, x, g), \varepsilon_0, p)\text{-saturated,} \quad \mathcal{A}_i(\nu_{g,i}, x) \text{ is } (V(i, x, g), \varepsilon_0)\text{-concentrated.} \quad (14A.1)$$

Then level-only planes $W_0, \dots, W_N$ and a measurable good event exist with probability at least $1 - \varepsilon$ on which

$$\mu^{x,i} \text{ is } (W_i, \varepsilon, m)\text{-saturated,} \quad \mathcal{A}_i(\nu_{g,i}, x) \text{ is } (W_i, \varepsilon)\text{-concentrated.} \quad (14A.2)$$

The constants $\varepsilon_0, p, k_0, N_0$ depend only on d, R, ε, m ; N_0 may be chosen independently of $k \geq k_0$. The small-root-cell hypothesis is essential: without it, translating the base point can move the component action by an uncontrolled amount. Thus the proposition is intentionally local in both the spatial and group variables.

Separated-input clause. For every d, R, ε in the same ranges there is $M = M(d, R, \varepsilon)$ such that for every $m \geq M$ there are $a, \beta > 0, p_0 > m$ with this property. For every $p \geq p_0$ there are $c > 0, k_0, N_0$ such that, under the same small-root-cell hypotheses, a measurable field certifying joint (V, a, p) -saturation and (V, c) -concentration with probability at least $1 - \beta$ gives the same level-only conclusion at ε, m . The exact statement is (14B.1). Here a, β, p_0 precede p , and only c, k_0, N_0 follow it.

Proof strategy

Freeze one spatial base point and one rotation on the root cells. This turns the component actions into uniformly bounded image laws. Finite menus then replace the random concentration and saturation planes by a deterministic intermediate plane, after which descendant resampling produces a level-only family. The final interval selection accounts for the endpoint loss.

Freezing both coordinate choices

Choose x_0 in the spatial root cell with $x_0 \in \text{supp } \mu$, and g_0 in the group root cell with $g_0 \in \text{supp } \nu$. Write $U_0 = U_{g_0}$. For each positive-mass level- i group cell D , choose $g_D \in D \cap \text{supp } \nu$ also in the root cell. Set

$$Z_{i,D}(h) = 2^i U_0^{-1}(hx_0 - g_D x_0), \quad \Xi_{i,D} = (Z_{i,D})_* \nu_D. \quad (14A.3)$$

Here ν_D is the raw normalized restriction to D . Put

$$C = d\sqrt{d} R + \sqrt{d}, \quad J_d = d^{3/2}, \quad K = dC + J_d.$$

The matrix-entry bounds give $\|hx_0 - g_D x_0\| \leq d2^{-i}\|x_0\| + \sqrt{d}2^{-i} \leq C2^{-i}$. Thus $\|Z_{i,D}(h)\| \leq C$, so these centered image laws are carried by $[-C, C]^d$, uniformly in the level and cell.

For sampled $g \in D$ and x in the spatial root cell, subtract a translation from $\mathcal{A}_i(\nu_D, x)$ and couple it with $\Xi_{i,D}$ using the same $h \sim \nu_D$. The essential difference is

$$\begin{aligned} & 2^i U_0^{-1}(hx_0 - gx_0) - 2^i U_g^{-1}(hx - gx) \\ &= 2^i (U_0^{-1} - U_g^{-1})(hx_0 - gx_0) + 2^i U_g^{-1}(U_h - U_g)(x_0 - x). \end{aligned} \quad (14A.4)$$

Its norm is at most

$$e_k = K2^{-k}. \quad (14A.5)$$

Indeed, $\|U_0 - U_g\| \leq d2^{-k}$, $\|hx_0 - gx_0\| \leq C2^{-i}$, $\|U_h - U_g\| \leq d2^{-i}$, and $\|x - x_0\| \leq \sqrt{d}2^{-k}$. The difference between using gx_0 and $g_D x_0$ is another fixed translation. These estimates hold also for $i < k$, since both samples still lie in the same level- i cell as well as the root cell.

Consequently concentration on any fixed plane transfers in either direction between $\mathcal{A}_i(v_D, x)$ and $\Xi_{i,D}$ with radius increase at most e_k and no decrease in the good mass. In particular a concentration tolerance $b > 0$ becomes at most $b + e_k$. This is a coupling assertion about affine tubes, not continuity of entropy under this change.

The bounded-image modification of the finite menus

For a probability ξ carried by $[-C, C]^d$, apply the concentrated finite-menu construction (11.3) to

$$T_*\xi, \quad T(z) = \frac{z}{4C} + (1/2, \dots, 1/2).$$

Its support lies in $[1/4, 3/4]^d$. If ξ is (W, t) -concentrated, then $T_*\xi$ is (W, t) -concentrated: the radius shrinks by $4C$ and the good mass remains at least $1 - t$. Pulling the output tube back shows that the same finite-valued Borel choice $V_c(\xi)$ satisfies

$$\begin{aligned} \xi \text{ is } (V_c(\xi), \gamma_c^C)\text{-concentrated,} \\ \xi \text{ is } (W, t)\text{-concentrated} \implies \alpha(V_c(\xi), W) \leq \gamma_c^C, \quad \gamma_c^C = 4CC_d^c t^{1/(d+1)}. \end{aligned} \quad (14A.6)$$

The affine witness is a finite-valued Borel choice as well. The map for a saturated probability Θ on the unit cube is unchanged: (11.5) gives $V_s(\Theta)$ and $\gamma_s = C_d^s \log(p+1)/p$.

For use after conditioning, the one-sided change-of-plane estimate in Section 11 holds on $[-C, C]^d$ with

$$s_C = 2\sqrt{d}C + 1$$

in place of s_d . For completeness, if a tube of radius $b < 1$ about a translate of U has positive mass, its perpendicular offset has norm less than $\sqrt{d}C + 1$. For a point in that tube, projection onto U and then onto W gives distance to the same translate of W less than $b + s_C\alpha(U, W)$. This proves the stated change, including its mass requirement.

Parameters and the independent intermediate plane

Let $\eta = \varepsilon/8$. Use the constants C_I, C_S from Section 11. Choose $m \geq M$ sufficiently large that

$$C_I/\sqrt{m} + 2C_S/m < \eta/12.$$

Then choose $p > m$ so large that

$$\begin{aligned} C_I\sqrt{\gamma_s} + C_I\sqrt{m/p} &< \eta/12, \quad 4\gamma_s < 1, \\ \Gamma = c_d(4\gamma_s)^{1/(d+1)} &< \min\{2^{-m}, \eta/(24s_C)\}. \end{aligned} \quad (14A.7)$$

As in Section 11, put $D_* = d^2$, $b_* = 1/(4(D_* + 1))$ and $\delta_c = t^{b_*}$. Choose $t > 0$ sufficiently small that

$$\begin{aligned}
t &\leq \min\{1/p, \varepsilon/8\}, & \gamma_c^C &\leq \delta_c \leq \gamma_s, & 2^p \delta_c &< \eta/12, \\
\delta_c + \sqrt{\delta_c} + \sqrt{t} &< \eta/6, & \frac{\delta_c^{D_*+1}}{C_d^{\text{net}}} &> \sqrt{t}.
\end{aligned} \tag{14A.8}$$

The finite-menu construction is required to be valid at this t . All choices are possible in this order: the last inequality compares $t^{1/4}$ with $t^{1/2}$, and $b_* < 1/(d+1)$. Finally take

$$\begin{aligned}
0 &< \varepsilon_0 \leq \min\{t^2/4, t/4, \varepsilon/16\}, \\
K2^{-k_0} &\leq \min\{t/2, \eta\}, & N_0 + 1 &\geq 8p/\varepsilon.
\end{aligned} \tag{14A.9}$$

This fixes the entire hierarchy before the measures are sampled.

At a fixed outer level i , the laws

$$\Theta = \mu^{x,i}, \quad \Xi = \Xi_{i, \mathcal{D}_i^{G_0}(g)}$$

are independent: the first depends only on x and the second only on the group cell of g . Suppose that with probability at least $1-t$ they share a plane on which Θ is (V, t, p) -saturated and Ξ is (V, t) -concentrated.

Here are the exact intermediate outputs of Section 11's argument in this bounded-image setting. Put $V_c = V_c(\Xi)$ and $V_s = V_s(\Theta)$. By (14A.6), (11.5) and the approximate-containment triangle inequality,

$$\mathbb{P}\{\alpha(V_c, V_s) \leq \delta_c + \gamma_s\} \geq 1 - t.$$

Take the δ_c -net and retain the planes whose balls have V_c -mass greater than δ_c/N_{net} , where $N_{\text{net}} \leq C_d^{\text{net}} \delta_c^{-D_*}$. Their union has mass at least $1 - \delta_c$. Its envelope W , with parameter $2(\delta_c + \gamma_s)$, gives

$$\mathbb{P}_{\Xi}\{\alpha(V_c, W) \leq 2\delta'\} \geq 1 - \delta_c, \quad \delta' = c_d(2(\delta_c + \gamma_s))^{1/(d+1)} \leq \Gamma. \tag{14A.10}$$

Independence implies that, except for $\sqrt{t}$ of the Θ law, the conditional probability of failure of $\alpha(V_c, V_s) \leq \delta_c + \gamma_s$ is at most $\sqrt{t}$. Every retained net ball has mass greater than $\sqrt{t}$ by (14A.8), so it meets the good inclusion event for each such Θ . Envelope minimality therefore yields

$$\mathbb{P}_{\Theta}\{\alpha(W, V_s) \leq \delta'\} \geq 1 - \sqrt{t}. \tag{14A.11}$$

This is the complete independence step; the different support box changes only the coefficient in (14A.6), not its logic.

For Θ in (14A.11), saturation inheritance and downward/nearby stability, exactly as in (11.13), show that its spatial descendants over $0 \leq j < p$ are (W, η, m) -saturated except for probability at most

$$a_s + \sqrt{t}, \quad a_s = C_I \sqrt{\gamma_s} + C_I / \sqrt{m} + C_I \sqrt{m/p}. \tag{14A.12}$$

The saturation tolerance actually obtained is $a_s + 2C_S/m < \eta/6$. The proof uses the plane $\pi_{V_s} W$, whose gap from W is at most $\delta' < 2^{-m}$; the unit-cube spatial descendants are unchanged.

Conditioning the image law on descendant group cells

It remains to obtain descendant **group** image concentration. This does not follow from the spatial component conclusion of P-II.5.11.

Fix an outer group cell D whose law lies in (14A.10), and its witness tube $(a_D + V_c)^{(\delta_c)}$ for $\Xi_{i,D}$. For any $0 \leq j < p$, disintegrate ν_D over its level- $i+j$ group cells E . The mean conditional mass outside that same tube, measured under $Z_{i,D}$, is at most δ_c . Markov shows that, except for conditional group-cell probability at most $\sqrt{\delta_c}$, the conditional mass inside it is at least $1 - \sqrt{\delta_c}$.

The centered descendant image is exactly

$$Z_{i+j,E}(h) = 2^j Z_{i,D}(h) + 2^{i+j} U_0^{-1}(g_D x_0 - g_E x_0). \quad (14A.13)$$

Thus $\Xi_{i+j,E}$ has good mass at least $1 - \sqrt{\delta_c}$ in a tube parallel to V_c of radius $2^j \delta_c$. It is (V_c, r) -concentrated with

$$r = \max\{\sqrt{\delta_c}, 2^p \delta_c\} < \eta/6.$$

Its support is again in $[-C, C]^d$. Using (14A.10) and the bounded-box change of plane, it is therefore concentrated on W with tolerance at most

$$r + 2s_C \delta' < \eta/6 + \eta/12 = \eta/4.$$

The failures are bounded by $\delta_c + \sqrt{\delta_c}$ after averaging over the outer cell and the descendant level. Combining with (14A.12) gives a joint good window of probability at least

$$1 - \delta_c - \sqrt{\delta_c} - \sqrt{t} - a_s > 1 - \eta. \quad (14A.14)$$

On that window the spatial descendants are (W, η, m) -saturated and the centered group images are (W, η) -concentrated. The component resampling identity identifies these descendants with the original measures at level $i+j$. By (14A.4)–(14A.5), the actual $\mathcal{A}_{i+j}$ images at the sampled spatial points are $(W, 2\eta)$ -concentrated. The same transfer holds uniformly over the spatial root cell.

All events used here are measurable. There are only countably many component cells; the menu planes and affine centers are finite-valued; and each conditional mass test is an integral of a Borel function divided by the positive cell mass. The good event for the actual actions is certified by the deterministic coupling bound, so no uncountable search over affine centers is used.

Selecting and covering the good outer levels

Let p_i be the failure probability of (14A.1) conditional on the outer level i . Its mean is at most ε_0 . Therefore

$$I = \{i : p_i \leq t/2\}, \quad |I| \geq (1 - 2\varepsilon_0/t)(N+1) \geq (1 - t/2)(N+1). \quad (14A.15)$$

For $i \in I$, freezing on the original good event gives Θ saturated at tolerance $\varepsilon_0 \leq t$ and Ξ concentrated at tolerance $\varepsilon_0 + e_k \leq 3t/4 < t$. The probability is at least $1 - t/2$. As

in Section 11, there are only countably many distinct pairs of these measures. For each successful pair choose one witnessing plane. This gives a measurable field depending only on the pair and cannot decrease the good probability. The independent intermediate-plane argument now applies at that outer level.

For each $i \in I$ choose the resulting plane $W(i)$ and its length- p window. Choose greedy disjoint such intervals as in Section 3, and set $W_q = W(i)$ on the interval starting at i . The total of the local window failures, uncovered levels and the final truncated interval is at most

$$\eta + t/2 + p/(N+1) \leq \varepsilon/8 + \varepsilon/16 + \varepsilon/8 < \varepsilon. \quad (14A.16)$$

The output tolerances are at most $2\eta = \varepsilon/4 < \varepsilon$. Assign an arbitrary plane on any remaining levels. The union of the certified good cell events on the selected intervals is the required E_1 . This proves (14A.2), and completes P-II.5.11-Loc. $\square$

II.5.17 Localized decoupling with separated input tolerances

We next separate the failure probability from the much smaller concentration radius needed after passing to descendants.

Proposition II.5.17.1 (Localized decoupling with separated input tolerances). *For every $d \geq 1, R \geq 1$ and $0 < \varepsilon < 1$ there is $M = M(d, R, \varepsilon)$ such that for every $m \geq M$ there are $a > 0, \beta > 0$ and an integer $p_0 > m$ with this property. For **every** $p \geq p_0$ there are $c > 0, k_0, N_0$ such that, if $k \geq k_0, N \geq N_0$ and μ, ν are carried by single level- k spatial/group cells as in the localized decoupling proposition above, the hypothesis*

$$\mathbb{P}_{\substack{0 \leq i \leq N \\ x \sim \mu, g \sim \nu}} \left\{ \begin{array}{l} \mu^{x,i} \text{ is } (V(i, x, g), a, p)\text{-saturated,} \\ \mathcal{A}_i(v_{g,i}, x) \text{ is } (V(i, x, g), c)\text{-concentrated} \end{array} \right\} \geq 1 - \beta \quad (14B.1)$$

implies level-only planes $W_0, \dots, W_N$ satisfying (14A.2). The hypothesis is certified by a measurable field and good event. The saturation tolerance a and failure allowance β are fixed before p ; only c, k_0, N_0 are chosen after p . The constants are independent of the particular measures, and N_0 is independent of $k \geq k_0$.

Proof strategy

Construct finite saturated and concentrated menus with independent error parameters. Independence supplies a popular compatible pair of menu cells; an approximate envelope merges the surviving planes. Descendant conditioning and disjoint scale windows then turn the intermediate plane into a level-only field with the desired tolerances.

Saturated menus with a separate tolerance

The finite saturated-menu proof in Section 11 has the following version. For $a > 0, p \geq 2$ with $\gamma_s = C_s(a + \log(p+1)/p)$ sufficiently small, there is a finite-valued Borel choice $V_s(\Theta)$ such that

$$\begin{aligned} \Theta &\text{ is } (V_s(\Theta), \gamma_s, p)\text{-saturated,} \\ \Theta &\text{ is } (V, a, p)\text{-saturated} \implies \alpha(V, V_s(\Theta)) \leq \gamma_s. \end{aligned} \quad (14B.2)$$

Here C_s is a dimension constant. To prove the version, use the same subspace net of mesh less than 2^{-p} , and set

$$\delta_j = A4^j \left(a + \frac{\log(p+1)}{p} \right), \quad 0 \leq j \leq d, \quad C_s = 4A4^d. \quad (14B.3)$$

Choose A above the downward, nearby-plane and transverse-sum constants as in Section 11, increasing it if necessary. Choose a net plane of maximal dimension among those saturated with error $\delta_{\dim V}$, with the fixed ordering breaking ties. The zero plane is admissible and all tests are Borel.

If a plane V saturated with error a escapes the selected V_s by more than $2\delta_d$, choose a line in V in an escaping direction. Downward saturation gives it error at most $a + C_0/p$. At positive selected dimension, the transverse-sum estimate makes its sum with V_s saturated with error at most $3\delta_{\dim V_s}$: the remaining transverse error is bounded by

$$\frac{C_1 \log(2/(2\delta_d))}{p} \leq \frac{2C_1 \log(p+1)}{p},$$

because $\delta_d \geq 1/p$ after the choice of A , while $2\delta_d < 1$ in the stated small-error range. A nearby net plane costs at most C_0/p more, leaving error at most $4\delta_{\dim V_s} = \delta_{\dim V_{s+1}}$. This contradicts maximality. At selected dimension zero, the line and its net approximation have error at most $a + 2C_0/p \leq \delta_1$, giving the same contradiction. Thus the escaped direction cannot exist. Since $2\delta_d \leq \gamma_s$ and the selected plane itself has saturation error at most $\delta_d \leq \gamma_s$, (14B.2) follows. This also proves its asserted measurability.

Parameter order

Use the bounded centered image laws $\Xi_{i,D}$ and constants C, K, s_C from the preceding localized construction. Their freezing error is $e_k = K2^{-k}$. Write $\eta = \varepsilon/8$. Require $m \geq M$ large enough that

$$C_I/\sqrt{m} + 2C_S/m \leq \eta/20. \quad (14B.4)$$

Choose $0 < \gamma < 1/4$ so small that

$$C_I\sqrt{\gamma} \leq \eta/20, \quad \Gamma = c_d(4\gamma)^{1/(d+1)} < \min\{2^{-m}, \eta/(40s_C)\}. \quad (14B.5)$$

Here c_d is the envelope constant from L-II.5.05. Choose $0 < h < \min(\gamma, \eta/20)$. Put $D_* = d^2$ and let C_{net} be the net-size constant in (11.2). Choose $b > 0$ so that

$$b < \eta/20, \quad \sqrt{b} < \min\{\eta/20, h^{D_*+1}/(2C_{\text{net}})\}, \quad \beta = b^2/2. \quad (14B.6)$$

Take $0 < a \leq \gamma/(2C_s)$, then choose $p_0 > m$ so large that for every $p \geq p_0$,

$$C_s \log(p+1)/p \leq \gamma/2, \quad C_I\sqrt{m/p} \leq \eta/20. \quad (14B.7)$$

Thus the menu error γ_s is at most γ . These choices of a, β, p_0 precede p .

For a specified $p \geq p_0$, choose $t > 0$ in the concentrated-menu range so small that, with $\rho = C_c t^{1/(d+1)}$, where C_c is the bounded-box constant from (14A.6),

$$\rho \leq h, \quad \sqrt{\rho} \leq \eta/20, \quad 2^p \rho \leq \eta/20. \quad (14B.8)$$

Finally set

$$c = t/2, \quad K2^{-k_0} \leq \min\{t/2, \eta/10\}, \quad N_0 + 1 \geq 4p/\varepsilon. \quad (14B.9)$$

This is a noncircular hierarchy.

The independent intermediate plane

At a fixed outer level, freeze the group-image law as in (14A.3). The random measures $\Theta = \mu^{x,i}$ and $\Xi = \Xi_{i, \mathcal{D}_i^{G_0}(g)}$ are independent. Suppose the conditional hypothesis (14B.1) has failure probability at most b . Freezing preserves saturation at error a and changes concentration error by at most e_k , so on the same good event the centered law is concentrated at tolerance t .

Use (14B.2) for $V_s(\Theta)$ and the concentrated menu (14A.6) for $V_c(\Xi)$, whose concentration and containment error is at most $\rho \leq h$. With probability at least $1 - b$,

$$\alpha(V_c, V_s) \leq h + \gamma. \quad (14B.10)$$

Take an h -net with at most $C_{\text{net}} h^{-D_*}$ points. Retain the net balls whose V_c -probability exceeds h/N_{net} . Their union has probability at least $1 - h$. Apply the envelope lemma to their centers, with parameter $2(h + \gamma) \leq 4\gamma$, and call the resulting plane W . Every retained center is approximately contained in W with tolerance at most Γ . Therefore

$$\mathbb{P}_{\Xi}\{\alpha(V_c, W) \leq 2\Gamma\} \geq 1 - h, \quad (14B.11)$$

using $h \leq \Gamma$.

By independence, (14B.10) and conditional Markov, except for $\sqrt{b}$ of the Θ law the Ξ -probability of failure of that containment is at most $\sqrt{b}$. Every retained ball has mass greater than

$$h/N_{\text{net}} \geq h^{D_*+1}/C_{\text{net}} > 2\sqrt{b}.$$

Hence, for each such Θ , every retained ball contains a plane satisfying the containment in (14B.10). Its center is approximately contained in V_s with tolerance at most $2(h + \gamma)$. Envelope minimality gives

$$\mathbb{P}_{\Theta}\{\alpha(W, V_s) \leq \Gamma\} \geq 1 - \sqrt{b}. \quad (14B.12)$$

No concentration radius was used to set the heavy-ball probability threshold. That separation is the key distinction from the single-tolerance proof.

Descendants and the level-only conclusion

For the Θ realizations in (14B.12), inherit the (V_s, γ_s, p) saturation to window m , then use downward and nearby-plane stability via $\pi_{V_s} W$. Because $\Gamma < 2^{-m}$, the saturation tolerance is at most

$$a_s + 2C_S/m \leq 3\eta/20, \quad a_s = C_I\sqrt{\gamma_s} + C_I/\sqrt{m} + C_I\sqrt{m/p}. \quad (14B.13)$$

The associated spatial failure is at most $a_s + \sqrt{b}$.

For a group law in (14B.11), its menu provides a fixed ρ -tube parallel to V_c with mass at least $1 - \rho$. Disintegrate the original group law over descendants, exactly as in (14A.13). Conditional Markov loses probability at most $\sqrt{\rho}$, and the rescaled tube has radius at most $2^p\rho$ and missing mass at most $\sqrt{\rho}$. By (14B.8) it is concentrated with tolerance $\eta/20$. Change its plane to W on the bounded centered support. The additional error is at most $2s_c\Gamma < \eta/20$, so centered descendant concentration has tolerance $\eta/10$. Transfer back to the actual action by (14A.4). Equation (14B.9) makes its concentration tolerance at most $\eta/5$.

The combined failure in the length- p descendant window is at most

$$h + \sqrt{\rho} + \sqrt{b} + a_s < \eta. \quad (14B.14)$$

All finite-menu choices, group conditional-mass events and couplings are measurable, by the same finite-menu construction used above.

The mean outer-level failure is at most β . Except for a fraction at most $\beta/b = b/2$ of the outer levels it is at most b . At each remaining level the preceding independent argument applies. If needed, choose one witnessing plane for each of the countably many successful pairs of centered laws, exactly as in the localized construction above; freezing has made the law pair independent without reducing success.

Use greedy disjoint length- p windows starting at these levels. Their assigned planes have global failure at most

$$\eta + b/2 + p/(N+1) < \varepsilon \quad (14B.15)$$

by (14B.6), (14B.9). Their saturation and concentration tolerances are both less than ε . This proves the separated-input clause of P-II.5.11-Loc. $\square$

II.5.18 Nondegenerate slices and finite-window smooth entropy

The remaining smooth-map argument has three distinct tasks. This section controls nonlinear slices on one fixed observation window; the next section returns from image components to components of the original input, and the following one removes the subspace's dependence on the sampled pair.

Definition II.5.18.1 (II.5.09: Two-input smooth maps and partial differential images). Let $d_1, d_2, q \geq 1$ be integers. Let $I \subset \mathbb{R}^{d_1}$ and $J \subset \mathbb{R}^{d_2}$ be closed balls. Fix a specified C^1

map on an open neighborhood of $I \times J$, and denote both that map and its restriction $I \times J \rightarrow \mathbb{R}^q$ by f . All differentials below are those of this specified extension, also when a ball has radius zero. For $z = (x, y)$, write

$$Df(z) = [A_z, B_z], \quad A_z = D_x f(z) : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^q, \quad B_z = D_y f(z) : \mathbb{R}^{d_2} \rightarrow \mathbb{R}^q.$$

For a linear map T and a probability θ , the notation $T\theta$ means $T_*\theta$. The order of the two measures in every f_* expression is the order of $I \times J$.

For a linear map $A : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^q$ its **least stretch** is

$$s_{\min}(A) = \inf_{\|v\|=1} \|Av\|.$$

It is injective exactly when this number is positive. The phrase **uniformly nondegenerate in the first variable on** $X \times Y$, for nonempty compact $X \subset I, Y \subset J$, means that some $c > 0$ satisfies $s_{\min}(A_{x,y}) \geq c$ for every $(x, y) \in X \times Y$. It does not require nondegeneracy at every point of the containing balls.

The **derivative modulus** used here is

$$\omega_f(t) = \sup_{\substack{z, z' \in I \times J \\ \|z - z'\| \leq t}} \|Df(z) - Df(z')\|_{\text{op}} \quad (t \geq 0).$$

It is finite, nondecreasing and tends to zero as t decreases to zero, by uniform continuity on the compact product. When needed, $\Lambda \geq 1$ is a fixed upper bound for $\|Df\|_{\text{op}}$ on that product.

For $f_y(x) = f(x, y)$, probabilities $\alpha \in \mathcal{P}(I)$, $\beta \in \mathcal{P}(J)$ and $N \geq 1$, the **nonlinear slice entropy increment** is

$$\Delta_N^f(\alpha, \beta) = h_N(f_*(\alpha \times \beta)) - \int h_N((f_y)_*\alpha) d\beta(y).$$

Its nonnegativity and coarse-component inequalities are proved in [L-II.5.14-ND](#).

Lemma II.5.18.2 (L-II.5.14-ND – Nondegenerate smooth slices and entropy localization). *Use [D-II.5.09](#). Let $X \subset I, Y \subset J$ be nonempty compact sets and assume, for $\Lambda \geq 1, 0 < c \leq \Lambda$,*

$$\sup_{I \times J} \|Df\|_{\text{op}} \leq \Lambda, \quad s_{\min}(A_z) \geq c \quad (z \in X \times Y).$$

1. *If $E \subset \mathbb{R}^r$ is Borel, $r \geq 1$, and $T : E \rightarrow \mathbb{R}^q$ is a Borel map with $0 < c_0 \leq L_0 < \infty$ and $c_0\|x - x'\| \leq \|T(x) - T(x')\| \leq L_0\|x - x'\|$, and ρ is a compactly supported probability carried by E , then for every integer n ,*

$$-\log([\sqrt{q}/c_0] + 2)^r \leq H_n(T_*\rho) - H_n(\rho) \leq \log([L_0\sqrt{r}] + 2)^q.$$

2. For all sufficiently large $k \geq k_{\text{bi}}(f, c)$, every f_y restricted to one level- k cell intersected with X is bi-Lipschitz with constants $c/2, \Lambda$, uniformly in $y \in Y$. For a probability ρ carried by that intersection and every $z \in X \times Y$,

$$|H_n((f_y)_*\rho) - H_n((A_z)_*\rho)| \leq C \quad \text{for every integer } n,$$

with $C = C(d_1, q, c, \Lambda)$.

3. For $\alpha \in \mathcal{P}(X)$, $\beta \in \mathcal{P}(Y)$, $k \geq 0$, $N \geq \max(k, 1)$, the nonlinear slice increment is nonnegative and satisfies

$$\mathbb{E}_{E,D} \Delta_N^f(\alpha_E, \beta_D) \leq \Delta_N^f(\alpha, \beta) + \frac{H_k(\alpha)}{N \log 2}.$$

Here the root components are independently mass-weighted. Consequently the probability that their increment exceeds $\tau > 0$ is at most the displayed right side divided by τ . This part does not require nondegeneracy.

4. There is $C = C(d_1, d_2, q, c, \Lambda)$ such that for every $w \geq 1$ there is $k_{\text{lin}}(f, c, w) \geq k_{\text{bi}}$ with the following property. If $k \geq k_{\text{lin}}$, $N > \max(k, w)$ and α, β are each carried by a level- k cell, choose representatives $z_{E,D} = (x_E, y_D) \in X \times Y$ in every positive-mass level- i pair, and set

$$\Theta_{i,E,D} = (A_{z_{E,D}})_* \alpha^E, \quad \Xi_{i,E,D} = (B_{z_{E,D}})_* \beta^D, \quad D_{i,w} = h_w(\Theta_{i,E,D} * \Xi_{i,E,D}) - h_w(\Theta_{i,E,D}).$$

Then $D_{i,w} \geq -q/w$ and

$$\Delta_N^f(\alpha, \beta) \geq \frac{1}{N} \sum_{i=k}^{N-1} \mathbb{E} D_{i,w} - C \left(\frac{1}{w} + \frac{w+k+1}{N} \right).$$

Its precise shifted Markov consequence is (14C.18). All constants and thresholds precede the measures.

5. For nonempty compact $C_K \subset \text{GL}_d(\mathbb{R})$, the maps $f(x, A) = Ax$ and $f(A, B) = AB$ meet the first-variable nondegeneracy condition on $C_x \times C_K$ and $C_K \times C_K$, respectively, with least stretch at least $(\max_{A \in C_K} \|A^{-1}\|_{\text{op}})^{-1}$. Matrix space uses Frobenius norm.

The conclusion is deliberately restricted to a fixed observation window; no fine-scale comparison of a nonlinear image with one frozen convolution is asserted.

Proof strategy

We begin with a uniform grid comparison for bi-Lipschitz maps. Uniform nondegeneracy makes every sufficiently small first-variable slice bi-Lipschitz. Coarse-component localization then reduces the global entropy increment to root cells, and a Taylor estimate controls each fixed finite window. The matrix examples follow by computing their partial derivatives.

Hypotheses and elementary grid comparison

Use D-II.5.09. Let $X \subset I$ and $Y \subset J$ be nonempty compact sets, where $I \subset \mathbb{R}^{d_1}$ and $J \subset \mathbb{R}^{d_2}$ are closed balls, and f is C^1 on a neighborhood of $I \times J$. Suppose $0 < c \leq \Lambda$, $\Lambda \geq 1$,

$$\|Df(z)\|_{\text{op}} \leq \Lambda \quad (z \in I \times J), \quad \|A_{x,y}v\| \geq c\|v\| \quad (x \in X, y \in Y, v \in \mathbb{R}^{d_1}). \quad (14C.1)$$

In particular $q \geq d_1$. Nondegeneracy is required only on $X \times Y$, not throughout the larger product of balls.

We first prove an entropy comparison used repeatedly below. Suppose a map $T : E \subset \mathbb{R}^r \rightarrow \mathbb{R}^q$ satisfies

$$c_0\|x - x'\| \leq \|T(x) - T(x')\| \leq L_0\|x - x'\| \quad (x, x' \in E), \quad c_0 > 0. \quad (14C.2)$$

Here $r \geq 1$ and $0 < c_0 \leq L_0 < \infty$. Assume E is Borel, T is Borel and ρ is a compactly supported probability with $\rho(E) = 1$. For every integer n ,

$$-\log J_- \leq H_n(T_*\rho) - H_n(\rho) \leq \log J_+, \quad J_+ = (\lceil L_0\sqrt{r} \rceil + 2)^q, \quad J_- = (\lceil \sqrt{q}/c_0 \rceil + 2)^r. \quad (14C.3)$$

Indeed, points in a source level- n cube have image diameter at most $L_0\sqrt{r}2^{-n}$, so their image meets at most J_+ target cells. Points whose images lie in one target cell have source diameter at most $\sqrt{q}2^{-n}/c_0$, so that preimage meets at most J_- source cells. The corresponding two conditional cell-label entropies are at most $\log J_+$ and $\log J_-$. The entropy chain rule proves (14C.3). No image density or positive-volume support is assumed. The estimate is independent of the support's location.

The same calculation applies to a linear map $A : \mathbb{R}^r \rightarrow \mathbb{R}^q$ with $s_{\min}(A) \geq c_0$ and $\|A\|_{\text{op}} \leq L_0$. Here $s_{\min}(A) = \inf_{\|v\|=1} \|Av\|$. The elementary inequality

$$|s_{\min}(A) - s_{\min}(A')| \leq \|A - A'\|_{\text{op}} \quad (14C.4)$$

follows by applying the triangle inequality to each unit vector and then taking infima in both directions. For an injective linear map the continuous norm of its image attains a positive minimum on the unit sphere; conversely, positive least stretch excludes a nonzero kernel.

Uniform small-cell slice embeddings

Let ω_f be the derivative modulus in D-II.5.09. Choose $k_{\text{bi}} \geq 0$ so that

$$\omega_f(\sqrt{d_1} 2^{-k_{\text{bi}}}) \leq c/2. \quad (14C.5)$$

For any $k \geq k_{\text{bi}}$, any level- k spatial cube E , any $x, x' \in E \cap X$ and any $y \in Y$, the segment from x to x' lies in I and in the closure of E . The fundamental theorem of calculus gives

$$f(x', y) - f(x, y) = A_{x,y}(x' - x) + \int_0^1 (A_{x+t(x'-x),y} - A_{x,y})(x' - x) dt.$$

The error has norm at most $\omega_f(\sqrt{d_1}2^{-k})\|x' - x\|$. By (14C.1), (14C.5),

$$\frac{c}{2}\|x' - x\| \leq \|f(x', y) - f(x, y)\| \leq \Lambda\|x' - x\|. \quad (14C.6)$$

The lower bound does not require that the segment itself lie in X : nondegeneracy is used at its endpoint in X , and derivative continuity controls the rest of the segment.

For every probability ρ carried by $E \cap X$, every $y \in Y$ and every integer n , (14C.3) therefore gives

$$|H_n((f_y)_*\rho) - H_n(\rho)| \leq C_{bi}, \quad f_y(x) = f(x, y), \quad (14C.7)$$

with $C_{bi} = C(d_1, q, c, \Lambda)$. For every $z \in X \times Y$, the linear map A_z satisfies the same assertion with another constant of those parameters. Consequently

$$|H_n((f_y)_*\rho) - H_n((A_z)_*\rho)| \leq C_{cmp}. \quad (14C.8)$$

This holds for **all** n simultaneously, not only for a bounded range after k . Its justification is injectivity and bounded distortion, not a Taylor remainder small at scale 2^{-n} .

Exact nonlinear coarse-component localization

For compactly supported $\alpha \in \mathcal{P}(X)$ and $\beta \in \mathcal{P}(Y)$ and $N \geq 1$ define

$$\Delta_N^f(\alpha, \beta) = h_N(f_*(\alpha \times \beta)) - \int h_N((f_y)_*\alpha) d\beta(y). \quad (14C.9)$$

Entropy concavity implies $\Delta_N^f \geq 0$. All entropies are finite because $f(I \times J)$ is compact. If $N \geq \max(k, 1)$ and $k \geq 0$, and α_E, β_D are independently sampled raw level- k components, then

$$0 \leq \Delta_N^f(\alpha_E, \beta_D), \quad \mathbb{E}_{E,D} \Delta_N^f(\alpha_E, \beta_D) \leq \Delta_N^f(\alpha, \beta) + \frac{H_k(\alpha)}{N \log 2}. \quad (14C.10)$$

To prove the mean bound, joint component disintegration and entropy concavity bound the mean root-pair output entropy above by the full output entropy. For each fixed y , the entropy of the mixture $(f_y)_*\alpha$ is at most the mean entropy of $(f_y)_*\alpha_E$ plus the mixing-label entropy $H_k(\alpha)$. Integrating in y and subtracting the two inequalities gives (14C.10). The averaging over the β root cells recovers the original β integral exactly.

Thus for every $\tau > 0$,

$$\mathbb{P}\{\Delta_N^f(\alpha_E, \beta_D) > \tau\} \leq \frac{\Delta_N^f(\alpha, \beta) + H_k(\alpha)/(N \log 2)}{\tau}. \quad (14C.11)$$

No signed-variable correction is needed here: the baseline in (14C.9) is the exact mixture of nonlinear slice entropies. This coarse-localization assertion itself does not use nondegeneracy.

Finite-window linearization and entropy growth

There is $C = C(d_1, d_2, q, c, \Lambda)$ with the following property. Fix $w \geq 1$ and choose $k_{\text{lin}}(w) \geq k_{\text{bi}}$ so that

$$\sqrt{d_1 + d_2} \omega_f(\sqrt{d_1 + d_2} 2^{-k_{\text{lin}}(w)}) \leq 2^{-w}. \quad (14C.12)$$

Let $k \geq k_{\text{lin}}(w)$ and $N > \max(k, w)$. Suppose α, β are carried by single level- k cells in their respective spaces. At each $i \geq k$ and each positive-mass cell pair E, D choose representatives $x_E \in E \cap X$, $y_D \in D \cap Y$ and set $z_{E,D} = (x_E, y_D)$. Put

$$\begin{aligned} \Theta_{i,E,D} &= (A_{z_{E,D}})_* \alpha^E, \\ \Xi_{i,E,D} &= (B_{z_{E,D}})_* \beta^D, \\ D_{i,w}(E, D) &= h_w(\Theta_{i,E,D} * \Xi_{i,E,D}) - h_w(\Theta_{i,E,D}). \end{aligned} \quad (14C.13)$$

Here α^E, β^D are canonical spatial components at level i , in dimensions d_1, d_2 . The image laws are probabilities in $\mathbb{R}^q$ with uniformly bounded support. Then

$$\Delta_N^f(\alpha, \beta) \geq \frac{1}{N} \sum_{i=k}^{N-1} \mathbb{E}_{E,D} D_{i,w}(E, D) - C \left(\frac{1}{w} + \frac{w+k+1}{N} \right). \quad (14C.14)$$

Proof. For $(x, y) \in (E \cap X) \times (D \cap Y)$, Taylor's formula on the segment within the convex set $I \times J$ gives

$$\|f(x, y) - f(x_E, y_D) - A_{z_{E,D}}(x - x_E) - B_{z_{E,D}}(y - y_D)\| \leq \sqrt{d_1 + d_2} 2^{-i} \omega_f(\sqrt{d_1 + d_2} 2^{-i}) \leq 2^{-(i+w)}. \quad (14C.15)$$

The linearized law, after dilation by 2^i , is a translate of $\Theta_{i,E,D} * \Xi_{i,E,D}$. Bounded-perturbation and translation entropy comparisons cost only C_q in unnormalized entropy. The nonlinear component image has diameter at most $\Lambda \sqrt{d_1 + d_2} 2^{-i}$; hence its level- i entropy is bounded by another fixed constant. Therefore

$$\frac{H_{f_*(\alpha_E \times \beta_D)}(\mathcal{D}_{i+w} \mid \mathcal{D}_i)}{w \log 2} \geq h_w(\Theta_{i,E,D} * \Xi_{i,E,D}) - C/w. \quad (14C.16)$$

Joint component disintegration and conditional entropy concavity compare the mean left side with the full output's conditional entropy. For the output-dimension entropy-increment count, each increment before N is counted at most w times, and those beyond N have total excess multiplicity at most $w(w-1)/2$. It follows that

$$h_N(f_*(\alpha \times \beta)) \geq \frac{1}{N} \sum_{i=k}^{N-1} \mathbb{E} h_w(\Theta_{i,E,D} * \Xi_{i,E,D}) - C/w - \frac{q(w-1)}{2N}. \quad (14C.17)$$

The dropped levels $i < k$ have nonnegative conditional entropy. For each linear first factor, (14C.3) gives

$$h_w(\Theta_{i,E,D}) \geq h_w(\alpha^E) - C/(w \log 2).$$

The reverse increment count in source dimension d_1 gives the mean of these source component entropies over all $0 \leq i < N$ at least

$$h_N(\alpha) - \frac{H_0(\alpha)}{N \log 2} - \frac{d_1(w-1)}{2N}.$$

Here $H_0(\alpha) = 0$ because α is carried by a level- $k \geq 0$ cell. Dropping its first k component entropies loses at most $d_1 k/N$. Substitute these bounds and (14C.13) into (14C.17). Finally (14C.7) yields

$$\int h_N((f_y)_* \alpha) d\beta(y) \leq h_N(\alpha) + C/(N \log 2).$$

Subtract this baseline. The explicit remaining endpoint error is bounded by

$$\frac{(d_1 + q)(w-1)}{2N} + \frac{d_1 k}{N} + \frac{C}{N \log 2};$$

enlarging C proves (14C.14). $\square$

The entropy increment in (14C.13) is at least $-q/w$: conditioning a convolution on its second summand and comparing a translated dyadic grid costs at most $q \log 2$. For uniform $k \leq i < N$, Markov thus gives

$$\mathbb{P}\{D_{i,w} > v\} \leq \frac{\frac{N}{N-k} \left(\Delta_N^f(\alpha, \beta) + C \left(\frac{1}{w} + \frac{w+k+1}{N} \right) \right) + q/w}{v + q/w} \quad (v > 0). \quad (14C.18)$$

Consequently one may prescribe an arbitrarily small local-growth threshold and exceptional probability, then choose w large, $k \geq k_{\text{lin}}(w)$, the permitted global increment small, and N large. All choices can be uniform over the probabilities on the fixed compact supports. This does not identify the local inverse theorem's image components with components in the input coordinates; that change requires the next lemma.

The matrix examples meet the nondegeneracy hypothesis

Let $C_K \subset \text{GL}_d(\mathbb{R})$ be compact and nonempty. Continuity of matrix inversion and compactness give

$$c_K = \left(\max_{A \in C_K} \|A^{-1}\|_{\text{op}} \right)^{-1} > 0, \quad L_K = \max_{A \in C_K} \|A\|_{\text{op}} < \infty. \quad (14C.19)$$

For $f(x, A) = Ax$ the first derivative is A , so its least stretch is at least c_K on any compact $C_x \times C_K$. The second derivative is $E_x : H \mapsto Hx$, with norm at most $\|x\|$ when matrices have Frobenius norm.

For $f(A, B) = AB$ the partial derivatives are $R_B : H \mapsto HB$ and $L_A : H \mapsto AH$. On $C_K \times C_K$,

$$\|HB\|_F \geq c_K \|H\|_F, \quad \|AH\|_F \leq L_K \|H\|_F. \quad (14C.20)$$

The first inequality follows from $H = (HB)B^{-1}$ and $\|MC\|_F \leq \|M\|_F \|C\|_{\text{op}}$; the latter bound follows by applying the operator norm to each row. The second displayed inequality follows by applying A to each column. Both maps are polynomial on the ambient vector spaces. Thus they are C^1 on any containing balls, and their first derivatives are uniformly injective on the stated compact supports. The containing balls themselves need not consist of invertible matrices.

These verifications clear the nondegeneracy hypotheses for the planned matrix specializations. They do not by themselves prove their structural conclusions. This completes L-II.5.14-ND. $\square$

II.5.19 Linear images and the input component partitions

Proposition II.5.19.1 (P-II.5.12-Lin – Linear-image component inheritance and transfer). *Fix $d_1, d_2, q \geq 1, \Lambda \geq 1$ and $c > 0$. Let $A : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^q$ satisfy $c\|v\| \leq \|Av\| \leq \Lambda\|v\|$, and let $B : \mathbb{R}^{d_2} \rightarrow \mathbb{R}^q$ have norm at most Λ . The constants below are uniform over these maps.*

1. *If $\mu \in \mathcal{P}([0, 1]^{d_1})$, $\varepsilon \geq 0$, $1 \leq m < p$ and $A_*\mu$ is (V, ε, p) -saturated, then*

$$\mathbb{E}_{0 \leq j < p, x \sim \mu} (\text{def}_{V, m}(A_*\mu^{x, j}))_+ \leq C(\varepsilon + 1/m + m/p),$$

where $C = C(d_1, q, c, \Lambda)$. Thus these input-component images are saturated with tolerance and failure at most $C'(\sqrt{\varepsilon} + 1/\sqrt{m} + \sqrt{m/p})$.

2. *For compactly supported μ , $\theta = A_*\mu$, $m \geq 1$, $0 < \delta < 1$, $N \geq 0$ and level-only planes V_i , if $\theta^{z, i}$ is (V_i, δ, m) -saturated outside joint level/component probability δ , then*

$$\mathbb{P}_{0 \leq i \leq N, x \sim \mu} \{A_*\mu^{x, i} \text{ is not } (V_i, a, m)\text{-saturated}\} \leq C(\delta + 1/m)/a \quad (a > 0).$$

3. *For every $0 < a < 1$ there is $M = M(d_1, d_2, q, c, \Lambda, a)$ such that for every $m \geq M$, $0 < \zeta < 1$ there are $\delta > 0, \ell_0 > m$ such that for every $\ell \geq \ell_0$ there are $z > 0, N_0$ with this property. For compactly supported μ, v , put $\theta = A_*\mu, \xi = B_*v$. Joint saturation (V_i, δ, ℓ) of θ image components and concentration (V_i, z) of ξ image components with probability at least $1 - \delta$ imply level-only planes W_i with*

$$\mathbb{P}_{0 \leq i \leq N, x \sim \mu, y \sim v} \left\{ \begin{array}{l} A_*\mu^{x, i} \text{ is } (W_i, a, m)\text{-saturated,} \\ B_*v^{y, i} \text{ is } (W_i, \zeta)\text{-concentrated} \end{array} \right\} \geq 1 - a \quad (N \geq N_0).$$

The saturation/failure parameter δ precedes ℓ ; the concentration input z follows ℓ . No injectivity of B is required.

The three parts are proved below using the original input filtrations.

Proof strategy

Conditional entropy first gives inheritance along the original input filtration. A bounded-overlap comparison transfers saturation from image components back to images of input components. Applying the same comparison to the second input and then merging the resulting planes proves the two-input assertion.

Let $A : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^q$ satisfy $c\|v\| \leq \|Av\| \leq \Lambda\|v\|$ with $c > 0$, $\Lambda \geq 1$. All constants in the first two clauses may depend on d_1, q, c, Λ , but not on the measure, plane or dyadic levels.

Write $\text{def}_{V,m}(\theta)$ for the saturation deficit in [D-II.5.03](#), and $r = \dim V$. If θ is carried by $[-S, S]^q$, $S \geq 1$, then

$$-C_{q,S}/m \leq \text{def}_{V,m}(\theta) \leq q + C_q/m. \quad (14D.1)$$

Indeed, the product-coordinate characterization of saturation compares the entropy difference with conditional entropy along V up to C_q/m . In the bounded box that conditional entropy is at most $rm \log 2 + C_{q,S}$. For the other inequality, projecting an ambient dyadic cube meets only boundedly many projected-grid cells, so $H_m(\pi_{V^\perp}\theta) \leq H_m(\theta) + C_q$.

Inheritance in the original input coordinates

There is $C = C(d_1, q, c, \Lambda)$ such that, for $\mu \in \mathcal{P}([0, 1]^{d_1})$, $\varepsilon \geq 0$ and $1 \leq m < p$, if $A_*\mu$ is (V, ε, p) -saturated, then

$$\mathbb{E}_{0 \leq j < p} \left(\text{def}_{V,m}(A_*\mu^{x,j}) \right)_+ \leq C \left(\varepsilon + \frac{1}{m} + \frac{m}{p} \right). \quad (14D.2)$$

In particular, with

$$a = C' \left(\sqrt{\varepsilon} + \frac{1}{\sqrt{m}} + \sqrt{\frac{m}{p}} \right),$$

the linear images $A_*\mu^{x,j}$ are (V, a, m) -saturated outside probability at most a . These are images of input components, not canonical components of $A_*\mu$.

Proof. Put $\theta = A_*\mu$. On its support use the intrinsic coordinate partitions $\mathcal{P}_n^V, \mathcal{P}_n^{V^\perp}$ from [D-II.5.01](#), and write

$$c_n = H_\theta(\mathcal{P}_n^V \mid \mathcal{P}_n^{V^\perp}), \quad \mathcal{J}_n = \mathcal{P}_n^V \vee \mathcal{P}_n^{V^\perp}.$$

Let F_j be the image under A of the source level- j partition. Because A is injective it is a genuine Borel partition of $A\mathbb{R}^{d_1}$. A J_j atom has diameter at most $\sqrt{q}2^{-j}$ in the image. Its preimage therefore meets at most $(\lceil \sqrt{q}/c \rceil + 2)^{d_1}$ source cubes. Consequently

$$H_\theta(\mathcal{F}_j \mid \mathcal{J}_j) \leq C. \quad (14D.3)$$

The canonical input rescaling commutes with A up to an image translation. The entropy comparison for translated coordinate grids, followed by component disintegration, gives

$$\left| \mathbb{E}_x H_{A_*\mu^{x,j}}(\mathcal{P}_m^V \mid \mathcal{P}_m^{V^\perp}) - H_\theta(\mathcal{P}_{j+m}^V \mid \mathcal{P}_{j+m}^{V^\perp} \vee \mathcal{F}_j) \right| \leq C_q. \quad (14D.4)$$

To see why its constant is uniform, on a source cube with integer address v_E the image rescaling is $2^j Ax - Av_E$. Translation changes each of the two coordinate-partition entropies, joint and marginal, by a bounded amount, independent of the address. Their difference is the conditional entropy in [\(14D.4\)](#).

For discrete labels, the chain rule gives $H(U|W, F) \geq H(U|W, J) - H(F|J)$. Apply this with the fine V label, fine perpendicular label, F_j and J_j . Using (14D.3) and refinement of both intrinsic grids,

$$\begin{aligned} H_\theta(\mathcal{P}_{j+m}^V \mid \mathcal{P}_{j+m}^{V^\perp} \vee \mathcal{F}_j) &\geq H_\theta(\mathcal{P}_{j+m}^V \mid \mathcal{P}_{j+m}^{V^\perp} \vee \mathcal{J}_j) - C \\ &= c_{j+m} - H_\theta(\mathcal{P}_j^V \mid \mathcal{P}_{j+m}^{V^\perp}) - C \\ &\geq c_{j+m} - c_j - C. \end{aligned} \quad (14D.5)$$

The support of θ lies in a fixed box depending on Λ, d_1 . Hence $c_j \leq rj \log 2 + C$ for $j \geq 0$. Refining the perpendicular grid over u levels can remove at most $(q-r)u \log 2$ of conditional entropy, so

$$c_{p+u} \geq c_p - (q-r)u \log 2 \quad (u \geq 0).$$

Therefore

$$\begin{aligned} \sum_{j=0}^{p-1} (c_{j+m} - c_j) &= \sum_{j=p}^{p+m-1} c_j - \sum_{j=0}^{m-1} c_j \\ &\geq mc_p - \frac{qm(m-1)}{2} \log 2 - Cm. \end{aligned} \quad (14D.6)$$

The saturation assumption implies $c_p \geq (r-\varepsilon)p \log 2 - Cq$. Average (14D.4)–(14D.6), divide by $m \log 2$, and apply the product-coordinate entropy comparison once more. This bounds the mean saturation deficit of $A_* \mu^{x,j}$ by $\varepsilon + C/m + Cm/p$. Each such image is in the same bounded box, so (14D.1) also bounds its negative deficit by C/m . Adding that bound gives (14D.2). Markov at the displayed square-root threshold, with C' sufficiently large, proves the probability claim. $\square$

Same-level transfer from image cells to input cells

Let μ be a compactly supported probability on $\mathbb{R}^{d_1}$, $\theta = A_* \mu$, $m \geq 1$ and $0 < \delta < 1$. For any $N \geq 0$ and planes $V_0, \dots, V_N$, if

$$\mathbb{P}_{0 \leq i \leq N, \theta^{z,i}} \{ \theta^{z,i} \text{ is } (V_i, \delta, m)\text{-saturated} \} \geq 1 - \delta, \quad (14D.7)$$

then, for every $a > 0$,

$$\mathbb{P}_{0 \leq i \leq N, x \sim \mu} \{ A_* \mu^{x,i} \text{ is not } (V_i, a, m)\text{-saturated} \} \leq \frac{C(\delta + 1/m)}{a}. \quad (14D.8)$$

If the right side exceeds one, the bound is simply vacuous.

Proof. At level i compare the image-source partition F_i from the previous proof with the ordinary ambient image partition $\mathcal{D}_i^q$. Injectivity and (14C.3)'s cell count give

$$H_\theta(\mathcal{F}_i \mid \mathcal{D}_i^q) \leq C.$$

The same conditional-label inequality used in (14D.5), now with $J = \mathcal{D}_i^q$, compares the fine coordinate conditional entropies given these two coarse partitions. Translating their rescaled components and using the product-coordinate comparison therefore gives

$$\mathbb{E}_x \text{def}_{V_i, m}(A_* \mu^{x, i}) \leq \mathbb{E}_z \text{def}_{V_i, m}(\theta^{z, i}) + C/m. \quad (14D.9)$$

The image components on the right are unit-cube probabilities, so their deficits are at most $q + C_q/m$ by (14D.1). On the event (14D.7), the deficit is at most δ . Average over levels: the right-hand mean in (14D.9) is at most $C\delta + C/m$. Every left-hand component is a linear image of a unit-cube probability; its deficit is bounded below by $-C/m$ uniformly. Thus the averaged positive deficit is at most $C(\delta + 1/m)$. Markov gives (14D.8). This argument uses the global failure probability in (14D.7); it does not assume that every individual level has that bound. $\square$

Two-input image transfer with separate concentration accuracy

Fix also $d_2 \geq 1$ and a linear map $B : \mathbb{R}^{d_2} \rightarrow \mathbb{R}^q$ with $\|B\|_{\text{op}} \leq \Lambda$; no injectivity is required of B . For every $0 < a < 1$ there is $M = M(d_1, d_2, q, c, \Lambda, a)$ such that, for every $m \geq M$ and $0 < \zeta < 1$, there are $\delta > 0, \ell_0 > m$ with this property. For every $\ell \geq \ell_0$ there are $z > 0, N_0$ such that, if μ, ν are compactly supported probabilities in their source spaces, $\theta = A_* \mu, \xi = B_* \nu$ and

$$\mathbb{P}_{\substack{0 \leq i \leq N \\ x \sim \theta, y \sim \xi}} \left\{ \begin{array}{l} \theta^{x, i} \text{ is } (V_i, \delta, \ell)\text{-saturated,} \\ \xi^{y, i} \text{ is } (V_i, z)\text{-concentrated} \end{array} \right\} \geq 1 - \delta, \quad (14D.10)$$

for $N \geq N_0$, then level-only planes W_i exist with

$$\mathbb{P}_{\substack{0 \leq i \leq N \\ x \sim \mu, y \sim \nu}} \left\{ \begin{array}{l} A_* \mu^{x, i} \text{ is } (W_i, a, m)\text{-saturated,} \\ B_* \nu^{y, i} \text{ is } (W_i, \zeta)\text{-concentrated} \end{array} \right\} \geq 1 - a. \quad (14D.11)$$

The constants are uniform over the displayed class of A, B . In particular, the input saturation/failure δ precedes the freely enlargable window ℓ , and the input concentration accuracy z follows ℓ .

Proof. We first verify the concentration-side transfer with an arbitrary ambient first measure θ , without changing its image components. A level- j source cube for ν has B -image diameter at most

$$C_B 2^{-j}, \quad C_B = \Lambda \sqrt{d_2}.$$

It meets at most $M_B = ([C_B] + 2)^q$ ambient image cells. The nested source partitions are precisely $\mathcal{D}_j^{d_2}$. These are the diameter and disintegration facts needed for the parallel-tube merging calculation (12A.6), which is a statement about arbitrary conditional supports and countable partitions.

Here are all its parameter choices and errors in this setting. For a target saturation/failure $0 < u < 1$ and concentration ζ require m large enough that $C_I / \sqrt{m} \leq u/20$, where C_I is the ambient q -dimensional inheritance constant. Set

$$\rho = \zeta/4, \quad 0 < v \leq \min\{\zeta^2/16, u^2/400\}, \quad T = M_B^2 \left(2 + \log_2 \frac{C_B + 2}{\rho} \right).$$

Choose δ small enough that

$$C_I\sqrt{\delta} \leq u/20, \quad 2\sqrt{\delta} \leq u/20, \quad \sqrt{\delta}/v \leq u/20.$$

Choose $\ell_0 > m$ so that every $\ell \geq \ell_0$ has $C_I\sqrt{m/\ell} \leq u/20$ and $T/\ell \leq u/20$. For the selected ℓ choose

$$0 < z \leq \min\{2^{-\ell}\zeta/4, uv/20\}, \quad N_0 + 1 \geq 20\ell/u. \quad (14D.12)$$

From (14D.10), at all but a $\sqrt{\delta}$ fraction of levels the two marginal failures are at most $\sqrt{\delta}$. At such a level choose the witnessing image tubes for ζ . Their cellwise union misses at most $\sqrt{\delta} + z$ of ζ . Conditional Markov over the original v cells leaves union mass at least $1 - v$ outside cell probability $(\sqrt{\delta} + z)/v$. Each good cell uses at most M_B parallel tubes.

Apply (12A.6) to its source descendants: the scaled tube radius is at most $\rho + 2^\ell z \leq \zeta/2$ and missing mass at most $\sqrt{v} \leq \zeta/4$. The additional descendant failure is at most $\sqrt{v} + T/\ell$. The spatial θ components inherit saturation with error and failure at most

$$a_s = C_I\sqrt{\delta} + C_I/\sqrt{m} + C_I\sqrt{m/\ell} \leq 3u/20.$$

The greedy disjoint length- ℓ windows have total joint failure at most

$$2\sqrt{\delta} + a_s + \frac{\sqrt{\delta} + z}{v} + \sqrt{v} + T/\ell + \frac{\ell}{N+1} \leq 9u/20 < u. \quad (14D.13)$$

The second laws produced by this argument are $(S_i)_*B_*(v_{y,i})$. These differ from $B_*v^{y,i}$ by a translation, which does not affect concentration. We have therefore obtained fixed planes W_i for which $\theta^{x,i}$ is (W_i, u, m) -saturated and $B_*v^{y,i}$ is (W_i, ζ) -concentrated, jointly outside probability at most u . All events are certified by countable source/image cell choices and tube-mass tests.

Now take $\theta = A_*\mu$. Let C be the constant in (14D.8), enlarged to be at least one. Before choosing m , set $u = a^2/(8C)$, and require

$$m \geq 8C/a^2, \quad C_I/\sqrt{m} \leq u/20. \quad (14D.14)$$

These are conditions only on $d_1, d_2, q, c, \Lambda, a$, not on ζ . After m, ζ are specified, make the choices above to obtain δ, ℓ_0 and then z, N_0 for every $\ell \geq \ell_0$.

The first marginal of the intermediate result satisfies (14D.7) with $\delta = u$. Its input-coordinate saturation failure is at most

$$C(u + 1/m)/a \leq a/4.$$

The second marginal's concentration failure is at most $u \leq a/8$. A union bound in the original product source law gives joint failure at most $3a/8 < a$, proving (14D.11). This completes P-II.5.12-Lin. $\square$

II.5.20 Independent linear-image decoupling

Proposition II.5.20.1 (P-II.5.11-Lin – Independent linear-image decoupling). *Fix integers $d_1, d_2, q \geq 1$, $\Lambda \geq 1$, $0 < c \leq \Lambda$ and $0 < \varepsilon < 1$. There is $M = M(d_1, d_2, q, c, \Lambda, \varepsilon)$ such that for every integer $m \geq M$ there are $a, \beta > 0$ and an integer $p_0 > m$ with the following property. For every integer $p \geq p_0$ there are $t > 0$ and an integer N_0 such that for $N \geq N_0$ the following assertion holds.*

Let $A : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^q$ and $B : \mathbb{R}^{d_2} \rightarrow \mathbb{R}^q$ be linear with

$$c\|v\| \leq \|Av\| \leq \Lambda\|v\| \quad (v \in \mathbb{R}^{d_1}), \quad \|B\|_{\text{op}} \leq \Lambda. \quad (14\text{E.1})$$

For compactly supported probabilities μ, ν in the two input spaces, suppose a measurable subspace field $V(i, x, y)$ and a measurable good event of probability at least $1 - \beta$ certify

$$A_*\mu^{x,i} \text{ is } (V(i, x, y), a, p)\text{-saturated}, \quad B_*\nu^{y,i} \text{ is } (V(i, x, y), t)\text{-concentrated}. \quad (14\text{E.2})$$

Here i is uniform on $\{0, \dots, N\}$, and $x \sim \mu, y \sim \nu$ are independent of it and each other. Then there are level-only subspaces $W_0, \dots, W_N$ and a measurable good event of probability at least $1 - \varepsilon$ certifying

$$A_*\mu^{x,i} \text{ is } (W_i, \varepsilon, m)\text{-saturated}, \quad B_*\nu^{y,i} \text{ is } (W_i, \varepsilon)\text{-concentrated}. \quad (14\text{E.3})$$

The constants are uniform over the maps and measures. The first-input map is injective; the second need not be. No small-root-cell assumption is required. The saturation and failure tolerances precede the freely enlargable input window; only concentration accuracy and the terminal level threshold follow it.

Proof strategy

Affine normalization puts every image law in one fixed box. Finite saturation and concentration menus then replace the random plane by a deterministic intermediate plane. Inheritance along the original input filtrations and exact dyadic resampling allow the intermediate conclusion to be spread over the required level range. The maps remain fixed throughout; no nonlinear approximation is used.

Bounded-box entropy and finite menus

Set

$$S = \max\{1, \Lambda\sqrt{d_1 + d_2}\}, \quad T(z) = \frac{z}{4S} + (1/2, \dots, 1/2), \quad s_S = 2\sqrt{q}S + 1. \quad (14\text{E.4})$$

Every linear image in (14E.2), and every image of a canonical descendant input component, is carried by $[-S, S]^q$. The map T sends that box into the interior of the unit cube.

L-II.5.01 in both directions gives, for every such probability θ , every subspace U and $n \geq 1$,

$$|\text{def}_{U,n}(T_*\theta) - \text{def}_{U,n}(\theta)| \leq C_T/n, \quad C_T = C(q, S). \quad (14E.5)$$

Indeed T, T^{-1} are similarities with logarithmic dilations $-\log_2(4S), \log_2(4S)$. Their linear parts preserve each subspace. The direct conditional-entropy comparison (1.2)–(1.3) bounds the signed difference of the two deficits.

Consequently the downward and nearby-subspace saturation bounds of **L-II.5.04** hold on this box with error C_S/n , where $C_S = C(q, S)$. For the nearby bound the gap must be at most 2^{-n} . To justify the extension, apply (14E.5), the unit-cube bound, and (14E.5) in reverse; the accumulated error is $(2C_T + C_q)/n$. For concentration, the proof following (14A.6) gives the one-sided implication

$$\theta \text{ is } (U, b)\text{-concentrated, } 0 < b < 1, \alpha(U, W) \leq v \implies \theta \text{ is } (W, b + s_S v)\text{-concentrated.} \quad (14E.6)$$

There are finite-valued Borel saturated-menu maps on this box with

$$\begin{aligned} \theta &\text{ is } (V_s(\theta), \gamma_s, p)\text{-saturated,} \\ \theta &\text{ is } (V, a, p)\text{-saturated} \implies \alpha(V, V_s(\theta)) \leq \gamma_s, \quad \gamma_s = C_s \left(a + \frac{\log(p+1)}{p} \right), \end{aligned} \quad (14E.7)$$

provided the displayed error is sufficiently small; $C_s = C(q, S)$. Here is the precise reduction to the proved menu (14B.2). An input error a becomes $a + C_T/p$ under T . Apply that unit-cube menu with this enlarged error, and then pull its saturation conclusion back by (14E.5). Its containment conclusion is unchanged. Since $1/p \leq \log(p+1)/p$ for $p \geq 2$, increasing C_s bounds both conclusions by (14E.7). Borel measurability and finite-valuedness are preserved under $\theta \mapsto T_*\theta$.

The concentrated menu (14A.6), now with $C = S, d = q$, gives finite-valued Borel choices of a plane $V_c(\xi)$ and a witnessing affine tube such that

$$\begin{aligned} \xi &\text{ is } (V_c(\xi), \rho)\text{-concentrated,} \\ \xi &\text{ is } (V, t)\text{-concentrated} \implies \alpha(V_c(\xi), V) \leq \rho, \quad \rho = C_c t^{1/(q+1)}, \quad C_c = C(q, S). \end{aligned} \quad (14E.8)$$

The finite-menu tests evaluate fixed Borel tube masses and finite-label entropies. Their measurability was proved in Section 11; no uncountable selection is invoked.

Parameter order

Let $C_I = C(d_1, q, c, \Lambda)$ be large enough for the probability-and-tolerance inheritance conclusion of **P-II.5.12-Lin**(1). Enlarge C_S, C_s, C_c if needed so all the preceding box bounds hold uniformly. Put $\eta = \varepsilon/8$. Require $m \geq M$ so large that

$$C_I/\sqrt{m} + 2C_S/m \leq \eta/20. \quad (14E.9)$$

Choose $\gamma > 0$ in the small-error range of (14E.7) such that $4\gamma < 1$ and

$$C_I \sqrt{\gamma} \leq \eta/20, \quad \Gamma = c_q(4\gamma)^{1/(q+1)} < \min\{2^{-m}, \eta/(40s_S)\}. \quad (14E.10)$$

The constant c_q is the envelope constant of L-II.5.05, enlarged to be at least one. Choose $0 < h < \min(\gamma, \eta/20)$. With the net constant from (11.2), choose $b > 0$ satisfying

$$b < \eta/20, \quad \sqrt{b} < \min\{\eta/20, h^{q^2+1}/(2C_q^{\text{net}})\}, \quad \beta = b^2/2. \quad (14E.11)$$

Choose $0 < a \leq \gamma/(2C_s)$ and $p_0 > \max(m, 1)$ such that every $p \geq p_0$ satisfies

$$C_s \log(p+1)/p \leq \gamma/2, \quad C_I \sqrt{m/p} \leq \eta/20. \quad (14E.12)$$

Thus $\gamma_s \leq \gamma$. These choices depend only on the parameters specified before p in the statement.

For the given $p \geq p_0$, take $t > 0$ sufficiently small for (14E.8), with

$$\rho \leq h, \quad \sqrt{\rho} \leq \eta/20, \quad 2^p \rho \leq \eta/20, \quad N_0 + 1 \geq 4p/\varepsilon. \quad (14E.13)$$

This is possible because $\rho = C_c t^{1/(q+1)}$ tends to zero with t . All choices precede the two measures. There is no root scale to be chosen in this lemma.

The deterministic intermediate plane at one outer level

Fix a level i where the conditional failure of (14E.2) is at most b . The random laws

$$\Theta = A_* \mu^{x,i}, \quad \Xi = B_* \nu^{y,i}$$

are independent, because their inputs x, y are independent and the maps A, B are fixed. By (14E.7), (14E.8) and the approximate-containment triangle inequality,

$$\mathbb{P}\{\alpha(V_c(\Xi), V_s(\Theta)) \leq h + \gamma\} \geq 1 - b. \quad (14E.14)$$

This implication holds on the given good event, whatever the additional point-dependence of its witness $V(i, x, y)$. It does not require that witness to be a function of the two laws alone.

Take an h -net of all subspaces in $\mathbb{R}^q$ with $J \leq C_q^{\text{net}} h^{-q^2}$ points. Retain its balls whose V_c -probability is greater than h/J . Their union has probability at least $1 - h$: the union of all discarded balls has probability at most h , and all balls cover the subspace space. Apply the envelope lemma to the retained centers with parameter $2(h + \gamma) \leq 4\gamma$, and denote its envelope by W . The retained family is nonempty, since $h < 1$. Every retained center is approximately contained in W with tolerance at most Γ . Since $h \leq \Gamma$,

$$\mathbb{P}_{\Xi}\{\alpha(V_c(\Xi), W) \leq 2\Gamma\} \geq 1 - h. \quad (14E.15)$$

Independence and conditional Markov in (14E.14) show that, outside Θ -probability $\sqrt{b}$, the Ξ -failure probability in (14E.14) is at most $\sqrt{b}$. Each retained ball has probability greater than

$$h/J \geq h^{q^2+1}/C_q^{\text{net}} > 2\sqrt{b}.$$

It therefore contains a V_c satisfying (14E.14) for each such Θ . By the triangle inequality its center is approximately contained in $V_s(\Theta)$ with tolerance at most $2(h + \gamma)$. The minimality property of the envelope gives

$$\mathbb{P}_\Theta\{\alpha(W, V_s(\Theta)) \leq \Gamma\} \geq 1 - \sqrt{b}. \quad (14E.16)$$

The plane $W = W(i)$ depends on the level and the two component distributions, but not on the sampled points.

Inheritance through the original input partitions

For a first-input outer cell E , put $\phi = \mu^E$. The law $A_*\phi$ has the saturated plane from (14E.7). Apply P-II.5.12-Lin(1) directly to ϕ , with the same fixed map A . Its conclusion concerns $A_*\phi^{\mu,j}$, not image-grid components of $A_*\phi$. With $0 \leq j < p$ uniform, the inherited saturation tolerance and failure are at most

$$a_s = C_I\sqrt{\gamma_s} + C_I/\sqrt{m} + C_I\sqrt{m/p}. \quad (14E.17)$$

For outer laws satisfying (14E.16), let $U = \pi_{Y_s}W$. The near-inclusion bound $\Gamma < 1$ makes this projection injective on W , so $\dim U = \dim W$ and $d_G(U, W) \leq \Gamma < 2^{-m}$ by L-II.5.04(1). Downward saturation to U and nearby transfer to W cost at most $2C_S/m$ on the bounded image box. Equations (14E.9)–(14E.12) give

$$a_s + 2C_S/m \leq 3\eta/20.$$

The first-input descendant failure, including the bad outer laws in (14E.16), is at most $a_s + \sqrt{b}$.

For a second-input outer cell D , put $\psi = \nu^D$. The menu (14E.8) fixes a tube

$$\psi\{u : Bu \in (z + V_c)^{(\rho)}\} \geq 1 - \rho. \quad (14E.18)$$

Disintegrate ψ over its original source level- j cells. At each j , conditional Markov shows that, outside cell probability $\sqrt{\rho}$, the conditional tube mass is at least $1 - \sqrt{\rho}$. For such a source cell F , with integer address v_F , the exact identity is

$$B_*\psi^F = (u \mapsto 2^j Bu - Bv_F)_*\psi_F. \quad (14E.19)$$

Thus the good set is in the tube about $2^j z - Bv_F + V_c$ of radius $2^j \rho$. For $j < p$, (14E.13) makes this radius and its missing mass at most $\eta/20$. The law in (14E.19) is itself carried by $[-S, S]^q$, even if the displayed affine offset is large. On the positive-mass tube choose its perpendicular offset; its norm is bounded by $\sqrt{q}S + 1$, exactly as in the proof of (14E.6).

For outer laws satisfying (14E.15), transfer this concentration from V_c to W . The additional radius is at most $2s_S\Gamma < \eta/20$. Consequently the second-input descendant concentration tolerance is at most $\eta/10$, with failure at most $h + \sqrt{\rho}$. No image partition or inverse of B was used.

A union bound, averaging over the common descendant level and the two independent input component kernels, now gives joint failure at most

$$h + \sqrt{\rho} + \sqrt{b} + a_s < \eta. \quad (14E.20)$$

Both output tolerances are less than ε .

Resampling, measurable events, and the final level count

The exact resampling identity for dyadic input components identifies $\phi^{u,j}$ with $\mu^{x,i+j}$ and $\psi^{v,j}$ with $\nu^{y,i+j}$ under the original input laws. The canonical affine maps compose exactly because the grids are dyadic. Applying the fixed maps A, B after this identity leaves it unchanged.

The mean outer-level failure in (14E.2) is at most β . Hence all but a fraction $\beta/b = b/2$ of the levels have conditional failure at most b , and each of these levels has the preceding proved window with its deterministic plane $W(i)$. Choose disjoint length- p intervals greedily starting at such levels, and assign their planes to the covered levels. Uncovered levels before the last truncated interval are bad starting levels. Discarding that last interval costs at most $p/(N+1)$. The total joint failure is at most

$$\eta + b/2 + p/(N+1) < \varepsilon. \quad (14E.21)$$

Assign arbitrary planes at the remaining levels.

There are only countably many positive-mass input cells. The menu planes, offsets, and heavy-ball families are finite-valued. All conditional good events in (14E.18)–(14E.19) are Borel mass tests divided by positive cell masses. Fix an envelope for each finite retained family. After the interval choices, the union of these certified good cell events is measurable and has the probability just estimated. It certifies (14E.3). This proves P-II.5.11-Lin in full. $\square$

II.5.21 The classical convex-set normal approximation input

Repeated independent addition accumulates covariance. A normal approximation then converts a spectral gap into components that are uniform along the top spectral plane, and uniformity yields saturation:

$$\begin{aligned} \text{spectral gap} &\implies \text{Gaussian approximation} \\ &\implies \text{uniform components} \implies \text{saturated components.} \end{aligned}$$

The sharp Rotar estimate below is included for reference. The later proof uses the qualitative bounded-array central limit theorem proved in this section, not that quantitative black box.

Cited result — B-II.5.15 — Berry–Esseen–Rotar convex-set estimate

Let $\mu_1, \dots, \mu_k$ be probabilities on $\mathbb{R}^d$ with $\rho_j = \int \|x\|^3 d\mu_j(x) < \infty$. Put $\theta = \mu_1 * \dots * \mu_k$, let γ be the Gaussian law with the same mean and covariance as θ , and assume $\lambda_d(\Sigma(\theta)) > 0$. Then

$$d_{\text{cvx}}(\theta, \gamma) \leq C_d \frac{\sum_{j=1}^k \rho_j}{\lambda_d(\Sigma(\theta))^{3/2}}. \quad (\text{II.5.28})$$

No proof is supplied. Source: V. I. Rotar', "On the Speed of Convergence in the Multidimensional Central Limit Theorem," *Theory Probab. Appl.* 15(2) (1970), 354–356, DOI 10.1137/1115044; Russian original *Teor. Veroyatnost. i Primenen.* 15(2) (1970), 370–372. The exact working statement is [Hoc20, Theorem 4.8, printed pp. 48–49].

Reference: [Hoc20].

II.5.21.1 Exact role of the estimate

If $\rho_j \leq B$ and $\lambda_d(\Sigma(\mu_j)) \geq b > 0$ for every j , then covariance additivity and (II.5.28) give

$$d_{\text{cvx}}(\theta, \gamma) \leq C_d B b^{-3/2} k^{-1/2}. \quad (15.2)$$

The same inequality holds after a common invertible affine change of coordinates, because inverse images of convex Borel sets are convex Borel sets. This observation does not identify convex-set discrepancy with total variation distance. For example, a lattice-supported sum and a nondegenerate Gaussian can have total variation distance one. All later entropy deductions require their own finite-cell and conditional-mass estimates.

The sharp estimate is not used below. It is retained to record the source's quantitative version, whereas the qualitative theorem proved next supplies every central-limit input used by G-II.5.16 and its descendants.

II.5.21.2 A proved qualitative substitute sufficient for every later use

The sharp rate in B-II.5.15 is not needed below. We prove the uniform bounded-array central limit statement that the smoothing argument actually consumes.

Lemma II.5.21.1 (Uniform convex-set central limit theorem for bounded arrays). *For every $d \geq 1$, $R \geq 1$, $\sigma > 0$, and $\varepsilon > 0$ there is $k_0 = k_0(d, R, \sigma, \varepsilon)$ with the following property. Let $k \geq k_0$, let $X_1, \dots, X_k$ be independent random vectors supported on $[-R, R]^d$, and put*

$$Z_k = s_k \sum_{j=1}^k (X_j - \mathbb{E}X_j), \quad k^{-1/2} \leq s_k < 2k^{-1/2}.$$

If every eigenvalue of $\Sigma(Z_k)$ is at least σ , and $G_k = N(0, \Sigma(Z_k))$, then

$$\sup_{C \subset \mathbb{R}^d \text{ convex Borel}} |\mathbb{P}(Z_k \in C) - G_k(C)| < \varepsilon. \quad (\text{II.5.21.1})$$

The conclusion is uniform over all choices of the summands.

Proof. Put $Y_j = X_j - \mathbb{E}X_j$ and $B = 2R\sqrt{d}$. For fixed $u \in \mathbb{R}^d$, Taylor's formula and $\mathbb{E}Y_j = 0$

give, uniformly in j ,

$$\mathbb{E} e^{is_k \langle u, Y_j \rangle} = 1 - \frac{s_k^2}{2} \mathbb{E} \langle u, Y_j \rangle^2 + O(B^3 |u|^3 s_k^3).$$

Since $s_k = O(k^{-1/2})$, taking logarithms and summing shows, uniformly for u in a fixed compact set,

$$\log \mathbb{E} e^{i \langle u, Z_k \rangle} = -\frac{1}{2} u^\top \Sigma(Z_k) u + O_{d,R,u}(k^{-1/2}). \quad (\text{II.5.21.2})$$

The covariance matrices lie in the compact set

$$\{A = A^\top \geq 0 : \sigma I \leq A \leq 4B^2 I\}.$$

Suppose uniformity in (II.5.21.1) failed. There would be a sequence of arrays with $k \rightarrow \infty$ and convex sets C_k for which the discrepancy is bounded below by some $\varepsilon_0 > 0$. Pass to a subsequence on which $\Sigma(Z_k) \rightarrow A$, where $A \geq \sigma I$. Equation (II.5.21.2) and Lévy's continuity theorem imply both

$$Z_k \Rightarrow N(0, A), \quad G_k \Rightarrow N(0, A).$$

We record why the varying convex sets do not obstruct the conclusion. The second moments are uniformly bounded, so for large M both laws give mass at most $O(B^2/M^2)$ to the complement of $B(0, M)$. By the Blaschke selection theorem, after another subsequence either $C_k \cap \overline{B}(0, M)$ is empty throughout, or the compact convex sets

$$K_k = \overline{C_k \cap \overline{B}(0, M)}$$

converge in Hausdorff distance to a nonempty compact convex set K . The empty case contributes no mass inside the ball. In the nonempty case, for every $\eta > 0$ and all large k , $C_k \cap \overline{B}(0, M) \subset K^{(\eta)}$. If K has empty interior, then $N(0, A)(K) = 0$; Portmanteau on a fixed closed η -neighborhood of K , followed by $\eta \downarrow 0$, shows that both laws give $C_k \cap \overline{B}(0, M)$ mass tending to zero.

Suppose K has nonempty interior. Hausdorff convergence of compact convex sets implies that, for every $\eta > 0$ and all large k ,

$$K_{-\eta} \subset \text{int } K_k \subset C_k \cap \overline{B}(0, M) \subset K^{(\eta)}.$$

Here $K_{-\eta} = \{x \in K : \text{dist}(x, K^c) > \eta\}$. The middle inclusion uses the elementary convex-set identity $\text{int } \overline{D} \subset D$ whenever D is full-dimensional and convex: a point in the interior of the closure lies inside a simplex with vertices in D , hence belongs to D . Thus no boundary mass of the arbitrary Borel sets C_k is silently discarded. The boundary of a full-dimensional convex body has Lebesgue, hence $N(0, A)$, measure zero. Portmanteau applied to fixed inner and outer parallel sets, then $\eta \downarrow 0$, shows that both probabilities of $C_k \cap \overline{B}(0, M)$ tend to $N(0, A)(K)$. The tail outside $B(0, M)$ is $O(B^2/M^2)$ uniformly. Letting $M \rightarrow \infty$ makes the discrepancy tend to zero, contradicting ε_0 . This proves the asserted uniform convergence. $\square$

Lemma II.5.21.2 (Finite-cell Gaussian entropy transfer). *Fix $1 \leq r \leq d$, $0 < a \leq b < \infty$, $0 < \eta < 1$, and an integer $m \geq 1$. There are $p = p(d, a, b, \eta, m)$ and $\varepsilon = \varepsilon(d, a, b, \eta, m) > 0$ such that the following holds. Let V be an r -plane, let γ be a Gaussian probability on V whose covariance eigenvalues lie in $[a, b]$, and let θ be a probability on V with*

$$d_{\text{cvx}}(\theta, \gamma) < \varepsilon.$$

Then, using any orthonormal dyadic grid on V ,

$$\mathbb{P}_{x \sim \theta} \left\{ \frac{1}{m \log 2} H(\theta^{x \cdot p}, \mathcal{D}_m^V) > r - \eta \right\} > 1 - \eta. \quad (\text{II.5.21.3})$$

The choices are uniform in V , in the mean of γ , and in the orientation and translation of the grid.

Proof. For a Gaussian with covariance spectrum in $[a, b]$, its density and first derivatives satisfy uniform bounds after translating the mean to the origin. Gaussian tails are also uniform. Choose a compact ball carrying mass $> 1 - \eta/16$. On the portion of that ball where the density is at least a sufficiently small fixed number, uniform continuity of the density shows that, once p is large, its normalized restriction to every positive-mass level- p cube is within total variation $o_{p \rightarrow \infty}(1)$ of normalized Lebesgue measure on that cube. Such Lebesgue measure has exactly $rm \log 2$ entropy at the next m intrinsic dyadic levels. Entropy continuity for a partition with 2^{rm} atoms therefore gives

$$\mathbb{P}_{x \sim \gamma} \{ (m \log 2)^{-1} H(\gamma^{x \cdot p}, \mathcal{D}_m^V) > r - \eta/4 \} > 1 - \eta/4. \quad (\text{II.5.21.4})$$

Compactness of the covariance interval, of the orthogonal group, and of a fundamental domain for translations of the dyadic grid makes this choice of p uniform.

Only finitely many level- p cubes in the chosen ball are involved. Discard those having Gaussian mass below a threshold chosen so that their total mass is $< \eta/8$. On each remaining cube the normalized entropy of the 2^{rm} descendants is a continuous function of the finitely many parent and descendant masses. Every parent and descendant cube is convex, so convex-set discrepancy smaller than a sufficiently small ε makes all these masses uniformly close to their Gaussian counterparts. Thus (II.5.21.4), with its margins reduced from $\eta/4$ to η , holds for θ . This is (II.5.21.3). $\square$

II.5.22 Gaussian smoothing and component saturation

Theorem II.5.22.1 (G-II.5.16 – Gaussian spectral smoothing and local saturation). *The result has two complementary parts.*

1. *For every $R \geq 1$ and $0 < \delta < 1$ there is $M_0 = M_0(d, R, \delta)$ such that for every $m \geq M_0$ and $\sigma > 0$ there are integers $p = p_0(d, R, \delta, m, \sigma)$, $k_0 = k_0(d, R, \delta, m, \sigma)$ and, for each $k \geq k_0$, $\rho_0 = \rho_0(d, R, \delta, m, \sigma, k) > 0$ with this property. Let $\mu_j \in \mathcal{P}([-R, R]^d)$, $\theta = \mu_1 * \dots * \mu_k$ and $1 \leq r \leq d$. If*

$$\lambda_r(\Sigma(\theta)) \geq \sigma k, \quad \lambda_{r+1}(\Sigma(\theta)) < \min\{\rho_0, \sigma k/2\}, \quad (16.1)$$

put $V = E_r(\Sigma(\theta))$ and $q = p - \lfloor \log_2 \sqrt{k} \rfloor$. Then

$$\mathbb{P}_{x \sim \theta} \{\theta^{x,q} \text{ is } (V, \delta, m)\text{-uniform}\} \geq 1 - \delta. \quad (\text{II.5.29})$$

Components at negative q are defined by D-II.2.01. The interior spectral plane has a strict gap; at $r = d$ the tail is zero.

2. For every $0 < \delta < 1$ there is $M_1 = M_1(d, \delta)$ such that for every $m \geq M_1$ and $\sigma > 0$ there are integers $p = p_1(d, \delta, m, \sigma)$, $k_1 = k_1(d, \delta, m, \sigma)$ and, for each $k \geq k_1$, $\rho_1 = \rho_1(d, \delta, m, \sigma, k) > 0$ with this property. For a Borel probability μ on $\mathbb{R}^d$, $i_0 \geq 0$ and $1 \leq r \leq d$, put $A = \bar{\Sigma}_{i_0}(\mu)$. If

$$\lambda_r(A) > \sigma, \quad \lambda_{r+1}(A) < \min\{\rho_1, \sigma/2\}, \quad (16.3)$$

then, with $V = E_r(A)$, $\theta = \mu^{*k}$ and $j_0 = i_0 - \lfloor \log_2 \sqrt{k} \rfloor + p$,

$$\mathbb{P}_{x \sim \theta} \{\theta^{x,j_0} \text{ is } (V, \delta, m)\text{-saturated}\} \geq 1 - \delta. \quad (\text{II.5.29a})$$

No global second moment of μ is assumed: A is the mean covariance of its rescaled components.

The proof uses the qualitative bounded-array central limit theorem and the finite-cell Gaussian transfer proved in the preceding section. The sharper quantitative estimate B-II.5.15 is not needed.

Proof strategy

For the first part, the spectral gap produces a Gaussian whose components are nearly uniform along the top spectral plane; finite-cell transfer carries this property back to the convolution. For the second part, sample independent rescaled components, use concentration of their empirical covariance around the mean covariance, and apply the first part conditionally. Mixture stability then returns the conclusion to components of μ^{*k} .

Proof

The spectral-gap sum statement

We first isolate the deterministic partition transfer used below. Let $\zeta > 0$, let $p, m \geq 1$, and let ϑ be a probability which gives mass at least $1 - \zeta$ to a tube of radius 2^{-p-m-c_d} about an affine translate of an r -plane V . Suppose that, for the intrinsic level- p components of $(\pi_V)_* \vartheta$,

$$\mathbb{P} \{(m \log 2)^{-1} H_m^V > r - \zeta\} > 1 - \zeta. \quad (\text{II.5.22.1})$$

Then, provided m is large in terms of d and a target error δ , and ζ is small in terms of d, δ, m , one has

$$\mathbb{P}_{x \sim \vartheta} \{\vartheta^{x,p} \text{ is } (V, \delta, m)\text{-uniform}\} > 1 - \delta. \quad (\text{II.5.22.2})$$

Here is the full partition argument. On the retained tube, every ambient level- p cube meets at most K_d intrinsic V -cells of that level, and every intrinsic cell meets at most K_d ambient cubes which carry retained mass; the same assertion holds at level $p + m$. This is the grid-comparison lemma from Section 1 after translating the affine plane. Hence the four conditional entropies obtained by conditioning first on either coarse grid and measuring either fine grid differ in average by at most $4 \log K_d$. Removing mass ζ changes each fine conditional entropy by at most

$$2b(\zeta) + 2\zeta(dm \log 2 + \log K_d)$$

by the finite conditional-entropy continuity estimate L-II.4.01. Thus (II.5.22.1) makes the mean normalized ambient component entropy at least $r - O_d(\zeta) - O_d(1/m)$. Component entropy is at most $r + O_d(\zeta) + O_d(1/m)$ on the tube; Markov applied to its nonnegative deficit gives entropy at least $r - \delta$ outside a set of mass $< \delta/2$ when m and then ζ have the stated sizes. Finally, the expected conditional mass outside the tube is at most ζ . Markov makes this conditional mass at most 2^{-m} outside a set of mass at most $2^m \zeta$. Inside a level- p cube the tube rescales to radius at most 2^{-m-c_d} , so the component is $(V, 2^{-m})$ -concentrated. Taking $2^m \zeta < \delta/2$ proves (II.5.22.2).

We apply this transfer on the grid corresponding to the desired components of θ . Fix d, R, δ and choose m large enough for the preceding argument, then choose its internal ζ . Let $\theta = \mu_1 * \cdots * \mu_k$ satisfy (16.1), put

$$\ell = \lfloor \log_2 \sqrt{k} \rfloor, \quad s = 2^{-\ell}, \quad V = E_r(\Sigma(\theta)),$$

and let $\vartheta = T_*\theta$, where $T(x) = s(x - \int x d\theta(x))$. The image under T of the standard level- $q = p - \ell$ dyadic grid is a translated level- p dyadic grid $\mathcal{G}_p$. Use $\mathcal{G}_p$ as the ambient coarse grid in the partition transfer above. Its bounded-intersection constants do not depend on the translation. For the projected measure $\eta = (\pi_V)_*\vartheta$, the r positive covariance eigenvalues lie in

$$[\sigma, 16dR^2]. \quad (\text{II.5.22.3})$$

Indeed $s^2k \in [1, 4)$, while every centered summand has norm at most $2R\sqrt{d}$. Lemma II.5.21.1, applied after an orthogonal identification of V with $\mathbb{R}^r$, and then Lemma II.5.21.2, give a level $p = p_0(d, R, \delta, m, \sigma)$ for which (II.5.22.1) holds for η on the intrinsic grid paired with $\mathcal{G}_p$, once $k \geq k_0$. The Gaussian-cell transfer is uniform over translations and orientations of its grid, so the thresholds do not depend on the mean of θ . Taking the maximum over $1 \leq r \leq d$ makes both thresholds independent of r .

Let $P = \pi_{V^\perp}$. By the covariance identity and (16.1),

$$\int \|Px\|^2 d\vartheta(x) = \text{tr}(P\Sigma(\vartheta)P) \leq 4d\rho_0/k.$$

Choose $\rho_0 > 0$, after p, k are fixed, so small that Chebyshev's inequality makes the mass outside the tube of radius 2^{-p-m-c_d} at most the ζ required above. Decrease it also so that $\rho_0 < \sigma k/2$. The partition transfer gives high-probability (V, δ, m) -uniformity for the $\mathcal{G}_p$ -components of ϑ . Since $\mathcal{G}_p = T(\mathcal{D}_q^{(d)})$, their canonical rescalings agree exactly with the standard level- q components of θ . The positive scalar s preserves V , so entropy and tube concentration transfer exactly. This proves the first assertion. The same proof applies with unchanged thresholds to every translated level- q grid on θ , since its image under T is a translated level- p grid and the preceding estimates are uniform over translations.

Proof of the mean-component covariance statement

Fix d, δ , take m sufficiently large, and choose an internal $0 < \eta \ll_d \delta^2$. We apply the first assertion with error η , window $m + 1$, support bound $R = 1$, and spectral lower bound $\sigma/2$. Let p and k_0 be its thresholds. Choose a number $t > 0$ so small that whenever positive semidefinite matrices B, A satisfy

$$\|B - A\|_{\text{op}} < t, \quad \lambda_r(A) > \sigma, \quad \lambda_{r+1}(A) < \sigma/4,$$

then

$$\lambda_r(B) > \sigma/2, \quad d_G(E_r(B), E_r(A)) < d^{-1/2} 2^{-(m+2)}. \quad (\text{II.5.22.4})$$

This follows from the proved spectral-continuity estimates (7.6) and the following plane estimate in Section 7. Choose $k \geq k_0$ so large that $d^2/(kt^2) < \eta$.

Sample independently k level- i_0 cells $Q_1, \dots, Q_k$ with probabilities $\mu(Q_j)$. Write $\hat{\Theta}_j = \mu_{Q_j}$ for the raw conditional probability and $\Theta_j = T_{Q_j^*} \hat{\Theta}_j$ for its canonical level- i_0 rescaling. Put

$$S = \sum_{j=1}^k \Sigma(\Theta_j), \quad V = E_r(A), \quad P = \pi_{V^\perp}.$$

Their covariance entries are bounded by one and have mean A . Therefore

$$\mathbb{P}\{\|S/k - A\|_{\text{op}} > t\} \leq \frac{d^2}{kt^2} < \eta. \quad (\text{II.5.22.5})$$

Indeed the expected squared Frobenius norm of $S/k - A$ is at most d^2/k . Also

$$\mathbb{E} \text{tr}(PSP) = k \text{tr}(PAP) \leq kd\lambda_{r+1}(A) \leq kd\rho_1. \quad (\text{II.5.22.6})$$

Let ρ_0 be the transverse threshold supplied by the first assertion for this k . Choose

$$\rho_1 < \min\{\sigma/4, \eta\rho_0/(kd)\}. \quad (\text{II.5.22.7})$$

Markov's inequality and (II.5.22.6) show that $\text{tr}(PSP) < \rho_0$ with probability $> 1 - \eta$. On the intersection with (II.5.22.5), put $W = E_r(S)$. Then (II.5.22.4) gives

$$\lambda_r(S) > \sigma k/2, \quad d_G(W, V) < d^{-1/2} 2^{-(m+2)},$$

and the min-max formula, tested on V , gives

$$\lambda_{r+1}(S) \leq \|PSP\|_{\text{op}} \leq \text{tr}(PSP) < \rho_0.$$

For every such component tuple, the first assertion applies to $\Theta_1 * \dots * \Theta_k$. With probability $> 1 - \eta$ over its level- $p - \lfloor \log_2 \sqrt{k} \rfloor$ component, that component is $(W, \eta, m + 1)$ -uniform. By L-II.5.02(1), the near-plane transfer in L-II.5.04, and the one-level entropy comparison, it is $(V, \delta/4, m)$ -saturated, after the initial choices of m and η .

It remains to pass from the sampled convolution to the component mixture; we give the bookkeeping explicitly. Let $\tilde{\mu} = \mu^{\times k}$ and $\pi(x_1, \dots, x_k) = x_1 + \dots + x_k$. Disintegrating each factor at level i_0 gives

$$\mu^{*k} = \pi_* \tilde{\mu} = \int \hat{\Theta}_1 * \dots * \hat{\Theta}_k d\mathbb{P}(Q_1, \dots, Q_k).$$

If $Q_j = 2^{-i_0}(a_j + [0, 1)^d)$, then

$$\widehat{\Theta}_1 * \cdots * \widehat{\Theta}_k = (x \mapsto 2^{-i_0}(x + a_1 + \cdots + a_k))_*(\Theta_1 * \cdots * \Theta_k).$$

Consequently, if

$$j_0 = i_0 - \lfloor \log_2 \sqrt{k} \rfloor + p,$$

then the inverse image of the standard level- j_0 grid under the displayed affine map is a translated level- $p - \lfloor \log_2 \sqrt{k} \rfloor$ grid. Apply the translation-uniform version of the first assertion to this exact grid. Its canonical component coordinates agree with those of the raw convolution, so the high-probability uniformity, and hence saturation by [L-II.5.02](#), pass to the raw components without a pointwise grid-comparison claim. Conditional entropy in the product grids $\mathcal{P}_{j_0+m}^V \mid \mathcal{P}_{j_0+m}^{V^\perp}$ is concave under this integral mixture, by the calculation [\(8.1\)](#). Thus the mean saturation deficit of level- j_0 components of μ^{*k} is $O_d(\eta) + O_d(1/m)$. Markov's inequality makes the deficit at most δ outside a set of mass $< \delta$, for our choices of m and η . This is [\(II.5.29a\)](#), and completes the proof of both parts of [G-II.5.16](#).

II.5.23 A bounded convolution power with subspace structure

Theorem II.5.23.1 (G-II.5.17 – Repeated self-convolution subspace structure). *For every $0 < \delta < 1$ there is $M = M(d, \delta)$ such that for every $m \geq M$ and $0 < \zeta < 1$ there are integers $K = K(d, \delta, m, \zeta) \geq 1$ and $N_0 = N_0(d, \delta, m, \zeta)$ such that every Borel probability μ on $\mathbb{R}^d$ and every $N \geq N_0$ admit $1 \leq k \leq K$ and level-only subspaces $V_0, \dots, V_N$ satisfying, with $\theta = \mu^{*k}$,*

$$\begin{aligned} \mathbb{P}_{0 \leq i \leq N} \{ \theta^{x,i} \text{ is } (V_i, \delta, m)\text{-saturated} \} &\geq 1 - \delta, \\ \mathbb{P}_{0 \leq i \leq N} \{ \mu^{y,i} \text{ is } (V_i, \zeta)\text{-concentrated} \} &\geq 1 - \delta. \end{aligned} \quad (\text{II.5.30})$$

The threshold M is independent of ζ . The selected power may depend on μ, N , but its finite menu and N_0 are fixed in advance.

The proof below combines G-II.5.16 with a finite covariance-band pigeonhole argument and aligns the resulting component levels by finitely many shifts.

Proof strategy

Choose in advance a finite descending list of covariance thresholds and one convolution power for each threshold. At most one spectral band can be occupied at most levels, so one power works on a large set of levels. Gaussian smoothing provides saturation on the corresponding spectral plane, covariance controls concentration of the original components, and slow variation plus finite shift alignment places both conclusions at the same levels.

Covariance-band selection and level alignment

Let $u = \delta/8$. Choose m so large that (II.5.29a) is available at accuracy u and that the downward-saturation error obeys

$$C_S/m \leq u,$$

where C_S is the constant in L-II.5.04(4). This lower bound depends only on d, δ .

We first choose all spectral thresholds and convolution powers without seeing μ or N . Set $S_0 = \lceil d/u \rceil + 1$ and $a_0 = d$. Suppose $a_{s-1} > 0$ is chosen, where $1 \leq s \leq S_0$. Apply (II.5.29a) at

$$\sigma_s = a_{s-1}/2, \quad \delta_{\text{loc}} = u$$

to obtain p_s and its lower power threshold. Choose a positive integer k_s above that threshold and large enough that

$$\ell_s = \lfloor \log_2 \sqrt{k_s} \rfloor - p_s \geq 1. \quad (17.2)$$

This is possible because p_s is chosen before k_s .

Let $C_{\text{cv}} \geq 1$ dominate the two constants in the random covariance concentration estimate (II.5.12), and put

$$b_s = C_{\text{cv}} a_s^{1/6}, \quad L_s = \left\lfloor \frac{1}{2} \log_2(1/b_s) \right\rfloor + 1.$$

Choose the next threshold $a_s > 0$ so small that all of the following hold:

$$\begin{aligned} a_s &< \min\{a_{s-1}/4, \rho_1(d, u, m, \sigma_s, k_s), 1\}, \\ b_s &< \min\{\varepsilon_d, 2^{-2\ell_s}\}, \\ \omega_d(b_s) &\leq \min\{u, \zeta/(1 + s_d)\}, \quad L_s \geq 2(d+1)\ell_s/u. \end{aligned} \quad (17.3)$$

Here ω_d and ε_d are the explicit modulus and small-parameter cutoff in (6.1), and $s_d = 2\sqrt{(d)+1}$ is the concentration stability constant. Every constraint can be met: k_s, p_s, ℓ_s are already fixed, $\omega_d(b_s) \rightarrow 0$ and $L_s \rightarrow \infty$ as $a_s \rightarrow 0$. This also displays precisely where the independently prescribed ζ enters.

After these finitely many steps, define

$$K = \max_{1 \leq s \leq S_0} k_s$$

and choose N_0 so that, for every s and $N \geq N_0$,

$$N \geq \ell_s, \quad N+1 \geq 2(d+1)\ell_s/u. \quad (17.4)$$

No threshold in this construction depends on the eventual measure.

Selecting an empty spectral band

Now fix μ and $N \geq N_0$ and write

$$A_q = \bar{\Sigma}_q(\mu), \quad 0 \leq q \leq N.$$

These positive semidefinite matrices have eigenvalues in $[0, d]$: every component is a unit-cube probability and $\text{tr } \Sigma \leq \int \|x\|^2 d\mu^{x,q} \leq d$. For each s put

$$B_s(A_q) = \#\{1 \leq j \leq d : \lambda_j(A_q) \in (a_s, a_{s-1}]\}.$$

The positive bands are disjoint, so

$$\sum_{s=1}^{S_0} B_s(A_q) \leq d, \quad \sum_{s=1}^{S_0} \mathbb{E}_q B_s(A_q) \leq d. \quad (17.5)$$

Zero eigenvalues contribute to no band. Some s satisfies $\mathbb{E}_q B_s(A_q) \leq d/S_0 < u$. Since this is an integer count, Markov gives

$$I = \{q : B_s(A_q) = 0\}, \quad |I| > (1-u)(N+1). \quad (17.6)$$

Fix this s , and abbreviate $k = k_s, \ell = \ell_s, a = a_s, b = b_s$ and $L = L_s$. For $q \in I$ let

$$r_q = \#\{j : \lambda_j(A_q) > a_s\}, \quad W_q = E_{r_q}(A_q).$$

If $r_q > 0$, the empty-band condition gives $\lambda_{r_q}(A_q) > a_{s-1} > \sigma_s$ and $\lambda_{r_q+1}(A_q) \leq a_s$. Thus (II.5.29a), including its strict threshold by (17.3), gives

$$\mathbb{P}_{x \sim \theta} \{\theta^{x, q-\ell} \text{ is } (W_q, u, m)\text{-saturated}\} \geq 1 - u, \quad \theta = \mu^{*k}. \quad (17.7)$$

If $r_q = 0$, use $W_q = \{0\}$. Then (17.7) holds automatically, because projection onto $W_q^\perp = \mathbb{R}^d$ leaves the entropy unchanged. No nondegenerate Gaussian claim at rank zero is needed.

For $q \notin I$ set $W_q = \mathbb{R}^d$. For every $q \in I$, (II.5.12) shows that the rescaled μ -component is (W_q, b) -concentrated outside probability at most $C_c \nu \sqrt{a} \leq b$. On the other levels the same assertion is automatic. Hence, for **every** q ,

$$\mathbb{P}_{y \sim \mu} \{\mu^{y, q} \text{ is } (W_q, b)\text{-concentrated}\} \geq 1 - b. \quad (17.8)$$

Aligning the two scale sequences

Apply P-II.5.06(3), in its proved form for arbitrary probabilities, to (17.8). It gives $U_q \leq W_q$ such that all level- q components of μ are $(U_q, \omega_d(b))$ -concentrated outside probability $\omega_d(b)$. It also gives a set

$$J = \{\ell \leq q \leq N : d_G(U_q, U_{q-\ell}) \leq \omega_d(b)\}$$

with, by (17.3)–(17.4),

$$\frac{|J|}{N+1} \geq 1 - (d+1)\ell \left(\frac{1}{L} + \frac{1}{N+1} \right) \geq 1 - u. \quad (17.9)$$

For $q \in I \cap J$ set $V_{q-\ell} = U_q$. There are more than $(1 - 2u)(N+1)$ such assigned levels, and $q - \ell$ lies in $[0, N]$. Assign arbitrary planes to the remaining levels.

For an assigned level, downward saturation from W_q to U_q in (17.7) has error at most $u + C_S/m \leq 2u$ and fails with probability at most u . On the concentration side, the original measure at level $q - \ell$ is concentrated on $U_{q-\ell}$. Changing to U_q by the gap bound in J gives concentration tolerance at most

$$(1 + s_d)\omega_d(b) \leq \zeta$$

and failure at most $\omega_d(b) \leq u$. Averaging over the assigned levels and allowing all remaining levels to fail gives total exceptional probability at most $3u < \delta$ for each of the two assertions in (II.5.30). The saturation tolerance $2u$ is smaller than δ .

This proves the theorem from (II.5.29a). In particular, the power menu, empty-band selection, zero-rank case, parameter separation and finite shift losses are all included in the estimate. $\square$

II.5.24 The Euclidean entropy inverse theorem

The main Euclidean deduction follows the chain

$$\begin{aligned} & \text{small one-step growth} \longrightarrow \text{small bounded-power growth} \\ & \longrightarrow \text{structured convolution power} \longrightarrow \text{structure of the original pair.} \end{aligned}$$

Each arrow has already been isolated above; the proof below orders the parameters so that the errors can be absorbed without circular choices.

Theorem II.5.24.1 (G-II.5.18 – Higher-dimensional Euclidean inverse theorem). *For every $0 < \varepsilon < 1$ there is $M = M(d, \varepsilon)$ such that for every $m \geq M$, $R \geq 1$ and $0 < \zeta < 1$ there are $\delta = \delta(d, R, \varepsilon, m, \zeta) > 0$ and $N_0 = N_0(d, R, \varepsilon, m, \zeta)$ such that, if $N \geq N_0$ and $\mu, \nu \in \mathcal{P}([-R, R]^d)$ satisfy*

$$h_N(\mu * \nu) < h_N(\mu) + \delta, \quad (18.1)$$

then level-only subspaces $V_0, \dots, V_N$ exist with

$$\mathbb{P}_{\substack{0 \leq i \leq N \\ x \sim \mu, y \sim \nu}} \{ \mu^{x,i} \text{ is } (V_i, \varepsilon, m)\text{-saturated and } \nu^{y,i} \text{ is } (V_i, \zeta)\text{-concentrated} \} \geq 1 - \varepsilon. \quad (\text{II.5.31})$$

The window threshold is independent of the concentration tolerance ζ . Taking $\zeta = \varepsilon$ gives the single-tolerance form.

The large-window hypothesis is essential for the fixed-grid formulation. The proof combines G-II.5.17 with the convolution-stability and Kaimanovich–Vershik estimates proved above.

Proof strategy

Average the entropy gain of component convolutions over scales. Small global growth forces most local gains to be small. Apply the bounded-power theorem to the first measure, use convolution stability to transfer its saturated planes back to the original components, and use the Kaimanovich–Vershik inequality to retain concentration for the second measure. The parameter order is chosen so that the prescribed concentration tolerance does not affect the lower bound on the observation window.

The bounded-support local-average comparison

We need T-II.3.08 beyond its originally stated unit-cube support. Here is the exact extension used, proved directly.

Let α, β be compactly supported probabilities and let $N, m \geq 1$. Conditional entropy concavity and the component convolution disintegration give

$$H_{\alpha * \beta}(\mathcal{D}_{i+m} \mid \mathcal{D}_i) \geq \mathbb{E}_{x,y} H_m(\alpha^{x,i} * \beta^{y,i}) - d \log 2. \quad (18.3)$$

Indeed the convolution of two raw level- i components is carried by the sum of two level- i cubes, which meets at most 2^d level- i cells. Its fine entropy is exactly that of the convolution of the canonical components, since the rescaling translations add to a dyadic lattice translation. Subtracting its coarse entropy costs at most $d \log 2$. Averaging the conditional entropy over component pairs can only decrease the conditional entropy of their mixture. All relevant entropies are finite by bounded support.

Put $c_j = H_j(\alpha * \beta) - H_{j-1}(\alpha * \beta)$. Since $0 \leq c_j \leq d \log 2$, counting each increment in the length- m windows gives

$$\sum_{i=0}^{N-1} H_{\alpha * \beta}(\mathcal{D}_{i+m} \mid \mathcal{D}_i) \leq m(H_N(\alpha * \beta) - H_0(\alpha * \beta)) + \frac{dm(m-1)}{2} \log 2.$$

Combining with (18.3), dividing by $Nm \log 2$ and discarding the nonnegative coarse entropy yields

$$h_N(\alpha * \beta) \geq \frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_{x,y} h_m(\alpha^{x,i} * \beta^{y,i}) - \frac{d}{m} - \frac{d(m-1)}{2N}. \quad (18.4)$$

The reverse count of missing initial increments, applied to α , gives

$$\frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_x h_m(\alpha^{x,i}) \geq h_N(\alpha) - \frac{H_0(\alpha)}{N \log 2} - \frac{d(m-1)}{2N}. \quad (18.5)$$

For (18.5), component disintegration identifies the numerator of each summand with $H_{i+m}(\alpha) - H_i(\alpha)$. The missing initial weights sum to at most $m(m-1)/2$, and terminal increments are nonnegative. Thus no support renormalization is hidden in (18.4)–(18.5).

Write

$$D_{i,x,y} = h_m(\alpha^{x,i} * \beta^{y,i}) - h_m(\alpha^{x,i}).$$

Subtracting (18.5) from (18.4) in the appropriate direction proves

$$h_N(\alpha * \beta) - h_N(\alpha) \geq \mathbb{E}_{\substack{0 \leq i < N \\ x,y}} D_{i,x,y} - \frac{d}{m} - \frac{d(m-1)}{N} - \frac{H_0(\alpha)}{N \log 2}. \quad (18.6)$$

The global support of β enters only to ensure finite entropy, not in the $1/m$ constant. This matters when β is a large but bounded convolution power.

Parameter order and the contradiction

Let $a = \varepsilon/8$. Let C_8 control both estimates (II.5.13)–(II.5.14). Choose m large enough for G-II.5.17 at accuracy a and for

$$(C_8 + d)/m \leq a^2/4. \quad (18.7)$$

Apply G-II.5.17 with concentration tolerance ζ to obtain a finite power bound K and its uniform scale threshold. Set

$$\delta = \frac{a^2}{8K}. \quad (18.8)$$

Choose N_0 above that scale threshold and large enough that for every $N \geq N_0$,

$$\begin{aligned} \frac{a + d(m-1) + d \log_2(2R+2)}{N} &\leq a^2/4, \\ \frac{Kd \log 5 + d \log(2K+3)}{N \log 2} &\leq a^2/8. \end{aligned} \quad (18.9)$$

All choices precede the measures. In particular, m is not chosen after the convolution power K .

Given μ, ν satisfying (18.1), use G-II.5.17 for ν to choose $1 \leq k \leq K$, $\tau = \nu^{*k}$ and planes V_i . Then typical τ -components are (V_i, a, m) -saturated with total failure at most a , and typical ν -components are (V_i, ζ) -concentrated with total failure at most a .

Suppose the probability, over uniform $0 \leq i \leq N$ and $x \sim \mu$, of failure of $(V_i, 2a, m)$ -saturation is at least $2a$. Combining with the τ saturation marginal, at least a of the independent component pairs satisfy both that failure and (V_i, a, m) -saturation for $\tau^{y,i}$. By (II.5.14), on this event the local entropy increment is at least $a - C_8/m$. It is at least $-C_8/m$ for every pair by (II.5.13).

Replacing the level range $0, \dots, N$ by $0, \dots, N-1$ loses at most $1/N$ in the probability of this event. Consequently

$$\mathbb{E}_{\substack{0 \leq i < N \\ x \sim \mu, y \sim \tau}} (h_m(\mu^{x,i} * \tau^{y,i}) - h_m(\mu^{x,i})) \geq a^2 - \frac{a}{N} - \frac{C_8}{m}. \quad (18.10)$$

The support bound gives $H_0(\mu) \leq d \log(2R + 2)$. Equations (18.6)–(18.10) therefore imply

$$h_N(\mu * \tau) - h_N(\mu) \geq a^2/2. \quad (18.11)$$

On the other hand, the proved dyadic Kaimanovich–Vershik inequality (II.5.16), with its stated subtracted term, gives

$$\begin{aligned} h_N(\mu * \tau) - h_N(\mu) &\leq k(h_N(\mu * \nu) - h_N(\mu)) + \frac{kd \log 5 + d \log(2k + 3)}{N \log 2} \\ &< K\delta + a^2/8 = a^2/4. \end{aligned} \quad (18.12)$$

This contradicts (18.11). Thus the saturation failure probability for μ is less than $2a$. Combining it with the concentration failure probability at most a for ν proves joint success at least $1 - 3a \geq 1 - \varepsilon$. Since $2a < \varepsilon$, the saturation tolerance is also sufficient. This proves (II.5.31).

Every concentration event used has a fixed plane at each of finitely many levels and only countably many positive-mass component cells. Choosing witnesses on the successful cells certifies these events measurably. No extra measurable-selection assumption is used. This completes the deduction of G-II.5.18 from G-II.5.17. $\square$

II.5.25 Self-convolution and uniform components

Theorem II.5.25.1 (G-II.5.19 – Self-convolution uniform-subspace corollary). *For every $0 < \varepsilon < 1$ there is $M = M(d, \varepsilon)$ such that for every $m \geq M$ there are $\delta > 0$ and N_0 such that, if $N \geq N_0$, $\mu \in \mathcal{P}([0, 1]^d)$ and*

$$h_N(\mu * \mu) < h_N(\mu) + \delta, \quad (19.1)$$

then there are level-only subspaces $V_0, \dots, V_N$ satisfying

$$\mathbb{P}_{0 \leq i \leq N} \{ \mu^{x,i} \text{ is } (V_i, \varepsilon, m)\text{-uniform} \} \geq 1 - \varepsilon. \quad (\text{II.5.32})$$

The large-window restriction is inherited from G-II.5.18 and cannot in general be replaced by a fixed short window.

Proof

Use (II.5.31) with the same measure in both positions, saturation/probability tolerance $\varepsilon/3$, and concentration tolerance

$$\zeta = 2^{-m}.$$

This is a permitted parameter order: the required lower bound on m depends on $d, \varepsilon/3$, not on ζ . Choose δ, N_0 from that theorem with $R = 1$.

For the resulting planes, each of the two marginal conditions has failure probability at most $\varepsilon/3$. Sample a single component $\mu^{x,i}$ instead of two independent ones. A union bound shows that, outside probability at most $2\varepsilon/3$, it is both $(V_i, \varepsilon/3, m)$ -saturated and $(V_i, 2^{-m})$ -concentrated. Nonnegativity of projected entropy gives

$$h_m(\mu^{x,i}) \geq \dim V_i - \varepsilon/3 \geq \dim V_i - \varepsilon.$$

These are exactly the two defining conditions for (V_i, ε, m) -uniformity. Their probability is at least $1 - 2\varepsilon/3 \geq 1 - \varepsilon$, proving (II.5.32). $\square$

II.5.26 The inverse theorem for the isometry action

To pass from Euclidean convolution to an isometry action, we localize both inputs, linearize on the chosen roots, apply the Euclidean inverse theorem, and then recover a level-only subspace by resampling.

Theorem II.5.26.1 (G-II.5.20 – Inverse theorem for the isometry action). *For every $R \geq 1$ and $0 < \varepsilon < 1$ there is $M = M(d, R, \varepsilon)$ such that for every $m \geq M$ there are $\delta > 0$, k_0 and, for each $k \geq k_0$, $N_0 = N_0(d, R, \varepsilon, m, k)$ such that the following holds for every $N \geq N_0$. Let $v \in \mathcal{P}(G_0(d))$ and $\mu \in \mathcal{P}([-R, R]^d)$ have supports contained in radius- R balls. Either*

$$h_N(v \cdot \mu) > h_N(\mu) + \delta, \quad (\text{II.5.33})$$

or, with probability at least $1 - \varepsilon$ over independent raw level- k components $\tilde{v} = v_{g,k}$ and $\tilde{\mu} = \mu_{x,k}$, there are subspaces $V_0(\tilde{v}, \tilde{\mu}), \dots, V_N(\tilde{v}, \tilde{\mu})$ and a measurable certifying event $\mathcal{E}$ such that

$$\mathbb{P}(\mathcal{E}) \geq 1 - \varepsilon, \quad \mathcal{E} \subseteq \left\{ (i, z, h) : \begin{array}{l} \tilde{\mu}^{z,i} \text{ is } (V_i, \varepsilon, m)\text{-saturated,} \\ \mathcal{A}_i(\tilde{v}_{h,i}, z) \text{ is } (V_i, \varepsilon)\text{-concentrated} \end{array} \right\}. \quad (\text{II.5.34})$$

Here i is uniform on $\{0, \dots, N\}$, $z \sim \tilde{\mu}$, $h \sim \tilde{\nu}$, and these samples are independent. The event convention is [D-II.5.08](#).

If, in addition, μ is $((\varepsilon/(5d))^{2(d+1)}, \sigma)$ -non-affine, impose the explicit additional threshold $m \geq M_{\dim}(d, R, \varepsilon, \sigma)$. The other parameters may also depend on σ , and a good set of component pairs of probability at least $1 - \varepsilon$ satisfies

$$\frac{1}{N+1} \sum_{i=0}^N \dim V_i > \frac{1}{d+1} h_N^{G_0}(\tilde{\nu}) - \varepsilon, \quad (\text{II.5.35})$$

For the averaged assertion the planes are assigned to every root pair, with $V_i = \mathbb{R}^d$ on the pairs discarded by the structural stage:

$$\mathbb{E}_{\text{roots}} \frac{1}{N+1} \sum_{i=0}^N \dim V_i > \frac{1}{d+1} h_N^{G_0}(\nu) - \varepsilon.$$

The local removal of point dependence uses the small-root-cell hypothesis in [P-II.5.11-Loc](#). This localization is essential: the corresponding unrestricted translation principle is false.

Proof strategy

First prove a local theorem for measures supported on one spatial and one group cell. A finite-window entropy estimate turns small action growth into small Euclidean convolution growth; the Euclidean inverse theorem and the localized decoupling propositions then produce level-only planes. Coarse localization lifts this conclusion to arbitrary bounded supports. Finally, affine-frame recovery converts the entropy of the group component into a lower bound for the average plane dimension.

Proof

The local structural theorem

Lemma II.5.26.2 (Local inverse theorem for isometry actions). *For $d \geq 1$, $R \geq 1$ and $0 < \eta < 1$, there is a threshold $M_{\text{loc}}(d, R, \eta)$ such that, for every $m \geq M_{\text{loc}}$, there are $\tau > 0$ and k_{loc} and, for each $k \geq k_{\text{loc}}$, a uniform $N_{\text{loc}}(k)$ with the following property. If $N \geq N_{\text{loc}}(k)$, $\mu \in \mathcal{P}([-R, R]^d)$ and $\nu \in \mathcal{P}(G_0(d))$ are each carried by a single level- k cell and*

$$\Delta_N(\nu, \mu) \leq \tau, \quad (20.1)$$

there are level-only planes $V_0, \dots, V_N$ and a measurable certifying event $\mathcal{E}$ such that

$$\mathbb{P}(\mathcal{E}) \geq 1 - \eta, \quad \mathcal{E} \subseteq \left\{ (i, x, g) : \begin{array}{l} \mu^{x,i} \text{ is } (V_i, \eta, m)\text{-saturated,} \\ \mathcal{A}_i(\nu_{g,i}, x) \text{ is } (V_i, \eta)\text{-concentrated} \end{array} \right\}. \quad (20.2)$$

Here i is uniform on $\{0, \dots, N\}$, $x \sim \mu$, $g \sim \nu$; these three samples are independent. All other dependence is the displayed component dependence. The planes may depend on

μ, ν, N but not on the sampled i -level component pair. A measurable good event is required as in P-II.5.11-Loc.

Proof of the local theorem

We first obtain a quantitative local convolution increment from the action increment. We then apply the Euclidean inverse theorem on most component pairs and remove the remaining point dependence by the localized decoupling results proved earlier in the chapter.

A finite-window entropy criterion

The following estimate supplies the local convolution-growth input for (20.1)–(20.2). Write $\Theta_i(x) = \mu^{x,i}$ and, for sampled x, g , define the centered, rotation-normalized group-image law

$$\Xi_i(g, x) = (h \mapsto 2^i U_g^{-1}(hx - gx))_* \nu_{g,i}. \quad (20.6)$$

It differs from $\mathcal{A}_i(\nu_{g,i}, x)$ only by a translation. Unlike Θ_i , it depends on both g and x . Its support has radius at most $C_R = d\sqrt{d}R + \sqrt{d}$, by the matrix-entry bounds in the localized action construction. For an integer $w \geq 1$ put

$$D_{i,w}(g, x) = h_w(\Theta_i(x) * \Xi_i(g, x)) - h_w(\Theta_i(x)). \quad (20.7)$$

There is $C = C(d, R) \geq 1$ such that, if $k \geq k_{\text{lin}}(d, R, w)$, $N > k$, and ν, μ have the stated bounded spatial support, then

$$\Delta_N(\nu, \mu) \geq \frac{1}{N} \sum_{i=k}^{N-1} \mathbb{E}_{g,x} D_{i,w}(g, x) - \frac{C}{w} - \frac{d(w-1+k)}{N} - \frac{H_0(\mu)}{N \log 2}. \quad (20.8)$$

Here the group measure need not be confined to a root cell; bounded support suffices to make all entropies finite.

There is also a cellwise frozen version. For each positive-mass level- i group/spatial cell D, E choose representatives $g_D \in D \cap \text{supp } \nu$ and $x_E \in E \cap \text{supp } \mu$. Put

$$\bar{\Xi}_{i,D,E} = (h \mapsto 2^i U_{g_D}^{-1}(hx_E - g_D x_E))_* \nu_D, \quad \bar{D}_{i,w}(D, E) = h_w(\mu^E * \bar{\Xi}_{i,D,E}) - h_w(\mu^E). \quad (20.8a)$$

Here μ^E is the canonically rescaled component. Both (20.8) and the signed Markov inequality (20.13) hold with $D_{i,w}$ replaced by $\bar{D}_{i,w}$, after increasing the same dimension/support constants if needed. The frozen increment depends only on the two cells, not on their sampled points.

Proof. Joint component disintegration and conditional entropy concavity give, for every i ,

$$H_{\nu,\mu}(\mathcal{D}_{i+w} \mid \mathcal{D}_i) \geq \mathbb{E}_{g,x} H_{\nu_{g,i}, \mu_{x,i}}(\mathcal{D}_{i+w} \mid \mathcal{D}_i). \quad (20.9)$$

For $i \geq k$, L-II.5.14 and its exact bilinear identity apply on the two level- i cells. After centering and rotating, the linearized image is, up to a translation, the scale- 2^{-i} copy of $\Theta_i * \Xi_i$. The error in the image location is $O_d(2^{-2i})$. Choose k_{lin} so that this is at most $2^{-(i+w)}$ for every $i \geq k_{\text{lin}}$.

The entropy perturbation, translation and rotation comparisons then cost at most a fixed $C(d, R) \log 2$ at level $i + w$. Moreover the entire component-action image has diameter at most $C'(d, R)2^{-i}$: for h, h' in the same group cell and z, z' in the same spatial cell,

$$\|hz - h'z'\| \leq \|z - z'\| + \|U_h - U_{h'}\|_{\text{op}}\|z'\| + \|a_h - a_{h'}\| \leq C'(d, R)2^{-i}.$$

Its level- i entropy is therefore at most another fixed $C(d, R) \log 2$. Enlarging C gives

$$\frac{H_{v_{g,i}, \mu_{x,i}}(\mathcal{D}_{i+w} \mid \mathcal{D}_i)}{w \log 2} \geq h_w(\Theta_i * \Xi_i) - C/w. \quad (20.10)$$

We record the two endpoint estimates rather than discard them. For any bounded probability ρ on $\mathbb{R}^d$, counting the multiplicity of its dyadic entropy increments yields

$$h_N(\rho) \geq \frac{1}{N} \sum_{i=0}^{N-1} \frac{H_\rho(\mathcal{D}_{i+w} \mid \mathcal{D}_i)}{w \log 2} - \frac{d(w-1)}{2N}. \quad (20.11)$$

Indeed, increments numbered at most N have multiplicity at most w , and those after N have total excess multiplicity $w(w-1)/2$. The term $H_0(\rho) \geq 0$ can be discarded. For μ the reverse counting bound gives

$$\frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}_x h_w(\Theta_i(x)) \geq h_N(\mu) - \frac{H_0(\mu)}{N \log 2} - \frac{d(w-1)}{2N}. \quad (20.12)$$

This time the total missing multiplicity at the initial endpoint is $w(w-1)/2$; final increments have nonnegative contribution. The equality between the mean rescaled-component entropy and the normalized conditional entropy uses exact dyadic rescaling.

Apply (20.11) to $v \cdot \mu$, use (20.9)–(20.10) on $k \leq i < N$, and use nonnegativity of conditional entropy on $i < k$. In (20.12), removing the first k component entropies loses at most dk/N because Θ_i is a unit-cube probability. Subtracting the baseline proves (20.8).

For the cellwise frozen version, linearize at the chosen (g_D, x_E) instead of the sampled (g, x) . The exact identity is unchanged and its remainder is still $O_d(2^{-2i})$ uniformly on the product cell. The centered linearized law is a translate of $(S_{-i})_*(\mu^E * \bar{\Xi}_{i,D,E})$ followed by the fixed rotation U_{g_D} . Thus (20.10), both endpoint counts and the uniform convolution lower bound all hold with the bars. This proves the frozen comparison as well; the probability estimate needed for the local theorem is the next step.

L-II.5.08 also gives $D_{i,w} \geq -C_8/w$ uniformly. Consequently, if I is uniform on $\{k, \dots, N-1\}$ and $s > 0$, Markov applied to $D_{I,w} + C_8/w$ gives

$$\mathbb{P}\{D_{I,w} > s\} \leq \frac{\frac{N}{N-k} \left(\Delta_N(v, \mu) + \frac{C}{w} + \frac{d(w-1+k)}{N} + \frac{H_0(\mu)}{N \log 2} \right) + \frac{C_8}{w}}{s + \frac{C_8}{w}}. \quad (20.13)$$

For any prescribed $s > 0$ and $0 < b < 1$, this is at most b after first taking w large in terms of d, R, s, b , then $k \geq k_{\text{lin}}(d, R, w)$, then N large in terms of these choices, and the global growth upper bound sufficiently small. For example require the two $1/w$ contributions to be at most $bs/8$ each, $N/(N - k) \leq 2$, and choose the remaining three terms inside parentheses to have sum at most $bs/8$. The numerator is then at most $5bs/8 < bs$. These parameter choices precede the measures, since $H_0(\mu) \leq d \log(2R + 2)$.

This is a statement about small local convolution growth, not yet the level-only action structure. It does not assert independence of the two random measures in (20.6)–(20.7).

Completion of the local structural assembly

We prove (20.1)–(20.2) from G-II.5.18, using the separated-accuracy transfer and separated-input decoupling propositions to maintain the required parameter order.

Parameters, in their actual order

Fix d, R, ε and let $m \geq M(d, R, \varepsilon)$ be in the range of the separated-input proposition. Obtain its saturation tolerance a , failure allowance β and lower window bound p_0 . Choose

$$u = \min\{a/2, \beta/8, 1/8\}, \quad p \geq \max\{p_0, M_{\text{tr}}(d, u)\}. \quad (20.29)$$

Here $M_{\text{tr}}(d, u)$ denotes the window threshold in the separated-accuracy transfer proposition.

The separated-input proposition now supplies $c, k_{\text{dec}}, N_{\text{dec}}$ for this chosen p . Apply the separated-accuracy transfer with target saturation/failure u , target window p and concentration tolerance $\zeta = c/4$. It supplies s, L_0 . Choose

$$L \geq \max\{L_0, M_{\text{Euc}}(d, s)\}, \quad (20.30)$$

Here $M_{\text{Euc}}(d, s)$ is the window threshold supplied by G-II.5.18.

Then take z, N_{tr} from the transfer proposition for this L . This is legal precisely because its s precedes L .

Let $C_R = d\sqrt{d}R + \sqrt{d}$ and choose $R_* = \max(1, C_R + 1)$. Apply G-II.5.18 with saturation/failure s , window L , support box $[-R_*, R_*]^d$ and concentration tolerance z . Obtain $\delta_E > 0, N_E$. Put

$$\kappa = \beta/4, \quad \nu = \delta_E/2, \quad \tau = \kappa\nu/16. \quad (20.31)$$

Choose an integer $w \geq \max(N_{\text{tr}}, N_E, p, 1)$ so large that

$$(C + C_8)/w \leq \kappa\nu/16, \quad (20.32)$$

where C, C_8 are the constants in the frozen version of (20.8), (20.13). Choose k_{loc} at least k_{dec} and $k_{\text{lin}}(d, R, w)$, and so large that

$$K_R 2^{-k_{\text{loc}}} \leq c/4, \quad K_R = dC_R + d^{3/2}. \quad (20.33)$$

For each $k \geq k_{\text{loc}}$ choose $N_{\text{loc}}(k) \geq N_{\text{dec}}$ large enough that for all $N \geq N_{\text{loc}}(k)$,

$$\frac{N}{N-k} \leq 2, \quad \frac{d(w-1+k)}{N} + \frac{d \log(2R+2)}{N \log 2} \leq \kappa v/16, \quad \frac{k+w+1}{N+1} \leq \beta/4. \quad (20.34)$$

Every choice precedes the measures. None requires enlarging a window in terms of a concentration tolerance that had to be chosen before that same window.

Small local convolution growth on most component pairs

Let μ, ν satisfy the small-root-cell hypotheses and $\Delta_N(\nu, \mu) \leq \tau$. Use the cellwise frozen increments $\bar{D}_{i,w}$. Equations (20.13), (20.31)–(20.34) give

$$\mathbb{P}_{\substack{k \leq i < N \\ x \sim \mu, g \sim \nu}} \{ \bar{D}_{i,w} > v \} < \kappa. \quad (20.35)$$

For example, the numerator of the Markov bound is at most $7\kappa v/16$, while its denominator is at least v . The strict inequality in (20.35) is therefore available even when the global growth bound is non-strict.

Fix a successful outer cell pair D, E at level i . Set $\theta = \mu^E$, $\xi = \bar{\Xi}_{i,D,E}$ and $\tau_D = \nu_D$. Both θ, ξ are carried by $[-R_*, R_*]^d$, and

$$h_w(\theta * \xi) - h_w(\theta) \leq v < \delta_E. \quad (20.36)$$

G-II.5.18 supplies level-only planes for the rescaled measures θ, ξ at levels $0, \dots, w$, with joint success at least $1 - s$, saturation error s at window L , and concentration tolerance z .

Apply the separated-accuracy transfer with base level $b = i$, orthogonal map $Q = U_{g_D}^{-1}$, point $x_0 = x_E$, translation $t = g_D x_E$, group law τ_D and spatial law θ . Its image law is exactly ξ , and its group partitions are the original $\mathcal{D}_{i+j}^{G_0}$. It gives planes $W_j(D, E, i)$ such that, for uniform $0 \leq j \leq w$, independent spatial descendants and group descendants, joint success is at least $1 - u$. The spatial descendants are saturated at window p with error u . The group image laws

$$(h' \mapsto 2^{i+j} U_{g_D}^{-1}(h' x_E - g_D x_E))_* (\tau_D)_{h,i+j} \quad (20.37)$$

are concentrated on W_j with tolerance $c/4$. Since τ_D is already in one level- i cell, $(\tau_D)_{h,i+j} = \nu_{h,i+j}$ for $j \geq 0$.

Return to the actual point and rotation

Let y be any point of the corresponding spatial descendant cell, h any point of the group descendant cell, and h' range over that group cell. After translations, the difference between the image in (20.37) and the actual normalized action at (h, y) is

$$\begin{aligned} & 2^{i+j} U_{g_D}^{-1}(h' x_E - h x_E) - 2^{i+j} U_h^{-1}(h' y - h y) \\ &= 2^{i+j} (U_{g_D}^{-1} - U_h^{-1})(h' x_E - h x_E) + 2^{i+j} U_h^{-1}(U_{h'} - U_h)(x_E - y). \end{aligned} \quad (20.38)$$

The first difference of rotations has norm at most $d2^{-i}$; the group image difference has norm at most $C_R 2^{-(i+j)}$. The second matrix difference has norm at most $d2^{-(i+j)}$, and $\|x_E - y\| \leq \sqrt{d} 2^{-i}$. Thus the norm of (20.38) is at most

$$K_R 2^{-i} \leq K_R 2^{-k} \leq c/4. \quad (20.39)$$

Concentration therefore transfers to $\mathcal{A}_{i+j}(v_{h,i+j}, y)$ with tolerance at most $c/2 < c$. The good mass is unchanged. Canonical spatial rescaling and resampling identify the spatial descendants with $\mu^{y,i+j}$, with no rotation or entropy comparison imposed on those spatial components.

Importantly, this certification is uniform over all y, h in their fine component cells. The successful spatial entropy test and frozen group-tube test depend only on those cells.

Resampling and a measurable field on the original levels

Sample i uniformly from $\{k, \dots, N-1\}$, then the outer cell pair with its product mass weights, then j uniformly from $\{0, \dots, w\}$, and finally independent descendants conditioned on that outer pair. By (20.35) and the preceding construction, this sampling gives a certified plane with the desired window- p saturation and actual-action concentration, with probability at least

$$1 - \kappa - u. \quad (20.40)$$

For each fine level $q = i + j$ and positive-mass fine spatial/group cell pair, there are finitely many possible outer levels i and a uniquely determined outer pair for each such i . Choose the first of their candidate planes that certifies success on the entire fine cell pair, if any; otherwise choose the zero plane. All witnesses in this choice have already been certified by cell entropy and tube-mass tests. Thus it defines a countable-cell measurable field $V(q, y, h)$. Its success event contains every success in (20.40); no measurable selection over an uncountable family of planes is invoked.

For each fixed i, j , component resampling gives the original product law $\mu \times \nu$ at level $q = i + j$. The only change needed is therefore the distribution of q . For a fixed j , it is uniform on the integer interval $\{k + j, \dots, N - 1 + j\}$. Its total variation distance from the uniform law on $\{0, \dots, N\}$ is at most

$$\frac{k + 1 + \max(j - 1, 0)}{N + 1} \leq \frac{k + w + 1}{N + 1}. \quad (20.41)$$

Indeed, the shorter interval has length $N - k$, and its overlap with $\{0, \dots, N\}$ has length $N - k - \max(j - 1, 0)$; the choices in (20.34) make this positive. Summing the minimum of the two uniform masses proves (20.41). Averaging over j preserves the bound.

Assigning arbitrary planes on levels not reached by this sampling and restricting the resulting field to $0, \dots, N$, its joint failure probability is at most

$$\kappa + u + \frac{k + w + 1}{N + 1} \leq \beta/4 + \beta/8 + \beta/4 = 5\beta/8 < \beta. \quad (20.42)$$

The spatial saturation tolerance is $u \leq a$ and the concentration tolerance is less than c . Thus (14B.1) holds. The original measures lie in the requisite root cells, $k \geq k_{\text{dec}}$ and $N \geq N_{\text{dec}}$, so the separated-input clause of P-II.5.11-Loc supplies level-only planes with the target ε, m properties (14A.2).

This proves the local structural theorem (20.1)–(20.2), with its local accuracy parameter named ε here. $\square$

From the local theorem to bounded supports

Fix a desired outer accuracy $0 < \varepsilon < 1$ and take $m \geq M_{\text{loc}}(d, R, \varepsilon)$. Let $\tau, k_{\text{loc}}, N_{\text{loc}}$ be the parameters of the local theorem. Set

$$\delta = \varepsilon\tau/4. \quad (20.3)$$

For each $k \geq k_{\text{loc}}$, increase $N_0(k)$ above $N_{\text{loc}}(k)$ so that

$$\frac{d \log(2^{k+1}R + 2) + c_{\text{iso},d}}{N \log 2} \leq \varepsilon\tau/4 \quad (N \geq N_0(k)). \quad (20.4)$$

The support of μ lies in $[-R, R]^d$, so the numerator bounds $H_k(\mu) + c_{\text{iso},d}$. If the global growth alternative fails, $\Delta_N(\nu, \mu) \leq \delta$. The joint localization estimate (10.8) gives

$$\mathbb{P}_{g,x}\{\Delta_N(\nu_{g,k}, \mu_{x,k}) > \tau\} \leq \varepsilon/2. \quad (20.5)$$

For every pair in the complementary event, the two conditioned measures have precisely the small supports required by the local theorem. Assign its family $V_i(\nu_{g,k}, \mu_{x,k})$ to that pair. There are only finitely many positive-mass root cells under the bounded-support assumptions, so these choices introduce no measurability issue.

Thus at least $1 - \varepsilon/2$ of the root pairs have the joint inner success probability $1 - \varepsilon$ in (II.5.34). Assign $V_i = \mathbb{R}^d$ on the other root pairs. This proves the bounded-support structural assertion. $\square$

The non-affine dimension deduction

The following deduction uses the global structural assertion at each requested accuracy. To keep its parameter order explicit, we permit the window threshold in this additional clause to depend on the non-affinity scale σ .

Fix $0 < \varepsilon < 1$ and $\sigma > 0$, and suppose that μ is (a, σ) -non-affine, where

$$a = \left(\frac{\varepsilon}{5d}\right)^{2(d+1)}, \quad q = ((d+1)a)^{1/(d+1)} \leq \frac{2\varepsilon^2}{25d^2}. \quad (20.14)$$

Let $D = d^2 + d$ be the dimension of the ambient matrix-entry and translation-coordinate space used by the group grid. For every raw level- i group component,

$$0 \leq h_{i,\ell}^{G_0}(\nu_{g,i}) \leq D, \quad 0 \leq \frac{1}{d+1} h_{i,\ell}^{G_0}(\nu_{g,i}) \leq d. \quad (20.15)$$

These bounds follow by counting at most $2^{D\ell}$ descendant grid cells. In general the last upper bound is not 1.

Choose an integer $\ell \geq 2$ so large that the sum of the tube, rotation and affine-frame errors satisfies

$$e_\ell := C_{d,R} \frac{\log(\ell+1)}{\ell} + \frac{C_{d,R,\sigma}}{\ell} < \varepsilon/8. \quad (20.16)$$

All constants here come from [L-II.5.02](#), the entropy grid comparison, and [L-II.5.13](#); enlarge them as needed once and for all. Next choose

$$0 < \eta \leq \min\{2^{-\ell}, \varepsilon^2/(400d)\}. \quad (20.17)$$

Take $m \geq M_{\text{dim}}(d, R, \varepsilon, \sigma)$, where M_{dim} is at least the structural window threshold $M_{\text{loc}}(d, R, \eta)$. Use the structural theorem at accuracy η , obtaining a good set of root pairs of probability at least $1 - \eta$. Assign $V_i = \mathbb{R}^d$ on the other root pairs.

For the simultaneous structural and dimension statements, take the growth constant no larger than the constants required at accuracies ε and η , and take the root and terminal thresholds at least their respective maxima. The planes from the η application already satisfy the structural ε conclusion, since $\eta \leq \varepsilon$. Thus both conclusions concern one family of planes; no compatibility of two independent choices is being assumed.

Increase $N_0(k)$ so that, for every $N \geq N_0(k)$,

$$\frac{k}{N+1} \leq \eta, \quad \frac{d(\ell+1)}{N} < \varepsilon/8, \quad \frac{H_k^{G_0}(v)}{(d+1)N \log 2} + \frac{d(\ell+1)}{N} < \varepsilon/4. \quad (20.18)$$

These choices can be uniform: a group support in a radius- R ball has at most $(2^{k+1}R + 2)^D$ occupied level- k cells. For the second and third inequalities one may impose strict inequalities on this uniform bound.

For $i \geq k$, every sampled fine component belongs to a unique root pair. Let

$$r_i(g, x) = \dim V_i(v_{g,k}, \mu_{x,k}).$$

Globally, the concentration conclusion fails with probability at most η : on a bad root pair it is automatic because $V_i = \mathbb{R}^d$; on a good pair the local failure is at most η . On the concentration event, translate $\mathcal{A}_i$ into the fixed bounded box supplied by [\(20.6\)](#). Since $\eta \leq 2^{-\ell}$, [L-II.5.02](#) and the translation/rotation comparisons give

$$h_{i,\ell}(v_{g,i} \cdot x) \leq r_i(g, x) + C_{d,R} \frac{\log(\ell+1)}{\ell}. \quad (20.19)$$

By [D-II.5.06](#) this raw image window entropy has numerator $H_{i+\ell}$, not a subtracted coarse entropy. Exact dyadic dilation identifies it with $h_\ell((S_i)_*(v_{g,i} \cdot x))$; the fixed rotation and centering change that entropy by at most a dimension constant divided by ℓ , already included above.

For each fixed i, g , the non-affinity estimate [L-II.5.13](#) applies with x sampled from the original μ , independently of the group component. Except for μ -probability at most q ,

$$h_{i,\ell}(v_{g,i} \cdot x) > \frac{1}{d+1} h_{i,\ell}^{G_0}(v_{g,i}) - C_{d,R,\sigma}/\ell. \quad (20.20)$$

This use is global, before conditioning on the spatial root: individual spatial root components need not be non-affine. Combining [\(20.19\)](#)–[\(20.20\)](#), and declaring the levels $i < k$ exceptional, yields a measurable failure event F with

$$\mathbb{P}(F) \leq \eta + q + \frac{k}{N+1} \leq 2\eta + q, \quad (20.21)$$

such that outside F

$$\frac{1}{d+1} h_{i,\ell}^{G_0}(v_{g,i}) < r_i(g, x) + e_\ell. \quad (20.22)$$

The probability here is over uniform $0 \leq i \leq N$ and independent $g \sim \nu, x \sim \mu$. Each entropy and dimension event is Borel: it uses a finite root-pair choice and the existing countable-cell entropy functions.

Condition on the root pair. Markov and (20.14), (20.17) give

$$\mathbb{P}_{\text{roots}} \left\{ \mathbb{P}(F \mid \text{root pair}) > \frac{\varepsilon}{4d} \right\} \leq \frac{4d(2\eta + q)}{\varepsilon} \leq \frac{\varepsilon}{50} + \frac{8\varepsilon}{25d} < \frac{\varepsilon}{2}. \quad (20.23)$$

Intersect the complementary root event with the structural good set. Its probability is at least $1 - \varepsilon/2 - \eta > 1 - \varepsilon$. For each pair in that intersection, use (20.15) on the failure event and (20.22) on its complement. Since the assigned plane at a level is fixed for this root pair,

$$\frac{1}{N+1} \sum_{i=0}^N \dim V_i > \frac{1}{d+1} \mathbb{E}_{i,g \sim \tilde{\nu}} h_{i,\ell}^{G_0}(\tilde{v}_{g,i}) - e_\ell - \varepsilon/4. \quad (20.24)$$

For $i < k$ the two group components in the global and conditioned notations may differ; those levels were included in F , and the bound (20.15) applies to the conditioned ones as well.

The increment count from (20.12) applies in ambient dimension D . The root measure $\tilde{\nu}$ has $H_0^{G_0} = 0$ because $k \geq 0$ and its support is in one level- k cell. Passing from $0, \dots, N-1$ to $0, \dots, N$ costs at most $D/(N+1)$. Consequently

$$\mathbb{E}_{i,g \sim \tilde{\nu}} h_{i,\ell}^{G_0}(\tilde{v}_{g,i}) \geq h_N^{G_0}(\tilde{\nu}) - \frac{D(\ell+1)}{N}. \quad (20.25)$$

Equations (20.16), (20.18), (20.24)–(20.25) imply the strictly stronger error $\varepsilon/2$, hence

$$\frac{1}{N+1} \sum_{i=0}^N \dim V_i > \frac{1}{d+1} h_N^{G_0}(\tilde{\nu}) - \varepsilon. \quad (20.26)$$

This proves the good-root dimension conclusion.

For the average over *all* root pairs, do not discard the bad pairs and lose an uncontrolled entropy contribution. Instead average (20.22) globally, using (20.15) on F , then apply (20.25) to every root. The exact conditional entropy identity is

$$\mathbb{E}_g h_N^{G_0}(v_{g,k}) = h_N^{G_0}(\nu) - \frac{H_k^{G_0}(\nu)}{N \log 2}. \quad (20.27)$$

It follows that

$$\begin{aligned}
& \mathbb{E}_{\text{roots}} \frac{1}{N+1} \sum_{i=0}^N \dim V_i \\
& \geq \frac{1}{d+1} h_N^{G_0}(v) - \frac{H_k^{G_0}(v)}{(d+1)N \log 2} - \frac{d(\ell+1)}{N} - e_\ell - d(2\eta + q).
\end{aligned} \tag{20.28}$$

Here $d(2\eta + q) \leq \varepsilon^2/200 + 2\varepsilon^2/(25d) < \varepsilon/8$. The choices in (20.16)–(20.18) make the total error strictly less than ε . This is the corresponding averaged dimension bound.

Thus the entire dimension deduction is complete relative to the structural theorem, with the explicitly σ -dependent large-window threshold just stated. It does not assert that an arbitrary preassigned short window can be used. $\square$

II.5.27 The nondegenerate smooth-map structural theorem

The smooth-map theorem uses only a finite observation window. Uniform nondegeneracy supplies an all-fine-level comparison on each slice; finite-window linearization handles the two-input increment. The theorem is stated with precisely those hypotheses, and no unrestricted nonlinear claim is being made.

Theorem II.5.27.1 (G-II.5.21 – Nondegenerate smooth-map structural inverse theorem). *The injectivity hypothesis on the first partial derivative and the large-window condition are essential: the conclusion is a fixed-window structural theorem on the prescribed compact supporting sets.*

Use D-II.5.09. Fix nonempty compact $X \subset I, Y \subset J$, where $I \subset \mathbb{R}^{d_1}, J \subset \mathbb{R}^{d_2}$ are closed balls and $f : I \times J \rightarrow \mathbb{R}^q$ is C^1 on a neighborhood of their product. Suppose $\Lambda \geq 1, 0 < c \leq \Lambda$, and

$$\sup_{I \times J} \|Df\|_{\text{op}} \leq \Lambda, \quad s_{\min}(A_{x,y}) \geq c \quad ((x, y) \in X \times Y). \tag{21.1}$$

For every $0 < \varepsilon < 1$ there is $M = M(d_1, d_2, q, c, \Lambda, \varepsilon)$ such that for every integer $m \geq M$ there are $\delta > 0$, a nonnegative integer k_0 , and for each integer $k \geq k_0$ an integer $N_0(k)$, with this property. For $\alpha \in \mathcal{P}(X), \beta \in \mathcal{P}(Y)$ and integer $N \geq N_0(k)$, either

$$h_N(f_*(\alpha \times \beta)) > \int h_N((f_y)_* \alpha) d\beta(y) + \delta, \tag{II.5.36}$$

or, with probability at least $1 - \varepsilon$ over the independently mass-weighted raw level- k components $(\tilde{\alpha}, \tilde{\beta})$, there are level-only subspaces $V_0, \dots, V_N \leq \mathbb{R}^q$ and a measurable event $\mathcal{E}$ with

$$\mathbb{P}(\mathcal{E}) \geq 1 - \varepsilon, \quad \mathcal{E} \subseteq \left\{ (i, x, y) : \begin{array}{l} (A_{x,y})_* \tilde{\alpha}^{x,i} \text{ is } (V_i, \varepsilon, m)\text{-saturated,} \\ (B_{x,y})_* \tilde{\beta}^{y,i} \text{ is } (V_i, \varepsilon)\text{-concentrated} \end{array} \right\}. \tag{II.5.37}$$

Here i is uniform on $\{0, \dots, N\}$, $x \sim \tilde{\alpha}, y \sim \tilde{\beta}$, and all three samples are independent. The parameters are uniform over α, β on the fixed supports. Except for the displayed

dependence of M , they may depend on the map, balls, supporting sets and constants above.

The proof uses finite-window linearization, the linear-image transfer and decoupling results above, and the Euclidean inverse theorem G-II.5.18. No dimension inequality beyond the displayed structural conclusion is asserted.

Proof strategy

First localize both inputs to one coarse cell. On such a cell the two partial derivatives vary little, so the finite-window increment is controlled by a convolution of their frozen linear images. Apply the Euclidean inverse theorem, transfer its structural information back to the source components, and use the decoupling proposition to choose one subspace at each level. An exact coarse entropy decomposition and Markov's inequality then pass from the local assertion to arbitrary input measures.

Proof

A local inverse theorem

Lemma II.5.27.2 (Local inverse theorem for a nondegenerate smooth map). *Under the hypotheses of G-II.5.21, for the same sufficiently large m , there are $\tau > 0$, k_{loc} and $N_{\text{loc}}(k)$ such that, whenever α, β are each carried by one level- k input cell, $k \geq k_{\text{loc}}$, $N \geq N_{\text{loc}}(k)$, and*

$$\Delta_N^f(\alpha, \beta) \leq \tau, \quad (21.4)$$

there are level-only planes satisfying (II.5.37) with α, β themselves in place of their tilded root components. This local claim is proved in the next five subsections, before it is used in the global deduction.

Finite-window derivative comparisons

We record the elementary comparison needed when freezing the two derivative matrices inside a root pair. If $\phi \in \mathcal{P}([0, 1]^{d_1})$ and linear maps $C, C' : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^q$ satisfy $\sqrt{d_1}\|C - C'\|_{\text{op}} \leq 2^{-n}$, then

$$|\text{def}_{V,n}(C_*\phi) - \text{def}_{V,n}(C'_*\phi)| \leq C_P/n, \quad C_P = 2q \log_2 5, \quad (21.5)$$

for every subspace V and integer $n \geq 1$. Indeed, the coupling $u \mapsto (Cu, C'u)$ moves points by at most 2^{-n} . Given one of their dyadic cell labels, the other has at most 5^q possible values. The two label entropies therefore differ by at most $q \log 5$. The same argument applies after orthogonal projection to $V^\perp$. Subtracting the two entropy comparisons and dividing by $n \log 2$ proves (21.5). The dimension term in the deficit is unchanged.

Similarly, for linear maps $D, D' : \mathbb{R}^{d_2} \rightarrow \mathbb{R}^q$, $b > 0$ and $r \geq 0$, if $\psi \in \mathcal{P}([0, 1]^{d_2})$, $\sqrt{d_2}\|D - D'\|_{\text{op}} \leq r$, and $D_*\psi$ is (V, b) -concentrated, then

$$D'_*\psi \text{ is } (V, b+r)\text{-concentrated.} \quad (21.6)$$

The same-point coupling sends the good open b -tube into its open $(b+r)$ -tube and preserves its mass. Neither comparison claims entropy continuity at resolutions finer than the stipulated n .

Choose m so that [P-II.5.11-Lin](#) applies at output accuracy $\varepsilon/4$, and require

$$m \geq 4C_P/\varepsilon. \quad (21.7)$$

These conditions define the promised threshold M .

All parameters in their order

[P-II.5.11-Lin](#), at accuracy $\varepsilon/4$ and window m , supplies a, β_*, p_0 . Shrink a, β_* , if necessary, to be at most $1/4$; the resulting hypothesis is only stronger. The subscript distinguishes the failure allowance β_* from the second probability β . Set

$$u = \min\{a/4, \beta_*/8, 1/8\}, \quad p \geq \max\{p_0, M_{12\text{Lin}}(d_1, d_2, q, c, \Lambda, u), 2C_P/a\}. \quad (21.8)$$

Take p to be an integer. It exceeds m . For this p , [P-II.5.11-Lin](#) supplies a concentration tolerance t and threshold N_{dec} . We may shrink t to be at most $1/4$.

Apply [P-II.5.12-Lin](#)(3) with target saturation/failure u , target window p and concentration accuracy $t/4$. It supplies an input saturation/failure $s > 0$ and an initial long-window threshold L_0 . Shrink s to be at most $1/4$, and choose an integer

$$L \geq \max\{L_0, M_{\text{Euc}}(q, s)\}. \quad (21.9)$$

Here $M_{\text{Euc}}(q, s)$ denotes the corresponding window threshold in [G-II.5.18](#).

For this L , take the concentration accuracy $z > 0$ and terminal threshold N_{tr} from that transfer lemma. We may shrink z to be less than one. These choices are legal because s, L_0 precede L and z follows it.

Put $R_* = \max\{1, \Lambda\sqrt{d_1 + d_2}\}$. Apply [G-II.5.18](#) in dimension q , with support box $[-R_*, R_*]^q$, saturation/failure s , window L and concentration accuracy z . Obtain $\delta_E > 0, N_E$, and set

$$\kappa = \beta_*/4, \quad v = \delta_E/2, \quad \tau = \kappa v/32. \quad (21.10)$$

Let $C \geq 1$ be valid in [\(14C.14\)](#) and [\(14C.18\)](#). Choose an integer $w \geq \max\{N_{\text{tr}}, N_E, p, 1\}$ so large that

$$(C + q)/w \leq \kappa v/32. \quad (21.11)$$

Use the derivative modulus of [D-II.5.09](#). Choose $k_{\text{loc}} \geq k_{\text{lin}}(w)$ from the finite-window linearization lemma, large enough that, for every $k \geq k_{\text{loc}}$,

$$\begin{aligned} \sqrt{d_1} \omega_f(\sqrt{d_1 + d_2} 2^{-k}) &\leq 2^{-p}, \\ \sqrt{d_2} \omega_f(\sqrt{d_1 + d_2} 2^{-k}) &\leq \min\{t/4, \varepsilon/4\}. \end{aligned} \quad (21.12)$$

Finally, for each such k , choose $N_{\text{loc}}(k)$ at least N_{dec} , and so large that for every $N \geq N_{\text{loc}}(k)$,

$$N > \max\{k, w\}, \quad \frac{N}{N-k} \leq 2, \quad C \frac{w+k+1}{N} \leq \kappa v/32, \quad \frac{k+w+1}{N+1} \leq \beta_*/4. \quad (21.13)$$

All choices are uniform over the probabilities. No step enlarges a window in terms of a tolerance which had to be chosen before that same window.

Euclidean structure on most fine input pairs

Suppose the local hypotheses and (21.4) hold. Choose representatives $x_E \in E \cap X$ and $y_D \in D \cap Y$ in each positive-mass outer cell pair at every $k \leq i < N$. Set $z_{E,D} = (x_E, y_D)$, $A_i = A_{z_{E,D}}$, $B_i = B_{z_{E,D}}$, and use the finite-window increment $D_{i,w}(E, D)$ of (14C.13).

The Markov estimate (14C.18), with (21.10)–(21.13), gives

$$\mathbb{P}_{k \leq i < N} \{D_{i,w}(E, D) > v\} < \kappa. \quad (21.14)$$

The level is uniform and the outer cells have their product mass weights. To check the numerical margin, the numerator in (14C.18) is at most $7\kappa v/32$, and its denominator is at least v .

For a successful outer pair, put $\phi = \alpha^E$, $\psi = \beta^D$. The measures $\theta = (A_i)_* \phi$ and $\xi = (B_i)_* \psi$ are in $[-R_*, R_*]^q$ and satisfy

$$h_w(\theta * \xi) - h_w(\theta) \leq v < \delta_E. \quad (21.15)$$

G-II.5.18 supplies image-grid component planes for θ, ξ at levels $0, \dots, w$, with joint failure at most s , saturation tolerance s at window L , and concentration tolerance z . Now apply P-II.5.12-Lin(3) to the fixed maps A_i, B_i and the source probabilities ϕ, ψ . Its hypotheses hold uniformly by (21.1), and w exceeds both required terminal thresholds. It yields planes $U_j(i, E, D)$ such that

$$\mathbb{P}_{\substack{0 \leq j \leq w \\ a' \sim \phi, b' \sim \psi}} \left\{ \begin{array}{l} (A_i)_* \phi^{a',j} \text{ is } (U_j, u, p)\text{-saturated,} \\ (B_i)_* \psi^{b',j} \text{ is } (U_j, t/4)\text{-concentrated} \end{array} \right\} \geq 1 - u. \quad (21.16)$$

The symbols a', b' here are sampled points in the canonical source cubes, not the tolerance a or the original points in X, Y . The conclusion uses the original source component partitions. It does not identify them with components of θ, ξ .

Fixed derivatives on one root pair

Choose $x_0 \in X, y_0 \in Y$ within the two carrying root cells and put $A_0 = A_{x_0, y_0}$, $B_0 = B_{x_0, y_0}$. For every point of $X \times Y$ in this root pair, and every representative $z_{E,D}$ used above, the distance to (x_0, y_0) is at most $\sqrt{d_1 + d_2} 2^{-k}$. Since a partial derivative difference is bounded by the full derivative difference, (21.12) applies to $A_i - A_0, B_i - B_0$ and also to $A_{x,y} - A_0, B_{x,y} - B_0$.

Under component resampling, the laws in (21.16) become $(A_i)_* \alpha^{x,i+j}$ and $(B_i)_* \beta^{y,i+j}$. Equations (21.5)–(21.6) therefore transfer their properties to

$$\begin{aligned} (A_0)_* \alpha^{x,i+j} &\text{ is } (U_j, a, p)\text{-saturated,} \\ (B_0)_* \beta^{y,i+j} &\text{ is } (U_j, t)\text{-concentrated.} \end{aligned} \quad (21.17)$$

Indeed the new saturation error is at most $u + C_P/p \leq 3a/4 < a$, and the concentration error is at most $t/2 < t$. This certification is constant on each fine input cell pair: the derivatives are now fixed and the canonical component laws depend only on those cells.

Choose a witnessing tube for every successful component test in (21.16), using the countable-cell witnesses in P-II.5.12-Lin's proof. Its coupling in (21.6) supplies a witnessing tube for (21.17). Thus this transfer gives certified Borel cell events, not an uncountable search over affine centers.

Resampling, selection and independent decoupling

Sample i uniformly from $\{k, \dots, N-1\}$, then the outer pair with its product mass weights, then j uniformly from $\{0, \dots, w\}$ and the two independent source descendants. The preceding construction succeeds with probability at least $1 - \kappa - u$. For each fixed i, j , resampling recovers $\alpha \times \beta$ at the fine level $r = i + j$.

For each positive-mass fine cell pair at level r , only finitely many outer levels can give this pair and this r ; for each such level the outer ancestor pair is unique. Choose the first candidate plane that certifies (21.17) on that cell pair, if any, and otherwise the zero plane. This defines a Borel countable-cell field $V(r, x, y)$ whose success event contains every success of the preceding sampling.

For fixed j , the distribution of r is uniform on $\{k + j, \dots, N - 1 + j\}$. Its total variation distance from the uniform law on $\{0, \dots, N\}$ is at most

$$\frac{k + 1 + \max(j - 1, 0)}{N + 1} \leq \frac{k + w + 1}{N + 1}. \quad (21.18)$$

To verify this count, the shorter interval has length $N - k$ and its overlap with $\{0, \dots, N\}$ has length $N - k - \max(j - 1, 0) > 0$ by (21.13). The sum of the minimum of the two uniform masses is the overlap length divided by $N + 1$; subtract it from one. Averaging in j preserves (21.18).

Consequently, in the original sampling over $0 \leq r \leq N$, the field (21.17) has joint failure at most

$$\kappa + u + \frac{k + w + 1}{N + 1} \leq 5\beta_*/8 < \beta_*. \quad (21.19)$$

At each fixed level the laws $(A_0)_* \alpha^{x,r}$ and $(B_0)_* \beta^{y,r}$ are independent. The maps satisfy the uniform bounds in P-II.5.11-Lin, including the least stretch of A_0 . Its separated-input form, with the choices in (21.8), now supplies level-only planes $W_0, \dots, W_N$ and a measurable event of probability at least $1 - \varepsilon/4$ on which

$$\begin{aligned} (A_0)_* \alpha^{x,r} &\text{ is } (W_r, \varepsilon/4, m)\text{-saturated,} \\ (B_0)_* \beta^{y,r} &\text{ is } (W_r, \varepsilon/4)\text{-concentrated.} \end{aligned} \quad (21.20)$$

Return to the actual derivatives at (x, y) . Since $p > m$, (21.12) also permits (21.5) at window m . The new saturation error is at most $\varepsilon/4 + C_P/m \leq \varepsilon/2$ by (21.7). Equation (21.6)

makes the concentration error at most $\varepsilon/2$. These comparisons hold for every sampled (x, y) in the root supports, not just for representatives. They preserve the good event, so (II.5.37) follows for the local pair with failure at most $\varepsilon/4 < \varepsilon$. This proves the local statement (21.4).

Exact coarse-to-global deduction

Choose

$$\delta = \varepsilon\tau/4, \quad k_0 = k_{\text{loc}}. \quad (21.21)$$

Fix $R_X \geq 1$ with $X \subset [-R_X, R_X]^{d_1}$. The number of its occupied level- k cells is at most $(2^{k+1}R_X + 2)^{d_1}$, so uniformly in $\alpha \in \mathcal{P}(X)$,

$$H_k(\alpha) \leq d_1 \log(2^{k+1}R_X + 2). \quad (21.22)$$

For each $k \geq k_0$, enlarge $N_0(k)$ above $N_{\text{loc}}(k)$ so that for every $N \geq N_0(k)$,

$$\frac{d_1 \log(2^{k+1}R_X + 2)}{N \log 2} \leq \varepsilon\tau/4. \quad (21.23)$$

If (II.5.36) fails, the exact nonlinear increment satisfies $\Delta_N^f(\alpha, \beta) \leq \delta$. The coarse Markov estimate (14C.11) now gives

$$\mathbb{P}\{\Delta_N^f(\tilde{\alpha}, \tilde{\beta}) > \tau\} \leq \frac{\delta + H_k(\alpha)/(N \log 2)}{\tau} \leq \varepsilon/2. \quad (21.24)$$

Every remaining root pair meets the local hypotheses with the same constants. Its local conclusion is (II.5.37), and the root-pair probability is at least $1 - \varepsilon/2 \geq 1 - \varepsilon$. There are only countably many root pairs, so their planes and certified events may be fixed pair by pair with no measurability issue. This proves G-II.5.21 under (21.1).

□

II.5.28 Matrix actions and matrix multiplication

For matrix action the first variable is the spatial point; for matrix multiplication it is the first matrix. This determines which partial derivative carries saturation and which carries concentration. The order of matrix multiplication is never exchanged.

Definition II.5.28.1 (II.5.10: Matrix actions, multiplication laws and invariant subspaces). Let $M_d(\mathbb{R})$ be the vector space of real d -by- d matrices. Identify it with $\mathbb{R}^{d^2}$ using its entries and the Frobenius norm of D-II.5.04. Write $\text{GL}_d(\mathbb{R})$ for the invertible real d -by- d matrices. A **matrix group** K here is a subgroup of $\text{GL}_d(\mathbb{R})$: it contains the identity I_d and satisfies $AB, A^{-1} \in K$ for all $A, B \in K$. Give it the inherited topology. All measures used below are carried by specified compact subsets of this ambient matrix space which lie in K .

For probabilities α, β on such compact matrix sets and a compactly supported proba-

bility μ on $\mathbb{R}^d$, define the **matrix action law** and the **matrix multiplication convolution** by

$$\alpha \cdot \mu = ((A, x) \mapsto Ax)_*(\alpha \times \mu), \quad \alpha *_K \beta = ((A, B) \mapsto AB)_*(\alpha \times \beta).$$

The order in $*_K$ is exactly the displayed matrix order; it is not the additive convolution on $\mathbb{R}^{d^2}$. The product support lies in a compact subset of K . Raw and canonical matrix components use the entrywise dyadic grid of $\mathbb{R}^{d^2}$ and D-II.2.01. A canonical component is a probability on the ambient vector space $M_d(\mathbb{R})$; it need not be carried by K , or even by invertible matrices.

For $x \in \mathbb{R}^d$ and $A, B \in M_d(\mathbb{R})$, define the linear maps

$$E_x(H) = Hx, \quad R_B(H) = HB, \quad L_A(H) = AH.$$

Thus $E_x : M_d(\mathbb{R}) \rightarrow \mathbb{R}^d$, while R_B, L_A act on $M_d(\mathbb{R})$. Their use on canonical matrix components is meaningful without any invertibility of the sampled component matrices.

For a set $K \subset GL_d(\mathbb{R})$, a linear subspace $V \leq \mathbb{R}^d$ is **common K -invariant** if $TV \subset V$ for every $T \in K$. The action of K is **reducible** if it has a common invariant subspace with $\{0\} < V < \mathbb{R}^d$, and **irreducible** otherwise.

For a family of similarities Φ , its **linearization** is the set of orthogonal linear parts $\{U_\phi : \phi \in \Phi\}$. The dimension-drop trichotomy involving this obstruction is not a theorem of II.5; it is a later II.7 result.

Theorem II.5.28.2 (G-II.5.22 – Structural matrix action and multiplication corollaries). *Use D-II.5.10, including the order $\alpha *_K \beta = ((A, B) \mapsto AB)_*(\alpha \times \beta)$. Fix a subgroup $K \leq GL_d(\mathbb{R})$ and nonempty compact sets $C_K \subset K, C_x \subset \mathbb{R}^d$.*

For every $0 < \varepsilon < 1$ there is $M = M(d, C_K, C_x, \varepsilon)$ such that for every integer $m \geq M$ there are $\delta > 0$, a nonnegative integer k_0 , and for each integer $k \geq k_0$ an integer $N_0(k)$, for which both assertions hold whenever $N \geq N_0(k)$ is an integer.

1. For $\alpha \in \mathcal{P}(C_K), \mu \in \mathcal{P}(C_x)$, either

$$h_N(\alpha \cdot \mu) > h_N(\mu) + \delta, \tag{II.5.38}$$

or, with probability at least $1 - \varepsilon$ over independent raw level- k components $(\tilde{\alpha}, \tilde{\mu})$, there are level-only $V_0, \dots, V_N \leq \mathbb{R}^d$ and a measurable event $\mathcal{E}_1$ with

$$\mathbb{P}(\mathcal{E}_1) \geq 1 - \varepsilon, \quad \mathcal{E}_1 \subseteq \left\{ (i, A, x) : \begin{array}{l} A_* \tilde{\mu}^{x,i} \text{ is } (V_i, \varepsilon, m)\text{-saturated,} \\ (E_x)_* \tilde{\alpha}^{A,i} \text{ is } (V_i, \varepsilon)\text{-concentrated} \end{array} \right\}. \tag{II.5.39}$$

Here $A \sim \tilde{\alpha}, x \sim \tilde{\mu}$. 2. For $\alpha, \beta \in \mathcal{P}(C_K)$, either

$$h_N(\alpha *_K \beta) > h_N(\alpha) + \delta, \tag{II.5.40}$$

or, with probability at least $1 - \varepsilon$ over independent raw level- k components $(\tilde{\alpha}, \tilde{\beta})$, there are level-only $W_0, \dots, W_N \leq M_d(\mathbb{R})$ and a measurable event $\mathcal{E}_2$ with

$$\mathbb{P}(\mathcal{E}_2) \geq 1 - \varepsilon, \quad \mathcal{E}_2 \subseteq \left\{ (i, A, B) : \begin{array}{l} (R_B)_* \tilde{\alpha}^{A,i} \text{ is } (W_i, \varepsilon, m)\text{-saturated,} \\ (L_A)_* \tilde{\beta}^{B,i} \text{ is } (W_i, \varepsilon)\text{-concentrated} \end{array} \right\}. \quad (\text{II.5.41})$$

Here $A \sim \tilde{\alpha}, B \sim \tilde{\beta}$.

In both conclusions i is uniform on $\{0, \dots, N\}$ and independent of the two independent point samples. Canonical matrix components are ambient vector-space probabilities; no claim places them inside K . All parameters are uniform over the probabilities on the fixed compact sets.

These assertions follow from [G-II.5.21](#) after verifying uniform nondegeneracy of the relevant partial derivatives and comparing each linear slice entropy with the entropy of its input.

Proof strategy

For each polynomial map, compute its two partial derivatives and verify that the first is uniformly injective on the fixed compact support. The smooth-map theorem then gives the stated structural alternative with a slice-entropy baseline. Uniform bi-Lipschitz bounds replace that baseline by the entropy of the first input, at a cost tending to zero as the observation length grows.

Uniform derivative hypotheses

As proved in [L-II.5.14-ND\(5\)](#), set

$$c_K = \left(\max_{A \in C_K} \|A^{-1}\|_{\text{op}} \right)^{-1} > 0, \quad L_K = \max_{A \in C_K} \|A\|_{\text{op}} < \infty. \quad (22.5)$$

Take closed origin-centered balls of radii $R_x, R_K \geq 1$ containing C_x, C_K , with the Frobenius norm in matrix space. For $f_1(x, A) = Ax$ on the spatial-by-matrix product,

$$D_x f_1(x, A) = A, \quad D_A f_1(x, A) = E_x.$$

The first derivative has least stretch at least c_K on $C_x \times C_K$. On the whole product of balls, $\|Df_1\|_{\text{op}} \leq R_K + R_x$. For $f_2(A, B) = AB$ on two matrix balls,

$$D_A f_2(A, B) = R_B, \quad D_B f_2(A, B) = L_A.$$

On $C_K \times C_K$ the first derivative has least stretch at least c_K in Frobenius norm; on the whole product $\|Df_2\|_{\text{op}} \leq 2R_K$. Both maps are polynomial on the ambient vector spaces. Thus [G-II.5.21](#) applies with

map	(d_1, d_2, q)	Λ
f_1	(d, d^2, d)	$1 + R_K + R_x$
f_2	(d^2, d^2, d^2)	$1 + 2R_K$

(22.6)

It is essential that the positive least-stretch bound is required only on the compact supports. Containing balls may contain singular matrices.

Replacing the slice baseline

The slice maps are globally linear and bi-Lipschitz with uniform constants c_K, L_K : they are $x \mapsto Ax$ for f_1 and $A \mapsto AB$ for f_2 . The grid comparison (14C.3) therefore gives constants $C_1, C_2 < \infty$, depending only on d, c_K, L_K , such that

$$\begin{aligned} \left| \int h_N(A_* \mu) d\alpha(A) - h_N(\mu) \right| &\leq C_1/(N \log 2), \\ \left| \int h_N((R_B)_* \alpha) d\beta(B) - h_N(\alpha) \right| &\leq C_2/(N \log 2). \end{aligned} \quad (22.7)$$

These estimates hold without conditioning the first input into a small cell; no nonlinear Taylor approximation is being used in this step.

Choose M to dominate the two window thresholds of G-II.5.21 given by (22.6). For $m \geq M$, denote its two growth constants by $\delta_1, \delta_2 > 0$, root thresholds by k_1, k_2 , and terminal thresholds by $N_1(k), N_2(k)$. Take

$$\delta = \frac{1}{2} \min\{\delta_1, \delta_2\}, \quad k_0 = \max\{k_1, k_2\}, \quad (22.8)$$

and choose $N_0(k)$ at least both terminal thresholds and so large that

$$C_j/(N \log 2) \leq \delta_j/2 \quad (j = 1, 2, N \geq N_0(k)). \quad (22.9)$$

If (II.5.38) fails, the first estimate in (22.7) shows

$$\Delta_N^{f_1}(\mu, \alpha) \leq h_N(\alpha \cdot \mu) - h_N(\mu) + C_1/(N \log 2) \leq \delta_1.$$

Theorem G-II.5.21 gives (II.5.39), because its first and second partial derivatives are exactly A and E_x . Permuting the order in which the two independent root components are listed does not change their product law.

If (II.5.40) fails, the second estimate in (22.7) gives

$$\Delta_N^{f_2}(\alpha, \beta) \leq h_N(\alpha *_K \beta) - h_N(\alpha) + C_2/(N \log 2) \leq \delta_2.$$

Now G-II.5.21 gives (II.5.41), with first derivative R_B and second derivative L_A . Neither law in the conclusion is interpreted as a convolution of canonical components inside K : the displayed derivatives act on the full ambient matrix space, as defined in D-II.5.10.

This proves both structural matrix corollaries, uniformly over their fixed compact supports. $\square$

Chapter II.6

Dimension Drop on the Line

Geometric intuition

The line dimension-drop theorem converts an asymptotic loss of dimension into super-exponentially close cylinders. Stopping words, natural-scale entropy, and the line inverse theorem are assembled carefully for signed and nonhomogeneous similarities, followed by the analytic-family consequences.

A self-similar measure can fall below its entropy–Lyapunov dimension only if many symbolic words become geometrically indistinguishable. The purpose of this chapter is to make *indistinguishable* quantitative: a strict dimension deficit forces equal-contraction cylinders to approach one another faster than every exponential rate. We first turn exact dimensionality into uniform information about dyadic components, then show that finite word laws flatten below their natural scale, and finally convert the lost entropy into a collision. The homogeneous and unequal-contraction arguments share this plan but require different bookkeeping.

Throughout, logarithms are natural. Raw components are normalized restrictions, superscript components are their canonical images in $[0, 1)$, and every sum over dyadic cells omits zero-mass components.

II.6.1 Affine coordinates and local entropy

A fixed change of affine coordinates alters a dyadic entropy by only a bounded amount after the appropriate shift of scale. The grid is not affine invariant: reflection reverses half-open cells, and a non-dyadic scale does not align them. The bounded-intersection proof is what lets those geometric discrepancies disappear after division by the observing length. Positive dyadic dilation is the special case with an exact component identity.

The level-zero entropy has a different role. A measure in the unit interval has only one initial cell, but a general bounded measure may already have several. The local-average calculation must start from $H_N - H_0$. Although H_0/N vanishes in the limit, retaining it prevents a unit-support theorem from being used outside its hypotheses.

Definition II.6.1.1 (II.6.01: Signed line systems, words and natural dimensions). A finite line similarity system is a labelled family $\Phi = (\varphi_i)_{i \in \Lambda}$, where Λ is a nonempty finite alphabet and

$$\varphi_i(x) = r_i x + a_i, \quad a_i \in \mathbb{R}, \quad 0 < |r_i| < 1.$$

Labels need not represent distinct maps. Put $r_{\min} = \min_i |r_i|$, $r_{\max} = \max_i |r_i|$. For a finite word $u = u_1 \cdots u_n$, its length is $|u| = n$, and

$$\varphi_u = \varphi_{u_1} \circ \cdots \circ \varphi_{u_n}, \quad r_u = \prod_{j=1}^n r_{u_j}, \quad a_u = \varphi_u(0) = \sum_{j=1}^n a_{u_j} \prod_{\ell < j} r_{u_\ell}.$$

The empty word $\emptyset$ has length zero and $\varphi_\emptyset = \text{id}$, $r_\emptyset = 1$, $a_\emptyset = 0$. Concatenation uv puts u before v . A nonempty word's immediate predecessor u^- deletes its last symbol. The attractor K_Φ and coding map $\pi_\Phi : \Lambda^\mathbb{N} \rightarrow K_\Phi$ are those proved to exist in I.7.

A strict probability vector means $p_i > 0$ and $\sum_i p_i = 1$. Its word weights are $p_u = \prod_{j=1}^{|u|} p_{u_j}$, with $p_\emptyset = 1$. Write $\mu_{\Phi, p}$, or μ , for the invariant Bernoulli probability of I.7. For a vector allowing zero entries, its active alphabet is $\Lambda_p = \{i : p_i > 0\}$; restrictions to this alphabet are stated explicitly when used.

Define

$$h(p) = - \sum_i p_i \log p_i, \quad \chi(\Phi, p) = - \sum_i p_i \log |r_i| > 0, \quad s(\Phi, p) = \frac{h(p)}{\chi(\Phi, p)}.$$

Here $0 \log 0 = 0$. The natural measure bound is $\min\{1, s(\Phi, p)\}$. The set similarity dimension s_Φ is the unique nonnegative solution of $\sum_i |r_i|^{s_\Phi} = 1$, including $s_\Phi = 0$ for a one-map system. The natural set bound is $\min\{1, s_\Phi\}$. Existence, uniqueness and the upper bounds are earlier results, not new definitional claims.

For later coordinate changes put $A_{c,b}(x) = cx + b$, where $c \neq 0$, and $S_c = A_{c,0}$. A negative c includes reflection. The conjugate system has maps $A_{c,b} \circ \varphi_i \circ A_{c,b}^{-1}$.

A negative contraction reverses the orientation of half-open dyadic cells, whereas a positive dyadic dilation preserves the component coordinates exactly. The next lemma records both facts and keeps the level-zero entropy term that is invisible only for unit-supported measures.

Lemma II.6.1.2 (L-II.6.01 — Signed affine entropy comparison). *Let θ be compactly supported, $c \neq 0$, $b \in \mathbb{R}$, $j \in \mathbb{Z}$, and $v = \lfloor j + \log_2 |c| \rfloor$. Then*

$$|H_j((A_{c,b})_* \theta) - H_v(\theta)| \leq \log 4. \quad (\text{II.6.01})$$

Consequently, for fixed $0 < c_- \leq c_+ < \infty$, the difference $|H_j((A_{c,b})_ \theta) - H_j(\theta)|$ is bounded uniformly in j, b and $c_- \leq |c| \leq c_+$, by a constant depending only on c_-, c_+ . For integer a , dyadic dilation gives the exact component identity, outside the usual null sets,*

$$((S_{2^{-a}})_*\theta)^{2^{-a}x,i} = \theta^{x,i-a} \quad (i \in \mathbb{Z}).$$

For any compactly supported θ and integers $N, m \geq 1$, the bounded-support local-average identity has the quantitative form

$$\left| \frac{H_N(\theta) - H_0(\theta)}{N \log 2} - \frac{1}{N} \sum_{i=0}^{N-1} \int h_m(\theta^{x,i}) d\theta(x) \right| \leq \frac{m-1}{N}. \quad (\text{II.6.01a})$$

Thus the local-average formula remains valid outside the unit interval, provided the level-zero entropy term is retained.

For comparison, see Hochman [Hoc14, Lemma 3.2, pp. 16–17]. The proof below also treats reflection, half-open endpoints, and the explicit comparison error.

Proof. Let $\mathcal{A} = A_{c,b}^{-1} \mathcal{D}_j$ be the pullback partition. Its cells are intervals of length

$$\lambda = 2^{-j}/|c|, \quad 2^{-v-1} < \lambda \leq 2^{-v}, \quad v = \lfloor j + \log_2 |c| \rfloor. \quad (\text{II.6.N01a})$$

If $c < 0$, their half-open orientation is reversed. An interval of length at most twice a grid cell length meets at most four cells of that grid, even if one counts an endpoint intersection. Thus each cell of $\mathcal{A}$ meets at most four cells of $\mathcal{D}_v$, and conversely. L-II.1.06 gives

$$|H_\theta(\mathcal{A}) - H_\theta(\mathcal{D}_v)| \leq \log 4.$$

The first entropy is $H_j((A_{c,b})_*\theta)$, proving (II.6.01). Compact support makes all these entropies finite. Since

$$|v - j| \leq 1 + \max\{|\log_2 c_-|, |\log_2 c_+|\} \quad (c_- \leq |c| \leq c_+),$$

the scale comparison T-II.1.07 proves the uniform same-level assertion. There is no dependence on the translation b .

For an integer a , let $\zeta = (S_{2^{-a}})_*\theta$. If $Q = [z2^{-(i-a)}, (z+1)2^{-(i-a)})$, then

$$S_{2^{-a}}Q = [z2^{-i}, (z+1)2^{-i}), \quad (2^i S_{2^{-a}}x - z) = 2^{i-a}x - z.$$

The masses, raw restrictions and canonical coordinate maps therefore give $\zeta^{2^{-a}x,i} = \theta^{x,i-a}$ on each positive-mass cell. This is an exact identity for positive dyadic dilation, not an assertion of exact reflection covariance.

For the final assertion put

$$a_i = H_{i+1}(\theta) - H_i(\theta) \in [0, \log 2] \quad (i \geq 0).$$

By L-II.2.01,

$$\int H_m(\theta^{x,i}) d\theta(x) = H_{i+m}(\theta) - H_i(\theta) = \sum_{\ell=0}^{m-1} a_{i+\ell}.$$

For each $\ell \geq 0$, the difference between $\sum_{i=0}^{N-1} a_{i+\ell}$ and $\sum_{i=0}^{N-1} a_i$ equals

$$\sum_{i=N}^{N+\ell-1} a_i - \sum_{i=0}^{\ell-1} a_i.$$

Its absolute value is at most $\ell \log 2$; the identity remains true when the two endpoint sums overlap. Average over $0 \leq \ell < m$ and divide by $N \log 2$. This gives an error at most $(m-1)/(2N)$, and in particular the claimed $(m-1)/N$ in (II.6.01a). The central sum is $H_N(\theta) - H_0(\theta)$, so its level-zero term cannot be discarded for a general bounded support. $\square$

Thus a fixed affine normalization costs only $O(1)$ unnormalized entropy and disappears after division by the observing length. The term H_0 , however, must be retained until that normalization has been made.

II.6.2 From exact dimension to typical components

Exact dimensionality describes almost every point as the radius tends to zero. The dimension-drop argument will instead need many finite components with controlled entropy at a common window length.

L-II.6.02 first connects pointwise dimension to global finite-scale entropy. The stronger uniform component statement then uses self-similarity, not exact dimensionality alone. The distinction matters: the proofs do not assert that every component has the same dimension, or that every point and scale is typical.

Lemma II.6.2.1 (L-II.6.02 — Exact dimension, entropy dimension and atoms). *Every compactly supported exact-dimensional probability of dimension α satisfies $h_N(\theta) \rightarrow \alpha$. For each fixed $0 < c_- \leq c_+ < \infty$, this convergence is uniform for $(A_{c,b})_*\theta$, over $c_- \leq |c| \leq c_+$ and all b .*

For a strict self-similar probability $\mu_{\Phi,p}$, either it has no atoms or $\mu = \delta_z$, every map fixes z , and $K_{\Phi} = \{z\}$. In the non-atomic case,

$$\lim_{\delta \downarrow 0} \sup_{x \in \mathbb{R}} \mu([x - \delta, x + \delta]) = 0.$$

For comparison, see Hochman [Hoc14, p. 31 and Lemma 5.3]. Exact dimensionality follows from T-I.9.08, while the entropy passage uses T-II.2.02, T-II.2.04 and T-II.2.08.

Proof. First verify the use of T-I.9.08 for the present signed systems. The Bernoulli law $m = p^{\mathbb{N}}$ is shift invariant. If an event E is invariant modulo null sets, then $E = \sigma^{-n}E$ modulo null sets for every n . The event $\sigma^{-n}E$ depends only on coordinates after n ; it is independent of every cylinder in the first n coordinates. Consequently E is independent of every finite cylinder. The monotone-class theorem extends that independence to the entire product sigma-algebra. Applying it to E itself yields

$$m(E) = m(E)^2, \quad m(E) \in \{0, 1\}.$$

This proves ergodicity. Every map $x \mapsto r_i x + a_i$ is a smooth diffeomorphism with nonzero conformal derivative of magnitude $|r_i|$. For a sufficiently large closed interval $X = [-M, M]$ and a slightly larger open interval U , all maps preserve X and map U inside U : it suffices that $M > \max_i |a_i|/(1 - r_{\max})$. The common Lipschitz constant is $r_{\max} < 1$. Thus the hypotheses of T-I.9.08 hold, including when some $r_i < 0$. It follows that the invariant probability is exact dimensional. C-I.4.08 identifies its four measure dimensions with that local dimension.

Now let θ be any compactly supported exact-dimensional probability, not necessarily self-similar. Choose an affine map A carrying its support into $[0, 1]$, and write $\theta' = A_*\theta$. If $A(x) = cx + b$, then

$$\theta'(B(Ax, r)) = \theta(B(x, r/|c|)),$$

so θ' has the same exact local dimension α . Fix any integer $m \geq 1$. T-II.2.04 and T-II.2.08 imply, for θ' -almost every x ,

$$\frac{1}{N} \sum_{i=0}^{N-1} h_m((\theta')^{x,i}) \rightarrow \alpha. \quad (\text{II.6.N02a})$$

Each summand belongs to $[0, 1]$, since there are 2^m dyadic descendants of a line component. Dominated convergence therefore gives the same limit after integration. T-II.2.02 then gives $h_N(\theta') \rightarrow \alpha$. In particular $0 \leq \alpha \leq 1$. L-II.6.01 transfers the conclusion to θ , and its same-level comparison gives

$$\sup_{\substack{c_- \leq |c| \leq c_+ \\ b \in \mathbb{R}}} |h_N((A_{c,b})_*\theta) - h_N(\theta)| \leq \frac{C(c_-, c_+)}{N \log 2}.$$

This proves the claimed uniform convergence.

Suppose next that the strict self-similar probability μ has an atom. Put $M_0 = \sup_x \mu\{x\} > 0$. There are only finitely many atoms of mass at least $M_0/2$, so this supremum is attained. The set $A = \{x : \mu\{x\} = M_0\}$ is finite and nonempty. For each $x \in A$, invariance gives

$$M_0 = \sum_i p_i \mu\{\varphi_i^{-1}(x)\}.$$

Every summand mass is at most M_0 , and every $p_i > 0$. Hence $\varphi_i^{-1}(x) \in A$ for all i . The inverse map is injective, so $\varphi_i^{-1}(A) \subset A$ has the same finite cardinality as A and therefore equals A . Thus $\varphi_i(A) = A$, and

$$\text{diam } A = \text{diam } \varphi_i(A) = |r_i| \text{diam } A.$$

It follows that $A = \{z\}$, and every map fixes z . The point mass δ_z is invariant; uniqueness in T-I.7.04 gives $\mu = \delta_z$. Attractor uniqueness T-I.7.02 gives $K_\Phi = \{z\}$.

Finally, assume that μ has no atoms and define

$$q_\mu(\delta) = \sup_x \mu([x - \delta, x + \delta]) \quad (\delta > 0).$$

If $q_\mu(\delta) \not\rightarrow 0$, there exist $\epsilon_0 > 0$, $\delta_j \downarrow 0$ and centers x_j for which these interval masses are at least ϵ_0 . The intervals meet the compact support, so the centers are bounded. Pass to

a subsequence $x_j \rightarrow x$. For every $r > 0$, all sufficiently late intervals lie in $[x - r, x + r]$, whose mass is consequently at least ϵ_0 . Continuity from above as $r \downarrow 0$ gives $\mu\{x\} \geq \epsilon_0$, a contradiction. $\square$

Definition II.6.2.2 (II.6.04: Stopping words and anchor-cell mixtures). For $0 < h < 1$, the stopping family at absolute contraction scale h is

$$\Lambda(h) = \{u \neq \emptyset : |r_u| \leq h < |r_{u-}|\}.$$

Set $\Lambda(1) = \{\emptyset\}$. Finiteness, complete symbolic coverage and the corresponding measure decomposition are proved in L-II.6.03.

Fix an anchor $b \in K_\Phi$, a dyadic level $k \geq 0$, a number $0 < \rho < 1$, and $h = \rho 2^{-k}$. For a level- k cell Q , define its anchor weight and whole-cylinder mixture by

$$w_Q = \sum_{\substack{u \in \Lambda(h) \\ \varphi_u(b) \in Q}} p_u, \quad \eta_Q = \begin{cases} \frac{1}{w_Q} \sum_{\substack{u \in \Lambda(h) \\ \varphi_u(b) \in Q}} p_u (\varphi_u)_* \mu, & w_Q > 0, \\ \delta_b, & w_Q = 0. \end{cases}$$

The fallback is a convention only. Any successful approximation event will require $w_Q > 0$. The whole-cylinder mixture is not the raw restriction μ_Q : individual cylinders can cross the boundary of Q , and w_Q need not equal $\mu(Q)$.

Total variation of probabilities is $\|\theta - \theta'\|_{TV} = \sup_B |\theta(B) - \theta'(B)|$, the half- L^1 convention of II.4. For a conjugate system the same definition uses its attractor, invariant measure, anchor and word maps.

With unequal contractions, words of one symbolic length can have very different geometric sizes. Stopping at a prescribed contraction scale puts all selected cylinders in a fixed multiplicative size range. That makes them suitable for comparison with one dyadic grid level.

Assigning each whole cylinder to the cell of an attractor anchor does not cut the cylinder at the cell boundary. This preserves its form as an affine copy of the original measure. The cost is that the resulting mixture need not equal the actual cell restriction.

L-II.6.03 quantifies exactly that cost. Couple a point in the cylinder with its anchor. A cell-label disagreement is possible only near a grid boundary. A coarser stopping decomposition converts the total mass of all those narrow boundary bands into the small-interval modulus of the original non-atomic measure. The subsequent conditional-law calculation is essential: closeness of the labels alone does not automatically compare two differently normalized restrictions.

Lemma II.6.2.3 (L-II.6.03 – Stopping decompositions and boundary approximation). For $0 < h < 1$, $\Lambda(h)$ is finite, each infinite word has exactly one prefix in it, and

$$r_{\min} h < |r_u| \leq h \quad (u \in \Lambda(h)), \quad \sum_{u \in \Lambda(h)} p_u = 1, \quad \mu = \sum_{u \in \Lambda(h)} p_u (\varphi_u)_* \mu. \quad (\text{II.6.02})$$

The last two identities also hold for $h = 1$.

For every $0 < \varepsilon < 1$ there is $\rho_0 > 0$ such that for all $0 < \rho \leq \rho_0, k \geq 0$, and $b \in K_\Phi$, the mixtures of [D-II.6.04](#) satisfy

$$\mu\{x : w_{\mathcal{D}_k(x)} > 0, \|\mu_{x,k} - \eta_{\mathcal{D}_k(x)}\|_{\text{TV}} \leq \varepsilon\} \geq 1 - \varepsilon. \quad (\text{II.6.03})$$

For $0 < c_- \leq c_+ < \infty, \rho_0$ can be chosen uniformly over all conjugate systems by A_{c,b_0} , $c_- \leq |c| \leq c_+, b_0 \in \mathbb{R}$, and all anchors in their attractors.

For comparison, see Hochman [[Hoc14](#), Lemma 5.3 and the stopping-word paragraph], pp. 31–32. The proof below includes the normalization of conditional mixtures, zero weights, signed products, and the uniform affine version.

Proof. We first stop cylinders at one geometric scale, then assign whole cylinders to anchor cells. A boundary coupling compares this anchor mixture with the genuine dyadic component, and finite-alphabet entropy continuity transfers the comparison.

Fix $0 < h < 1$. Along each infinite word the products $|r_{\omega_1 \dots \omega_n}|$ strictly decrease to zero, and are at most $r_{\max}^n$. Therefore each word has a unique first prefix crossing h , whose length is bounded by any integer N with $r_{\max}^N \leq h$. The stopping set is finite and its symbolic cylinders partition the full shift. The crossing inequalities give

$$|r_u| \leq h, \quad |r_u| = |r_{u-}| |r_{u_{|u|}}| > r_{\min} h.$$

Its cylinders have masses p_u , so their masses sum to one. Conditioning a Bernoulli sequence on such a cylinder leaves its tail with the original Bernoulli law; T-I.7.03 then gives $\mu = \sum_{u \in \Lambda(h)} p_u (\varphi_u)_* \mu$. At $h = 1$, the empty word makes the two identities immediate.

For the approximation, first suppose μ is non-atomic. Put $D = \text{diam } K_\Phi > 0$ and $t = 2^{-k}$. Sample $U \in \Lambda(\rho t)$ with law p_u and, independently, $X \sim \mu$. For a fixed $b \in K_\Phi$, let

$$Y = \varphi_U(X), \quad Z = \varphi_U(b).$$

The stopping decomposition gives $Y \sim \mu$, and $|Y - Z| \leq D\rho t$. Write $A = \mathcal{D}_k(Y)$, $B = \mathcal{D}_k(Z)$ for their cell labels, and $d = \mathbb{P}(A \neq B)$. A disagreement forces

$$Y \in \mathcal{B}_{k,\rho} := \bigcup_{\ell \in \mathbb{Z}} [\ell t - D\rho t, \ell t + D\rho t]. \quad (\text{II.6.N03a})$$

This includes a crossing at a dyadic endpoint.

Use the coarser stopping decomposition at scale t . When $k = 0$ this is the empty-word decomposition. In both cases each coarse word v satisfies

$$r_{\min} t < |r_v| \leq t, \quad \text{diam } \varphi_v(K_\Phi) \leq Dt.$$

For $0 < \rho < 1$, at most $C_D = \lceil 3D \rceil + 3$ of the intervals in [\(II.6.N03a\)](#) can meet the convex hull of $\varphi_v(K_\Phi)$. Indeed, their centers must lie in an interval of length at most $D(1 + 2\rho)t$.

The inverse image under φ_v of each band has radius at most $D\rho/r_{\min}$. Reflection does not change that radius. Consequently

$$d \leq \mu(\mathcal{B}_{k,\rho}) \leq \sum_{v \in \Lambda(t)} p_v C_D q_\mu(D\rho/r_{\min}) = C_D q_\mu(D\rho/r_{\min}). \quad (\text{II.6.N03b})$$

L-II.6.02 makes the last quantity tend to zero as $\rho \downarrow 0$, uniformly in k and b .

We next justify the conditioning step, including its normalization. For each cell Q , let α_Q, β_Q be the finite measures

$$\alpha_Q(C) = \mathbb{P}(Y \in C, A = Q), \quad \beta_Q(C) = \mathbb{P}(Y \in C, B = Q)$$

on Borel sets C . Put $p_Q = \mu(Q)$ and $w_Q = \mathbb{P}(B = Q)$. For positive masses, $\alpha_Q/p_Q = \mu_Q$, whereas $\beta_Q/w_Q = \eta_Q$ by the independent-tail construction. If

$$d_Q = \mathbb{P}(A = Q, B \neq Q) + \mathbb{P}(B = Q, A \neq Q),$$

then the total variation mass of the signed measure $\alpha_Q - \beta_Q$ is at most d_Q , and $|p_Q - w_Q| \leq d_Q$. For $p_Q, w_Q > 0$, the half- L^1 convention gives

$$\begin{aligned} p_Q \|\mu_Q - \eta_Q\|_{\text{TV}} &\leq \frac{1}{2}(|\alpha_Q - \beta_Q|(\mathbb{R}) + |p_Q - w_Q|) \\ &\leq d_Q. \end{aligned}$$

Here $|\sigma|(\mathbb{R})$ denotes the total variation mass of a finite signed measure σ , as in the permitted measure-theory roots. If $p_Q > 0 = w_Q$, the left side is at most $p_Q = d_Q$, whatever the specified fallback probability is. Summing gives

$$\sum_Q p_Q \|\mu_Q - \eta_Q\|_{\text{TV}} \leq 2d, \quad \sum_{Q: w_Q=0} p_Q \leq d. \quad (\text{II.6.N03c})$$

The second inequality holds because $A = Q$ with $w_Q = 0$ necessarily has a different B -label, up to a null event.

Choose $\rho_0 \in (0, 1)$ so that (II.6.N03b) is at most $\varepsilon^2/4$ for all $\rho \leq \rho_0$. Markov's inequality and (II.6.N03c) bound the mass of the complement of the desired event by

$$d + \frac{2d}{\varepsilon} \leq \frac{\varepsilon^2}{4} + \frac{\varepsilon}{2} < \varepsilon.$$

This proves (II.6.03). If μ has an atom, L-II.6.02 instead gives $K_\Phi = \{z\}$, $b = z$, and $Y = Z = z$ surely. The unique positive-mass cell has $w_Q = 1$ and both conditional laws δ_z ; the assertion holds with no error.

For uniform conjugacies $\theta = (A_{c,b_0})_* \mu$, the contractions and stopping words are unchanged and $\text{diam } A_{c,b_0} K_\Phi \leq D_* = c_+ D$. The conjugate non-atomic modulus satisfies

$$q_\theta(s) \leq q_\mu(s/c_-).$$

Thus the preceding proof gives the uniform disagreement bound

$$([3D_*] + 3) q_\mu(D_* \rho / (r_{\min} c_-)).$$

It tends to zero independently of c, b_0, k and the anchor. The atomic argument is also unchanged. This proves the full affine-uniform clause. $\square$

Definition II.6.2.4 (II.6.05: Uniform entropy dimension and component profiles). A compactly supported probability θ has entropy dimension α if $\lim_{N \rightarrow \infty} h_N(\theta) = \alpha$. It has uniform entropy dimension α if

$$\forall 0 < \varepsilon < 1 \exists M \geq 1 \forall m \geq M \exists N_0(m) \forall N \geq N_0(m),$$

$$\frac{1}{N} \sum_{i=0}^{N-1} \theta \{x : |h_m(\theta^{x,i}) - \alpha| < \varepsilon\} \geq 1 - \varepsilon.$$

Here M, m, N_0, N are integers. This is a joint level-component condition; it does not assert that every component is exact dimensional.

For an entropy threshold $a \in \mathbb{R}$ define the upper component profile

$$\mathcal{U}_{N,m}(\theta; a) = \frac{1}{N} \sum_{i=0}^{N-1} \theta \{x : h_m(\theta^{x,i}) > a\}.$$

The formula is meaningful for any Borel probability; its components lie in the unit interval. The finite-scale porosity condition of II.4 is $\mathcal{U}_{N,m}(\theta; 1 - \gamma) \leq \rho$. When θ is not carried by $[0, 1)$, we use this displayed condition without silently applying the unit-interval theorem; P-II.6.06 proves the required bounded-support extension.

The whole-cylinder mixture retains a uniform lower entropy bound because all its affine scales stay in a fixed nonzero range once the stopping tolerance has been chosen. Finite-alphabet entropy continuity then transfers this lower bound to most actual components.

The mean component entropy is already close to α . If very few components are much below α , a substantial population much above α would push the mean too high. T-II.6.05 turns this simple balance into a uniform two-sided probability estimate.

An extra dyadic contraction only delays the observation. Its first few levels may be uninformative; after those levels, the components coincide with the original ones. The delay term a/L records that initial part of the observation interval explicitly.

Proposition II.6.2.5 (P-II.6.04 – Uniform lower component entropy). *Let $\mu = \mu_{\Phi,p}$ be strict and $\alpha = \dim \mu$. Fix $0 < c_- \leq c_+ < \infty$ and $0 < \varepsilon < 1$. There is an integer M such that for every $m \geq M$, every $k \geq 0$, and every $\theta = (A_{c,b})_* \mu$ with $c_- \leq |c| \leq c_+$,*

$$\theta \{x : h_m(\theta^{x,k}) > \alpha - \varepsilon\} \geq 1 - \varepsilon. \quad (\text{II.6.04})$$

The threshold is uniform in b , the scale k and the affine map in the declared range. For comparison, see Hochman [Hoc14, Lemma 5.4, p. 32]. The scaling factors in the proof stay in a fixed compact nonzero range.

Proof. Let $D = \text{diam } K_\Phi$ and $D_* = c_+ D$. If $\alpha = 0$, every component entropy is nonnegative and the assertion is immediate. Otherwise choose a number

$$0 < \tau < \min\{\varepsilon/4, 1/4\}.$$

Apply the uniform clause of L-II.6.03 with tolerance τ , and fix one $0 < \rho < \min\{1, \rho_0\}$. This choice precedes m . For any $\theta = (A_{c,b})_* \mu$ in the stated range and any level k , at least $1 - \tau$ of its mass lies in cells Q with positive anchor weight and TV distance at most τ between the raw component and the whole-cylinder mixture.

Let T_Q be the canonical affine map from Q onto $[0, 1]$, and set $\hat{\eta}_Q = (T_Q)_* \eta_Q$. Push-forward cannot increase TV distance. Each summand of $\hat{\eta}_Q$, expressed as an affine image of the original μ , has slope $s = 2^k c r_u$, where $u \in \Lambda(\rho 2^{-k})$. Hence

$$c_- r_{\min} \rho < |s| \leq c_+ \rho. \quad (\text{II.6.N04a})$$

L-II.6.02 shows, uniformly over these slopes and all translations, that its normalized level- m entropy is at least $\alpha - \varepsilon/4$ for all sufficiently large m . Entropy concavity T-II.1.05 then gives

$$h_m(\hat{\eta}_Q) \geq \alpha - \varepsilon/4. \quad (\text{II.6.N04b})$$

The mixture is not necessarily supported in the unit interval. But its anchor points lie in Q , and each cylinder has diameter at most $D_* \rho 2^{-k}$. Therefore both $\hat{\eta}_Q$ and θ^Q are carried by $[-D_* \rho, 1 + D_* \rho]$. Their level- m laws have a common alphabet of size at most $C 2^m$, where $C = 4 + 2D_*$ is sufficient. Use at least two symbols, adding zero-probability symbols if necessary. Use the binary entropy function $b(t) = -t \log t - (1 - t) \log(1 - t)$, with $b(0) = b(1) = 0$, as in L-II.4.01. That lemma, with actual TV distance $t \leq \tau < 1/4$, yields

$$|h_m(\theta^Q) - h_m(\hat{\eta}_Q)| \leq \tau + \frac{b(\tau) + \tau \log C}{m \log 2}. \quad (\text{II.6.N04c})$$

We used the monotonicity of the binary entropy b on $[0, 1/2]$. Choose M large enough that (II.6.N04b) holds and the right side of (II.6.N04c) is less than $\varepsilon/2$ for every $m \geq M$. On the good cells,

$$h_m(\theta^Q) > \alpha - 3\varepsilon/4 > \alpha - \varepsilon.$$

Their mass is at least $1 - \tau \geq 1 - \varepsilon$, proving the result. All choices depended only on the fixed system, weights, affine range and ε , not on k, b or the cell. □

Theorem II.6.2.6 (T-II.6.05 — Uniform entropy dimension and delayed profiles). *Every strict self-similar probability μ has uniform entropy dimension $\alpha = \dim \mu$. More precisely, for fixed $0 < c_- \leq c_+ < \infty$ and $0 < \varepsilon < 1$, there is M such that for every $m \geq M$ there is $N_0(m)$ for which*

$$\sup_{\substack{c_- \leq |c| \leq c_+ \\ b \in \mathbb{R}}} \frac{1}{N} \sum_{i=0}^{N-1} \theta\{x : |h_m(\theta^{x,i}) - \alpha| \geq \varepsilon\} \leq \varepsilon \quad (N \geq N_0(m)), \quad \theta = (A_{c,b})_* \mu. \quad (\text{II.6.05})$$

For such m , for integers $a \geq 0$, $L > a$ with $L - a \geq N_0(m)$, and $\zeta = (S_{2^{-a}})_* \theta$,

$$\mathcal{U}_{L,m}(\zeta; \alpha + \varepsilon) \leq \varepsilon + \frac{a}{L}. \quad (\text{II.6.06})$$

For comparison, see Hochman [Hoc14, Proposition 5.2, pp. 30–32]. The delayed, affine-uniform clause follows from the same component argument.

Proof. Fix the affine range and $0 < \varepsilon < 1$, and put $\delta = \varepsilon^2/16$. P-II.6.04 gives an integer M such that for every $m \geq M$, every $i \geq 0$, and every permitted $\theta = (A_{c,b})_* \mu$,

$$\theta\{x : h_m(\theta^{x,i}) > \alpha - \delta\} \geq 1 - \delta. \quad (\text{II.6.N05a})$$

Fix such an m . All these measures have support diameter at most $D_* = c_+ \text{diam } K_\Phi$; their level-zero entropies are at most $\log([D_*] + 2)$, independently of translation. L-II.6.01 and the uniform entropy convergence in L-II.6.02 imply

$$\sup_{\theta} \left| \frac{1}{N} \sum_{i=0}^{N-1} \int h_m(\theta^{x,i}) d\theta(x) - \alpha \right| \rightarrow 0 \quad (N \rightarrow \infty). \quad (\text{II.6.N05b})$$

Choose $N_0(m)$ so that the absolute value is at most δ for $N \geq N_0(m)$. For such an N , sample i uniformly from $\{0, \dots, N-1\}$ and x from θ , independently, and write $Z = h_m(\theta^{x,i}) \in [0, 1]$. By (II.6.N05a),

$$\mathbb{E}(\alpha - Z)_+ \leq 2\delta.$$

Indeed the contribution where $Z > \alpha - \delta$ is at most δ , and the complementary event has probability at most δ and deficit at most one. Since $\mathbb{E}(Z - \alpha) \leq \delta$, it follows that

$$\mathbb{E}(Z - \alpha)_+ = \mathbb{E}(Z - \alpha) + \mathbb{E}(\alpha - Z)_+ \leq 3\delta.$$

Therefore

$$\begin{aligned} \mathbb{P}(|Z - \alpha| \geq \varepsilon) &\leq \mathbb{P}(Z \leq \alpha - \varepsilon) + \mathbb{P}(Z \geq \alpha + \varepsilon) \\ &\leq \delta + \frac{3\delta}{\varepsilon} \leq \varepsilon/4 < \varepsilon. \end{aligned}$$

This proves (II.6.05), including the advertised uniformity.

Finally let $\zeta = (S_{2^{-a}})_* \theta$, where $a \geq 0$ is an integer and $L - a \geq N_0(m)$. The first a terms of its length- L upper profile contribute at most a/L . For $i \geq a$, the exact identity of L-II.6.01 identifies the component with $\theta^{x,i-a}$. Hence

$$\begin{aligned}\mathcal{U}_{L,m}(\zeta; \alpha + \varepsilon) &\leq \frac{a}{L} + \frac{L-a}{L} \mathcal{U}_{L-a,m}(\theta; \alpha + \varepsilon) \\ &\leq \frac{a}{L} + \varepsilon,\end{aligned}$$

as claimed. □

The entropy-growth theorem T-II.4.16 expects probabilities in the unit interval. Translating by an integer and scaling by a dyadic factor makes that hypothesis exact and changes the number of observed initial levels by a fixed integer.

Those new levels can slightly worsen the porosity proportion and the normalized entropy threshold. The strict gap $\rho < \beta$ leaves room to absorb both changes. The proof first fixes two interior thresholds and only then chooses how large the observing length must be. That order will be important again for unequal contraction classes.

Proposition II.6.2.7 (P-II.6.06 – Bounded-support porosity forces entropy growth). *Fix $R \geq 1$, $0 < \gamma \leq 1$, an integer $m \geq 1$, and $0 \leq \rho < \beta \leq 1$. There are $\delta > 0$ and N_0 such that, for probabilities θ, ξ on $[-R, R]$, $N \geq N_0$,*

$$\mathcal{U}_{N,m}(\theta; 1 - \gamma) \leq \rho, \quad h_N(\xi) \geq \beta \quad \implies \quad h_N(\theta * \xi) \geq h_N(\theta) + \delta. \quad (\text{II.6.07})$$

All thresholds depend only on the displayed parameters. The proof transfers T-II.4.16 by a fixed dyadic affine normalization, charging the finitely many added initial levels. For comparison, see Hochman [Hoc14], Corollary 5.5, p. 33.

Proof. Choose an integer $R_0 > R$ and an integer $d \geq 1$ with $2^{-d}(R_0 + R) < 1$. Define

$$A(x) = 2^{-d}(x + R_0), \quad \theta' = A_*\theta, \quad \xi' = A_*\xi.$$

Both image measures are carried by $[0, 1)$. An integer translation preserves every nonnegative-level dyadic partition, and the following dyadic dilation shifts its levels exactly. Consequently

$$H_{N+d}(\theta') = H_N(\theta), \quad H_{N+d}(\xi') = H_N(\xi). \quad (\text{II.6.N06a})$$

For $i \geq d$, their canonical components are exactly those of the original measure at level $i - d$. The first d levels contribute at most d , so

$$\mathcal{U}_{N+d,m}(\theta'; 1 - \gamma) \leq \frac{d + N\rho}{N + d}, \quad h_{N+d}(\xi') \geq \frac{N\beta}{N + d}. \quad (\text{II.6.N06b})$$

Fix numbers ρ', β' with $\rho < \rho' < \beta' < \beta$. The exact theorem T-II.4.16 with parameters γ, m, ρ', β' supplies a gain $\delta' > 0$ and a scale threshold. Choose N_0 large enough that $N + d$ exceeds that threshold and both

$$\frac{d + N\rho}{N + d} \leq \rho', \quad \frac{N\beta}{N + d} \geq \beta'$$

hold whenever $N \geq N_0$. T-II.4.16 gives

$$H_{N+d}(\theta' * \xi') - H_{N+d}(\theta') \geq \delta'(N + d) \log 2.$$

The convolution is $(S_{2^{-d}})_*(T_{2R_0})_*(\theta * \xi)$, where $T_{2R_0}(x) = x + 2R_0$. Since $2R_0$ is an integer, its level- $(N + d)$ entropy is $H_N(\theta * \xi)$. Using (II.6.N06a) and dividing by $N \log 2$ gives

$$h_N(\theta * \xi) - h_N(\theta) \geq \delta'(1 + d/N) \geq \delta'.$$

Take $\delta = \delta'$. All choices depended only on the displayed parameters of P-II.6.06. This argument uses the Berry–Esseen interface X-BE-II.4 only through T-II.4.16. □

Theorem T-II.6.05 supplies porous tails with a uniform entropy profile. Proposition P-II.6.06 then says that any component carrying positive entropy must create a fixed entropy gain when convolved with such a tail. This is the contradiction mechanism used in the finite word laws below.

II.6.3 Finite word laws and conditional-entropy flattening

Definition II.6.3.1 (II.6.03: Finite cylinder laws, contraction labels and dyadic lengths). For a strict vector p and integer $n \geq 1$, define the location law, the lifted law and the finite set of contraction labels by

$$\nu_n = \sum_{u \in \Lambda^n} p_u \delta_{a_u}, \quad \tilde{\nu}_n = \sum_{u \in \Lambda^n} p_u \delta_{(a_u, r_u)}, \quad \mathcal{R}_n = \{r_u : u \in \Lambda^n\}.$$

Coincident locations or lifted locations are aggregated as atoms. For $r \in \mathcal{R}_n$, define

$$\beta_n(r) = \sum_{\substack{u \in \Lambda^n \\ r_u = r}} p_u, \quad \nu_{n,r} = \frac{1}{\beta_n(r)} \sum_{\substack{u \in \Lambda^n \\ r_u = r}} p_u \delta_{a_u}, \quad \tau_r = (S_r)_* \mu.$$

All $\beta_n(r)$ are positive and sum to one. For this fixed n , the countable labelled partition on $\mathbb{R} \times \mathcal{R}_n$ is

$$\tilde{\mathcal{D}}_{j,n} = \{Q \times \{r\} : Q \in \mathcal{D}_j, r \in \mathcal{R}_n\}.$$

Its entropy is computed under $\tilde{\nu}_n$. This is not an undeclared uncountable point partition of the full plane.

The natural dyadic length and a finer observing level are

$$\ell_n = \left\lfloor \frac{n\chi(\Phi, p)}{\log 2} \right\rfloor, \quad b_n(q) = \lfloor q\ell_n \rfloor \quad (q > 1).$$

Any expression divided by ℓ_n is restricted to $\ell_n \geq 1$. For fixed $q > 1$, its asymptotic assertions concern large n , so $b_n(q) > \ell_n$. In a homogeneous system all signed r_i equal one $r \neq 0$; then $\chi = -\log |r|$, not $-\log r$.

For $0 < \varepsilon < 1$, the typical contraction labels are

$$\mathcal{T}_n(\varepsilon) = \{r \in \mathcal{R}_n : 2^{-(1+\varepsilon)\ell_n} < |r| \leq 2^{-(1-\varepsilon)\ell_n}\}.$$

The word typical is only a name for this displayed set. Its probability convergence is a result, not part of the definition.

Different words can send zero to the same location while having different slopes. Such a positional coincidence does not identify their affine maps. The contraction label records precisely the extra datum needed to turn a collision of finite observations into two close maps with the same signed contraction.

There are $|\Lambda|^n$ words and at most $(n+1)^{|\Lambda|}$ contraction labels. Multiplication on the line depends on symbol counts, including the parity information already carried by the signed factors. The labels therefore cost only $O(\log n)$ entropy. At an observing length proportional to n , that cost vanishes after normalization.

This does not mean labels can be discarded at the start. They are needed both to express the invariant probability as a mixture of convolutions with fixed tails and to extract equal-contraction pairs at the end.

At its natural contraction scale, the finite location law has essentially the same entropy as the full invariant measure. Couple the finite prefix with the corresponding infinite coded point. For most words, their distance is small compared with a slightly coarser observing cell.

The exceptional words need not be geometrically well behaved. Their conditional laws still have a fixed bounded support, so their entropy cost is at most a constant times the observing length multiplied by their vanishing probability. L-II.6.07 keeps that cost until after normalization. Weak convergence alone would not supply this estimate.

Lemma II.6.3.2 (L-II.6.07 – Contraction labels and the natural-scale approximation).

Let $\mu = \mu_{\Phi,p}$ and $\alpha = \dim \mu$, with the data of D-II.6.03.

$$\begin{aligned} |\mathcal{R}_n| &\leq (n+1)^{|\Lambda|}, & H(\beta_n) &\leq |\Lambda| \log(n+1), \\ v_n &= \sum_r \beta_n(r) v_{n,r}, & \mu &= \sum_r \beta_n(r) (v_{n,r} * \tau_r). \end{aligned} \quad (\text{II.6.08})$$

For every integer j ,

$$0 \leq H_{\tilde{v}_n}(\tilde{\mathcal{D}}_{j,n}) - H_j(v_n) \leq H(\beta_n). \quad (\text{II.6.09})$$

For each $0 < \varepsilon < 1$, $\sum_{r \in \mathcal{T}_n(\varepsilon)} \beta_n(r) \rightarrow 1$. Moreover,

$$\frac{H_{\ell_n}(v_n)}{\ell_n \log 2} \rightarrow \alpha, \quad \frac{H_{\tilde{v}_n}(\tilde{\mathcal{D}}_{\ell_n,n})}{\ell_n \log 2} \rightarrow \alpha. \quad (\text{II.6.10})$$

For comparison, see Hochman [Hoc14, Theorems 1.3–1.4 and Sections 5.2–5.3], pp. 4–5 and 33–36. The natural-scale approximation is established independently of the fine-scale flattening theorem.

Proof. The proof compares a finite prefix with the corresponding infinite coded point,

controls atypical words by bounded support, and keeps the contraction label so that a later collision identifies affine maps rather than translations alone.

Write $d = |\Lambda|$. For a word u , let $N_i(u)$ be the number of occurrences of its symbol i . Then

$$r_u = \prod_{i \in \Lambda} r_i^{N_i(u)}, \quad 0 \leq N_i(u) \leq n.$$

This formula retains the sign. There are at most $(n+1)^d$ count vectors, so $|\mathcal{R}_n| \leq (n+1)^d$. L-II.1.02 gives $H(\beta_n) \leq d \log(n+1)$.

Let U be a length- n word with law p_u , and independently let $X \sim \mu$. Define

$$A_n = a_U, \quad R_n = r_U, \quad Y_n = a_U + r_U X.$$

Here R_n is a random real contraction label, whereas $\mathcal{R}_n$ denotes its finite range. The iterated invariance identity T-I.7.04 gives $\text{Law}(Y_n) = \mu$. Conditioning on $R_n = r$ leaves X independent of A_n , with their laws respectively μ and $v_{n,r}$. Hence

$$\text{Law}(A_n) = \sum_r \beta_n(r) v_{n,r} = v_n, \quad \mu = \sum_r \beta_n(r) (v_{n,r} * \tau_r). \quad (\text{II.6.N07a})$$

The labelled chain rule and the mixture inequalities T-II.1.03/T-II.1.05 give

$$\begin{aligned} H_{\tilde{v}_n}(\tilde{\mathcal{D}}_{j,n}) &= H(\beta_n) + \sum_r \beta_n(r) H_j(v_{n,r}), \\ \sum_r \beta_n(r) H_j(v_{n,r}) &\leq H_j(v_n) \leq H(\beta_n) + \sum_r \beta_n(r) H_j(v_{n,r}). \end{aligned} \quad (\text{II.6.N07b})$$

Subtracting proves (II.6.09), for every integer j .

Set $t_i = -\log_2 |r_i|$ and $\chi_2 = \sum_i p_i t_i = \chi / \log 2 > 0$. Realize the words U as initial segments of one infinite Bernoulli sequence. L-I.8.01 and the definition of ℓ_n give, almost surely,

$$\frac{-\log_2 |R_n|}{n} \rightarrow \chi_2, \quad \frac{\ell_n}{n} \rightarrow \chi_2, \quad \frac{-\log_2 |R_n|}{\ell_n} \rightarrow 1.$$

Thus, for every fixed $0 < \varepsilon < 1$, the event

$$(1 - \varepsilon)\ell_n \leq -\log_2 |R_n| < (1 + \varepsilon)\ell_n$$

has probability tending to one. This event is precisely $R_n \in \mathcal{T}_n(\varepsilon)$, including the stated endpoint conventions.

We prove the natural-scale entropy approximation without assuming $0 \in K_\Phi$. The word formula and the coding series imply

$$|a_u| \leq B_0, \quad K_\Phi \subset [-B_0, B_0], \quad B_0 = \frac{\max_i |a_i|}{1 - r_{\max}}. \quad (\text{II.6.N07c})$$

Indeed each series is bounded in absolute value by the geometric series with terms $\max_i |a_i| r_{\max}^{j-1}$. In particular A_n, Y_n and all their conditional laws have a common fixed bounded support. Put $C_0 = 2B_0 + 3$. For every $k \geq 0$, any probability on that support has

$$0 \leq H_k \leq k \log 2 + \log C_0, \quad (\text{II.6.N07d})$$

because the interval meets at most $C_0 2^k$ grid cells.

Fix $0 < \eta < 1/4$, put $k_n = \lfloor (1 - 2\eta)\ell_n \rfloor$, and let $E_n = \{R_n \in \mathcal{T}_n(\eta)\}$. Write $s_n = \mathbb{P}(E_n) \rightarrow 1$ and $e_n = 1 - s_n$. On this event,

$$|Y_n - A_n| \leq B_0 2^{-(1-\eta)\ell_n} \leq 2^{-k_n} \quad (\text{II.6.N07e})$$

for all sufficiently large n , since $B_0 2^{k_n - (1-\eta)\ell_n} \leq B_0 2^{-\eta\ell_n} \rightarrow 0$. Let ν_n^+, μ_n^+ denote the conditional laws of A_n, Y_n given E_n , and let ν_n^-, μ_n^- be the corresponding laws on its complement. A law on an event of zero probability can be chosen arbitrarily on the fixed interval; its coefficient is zero. For all sufficiently large n , $s_n > 0$, and P-II.1.09 applied to the conditional coupling in (II.6.N07e) yields

$$|H_{k_n}(\nu_n^+) - H_{k_n}(\mu_n^+)| \leq \log 3.$$

Use the binary entropy $b(e) = -e \log e - (1 - e) \log(1 - e)$, extended by zero at the endpoints. The two binary-mixture estimates T-II.1.05 and (II.6.N07d) now give

$$|H_{k_n}(\nu_n) - H_{k_n}(\mu)| \leq \log 3 + e_n(k_n \log 2 + \log C_0) + b(e_n). \quad (\text{II.6.N07f})$$

In this inequality the two mixture errors each lie in the same interval $[0, b(e_n)]$, so their difference has absolute value at most $b(e_n)$. The displayed right side divided by ℓ_n tends to zero. By T-II.1.07 the two increments from k_n to ℓ_n belong to $[0, (\ell_n - k_n) \log 2]$. Therefore

$$\limsup_{n \rightarrow \infty} \frac{|H_{\ell_n}(\nu_n) - H_{\ell_n}(\mu)|}{\ell_n \log 2} \leq 2\eta.$$

The left side does not depend on η . Letting $\eta \downarrow 0$ and using L-II.6.02 proves $H_{\ell_n}(\nu_n)/(\ell_n \log 2) \rightarrow \alpha$. Finally (II.6.N07b) bounds the difference between lifted and unlabelled natural entropies by $d \log(n + 1) = o(\ell_n)$. This proves both limits in (II.6.10). $\square$

In the homogeneous case the invariant measure is the convolution of the location law with one scaled copy of itself. At the natural level, decompose the location law into components and compare their convolutions with that common tail.

The tail has normalized fine entropy close to α . The full measure's conditional entropy budget over the same range is also close to α . Conditional concavity therefore leaves asymptotically no average room for the component convolutions to increase that entropy.

The porosity-growth theorem says that a location component with substantial entropy would produce a fixed positive increase. Hence such components must have vanishing total weight. This is the conditional-entropy flattening conclusion. The proof shifts each gain by $1/L$ before using a nonnegative-mean argument; a raw dyadic convolution gain is not always nonnegative.

Theorem II.6.3.3 (T-II.6.08 – Homogeneous conditional-entropy flattening). *Suppose all signed contractions equal r , $0 < |r| < 1$, and $\alpha = \dim \mu < 1$. For every $q > 1$,*

$$\lim_{n \rightarrow \infty} \frac{H_{\nu_n}(\mathcal{D}_{b_n(q)} \mid \mathcal{D}_{\ell_n})}{\ell_n \log 2} = 0. \quad (\text{II.6.11})$$

For comparison, see Hochman [Hoc14, Theorem 1.3 and Section 5.2, pp. 4 and 33–35]. The theorem includes a negative common contraction and every fixed $q > 1$, with all floor and signed-scaling errors retained.

Proof. Fix $q > 1$, and abbreviate $\ell = \ell_n$, $b = b_n(q)$, $L = b - \ell$. We work at sufficiently large n , so $L \geq 1$ and $L/\ell \rightarrow q - 1$. By L-II.6.07 in the homogeneous case,

$$\mu = \nu_n * \tau_n, \quad \tau_n = (S_r^n)_* \mu.$$

Put $c_n = 2^\ell r^n$ and $\theta_n = (S_{c_n})_* \mu$; then $1/2 < |c_n| \leq 1$. For a positive-mass level- ℓ cell Q , define

$$w_Q = \nu_n(Q), \quad \xi_Q = (\nu_n)^Q, \quad \lambda_Q = (\nu_n)_Q * \tau_n.$$

There are finitely many such cells. If $Q = 2^{-\ell}(z + [0, 1))$, $z \in \mathbb{Z}$, then $x \mapsto 2^\ell x - z$ sends λ_Q to $\xi_Q * \theta_n$. Since an integer translation preserves every nonnegative dyadic grid,

$$H_b(\lambda_Q) = H_L(\xi_Q * \theta_n). \quad (\text{II.6.N08a})$$

Let $D = \text{diam } K_\Phi$ and $C_D = \log([D] + 3)$. The support of λ_Q has diameter at most $2^{-\ell} + |r|^n D \leq (1 + D)2^{-\ell}$; hence $H_\ell(\lambda_Q) \leq C_D$. By conditional-entropy concavity and $\mu = \sum_Q w_Q \lambda_Q$,

$$\sum_Q w_Q H_L(\xi_Q * \theta_n) \leq H_\mu(\mathcal{D}_b \mid \mathcal{D}_\ell) + C_D. \quad (\text{II.6.N08b})$$

L-II.6.02 gives

$$\frac{H_\mu(\mathcal{D}_b \mid \mathcal{D}_\ell)}{L \log 2} = \frac{H_b(\mu) - H_\ell(\mu)}{L \log 2} \longrightarrow \alpha, \quad h_L(\theta_n) \longrightarrow \alpha. \quad (\text{II.6.N08c})$$

For the first limit, both b, ℓ are comparable to L ; for the second, use the uniform affine range $1/2 \leq |c| \leq 1$. Define $g_n(Q) = h_L(\xi_Q * \theta_n) - h_L(\theta_n)$. T-II.3.03 gives $g_n(Q) \geq -1/L$ for every Q . Combining this with (II.6.N08b)–(II.6.N08c) gives the nonnegative mean convergence

$$G_n := \sum_Q w_Q (g_n(Q) + 1/L) \longrightarrow 0. \quad (\text{II.6.N08d})$$

Fix $0 < \beta < 1$. Choose $0 < \eta < \min\{\beta, 1 - \alpha\}/4$ and put $\gamma = 1 - \alpha - \eta > 0$. T-II.6.05, for the affine range $1/2 \leq |c| \leq 1$, supplies a window threshold. Fix one window m above it. For all sufficiently large n ,

$$\mathcal{U}_{L,m}(\theta_n; 1 - \gamma) = \mathcal{U}_{L,m}(\theta_n; \alpha + \eta) \leq \eta.$$

The ξ_Q lie in $[0, 1)$, and all θ_n lie in one fixed $[-R, R]$ with $R \geq 1$, depending only on the system. Apply P-II.6.06 with $\rho = \eta < \beta$. It supplies $\delta > 0$ such that, for all large n ,

$$h_L(\xi_Q) \geq \beta \implies g_n(Q) \geq \delta.$$

Consequently (II.6.N08d) gives

$$\sum_{Q: h_L(\xi_Q) \geq \beta} w_Q \leq G_n / \delta \rightarrow 0.$$

Every $h_L(\xi_Q)$ is between zero and one, so $\limsup_n \sum_Q w_Q h_L(\xi_Q) \leq \beta$. As $\beta > 0$ was arbitrary, that mean tends to zero. The component identity L-II.2.01 finally gives

$$\frac{H_{v_n}(\mathcal{D}_b | \mathcal{D}_\ell)}{\ell \log 2} = \frac{L}{\ell} \sum_Q w_Q h_L(\xi_Q) \rightarrow 0.$$

All signed factors were retained in θ_n ; thus the proof also covers negative r , whose powers may alternate orientation. □

Each contraction class has its own tail. Observing slightly before the natural level makes every typical tail fit inside a uniformly bounded interval after rescaling. This costs a small part of the observation range, which P-II.6.09 records explicitly.

Two separate estimates are required. One bounds the average entropy gain after mixing over classes and coarse cells. The other shows that every typical rescaled tail remains porous for the inverse theorem. The second statement is uniform in the later contraction tolerance. Without that uniformity, choosing a smaller tolerance might change the window length, which could change the inverse gain and invalidate the required error bound.

T-II.6.10 therefore chooses the entropy threshold and profile tolerance, then the window, then the inverse gain, and only afterward the contraction tolerance. It first lets the word length tend to infinity, then the contraction tolerance tend to zero, and finally the entropy threshold tend to zero. No rate for the inverse theorem is needed.

Proposition II.6.3.4 (P-II.6.09 — Typical contraction-class localization). *Let $\mu = \mu_{\Phi, p}$ and $\alpha = \dim \mu$, with the notation of D-II.6.03. Fix $q > 1$ and $0 < \varepsilon < \min\{1/4, (q-1)/4\}$. Put*

$$k_n = \lfloor (1 - \varepsilon)\ell_n \rfloor, \quad b_n = \lfloor q\ell_n \rfloor, \quad L_n = b_n - k_n.$$

For $r \in \mathcal{R}_n$ and a positive-mass level- k_n cell Q of $v_{n,r}$, put

$$\theta_{n,r} = (S_{2^{k_n r}})_* \mu, \quad \xi_{n,r,Q} = (v_{n,r})^Q, \quad g_n(r, Q) = h_{L_n}(\xi_{n,r,Q} * \theta_{n,r}) - h_{L_n}(\theta_{n,r}).$$

These expressions are used for large n , so $L_n \geq 1$. There is a fixed R , depending only on Φ , such that for $r \in \mathcal{T}_n(\varepsilon)$, both measures are carried by $[-R, R]$. All $g_n(r, Q) \geq -1/L_n$, and

$$\limsup_{n \rightarrow \infty} \sum_{r \in \mathcal{T}_n(\varepsilon)} \beta_n(r) \sum_{Q \in \mathcal{D}_{k_n}} v_{n,r}(Q) \left(g_n(r, Q) + \frac{1}{L_n} \right) \leq \frac{4\varepsilon}{q-1}. \quad (\text{II.6.12})$$

For every $0 < \eta < 1$, there is $M = M(\Phi, p, \eta)$, independent of q, ε , such that for every $m \geq M$,

$$\limsup_{n \rightarrow \infty} \sup_{r \in \mathcal{T}_n(\varepsilon)} \mathcal{U}_{L_n, m}(\theta_{n, r}; \alpha + \eta) \leq \eta + \frac{3\varepsilon}{q-1}. \quad (\text{II.6.13})$$

The supremum over an empty typical set is assigned zero. For comparison, see Hochman [Hoc14, Section 5.3, pp. 35–36]. The proof below gives the explicit bounds and shows that M is independent of the later contraction tolerance.

Proof. We localize to typical contraction classes, move slightly before their natural scale, and retain a margin that is uniform in the contraction tolerance.

Fix the stated q, ε , and abbreviate $\ell = \ell_n, k = k_n, b = b_n, L = L_n$. For a typical label r ,

$$2^k |r| \leq 1, \quad 2^{-(1+\varepsilon)\ell} < |r| \leq 2^{-(1-\varepsilon)\ell}. \quad (\text{II.6.N09a})$$

Since $\xi_{n, r, Q}$ is supported in $[0, 1)$ and μ in the fixed compact K_Φ , the required common $[-R, R]$ follows, with R depending only on Φ . For every label, including atypical ones, its two measures are compactly supported; T-II.3.03 gives $g_n(r, Q) \geq -1/L$.

Write

$$w_{r, Q} = \beta_n(r) v_{n, r}(Q), \quad \lambda_{r, Q} = (v_{n, r})_Q * \tau_r, \quad s_n = \sum_{r \in \mathcal{T}_n(\varepsilon)} \beta_n(r) \rightarrow 1.$$

The weights over all r, Q sum to one, and $\mu = \sum_{r, Q} w_{r, Q} \lambda_{r, Q}$ by L-II.6.07. For typical r , the support diameter of $\lambda_{r, Q}$ is at most $(1+D)2^{-k}$, $D = \text{diam } K_\Phi$. Hence $H_k(\lambda_{r, Q}) \leq C_D = \log([D] + 3)$. Concavity of conditional entropy gives

$$\begin{aligned} H_\mu(\mathcal{D}_b \mid \mathcal{D}_k) &\geq \sum_{r, Q} w_{r, Q} H_{\lambda_{r, Q}}(\mathcal{D}_b \mid \mathcal{D}_k) \\ &\geq \sum_{\substack{r \in \mathcal{T}_n(\varepsilon) \\ Q}} w_{r, Q} (H_b(\lambda_{r, Q}) - C_D). \end{aligned} \quad (\text{II.6.N09b})$$

The second inequality discards nonnegative atypical conditional entropies. It does not discard atypical unnormalized gains. For $Q = 2^{-k}(z + [0, 1))$, $z \in \mathbb{Z}$, the map $x \mapsto 2^k x - z$ sends $\lambda_{r, Q}$ to $\xi_{n, r, Q} * \theta_{n, r}$; dilation by 2^k sends τ_r to $\theta_{n, r}$. These maps give

$$H_b(\lambda_{r, Q}) = H_L(\xi_{n, r, Q} * \theta_{n, r}), \quad H_b(\tau_r) = H_L(\theta_{n, r}). \quad (\text{II.6.N09c})$$

Both identities are exact; the cell translation is an integer after rescaling. Thus the weighted typical gain satisfies

$$\sum_{\substack{r \in \mathcal{T}_n(\varepsilon) \\ Q}} w_{r, Q} g_n(r, Q) \leq \frac{H_\mu(\mathcal{D}_b \mid \mathcal{D}_k) + C_D}{L \log 2} - \sum_{r \in \mathcal{T}_n(\varepsilon)} \beta_n(r) \frac{H_b(\tau_r)}{L \log 2}. \quad (\text{II.6.N09d})$$

We bound its second term uniformly before taking the limit. Let $v_r = \lfloor b + \log_2 |r| \rfloor$. By (II.6.N09a),

$$b - (1 + \varepsilon)\ell - 1 \leq v_r \leq b - (1 - \varepsilon)\ell \leq L. \quad (\text{II.6.N09e})$$

The lower bound tends to infinity linearly with ℓ , because $q - 1 - \varepsilon > 0$. L-II.6.01 gives $|H_b(\tau_r) - H_{v_r}(\mu)| \leq \log 4$. L-II.6.02 therefore implies, uniformly over typical labels,

$$H_b(\tau_r) \geq \alpha(b - (1 + \varepsilon)\ell - 1) \log 2 - o(\ell). \quad (\text{II.6.N09f})$$

For clarity, the uniform error follows by taking the supremum of $|h_v(\mu) - \alpha|$ over all integers v above the lower bound in (II.6.N09e); this supremum tends to zero, and $v_r \leq L = O(\ell)$. Also

$$H_\mu(\mathcal{D}_b \mid \mathcal{D}_k) = \alpha L \log 2 + o(\ell), \quad \frac{L}{\ell} \longrightarrow q - 1 + \varepsilon. \quad (\text{II.6.N09g})$$

Since $s_n \rightarrow 1$, substituting (II.6.N09f)–(II.6.N09g) in (II.6.N09d), and adding the vanishing term s_n/L , yields

$$\begin{aligned} \limsup_n \sum_{\substack{r \in \mathcal{T}_n(\varepsilon) \\ Q}} w_{r,Q}(g_n(r, Q) + 1/L) &\leq \frac{2\alpha\varepsilon}{q - 1 + \varepsilon} \\ &\leq \frac{4\varepsilon}{q - 1}. \end{aligned}$$

Here $0 \leq \alpha \leq 1$. This proves (II.6.12), with slack.

For the profile assertion put $j_r = \lfloor \log_2(1/|r|) \rfloor$, $a_r = j_r - k$. For typical labels, the exact bounds are

$$0 \leq a_r \leq 2\varepsilon\ell + 1, \quad L - a_r = b - j_r \geq (q - 1 - \varepsilon)\ell - 1, \quad 1/2 < 2^{j_r}|r| \leq 1. \quad (\text{II.6.N09h})$$

Define $\zeta_r = (S_{2^{j_r}})_* \mu$. Then $\theta_{n,r} = (S_{2^{-a_r}})_* \zeta_r$. For any $0 < \eta < 1$, T-II.6.05 on the fixed affine range $1/2 \leq |c| \leq 1$ supplies $M = M(\Phi, p, \eta)$. This choice involves neither q nor ε . Fix $m \geq M$. The lower bound for $L - a_r$ tends to infinity uniformly, so the delayed profile clause of T-II.6.05 gives, for all sufficiently large n and every typical r ,

$$\mathcal{U}_{L,m}(\theta_{n,r}; \alpha + \eta) \leq \eta + \frac{a_r}{L} \leq \eta + \frac{2\varepsilon\ell + 1}{L}.$$

Taking the supremum and limsup proves the stronger bound $\eta + 2\varepsilon/(q - 1 + \varepsilon)$, hence (II.6.13). The assigned zero supremum handles an empty typical set. $\square$

Theorem II.6.3.5 (T-II.6.10 – General labelled conditional-entropy flattening). *For every strict line system with $\alpha = \dim \mu < 1$, including unequal signed contractions, and every $q > 1$,*

$$\lim_{n \rightarrow \infty} \frac{H_{v_n}(\tilde{\mathcal{D}}_{b_n(q),n} \mid \tilde{\mathcal{D}}_{\ell_n,n})}{\ell_n \log 2} = 0. \quad (\text{II.6.14})$$

Consequently the same limit is zero without the contraction labels, with numerator $H_{v_n}(\mathcal{D}_{b_n(q)} \mid \mathcal{D}_{\ell_n})$.

For comparison, see Hochman [Hoc14, Theorem 1.4 and Section 5.3, pp. 5 and 35–36].

The proof fixes the inverse-growth parameters before choosing the typical-contraction tolerance. No homogeneous-only reduction is assumed.

Proof. The parameter choices are made in the only order compatible with the argument: first the entropy and profile tolerances, then the observation window and inverse-theorem gain, and only afterward the contraction-class tolerance. The word length tends to infinity before the latter tolerances are removed.

Fix $q > 1$. The quantity whose limit we must prove is

$$A_n = \frac{H_{\tilde{v}_n}(\tilde{\mathcal{D}}_{b_n(q),n} \mid \tilde{\mathcal{D}}_{\ell_n,n})}{\ell_n \log 2} \geq 0.$$

Fix $0 < \beta < 1$, and then choose $0 < \eta < \min\{\beta/8, (1 - \alpha)/4, 1/8\}$. Put $\gamma = 1 - \alpha - \eta > 0$, $\rho = 3\eta < \beta$. P-II.6.09 supplies a threshold $M(\Phi, p, \eta)$; fix one integer $m \geq M$. Let $\delta > 0$ and a length threshold be the constants in P-II.6.06 for $R, \gamma, m, \rho, \beta$, with the common support bound R from P-II.6.09. All these choices precede the contraction tolerance ε .

Now take any

$$0 < \varepsilon < \min\{1/4, (q - 1)/4, \eta(q - 1)/6\}, \quad (\text{II.6.N10a})$$

and use $k, b, L, \theta_{n,r}, \xi_{n,r,Q}, g_n(r, Q)$ from P-II.6.09. Its profile conclusion implies

$$\limsup_n \sup_{r \in \mathcal{T}_n(\varepsilon)} \mathcal{U}_{L,m}(\theta_{n,r}; 1 - \gamma) \leq \eta + \frac{3\varepsilon}{q - 1} < \frac{3}{2}\eta < \rho.$$

Thus the porosity hypothesis holds for every typical label at every sufficiently large n ; also L exceeds the inverse-growth length threshold. For those labels P-II.6.06 gives

$$h_L(\xi_{n,r,Q}) \geq \beta \implies g_n(r, Q) \geq \delta.$$

Let B_n be the sum of weights $\beta_n(r)v_{n,r}(Q)$ over typical r and cells satisfying this entropy inequality. The nonnegative shifted gains in P-II.6.09 give

$$\limsup_n B_n \leq \frac{4\varepsilon}{(q - 1)\delta}. \quad (\text{II.6.N10b})$$

Let $s_n = \sum_{r \in \mathcal{T}_n(\varepsilon)} \beta_n(r) \rightarrow 1$. All canonical component entropies belong to $[0, 1]$, including the components for atypical labels. Hence

$$\begin{aligned} E_n &:= \sum_{r \in \mathcal{R}_n} \beta_n(r) \sum_{Q \in \mathcal{D}_k} v_{n,r}(Q) h_L(\xi_{n,r,Q}) \\ &\leq \beta + B_n + (1 - s_n), \\ \limsup_n E_n &\leq \beta + \frac{4\varepsilon}{(q - 1)\delta}. \end{aligned} \quad (\text{II.6.N10c})$$

The component identity and the labelled chain rule give exactly

$$H_{\tilde{v}_n}(\tilde{\mathcal{D}}_{b,n} \mid \tilde{\mathcal{D}}_{k,n}) = \sum_r \beta_n(r) H_{v_{n,r}}(\mathcal{D}_b \mid \mathcal{D}_k) = L \log 2 E_n. \quad (\text{II.6.N10d})$$

Since $k \leq \ell_n \leq b$, increasing the coarse level to ℓ_n decreases this conditional entropy. We obtain

$$\limsup_n A_n \leq (q - 1 + \varepsilon) \left(\beta + \frac{4\varepsilon}{(q - 1)\delta} \right). \quad (\text{II.6.N10e})$$

The left side is independent of ε . For the fixed β, η, m, δ , (II.6.N10e) holds for every sufficiently small $\varepsilon > 0$. Letting $\varepsilon \downarrow 0$ therefore gives $\limsup_n A_n \leq (q - 1)\beta$. Now let $\beta \downarrow 0$. Because $A_n \geq 0$, the desired labelled limit is zero. This order of limits does not require any lower bound on how the inverse gain depends on β or m .

Finally the conditional mixture inequalities T-II.1.05 give

$$\begin{aligned} 0 &\leq H_{V_n}(\mathcal{D}_{b_n(q)} \mid \mathcal{D}_{\ell_n}) - H_{\tilde{V}_n}(\tilde{\mathcal{D}}_{b_n(q),n} \mid \tilde{\mathcal{D}}_{\ell_n,n}) \\ &\leq H(\beta_n) \leq |\Lambda| \log(n + 1). \end{aligned}$$

Divide by $\ell_n \log 2$ and use $\ell_n/n \rightarrow \chi / \log 2 > 0$. This proves the unlabelled assertion as well. □

Definition II.6.3.6 (II.6.02: Equal-contraction cylinder distance and decay regimes). For distinct words of the same length define

$$d_\Phi(u, v) = \begin{cases} |a_u - a_v|, & r_u = r_v, \\ +\infty, & r_u \neq r_v, \end{cases} \quad \Delta_n(\Phi) = \min_{\substack{u, v \in \Lambda^n \\ u \neq v}} d_\Phi(u, v).$$

The equality of contractions is equality of signed real products. The minimum of an empty collection is $+\infty$. An exact overlap means $\varphi_u = \varphi_v$ for distinct equal-length words, as in I.7. The connection with zero cylinder distance and with different word lengths is proved in this chapter.

The system has super-exponential cylinder concentration if

$$\forall A > 0 \exists n_0 \forall n \geq n_0, \quad \Delta_n(\Phi) < e^{-An}.$$

Thus the statement concerns every sufficiently large generation, not merely a subsequence. Equivalently the extended quantities $-n^{-1} \log \Delta_n$ tend to $+\infty$, with $-\log 0 = +\infty$ and $-\log(+\infty) = -\infty$.

The line exponential-separation condition used here is

$$\exists C > 0 \quad \Delta_n(\Phi) \geq e^{-Cn} \quad \text{for infinitely many integers } n.$$

This is the precise same-contraction convention used in this chapter; a lower bound at all generations is a stronger statement. Negation of super-exponential concentration is equivalent to this subsequence condition. No assertion of positive Lebesgue measure or absolute continuity is included in either definition.

The word distribution carries entropy $nh(p)$. If $\alpha < 1$, flattening says that the lifted location law has entropy $\alpha \ell_n \log 2 + o(n)$ even when observed at the finer level $b_n(q)$, for

any fixed $q > 1$. The normalization here remains the natural length ℓ_n , not the finer length $b_n(q)$.

When the dimension is also below $h(p)/\chi$, this observation has strictly less entropy than the word distribution. It cannot distinguish all words. A repeated labelled cell supplies two equal signed slopes and two translations inside one very small dyadic interval.

Taking the observing multiple as large as needed makes that interval smaller than any prescribed exponential rate. The entropy convergence holds along all large word lengths, which is why the conclusion is not merely a statement about a favorable subsequence.

Theorem II.6.3.7 (T-II.6.11 – Dimension drop forces super-exponential concentration). *For a strict probability vector,*

$$\dim \mu_{\Phi,p} < \min\{1, s(\Phi, p)\} \implies \forall A > 0 \exists n_0 \forall n \geq n_0, \Delta_n(\Phi) < e^{-An}. \quad (\text{II.6.15})$$

The same conclusion follows if $\dim_{\text{H}} K_{\Phi} < \min\{1, s_{\Phi}\}$. For comparison, see Hochman [Hoc14, Theorem 1.1 and Corollary 1.2, pp. 3–5]. The two words extracted at each length have exactly equal signed contractions. A same-position pair with unequal contractions is not sufficient.

Proof. Assume $\alpha = \dim \mu < \min\{1, s(\Phi, p)\}$ and fix $q > 1$. For all sufficiently large n , the labelled chain rule gives

$$\begin{aligned} H_{\tilde{V}_n}(\tilde{\mathcal{D}}_{b_n(q),n}) &= H_{\tilde{V}_n}(\tilde{\mathcal{D}}_{\ell_n,n}) \\ &\quad + H_{\tilde{V}_n}(\tilde{\mathcal{D}}_{b_n(q),n} \mid \tilde{\mathcal{D}}_{\ell_n,n}). \end{aligned}$$

L-II.6.07 and T-II.6.10 therefore imply

$$\frac{H_{\tilde{V}_n}(\tilde{\mathcal{D}}_{b_n(q),n})}{\ell_n \log 2} \longrightarrow \alpha. \quad (\text{II.6.N11a})$$

For the word law p_u , expansion of $\log p_u$ gives

$$- \sum_{u \in \Lambda^n} p_u \log p_u = - \sum_{j=1}^n \sum_{i \in \Lambda} p_i \log p_i = nh(p). \quad (\text{II.6.N11b})$$

Moreover $nh(p)/(\ell_n \log 2) \rightarrow h(p)/\chi = s(\Phi, p) > \alpha$. Thus, for every sufficiently large n ,

$$H_{\tilde{V}_n}(\tilde{\mathcal{D}}_{b_n(q),n}) < nh(p).$$

Consider the map from the finite word set to labelled grid cells,

$$u \mapsto (\mathcal{D}_{b_n(q)}(a_u), r_u).$$

If it were injective, the positive masses of its image cells would be exactly the word masses p_u , so its entropy would equal (II.6.N11b). Consequently there are distinct length- n words u, v with

$$r_u = r_v, \quad \mathcal{D}_{b_n(q)}(a_u) = \mathcal{D}_{b_n(q)}(a_v).$$

Their locations belong to the same half-open interval of length $2^{-b_n(q)}$, whence

$$\Delta_n(\Phi) \leq |a_u - a_v| < 2^{-b_n(q)}. \quad (\text{II.6.N11c})$$

Given any $A > 0$, choose $q > 1$ so large that $q\chi > A$. The floors in $\ell_n, b_n(q)$ do not change the limit $b_n(q) \log 2/n \rightarrow q\chi$. Hence $2^{-b_n(q)} < e^{-An}$ for every sufficiently large n . Combining this with (II.6.N11c) proves the asserted all-large-length super-exponential concentration. The colliding labels are equal signed products; equality of positions alone was not used.

For the set conclusion, put $s = s_\Phi$ and $p_i = |r_i|^s$. L-I.7.05 gives a strict probability vector, and

$$h(p) = - \sum_i |r_i|^s \log |r_i|^s = s\chi(\Phi, p).$$

The invariant probability is carried by K_Φ . The measure-dimension definition D-I.4.03, together with its exact dimensionality, gives $\dim \mu_{\Phi, p} \leq \dim_H K_\Phi$. Thus a strict set drop below $\min\{1, s_\Phi\}$ implies the strict measure drop already handled. In a one-map system the purported strict drop below the natural bound zero is impossible, so no exceptional case has been omitted. □

The separation corollary is the contrapositive of the proved dimension-drop implication, combined with the natural upper bounds. Even an exponential lower bound on an infinite subsequence of lengths suffices.

Zero weights are handled by restricting to their active maps. A fixed affine change of coordinates multiplies every finite equal-contraction gap by the same positive factor and cannot change the decay regime.

An exact overlap forces zero gaps at every later generation. The corollary does not reverse this into a claim that every exact overlap must cause dimension drop. In particular, the two directions should not be conflated with an unrestricted exact-overlaps conjecture.

Corollary II.6.3.8 (C-II.6.12 — Distance conventions and separation consequences).

The following are equivalent: an exact overlap; $\Delta_n = 0$ at some generation; and $\Delta_n = 0$ at every sufficiently large generation. Equality of maps at two different word lengths also implies an equal-length exact overlap.

For every nonzero affine conjugacy $A_{c,b}$, $\Delta_n(A_{c,b}\Phi A_{c,b}^{-1}) = |c|\Delta_n(\Phi)$, with the usual extended-value convention. Super-exponential concentration and the declared exponential-separation condition are therefore invariant.

If Φ is exponentially separated in the sense of D-II.6.02, then

$$\dim \mu_{\Phi, p} = \min\{1, s(\Phi, p)\} \quad \text{for every probability vector } p, \quad \dim_H K_\Phi = \min\{1, s_\Phi\}. \quad (\text{II.6.16})$$

The vector may have zero entries; a single active map is included. No converse asserting

that an exact overlap forces dimension drop is claimed. For comparison, see Hochman [Hoc14, pp. 3–5] and the conventions of I.7.

Proof. Two affine word maps coincide exactly when their slopes and translations coincide. Because the minimum in Δ_n ranges over finitely many pairs, $\Delta_n = 0$ is equivalent to an equal-length exact overlap. If $\varphi_u = \varphi_v$ for distinct length- N words, then, for any word w , also $\varphi_{uw} = \varphi_{vw}$. The concatenations remain distinct. The alphabet is nonempty, so one can choose w of every required length and obtain $\Delta_n = 0$ for all $n \geq N$. This proves all three equivalences.

Suppose now that distinct-length words u, v define the same map. Neither can be empty, since a nonempty word has contraction magnitude less than one and cannot be the identity. Neither is a proper prefix of the other: if, for example, $v = uw$ with $w \neq \emptyset$, then

$$\varphi_u = \varphi_v = \varphi_u \circ \varphi_w \implies \varphi_w = \text{id},$$

by invertibility of φ_u , a contradiction. The words uv, vu must be distinct. Indeed, if they were equal and $|u| < |v|$, comparison of their first $|u|$ symbols would make u a proper prefix of v ; the other length order is symmetric. They have equal length and

$$\varphi_{uv} = \varphi_u \circ \varphi_v = \varphi_u \circ \varphi_u = \varphi_v \circ \varphi_u = \varphi_{vu}.$$

This gives an equal-length exact overlap.

For $A_{c,b}(x) = cx + b$, direct composition gives

$$A_{c,b} \circ \varphi_u \circ A_{c,b}^{-1}(x) = r_u x + ca_u + (1 - r_u)b. \quad (\text{II.6.N12a})$$

The signed products are unchanged. For a pair with $r_u = r_v$, the translation difference is $c(a_u - a_v)$; pairs with unequal products still have infinite distance. Taking the minimum proves $\Delta_n(A_{c,b}\Phi A_{c,b}^{-1}) = |c|\Delta_n(\Phi)$, also when the minimum is zero or infinite. A fixed positive multiplicative factor preserves super-exponential concentration: for a prescribed $A > 0$, an eventual bound at exponent $A + 1$ absorbs that factor for all large n . Apply this in both directions, using $|c|$ and $1/|c|$.

The exact logical negation of the super-exponential condition is

$$\exists C > 0 \quad \forall N \quad \exists n \geq N, \quad \Delta_n(\Phi) \geq e^{-Cn}. \quad (\text{II.6.N12b})$$

This is precisely the declared subsequence exponential-separation condition. It too is therefore invariant under affine conjugacy.

Assume this separation condition. For a strict probability vector, T-II.6.11 contrapositively gives $\dim \mu_{\Phi,p} \geq \min\{1, s(\Phi, p)\}$. The opposite inequality is T-I.8.03, so equality follows. Likewise the set part of T-II.6.11 gives $\dim_{\text{H}} K_{\Phi} \geq \min\{1, s_{\Phi}\}$, and T-I.7.06 gives the matching upper bound.

For a vector with zero entries, restrict to its nonempty active alphabet Λ_p . Its invariant probability, entropy and Lyapunov exponent are unchanged. Every active pair was also a pair in the original system, so

$$\Delta_n(\Phi|_{\Lambda_p}) \geq \Delta_n(\Phi) \quad (n \geq 1).$$

The same subsequence separation bound holds, and the strict-vector argument applies on the active alphabet. If there is only one active map $x \mapsto r_i x + a_i$, its invariant probability is the point mass at $a_i/(1 - r_i)$, by T-I.7.04; its entropy quotient is zero, as required. This completes every assertion in (II.6.16). Nothing in this argument states that an exact overlap necessarily causes a strict dimension drop. $\square$

Main theorem achieved. A dimension deficit has now produced super-exponentially close cylinders with the same signed contraction. The contraction label is exactly what turns a collision of positions into a collision of affine maps; without it, different slopes could share the same observed translation.

II.6.4 Arithmetic rigidity

Definition II.6.4.1 (II.6.06: Algebraic data, coefficient height and multiplication). A real number is algebraic over $\mathbb{Q}$ if it is a root of a nonzero polynomial with integer coefficients. A subfield of $\mathbb{R}$ is a subset containing 0, 1 and closed under addition, subtraction, multiplication and division by its nonzero elements. For a finite tuple $A = (\alpha_1, \dots, \alpha_s)$ of such numbers, $\mathbb{Q}(A)$ denotes the smallest subfield of $\mathbb{R}$ containing $\mathbb{Q}$ and all α_j . Its finite-dimensionality is part of L-II.6.13.

For a polynomial in s formal variables with integer coefficients, write

$$P(X) = \sum_{\gamma \in \mathbb{N}_0^s} c_\gamma X^\gamma, \quad X^\gamma = \prod_{j=1}^s X_j^{\gamma_j}, \quad |\gamma| = \sum_j \gamma_j, \quad \|P\|_1 = \sum_\gamma |c_\gamma|.$$

Only finitely many coefficients are nonzero. For nonzero P , its total degree is $\max\{|\gamma| : c_\gamma \neq 0\}$; the zero polynomial satisfies every nonnegative degree bound. The quantity $\|P\|_1$ is the coefficient L^1 height used here. It is not a logarithmic height or a height of an algebraic number. An exponential-height class at length n means $\deg P \leq n$, $\|P\|_1 \leq L^n$, with a fixed $L \geq 1$.

A subfield $K \subset \mathbb{R}$ containing $\mathbb{Q}$ is a field extension of $\mathbb{Q}$; its dimension over $\mathbb{Q}$ is its dimension as a rational vector space. If that dimension is finite, multiplication by $x \in K$ is the rational linear map

$$M_x : K \longrightarrow K, \quad M_x(y) = xy.$$

Its regular-representation matrix means its matrix in a specified $\mathbb{Q}$ -basis. A change of basis does not change its determinant. No assertion about field automorphisms or algebraic conjugates is implicit.

The word translations are polynomial expressions in finitely many fixed algebraic parameters. Two controls matter: their degrees grow at most linearly with word length,

and their coefficient sums grow at most exponentially. A degree restriction without a coefficient restriction would not give the needed lower bound.

L-II.6.13 realizes these algebraic expressions as rational multiplication matrices in one fixed finite-dimensional field. A nonzero expression gives an invertible matrix. Clearing the generator denominators bounds its determinant away from zero exponentially; the coefficient-height bound controls its matrix norm exponentially. The real value of the expression is an eigenvalue, so expanding the determinant isolates that value and gives an exponential lower bound for it.

Thus a cylinder gap is either exactly zero or not too small. Combined with the dimension-drop theorem, this shows that an algebraic system without exact overlaps has its natural dimensions. This implication does not claim that every exact overlap must lower dimension.

Lemma II.6.4.2 (L-II.6.13 – Exponential-height algebraic separation). *For every finite tuple A of real algebraic numbers and every $L \geq 1$, there is $c = c(A, L) \in (0, 1)$ such that for every integer $n \geq 1$ and every integer polynomial P with $\deg P \leq n$, $\|P\|_1 \leq L^n$,*

$$P(A) = 0 \quad \text{or} \quad |P(A)| \geq c^n. \quad (\text{II.6.17})$$

The proof also establishes finite-dimensionality of $\mathbb{Q}(A)$ and the multiplication-matrix determinant estimate it uses. No number-field or Galois theorem is invoked without proof. For comparison, see the bounded-height argument in Hochman [Hoc14, Lemma 5.10, p. 40]. No bound for unrestricted coefficients at fixed degree is asserted.

Proof. Let $A = (\alpha_1, \dots, \alpha_s)$ be the fixed algebraic tuple. For each j , choose an integer polynomial vanishing at α_j , of degree $d_j \geq 1$ with nonzero leading coefficient. Solving its relation for $\alpha_j^{d_j}$ expresses that power as a rational linear combination of lower powers. Repeated substitution expresses every nonnegative power of α_j as such a combination. Consequently

$$V = \text{span}_{\mathbb{Q}}\{\alpha_1^{e_1} \cdots \alpha_s^{e_s} : 0 \leq e_j < d_j\}$$

is a finite-dimensional rational vector subspace of $\mathbb{R}$, contains 1 and all the α_j , and is closed under multiplication. If $x \in V \setminus \{0\}$, multiplication by x is injective on V , since $V \subset \mathbb{R}$. An injective linear map from a finite-dimensional vector space to itself is surjective. Thus $xy = 1$ for some $y \in V$. It follows that V is a field. Every spanning monomial belongs to $\mathbb{Q}(A)$, whereas V is a field containing $\mathbb{Q}$ and A . These two inclusions give $V = \mathbb{Q}(A)$.

Put $d = \dim_{\mathbb{Q}} V$, and choose a rational basis $b_1 = 1, b_2, \dots, b_d$. Let T_j be the matrix of multiplication by α_j in this basis, with column vectors as coordinates. All its entries are rational. Choose one integer $D \geq 1$ so that every entry of every DT_j is an integer, and put

$$B_1 = \max\{1, \|T_1\|, \dots, \|T_s\|\}, \quad B = LB_1.$$

Here $\|T\| = \sup_{|z|=1} |Tz|$ is the real Euclidean operator norm. It satisfies $\|ST\| \leq \|S\|\|T\|$, by its definition. Multiplication maps satisfy $M_x M_y = M_{xy}$ and are rational linear in x . Therefore, for a polynomial P in the stated degree and height class and $x = P(A)$, its multiplication matrix is $M = P(T_1, \dots, T_s)$, and

$$(D^n M)_{ab} \in \mathbb{Z} \quad (1 \leq a, b \leq d), \quad \|M\| \leq \sum_{\gamma} |c_{\gamma}| B_1^{|\gamma|} \leq B^n. \quad (\text{II.6.N13a})$$

The integer-entry assertion holds because the term of degree $|\gamma| \leq n$ acquires the integer factor $D^{n-|\gamma|}$ after the denominators in its product of generators are cleared. If $x \neq 0$, multiplication by x is invertible. Its determinant is consequently nonzero, and (II.6.N13a) gives

$$|\det M| \geq D^{-nd}. \quad (\text{II.6.N13b})$$

For the opposite determinant estimate use the nonzero real row $v = (b_1, \dots, b_d)$. Evaluation of the coordinate identity for multiplication by x gives $vM = xv$. Extend $v/|v|$ to an orthonormal basis of real row vectors, and let O have these rows. Then OMO^T has first row $(x, 0, \dots, 0)$, has the same determinant as M , and has operator norm $\|M\|$. Every entry of its lower-right $(d-1)$ -by- $(d-1)$ submatrix has magnitude at most B^n . Expansion of that submatrix determinant into its $(d-1)!$ signed products, followed by expansion along the first row, gives

$$|\det M| \leq |x| (d-1)! B^{n(d-1)}. \quad (\text{II.6.N13c})$$

For $d = 1$, the empty submatrix determinant and $0!$ both mean one. Combining (II.6.N13b) and (II.6.N13c), and using $n \geq 1$, we may take

$$c = (2(d-1)! D^d B^{d-1})^{-1} \in (0, 1), \quad |P(A)| \geq c^n \quad \text{if } P(A) \neq 0. \quad (\text{II.6.N13d})$$

All constants depend only on the fixed tuple and L , through a basis and denominator bound chosen once. If the tuple is empty, the polynomial is an integer constant; a nonzero one has magnitude at least one, and $c = 1/2$ works. This also proves the required finite-dimensionality and matrix assertions, without a premise about algebraic conjugates or field automorphisms. □

Theorem II.6.4.3 (T-II.6.14 – Algebraic cylinder gaps and natural dimension). *If every r_i, a_i is real algebraic, there is $c \in (0, 1)$ such that*

$$\Delta_n(\Phi) \in \{0\} \cup [c^n, +\infty] \quad (n \geq 1). \quad (\text{II.6.18})$$

If there are no exact overlaps, all set and measure conclusions (II.6.16) hold, including zero weights. In particular any strict drop below the relevant natural bound forces an exact overlap. For comparison, see Hochman [Hoc14, Theorem 1.5, p. 6, and Lemma 5.10, p. 40]. The proof below includes the word-polynomial coefficient bound.

Proof. Use a formal variable R_i and a formal variable A_i for each map. For a word $u = u_1 \cdots u_n$, its location is the evaluation of the integer polynomial

$$Q_u(R, A) = \sum_{j=1}^n A_{u_j} \prod_{\ell < j} R_{u_\ell}. \quad (\text{II.6.N14a})$$

Every displayed monomial has total degree $j \leq n$. Even when some monomials coincide, their combined coefficient L^1 height is at most n . Thus $Q_u - Q_v$ has degree at most n and height at most $2n \leq 2^n$, for every $n \geq 1$. The last inequality holds at $n = 1$, and its inductive step follows from $2(n+1) \leq 4n$.

Apply [L-II.6.13](#) to the fixed tuple of all r_i, a_i , with $L = 2$. There is a single $c \in (0, 1)$, independent of n, u, v , such that

$$a_u - a_v = 0 \quad \text{or} \quad |a_u - a_v| \geq c^n.$$

Only pairs with equal signed contraction products enter Δ_n . Taking their finite minimum, with the declared empty-minimum convention, gives

$$\Delta_n \in \{0\} \cup [c^n, \infty) \quad (n \geq 1). \quad (\text{II.6.N14b})$$

If there is no exact overlap, [C-II.6.12](#) excludes the zero alternative at every length. The resulting lower bound is exponential separation, so that same corollary proves the natural dimension formula for every probability vector, including vectors with zero entries, and the natural dimension formula for the attractor. Contrapositively, a strict dimension drop for one of these measures or for the attractor forces an exact overlap. The theorem does not assert the reverse implication from an exact overlap to a strict drop. □

Nonzero algebraic cylinder gaps are bounded below exponentially. The super-exponential alternative can therefore occur only through an exact overlap, giving the natural-dimension formulas for algebraic systems.

II.6.5 Irrational gasket projections

Definition II.6.5.1 (II.6.11: Gasket projections and the symmetric two-map family). Let $\mathcal{V} = \{(0, 0), (1, 0), (0, 1)\}$, and define

$$F = \left\{ \sum_{j=1}^{\infty} 3^{-j} v_j : v_j \in \mathcal{V} \right\}, \quad \Pi_t(x, y) = tx + y, \quad F_t = \Pi_t(F).$$

For a probability vector p on $\mathcal{V}$, let γ_p be the coding probability on F , and $\gamma_{t,p} = (\Pi_t)_* \gamma_p$. Equivalently F_t has the three labelled maps $x \mapsto x/3$, $x \mapsto (x+t)/3$, $x \mapsto (x+1)/3$, with the corresponding vector p .

For $0 < \lambda < 1$, the symmetric Bernoulli convolution convention used in the II.6 application is

$$\nu_\lambda = \text{Law} \left(\sum_{j=0}^{\infty} \epsilon_j \lambda^j \right), \quad \mathbb{P}(\epsilon_j = 1) = \mathbb{P}(\epsilon_j = -1) = \frac{1}{2},$$

where the signs are mutually independent. It is the invariant probability of $x \mapsto \lambda x + 1$, $x \mapsto \lambda x - 1$, with equal weights. This convention is exported to II.8; other normalizations must be related by an explicit affine map.

Finally

$$\mathcal{P}_n = \left\{ \sum_{j=0}^n c_j T^j : c_j \in \{-1, 0, 1\} \right\} \quad (n \geq 0)$$

is a set of polynomials, including zero. A theorem asking for a nonzero polynomial must explicitly exclude zero.

Its word gaps have the form $x_n + t y_n$, with x_n, y_n lying on a rational grid of mesh 3^{-n} . A very small nonzero gap produces a very accurate rational approximation to the slope t . Distinct rational numbers with controlled denominators cannot be too close to one another. Approximants supplied at two consecutive lengths are closer than this separation permits, so they must be equal.

The dimension-drop theorem supplies the small gaps at every sufficiently large length. That is the reason the consecutive comparison applies. An arbitrary sparse sequence of excellent rational approximations would not justify the conclusion that t is rational. The all-weight dimension statement follows from the same separation argument; no general projection theorem is used.

Lemma II.6.5.2 (L-II.6.15 — Consecutive rational approximants become constant). *Let $B > 1$, $C \geq 1$, $0 < a < B^{-2}$, and $t \in \mathbb{R}$. Suppose that for every sufficiently large integer n there are integers p_n , $1 \leq q_n \leq CB^n$, with*

$$\left| t - \frac{p_n}{q_n} \right| \leq a^n.$$

Then p_n/q_n is eventually constant and t is rational. The hypothesis is at every sufficiently large n , not on an arbitrary subsequence. Compare the Solomyak–Shmerkin argument presented with Hochman [Hoc14, Theorem 1.6, pp. 7–8].

Proof. Let $z_n = p_n/q_n$, where $p_n \in \mathbb{Z}$ and $1 \leq q_n \leq CB^n$, for every sufficiently large integer n . If $z_{n+1} \neq z_n$, the numerator of their difference over the common denominator $q_n q_{n+1}$ is a nonzero integer. Hence

$$|z_{n+1} - z_n| \geq \frac{1}{q_n q_{n+1}} \geq C^{-2} B^{-(2n+1)}. \quad (\text{II.6.N15a})$$

The assumed approximation estimates, on the other hand, give

$$|z_{n+1} - z_n| \leq |z_{n+1} - t| + |z_n - t| \leq (1 + a)a^n. \quad (\text{II.6.N15b})$$

The ratio of the last bound to the last lower bound in (II.6.N15a) is $C^2 B(1 + a)(aB^2)^n$, which tends to zero because $aB^2 < 1$. Thus $z_{n+1} = z_n$ for every sufficiently large n . All the z_n from that point onward equal a single rational number. Since $0 < a < 1$, their errors

tend to zero, so that number is t . The use of two consecutive indices is essential; this proof does not make the corresponding assertion along an arbitrary subsequence. $\square$

Theorem II.6.5.3 (T-II.6.16 — Irrational projections of the one-dimensional gasket).
For the objects of D-II.6.11, every irrational t satisfies

$$\dim_{\text{H}} F_t = 1, \quad \dim \gamma_{t,p} = \frac{h(p)}{\log 3} \quad \text{for every probability vector } p. \quad (\text{II.6.19})$$

For comparison, see Hochman [Hoc14, Theorem 1.6, pp. 7–8], with its explicit Solomyak–Shmerkin attribution. The all-weight measure clause follows from the same cylinder argument and T-II.6.11; no general projection theorem is used.

Proof. Let $V = \{(0, 0), (1, 0), (0, 1)\}$. The series defining the gasket converges absolutely, and its image under $\Pi_t(x, y) = tx + y$ is

$$F_t = \left\{ \sum_{j \geq 1} 3^{-j}(tx_j + y_j) : (x_j, y_j) \in V \right\}. \quad (\text{II.6.N16a})$$

The coding theorem T-I.7.03 identifies this set with the attractor of the three labelled maps $z \mapsto z/3, (z+t)/3, (z+1)/3$. The same operation on the Bernoulli coding law identifies $\gamma_{t,p}$ with the self-similar probability for these maps and weights p . All three signed contraction ratios are $1/3$.

For two distinct words u, v of length n , their location difference has the form

$$a_u - a_v = x_n + ty_n, \quad x_n = \sum_{j=1}^n 3^{-j}(u_{j,2} - v_{j,2}), \quad y_n = \sum_{j=1}^n 3^{-j}(u_{j,1} - v_{j,1}). \quad (\text{II.6.N16b})$$

Both x_n, y_n belong to $3^{-n}\mathbb{Z}$ and have magnitude less than $1/2$. A finite series with coefficients in $\{-1, 0, 1\}$ cannot vanish unless all its coefficients vanish: if its first nonzero coefficient has index j , the magnitude of that term is 3^{-j} , whereas the sum of magnitudes of all remaining terms is less than $3^{-j}/2$. Consequently $x_n = y_n = 0$ would force the coordinate digits of the two words to agree at every position, contrary to $u \neq v$.

Suppose that this system has super-exponential concentration. At every sufficiently large n , choose a distinct pair with $|x_n + ty_n| < 30^{-n}$. If $y_n = 0$, the preceding paragraph gives $x_n \neq 0$ and $|x_n| \geq 3^{-n}$, a contradiction. Thus $|y_n| \geq 3^{-n}$, and

$$\left| t - \frac{-x_n}{y_n} \right| < 10^{-n}. \quad (\text{II.6.N16c})$$

Write the fraction in (II.6.N16c) with a positive integer denominator after cancellation. Its denominator is at most $|3^n y_n| < 3^n/2$, and in particular is at most 3^n . L-II.6.15, with $B = 3, C = 1, a = 1/10 < 1/9$, now implies that t is rational.

For irrational t , super-exponential concentration is therefore impossible. C-II.6.12 gives exponential separation and both natural dimension formulas. The similarity dimension of the three-map system is 1, since $3(1/3)^1 = 1$; and $h(p) \leq \log 3$ by L-II.1.02. We obtain

$$\dim_{\text{H}} F_t = 1, \quad \dim \gamma_{t,p} = \frac{h(p)}{\log 3}. \quad (\text{II.6.N16d})$$

This holds for every probability vector on the three labels. Restriction to the active alphabet, including the single-map case, is already part of C-II.6.12. No general projection theorem is used in this application. $\square$

The phrase *for every sufficiently large generation* is decisive here. Consecutive rational approximants must then stabilize, whereas an estimate on an arbitrary subsequence would not force that rigidity.

II.6.6 Analytic families and exceptional parameters

Definition II.6.6.1 (II.6.07: Real analytic functions and weighted power-series spaces). A real-valued function on an open real interval J is real analytic if at every $t_0 \in J$ there is $R > 0$ and real coefficients c_j such that

$$(t_0 - R, t_0 + R) \subset J, \quad f(t_0 + x) = \sum_{j=0}^{\infty} c_j x^j \quad (|x| < R),$$

where the power series converges absolutely on every smaller closed interval. A function on a compact interval is called analytic near it only when a specified such extension is given on an open interval containing it. Identity and limit assertions are proved in L-II.6.17.

For $R > 0$, let $\mathcal{A}_R$ be the vector space of real coefficient sequences, written as formal power series $F(z) = \sum_{j \geq 0} c_j z^j$, with

$$\|F\|_R = \sum_{j \geq 0} |c_j| R^j < \infty.$$

Addition is coefficientwise and multiplication is the Cauchy product. Evaluation means the convergent real power series for $|z| \leq R$; termwise derivatives are considered only for $|z| < R$. A real normed algebra here means a real normed vector space with an associative bilinear multiplication satisfying $\|FG\| \leq \|F\|\|G\|$. It is complete if every norm-Cauchy sequence converges in that norm to an element of the space. Completeness, multiplication bounds and derivative estimates are proved rather than assumed here.

For an integer $k \geq 0$ and a function f of class C^k near a compact interval I , put

$$\|f\|_{I, C^k} = \max_{0 \leq j \leq k} \sup_{t \in I} |f^{(j)}(t)|.$$

The same notation applies to restrictions with one-sided endpoint derivatives. All analytic functions used below have specified open extensions, so their endpoint derivatives are unambiguous. The order- k jet at t means the tuple $(f(t), f'(t), \dots, f^{(k)}(t))$. A zero infinite jet means $f^{(j)}(t) = 0$ for every integer $j \geq 0$.

For distinct real nodes $x_0, \dots, x_p$ and a function specified at these nodes, its order- p divided difference is

$$[x_0, \dots, x_p]f = \sum_{j=0}^p \frac{f(x_j)}{\prod_{\ell \neq j} (x_j - x_\ell)}.$$

For $p = 0$ the empty denominator product is one. The mean-value identity used for smooth functions is proved in L-II.6.19; it is not assumed by this definition.

Definition II.6.6.2 (II.6.08: Analytic families and infinite coding nondegeneracy). Fix a nondegenerate compact interval $I = [a, b]$, $a < b$, in a connected open real interval J . An analytic similarity family consists of finitely many real analytic functions $r_i, a_i : J \rightarrow \mathbb{R}$, $i \in \Lambda$, with $0 < |r_i(t)| < 1$ for every $t \in J$, and maps

$$\varphi_{i,t}(x) = r_i(t)x + a_i(t).$$

The attractor K_t , coding map π_t and invariant probability $\mu_{t,p}$ are the I.7 objects for this system. Put

$$\underline{r} = \min_{\substack{i \in \Lambda \\ t \in I}} |r_i(t)| > 0, \quad \bar{r} = \max_{\substack{i \in \Lambda \\ t \in I}} |r_i(t)| < 1.$$

For $\omega, \tau \in \Lambda^{\mathbb{N}}$, their coding difference is $F_{\omega,\tau}(t) = \pi_t(\omega) - \pi_t(\tau)$. The family is infinitely coding-nondegenerate on I if

$$\omega \neq \tau \implies \exists t \in I \quad F_{\omega,\tau}(t) \neq 0.$$

This permits overlaps at individual parameters. It forbids identical coding functions on the entire interval. It does not require a single parameter at which every pair is separated simultaneously. Analyticity of these coding functions is L-II.6.17, not a definitional consequence of their pointwise existence.

Pointwise convergence of coding series alone does not prove that their parameter derivatives converge. Even uniform convergence of real analytic functions alone is not the required analytic-limit theorem. The weighted power-series norm in L-II.6.17 controls the coefficients, multiplication and every derivative on a smaller radius.

Near any fixed parameter, all contraction series have norm less than one after the radius is made small enough. The coding series then converges geometrically in that norm, and so do its derivative series on a smaller interval. A finite cover makes these estimates uniform on the chosen compact parameter interval.

Nondegeneracy says that no two distinct infinite codes give identical functions through-

out the interval. Analyticity turns this into a nonzero derivative of some finite order at every parameter. Compactness of the different-first-symbol code pairs makes one finite maximum order suffice uniformly. Removing a long common prefix costs only a polynomial factor in its length in addition to its contraction product. The reciprocal-product calculation proves that cost, including the empty prefix and negative contractions.

Lemma II.6.6.3 (L-II.6.17 — Power-series control and analytic coding). *The spaces $\mathcal{A}_R$ are complete normed algebras, with $\|FG\|_R \leq \|F\|_R \|G\|_R$. For $0 < r < R$ and $j \geq 0$, there is $C(j, r, R) < \infty$ such that*

$$\sup_{|x| \leq r} |F^{(j)}(x)| \leq C(j, r, R) \|F\|_R.$$

A real analytic function on a connected open interval which has all derivatives zero at a point of that interval is identically zero there. A function zero on a nondegenerate subinterval satisfies that hypothesis.

For an analytic family of D-II.6.08, every coding function $t \mapsto \pi_t(\omega)$ is analytic on J . On each compact $I \subset J$, for every $j \geq 0$ there are $C_j < \infty$ and $0 < q_j < 1$ such that

$$\sup_{\omega} \|\pi(\omega)\|_{I, C^j} \leq C_j, \quad \|\pi(\omega) - \pi(\tau)\|_{I, C^j} \leq C_j q_j^n$$

whenever ω, τ agree in their first n symbols. All anchored finite differences consequently have uniformly bounded C^j norms, independently of word length.

For comparison, see the analytic and derivative convergence step in Hochman [Hoc14, Proposition 5.7, p. 38]. The real power-series proof below supplies all analytic machinery needed here.

Proof. We place the coding series in a weighted Banach space, prove uniform derivative bounds on compact parameter intervals, and then invoke the identity theorem only after analyticity has been established.

Fix $R > 0$. If $F_m(z) = \sum_{\ell \geq 0} c_{m, \ell} z^\ell$ is Cauchy in $\mathcal{A}_R$, every coefficient sequence $(c_{m, \ell})_m$ is Cauchy in $\mathbb{R}$, since its difference is bounded in magnitude by $R^{-\ell} \|F_m - F_{m'}\|_R$. Write its limit as c_ℓ . The norms of F_m are bounded, and Fatou's inequality for a series of nonnegative numbers gives

$$\sum_{\ell \geq 0} |c_\ell| R^\ell \leq \liminf_m \|F_m\|_R < \infty, \quad \sum_{\ell \geq 0} |c_{m, \ell} - c_\ell| R^\ell \leq \liminf_{m' \rightarrow \infty} \|F_m - F_{m'}\|_R. \quad (\text{II.6.N17a})$$

The Cauchy property makes the last expression tend to zero as $m \rightarrow \infty$. Thus $F = \sum c_\ell z^\ell$ belongs to $\mathcal{A}_R$, and $F_m \rightarrow F$ in its norm. This proves completeness. The ordinary norm axioms follow immediately from the coefficient formula.

For $F = \sum c_\ell z^\ell$ and $G = \sum d_\ell z^\ell$, absolute summation gives

$$\|FG\|_R \leq \sum_{n \geq 0} \sum_{\ell=0}^n |c_\ell| |d_{n-\ell}| R^n = \left(\sum_{\ell \geq 0} |c_\ell| R^\ell \right) \left(\sum_{m \geq 0} |d_m| R^m \right). \quad (\text{II.6.N17b})$$

The Cauchy product is bilinear and associative: coefficientwise this is the equality of the finite sums indexed by the decompositions of an integer into two or three nonnegative integers. The constant series 1 is an identity for multiplication. Formula (II.6.N17b) proves the asserted normed-algebra property; it also justifies evaluation of products by absolutely convergent series.

For $j \geq 0$ and $\ell \geq j$, put $(\ell)_j = \ell(\ell-1) \cdots (\ell-j+1)$, with $(\ell)_0 = 1$. Given $0 < r < R$, define

$$C(j, r, R) = r^{-j} \sup_{\ell \geq j} (\ell)_j (r/R)^\ell < \infty.$$

The supremum is finite because a fixed polynomial in ℓ times a geometric sequence with ratio less than one is bounded. For every $|x| \leq r$,

$$\sum_{\ell \geq j} |c_\ell| (\ell)_j |x|^{\ell-j} \leq C(j, r, R) \|F\|_R. \quad (\text{II.6.N17c})$$

The same estimate on the coefficient tail proves uniform absolute convergence of the formal j -th derivative on $[-r, r]$. These are actual derivatives: the polynomial partial sums and their first formal derivatives converge uniformly there, so passing to the limit in their fundamental-theorem-of-calculus identities gives $F(x) - F(0) = \int_0^x F_1(u) du$, where F_1 is the first formal derivative series. Repeating this argument for successive formal derivatives proves the claim for every j . In particular Taylor coefficients are $F^{(j)}(0)/j!$. Only real integration and uniform convergence from the allowed analysis roots have been used.

Now let f be real analytic on a connected open interval J . The preceding argument on smaller radii makes f infinitely differentiable and identifies its local power-series coefficients with its derivatives divided by factorials. The set

$$Z = \{t \in J : f^{(j)}(t) = 0 \text{ for every } j \geq 0\}$$

is relatively closed, by continuity of all derivatives. It is also relatively open: at a point of Z , the analytic power series is identically zero, so f is zero on a neighborhood and all its derivatives are zero there. If Z is nonempty, connectedness of J gives $Z = J$, hence $f = 0$ on J . If f vanishes on a nondegenerate subinterval, a point in its interior belongs to Z . This proves both identity assertions.

Consider next the analytic family and fix $t_0 \in J$. Finitely many local power-series expansions give a common positive radius on which every $r_i(t_0 + z)$ and $a_i(t_0 + z)$ belongs to $\mathcal{A}_R$, after shrinking R if necessary. As $R \downarrow 0$, the norm of the series for r_i decreases to $|r_i(t_0)|$: the constant coefficient stays fixed and the sum of positive-degree terms tends to zero by domination at any previous admissible radius. Since $\max_i |r_i(t_0)| < 1$, we can choose this radius and a number $q \in (0, 1)$ so that

$$\max_i \|r_i(t_0 + \cdot)\|_R \leq q, \quad A_0 = \max_i \|a_i(t_0 + \cdot)\|_R < \infty.$$

Take the radius small enough also that $[t_0 - R, t_0 + R] \subset J$. For a code ω , the series in the complete space $\mathcal{A}_R$

$$G_\omega(z) = \sum_{m=1}^{\infty} a_{\omega_m}(t_0 + z) \prod_{\ell < m} r_{\omega_\ell}(t_0 + z) \quad (\text{II.6.N17d})$$

converges, since the norm of its m -th term is at most $A_0 q^{m-1}$. Evaluation at each $|z| \leq R$ is a continuous linear functional of norm at most one, by the norm definition. It therefore commutes with this convergent sum. The evaluated sum is the coding formula of T-I.7.03, so $G_\omega(z) = \pi_{t_0+z}(\omega)$. It is a convergent real power series and proves analyticity of every coding function near t_0 , hence on J .

The same construction gives the uniform bounds

$$\|G_\omega\|_R \leq \frac{A_0}{1-q}, \quad \|G_\omega - G_\tau\|_R \leq \frac{2A_0 q^n}{1-q} \quad \text{if } \omega_\ell = \tau_\ell \ (1 \leq \ell \leq n). \quad (\text{II.6.N17e})$$

The difference estimate follows by cancellation of the first n terms and summing the two tails. Apply (II.6.N17c), for each derivative order up to j , on the smaller radius $R/2$. This bounds the C^j norms there by a fixed multiple of the corresponding quantities in (II.6.N17e).

Finally cover a given compact $I \subset J$ by finitely many of these smaller open intervals. The maximum of their finitely many geometric ratios is still less than one, and the maximum of their constants is finite. Enlarging that constant to cover both estimates gives the chosen C_j, q_j . Each anchored finite difference is the difference of the coding functions for $u\omega_*, v\omega_*$. The uniform individual C^j bound consequently bounds it by $2C_j$, independently of the word length. □

Definition II.6.6.4 (II.6.09: Anchored differences and higher-order transversality). Fix one infinite tail $\omega_* \in \Lambda^\mathbb{N}$, once for the entire family, and put $b_*(t) = \pi_t(\omega_*)$. For distinct equal-length words u, v , define

$$\Delta_{u,v}^*(t) = \varphi_{u,t}(b_*(t)) - \varphi_{v,t}(b_*(t)) = \pi_t(u\omega_*) - \pi_t(v\omega_*).$$

Let $w = u \wedge v$ be their longest common prefix; it can be empty, and $|w| \leq |u| - 1$. Products $r_{w,t}$ use D-II.6.01, including $r_{\emptyset,t} = 1$. These anchored differences are distinct from the equal-contraction distance, but if $r_{u,t} = r_{v,t}$ then $\Delta_{u,v}^*(t) = a_{u,t} - a_{v,t}$. The anchored family has prefix-controlled transversality of order $k \geq 1$ on I if its anchored differences are C^k near I and there is $c > 0$ such that

$$\max_{0 \leq j \leq k} |(\Delta_{u,v}^*)^{(j)}(t)| \geq \frac{c |r_{w,t}|}{(1 + |w|)^k}$$

for every length, every distinct equal-length pair and every $t \in I$. The maximum includes $j = 0$. This deliberately uses a uniform prefix loss sufficient for the dimension argument; it does not assert the sharper derivative-by-derivative convention in the source. The case of a zero-length common prefix has no negative power of zero.

Proposition II.6.6.5 (P-II.6.18 — Nondegenerate coding gives finite-order transversality). *If the analytic family is infinitely coding-nondegenerate on I , then for every fixed*

tail ω_* there are an integer $k \geq 1$ and $c > 0$ such that it satisfies D-II.6.09. In particular, for every $n \geq 1$, every distinct $u, v \in \Lambda^n$ and every $t \in I$,

$$\max_{0 \leq j \leq k} |(\Delta_{u,v}^*)^{(j)}(t)| \geq \frac{c \underline{r}^n}{(n+1)^k}. \quad (\text{II.6.20})$$

For comparison, see Hochman [Hoc14, Proposition 5.7 and its preceding calculation], pp. 37–38. The proof below uses anchored finite words, absolute products, and the explicitly weakened uniform prefix loss.

Proof. Compactness first converts analytic nonidentity into a finite derivative order. Prefix factorization then makes the lower bound uniform over all word pairs.

If the alphabet has one symbol, there are no distinct equal-length word pairs, and the conclusion is vacuous with $k = 1, c = 1$. Assume there are at least two symbols, and write $\Sigma = \Lambda^{\mathbb{N}}$, with the prefix topology of I.7. Consider

$$\mathcal{X} = \{(\omega, \tau, t) \in \Sigma \times \Sigma \times I : \omega_1 \neq \tau_1\}.$$

This is compact: Σ is compact by the diagonal-subsequence argument in the proof of T-I.7.03, and the first-symbol restriction is closed. For each derivative order j , the function $(\omega, \tau, t) \mapsto F_{\omega, \tau}^{(j)}(t)$ is continuous. Indeed L-II.6.17 bounds its change under fixing longer and longer prefixes uniformly by a quantity tending to zero. It also bounds its parameter variation by a constant times $|t - s|$, using the uniform derivative bound of order $j + 1$ and the mean-value theorem on I . These two bounds together prove joint continuity.

For every point of $\mathcal{X}$, some derivative of the corresponding coding difference is nonzero there. Otherwise the identity theorem in L-II.6.17 would make that difference identically zero on J , contrary to infinite coding nondegeneracy on I . Continuity supplies a neighborhood on which the magnitude of that chosen derivative exceeds half its positive value at the point. A finite subcover of $\mathcal{X}$ thus supplies $k \geq 1$ and $c_0 > 0$ such that

$$\max_{0 \leq j \leq k} |F_{\omega, \tau}^{(j)}(t)| \geq c_0 \quad ((\omega, \tau, t) \in \mathcal{X}). \quad (\text{II.6.N18a})$$

Here k can be increased to one if the chosen derivative orders are all zero, without weakening the bound.

We next account for an arbitrarily long common prefix, including the empty one. For every symbol put $q_i = 1/r_i$, which is C^k on J since r_i is nonzero there. On the compact I , choose $B \geq 1$ so that

$$|q_i^{(j)}(t)| \leq B|q_i(t)| \quad (i \in \Lambda, 1 \leq j \leq k, t \in I).$$

Such a constant exists because the finitely many ratios are continuous and their denominators do not vanish. For a word w of length $m \geq 1$, the product rule, iterated j times, is

$$\left(\prod_{\ell=1}^m q_{w_\ell} \right)^{(j)} = \sum_{\substack{\eta_1 + \dots + \eta_m = j \\ \eta_\ell \geq 0}} \frac{j!}{\eta_1! \dots \eta_m!} \prod_{\ell=1}^m q_{w_\ell}^{(\eta_\ell)}.$$

At most j factors in each term are differentiated. The sum of the displayed multinomial coefficients is m^j , obtained by expanding $(1 + \dots + 1)^j$. For $j = 0$ the sum is one. We obtain, also for $m = 0$ by direct differentiation of the constant empty product,

$$\left| \left(\frac{1}{r_w} \right)^{(j)}(t) \right| \leq \frac{B^j(1+m)^j}{|r_w(t)|} \quad (0 \leq j \leq k). \quad (\text{II.6.N18b})$$

Fix distinct $u, v \in \Lambda^n$, and let w be their longest common prefix of length $m < n$. On removing that prefix and appending the fixed tail, obtain two codes ω, τ with different first symbols. The affine coding identity gives

$$\Delta_{u,v}^*(t) = r_w(t) F_{\omega,\tau}(t). \quad (\text{II.6.N18c})$$

At a fixed t , let $H = \max_{0 \leq j \leq k} |(\Delta_{u,v}^*)^{(j)}(t)|$. Apply the product rule to $F_{\omega,\tau} = (1/r_w)\Delta_{u,v}^*$, using (II.6.N18b). For $0 \leq j \leq k$,

$$|F_{\omega,\tau}^{(j)}(t)| \leq \frac{H}{|r_w(t)|} \sum_{\ell=0}^j \binom{j}{\ell} B^\ell (1+m)^\ell \leq \frac{H(1+B)^k(1+m)^k}{|r_w(t)|}. \quad (\text{II.6.N18d})$$

Combining (II.6.N18a) with (II.6.N18d) proves D-II.6.09 with $c = c_0/(1+B)^k$. Finally $|r_w(t)| \geq \underline{r}^m \geq \underline{r}^n$ and $1+m \leq n+1$, so (II.6.20) follows. Every contraction product in the estimate is in absolute value; negative ratios and the empty common prefix require no further convention. □

The sublevel lemma first divides the parameter interval into pieces on which one derivative stays bounded away from zero. Divided differences then rule out too many well-separated points at which the function is small. This produces a finite cover by short intervals. The extra derivative in the hypothesis is used to keep the chosen derivative stable on each piece.

At word length n , all pairs contribute at most an exponential number of intervals, while their lengths decrease at a rate that can be made arbitrarily fast by shrinking the prescribed gap threshold. The eventual-intersection sets lie in these covers at every late length. Bounded ratios between consecutive covering scales give upper box bounds, and the countable-cover characterization of packing dimension then controls the eventual unions. This is stronger than only showing a Hausdorff-dimension bound along a sparse scale sequence.

The coding-point anchor ensures that a finite-word difference is itself an infinite-coding difference. For equal contraction products, that anchor cancels exactly. Hence any super-exponential cylinder concentration lies inside the one constructed Borel exceptional set, and every weight vector has its natural dimension outside that set. The exceptional set is a cover of possible dimension-drop parameters, not a claim to identify them exactly.

Lemma II.6.6.6 (L-II.6.19 – Quantitative sublevel covers with one extra derivative).
 Let I be a compact interval, $k \geq 1$, $M \geq 1$, $0 < c \leq 1$, and f be C^{k+1} near I , with

$$\|f\|_{I, C^{k+1}} \leq M, \quad \max_{0 \leq j \leq k} |f^{(j)}(t)| \geq c \quad (t \in I).$$

For every $0 < \rho < c/4$, the set $\{t \in I : |f(t)| < \rho\}$ can be covered by at most $C_k(1 + M|I|/c)$ closed intervals, each of length at most $C_k(\rho/c)^{1/k}$, where $C_k = 32(k+1)$ is admissible. Here $|I|$ is interval length. A singleton interval is allowed. The proof uses a derivative-stability subdivision and a Rolle/divided-difference estimate, both written in full. Compare Hochman [Hoc14], Lemma 5.8, pp. 38–39; the present C^{k+1} lemma is sufficient for the analytic theorem and does not claim the source's exact C^k bound.

Proof. If I is a singleton, its sublevel set is empty or is covered by that singleton interval, so the bounds hold. Assume $|I| > 0$. Partition I into N closed intervals of equal length, where

$$N = \max\{1, \lceil 4M|I|/c \rceil\} \leq 1 + 4M|I|/c.$$

Their lengths are at most $c/(4M)$. In one such interval J_0 , choose at its left endpoint an order $p \in \{0, \dots, k\}$ with derivative magnitude at least c . The bound on $f^{(p+1)}$ and the mean-value theorem give

$$|f^{(p)}(t)| \geq 3c/4 \quad (t \in J_0). \quad (\text{II.6.N19a})$$

If $p = 0$, the interval contains no point where $|f| < \rho$, since $\rho < c/4$. Suppose $p \geq 1$. For ordered distinct nodes $x_0 < \dots < x_p$ in J_0 , the polynomial

$$Q(x) = \sum_{j=0}^p f(x_j) \frac{\prod_{\ell \neq j} (x - x_\ell)}{\prod_{\ell \neq j} (x_j - x_\ell)}$$

has degree at most p , agrees with f at those nodes, and has coefficient $[x_0, \dots, x_p]f$ at x^p . The function $f - Q$ has $p + 1$ zeros. Successive applications of Rolle's theorem give at least $p, p - 1, \dots, 1$ intervening zeros for its first through p -th derivatives. In particular there is $\xi \in (x_0, x_p)$ such that

$$f^{(p)}(\xi) = p![x_0, \dots, x_p]f. \quad (\text{II.6.N19b})$$

This proves the divided-difference identity used here.

Suppose now that all these nodes are sublevel points and that successive gaps are at least $h > 0$. Then $|x_j - x_\ell| \geq |j - \ell|h$, and hence

$$\prod_{\ell \neq j} |x_j - x_\ell| \geq h^p j!(p-j)!, \quad |f^{(p)}(\xi)| \leq \frac{p! \rho}{h^p} \sum_{j=0}^p \frac{1}{j!(p-j)!} = \frac{2^p \rho}{h^p}. \quad (\text{II.6.N19c})$$

Choose $h = (2^{p+2}\rho/c)^{1/p}$. The final bound is $c/4$, contradicting (II.6.N19a). Thus the sublevel set in J_0 has no collection of $p + 1$ points with all mutual distances at least h .

Choose a point of this set, if nonempty, and put a closed radius- h interval around it. If the set is not covered, choose an uncovered point and repeat. Each new point has distance greater than h from all previously chosen points. The preceding obstruction makes this procedure stop after at most p choices. The resulting intervals cover the sublevel set in J_0 , and each has length

$$2h = 2^{2+2/p}(\rho/c)^{1/p} \leq 16(\rho/c)^{1/k}. \quad (\text{II.6.N19d})$$

For the inequality use $1 \leq p \leq k$ and $0 < \rho/c < 1$. The covering intervals are allowed to extend past J_0 ; this does not affect the cover or their lengths. There are at most k intervals for each of the N pieces. Their total number is bounded by

$$k(1 + 4M|I|/c) \leq 4k(1 + M|I|/c).$$

Both the number bound and the length bound in (II.6.N19d) are therefore valid with the single constant $C_k = 32(k + 1)$. The derivative-stability step used the $(k + 1)$ -st derivative when $p = k$; it has not asserted an auxiliary theorem with only C^k hypotheses. $\square$

Definition II.6.6.7 (II.6.10: Eventual anchored sublevel sets). In an anchored family, for $0 < \varepsilon < 1$ and integer $n \geq 1$, define

$$E_{\varepsilon,n}^* = \bigcup_{\substack{u,v \in \Lambda^n \\ u \neq v}} \{t \in I : |\Delta_{u,v}^*(t)| < \varepsilon^n\}, \quad F_{\varepsilon,N}^* = \bigcap_{n \geq N} E_{\varepsilon,n}^*.$$

The distinct letters denote respectively the length- n sublevel union and the eventual sublevel intersection. Put

$$E_\varepsilon^* = \bigcup_{N \geq 1} F_{\varepsilon,N}^*, \quad E^* = \bigcap_{j \geq 1} E_{e^{-j}}^*.$$

Thus E^* is the set where the minimum anchored positional gap is smaller than every prescribed exponential rate at every sufficiently large length. It may contain parameters at which there is no dimension drop; it is a covering exceptional set, not a definition of dimension drop. Empty sets use the dimension conventions of I.1–I.2.

Theorem II.6.6.8 (T-II.6.20 — Anchored transversality gives packing-zero exceptions). Consider an analytic family with prefix-controlled transversality of order k . For every $0 < \varepsilon < \underline{r}$ and $N \geq 1$,

$$\overline{\dim}_B F_{\varepsilon,N}^* \leq \min \left\{ 1, \frac{k(2 \log |\Lambda| + \log(1/r))}{\log(r/\varepsilon)} \right\}. \quad (\text{II.6.21})$$

The same bound holds for $\dim_p E_\varepsilon^*$. Consequently

$$\dim_H E^* = \dim_p E^* = 0. \quad (\text{II.6.22})$$

The covering proof counts all ordered word pairs and uses the earlier modified-upper-box characterization of packing dimension. For comparison, see Hochman [Hoc14, Theorem 5.9, pp. 39–40]. The proof below uses the explicitly anchored exceptional set and the C^{k+1} sublevel bound.

Proof. For each word pair we apply the sublevel lemma, count the pairs, and estimate the upper box dimension of every fixed-onset exceptional set. Countable stability then yields packing dimension zero for the eventual exceptional set.

We prove T-II.6.20.

Write $m = |\Lambda|$ and $r = \underline{r} \in (0, 1)$. If $m = 1$, every union over distinct equal-length word pairs is empty, and the conclusions hold with the zero-dimension convention for the empty set. Assume $m \geq 2$. Prefix-controlled transversality implies, for every distinct pair of length n ,

$$\max_{0 \leq j \leq k} |(\Delta_{u,v}^*)^{(j)}(t)| \geq c_n, \quad c_n = \frac{cr^n}{(n+1)^k}, \quad t \in I. \quad (\text{II.6.N20a})$$

Indeed its prefix has length at most $n - 1$, its absolute product is at least r^n , and its length plus one is at most $n + 1$. Decrease the transversality constant c , if necessary, to have $0 < c \leq 1$. L-II.6.17 supplies $M \geq 1$, independent of n, u, v , with $\|\Delta_{u,v}^*\|_{I, C^{k+1}} \leq M$.

Fix $0 < \varepsilon < r$. Since

$$\frac{\varepsilon^n}{c_n} = c^{-1}(n+1)^k(\varepsilon/r)^n \longrightarrow 0,$$

we have $\varepsilon^n < c_n/4$ for all sufficiently large n . For each pair, L-II.6.19 covers its sublevel set $\{t \in I : |\Delta_{u,v}^*(t)| < \varepsilon^n\}$ by at most $C_k(1 + M|I|/c_n)$ intervals of length at most $C_k(\varepsilon^n/c_n)^{1/k}$. There are at most m^{2n} ordered pairs. Thus, for a fixed constant $C \geq 1$, all sufficiently large n admit a cover of $E_{\varepsilon,n}^*$ by at most

$$K_n = C(n+1)^k(m^2/r)^n \quad \text{intervals, each of length at most} \quad \rho_n = C(n+1)(\varepsilon/r)^{n/k}. \quad (\text{II.6.N20b})$$

The constant may be enlarged to serve both bounds and does not depend on n . Counts need not be integers in this upper bound.

Fix $N \geq 1$ and put $F = F_{\varepsilon,N}^*$. This set is contained in $E_{\varepsilon,n}^*$ for every $n \geq N$. If F is empty there is nothing to prove. Otherwise discard covering intervals which do not meet F , and choose a point of F in each remaining one. Its open radius- $2\rho_n$ ball contains that interval's intersection with F . In the covering-number convention of I.2 this gives

$$N_{2\rho_n}(F) \leq K_n. \quad (\text{II.6.N20c})$$

Put $a = (\varepsilon/r)^{1/k} \in (0, 1)$. The scales $2\rho_n$ tend to zero, and their successive ratios are $a(n+2)/(n+1)$. They eventually decrease and those ratios are bounded below by $a > 0$. L-I.2.02 therefore permits the box-dimension limit along these scales. Taking logarithms in (II.6.N20c) yields

$$\begin{aligned} \overline{\dim}_B F &\leq \lim_{n \rightarrow \infty} \frac{\log C + k \log(n+1) + n \log(m^2/r)}{-\log(2C) - \log(n+1) + (n/k) \log(r/\varepsilon)} \\ &= \frac{k(2 \log m + \log(1/r))}{\log(r/\varepsilon)}. \end{aligned} \quad (\text{II.6.N20d})$$

Also $F \subset I$, and an interval of length $|I|$ has a cover by at most $2 + 2|I|/\delta$ radius- δ balls with centers in the interval. By monotonicity and the box-dimension definition its subsets have upper box dimension at most one. This gives the minimum with one in (II.6.21).

By T-I.2.09, a bounded set's packing dimension is at most its upper box dimension, by using that set itself in a countable cover for modified upper box dimension. Countable stability T-I.2.07 and $E_\varepsilon^* = \bigcup_{N \geq 1} F_{\varepsilon, N}^*$ now give the same bound for $\dim_P E_\varepsilon^*$. For every sufficiently large integer j , $e^{-j} < r$, and $E^* \subset E_{e^{-j}}^*$. Packing-dimension monotonicity and the preceding bounds imply

$$0 \leq \dim_P E^* \leq \frac{k(2 \log m + \log(1/r))}{j + \log r} \rightarrow 0. \quad (\text{II.6.N20e})$$

The denominator is positive for these j . Finally $\dim_H E^* \leq \dim_P E^*$ by T-I.2.11. This proves (II.6.22), including the case of an empty exceptional set. $\square$

Corollary II.6.6.9 (C-II.6.21 — Analytic-family natural-dimension theorem). *If the analytic family of D-II.6.08 is infinitely coding-nondegenerate, then the Borel set E^* of D-II.6.10 has Hausdorff and packing dimension zero, and for every $t \in I \setminus E^*$,*

$$\dim \mu_{t,p} = \min\{1, s(\Phi_t, p)\} \quad \text{for every probability vector } p, \quad \dim_H K_t = \min\{1, s_{\Phi_t}\}. \quad (\text{II.6.23})$$

The same exceptional set works for all weights. We assert a zero-dimensional covering set for dimension-drop parameters, not that an arbitrary exterior-anchor positional collision set is small. For comparison, see Hochman [Hoc14, Theorems 1.7–1.8 and Section 5.4, pp. 8 and 37–40].

Proof. Fix the distinguished tail in D-II.6.09. Each anchored difference is continuous by L-II.6.17. Consequently every $E_{\varepsilon, n}^*$ is relatively open in I , being a finite union of inverse images of open intervals under continuous functions. The sets

$$E_\varepsilon^* = \bigcup_{N \geq 1} \bigcap_{n \geq N} E_{\varepsilon, n}^*, \quad E^* = \bigcap_{j \geq 1} E_{e^{-j}}^* \quad (\text{II.6.N21a})$$

are Borel subsets of I , hence Borel subsets of $\mathbb{R}$ because I is closed. Nondegeneracy gives the prefix-controlled transversality of P-II.6.18, and T-II.6.20 gives $\dim_H E^* = \dim_P E^* = 0$.

At any parameter t for which the full system has super-exponential concentration, each integer $j \geq 1$ has an N_j such that for every $n \geq N_j$ some distinct words $u, v \in \Lambda^n$ have

$$r_{u,t} = r_{v,t}, \quad |a_{u,t} - a_{v,t}| < e^{-jn}. \quad (\text{II.6.N21b})$$

For these pairs the anchor cancels exactly:

$$\Delta_{u,v}^*(t) = a_{u,t} - a_{v,t} + (r_{u,t} - r_{v,t})b_*(t) = a_{u,t} - a_{v,t}. \quad (\text{II.6.N21c})$$

It follows that $t \in E_{e^{-j}}^*$ for every j , and hence $t \in E^*$. Thus a parameter outside E^* cannot have super-exponential concentration. The quantified equivalence in C-II.6.12 gives exponential separation there, and its natural dimension conclusions give all of (II.6.23), simultaneously for every weight vector and for the attractor.

In particular zero entries in a vector cause no difficulty: the active subalphabet inherits separation, as proved in C-II.6.12. There is no union over weight vectors in constructing E^* ; the exclusion of concentration was for the full labelled system. Equivalently, T-II.6.11 places any strict dimension drop inside this same covering set. We have not asserted that the covering set equals the set of parameters with a dimension drop. $\square$

The logical chain is now complete:

$$\begin{aligned} \text{analytic nonidentity} &\implies \text{finite-order transversality} \\ &\implies \text{sublevel covers} \implies \text{packing-zero exceptional set.} \end{aligned}$$

II.6.7 Bernoulli convolutions

The family $x \mapsto \lambda x \pm 1$ is analytically nondegenerate: identical coding functions on an interval would be identical at $\lambda = 1/4$, where separated first-level pieces distinguish every code. This lets the analytic-family theorem give dimension one outside a packing-zero parameter set in $(1/2, 1)$.

A dimension drop also gives a nonzero polynomial with coefficients in $\{-1, 0, 1\}$ and a super-exponentially small value at the parameter. The polynomial is nonzero as a formal polynomial; its evaluated value is permitted to be zero. Neither this small-value property nor full dimension alone proves absolute continuity or Fourier decay; those are separate questions requiring additional hypotheses and arguments.

Corollary II.6.7.1 (C-II.6.22 – Bernoulli-convolution dimension exceptions and small polynomials). *There is a Borel set $E_{\text{BC}} \subset (1/2, 1)$ with*

$$\dim_{\text{H}} E_{\text{BC}} = \dim_{\text{P}} E_{\text{BC}} = 0, \quad \dim v_{\lambda} = 1 \quad (\lambda \in (1/2, 1) \setminus E_{\text{BC}}). \quad (\text{II.6.24})$$

If $\lambda \in (1/2, 1)$ and $\dim v_{\lambda} < 1$, then for every $0 < \varepsilon < 1$ there is n_0 such that for every $n \geq n_0$ some nonzero $P_n \in \mathcal{P}_{n-1}$ satisfies

$$|P_n(\lambda)| < \varepsilon^n. \quad (\text{II.6.25})$$

The polynomial may vanish at λ ; nonzero refers to the polynomial itself. No assertion about proximity to a root, Fourier decay, absolute continuity or the full transcendental-

parameter theorem is included. For comparison, see Hochman [Hoc14, Theorem 1.9, p. 9].

Proof. On the connected open interval $J = (0, 1)$, consider the analytic family $\varphi_{\pm, \lambda}(x) = \lambda x \pm 1$. Its coding function for a sign sequence $\omega = (\omega_1, \omega_2, \dots) \in \{-1, 1\}^{\mathbb{N}}$ is

$$\pi_{\lambda}(\omega) = \sum_{j=1}^{\infty} \omega_j \lambda^{j-1}. \quad (\text{II.6.N22a})$$

The series is absolutely convergent, and agrees with the convention for ν_{λ} in D-II.6.11 under equal Bernoulli weights. L-II.6.17 makes every coding function analytic on J .

We check infinite coding nondegeneracy on every nondegenerate compact subinterval $I \subset J$. If two distinct coding functions agreed on I , their difference would be zero throughout J , by the identity theorem in L-II.6.17. At $\lambda = 1/4$, however, (II.6.N22a) places the attractor inside $[-4/3, 4/3]$. Its two first-level pieces lie respectively inside

$$[-4/3, -2/3] \quad \text{and} \quad [2/3, 4/3].$$

They are disjoint. Strong separation therefore gives injective coding by L-I.7.07, contradicting equality for the two distinct codes at this parameter. This proves the required nondegeneracy. The parameter $1/4$ is used only for this analytic identity test; it need not belong to the interval I .

For integers $m \geq 5$, let $I_m = [1/2 + 1/m, 1 - 1/m]$. These nondegenerate compact intervals increase to $(1/2, 1)$. Fix the all-positive distinguished tail on the whole family. Apply C-II.6.21 on each I_m to obtain a Borel set $E_m \subset I_m$ of Hausdorff and packing dimension zero. Put

$$E_{\text{BC}} = \bigcup_{m \geq 5} E_m. \quad (\text{II.6.N22b})$$

This is a Borel subset of $(1/2, 1)$, and T-I.2.07 gives $\dim_{\text{p}} E_{\text{BC}} = 0$. The hierarchy T-I.2.11 gives its Hausdorff dimension zero as well. If $\lambda \in (1/2, 1) \setminus E_{\text{BC}}$, choose an m with $\lambda \in I_m$. The equal-weight measure has $h = \log 2$ and $\chi = -\log \lambda$, so that

$$\min\{1, h/\chi\} = 1 \quad (1/2 < \lambda < 1).$$

C-II.6.21 gives $\dim \nu_{\lambda} = 1$, proving (II.6.24).

Finally suppose $\dim \nu_{\lambda} < 1$ at a fixed $\lambda \in (1/2, 1)$, and fix $0 < \varepsilon < 1$. T-II.6.11 applies because the natural bound is one. For every sufficiently large n , it supplies distinct sign words u, v of length n with $|a_u - a_v| < \varepsilon^n$; their contraction products both equal λ^n . Their translations satisfy

$$a_u - a_v = \sum_{j=1}^n (u_j - v_j) \lambda^{j-1} = 2P_n(\lambda), \quad P_n(X) = \sum_{j=1}^n \frac{u_j - v_j}{2} X^{j-1}. \quad (\text{II.6.N22c})$$

Each coefficient of P_n is in $\{-1, 0, 1\}$, and its degree is at most $n - 1$. Since the two words are distinct, at least one coefficient is nonzero, so P_n is not the zero polynomial. We

conclude $|P_n(\lambda)| < \varepsilon^n/2 < \varepsilon^n$, as in (II.6.25). A nonzero polynomial can still have $P_n(\lambda) = 0$, which is allowed. No estimate for distance to a root, absolute continuity or Fourier decay follows merely from this small-value statement.

□

This last corollary proves a full-dimension statement outside a packing-zero set and produces small polynomials under dimension drop. It does not by itself prove absolute continuity or Fourier decay; neither conclusion follows from the stated hypotheses alone.

Chapter II.7

Self-Similar Measures in Higher Dimensions

Geometric intuition

Higher-dimensional self-similar measures may hide entropy in invariant directions. The chapter derives the corresponding dimension trichotomy, irreducibility consequences, projection statements, and the geometric fiber estimates that accompany them. Its central theme is that failure of entropy growth reveals a persistent invariant direction, which can then be interpreted through conditional measures on affine fibers.

The line argument has a new obstruction in higher dimensions. A measure may already fill one family of directions, so moving it along those directions creates almost no new entropy even though distinct cylinder maps remain separated. We encode those directions by finite-scale saturation, extract a maximal invariant persistent subspace, and identify it with full-dimensional conditional measures on affine fibers. This yields the higher-dimensional trichotomy: natural dimension, super-exponential cylinder concentration, or a full-dimensional invariant fiber.

Entropy is computed with natural logarithms. An entropy of a random variable relative to a grid means the entropy of its cell label.

II.7.1 Geometry, group coordinates, and finite words

In one dimension, a measure can absorb translation entropy by already filling an interval at the relevant scales. In higher dimensions it may fill only certain directions. An interval times a thin Cantor set is the basic picture: motion along the interval can be hard to detect by entropy, while motion across the Cantor set can still create new information.

We begin by fixing the decomposition of each similarity into a positive scalar ratio, an orthogonal part, and a translation. This convention keeps reflections in the orthogonal factor and makes the scalar Lyapunov exponent unambiguous.

Definition II.7.1.1 (II.7.01: Positive-ratio similarity systems). Fix $d \geq 1$, a finite labelled alphabet Λ with at least two symbols, and

$$\Phi = (\varphi_i)_{i \in \Lambda}, \quad \varphi_i(x) = r_i U_i x + a_i, \quad 0 < r_i < 1, \quad U_i \in O(d), \quad a_i \in \mathbb{R}^d.$$

Repeated labelled maps are permitted. The scalar ratio is positive; reflections are part of U_i . Put $r_{\min} = \min_i r_i$, $r_{\max} = \max_i r_i$. For a finite word $u = u_1 \cdots u_n$, let

$$\varphi_u = \varphi_{u_1} \circ \cdots \circ \varphi_{u_n}, \quad r_u = \prod_{j=1}^n r_{u_j}, \quad U_u = U_{u_1} \cdots U_{u_n}, \quad a_u = \varphi_u(0).$$

The empty word represents the identity; its ratio, orthogonal part and translation are 1, I_d , 0. Concatenation uv puts u before v . The predecessor u^- deletes the final symbol of a nonempty word. Attractor, coding map and Bernoulli invariant measure retain their I.7 meanings and are denoted K_Φ , π_Φ , $\mu_{\Phi,p}$.

A strict probability vector has $p_i > 0$ and $\sum_i p_i = 1$. Put $p_u = \prod_j p_{u_j}$, with $p_\emptyset = 1$, and

$$h(p) = - \sum_i p_i \log p_i, \quad \chi(\Phi, p) = - \sum_i p_i \log r_i > 0, \quad s(\Phi, p) = h(p)/\chi(\Phi, p).$$

The similarity dimension s_Φ satisfies $\sum_i r_i^{s_\Phi} = 1$. The natural measure and set bounds mean $\min\{d, s(\Phi, p)\}$ and $\min\{d, s_\Phi\}$, respectively. For a probability vector allowing zero entries, its active alphabet is $\Lambda_p = \{i : p_i > 0\}$, and $0 \log 0 = 0$. Every irreducibility assertion specifies which alphabet is used. A one-symbol active subsystem is treated separately as a point measure.

Unless a result explicitly introduces a different probability vector, we henceforth fix a strict vector p for Φ and write $\mu = \mu_{\Phi,p}$ and $\alpha = \dim \mu$. The existence and almost-sure constancy of α follow from the exact-dimensionality theorem used below. Results that permit zero weights first restrict to the stated active alphabet.

Two irreducibility assumptions answer different questions. Affine irreducibility says that the entire problem does not secretly live in a smaller affine space. Linear irreducibility says that the rotations do not preserve a possible family of flat directions. Neither implies the other. The algebraic example in L-II.7.24 will make the second obstruction visible even when the first assumption holds.

Definition II.7.1.2 (II.7.02: Orthogonal group and two irreducibility conditions). The generated orthogonal group is

$$G_\Phi = \overline{\langle U_i : i \in \Lambda \rangle} \subseteq O(d),$$

where angle brackets mean all finite products of generators and their inverses, including the identity, and the closure is in the operator norm. A linear subspace V is Φ -invariant if $U_i V = V$ for every i . Linear irreducibility means that the only such subspaces are $\{0\}$ and $\mathbb{R}^d$.

Affine irreducibility means that there is no nonempty proper affine subspace $A \subset \mathbb{R}^d$ with $\varphi_i(A) \subseteq A$ for every i . Here $A = x + V$; its dimension is $\dim V$. The affine hull of a nonempty set is the intersection of its containing affine subspaces. Full affine span means $\text{aff } K = \mathbb{R}^d$.

A nonzero subspace can equal $\mathbb{R}^d$; a nontrivial subspace in this chapter means $0 < \dim V < d$. An orientation-preserving orthogonal map belongs to $SO(d) = \{U \in O(d) : \det U = 1\}$.

Dimension drop will eventually be converted into pairs of cylinder maps that are extremely close. To state that conclusion one must compare the scalar, orthogonal, and translation coordinates simultaneously; comparing only the images of the origin would lose two of those coordinates.

Definition II.7.1.3 (II.7.03: Similarity distance and cylinder concentration). On the similarity group $G(d)$ of D-II.5.05 use

$$\rho_{\text{sim}}(rU + a, sW + b) = |\log r - \log s| + \|U - W\|_{\text{op}} + \|a - b\|,$$

and also the matrix-coordinate distance

$$\rho_{\text{mat}}(rU + a, sW + b) = \|rU - sW\|_{\text{op}} + \|a - b\|.$$

Define

$$\Delta_n^{\text{sim}}(\Phi) = \min_{\substack{u, v \in \Lambda^n \\ u \neq v}} \rho_{\text{sim}}(\varphi_u, \varphi_v), \quad n \geq 1.$$

Super-exponential cylinder concentration means

$$\forall A > 0 \exists n_0 \forall n \geq n_0, \Delta_n^{\text{sim}}(\Phi) < e^{-An}.$$

Exponential separation means that some $A > 0$ satisfies $\Delta_n^{\text{sim}}(\Phi) \geq e^{-An}$ for infinitely many n . An exact overlap retains D-I.7.07's convention: equality of maps for two distinct finite words. The same-length witness is proved locally. Neither of these distances is the equal-signed-contraction distance of II.6; here unequal ratios have finite distance.

There is also a coordinate issue before the geometry can be used. A similarity has a contraction, an orthogonal part and a translation. Observing only its value at zero loses the first two pieces. The [similarity-chart lemma](#) below separates three ways of observing a group element. Left dilation scales every coordinate of its affine matrix chart, but does not scale the rotation coordinate in the logarithmic chart. The proof must choose the chart suited to the operation, not treat these observations as identical.

The [finite-label lemma](#) shows that random word contractions fluctuate by a sublinear amount around their mean logarithmic scale. The error need not stay bounded. There can be exponentially many words while the number of distinct ratio labels grows only polynomially. Recording those labels is consequently cheap enough for a normalized

entropy argument.

The [uniform non-affinity lemma](#) and [stopping-mixture lemma](#) then compare a dyadic component with a mixture of whole self-similar cylinders. Keeping cylinders whole preserves their geometry, but may put a small amount of mass on the wrong side of a grid boundary. Uniform small mass near affine hyperplanes controls this error. Mere absence of point atoms would not suffice in higher dimensions.

Lemma II.7.1.4 (L-II.7.01 — Basic system and overlap concordance). *The measure μ is exact-dimensional and*

$$\lim_{n \rightarrow \infty} h_n(\mu) = \alpha \leq \min\{d, s(\Phi, p)\}, \quad \dim_{\text{H}} K_{\Phi} \leq \min\{d, s_{\Phi}\}.$$

Affine irreducibility is equivalent to $\text{aff } K_{\Phi} = \mathbb{R}^d$. Any exact overlap has a witness of equal word length. If one exists, then $\Delta_n^{\text{sim}} = 0$ for every sufficiently large n . Exponential separation is precisely the negation of super-exponential cylinder concentration. These assertions do not impose linear irreducibility.

Exact dimensionality follows from T-I.9.08; the proof below verifies its hypotheses in the present similarity setting.

Proof. We first verify exact dimensionality and the entropy-dimension limit, then separate affine irreducibility from linear irreducibility, and finally align exact overlaps at equal symbolic lengths.

Exact dimensionality and entropy dimension

Let $\mathbb{P}_p$ be the Bernoulli probability on $\Lambda^{\mathbb{N}}$ and σ the left shift. We first check the ergodicity hypothesis in T-I.9.08. If a Borel event E satisfies $E = \sigma^{-1}E$ modulo null sets, it agrees modulo null sets with $\sigma^{-n}E$ for every n . The latter is independent of every event depending on the first n coordinates. Thus

$$\mathbb{P}_p(E \cap C) = \mathbb{P}_p(E)\mathbb{P}_p(C) \tag{II.7.N01a}$$

for every finite-cylinder event C . The finite-cylinder algebra generates the product sigma algebra; its elements approximate every Borel event in probability, by the elementary monotone-class argument for approximation by a generating algebra. Approximate E in (II.7.N01a) to obtain $\mathbb{P}_p(E) = \mathbb{P}_p(E)^2$. This proves ergodicity.

Each similarity is a smooth conformal contraction, its derivative being $r_i U_i$ with $r_i > 0$. It is defined on all of $\mathbb{R}^d$; a sufficiently large closed ball contains the attractor and is mapped into itself by every map. The finite-family contraction and distortion hypotheses of T-I.9.08 hold, with distortion one. That theorem applied to $(\pi_{\Phi})_* \mathbb{P}_p$ shows that μ is exact-dimensional, of some dimension α .

For clarity we supply the entropy-dimension passage. Choose $b \in \mathbb{Z}^d$ and an integer $a \geq 0$ such that $x + b \in (0, 2^a)^d$ for every $x \in K_{\Phi}$. Let $F(x) = 2^{-a}(x + b)$ and $\theta = F_* \mu$. The exact local dimension of θ is α , since

$$\theta(B(F(x), \rho)) = \mu(B(x, 2^a \rho)). \quad (\text{II.7.N01b})$$

The measure θ is carried by $[0, 1]^d$. For the one-step local entropy average $\mathcal{A}_{N,1}$ of II.2, C-II.2.09 gives $\mathcal{A}_{N,1}(\theta, x) \rightarrow \alpha$ almost everywhere. This average lies between zero and d . Dominated convergence and L-II.2.01 therefore give

$$\alpha = \lim_{N \rightarrow \infty} \int \mathcal{A}_{N,1}(\theta, x) d\theta(x) = \lim_{N \rightarrow \infty} \frac{H_N(\theta) - H_0(\theta)}{N \log 2} = \lim_{N \rightarrow \infty} h_N(\theta). \quad (\text{II.7.N01c})$$

Here $H_0(\theta) = 0$. For $N \geq a$, pulling back the dyadic grid under F gives exactly the level- $N - a$ grid, translated by the integer vector $-b$. Integer translations preserve that grid, so

$$H_N(\theta) = H_{N-a}(\mu). \quad (\text{II.7.N01d})$$

The fixed shift does not change the normalized limit. This proves $h_n(\mu) \rightarrow \alpha$. The natural measure upper bound from T-I.8.03 and the set upper bound from T-I.7.06 give $\alpha \leq \min\{d, h(p)/\chi\}$ and $\dim_{\text{H}} K_{\Phi} \leq \min\{d, s_{\Phi}\}$.

Affine irreducibility

For an invertible affine map g and a nonempty set E ,

$$g(\text{aff } E) = \text{aff}(gE). \quad (\text{II.7.N01e})$$

Indeed each side is an affine subspace containing gE ; applying g^{-1} to any containing affine subspace gives the reverse minimality assertion. Since $\varphi_i(K_{\Phi}) \subseteq K_{\Phi}$, the affine hull of K_{Φ} is invariant under every generator. Thus affine irreducibility implies full affine span.

Conversely suppose a nonempty proper affine subspace A is invariant under all generators. Choose $z \in A$. Every finite word image $\varphi_u(z)$ belongs to A , and every infinite coding limit belongs to A , since a finite-dimensional affine subspace is closed. The coding limits fill K_{Φ} , by I.7. Hence $\text{aff } K_{\Phi} \subseteq A$, so the attractor cannot have full affine span. This proves the equivalence.

Equal-length witnesses

We need the following elementary fact about finite nonempty strings: if $uv = vu$, then $u = w^a, v = w^b$ for a nonempty word w and positive integers a, b . Here w^a denotes concatenation of a copies. Prove this by induction on $|u| + |v|$. If the lengths agree, equality of the initial blocks gives $u = v$. Otherwise interchange the two words if necessary so that $|u| < |v|$. The initial $|u|$ letters show that $v = uz$ with $z \neq \emptyset$. Cancelling the initial u from $uuz = uzu$ gives $uz = zu$. The induction hypothesis applied to u, z gives the assertion for u, v .

Now suppose $\varphi_u = \varphi_v$ with $u \neq v$. Neither word can be empty unless both are: a nonempty word has ratio less than one, whereas the empty word has ratio one. If $uv \neq vu$, then

$$\varphi_{uv} = \varphi_u \varphi_v = \varphi_v \varphi_u = \varphi_{vu} \quad (\text{II.7.N01f})$$

is an equal-length witness. If $uv = vu$, the string fact gives $u = w^a, v = w^b$. Equality of maps implies equality of their positive ratios, so $r_w^a = r_w^b$. Since $0 < r_w < 1$, this forces $a = b$ and $u = v$, a contradiction. Thus the equal-length witness always exists.

If it has length N , append any fixed word of length $n - N$ to both witnesses. The words remain distinct and their maps remain equal. Therefore $\Delta_n^{\text{sim}} = 0$ for every $n \geq N$. Finally the negation of

$$\forall A > 0 \exists n_0 \forall n \geq n_0, \Delta_n^{\text{sim}} < e^{-An}$$

is

$$\exists A > 0 \forall n_0 \exists n \geq n_0, \Delta_n^{\text{sim}} \geq e^{-An}.$$

The latter says exactly that the required lower bound holds at infinitely many generations. This completes the proof. $\square$

Definition II.7.1.5 (II.7.04: Three group grids and left dilation). Flatten a matrix into its ordered list of entries, in row order. For $j \in \mathbb{Z}$, let $\mathcal{E}_j, \mathcal{L}_j, \mathcal{M}_j$ be the pullbacks of the standard half-open level- j dyadic grids under, respectively,

$$(r, U, a) \mapsto a, \quad (r, U, a) \mapsto (-\log_2 r, U, a), \quad (r, U, a) \mapsto (rU, a).$$

Empty atoms are deleted. These are countable partitions, not point partitions of an uncountable group. Conditional entropy always means $H(\nu, \mathcal{P} \mid \mathcal{Q}) = H(\nu, \mathcal{P} \vee \mathcal{Q}) - H(\nu, \mathcal{Q})$.

For $b \in \mathbb{R}$, put $S_b(x) = 2^b x$ and let $T_b(g) = S_b \circ g$ be left dilation on the group. For $z \in \mathbb{R}^d$, let $L_z(g)(x) = g(x) + z$. Thus left translation is not conjugation. On probabilities, $(T_b)_* \nu$ and $(L_z)_* \nu$ mean pushforwards of the group law. On $G_0(d)$, $\mathcal{M}_j$ agrees with $\mathcal{D}_j^{G_0}$ from D-II.5.05.

The entropy argument must keep track of three pieces of a word map: its translation, its contraction ratio, and its orthogonal part. The following finite word law and scale convention will be used throughout the comparison of the logarithmic and matrix-coordinate grids.

Definition II.7.1.6 (II.7.05: Word laws, finite labels and observing levels). For a strict vector and $n \geq 1$, put

$$\nu_n = \sum_{u \in \Lambda^n} p_u \delta_{\varphi_u}, \quad \mathcal{R}_n = \{r_u : u \in \Lambda^n\}, \quad t_u = -\log_2 r_u.$$

Coincident maps are aggregated as atoms. On the finite carrier $\{\varphi_u : u \in \Lambda^n\}$, $\mathcal{F}_n$ is the partition into equal-ratio classes. This notation is used only on that finite carrier.

It is not an uncountable measurable partition on $G(d)$. For a positive-mass atom C of any displayed join of these partitions, $(v_n)_C = v_n|_C / v_n(C)$.

Define

$$\ell_n = \left\lfloor \frac{n\chi(\Phi, p)}{\log 2} \right\rfloor, \quad b_n(q) = \lfloor q\ell_n \rfloor, \quad L_n(q) = b_n(q) - \ell_n \quad (q > 1).$$

Statements dividing by ℓ_n or $L_n(q)$ use only large n for which these integers are positive. Logarithmic-grid flattening denotes the property

$$\forall q > 1, \quad \frac{H(v_n, \mathcal{L}_{b_n(q)} \mid \mathcal{E}_{\ell_n})}{\ell_n \log 2} \rightarrow 0.$$

Matrix-grid flattening is the same property with $\mathcal{M}$ in place of $\mathcal{L}$. Their equivalence is proved in the next lemma, with a linear change in the fine observation level.

Lemma II.7.1.7 (L-II.7.02 – Similarity charts and dilation calculus). *For maps with ratios $r, s \in [r_{\min}^n, 1]$,*

$$\rho_{\text{mat}} \leq \rho_{\text{sim}} \leq 3r_{\min}^{-n} \rho_{\text{mat}}.$$

For each $c_0 > \log_2(1/r_{\min})$, there is an integer $C = C(d, c_0, r_{\min})$ such that, for $n \geq 1$, $k, j \geq 0$,

$$\begin{aligned} H(v_n, \mathcal{L}_k \mid \mathcal{E}_j) &\leq H(v_n, \mathcal{M}_{k+[c_0 n]+C} \mid \mathcal{E}_j) + C, \\ H(v_n, \mathcal{M}_k \mid \mathcal{E}_j) &\leq H(v_n, \mathcal{L}_{k+C} \mid \mathcal{E}_j) + C. \end{aligned} \tag{II.7.1.1}$$

Furthermore T_b acts as

$$(t, U, a) \mapsto (t - b, U, 2^b a), \quad (B, a) \mapsto (2^b B, 2^b a), \quad t = -\log_2 r, B = rU.$$

Consequently matrix-grid entropies before and after T_b agree up to C_d after shifting the grid level by b and rounding to an integer. Precisely, for every $j \in \mathbb{Z}$ and every group probability for which these entropies are finite,

$$|H((T_b)_* v, \mathcal{M}_j) - H(v, \mathcal{M}_{[j+b]})| \leq C_d.$$

The same comparison holds after conditioning on any countable partition $\mathcal{Q}$ on the original group and its transported partition $T_b \mathcal{Q}$ on the image group, whenever the conditional entropies are finite. On a ratio class $r = 2^{-t}$, T_t takes values in $G_0(d)$. Flattening in the two charts is equivalent when quantified over every $q > 1$, not necessarily at the same q .

Proof. The proof compares matrix and similarity coordinates, checks their dyadic grids, and records exactly which coordinates change under left dilation.

The two distances

Write $B = rU$, $C = sW$, and $\eta = r_{\min}^n$, so $\eta \leq r, s \leq 1$. The reverse triangle inequality for operator norms gives

$$|r - s| = |\|B\|_{\text{op}} - \|C\|_{\text{op}}| \leq \|B - C\|_{\text{op}}. \quad (\text{II.7.N02a})$$

The derivative of the logarithm on $[\eta, 1]$ is at most η^{-1} . Also

$$\begin{aligned} |\log r - \log s| &\leq \eta^{-1} \|B - C\|_{\text{op}}, \\ \|U - W\|_{\text{op}} &= \|B/r - C/s\|_{\text{op}} \\ &\leq r^{-1} \|B - C\|_{\text{op}} + \|C\|_{\text{op}} |r^{-1} - s^{-1}| \\ &= r^{-1} (\|B - C\|_{\text{op}} + |r - s|) \leq 2\eta^{-1} \|B - C\|_{\text{op}}. \end{aligned} \quad (\text{II.7.N02b})$$

Adding the translation distance proves $\rho_{\text{sim}} \leq 3\eta^{-1} \rho_{\text{mat}}$. In the other direction, $|r - s| \leq |\log r - \log s|$ on $(0, 1]$, and

$$\|rU - sW\|_{\text{op}} \leq r\|U - W\|_{\text{op}} + |r - s| \leq \|U - W\|_{\text{op}} + |\log r - \log s|. \quad (\text{II.7.N02c})$$

Adding the translation distance gives $\rho_{\text{mat}} \leq \rho_{\text{sim}}$.

Dyadic comparisons, including conditional entropy

Let $D = d^2 + d$, the dimension of the matrix/translation coordinate space. All coordinate vectors carry their ordinary Euclidean norm. The Frobenius/operator norm inequalities and (II.7.N02a)–(II.7.N02b) show that the inverse coordinate map

$$(B, a) \mapsto (-\log_2 \|B\|_{\text{op}}, B/\|B\|_{\text{op}}, a) \quad (\text{II.7.N02d})$$

is $C_d \eta^{-1}$ -Lipschitz on the set of similarity matrices with ratios in $[\eta, 1]$. This is a comparison on that subset, not an assertion that every matrix is a similarity.

Fix $c_0 > \log_2(1/r_{\min})$. Choose a nonnegative integer $C_0 = C_0(d)$ large enough for the following estimate. An atom of $\mathcal{M}_{k+[c_0 n]+C_0}$, intersected with this subset, has inverse-image diameter at most

$$C_d \sqrt{D} r_{\min}^{-n} 2^{-k-[c_0 n]-C_0} \leq 2^{-k}. \quad (\text{II.7.N02e})$$

A subset of Euclidean diameter at most 2^{-k} meets at most three level- k intervals in each coordinate, even when endpoints are counted. Therefore it meets at most 3^{D+1} logarithmic-grid atoms. The conditional entropy of $\mathcal{L}_k$ given this finer matrix grid is at most $(D+1) \log 3$. For countable partitions $\mathcal{P}, \mathcal{R}, \mathcal{Q}$ with the relevant entropies finite, the entropy chain rule gives

$$H(\mathcal{P} \mid \mathcal{Q}) \leq H(\mathcal{R} \mid \mathcal{Q}) + H(\mathcal{P} \mid \mathcal{R}). \quad (\text{II.7.N02f})$$

Apply this with $\mathcal{P} = \mathcal{L}_k$, the preceding matrix grid as $\mathcal{R}$, and $\mathcal{Q} = \mathcal{E}_j$. Increasing one integer constant C to dominate both C_0 and $(D+1)\log 3$ proves the first inequality. No ratio-label entropy is required.

For the reverse inequality, the coordinate map $(t, U, a) \mapsto (2^{-t}U, a)$ is C_d -Lipschitz on $t \geq 0, U \in O(d)$: use $|2^{-t} - 2^{-s}| \leq (\log 2)|t - s|$ and the bounded matrix norms of U, W . The same diameter/intersection argument, followed by (II.7.N02f), gives

$$H(v_n, \mathcal{M}_k \mid \mathcal{E}_j) \leq H(v_n, \mathcal{L}_{k+C} \mid \mathcal{E}_j) + C. \quad (\text{II.7.N02g})$$

All word maps lie in the ratio range used above. Their finite carrier also ensures that every entropy in these comparisons is finite.

Left dilation and its direction of scale shift

For $g(x) = rUx + a$, direct substitution gives

$$(T_b g)(x) = 2^b rUx + 2^b a.$$

Thus its logarithmic coordinates are $(t - b, U, 2^b a)$, and its matrix coordinates are $(2^b B, 2^b a)$. In the matrix coordinate space, the pullback of a level- j grid cube under this map is a cube of side 2^{-j-b} . Put $v = \lfloor j + b \rfloor$. The two side lengths satisfy

$$2^{-v-1} < 2^{-j-b} \leq 2^{-v}. \quad (\text{II.7.N02h})$$

Each cube in either grid meets at most 4^D cubes of the other. Intersecting with the similarity coordinate image cannot increase this bound. The bounded-intersection entropy comparison L-II.1.06 therefore gives

$$|H((T_b)_* v, \mathcal{M}_j) - H(v, \mathcal{M}_v)| \leq D \log 4. \quad (\text{II.7.N02i})$$

Conditioning on an atom of any countable original partition and averaging gives the same inequality for the conditional entropies and transported partition. If b is an integer the pullback grids agree exactly. On a ratio class $r = 2^{-t}$, T_t sets the ratio equal to one, so its image is in $G_0(d)$. The matrix grid there is exactly the previously defined isometry grid.

Equivalence of all-factor flattening

Let $c = \chi / \log 2 > 0$, so $\ell_n/n \rightarrow c$. For any fixed $q > 1$, choose $q' > q + c_0/c + 1$. For all sufficiently large n ,

$$b_n(q) + \lceil c_0 n \rceil + C \leq b_n(q'). \quad (\text{II.7.N02j})$$

The first grid inequality and monotonicity under dyadic refinement show that matrix flattening at q' implies logarithmic flattening at q , after division by $\ell_n \log 2$. Conversely for any $q' > q$, eventually $b_n(q) + C \leq b_n(q')$; the reverse grid inequality gives matrix flattening at q from logarithmic flattening at q' . Since each property quantifies over all $q > 1$, they are equivalent. The argument does not identify their entropies at the same observing factor. This proves both comparison inequalities. □

Lemma II.7.1.8 (L-II.7.03 – Finite labels and the natural observation scale). *The law ν_n satisfies*

$$H(\nu_n, \mathcal{F}_n) \leq |\Lambda| \log(n+1), \quad \sum_{u \in \Lambda^n} p_u |t_u - \ell_n| = o(n), \quad \frac{H(\nu_n, \mathcal{E}_{\ell_n})}{\ell_n \log 2} \rightarrow \alpha.$$

In particular the last entropy approximation has error $o(\ell_n)$; no uniform $O(1)$ approximation is asserted in the nonhomogeneous case.

Proof. Let $U = I_1 \cdots I_n$ be a random word with independent labels of law p . Put $R = r_U$, $T = -\log_2 R$, and $c = \chi / \log 2$. The ratio depends only on the counts of the symbols in the word. Each count is an integer in $\{0, \dots, n\}$, so

$$|\mathcal{R}_n| \leq (n+1)^{|\Lambda|}, \quad H(R) = H(\nu_n, \mathcal{F}_n) \leq |\Lambda| \log(n+1). \quad (\text{II.7.N03a})$$

Different count vectors may have the same ratio; this only decreases the number of labels and does not affect the estimate. The random variables $Z_j = -\log_2 r_{I_j}$ are bounded iid, with mean c and a finite variance $\nu \geq 0$. Hence

$$\mathbb{E}|T - \ell_n| \leq \mathbb{E}|T - nc| + |nc - \lfloor nc \rfloor| \leq \sqrt{n\nu} + 1 = o(n). \quad (\text{II.7.N03b})$$

This elementary second-moment bound uses no Berry–Esseen theorem.

Choose $X \sim \mu$ independently of U , and define

$$Y = \varphi_U(X), \quad Z = a_U.$$

Iterating the self-similar identity n times shows that Y has law μ ; Z has the translation marginal of ν_n . Let $R_K = \sup_{x \in K_\Phi} \|x\| < \infty$. Conditional on $R = 2^{-t}$,

$$\|Y - Z\| = r_U \|U_U X\| \leq R_K 2^{-t}. \quad (\text{II.7.N03c})$$

Write Q_Y, Q_Z for their level- ℓ_n dyadic cell labels. Given Q_Z and $R = 2^{-t}$, the point Y lies in the $R_K 2^{-t}$ -neighborhood of that cube. The number of possible cells Q_Y is at most

$$N(t) = (3 + 2R_K 2^{\ell_n - t})^d.$$

The same bound applies to Q_Z given Q_Y, R . If needed replace this real upper bound by its ceiling; this increases its logarithm by at most $\log 2$. There is $C = C(d, R_K)$ such that both conditional entropies satisfy

$$H(Q_Y | Q_Z, R), H(Q_Z | Q_Y, R) \leq C + d \log 2 \mathbb{E}(\ell_n - T)_+. \quad (\text{II.7.N03d})$$

The entropy chain rule now gives, in both directions,

$$|H(Q_Y) - H(Q_Z)| \leq H(R) + C + d \log 2 \mathbb{E}(\ell_n - T)_+ = o(n). \quad (\text{II.7.N03e})$$

For example, $H(Q_Y) \leq H(Q_Z, R) + H(Q_Y \mid Q_Z, R) \leq H(Q_Z) + H(R) + H(Q_Y \mid Q_Z, R)$; interchange the labels for the opposite inequality. Every entropy here is finite: translations of word maps are bounded by $\max_i \|a_i\|/(1 - r_{\max})$, and Y is in the compact attractor. Finally

$$H(Q_Y) = H_{\ell_n}(\mu), \quad H(Q_Z) = H(\nu_n, \mathcal{E}_{\ell_n}), \quad \ell_n/n \longrightarrow c > 0.$$

Divide (II.7.N03e) by $\ell_n \log 2$ and use the entropy-dimension limit in L-II.7.01. This yields the final assertion and completes the proof. $\square$

Lemma II.7.1.9 (L-II.7.04 — Uniform non-affinity). *If Φ is affinely irreducible, then $\mu(A) = 0$ for every proper affine subspace A , and*

$$\lim_{\rho \downarrow 0} \sup_{A \text{ proper affine}} \mu(A^{(\rho)}) = 0.$$

Thus for every $0 < \varepsilon < 1$, some $\sigma > 0$ makes μ (ε, σ) -non-affine in D-II.5.07's sense. The constants depend on this fixed system and measure.

Proof. **Every proper affine subspace has zero mass**

Suppose some proper affine subspace has positive μ -mass. Choose the smallest possible dimension $q \in \{0, \dots, d-1\}$ of such a subspace, and let $\mathcal{A}_q$ denote all affine q -planes. Distinct members of $\mathcal{A}_q$ intersect either in the empty set or in an affine subspace of dimension less than q . Their intersection therefore has zero μ -mass by minimality of q . In particular, for any finite family of distinct such planes,

$$\sum_{j=1}^N \mu(A_j) = \mu\left(\bigcup_{j=1}^N A_j\right) \leq 1. \quad (\text{II.7.N04a})$$

Put $s = \sup_{A \in \mathcal{A}_q} \mu(A) > 0$. This supremum is attained. Indeed there are at most $2/s$ planes of mass at least $s/2$, by (II.7.N04a), and every sufficiently accurate maximizing sequence lies in this finite family. The collection

$$\mathcal{F} = \{A \in \mathcal{A}_q : \mu(A) = s\} \quad (\text{II.7.N04b})$$

is nonempty and finite, with cardinality at most $1/s$. For $A \in \mathcal{F}$, the self-similar identity gives

$$s = \mu(A) = \sum_{i \in \Lambda} p_i \mu(\varphi_i^{-1} A). \quad (\text{II.7.N04c})$$

Each inverse image is an affine q -plane, so every summand's mass is at most s . Since every p_i is strictly positive, equality in this convex average forces $\mu(\varphi_i^{-1} A) = s$ for every i . For a fixed i , the inverse-image map is injective on the finite family $\mathcal{F}$. It therefore permutes $\mathcal{F}$, as does the forward map.

Let $m = |\mathcal{F}|$ and $N = m!$. Every permutation of m objects has order dividing $m!$, so $\varphi_i^N(A) = A$ for all i and all $A \in \mathcal{F}$. Fix one generator i_0 . The contraction $\varphi_{i_0}^N$ has a unique fixed point z in $\mathbb{R}^d$. Its restriction to any $A \in \mathcal{F}$ is also a contraction of a nonempty complete metric space into itself. The fixed point obtained there must be z , by uniqueness in the ambient space. Thus

$$A_* := \bigcap_{A \in \mathcal{F}} A \quad (\text{II.7.N04d})$$

is a nonempty proper affine subspace. Every φ_i is a bijection and permutes the family, so

$$\varphi_i(A_*) = \bigcap_{A \in \mathcal{F}} \varphi_i(A) = \bigcap_{A \in \mathcal{F}} A = A_*.$$

This contradicts affine irreducibility. Therefore every proper affine subspace has zero mass, including points and hyperplanes.

Uniformity over all affine planes

Every proper affine subspace is contained in a hyperplane. Its ρ -neighborhood is contained in that hyperplane's ρ -neighborhood. Consequently it suffices to prove

$$\omega_\mu(\rho) := \sup_{\substack{v \in \mathbb{R}^d, \|v\|=1 \\ t \in \mathbb{R}}} \mu\{x : |v \cdot x - t| < \rho\} \longrightarrow 0 \quad (\rho \downarrow 0). \quad (\text{II.7.N04e})$$

Suppose instead that there are $\eta > 0$, $\rho_j \downarrow 0$ and hyperplanes with unit normals v_j and offsets t_j such that the masses in (II.7.N04e) are at least η . Let $R = \sup_{x \in K_\Phi} \|x\|$. For $\rho_j \leq 1$, positive mass implies $|t_j| \leq R + 1$, since the corresponding tube meets K_Φ . Compactness gives a subsequence for which $v_j \rightarrow v$, $t_j \rightarrow t$, and $\|v\| = 1$. For every $x \in K_\Phi$ in the j -th tube,

$$|v \cdot x - t| \leq \rho_j + R\|v_j - v\| + |t_j - t| =: e_j \longrightarrow 0. \quad (\text{II.7.N04f})$$

For each $e > 0$, sufficiently large j then give $\mu\{x : |v \cdot x - t| \leq e\} \geq \eta$. Letting $e \downarrow 0$ and using continuity from above of a finite measure gives $\mu\{x : v \cdot x = t\} \geq \eta$, contradicting the first part. This proves (II.7.N04e). The supremum over proper affine subspaces has the same upper bound and also tends to zero. Choose $\sigma > 0$ so that this supremum is strictly below a prescribed ε ; this is exactly D-II.5.07's non-affinity condition and completes the proof. □

Definition II.7.1.10 (II.7.06: Stopping families and anchor mixtures). For $0 < h < 1$, define

$$\Lambda(h) = \{u \neq \emptyset : r_u \leq h < r_{u^-}\}, \quad \Lambda(1) = \{\emptyset\}.$$

Fix $b \in K_\Phi$, $k \geq 0$, $0 < \rho < 1$. For a level- k dyadic cell Q , put

$$w_Q = \sum_{\substack{u \in \Lambda(\rho 2^{-k}) \\ \varphi_u(b) \in Q}} p_u, \quad \eta_Q = \frac{1}{w_Q} \sum_{\substack{u \in \Lambda(\rho 2^{-k}) \\ \varphi_u(b) \in Q}} p_u(\varphi_u)_* \mu \quad (w_Q > 0),$$

and $\eta_Q = \delta_b$ when $w_Q = 0$. This is a mixture of whole cylinders, not $\mu|_Q/\mu(Q)$. Total variation is the half- L^1 convention $\|\theta - \eta\|_{TV} = \sup_B |\theta(B) - \eta(B)|$, as in II.4 and II.6.

Lemma II.7.1.11 (L-II.7.05 – Stopping decomposition and boundary-safe mixtures). *For $0 < h \leq 1$, the stopping family is finite, its symbolic cylinders partition the full symbolic space, and*

$$\mu = \sum_{u \in \Lambda(h)} p_u(\varphi_u)_* \mu, \quad r_{\min} h < r_u \leq h \quad (0 < h < 1).$$

If the system is affinely irreducible, then for every $0 < \varepsilon < 1$ there is $\rho_0 > 0$ such that, for all $0 < \rho < \rho_0$, all $k \geq 0$, and all anchors $b \in K_\Phi$,

$$\mu\{x : w_{\mathcal{D}_k(x)} > 0, \|\mu_{x,k} - \eta_{\mathcal{D}_k(x)}\|_{TV} < \varepsilon\} > 1 - \varepsilon.$$

Here $\mu_{x,k}$ is the raw component. No boundary-null assertion for an arbitrary non-atomic measure replaces L04's hyperplane control.

Proof. Stopping words put cylinders at comparable geometric scales. Anchor mixtures keep cylinders whole, and uniform hyperplane control bounds the boundary error when those mixtures are compared with actual dyadic components.

The stopping identity, without irreducibility

Fix $0 < h < 1$. Choose $N \geq 1$ with $r_{\max}^N \leq h$. Along every infinite word the positive ratio products decrease strictly from one, and have fallen below h by time N . There is a unique first crossing. Its word u satisfies

$$r_u \leq h < r_{u^-}, \quad r_u = r_{u^-} r_{u|u|} > h r_{\min}. \quad (\text{II.7.N05a})$$

Thus the stopping family is finite, consists of words of length at most N , and its symbolic cylinders partition the full symbolic space. No stopping word is a proper prefix of another. Their Bernoulli weights sum to one, and conditional on any such prefix the remaining symbols retain their independent law. Since $\pi_\Phi(u\omega) = \varphi_u(\pi_\Phi(\omega))$,

$$\mu = \sum_{u \in \Lambda(h)} p_u(\varphi_u)_* \mu. \quad (\text{II.7.N05b})$$

For $h = 1$ the family is the singleton containing the empty word, so the same identity is immediate. No affine irreducibility has been used in this part.

A uniform grid-boundary estimate

Assume henceforth affine irreducibility. Put $D_0 = \max\{1, \text{diam } K_\Phi\}$ and use the function ω_μ of (II.7.N04e). For $k \geq 0$, let $\mathcal{H}_k$ be the family of coordinate hyperplanes

$$H_{j,z} = \{x : x_j = z2^{-k}\}, \quad 1 \leq j \leq d, \quad z \in \mathbb{Z}.$$

For $0 < \rho < 1$, define the open boundary band

$$E_{k,\rho} = \bigcup_{H \in \mathcal{H}_k} H^{(2D_0\rho 2^{-k})}. \quad (\text{II.7.N05c})$$

Use (II.7.N05b) at the coarser stopping scale $h = 2^{-k}$. For each of its words v ,

$$r_{\min} 2^{-k} \leq r_v \leq 2^{-k}, \quad \text{diam } \varphi_v(K_\Phi) \leq D_0 2^{-k}. \quad (\text{II.7.N05d})$$

These non-strict inequalities also hold at $k = 0$, where the word is empty. For any fixed coordinate, the hyperplanes whose bands in (II.7.N05c) meet $\varphi_v(K_\Phi)$ have offsets in an interval of length at most $D_0(1 + 4\rho)2^{-k} \leq 5D_0 2^{-k}$. Their number is at most $[5D_0] + 2$. Across all coordinates there are at most

$$M_0 = d([5D_0] + 2) \quad (\text{II.7.N05e})$$

relevant bands, independently of v, k, ρ . The inverse image of each band under the similarity φ_v is a band about a proper affine hyperplane of width

$$\frac{2D_0\rho 2^{-k}}{r_v} \leq \frac{2D_0\rho}{r_{\min}}.$$

Using the union bound in each cylinder and averaging (II.7.N05b),

$$\mu(E_{k,\rho}) \leq M_0 \omega_\mu(2D_0\rho/r_{\min}) \longrightarrow 0 \quad (\rho \downarrow 0), \quad (\text{II.7.N05f})$$

uniformly in k . This uses hyperplane nullity, not merely the absence of atoms.

Coupling and conditional total variation

Fix an anchor $b \in K_\Phi$. Choose a stopping word U at scale $\rho 2^{-k}$ with probabilities p_u , independently of $X \sim \mu$, and put

$$Y = \varphi_U(X), \quad Z = \varphi_U(b), \quad J = \mathcal{D}_k(Y), \quad J' = \mathcal{D}_k(Z). \quad (\text{II.7.N05g})$$

Both points lie in K_Φ , Y has law μ , and $\|Y - Z\| \leq D_0\rho 2^{-k}$. If $J \neq J'$, the segment from Y to Z meets a coordinate grid boundary. This also covers a point on a boundary, with the half-open grid convention. It follows that $Y \in E_{k,\rho}$, so

$$p_{\text{bad}} := \mathbb{P}(J \neq J') \leq M_0 \omega_\mu(2D_0\rho/r_{\min}). \quad (\text{II.7.N05h})$$

Let $q_Q = \mathbb{P}(J = Q) = \mu(Q)$ and $w_Q = \mathbb{P}(J' = Q)$. There are only finitely many relevant cells, since K_Φ is compact. For $q_Q > 0$, the conditional law of Y given $J = Q$ is the raw

component μ_Q . For $w_Q > 0$, the conditional law of Y given $J' = Q$ is exactly η_Q of D-II.7.06, by independence of X, U .

We prove the precise comparison needed here. On the product of the ambient space and the finite cell set let P and Q' be the laws of (Y, J) and (Y, J') , respectively, and define a third probability R by

$$R(B \times \{Q\}) = q_Q \eta_Q(B).$$

Use the half- L^1 convention for all these total variations. The coupling with the same Y shows $\|P - Q'\|_{TV} \leq p_{\text{bad}}$. Since $Q'(B \times \{Q\}) = w_Q \eta_Q(B)$, including the cells with $w_Q = 0$,

$$\|Q' - R\|_{TV} = \frac{1}{2} \sum_Q |w_Q - q_Q| \leq p_{\text{bad}}. \quad (\text{II.7.N05i})$$

The last bound follows from the same coupling of J, J' . The first and third probabilities have the common cell marginal (q_Q) . Taking the total variation separately on their disjoint cell blocks gives

$$\sum_{Q: q_Q > 0} q_Q \|\mu_Q - \eta_Q\|_{TV} = \|P - R\|_{TV} \leq 2p_{\text{bad}}. \quad (\text{II.7.N05j})$$

For completeness, the equality can be seen directly from the supremum definition: on each block choose a Borel set approximating the positive variation supremum and take their finite disjoint union. The opposite inequality is the sum of the blockwise bounds. No density or domination assumption between μ_Q and η_Q is used. Moreover,

$$\sum_{Q: w_Q = 0} q_Q \leq p_{\text{bad}}, \quad (\text{II.7.N05k})$$

because $J' = Q$ has probability zero on each of those cells.

Select a scale uniformly in the observing level

Given $0 < \varepsilon < 1$, choose $0 < \rho_0 < 1$ so small that

$$M_0 \omega_\mu (2D_0 \rho_0 / r_{\min}) < \varepsilon^2 / 8.$$

This is possible by L04. Fix $0 < \rho < \rho_0$, $k \geq 0$, and an anchor. The estimates (II.7.N05h)–(II.7.N05k), together with Markov's inequality, show that the mass of cells failing the desired conclusion is at most

$$p_{\text{bad}} + \frac{2p_{\text{bad}}}{\varepsilon} < \frac{\varepsilon^2}{8} + \frac{\varepsilon}{4} < \varepsilon.$$

Thus with μ -probability strictly greater than $1 - \varepsilon$ the anchor weight is positive and the raw component is within ε of the whole-cylinder mixture in total variation. This completes the proof, including the zero anchor-weight convention and the case $k = 0$. □

Whole-cylinder mixtures, group charts, and hyperplane control are the three geometric inputs used in the entropy arguments below: they preserve self-similar structure, retain all coordinates of a similarity, and make the replacement of cell restrictions quantitative.

II.7.2 From finite-scale saturation to a persistent direction

Definition II.7.2.1 (II.7.08: Saturation at a component level). For $\varepsilon > 0$, integers $m \geq 1, n \geq 0$, define

$$\text{Sat}(\mu; \varepsilon, m, n) = \{V \leq \mathbb{R}^d : \mu\{x : \text{def}_{V,m}(\mu^{x,n}) \leq \varepsilon\} \geq 1 - \varepsilon\},$$

where the saturation defect of [D-II.5.03](#) is

$$\text{def}_{V,m}(\theta) = \dim V - h_m(\theta) + h_m((\pi_{V^\perp})_*\theta).$$

It may be negative. Probabilities in the definition are for fixed V ; any later random choice of subspaces must have the measurable certifying events required by [D-II.5.08](#).

Definition II.7.2.2 (II.7.09: Persistent saturation). A subspace V is persistently saturated for μ if

$$\forall \varepsilon \in (0, 1) \exists M \geq 1 \forall m \geq M \exists N \geq 0 \forall n \geq N, V \in \text{Sat}(\mu; \varepsilon, m, n).$$

All four scale variables are integers when they denote levels or window lengths. In particular N may depend on m ; exchanging these quantifiers strengthens the property and is not permitted. Let $\mathcal{S}(\mu)$ be the family of such subspaces. A maximal persistent subspace means a member containing every member of $\mathcal{S}(\mu)$. Its existence and uniqueness are results.

Saturation measures how much of the available entropy in a direction is already present after the complementary projection has been accounted for. A small defect is a finite-scale assertion, not a claim that the measure contains a plane or has a smooth density.

The transport-and-mixture lemma [L-II.7.08](#) explains two ways to move such information. A similarity rotates the direction and shifts the observation level. A finite mixture can inherit saturation from all its summands; descent to summands needs fine enough conditioning so that the label densities are almost constant in most cells. The mass-weighted version is crucial when some cylinder weights are tiny: choosing a tolerance from the smallest weight would make the later scale choices circular.

The maximal-persistent-subspace proposition [P-II.7.09](#) takes finite sums of genuinely persistent directions to obtain a maximal one. Self-similarity then forces this maximal space to be invariant under the rotation group. The compact-group approximation lemma [L-II.7.07](#) makes this invariance quantitative at finite scales.

The almost-invariant enlargement proposition [P-II.7.10](#) finds an almost-invariant space containing all sufficiently saturated directions at a fixed fine level. The propagation proposition [P-II.7.11](#) then transfers approximate saturation near a truly invariant space to every sufficiently late level. The stopping decomposition is what reaches all of those levels. Finally, the compactness proposition [P-II.7.12](#) packages the parameter choices so that a convergent sequence of tuned, vanishing-error directions really lands inside the persistent space. An arbitrary sequence of isolated favorable windows would not establish this

conclusion.

Lemma II.7.2.3 (L-II.7.07 — Positive words and compact-group approximation). *The closure of the positive-word semigroup generated by the U_i equals G_Φ . For every $\eta > 0$, finitely many positive words form an η -net of this group in operator norm. For each $\eta > 0$ there is $\delta > 0$ such that every δ -almost invariant subspace is within gap distance η of an exactly G_Φ -invariant subspace of the same dimension. The latter assertion is uniform over the finite union of Grassmannians for this fixed group.*

Proof. Recovering inverses by positive powers

Let $U \in O(d)$. Compactness gives a convergent subsequence of $(U^n)_{n \geq 1}$. For each j , choose two sufficiently late terms of this subsequence, with indices $a_j < b_j$ and $b_j - a_j \geq \max\{2, j\}$, whose distance is below $1/j$. Multiplication by an orthogonal matrix preserves operator norm, so

$$\|U^{b_j-a_j} - I_d\|_{\text{op}} = \|U^{b_j} - U^{a_j}\|_{\text{op}} < 1/j, \quad U^{b_j-a_j-1} \longrightarrow U^{-1}. \quad (\text{II.7.N07a})$$

All powers displayed on the left and approximating the inverse are positive powers. Let S be the set of positive-word products of the generators U_i . Its closure $\bar{S}$ is closed under multiplication: approximate each of two factors by words and use continuity of matrix multiplication. By (II.7.N07a), it contains the identity and each U_i^{-1} . It consequently contains all products of generators and inverses, and hence their closure G_Φ . Conversely $S \subseteq G_\Phi$, which is closed, so $\bar{S} = G_\Phi$.

For $\eta > 0$, the open operator-norm balls of radius η about members of S cover G_Φ . A finite subcover exists by compactness. Their centers are the required finite positive-word η -net; no inverse symbol is introduced into those words.

Almost invariance for this fixed group

Write P_V for the orthogonal projection onto V . The Grassmannian of fixed dimension q is compact, as proved in L-II.5.04; directly its projection matrices satisfy $P^\Gamma = P$, $P^2 = P$, $\text{tr } P = q$ and form a closed bounded subset of the finite-dimensional matrix space. For $g \in O(d)$,

$$P_{gV} = gP_Vg^{-1}, \quad d_G(gV, gW) = d_G(V, W). \quad (\text{II.7.N07b})$$

Fix $\eta > 0$. If the claim failed, then for every integer $j \geq 1$ there would be a $1/j$ -almost invariant subspace V_j at distance at least η from every exactly G_Φ -invariant subspace of the same dimension. There are only $d + 1$ possible dimensions, so pass to a fixed-dimension subsequence and then to a gap-metric limit V . For each $g \in G_\Phi$, (II.7.N07b) and the triangle inequality give

$$d_G(gV, V) \leq 2d_G(V, V_j) + d_G(gV_j, V_j) \leq 2d_G(V, V_j) + 1/j \longrightarrow 0. \quad (\text{II.7.N07c})$$

Thus V is invariant under every g , and has the same dimension as V_j . But $d_G(V_j, V) \rightarrow 0$, a contradiction. This proves the existence of $\delta > 0$, uniformly over all dimensions, for the

fixed group and the prescribed tolerance. The zero and full subspaces cause no exception: they are already invariant. □

Lemma II.7.2.4 (L-II.7.08 — Saturation under mixtures and similarity transport). *The following two finite-level statements, and their persistent consequences, are owned here.*

1. *For every $0 < \varepsilon < 1$ there are $\delta > 0, M \geq 1$, depending only on d, ε , such that, for any compactly supported probability θ , any similarity $g(x) = rUx + a$, $m \geq M$, and integers $n, n_g = \lfloor n - \log_2 r \rfloor \geq 0$,*

$$V \in \text{Sat}(\theta; \delta, m, n) \implies UV \in \text{Sat}(g_*\theta; \varepsilon, m, n_g).$$

For a fixed g , the same implication holds with target level n instead of n_g , after allowing δ, M to depend on $|\log_2 r|$ and restricting to sufficiently large n . This fixed-map corollary is not uniform over shrinking word maps.

2. *Fix a finite mixture $\theta = \sum_{j=1}^J a_j \theta_j$ with $a_j > 0, \sum_j a_j = 1$, and compactly supported probabilities. For every $\varepsilon > 0$ there are $\delta > 0, M \geq 1$ such that for each $m \geq M$, some N makes both implications valid for all $n \geq N$ and all subspaces V :*

$$\begin{aligned} V \in \text{Sat}(\theta; \delta, m, n) &\implies V \in \bigcap_j \text{Sat}(\theta_j; \varepsilon, m, n), \\ V \in \bigcap_j \text{Sat}(\theta_j; \delta, m, n) &\implies V \in \text{Sat}(\theta; \varepsilon, m, n). \end{aligned} \tag{II.7.2.1}$$

The second implication has thresholds δ, M independent of the number and weights of summands and may take $N = 0$. The first is for a fixed mixture; its fine-level threshold is not asserted uniformly for a mixture changing with n .

3. *There is also a mass-weighted descent with δ, M depending only on d, ε , not on J or the weights. For each fixed mixture and $m \geq M$, some N makes, for all $n \geq N$ and all V ,*

$$V \in \text{Sat}(\theta; \delta, m, n) \implies \sum_{j=1}^J a_j \theta_j \{x : \text{def}_{V,m}(\theta_j^{x,n}) \leq \varepsilon\} \geq 1 - \varepsilon.$$

The threshold N still depends on the fixed mixture. This weighted statement, not a tolerance depending on the smallest word weight, is used in G18's root/word selection.

Consequently $\mathcal{S}(g_\theta) = \{UV : V \in \mathcal{S}(\theta)\}$ and $\mathcal{S}(\sum a_j \theta_j) = \bigcap_j \mathcal{S}(\theta_j)$. In the first finite-level assertion a contracting map shifts to a finer, not a coarser, target level.*

Proof. There are four steps: transport saturation under similarities, ascend through mixtures, descend to mass-weighted summands at fine scales, and then pass these finite-scale statements to persistent directions.

All auxiliary constants below depend only on d , unless an additional dependence is explicitly displayed. We prove estimates for $0 < \varepsilon < 1$; larger tolerances follow by monotonicity. Every component is sampled with its mass, not uniformly over its cells.

Expected defects and a conditional-entropy formula

For a compactly supported probability θ , put

$$e_{V,m,n}(\theta) = \int \text{def}_{V,m}(\theta^{x,n}) d\theta(x). \quad (\text{II.7.N08a})$$

Write $q = \dim V$. For a probability ζ carried by $[0, 1]^d$, [L-II.5.01](#) and the finite-alphabet entropy bound give

$$-C_d/m \leq \text{def}_{V,m}(\zeta) \leq d + C_d/m. \quad (\text{II.7.N08b})$$

For the lower bound use the product-coordinate formula in [L-II.5.01](#): the projection of the unit cube to V meets at most $C_d 2^{qm}$ intrinsic level- m cubes, so its conditional entropy is at most $qm \log 2 + \log C_d$. For the upper bound use $H_m(\zeta) \geq 0$ and the analogous count of at most $C_d 2^{(d-q)m}$ projected cubes in $V^\perp$. In particular, if $V \in \text{Sat}(\theta; \delta, m, n)$ and $0 < \delta < 1$, then

$$e_{V,m,n}(\theta) \leq (d+1)\delta + C_d/m. \quad (\text{II.7.N08c})$$

Conversely Markov's inequality applied to the nonnegative defect plus C_d/m gives, for $t > 0$,

$$\theta\{x : \text{def}_{V,m}(\theta^{x,n}) > t\} \leq \frac{e_{V,m,n}(\theta) + C_d/m}{t}. \quad (\text{II.7.N08d})$$

Let $\mathcal{Q}_s^V = \pi_V^{-1} \mathcal{D}_s^V$, using the intrinsic grid of II.5; define $\mathcal{Q}_s^{V^\perp}$ likewise. For integer $s = n + m$,

$$\left| e_{V,m,n}(\theta) - q + \frac{H_\theta(\mathcal{Q}_{n+m}^V \mid \mathcal{Q}_{n+m}^{V^\perp} \vee \mathcal{D}_n)}{m \log 2} \right| \leq C_d/m. \quad (\text{II.7.N08e})$$

Here all entropies are finite. To verify the formula, condition first on a positive-mass cell $Q = 2^{-n}(k + [0, 1]^d)$. Its canonical map is $x \mapsto 2^n x - k$. Pulling back a level- m grid in either orthogonal factor yields a translate of its level- $n + m$ grid. Two translates of a grid in a fixed a -dimensional space have mutual intersection multiplicity at most 2^a . The conditional entropy comparison of II.1 therefore changes the two-factor conditional entropy by at most C_d . Apply [L-II.5.01](#) to the canonical component and average over Q . This proves [\(II.7.N08e\)](#), without assuming that the projected grid refines the ambient grid.

We shall also use the following consequence of the chain rule. For any four finite-on-the-support partitions $\mathcal{A}, \mathcal{B}, \mathcal{F}, \mathcal{P}$,

$$\begin{aligned} H(\mathcal{F} \mid \mathcal{P} \vee \mathcal{B}) &\geq H(\mathcal{F} \mid \mathcal{P} \vee \mathcal{A} \vee \mathcal{B}) \\ &\geq H(\mathcal{F} \mid \mathcal{P} \vee \mathcal{A}) - H(\mathcal{B} \mid \mathcal{A}). \end{aligned} \quad (\text{II.7.N08f})$$

The second inequality follows by writing the conditional entropy of $\mathcal{F} \vee \mathcal{B}$ given $\mathcal{P} \vee \mathcal{A}$ in both orders and using nonnegativity. This is the only refinement loss needed for transport.

Similarity transport at matched levels

Let $g(x) = rUx + a$, $n' = \lfloor n - \log_2 r \rfloor \geq 0$. On the original space put $\mathcal{A} = \mathcal{D}_n$ and $\mathcal{B} = g^{-1}\mathcal{D}_{n'}$. A cell of $\mathcal{B}$ is a rotated cube with side in $[2^{-n}, 2^{1-n})$. Therefore every cell of $\mathcal{A}$ meets at most a dimension-dependent number of cells of $\mathcal{B}$, and

$$H_\theta(\mathcal{B} \mid \mathcal{A}) \leq C_d. \quad (\text{II.7.N08g})$$

For completeness this counting claim follows by comparing volumes: all cells of a fixed rotated grid meeting a cube of side 2^{-n} lie in a ball of radius $C_d 2^{-n}$; their disjoint interiors each have volume at least 2^{-dn} . Boundaries do not change the count, since the diameter enlargement includes their closures.

Pull back the level- $n' + m$ intrinsic factor grids for UV and $(UV)^\perp$ by g . They are product-coordinate partitions on $V, V^\perp$, with side between $2^{-(n+m)}$ and $2^{1-(n+m)}$, up to independent translations and orthogonal choices of coordinates within the factors. Their mutual intersection multiplicity with $\mathcal{Q}_{n+m}^V, \mathcal{Q}_{n+m}^{V^\perp}$ is bounded by the same volume argument in the two factor spaces. Replacing these fine partitions changes their conditional entropy, even after conditioning on $\mathcal{B}$, by at most C_d . This follows by bounding each of the two conditional entropies of a partition given its comparison partition, and then applying the chain rule to their joins.

Apply (II.7.N08e) on each side of the change of variables, replace the fine partitions as above, and apply (II.7.N08f)–(II.7.N08g). We obtain

$$e_{UV,m,n'}(g_*\theta) \leq e_{V,m,n}(\theta) + C_d/m. \quad (\text{II.7.N08h})$$

There is no dependence here on r , translation, support diameter, or observing level. In particular a contracting map gives the finer target level required in the statement. Choose $\delta > 0$ small and then $M \geq 1$ large so that $(d+1)\delta + 3C_d/M \leq \varepsilon^2/2$. Equations (II.7.N08c), (II.7.N08h) and (II.7.N08d) show that target bad-component mass is at most $\varepsilon/2$, with defect threshold ε . This proves the first implication.

For the fixed-map assertion use $\mathcal{B} = g^{-1}\mathcal{D}_n$ instead. Its cube side is $r^{-1}2^{-n}$, and the analogous coarse and fine grid-intersection counts have logarithms bounded by $C_d(1 + |\log_2 r|)$. The same chain-rule argument gives

$$e_{UV,m,n}(g_*\theta) \leq e_{V,m,n}(\theta) + \frac{C_d(1 + |\log_2 r|)}{m}. \quad (\text{II.7.N08i})$$

Selecting M with this additional dependence proves the fixed-map version. This version actually holds for every $n \geq 0$; no uniformity over an unbounded family of ratios is claimed.

Mixture ascent, with no least-weight dependence

Let $\theta = \sum_{j=1}^J a_j \theta_j$ be the fixed mixture of the statement. Given a positive-mass ambient cell Q , its canonical component satisfies the exact identity

$$\theta^Q = \sum_{j: \theta_j(Q) > 0} \frac{a_j \theta_j(Q)}{\theta(Q)} \theta_j^Q. \quad (\text{II.7.N08j})$$

Conditional entropy is concave under such mixtures. One proof is to introduce the finite label j : conditioning also on this label cannot increase entropy, and its conditional entropy is the weighted sum of those of the summands. Apply this to the two factor partitions in L-II.5.01. With its C_d/m comparison error, (II.7.N08j) therefore gives

$$e_{V,m,n}(\theta) \leq \sum_{j=1}^J a_j e_{V,m,n}(\theta_j) + C_d/m. \quad (\text{II.7.N08k})$$

If every summand is in $\text{Sat}(\cdot; \delta, m, n)$, (II.7.N08c)–(II.7.N08d) and (II.7.N08k) prove ascent by taking δ small and M large in terms of d, ε . Neither J nor the weights appear, and $N = 0$ is valid.

Fine components of a fixed absolutely continuous summand

Set $f_j = d\theta_j/d\theta$; choose a Borel version with $0 \leq f_j \leq 1/a_j$. For a level- n cell Q of positive θ -mass, set

$$f_{j,n}|_Q = \frac{1}{\theta(Q)} \int_Q f_j d\theta. \quad (\text{II.7.N08l})$$

Give zero-mass cells value zero. These finite-on-the-support conditional averages satisfy $\|f_j - f_{j,n}\|_{L^1(\theta)} \rightarrow 0$. Here is a direct justification sufficient for the asserted probability bounds. The refining dyadic partitions generate the Borel sigma algebra on the compact support. Their finite algebras approximate each Borel set in measure: the class of approximable sets is closed under complement and countable union, using finite truncation of the union and finite measure. Thus grid-simple functions are dense in $L^1(\theta)$. Given such an approximation s measurable at level N , for $n \geq N$ conditional averaging fixes s and is an L^1 contraction. Consequently

$$\|f_j - f_{j,n}\|_1 \leq 2\|f_j - s\|_1 \rightarrow 0 \quad (n \rightarrow \infty). \quad (\text{II.7.N08m})$$

The displayed limit means to choose s to any prescribed accuracy first. If $\theta_j(Q) > 0$, then $f_{j,n}|_Q > 0$, and the Radon–Nikodym derivative of the raw θ_j -component relative to the raw θ -component is $f_j/f_{j,n}|_Q$. The common canonical bijection preserves total variation. Hence exactly

$$\begin{aligned} \int \|\theta_j^{x,n} - \theta^{x,n}\|_{\text{TV}} d\theta_j(x) &= \frac{1}{2} \sum_{Q: \theta_j(Q) > 0} \int_Q |f_j - f_{j,n}| d\theta \\ &= \frac{1}{2} \|f_j - f_{j,n}\|_{L^1(\theta)} \rightarrow 0. \end{aligned} \quad (\text{II.7.N08n})$$

On omitted cells $f_j = f_{j,n} = 0$ almost everywhere, so their integrals are zero. No division by a zero density and no pointwise convex domination is needed. This averaged identity is sufficient; an almost-sure differentiation theorem is not being imported.

Entropy continuity and the two descent statements

For two canonical components ζ, ξ at total variation distance at most $0 < t < 1/2$, uniformly in V ,

$$|\text{def}_{V,m}(\zeta) - \text{def}_{V,m}(\xi)| \leq C_d t + \frac{C_d(t + b(t))}{m}, \quad b(t) = -t \log t - (1-t) \log(1-t). \quad (\text{II.7.N08o})$$

Indeed the ambient grid has at most 2^{dm} possible cells on $[0, 1)^d$, and each projected grid has at most $C_d 2^{dm}$ cells, independently of orientation. Pushforward does not increase total variation. The finite-alphabet continuity bound of L-II.4.01 applied to both entropies gives (II.7.N08o), after division by $m \log 2$. This is a uniform numerical bound, not a continuity assertion about the choice of subspace coordinates.

Choose $t \in (0, 1/2)$ so small and M so large that the right side of (II.7.N08o) is at most $\varepsilon/2$ whenever $m \geq M$. For weighted descent take $0 < \delta \leq \varepsilon/4$, independent of the mixture. By (II.7.N08n) choose N so that, for every $n \geq N$,

$$\sum_j a_j \theta_j \{x : \|\theta_j^{x,n} - \theta^{x,n}\|_{\text{TV}} > t\} \leq \frac{1}{2t} \sum_j a_j \|f_j - f_{j,n}\|_1 < \varepsilon/4. \quad (\text{II.7.N08p})$$

The joint experiment j with weight a_j , followed by $x \sim \theta_j$, has point marginal θ . Thus if $V \in \text{Sat}(\theta; \delta, m, n)$, the probability in this experiment that the θ -component has defect greater than δ is at most δ . Outside that event and the event in (II.7.N08p), (II.7.N08o) makes the summand defect at most $\delta + \varepsilon/2 < \varepsilon$. Its total bad mass is less than $\varepsilon/2$, proving weighted descent. The same N works for all V , because neither TV event depends on V . It may depend on the fixed mixture; that dependence is essential.

For individual descent let $a_* = \min_j a_j > 0$, and require also $\delta \leq a_* \varepsilon/4$. The bad set for the θ -component then has θ_j -mass at most $\delta/a_j \leq \varepsilon/4$. Increase N , if necessary, so that each individual TV bad event has θ_j -mass below $\varepsilon/4$, using (II.7.N08n) and the finiteness of the mixture. The preceding defect comparison now proves that every θ_j is in $\text{Sat}(\cdot; \varepsilon, m, n)$. Together with ascent this proves both implications in part 2, with the stated parameter dependences. Taking minima of the input tolerances and maxima of the window thresholds makes them simultaneous.

Persistent subspaces

Fix an output tolerance ε first. For a fixed similarity, choose the transport input tolerance δ and threshold M_0 . Persistence of V for θ supplies M_1 for this δ , and then, for each $m \geq \max\{M_0, M_1\}$, a level threshold N . Apply the fixed-map implication at every $n \geq N$. This proves persistence of UV for $g_*\theta$. Apply the same argument to g^{-1} for the reverse inclusion.

For the fixed finite mixture use individual descent in the same order: ε , then its δ, M_0 , then the persistence window threshold for θ , then the maximum of the two level thresholds for each chosen window. This puts every persistent subspace of the mixture in every summand's persistent family. Conversely, if it is persistent for every summand, use the mixture-independent ascent tolerance, take the maximum of their finitely many window

thresholds, and then their finitely many level thresholds for the chosen window. Ascent applies at every subsequent level. We have proved exactly

$$\mathcal{S}(g_*\theta) = \{UV : V \in \mathcal{S}(\theta)\}, \quad \mathcal{S}\left(\sum_j a_j \theta_j\right) = \bigcap_j \mathcal{S}(\theta_j). \quad (\text{II.7.N08q})$$

No interchange of the window and observing-level quantifiers is used. □

Proposition II.7.2.5 (P-II.7.09 — Maximal persistent subspace). *The family $\mathcal{S}(\mu)$ contains $\{0\}$, is closed under taking subspaces and finite sums, and has a unique maximal member $V_{\max}(\mu)$. This member is G_Φ -invariant. No topological closedness of $\mathcal{S}(\mu)$ is assumed.*

Proof. Zero, subspaces and finite sums

For $V = \{0\}$, the projected measure in the defect formula is the measure itself, so the defect is exactly zero. Thus $\{0\} \in \mathcal{S}(\mu)$. If $W \subseteq V$ and $V \in \mathcal{S}(\mu)$, fix $0 < \varepsilon < 1$. By L-II.5.04(4), a $(V, \varepsilon/2, m)$ -saturated canonical component is $(W, \varepsilon/2 + C_d/m, m)$ -saturated. First make m at least the persistence threshold for V with error $\varepsilon/2$, and at least $2C_d/\varepsilon$; then take its eventual level threshold. At every subsequent level the required W -components have probability at least $1 - \varepsilon/2$. This proves $W \in \mathcal{S}(\mu)$.

Now let $V_1, V_2 \in \mathcal{S}(\mu)$. If one contains the other, their sum is already persistent. Otherwise, put $E = V_1 \cap V_2$ and use the transverse angle of D-II.5.01. The two unit spheres in $V_i \cap E^\perp$ are nonempty compact and disjoint: a common unit vector would belong both to E and $E^\perp$. Hence their mutual distance γ is positive. Set $\gamma_0 = \min\{\gamma, 1/2\}$. For a prescribed $0 < \varepsilon < 1$, use persistence of both spaces with input tolerance $\delta = \varepsilon/4$, and require also

$$m \geq \frac{2C_d(1 + \log(1/\gamma_0))}{\varepsilon}. \quad (\text{II.7.N09a})$$

For each such m , take the maximum of their eventual level thresholds. At every larger n , with probability at least $1 - 2\delta$, the same component is saturated on both spaces. L-II.5.04(5) and (II.7.N09a) make that component $(V_1 + V_2, \varepsilon, m)$ -saturated. Since $2\delta \leq \varepsilon$, the sum is persistent. Induction proves closure under every finite sum.

Constructing the maximal member in finite dimension

Begin with $W_0 = \{0\}$. If W_j does not contain every member of $\mathcal{S}(\mu)$, choose one such member V_j and set $W_{j+1} = W_j + V_j$. Each W_j is persistent by the finite-sum property, and a nonterminal step strictly increases its dimension. After at most d steps the construction stops at a persistent space containing every persistent space. Denote it by $V_{\max}(\mu)$. Two such members contain one another, so are equal. No closure of an infinite union, or topological closedness of $\mathcal{S}(\mu)$, was used.

Invariance under the generated orthogonal group

The self-similar identity and both persistent identities of L08 give

$$\mathcal{S}(\mu) = \mathcal{S}\left(\sum_i p_i(\varphi_i)_*\mu\right) = \bigcap_i \mathcal{S}((\varphi_i)_*\mu) = \bigcap_i \{U_i V : V \in \mathcal{S}(\mu)\}. \quad (\text{II.7.N09b})$$

All weights here are strictly positive and the mixture is fixed. Since $V_{\max}$ belongs to the left side, for each i , $U_i^{-1}V_{\max}$ is persistent, hence contained in $V_{\max}$. Equal dimensions force equality. Therefore $U_i V_{\max} = V_{\max}$ for every i , and every product of generators and inverses preserves it. Passing to the operator-norm closure preserves this property: if $g_j \rightarrow g$ and $g_j V_{\max} = V_{\max}$, then $g_j P_{V_{\max}} g_j^{-1} \rightarrow g P_{V_{\max}} g^{-1}$, and all matrices in the first sequence equal $P_{V_{\max}}$. Thus $V_{\max}$ is G_Φ -invariant, as claimed. $\square$

Definition II.7.2.6 (II.7.10: Almost invariance and projected component averages). For a compact subgroup $G \subset O(d)$, a subspace V is ε -almost invariant if

$$\sup_{g \in G} d_G(gV, V) \leq \varepsilon.$$

For $V \leq \mathbb{R}^d$ define

$$A_{V,m,n}(\mu) = \int h_m((\pi_{V^\perp})_* \mu^{x,n}) d\mu(x), \quad m \geq 1, \quad n \geq 0.$$

Projection is applied after conditioning in the ambient grid. Thus this is not the mean entropy of components of $(\pi_{V^\perp})_* \mu$. The order $N \rightarrow \infty$ followed by $m \rightarrow \infty$ in later averages is part of each theorem, not interchangeable by convention.

Proposition II.7.2.7 (P-II.7.10 – Almost-maximal subspaces at a fine level). *For every $0 < \varepsilon < 1/10$, there are $\delta > 0$, $M \geq 1$ such that for each $m \geq M$, some N has the following property. For every $n \geq N$ there is $V^* = V^*(m, n)$ with*

$$V^* \in \text{Sat}(\mu; \varepsilon, m, n), \quad \sup_{g \in G_\Phi} d_G(gV^*, V^*) \leq \varepsilon,$$

$$W \in \text{Sat}(\mu; \delta, m, n) \implies W \subseteq (V^*)^{(\varepsilon)}.$$

All parameters may depend on Φ , p . Directed containment, not the symmetric transverse angle, is the last conclusion.

Proof. A finite Grassmannian net replaces arbitrary saturated directions by a common almost-maximal plane. Stability under sums and small rotations then yields almost invariance.

First choose the finite rotation family

Fix $0 < \varepsilon < 1/10$. Choose $0 < \tau < 1/2$ so that $2d2^d\tau < \varepsilon/2$ and $\tau < \varepsilon/2$. For $1 \leq j \leq d+1$, set

$$a_j = \tau^{j!}, \quad \zeta = a_{d+1}/2, \quad \sigma = \frac{\varepsilon}{2^{d+4}(d+1)}. \quad (\text{II.7.N10a})$$

By L07 choose a finite collection $\mathcal{U}$ of positive words whose inverse orthogonal parts U_u^{-1} form a ζ -net of G_Φ . Inversion is an isometry on $O(d)$, so L07 gives exactly such a net. Add the empty word. This entire family is fixed before any window or observing level is chosen.

For each nonempty $u \in \mathcal{U}$, the fixed-mixture identity at depth $|u|$ contains $(\varphi_u)_*\mu$ with its positive weight p_u . Individual mixture descent in L08, followed by its fixed-map transport for φ_u^{-1} , gives thresholds $\delta > 0, M_0 \geq 1$ such that, for each $m \geq M_0$, some $N_0(m)$ makes the implication

$$V \in \text{Sat}(\mu; \delta, m, n) \implies U_u^{-1}V \in \text{Sat}(\mu; \sigma/2, m, n) \quad (u \in \mathcal{U}) \quad (\text{II.7.N10b})$$

valid for all $n \geq N_0(m)$ and all V . Explicitly, choose the inverse-transport tolerance for output $\sigma/2$ for each of the finitely many words first; choose a smaller common output tolerance for all their mixture descents; take the minimum of the resulting input tolerances and the maximum of all window thresholds. After m is chosen, take the maximum of their finitely many level thresholds. For the empty word require only $\delta \leq \sigma/2$. All these choices depend on the fixed system and ε , not on V or n .

Increase M_0 so that $C_d/m \leq \sigma/2$, where C_d is the subspace-descent constant in L-II.5.04(4). Every line in a space on the right of (II.7.N10b) is then in $\text{Sat}(\mu; \sigma, m, n)$.

An approximately rotation-invariant set of unit vectors

Fix an allowed m, n and let $\mathcal{A} = \text{Sat}(\mu; \delta, m, n)$. Form the set

$$E = \{e : \|e\| = 1, e \in U_u^{-1}V \text{ for some } u \in \mathcal{U}, V \in \mathcal{A}\}. \quad (\text{II.7.N10c})$$

If $E = \emptyset$, every space in $\mathcal{A}$ is zero, and take $V^* = \{0\}$. All conclusions hold. Otherwise every line Re with $e \in E$ is saturated with tolerance σ . Moreover, for every $g \in G_\Phi$ and $e \in E$, there exists $e' \in E$ such that

$$\|ge - e'\| < \zeta. \quad (\text{II.7.N10d})$$

Indeed write $e = U_u^{-1}v$ with $v \in V$, $\|v\| = 1$. Choose $u' \in \mathcal{U}$ with $\|gU_u^{-1} - U_{u'}^{-1}\|_{\text{op}} < \zeta$, and use $e' = U_{u'}^{-1}v$. No measurable selection from E is needed: the construction below chooses finitely many deterministic vectors for the fixed m, n .

A finite greedy span and its invariance estimate

Start with $W_0 = \{0\}$. At step j , if there is $v_j \in E$ with $\text{dist}(v_j, W_{j-1}) > a_j$, choose one and put $W_j = W_{j-1} + \mathbb{R}v_j$; otherwise stop. At most d choices can occur. Write k for their number,

and put $W = W_k$. If $k = d$, then $W = \mathbb{R}^d$ and the invariance and containment assertions are immediate. If $k < d$, the stopping rule gives

$$\text{dist}(e, W) \leq a_{k+1} \quad (e \in E). \quad (\text{II.7.N10e})$$

For a unit $w \in W$, write $w = \sum_{i=1}^k c_i v_i$. The coefficient estimate L-II.5.05(1), since all successive distances are greater than a_k , gives

$$\sum_i |c_i| \leq k 2^k a_k^{-k}. \quad (\text{II.7.N10f})$$

By (II.7.N10d)–(II.7.N10e), $\text{dist}(g v_i, W) \leq \zeta + a_{k+1} \leq 2a_{k+1}$. It follows that

$$\begin{aligned} \text{dist}(gw, W) &\leq 2k 2^k a_{k+1} a_k^{-k} = 2k 2^k \tau^{(k+1)!-k k!} \\ &= 2k 2^k \tau^{k!} \leq 2d 2^d \tau < \varepsilon/2. \end{aligned} \quad (\text{II.7.N10g})$$

Equal dimensions and the gap characterization in L-II.5.04(1) turn this directed containment into $d_G(gW, W) \leq \varepsilon/2$, for every g . If $V \in \mathcal{A}$, all its unit vectors lie in E , since the empty word was included. Equation (II.7.N10e), with $a_{k+1} \leq \tau < \varepsilon$, gives $V \sqsubseteq W^{(\varepsilon)}$. This also holds in the full-dimensional case.

Saturation of the span and final parameter order

The event that one component is saturated on all k chosen lines has probability at least $1 - k\sigma$. When $i \geq 2$, the line $\mathbb{R}v_i$ is disjoint from W_{i-1} except at zero. Its transverse angle with W_{i-1} is at least $\text{dist}(v_i, W_{i-1}) > a_i \geq a_d$: the distance between unit vectors is no smaller than the distance to the whole subspace. Repeated use of L-II.5.04(5) therefore bounds the defect on W_i on this common event by the numbers

$$b_1 = \sigma, \quad b_i = 2b_{i-1} + \frac{C_d(1 + \log(1/a_d))}{m}, \quad b_i \leq 2^{i-1}\sigma + \frac{(2^{i-1} - 1)C_d(1 + \log(1/a_d))}{m}. \quad (\text{II.7.N10h})$$

Choose $M \geq M_0$ also so large that the last term for $i = d$ is at most $\varepsilon/4$. Our choice of σ makes $2^{d-1}\sigma < \varepsilon/4$ and $d\sigma < \varepsilon$. Consequently W is in $\text{Sat}(\mu; \varepsilon, m, n)$. Set $V^* = W$. The construction works also when $k = d$; saturation was never assumed merely because the span filled the ambient space.

To summarize the order, choose ε , then τ, ζ, σ and the finite word net, then δ, M , then $m \geq M$, then $N_0(m)$, and finally any $n \geq N_0(m)$. This is exactly the quantifier order of P10. No assertion that an arbitrary small perturbation of a saturated subspace remains saturated was used.

□

Proposition II.7.2.8 (P-II.7.11 – Fixed short-window persistence). *For every $0 < \varepsilon < 1$ there is $P_0 \geq 1$ such that for each integer $p \geq P_0$ there are $\delta > 0, M > p$ with the following property. If $m \geq M, n \geq 0$, W is exactly G_Φ -invariant, $d_G(V, W) \leq \delta$, and*

$V \in \text{Sat}(\mu; \delta, m, n)$, then

$$W \in \text{Sat}(\mu; \varepsilon, p, n') \quad \text{for every } n' \geq n + m + C_\Phi,$$

where C_Φ is an integer depending only on the system. The short window p is selected before δ, M . The proof must reach every later level using stopping words; a sequence of intervals with sublinear widths is not sufficient justification.

Proof. The stopping decomposition reaches every sufficiently late level; mixture stability transports the fixed-window saturation while keeping the error independent of small cylinder weights.

Uniform transport across a stopping family

We first record the bounded-mismatch version of the grid argument in L08. Suppose $l, n' \geq 0, n' > l$, and

$$r_{\min} 2^{-(n'-l)} < r \leq 2^{-(n'-l)}, \quad g(x) = rUx + a. \quad (\text{II.7.N11a})$$

The pullback of the target level- n' grid has cube side between 2^{-l} and $r_{\min}^{-1} 2^{-l}$. The corresponding fine grids at window length p have the same side ratio. The intersection counts and conditional chain-rule inequality (II.7.N08e)–(II.7.N08f) therefore give

$$e_{UW, p, n'}(g_*\theta) \leq e_{W, p, l}(\theta) + K_\Phi/p, \quad K_\Phi = C_d(1 + \log(1/r_{\min})), \quad (\text{II.7.N11b})$$

after increasing the dimension constant. This estimate is uniform over all maps satisfying (II.7.N11a), regardless of their word length or weight. Its justification is the same direct coarse/fine cube count, not the fixed-map assertion with a constant growing with $|\log r|$.

Assume now that W is exactly G_Φ -invariant. Use the stopping identity L05 at $h = 2^{-(n'-l)}$. Every stopping map has (II.7.N11a), and $U_u W = W$. Average (II.7.N11b) with its weights p_u , then apply the expectation form of mixture ascent, (II.7.N08k). For a larger fixed K_Φ we obtain

$$e_{W, p, n'}(\mu) \leq e_{W, p, l}(\mu) + K_\Phi/p \quad (n' > l). \quad (\text{II.7.N11c})$$

The finite stopping family changes with n' , but neither bound contains its cardinality or least weight. Thus (II.7.N11c) holds for every later integer level, not just a subsequence of aligned levels.

Selecting a short saturated window before retuning

Fix $0 < \varepsilon < 1$ and put $\eta = \varepsilon^2/[8(d+1)]$. Let $C \geq 1$ dominate the dimension constants in P-II.5.03(1), L-II.5.04(4) and (II.7.N08b). Choose an integer $P_0 \geq 1$ so large that for every $p \geq P_0$,

$$\frac{C}{\sqrt{p}} \leq \eta/8, \quad \frac{C}{p} \leq \eta/8, \quad \frac{K_\Phi + 2C}{p} \leq \varepsilon^2/8. \quad (\text{II.7.N11d})$$

Fix such a short window p . Next choose $0 < \delta < 1$ and an integer $M > p$ satisfying

$$\delta \leq \min\{2^{-p}, \eta/8\}, \quad C\sqrt{\delta} \leq \eta/8, \quad C\sqrt{p/M} \leq \eta/8. \quad (\text{II.7.N11e})$$

For $m \geq M$, define

$$\alpha = C\sqrt{\delta} + C/\sqrt{p} + C\sqrt{p/m} \leq 3\eta/8. \quad (\text{II.7.N11f})$$

Suppose the hypotheses of P11 hold at the given level n . With μ -probability at least $1 - \delta$, the canonical level- n component is (V, δ, m) -saturated. Apply P-II.5.03(1) to each such component with long window m and short window p . Its level average of good short components is at least $1 - \alpha$. Composition of canonical components and mass-weighted sampling give

$$\frac{1}{m} \sum_{k=0}^{m-1} \mu\{z : \text{def}_{V,p}(\mu^{z,n+k}) \leq \alpha\} \geq (1 - \delta)(1 - \alpha) \geq 1 - \delta - \alpha. \quad (\text{II.7.N11g})$$

The composition used here is the dyadic conditioning identity of II.2: first sample a coarse cell with its μ -mass, then a point in its conditional measure; the point's unconditional law is μ , and its further k -level canonical component is the direct level- $n + k$ component. This is why no unweighted cell average can replace (II.7.N11g).

For some $k \in \{0, \dots, m-1\}$ the corresponding summand is at least $1 - \delta - \alpha$. Set $l = n + k$. Since $d_G(V, W) \leq \delta \leq 2^{-p}$, the nearby-subspace assertion L-II.5.04(4) makes every good component there $(W, \alpha + C/p, p)$ -saturated. Both its new defect bound and its bad probability are at most η , by (II.7.N11d)–(II.7.N11f). Therefore

$$W \in \text{Sat}(\mu; \eta, p, l), \quad e_{W,p,l}(\mu) \leq (d+1)\eta + C/p. \quad (\text{II.7.N11h})$$

Reaching every later level

Take $C_\Phi = 1 + \lceil \log_2(1/r_{\min}) \rceil$. For every $n' \geq n + m + C_\Phi$, one has $n' > l$. Equations (II.7.N11c)–(II.7.N11h), followed by (II.7.N08d), give

$$\begin{aligned} \mu\{z : \text{def}_{W,p}(\mu^{z,n'}) > \varepsilon\} &\leq \frac{(d+1)\eta + (K_\Phi + 2C)/p}{\varepsilon} \\ &\leq \frac{\varepsilon^2/8 + \varepsilon^2/8}{\varepsilon} = \varepsilon/4. \end{aligned} \quad (\text{II.7.N11i})$$

This proves the claimed $\text{Sat}(\mu; \varepsilon, p, n')$ membership at every such n' . The short window was chosen before δ, M ; the latter depend on p . Exact invariance is used only in the stopping-family transport. No convergence of a changing mixture, and no comparison at a fixed ratio-dependent error, enters the all-later-level conclusion. □

Proposition II.7.2.9 (P-II.7.12 – Capture of vanishing-error saturation limits). *There are integer-valued functions $M_*(\varepsilon) \geq 1$ and $N_*(\varepsilon, m) \geq 0$, with $M_*(\varepsilon) \rightarrow \infty$ as $\varepsilon \downarrow 0$, such*

that the following holds. If

$$\varepsilon_j \rightarrow 0, \quad m_j \geq M_*(\varepsilon_j), \quad n_j \geq N_*(\varepsilon_j, m_j), \quad V_j \in \text{Sat}(\mu; \varepsilon_j, m_j, n_j), \quad d_G(V_j, V) \rightarrow 0,$$

then $V \subseteq V_{\max}(\mu)$. Only this tuned family of limits is asserted; saturation at arbitrary isolated windows need not be persistent.

Proof. An explicit diagonal choice of threshold functions

For each integer $k \geq 1$, set $t_k = 2^{-k-4}$. Apply P10 with output error t_k , and denote its input tolerance and thresholds by $\delta_k, M_k, N_k(m)$. Enlarge M_k to an integer at least one. Define the positive decreasing sequence

$$a_k = \min\{2^{-k}, \delta_1, \dots, \delta_k\}. \quad (\text{II.7.N12a})$$

For $0 < \varepsilon \leq a_1$, let $k(\varepsilon) = \max\{k \geq 1 : \varepsilon \leq a_k\}$. This maximum is finite because $a_k \leq 2^{-k}$, and $k(\varepsilon) \rightarrow \infty$ as $\varepsilon \downarrow 0$. Set

$$\begin{aligned} M_*(\varepsilon) &= \max\{[1/\varepsilon], M_{k(\varepsilon)}\}, \\ N_*(\varepsilon, m) &= N_{k(\varepsilon)}(m) \quad (m \geq M_*(\varepsilon)). \end{aligned} \quad (\text{II.7.N12b})$$

On unused values $m < M_*(\varepsilon)$ set $N_* = 0$. For $\varepsilon > a_1$, set $M_*(\varepsilon) = \max\{1, [1/\varepsilon]\}$ and $N_*(\varepsilon, m) = 0$. These choices define the requested integer-valued functions for every positive tolerance, and $M_*(\varepsilon) \rightarrow \infty$.

Passing to an invariant containing space

Consider a sequence in the statement. Discard a finite prefix so that $\varepsilon_j \leq a_1$, and put $k_j = k(\varepsilon_j)$. Since $\varepsilon_j \leq \delta_{k_j}$, monotonicity of the definition of Sat and P10 give spaces W_j such that

$$\begin{aligned} W_j &\in \text{Sat}(\mu; t_{k_j}, m_j, n_j), & \sup_{g \in G_\Phi} d_G(gW_j, W_j) &\leq t_{k_j}, \\ V_j &\subseteq W_j^{(t_{k_j})}, & t_{k_j} &\longrightarrow 0. \end{aligned} \quad (\text{II.7.N12c})$$

Extract a fixed-dimension subsequence of W_j converging in the gap metric to a subspace W . For every $g \in G_\Phi$,

$$d_G(gW, W) \leq 2d_G(W, W_j) + t_{k_j} \longrightarrow 0. \quad (\text{II.7.N12d})$$

Thus W is exactly invariant. Directed containment, rather than a transverse-angle estimate, gives

$$\|(I_d - P_{W_j})P_{V_j}\|_{\text{op}} \leq t_{k_j}. \quad (\text{II.7.N12e})$$

Pass to the limit using $P_{V_j} \rightarrow P_V$ and $P_{W_j} \rightarrow P_W$. We get $(I_d - P_W)P_V = 0$, equivalently $V \subseteq W$.

Testing each fixed short window

Fix $0 < \eta < 1$. P11 gives a threshold $P_0(\eta)$. Fix any integer $p \geq P_0(\eta)$; then P11 gives its tolerance $\delta(\eta, p) > 0$ and long-window threshold $M(\eta, p) > p$. Choose a sufficiently late index in the extracted subsequence so that

$$t_{k_j} \leq \delta(\eta, p), \quad d_G(W_j, W) \leq \delta(\eta, p), \quad m_j \geq M(\eta, p). \quad (\text{II.7.N12f})$$

The last requirement is possible because $m_j \geq 1/\varepsilon_j \rightarrow \infty$. P11 applied to W_j, W now yields

$$W \in \text{Sat}(\mu; \eta, p, n') \quad \text{for every } n' \geq n_j + m_j + C_\Phi. \quad (\text{II.7.N12g})$$

The chosen index may depend on the fixed short window p , exactly as its eventual level threshold is allowed to. Since η was arbitrary and this argument works for every $p \geq P_0(\eta)$, it proves $W \in \mathcal{S}(\mu)$. P09 gives $W \subseteq V_{\max}(\mu)$, and (II.7.N12e) gives the desired $V \subseteq V_{\max}(\mu)$. There is no assertion about an untuned sequence of isolated scales. $\square$

Projecting after conditioning is not the same as conditioning after projecting. The grid cells have different shapes and the first operation remembers more location information. The quotient-entropy lemma L-II.7.13 bounds their discrepancy by using self-similar cylinders at the original observation scale. Its lower bound is special to an invariant quotient; it does not invoke a general projection theorem.

The two limits have distinct jobs. First average over many locations in the scale sequence with a fixed window. Then let the window grow. The full-fiber proposition G-II.7.16 uses these ordered limits to turn persistent saturation into an equality between the dimension lost by projection and the dimension of the persistent space. The Falconer–Jin theorem B-II.7.14 and its invariant-fiber specialization G-II.7.15 then identify that loss with the exact dimension of almost every fiber.

Lemma II.7.2.10 (L-II.7.13 – Invariant quotient and projected entropy averages). *If V is G_Φ -invariant and $\pi = \pi_{V^\perp}$, then $\pi_*\mu$ is the self-similar probability for $y \mapsto r_i(U_i|_{V^\perp})y + \pi a_i$, and is exact-dimensional (the zero-dimensional case is a point measure). Put $\alpha_\pi = \dim \pi_*\mu$. There is $C = C(\Phi, d)$ such that for all $m \geq 1, n \geq 0$,*

$$H_m(\pi_*\mu) - C \leq m \log 2 A_{V,m,n}(\mu) \leq H_{n+m}(\pi_*\mu) - H_n(\pi_*\mu) + C.$$

It follows that

$$\lim_{m \rightarrow \infty} \liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} A_{V,m,n}(\mu) = \lim_{m \rightarrow \infty} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} A_{V,m,n}(\mu) = \alpha_\pi.$$

This is a quotient-specific argument; no general projection/local-entropy theorem is used.

Proof. We compare quotient components with projected original components at a fixed window, average first over locations and scales, and only then let the window grow.

The quotient is again a similarity system

Let V be invariant and $\pi = \pi_{V^\perp}$. Since U_i is orthogonal and $U_i V = V$, it also preserves $V^\perp$: for $w \in V^\perp, v \in V, \langle U_i w, v \rangle = \langle w, U_i^{-1} v \rangle = 0$. Consequently $\pi U_i = U_i \pi$, and

$$\pi \varphi_i(x) = r_i(U_i|_{V^\perp}) \pi x + \pi a_i. \quad (\text{II.7.N13a})$$

Pushing forward the self-similar identity gives exactly the invariant probability for these quotient similarities, with the same labelled weights. If $q = \dim V^\perp > 0$, identify $V^\perp$ with $\mathbb{R}^q$ by its chosen coordinate isometry. L01, applied to this similarity system, gives exact dimensionality of $\nu = \pi_* \mu$ and

$$\frac{H_s(\nu)}{s \log 2} \longrightarrow \alpha_\pi = \dim \nu. \quad (\text{II.7.N13b})$$

The quotient may have repeated maps, exact overlaps or smaller affine span: L01 imposes no irreducibility for this conclusion. If $q = 0$, the quotient is a point probability and all entropies and claimed dimensions are zero. All inequalities below are then immediate, so assume $q > 0$.

Comparing ambient components with projected grid information

Put $\mathcal{A}_n = \mathcal{D}_n$ on $\mathbb{R}^d$, and let $\mathcal{P}_s = \pi^{-1} \mathcal{D}_s^{V^\perp}$. Conditioning on a positive-mass cell $Q = 2^{-n}(k + [0, 1]^d)$, its canonical map sends the projected coordinate to $2^n \pi x - \pi k$. Translated intrinsic grids have bounded intersection multiplicity, so for some dimension constant,

$$|m \log 2 A_{V,m,n}(\mu) - H_\mu(\mathcal{P}_{n+m} | \mathcal{A}_n)| \leq C_d. \quad (\text{II.7.N13c})$$

Indeed the pulled-back grid on each cell is a translate of $\mathcal{P}_{n+m}$; the two grid labels have mutual conditional entropies at most $q \log 2$. Apply the chain rule and average over the coarse cells. This gives the additive, not normalized, constant in (II.7.N13c). The projection of an ambient level- n cube has diameter at most $\sqrt{d} 2^{-n}$, so it meets at most C_d intrinsic level- n cubes. Hence $H_\mu(\mathcal{P}_n | \mathcal{A}_n) \leq C_d$. The chain rule and decreasing entropy under conditioning imply

$$\begin{aligned} H_\mu(\mathcal{P}_{n+m} | \mathcal{A}_n) &\leq H_\mu(\mathcal{P}_n | \mathcal{A}_n) + H_\mu(\mathcal{P}_{n+m} | \mathcal{P}_n \vee \mathcal{A}_n) \\ &\leq C_d + H_\mu(\mathcal{P}_{n+m} | \mathcal{P}_n) \\ &= C_d + H_{n+m}(\nu) - H_n(\nu). \end{aligned} \quad (\text{II.7.N13d})$$

The last equality uses nested intrinsic grids. Combining with (II.7.N13c) proves the upper bound in the statement.

The lower bound from stopping cylinders

Use L05 at $h = 2^{-n}$, including its empty word at $n = 0$, and write $\mu_u = (\varphi_u)_* \mu$. Its support has diameter at most $D 2^{-n}$, where $D = \text{diam } K_\Phi$. Every such support meets at most $C(d, D)$ ambient level- n cubes, by the elementary bounded-diameter grid count. Thus

$H_{\mu_u}(\mathcal{A}_n) \leq \log C(d, D)$, uniformly in u, n . Conditional-entropy concavity and the chain rule now give

$$\begin{aligned} H_{\mu}(\mathcal{P}_{n+m} \mid \mathcal{A}_n) &\geq \sum_{u \in \Lambda(2^{-n})} p_u H_{\mu_u}(\mathcal{P}_{n+m} \mid \mathcal{A}_n) \\ &\geq \sum_{u \in \Lambda(2^{-n})} p_u H_{\mu_u}(\mathcal{P}_{n+m}) - \log C(d, D). \end{aligned} \quad (\text{II.7.N13e})$$

Invariance of V makes $\pi_* \mu_u$ a similarity image of ν , with ratio r_u and orthogonal part $U_u|_{V^\perp}$. After dilation by 2^n , its ratio is in $[r_{\min}, 1]$, also at $n = 0$. Dilation by this integer power of two shifts intrinsic grid levels exactly. Uniform grid intersection counts for a rotation, translation and a ratio in that fixed interval yield

$$|H_{\mu_u}(\mathcal{P}_{n+m}) - H_m(\nu)| \leq C_d(1 + \log(1/r_{\min})). \quad (\text{II.7.N13f})$$

The constant does not depend on the translation of the cylinder or on V : in each quotient dimension at most d , a rotated cube of comparable side meets boundedly many cubes of the other grid. Equations (II.7.N13c), (II.7.N13e) and (II.7.N13f) prove the lower bound. Increasing a common constant gives, for every $m \geq 1, n \geq 0$,

$$H_m(\nu) - C \leq m \log 2 A_{V,m,n}(\mu) \leq H_{n+m}(\nu) - H_n(\nu) + C, \quad C = C(\Phi, d). \quad (\text{II.7.N13g})$$

Averaging in the correct order

Fix m . For $N \geq m$, exact telescoping gives

$$\sum_{n=0}^{N-1} (H_{n+m}(\nu) - H_n(\nu)) = \sum_{s=N}^{N+m-1} H_s(\nu) - \sum_{s=0}^{m-1} H_s(\nu). \quad (\text{II.7.N13h})$$

Divide by N . By (II.7.N13b), the first sum on the right, after this division, tends to $m\alpha_\pi \log 2$; the fixed finite second sum contributes zero. Average (II.7.N13g) in n , divide by $m \log 2$, and take the lower and upper limits in N . We obtain

$$\begin{aligned} \frac{H_m(\nu) - C}{m \log 2} &\leq \liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} A_{V,m,n}(\mu) \\ &\leq \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} A_{V,m,n}(\mu) \leq \alpha_\pi + \frac{C}{m \log 2}. \end{aligned} \quad (\text{II.7.N13i})$$

Finally let $m \rightarrow \infty$ and use (II.7.N13b) again. Both outer limits equal α_π , proving the assertion. The argument uses the invariant quotient and self-similarity; no later general projection/local-entropy theorem is a premise. □

The compactness proposition P-II.7.12 turns tuned finite-scale directions into the maximal persistent subspace. The quotient-entropy lemma L-II.7.13 identifies the quotient entropy seen by that subspace.

II.7.3 Conditional measures on invariant fibers

A typical fiber has zero mass under the original measure. Dividing its restriction by its mass is therefore not a definition of a conditional probability. The disintegration lemma [L-II.7.06](#) constructs a kernel by conditioning on successively finer finite partitions of the target, and proves uniqueness through its integral identity. That identity is what makes later changes of source coordinates and target labels legitimate.

Exact-dimensional fibers and dimension conservation require more than the existence of this kernel. The Falconer–Jin theorem [B-II.7.14](#) provides those properties in the required self-similar setting. Its invariant-fiber specialization [G-II.7.15](#) explains how invariance removes the extra group element in that interface. Reflections are handled by adding one zero coordinate and a matching determinant sign. The construction changes orientation without changing the measure’s geometry.

Definition II.7.3.1 (II.7.07: Conditional probabilities on fibers). Let K, Y be compact metric spaces, $f : K \rightarrow Y$ continuous, and $\mu \in \mathcal{P}(K)$. Write $\nu = f_*\mu$. A disintegration of μ over f is a Borel probability kernel $y \mapsto \mu_y^f \in \mathcal{P}(K)$, in the sense of [D-II.5.08](#), such that for all Borel $A \subseteq K, B \subseteq Y$,

$$\mu(A \cap f^{-1}B) = \int_B \mu_y^f(A) d\nu(y), \quad \mu_y^f(f^{-1}\{y\}) = 1 \quad \text{for } \nu\text{-a.e. } y.$$

The kernel can be chosen arbitrarily on a ν -null set; existence and almost-everywhere uniqueness are [L-II.7.06](#). For $f = \pi_{V^\perp}|_K$, write $\mu_{V+x} = \mu_{f(x)}^f$. This is never shorthand for restriction to a possibly null affine plane.

The phrase constant exact fiber dimension β means that μ_y^f is exact-dimensional of dimension β for ν -almost every y ; exact dimensionality is [D-I.4.06](#), with the inherited ambient metric. Dimension conservation over f means that μ, ν , and almost every conditional measure are exact-dimensional with constant fiber dimension and $\dim \mu = \dim \nu + \beta$. It is a property, not automatic for an arbitrary continuous map.

Lemma II.7.3.2 (L-II.7.06 — Compact-space disintegration and relabelling). *Every (K, Y, f, μ) in [D-II.7.07](#) admits a disintegration, unique outside a single $f_*\mu$ -null set as a probability-valued kernel. If $J : f(K) \rightarrow Y'$ is a continuous injection into a compact metric space, the conditional measures over Jf are the same measures after relabelling $y \mapsto J(y)$. Isometric changes of the source push the conditional measures forward and preserve their dimensions. This includes orthogonal target relabelling and embedding into one extra zero coordinate. No unrestricted dimension-conservation assertion is made.*

Proof. Finite target partitions produce martingale conditional measures. Compactness gives a limiting kernel, and the rectangle identity supplies uniqueness, fiber support, and covariance under relabelling.

Finite observations of the target

Put $Y_0 = f(K)$. This is compact, and $\nu(Y_0) = 1$. Choose $x_0 \in K$; such a point exists because $\mu(K) = 1$. For each $n \geq 1$, cover Y_0 by finitely many open balls of radius 2^{-n-2} , replace that cover by its successive ordered set differences, delete empty sets, and refine by the preceding partitions. Denote the resulting finite Borel partition by $\mathcal{P}_n$. Its atoms have diameter at most 2^{-n-1} , and $\mathcal{P}_{n+1}$ refines $\mathcal{P}_n$. These partitions generate the Borel sigma algebra of Y_0 . Indeed, if U is relatively open and $y \in U$, then for large n the atom containing y is contained in U . Thus U is a union from the countable collection of all atoms that are contained in U .

For $y \in Y_0$, write $P_n(y)$ for its atom and set

$$\kappa_n(y) = \begin{cases} \frac{\mu|_{f^{-1}P_n(y)}}{\nu(P_n(y))}, & \nu(P_n(y)) > 0, \\ \delta_{x_0}, & \nu(P_n(y)) = 0. \end{cases} \quad (\text{II.7.N06a})$$

This is a simple Borel map into $\mathcal{P}(K)$. The union of all zero-mass atoms at all levels is a ν -null Borel set. For a real $g \in C(K)$, let $M_n^g(y) = \int g d\kappa_n(y)$. On (Y_0, ν) it is a martingale for the finite sigma algebras generated by $\mathcal{P}_n$. To check this, sum the numerator in (II.7.N06a) over the children of a positive-mass parent. One obtains

$$\int_P M_{n+1}^g d\nu = \int_{f^{-1}P} g d\mu = \int_P M_n^g d\nu \quad (P \in \mathcal{P}_n). \quad (\text{II.7.N06b})$$

Also $|M_n^g| \leq \|g\|_\infty$.

The bounded martingale limit needed here

Here is the convergence argument, so no disintegration or general martingale-convergence theorem is an additional input. For any real bounded martingale (M_n) , distinct increments are orthogonal in L^2 : condition the later increment on the earlier sigma algebra. Consequently, for $N > n$,

$$\mathbb{E}|M_N - M_n|^2 = \sum_{j=n}^{N-1} \mathbb{E}|M_{j+1} - M_j|^2, \quad \sum_{j=1}^{\infty} \mathbb{E}|M_{j+1} - M_j|^2 < \infty. \quad (\text{II.7.N06c})$$

The last bound follows by adding $\mathbb{E}|M_1|^2$ and using $\mathbb{E}|M_N|^2 \leq \sup_j \|M_j\|_\infty^2$. Apply L-II.2.05 to the increments after n , and let the finite terminal time increase. It gives

$$\mathbb{P}\left(\sup_{k \geq n} |M_k - M_n| > t\right) \leq \frac{4}{t^2} \sum_{j=n}^{\infty} \mathbb{E}|M_{j+1} - M_j|^2. \quad (\text{II.7.N06d})$$

Choose increasing n_ℓ with the tail sum at most $2^{-4\ell}$, and use $t = 2^{-\ell}$. The probabilities in (II.7.N06d) are summable. Borel–Cantelli shows that, eventually, all terms after n_ℓ differ from M_{n_ℓ} by at most $2^{-\ell}$. The sequence is therefore almost surely Cauchy. It is Cauchy in L^2 by (II.7.N06c); bounded convergence identifies the two limits. This proves the needed almost-sure and L^2 limit.

A probability-valued limit and its support

By the compact-probability construction in L-I.12.01, choose a countable uniformly dense family (g_j) in $C(K)$. On one Borel set $E \subseteq Y_0$ of full ν -measure, every $M_n^{g_j}(y)$ converges, and all cells $P_n(y)$ have positive mass. Compactness of $\mathcal{P}(K)$ gives an accumulation point of $\kappa_n(y)$. Every accumulation point has the same integrals of every g_j , and hence of every continuous function by uniform approximation. Uniqueness in Riesz representation makes these points equal. Sequential compactness then implies convergence of the full sequence. Write its limit as $\kappa(y)$ on E and put $\kappa(y) = \delta_{x_0}$ on $Y \setminus E$.

This is a Borel map into $\mathcal{P}(K)$: the weak topology is metrizable by L-I.12.01, and a pointwise limit of Borel maps to a metric space is Borel on its Borel convergence set. Equivalently its inverse images of closed sets are determined by limits of their distance functions. Evaluation $\theta \mapsto \theta(A)$ is Borel for every Borel $A \subseteq K$. For open A , approximate its indicator increasingly by the continuous functions $\min\{1, m \operatorname{dist}(x, K \setminus A)\}$, treating $A = K$ separately. Complements and disjoint countable unions then give all Borel sets by the monotone-class theorem. Thus κ is a probability kernel in the precise sense of D-II.5.08.

For fixed $y \in E$, the continuous nonnegative function $x \mapsto d_Y(f(x), y)$ has integral at most 2^{-n-1} against $\kappa_n(y)$, since that measure is carried by $f^{-1}P_n(y)$. Passing to the weak limit gives

$$\int_K d_Y(f(x), y) d\kappa(y)(x) = 0, \quad \kappa(y)(f^{-1}\{y\}) = 1. \quad (\text{II.7.N06e})$$

The second equality follows because a nonnegative function with zero integral is zero almost everywhere.

The integral identity and one common uniqueness set

If B is a union of atoms of $\mathcal{P}_N$, then for $n \geq N$, (II.7.N06b) gives

$$\int_B \int_K g d\kappa_n(y) d\nu(y) = \int_{f^{-1}B} g d\mu \quad (g \in C(K)).$$

Bounded convergence passes this identity to κ . The union of these finite algebras is an algebra generating the Borel sets of Y_0 . For fixed g , both sides are finite signed measures in B ; the elementary uniqueness theorem for measures extends the identity to every Borel $B \subseteq Y$. For such a fixed B , the two finite measures in the variable $A \subseteq K$ have identical continuous integrals. Riesz uniqueness yields

$$\mu(A \cap f^{-1}B) = \int_B \kappa(y)(A) d\nu(y) \quad (A \in \mathcal{B}(K), B \in \mathcal{B}(Y)). \quad (\text{II.7.N06f})$$

The right side is a measure in A by the kernel property and monotone convergence. This justifies that application of uniqueness.

Suppose $\tilde{\kappa}$ is another disintegration. For each g_j , (II.7.N06f) for both kernels implies that $\int g_j d\kappa(y) = \int g_j d\tilde{\kappa}(y)$ for almost every y : their difference has zero integral on every Borel set. Remove the union of these countably many null sets. On what remains the probabilities agree on the determining family, so they are equal. This is one exceptional null set for the whole probability-valued kernel, not a separate set for each Borel A .

Relabelling and isometric changes of source

A continuous injection $J : Y_0 \rightarrow Y'$ is a homeomorphism onto its compact image: images of closed sets are compact and hence closed. Define the relabelled kernel at $z \in J(Y_0)$ by $\kappa'(z) = \kappa(J^{-1}z)$, and fill the complement by δ_{x_0} . It is Borel and, for Borel $B' \subseteq Y'$,

$$\int_{B'} \kappa'(z)(A) d(J_*\nu)(z) = \int_{J^{-1}B'} \kappa(y)(A) d\nu(y) = \mu(A \cap (Jf)^{-1}B'). \quad (\text{II.7.N06g})$$

Its fiber support is (II.7.N06e); uniqueness proves the assertion. If $I : K \rightarrow K'$ is an isometry onto its image, give the image the compact inherited metric and replace μ, f by $I_*\mu, f \circ I^{-1}$. The kernels $I_*\kappa(y)$ satisfy (II.7.N06f) by taking inverse images under I , so are the required conditional probabilities. Balls in the image pull back to balls of the same radius. Therefore all local dimensions, and the corresponding dimensions of measures, are preserved by I.4. This includes the orthogonal and zero-coordinate embeddings in the statement. No dimension-conservation identity was used or proved for a general map. $\square$

Cited result — B-II.7.14 — Falconer–Jin deterministic fiber interface

For an orientation-preserving similarity system in $\mathbb{R}^D$, $D \geq 2$, and a strict probability vector on at least two labels, let θ be its self-similar probability and G its generated orthogonal group. For every orthogonal projection π of rank $1 \leq k < D$, there exists $g \in G$ such that $(\pi g)_*\theta$ and almost all conditional probabilities of θ over πg are exact-dimensional, the conditional dimension is a common constant β , and

$$\dim \theta = \dim((\pi g)_*\theta) + \beta.$$

Source: [FJ14, Theorem 3.1(ii)–(iii) and Corollary 3.1, pp. 10–11, group and standing assumptions pp. 4, 7]. Only this deterministic existential specialization is used. Its proof is not reproduced here; the orientation and invariant-fiber reductions are proved in G-II.7.15.

Primary reference: [FJ14].

Verification of the specialization.

Here is the exact specialization, not a substitute proof. Set each source cascade weight equal to the deterministic number p_i . All level total masses are then one. The nontrivial-branch condition holds because at least two weights are positive, and for every $t > 1$,

$$\sum_i p_i^t < \sum_i p_i = 1. \quad (\text{II.7.N14b})$$

Thus the source's standing weight hypotheses hold, without a new random model here. Its geometric conditional probabilities are the pushforwards of the symbolic conditionals in source equation (3.1). Testing that identity on inverse images of geometric Borel sets gives

the rectangle identity of D-II.7.07; uniqueness in L06 identifies them with the conditional probabilities used here. The source theorem implies the existence of the displayed g ; we do not assert that every group element or every projection is dimension conserving. No separation condition is added.

Guided result – G-II.7.15 – Dimension conservation on invariant fibers

For every G_Φ -invariant subspace V , the projection $(\pi_{V^\perp})_*\mu$ and almost all μ_{V+x} are exact-dimensional, with a constant fiber dimension β_V , and

$$\alpha = \dim((\pi_{V^\perp})_*\mu) + \beta_V, \quad 0 \leq \beta_V \leq \dim V.$$

Reflections are allowed. Both $V = 0$ and $V = \mathbb{R}^d$ are included and proved directly. For proper nonzero V , the sole external input is B-II.7.14. The lift $\widehat{U}_i = \text{diag}(U_i, \det U_i)$ and target relabeling are proved below.

Primary references: [FJ14; Hoc20].

Proof. The two endpoint subspaces

If $V = \{0\}$, the projection is the identity, and its conditional probability at x is δ_x : L06's fiber support property leaves no other probability on that singleton. Its dimension is zero, while the projection has dimension α . If $V = \mathbb{R}^d$, the target is one point and its only conditional probability is μ , by the rectangle identity. Then $\beta_V = \alpha \leq d$, and the projected dimension is zero. These endpoint assertions use L01 and L06, not B14.

An orientation-preserving lift, including reflections

Assume $0 < \dim V < d$. For every generator define on $\mathbb{R}^{d+1}$

$$\iota(x) = (x, 0), \quad \widehat{U}_i = \text{diag}(U_i, \det U_i), \quad \widehat{\phi}_i(z) = r_i \widehat{U}_i z + (a_i, 0). \quad (\text{II.7.N15a})$$

The matrix $\widehat{U}_i$ is orthogonal and has determinant $(\det U_i)^2 = 1$, so these are orientation-preserving similarities. They obey $\widehat{\phi}_i \iota = \iota \phi_i$. Uniqueness of the compact attractor and invariant probability from I.7 gives

$$K_{\widehat{\Phi}} = \iota(K_\Phi), \quad \widehat{\mu} = \iota_* \mu. \quad (\text{II.7.N15b})$$

The map $g \mapsto \text{diag}(g, \det g)$ is a continuous group homomorphism. Its image of the compact group G_Φ is compact and hence closed, and its images of the generators are precisely $\widehat{U}_i$. Taking products, inverses and closures shows that the generated lifted group is exactly

$$\widehat{G} = \{\text{diag}(g, \det g) : g \in G_\Phi\}. \quad (\text{II.7.N15c})$$

In particular every lifted group element preserves the original hyperplane $\iota(\mathbb{R}^d)$.

Removing the group element by target relabelling

Let $\hat{\pi}$ be orthogonal projection onto $V^\perp \times \{0\}$. Its rank is $d - \dim V$, strictly between zero and $d + 1$, so B14 applies to $\hat{\mu}$ and $\hat{\pi}$. It supplies some $\hat{g} = \text{diag}(g, \det g) \in \hat{G}$. Invariance of V , and orthogonality, give $\pi_{V^\perp} g = (g|_{V^\perp}) \pi_{V^\perp}$, as in L13. The map $J : V^\perp \rightarrow V^\perp \times \{0\}$, $J(y) = \iota(gy)$, is an isometry, and

$$\hat{\pi} \hat{g} \iota = J \pi_{V^\perp}, \quad (\hat{\pi} \hat{g})_* \hat{\mu} = J_*((\pi_{V^\perp})_* \mu). \quad (\text{II.7.N15d})$$

L06 first transfers conditional probabilities under the source isometry ι , then relabels the target by the injection J . It follows that B14's conditional probabilities, at target $J(y)$, are exactly $\iota_* \mu_y^{\pi_{V^\perp}}$, outside one null set. Both isometries preserve local dimensions. Thus B14 proves that these original fiber probabilities are exact-dimensional with a common dimension β_V , and that

$$\alpha = \dim((\pi_{V^\perp})_* \mu) + \beta_V. \quad (\text{II.7.N15e})$$

The projected exact dimensionality also follows independently from L13. By L06 each conditional probability is carried by its affine fiber, which is isometric to V . The dimension bound for a measure carried by a set, proved in I.4, therefore gives $0 \leq \beta_V \leq \dim V$. This completes the reductions, including reflections. The only external theorem used in this proof is B-II.7.14. □

Guided result — G-II.7.16 — Persistent saturation gives full-dimensional fibers

If V is invariant and persistently saturated, then $\beta_V = \dim V$. In particular

$$\mu_{V+x} \text{ is exact-dimensional of dimension } \dim V \quad \text{for } \mu\text{-a.e. } x, \quad \dim V \leq \alpha.$$

Only the implication displayed is claimed. It follows from P-II.7.09, L-II.7.13, and G-II.7.15, including the two ordered entropy limits.

Primary references: [FJ14; Hoc20].

Proof. Put $q = \dim V$, $\nu = (\pi_{V^\perp})_* \mu$, and $\alpha_\pi = \dim \nu$. For each fixed m , the dyadic chain rule of II.2 gives

$$\int h_m(\mu^{x,n}) d\mu(x) = \frac{H_{n+m}(\mu) - H_n(\mu)}{m \log 2}. \quad (\text{II.7.N16a})$$

Average over $0 \leq n < N$. The exact telescoping calculation in L13, now using L01's entropy-dimension limit for μ , shows that this average tends to α as $N \rightarrow \infty$, for every fixed m . Define

$$B_{m,N} = \frac{1}{N} \sum_{n=0}^{N-1} \int h_m(\mu^{x,n}) d\mu(x), \quad A_{m,N} = \frac{1}{N} \sum_{n=0}^{N-1} A_{V,m,n}(\mu). \quad (\text{II.7.N16b})$$

Their defect identity is

$$\frac{1}{N} \sum_{n=0}^{N-1} e_{V,m,n}(\mu) = q - B_{m,N} + A_{m,N}. \quad (\text{II.7.N16c})$$

Fix $0 < \delta < 1$. Persistence supplies $M(\delta)$ such that for every $m \geq M(\delta)$ and every sufficiently large n , $V \in \text{Sat}(\mu; \delta, m, n)$. The bounds (II.7.N08b)–(II.7.N08c) show that $e_{V,m,n}(\mu) \leq (d+1)\delta + C_d/m$ at those levels. For fixed m , the finitely many preceding terms are bounded and contribute zero to the average as $N \rightarrow \infty$. Since $B_{m,N} \rightarrow \alpha$, (II.7.N16c) implies

$$\limsup_{N \rightarrow \infty} A_{m,N} \leq \alpha - q + (d+1)\delta + C_d/m \quad (m \geq M(\delta)). \quad (\text{II.7.N16d})$$

L13 states that the left side tends to α_π when $m \rightarrow \infty$. Thus, first taking that limit with δ fixed and then letting $\delta \downarrow 0$, we obtain

$$\alpha - \alpha_\pi \geq q. \quad (\text{II.7.N16e})$$

G15 identifies the left side with β_V and bounds it above by q . Hence $\beta_V = q$. Its exact-dimensionality statement and the nonnegativity of α_π give the full conclusion, including $\dim V \leq \alpha$. The same argument includes the zero and full-space cases with their already established point conventions. It uses the ordered entropy limits, not a pointwise dimension assertion for arbitrary projections. □

Persistent saturation has now become a geometric statement: almost every conditional measure along the invariant plane has the full dimension of that plane. The imported Falconer–Jin theorem is used only to obtain exact-dimensional fibers and dimension conservation; the invariance and saturation deductions are local.

II.7.4 Action entropy and the main alternative

The localization lemma L-II.7.17 removes the typical contraction from a word map and recenters its translation. The resulting random maps are bounded isometries. The selection has to retain positive group entropy while the action creates asymptotically no new entropy.

A subtlety is that finite-grid entropy gains can be slightly negative. Small average gain alone does not force small average absolute gain. The isometry grid comparison supplies a uniform lower bound tending to zero after normalization; this is the missing ingredient in that passage.

The inverse-action consequence G-II.7.18 applies the earlier localized inverse theorem. Its useful directions first belong to coarse root components, not directly to the original measure. The proof therefore passes through a fixed whole-cylinder mixture, selects a cylinder by its mass, and pulls it back. The observation length is chosen only after those mixtures and their thresholds are fixed. The compactness proposition P-II.7.12 captures the resulting limiting direction. The entropy-flattening alternative G-II.7.19 combines this with the fiber conclusion; a dimension deficit makes the nonzero persistent space proper.

Lemma II.7.4.1 (L-II.7.17 – Localization to a bounded isometry law). *Suppose for some $q > 1$,*

$$\limsup_{n \rightarrow \infty} \frac{H(\nu_n, \mathcal{M}_{b_n(q)} \mid \mathcal{E}_{\ell_n})}{\ell_n \log 2} > 0.$$

Then there exist $\delta_ > 0$, integers $n_j \rightarrow \infty$, and finitely supported probabilities σ_j on $G_0(d)$ such that, with $L_j = L_{n_j}(q)$,*

$$\text{supp } \sigma_j \subset \{g : d_0(g, \text{id}) \leq 2 + \sqrt{d}\}, \quad h_{L_j}^{G_0}(\sigma_j) \geq \delta_*, \quad h_{L_j}(\sigma_j \cdot \mu) - h_{L_j}(\mu) \rightarrow 0.$$

The normalization conditions on a finite ratio class, dilates in the matrix chart, and recenters a unit translation cell. The proof owns the mean-to- L^1 gain estimate using the uniform lower bound $H_L(\sigma \cdot \mu) - H_L(\mu) \geq -C_d$. Boundedness alone does not give this estimate. No inverse theorem is used in L17.

Proof. We normalize the typical contraction, localize translations to a bounded set, and retain a positive amount of group entropy while showing that the action creates asymptotically no spatial entropy.

Fix the $q > 1$ in the hypothesis. Write $\ell = \ell_n$, $b = b_n(q)$, and $L = b - \ell$; we use only sufficiently large n so that $\ell, L \geq 1$. Thus $\ell/n \rightarrow \chi / \log 2 > 0$ and $L/\ell \rightarrow q - 1$. Let Ψ have law ν_n , write $\Psi(x) = r_\Psi U_\Psi x + a_\Psi$, and put $T = -\log_2 r_\Psi$. Take $X \sim \mu$ independently and set $Y = \Psi(X)$. The iterated invariance identity gives $Y \sim \mu$. All partitions below have finitely many positive cells for these laws.

Conditioning costs only a sublinear amount of entropy

Let I be the cell label of a_Ψ in $\mathcal{D}_\ell$ and R its finite ratio label, equivalently the value of T . Let $J = \lfloor 2^T a_\Psi \rfloor \in \mathbb{Z}^d$, with the floor taken coordinatewise, and let $C = (I, R, J)$ be the combined label. Write $w_c = \mathbb{P}(C = c)$ for its positive weights and ν_c for the conditional group law.

L03 gives

$$H(R) \leq |\Lambda| \log(n+1), \quad \mathbb{E}|T - \ell| = o(n), \quad H(I) = \alpha \ell \log 2 + o(n). \quad (\text{II.7.N17a})$$

Conditional on I, R with $T = t$, the vector $2^t a_\Psi$ lies in a cube of side $2^{t-\ell}$. It meets at most $(2^{(t-\ell)_+} + 2)^d$ unit grid cells. Therefore

$$H(J \mid I, R) \leq C_d + d \log 2 \mathbb{E}(T - \ell)_+ = o(n), \quad H(C \mid I) = o(n). \quad (\text{II.7.N17b})$$

The chain rule for adding the labels gives

$$\sum_c w_c H(\nu_c, \mathcal{M}_b) = H(\nu_n, \mathcal{M}_b \mid C) \geq H(\nu_n, \mathcal{M}_b \mid \mathcal{E}_\ell) - o(n). \quad (\text{II.7.N17c})$$

Here C refines I , so the possible entropy loss is at most $H(C \mid I)$, not an equality obtained by dropping a conditional mutual-information term.

Normalize each class in the matrix chart

On a class c , let t_c and $z_c \in \mathbb{Z}^d$ be the fixed values of T, J . Define the probability on isometries

$$\sigma_c = (L_{-z_c} \circ T_{t_c})_* \nu_c. \quad (\text{II.7.N17d})$$

In these maps T_{t_c} and L_{-z_c} have the group-action meanings in D-II.7.04, not conjugation. A map in this class becomes

$$x \mapsto U_\Psi x + (2^{t_c} a_\Psi - z_c), \quad 2^{t_c} a_\Psi - z_c \in [0, 1)^d. \quad (\text{II.7.N17e})$$

Thus each σ_c is finitely supported and $d_0(g, \text{id}) \leq 2 + \sqrt{d}$ on its support.

Put $k_c = b - \lfloor t_c \rfloor$, an integer which may be negative. In the matrix chart, L02's all-integer dilation formula gives

$$|H(\sigma_c, \mathcal{M}_{k_c}) - H(\nu_c, \mathcal{M}_b)| \leq C_d. \quad (\text{II.7.N17f})$$

Indeed $\lfloor k_c + t_c \rfloor = b$; the subsequent fixed translation changes grid entropy by only a bounded amount, at any integer level. Matrix grids on $G_0(d)$ are precisely the group grids of II.5. There are $D_0 = d^2 + d$ matrix/translation coordinates, so the dyadic scale chain rule gives, for any two integer levels, an entropy difference of at most their distance times $D_0 \log 2$. Since $|k_c - L| \leq |t_c - \ell| + 1$,

$$\left| H(\sigma_c, \mathcal{D}_L^{G_0}) - H(\nu_c, \mathcal{M}_b) \right| \leq C_d + D_0 \log 2 |t_c - \ell|. \quad (\text{II.7.N17g})$$

The average of these errors is $o(n)$, by (II.7.N17a). This calculation uses scaling of all matrix-chart coordinates; it is not valid if one silently substitutes the logarithmic ratio chart.

Bounding the average output entropy

Let Y_b denote the level- b cell label of Y . Given Y_b and R with $T = t$, the vector a_Ψ is within distance $2^{-t} R_0$ of Y , where $R_0 = \sup_{x \in K_\Phi} \|x\|$. Since $b \geq \ell$, the possible level- ℓ cells of a_Ψ number at most $C_{d,R_0} (1 + 2^{(\ell-t)_+})^d$. This is a support count conditional on the displayed labels, so

$$H(I \mid Y_b, R) \leq C_{d,R_0} + d \log 2 \mathbb{E}(\ell - T)_+ = o(n). \quad (\text{II.7.N17h})$$

By conditioning monotonicity, the chain rule, and (II.7.N17a),

$$\begin{aligned} H(Y_b \mid C) &\leq H(Y_b \mid I, R) \\ &= H(Y_b) + H(I, R \mid Y_b) - H(I, R) \\ &\leq H_b(\mu) - H(I) + H(R) + H(I \mid Y_b, R) \\ &\leq H_b(\mu) - H(I) + o(n). \end{aligned} \quad (\text{II.7.N17i})$$

Conditional on $C = c$, X still has law μ and is independent of the selected group map, because the label depends only on Ψ . Its transformed output $2^{t_c} Y - z_c$ has law $\sigma_c \cdot \mu$. The ordinary spatial grid scaling rule, first at k_c and then at L , gives

$$|H_L(\sigma_c \cdot \mu) - H(Y_b \mid C = c)| \leq C_d + d \log 2 |t_c - \ell|. \quad (\text{II.7.N17j})$$

Average, use (II.7.N17i), and apply L01 and (II.7.N17a). Since b, L, ℓ are each comparable to n , we get

$$\begin{aligned} \sum_c w_c H_L(\sigma_c \cdot \mu) &\leq H_b(\mu) - H(I) + o(n) \\ &= \alpha(b - \ell) \log 2 + o(n) = H_L(\mu) + o(n). \end{aligned} \quad (\text{II.7.N17k})$$

From the mean gain to its mean absolute value

Every isometry preserves level- L spatial grid entropy up to C_d , uniformly over its rotation and translation: a rotated unit-scale cube meets boundedly many cubes of the other grid, and the same holds in reverse. Concavity of entropy therefore implies, for every finitely supported isometry law σ ,

$$H_L(\sigma \cdot \mu) \geq \int H_L(g_* \mu) d\sigma(g) \geq H_L(\mu) - C_d. \quad (\text{II.7.N17l})$$

Set $G_c = [H_L(\sigma_c \cdot \mu) - H_L(\mu)] / (L \log 2)$. It follows from (II.7.N17k)–(II.7.N17l) that

$$G_c \geq -C_d / (L \log 2), \quad \sum_c w_c |G_c| \leq \sum_c w_c G_c + \frac{2C_d}{L \log 2} \rightarrow 0. \quad (\text{II.7.N17m})$$

This is the needed mean absolute-gain estimate. The upper mean bound alone, even with bounded variables, would not imply it.

Select a positive-entropy class with vanishing gain

The hypothesis supplies $c_0 > 0$ and infinitely many increasing n for which $H(v_n, \mathcal{M}_b \mid \mathcal{E}_\ell) \geq c_0 \ell \log 2$. By (II.7.N17c) and (II.7.N17g), along these n , for some fixed $a > 0$ and all sufficiently late terms,

$$\sum_c w_c h_L^{G_0}(\sigma_c) \geq a. \quad (\text{II.7.N17n})$$

One may take any $a < c_0 / (q - 1)$ after discarding sufficiently many terms. On the bounded support in (II.7.N17e), the number of level- L group cells is at most $C_d 2^{D_0 L}$. Hence $0 \leq h_L^{G_0}(\sigma_c) \leq K_d$ with a fixed $K_d \geq 1$, for every $L \geq 1$. Equation (II.7.N17n) implies

$$\sum_{c: h_L^{G_0}(\sigma_c) \geq a/2} w_c \geq a / (2K_d) > 0. \quad (\text{II.7.N17o})$$

Enumerate this subsequence by j and let $s_j = \sum_c w_c |G_c| \rightarrow 0$. Take $\eta_j = \sqrt{s_j} + 1/j$, so $\eta_j \rightarrow 0$ and Markov gives $\sum_{|G_c| > \eta_j} w_c \leq s_j / \eta_j \rightarrow 0$. For every sufficiently large j , (II.7.N17o) therefore has a positive-weight class c_j with $|G_{c_j}| \leq \eta_j$. Set $\sigma_j = \sigma_{c_j}$ and $\delta_* = a/2$, relabelling the increasing n_j after discarding the finite prefix. These probabilities have exactly the asserted support, group entropy and vanishing action gain. No inverse theorem was used. $\square$

Theorem II.7.4.2 (G-II.7.18 — Action entropy forces a nonzero persistent subspace). Suppose Φ is affinely irreducible. If $N_j \rightarrow \infty$, $\sigma_j \in \mathcal{P}(G_0(d))$ have supports in a fixed bounded ball, and

$$\liminf_j h_{N_j}^{G_0}(\sigma_j) > 0, \quad h_{N_j}(\sigma_j \cdot \mu) - h_{N_j}(\mu) \rightarrow 0,$$

then $\dim V_{\max}(\mu) > 0$. The only inverse-theorem input is the localized, large-window G-II.5.20. The proof below supplies the root-cell selection, whole-cylinder approximation, fixed-mixture descent, parameter retuning, and the limiting argument from P-II.7.12.

Primary source: [Hoc20].

Proof. The deduction follows six stages:

local inverse theorem $\rightarrow$ root component $\rightarrow$ whole-cylinder replacement
 $\rightarrow$ selected cylinder $\rightarrow$ pulled-back subspace $\rightarrow$ capture by P-II.7.12.

The inverse-theorem input in this deduction is the fully proved G-II.5.20. We use its localized, large-window statement, including the averaged dimension estimate and the measurable certifying events. No unrestricted isometry-action theorem is being substituted.

Discard a finite prefix and fix $a > 0$ so that $h_{N_j}^{G_0}(\sigma_j) \geq a$ thereafter. Choose $R \geq 1$ large enough for the fixed supporting sets in G-II.5.20, both for μ and for every σ_j . Put

$$c = \frac{a}{2d(d+1)} > 0. \quad (\text{II.7.N18a})$$

We will obtain a nonzero subspace in $\text{Sat}(\mu; \xi, m, n)$, with the thresholds of P12, for each sufficiently small $\xi > 0$. The subspace and levels may depend on ξ .

A uniform continuity estimate for averaged component defects

We first justify the approximation step needed below. For compactly supported probabilities θ, η with $\|\theta - \eta\|_{\text{TV}} \leq t < 1/16$, there is a function $\omega_d(t) \downarrow 0$ as $t \downarrow 0$ such that, uniformly in $i \geq 0, m \geq 1, V$,

$$|e_{V,m,i}(\theta) - e_{V,m,i}(\eta)| \leq \omega_d(t). \quad (\text{II.7.N18b})$$

Here e is the expected defect defined in the proof of L08. To prove the assertion, for each level- i cube A put $a_A = \theta(A), b_A = \eta(A)$. Give a zero-mass conditional measure an arbitrary probability value supported in A . The triangle inequality for the total variation norm of signed measures gives

$$\sum_A |a_A - b_A| \leq 2t, \quad \sum_A a_A \|\theta_A - \eta_A\|_{\text{TV}} \leq 2t. \quad (\text{II.7.N18c})$$

For the second inequality, use $a_A(\theta_A - \eta_A) = \theta|_A - \eta|_A + (b_A - a_A)\eta_A$, sum norms, and divide by two. Apply the same canonical affine normalization on both measures of a cell. The total a_A -weight of cells with normalized TV distance greater than $\sqrt{t}$ is at most $2\sqrt{t}$. On

every other cell use (II.7.N08o); on the exceptional cells use the uniform bound on absolute component defects from (II.7.N08b). Finally changing the weights from a_A to b_A costs at most $C_d t$. This proves (II.7.N18b), for example with

$$\omega_d(t) = C_d(\sqrt{t} + b(\sqrt{t})), \quad b(s) = -s \log s - (1-s) \log(1-s), \quad (\text{II.7.N18d})$$

after enlarging C_d . The estimate is uniform over the subspace, although no continuously varying choice of its grid basis is asserted. It does not say that close measures dominate one another.

Choose accuracy and window before choosing the root mixtures

Fix $0 < \xi < 1/10$. Apply the matched-level transport clause of L08 with output tolerance ξ , and denote its input tolerance and window threshold by δ_t, M_t . Decrease δ_t so that $0 < \delta_t < 1/4$. Apply the mass-weighted descent clause of L08 with output tolerance $\varepsilon_w = \delta_t/2$. Write its input tolerance and window threshold as δ_w, M_w , decreasing δ_w to lie in $(0, 1/4)$. These four constants do not depend on any mixture weights or word lengths.

Choose $0 < \tau < 1$, then $0 < t < 1/16$, and then the inverse-theorem tolerance $0 < \varepsilon < 1$, with

$$\begin{aligned} (d+1)\tau &\leq \delta_w^2/8, & t &< c/8, & \omega_d(t) &\leq \delta_w^2/8, \\ \varepsilon &< \min\{\tau, c/8, c\tau/8, a/(2(d+1))\}. \end{aligned} \quad (\text{II.7.N18e})$$

By L04 choose $\sigma > 0$ making $\mu((\varepsilon/(5d))^{2(d+1)}, \sigma)$ -non-affine. Now choose one integer m satisfying all of

$$\begin{aligned} m &\geq M_*(\xi), \quad m \geq M_t, M_w, \quad m \geq \xi^{-1}, \\ m &\geq M(d, R, \varepsilon), \quad m \geq M_{\dim}(d, R, \varepsilon, \sigma), \quad 2C_d/m \leq \delta_w^2/4. \end{aligned} \quad (\text{II.7.N18f})$$

The last constant is large enough for the defect bounds of L08. All the thresholds here have already been determined. G-II.5.20 now gives $\delta > 0, k_0$, and we fix an integer $k \geq k_0$. This ordering is important: no later smallest cylinder weight enters the choice of m or ε .

Use L05 with accuracy t , this fixed k , a fixed anchor in K_Φ , and a sufficiently small fixed stopping parameter ρ . Call a positive-mass level- k cube Q good if its whole-cylinder mixture η_Q exists and, with $\theta_Q = \mu_Q$ the raw component,

$$\|\theta_Q - \eta_Q\|_{\text{TV}} < t. \quad (\text{II.7.N18g})$$

The total μ -mass of the good cubes exceeds $1 - t$. There are finitely many such cubes and finitely many stopping words. For each good cube apply L08's weighted descent at the already fixed window m to its fixed finite mixture $\eta_Q = \sum_{u \in \mathcal{U}_Q} a_{Q,u}(\varphi_u)_* \mu$. Let N_{mix} be the maximum of the resulting finitely many fine-level thresholds. Set $S = \max_u [-\log_2 r_u]$ over all these words. Choose an integer

$$B \geq \max\{k, N_{\text{mix}}, S + N_*(\xi, m), S + \xi^{-1}\}. \quad (\text{II.7.N18h})$$

These mixtures, S , and B are now fixed before choosing a further index of the given sequence.

Select a nonzero saturated subspace on a good root cube

Take j sufficiently large that G-II.5.20 applies at $N = N_j$, that the action gain is less than its δ , and that $B/(N+1) < c/8$. The growth branch is impossible. Sample a raw group root $(\sigma_j)_{g,k}$, independently a spatial root θ_Q with weight $\mu(Q)$, and a uniform $i \in \{0, \dots, N\}$. Write V_i for the subspace assigned by the theorem to this root pair and level, extending the assignment by $\mathbb{R}^d$ on discarded root pairs as specified in that theorem. Its averaged dimension assertion gives

$$\mathbb{E} \dim V_i > \frac{a}{d+1} - \varepsilon > \frac{a}{2(d+1)}, \quad \mathbb{P}\{\dim V_i \geq 1\} > c. \quad (\text{II.7.N18i})$$

The structural conclusion holds on a set of root pairs of probability at least $1 - \varepsilon$. On each such pair define

$$b_i = \theta_Q\{z : \text{def}_{V_i,m}(\theta_Q^{z,i}) > \varepsilon\}. \quad (\text{II.7.N18j})$$

Its certifying event implies $(N+1)^{-1} \sum_{i=0}^N b_i \leq \varepsilon$: discard the group-point coordinate from that event, keeping its saturation condition. Consequently the global mass of structural root/level choices with $b_i > \tau$ is at most ε/τ . The mass of choices with a bad spatial cube is at most t ; the mass with $i < B$ is at most $B/(N+1)$. Thus removing these choices and the nonstructural root pairs from the event in (II.7.N18i) costs at most

$$\varepsilon + \varepsilon/\tau + t + B/(N+1) < c/2. \quad (\text{II.7.N18k})$$

A choice therefore remains with $\dim V_i \geq 1$, good Q , $i \geq B$, and $b_i \leq \tau$. Since $\varepsilon < \tau$, this choice satisfies $V_i \in \text{Sat}(\theta_Q; \tau, m, i)$. The root partitions on the bounded supports are finite; selection here is from a positive-probability collection of finitely many root/level choices. The only event measurability used is that certified by the inverse theorem, not an unstated measurable Grassmannian selection.

Descend to a whole cylinder and pull back to the original measure

By the expected-defect bounds of L08 and (II.7.N18b),

$$e_{V_i,m,i}(\eta_Q) + C_d/m \leq (d+1)\tau + \omega_d(t) + 2C_d/m \leq \delta_w^2/2. \quad (\text{II.7.N18l})$$

Each component defect is at least $-C_d/m$. Markov's inequality applied to this non-negative shifted defect yields $\eta_Q\{z : \text{def}_{V_i,m}(\eta_Q^{z,i}) > \delta_w\} \leq \delta_w/2$. In particular $V_i \in \text{Sat}(\eta_Q; \delta_w, m, i)$. Since $i \geq N_{\text{mix}}$, weighted descent gives

$$\sum_{u \in \mathcal{U}_Q} a_{Q,u}(\varphi_u)_* \mu\{z : \text{def}_{V_i,m}(((\varphi_u)_*\mu)^{z,i}) \leq \varepsilon_w\} \geq 1 - \varepsilon_w. \quad (\text{II.7.N18m})$$

At least one positive-weight word u has its displayed probability at least $1 - \varepsilon_w$. Hence $V_i \in \text{Sat}((\varphi_u)_*\mu; \varepsilon_w, m, i)$, and also with tolerance δ_t . Apply matched-level transport to the inverse similarity φ_u^{-1} . Its ratio is r_u^{-1} , so the resulting level and subspace are precisely

$$n = [i + \log_2 r_u] = i - \lceil -\log_2 r_u \rceil, \quad W = U_u^{-1}V_i \in \text{Sat}(\mu; \xi, m, n). \quad (\text{II.7.N18n})$$

By (II.7.N18h), $n \geq N_*(\xi, m)$ and $n \geq \xi^{-1}$. Also $\dim W = \dim V_i \geq 1$. The scale shift has been subtracted, not added. Uniform matched-level transport, rather than a fixed-level bound depending on the shrinking word ratio, is what preserves the parameter order.

Finally perform this construction for $\xi_l = 2^{-l-4}$. The resulting subspaces have a subsequence of a common dimension $r \in \{1, \dots, d\}$. Compactness of the rank- r Grassmannian gives a further subsequence converging to a rank- r subspace W_∞ . Its tolerances, windows and levels meet exactly P12's hypotheses. Therefore

$$W_\infty \subseteq V_{\max}(\mu), \quad \dim V_{\max}(\mu) \geq r \geq 1. \quad (\text{II.7.N18o})$$

This proves the claimed implication in full. □

Guided result — G-II.7.19 — Higher-dimensional entropy alternative

If Φ is affinely irreducible and $\alpha < \min\{d, s(\Phi, p)\}$, then either logarithmic-grid flattening holds or there is an invariant subspace V with

$$0 < \dim V < d, \quad \mu_{V+x} \text{ exact-dimensional of dimension } \dim V \quad \text{for } \mu\text{-a.e. } x.$$

For comparison, see [Hoc20, Theorems 1.5 and 6.15, pp. 8, 84–88]. The proof uses the chart comparison L-II.7.02, the localization lemma L-II.7.17, the action-entropy theorem G-II.7.18, and the fiber conclusion G-II.7.16.

Primary reference: [Hoc20].

Proof. Assume the stated affine irreducibility and $\alpha < \min\{d, s(\Phi, p)\}$. If logarithmic-grid flattening holds there is nothing further to prove. Otherwise its equivalence with matrix-grid flattening in L02 gives a fixed $q > 1$ for which the nonnegative normalized matrix conditional entropies do not tend to zero. Their limsup is therefore positive. L17 provides bounded isometry laws with positive normalized group entropy and action gain tending to zero. The hypotheses of G18 are met, so $\dim V_{\max}(\mu) > 0$.

By P09 this subspace is invariant; G16 applies to its persistent saturation and gives

$$\begin{aligned} \mu_{V_{\max}+x} \text{ is exact-dimensional of dimension } \dim V_{\max} \quad \text{for } \mu\text{-a.e. } x, \\ 0 < \dim V_{\max} \leq \alpha < d. \end{aligned} \quad (\text{II.7.N19a})$$

Thus it is a proper subspace and has all the required properties. The argument uses G-II.7.18 and the Falconer–Jin fiber input carried by G-II.7.16. □

This is the chapter's main measure alternative. A strict dimension deficit either forces logarithmic-grid flattening or produces a nonzero proper invariant plane with full-dimensional fibers.

II.7.5 Separation, irreducibility, and arithmetic

If cylinder maps remain exponentially distinguishable along a subsequence, a sufficiently fine group grid records the full word entropy. Translation entropy records only the dimension of the measure at its natural scale. A deficit between the two prevents flattening and forces the invariant fiber alternative. This is the mechanism of the measure-and-set trichotomy [G-II.7.20](#); it does not assert that the branches are mutually exclusive.

The linear-irreducibility criterion [G-II.7.21](#) excludes the fiber branch. The algebraic-separation lemma [L-II.7.22](#) obtains exponential separation by finding one nonzero coordinate of a word-map difference and bounding that algebraic number from below. The arithmetic statement itself is independent of the inverse theorem; the algebraic dimension formula [G-II.7.23](#) combines them only afterward.

The reducible example [L-II.7.24](#) shows why this separation of roles matters. It builds an algebraic system with no exact overlaps and full affine span, but with an interval factor and a Cantor factor. Its dimension is below the natural ambient bound. The preserved horizontal direction explains the obstruction.

Guided result – G-II.7.20 – Measure and set dimension trichotomies

For an affinely irreducible system and strict weights, at least one of the following holds:

$$\alpha = \min\{d, s(\Phi, p)\}; \quad \Delta_n^{\text{sim}} \text{ is super-exponentially small}; \quad \exists V \text{ as in G19.}$$

For the attractor of an affinely irreducible system, at least one holds:

$$\dim_{\text{H}} K_{\Phi} = \min\{d, s_{\Phi}\};$$

or Δ_n^{sim} is super-exponentially small;

or there exist V, x with $0 < \dim V < d$, V invariant, $x \in K_{\Phi}$, and

$$\dim_{\text{H}}(K_{\Phi} \cap (x + V)) = \dim V.$$

The branches are not asserted mutually exclusive. Source: [[Hoc20](#), Theorems 1.4–1.5, pp. 6–8]. The set deduction uses $p_i = r_i^{s_{\Phi}}$, not an assumed dimension-maximizing measure.

Primary reference: [[Hoc20](#)].

Proof. The upper bound in L01 reduces the measure assertion to the case $\alpha < \min\{d, s(\Phi, p)\}$. Suppose in addition that super-exponential cylinder concentration fails. By L01 there are $C > 0$ and infinitely many increasing n with $\Delta_n^{\text{sim}} \geq e^{-Cn}$. No exact overlap can exist, since L01 would then make every sufficiently late minimum zero. In particular, the labelled word maps of each length are all distinct.

For this subsequence, L02 gives the matrix-coordinate separation

$$\rho_{\text{mat}}(\varphi_u, \varphi_v) \geq \frac{1}{3} r_{\min}^n e^{-Cn} \quad (u \neq v, u, v \in \Lambda^n). \quad (\text{II.7.N20a})$$

Choose a fixed $q > 1$ with $q\chi > C + \log(1/r_{\min})$. A level- $b_n(q)$ matrix cell has ρ_{mat} -diameter at most $C_d 2^{-b_n(q)}$. Since $b_n(q)/n \rightarrow q\chi/\log 2$, it contains at most one word map for every sufficiently late n in this subsequence. Its entropy is consequently exactly the word-label entropy $nh(p)$. The matrix partition at that level refines the translation partition at level ℓ_n , and hence

$$\begin{aligned} \frac{H(v_n, \mathcal{M}_{b_n(q)} \mid \mathcal{E}_{\ell_n})}{\ell_n \log 2} &= \frac{nh(p) - H(v_n, \mathcal{E}_{\ell_n})}{\ell_n \log 2} \\ &\rightarrow s(\Phi, p) - \alpha > 0. \end{aligned} \quad (\text{II.7.N20b})$$

Here L03 supplies the translation-entropy limit. Thus matrix flattening fails and, by L02, so does logarithmic flattening. G19 must give the invariant full-dimensional fiber branch. This proves the measure trichotomy without asserting that its alternatives are disjoint.

For the set assertion let $p_i = r_i^{s_\Phi}$, a strict probability vector. Its entropy obeys $h(p) = s_\Phi \chi(\Phi, p)$. If $\dim_{\text{H}} K_\Phi = \min\{d, s_\Phi\}$, the first branch already holds. Otherwise the supporting-set bound and L01 give

$$\dim \mu \leq \dim_{\text{H}} K_\Phi < \min\{d, s_\Phi\} = \min\{d, s(\Phi, p)\}. \quad (\text{II.7.N20c})$$

Apply the measure assertion to this vector. Its cylinder-concentration branch is the same system property as in the set statement. In the remaining branch choose $x \in K_\Phi$ in the full-measure set where the conditional probability is exact-dimensional of dimension $\dim V$ and is carried by $K_\Phi \cap (x + V)$. These support properties are part of the disintegration on K_Φ . The measure-to-set dimension bound of I.4 and the affine-plane upper bound give

$$\dim V \leq \dim_{\text{H}}(K_\Phi \cap (x + V)) \leq \dim V. \quad (\text{II.7.N20d})$$

This is the required section. The argument did not assume the existence of a dimension-maximizing self-similar measure. □

Guided result — G-II.7.21 — Irreducible separation consequences

If Φ is both affinely and linearly irreducible and has exponential separation, then

$$\dim \mu_{\Phi, p} = \min\{d, s(\Phi, p)\} \quad \text{for every strict } p, \quad \dim_{\text{H}} K_\Phi = \min\{d, s_\Phi\}.$$

For weights with zeros this assertion applies only if the active subsystem has the required irreducibility hypotheses, or is a singleton treated directly. Deleting labels is not an irreducibility argument.

Primary reference: [Hoc20].

Proof. Fix any strict probability vector. Exponential separation is the negation of super-exponential concentration by L01, so the second branch of G20 is excluded. Linear irreducibility excludes the proper nonzero invariant subspace in its third branch. Its first branch gives

$$\dim \mu_{\Phi, p} = \min\{d, s(\Phi, p)\}. \quad (\text{II.7.N21a})$$

The same exclusions in the set trichotomy give $\dim_{\mathbb{H}} K_{\Phi} = \min\{d, s_{\Phi}\}$. For a vector with zero weights, discard exactly those labels and apply this argument only after the active subsystem's affine and linear irreducibility have been checked. Exponential separation passes to that subsystem because its distinct word pairs form a subset of the original pairs at each length. Irreducibility need not pass. If the active alphabet is a singleton, its probability is the point mass at the contraction's fixed point, of dimension zero; this case is direct and does not satisfy an artificial full-span hypothesis. $\square$

Definition II.7.5.1 (II.7.11: Algebraic systems and homotheties). A similarity system is algebraic if every r_i , every entry of U_i , and every coordinate of a_i is a real algebraic number in the sense of D-II.6.06. The field generated by these finitely many numbers, polynomial coefficient height and multiplication matrices retain D-II.6.06's definitions. A positive homothety is $x \mapsto rx + a$ with $r > 0$; in a contracting homothetic system all $U_i = I_d$. The strong separation condition (SSC) is D-I.7.06: the compact sets $\varphi_i(K_{\Phi})$ are pairwise disjoint. Homogeneity additionally means that every ratio is the same.

Lemma II.7.5.2 (L-II.7.22 – Algebraic exponential separation). *If Φ is algebraic and has no exact overlaps, there exists $C > 0$ such that*

$$\Delta_n^{\text{sim}}(\Phi) \geq e^{-Cn} \quad (n \geq 1).$$

No irreducibility assumption is needed. The proof applies L-II.6.13 to a nonzero entry of the matrix or translation difference of two word maps, after proving degree $O(n)$ and coefficient height $e^{O(n)}$. Neither a norm nor a logarithm is called an integer polynomial.

Proof. Write $B_i = r_i U_i$. All its entries and all coordinates of a_i are algebraic. Form one finite tuple $\mathcal{A}$ of these real numbers, retaining labelled coordinates as separate polynomial variables if convenient. For a word of length n ,

$$B_u = B_{u_1} \cdots B_{u_n}, \quad a_u = \sum_{j=1}^n B_{u_1} \cdots B_{u_{j-1}} a_{u_j}. \quad (\text{II.7.N22a})$$

The empty matrix product is the identity. Each matrix entry on the right is an integer polynomial in $\mathcal{A}$ of degree n , before cancellation with at most d^{n-1} monomials. Each translation coordinate is a sum of at most $\sum_{j=1}^n d^{j-1}$ monomials of degrees at most n . Thus every entry of $B_u - B_v$ or coordinate of $a_u - a_v$ is an integer polynomial P with

$$\deg P \leq n, \quad \|P\|_1 \leq 2n \max\{1, d\}^{n-1} \leq (4 \max\{1, d\})^n. \quad (\text{II.7.N22b})$$

Cancellation can only decrease this coefficient-norm bound. Apply L-II.6.13 to the fixed tuple and $L = 4 \max\{1, d\}$. It gives $c \in (0, 1)$ such that each such evaluated coordinate is either zero or has absolute value at least c^n .

There are no exact overlaps, so two distinct length- n words have different affine maps. At least one of these coordinates is nonzero. The operator norm of a matrix bounds each of its entries and the Euclidean norm bounds each vector coordinate. Therefore, using L02 for these contracting word maps,

$$\rho_{\text{sim}}(\varphi_u, \varphi_v) \geq \rho_{\text{mat}}(\varphi_u, \varphi_v) \geq c^n. \quad (\text{II.7.N22c})$$

Taking the minimum and $C = -\log c$ proves the result for every $n \geq 1$. The polynomial estimate was applied to a single nonzero coordinate, not to a norm, a logarithm, or a minimum. Neither irreducibility nor an entropy inverse theorem was used. $\square$

Guided result — G-II.7.23 — Algebraic natural dimensions

An algebraic system with no exact overlaps, both affinely and linearly irreducible, satisfies the measure and set formulas of G-II.7.21. Source: [Hoc20, Corollaries 1.6–1.7 and Lemma 6.30, pp. 8, 95–96]. The required exponential separation is proved in L-II.7.22.

Primary reference: [Hoc20].

Proof. The algebraic separation lemma L-II.7.22 verifies exponential separation at every length, hence in particular along infinitely many lengths. The two irreducibility assumptions are those of G-II.7.21 and require no arithmetic deduction. Applying that result gives both asserted natural-dimension formulas. $\square$

Lemma II.7.5.3 (L-II.7.24 — An algebraic reducible obstruction). *Let $r = 2^{-1/2}$, $\Phi_0 = \{x \mapsto rx, x \mapsto rx + 1\}$, and let Φ_0^3 be its eight labelled length-three maps. Put $q = r^3$, $\Psi_0 = \{y \mapsto qy, y \mapsto qy + 1\}$, and let Γ be the sixteen product maps from $\Phi_0^3 \times \Psi_0$ on $\mathbb{R}^2$. Then Γ is algebraic, affinely irreducible, has no exact overlaps, and*

$$s_\Gamma = \frac{8}{3}, \quad \dim_{\text{H}} K_\Gamma = \frac{5}{3} < \min\{2, s_\Gamma\} = 2.$$

It is linearly reducible and has interval horizontal fibers. All claims, including non-overlap, are proved directly; G20 is explanatory, not a premise.

Proof. Set $r = 2^{-1/2}$ and $I = [0, (1 - r)^{-1}]$. The two images rI and $rI + 1$ have union I : their outside endpoints are those of I , and they overlap because $r/(1 - r) \geq 1$. Uniqueness of the compact IFS attractor in I.7 gives $K_{\Phi_0} = I$. Iterating the same invariance identity

three times shows that its length-three system has the same attractor and common ratio $q = r^3 = 2^{-3/2} < 1/2$.

We first check that no two different words of Φ_0 give the same map. Different lengths have different ratios. For words of the same length their translation difference has the form $P(r)$, where P is a nonzero polynomial with coefficients in $\{-1, 0, 1\}$. Split even and odd powers:

$$P(r) = E(1/2) + rO(1/2). \quad (\text{II.7.N24a})$$

Both evaluations on the right are rational. Since r is irrational, $P(r) = 0$ would imply they both vanish. But any nonzero polynomial $Q(x) = \sum_{j=0}^M e_j x^j$ with these coefficients has $Q(1/2) \neq 0$: if j_0 is its least nonzero exponent, then

$$|Q(1/2)| \geq 2^{-j_0} - \sum_{j=j_0+1}^M 2^{-j} = 2^{-M} > 0. \quad (\text{II.7.N24b})$$

At least one of E, O is nonzero, proving the contradiction. Thus Φ_0^3 also has no exact overlaps.

Let C be the attractor of the vertical two-map system Ψ_0 . It is contained in $[0, (1-q)^{-1}]$, and its two first-level images lie in the disjoint intervals $[0, q/(1-q)]$ and $[1, (1-q)^{-1}]$. Hence it satisfies SSC. The I.8 separated-system dimension theorem yields

$$\dim_H C = \dim_B C = \frac{\log 2}{-\log q} = \frac{2}{3}. \quad (\text{II.7.N24c})$$

Here the equality of upper and lower box dimensions is included in the SSC conclusion. This also gives no exact overlaps in Ψ_0 : distinct words of the same length have disjoint compact cylinders, and different lengths have distinct ratios.

The sixteen maps of Γ are the Cartesian products of the eight horizontal maps and the two vertical maps. Independent choice of the two labels at each step and the compact invariance identity give $K_\Gamma = I \times C$. Two equal-length product word maps can agree only if their horizontal length-three block sequences agree, by the digit-polynomial argument, and their vertical sequences agree, by SSC. They are then the same product word. Unequal lengths again have different ratios. This proves the asserted absence of all exact overlaps, not merely distinctness of the sixteen generators.

The product dimension bounds of I.2 give

$$1 + \dim_H C \leq \dim_H(I \times C) \leq \dim_B(I \times C) \leq 1 + \dim_B C, \quad \dim_H K_\Gamma = \frac{5}{3}. \quad (\text{II.7.N24d})$$

There are sixteen maps of ratio q ; hence $s_\Gamma = \log 16 / (-\log q) = 8/3$. Their coefficients are algebraic. The set $I \times C$ contains three noncollinear points, so its affine hull is $\mathbb{R}^2$; L01 gives affine irreducibility. All orthogonal parts are the identity, making every line invariant and the system linearly reducible. For each $y \in C$, its horizontal section is exactly $I \times \{y\}$, an interval. These direct calculations establish the obstruction and its fibers without using G20.

□

The last example explains why affine irreducibility is not enough. Full affine span does not prevent the orthogonal parts from preserving a flat direction; linear irreducibility is the hypothesis that removes that obstruction.

II.7.6 Projection parameters

Definition II.7.6.1 (II.7.12: Projection charts and dimension-drop directions). Fix $1 \leq k < d$. For $A \in \mathbb{R}^{(d-k) \times k}$, define

$$W_A = \{(x, Ax) : x \in \mathbb{R}^k\}, \quad T_A(x, y) = x + A^\top y, \quad (x, y) \in \mathbb{R}^k \times \mathbb{R}^{d-k}.$$

The graph chart is $A \mapsto W_A \in \text{Gr}(d, k)$. Coordinate-permuted charts use the same formula after permuting the standard coordinate axes. The Euclidean matrix norm here is Frobenius norm. Chart dimension, metric comparison and the relation of T_A to orthogonal projection are proved, not presumed.

For compact $K \subset \mathbb{R}^d$, its dimension-drop directions are

$$\mathcal{B}_k(K) = \{W \in \text{Gr}(d, k) : \dim_{\text{H}} \pi_W K < \min\{k, \dim_{\text{H}} K\}\}.$$

Hausdorff and packing dimensions of direction sets use the gap metric of [D-II.5.01](#) and the definitions of I.1–I.2. No assertion about their size is included here.

Definition II.7.6.2 (II.7.13: Anchored projection gaps and eventual exceptional sets). For a contracting homothetic system and a fixed $b \in K_\Phi$, put $x_u = \varphi_u(b)$. For $A > 0$ and $n \geq 1$, define

$$\mathcal{U}_{n,A}^{(k)} = \bigcup_{\substack{u,v \in \Lambda^n \\ u \neq v}} \{W \in \text{Gr}(d, k) : \|\pi_W(x_u - x_v)\| < e^{-An}\},$$

and

$$\mathcal{E}_A^{(k)} = \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \mathcal{U}_{n,A}^{(k)}, \quad \mathcal{E}_\infty^{(k)} = \bigcap_{j=1}^{\infty} \mathcal{E}_j^{(k)}.$$

The parameter A in these sets is a positive scalar, not the matrix in a graph chart. The eventual intersection requires every sufficiently large generation; an infinitely-often condition is weaker. These anchored sets do not impose equality of cylinder ratios.

Near a graph plane, orthogonal projection of a fixed vector is controlled by the simple expression $x + A^\top y$. If that vector is not small and its projection is small, its transverse part y cannot vanish. Varying the graph matrix then gives exactly k independent constraints. The [Grassmannian tube lemma](#) turns this calculation into a uniform covering estimate with codimension k .

SSC makes distinct anchored cylinders differ by a contraction times a vector in a fixed annulus. The [exceptional-direction covering lemma](#) can therefore apply the same tube

estimate to every word pair. The exceptional set requires close pairs at every sufficiently large generation. This permits an upper box estimate for each fixed-onset piece, followed by a packing-dimension countable union.

The [homothetic dimension-drop theorem](#) gives the below-one criterion even when the attractor has smaller affine span. The [line-projection theorem](#) uses the earlier one-dimensional result instead and is independent of the higher-dimensional inverse-action input. The [higher-rank theorem](#) applies the below-one criterion only when the original similarity dimension is at most one; that hypothesis is part of its scope.

Lemma II.7.6.3 (L-II.7.25 – Grassmannian charts and uniform projection tubes). *The coordinate-permuted graph charts of $D12$ cover $\text{Gr}(d, k)$. On each bounded matrix cube the graph map is bi-Lipschitz, with*

$$P_A = \begin{pmatrix} I \\ A \end{pmatrix} (I + A^\top A)^{-1} \begin{pmatrix} I & A^\top \end{pmatrix}, \quad \dim_{\text{H}} \text{Gr}(d, k) = \dim_{\text{p}} \text{Gr}(d, k) = k(d - k).$$

For fixed $R, D, \delta > 0$, there is C such that for every $v \in \mathbb{R}^d$ with $\delta \leq \|v\| \leq D$ and $0 < \rho < 1$,

$$\{A : \|A\|_F \leq R, \|\pi_{W_A} v\| < \rho\}$$

can be covered by at most $C\rho^{-[k(d-k)-k]}$ matrix balls of radius $C\rho$. This is a uniform linear projection-tube lemma; no general nonlinear parameter-rank statement is needed here.

Proof. Graph charts reduce projection-smallness to k independent linear constraints. Uniform tube covers in the matrix parameter then give the required codimension.

Fix $1 \leq k < d$, and write $D_k = k(d - k)$ for the dimension of the Euclidean matrix space $\mathbb{R}^{(d-k) \times k}$. This is a notation for the number of entries, before any dimension assertion about the Grassmannian. For a graph matrix A , put $F_A = \begin{pmatrix} I_k \\ A \end{pmatrix}$ and $G_A = I_k + A^\top A$. Solving the orthogonality equation $F_A^\top(v - F_A x) = 0$ gives

$$P_A = F_A G_A^{-1} F_A^\top, \quad P_A v = F_A G_A^{-1} T_A v. \quad (\text{II.7.N25a})$$

The matrix G_A is positive definite, so the solution is unique. This proves the displayed formula in the statement and its interpretation as the orthogonal projection onto W_A .

Metric comparison and dimension

On $\|A\|_F, \|B\|_F \leq R$, we have $\|G_A^{-1}\|_{\text{op}} \leq 1$ and $\|G_A - G_B\|_{\text{op}} \leq 2R\|A - B\|_F$. The identity $G_A^{-1} - G_B^{-1} = G_A^{-1}(G_B - G_A)G_B^{-1}$ and (II.7.N25a) therefore show $\|P_A - P_B\|_{\text{op}} \leq C_{d,R}\|A - B\|_F$. In the indicated two-by-two block decomposition,

$$(P_A)_{11} = G_A^{-1}, \quad (P_A)_{21} = A G_A^{-1}, \quad A = (P_A)_{21}((P_A)_{11})^{-1}. \quad (\text{II.7.N25b})$$

The inverse in the last expression has norm at most $1 + R^2$. The same inverse-difference identity, now applied to these blocks, and equivalence of matrix entry, Frobenius and operator norms in the fixed dimensions give the reverse bound. Consequently

$$C_{d,R}^{-1} \|A - B\|_F \leq d_G(W_A, W_B) = \|P_A - P_B\|_{\text{op}} \leq C_{d,R} \|A - B\|_F. \quad (\text{II.7.N25c})$$

For every k -plane, a matrix whose columns form a basis has some nonsingular k -row minor. Permute these rows to the first k positions and change the column basis to make that minor the identity. The plane is then a graph. Thus the finitely many coordinate-permuted charts cover the Grassmannian; countably many bounded matrix cubes cover each chart. Bi-Lipschitz invariance and countable stability of Hausdorff and packing dimension, proved in I.1–I.2, bound both dimensions above by D_k . The image of one matrix cube with nonempty interior has both dimensions D_k , by (II.7.N25c). Hence

$$\dim_H \text{Gr}(d, k) = \dim_P \text{Gr}(d, k) = D_k. \quad (\text{II.7.N25d})$$

Uniform tubes for a nonzero vector

Fix the $R, D, \delta > 0$ in the statement. Write $v = (x, y) \in \mathbb{R}^k \times \mathbb{R}^{d-k}$ with $\delta \leq \|v\| \leq D$. Formula (II.7.N25a) gives

$$\|P_A v\|^2 = (T_A v)^T G_A^{-1} (T_A v), \quad \frac{\|x + A^T y\|}{\sqrt{1 + R^2}} \leq \|P_A v\| \leq \|x + A^T y\|. \quad (\text{II.7.N25e})$$

Suppose first that $0 < \rho < \rho_0$, where $\rho_0 = \min\{1, \delta/(4\sqrt{1 + R^2})\}$, and that the tube contains some A with $\|A\|_F \leq R$. Then $\|x + A^T y\| < \delta/4$, while

$$\delta \leq \|v\| \leq \|x\| + \|y\| < \delta/4 + (R + 1)\|y\|. \quad (\text{II.7.N25f})$$

In particular $\|y\| \geq \gamma = \delta/(2(R + 1)) > 0$. If this fails the small tube is empty and requires no balls.

For $y \neq 0$, the linear map $L_y : A \mapsto A^T y$ onto $\mathbb{R}^k$ has adjoint $L_y^* : z \mapsto y z^T$, and $L_y L_y^* = \|y\|^2 I_k$. Therefore its kernel has dimension $D_k - k$, and on the kernel's orthogonal complement it multiplies Euclidean lengths by $\|y\|$. The matrix $A_0 = -y x^T / \|y\|^2$ belongs to this complement and solves $L_y A_0 = -x$. Decompose $A = A_K + A_\perp$ according to these orthogonal subspaces. Every matrix in the tube obeys

$$\|A_K\|_F \leq R, \quad \|A_\perp - A_0\|_F < \frac{\sqrt{1 + R^2}}{\gamma} \rho. \quad (\text{II.7.N25g})$$

A Euclidean ball of radius R in dimension $D_k - k$ has a cover by at most $C\rho^{-(D_k - k)}$ balls of radius ρ : place a grid of mesh $\rho/\sqrt{D_k - k}$ in orthonormal coordinates, with the zero-dimensional case requiring one point. Enlarging the radii by a fixed factor covers the additional thickness in (II.7.N25g). This proves the desired bound for small ρ , uniformly in v . For $\rho \geq \rho_0$ one ball of radius $C\rho$ covers the original radius- R matrix ball after increasing

$C \geq R/\rho_0$. Since $D_k - k = k(d - k - 1) \geq 0$, the same stated count holds. Coordinate permutations preserve all these estimates. $\square$

Lemma II.7.6.4 (L-II.7.26 — Anchored exceptional-direction covering). *For every contracting homothetic SSC system in $\mathbb{R}^d$, every anchor $b \in K_\Phi$ and every $1 \leq k < d$,*

$$\dim_H \mathcal{E}_\infty^{(k)} \leq \dim_P \mathcal{E}_\infty^{(k)} \leq k(d - k) - k.$$

More precisely, on each bounded chart, for $A > \log(1/r_{\min})$, each eventual-intersection piece of $\mathcal{E}_A^{(k)}$ has upper box dimension at most

$$k(d - k) - k + \frac{2 \log |\Lambda|}{A - \log(1/r_{\min})}.$$

The displayed upper box bound concerns the fixed-onset pieces, not their countable union. Packing dimension is the countable-union step.

Proof. If the system has only one label, there are no pairs of distinct words of any fixed length. Thus every exceptional-direction set in the statement is empty, and both dimension bounds are immediate. Assume henceforth that $|\Lambda| \geq 2$.

Let $K = K_\Phi$, and let $\delta > 0$ be the minimum distance between two distinct first-level compact pieces. It is positive by SSC. Fix an anchor $b \in K$. If $u, v \in \Lambda^n$ are distinct and w is their longest common prefix, the suffixes begin with different labels. Homothety gives

$$x_u - x_v = r_w z_{u,v}, \quad \delta \leq \|z_{u,v}\| \leq \text{diam } K, \quad r_w \geq r_{\min}^{n-1}. \quad (\text{II.7.N26a})$$

The suffix images of the anchor belong to the respective first-level pieces, which justifies both annulus bounds even if the word ratios are unequal.

To distinguish the scalar decay rate from a chart matrix, denote that rate by $a > \log(1/r_{\min})$ in this proof. Put $c_a = a - \log(1/r_{\min}) > 0$ and $\rho_n = e^{-c_a n}$. If $\|\pi_W(x_u - x_v)\| < e^{-an}$, then (II.7.N26a) implies $\|\pi_W z_{u,v}\| < \rho_n$. On any fixed bounded chart, L25 thus covers the exceptional matrices for this pair by at most $C\rho_n^{-(D_k - k)}$ matrix balls of radius $C\rho_n$. There are at most $|\Lambda|^{2n}$ ordered pairs. For the set

$$F_{a,N,R} = \{A : \|A\|_F \leq R, \quad W_A \in \bigcap_{n \geq N} \mathcal{U}_{n,a}^{(k)}\}, \quad (\text{II.7.N26b})$$

we therefore have, at every $n \geq N$, a cover with

$$\# \text{ balls} \leq C|\Lambda|^{2n} \rho_n^{-(D_k - k)}, \quad \text{radius} \leq C\rho_n. \quad (\text{II.7.N26c})$$

The same set is being covered at every such scale; replacing its eventual intersection by an infinitely-often union would lose this upper box conclusion. For any sufficiently small radius between two successive $C\rho_n$, use the finer-scale cover. The ratio of these successive radii is the fixed number e^{c_a} , so interpolation changes the quotient defining upper box dimension by a quantity tending to zero. Taking logarithms of (II.7.N26c) yields

$$\overline{\dim}_B F_{a,N,R} \leq D_k - k + \frac{2 \log |\Lambda|}{c_a}. \quad (\text{II.7.N26d})$$

The graph chart is bi-Lipschitz on the bounded matrix ball by L25. Take the countable union over integer radii R , over onsets N , and over the finitely many coordinate-permuted charts. Packing dimension is bounded by upper box dimension on each bounded piece and is countably stable. Thus

$$\dim_P \mathcal{E}_a^{(k)} \leq D_k - k + \frac{2 \log |\Lambda|}{a - \log(1/r_{\min})}. \quad (\text{II.7.N26e})$$

For every positive integer j , $\mathcal{E}_\infty^{(k)} \subseteq \mathcal{E}_j^{(k)}$. Use (II.7.N26e) for arbitrarily large integers j and let $j \rightarrow \infty$. Monotonicity of packing dimension and the inequality $\dim_H \leq \dim_P$ give both claims. No countable stability of upper box dimension was asserted. $\square$

Guided result – G-II.7.27 – Homothetic dimension drop below one

For a contracting homothetic system, without affine irreducibility, if a strict self-similar probability satisfies

$$\dim \mu < \min\{1, s(\Phi, p)\},$$

then Δ_n^{sim} is super-exponentially small. This follows from the main measure alternative G-II.7.20 after restriction to the affine hull; positive-dimensional full fibers would have dimension at least one. The zero-dimensional affine hull is handled by commuting homotheties. The statement is not extended to arbitrary orthogonal parts.

Primary reference: [Hoc20].

Proof. Let $A = \text{aff } K_\Phi = b + L$ with $b \in K_\Phi$, and write $k = \dim L$. Each generator sends A into itself: it sends K_Φ into K_Φ , and affine maps commute with affine combinations. If $k \geq 1$, choose a linear isometry $J : \mathbb{R}^k \rightarrow L$. Conjugating the restrictions by $x \mapsto b + Jx$ gives homotheties of $\mathbb{R}^k$ with the same ratios and translations

$$c_i = J^T(a_i + (r_i - 1)b). \quad (\text{II.7.N27a})$$

Their attractor has full affine span and hence is affinely irreducible by L01. The conjugated probability has the same dimension α and the same similarity value $s(\Phi, p)$. The strict hypothesis $\alpha < \min\{1, s(\Phi, p)\}$ implies $\alpha < \min\{k, s(\Phi, p)\}$, so the natural-dimension branch of G20 is impossible. Its fiber branch is impossible as well: G15, applied to the invariant subspace in that branch, gives $\alpha = \dim \pi_{V^\perp} \mu + \dim V \geq 1$, whereas $\alpha < 1$. Thus the restricted system has super-exponential concentration.

For word maps, the same conjugacy gives $Jc_u = a_u + (r_u - 1)b$. Consequently

$$\begin{aligned} \|a_u - a_v\| &\leq \|c_u - c_v\| + \|b\| |r_u - r_v|, \\ \rho_{\text{sim}}(\varphi_u, \varphi_v) &\leq (1 + \|b\|)(|\log r_u - \log r_v| + \|c_u - c_v\|). \end{aligned} \quad (\text{II.7.N27b})$$

We used $|r - s| \leq |\log r - \log s|$ for $0 < r, s \leq 1$; the orthogonal parts in both systems are the identity. The fixed multiplicative constant in (II.7.N27b) is absorbed by increasing any prescribed exponential rate. Hence concentration transfers to the original system, at every sufficiently large length.

If $k = 0$, the attractor is the point b . All maps are $x \mapsto b + r_i(x - b)$, and any two distinct labels satisfy $\varphi_i \varphi_j = \varphi_j \varphi_i$. The distinct words ij and ji are an exact overlap. L01 gives eventually zero cylinder distance and therefore super-exponential concentration. This use of commutation is specific to homotheties.

□

Theorem II.7.6.5 (T-II.7.28 – Line projections of homothetic SSC sets). *If Φ is a contracting homothetic SSC system in $\mathbb{R}^d$, $d \geq 2$, then*

$$\dim_{\text{H}} \mathcal{B}_1(K_\Phi) \leq \dim_{\text{P}} \mathcal{B}_1(K_\Phi) \leq d - 2.$$

The proof uses the line theorem T-II.6.11 and the exceptional-subspace estimate L-II.7.26, rather than the homothetic dimension-drop criterion G-II.7.27. Through T-II.6.11 it inherits the Berry–Esseen interface X-BE-II.4.

Proof. The SSC dimension theorem of I.8 gives $\dim_{\text{H}} K_\Phi = s_\Phi$. Use the strict vector $p_i = r_i^{s_\Phi}$, whose measure similarity value is also s_Φ . Fix a line $W \in \mathcal{B}_1(K_\Phi)$, and identify it isometrically with $\mathbb{R}$. Its projected maps are the homotheties $y \mapsto r_i y + \pi_W a_i$, in this coordinate. Their invariant probability is $(\pi_W)_* \mu$, by the invariance identity and uniqueness. It is carried by $\pi_W K_\Phi$, so

$$\dim((\pi_W)_* \mu) \leq \dim_{\text{H}} \pi_W K_\Phi < \min\{1, s_\Phi\}. \quad (\text{II.7.N28a})$$

Apply the line theorem T-II.6.11 to this labelled projected system. For every $a > 0$, at every sufficiently large n there are distinct words $u, v \in \Lambda^n$ with exactly equal ratios and $\|\pi_W(a_u - a_v)\| < e^{-an}$. Choose any fixed $b \in K_\Phi$. Equal ratios cancel the anchor term, giving

$$\pi_W(x_u - x_v) = \pi_W(a_u - a_v), \quad W \in \mathcal{E}_\infty^{(1)}. \quad (\text{II.7.N28b})$$

Thus $\mathcal{B}_1(K_\Phi) \subseteq \mathcal{E}_\infty^{(1)}$. L26 with $k = 1$ gives the asserted Hausdorff and packing bounds $d - 2$. No higher-dimensional inverse-action theorem was used. The only inherited analytic exception is X-BE-II.4 in the proof chain of T-II.6.11.

□

Guided result – G-II.7.29 – Higher-rank projections in the small-dimension regime

If Φ is a contracting homothetic SSC system in $\mathbb{R}^d$ with $s_\Phi \leq 1$, then for $1 \leq k < d$,

$$\dim_{\text{H}} \mathcal{B}_k(K_\Phi) \leq \dim_{\text{P}} \mathcal{B}_k(K_\Phi) \leq k(d - k) - k.$$

The additional hypothesis $s_\Phi \leq 1$ is essential to this argument. The deduction uses the exceptional-subspace estimate L-II.7.26 and the homothetic dimension-drop criterion G-II.7.27. No assertion is made here for arbitrary orthogonal parts or without the small-dimension hypothesis.

Primary reference: [Hoc20].

Proof. Again SSC gives $\dim_{\mathbb{H}} K_\Phi = s_\Phi$; choose the same strict vector $p_i = r_i^{s_\Phi}$. For $W \in \mathcal{B}_k(K_\Phi)$, the projected homothetic system on $W \cong \mathbb{R}^k$ has this vector and the unchanged measure similarity value s_Φ . The essential use of the additional hypothesis is the equality

$$\dim((\pi_W)_* \mu) \leq \dim_{\mathbb{H}} \pi_W K_\Phi < \min\{k, s_\Phi\} = s_\Phi = \min\{1, s_\Phi\}. \quad (\text{II.7.N29a})$$

Here $1 \leq k$ and $s_\Phi \leq 1$. Thus G27 applies, without any assumption of full affine span of the projected system. For each $a > 0$, at every sufficiently large n there are distinct words whose projected full-similarity distance is less than $e^{-(a+1)n}$. For a fixed anchor $b \in K_\Phi$,

$$\begin{aligned} \|\pi_W(x_u - x_v)\| &\leq \|\pi_W(a_u - a_v)\| + \|b\| |r_u - r_v| \\ &\leq (1 + \|b\|) \rho_{\text{sim}}(\pi_W \varphi_u, \pi_W \varphi_v) < e^{-an} \end{aligned} \quad (\text{II.7.N29b})$$

for all sufficiently large n . In this notation the two arguments of ρ_{sim} are the induced word maps on W , not maps from $\mathbb{R}^d$ to a different-dimensional space. The logarithmic ratio term controls $|r_u - r_v|$, and their orthogonal parts are the identity. Therefore $\mathcal{B}_k(K_\Phi) \subseteq \mathcal{E}_\infty^{(k)}$. L26 proves the required bound $k(d - k) - k$. Without $s_\Phi \leq 1$, the last equality in (II.7.N29a) fails in general, so this proof supplies no unrestricted higher-rank conclusion. $\square$

The line-projection theorem T-II.7.28 is a full consequence of the line theory in II.6. The higher-rank statement G-II.7.29 has guided ancestry and requires the displayed small-dimension hypothesis; the two scopes should not be conflated.

II.7.7 A diagonal nonconformal model

Definition II.7.7.1 (II.7.14: Homogeneous diagonal systems). Fix $d \geq 1$ and $1 > \lambda_1 > \dots > \lambda_d > 0$. Put $A_\lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$. A homogeneous diagonal system here is

$$\Psi = (\psi_i)_{i \in \Lambda}, \quad \psi_i(x) = A_\lambda x + a_i, \quad a_i \in \mathbb{R}^d.$$

Its attractor and Bernoulli invariant probability are the contraction constructions of I.7. Its j -th coordinate system, on the same labelled alphabet, is

$$\Psi_j = (t \mapsto \lambda_j t + a_{i,j})_{i \in \Lambda}.$$

Coordinate non-overlap means absence of exact overlaps in every Ψ_j . Rational translations mean $a_i \in \mathbb{Q}^d$ for every symbol. No algebraicity of the λ_j is included. This is a

diagonal affine system, not generally a similarity system.

Definition II.7.7.2 (II.7.15: Capped diagonal dimension value). Set $\chi_j = -\log \lambda_j$, so $0 < \chi_1 < \dots < \chi_d$. For $h \geq 0$, put

$$m(h) = \max \left\{ 0 \leq j \leq d : \sum_{i=1}^j \chi_i \leq h \right\}, \quad \gamma_\lambda(h) = \begin{cases} m(h) + \frac{h - \sum_{i=1}^{m(h)} \chi_i}{\chi_{m(h)+1}}, & m(h) < d, \\ d, & m(h) = d. \end{cases}$$

The empty sum is zero. For this homogeneous model only, call $\gamma_\lambda(h(p))$ its capped Lyapunov value and $\gamma_\lambda(\log |\Lambda|)$ its capped affinity value. These are local names for the displayed formula, not an import of the general singular-value-pressure formalism. Their identification with measure or set dimension requires the diagonal theorem below.

Definition II.7.7.3 (II.7.16: Coupled Bernoulli convolutions and digit polynomials). Let $(\varepsilon_n)_{n \geq 0}$ be independent random signs with $\mathbb{P}(\varepsilon_n = 1) = \mathbb{P}(\varepsilon_n = -1) = 1/2$. For the ordered vector of D-II.7.14 define

$$X_\lambda = \sum_{n=0}^{\infty} \varepsilon_n (\lambda_1^n, \dots, \lambda_d^n), \quad \mu_\lambda = \text{Law}(X_\lambda).$$

There is one shared sign at each time, not independent signs in different coordinates. Uniform absolute convergence is a local result. Let

$$\mathcal{P}_\pm = \left\{ \sum_{j=0}^N c_j t^j : N \geq 0, c_j \in \{-1, 0, 1\}, (c_0, \dots, c_N) \neq 0 \right\}.$$

This is the same finite-polynomial class as in II.6, explicitly repeated to distinguish it from infinite power series. The non-root condition is $P(\lambda_j) \neq 0$ for every $P \in \mathcal{P}_\pm$ and every $1 \leq j \leq d$.

The diagonal system contracts different coordinates at different rates. It is generally not a similarity system, so the preceding alternative does not prove its dimension formula. The coding-and-covering lemma L-II.7.30 gives an elementary rectangle-covering upper bound. Its piecewise linear capped value records how successive contraction directions contribute to that bound.

The Rapoport–Ren measure theorem B-II.7.31 supplies the matching measure dimension. Its set and coupled-measure consequences G-II.7.32 follow from the uniform measure and the elementary upper bound. The same random sign is used in all coordinates at each step. Replacing it by independent coordinate signs would change both the model and its entropy. The result illustrates a nonconformal extension; the cited measure theorem is not replaced

by a similarity argument.

Lemma II.7.7.4 (L-II.7.30 – Diagonal coding, covering and polynomial concordance). *For a homogeneous diagonal system, its invariant probability is the law of $\sum_{n \geq 0} A_\lambda^n a_{I_n}$, where the independent labels I_n have law p . The series converges uniformly over label sequences. Deleting inactive labels preserves coordinate non-overlap. For the two translations $\pm(1, \dots, 1)$, coordinate non-overlap is equivalent to D16's non-root condition, and the invariant law with fair weights is μ_λ .*

The function γ_λ is continuous, increasing until it reaches d , and constant thereafter. Its endpoint values and all breakpoints follow from D15's formula, including $h = 0$ and $h = \sum_{j=1}^d \chi_j$. For arbitrary translations, without a non-overlap assumption, elementary cylinder covering gives

$$\dim_H K_\Psi \leq \overline{\dim}_B K_\Psi \leq \gamma_\lambda(\log |\Lambda|).$$

No dimension equality is inferred here, and no general affinity-pressure theorem is needed for this upper bound.

Proof. Let $A_\lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ and $R_a = \max_i \|a_i\|$. For any label sequence,

$$\left\| \sum_{j=n}^{\infty} A_\lambda^j a_{i_j} \right\| \leq R_a \sum_{j=n}^{\infty} \lambda_1^j = \frac{R_a \lambda_1^n}{1 - \lambda_1}. \quad (\text{II.7.N30a})$$

This proves absolute convergence, uniformly over sequences, and continuity of the coding map. For independent labels with probabilities p_i , separate the first term: the random sum has the same law as $a_{I_0} + A_\lambda X'$, where X' is an independent copy of the sum. Its probability law satisfies the invariant-measure identity and is the unique law constructed in I.7. The deterministic image of all sequences is the attractor, by the same compact invariance identity. A restriction to an active alphabet merely restricts the possible word pairs in each coordinate; it cannot create a new exact overlap.

For translations $\pm(1, \dots, 1)$, the series is precisely that of D16 with one common sign at each time. In coordinate j , a word of length n with signs $s_1, \dots, s_n$ has translation $\sum_{l=0}^{n-1} s_{l+1} \lambda_j^l$ and ratio λ_j^n . Different lengths have different ratios. For two distinct words of equal length their translation difference is

$$2P(\lambda_j), \quad P(t) = \sum_{l=0}^{n-1} \frac{s_{l+1} - s'_{l+1}}{2} t^l \in \mathcal{P}_\pm. \quad (\text{II.7.N30b})$$

Here $s'_1, \dots, s'_n$ is the second sign word, and t is the polynomial variable. Conversely, for each coefficient c_l of any $P \in \mathcal{P}_\pm$, choose the sign pair $(1, -1)$ if $c_l = 1$, $(-1, 1)$ if $c_l = -1$, and $(1, 1)$ if $c_l = 0$. This realizes $2P$ as a word-translation difference of length one more than the largest displayed exponent. At least one coefficient is nonzero, so the words are distinct. This proves the equivalence of coordinate non-overlap and the non-root condition, including the order of composition and powers. Fair weights identify the invariant law with μ_λ .

For the dimension-value formula, write $S_k = \sum_{i=1}^k \chi_i$, with $S_0 = 0$. On each interval $[S_k, S_{k+1}]$, $0 \leq k < d$, the formula agrees with

$$\gamma_\lambda(h) = k + \frac{h - S_k}{\chi_{k+1}}. \quad (\text{II.7.N30c})$$

Its value at the left endpoint is k and at the right endpoint is $k + 1$. Adjacent formulas coincide there, including at S_d , after which the value is constantly d . The slopes $1/\chi_{k+1}$ are positive. This proves continuity, strict increase up to the cap, $\gamma_\lambda(0) = 0$, and all asserted breakpoint values.

Finally set $M = |\Lambda|$, $h = \log M$, and $k = m(h)$. Choose $R \geq 1$ with $K_\Psi \subset [-R, R]^d$. If $k = d$, the ambient upper box bound already gives the claim. Otherwise every level- n cylinder lies in a translate of a rectangle with side lengths $2R\lambda_i^n$. It can be covered by at most

$$C_{d,R} \prod_{i=1}^k \left(\frac{\lambda_i}{\lambda_{k+1}} \right)^n \quad (\text{II.7.N30d})$$

cubes of side λ_{k+1}^n , by using a product grid; the remaining coordinate lengths are at most $2R\lambda_{k+1}^n$ and require only a fixed number of intervals. The empty product when $k = 0$ is one. Multiplying by the number M^n of labelled cylinders covers the entire attractor, irrespective of overlaps. Division of the logarithmic covering count by $n\chi_{k+1}$ gives

$$\frac{h + \sum_{i=1}^k (\log \lambda_i - \log \lambda_{k+1})}{\chi_{k+1}} = k + \frac{h - S_k}{\chi_{k+1}} = \gamma_\lambda(h). \quad (\text{II.7.N30e})$$

The successive geometric scales have fixed ratio λ_{k+1} , so monotonicity of covering numbers interpolates the bound to all sufficiently small scales. Hausdorff dimension is at most upper box dimension. This proves the full upper bound without a separation assumption or any general affinity-pressure theorem. □

Cited result – B-II.7.31 – Rapaport–Ren diagonal measure dimension theorem

For D14's strictly ordered positive contractions, rational translations and coordinate non-overlap on the full alphabet, every Bernoulli invariant probability is exact-dimensional and

$$\dim \mu_{\Psi,p} = \gamma_\lambda(h(p)).$$

Zero weights are allowed, with a singleton active alphabet treated as a point probability. The set formula below is deduced in [G-II.7.32](#) from this measure theorem and the elementary covering bound [L-II.7.30](#); only the measure theorem is cited here. Source: [\[RR24, Theorem 1.2, p. 4, exact-dimensional measure conventions pp. 1–4 and 9\]](#). The self-affine exact-dimensionality input is included in this cited package, not attributed to the self-similar theorem T-I.9.08. No algebraicity assumption on λ_j is added.

Primary reference: [\[RR24\]](#).

Verification of the specialization.

The source takes the minimum of its Lyapunov value and d . Below S_d its formula is (II.7.N30c); for $h \geq S_d$ its uncapped expression is $dh/S_d \geq d$. Thus its minimum is exactly the locally defined γ_λ , including the endpoint. The source allows a probability vector without a strictness condition. Alternatively, deleting inactive labels preserves all its hypotheses by L30; if only one remains, the measure is directly a point mass and $\gamma_\lambda(0) = 0$.

Only the measure theorem is imported. Its proof, including the nonconformal entropy-growth and approximation work, is not supplied in this chapter. The set and coupled-measure consequences below are proved from this statement and L-II.7.30.

Guided result — G-II.7.32 — Diagonal set and coupled Bernoulli-convolution dimensions

For a diagonal system satisfying the hypotheses of the Rapaport–Ren measure theorem B-II.7.31,

$$\dim_{\text{H}} K_\Psi = \gamma_\lambda(\log |\Lambda|).$$

This follows using the uniform probability vector and the elementary upper bound L-II.7.30.

For every ordered vector $1 > \lambda_1 > \dots > \lambda_d > 0$ satisfying D16’s non-root condition,

$$\mu_\lambda \text{ is exact-dimensional,} \quad \dim \mu_\lambda = \gamma_\lambda(\log 2).$$

The support has the same Hausdorff dimension. This includes the complete deduction of [RR24, Theorem 1.1] from B-II.7.31 and L-II.7.30. It is a theorem about a coupled measure, not the product of scalar Bernoulli convolutions.

Primary reference: [RR24].

Proof. For a system satisfying B31, take the uniform probability vector on $M = |\Lambda|$ labels. B31 and the supporting-set bound in I.4 give the lower bound $\dim_{\text{H}} K_\Psi \geq \gamma_\lambda(\log M)$. L30 proves the matching upper bound. Therefore

$$\dim_{\text{H}} K_\Psi = \gamma_\lambda(\log M). \quad (\text{II.7.N32a})$$

For the coupled Bernoulli convolution, L30 identifies the homogeneous diagonal system with translations $\pm(1, \dots, 1)$, which are rational, and identifies coordinate non-overlap with the hypothesis in D16. The contractions are strictly ordered as required. B31 applied to its fair vector gives exact dimensionality and

$$\dim \mu_\lambda = \gamma_\lambda(\log 2) = \dim_{\text{H}} K_\Psi. \quad (\text{II.7.N32b})$$

The topological support of this measure is the whole attractor: for every coded $x \in K_\Psi$ and every neighborhood of x , some sufficiently long cylinder containing x lies inside that neighborhood, since cylinder diameters are at most $\lambda_1^n \text{diam } K_\Psi$. Its symbolic probability is $2^{-n} > 0$; overlaps cannot remove this mass from its image. The measure is already carried by K_Ψ , so the support assertion follows.

This law uses one sign for all coordinates at a given time. Independent coordinate signs would instead give a system with 2^d sign vectors per step, generally a different measure. No such replacement has occurred in the argument.

□

The coupled Bernoulli convolution uses one shared sign across all coordinates at each step; it is not a product of scalar Bernoulli convolutions. This model is genuinely nonconformal and does not follow from the similarity alternative above.

Chapter II.8

Bernoulli Convolutions: Models and Classical Regimes

Geometric intuition

Bernoulli convolutions provide the testing ground where Fourier analysis, algebraic separation, transversality, and entropy meet. We establish the model and use its exact-dimensional structure, then present the classical separated, Pisot, almost-everywhere, and algebraic regimes needed by the modern theory.

The same random series exhibits three different mechanisms. Below $1/2$, geometry settles the question because the cylinders are disjoint; at $1/2$, binary coding gives Lebesgue measure explicitly; above $1/2$, the support is already an interval, so regularity is governed by arithmetic separation, Fourier alignment, or cancellation for typical parameters. We develop those mechanisms in that order.

II.8.1 The model and its decompositions

Definition II.8.1.1 (II.8.01: Symmetric Bernoulli convolution and its affine system). Let $(\varepsilon_n)_{n \geq 0}$ be independent random variables with $\mathbb{P}(\varepsilon_n = 1) = \mathbb{P}(\varepsilon_n = -1) = 1/2$. For $0 < \lambda < 1$, the symmetric Bernoulli convolution with parameter λ is

$$\nu_\lambda = \text{Law}(Y_\lambda), \quad Y_\lambda = \sum_{n=0}^{\infty} \varepsilon_n \lambda^n. \quad (\text{II.8.D01})$$

The series converges absolutely. Equivalently, ν_λ is the equal-weight self-similar probability for $f_{\lambda,+}(x) = 1 + \lambda x$ and $f_{\lambda,-}(x) = -1 + \lambda x$. This is the normalization already used in [D-II.6.11](#); no factor $1 - \lambda$ is inserted.

Definition II.8.1.2 (II.8.02: Prefix, tail and residue-class laws). For $N \geq 1$, set

$$Y_{\lambda,N} = \sum_{n=0}^{N-1} \varepsilon_n \lambda^n, \quad \nu_{\lambda,N} = \text{Law}(Y_{\lambda,N}).$$

The tail after level N is $\lambda^N Y'_\lambda$, where Y'_λ is an independent copy of Y_λ . For $k \geq 1$ and $0 \leq j < k$, the residue-class law is the law of $\sum_{m \geq 0} \varepsilon_{j+mk} \lambda^{j+mk}$. Convolution always means convolution of Borel probabilities on $\mathbb{R}$.

Definition II.8.1.3 (II.8.03: Fourier transform and the Rajchman property). For a finite Borel measure μ on $\mathbb{R}$, use

$$\hat{\mu}(\xi) = \int_{\mathbb{R}} e^{-2\pi i \xi x} d\mu(x).$$

The measure is Rajchman if $\hat{\mu}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$. Fourier nondecay means that there are $|\xi_j| \rightarrow \infty$ and $c > 0$ with $|\hat{\mu}(\xi_j)| \geq c$ for every j . These are properties of the measure, not definitions of absolute continuity or singularity.

Lemma II.8.1.4 (L-II.8.01 – Fixed point, support, Fourier product and splitting). For $0 < \lambda < 1$,

$$\nu_\lambda = \frac{1}{2}(f_{\lambda,+})_* \nu_\lambda + \frac{1}{2}(f_{\lambda,-})_* \nu_\lambda, \quad \text{spt } \nu_\lambda = K_\lambda,$$

where K_λ is the unique nonempty compact set satisfying $K_\lambda = (1 + \lambda K_\lambda) \cup (-1 + \lambda K_\lambda)$. Moreover

$$\hat{\nu}_\lambda(\xi) = \prod_{n=0}^{\infty} \cos(2\pi \lambda^n \xi), \quad \nu_\lambda = \nu_{\lambda,N} * (S_{\lambda^N})_* \nu_\lambda. \quad (\text{II.8.01})$$

For every $k \geq 1$, ν_λ is the convolution of its k residue-class laws, each an affine image of ν_{λ^k} .

Proof. Absolute convergence of Y_λ follows from $\sum_n |\varepsilon_n| \lambda^n = (1 - \lambda)^{-1}$. Separating the first digit gives

$$Y_\lambda = \varepsilon_0 + \lambda Y'_\lambda,$$

where Y'_λ is an independent copy of Y_λ . Taking laws proves the fixed-point identity. Volume I, Theorem T-I.7.02 gives uniqueness of the nonempty compact invariant set, and Volume I, Theorem T-I.7.04 gives uniqueness and support of the invariant probability.

Independence gives, for each N ,

$$\mathbb{E} e^{-2\pi i \xi Y_{\lambda,N}} = \prod_{n=0}^{N-1} \frac{e^{-2\pi i \xi \lambda^n} + e^{2\pi i \xi \lambda^n}}{2} = \prod_{n=0}^{N-1} \cos(2\pi \lambda^n \xi).$$

Since $Y_{\lambda,N} \rightarrow Y_\lambda$ almost surely and the integrands have modulus one, dominated convergence gives the infinite product formula. The identity

$$Y_\lambda = Y_{\lambda,N} + \lambda^N Y_\lambda''$$

with independent summands gives the prefix-tail convolution. Finally, partitioning the indices by their residue modulo k gives

$$Y_\lambda = \sum_{j=0}^{k-1} \lambda^j \sum_{m=0}^{\infty} \varepsilon_{j+mk} (\lambda^k)^m.$$

The inner random series are independent copies of Y_{λ^k} , which proves the residue-class assertion. □

These identities are the reusable toolkit for the chapter. The fixed-point and support equations control geometry, the product exposes exceptional Fourier behavior, and the two splitting formulas let us transfer regularity from a finite prefix or a residue-class factor to the full measure.

Definition II.8.1.5 (II.8.04: Regularity and dimension terminology). A probability μ is absolutely continuous if $d\mu = f dx$ for some $f \in L^1(\mathbb{R})$, has an L^2 density if such an f can be chosen in L^2 , and has a bounded density if it can be chosen in L^∞ . It is singular if it is carried by a Borel set of Lebesgue measure zero. A measure is of pure type if it is either absolutely continuous or singular. A measure is exact-dimensional if its lower and upper local dimensions agree and are equal to a constant at μ -almost every point. That constant is its exact dimension, denoted by $\dim \mu$. Exact dimensionality and the definitions of the local dimensions are given in Chapter I.9. Exact dimension must not be confused with any of the regularity classes above.

II.8.2 The geometric threshold

Theorem II.8.2.1 (T-II.8.02 – The separated Cantor regime). *If $0 < \lambda < 1/2$, the two first-level pieces of K_λ are disjoint,*

$$\dim_{\text{H}} K_\lambda = \dim v_\lambda = \frac{\log 2}{\log(1/\lambda)} < 1,$$

and v_λ is singular. The measure is the equal-weight Cantor measure for the displayed two-map system.

Proof. The convex hull of K_λ is contained in $[-R, R]$, $R = (1 - \lambda)^{-1}$. The two first-level hulls are

$$[-1 - \lambda R, -1 + \lambda R], \quad [1 - \lambda R, 1 + \lambda R].$$

Their gap is $2(1 - 2\lambda)/(1 - \lambda) > 0$, so the strong separation condition holds. The fully proved SSC measure-dimension theorem T-I.8.05 in Volume I and SSC Moran set-dimension Theorem T-I.8.06 there, applied to two contractions of common ratio λ and canonical weights $1/2, 1/2$, give

$$\dim_{\text{H}} K_{\lambda} = \frac{\log 2}{\log(1/\lambda)}, \quad \dim \nu_{\lambda} = \frac{-2(\frac{1}{2} \log \frac{1}{2})}{-\log \lambda} = \frac{\log 2}{\log(1/\lambda)}.$$

This number is less than one. Hence K_{λ} has Lebesgue measure zero, and the supported measure ν_{λ} is singular. □

Theorem II.8.2.2 (T-II.8.03 — The boundary parameter). *At $\lambda = 1/2$,*

$$K_{1/2} = [-2, 2], \quad d\nu_{1/2}(x) = \frac{1}{4} \mathbf{1}_{[-2, 2]}(x) dx.$$

Proof. Put $B_n = (\varepsilon_n + 1)/2$. Then the B_n are independent fair $\{0, 1\}$ -digits and

$$U = \sum_{n=0}^{\infty} B_n 2^{-(n+1)}$$

is uniform on $[0, 1]$: every half-open dyadic interval of generation N has one prescribed length- N digit word and probability 2^{-N} ; the dyadic intervals generate the Borel sets. Therefore

$$Y_{1/2} = \sum_{n \geq 0} (2B_n - 1)2^{-n} = 4U - 2.$$

Its law is uniform on $[-2, 2]$, with density $1/4$. Its support is that interval. □

The threshold $\lambda = 1/2$ changes the geometry of the support. Above it, however, support alone no longer distinguishes a singular measure from an absolutely continuous one.

II.8.3 A measure-theoretic bridge

Lemma II.8.3.1 (L-II.8.04 — Pure type for a strict affine self-similar probability). *Every strict finite affine self-similar probability on $\mathbb{R}$ is of pure type. In particular, every ν_{λ} is either absolutely continuous or singular.*

Proof. Let μ be the invariant probability for a strict finite affine IFS and write its Lebesgue decomposition as $\mu = \mu_{\text{ac}} + \mu_{\text{s}}$. A nondegenerate affine map and its inverse preserve null sets; consequently it sends absolutely continuous measures to absolutely continuous

measures and singular measures to singular measures. Applying the Hutchinson operator $\mathcal{T}$ and using uniqueness of the Lebesgue decomposition gives

$$\mu_{ac} = \mathcal{T} \mu_{ac}, \quad \mu_s = \mathcal{T} \mu_s. \quad (\text{II.8.N04})$$

The operator preserves total mass. If both terms were nonzero, their normalizations would be two invariant probabilities. By Volume I, Theorem T-I.7.04 each would equal the unique invariant probability μ . This is impossible because one is absolutely continuous and the other singular. Thus exactly one part has total mass one. Applying this to [D-II.8.01](#) proves the final assertion. $\square$

This lemma will be used in the Pisot regime: Fourier nondecay rules out absolute continuity, and pure type upgrades that failure to singularity.

II.8.4 Uniform separation and Garsia parameters

A prefix atom has mass 2^{-N} . Thus a separation bound at the same exponential scale as this mass is exactly what turns discrete information into a uniform interval estimate. The next lemma isolates that analytic step so that the algebraic argument can be confined to proving separation.

Lemma II.8.4.1 (L-II.8.05 – Uniform separation implies bounded density). *Suppose that for some $c > 0$, every $N \geq 1$ gives 2^N distinct atoms of $\nu_{\lambda,N}$, and distinct atoms are separated by at least $c2^{-N}$. Then*

$$\nu_{\lambda}(I) \leq C_c |I|$$

for every interval I , where $C_c < \infty$ depends only on c . Consequently ν_{λ} has an L^{∞} density.

Proof. Let I be an interval of length r . The assertion is trivial for $r \geq c/2$, after enlarging the constant. Otherwise choose $N \geq 1$ so that

$$c2^{-(N+1)} < r \leq c2^{-N}. \quad (\text{II.8.N05})$$

An interval of length r contains at most two atoms of the $c2^{-N}$ -separated probability $\nu_{\lambda,N}$. Every atom has mass 2^{-N} , because all prefix sums are distinct. Hence, for every translate $I - t$,

$$\nu_{\lambda,N}(I - t) \leq 2^{1-N} < \frac{4r}{c}.$$

Average this inequality against the independent tail in [L-II.8.01](#) to obtain $\nu_{\lambda}(I) \leq 4r/c$. Interval outer approximation gives $\nu_{\lambda}(E) \leq (4/c)\mathcal{L}^1(E)$ for every Borel set E . Radon–Nikodym and differentiation then give a density bounded by $4/c$ almost everywhere. $\square$

Definition II.8.4.2 (II.8.05: Reciprocal parameter). The reciprocal parameter is $\beta = \lambda^{-1} > 1$. Thus $1/2 < \lambda < 1$ corresponds to $1 < \beta < 2$.

Definition II.8.4.3 (II.8.06: Algebraic integer, conjugates, minimal polynomial and norm). A complex number α is an algebraic integer if it is a root of a monic polynomial in $\mathbb{Z}[X]$. Its minimal polynomial is the unique monic irreducible polynomial $m_\alpha \in \mathbb{Q}[X]$ with $m_\alpha(\alpha) = 0$. The roots $\alpha = \alpha_1, \dots, \alpha_d$ of m_α are its conjugates. Its algebraic norm is

$$N(\alpha) = \prod_{j=1}^d \alpha_j.$$

The next elementary lemma is the arithmetic reason that conjugate products give useful lower bounds. We include the proof so that these facts do not enter later separation arguments as hidden algebraic assumptions.

Lemma II.8.4.4 (L-II.8.07 – Integral conjugate products). *If α is an algebraic integer of degree d , then $m_\alpha \in \mathbb{Z}[X]$ and*

$$N(\alpha) = (-1)^d m_\alpha(0) \in \mathbb{Z}.$$

For every $P \in \mathbb{Z}[X]$, either $P(\alpha) = 0$, or $\prod_{j=1}^d P(\alpha_j)$ is a nonzero rational integer.

Proof. Choose a monic $f \in \mathbb{Z}[X]$ with $f(\alpha) = 0$. Minimality gives $f = m_\alpha q$ in $\mathbb{Q}[X]$, with both factors monic. Choose positive integers a, b so that am_α and bq are primitive integer polynomials. The product of two primitive integer polynomials is primitive: if a prime divided all its coefficients, reduction modulo that prime would make the product of two nonzero polynomials in $\mathbb{F}_p[X]$ vanish. Thus the content of $(am_\alpha)(bq) = abf$ is one. Since the monic polynomial f is primitive, the content of abf is ab , so $a = b = 1$. Hence $m_\alpha \in \mathbb{Z}[X]$.

Expanding $m_\alpha(X) = \prod_{j=1}^d (X - \alpha_j)$ and comparing constant terms gives the norm formula and its integrality. Finally, $\prod_j P(\alpha_j)$ is a symmetric polynomial with integer coefficients in the conjugates. The usual leading-monomial elimination writes every such symmetric polynomial as an integer polynomial in the elementary symmetric functions; by the coefficient identities for m_α , those elementary symmetric functions are integers. The product is therefore an integer. If it vanished, some conjugate α_j would be a common root of m_α and P . Irreducibility of m_α over $\mathbb{Q}$ would then give $m_\alpha \mid P$, and hence $P(\alpha) = 0$. This proves the stated alternative. $\square$

Definition II.8.4.5 (II.8.08: Mahler measure and root notation). If an algebraic number α has primitive minimal polynomial over $\mathbb{Z}$

$$a_d \prod_{j=1}^d (X - \alpha_j), \quad a_d > 0,$$

its (multiplicative) Mahler measure is

$$M(\alpha) = a_d \prod_{j=1}^d \max\{1, |\alpha_j|\}. \quad (\text{II.8.D08})$$

For a nonzero polynomial $P(X) = a_d \prod_{j=1}^d (X - z_j)$, define $M(P) = |a_d| \prod_j \max\{1, |z_j|\}$. Root multiplicities are retained.

Definition II.8.4.6 (II.8.09: Garsia number and Garsia parameter). A Garsia number is a real algebraic integer $\beta \in (1, 2)$ such that $|N(\beta)| = 2$ and every conjugate of β other than β has modulus strictly greater than one. A Garsia parameter is $\lambda = \beta^{-1}$. The modulus condition is part of the definition; it is not replaced by the weaker statement $M(\beta) = 2$.

Lemma II.8.4.7 (L-II.8.06 – Garsia separation estimate). *If β is a Garsia number and $\lambda = \beta^{-1}$, there is $c_\beta > 0$ such that for every $N \geq 1$ and every distinct $\varepsilon, \varepsilon' \in \{-1, 1\}^N$,*

$$\left| \sum_{n=0}^{N-1} (\varepsilon_n - \varepsilon'_n) \lambda^n \right| \geq c_\beta 2^{-N}. \quad (\text{II.8.02})$$

Proof. Let $\beta = \beta_1, \dots, \beta_d$ be the conjugates of the Garsia number and put $\lambda = \beta^{-1}$. For two sign words define $a_n = (\varepsilon_n - \varepsilon'_n)/2 \in \{-1, 0, 1\}$ and

$$Q(X) = \sum_{n=0}^{N-1} a_n X^{N-1-n}.$$

The polynomial is nonzero. It also satisfies $Q(\beta) \neq 0$. Indeed, if $Q(\beta) = 0$, remove a power of X from Q to obtain $R(\beta) = 0$ with $R(0) = \pm 1$. The monic minimal polynomial of β , whose constant term has modulus two, would divide R in $\mathbb{Z}[X]$ by Gauss's lemma, forcing $2 \mid R(0)$, a contradiction.

The algebraic integer $Q(\beta)$ has nonzero integer norm, so

$$1 \leq \prod_{j=1}^d |Q(\beta_j)|.$$

For $j \geq 2$, $|\beta_j| > 1$ and

$$|Q(\beta_j)| \leq \sum_{m=0}^{N-1} |\beta_j|^m \leq \frac{|\beta_j|^N}{|\beta_j| - 1}.$$

Writing $c_0 = \prod_{j=2}^d (|\beta_j| - 1) > 0$, we obtain

$$|Q(\beta)| \geq c_0 \prod_{j=2}^d |\beta_j|^{-N} = c_0 (2/\beta)^{-N} = c_0 (2\lambda)^{-N},$$

because $\prod_j |\beta_j| = |N(\beta)| = 2$. Therefore

$$\left| \sum_{n=0}^{N-1} (\varepsilon_n - \varepsilon'_n) \lambda^n \right| = 2\lambda^{N-1} |Q(\beta)| \geq \frac{2c_0}{\lambda} 2^{-N}.$$

This is (II.8.02) with $c_\beta = 2c_0/\lambda$.

□

Theorem II.8.4.8 (T-II.8.07 — Garsia’s bounded-density theorem). *For every Garsia parameter $\lambda = \beta^{-1}$, the Bernoulli convolution ν_λ is absolutely continuous with a bounded density.*

Proof. L-II.8.06 shows that all 2^N prefix sums are distinct and uniformly $c_\beta 2^{-N}$ -separated. L-II.8.05 therefore gives a bounded density.

□

The mechanism is now transparent:

integer norm $\implies$ exponential separation $\implies$ interval-mass bound $\implies L^\infty$ density.

II.8.5 Pisot parameters and Fourier nondecay

Definition II.8.5.1 (II.8.07: Pisot number). A Pisot number is a real algebraic integer $\theta > 1$ whose other conjugates all have modulus strictly less than one. In the Pisot regime for Bernoulli convolutions we require $1 < \theta < 2$ and put $\lambda = \theta^{-1} \in (1/2, 1)$.

The Pisot mechanism points in the opposite direction. Instead of keeping different sums apart, it makes the factors in the Fourier product nearly periodic along the frequencies θ^k . The trace lemma supplies the required near-integrality.

Lemma II.8.5.2 (L-II.8.08 — Pisot trace approximation). *If θ is Pisot, there are $C < \infty$ and $0 < \rho < 1$ such that*

$$\text{dist}(\theta^n, \mathbb{Z}) \leq C\rho^n \quad (n \geq 0). \quad (\text{II.8.03})$$

Proof. If $d = [\mathbb{Q}(\theta) : \mathbb{Q}] = 1$, then θ is an integer and $\text{dist}(\theta^n, \mathbb{Z}) = 0$ for every n ; take any $0 < \rho < 1$ and $C = 1$. Assume $d \geq 2$. Let $\theta = \theta_1, \dots, \theta_d$ be the conjugates and choose ρ with $\max_{j \geq 2} |\theta_j| < \rho < 1$. The trace

$$q_n = \text{Tr}_{\mathbb{Q}(\theta)/\mathbb{Q}}(\theta^n) = \sum_{j=1}^d \theta_j^n$$

is a rational algebraic integer, hence an integer. Thus

$$\text{dist}(\theta^n, \mathbb{Z}) \leq |\theta^n - q_n| \leq \sum_{j=2}^d |\theta_j|^n \leq (d-1)\rho^n.$$

Take $C = d - 1$.

□

Theorem II.8.5.3 (T-II.8.09 – Erdős’s Pisot nondecay and singularity theorem). *Let $1 < \theta < 2$ be Pisot and $\lambda = \theta^{-1}$. There is $c_\theta > 0$ such that*

$$|\hat{v}_\lambda(\theta^k)| \geq c_\theta \quad (k \geq 1). \quad (\text{II.8.04})$$

Thus v_λ is not Rajchman, is not absolutely continuous, and is singular.

Proof. Put $\lambda = \theta^{-1}$. From L-II.8.01,

$$|\hat{v}_\lambda(\theta^k)| = \prod_{m=0}^k |\cos(2\pi\theta^m)| \prod_{m=1}^{\infty} |\cos(2\pi\theta^{-m})|. \quad (\text{II.8.N09})$$

The first infinite product obtained by allowing $m \geq 0$ is positive. Indeed, L-II.8.08 and $1 - |\cos(2\pi t)| \leq 2\pi^2 \text{dist}(t, \mathbb{Z})^2$ show summability of the deficits for large m . No factor vanishes: if $\cos(2\pi\theta^m) = 0$, then $\theta^m \in \frac{1}{4} + \frac{1}{2}\mathbb{Z}$; this is rational and an algebraic integer, hence an integer, a contradiction.

The second product in (II.8.N09) is also positive because $\sum_m \theta^{-2m} < \infty$. It has no zero factor. Otherwise $\theta^{-m} = (2q+1)/4$, so the rational algebraic integer $\theta^m = 4/(2q+1)$ is an integer, necessarily 4. If $\theta^m = 4$, every conjugate of θ is a root of $X^m - 4$ and has modulus $\theta > 1$. The Pisot condition therefore forces degree one, which would make θ an integer in $(1, 2)$, impossible.

Let A and B be the two positive infinite products just proved. A finite initial product is at least its infinite continuation, so $|\hat{v}_\lambda(\theta^k)| \geq AB =: c_\theta > 0$. Since $\theta^k \rightarrow \infty$, the measure is not Rajchman. The Riemann–Lebesgue lemma excludes absolute continuity. L-II.8.04 then forces singularity.

□

Garsia parameters separate cylinders; Pisot parameters align cosine factors. The two arithmetic hypotheses therefore control different objects and lead to opposite regularity conclusions.

II.8.6 Parameter averaging and transversality

For a typical parameter there is no useful uniform algebraic separation. We average over λ instead. The lower-density criterion reduces L^2 regularity to an energy integral, and the parameter-pair lemma rewrites that energy as the average size of a small-value set for a power series.

Definition II.8.6.1 (II.8.13: Lower symmetric density). For a finite Borel measure μ , define

$$\underline{D}\mu(x) = \liminf_{r \downarrow 0} \frac{\mu([x-r, x+r])}{2r} \in [0, +\infty].$$

By the Lebesgue differentiation theorem, $\mu \ll dx$ if $\underline{D}\mu < \infty$ at μ -almost every point. If $d\mu = f dx$, then $\underline{D}\mu = f$ almost everywhere with respect to both dx on $\{f > 0\}$ and μ .

Definition II.8.6.2 (II.8.12: Thinned Bernoulli convolution). Let

$$J = \{n \in \mathbb{N}_0 : n \not\equiv 2 \pmod{3}\}, \quad \widetilde{\Omega} = \{-1, 1\}^J,$$

and equip $\widetilde{\Omega}$ with the fair product probability $\widetilde{\mathbb{P}}$. Define

$$\widetilde{\Pi}_\lambda(\omega) = \sum_{n \in J} \omega_n \lambda^n, \quad \widetilde{\nu}_\lambda = (\widetilde{\Pi}_\lambda)_* \widetilde{\mathbb{P}}. \quad (\text{II.8.D12})$$

Equivalently, $\widetilde{\nu}_\lambda$ is the law of the random series obtained from Y_λ by omitting the digits with indices congruent to 2 (mod 3). The omitted digits form an independent convolution factor of ν_λ .

Lemma II.8.6.3 (L-II.8.10 – Lower-density criterion with the exact L^2 conclusion). *Let μ be a compactly supported probability. If*

$$\int \underline{D}\mu(x) d\mu(x) < \infty, \quad (\text{II.8.05})$$

then $d\mu = f dx$ for some $f \in L^2$, and the integral in (II.8.05) equals $\|f\|_2^2$.

Proof. The differentiation criterion in D-II.8.13 and (II.8.05) imply $\mu \ll dx$. Write $d\mu = f dx$. Differentiation gives $\underline{D}\mu(x) = f(x)$ for μ -almost every x . Hence

$$\int \underline{D}\mu d\mu = \int f d\mu = \int f(x)^2 dx < \infty.$$

Thus $f \in L^2$, and the claimed equality holds. □

Lemma II.8.6.4 (L-II.8.11 – Parameter-pair energy reduction). *Let $I = [\lambda_0, \lambda_1] \subset (1/2, 1)$. If $\Pi_\lambda(\omega) = \sum_{n \geq 0} \omega_n \lambda^n$ on the fair sign space Ω , then*

$$\int_I \int \underline{D}v_\lambda(x) dv_\lambda(x) d\lambda \leq \liminf_{r \downarrow 0} \frac{1}{2r} \iint_{\Omega^2} \mathcal{L}^1\{\lambda \in I : |\Pi_\lambda(\omega) - \Pi_\lambda(\tau)| \leq r\} d\mathbb{P}^2. \quad (\text{II.8.06})$$

For the thinned coding map of D-II.8.12, the corresponding inequality is

$$\int_I \int \underline{D}\tilde{v}_\lambda(x) d\tilde{v}_\lambda(x) d\lambda \leq \liminf_{r \downarrow 0} \frac{1}{2r} \iint_{\tilde{\Omega}^2} \mathcal{L}^1\{\lambda \in I : |\tilde{\Pi}_\lambda(\omega) - \tilde{\Pi}_\lambda(\tau)| \leq r\} d(\tilde{\mathbb{P}} \otimes \tilde{\mathbb{P}}). \quad (\text{II.8.06a})$$

Proof. For fixed λ , Fatou's lemma gives

$$\int \underline{D}v_\lambda(x) dv_\lambda(x) \leq \liminf_{r \downarrow 0} \frac{1}{2r} \int v_\lambda([x-r, x+r]) dv_\lambda(x).$$

The map $(\lambda, \omega) \mapsto \Pi_\lambda(\omega)$ is continuous on compact parameter intervals, so every integrand is measurable. Apply Fatou again in λ , use $v_\lambda = (\Pi_\lambda)_* \mathbb{P}$, and then use Tonelli:

$$\begin{aligned} & \int_I \int \underline{D}v_\lambda dv_\lambda d\lambda \\ & \leq \liminf_{r \downarrow 0} \frac{1}{2r} \int_I \iint_{\Omega^2} \mathbf{1}_{\{|\Pi_\lambda(\omega) - \Pi_\lambda(\tau)| \leq r\}} d\mathbb{P}^2 d\lambda, \end{aligned}$$

which is (II.8.06) after interchanging integrals. Replacing the full product space and coding map by their thinned versions proves the final assertion. $\square$

Definition II.8.6.5 (II.8.10: Small-coefficient polynomial classes). Let

$$\mathcal{Q}_n = \left\{ \sum_{j=0}^n a_j X^j : a_j \in \{-1, 0, 1\} \right\}.$$

The zero polynomial belongs to $\mathcal{Q}_n$, so a nonzero condition is always stated explicitly. Let $\mathcal{P}$ be the power series $1 + \sum_{j \geq 1} a_j X^j$ with $a_j \in \{-1, 0, 1\}$. For $i = 1, 2$, let $\mathcal{P}_i$ be the subclass in which $a_j = 0$ whenever $j \equiv i \pmod{3}$.

Definition II.8.6.6 (II.8.11: Transversality and star envelopes). Let $J \subset [0, 1)$ be a compact interval and let $\mathcal{F}$ be a family of functions differentiable on a neighborhood of J . We call J an interval of δ -transversality for $\mathcal{F}$ if

$$f(x) < \delta \implies f'(x) < -\delta \quad (f \in \mathcal{F}, x \in J). \quad (\text{II.8.D11})$$

The property passes to every subinterval of J .

For an index set $A \subset \mathbb{N}$, an A -star envelope is a power series $h = 1 + \sum_{n \in A} c_n X^n$, $-1 \leq c_n \leq 1$, for which there is a cut index k such that $c_n = -1$ below the cut, at most one coefficient at the cut is intermediate, and $c_n = 1$ above it. Missing indices have coefficient zero.

If two sign sequences first differ at index k , their difference is $2\lambda^k$ times a normalized small-coefficient series. Transversality prevents that normalized series from remaining close to zero over a long parameter interval.

Lemma II.8.6.7 (L-II.8.12 – Star-envelope transversality lemma). *Let $A \subset \mathbb{N}$, let h be an A -star envelope, and let $0 \leq a < x_0 < 1$, $0 < \delta < 1$. Suppose*

$$h(x) > \delta, \quad h'(x) < -\delta \quad (a \leq x \leq x_0), \quad 1 - \sum_{n \in A} x^n \geq \delta \quad (0 \leq x \leq a). \quad (\text{II.8.07a})$$

Then $[0, x_0]$ is an interval of δ -transversality for all $1 + \sum_{n \in A} a_n x^n$, $a_n \in [-1, 1]$. Consequently, for every such series g and every $\rho > 0$,

$$\mathcal{L}^1\{x \in [0, x_0] : |g(x)| \leq \rho\} \leq 2\rho/\delta. \quad (\text{II.8.07})$$

Proof. Let $g = 1 + \sum_{n \in A} b_n x^n$, $-1 \leq b_n \leq 1$, and set $f = g - h$. By the star form, the nonzero coefficients of f have at most one sign change, from nonnegative to nonpositive. Thus for a suitable cut k ,

$$f(x) = \sum_{n \leq k} u_n x^n - \sum_{n > k} v_n x^n, \quad u_n, v_n \geq 0.$$

If $x > 0$ and $f(x) < 0$, then

$$\sum_{n \leq k} u_n x^n < \sum_{n > k} v_n x^n.$$

Multiplying the terms on the left by at most k , and those on the right by at least $k+1$, gives $xf'(x) < 0$, hence $f'(x) < 0$.

If $g(x) < \delta$, the lower-envelope condition in (II.8.07a) forces $x > a$. There $h(x) > \delta$, so $f(x) < 0$ and therefore $g'(x) = h'(x) + f'(x) < -\delta$. This is transversality.

For the sublevel estimate, the claim is immediate when $\rho \geq \delta$, because $x_0 < 1 < 2\rho/\delta$. If $0 < \rho < \delta$, then $g' < -\delta$ at every point of $\{|g| \leq \rho\}$. This set is an interval: after g first enters it, the negative derivative prevents a later upward re-entry. The mean value theorem bounds its length by $2\rho/\delta$.

□

Lemma II.8.6.8 (L-II.8.13 – Full-series transversality interval). *There is an absolute $\delta_0 > 0$ such that $[0, 2^{-2/3}]$ is an interval of δ_0 -transversality for $\mathcal{P}$.*

Proof. Let $t = 2^{-2/3}$ and

$$h(x) = 1 - x - x^2 - x^3 + \frac{1}{2}x^4 + \frac{x^5}{1-x}. \quad (\text{II.8.N13})$$

This is a full-index star envelope. Its second derivative has coefficients with one sign change, so it has at most one positive zero: after division by the lowest power, the ratio of the positive-coefficient part to the negative-coefficient part is strictly increasing. Hence h' first decreases and then increases. Since $h'(0) = -1$, it is enough to check the right endpoint.

Using $t^3 = 1/4$, direct substitution gives

$$h(t) = \frac{3}{4} - \frac{7t}{8} - t^2 + \frac{t^2}{4(1-t)},$$

$$h'(t) = -1 - 2t - 3t^2 + 2t^3 + \frac{5t^4 - 4t^5}{(1-t)^2}.$$

The rational enclosure $6299/10000 < t < 63/100$, cubed to verify the two inequalities, gives

$$0.0700 < h(t) < 0.0701, \quad -0.0980 < h'(t) < -0.0978.$$

Thus on $[0, t]$, $h' < -1/1000$ and $h(x) \geq h(t) > 1/1000$. Apply L-II.8.12 with $a = 0$ and $\delta_0 = 1/1000$. □

After the first differing retained digit is removed from a pair of thinned sign sequences, the normalized difference belongs to $\mathcal{P}_1$ or $\mathcal{P}_2$, according to the residue class of that digit.

Lemma II.8.6.9 (L-II.8.14 – Thinned transversality interval). *There is an absolute $\delta_1 > 0$ such that $[0, 2^{-1/2}]$ is an interval of δ_1 -transversality for both $\mathcal{P}_1$ and $\mathcal{P}_2$.*

Proof. Put $s = 2^{-1/2}$ and consider

$$h_1(x) = 1 - x - x^3 - x^4 + \frac{x^6 + x^7}{1-x^3},$$

$$h_2(x) = 1 - x^2 - x^3 - x^5 - x^6 + \frac{x^8 + x^9}{1-x^3}. \quad (\text{II.8.N14})$$

The first is a star envelope for $\mathcal{P}_2$, and the second for $\mathcal{P}_1$. Their second derivatives have one coefficient sign change, so each first derivative decreases and then increases. From $7071/10000 < s < 7072/10000$, rational interval arithmetic gives

	$h_i(s)$	$h'_i(s)$	
$i = 1$	(0.0194, 0.0195)	(-0.1541, -0.1538)	(II.8.N15)
$i = 2$	(0.0097, 0.0098)	(-2.879, -2.876).	

For h_1 , $h'_1(0) = -1$; unimodality and (II.8.N15) show that $h'_1 < -1/1000$ and $h_1 > 1/1000$ on $[0, s]$.

For h_2 , direct rational evaluation at $1/2$ gives

$$h_2(1/2) = \frac{131}{224}, \quad h'_2(1/2) < -2.$$

Unimodality and (II.8.N15) imply $h'_2 < -1/1000$ on $[1/2, s]$, so $h_2 > h_2(s) > 1/1000$ there. On $[0, 1/2]$, the smallest possible member of $\mathcal{P}_1$ is bounded below by

$$1 - \sum_{\substack{n \geq 1 \\ n \neq 1 \pmod{3}}} x^n = 1 - \frac{x^2 + x^3}{1 - x^3} \geq 1 - \frac{3}{7} = \frac{4}{7}.$$

L-II.8.12, with $a = 0$ for h_1 and $a = 1/2$ for h_2 , proves the result with $\delta_1 = 1/1000$. □

Proposition II.8.6.10 (P-II.8.15 – Transversality gives almost-everywhere L^2 density). *If $I = [\lambda_0, \lambda_1]$, $\lambda_0 > 1/2$, is an interval of δ -transversality for $\mathcal{P}$, then v_λ has an L^2 density for almost every $\lambda \in I$. If $\lambda_0 > 2^{-2/3}$ and there is a common $\delta > 0$ for which I is an interval of δ -transversality for $\mathcal{P}_1$ and $\mathcal{P}_2$, the same conclusion follows by the thinned law.*

Proof. First use the full series. If k is the first index at which ω and τ differ, then

$$\Pi_\lambda(\omega) - \Pi_\lambda(\tau) = 2\sigma\lambda^k g(\lambda), \quad \sigma \in \{-1, 1\}, \quad g \in \mathcal{P}.$$

For $\lambda \in I$, the event that the absolute value is at most r implies $|g(\lambda)| \leq r/(2\lambda_0^k)$. L-II.8.12 therefore gives

$$\mathcal{L}^1\{\lambda \in I : |\Pi_\lambda(\omega) - \Pi_\lambda(\tau)| \leq r\} \leq \delta^{-1} r \lambda_0^{-k}. \quad (\text{II.8.N16})$$

The probability that the first differing index is k equals 2^{-k-1} . Insert (II.8.N16) into L-II.8.11:

$$\int_I \int \underline{D}v_\lambda dv_\lambda d\lambda \leq \frac{1}{2\delta} \sum_{k=0}^{\infty} 2^{-k-1} \lambda_0^{-k} < \infty,$$

because $2\lambda_0 > 1$. L-II.8.10 and Tonelli show that v_λ has an L^2 density for almost every $\lambda \in I$.

For the thinned law, the same factorization holds with $g \in \mathcal{P}_1 \cup \mathcal{P}_2$. Among the first k indices at least $2k/3 - 1$ are retained. Hence the probability of agreement before the first differing retained index k is at most $2^{1-2k/3}$. The analogue of the last series is bounded by a constant times

$$\sum_{k=0}^{\infty} \lambda_0^{-k} 2^{-2k/3},$$

which converges exactly when $\lambda_0 > 2^{-2/3}$. Thus $\tilde{v}_\lambda$ has an L^2 density almost everywhere on the stated interval. Since v_λ is its convolution with the independent omitted-digit law, Young's inequality preserves L^2 and proves the claim for v_λ . $\square$

Two transversality ranges are now available. The full digit series treats the lower overlap interval; deleting one residue class enlarges the range, and convolution restores the omitted digits. A final power decomposition propagates the conclusion from λ^k back to λ and covers the whole interval $(1/2, 1)$.

Theorem II.8.6.11 (T-II.8.16 – Peres–Solomyak theorem). *For Lebesgue-almost every $\lambda \in (1/2, 1)$, the Bernoulli convolution v_λ is absolutely continuous and has a density in $L^2(\mathbb{R})$.*

Proof. L-II.8.13 and P-II.8.15 cover every compact subinterval of $(1/2, 2^{-2/3}]$. L-II.8.14 and the thinned part of P-II.8.15 cover every compact subinterval of $(2^{-2/3}, 2^{-1/2}]$. A countable exhaustion therefore proves the theorem for almost every $\lambda \in (1/2, 2^{-1/2}]$.

For each $\lambda \in (1/2, 1)$, there is an integer $k \geq 1$ such that

$$\frac{1}{2} < \lambda^k \leq 2^{-1/2}. \quad (\text{II.8.N17})$$

For $\lambda > 2^{-1/2}$, choose the first power below or equal to $2^{-1/2}$; the preceding power is larger than $2^{-1/2}$, and $\lambda > 2^{-1/2}$ makes the selected power larger than $1/2$. For smaller λ , take $k = 1$.

For fixed k , the map $\lambda \mapsto \lambda^k$ is a C^1 diffeomorphism with locally Lipschitz inverse on the relevant parameter interval. It therefore pulls null sets back to null sets. For almost every λ satisfying (II.8.N17), v_{λ^k} has an L^2 density. L-II.8.01 writes v_λ as the convolution of k scaled copies of v_{λ^k} ; one L^2 factor convolved with probabilities remains in L^2 . Taking the countable union over k finishes the proof. $\square$

II.8.7 What the classical theory does and does not imply

Corollary II.8.7.1 (C-II.8.17 – Regularity implications). *For Bernoulli convolutions,*

$$L^\infty \text{ density} \implies L^2 \text{ density} \implies \text{absolute continuity} \implies \dim v_\lambda = 1, \quad (\text{II.8.08})$$

and Fourier nondecay implies failure of absolute continuity. The last implication uses exact dimensionality from T-I.9.08. No reverse implication in (II.8.08), and in particular no implication $\dim v_\lambda = 1 \implies v_\lambda \ll dx$, is asserted.

Proof. The first arrow is the inclusion $L^\infty \subset L^2$ on the compact support $[-(1-\lambda)^{-1}, (1-\lambda)^{-1}]$, and the second follows from the definition of an L^2 density. For the third, T-I.9.08 says that ν_λ is exact-dimensional. At almost every point where an absolutely continuous density is positive and finite, differentiation gives $\nu_\lambda([x-r, x+r]) = 2rf(x) + o(r)$, so the local dimension is one. Exact dimensionality therefore gives $\dim \nu_\lambda = 1$.

If a finite measure is absolutely continuous, the Riemann–Lebesgue lemma forces its Fourier transform to vanish at infinity. Thus Fourier nondecay excludes absolute continuity. T-II.8.07, T-II.8.09 and T-II.8.16 instantiate the three regimes without adding any converse. $\square$

The implications are deliberately one-way. Dimension one alone does not imply absolute continuity; absolute continuity does not imply any decay rate beyond the Riemann–Lebesgue conclusion; and the special arithmetic examples do not describe every parameter.

Chapter II.9

Varjú's Arithmetic Entropy Program

Geometric intuition

Varjú's arithmetic entropy program replaces symbolic entropy by the entropy remaining after exact collisions. Algebraic approximation, Mahler measure, scale entropy, and annular separation are combined to prove full dimension for transcendental Bernoulli parameters and to prepare the absolute-continuity arguments of Chapter II.10. The final sections distinguish dimension from measure type through an arithmetic classification criterion, and give direct singularity proofs for quartic-Salem parameters and a degree-31 non-Pisot example.

Assume for orientation that $\dim \nu_\lambda < 1$. The proof proceeds through five conversions:

dimension loss $\longrightarrow$ digit entropy deficit $\longrightarrow$ algebraic approximants
 $\longrightarrow$ bounded Mahler height $\longrightarrow$ annular contradiction.

All logarithms are natural. Our constant convention is as follows: c_α, C_α depend only on the displayed deficit parameter, while $c_{\lambda,\varepsilon}, C_{\lambda,\varepsilon}$ may also depend on the fixed Bernoulli parameter and error tolerance. No such constant depends on the running degree, scale, polynomial, or approximating algebraic number unless explicitly stated.

II.9.1 Exact collisions and Garsia entropy

Definition II.9.1.1 (II.9.01: Prefix laws and Garsia entropy). For $\lambda \in \mathbb{C}$, let

$$S_n(\lambda) = \sum_{j=0}^{n-1} \varepsilon_j \lambda^j, \quad \nu_\lambda^{(n)} = \text{Law}(S_n(\lambda)).$$

Entropy is computed after viewing $\mathbb{C}$ as $\mathbb{R}^2$. Its Garsia entropy is

$$h_\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} H(v_\lambda^{(n)}), \quad (\text{II.9.D01})$$

once existence is proved. Coincident sums are aggregated before entropy is computed. Thus $0 \leq h_\lambda \leq \log 2$. For real $0 < \lambda < 1$, the infinite law is the Bernoulli convolution v_λ of D-II.8.01.

Symbolic entropy counts words, whereas Garsia entropy counts the distinct values left after exact collisions have been merged. In the collision-free case every one of the 2^n words survives, so the entropy rate is $\log 2$.

Lemma II.9.1.2 (L-II.9.01 — Entropy-rate existence and elementary bounds). *The sequence $H(v_\lambda^{(n)})$ is subadditive, so the limit in D-II.9.01 exists and*

$$h_\lambda = \inf_{n \geq 1} \frac{1}{n} H(v_\lambda^{(n)}), \quad 0 \leq h_\lambda \leq \log 2. \quad (\text{II.9.01})$$

Proof. Let $S'_m(\lambda)$ be an independent copy of $S_m(\lambda)$. Then

$$S_{n+m}(\lambda) = S_n(\lambda) + \lambda^n S'_m(\lambda).$$

The value on the right is a function of the pair (S_n, S'_m) . The entropy of a function cannot exceed the entropy of its input, and independence makes the entropy of the pair additive. Consequently

$$H(v_\lambda^{(n+m)}) \leq H(v_\lambda^{(n)}) + H(v_\lambda^{(m)}). \quad (\text{II.9.N01})$$

Fekete's subadditive lemma now gives

$$\lim_{n \rightarrow \infty} \frac{H(v_\lambda^{(n)})}{n} = \inf_{n \geq 1} \frac{H(v_\lambda^{(n)})}{n}.$$

There are at most 2^n sign words, so $0 \leq H(v_\lambda^{(n)}) \leq n \log 2$. Division by n proves all assertions. □

II.9.2 Entropy across geometric scales

Definition II.9.2.1 (II.9.02: Translation-averaged entropy at a scale). For a bounded real random variable X and $r > 0$, define

$$\mathcal{H}(X; r) = \int_0^1 H(\lfloor X/r + t \rfloor) dt, \quad \mathcal{H}(X; r_1 | r_2) = \mathcal{H}(X; r_1) - \mathcal{H}(X; r_2). \quad (\text{II.9.D02})$$

For a law μ , write $\mathcal{H}(\mu; \cdot) = \mathcal{H}(X; \cdot)$. The base-two normalized form is $\mathcal{H} = \mathcal{H} / \log 2$. Consequently one dyadic digit has maximal normalized entropy one.

If a random vector Z has density f and the integral is defined, its differential entropy is

$$h_{\text{diff}}(Z) = - \int f(x) \log f(x) dx.$$

We also write $h_{\text{diff}}(\mu)$ when μ is the law of Z . Every use below is made after convolution with a nondegenerate interval or Gaussian law, in which case the displayed differential entropies are finite.

If θ is a finite nonnegative measure of mass $m > 0$, its conditional scale entropy is extended homogeneously by $\mathcal{H}(\theta; r_1 | r_2) = m\mathcal{H}(m^{-1}\theta; r_1 | r_2)$; it is zero for the zero measure. This convention is used only for decompositions into subprobabilities.

Definition II.9.2.2 (II.9.03: Scale-window Bernoulli law). Fix $0 < \lambda < 1$, let $(\varepsilon_j)_{j \in \mathbb{Z}}$ be a two-sided sequence of independent fair signs, and let $I \subset (0, \infty)$ be an interval that is bounded above. Define

$$\nu_\lambda^I = \text{Law} \left(\sum_{j \in \mathbb{Z}: \lambda^j \in I} \varepsilon_j \lambda^j \right). \quad (\text{II.9.D03})$$

The series converges absolutely: only finitely many negative indices occur, and the positive-index tail is geometric. The index set is finite when $\inf I > 0$. Disjoint scale windows give independent convolution factors. In particular $\nu_\lambda^{(n)} = \nu_\lambda^{(\lambda^n, 1]}$.

Lemma II.9.2.3 (L-II.9.02 – Averaged scale-entropy calculus). For bounded X , an independent uniform U_r on $[0, r]$, and $0 < r_1 \leq r_2$,

$$\mathcal{H}(X; r) = h_{\text{diff}}(X + U_r) - \log r, \quad 0 \leq \mathcal{H}(X; r_1 | r_2) \leq 2 \log(r_2/r_1). \quad (\text{II.9.02})$$

More precisely, if $c > 0$, $r_1 \leq r_2 \leq r_3$, and X, Y are independent, then

$$\mathcal{H}(cX; cr_1 | cr_2) = \mathcal{H}(X; r_1 | r_2),$$

$$\mathcal{H}(X; r_1 | r_3) = \mathcal{H}(X; r_1 | r_2) + \mathcal{H}(X; r_2 | r_3),$$

and, whenever $r_2/r_1 \in \mathbb{N}$,

$$\mathcal{H}(X + Y; r_1 | r_2) \geq \mathcal{H}(X; r_1 | r_2).$$

The scale-window laws obey

$$\mathcal{H}(\nu_\lambda^{\lambda^k I}; \lambda^k r_1 | \lambda^k r_2) = \mathcal{H}(\nu_\lambda^I; r_1 | r_2). \quad (\text{II.9.03})$$

Proof. Let $p_k(t) = \mathbb{P}(\lfloor X/r + t \rfloor = k)$. If f_r is the density of $X + U_r$, then

$$p_k(t) = \mathbb{P}\{X \in [r(k-t), r(k+1-t))\} = r f_r(r(k+1-t))$$

at every Lebesgue point. As $k \in \mathbb{Z}$ and $0 \leq t < 1$, the points $r(k+1-t)$ tile $\mathbb{R}$, each with Jacobian $r dt$. Thus Tonelli's theorem gives

$$\begin{aligned} \mathcal{H}(X; r) &= \int_0^1 \sum_k -p_k(t) \log p_k(t) dt \\ &= \int_{\mathbb{R}} -f_r(y) \log(r f_r(y)) dy = h_{\text{diff}}(X + U_r) - \log r. \end{aligned} \quad (\text{II.9.N02})$$

For completeness, we record the scale comparison behind the remaining claims. If $0 < a \leq b$, refinement of translated interval partitions, followed by conditioning, gives

$$0 \leq \mathcal{H}(X; a) - \mathcal{H}(X; b) \leq 2 \log(b/a). \quad (\text{II.9.N03})$$

Here is a direct verification. If μ is the law of X , Fubini applied to (II.9.N02) gives

$$\mathcal{H}(X; r) = \int_{\mathbb{R}} \int_0^1 -\log \mu([y + (t-1)r, y + tr]) dt d\mu(y). \quad (\text{II.9.N03a})$$

For fixed y, t , the interval in (II.9.N03a) grows with r , proving the lower bound in (II.9.N03). For the upper bound, first convolve μ with a tiny interval and write f for the resulting density. Differentiation at $r = 1$ gives

$$\begin{aligned} \left. \frac{d}{dr} \mathcal{H}(X; r) \right|_{r=1} &= - \int_{\mathbb{R}} \int_0^1 \frac{(1-t)f(y+t-1) + tf(y+t)}{\mu([y+t-1, y+t])} dt d\mu(y) \\ &\geq - \int_{\mathbb{R}} \int_y^{y+1} \frac{f(z-1) + f(z)}{\mu([z-1, z])} dz d\mu(y) = -2. \end{aligned} \quad (\text{II.9.N03b})$$

In the last equality interchange y, z ; the inner y -mass is $\mu([z-1, z])$. Dilation gives $0 \leq d\mathcal{H}(X; e^{-u})/du \leq 2$ at every differentiability point. Integration proves (II.9.N03). Letting the preliminary smoothing tend to zero is justified by (II.9.N02) and L^1 -continuity of convolution.

We shall use the following differential-entropy submodularity inequality. If A, B, C are independent random vectors and the displayed entropies are finite, then

$$h_{\text{diff}}(A + B + C) + h_{\text{diff}}(B) \leq h_{\text{diff}}(A + B) + h_{\text{diff}}(B + C). \quad (\text{II.9.N03c})$$

The data-processing inequality for the Markov chain $A \rightarrow A + B \rightarrow A + B + C$ gives

$$I(A; A + B + C) \leq I(A; A + B).$$

Because B is absolutely continuous and independent of A, C , the two sides are respectively $h_{\text{diff}}(A + B + C) - h_{\text{diff}}(B + C)$ and $h_{\text{diff}}(A + B) - h_{\text{diff}}(B)$. This proves (II.9.N03c) even when A or C is discrete; finite displayed differential entropies are understood throughout.

If $b/a = N \in \mathbb{N}$, write $U_b = U_a + Z$, where Z is independent and uniform on $\{0, a, \dots, (N-1)a\}$. Applying (II.9.N03c) to $Y, X + U_a, Z$ shows that

$$\mathcal{H}(X + Y; a | b) \geq \mathcal{H}(X; a | b) \quad (\text{II.9.N04})$$

when X, Y are independent. This proves the asserted integral-ratio convolution monotonicity.

Replacing X, r by cX, cr in the definition proves scale covariance. Subtracting three values of $\mathcal{H}$ proves additivity over nested scale intervals. Finally,

$$\sum_{\lambda^j \in \lambda^k I} \varepsilon_j \lambda^j \stackrel{d}{=} \lambda^k \sum_{\lambda^i \in I} \varepsilon_i \lambda^i,$$

after the index change $j = i + k$. Scale covariance gives (II.9.03). □

There are two entropies in play. Garsia entropy records information surviving exact symbolic collisions, whereas scale entropy records information visible at geometric resolution. Algebraicity links them because distinct algebraic prefix values cannot remain arbitrarily close without coinciding. The entropy-flattening theorem can therefore read the discrete prefix entropy at a sufficiently fine scale, producing the algebraic formula $\dim v_\eta = \min(1, h_\eta / \log(1/\eta))$.

Theorem II.9.2.4 (T-II.9.03 — Algebraic entropy–dimension formula). *For every algebraic $\eta \in (1/2, 1)$,*

$$\dim v_\eta = \min \left\{ 1, \frac{h_\eta}{\log(1/\eta)} \right\}. \quad (\text{II.9.04})$$

Proof. Put $\chi = \log(1/\eta)$, $\mu = v_\eta$, and $\alpha = \dim \mu$. Exact dimensionality is T-I.9.08, so L-II.6.02 identifies α with the entropy dimension.

Write $\mu = v_\eta^{(n)} * (S_{\eta^n})_* \mu$. Entropy of a mixture is at most the entropy of its mixing variable plus the averaged entropy of its components. Since the tail and the smoothing scale are both dilated by η^n , scale covariance and bounded support give a constant C_η , independent of n , such that

$$\mathcal{H}(\mu; \eta^n) \leq H(v_\eta^{(n)}) + C_\eta. \quad (\text{II.9.N05})$$

Divide by $n\chi$, use entropy dimension, and let $n \rightarrow \infty$. This proves

$$\alpha \leq \min\{1, h_\eta / \chi\}. \quad (\text{II.9.N06})$$

It remains to prove the reverse inequality. If $\alpha = 1$, (II.9.N06) already implies $h_\eta / \chi \geq 1$. Suppose $\alpha < 1$. For the two-map homogeneous system of ratio η , the natural dyadic level is $\ell_n = \lfloor n\chi / \log 2 \rfloor$. The natural-scale approximation L-II.6.07 and homogeneous flattening T-II.6.08 say, for each fixed $q > 1$, that

$$H_{\ell_n}(v_\eta^{(n)}) = \alpha n\chi + o(n), \quad H_{\lfloor q\ell_n \rfloor}(v_\eta^{(n)}) - H_{\ell_n}(v_\eta^{(n)}) = o(n). \quad (\text{II.9.N07})$$

Because η is algebraic, L-II.6.13 applied to the fixed tuple (η) gives $c_0 \in (0, 1)$ such that two distinct values in the support of $v_\eta^{(n)}$ are separated by at least c_0^n . Indeed, half their difference is a nonzero polynomial in η of degree below n with coefficients in $\{-1, 0, 1\}$. Choose q so large that $q\chi > \log(1/c_0)$. For all large n , every dyadic cell of level $[q\ell_n]$ then contains at most one distinct support point. Exact coincidences have already been aggregated in $v_\eta^{(n)}$; hence

$$H_{[q\ell_n]}(v_\eta^{(n)}) = H(v_\eta^{(n)}). \quad (\text{II.9.N08})$$

Equations (II.9.N07)–(II.9.N08), divided by n , yield $h_\eta = \alpha\chi$. Together with the case $\alpha = 1$, this is exactly (II.9.04). □

Algebraicity enters only when the discrete prefix law is read at a sufficiently fine scale: exponential separation ensures that different surviving values occupy different cells.

Lemma II.9.2.5 (L-II.9.04 – Lower semicontinuity of Bernoulli dimension). *The map $\lambda \mapsto \dim v_\lambda$ is lower semicontinuous on $(0, 1)$.*

Proof. By exact dimensionality and L-II.9.02,

$$\dim v_\lambda = \lim_{n \rightarrow \infty} \frac{\mathcal{H}(v_\lambda; \lambda^n \mid 1)}{n \log(1/\lambda)}. \quad (\text{II.9.N09})$$

For fixed n , the numerator is continuous in λ : truncate the series after N digits, use uniform convergence of the remaining tail on compact parameter intervals, and apply (II.9.N02) and L^1 continuity of convolution with an interval. The convergence is uniform for the two displayed positive scales.

Define

$$f_k(\lambda) = \frac{\mathcal{H}(v_\lambda; \lambda^{2^k} \mid 1) - 2 \log 2}{2^k \log(1/\lambda)}. \quad (\text{II.9.N10})$$

For every sufficiently large k , we show $f_{k+1} \geq f_k$. Choose $\lambda^{2^k}/2 \leq r \leq \lambda^{2^k}$ with $r/\lambda^{2^{k+1}} \in \mathbb{N}$. Nesting, convolution monotonicity (II.9.N04), the two-log scale bound (II.9.N03), and self-similar scaling give

$$\begin{aligned} \mathcal{H}(v_\lambda; \lambda^{2^{k+1}} \mid \lambda^{2^k}) &\geq \mathcal{H}(v_\lambda; \lambda^{2^{k+1}} \mid r) \\ &\geq \mathcal{H}((S_{\lambda^{2^k}})_* v_\lambda; \lambda^{2^{k+1}} \mid r) \\ &\geq \mathcal{H}((S_{\lambda^{2^k}})_* v_\lambda; \lambda^{2^{k+1}} \mid \lambda^{2^k}) - 2 \log 2 \\ &= \mathcal{H}(v_\lambda; \lambda^{2^k} \mid 1) - 2 \log 2. \end{aligned} \quad (\text{II.9.N11})$$

Adding the entropy from λ^{2^k} to 1 proves eventual monotonicity. Equation (II.9.N09) says $\dim v_\lambda = \sup_{k \geq k_0} f_k(\lambda)$ for every sufficiently large fixed k_0 . Each f_k is continuous, and a supremum of continuous functions is lower semicontinuous.

□

Lower semicontinuity is the bridge from a fixed exceptional parameter to nearby algebraic models: a nearby model may inherit an upper dimension bound without assuming that dimension is continuous.

II.9.3 Dimension drop as a uniform local deficit

Definition II.9.3.1 (II.9.09: Entropy deficit and good scale). For a probability μ , its normalized digit entropy at scale r is $\mathcal{H}(\mu; r \mid 2r)$. Given $0 < \alpha < 1$, the scale is α -deficient if this number is at most $1 - \alpha$.

For $E \subset \mathbb{R}$, let $N_1(E)$ be the largest cardinality of a subset of E whose distinct points have mutual distance at least one. This is the unit packing number of E .

Lemma II.9.3.2 (L-II.9.05 – Uniform one-digit deficit under dimension drop). *If $\lambda \in (1/2, 1)$ and $\dim v_\lambda < 1$, there is $\alpha > 0$ such that*

$$\mathcal{H}(v_\lambda^I; r \mid 2r) \leq 1 - \alpha \quad (\text{II.9.05})$$

for every $r > 0$ and every bounded interval $I \subset (0, \infty)$.

Proof. We first treat the full measure. Put $s = \dim v_\lambda < 1$. By (II.9.N09), choose $\alpha_0 > 0$ and N so that

$$\mathcal{H}(v_\lambda; 2^{-n}) < n(1 - \alpha_0) \quad (n \geq N). \quad (\text{II.9.N12})$$

We need one elementary doubling estimate. With χ_r the uniform probability on $[0, r]$ and $\eta_r = (\delta_0 + \delta_r)/2$, one has $\chi_{2r} = \chi_r * \eta_r$. Differential-entropy submodularity and concavity under the identity $\eta_r * \eta_r = (\eta_{2r} + \delta_r)/2$ give

$$1 - \mathcal{H}(\mu; 2r \mid 4r) \leq 4(1 - \mathcal{H}(\mu; r \mid 2r)). \quad (\text{II.9.N13})$$

Explicitly, L-II.9.02 turns the right deficit into $[h_{\text{diff}}(\mu * \eta_r * \chi_r) - h_{\text{diff}}(\mu * \chi_r)]/\log 2$. Submodularity bounds the twofold gain by twice the onefold gain; concavity of $-x \log x$ then bounds the η_{2r} -gain by twice the twofold gain. Replacing χ_r by $\chi_{2r} = \eta_r * \chi_r$ gives (II.9.N13).

Suppose for some $r > 0$ that $\mathcal{H}(v_\lambda; r \mid 2r) \geq 1 - \alpha$. Iterating (II.9.N13) yields

$$\mathcal{H}(v_\lambda; 2^j r \mid 2^{j+1} r) \geq 1 - 4^j \alpha. \quad (\text{II.9.N14})$$

Choose an integer K with $(K + 1)^{-1} < \alpha_0/3$, and then $\alpha > 0$ so small that $4^K \alpha < \alpha_0/3$. Summing (II.9.N14) gives

$$\mathcal{H}(v_\lambda; r \mid 2^K r) \geq K - 4^K \alpha. \quad (\text{II.9.N15})$$

Choose m with $2^{-K-1} \leq \lambda^m \leq 2^{-K}$. The scaled tail $(S_{\lambda^m})_* v_\lambda$ is a convolution factor of v_λ ; (II.9.N04), scaling and (II.9.N15) therefore give the same lower bound on each interval $[\lambda^{jm}r, 2^K \lambda^{jm}r]$. Summing J such nested blocks and absorbing at most one digit at each endpoint gives

$$\mathcal{H}(v_\lambda; 2^{-J(K+1)+\lceil \log_2 r \rceil}) \geq J(K+1)(1 - 4^K \alpha - (K+1)^{-1}). \quad (\text{II.9.N16})$$

For large J , (II.9.N16) contradicts (II.9.N12). Hence the full measure has a uniform deficit α .

For a bounded scale window I , choose an integer q so large that $\lambda^{qI} \subset (0, 1]$. The law $v_\lambda^{\lambda^{qI}}$ is then a convolution factor of v_λ . The ratio 2 is integral, so (II.9.N04), followed by window scaling, gives

$$\mathcal{H}(v_\lambda^I; r \mid 2r) = \mathcal{H}(v_\lambda^{\lambda^{qI}}; \lambda^q r \mid 2\lambda^q r) \leq 1 - \alpha.$$

This proves (II.9.05). □

This is the form of dimension drop used from now on: not merely an asymptotic inequality, but a fixed loss from one full bit at every location in the digit expansion and at every physical scale.

II.9.4 Low entropy forces algebra

Definition II.9.4.1 (II.9.10: Root families and rational-polynomial height). Let $\mathcal{X}_n$ be the set of complex roots of nonzero polynomials in $\mathcal{P}_n$. Repeated roots are listed once. A common root of a family $\mathcal{A} \subset \mathcal{P}_n$ is a zero of every member of $\mathcal{A}$. For $Q \in \mathbb{Q}[X]$, write uniquely $Q = a^{-1}\tilde{Q}$, where $a \in \mathbb{N}$, $\tilde{Q} = \sum_k q_k X^k \in \mathbb{Z}[X]$, and the integer tuple $(a, q_0, q_1, \dots)$ is primitive. Its coefficient height is

$$H_{\text{coeff}}(Q) = \max \left\{ a, \max_k |q_k| \right\}.$$

This convention also gives $H_{\text{coeff}}(0) = 1$, by taking $a = 1$ and $\tilde{Q} = 0$.

If many sign words fall into the same tiny scale cell, all their difference polynomials are tiny at λ . A controlled Bézout identity forces those polynomials to share a root near λ . The cell partition also shows that the root has low prefix entropy. Root separation makes this nearby root unique at a still finer scale.

Lemma II.9.4.2 (L-II.9.06 – Quantitative common-root lemma). *For every $0 < \varepsilon < 1/2$ there are $c > 0$ and n_0 such that: if $n \geq n_0$, $0 < r < n^{-3n}$, $\varepsilon < |\lambda| < 1 - \varepsilon$, and $\mathcal{A} \subset \mathcal{P}_n$ contains a nonzero polynomial and satisfies $|P(\lambda)| \leq r$ for all $P \in \mathcal{A}$, then $\mathcal{A}$ has a common root η with*

$$|\eta - \lambda| \leq r^c. \quad (\text{II.9.06})$$

Proof. We give the quantitative algebra in full. Denote by $\mathcal{P}_n$ the integer polynomials of degree at most n with coefficients in $\{-1, 0, 1\}$.

Controlled Bézout lemma. If $\mathcal{A} \subset \mathcal{P}_n$ and $D \in \mathbb{Z}[X]$ is its primitive greatest common divisor, then there are $m \leq n + 1$, polynomials $P_1, \dots, P_m \in \mathcal{A}$, and $Q_1, \dots, Q_m \in \mathbb{Q}[X]$ such that

$$D = \sum_{j=1}^m Q_j P_j, \quad \deg Q_j < n, \quad H_{\text{coeff}}(Q_j) \leq \exp(Cn \log n). \quad (\text{II.9.N17})$$

To prove this, choose a linearly independent subfamily that generates the same ideal over $\mathbb{Q}[X]$; it has at most $n + 1$ members. Euclidean division reduces every multiplier modulo one chosen polynomial, so all multiplier degrees are below n . Equating the $2n + 1$ coefficients in the identity gives a linear system with entries in $\{-1, 0, 1\}$. A divisor D of a member of $\mathcal{P}_n$ satisfies

$$\|D\|_1 \leq 2^n M(D) \leq 2^n M(P) \leq 2^n \|P\|_2 \leq 2^n \sqrt{n+1};$$

the first inequality follows by expanding $\prod (X - \zeta_i)$, and the last Mahler bound follows from Jensen's formula. Cramer's rule and Hadamard's determinant inequality now bound all numerators and denominators by $(2n + 1)! \|D\|_1$, which is $\exp(Cn \log n)$. This proves (II.9.N17).

Apply (II.9.N17) to the family in the lemma. Since $\varepsilon < |\lambda| < 1 - \varepsilon$,

$$|D(\lambda)| \leq r \exp(C_\varepsilon n \log n). \quad (\text{II.9.N18})$$

For $r < n^{-3n}$ and all sufficiently large n , the right side is at most $r^{1/2} < 1$. Thus D is nonconstant.

It remains to locate one of its roots. A polynomial in $\mathcal{P}_n$, after removal of its initial zero coefficients, has constant coefficient of modulus one. Jensen's formula on $|z| = \rho < 1$ shows that the number of its nonzero roots in $|z| \leq a$ is at most

$$\frac{\log(1/(1 - \rho))}{\log(\rho/a)} \quad (\text{II.9.N19})$$

whenever $a < \rho < 1$: on the circle, $|P(z)| \leq (1 - \rho)^{-1}$. Choose $a = 1 - \varepsilon/2$ and a fixed $\rho \in (a, 1)$. Hence at most K_ε nonzero roots of D lie in $|z| \leq 1 - \varepsilon/2$, because D divides a member of $\mathcal{A}$.

Write the roots of D , with multiplicity, as ζ_j . The zero roots have distance at least ε from λ , and every root outside $|z| \leq 1 - \varepsilon/2$ has distance at least $\varepsilon/2$. Since the leading coefficient has modulus at least one, (II.9.N18) implies

$$\prod_{j: 0 < |\zeta_j| \leq 1 - \varepsilon/2} |\lambda - \zeta_j| \leq r^{1/2} (2/\varepsilon)^n. \quad (\text{II.9.N20})$$

The right side is at most $r^{1/3}$ for large n . One of the at most K_ε factors is therefore at most $r^{1/(3K_\varepsilon)}$. Taking this root as η and $c = (3K_\varepsilon)^{-1}$, we have $D(\eta) = 0$, so every member of $\mathcal{A}$ vanishes at η . This is (II.9.06). □

Lemma II.9.4.3 (L-II.9.07 — Mahler separation for small-polynomial roots). *If $n \geq 9$ and distinct $\eta, \eta' \in \mathcal{X}_n$, then*

$$|\eta - \eta'| > 2n^{-4n}. \quad (\text{II.9.07})$$

Proof. We first prove the root-separation estimate used here. If $Q \in \mathbb{Z}[X]$ is square-free of degree $D \geq 2$, with distinct roots $\zeta_1, \dots, \zeta_D$, the discriminant identity

$$1 \leq |\text{disc } Q| = |a_D|^{2D-2} \prod_{i < j} |\zeta_i - \zeta_j|^2 \quad (\text{II.9.N21})$$

and Hadamard's inequality applied to the Vandermonde matrix after normalizing each column by $\max(1, |\zeta_j|)$ give, for every $i \neq j$,

$$|\zeta_i - \zeta_j| \geq \sqrt{3} D^{-(D+2)/2} M(Q)^{-(D-1)}. \quad (\text{II.9.N22})$$

Indeed, remove the factor $\zeta_i - \zeta_j$ from the Vandermonde product in (II.9.N21); each remaining row norm is at most $\sqrt{D}$, while the two removed columns have determinant norm at most $M(Q)^{D-1}/\sqrt{3}$. Substitution and taking square roots gives (II.9.N22). Replacing Q by its square-free part only decreases its degree and Mahler measure, so the estimate applies to distinct roots of an arbitrary integer polynomial.

If $\eta, \eta' \in \mathcal{X}_n$ are conjugate, take Q to be their minimal polynomial. Otherwise take the product of their two distinct minimal polynomials. In either case $D \leq 2n$. Each minimal polynomial divides a polynomial in $\mathcal{P}_n$, and $M(P) \leq \|P\|_2 \leq \sqrt{n+1}$; hence $M(Q) \leq n+1$. Equation (II.9.N22) yields

$$|\eta - \eta'| \geq \sqrt{3}(2n)^{-n-1}(n+1)^{-2n+1} > (2n)^{-3n} > 2n^{-4n}$$

for $n \geq 9$, proving (II.9.07). □

A strict entropy loss puts several sign words in the same small cell. Their differences are small-coefficient polynomials that are simultaneously tiny at λ . The common-root lemma converts simultaneous smallness into a nearby algebraic parameter, while root separation prevents competing parameters at the finer scale.

Proposition II.9.4.4 (P-II.9.08 — Low scale entropy produces an exceptional algebraic root). *For every $0 < \varepsilon < 1/2$ there are $c > 0, n_0$ such that, if $n \geq n_0$, $0 < r < n^{-3n}$, $\varepsilon < \lambda < 1 - \varepsilon$, and $\mathcal{H}(v_\lambda^{(n)}; r) < n \log 2$, then some $\eta \in \mathcal{X}_n$ satisfies*

$$|\eta - \lambda| < r^c, \quad h_\eta \leq \frac{1}{n} \mathcal{H}(v_\lambda^{(n)}; r). \quad (\text{II.9.08})$$

Proof. Let $\xi_0, \dots, \xi_{n-1}$ be independent fair signs. By the definition of averaged entropy, there is $t \in [0, 1]$ such that

$$H\left(\left[r^{-1} \sum_{j=0}^{n-1} \xi_j \lambda^j + t\right]\right) \leq \mathcal{H}(v_\lambda^{(n)}; r) < n \log 2. \quad (\text{II.9.N23})$$

For $a \in \mathbb{Z}$, let Ω_a be the set of sign words giving the cell a in (II.9.N23). At least one cell contains two words. For every pair ω, ω' in the same cell form

$$P_{\omega, \omega'}(X) = \frac{1}{2} \sum_{j=0}^{n-1} (\omega_j - \omega'_j) X^j \in \mathcal{P}_n.$$

Every such polynomial has $|P_{\omega, \omega'}(\lambda)| < r$. L-II.9.06 gives a common root η with $|\eta - \lambda| < r^c$.

Within each Ω_a , all sums $\sum \omega_j \eta^j$ coincide. Therefore the η -prefix sum is a function of the cell label, and

$$H(v_\eta^{(n)}) \leq \sum_a \frac{|\Omega_a|}{2^n} \log \frac{2^n}{|\Omega_a|} \leq \mathcal{H}(v_\lambda^{(n)}; r).$$

L-II.9.01 gives $h_\eta \leq n^{-1} H(v_\eta^{(n)})$, proving (II.9.08). □

Proposition II.9.4.5 (P-II.9.09 — A close small-polynomial root forces full prefix entropy). *With the same c, n_0 , if $\varepsilon < \lambda < 1 - \varepsilon$, $\eta \in \mathcal{X}_n$, and $|\lambda - \eta| < n^{-4n}$, then*

$$\mathcal{H}(v_\lambda^{(n)}; r) = n \log 2 \quad \text{for every } r \leq |\lambda - \eta|^{1/c}. \quad (\text{II.9.09})$$

Proof. Suppose for some $r \leq |\lambda - \eta|^{1/c}$ that the asserted entropy is below $n \log 2$. For large n , $r < n^{-3n}$, so P-II.9.08 produces $\eta' \in \mathcal{X}_n$ with

$$|\lambda - \eta'| < r^c \leq |\lambda - \eta|.$$

Thus $\eta' \neq \eta$, while

$$|\eta - \eta'| \leq |\eta - \lambda| + |\lambda - \eta'| < 2|\eta - \lambda| < 2n^{-4n},$$

contradicting L-II.9.07. Hence the entropy equals its maximum $n \log 2$ for every displayed r . □

Common-root extraction supplies existence; root separation supplies uniqueness and stability. Together they form the algebraic-approximation engine used in the multiscale argument.

II.9.5 Entropy growth and the multiscale budget

A dimension-deficient Bernoulli convolution is uniformly deficient in every binary digit. Convolving it with a measure that carries entropy across many scales must therefore create a definite entropy gain. Separated digit windows supply independent gains. Because total entropy across a fixed scale interval is bounded by its length, too many reservoirs are impossible. This restriction is what forces an exceptionally accurate algebraic approximation.

Theorem II.9.5.1 (T-II.9.10 – Quantitative entropy increment). *For every $0 < \alpha < 1/2$ there is $c_\alpha > 0$ such that the following holds. Let μ, ν be compactly supported probabilities, $\sigma_2 < \sigma_1 < 0$, and $0 < \beta \leq 1/2$. If*

$$N_1\{\sigma \in [\sigma_2, \sigma_1] : \mathcal{H}(\mu; 2^\sigma | 2^{\sigma+1}) > 1 - \alpha\} < c_\alpha \beta (\sigma_1 - \sigma_2)$$

and

$$\mathcal{H}(\nu; 2^{\sigma_2} | 2^{\sigma_1}) > \beta (\sigma_1 - \sigma_2),$$

then

$$\mathcal{H}(\mu * \nu; 2^{\sigma_2} | 2^{\sigma_1}) > \mathcal{H}(\mu; 2^{\sigma_2} | 2^{\sigma_1}) + \frac{c_\alpha \beta}{\log(1/\beta)} (\sigma_1 - \sigma_2) - 3. \quad (\text{II.9.10})$$

Proof. The proof has four stages. We first quantify the gain from adjoining one separated Bernoulli digit. We then decompose a measure with positive scale entropy into many such two-point pieces. A selection argument finds well-separated scales on which μ is deficient, and summing the independent gains over those scales gives the asserted increment.

We work with normalized entropy $\mathcal{H}$, so a binary digit has entropy one. Four elementary ingredients are proved first.

1. A Bernoulli increment. Let θ_t be the probability equally supported on two points at distance t , and put

$$\delta = 1 - \mathcal{H}(\mu; t | 2t).$$

If

$$r_2 \leq t\delta/10, \quad r_1 \geq 144t\delta^{-2}, \quad (\text{II.9.N24})$$

then

$$\mathcal{H}(\mu * \theta_t; r_2 | r_1) \geq \mathcal{H}(\mu; r_2 | r_1) + \delta/3. \quad (\text{II.9.N25})$$

After translation, take $\theta_t = (\delta_0 + \delta_t)/2$. Since $\chi_{2t} = \chi_t * \theta_t$, formula (II.9.N02) gives the exact identity

$$\mathcal{H}(\mu * \theta_t; t) = \mathcal{H}(\mu; t) + \delta. \quad (\text{II.9.N26})$$

We use two smoothing estimates:

$$|\mathcal{H}(\mu * \theta_t; r) - \mathcal{H}(\mu; r)| \leq 4(t/r)^{1/2} \quad (r \geq t), \quad (\text{II.9.N27})$$

and

$$\mathcal{H}(\mu * \theta_t; r | t) \geq \mathcal{H}(\mu; r | t) - \frac{2}{(\log 2)(t/r - 1)} \quad (0 < r < t). \quad (\text{II.9.N28})$$

Here is a proof of both estimates. For (II.9.N27), couple $X \sim \mu$ and $Y = X + tB$, where B is an independent fair bit, and let $K_u = \lfloor X/r + u \rfloor$, $L_u = \lfloor Y/r + u \rfloor$, $0 \leq u < 1$. If $r \geq 2t$, $D_u = L_u - K_u \in \{0, 1\}$, so the entropy chain rule gives

$$\frac{|H(L_u) - H(K_u)|}{\log 2} \leq \frac{H(D_u)}{\log 2} = h_2(p_u), \quad p_u = \mathbb{P}(D_u = 1).$$

Here $h_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$, with $0 \log_2 0 = 0$. For every realization $0 \leq Y - X \leq t$, the set of shifts $u \in [0, 1)$ for which $D_u = 1$ has length $(Y - X)/r$. Consequently $\int_0^1 p_u du \leq t/r$. Jensen's inequality and monotonicity of binary entropy on $[0, 1/2]$ yield

$$|\mathcal{H}(\mu * \theta_t; r) - \mathcal{H}(\mu; r)| \leq h_2(t/r) \leq 2(t/r) \log_2(r/t) \leq 4\sqrt{t/r}.$$

If $t \leq r < 2t$, conditioning on B and observing that a translation by at most one cell changes a cell label by at most one binary choice give $|H(L_u) - H(K_u)|/\log 2 \leq 2$; this is at most $4\sqrt{t/r}$.

For (II.9.N28), set $N = \lfloor t/r \rfloor \geq 1$. Integral-ratio convolution monotonicity (II.9.N04) and additivity of scale entropy give

$$\begin{aligned} \mathcal{H}(\mu * \theta_t; r | t) &= \mathcal{H}(\mu * \theta_t; r | Nr) + \mathcal{H}(\mu * \theta_t; Nr | t) \\ &\geq \mathcal{H}(\mu; r | Nr) \\ &= \mathcal{H}(\mu; r | t) - \mathcal{H}(\mu; Nr | t). \end{aligned}$$

The scale bound (II.9.N03) implies $\mathcal{H}(\mu; Nr | t) \leq 2 \log_2(t/(Nr))$. Since $1 \leq t/(Nr) < 1 + 1/N < 1 + (t/r - 1)^{-1}$, the inequality $\log_2(1 + u) \leq u/\log 2$ gives precisely (II.9.N28). These arguments apply directly to compactly supported probabilities; no discreteness approximation is needed.

The first condition in (II.9.N24) makes the loss in (II.9.N28) at most $\delta/3$, and the second makes (II.9.N27) at r_1 at most $\delta/3$. Add the ranges $[r_2, t]$ and $[t, r_1]$ and use (II.9.N26). This proves (II.9.N25).

2. Extraction of Bernoulli pieces. Let θ be a finitely supported probability and suppose

$$\mathcal{H}(\theta; r/2 | r) \leq \frac{3}{2} \mathcal{H}(\theta; r | 2r). \quad (\text{II.9.N29})$$

Then

$$\theta = \theta_0 + \theta_1 + \cdots + \theta_N \quad (\text{II.9.N30})$$

with nonnegative θ_0 , with every θ_i , $i \geq 1$, an equal-weight two-point measure whose support distance lies in $[r/2, 2r]$, and

$$\sum_{i=1}^N \|\theta_i\| \geq \frac{1}{128} \frac{\mathcal{H}(\theta; r \mid 2r)}{\log_2(1/\mathcal{H}(\theta; r \mid 2r)) + 1}. \quad (\text{II.9.N31})$$

If the numerator is zero, the right side is defined to be zero.

To prove this, first note the following separation identity. If a finite measure ρ is supported on disjoint intervals of length at most $r/2$, with mutual gaps exceeding $2r$, then direct decomposition of the smoothed densities gives

$$\mathcal{H}(\rho; r/2 \mid r) = 2\mathcal{H}(\rho; r \mid 2r). \quad (\text{II.9.N32})$$

For $a, b \geq 0$, set $H_2(a, b) = (a + b)h_2(a/(a + b))$, with value 0 if $a + b = 0$. Indeed, for a probability supported on one interval of length $r/2$,

$$\mathcal{H}(\rho; s \mid 2s) = \frac{1}{2s} \int_{\mathbb{R}} H_2(\rho(-\infty, x), \rho[x, \infty)) dx$$

after translating the interval; the right side is proportional to s^{-1} for $s \geq r/2$. Separated components have disjoint smoothed supports, so their homogeneous entropies add, proving (II.9.N32).

We also need the restriction estimate. If a probability $\rho = \rho_0 + \zeta$ with $\|\zeta\| = u \leq 1/2$, the conditional entropy chain rule applied to the indicator of the selected component gives

$$\mathcal{H}(\rho_0; r \mid 2r) \leq \mathcal{H}(\rho; r \mid 2r) \leq \mathcal{H}(\rho_0; r \mid 2r) + 3u \log_2(1/u). \quad (\text{II.9.N33})$$

The left inequality follows from concavity and homogeneity of conditional entropy for the coupled nested grids: with $K_t = \lfloor X/r + 2t \rfloor$ and $J_t = \lfloor X/(2r) + t \rfloor = \lfloor K_t/2 \rfloor$, one has

$$\mathcal{H}(X; r \mid 2r) = \int_0^1 H(K_t \mid J_t) dt / \log 2.$$

Conditional entropy is concave in the joint cell law, so the homogeneous conditional entropy of a sum of measures is at least the sum of the homogeneous conditional entropies. More precisely for the upper inequality, at each of the two scales the mixture entropy exceeds the sum of the homogeneous component entropies by a number in $[0, h_2(u)]$. Subtracting the scales shows that the conditional entropy differs from that sum by at most $h_2(u)$, not twice this quantity. The discarded component contributes at most u ; since $h_2(u) \leq 2u \log_2(1/u)$ and $u \leq u \log_2(1/u)$, the right inequality follows.

Choose a decomposition (II.9.N30) minimizing $\|\theta_0\|$. Then the support of θ_0 has no pair at distance in $[r/2, 2r]$, and hence is covered by intervals of length below $r/2$ separated by gaps above $2r$. Put $u = 1 - \|\theta_0\| = \sum_{i \geq 1} \|\theta_i\|$. If $u \geq 1/2$, (II.9.N31) is immediate. Otherwise (II.9.N32)–(II.9.N33) and (II.9.N29), with $h = \mathcal{H}(\theta; r \mid 2r)$, give

$$8u \log_2(1/u) \geq h/2, \quad u \log_2(1/u) \geq h/16. \quad (\text{II.9.N34})$$

If (II.9.N31) failed, put $L = \log_2(1/h) \geq 0$. Monotonicity of $u \log_2(1/u)$ on $(0, e^{-1}]$ would give

$$u \log_2(1/u) < \frac{h}{128(L+1)} (L+7 + \log_2(L+1)) \leq \frac{7h}{128} < \frac{h}{16},$$

a contradiction. This proves the extraction claim.

3. Selection of scales. Set

$$K = \lceil \log_2(144/\alpha^2) \rceil + 2, \quad L = \sigma_1 - \sigma_2.$$

There is a $2K$ -separated set $B \subset \mathbb{Z} \cap [\sigma_2 + K, \sigma_1 - K]$ such that, for every $n \in B$,

$$\begin{aligned} \frac{3}{2} \mathcal{H}(v; 2^n \mid 2^{n+1}) &> \mathcal{H}(v; 2^{n-1} \mid 2^n), \\ \mathcal{H}(v; 2^n \mid 2^{n+1}) &\geq \beta/12, \\ \mathcal{H}(\mu; t \mid 2t) &\leq 1 - \alpha \quad (2^{n-1} \leq t \leq 2^{n+1}), \end{aligned} \tag{II.9.N35}$$

and

$$\sum_{n \in B} \mathcal{H}(v; 2^n \mid 2^{n+1}) \geq c'_\alpha \beta L - 1. \tag{II.9.N36}$$

Here are the counting details. Let B_1 contain the integers in the interior interval satisfying the first line of (II.9.N35). Between two consecutive members of B_1 , failure of that inequality makes the digit entropies decay by successive factors $2/3$; their sum is at most three times the entropy at the lower selected scale. The two end intervals cost at most $2K + 5$. Hence

$$\beta L \leq 3 \sum_{n \in B_1} \mathcal{H}(v; 2^n \mid 2^{n+1}) + 2K + 5. \tag{II.9.N37}$$

Greedy retain a maximal-weight $2K$ -separated subfamily B_2 . Each retained scale is charged by fewer than $4K$ deleted scales. Delete from B_2 the terms below $\beta/12$; their total is at most $\beta L/(24K)$. Finally delete every n for which the last line of (II.9.N35) fails. Choosing one violating exponent for each deleted n gives, because B_2 is $2K$ -separated, a unit-separated subset of the exceptional-scale set. The N_1 hypothesis and a sufficiently small c_α make the total charge at most half the remaining lower bound. Equations (II.9.N37) and these three losses give (II.9.N36).

4. Summation. Approximate v by a finitely supported measure. Formula (II.9.N02) makes all entropies on the fixed scale interval converge, so it is enough to prove the result for this approximation. For $n \in B$, apply (II.9.N30)–(II.9.N31) at $r = 2^n$. Every Bernoulli piece has separation $t \in [2^{n-1}, 2^{n+1}]$. The choice of K makes (II.9.N24) valid with $r_2 = 2^{n-K}$, $r_1 = 2^{n+K}$, and $\delta \geq \alpha$. Homogeneity of conditional entropy, (II.9.N25), (II.9.N31), and the lower bound $\mathcal{H}(v; 2^n \mid 2^{n+1}) \geq \beta/12$ yield

$$\begin{aligned} \mathcal{H}(\mu * v; 2^{n-K} \mid 2^{n+K}) &\geq \mathcal{H}(\mu; 2^{n-K} \mid 2^{n+K}) \\ &\quad + \frac{c_\alpha}{\log(1/\beta)} \mathcal{H}(v; 2^n \mid 2^{n+1}). \end{aligned} \tag{II.9.N38}$$

The intervals $[n - K, n + K]$, $n \in B$, are disjoint. On every remaining integral digit, convolution monotonicity gives no negative gain. Sum (II.9.N38), use (II.9.N36), and then move the two endpoint scales by less than one dyadic digit each. The endpoint loss is at most two and the constant in (II.9.N36) costs one more. We obtain

$$\mathcal{H}(\mu * \nu; 2^{\sigma_2} \mid 2^{\sigma_1}) > \mathcal{H}(\mu; 2^{\sigma_2} \mid 2^{\sigma_1}) + \frac{c_\alpha \beta}{\log(1/\beta)} L - 3,$$

which is (II.9.10). □

The increment theorem is local in scale. The next proposition packages repeated local gains into a global budget: scale windows provide independent convolution factors, but total entropy across the ambient interval is bounded.

Proposition II.9.5.2 (P-II.9.11 — Multiscale bootstrap constraint). *Fix $\lambda \in (1/2, 1)$ satisfying (II.9.05). There is $C = C(\lambda, \alpha)$ such that any finite sequences $n_1 < \dots < n_N$ and $K_j \geq 11$ satisfying*

$$n_{j+1} \geq K_j n_j \quad (1 \leq j < N), \quad \mathcal{H}(v_\lambda^{(n_j)}; \lambda^{K_j n_j} \mid \lambda^{10n_j}) \geq \alpha n_j, \quad n_j \geq C(\log K_j)^2$$

where the second and third conditions hold for $1 \leq j \leq N$, obey

$$\sum_{j=1}^N \frac{1}{\log K_j \log \log K_j} \leq C \left(1 + \frac{\sum_{j=1}^N \log K_j}{n_1} \right). \quad (\text{II.9.11})$$

Proof. We select disjoint reservoirs, apply the increment theorem to each one, and compare their accumulated gain with the total available scale entropy.

All entropies in this proof are normalized. We first isolate entropy inside separated digit windows.

Window-transfer lemma. Suppose $0 \leq a_i < b_i \leq 1$ and $0 < r_i < s_i$, $i = 1, 2$, satisfy

$$\lambda^{-1} s_1 / s_2 \leq a_1 / a_2, \quad \lambda r_1 / r_2 \geq b_1 / b_2, \quad \max(2, \lambda^{-2}) \leq s_2 / r_2 \leq s_1 / r_1. \quad (\text{II.9.N39})$$

If

$$\mathcal{H}(v_\lambda^{(a_1, b_1]}; r_1 \mid s_1) \geq \gamma \log_2(s_1 / r_1),$$

then

$$\mathcal{H}(v_\lambda^{(a_2, b_2]}; r_2 \mid s_2) \geq \frac{\gamma}{6} \log_2(s_2 / r_2). \quad (\text{II.9.N40})$$

Choose powers λ^{k_i} so that the intervals $[\lambda^{k_i} r_2, \lambda^{k_i} s_2]$ minimally cover $[r_1, s_1]$. Minimality implies that the number of intervals is at most $3 \log(s_1 / r_1) / \log(s_2 / r_2)$. Additivity of scale entropy therefore gives one translated interval carrying at least the corresponding average. The first two inequalities in (II.9.N39) ensure $(a_1, b_1] \subset \lambda^{k_i}(a_2, b_2]$. Window scaling, convolution monotonicity, and the harmless factor two for nonintegral endpoint ratios give (II.9.N40).

Now fix one pair n, K from the proposition and put

$$J = \left\lceil \frac{\log(K/10)}{\log(11/10)} \right\rceil.$$

The resulting J adjacent scale intervals cover the assumed reservoir and may extend its fine endpoint by less than one interval. Since refining the fine endpoint can only increase conditional scale entropy, write

$$\mathcal{H}(v_\lambda^{(\lambda^n, 1]}; \lambda^{11(11/10)^j n} \mid \lambda^{10(11/10)^j n}) = b_j n. \quad (\text{II.9.N41})$$

Then $\sum_{j=0}^{J-1} b_j \geq \alpha$. Retain indices with $b_j \geq \alpha/(2J)$; they still carry total mass at least $\alpha/2$. For such a j , set $T = \lfloor (11/10)^j \rfloor$. The law $v_\lambda^{(\lambda^{Tn}, 1]}$ is the convolution of T scaled copies of $v_\lambda^{(\lambda^n, 1]}$. Apply T-II.9.10 successively to these copies on the scale interval in (II.9.N41). L-II.9.05 supplies the same deficit α to every partial convolution, while $b_j \gg 1/\log K$ and $n \geq C(\log K)^2$ absorb the additive constant. This gives

$$\mathcal{H}(v_\lambda^{(\lambda^{Tn}, 1]}; \lambda^{12(11/10)^j n} \mid \lambda^{10(11/10)^j n}) \geq \frac{cb_j}{\log K} \log_2(\lambda^{-2(11/10)^j n}). \quad (\text{II.9.N42})$$

Apply the window-transfer lemma with

$$Q_j = \lambda^{-9(11/10)^j n}.$$

For every $r \leq \lambda^{2Kn}$ and $\max(2, \lambda^{-2}) \leq A \leq \lambda^{-n}$, it yields

$$\mathcal{H}(v_\lambda^{(Q_j r, Q_j^2 r]}; r \mid Ar) \geq \frac{cb_j}{\log K} \log_2 A. \quad (\text{II.9.N43})$$

Select one residue class modulo 10 whose retained b_j -mass is maximal. Relabel its members by i . Then

$$\lambda^{-9n} \leq R_i := Q_{j_i} \leq \lambda^{-Kn}, \quad R_{i+1} \geq R_i^2, \quad a_i := b_{j_i} \geq c/\log K, \quad \sum_i a_i \geq c, \quad (\text{II.9.N44})$$

and (II.9.N43) holds with R_i, a_i .

Apply (II.9.N44) to every pair (n_j, K_j) , using the common choices

$$A = \lambda^{-n_1}, \quad r = \lambda^{2K_N n_N}.$$

The hypotheses $n_{j+1} \geq K_j n_j$ and $R_{i+1} \geq R_i^2$ make all digit windows $(R_{j,i} r, R_{j,i}^2 r]$ pairwise disjoint. Therefore v_λ is the convolution of all these window laws and a remaining probability. Apply T-II.9.10 successively with

$$\beta_{j,i} = ca_{j,i} / \log K_j.$$

Since $a_{j,i} \geq c/\log K_j$, $\log(1/\beta_{j,i}) \leq C \log \log K_j$. We obtain

$$\mathcal{H}(v_\lambda; r \mid Ar) \geq \sum_{j,i} \left(\frac{ca_{j,i}}{\log K_j \log \log K_j} \log_2 A - C \right). \quad (\text{II.9.N45})$$

There are $O(\log K_j)$ windows for the j -th pair, while $\sum_i a_{j,i} \geq c$. The left side of (II.9.N45) is at most $2 \log_2 A$ by L-II.9.02. Hence

$$\sum_{j=1}^N \frac{1}{\log K_j \log \log K_j} \leq C \left(1 + \frac{\sum_{j=1}^N \log K_j}{\log A} \right).$$

Finally $\log A = n_1 \log(1/\lambda)$; absorb this fixed factor into $C(\lambda, \alpha)$. This is (II.9.11). $\square$

Hence a dimension-deficient measure cannot possess too many large, well-separated entropy reservoirs.

II.9.6 Exceptional algebraic approximation

Definition II.9.6.1 (II.9.04: Algebraic exceptional sets). For $d \geq 1$ and $0 \leq a \leq 1$, let

$$E_{d,a} = \{\eta \in (1/2, 1) : \dim v_\eta < a, P(\eta) = 0 \text{ for some nonzero } P \in \mathcal{P}_d\}.$$

The degree bound is on the witnessing small-coefficient polynomial, not only on the algebraic degree of η .

Definition II.9.6.2 (II.9.05: Iterated logarithms and approximation schedules). Write $\log^{(k)}$ and $\exp^{(k)}$ for k -fold iterates of the natural logarithm and exponential, on ranges where they are defined. A superquadratic algebraic approximation schedule is a sequence (d_j, η_j) with $d_j \rightarrow \infty$, $\deg \eta_j \leq d_j$, and

$$|\lambda - \eta_j| \leq e^{-d_j^2}.$$

Theorem II.9.6.3 (T-II.9.12 – Breuillard–Varjú exceptional approximation theorem). *If $\lambda \in (1/2, 1)$ and $s = \dim v_\lambda < 1$, then for every $\varepsilon > 0$ there are arbitrarily large integers d and $\eta \in E_{d,s+\varepsilon}$ such that*

$$|\lambda - \eta| \leq \exp(-d^{\log^{(3)} d}). \quad (\text{II.9.12})$$

Proof. We argue on a long finite scale range. At each stage either the prefix already contains a usable entropy reservoir, or low entropy yields a nearby algebraic root. In the latter case maximal closeness determines the next degree and forces full entropy at a finer scale, again producing a reservoir. If no exceptional approximation occurs, the recursion creates more reservoirs than the multiscale budget permits.

Put $s = \dim v_\lambda < 1$, and let $\varepsilon > 0$ be the tolerance in the theorem. Choose $\delta > 0$ so small that

$$s + 4\delta < \min(1, s + \varepsilon). \quad (\text{II.9.N46})$$

Write $\chi = \log(1/\lambda)$. [L-II.9.05](#) supplies $\alpha > 0$ with

$$\mathcal{H}(v_\lambda^I; r \mid 2r) \leq 1 - \alpha \quad (\text{II.9.N47})$$

for every digit window I and every $r > 0$. Entropy dimension and convolution monotonicity also give, for every $I \subset (0, 1]$ and all sufficiently small r ,

$$\mathcal{H}(v_\lambda^I; r) \leq (s + \delta) \log(1/r). \quad (\text{II.9.N48})$$

Finally [T-II.6.08](#), used with the fixed factor 10, gives

$$\mathcal{H}(v_\lambda^{(n)}; \lambda^{10n} \mid \lambda^n) < \delta \chi n \quad (\text{II.9.N49})$$

for all sufficiently large n .

Let $c_0 > 0$ be no larger than the constants in [P-II.9.08](#) and [P-II.9.09](#) on a fixed compact parameter interval containing λ . We may decrease it so that $c_0 < 1$.

We prove the theorem by contradiction on a long finite range. Suppose that, for arbitrarily large n_0 , no d in

$$[n_0, \exp^{(5)}(\log^{(5)} n_0 + A)] \quad (\text{II.9.N50})$$

admits $\eta \in E_{d, s+4\delta}$ with

$$|\lambda - \eta| \leq \exp\{-\exp(\log d \log^{(3)} d)\} = \exp\{-d^{\log^{(3)} d}\}. \quad (\text{II.9.N51})$$

The constant A will be fixed below, independently of n_0 .

Starting with this n_0 , construct $n_1, n_2, \dots$. Given n_j , put

$$m = \left\lceil \frac{4n_j \log n_j}{c_0 \chi} \right\rceil. \quad (\text{II.9.N52})$$

There are two cases.

If

$$\mathcal{H}(v_\lambda^{(m)}; m^{-4m/c_0}) \geq m\chi(s + 3\delta), \quad (\text{II.9.N53})$$

then [\(II.9.N48\)](#)–[\(II.9.N49\)](#), applied at m , give

$$\mathcal{H}(v_\lambda^{(m)}; m^{-4m/c_0} \mid \lambda^{10m}) \geq \delta \chi m. \quad (\text{II.9.N54})$$

Set $n_{j+1} = m$.

If [\(II.9.N53\)](#) fails, [P-II.9.08](#) gives a root $\eta \in \mathcal{L}_m$ with

$$|\lambda - \eta| < m^{-4m}, \quad h_\eta < \chi(s + 3\delta). \quad (\text{II.9.N55})$$

The conjugate $\bar{\eta}$ is also in $\mathcal{L}_m$, and both are within m^{-4m} of the real number λ . [L-II.9.07](#) therefore forces $\eta = \bar{\eta}$, so η is real. For large m , it lies in $(1/2, 1)$, and

$$h_\eta < (s + 4\delta) \log(1/\eta).$$

T-II.9.03 gives $\dim v_\eta < s + 4\delta$; hence $\eta \in E_{m,s+4\delta}$.

If $\eta = \lambda$, **(II.9.N51)** holds and contradicts the contrary assumption. Hence $\eta \neq \lambda$. Let n_{j+1} be the largest integer n for which $|\lambda - \eta| < n^{-4n}$. Then $n_{j+1} \geq m$ and

$$|\lambda - \eta| \geq (n_{j+1} + 1)^{-4(n_{j+1}+1)}. \quad (\text{II.9.N56})$$

P-II.9.09, with the scale $(n_{j+1} + 1)^{-4(n_{j+1}+1)/c_0}$, says that the prefix law has maximal entropy $n_{j+1} \log 2$ there. **L-II.9.02** moves to $n_{j+1}^{-4n_{j+1}/c_0}$ at a cost $O(\log n_{j+1})$. Subtracting **(II.9.N48)**–**(II.9.N49)** at the coarser scale $\lambda^{10n_{j+1}}$ yields, for all large n_0 ,

$$\mathcal{H}(v_\lambda^{(n_{j+1})}; n_{j+1}^{-4n_{j+1}/c_0} \mid \lambda^{10n_{j+1}}) \geq \frac{1}{2}(\log 2 - \chi)n_{j+1}. \quad (\text{II.9.N57})$$

In either case the current stage produces an entropy reservoir at n_{j+1} , while **(II.9.N52)** gives the required growth from n_j to n_{j+1} . Consequently, for every $j \geq 1$, the preceding stage supplies the reservoir at n_j . With

$$K_j = \frac{4 \log n_j}{c_0 \chi}, \quad \gamma = \frac{1}{2 \log 2} \min\{\delta \chi, \log 2 - \chi\} > 0, \quad (\text{II.9.N58})$$

we therefore have

$$n_{j+1} \geq K_j n_j, \quad \mathcal{H}(v_\lambda^{(n_j)}; \lambda^{K_j n_j} \mid \lambda^{10n_j}) \geq \gamma n_j. \quad (\text{II.9.N59})$$

Set $\alpha' = \min\{\alpha, \gamma\} > 0$. The one-digit deficit **(II.9.N47)** remains valid with α' , and the reservoir in **(II.9.N59)** is at least $\alpha' n_j$, matching the *same* parameter in **P-II.9.11**. Its constant is now $C(\lambda, \alpha')$. Also $K_j \geq 11$ and $n_j \geq C(\lambda, \alpha')(\log K_j)^2$ once n_0 is large. Hence that proposition applies to the sequence beginning with n_1 .

We now bound how fast the recursion can grow under the contrary assumption. In the second case, **(II.9.N51)**, **(II.9.N55)**, and maximality in **(II.9.N56)** imply

$$\exp(n_{j+1}) < n_{j+1}^{4n_{j+1}} < |\lambda - \eta|^{-1} < \exp^{(2)}(\log m \log^{(3)} m).$$

Since $m \leq C_\lambda n_j \log n_j$, both cases obey

$$n_{j+1} < \exp(2 \log n_j \log^{(3)} n_j) \quad (\text{II.9.N60})$$

for large n_0 . If $j_0 = \log^{(2)} n_0$, induction in **(II.9.N60)** gives

$$n_j < \exp^{(2)}((2j + j_0) \log^{(2)}(2j + j_0)). \quad (\text{II.9.N61})$$

Indeed, take three logarithms in **(II.9.N60)**, use $\log^{(3)} n_j < 2 \log(2j + j_0)$, and then exponentiate back. Consequently

$$\frac{1}{\log K_j \log \log K_j} \geq \frac{1}{2(2j + j_0) \log(2j + j_0) \log^{(2)}(2j + j_0)}. \quad (\text{II.9.N62})$$

Summing and comparing with the integral of $[x \log x \log^{(2)} x]^{-1}$ gives

$$\sum_{j=1}^N \frac{1}{\log K_j \log \log K_j} \geq c(\log^{(3)}(N + j_0) - \log^{(3)} j_0). \quad (\text{II.9.N63})$$

Let C be the constant in P-II.9.11 and choose

$$N = \left\lfloor \exp^{(3)}(\log^{(3)} j_0 + 2C/c) \right\rfloor.$$

Then the left side of (II.9.N63) is at least $2C$. On the other hand, (II.9.N61) implies

$$\sum_{j=1}^N \log K_j \leq 10N^3, \quad N \leq \log n_0$$

for large n_0 . Since $n_1 > n_0$, P-II.9.11 bounds the same sum by

$$C \left(1 + \frac{10N^3}{n_1} \right) < 2C,$$

a contradiction.

It remains only to justify that every use of (II.9.N51) stayed inside (II.9.N50). Applying (II.9.N61) at the displayed value of N , and successively taking five logarithms, gives

$$n_N \leq \exp^{(5)}(\log^{(5)} n_0 + A)$$

provided $A \geq 2C/c+3$. This verifies the range condition and completes the contradiction. Therefore arbitrarily large d satisfy (II.9.N51) with an element of $E_{d,s+4\delta}$. By (II.9.N46), that set is contained in $E_{d,s+\varepsilon}$, proving the theorem. $\square$

Corollary II.9.6.4 (C-II.9.13 — Superquadratic exceptional approximation). *Under the hypotheses of T-II.9.12 there are arbitrarily large d and algebraic η with*

$$\deg \eta \leq d, \quad \dim v_\eta < 1, \quad |\lambda - \eta| \leq e^{-d^2}. \quad (\text{II.9.13})$$

Proof. For large d , $\log^{(3)} d > 2$, so

$$\exp(-d^{\log^{(3)} d}) \leq e^{-d^2}.$$

Take the arbitrarily large approximants supplied by T-II.9.12 with $\varepsilon < (1-s)/2$. They have $\deg \eta \leq d$ and $\dim v_\eta < s + \varepsilon < 1$, giving (II.9.13). $\square$

We have obtained approximants far closer than ordinary exponential Diophantine approximation. Their degrees are controlled, but their arithmetic complexity is not; the next task is to bound their Mahler measures uniformly.

II.9.7 Repulsion near an algebraic center

Definition II.9.7.1 (II.9.12: Algebraic annulus at degree scale). For $M \geq 1$, $n \geq 1$, and an algebraic center ξ , the (M, n) -annulus is

$$(5M)^{-n} \leq |x - \xi| \leq (5M)^{-n+1}. \quad (\text{II.9.D05})$$

Lemma II.9.7.2 (L-II.9.14 — Algebraic small-polynomial value bound). *If ξ is algebraic of degree d , $P \in \mathcal{P}_n$, and $P(\xi) \neq 0$, then*

$$|P(\xi)| \geq M(\xi)^{-n}(n+1)^{1-d}. \quad (\text{II.9.14})$$

Proof. Let

$$Q(X) = a_d \prod_{j=1}^d (X - \xi_j) \in \mathbb{Z}[X]$$

be the primitive minimal polynomial of $\xi = \xi_1$, with $a_d > 0$, and let $m = \deg P \leq n$. The resultant identity gives

$$a_d^m \prod_{j=1}^d P(\xi_j) = \text{Res}(Q, P) \in \mathbb{Z}.$$

It is nonzero because $P(\xi) \neq 0$, so its absolute value is at least one. For every j ,

$$|P(\xi_j)| \leq (n+1) \max(1, |\xi_j|)^n.$$

Consequently

$$\begin{aligned} |P(\xi)| &\geq a_d^{-m}(n+1)^{1-d} \prod_{j=2}^d \max(1, |\xi_j|)^{-n} \\ &\geq (n+1)^{1-d} \left(a_d \prod_{j=1}^d \max(1, |\xi_j|) \right)^{-n} \\ &= (n+1)^{1-d} M(\xi)^{-n}. \end{aligned}$$

This is (II.9.14). □

Proposition II.9.7.3 (P-II.9.15 — Transversality isolates a bounded-Mahler center). *There is an absolute d_0 such that the following holds. Let $\xi \in (1/2, 2^{-1/2})$ be algebraic with $\deg \xi \leq d$, $M(\xi) \leq M$, and let $n > 10d \log d$. Assume $d \geq d_0$. If $P \neq 0$ has degree at most n , coefficients in $\{-1, 0, 1\}$, and, when $\xi > 2^{-2/3}$, its coefficient of X^j vanishes for*

every $j \equiv 2 \pmod{3}$, then

$$|P(x)| > (20M)^{-n} \quad (\text{II.9.15})$$

for every real x in the annulus (II.9.D05).

Proof. Assume, toward a contradiction, that x lies in the annulus and $|P(x)| \leq (20M)^{-n}$. Remove the initial zero coefficients from P , divide by the corresponding power of X , and change its sign so the constant coefficient is 1. Since $x > 1/2$, the new bound is $|P(x)| \leq (10M)^{-n}$. The normalized polynomial belongs to the full transversality class of L-II.8.13; in the thinned case it belongs to one of the two classes of L-II.8.14.

Let $\delta > 0$ be the common transversality constant on the relevant interval. Uniform convergence of the coefficient-bounded power series and their derivatives on compact subintervals of $(0, 1)$ shows that, after halving δ , transversality persists on a fixed right-hand neighborhood of each stated endpoint. Since the annular radius tends to zero super-exponentially in d , both ξ and x lie in this slightly enlarged interval. We first show $P(\xi) \neq 0$. Otherwise, for large n , both $P(\xi)$ and $P(x)$ are below δ . Transversality gives $P'(t) < -\delta$ throughout the interval between them, and the mean value theorem yields

$$|P(x) - P(\xi)| > \delta|x - \xi| \geq \delta(5M)^{-n} > (10M)^{-n},$$

a contradiction.

Therefore L-II.9.14 applies:

$$|P(\xi)| \geq (n+1)^{1-d}M^{-n} \geq (2M)^{-n}. \quad (\text{II.9.N64})$$

The last inequality follows from $n > 10d \log d$, after increasing the absolute threshold d_0 . On $[0, 1]$,

$$|P'(t)| \leq \sum_{j=1}^n j \leq n^2.$$

Hence the annular upper bound gives

$$|P(\xi)| \leq |P(x)| + n^2|x - \xi| \leq (10M)^{-n} + n^2(5M)^{-n+1} < (2M)^{-n}$$

for large d , contradicting (II.9.N64). This proves (II.9.15). □

Once an approximant has bounded Mahler measure, its surrounding degree-scale annulus cannot contain another excessively small full or thinned polynomial value. Proposition P-II.9.15 is the final contradiction tool.

II.9.8 Mahler measure as an entropy barrier

Definition II.9.8.1 (II.9.06: Algebraic unit and contracting conjugate product). An algebraic number is a unit if it and its inverse are algebraic integers. For algebraic λ , put

$$\tilde{M}(\lambda) = \prod_{\lambda_j: |\lambda_j| < 1} |\lambda_j|^{-1},$$

where complex conjugates are counted separately. If λ is a unit, none of its conjugates is zero.

Lemma II.9.8.2 (L-II.9.15 — Mahler measure of an algebraic unit). *If λ is an algebraic unit, then*

$$\tilde{M}(\lambda) = M(\lambda).$$

Proof. Let $\lambda_1, \dots, \lambda_d$ be the conjugates of λ . Because λ is an algebraic integer, its minimal polynomial is monic with integer coefficients by L-II.8.07. Because λ^{-1} is also an algebraic integer, the rational integer $N(\lambda)$ has an integer inverse, namely $N(\lambda^{-1})$. Consequently

$$\prod_{j=1}^d |\lambda_j| = |N(\lambda)| = 1.$$

The leading coefficient in the Mahler-measure definition is therefore one, and conjugates on the unit circle contribute one. Splitting the last product between conjugates inside and outside the unit circle gives

$$M(\lambda) = \prod_{|\lambda_j| > 1} |\lambda_j| = \left(\prod_{|\lambda_j| < 1} |\lambda_j| \right)^{-1} = \tilde{M}(\lambda).$$

□

Definition II.9.8.3 (II.9.07: Gaussian entropy between covariance scales). Let G be a standard Gaussian in $\mathbb{R}^m$. For a bounded random vector X and invertible real $m \times m$ matrices B, B_1, B_2 , put

$$\mathbf{H}(X; B) = h_{\text{diff}}(X + BG) - h_{\text{diff}}(BG), \quad \mathbf{H}(X; B_1 \mid B_2) = \mathbf{H}(X; B_1) - \mathbf{H}(X; B_2).$$

Write $B_1 \preccurlyeq B_2$ if $B_2 B_2^\top - B_1 B_1^\top$ is positive semidefinite.

Definition II.9.8.4 (II.9.08: The one-dimensional Mahler entropy function). Let ε be a fair sign and let G be a standard real Gaussian independent of ε . For $a \geq 1$, define

$$\Phi(a) = \sup_{t>0} (h_{\text{diff}}(ta\varepsilon + G) - h_{\text{diff}}(t\varepsilon + G)). \quad (\text{II.9.D04})$$

Contracting Galois conjugates provide a finite-dimensional linear model for the Bernoulli walk. Gaussian smoothing turns determinant contraction into an entropy difference. Optimizing that difference gives $\Phi(M(\eta))$, which approaches one full binary digit as $M(\eta) \rightarrow \infty$. Therefore algebraic parameters with dimension uniformly below one must have uniformly bounded Mahler measure.

Lemma II.9.8.5 (L-II.9.16 — Gaussian entropy comparison). *If X, Z are independent bounded random vectors, then*

$$\mathbf{H}(X + Z; B) \leq \mathbf{H}(X; B) + \mathbf{H}(Z; B).$$

If $B_1 \preccurlyeq B_2$, then

$$\mathbf{H}(X; B_1) \geq \mathbf{H}(X; B_2)$$

and

$$\mathbf{H}(X + Z; B_1 \mid B_2) \geq \mathbf{H}(X; B_1 \mid B_2).$$

Proof. We use the differential-entropy submodularity inequality (II.9.N03c), established in the proof of L-II.9.02:

$$h_{\text{diff}}(X + Y + Z) + h_{\text{diff}}(Y) \leq h_{\text{diff}}(X + Y) + h_{\text{diff}}(Y + Z) \quad (\text{II.9.N65})$$

for independent random vectors whenever the displayed entropies are finite. Gaussian regularization and truncation extend the inequality from bounded densities to the present bounded random vectors.

Taking $Y = BG$ in (II.9.N65) gives, for independent X, Z ,

$$\mathbf{H}(X + Z; B) \leq \mathbf{H}(X; B) + \mathbf{H}(Z; B). \quad (\text{II.9.N66})$$

If $B_1 \preccurlyeq B_2$, choose C with $CC^\top = B_2B_2^\top - B_1B_1^\top$. Then B_2G has the law of $B_1G_1 + CG_2$ for independent standard Gaussians. Applying (II.9.N65) first with X, B_1G_1, CG_2 shows

$$\mathbf{H}(X; B_1) \geq \mathbf{H}(X; B_2). \quad (\text{II.9.N67})$$

Apply (II.9.N65) once more to the independent triple $Z, X + B_1G_1, CG_2$. After cancelling the Gaussian entropies, it gives

$$\mathbf{H}(X + Z; B_1 \mid B_2) \geq \mathbf{H}(X; B_1 \mid B_2). \quad (\text{II.9.N68})$$

Equations (II.9.N66)–(II.9.N68) are precisely the asserted subadditivity, covariance monotonicity, and monotonicity of the between-covariance entropy under independent addition. $\square$

Embed the scalar Bernoulli walk simultaneously along the contracting conjugates of η . After a change of norm, the associated matrix has all but one singular value equal to one, while the remaining singular value records the contracting-conjugate product. Gaussian smoothing converts this determinant defect into the one-dimensional entropy gain Φ .

Theorem II.9.8.6 (T-II.9.17 – Mahler measure controls Garsia entropy). *For every algebraic $\eta \in (0, 1)$,*

$$h_\eta \geq \Phi(M(\eta)). \quad (\text{II.9.16})$$

Moreover Φ is nondecreasing, $\Phi(a^2) \leq 2\Phi(a)$,

$$\lim_{a \rightarrow \infty} \Phi(a) = \log 2, \quad (\text{II.9.17})$$

and there is an absolute $c > 0$ such that $h_\eta \geq c \min\{\log M(\eta), \log 2\}$.

Proof. The proof builds the finite-dimensional conjugate walk, compares its entropy before and after Gaussian smoothing, reads the determinant contraction, and optimizes the resulting one-dimensional inequality.

We first prove the Gaussian lower bound for an algebraic unit η . Let $\eta_1, \dots, \eta_d$ be all conjugates of η inside the open unit disk, counted with complex conjugates separately. Use real 2×2 blocks for nonreal pairs, and let A_0 be a real diagonalizable matrix with precisely these eigenvalues.

There is a real matrix A , similar to A_0 , with operator norm one and singular values

$$1, \dots, 1, \tilde{M}(\eta)^{-1}. \quad (\text{II.9.N69})$$

Here is the construction. Choose v such that the covectors $(A_0^T)^n v$, $n \geq 0$, span $\mathbb{R}^d$; such v exists because A_0 has distinct nonzero eigenvalues. The quadratic form

$$q(x) = \sum_{n \geq 0} \langle A_0^n x, v \rangle^2$$

is positive definite and convergent. Write $q(x) = \|Bx\|^2$. Then

$$\|BA_0 B^{-1}y\|^2 = \|y\|^2 - \langle B^{-1}y, v \rangle^2.$$

Thus $A = BA_0 B^{-1}$ has $d - 1$ singular values equal to one and the last equal to $|\det A| = \prod_{j=1}^d |\eta_j| = \tilde{M}(\eta)^{-1}$. This proves (II.9.N69).

Let ξ_j be fair signs and, for $x \in \mathbb{R}^d$, put

$$X_{A,x} = \sum_{j=0}^{\infty} \xi_j A^j x, \quad X_{A,x}^{(\ell)} = \sum_{j=0}^{\ell-1} \xi_j A^j x.$$

The scalar prefix sum determines the vector prefix sum: if two sign polynomials agree at η , their difference is divisible by the minimal polynomial and therefore vanishes at A . Hence

$$H(v_\eta^{(\ell)}) \geq H(X_{A,x}^{(\ell)}). \quad (\text{II.9.N70})$$

The self-similarity $X_{A,x} = X_{A,x}^{(\ell)} + A^\ell X'_{A,x}$, L-II.9.16, and entropy of a discrete mixture give

$$\mathbf{H}(X_{A,x}; A^\ell \mid I) \leq H(X_{A,x}^{(\ell)}). \quad (\text{II.9.N71})$$

On the other hand,

$$\mathbf{H}(X_{A,x}; A^\ell \mid I) = \sum_{i=1}^{\ell} \mathbf{H}(X_{A,x}; A^i \mid A^{i-1}).$$

Since $\|A\| \leq 1$, $A^i \preceq A^{i-1}$. Apply (II.9.N68) to the first $i-1$ digits and then scale by A^{i-1} ; every summand is at least $\mathbf{H}(X_{A,x}; A \mid I)$. Divide (II.9.N70)–(II.9.N71) by ℓ and let $\ell \rightarrow \infty$:

$$h_\eta \geq \mathbf{H}(X_{A,x}; A \mid I) \geq \mathbf{H}(\xi_0 x; A \mid I). \quad (\text{II.9.N72})$$

The second inequality is (II.9.N68), because the remaining tail is independent.

Orthogonal changes on the left and right of A preserve the Gaussian law and differential entropy. By (II.9.N69), (II.9.N72) may therefore be evaluated with

$$D = \text{diag}(1, \dots, 1, \tilde{M}(\eta)^{-1})$$

and $x = te_d$. Integrating out the first $d-1$ Gaussian coordinates and scaling the last gives

$$\mathbf{H}(\xi_0 te_d; D \mid I) = h_{\text{diff}}(t\tilde{M}(\eta)\xi_0 + G) - h_{\text{diff}}(t\xi_0 + G).$$

Taking the supremum over $t > 0$ proves

$$h_\eta \geq \Phi(\tilde{M}(\eta)). \quad (\text{II.9.N73})$$

For a unit, the product formula gives $\tilde{M}(\eta) = M(\eta)$.

Now let $\eta \in (0, 1)$ be arbitrary algebraic. If no nonzero polynomial with coefficients in $\{-1, 0, 1\}$ vanishes at η , all 2^n prefix sums are distinct, so $h_\eta = \log 2$, and the desired estimate follows from $\Phi \leq \log 2$. Otherwise, remove the initial and terminal zero coefficients from such a relation. Its constant and leading coefficients are both ± 1 . Gauss's lemma then makes the primitive minimal polynomial of η monic with constant coefficient ± 1 ; hence η is a unit. Equation (II.9.N73) applies and proves (II.9.16) in all cases.

It remains to establish the assertions about Φ . The change of variables formula gives

$$h_{\text{diff}}(ta\varepsilon + G) - h_{\text{diff}}(t\varepsilon + G) = \mathbf{H}(t\varepsilon; a^{-1} \mid 1).$$

Covariance monotonicity (II.9.N67) shows that this is nondecreasing in a , hence so is Φ . For any $a > 1$, split the difference from t to ta^2 at ta ; taking the supremum gives

$$\Phi(a^2) \leq 2\Phi(a). \quad (\text{II.9.N74})$$

Also $\Phi(a) \leq H(\varepsilon) = \log 2$, since the entropy increase caused by revealing an independent binary variable is at most its Shannon entropy. With $t = a^{-1/2}$,

$$\Phi(a) \geq h_{\text{diff}}(a^{1/2}\varepsilon + G) - h_{\text{diff}}(a^{-1/2}\varepsilon + G).$$

The second term tends to $h_{\text{diff}}(G)$. The first tends to $h_{\text{diff}}(G) + \log 2$, as follows from the binary Fano bound. We record the bound explicitly. If B is a fair sign, W is an observation, and $\hat{B}(W)$ is an estimator with error probability p , then

$$I(B; W) \geq \log 2 - h_{\text{bin}}(p), \quad h_{\text{bin}}(p) = -p \log p - (1-p) \log(1-p). \quad (\text{II.9.N74a})$$

Indeed, if $E = \mathbf{1}_{\{B \neq \hat{B}(W)\}}$, then B is determined by (E, W) . Hence

$$H(B | W) \leq H(E | W) \leq H(E) = h_{\text{bin}}(p),$$

which proves (II.9.N74a). Apply this with $W = a^{1/2}\varepsilon + G$ and $\hat{e}(W) = \text{sgn } W$. Its error probability is $p_a = \mathbb{P}(G > a^{1/2}) \rightarrow 0$, while translation invariance gives

$$h_{\text{diff}}(W) = h_{\text{diff}}(G) + I(\varepsilon; W).$$

Thus $h_{\text{diff}}(W) \rightarrow h_{\text{diff}}(G) + \log 2$, and therefore $\Phi(a) \rightarrow \log 2$.

Finally $\Phi(a) > 0$ for $a > 1$. Indeed, differentiating the mutual information $I(\varepsilon; t\varepsilon + G)$ and integrating by parts gives

$$\frac{d}{dt} I(\varepsilon; t\varepsilon + G) = t \mathbb{E}[(\varepsilon - \mathbb{E}[\varepsilon | t\varepsilon + G])^2] > 0. \quad (\text{II.9.N75})$$

Continuity and compactness make

$$c_0 = \min_{\sqrt{2} \leq a \leq 2} \frac{\Phi(a)}{\log a} > 0.$$

For $1 < a < 2$, choose k with $\sqrt{2} \leq a^{2^k} < 2$ and iterate (II.9.N74); for $a \geq 2$, use monotonicity. After reducing c_0 if necessary,

$$\Phi(a) \geq c_0 \min\{\log a, \log 2\}.$$

Together with (II.9.16), this completes the theorem. □

Corollary II.9.8.7 (C-II.9.18 — Exceptional algebraic approximants have bounded Mahler measure). *Fix $\lambda \in (1/2, 1)$ and $s < 1$. There are a neighborhood U of λ and $M_0 < \infty$ such that every algebraic $\eta \in U$ with $\dim v_\eta \leq s$ satisfies $M(\eta) \leq M_0$.*

Proof. Choose a neighborhood U of λ so small that

$$s \sup_{\eta \in U} \log(1/\eta) < \log 2. \quad (\text{II.9.N76})$$

If $\eta \in U$ is algebraic and $\dim v_\eta \leq s$, T-II.9.03 gives

$$h_\eta \leq s \log(1/\eta) \leq s \sup_{\zeta \in U} \log(1/\zeta) =: h_0 < \log 2.$$

But T-II.9.17 says $h_\eta \geq \Phi(M(\eta))$, and $\Phi(a) \rightarrow \log 2$. Choose M_0 with $\Phi(a) > h_0$ for $a > M_0$. Then necessarily $M(\eta) \leq M_0$. □

All exceptional approximants near the fixed parameter now have a uniform Mahler bound. This supplies the height control missing after the exceptional-approximation theorem.

II.9.9 Thinning and the final contradiction

Definition II.9.9.1 (II.9.11: Trimmed block system). The thinned law $\tilde{v}_\lambda$ is D-II.8.12. Grouping digits in blocks of three makes it the homogeneous four-map self-similar probability with ratio λ^3 , equal weights, and translations $\{\pm 1 \pm \lambda\}$. Its level- m cylinder differences are polynomials with coefficients in $\{-2, 0, 2\}$ and zero coefficients at degrees congruent to 2 (mod 3).

Lemma II.9.9.2 (L-II.9.19 — Dimension monotonicity under splitting and trimming). For every $k \geq 1$,

$$\dim v_{\lambda^k} \leq \dim v_\lambda, \quad \dim \tilde{v}_\lambda \leq \dim v_\lambda. \quad (\text{II.9.18})$$

Proof. If $\mu = \theta * \kappa$ and E is Borel with $\mu(E) > 0$, then

$$\mu(E) = \int \theta(E - y) d\kappa(y) > 0.$$

Thus $\theta(E - y) > 0$ for some y , and $\dim_{\text{H}} E = \dim_{\text{H}}(E - y) \geq \dim \theta$. Taking the infimum over positive- μ -measure sets yields

$$\dim(\theta * \kappa) \geq \dim \theta. \quad (\text{II.9.N77})$$

The digits whose indices are divisible by k have law v_{λ^k} , and all remaining digits form an independent convolution factor. Likewise $\tilde{v}_\lambda$ is the factor formed by retaining the indices not congruent to 2 (mod 3). Apply (II.9.N77) to these two splittings to obtain (II.9.18). □

Lemma II.9.9.3 (L-II.9.20 — Trimmed small-polynomial approximation at every scale). If $\lambda \in [2^{-2/3}, 1)$ and $\dim \tilde{v}_\lambda < 1$, then for every $0 < \theta < 1$ and every sufficiently large n , there is a nonzero $P \in \mathcal{P}_n$ whose coefficient of X^j vanishes for $j \equiv 2 \pmod{3}$, and

$$|P(\lambda)| < \theta^n. \quad (\text{II.9.19})$$

Proof. Group the retained signs into blocks of three. By D-II.9.11, $\tilde{\nu}_\lambda$ is the equal-weight four-map homogeneous self-similar measure of contraction λ^3 . Its symbolic entropy is $2 \log 2$, while its Lyapunov exponent is $3 \log(1/\lambda) \leq 2 \log 2$ for $\lambda \geq 2^{-2/3}$. Its natural dimension is therefore at least one.

Because the actual dimension is below one, T-II.6.11 gives super-exponential cylinder concentration at every sufficiently large block length m : for every $A > 0$, two distinct length- m cylinders have translation difference below e^{-Am} . Dividing that difference by 2 produces a nonzero polynomial P_m with coefficients in $\{-1, 0, 1\}$, degree at most $3m - 1$, and zero coefficient in every degree $2 \pmod{3}$, such that

$$|P_m(\lambda)| < e^{-Am}. \quad (\text{II.9.N78})$$

Given a sufficiently large integer n , take $m = \lfloor (n+1)/3 \rfloor$; then $3m-1 \leq n$ and $m \geq (n-2)/3$. Choose $A > 4 \log(1/\theta)$. For all large n , (II.9.N78) is below θ^n . This proves (II.9.19) at every sufficiently large degree scale. □

All components of the contradiction are now available. Exceptional approximation gives algebraic centers of growing degree; the Mahler argument bounds their height; dimension loss for the full or thinned system supplies a tiny polynomial at the annular degree scale; and annular isolation forbids that value. The split at $2^{-2/3}$ merely decides which transversality class is used.

Theorem II.9.9.4 (T-II.9.21 — Varjú's transcendental full-dimension theorem). *For every transcendental $\lambda \in (1/2, 1)$,*

$$\dim \nu_\lambda = 1. \quad (\text{II.9.20})$$

Proof. Suppose, toward a contradiction, that a transcendental $\lambda \in (1/2, 1)$ satisfies $s = \dim \nu_\lambda < 1$. By L-II.9.19, $\dim \nu_{\lambda^k} \leq s$. If necessary choose k so that $\lambda^k \in (1/2, 2^{-1/2})$. A positive integral power of a transcendental number is transcendental, so after this replacement we may assume

$$\lambda \in (1/2, 2^{-1/2}). \quad (\text{II.9.N79})$$

Apply T-II.9.12 with $\varepsilon = (1-s)/2$. There are arbitrarily large d and algebraic ξ such that

$$\deg \xi \leq d, \quad \dim \nu_\xi < (1+s)/2, \quad |\lambda - \xi| \leq \exp(-d^{\log^{(3)} d}). \quad (\text{II.9.N80})$$

For large d , ξ lies in a fixed neighborhood of λ . C-II.9.18 supplies a constant M , independent of d , with $M(\xi) \leq M$. Since λ is transcendental, $\lambda \neq \xi$. Choose n by

$$(5M)^{-n} \leq |\lambda - \xi| \leq (5M)^{-n+1}. \quad (\text{II.9.N81})$$

Equation (II.9.N80) implies $n \gg d^{\log^{(3)} d}$, so $n > 10d \log d$ for large d .

If $\lambda < 2^{-2/3}$, then also $\xi \leq 2^{-2/3}$ for all large d . C-II.6.22, applied with $\theta = (20M)^{-1}$, supplies at every sufficiently large n a nonzero $P \in \mathcal{P}_n$ with

$$|P(\lambda)| < (20M)^{-n}. \quad (\text{II.9.N82})$$

If $\lambda > 2^{-2/3}$, L-II.9.19 gives $\dim \tilde{v}_\lambda < 1$. L-II.9.20 supplies (II.9.N82) with P in the thinned polynomial class. The equality $\lambda = 2^{-2/3}$ is impossible because that number is algebraic.

In either case, $\xi \in (1/2, 2^{-1/2})$ for large d , and (II.9.N81)–(II.9.N82) contradict P-II.9.15. The assumption $s < 1$ was false, proving $\dim v_\lambda = 1$. □

The proof can be read as the following pipeline:

$$\begin{aligned} &\text{exceptional transcendental parameter} \xrightarrow{\text{T-II.9.12}} \text{algebraic approximants} \\ &\xrightarrow{\text{C-II.9.18}} \text{bounded Mahler measure} \longrightarrow \text{degree-scale annulus} \\ &\longrightarrow \text{tiny full or thinned polynomial} \xrightarrow{\text{P-II.9.15}} \perp. \end{aligned}$$

II.9.10 Consequences for exceptional sets and affine growth

Definition II.9.10.1 (II.9.13: Affine Bernoulli semigroup and growth entropy). For algebraic $\lambda \in (0, 1)$, let

$$S_\lambda = \{x \mapsto \lambda x + 1, x \mapsto \lambda x - 1\}.$$

Let S_λ^n be the set of distinct length- n compositions and define

$$\rho_\lambda = \inf_{n \geq 1} \frac{1}{n} \log |S_\lambda^n|.$$

The equality of this infimum with the asymptotic growth limit, and its relation to h_λ , are proved below.

Corollary II.9.10.2 (C-II.9.22 – Arithmetic exceptional set and affine growth). *The set $\{\lambda \in (1/2, 1) : \dim v_\lambda < 1\}$ is countable and consists entirely of algebraic numbers. For algebraic $\lambda \in (0, 1)$, the limit in D-II.9.13 exists and*

$$\rho_\lambda \geq h_\lambda \geq c \min\{\log M(\lambda), \log 2\}. \quad (\text{II.9.21})$$

Proof. T-II.9.21 says every member of $\{\lambda \in (1/2, 1) : \dim v_\lambda < 1\}$ is algebraic. The algebraic numbers are countable, proving the first assertion.

For the growth statement, a length- n word in the two affine generators has the form

$$x \mapsto \lambda^n x + \sum_{j=0}^{n-1} \varepsilon_j \lambda^j. \quad (\text{II.9.N83})$$

Thus two words define the same affine map exactly when their prefix sums agree, and

$$|S_\lambda^n| = |\text{supp } \nu_\lambda^{(n)}|. \quad (\text{II.9.N84})$$

Composition gives $|S_\lambda^{n+m}| \leq |S_\lambda^n| |S_\lambda^m|$. Fekete's lemma therefore proves existence of ρ_λ . Entropy is at most the logarithm of support cardinality, so L-II.9.01 and (II.9.N84) give

$$h_\lambda \leq \rho_\lambda.$$

The lower bound $h_\lambda \geq c \min\{\log M(\lambda), \log 2\}$ is T-II.9.17. This proves (II.9.21). □

The theorem is a dimension statement, not an absolute-continuity theorem. It removes transcendental parameters from the dimension-drop exceptional set and yields an arithmetic growth bound, while stronger regularity questions require the Fourier-decay and convolution mechanisms developed later.

II.9.11 From dimension to arithmetic classification

The preceding entropy arguments determine dimension, but full dimension does not imply absolute continuity. We now address measure type. The results and proofs in the following sections are adapted from [Ope26a], with the normalization of Definition II.8.1 unchanged. The arithmetic criterion is an infinite approximation condition, not a finite algorithm deciding every real parameter. Its square-root character estimate is proved below for the exact binary polynomial modulus required by the argument. The quartic-Salem and degree-31 singularity arguments are separate direct proofs and do not use that estimate. All logarithms and entropies in these arithmetic-classification and singularity sections use the natural logarithm, unless explicitly stated otherwise. Thus the entropy of a fair bit is $\log 2$, not 1.

Definition II.9.11.1 (Algebraic terminology). A nonconstant polynomial over a field is *irreducible* if it is not a product of two nonconstant polynomials over that field. It is *monic* if its leading coefficient is 1. An algebraic number is a complex root of a nonzero polynomial in $\mathbb{Q}[t]$. Its minimal polynomial is the monic polynomial of least positive degree over $\mathbb{Q}$ vanishing at that number; it is irreducible. Its degree is the degree of this minimal polynomial over $\mathbb{Q}$; its conjugates are the roots of that polynomial. An algebraic integer is a root of a monic polynomial in $\mathbb{Z}[t]$. An algebraic unit is an algebraic integer whose inverse is also an algebraic integer. In particular its monic minimal polynomial has constant coefficient ± 1 . A Pisot number is a real algebraic integer $\beta > 1$ whose other conjugates have modulus less than 1. A Salem number is a real algebraic integer $\beta > 1$ whose conjugates other than β consist of β^{-1} and numbers of modulus 1, with at least one on the unit circle. A *number field* is a subfield

$K \subset \mathbb{C}$ containing $\mathbb{Q}$ and finite-dimensional as a vector space over $\mathbb{Q}$. An embedding $\sigma : K \rightarrow \mathbb{C}$ over $\mathbb{Q}$ is an injective map preserving addition, multiplication and 1, and fixing every rational number. For $z \in K$, the field trace and field norm are

$$\mathrm{Tr}_{K/\mathbb{Q}}(z) = \sum_{\sigma} \sigma(z), \quad \mathrm{N}_{K/\mathbb{Q}}(z) = \prod_{\sigma} \sigma(z),$$

with the sum and product over all such embeddings.

Definition II.9.11.2 (Polynomial characters). The field $\mathbb{F}_2 = \{0, 1\}$ has addition and multiplication modulo 2. Reduction modulo 2 of an integer polynomial reduces each coefficient. For $\ell \geq 1$, two polynomials represent the same residue class modulo t^ℓ when their difference is divisible by t^ℓ . The quotient $\mathbb{F}_2[t]/(t^\ell)$ has addition and multiplication of these classes. Its *units*, namely classes with a multiplicative inverse, are precisely those with constant coefficient 1. A Dirichlet character modulo t^ℓ is a map from this finite unit group to $\{z \in \mathbb{C} : |z| = 1\}$, with $\chi(ab) = \chi(a)\chi(b)$ and $\chi(1) = 1$, extended by zero on nonunits. It is *imprimitive* if its values on units factor through reduction modulo t^m for some $1 \leq m < \ell$. The principal character takes value 1 on every unit. For a character χ , set

$$L(u, \chi) = \sum_{f \text{ monic}} \chi(f) u^{\deg f} = \prod_{P \text{ monic irreducible}} (1 - \chi(P) u^{\deg P})^{-1};$$

The equality is initially an identity of formal power series: coefficients are compared at each degree, with the usual finite coefficient convolutions for multiplication, without an analytic convergence assertion. If this is a polynomial, its inverse roots are the α_j in $L(u, \chi) = \prod_j (1 - \alpha_j u)$. The factor $1 - u$, when present, is called a trivial factor here; its inverse root is 1.

II.9.12 An arithmetic criterion for the measure type

The criterion in this section is expressed through finite sets of algebraic units and the coefficient vectors of their minimal polynomials. Its proof has two parts. A sufficiently large cluster of coefficient vectors forces concentration of the Bernoulli convolution. Conversely, concentration produces many short polynomial prefixes, each of which can be completed to an irreducible polynomial with a root just above the parameter.

Definition II.9.12.1 (Retained algebraic-unit clusters). For integers $n \geq 1$ and $d \geq 2$, let $R(n, d)$ be the set of real algebraic units $r \in (1/2, 1)$ of degree dn whose minimal polynomial has all coefficients in $\{-1, 0, 1\}$ and is irreducible modulo 2. An algebraic unit here means an algebraic integer with algebraic-integer inverse. Write p_r for the minimal polynomial with its sign chosen so that its constant coefficient is -1 , and set

$$\pi_n(p_r) = ([t^0]p_r, \dots, [t^{n-1}]p_r) \in \{-1, 0, 1\}^n.$$

For $r \in R(n, d)$, define the finite set

$$W_n(r; d) = \{\pi_n(p_q) : q \in R(n, d), |q - r| \leq 4^{-n}\}. \quad (\text{II.9.12.1})$$

This is a set of vectors: repeated projections, including those arising from conjugate roots of one polynomial, are counted only once.

For a bit word $u = (u_0, \dots, u_{n-1}) \in \{0, 1\}^n$, let

$$\mathcal{O}_u = \{w \in \mathbb{R}^n : (2u_i - 1)w_i \geq 0 \text{ for } 0 \leq i < n\}$$

be the associated closed orthant. A vector with a zero coordinate belongs to both choices of sign in that coordinate. For $j \geq 1$, put

$$R_*(n, d, j) = \left\{ r \in R(n, d) : \#\{u \in \{0, 1\}^n : \#(W_n(r; d) \cap \mathcal{O}_u) \geq j(2r)^n\} \geq \frac{2^n}{d} \right\}. \quad (\text{II.9.12.2})$$

All comparisons in this definition involve algebraic numbers and finite sets. In particular, $R(n, d)$ and $R_*(n, d, j)$ can be obtained by finite enumeration of integer polynomials and exact algebraic comparisons.

Theorem II.9.12.2 (Arithmetic criterion, OpenAI). *The proof below includes the function-field estimate in Lemma II.9.13.13 and its irreducible-polynomial consequence; both are proved in this chapter. The criterion is from [Ope26a, Theorem 2.1]. Define the parameter set*

$$\mathcal{C} = (0, 1/2) \cup \left[(1/2, 1) \cap \bigcup_{d \geq 2} \bigcap_{j \geq 1} \limsup_{n \rightarrow \infty} \bigcup_{r \in R_*(n, d, j)} [r - 4^{-n}, r) \right]. \quad (\text{II.9.12.3})$$

For every $\lambda \in (0, 1)$, the measure ν_λ is singular if and only if $\lambda \in \mathcal{C}$, and is absolutely continuous if and only if $\lambda \notin \mathcal{C}$. Replacing $\limsup$ by $\liminf$ in (II.9.12.3) gives the same set.

Thus the degree multiplier d may be fixed before the crowding level j is chosen. For a singular parameter in $(1/2, 1)$, each level j is attained at every sufficiently large n , with the threshold in n allowed to depend on j . The predicate uses only polynomial arithmetic, one-sided approximation, and counts of coefficient vectors. Determining membership for a particular parameter can nevertheless be difficult; the theorem does not supply a finite decision procedure for arbitrary real inputs.

II.9.12.1 Measure-theoretic preliminaries

Use the independent fair bits $u_i = (\varepsilon_i + 1)/2$ obtained from the independent signs of Definition II.8.1, and set

$$X_\lambda = \sum_{i=0}^{\infty} u_i \lambda^i, \quad \mathbb{P}(u_i = 0) = \mathbb{P}(u_i = 1) = \frac{1}{2},$$

with law μ_λ , and write $L = (1 - \lambda)^{-1}$. The affine identity $Y_\lambda = 2X_\lambda - L$ shows that μ_λ and ν_λ have the same measure type. For a word $u \in \{0, 1\}^n$, put

$$s_u(\lambda) = \sum_{i=0}^{n-1} u_i \lambda^i. \quad (\text{II.9.12.4})$$

Conditioned on this prefix, X_λ lies in $s_u(\lambda) + \lambda^n[0, L]$.

Lemma II.9.12.3 (Pure type and the baseline parameters). *For $0 < \lambda < 1$, the measure μ_λ is either singular or absolutely continuous. It is singular for $\lambda < 1/2$ and uniform on $[0, 2]$ for $\lambda = 1/2$. Consequently, $\nu_{1/2}$ is uniform on $[-2, 2]$.*

Proof. Let $T_i(x) = i + \lambda x$ for $i = 0, 1$, and let $\mathcal{T}\rho = \frac{1}{2}(T_0)_*\rho + \frac{1}{2}(T_1)_*\rho$. Iteration from any probability measure ρ gives the law of

$$\sum_{i=0}^{n-1} u_i \lambda^i + \lambda^n Z,$$

where Z has law ρ and is independent of the bits. Since $\lambda^n Z \rightarrow 0$ in probability, these laws converge weakly to μ_λ . Hence μ_λ is the unique invariant probability of $\mathcal{T}$.

The operator $\mathcal{T}$ preserves absolute continuity and singularity. By uniqueness of the Lebesgue decomposition, the absolutely continuous and singular parts of its invariant probability are separately invariant. If both were nonzero, normalizing them would give two distinct invariant probabilities, a contradiction.

For $\lambda < 1/2$, the support is covered, for every n , by 2^n intervals of length $L\lambda^n$. Its Lebesgue measure is therefore zero. For $\lambda = 1/2$, the usual fair binary expansion gives the uniform law on $[0, 2]$. $\square$

We shall also use the following elementary formulation of concentration. It isolates the measure-theoretic input needed to establish membership in (II.9.12.3).

Lemma II.9.12.4 (Compact concentration). *A Borel probability measure ρ on $\mathbb{R}$ is singular if and only if, for every $\varepsilon > 0$, there is a compact set K with*

$$\text{Leb}(K) < \varepsilon, \quad \rho(K) > 1 - \varepsilon.$$

Proof. If ρ is singular, apply inner regularity to a Borel null set of full ρ -measure. Conversely, choose such compact sets K_m with $\varepsilon = 2^{-m}$. The set $\limsup_m K_m$ has Lebesgue measure zero, since $\sum_m \text{Leb}(K_m) < \infty$, and has ρ -measure one, since every union $\bigcup_{m \geq M} K_m$ has ρ -measure one. $\square$

II.9.12.2 Algebraic clusters imply singularity

Proof of the cluster-to-singularity implication. Suppose $\lambda \in (1/2, 1)$ satisfies the condition in (II.9.12.3), and fix a degree multiplier d witnessing it. For each j , there are arbitrarily large n and $r \in R_*(n, d, j)$ such that

$$r - 4^{-n} \leq \lambda < r.$$

If $w \in W_n(r; d)$, choose a polynomial p_q giving this projection. Its root q satisfies $|q - \lambda| \leq 2 \cdot 4^{-n}$. On any fixed compact subinterval of $(0, 1)$ containing λ in its interior,

$$|p'_q(t)| \leq \sum_{k \geq 1} k t^{k-1} = \frac{1}{(1-t)^2}.$$

The coefficients of index at least n contribute at most $L\lambda^n$ at λ . Thus there is a constant $B = B(\lambda)$, independent of n, d, j, w , such that for all sufficiently large n ,

$$\left| \sum_{i=0}^{n-1} w_i \lambda^i \right| \leq B\lambda^n. \quad (\text{II.9.12.5})$$

Here we used $4^{-n} = o(\lambda^n)$.

If $w \in \mathcal{O}_u \cap \{-1, 0, 1\}^n$, then $v = u - w$ belongs to $\{0, 1\}^n$. Different w give different v . By (II.9.12.2), at least $2^n/d$ words u consequently have at least $j(2r)^n \geq j(2\lambda)^n$ neighbors v satisfying

$$|s_u(\lambda) - s_v(\lambda)| \leq B\lambda^n.$$

Partition the line into half-open bins of length λ^n , and let $\mathcal{J}$ be the bins containing at least one of these words u , where a word occupies the bin containing its prefix sum. For each $J \in \mathcal{J}$, choose one such word. Its neighbors lie in the $B\lambda^n$ -enlargement of J . Each prefix sum can belong to the enlargements of at most $N_B = 2[B] + 3$ bins. Counting word-bin incidences yields

$$\#\mathcal{J} j(2\lambda)^n \leq N_B 2^n, \quad \#\mathcal{J} \leq \frac{N_B \lambda^{-n}}{j}.$$

Enlarge each bin $J = [a, a + \lambda^n)$ to $[a, a + (1 + L)\lambda^n]$, and denote their union by E_j . The tail bound following (II.9.12.4) gives

$$\text{Leb}(E_j) \leq \frac{(1 + L)N_B}{j}, \quad \mu_\lambda(E_j) \geq \frac{1}{d}. \quad (\text{II.9.12.6})$$

Absolute continuity of a finite measure makes its mass uniformly small on sets of sufficiently small Lebesgue measure. Therefore (II.9.12.6), as $j \rightarrow \infty$, rules out absolute continuity. Lemma II.9.12.3 proves singularity. □

II.9.13 Constructing the algebraic approximants

We now prove the converse in Theorem II.9.12.2. Throughout this section, $\lambda \in (1/2, 1)$ is fixed, and we abbreviate $\mu = \mu_\lambda$, $X = X_\lambda$, and $s_u = s_u(\lambda)$. The first step is to turn concentration of μ into many close prefix sums with opposite first bits. Opposite first bits ensure that their difference polynomial has constant coefficient -1 , as required by the algebraic criterion.

II.9.13.1 Overlap of the first-bit measures

Define subprobability measures

$$\mu_i(A) = \mathbb{P}(X \in A, u_0 = i), \quad i = 0, 1.$$

Their sum is μ . The following fact uses only $\lambda > 1/2$, not singularity.

Lemma II.9.13.1. *There is a finite measure $\xi \neq 0$ with $\xi \leq \mu_0$ and $\xi \leq \mu_1$.*

Proof. If no such measure exists, the densities of μ_0 and μ_1 with respect to μ have disjoint supports almost everywhere. Thus μ_0 and μ_1 are mutually singular. We will show that this makes the conditional entropy of the first n bits, given a λ^n -quantization of X , arbitrarily small compared with n . There are only $O(\lambda^{-n})$ quantization cells, too few to carry the full entropy $n \log 2$ of those bits. For every $\eta \in (0, 1/2)$, inner regularity then gives disjoint compact sets K_0, K_1 such that

$$\mathbb{P}(X \in K_{u_0}) > 1 - \eta.$$

Let $\delta = \text{dist}(K_0, K_1) > 0$, and choose an integer $h \geq 1$ with $\lambda^h < \delta/3$.

Set $Q_n = \lfloor X/\lambda^n \rfloor$. Given Q_n and the bits $u_0, \dots, u_{i-1}$, one can approximate

$$X^{(i)} = \frac{X - \sum_{k < i} u_k \lambda^k}{\lambda^i}$$

to within λ^{n-i} . The pair $(X^{(i)}, u_i)$ has the same law as (X, u_0) . For $i \leq n-h$, the approximation therefore determines an estimator of u_i with error probability at most η : when $X^{(i)} \in K_{u_i}$, choose the nearer of the two compact sets.

Write H for Shannon entropy using natural logarithms, and put $h_2(\eta) = -\eta \log \eta - (1 - \eta) \log(1 - \eta)$. The error-indicator argument gives

$$H(u_i \mid Q_n, u_0, \dots, u_{i-1}) \leq h_2(\eta) \quad (i \leq n - h).$$

Indeed, the estimator and its binary error indicator recover u_i ; the entropy of that indicator is at most $h_2(\eta)$. The remaining at most h bits have conditional entropy at most $h \log 2$. Since $X \in [0, L]$, the variable Q_n takes at most $L\lambda^{-n} + 2$ values. The chain rule now gives

$$n \log 2 \leq H(Q_n) + H(u_0, \dots, u_{n-1} \mid Q_n) \leq n \log(1/\lambda) + nh_2(\eta) + O_{\lambda, \eta}(1).$$

Choose η so small that $h_2(\eta) < \log(2\lambda)$ and let $n \rightarrow \infty$. This is a contradiction. $\square$

Proposition II.9.13.2 (Many close prefixes with opposite first bits). *If μ_λ is singular, there are constants $a \in (0, L/2)$ and $c > 0$, depending only on λ , with the following property. For every $Q > 1$ and all sufficiently large n , at least $c2^n$ words $u \in \{0, 1\}^n$ with $u_0 = 0$ each have at least $Q(2\lambda)^n$ distinct neighbors $v \in \{0, 1\}^n$ such that*

$$v_0 = 1, \quad |s_u - s_v| \leq (L - a)\lambda^n. \quad (\text{II.9.13.1})$$

The constants a, c are independent of Q and n .

Proof. Fix ξ from Lemma II.9.13.1, and write $b = \xi(\mathbb{R}) > 0$. The endpoints $0, L$ have μ -measure zero, because each requires all the bits to have one prescribed value. Choose $a \in (0, L/2)$ so that $\mu([a, L-a]) > 1 - b/8$.

For each n , let $X^{(n)} = \sum_{k \geq 0} u_{n+k} \lambda^k$, and restrict the first-bit measures by defining

$$\mu'_{i,n}(A) = \mathbb{P}(X \in A, u_0 = i, X^{(n)} \in [a, L-a]).$$

The sum of the masses discarded from μ_0, μ_1 is less than $b/8$, since $X^{(n)}$ has law μ .

Fix $Q > 1$. By Lemma II.9.12.4, choose a compact set K with $\text{Leb}(K) < \varepsilon$ and $\mu(K) > 1 - \varepsilon$, where

$$\varepsilon < b/8, \quad \frac{2\varepsilon Q}{a} < \frac{b}{4}.$$

Then $\xi(K) > 7b/8$. Partition the line into half-open intervals of length $a\lambda^n$, and let $\mathcal{J}_n$ consist of those meeting K . Compactness of K implies, for all sufficiently large n ,

$$\#\mathcal{J}_n \leq \frac{2\varepsilon}{a\lambda^n}.$$

To see this, their union is contained in the closed $a\lambda^n$ -neighborhood of K , whose Lebesgue measure tends to $\text{Leb}(K)$. Moreover,

$$\sum_{J \in \mathcal{J}_n} \min_{i=0,1} \mu'_{i,n}(J) \geq \frac{3b}{4}. \quad (\text{II.9.13.2})$$

Indeed, the corresponding sum with μ_i in place of $\mu'_{i,n}$ is at least $\xi(K)$, and restricting the two measures loses less than $b/8$ in total.

Let $U_i(J)$ be the set of words u of length n with $u_0 = i$ for which

$$(s_u + \lambda^n[a, L-a]) \cap J \neq \emptyset.$$

Each prefix has probability 2^{-n} , so $\mu'_{i,n}(J) \leq 2^{-n} \#U_i(J)$. Discard the intervals for which $\#U_1(J) < Q(2\lambda)^n$. Their contribution to (II.9.13.2) is at most

$$\frac{2\varepsilon}{a\lambda^n} 2^{-n} Q(2\lambda)^n = \frac{2\varepsilon Q}{a} < \frac{b}{4}.$$

Thus the sum of the minima over the remaining intervals is at least $b/2$.

A fixed interval $s_u + \lambda^n[a, L-a]$ meets at most

$$C_a = \left\lceil \frac{L-2a}{a} \right\rceil + 2$$

of the bins. If $\mathcal{U}$ is the union of $U_0(J)$ over the retained bins, it follows that

$$\frac{b}{2} \leq 2^{-n} \sum_{J \text{ retained}} \#U_0(J) \leq 2^{-n} C_a \#\mathcal{U}.$$

Take $c = b/(2C_a)$. For every $u \in \mathcal{U}$, choose one retained bin J with $u \in U_0(J)$. All $v \in U_1(J)$ satisfy (II.9.13.1): two points in J differ by less than $a\lambda^n$, and two tails in $\lambda^n[a, L-a]$ differ by at most $(L-2a)\lambda^n$. There are at least $Q(2\lambda)^n$ such neighbors. $\square$

II.9.13.2 Completion to irreducible polynomials

For every pair in Proposition II.9.13.2, the difference $w = u - v$ has $w_0 = -1$ and $|\sum_{i < n} w_i \lambda^i| \leq (L - a)\lambda^n$. We next extend any such coefficient vector to a polynomial whose root lies in $(\lambda, \lambda + 4^{-n}/2]$. The first lemma ensures that a bounded-coefficient polynomial cannot remain too flat throughout this short interval.

Lemma II.9.13.3 (Uniform finite-order nonvanishing). *Fix $\lambda \in (0, 1)$. There are an integer $M \geq 1$ and constants $c_0, x_0 > 0$ such that every real polynomial $p(t) = -1 + \sum_{k \geq 1} a_k t^k$ with $|a_k| \leq 1$ satisfies*

$$\sup_{t \in [\lambda, \lambda + x]} |p(t)| \geq c_0 x^M \quad (0 < x < x_0).$$

Proof. There exist $M \geq 1$ and $\delta > 0$ such that

$$\max_{0 \leq k \leq M} \left| \frac{p^{(k)}(\lambda)}{k!} \right| \geq \delta \quad (\text{II.9.13.3})$$

for every polynomial in the stated class. Otherwise, for each m one could choose a polynomial for which all these quantities through order m are less than $1/m$. Extend its coefficient sequence by zeros. A coefficientwise convergent subsequence exists by compactness of $[-1, 1]^{\mathbb{N}}$. The associated series converge, with every derivative, uniformly on compact subsets of the unit disk. Their limit would have all derivatives zero at λ and constant coefficient -1 , contradicting the identity theorem.

For fixed M , equivalence of norms on polynomials of degree at most M gives $\kappa_M > 0$ such that

$$\sup_{0 \leq t \leq 1} \left| \sum_{k=0}^M b_k t^k \right| \geq \kappa_M \max_{k \leq M} |b_k|.$$

Apply this to the Taylor polynomial of $p(\lambda + xt)$. For $x \leq 1$, (II.9.13.3) makes the right-hand side at least $\kappa_M \delta x^M$. The coefficient bound gives a uniform bound on the $(M + 1)$ st derivative in a fixed neighborhood of λ , so the Taylor remainder is $O_\lambda(x^{M+1})$, with M fixed. Reducing x_0 proves the assertion with $c_0 = \kappa_M \delta / 2$. $\square$

To impose irreducibility without disturbing the prescribed coefficients, we use the prime-polynomial theorem in arithmetic progressions. First we prove its square-root character input in the precise binary, power-of- t case required here; the progression deduction then follows immediately.

II.9.13.2.1 A finite-field interlude: the character bound

The square-root character estimate is a genuine theorem, not an elementary consequence of the Euler product. We prove the exact special case needed here. The strategy is to count fixed points on an explicit polynomial cover, use a high-power auxiliary function to obtain a square-root error, and finally recover the inverse roots from their power sums. The auxiliary function is the Bombieri–Stepanov mechanism [Bom74]; the polynomial

cover is the Carlitz construction [Con]. Everything about this particular cover that the argument needs is proved below. In particular we do not assume a general Riemann–Roch or Weil theorem.

Definition II.9.13.4 (The binary polynomial cover and its weights). Fix $\ell \geq 2$, put $N = 2^{\ell-1}$, and let k be an algebraic closure of $\mathbb{F}_2$, that is, an algebraically closed field algebraic over $\mathbb{F}_2$. Algebraically closed means that every nonconstant polynomial over the field has a root in the field; algebraic over $\mathbb{F}_2$ means that each element satisfies a nonzero polynomial over $\mathbb{F}_2$. Its subfield $\mathbb{F}_{2^n}$ consists of the roots of $X^{2^n} - X$. Let

$$A_\ell = k[x, z_2, \dots, z_\ell] / (z_j^2 + z_j - xz_{j-1} : 2 \leq j \leq \ell), \quad z_1 = 1.$$

This quotient identifies two polynomials when their difference is a sum of polynomial multiples of the displayed relations. An integral domain is a nonzero commutative ring in which a product of two nonzero elements is nonzero. Its fraction field consists of quotients f/g , $g \neq 0$, with $f/g = f'/g'$ when $fg' = f'g$. The degree of a field extension is its vector-space dimension over the smaller field. A free basis over $k[x]$ means a family giving a unique finite linear combination with coefficients in $k[x]$. These are the algebraic terms used below. A finite point is a tuple $P = (a, b_2, \dots, b_\ell) \in k^\ell$ satisfying these equations. The fibre over a is the set of points with $x = a$. For $\varepsilon = (\varepsilon_2, \dots, \varepsilon_\ell) \in \{0, 1\}^{\ell-1}$, put

$$b_\varepsilon = \prod_{j=2}^{\ell} z_j^{\varepsilon_j}, \quad w_j = N - 2^{\ell-j}, \quad w_\varepsilon = \sum_{j=2}^{\ell} \varepsilon_j w_j.$$

The weight of a nonzero normal polynomial $f = \sum_\varepsilon f_\varepsilon(x) b_\varepsilon$ is

$$w(f) = \max_{f_\varepsilon \neq 0} \{N \deg f_\varepsilon + w_\varepsilon\}.$$

Here normal means that each exponent of a z_j is at most one.

Lemma II.9.13.5 (Normal forms, a field and a determinant norm). *The normal monomials form a free $k[x]$ -basis of A_ℓ . The ring is an integral domain and $w(fg) = w(f) + w(g)$ for nonzero f, g . Its fraction field K_ℓ has degree N over $k(x)$. For $f \in A_\ell \setminus \{0\}$, let $\mathcal{N}(f)$ be the determinant of multiplication by f in this basis. Then*

$$\mathcal{N}(f) \in k[x] \setminus \{0\}, \quad \deg \mathcal{N}(f) = w(f).$$

Proof. Successive division by the monic quadratic in z_j gives existence and uniqueness of the normal form: at each stage adjoining z_j produces a free module with basis $1, z_j$. The residues $w_\varepsilon \pmod{N}$ are all distinct. Indeed, $w_\varepsilon \equiv -\sum_j \varepsilon_j 2^{\ell-j} \pmod{N}$, and the sums on the right run once through $0, \dots, N-1$. Consequently every nonzero normal polynomial has exactly one highest weight monomial.

In reducing a product, replace z_j^2 by $xz_{j-1} + z_j$. The first term has weight $2w_j$, since $2w_j = N + w_{j-1}$, with $w_1 = 0$; the second has strictly smaller weight. Following the first

term at every reduction gives one normal monomial with the original weight and coefficient one. All other terms have smaller weight. Thus the product of the highest monomials cannot cancel and $w(fg) = w(f) + w(g)$. This proves the domain assertion. Localization in $k[x] \setminus \{0\}$ is a domain of dimension N over $k(x)$; a finite-dimensional domain over a field is a field, since multiplication by a nonzero element is injective and hence surjective. It is therefore K_ℓ .

Write $fb_\eta = \sum_\varepsilon M_{\varepsilon\eta}(x)b_\varepsilon$. Every nonzero entry satisfies

$$N \deg M_{\varepsilon\eta} + w_\varepsilon \leq w(f) + w_\eta.$$

In each column exactly one entry attains equality, by the unique highest monomial just proved. Its row is determined by $w_\varepsilon \equiv w(f) + w_\eta \pmod{N}$, so these rows form a permutation. In the determinant that permutation has degree $N^{-1} \sum_\eta (w(f) + w_\eta - w_{\varepsilon(\eta)}) = w(f)$. Its leading coefficient is nonzero. Every other permutation loses weight in at least one column and has smaller degree. This proves the norm formula and excludes cancellation in its leading term. $\square$

Definition II.9.13.6 (Orders, poles and one-pole spaces). For a finite point $P = (a, b_2, \dots, b_\ell)$, expand functions locally with parameter $u = x - a$. A formal power series is a sequence of coefficients with coefficientwise addition and convolution multiplication; a Laurent series permits finitely many negative exponents. Their spaces are denoted $k[[u]]$ and $k((u))$, respectively. The order $v_P(f)$ is the least exponent with nonzero coefficient in the local Laurent expansion of $f \neq 0$. Extend the weight to fractions by $w(f/g) = w(f) - w(g)$, and set $v_\infty(f) = -w(f)$. The symbol ∞ denotes this additional order, not a new affine tuple. A pole of order r at P , finite or infinite, means $v_P(f) = -r < 0$; a zero of order r means $v_P(f) = r > 0$. For $m \geq 0$, let $L(mP)$ consist of zero and all functions in K_ℓ with no poles except possibly a pole of order at most m at P . All these are vector spaces over k .

Lemma II.9.13.7 (Local expansions and equality of total zero and pole orders). *Each fibre has N points. The local expansions in the definition exist uniquely and give injective field maps to $k((u))$. Every nonzero $f \in K_\ell$ has only finitely many nonzero finite orders, and*

$$\sum_{P \text{ } m \text{ finite}} v_P(f) + v_\infty(f) = 0. \quad (\text{FF.1})$$

Functions having no poles anywhere are constant. For a finite P , nonzero $f \in L(mP)$ has $-m \leq v_P(f) \leq 0$.

Proof. Given a , at each stage $z_j^2 + z_j = az_{j-1}$ has two distinct roots, because its derivative is one. This gives N tuples. For prescribed constant terms at one tuple, solve

$$Z_j(u)^2 + Z_j(u) = (a + u)Z_{j-1}(u), \quad Z_1 = 1.$$

The coefficient of degree $r > 0$ determines the new coefficient of Z_j ; the only other contribution from its square is a coefficient of degree $r/2 < r$. This proves unique existence

by induction in j and r . The two solutions at each quadratic stage differ by one. Their difference is invertible in $k[[u]]$, so polynomial division and the Chinese remainder theorem yield

$$A_\ell \otimes_{k[x]} k[[u]] \simeq \prod_{P: x(P)=a} k[[u]].$$

Here the tensor notation means extending the coefficients in the free normal form from $k[x]$ to $k[[u]]$ by $x \mapsto a + u$, with the same defining multiplication rules. The isomorphism sends a function to all of its branch expansions. The determinant norm in Lemma II.9.13.5 becomes the product of the branch expansions in this decomposition. A nonzero function cannot have an identically zero expansion, since that would make its nonzero polynomial norm zero. Hence fractions have well-defined Laurent expansions, and

$$\text{ord}_{x=a} \mathcal{N}(f) = \sum_{P: x(P)=a} v_P(f) \quad (f \in A_\ell \setminus \{0\}).$$

Summing over the roots of the norm gives $w(f)$; applying this to numerator and denominator proves (FF.1) for fractions. It also proves finiteness of the nonzero finite orders.

If a fraction has no finite poles, write it as $A/b(x)$ with $A \in A_\ell$, $b \in k[x]$: such a denominator exists by the field localization in Lemma II.9.13.5. At a root a of b of multiplicity r , every branch of A is divisible by u^r . The displayed direct-product decomposition, reduced modulo u^r , then says that $A \in (x - a)^r A_\ell$. Uniqueness of the free normal form shows that every coefficient of A is divisible by $(x - a)^r$. Doing this at each root cancels b and places the fraction in A_ℓ . If it has no infinite pole, its weight is at most zero; the only normal monomial of weight zero is one. It is therefore constant.

A nonzero member of $L(mP)$ cannot have positive order at P : it would then have no poles and at least one zero, contradicting the constant assertion. The remaining order bounds follow from the definition. $\square$

We need only a dimension estimate for one-pole functions. For this explicit cover it is obtained by counting monomials and imposing linear conditions; no general theorem about algebraic curves is involved.

Lemma II.9.13.8 (Elementary one-pole dimension bounds). *Put $g_\ell = \sum_\varepsilon \lfloor w_\varepsilon / N \rfloor$. For every finite point P and every integer $m \geq 0$,*

$$m + 1 - g_\ell \leq \dim_k L(mP) \leq m + 1. \quad (\text{FF.2})$$

There is a basis of $L(mP)$ whose members have distinct pole orders at P , one of them being the constant function one.

Proof. A function regular at every finite point is a normal polynomial, as proved in Lemma II.9.13.7. Hence a basis of $L(r\infty)$ has one normal monomial for each available weight at most r . For a fixed ε , these weights form the progression

$$w_\varepsilon, w_\varepsilon + N, w_\varepsilon + 2N, \dots$$

Because the residues are distinct, among all nonnegative integers exactly g_ℓ do not occur as weights. Therefore $\dim L(r\infty) \geq r + 1 - g_\ell$ for all $r \geq 0$.

Let $x(P) = a$ and consider numerators $A \in L(Nm\infty)$. Demand that at the other $N - 1$ points above a , their first m local coefficients vanish. These are at most $(N - 1)m$ linear conditions. The resulting fractions $A/(x - a)^m$ have no poles except at P , and at infinity their order is $Nm - w(A) \geq 0$. The injective numerator map consequently gives

$$\dim L(mP) \geq Nm + 1 - g_\ell - (N - 1)m = m + 1 - g_\ell.$$

For the upper bound, the m negative Laurent coefficients at P define a map into k^m . Its kernel has no poles and consists of constants. Thus $\dim L(mP) \leq m + 1$. Each successive filtration quotient $L(rP)/L((r - 1)P)$ has dimension at most one, since its coefficient of u^{-r} is scalar. Choosing a vector at each nonzero quotient produces the asserted distinct-order basis. $\square$

Definition II.9.13.9 (Carlitz operators and the fibre action). Put $t = x^{-1}$, $y_j = tz_j$ and $y_1 = t$ in K_ℓ . Define the additive polynomial $C_t(Y) = Y^2 + tY$, its iterates $C_{t^r} = C_t \circ \cdots \circ C_t$, $C_1(Y) = Y$, and, for $a(t) = \sum a_r t^r \in \mathbb{F}_2[t]$, $C_a(Y) = \sum a_r C_{t^r}(Y)$. Let $G_\ell = (\mathbb{F}_2[t]/(t^\ell))^\times$. For $a \in G_\ell$, represented by $a(t) = 1 + \sum_{r=1}^{\ell-1} a_r t^r$, define

$$\sigma_a(x) = x, \quad \sigma_a(z_j) = \sum_{r=0}^{j-1} a_r z_{j-r}, \quad a_0 = 1.$$

The same formulas describe an action on finite tuples. For $q = 2^n$, F_q sends a tuple to its coordinatewise q -th powers; it is the Frobenius map. A group action here means $\sigma_a \sigma_b = \sigma_{ab}$ and $\sigma_1 = \text{id}$. It is free if no nonidentity group element fixes a point, and transitive on a fibre if any two of its points are related by some σ_a . For a map Φ given by these coordinate formulas, its pullback Φ^* substitutes those formulas in a rational function and leaves its coefficients in k fixed. In particular the pullback of F_q is k -linear; it is not the operation that also raises coefficients to their q -th powers. Write

$$U_a(q) = \#\{P \text{ finite} : F_q(P) = \sigma_a(P)\}.$$

Lemma II.9.13.10 (The regular action and the prime dictionary). *These maps form a group action of G_ℓ , defined over $\mathbb{F}_2$, and act freely and transitively on every finite fibre. For every monic irreducible $P(t) \neq t$ of degree d , over any root α of P the map F_{2^d} on the fibre $x = \alpha^{-1}$ is $\sigma_P \bmod t^\ell$. The fibre at $x = 0$ consists of N points defined over $\mathbb{F}_2$.*

Proof. The defining recurrences say $C_t(y_j) = y_{j-1}$ and $C_t(y_1) = 0$. Linearity over $\mathbb{F}_2$ gives $C_a \circ C_b = C_{ab}$. Applied to y_ℓ , the formula for σ_a is $y_\ell \mapsto C_a(y_\ell)$. Substitution in the triangular equations verifies that each displayed σ_a preserves them. The product formula shows that its inverse is $\sigma_{a^{-1}}$ and that the action depends only on the residue modulo t^ℓ .

If $a \neq 1$, let $r \geq 1$ be its first nonzero coefficient after the constant one. At every finite tuple, $\sigma_a(z_{r+1}) = z_{r+1} + 1$. No point is fixed. Since $|G_\ell| = N$ and each fibre has N points,

the action is free and transitive. At $x = 0$, each equation reduces to $z_j^2 + z_j = 0$, proving the last assertion.

For the prime dictionary reduce coefficients of C_P modulo P , so $t = \alpha$. Write the resulting additive polynomial as $D(Y) = \sum_{i=0}^d c_i Y^{2^i}$. Recursion shows $c_d = 1$ and $c_0 = P(\alpha) = 0$. Since it commutes with $Y^2 + \alpha Y$, comparison of the coefficient of Y^{2^i} gives, for $1 \leq i < d$,

$$(\alpha^{2^i} - \alpha)c_i = c_{i-1}^2 - c_{i-1}.$$

The coefficient on the left is nonzero: otherwise α would lie in a proper subfield of $\mathbb{F}_{2^d}$, contrary to its degree d . Induction gives $c_i = 0$ for all $i < d$, and hence $C_P(Y) \equiv Y^{2^d} \pmod{P}$. On the fibre in question $y_\ell = \alpha z_\ell$; thus this identity identifies its 2^d -th power with $C_P(y_\ell)$. The coordinates $y_j = C_t^{\ell-j}(y_\ell)$ determine the entire tuple, so the maps agree on the fibre, as asserted. $\square$

Lemma II.9.13.11 (Square-root errors for every fibre action). *There is a constant B_ℓ such that for every sufficiently large even n , every $a \in G_\ell$, and $q = 2^n$,*

$$|U_a(q) - q| \leq B_\ell \sqrt{q}. \quad (\text{FF.3})$$

The threshold and constant depend on ℓ , not on a .

Proof. Write $R = \sqrt{q}$, $g = g_\ell$, and $\Phi = \sigma_a^{-1} \circ F_q$. If it has no finite fixed point, the upper bound $U_a(q) \leq q + (2g + 1)R$ is automatic. Otherwise choose a finite fixed point P . Pullback Φ^* is an injective k -linear map on K_ℓ . It takes $L(mP)$ into $L(qmP)$, multiplies the pole order by q , and each of its values is a q -th power in K_ℓ . Indeed σ_a is an automorphism preserving the set of orders; Frobenius is a bijection on geometric tuples and substitutes q -th powers in every coordinate. Its unique preimage of the fixed point is that point. In the finite local parameter $x - x(P)$ the pullback is its q -th power. Coefficients have q -th roots in k , which also proves the perfect-power assertion. The order at infinity is preserved by σ_a , because $\sigma_a(z_j)$ has the same highest weight term z_j , and is multiplied by q by Frobenius. These facts check both finite and infinite poles.

For nonnegative integers l, m , choose a distinct-pole-order basis $s_1, \dots, s_h$ of $L(mP)$. Consider the linear multiplication map on tuples $(v_1, \dots, v_h)$:

$$(L(lP)^R)^h \longrightarrow K_\ell, \quad (v_i)_{i=1}^h \longmapsto \sum_i v_i \Phi^* s_i, \quad L(lP)^R = \{v^R : v \in L(lP)\}.$$

When $lR < q$, this map is injective. In fact a nonzero coefficient in $L(lP)^R$ has pole order between 0 and lR by Lemma II.9.13.7. If s_i has pole order d_i , its product with such a coefficient has pole order in $[qd_i, qd_i + lR]$. The intervals are disjoint for distinct integer d_i , so the largest pole of a nonzero linear combination cannot cancel. The image V therefore has dimension $\dim L(lP) \dim L(mP)$. Define on this image the linear map

$$\Delta \left(\sum_i v_i \Phi^* s_i \right) = \sum_i v_i s_i \quad \text{into } L((lR + m)P).$$

It is well-defined by that injectivity. If $l, m \geq g - 1$, both dimension lower bounds in Lemma II.9.13.8 are nonnegative. Their product is then a lower bound for $\dim V$, so Δ has a nonzero kernel whenever

$$(l + 1 - g)(m + 1 - g) > lR + m + 1.$$

Take $l = R - 1$ and $m = R + 2g$, with $R > g^2 + 3g + 1$. In particular $l, m \geq g - 1$, so the dimension comparison applies. The difference between these two sides is $R - g(g+1) - 2g - 1$, positive for sufficiently large R ; also $lR < q$. Pick a nonzero $f \in \ker \Delta$. It has no pole except at P , with order at most $lR + qm$. At each finite fixed point other than P , $\Phi^* s_i = s_i$ after evaluation, and hence $f = 0$. Every product $v_i \Phi^* s_i$, and thus f , is an R -th power, since R divides q . Each such zero has order at least R . By (FF.1),

$$R(U_a(q) - 1) \leq lR + qm, \quad U_a(q) \leq 1 + l + qm/R = q + (2g + 1)R.$$

For every $x \in \mathbb{F}_q$, Frobenius permutes its regular G_ℓ -fibre and commutes with the group action. It agrees with exactly one σ_a on that fibre. If $x \notin \mathbb{F}_q$, no point above it can be fixed by Φ . Consequently $\sum_a U_a(q) = Nq$. Applying the upper bound to the other $N - 1$ actions gives

$$U_a(q) \geq q - (N - 1)(2g + 1)R.$$

Thus (FF.3) holds, for example with $B_\ell = (N - 1)(2g_\ell + 1)$, since $N \geq 2$. □

Lemma II.9.13.12 (Character power sums from the finite-point count). *For every nonprincipal character χ of G_ℓ , $L(u, \chi)$ has degree at most $\ell - 1$. If $L(u, \chi) = \prod_j (1 - \alpha_j u)$, then for all sufficiently large even n ,*

$$\left| \sum_j \alpha_j^n \right| \leq B_\ell 2^{n/2} + 1.$$

Proof. For $r \geq \ell$, the monic polynomials of degree r represent each residue class modulo t^ℓ exactly $2^{r-\ell}$ times. Summing the nonprincipal character over the unit group is zero: translate the sum by a unit on which the character is not one. The nonunit terms are zero by definition. Hence all coefficients of L in degrees $r \geq \ell$ vanish, while its constant term is one. Factorization over $\mathbb{C}$ gives the stated inverse roots.

Let $\Lambda(f) = \deg P$ for $f = P^j$ with P monic irreducible and $j \geq 1$, and zero otherwise. The formal logarithmic derivative of the Euler product is

$$S_n := \sum_{\substack{f \text{ monic} \\ \deg f = n}} \chi(f) \Lambda(f) = - \sum_j \alpha_j^n.$$

Each nonzero $t \in \mathbb{F}_{2^n}$ has a minimal irreducible polynomial $P \neq t$, of degree $d \mid n$; each such polynomial has d roots in that field. By Lemma II.9.13.10, Frobenius on the associated fibre is $\sigma_{Pn/d}$. The fibre at $x = 0$, omitted by the substitution $x = 1/t$, contributes the identity action at its N points. Thus

$$\frac{1}{N} \sum_{a \in G_\ell} \chi(a) U_a(2^n) = 1 + \sum_{\substack{d \mid n \\ P \neq t, \deg P = d}} d \chi(P)^{n/d} = 1 + S_n.$$

This identity counts prime powers, not merely primes. For even n in the range of Lemma II.9.13.11, subtract 2^n from each U_a , using $\sum_a \chi(a) = 0$, and apply the triangle inequality. It gives $|S_n| \leq B_\ell 2^{n/2} + 1$. $\square$

Lemma II.9.13.13 (Function-field character bound). *For every nonprincipal Dirichlet character modulo t^ℓ over $\mathbb{F}_2$, $L(u, \chi)$ is a polynomial of degree at most $\ell - 1$, and every inverse root satisfies $|\alpha_j| \leq \sqrt{2}$. The statement includes imprimitive characters and trivial factors.*

Proof. For $\ell = 1$ the unit group is trivial, so there is no nonprincipal character. Let $\ell \geq 2$. Polynomiality and the degree bound are proved in Lemma II.9.13.12. Put $\beta_j = \alpha_j^2$. That lemma, with $n = 2r$, says $|\sum_j \beta_j^r| \leq C_\ell 2^r$ for all large r ; enlarging C_ℓ covers the finitely many remaining positive r . Therefore

$$\sum_{r \geq 1} \left(\sum_j \beta_j^r \right) v^r = \sum_j \frac{\beta_j v}{1 - \beta_j v}$$

is holomorphic for $|v| < 1/2$. If any $|\beta_j| > 2$, the rational expression on the right has a pole there at $1/\beta_j$. Repeated equal β_j 's add with their positive integer multiplicity, so that pole cannot cancel; distinct β_j 's have distinct poles. This is a contradiction. Thus $|\alpha_j|^2 = |\beta_j| \leq 2$. No primitivity hypothesis was used: the action and orthogonality apply to every nonprincipal character of G_ℓ . In particular an imprimitive character and a factor $1 - u$, with inverse root one, are already covered. $\square$

This completes the exact character theorem used in the arithmetic classification. It agrees with the consequence of Weil's theorem recorded in [GK24, Theorem 3.2 and Lemma 3.3], but that theorem is now provenance rather than an omitted proof input. The constant B_ℓ need not be sharp or uniform in ℓ : it disappears when the inverse roots are recovered from all even power sums.

Lemma II.9.13.14 (Irreducibles with prescribed initial coefficients). *Fix integers $D \geq 2$ and $d > 2D$. For all sufficiently large n , every polynomial $F \in \mathbb{F}_2[t]$ with $\deg F < Dn$ and $F(0) = 1$ has a monic irreducible completion $P \in \mathbb{F}_2[t]$ satisfying*

$$\deg P = dn, \quad P \equiv F + t^{Dn} \pmod{t^{Dn+1}}.$$

The threshold in n is uniform in F .

Proof. For a monic polynomial f , write $\Lambda(f) = \deg P$ if $f = P^k$ for a monic irreducible P , and $\Lambda(f) = 0$ otherwise. Let χ be a nonprincipal Dirichlet character modulo t^ℓ , extended by zero to nonunits. By Lemma II.9.13.13, its L -function is a polynomial of degree at most $\ell - 1$, with inverse roots α_j satisfying $|\alpha_j| \leq \sqrt{2}$; this bound includes trivial factors. Taking

the logarithmic derivative of the Euler product gives

$$\left| \sum_{\substack{f \text{ monic} \\ \deg f = s}} \Lambda(f) \chi(f) \right| = \left| \sum_j \alpha_j^s \right| \leq \ell 2^{s/2}.$$

For the principal character, the same sum is $2^s - 1$, by the zeta function of $\mathbb{F}_2[t]$ after removal of the prime t . Character orthogonality therefore gives, in every invertible residue class A modulo t^ℓ ,

$$\sum_{\substack{f \text{ monic, } \deg f = s \\ f \equiv A \pmod{t^\ell}}} \Lambda(f) \geq \frac{2^s - 1}{2^{\ell-1}} - \ell 2^{s/2}. \quad (\text{II.9.13.4})$$

The contribution from proper prime powers, even without the congruence restriction, is at most

$$\sum_{\substack{k|s \\ k \geq 2}} 2^{s/k} \leq s 2^{s/2}.$$

Take $\ell = Dn + 1$, $s = dn$, and $A = F + t^{Dn}$. This is an invertible class. The main term in (II.9.13.4) has order $2^{(d-D)n}$, whereas the character error and proper-power contribution are $O(n 2^{dn/2})$. Since $d > 2D$, at least one irreducible polynomial remains for all sufficiently large n , uniformly in A . $\square$

Proposition II.9.13.15 (Realization of a polynomial prefix). *Fix $\lambda \in (1/2, 1)$ and $a \in (0, L)$, where $L = (1 - \lambda)^{-1}$. There is an integer d_0 such that, for each integer $d \geq d_0$ and all sufficiently large n , the following assertion holds uniformly over $w \in \{-1, 0, 1\}^n$. If*

$$w_0 = -1, \quad \left| \sum_{i=0}^{n-1} w_i \lambda^i \right| \leq (L - a) \lambda^n, \quad (\text{II.9.13.5})$$

then some $r \in R(n, d)$ satisfies

$$\lambda < r \leq \lambda + \frac{1}{2} 4^{-n}, \quad \pi_n(p_r) = w.$$

Proof. We first append digits to make the polynomial small at λ . We then construct two irreducible completions with opposite signs there but the same sign at a nearby point to its right. Uniform finite-order nonvanishing supplies that second point, and the intermediate value theorem supplies a root between the two points.

Let M, c_0, x_0 be supplied by Lemma II.9.13.3. Choose an integer $D \geq 2$ with

$$\lambda^D < 4^{-M}, \quad (\text{II.9.13.6})$$

and then take $d_0 = 2D + 1$.

Start with the polynomial in (II.9.13.5) and append coefficients through index $Dn - 1$. If the next index is k , let e be the negative of the current value at λ , divided by λ^k . Appending $e_0 \in \{-1, 0, 1\}$ changes this residual to

$$e \mapsto \frac{e - e_0}{\lambda}.$$

Choose a nearest digit e_0 . Once $|e| \leq 1$, the next residual has absolute value at most $1/(2\lambda) < 1$, so this bound is preserved. Before that time, the residual keeps its sign and its absolute value updates by

$$|e| \mapsto \frac{|e| - 1}{\lambda} = L + \frac{|e| - L}{\lambda}.$$

The initial bound $|e| \leq L - a$ implies that $|e| \leq 1$ is reached within a number of steps depending only on λ, a . Indeed, after k steps without reaching this interval the absolute value is at most $L - a\lambda^{-k}$, which eventually falls below 1. Consequently, for all sufficiently large n , the polynomial p_0 constructed through index $Dn - 1$ satisfies

$$|p_0(\lambda)| \leq \frac{\lambda^{Dn}}{2\lambda}. \quad (\text{II.9.13.7})$$

Put $x_n = 4^{-n}/2$. By Lemma II.9.13.3, there is $t_n \in [\lambda, \lambda + x_n]$ with $|p_0(t_n)| \geq c_0 x_n^M$. Meanwhile every possible appendage with coefficients of absolute value at most one, starting at index Dn , is bounded throughout this interval by

$$\frac{(\lambda + x_n)^{Dn}}{1 - \lambda - x_n} = o(x_n^M).$$

This follows from (II.9.13.6), since $Dnx_n \rightarrow 0$. The estimate is uniform in the prefix and in the eventual length of the appendage.

Fix $d \geq d_0$. Apply Lemma II.9.13.14 to the reduction of p_0 modulo 2, obtaining a monic irreducible polynomial P of degree dn with $P \equiv p_0 + t^{Dn} \pmod{t^{Dn+1}}$. Represent its coefficients of indices $Dn, \dots, dn$ by integers $b_k \in \{0, 1\}$, and let

$$T(t) = \sum_{k=Dn}^{dn} b_k t^k, \quad p_{\pm}(t) = p_0(t) \pm T(t).$$

Here $b_{Dn} = b_{dn} = 1$. Thus $T(\lambda) \geq \lambda^{Dn}$, and (II.9.13.7) implies $p_-(\lambda) < 0 < p_+(\lambda)$. At t_n , the negligible-tail estimate shows that both polynomials have the same nonzero sign as $p_0(t_n)$. One of them therefore has a real root

$$r \in (\lambda, t_n] \subset (\lambda, \lambda + x_n].$$

Both $p_{\pm}$ reduce to P modulo 2, so they are irreducible over $\mathbb{Q}$. They have degree dn , leading coefficient ± 1 , constant coefficient -1 , and all coefficients in $\{-1, 0, 1\}$. Their roots are algebraic units. For sufficiently large n , the chosen r is in $(1/2, 1)$, belongs to $R(n, d)$, and its signed minimal polynomial is the chosen $p_{\pm}$. Its first n coefficients are w . $\square$

II.9.13.3 Proof of the converse and the quantifiers

Converse and conclusion. Suppose μ_{λ} is singular. Proposition II.9.13.2 gives constants $a, c > 0$. Choose one integer d large enough for Proposition II.9.13.15 and also satisfying $1/d < c$. This choice depends on λ alone.

Fix any integer $j \geq 1$ and choose $Q > 2j$. For all sufficiently large n , each of at least $c2^n$ words u has at least $Q(2\lambda)^n$ neighbors v supplied by Proposition II.9.13.2. For each

difference $w = u - v$, Proposition II.9.13.15 supplies a root $r_w \in R(n, d)$ with signed minimal polynomial projecting to w , and

$$\lambda < r_w \leq \lambda + \frac{1}{2}4^{-n}.$$

Choose any one of these roots as a center r . All of them lie within 4^{-n} of r , so all their projected vectors belong to $W_n(r; d)$. For fixed u , its different neighbors give distinct vectors $u - v$, each in $\mathcal{O}_u$. Furthermore,

$$1 \leq \left(\frac{r}{\lambda}\right)^n \leq \left(1 + \frac{4^{-n}}{2\lambda}\right)^n \rightarrow 1$$

uniformly in the choice of center. Hence, for all sufficiently large n , each of these c^{2n} orthants contains at least $j(2r)^n$ distinct vectors of $W_n(r; d)$. Since $c > 1/d$, we have $r \in R_*(n, d, j)$ and $\lambda \in [r - 4^{-n}, r)$.

We have proved that a single d works for every j , at every sufficiently large n . This proves membership in the expression (II.9.12.3) with $\liminf$. Membership in that expression implies membership with $\limsup$, which implies singularity by Section II.9.12.2. Lemma II.9.12.3 handles $\lambda \leq 1/2$ and supplies the absolute continuity of every remaining parameter. This completes the proof of Theorem II.9.12.2. $\square$

II.9.14 Recovery of classical parameter classes

The arithmetic criterion is compatible with the classical norm separation argument for Garsia parameters and the Fourier argument for Pisot parameters. The first example directly bounds the crowding in (II.9.12.2); the second establishes singularity and then uses Theorem II.9.12.2 to obtain the approximation property.

Proposition II.9.14.1 (Garsia parameters, [Gar62]). *Let $\beta \in (1, 2)$ be an algebraic integer of absolute norm 2 whose conjugates all have modulus greater than 1. Then $1/\beta \notin \mathcal{C}$, and $v_{1/\beta}$ is absolutely continuous.*

Proof. Put $\lambda = 1/\beta$. Consider distinct vectors $w, w' \in \{-1, 0, 1\}^n$ lying in one closed orthant. Their difference has coefficients in $\{-1, 0, 1\}$, so

$$P(t) = \sum_{i=0}^{n-1} (w_i - w'_i)t^{n-1-i}$$

is a nonzero polynomial of that coefficient type. It cannot vanish at β . Otherwise, after removing any power of t dividing P , its constant coefficient would be ± 1 , while divisibility by the monic minimal polynomial of β would force that coefficient to be divisible by 2.

For every other conjugate γ of β ,

$$|P(\gamma)| \leq \frac{|\gamma|^n}{|\gamma| - 1}.$$

The nonzero algebraic integer $P(\beta)$ has absolute norm at least one. Since the product of the moduli of all conjugates of β is 2, the preceding estimate gives a constant $c_\beta > 0$ such that

$$\left| \sum_{i=0}^{n-1} (w_i - w'_i) \lambda^i \right| = \beta^{-(n-1)} |P(\beta)| \geq c_\beta 2^{-n}. \quad (\text{II.9.14.1})$$

For any potential hit of (II.9.12.3) at a sufficiently large n , all projected vectors satisfy the slab bound (II.9.12.5). Within any one orthant, (II.9.14.1) bounds their number by

$$1 + \frac{2B\lambda^n}{c_\beta 2^{-n}} \leq C_\beta (2\lambda)^n.$$

Choose an integer $j > C_\beta$. Since a hit has $r > \lambda$, no orthant can then meet the threshold $j(2r)^n$. This excludes all sufficiently large n , for every degree multiplier d . Thus $\lambda \notin \mathcal{C}$. Absolute continuity, in fact a bounded density, follows independently from the already proved Theorem II.8.4. This conclusion does not use the function-field character input. $\square$

Proposition II.9.14.2 (Reciprocal Pisot parameters, [Erd39]). *If $\beta \in (1, 2)$ is a Pisot number, then $v_{1/\beta}$ is singular.*

Proof. A Pisot number is an algebraic integer greater than one whose other conjugates have modulus less than one. Here its nonzero integer norm has absolute value less than $\beta < 2$, so β is a unit. Consequently β^ℓ is an algebraic integer for every $\ell \in \mathbb{Z}$, and it can never be a half-integer of odd numerator. In particular, all factors $\cos(\pi\beta^\ell)$ are nonzero.

For negative ℓ , the quantity β^ℓ tends exponentially to zero. For positive ℓ , its distance to an integer tends exponentially to zero: the trace of β^ℓ is an integer and the contributions of the other conjugates decay exponentially. The quadratic behavior of $-\log |\cos(\pi t)|$ near the integers therefore gives

$$\sum_{\ell \in \mathbb{Z}} -\log |\cos(\pi\beta^\ell)| < \infty.$$

With $\lambda = 1/\beta$, independence of the signs yields

$$|\mathbb{E} \exp(i\pi\beta^n Y_\lambda)| = \prod_{\ell \leq n} |\cos(\pi\beta^\ell)| \geq \prod_{\ell \in \mathbb{Z}} |\cos(\pi\beta^\ell)| > 0.$$

The Riemann–Lebesgue lemma rules out absolute continuity. Lemma II.9.12.3 gives singularity. $\square$

Corollary II.9.14.3 (Pisot parameters in the arithmetic criterion). *If $\beta \in (1, 2)$ is Pisot, then $1/\beta \in \mathcal{C}$. This arithmetic conclusion uses the classification theorem, including its locally proved character estimate, whereas the preceding singularity proof is independent of that route.*

Proof. Proposition II.9.14.2 gives singularity. Apply the singular-to-cluster implication of Theorem II.9.12.2. $\square$

II.9.15 Degree-four Salem parameters

For a degree-four Salem number β , the two conjugates on the unit circle form a single complex-conjugate pair. We use simultaneous control of this pair to construct many Fourier phases with appreciable expectation. After removing integer trace terms, widely separated phases depend, up to a small error, on disjoint blocks of the Bernoulli signs. Their averages therefore distinguish the Bernoulli convolution from Lebesgue measure.

Theorem II.9.15.1. *Let $\beta \in (1, 2)$ be a Salem number of degree four. Then the unbiased Bernoulli convolution $\nu_{1/\beta}$ is singular with respect to Lebesgue measure.*

The proof has two quantitative requirements. The Fourier products must lose little mass before the algebraic trace becomes an accurate integer approximation; the remaining trace errors must then stay small over many disjoint blocks. The multiplier construction below supplies both requirements. The trace approximation is the familiar arithmetic ingredient in the Fourier analysis of Pisot and Salem parameters [Erd39]; see in particular Salem's small-conjugate construction [Sal63, Chapter III, Section 3, Theorem III, pp. 28–29]. The estimates here also control the losses accumulated while the multiplier changes.

II.9.15.1 Multipliers with small unit-circle conjugates

Fix β as in Theorem II.9.15.1, and write its conjugates as $\beta^{-1}, z, \bar{z}$, where $z = e^{i\theta}$. The number β is a unit: the product of its four conjugates is 1. Moreover, θ/π is irrational, since otherwise z , and hence its conjugate β , would be a root of unity.

For positive integers $k_1, k_2, \dots$, to be chosen, define

$$\begin{aligned} A_m &= \prod_{h=1}^m (\beta^{k_h} - \beta^{-k_h}), & b_m &= \sum_{h=1}^m k_h, \\ c_m &= \prod_{h=1}^m (z^{k_h} - z^{-k_h}), & \delta_m &= |c_m|, & L_m &= \log(1/\delta_m). \end{aligned} \tag{II.9.15.1}$$

Here and throughout this section logarithms are natural. Irrationality ensures $\delta_m > 0$, and $0 < A_m \leq \beta^{b_m}$. For $j \in \mathbb{Z}$, put $u_{m,j} = A_m \beta^j$. All these numbers are algebraic integers, because β is a unit. Their field traces are the integers

$$T_{m,j} = u_{m,j} + (-1)^m u_{m,-j} + 2 \operatorname{Re}(c_m z^j) \in \mathbb{Z}. \tag{II.9.15.2}$$

Indeed, the embedding taking β to β^{-1} takes A_m to $(-1)^m A_m$. Also $u_{m,j} \notin \mathbb{Z} + \frac{1}{2}$, since a rational algebraic integer is an integer. Thus none of the cosine factors used below vanishes.

The construction below balances the decrease of δ_m against the cosine losses generated by A_m . Eventually L_m grows by a factor greater than two while $k_{m+1} \delta_m^2$ tends to zero; these are the margins used in the loss estimate below. It also keeps b_m small compared with δ_m^{-1} , leaving room for many separated blocks of signs. A uniform lower bound on the cosines will permit a quadratic estimate for perturbations of their logarithms.

Lemma II.9.15.2. *There are positive integers k_m and a constant $\gamma > 0$ for which the numbers L_m in (II.9.15.1) are positive and satisfy the following properties. Set*

$$D = \frac{10}{\log \beta}, \quad K_m = \lceil 2b_m + DL_m \rceil \quad (m \geq 2).$$

Then $A_m \geq 1$ for every m , and

$$K_m \leq k_{m+1} \leq e^{1.9L_m}, \quad L_{m+1} \geq 1.75L_m \quad (m \geq 2), \quad (\text{II.9.15.3})$$

$$\log K_m \leq .75L_m, \quad L_{m+1} \geq 2.1L_m \quad (m \geq 3). \quad (\text{II.9.15.4})$$

Moreover,

$$|\cos(\pi u_{m,j})| \geq \gamma \quad (m \geq 1, j \in \mathbb{Z}), \quad (\text{II.9.15.5})$$

and, for $m \geq 2$,

$$2\delta_m \leq \frac{\gamma}{2\pi}, \quad \beta^{b_m - k_{m+1}} \leq e^{-10L_m} \leq \frac{\gamma}{2\pi}. \quad (\text{II.9.15.6})$$

Proof. Write $\|x\|_{\mathbb{R}/\mathbb{Z}}$ for distance to the nearest integer. The pigeonhole principle gives, for every integer $N \geq 1$, an integer $q \in [1, N]$ such that

$$\|q\theta/\pi\|_{\mathbb{R}/\mathbb{Z}} \leq N^{-1}. \quad (\text{II.9.15.7})$$

For example, place the $N + 1$ fractional parts of $0, \theta/\pi, \dots, N\theta/\pi$ in N intervals of length $1/N$. Since θ/π is irrational, the possible q in (II.9.15.7) are unbounded as $N \rightarrow \infty$. In particular there are arbitrarily large q with $\|q\theta/\pi\|_{\mathbb{R}/\mathbb{Z}} \leq 1/q$.

Choose and fix k_1 so that $A_1 > 1$ and $\delta_1 < 1/16$. Define

$$\gamma_m = \inf_{j \in \mathbb{Z}} |\cos(\pi u_{m,j})|.$$

At this point $\gamma_1 > 0$: as $j \rightarrow -\infty$, $u_{1,j} \rightarrow 0$; as $j \rightarrow +\infty$, the trace identity shows that the absolute cosine differs from $|\cos(\pi u_{1,-j})|$ by at most $2\pi\delta_1 < \pi/8$. The remaining finite set of factors contains no zero.

Next choose $k_2 > k_1$ from the unbounded sequence satisfying $\|k_2\theta/\pi\|_{\mathbb{R}/\mathbb{Z}} \leq 1/k_2$. We will impose finitely many lower thresholds on this choice. Since

$$\delta_2 \leq \frac{2\pi\delta_1}{k_2}, \quad L_2 \geq \log k_2 - \log(2\pi\delta_1),$$

both k_2 and L_2 can be made arbitrarily large. In particular, $\log K_2 \leq 1.1L_2$ for every sufficiently large such choice: the last inequality bounds k_2 by a fixed multiple of e^{L_2} , whereas the other contribution DL_2 to K_2 is only linear in L_2 .

For $m \geq 2$, suppose $L = L_m$ is large and $\log K_m \leq 1.1L$. Take $N = \lfloor e^{1.9L} \rfloor$ and choose q by (II.9.15.7). Define

$$k_{m+1} = \begin{cases} q, & q \geq K_m, \\ \lceil K_m/q \rceil q, & q < K_m. \end{cases} \quad (\text{II.9.15.8})$$

Let $L' = L_{m+1}$. If $q \geq K_m$, then

$$\delta_{m+1} \leq 2\pi\delta_m/N, \quad L' \geq 2.9L - O(1) \geq 2.8L, \quad b_{m+1} \leq 2e^{1.9L}.$$

If $q < K_m$, the multiplier $[K_m/q]$ is at most $2K_m$ and $K_m \leq k_{m+1} < 2K_m$. Using $|\sin(\pi x)| \leq \pi \|x\|_{\mathbb{R}/\mathbb{Z}}$ gives

$$\delta_{m+1} \leq 4\pi\delta_m K_m/N, \quad L' \geq 2.9L - \log K_m - O(1) \geq 1.75L,$$

and $b_{m+1} \leq 3e^{1.1L}$. In both cases $K_m \leq k_{m+1} \leq e^{1.9L}$ when L is sufficiently large.

These estimates also give $\log K_{m+1} \leq .75L'$. To see this without an upper bound on L' , write

$$K_{m+1} \leq 2b_{m+1} + DL' + 1.$$

The bound on b_{m+1} makes the first term at most $\frac{1}{2}e^{.75L'}$ for large L , since $1.9 < .75 \cdot 2.8$ in the first case and $1.1 < .75 \cdot 1.75$ in the second. The remaining term satisfies the same bound for every sufficiently large L' . Thus the construction iterates. From $m = 3$ onward, the improved bound $\log K_m \leq .75L_m$ gives $L_{m+1} \geq (2.9 - .75)L_m - O(1) \geq 2.1L_m$ in the second case; the first case already gives more. This proves (II.9.15.3)–(II.9.15.4) after a lower threshold on L_2 depending only on β .

It remains to choose k_2 large enough to obtain the cosine floor. For $j \leq 0$ and $k = k_{m+1}$, the exact identity

$$u_{m+1,j} = u_{m,j+k} - u_{m,j-k}, \quad |u_{m,j-k}| \leq \beta^{b_m-k} \beta^j \quad (\text{II.9.15.9})$$

and the Lipschitz bound for the cosine give a lower bound $\gamma_m - \pi\beta^{b_m-k}$. For positive j , reflect to $-j$ using (II.9.15.2) at level $m + 1$. Consequently

$$\gamma_{m+1} \geq \gamma_m - \pi\beta^{b_m-k_{m+1}} - 2\pi\delta_{m+1}. \quad (\text{II.9.15.10})$$

For $m \geq 2$, the inequality $k_{m+1} \geq 2b_m + DL_m$ implies $\beta^{b_m-k_{m+1}} \leq e^{-10L_m}$. Also $L_m \geq 1.75^{m-2}L_2$. Thus the sum of the decrements in (II.9.15.10) is at most

$$\pi\beta^{k_1-k_2} + 2\pi e^{-L_2} + \sum_{r=0}^{\infty} \left(\pi e^{-10 \cdot 1.75^r L_2} + 2\pi e^{-1.75^{r+1} L_2} \right).$$

This tends to zero as $k_2 \rightarrow \infty$ along the chosen sequence. Choose k_2 so that this sum is less than $\gamma_1/10$, and so that $2e^{-L_2}$ and e^{-10L_2} are at most $\gamma_1/(4\pi)$. These requirements are compatible with the earlier lower thresholds. Now fix k_2 and the resulting sequence, and put $\gamma = \gamma_1/2$. Equations (II.9.15.5)–(II.9.15.6) follow. Finally, k_m is increasing and every factor $\beta^{k_m} - \beta^{-k_m}$ is at least its first factor $A_1 > 1$, so $A_m \geq 1$. $\square$

II.9.15.2 The loss in the cosine products

Fix the sequence furnished by Lemma II.9.15.2. Constants denoted by C below may change from line to line, but are independent of m and of the phase index. Define

$$f(u) = -\log |\cos(\pi u)| \quad \text{for } u \notin \mathbb{Z} + \frac{1}{2}, \quad S_m = \sum_{j \leq 0} f(u_{m,j}).$$

Each S_m is finite: its terms have a geometrically decaying tail because $u_{m,j} \rightarrow 0$ as $j \rightarrow -\infty$ and $f(u) = O(u^2)$ near zero. The next estimate controls the entire negative-power contribution even though A_m becomes large.

Lemma II.9.15.3. *For the sequence in Lemma II.9.15.2, $S_m = o(L_m)$. Furthermore, with $\eta = .01$, for $m \geq 2$ and $n \geq 0$,*

$$\sum_{j \leq n} f(u_{m,j}) \leq (2 + \eta)S_m + Cn\delta_m^2. \quad (\text{II.9.15.11})$$

Proof. We first record a perturbation estimate. If $|\cos(\pi u)| \geq \gamma$ and $|v| \leq \gamma/(2\pi)$, then

$$f(u + v) \leq (1 + \eta)f(u) + Cv^2. \quad (\text{II.9.15.12})$$

Indeed, the absolute cosine is at least $\gamma/2$ along the segment from u to $u + v$, so f'' is bounded there. Also

$$|f'(u)| \leq \frac{\pi}{\gamma} \sqrt{2f(u)},$$

because $1 - |\cos(\pi u)|^2 \leq 2f(u)$. Taylor's formula and $|f'(u)v| \leq \eta f(u) + Cv^2$ prove (II.9.15.12).

The function f is even and 1-periodic. Reflecting positive indices by (II.9.15.2), and using (II.9.15.6), gives

$$f(u_{m,i}) \leq (1 + \eta)f(u_{m,-i}) + C\delta_m^2 \quad (m \geq 2, i \geq 1). \quad (\text{II.9.15.13})$$

Summing this inequality proves (II.9.15.11).

To estimate S_{m+1} , apply (II.9.15.12) to (II.9.15.9), with base point $u_{m,j+k}$ and $v = -u_{m,j-k}$, where $k = k_{m+1}$. The cosine floor holds at every base point, including those with positive index. The squared perturbations have a geometric sum, so

$$\begin{aligned} S_{m+1} &\leq (1 + \eta) \left(S_m + \sum_{i=1}^k f(u_{m,i}) \right) + C\beta^{2(b_m-k)} \\ &\leq (1 + \eta)(2 + \eta)S_m + C(1 + k\delta_m^2). \end{aligned}$$

By (II.9.15.3), $k\delta_m^2 \leq e^{-.1L_m} \leq 1$. Therefore $S_{m+1} \leq rS_m + C$, where $r = (1.01)(2.01) = 2.0301$. The already fixed value S_2 is finite, so iteration gives $S_m = O(r^{m-2})$. In contrast, (II.9.15.4) gives $L_m \geq 2.1^{m-3}L_3$ for $m \geq 3$. Since $r < 2.1$, their ratio tends to zero. $\square$

II.9.15.3 Independent blocks of Fourier phases

We now turn the product estimates into sets of almost full $\nu_{1/\beta}$ -measure and vanishing Lebesgue measure. The comparison of phase averages under two measures is analogous to the separation argument for Riesz products in [Pey75, Section 1.1, Theorem 1.2]. Here the needed estimates follow from the independent signs, as we now show. Work on the probability space of independent uniform signs $(\epsilon_\ell)_{\ell \geq 0}$, and write

$$Y = \sum_{\ell \geq 0} \epsilon_\ell \beta^{-\ell}, \quad W_{m,n} = \exp(i\pi A_m \beta^n Y) \quad (n \geq 0).$$

The law of Y is $\nu_{1/\beta}$, supported on $I = [-(1 - \beta^{-1})^{-1}, (1 - \beta^{-1})^{-1}]$. Independence, followed by passage to the limit in the convergent sign series, gives

$$\mathbb{E}W_{m,n} = \prod_{j=-\infty}^n \cos(\pi u_{m,j}). \quad (\text{II.9.15.14})$$

This is a nonzero real number; the negative-index tail has a convergent sum of logarithmic losses.

Set

$$J_m = \lfloor e^{1.5L_m} \rfloor, \quad H_m = \lceil b_m + DL_m \rceil,$$

$$\mathcal{N}_m = \{H_m + s(2H_m + 1) : s \in \mathbb{Z}_{\geq 0}\} \cap [0, J_m],$$

and let $M_m = |\mathcal{N}_m|$. All subsequent estimates concern sufficiently large m . Since $H_m \leq K_m \leq e^{.75L_m}$,

$$e^{.7L_m} \leq M_m \leq J_m + 1. \quad (\text{II.9.15.15})$$

Lemma II.9.15.3 and $J_m \delta_m^2 \leq e^{-.5L_m}$ show that

$$|\mathbb{E}W_{m,n}| \geq e^{-L_m/20} \quad (0 \leq n \leq J_m). \quad (\text{II.9.15.16})$$

For $n \in \mathcal{N}_m$, retain only the signs whose indices lie in $[n - H_m, n + H_m]$, and define

$$\tilde{W}_{m,n} = (-1)^{\sum_{j=H_m+1}^n T_{m,j}} \exp \left(i\pi \sum_{\ell=n-H_m}^{n+H_m} u_{m,n-\ell} \epsilon_\ell \right). \quad (\text{II.9.15.17})$$

The integer sum in the prefactor is empty when $n = H_m$. To explain this approximation, expand the exponent of $W_{m,n}$ as $\pi \sum_{j \leq n} u_{m,j} \epsilon_{n-j}$. For $j > H_m$, replace $u_{m,j}$ by its integer trace $T_{m,j}$; the resulting factor is deterministic because $e^{i\pi T_{m,j} \epsilon_{n-j}} = (-1)^{T_{m,j}}$. For $j < -H_m$, drop the term. The remaining indices are exactly those in (II.9.15.17).

The difference of the two phase angles, after these replacements, is a sum of independent mean-zero signs. The inequality $|e^{ix} - e^{iy}| \leq |x - y|$ therefore gives

$$\begin{aligned} \|W_{m,n} - \tilde{W}_{m,n}\|_2^2 &\leq \pi^2 \left(\sum_{j=H_m+1}^n |u_{m,j} - T_{m,j}|^2 + \sum_{j < -H_m} |u_{m,j}|^2 \right) \\ &\leq C \left(\beta^{2(b_m - H_m)} + n \delta_m^2 \right) \leq C e^{-L_m/2}. \end{aligned} \quad (\text{II.9.15.18})$$

Here (II.9.15.2) bounds the first summand by a geometric tail and a contribution $Cn\delta_m^2$; the other sum is itself a geometric tail. Finally, $D \log \beta = 10$ and $n \leq J_m$ give the last inequality. All norms in (II.9.15.18) refer to the sign probability space. The windows belonging to consecutive members of $\mathcal{N}_m$ are disjoint, so the variables $\tilde{W}_{m,n}$ are independent.

Proof of Theorem II.9.15.1. For $n \in \mathcal{N}_m$, let $\sigma_{m,n} \in \{-1, 1\}$ be the sign of the real expectation in (II.9.15.14). Define the continuous function

$$F_m(y) = \frac{1}{M_m} \sum_{n \in \mathcal{N}_m} \sigma_{m,n} e^{i\pi A_m \beta^n y}.$$

Its mean at Y is real and, by (II.9.15.16), $a_m := \mathbb{E}F_m(Y) \geq e^{-L_m/20}$. The corresponding average of the independent variables $\sigma_{m,n} \tilde{W}_{m,n}$ has centered L^2 norm at most $M_m^{-1/2}$, since each variable has modulus one. By the triangle inequality and (II.9.15.18),

$$\|F_m(Y) - a_m\|_2 \leq M_m^{-1/2} + C e^{-L_m/4}. \quad (\text{II.9.15.19})$$

Subtracting the means of the approximation errors introduces at most a factor two, included in C .

On the other hand, the same average is small in Lebesgue $L^2(I)$. Write its frequencies in increasing order as $\xi_1 < \dots < \xi_{M_m}$. Since $A_m \geq 1$ and their exponents are distinct nonnegative integers, $\xi_{r+1} - \xi_r \geq \pi(\beta - 1)$. Thus $|\xi_r - \xi_s| \geq \pi(\beta - 1)|r - s|$ and

$$\left| \int_I e^{i(\xi_r - \xi_s)y} dy \right| \leq \frac{2}{|\xi_r - \xi_s|} \quad (r \neq s).$$

Expanding the square and summing by the rank difference yields

$$\int_I |F_m(y)|^2 dy \leq C \frac{1 + \log M_m}{M_m}. \quad (\text{II.9.15.20})$$

Consider the compact sets

$$E_m = \{y \in I : \operatorname{Re} F_m(y) \geq \frac{1}{2} e^{-L_m/20}\}.$$

Chebyshev's inequality, (II.9.15.19), and (II.9.15.15) imply

$$1 - \nu_{1/\beta}(E_m) \leq 4e^{L_m/10} \left(M_m^{-1/2} + Ce^{-L_m/4} \right)^2 \rightarrow 0.$$

Likewise, writing $|E_m|$ for Lebesgue measure,

$$|E_m| \leq 4e^{L_m/10} \int_I |F_m(y)|^2 dy \leq Ce^{L_m/10} \frac{1 + \log M_m}{M_m} \rightarrow 0,$$

because $M_m \geq e^{7L_m}$ and $\log M_m \leq 1.5L_m + O(1)$. Choose a subsequence (m_r) along which $\sum_r (|E_{m_r}| + 1 - \nu_{1/\beta}(E_{m_r})) < \infty$. Then $E = \limsup_r E_{m_r}$ is Borel and Lebesgue null, since $|\bigcup_{r \geq R} E_{m_r}| \leq \sum_{r \geq R} |E_{m_r}| \rightarrow 0$. Moreover, the summability of the complementary probabilities shows that $\nu_{1/\beta}$ -almost every point lies in all sufficiently late E_{m_r} , and hence in E . Therefore $\nu_{1/\beta}(E) = 1$. $\square$

II.9.15.4 Two explicit parameters

Corollary II.9.15.4. *Let β_1 and β_2 be the respective real roots greater than one of*

$$P_1(x) = x^4 - x^3 - x^2 - x + 1, \quad P_2(x) = x^4 - 2x^3 + x^2 - 2x + 1.$$

Then $\beta_1, \beta_2 \in (1, 2)$ and both ν_{1/β_1} and ν_{1/β_2} are singular.

Proof. For $y = x + x^{-1}$, the equations $P_i(x)/x^2 = 0$ become, respectively,

$$y^2 - y - 3 = 0, \quad y^2 - 2y - 1 = 0.$$

Their larger roots are $y_1 = (1 + \sqrt{13})/2$ and $y_2 = 1 + \sqrt{2}$, both in $(2, 5/2)$. For each i , the equation $x + x^{-1} = y_i$ has one root $\beta_i \in (1, 2)$ and its reciprocal. The other quadratic root

y'_i lies in $(-2, 2)$, so the remaining two roots of P_i form a nonreal conjugate pair on the unit circle.

It remains to check that these four roots are algebraic conjugates. The quadratic field $\mathbb{Q}(y_i)$ is real, and the discriminant $y_i^2 - 4$ has negative image $(y'_i)^2 - 4$ under its nontrivial embedding. It cannot be a square in that field, since the image of a square under a real embedding is nonnegative. Hence $x^2 - y_i x + 1$ is irreducible over $\mathbb{Q}(y_i)$, and $[\mathbb{Q}(\beta_i) : \mathbb{Q}] = 4$. Thus β_1 and β_2 are degree-four Salem numbers, and Theorem II.9.15.1 applies. $\square$

II.9.16 A polynomial with a weakly expanding conjugate pair

The final example uses a polynomial with one conjugate pair just outside the unit circle. We will prove singularity directly by counting typical prefixes. Counting all polynomial values in their conjugate coordinates therefore gives slightly too many possible prefixes. We obtain the required improvement by a product of positive trigonometric factors: its integral in the expanding coordinates stays bounded, whereas it grows exponentially on a set of words whose probability tends to one.

Throughout this section, every terminating decimal denotes the corresponding exact rational number. Put

$$p(t) = t^{31} - 2t^{30} + \sum_{j=0}^9 (t^{3j+1} - t^{3j}). \quad (\text{II.9.16.1})$$

The identity

$$(t^2 + t + 1)p(t) = t^{33} - t^{32} - t^{31} - t^{30} - 1 \quad (\text{II.9.16.2})$$

will control the near-cancellations in the trigonometric product.

Theorem II.9.16.1. *The polynomial p in (II.9.16.1) has a unique real root $\beta \in (1.8392, 1.8393)$. This root is not a Pisot number, and $v_{1/\beta}$ is singular with respect to Lebesgue measure.*

We first record the root bounds needed in the proof. The proposition is proved by rational root disks in Section II.9.17.1.

Proposition II.9.16.2 (Root bounds). *All roots of p are simple. Its roots outside the unit circle are $\beta, \alpha, \bar{\alpha}$, where β is real, $\text{Im } \alpha > 0$, and*

$$\begin{aligned} 1.8392 < \beta < 1.8393, \quad 1.0002089 < |\alpha|^2 < 1.0002094, \\ |\alpha|^{-1} < 0.9999. \end{aligned} \quad (\text{II.9.16.3})$$

The remaining roots are fourteen nonreal conjugate pairs, each of modulus less than 0.994.

Write $D = \{\beta, \alpha, \bar{\alpha}\}$ and let I be the set of remaining roots. The expanding volume factor is

$$\mathcal{M} = \prod_{r \in D} |r| = \beta |\alpha|^2, \quad \log \mathcal{M} < \log \beta + 0.0002094. \quad (\text{II.9.16.4})$$

The last inequality uses $\log(1+u) < u$ for $u > 0$.

The root bounds already show that β is not Pisot. Indeed, suppose β were Pisot, and divide p by its monic minimal polynomial. The quotient is monic and integral, has constant term of absolute value one, and contains $\alpha, \bar{\alpha}$ among its roots but not β . The product of the moduli of its roots is greater than one if it has no roots in I , and otherwise is at most $1.0002094 \cdot 0.994 < 1$. Both conclusions contradict the constant term. It remains to prove singularity.

II.9.16.1 Word vectors and characters

A full lattice in a k -dimensional real vector space is the image of $\mathbb{Z}^k$ under a linear isomorphism from $\mathbb{R}^k$. By a modulated character we mean a function $c \exp(i\ell(x))$, with $|c| = 1$ and ℓ a real-linear functional; the constant is a centering phase. The characters below are of this modulated form.

For a word $a = (a_0, \dots, a_{N-1}) \in \{0, 1\}^N$, define

$$y_r(a) = \sum_{j=0}^{N-1} a_j r^j \quad (r \in D \cup I), \quad x_r(a) = r^{-N} y_r(a) \quad (r \in D). \quad (\text{II.9.16.5})$$

Vectors indexed by roots are always constrained by $y_{\bar{r}} = \overline{y_r}$, and a conjugate pair counts as one complex coordinate, or two real coordinates. In particular, the space of all y -vectors has real dimension 31.

Lemma II.9.16.3. *All vectors $y(a)$, for all N and all words a , belong to a fixed full lattice in this real vector space. Their contracting coordinates $(y_r)_{r \in I}$ and normalized expanding coordinates $(x_r)_{r \in D}$ lie in fixed bounded boxes. The change from (x_D, y_I) to (y_D, y_I) has real Jacobian determinant of absolute value $\mathcal{M}^N$.*

Proof. Reduction of $\sum a_j t^j$ modulo the monic polynomial p gives an integer coefficient vector of length 31. Evaluation at the roots is an invertible real-linear map: a polynomial of degree at most 30 that vanishes at all 31 distinct roots is zero. The image of $\mathbb{Z}^{31}$ under this map is the required lattice. No irreducibility assumption is needed. The bounds follow from

$$|y_r(a)| \leq \frac{1}{1-|r|} \quad (r \in I), \quad |x_r(a)| \leq \frac{1}{|r|-1} \quad (r \in D).$$

Multiplication of the real coordinate by β^N and of the complex coordinate by α^N has determinant $\beta^N |\alpha|^{2N} = \mathcal{M}^N$. $\square$

To distinguish the word vectors from typical points of these boxes, define the real sequences

$$C_0(m) = \begin{cases} \sum_{r \in I} \frac{r^{-1-m}}{p'(r)}, & m < 0, \\ -\sum_{r \in D} \frac{r^{-1-m}}{p'(r)}, & m \geq 0, \end{cases} \quad C_1(m) = C_0(m+3) - C_0(m) + C_0(m-1). \quad (\text{II.9.16.6})$$

Partial fractions give

$$\frac{1}{p(t)} = \sum_{m \in \mathbb{Z}} C_0(m) t^m, \quad \max_{r \in I} |r| < |t| < \min_{r \in D} |r|.$$

Thus C_0 is the bilateral Laurent coefficient sequence of $1/p$, and both C_0, C_1 decay exponentially at both ends. The same inversion by an exponentially decaying two-sided sequence appears in [LS99, Lemma 4.5 and Example 4.7]. Here the explicit root formula will give characters adapted to the word vectors.

For arbitrary expanding coordinates x_D , with $x_\beta \in \mathbb{R}$ and $x_{\bar{\alpha}} = \overline{x_\alpha}$, put

$$\begin{aligned} Z_i^{(0,N)}(x) &= \exp\left(-2\pi i \sum_{r \in D} \frac{x_r r^i}{p'(r)}\right) \exp\left(-\pi i \sum_{k=0}^{N-1} C_0(k-i)\right), \\ Z_i^{(1,N)}(x) &= Z_{i-3}^{(0,N)}(x) \overline{Z_i^{(0,N)}(x)} Z_{i+1}^{(0,N)}(x), \quad i \in \mathbb{Z}. \end{aligned} \quad (\text{II.9.16.7})$$

These functions have modulus one.

Lemma II.9.16.4 (Characters at word vectors). *If $b_k = a_{N-1-k}$, then for $A \in \{0, 1\}$ and $i \in \mathbb{Z}$,*

$$Z_i^{(A,N)}(x(a)) = \exp\left(\pi i \sum_{k=0}^{N-1} C_A(k-i)(2b_k - 1)\right). \quad (\text{II.9.16.8})$$

Proof. The reversed word gives $x_r(a) = \sum_{k=0}^{N-1} b_k r^{-1-k}$. For each integer $l \geq 0$, Lagrange interpolation shows that

$$\sum_{r \in D \cup I} \frac{r^l}{p'(r)}$$

is the coefficient of t^{30} in the remainder of t^l modulo p , and hence is an integer. Consequently, when $m < 0$, the coefficient $-\sum_{r \in D} r^{-1-m}/p'(r)$ differs from $C_0(m)$ by an integer. Its contribution to the exponent with coefficient $2\pi i b_k$ is unchanged by this replacement. When $m \geq 0$, the coefficients already agree. The second factor in (II.9.16.7) supplies the centering by -1 , proving the formula for $A = 0$. Multiplication of the three characters proves it for $A = 1$. $\square$

The corresponding stationary model uses independent fair bits $(b_k)_{k \in \mathbb{Z}}$ and the unit random variables

$$\mathcal{Z}_i^{(A)} = \exp\left(\pi i \sum_{k \in \mathbb{Z}} C_A(k-i)(2b_k - 1)\right). \quad (\text{II.9.16.9})$$

The series converges absolutely. Independence gives the real moments

$$\begin{aligned} P_A &= \mathbb{E} \mathcal{Z}_i^{(A)} = \prod_{m \in \mathbb{Z}} \cos(\pi C_A(m)), \\ P_{AB}^\pm(k) &= \mathbb{E}[\mathcal{Z}_i^{(A)} (\mathcal{Z}_{i-k}^{(B)})^{\pm 1}] \\ &= \prod_{m \in \mathbb{Z}} \cos(\pi(C_A(m) \pm C_B(m+k))), \quad k \geq 0. \end{aligned} \quad (\text{II.9.16.10})$$

II.9.16.2 A predictor with a positive mean gain

The weakly expanding pair produces long decaying tails in C_0 . Some shifted tails nearly cancel in the signed products $P_{00}^\pm(k)$. A linear combination of the corresponding phases will use these correlations. For rational weights w_{ks} , define

$$\begin{aligned} G_i^{(N)}(x) &= -0.001 + 0.0055 Z_{i-33}^{(1,N)}(x) + \sum_{k=30}^{2000} \sum_{s=0}^1 w_{ks} (Z_{i-k}^{(0,N)}(x))^{1-2s}, \\ \mathcal{G}_i &= -0.001 + 0.0055 \mathcal{Z}_{i-33}^{(1)} + \sum_{k=30}^{2000} \sum_{s=0}^1 w_{ks} (\mathcal{Z}_{i-k}^{(0)})^{1-2s}, \\ \mathcal{L}_i &= \operatorname{Re}(\mathcal{Z}_i^{(0)} \mathcal{G}_i). \end{aligned} \tag{II.9.16.11}$$

Proposition II.9.16.5 (Predictor estimates). *There exist weights $w_{ks} \in 10^{-9}\mathbb{Z}$, indexed by $30 \leq k \leq 2000$ and $s \in \{0, 1\}$, such that*

$$0.001 + 0.0055 + \sum_{k,s} |w_{ks}| = \frac{198442946}{10^9} < x_* := 0.199 \tag{II.9.16.12}$$

and, in the stationary model,

$$\begin{aligned} \mathbb{E} \mathcal{L}_i &> 0.000210, & \mathbb{E} |\mathcal{G}_i|^2 &< 0.000145, \\ \left| \mathbb{E} [(\mathcal{Z}_i^{(0)})^2 \mathcal{G}_i^2] \right| &< 0.000007. \end{aligned} \tag{II.9.16.13}$$

In particular, $|G_i^{(N)}(x)| \leq x_*$ for every N, i, x , and $|\mathcal{G}_i| \leq x_*$ for every bit sequence.

We fix weights supplied by Proposition II.9.16.5. Their exact rational construction is given in Section II.9.17.2, and the moment bounds are proved in Section II.9.17.4.

The next two subsections construct a positive product whose logarithm has a positive mean under this stationary model and whose integral is bounded in the expanding coordinates. These two estimates will remove an exponential proportion of the lattice points allowed by Lemma II.9.16.3.

II.9.16.3 Damping the possible frequency cancellations

Choose a fixed integer $i_0 \geq 1$ so that

$$\frac{0.1\beta^{i_0}}{|p'(\beta)|} > 1. \tag{II.9.16.14}$$

For a word length N , use every complete block

$$B_q = \{i_0 + 22q, \dots, i_0 + 22q + 20\} \subseteq \{0, \dots, N-1\}, \quad q \geq 0,$$

and write J_N for their union. Thus each block has 21 indices and consecutive blocks are separated by one unused index.

The near-relation (II.9.16.2) dictates which signed digit patterns need to be retained when integrating a block. For a pattern $(d_i)_{i \in B_q} \in \{-1, 0, 1\}^{B_q}$, let $b = \max B_q$ and set

$$v_i = \beta^{-i} \sum_{j=i}^b d_j \beta^j, \quad v_{b+1} = 0.$$

Reading from the largest index downwards gives $v_i = \beta v_{i+1} + d_i$. At the root β , (II.9.16.2) becomes

$$\beta^3 - \beta^2 - \beta - 1 = \beta^{-30}. \quad (\text{II.9.16.15})$$

In particular, the four digits $[\sigma, -\sigma, -\sigma, -\sigma]$ leave the small normalized remainder $\sigma\beta^{-30}$. We encode these approximate returns by the transitions in Figure II.9.1. The initial state is 0, and the permitted terminal states are 0, C_+ , C_- . For each sign $\sigma \in \{-1, 1\}$, the numerical values assigned to the states are

$$0, \quad A_\sigma : \sigma, \quad B_\sigma : \sigma(\beta - 1), \quad C_\sigma : \sigma(\beta^2 - \beta - 1). \quad (\text{II.9.16.16})$$

A pattern is called *admissible* if every transition is allowed and its terminal state is permitted.

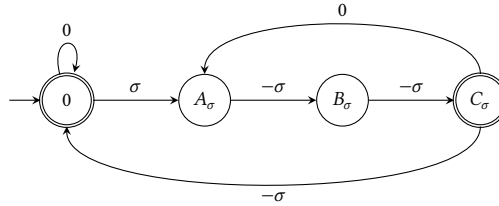


Figure II.9.1: Admissible digits, read from high exponents to low exponents. There are two copies of the three nonzero states, one for each sign σ , sharing state 0. Edge labels are digits; double circles mark permitted terminal states. A return to 0 may be followed by either sign.

Let $o(d) = \#\{i : d_i \neq 0\}$. At a position i of a block, define

$$\eta_i = \sum_{\substack{d \text{ admissible} \\ d_i \neq 0}} \frac{x_*^{o(d)-2}}{o(d)}. \quad (\text{II.9.16.17})$$

The same coefficients are repeated in every block. An admissible nonzero pattern has at least three nonzero digits, so all summands are well defined.

Lemma II.9.16.6 (Cost of admissible patterns). *Set*

$$S_* = \left(1 + \frac{2x_*^4}{1 - x_*^2}\right)^{18} \left(1 + \frac{2x_*^3}{1 - x_*^2}\right) - 1. \quad (\text{II.9.16.18})$$

Then $S_* < 0.081$, and

$$\begin{aligned} \sum_{\substack{d \text{ admissible} \\ o(d) > 0}} x_*^{o(d)} &\leq S_*, & \eta_i x_*^2 &\leq S_*/3, \\ \frac{\bar{\eta}}{1 - S_*/3} &< 0.101, & \bar{\eta} &= \frac{1}{21} \sum_{i \in B_q} \eta_i. \end{aligned} \quad (\text{II.9.16.19})$$

Proof. A completed excursion from 0 to 0 has the digit string

$$[\sigma, -\sigma, -\sigma, (0, -\sigma, -\sigma)^l, -\sigma], \quad l \geq 0.$$

It has $4 + 2l$ nonzero digits and can begin in at most 18 positions of a block. An optional final excursion ending at C_σ has the same form without the last digit; its sign and length determine its starting position. Ignoring incompatibilities between these choices overcounts the admissible patterns and gives (II.9.16.18). At $x_* = 0.199$, the two fractions in that expression are less than 0.00327 and 0.01642. The inequality $(1 + a)^{18} \leq (1 - 18a)^{-1}$, valid for $0 \leq 18a < 1$, proves $S_* < 0.081$ by rational arithmetic.

Since $o(d) \geq 3$, each $\eta_i x_*^2 \leq S_*/3$. Summing (II.9.16.17) over positions gives $\sum_i \eta_i \leq S_*/x_*^2$. Thus

$$\frac{\bar{\eta}}{1 - S_*/3} \leq \frac{0.081}{21(0.199)^2(1 - 0.081/3)} < 0.101.$$

□

For the moment fix N , and abbreviate $Z_i = Z_i^{(0,N)}$, $G_i = G_i^{(N)}$, and $x_i = |G_i|$. Define

$$h_i = 1 - \eta_i x_i^2, \quad F_i = h_i(1 + (1 - x_i^2)(Z_i G_i + \overline{Z_i G_i})), \quad i \in J_N. \quad (\text{II.9.16.20})$$

These factors are bounded above and bounded away from zero uniformly in N, i, x_D . Indeed, $h_i \geq 1 - S_*/3$, and the other factor is at least $1 - 2x_* > 0$.

Proposition II.9.16.7 (Bounded integral). *For every N and every fixed complex coordinate x_α ,*

$$\int_{\mathbb{R}} \psi(x_\beta) \prod_{i \in J_N} F_i(x_D) dx_\beta \leq \int_{\mathbb{R}} \psi(x_\beta) dx_\beta, \quad \psi(v) = \left(\frac{\sin(\pi v/10)}{\pi v/10} \right)^2, \quad (\text{II.9.16.21})$$

where $\psi(0) = 1$.

Proof. We use the Fourier convention $e^{2\pi i \xi x_\beta}$. The nonnegative integrable function ψ has Fourier transform supported in $[-0.1, 0.1]$, hence in $[-1, 1]$. On suppressing the common frequency factor $-1/p'(\beta)$, the frequency of Z_i is β^i . Every phase appearing in G_i has frequency of absolute value at most β^{i-30} . For the type-1 term this follows from

$$|\beta^{i-33}(\beta^{-3} - 1 + \beta)| \leq \beta^{i-30}.$$

We call these predictor frequencies slow. Expanding a factor F_i gives monomials with one fast digit $d_i \in \{-1, 0, 1\}$, corresponding to $Z_i^{d_i}$, and at most five slow frequencies: two from h_i , two from $1 - |G_i|^2$, and one from G_i or its conjugate.

Expand the last block according to its fast digits, keeping all earlier factors for the moment. We first show that every inadmissible pattern has zero integral, even after multiplication by those earlier factors. Expand the latter and all slow pieces into monomials. If the current index is i , the fast sum accumulated from the top of the last block, divided by β^i , follows the recurrence $v \mapsto \beta v + d_i$. By (II.9.16.15), each allowed transition agrees with the state values (II.9.16.16) up to an injected error of absolute value at most β^{-30} . The propagated error, including at a first forbidden step, is bounded by

$$\frac{\beta^{-30} \beta^{21}}{\beta - 1} < 0.005. \quad (\text{II.9.16.22})$$

At a first forbidden step, the ideal absolute value is at least $\beta(\beta - 1)$ for a departure from A_σ or B_σ , and at least 2 for a departure from C_σ . After (II.9.16.22), it is greater than 1.5. All lower fast digits together contribute at most $(\beta - 1)^{-1} < 1.192$ in the same units. If instead all transitions are allowed but the terminal state is A_σ or B_σ , the absolute value at the bottom index i is at least $\beta - 1 - 0.005$. The one-index gap then bounds the earlier fast terms by $1/(\beta(\beta - 1)) < 0.649$.

In either case, all slow terms, including those from earlier blocks, contribute at most

$$\frac{5\beta^{-30} \beta^{21}}{\beta - 1} < 0.025 \quad (\text{II.9.16.23})$$

relative to β^i . To see this, sum at most five frequencies of size β^{j-30} over indices j no larger than the top of the last block; that top is at most $i + 20$. Thus the full frequency has absolute value greater than $0.1\beta^i/|p'(\beta)| > 1$. Its integral against ψ is zero.

After removing these zero integrals, the last block contributes $\prod_{i \in B_q} h_i$ times the sum over admissible fast patterns. The zero pattern contributes one. A nonzero pattern has absolute value at most $\prod_{d_i \neq 0} x_i$, since $0 \leq 1 - x_i^2 \leq 1$. The arithmetic-geometric mean inequality gives

$$\prod_{d_i \neq 0} x_i \leq x_*^{o(d)-2} \frac{1}{o(d)} \sum_{d_i \neq 0} x_i^2.$$

The absolute value of the retained pattern sum is therefore at most $1 + \sum_{i \in B_q} \eta_i x_i^2$. Writing $t_i = \eta_i x_i^2$, we have $0 \leq t_i < 1$ and

$$\prod_{i \in B_q} (1 - t_i) \left(1 + \sum_{i \in B_q} t_i \right) \leq 1;$$

for example, use $\prod(1 - t_i) \leq \exp(-\sum t_i)$. Since ψ and all earlier factors are nonnegative, removing the last block can only increase the upper bound for the integral. Repeat with the preceding blocks to obtain (II.9.16.21). $\square$

II.9.16.4 A typical exponential gain

We now evaluate the same factors in the stationary model, replacing Z_i, G_i in (II.9.16.20) by $\mathcal{Z}_i^{(0)}, \mathcal{G}_i$; denote the resulting factors by $\mathcal{F}_i$. The damping coefficients remain periodic with period 22, and are used at the same 21 positions per period.

Lemma II.9.16.8 (A logarithmic estimate). *For $0 \leq x \leq x_*$ and $|u| \leq 2x$,*

$$\log(1 + (1 - x^2)u) \geq u - \frac{u^2}{2} - C_* x^3, \quad C_* := \frac{2}{3\sqrt{1 - x_*^2}}. \quad (\text{II.9.16.24})$$

Proof. The assertion is immediate for $x = 0$. Otherwise let $D_x(u) = u - u^2/2 - \log(1 + (1 - x^2)u)$. Its derivative is

$$D'_x(u) = \frac{x^2 - x^2 u - (1 - x^2)u^2}{1 + (1 - x^2)u}.$$

For $u \geq 0$, this is at most the positive part of $x^2 - (1 - x^2)u^2$. Integrating that positive part on $[0, \infty)$ gives $2x^3/(3\sqrt{1 - x^2}) \leq C_* x^3$. On $[-2x, 0]$, the only possible interior critical point is a minimum, so the maximum is at an endpoint. We have $D_x(0) = 0$, whereas

$$\frac{d}{dx} D_x(-2x) = \frac{2x^2(1 - 2x - 4x^2)}{1 - 2x + 2x^3} \leq 2x^2.$$

As $D_0(0) = 0$, integration gives $D_x(-2x) \leq 2x^3/3 \leq C_* x^3$. □

Proposition II.9.16.9 (Growth on typical words). *For uniformly distributed words $a \in \{0, 1\}^N$,*

$$\Pr \left\{ \sum_{i \in J_N} \log F_i(x(a)) > 0.000220N \right\} \longrightarrow 1. \quad (\text{II.9.16.25})$$

Proof. Apply Lemma II.9.16.8 with $x = |\mathcal{G}_i|$ and $u = 2\mathcal{L}_i$. The identity

$$2\mathbb{E}\mathcal{L}_i^2 = \mathbb{E}|\mathcal{G}_i|^2 + \operatorname{Re} \mathbb{E}[(\mathcal{Z}_i^{(0)})^2 \mathcal{G}_i^2]$$

and the bound $\mathbb{E}|\mathcal{G}_i|^3 \leq x_* \mathbb{E}|\mathcal{G}_i|^2$ control the second and third order terms. Also

$$\log(1 - \eta_i |\mathcal{G}_i|^2) \geq -\frac{\eta_i |\mathcal{G}_i|^2}{1 - S_*/3}.$$

Set $\gamma = 22^{-1} \sum_{i \in B_0} \mathbb{E} \log \mathcal{F}_i$, where the division by 22 includes the unused index. Since $C_* x_* < 0.136$, Proposition II.9.16.5 and Lemma II.9.16.6 give the stationary mean rate γ with

$$\begin{aligned} \gamma &\geq \frac{21}{22} \left(2\mathbb{E}\mathcal{L}_0 - \left(1 + C_* x_* + \frac{\bar{\eta}}{1 - S_*/3} \right) \mathbb{E}|\mathcal{G}_0|^2 - \left| \mathbb{E}[(\mathcal{Z}_0^{(0)})^2 \mathcal{G}_0^2] \right| \right) \\ &> \frac{21}{22} (2(0.000210) - 1.237(0.000145) - 0.000007) > 0.000222. \end{aligned} \quad (\text{II.9.16.26})$$

For completeness, we justify passage from this mean to finite words. Truncate each sequence C_A to $[-H, H]$, for a fixed integer H . The stationary logarithmic block sums formed from these truncated sequences depend on finitely many bits. They are uniformly bounded, have the same mean from block to block, and are independent when their blocks are sufficiently far apart. The variance of an average of Q such block sums is therefore $O_H(Q^{-1})$. Their averages converge in probability to their means.

Absolute summability of C_0, C_1 makes the truncated characters uniformly close to the original characters as $H \rightarrow \infty$. The predictors are finite sums with bounded total absolute weight, and the factors are uniformly bounded away from zero. Consequently the logarithms also converge uniformly under this truncation. Letting $H \rightarrow \infty$ after $N \rightarrow \infty$ proves

$$\frac{1}{N} \sum_{i \in J_N} \log \mathcal{F}_i \rightarrow \gamma \quad \text{in probability.}$$

Couple the reversed word bits to $(b_0, \dots, b_{N-1})$ in this stationary sequence. For fixed H , the truncated finite and stationary characters used in a factor agree whenever $H + 2000 \leq i \leq N - 1 - H$. Only a number of boundary indices depending on H , not on N , remain. The same uniform approximation thus shows

$$\frac{1}{N} \left(\sum_{i \in J_N} \log F_i(x(a)) - \sum_{i \in J_N} \log \mathcal{F}_i \right) \rightarrow 0$$

uniformly over the coupled bit sequences. Together with (II.9.16.26), this proves (II.9.16.25). $\square$

II.9.16.5 Counting typical vectors and proving singularity

The integral bound applies to continuous coordinates, whereas Proposition II.9.16.9 concerns lattice points. The remaining step is to thicken each such point by a fixed radius. Normalization in the expanding directions makes the resulting change in the whole logarithmic product bounded independently of the word length.

Lemma II.9.16.10 (Thickening word vectors). *There are constants $\rho > 0$ and $C_{\text{thick}} < \infty$, independent of N , such that the radius- ρ balls about distinct lattice points in Lemma II.9.16.3 are disjoint, and, whenever y' lies in the ball about $y(a)$,*

$$\left| \sum_{i \in J_N} \log F_i(x'_D) - \sum_{i \in J_N} \log F_i(x_D(a)) \right| \leq C_{\text{thick}}, \quad x'_r = r^{-N} y'_r \quad (r \in D). \quad (\text{II.9.16.27})$$

Proof. Choose $\rho \leq 1$ smaller than half the minimum distance between distinct points of the lattice. A coordinate change of size at most one in y changes the exponent of $Z_j^{(0,N)}$ by at most

$$2\pi \sum_{r \in D} \frac{|r|^{j-N}}{|p'(r)|}.$$

Every character occurring in F_i , including those in the type-1 term of the predictor, has index at most i . The logarithm of F_i is a uniformly Lipschitz function of these unit phases,

since the coefficient sums and damping coefficients are bounded and F_i is uniformly positive. Its variation is therefore at most a fixed constant times $\sum_{r \in D} |r|^{i-N}$. Finally,

$$\sum_{i \in J_N} |r|^{i-N} \leq \sum_{i=0}^{N-1} |r|^{i-N} < \frac{1}{|r| - 1} \quad (r \in D).$$

This proves a bound independent of N . $\square$

Proof of Theorem II.9.16.1. The root assertion and the fact that β is not Pisot were proved above. Call a word typical if it satisfies the inequality in (II.9.16.25), and let $\mathcal{Y}_N$ be the set of distinct vectors $y(a)$ obtained from typical words. Typical words have probability tending to one. Lemma II.9.16.10 implies that, for all sufficiently large N , the product $\prod_{i \in J_N} F_i$ is at least $\exp(0.000216N)$ throughout each of the disjoint balls about $\mathcal{Y}_N$.

All these balls lie in fixed bounded boxes in (x_D, y_I) coordinates, enlarged from those of Lemma II.9.16.3. In the real expanding coordinate one may take $|x_\beta| \leq 3$. The function ψ has a positive minimum on $[-3, 3]$. Integrating first in x_β , Proposition II.9.16.7 bounds the integral of the product over this fixed box by a constant independent of N . The product does not depend on y_I . Changing back to y coordinates contributes the Jacobian $\mathcal{M}^N$. Comparing with the disjoint balls, each of the same positive volume, gives

$$\#\mathcal{Y}_N \leq C_{\text{count}} \exp((\log \mathcal{M} - 0.000216)N) \quad (\text{II.9.16.28})$$

for a constant C_{count} .

Let μ be the law of $\sum_{k \geq 0} b_k \beta^{-k}$ for independent fair bits. By reversal in (II.9.16.5), its length- N prefix has the distribution of $\beta x_\beta(a)$. Projecting $\mathcal{Y}_N$ to this coordinate cannot increase its cardinality. For each resulting prefix value, attach an interval of length $\beta^{-N}/(1 - \beta^{-1})$, which contains all possible remaining tails. Their union K_N is compact, $\mu(K_N) \rightarrow 1$, and (II.9.16.4) and (II.9.16.28) give

$$|K_N| \leq C_{\text{cover}} \exp((0.0002094 - 0.000216)N) = C_{\text{cover}} e^{-0.0000066N}. \quad (\text{II.9.16.29})$$

Choose a subsequence N_j with $\mu(K_{N_j}) > 1 - 2^{-j}$. Then $\liminf_j K_{N_j}$ has full μ -measure and zero Lebesgue measure. Thus μ is singular. Finally,

$$\sum_{k \geq 0} (2b_k - 1)\beta^{-k} = 2 \sum_{k \geq 0} b_k \beta^{-k} - \frac{1}{1 - \beta^{-1}},$$

so its law $\nu_{1/\beta}$ is an affine image of μ and is singular as well. $\square$

II.9.17 A rational certificate for the degree-31 example

This certificate section proves the root bounds in Proposition II.9.16.2 and the predictor estimates in Proposition II.9.16.5. The root calculation uses exact rational arithmetic. The moment calculation combines finite rational products with an analytic estimate for their infinite tails. We specify the rounding and the order of operations so that the rational weights in (II.9.17.7) are completely determined. The accompanying script `volume-II/checks/verify_degree31.py` implements these instructions using only Python integers and rational fractions; the recurrences below give the complete certificate independently of that implementation. Every terminating decimal in this certificate section denotes its exact rational value.

II.9.17.1 Locating all the roots

Recall the polynomial

$$p(t) = \sum_{j=0}^9 (-t^{3j} + t^{3j+1}) - 2t^{30} + t^{31}.$$

Put $E = 10^{40}$. The entries in Table II.9.1 are the real and imaginary numerators of centers $z = (a + ib)/E$. Include the conjugate of each nonreal center. Thus the table specifies 31 disks, all of radius $\delta = 10^{-35}$. Denote its first and sixth centers by z_β and z_α , respectively.

Real numerator	Imaginary numerator
18392867573155250211326543340571993487329	0
-9741384652837567528189111907762645564212	960611633124778624015947511546714789832
-9390979934023233037686839914141181659013	2848063934279278841200845525848576791934
-8707010393241200595697704637503228167256	4634358961359777463662031763152574795910
-7723118558199735333585947045021501410451	6250214494152701297733561366714273855663
-6480873022754265587670654805348739312397	7617033927823246411701558316834470378693
-3296776282962928450195355562079974291919	9362810613239308448752821856735617305389
-1442513481645415489053007250852371013818	9733039088635544125591234293715446465839
473579005983151143865690217073943282984	9759329295153416086962240697595385836176
2372858204750359429049351486827380431084	9430908396110323024329804770889268443510
418406166245444789699337418151299054853	8750376307352230997951514116951420713869
5840543823426662247726359012202878127980	7733999914049269380846159864123679516166
7280042164772721585016920411987150804420	6409668313572020921539025901166241650504
8444162922855313595050499086097221186662	4814843758334683329011653039104490553475
9275649862302447872891854755866108397375	2996533501299661135726051235425500195711
9715324892541617145821300300817663390044	1019107767839771580294306582422817927060

Table II.9.1: Rational centers for the roots; both columns have denominator 10^{40} .

For a complex number write $\|z\|_1 = |\Re z| + |\Im z|$ and $\|z\|_\infty = \max(|\Re z|, |\Im z|)$. Exact Horner evaluation gives, at every displayed center,

$$|z| < 2, \quad \|p'(z)\|_\infty > 1, \quad 2 \cdot 10^{35} \|p(z)\|_1 < \|p'(z)\|_\infty. \quad (\text{II.9.17.1})$$

Here is an integer prescription for these comparisons. For $q(t) = \sum_{j=0}^d q_j t^j$ and the Gaussian integer $Z = Ez$, initialize $(V, h) = (q_d, 1)$ and, for $j = d - 1, d - 2, \dots, 0$, replace

$$h \leftarrow Eh, \quad V \leftarrow ZV + hq_j.$$

The final value is $q(z) = V/h$. Apply this to p and p' , and clear the positive denominators in (II.9.17.1). Comparisons of squared distances give $|z - z'| > .01$ for distinct centers, including conjugates. Comparisons of coordinates and squared moduli also give

$$1.83928 < z_\beta < 1.83929, \quad 1.0002090 < |z_\alpha|^2 < 1.0002093, \quad (\text{II.9.17.2})$$

$$|z| < .9939 \quad \text{at all other displayed centers.}$$

For $|h| = \delta$, the terms of degree at least two in $p(z + h) - p(z) - p'(z)h$ have total modulus at most

$$\delta^2 \sum_{j=0}^{31} |p_j| 3^j < 10^{17} \delta^2.$$

Indeed, $|z| < 2$ and $\delta < 1$, so the binomial expansion gives this bound term by term. By (II.9.17.1), the constant term is less than $|p'(z)|\delta/2$, whereas the displayed remainder

is less than $|p'(z)|\delta/2$. Rouché's theorem therefore puts exactly one root, counted with multiplicity, in each disk. The disks are disjoint and their number is the degree, so these are all the roots and all are simple. The root in the real-centered disk is real by conjugation. Every other disk misses the real axis, so the remaining roots form fifteen nonreal conjugate pairs. The radius 10^{-35} and (II.9.17.2) imply every bound in Proposition II.9.16.2, including $|1/\alpha| < .9999$.

II.9.17.2 Rounded coefficient arrays and finite products

The numerical inputs are the sequences C_0, C_1 defined in the main text in (II.9.16.6). We approximate their single and paired moments

$$P_A = \prod_{m \in \mathbb{Z}} \cos(\pi C_A(m)), \quad P_{AB}^\sigma(k) = \prod_{m \in \mathbb{Z}} \cos(\pi(C_A(m) + \sigma C_B(m+k))), \quad (\text{II.9.17.3})$$

where $A, B \in \{0, 1\}$, $\sigma \in \{1, -1\}$, and $k \geq 0$. Both products converge by exponential decay of the coefficients. The sign notation $+$ and $-$ means $\sigma = 1$ and $\sigma = -1$.

We now specify the approximations used to define the weights. Use

$$S = 10^{30}, \quad R = 3000, \quad H = 100, \quad K = 2000, \quad W = 10^9.$$

For a real rational x set

$$[x] = \frac{\lfloor Sx + 1/2 \rfloor}{S},$$

and round complex numbers componentwise. Thus, if $x = u/v$ with integers u, v and $v > 0$, its rounded numerator is $\lfloor (2Su + v)/(2v) \rfloor$. Additions, subtractions, conjugation, and multiplication by an integer are exact unless brackets are explicitly displayed. Set

$$\pi' = \frac{3141592653589793238462643383279}{S}.$$

The inequality $|\pi' - \pi| < 1/S$ follows by using 30 terms of the alternating series for $\arctan(1/5)$ and 10 for $\arctan(1/239)$ in Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$. The total remainder bound is

$$\frac{16}{61 \cdot 5^{61}} + \frac{4}{21 \cdot 239^{21}}.$$

In particular this verification, too, uses rational arithmetic only.

The contracting and expanding roots are denoted by I and D , as in the main text. For each representative center z first form $r_z = \lfloor 1/p'(z) \rfloor$, with $p'(z)$ evaluated exactly. Store a base and a coefficient as follows:

$$(a_z, v_z) = \begin{cases} ([z], r_z), & |z| < 1, \\ ([1/z], -\lfloor [1/z]r_z \rfloor), & |z| > 1. \end{cases} \quad (\text{II.9.17.4})$$

For a contracting representative, start with $v = v_z$. Add $2\Re v$ to the successive entries $c_0(-1), c_0(-2), c_0(-3), \dots$. After each addition replace v by $\lfloor va_z \rfloor$. For an expanding representative do the same on $c_0(0), c_0(1), \dots$; the multiplier is 1 for the real root and 2 for the nonreal representative. All arrays start at zero. Compute $c_0(m)$ on

$$-R - K - 5 \leq m \leq H + K + 5$$

and define, wherever these three entries are present,

$$c_1(m) = c_0(m+3) - c_0(m) + c_0(m-1). \quad (\text{II.9.17.5})$$

These extra endpoint entries ensure that every subsequent use of c_1 lies in its defined range. The arrays approximate the sequences C_0, C_1 in the main text.

For a rational input c let $g(c), h(c)$ be the following rounded cosine and sine values. The displayed loop fixes the order of the summation.

```
x = [pi' c]; y = [x*x]
t = 1; u = x; g = t; h = u
for j = 1,...,50:
    t = [-t*y/(2*j*(2*j-1))]
    u = [-u*y/(2*j*(2*j+1))]
    g = g+t; h = h+u
return (g,h)
```

Write $g_A(m) = g(c_A(m))$ and $h_A(m) = h(c_A(m))$.

The only expanding root close to the unit circle is α and its conjugate. To treat their long tail, set

$$a = 1/\alpha, \quad u_0 = -a/p'(\alpha), \quad u_1 = u_0(a^3 - 1 + 1/a). \quad (\text{II.9.17.6})$$

Their contribution to $C_A(m)$ for $m \geq H$ is $u_A a^m + \overline{u_A a^m}$. Let (a', u'_0) be the pair in (II.9.17.4) belonging to z_α . Compute

$$a'_0 = 1, \quad a'_{n+1} = [a' a'_n] \quad (0 \leq n < K), \quad u'_1 = [u'_0(a'_3 - 1 + [1/a'])].$$

The subscript in a'_n is a power-array index.

For a single moment use $w = u'_A$ and $k = 0$. For a paired moment of types $A, B \in \{0, 1\}$, sign $\sigma \in \{1, -1\}$, and lag $0 \leq k \leq K$, use $w = u'_A + \sigma[u'_B a'_k]$. In either case the following calculation returns an approximate moment. It retains the finite factors $-R - k \leq m < H$, omits the rapidly convergent negative tail, and treats the positive tail by summing the degree-2, 4, and 6 terms of $-\log \cos$ through geometric series. Their sum is the variable `pen`; the divisors 2, 12, 45 are the corresponding Taylor denominators. To compute $e^{-\text{pen}}$, the algorithm first approximates $e^{-\text{pen}/32}$ and then squares five times. This tail value is computed first, and the finite factors are then multiplied in increasing order of m . Section II.9.17.3 bounds all omissions and rounding errors.

```
U = [w*a'_H]
q_0 = 1; q_i = [q_{i-1}*U]          for i = 1,...,6
b_0 = 1; b_i = [b_{i-1}*pi']       for i = 1,...,6
pen = 0
for d = 2,4,6:
    mom = 0
    for j = 0,...,d:
        num = [q_j*conj(q_{d-j})]
        den = 1-[a'_j*conj(a'_{d-j})]
```

```

mom = mom+binomial(d,j)*Re([num/den])
pen = pen+[b_d*mom/f_d]          (f_2,f_4,f_6) = (2,12,45)
x = [-pen/32]; t = 1; value = t
for j = 1,...,40:
    t = [t*x/j]; value = value+t
repeat five times: value = [value*value]
for m = -R-k,...,H-1, in increasing order:
    v = g_A(m)                    (single)
    v = [g_A(m)*g_B(m+k)-sigma*h_A(m)*h_B(m+k)] (pair)
    value = [value*v]
return value

```

Denote the outputs by $\tilde{P}_A$ and $\tilde{P}_{AB}^\sigma(k)$, respectively.

We use only lags $30 \leq k \leq 2000$. With $s = 0$ for $\sigma = 1$ and $s = 1$ for $\sigma = -1$, put $r = \tilde{P}_{00}^\sigma(k)$ and set

$$w_{ks} = \frac{\text{sgn}(r)}{W} \left[.68W \max(0, |r| - .00026) + \frac{1}{2} \right], \quad 30 \leq k \leq 2000. \quad (\text{II.9.17.7})$$

Take $\text{sgn}(0) = 0$ and omit zero weights from subsequent sums. The recurrences above and this formula define the weights without any choices or numerical tolerances.

II.9.17.3 A uniform error bound for the moment calculation

We prove

$$|\tilde{P}_A - P_A| < \Delta, \quad |\tilde{P}_{AB}^\sigma(k) - P_{AB}^\sigma(k)| < \Delta, \quad \Delta = 5 \cdot 10^{-6}, \quad (\text{II.9.17.8})$$

for all types, signs, and lags used above. The important issue is the weakly contracting base $a = 1/\alpha$: truncating its powers at $H = 100$ would not be accurate. The geometric sums in the tail calculation retain all of these powers.

Coefficient and trigonometric errors. For $|z| < 2.001$, termwise differentiation gives

$$|p''(z)| < 4 \cdot 10^{13}.$$

Hence p' changes by less than $4 \cdot 10^{-22}$ between a center and its root. By (II.9.17.1), reciprocal differentiation and rounding in (II.9.17.4) give base errors less than $2/S$ and coefficient errors less than 10^{-21} . Direct squared modulus comparisons give $|a_z| < 1$ and $|v_z| \leq 1$ for every stored pair. The corresponding exact coefficients have modulus less than 1.001. Thus each recurrence step $v \leftarrow [va_z]$ propagates the existing error with factor at most 1 and adds less than $4/S$. There are fewer than 5100 such steps in any needed entry. Each coefficient-times-power term consequently has error less than $2 \cdot 10^{-21}$, and

$$|c_0(m) - C_0(m)| < 7 \cdot 10^{-20}, \quad |c_1(m) - C_1(m)| < 2.1 \cdot 10^{-19}. \quad (\text{II.9.17.9})$$

The finite recurrence also gives the following rational comparisons:

$$\begin{aligned} \max_{A,m} |c_A(m)| &< .847, & \max_{m \geq -20} |c_0(m)| &< .018, & \max_A |u'_A| &< .0087, \\ .8460 &< \max_{A,m} |c_A(m)| &< .8462. \end{aligned} \quad (\text{II.9.17.10})$$

The maxima here range over the computed entries; use squared moduli for the complex coefficients u'_A . To reproduce them, perform the coefficient recurrences in (II.9.17.4), the additions in (II.9.17.5), and the displayed definition of u'_1 . These comparisons require neither evaluation of trigonometric functions nor a root-finding routine.

Every later trigonometric input is a sum of at most four of these coefficients, counted with multiplicity. Its error is less than 10^{-18} , and its rounded angle satisfies $|x| \leq 13$ and $|y| \leq 170$. The angle error is less than $4 \cdot 10^{-18}$. For the Taylor recurrences at a fixed value of y , each term-rounding error contributes at most e^{14}/S to the final sum: the subsequent factorial denominators bound the amplification by $\sum_{j \geq 0} 170^j/(2j)! < e^{14}$. Replacing x^2 by y costs at most $100e^{14}/S$, by differentiating the absolute term sums. The omitted Taylor tails at $|x| \leq 13$ are less than 10^{-30} . Together these effects give errors below 10^{-20} relative to the computed angle, and below 10^{-16} relative to the true cosine and sine. In particular every single or paired finite factor has error less than 10^{-15} .

There are at most $R + K + H = 5100$ finite factors. The true factors have modulus at most 1, and their approximations have modulus at most $1 + 10^{-15}$. Telescoping the product, including its roundings, shows that with an initial value of modulus at most 2 these errors contribute less than $3 \cdot 10^{-11}$.

Terms omitted at the two ends. On the negative side, $|C_0(m)| \leq 32(.994)^{-1-m}$ for $m < 0$, by summing the contracting-root terms. Including the three shifts for type 1, a single or paired coefficient at $m < -R - k$ is bounded by

$$200(.994)^{-4-(m+k)}.$$

Using $1 - \cos u \leq u^2/2$ and telescoping products, the total omitted product costs at most

$$\frac{3.142^2 200^2}{2} \frac{(.994)^{5994}}{1 - (.994)^2} < 5 \cdot 10^{-8}.$$

For example, the rational inequality $(.994)^{2996} < 3 \cdot 10^{-8}$ already suffices for this estimate. On the positive side the only terms omitted from the coefficients are those from β . Their bases are less than .545, and their type-0 and type-1 coefficients have modulus at most 4. The cosine Lipschitz bound therefore gives a total error at most $3.142 \cdot 8 \cdot (.545)^{100}/(1 - .545) < 10^{-20}$.

The geometric tail and its rounding. Let $w_* = u_A$ in the single case and $w_* = u_A + \sigma u_B a^k$ in the paired case. Put $U_* = w_* a^H$. The exact sums of the even powers of the weak-root tail are

$$\mu_d = \sum_{n \geq 0} (U_* a^n + \overline{U_* a^n})^d = \sum_{j=0}^d \binom{d}{j} \Re \frac{U_*^j \overline{U_*}^{d-j}}{1 - a^j \bar{a}^{d-j}} \quad (d = 2, 4, 6). \quad (\text{II.9.17.11})$$

The equality follows by the binomial theorem and absolutely convergent geometric series. In particular $\mu_d \geq 0$.

The power-array errors satisfy $|a'_n - a^n| \leq 3n/S$. Because $|a|, |a'| > .99$, the formula for u'_1 gives errors below $5 \cdot 10^{-21}$ for both u'_A , and $|u_A| \leq .00871$. It follows that the computed

U has error below 10^{-19} ; its powers through degree 6 have errors below 10^{-18} . In each quotient in the algorithm, the numerator error is below $3 \cdot 10^{-18}$ and the denominator error is below 10^{-28} . The true and computed denominators have modulus greater than .00019: use

$$|1 - a^j \bar{a}^{d-j}| \geq 1 - |a|^d > 1 - (.9999)^2 > .00019$$

and the stated rounding error. Hence each quotient error is below $2 \cdot 10^{-14}$, the error in each computed μ_d is below $2 \cdot 10^{-12}$, and $|\mu_d| < 4 \cdot 10^5$ by the same denominator bound. The rounded powers of π' through degree 6 have errors below 10^{-22} . Consequently pen approximates the nonnegative number

$$P = \frac{\pi^2 \mu_2}{2} + \frac{\pi^4 \mu_4}{12} + \frac{\pi^6 \mu_6}{45} \quad (\text{II.9.17.12})$$

to within 10^{-9} .

For $m \geq H$, the weak coefficient has angle bounded by

$$3.142 \cdot 4 \cdot .00871 \cdot (.9999)^m < .11(.9999)^m.$$

Summing the powers of this bound gives

$$P \leq \frac{.11^2}{2(1 - .9999^2)} + \frac{.11^4}{12(1 - .9999^4)} + \frac{.11^6}{45(1 - .9999^6)} < 31.$$

Thus $|\text{pen}| < 32$ and $|x| < 1.1$ in the exponential calculation. The 40-term exponential recurrence has Taylor remainder less than 10^{-40} . Its rounding errors, bounded using $e^{1.1}$, are negligible compared with 10^{-9} . Each squaring amplifies its existing error by at most 3, since the true intermediate values for the computed penalty are at most $e^{10^{-9}}$. Including the error in (II.9.17.12), the returned exponential is within $4 \cdot 10^{-9}$ of e^{-P} .

The analytic remainder of the tail. For real z with $|z| \leq .11$ one has

$$\left| -\log \cos z - \frac{z^2}{2} - \frac{z^4}{12} - \frac{z^6}{45} \right| \leq .1z^8. \quad (\text{II.9.17.13})$$

Here are elementary bounds proving the stated constant. Set $y = z^2$, $h = 1 - \cos z$, and $h_0 = y/2 - y^2/24 + y^3/720$. Then $0 \leq h \leq y/2$ and $|h - h_0| \leq y^4/40320$. In $-\log(1 - h) = h + h^2/2 + h^3/3 + \dots$, the terms after the third are bounded by $h^4/(4(1 - h)) < .016y^4$. Substitution of h_0 in the first three terms costs less than $.00004y^4$. The terms through degree 3 are $y/2 + y^2/12 + y^3/45$; the sum of the absolute values of the remaining coefficients is less than .03, so their contribution is less than $.03y^4$ for $0 \leq y \leq .0121$. These bounds imply (II.9.17.13).

Applying (II.9.17.13) to all weak tail angles gives

$$\left| \sum_{m \geq H} -\log \cos(\pi(w_* a^m + \overline{w_* a^m})) - P \right| \leq \frac{.1(.11)^8}{1 - (.9999)^8} < 3 \cdot 10^{-6}.$$

Both penalties are nonnegative, so the same bound applies to the difference of their exponentials. Combining this with the rounding error, the finite-factor error, and the two omitted ends proves (II.9.17.8); indeed the sum of the stated errors is below $3.055 \cdot 10^{-6}$.

II.9.17.4 Correlation, variance, and the absolute Gram sum

For the rest of the certificate section, abbreviate $Z_i = \mathcal{Z}_i^{(0)}$ and $G_i = \mathcal{G}_i$ from (II.9.16.9) and (II.9.16.11). All expectations are under the stationary law of the two-sided independent bits.

The preceding calculation defines exact rational weights and bounds each moment error uniformly. We next sum those moments to prove the first three estimates of Proposition II.9.16.5. A final finite-product estimate will prove the fourth.

Write $X = -.001$, $Y = .0055$, and let $\mathcal{R}$ be the nonzero rows (w_{ks}, k, s) with $30 \leq k \leq 2000$. Define

$$b_A = \tilde{P}_A, \quad m_{k0} = \tilde{P}_{00}^+(k), \quad m_{k1} = \tilde{P}_{00}^-(k), \quad e = \tilde{P}_{01}^+(33).$$

The approximated correlation is

$$\tilde{c} = Xb_0 + Ye + \sum_{(w,k,s) \in \mathcal{R}} w m_{ks}. \quad (\text{II.9.17.14})$$

For each lag k , let J_{k0} and J_{k1} be respectively the second (minus) and first (plus) entries of the following pair:

$$\begin{cases} (\tilde{P}_{10}^+(k-33), \tilde{P}_{10}^-(k-33)), & k \geq 33, \\ (\tilde{P}_{01}^+(33-k), \tilde{P}_{01}^-(33-k)), & k < 33. \end{cases}$$

The approximated squared norm and absolute Gram sum are

$$\begin{aligned} \tilde{v} = X^2 + Y^2 + 2XYb_1 + 2 \sum_{(w,k,s) \in \mathcal{R}} w(Xb_0 + YJ_{ks}) \\ + \sum_{(w,k,s),(v,l,t) \in \mathcal{R}} wv m_{|k-l|, \mathbf{1}_{s=t}}, \end{aligned} \quad (\text{II.9.17.15})$$

$$\tilde{T} = \sum_{\substack{(w,k,s),(v,l,t) \in \mathcal{R} \\ k,l \geq 50}} |wv m_{|k-l|, \mathbf{1}_{s=t}}|. \quad (\text{II.9.17.16})$$

Both double sums are over ordered pairs. Equal signs require the minus moment in a conjugated phase pair; opposite signs require the plus moment. This explains the index $\mathbf{1}_{s=t}$, including the diagonal cases.

The exact finite calculations give

$$\begin{aligned} |\mathcal{R}| &= 753, \\ |X| + |Y| + \sum_{\mathcal{R}} |w| &= 198442946/W, \\ |X| + |Y| + \sum_{\substack{(w,k,s) \in \mathcal{R} \\ k < 50}} |w| &= 11816095/W, \\ WS\tilde{c} &= 211815537786494774284357596751188225, \\ W^2S\tilde{v} &= 143920884250294630908494564120361905955993456, \\ W^2S\tilde{T} &= 105303788239450118397416374596041413014338334. \end{aligned} \quad (\text{II.9.17.17})$$

These are integer equalities: compute the arrays and products in Section II.9.17.2, round the weights by (II.9.17.7), and use the unrounded rational sums (II.9.17.14)–(II.9.17.16). In particular

$$211810 < 10^9 \tilde{c} < 211820, \quad 143910 < 10^9 \tilde{v} < 143930, \quad 105300 < 10^9 \tilde{T} < 105310.$$

This gives $|G_i| < .199$ and $|G_i^{(N)}(x)| < .199$ by the triangle inequality. The total correlation error is at most $\Delta \cdot .199$, and the squared-norm error is at most $\Delta \cdot .199^2$. Thus

$$\mathbb{E}\Re(Z_i G_i) > .0002108205 > .000210, \quad \mathbb{E}|G_i|^2 < .000144119 < .000145.$$

To prepare for the last estimate, call the constant row, the type-1 row, and the rows with $k < 50$ the *small rows*; the remaining rows are the *big rows*. A row of weight w_v has phase $U_v = \exp(\pi i \sum_m f_v(m) e_m)$, where $e_m = 2b_{i+m} - 1$. Its sequence f_v is zero for the constant, $C_1(m + 33)$ for the type-1 row, and $(1 - 2s)C_0(m + k)$ for a row (w, k, s) . Define the true absolute Gram sum

$$T = \sum_{v, v' \text{ big}} |w_v w_{v'} \mathbb{E} U_v \overline{U_{v'}}|. \quad (\text{II.9.17.18})$$

The error between this sum and (II.9.17.16) is at most $\Delta \cdot .199^2$, by $\|x\| - \|y\| \leq |x - y|$. Hence

$$T < .000116, \quad \sqrt{T} < .011. \quad (\text{II.9.17.19})$$

The total absolute weight of the small rows is less than .012.

II.9.17.5 Finite products controlling the remaining moment

We now bound $\mathbb{E}(Z_i^2 G_i^2)$. The big rows have small coefficients on the 71 positions $-70 \leq m \leq 0$. More precisely, with $t = .06$, (II.9.17.9) and (II.9.17.10) give

$$|\tan(\pi f_v(m))| \leq t \quad (v \text{ big}, -70 \leq m \leq 0). \quad (\text{II.9.17.20})$$

Indeed $m + k \geq -20$ for these rows, and the coefficient bound .018, together with $\sin u \leq u$ and $\cos u \geq 1 - u^2/2$, is sufficient.

For any real sequence V put $c_m = |\cos(\pi V(m))|$ and $s_m = |\sin(\pi V(m))|$. The finite products needed below are

$$\begin{aligned} B_2(V) &= \prod_{m=-70}^0 \min\left(1, \frac{(1+t^2)c_m + 2ts_m}{1-t^2}\right), \\ B_1(V) &= \prod_{m=-70}^0 \min\left(1, \frac{c_m + ts_m}{\sqrt{1-t^2}}\right), \\ B_0(V) &= \prod_{m=-200}^0 c_m. \end{aligned} \quad (\text{II.9.17.21})$$

Their bounds are

$$B_2(2C_0) < .011, \quad \max_{f \text{ small}} B_1(2C_0 + f) < .015, \quad \max_{f, g \text{ small}} B_0(2C_0 + f + g) < .0032. \quad (\text{II.9.17.22})$$

Here the maxima range over the actual phase sequences of the small rows. Besides the constant and the type-1 row, their eight nonzero rows are as follows:

k	32	36	37	39	40	43	44	47
s	0	1	0	1	0	0	1	1
Ww_{ks}	-536932	1265024	331701	47119	1483338	436861	458216	756904

For completeness, the following upper-rounding calculation verifies all of (II.9.17.22). For each of its three cases and each indicated choice of small rows, form the approximate sequence V' by replacing C_A by c_A . Initialize $v = 1$. For increasing m in the relevant interval, compute

$$\begin{aligned}
 b &= |g(V'(m))| + 10^{-15}, & y &= |h(V'(m))| + 10^{-15}, \\
 F_0 &= b, & F_1 &= S^{-1} + [1.002(b + .06y)], \\
 & & F_2 &= S^{-1} + [1.004(1.0036b + .12y)], \\
 v &\leftarrow S^{-1} + [v \min(1, F_j)] & & \text{in case } j.
 \end{aligned}$$

The factors 1.002 and 1.004 bound $(1 - t^2)^{-1/2}$ and $(1 - t^2)^{-1}$, respectively. The added 10^{-15} exceeds the trigonometric error proved above, and each added S^{-1} compensates for a possibly downward rounding. Thus this calculation gives upper bounds, including for B_0 because its true factors are at most 1. The maximum final numerators at scale S , in the order B_2, B_1, B_0 , are

$$\begin{aligned}
 &10861284406435197401851134503, \\
 &14731866010915969694738277129, & \text{(II.9.17.23)} \\
 &3155364376927094122833206055.
 \end{aligned}$$

Multiplication by $10^6/S$ bounds them respectively by 10862, 14732, and 3156, proving (II.9.17.22).

II.9.17.6 From the finite products to the anisotropy bound

The estimates just obtained concern individual coefficients and finite products. We finish by showing how they control the full sum of phases without expanding every three-phase moment. Write $G_i = G_{\text{small}} + G_{\text{big}}$ according to the preceding partition, and fix any subset $J \subset \{-70, \dots, 0\}$. Expanding the big phases in the independent signs e_m on J gives

$$G_{\text{big}} = \sum_{A \subset J} i^{|A|} e_A H_A, \quad e_A = \prod_{m \in A} e_m, \quad \text{(II.9.17.24)}$$

where

$$H_A = \sum_{v \text{ big}} w_v U_v^{\text{out}} \prod_{m \in J} \cos(\pi f_v(m)) \prod_{m \in A} \tan(\pi f_v(m)).$$

The phase U_v^{out} depends only on signs outside J , so each H_A is independent of the signs on J .

We claim that

$$\|H_A\|_2 \leq \sqrt{T} t^{|A|} (1 - t^2)^{-|J|/2}. \quad \text{(II.9.17.25)}$$

For two big rows, independence gives their full Gram entry as the outside Gram entry times

$$\prod_{m \in J} \cos(\pi f_v(m)) \cos(\pi f_{v'}(m)) (1 + \tan(\pi f_v(m)) \tan(\pi f_{v'}(m))).$$

By (II.9.17.20), each last factor is at least $1 - t^2$. In the expansion of $\|H_A\|_2^2$, the extra tangent products have modulus at most $t^{2|A|}$. Taking absolute values term by term therefore bounds the squared norm by $Tt^{2|A|}(1 - t^2)^{-|J|}$, proving (II.9.17.25).

First consider $\mathbb{E}(Z_i^2 G_{\text{big}}^2)$ and set $V = 2C_0$. For a pair $A, A' \subset J$ in (II.9.17.24), averaging the signs on J contributes a factor of modulus c_m where membership in A, A' agrees, and a factor of modulus s_m where it differs. The outside expectation is bounded by $\|H_A\|_2 \|H_{A'}\|_2$ using Cauchy–Schwarz and the unit modulus of the outside part of Z_i^2 . Summing over the four membership choices at each site gives

$$|\mathbb{E}(Z_i^2 G_{\text{big}}^2)| \leq T \prod_{m \in J} \frac{(1 + t^2)c_m + 2ts_m}{1 - t^2}.$$

Choose J to contain precisely those positions whose displayed factor is less than 1. This yields

$$|\mathbb{E}(Z_i^2 G_{\text{big}}^2)| \leq TB_2(2C_0). \quad (\text{II.9.17.26})$$

For a fixed small phase U with sequence f , use the same expansion with $V = 2C_0 + f$. There is now one subset A ; summing its two membership choices produces $c_m + ts_m$. The outside expectation is bounded by $\|H_A\|_2$. Optimizing J as before gives

$$|\mathbb{E}(Z_i^2 U G_{\text{big}})| \leq \sqrt{T} B_1(2C_0 + f).$$

For two small phases with sequences f, g , independence directly bounds their full moment by $B_0(2C_0 + f + g)$: discard all the other absolute cosine factors, which are at most 1.

Finally sum these bounds with the absolute row weights. Using (II.9.17.19), (II.9.17.22), and the small-weight bound .012, we obtain

$$\begin{aligned} |\mathbb{E}(Z_i^2 G_i^2)| &\leq T \cdot .011 + 2(.012)\sqrt{T} \cdot .015 + (.012)^2 \cdot .0032 \\ &< .000116 \cdot .011 + 2 \cdot .012 \cdot .011 \cdot .015 + .012^2 \cdot .0032 \\ &= .0000056968 < .000007. \end{aligned}$$

This proves the remaining estimate of Proposition II.9.16.5.

Chapter II.10

From Full Dimension to Absolute Continuity

II.10.1 Why full dimension is not enough

The purpose of this chapter is to explain two complementary mechanisms for proving absolute continuity of overlapping self-similar measures. The first, due to Shmerkin, combines a dimension-one factor with a Fourier-decaying factor. The second, due to Varjú, turns a quantitative entropy deficit estimate into an $L \log L$ density. We begin with the obstruction that forces one to use genuinely stronger information than Hausdorff dimension.

Example II.10.1.1 (A singular probability of exact dimension one). Let $Q = \{1, 4, 9, 16, \dots\}$ and let

$$K = \left\{ \sum_{j \geq 1} \omega_j 2^{-j} : \omega_j \in \{0, 1\}, \omega_j = 0 \text{ for } j \in Q \right\}.$$

Choose all non-square digits independently and uniformly in $\{0, 1\}$, and denote the resulting probability by μ . Then K is compact, $\mathcal{L}(K) = 0$, $\mu(K) = 1$, and

$$\lim_{r \downarrow 0} \frac{\log \mu(B(x, r))}{\log r} = 1 \quad (x \in K).$$

In particular $\dim_{\text{H}} K = 1$, while μ is singular.

Proof. Let $F(n) = n - \lfloor \sqrt{n} \rfloor$, the number of free positions among the first n binary digits. There are exactly $2^{F(n)}$ admissible level- n dyadic intervals, each of length 2^{-n} and μ -mass $2^{-F(n)}$. Hence

$$\mathcal{L}(K) \leq 2^{F(n)} 2^{-n} = 2^{-\lfloor \sqrt{n} \rfloor} \longrightarrow 0.$$

Compactness follows from the usual diagonal subsequence argument on binary digits; the infinitely many forced zeros eliminate the only endpoint ambiguity relevant here.

If $2^{-n} \leq r < 2^{-n+1}$, the level- n interval containing x lies in $B(x, r)$, whereas an interval of length $2r$ meets at most five level- n dyadic intervals. Thus

$$2^{-F(n)} \leq \mu(B(x, r)) \leq 5 \cdot 2^{-F(n)}.$$

Since $F(n)/n \rightarrow 1$, the local-dimension limit is one. For $0 < s < 1$, the same estimate gives $\mu(B(y, r)) \leq C_s r^s$ for all sufficiently small r and all $y \in \mathbb{R}$, so the elementary mass-distribution argument yields $\dim_{\text{H}} K \geq s$. Letting $s \uparrow 1$ and using $K \subset [0, 1]$ proves $\dim_{\text{H}} K = 1$. $\square$

II.10.2 Splitting infinite convolutions

The first absolute-continuity mechanism begins by decomposing one digit series into independent residue classes. One factor will later supply dimension, while another supplies Fourier decay. We first fix the probabilistic notation and identify it with the usual self-similar measure.

Definition II.10.2.1. Let $0 < \lambda < 1$ and let $X_0, X_1, \dots$ be independent and identically distributed bounded real random variables with common law ρ . Write

$$\mu_{\lambda, \rho} = \text{Law} \left(\sum_{n \geq 0} \lambda^n X_n \right).$$

For $a \in \mathbb{R}$, $S_a(x) = ax$.

The next elementary identification explains in particular why coincident block maps must be merged by adding their weights.

Proposition II.10.2.2 (Digit-series and self-similar measures). *Let $0 < r < 1$. Let $b_1, \dots, b_m$ be real digits with positive weights $q_1, \dots, q_m$ summing to one. Let $Y_0, Y_1, \dots$ be independent and identically distributed with $\mathbb{P}(Y_n = b_i) = q_i$, and let μ be the law of*

$$Y_0 + rY_1 + r^2Y_2 + \dots.$$

If $S_i(x) = rx + b_i$, then

$$\mu = q_1(S_1)_*\mu + \dots + q_m(S_m)_*\mu.$$

Moreover, μ is the unique Borel probability satisfying this identity. Hence coincident affine maps arising from different block words may be merged by adding their weights.

Proof. Absolute convergence follows from boundedness of the digits. Set $X = Y_1 + rY_2 + r^2Y_3 + \dots$. Then X has law μ and is independent of Y_0 . Conditioning on Y_0 gives the fixed-point identity.

For uniqueness, let ν be any fixed point. Take Z with law ν , independent of independent indices $I_0, I_1, \dots$ with $\mathbb{P}(I_j = i) = q_i$. Repeatedly applying the fixed-point identity shows that the N -step random affine iterate applied to Z still has law ν . Expanding the iterate

gives the first N digit terms plus $r^N Z$. Since $r^N Z \rightarrow 0$ almost surely and the partial digit sums converge almost surely to the full series, the iterates converge almost surely to a variable with law μ . Dominated convergence against bounded continuous test functions gives $\nu = \mu$. Merging coincident maps leaves the mixture operator unchanged. $\square$

Proposition II.10.2.3 (Residue-class splitting). *For every integer $k \geq 1$,*

$$\mu_{\lambda, \rho} = \bigstar_{r=0}^{k-1} (S_{\lambda^r})_* \mu_{\lambda^k, \rho}.$$

If the digits are Bernoulli, the complementary “dense” factor obtained by deleting positions divisible by k is itself a homogeneous self-similar measure with contraction λ^k and block weights obtained by merging equal block translations and adding their probabilities.

Proof. Absolute convergence is immediate from boundedness of the digits. Split the series according to $n \equiv r \pmod{k}$ and use independence of the residue classes. For the dense factor, group the $k - 1$ retained digits in one block. Distinct words producing the same block translation define the same affine map; their masses must therefore be added before entropy or similarity dimension is computed. $\square$

Proposition II.10.2.4 (Fourier factorization). *For finite Borel probabilities,*

$$\widehat{\mu * \nu}(\xi) = \hat{\mu}(\xi) \hat{\nu}(\xi), \quad \widehat{(S_a)_* \mu}(\xi) = \hat{\mu}(a\xi).$$

Moreover

$$\widehat{\mu_{\lambda, \rho}}(\xi) = \lim_{N \rightarrow \infty} \prod_{n=0}^{N-1} \hat{\rho}(\lambda^n \xi)$$

pointwise in ξ .

Proof. The finite identities are Fubini’s theorem. The partial sums of the random series converge almost surely and are uniformly bounded, so dominated convergence applies to their unit-modulus exponentials. Independence factorizes each finite characteristic function. $\square$

II.10.3 Fourier energy and a smoothing criterion

After the residue-class decomposition, absolute continuity will follow if one factor has finite energy and the other has power Fourier decay. This section proves the precise smoothing statement that joins those two kinds of information.

We use the Fourier convention $\hat{f}(\xi) = \int f(x) e^{-2\pi i x \xi} dx$.

Lemma II.10.3.1 (Gaussian transform). *For $t > 0$,*

$$\int_{\mathbb{R}} e^{-\pi t x^2} e^{-2\pi i x \xi} dx = t^{-1/2} e^{-\pi \xi^2 / t}.$$

Proof. For $t = 1$, differentiation under the integral and integration by parts show that the transform F satisfies $F'(\xi) = -2\pi\xi F(\xi)$ and $F(0) = 1$. The general formula follows by scaling. $\square$

Theorem II.10.3.2 (Plancherel and the finite-measure L^2 criterion). *The Fourier transform extends unitarily to $L^2(\mathbb{R})$. If a finite positive Borel measure η has Fourier–Stieltjes transform $\hat{\eta} \in L^2(\mathbb{R})$, then $\eta = f \, dx$ for a nonnegative $f \in L^1 \cap L^2$, and the L^2 Fourier transform of f equals $\hat{\eta}$ almost everywhere.*

Proof. For Schwartz inversion, multiply $\hat{f}$ by a Gaussian, apply Lemma II.10.3.1, and let the Gaussian scale tend to zero. Parseval follows from Fubini and inversion; density of $\mathcal{S}(\mathbb{R})$ gives the unitary extension.

If $\hat{\eta} \in L^2$, choose $f \in L^2$ with $\hat{f} = \hat{\eta}$. For every Schwartz ϕ , inversion and Fubini give

$$\int \phi \, d\eta = \int \hat{\phi}(\xi) \hat{\eta}(-\xi) \, d\xi = \int \phi(x) f(x) \, dx.$$

Both sides define Radon measures locally; smooth approximation therefore identifies them. Positivity and finite total mass give $f \geq 0$ and $f \in L^1$. $\square$

Proposition II.10.3.3 (Energy identity). *For $0 < s < 1$ and every finite positive Borel measure μ ,*

$$I_s(\mu) := \iint |x - y|^{-s} \, d\mu(x) \, d\mu(y) = c_s \int_{\mathbb{R}} |\hat{\mu}(\xi)|^2 |\xi|^{s-1} \, d\xi,$$

with both sides allowed to be infinite and

$$c_s = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}.$$

Proof. Insert the Gamma-integral representation of r^{-s} , use Tonelli, then use the Gaussian inverse representation from Lemma II.10.3.1. A second Tonelli interchange and the substitution $u = \pi\xi^2/t$ give the stated constant. The diagonal is retained throughout, so atomic measures correctly have infinite energy. $\square$

Theorem II.10.3.4 (Finite energy plus Fourier decay). *Let μ, ν be compactly supported probabilities, $0 < s < 1$, $I_s(\mu) < \infty$, and*

$$|\hat{\nu}(\xi)| \leq C(1 + |\xi|)^{-\beta}$$

*for some $\beta > 0$ with $s + 2\beta > 1$. Then $\mu * \nu$ has a density in $H^t(\mathbb{R})$ for every*

$$0 \leq t < \frac{s + 2\beta - 1}{2}.$$

Proof. At low frequencies the product transform is bounded by one. At $|\xi| > 1$,

$$(1 + |\xi|^2)^t |\hat{v}(\xi)|^2 \lesssim |\xi|^{2t-2\beta} \leq |\xi|^{s-1}.$$

The energy identity therefore gives a finite weighted L^2 norm for $\hat{\mu}\hat{v}$. Apply Theorem II.10.3.2 to the convolution measure. $\square$

Proposition II.10.3.5 (Finite-energy restrictions of a dimension-one measure). *Suppose μ is compactly supported and exact dimensional of value one. For every $0 < s < 1$ there are increasing Borel sets E_j with $\mu(\bigcup_j E_j) = 1$ and $I_s(\mu|_{E_j}) < \infty$. Consequently, if v has any positive power Fourier decay, then $\mu * v \ll \mathcal{L}$.*

Proof. Choose $s < a < 1$ and put

$$E_j = \{x : \mu(B(x, r)) \leq r^a \text{ for every rational } 0 < r < 1/j\}.$$

Each E_j is Borel, $E_j \subset E_{j+1}$, and exact dimensionality gives $\mu(\bigcup_j E_j) = 1$. If $0 < r < 1/(4j)$ and $B(z, r)$ meets E_j , choose $x \in E_j \cap B(z, r)$. Then $B(z, r) \subset B(x, 2r)$; approximation of $2r$ from above by rational radii below $1/j$ gives $\mu|_{E_j}(B(z, r)) \leq (2r)^a$. For larger r , the same bound holds with a constant depending on j , since the restriction has mass at most one. Thus $\mu|_{E_j}(B(z, r)) \leq C_j r^a$ uniformly in z, r . The layer-cake formula for $|x - y|^{-s}$ gives finite s -energy. Normalize positive restrictions and apply Theorem II.10.3.4; monotone exhaustion of μ yields absolute continuity of the full convolution. $\square$

II.10.4 Entropy deficit and $L \log L$

The second absolute-continuity mechanism is entropic. Its endpoint is an $L \log L$ criterion: a uniformly bounded loss from the maximal dyadic entropy is exactly the integrability condition needed for the density.

Definition II.10.4.1 (Averaged scale entropy). For a bounded real random variable X and $r > 0$, define

$$H(X; r) = \int_0^1 H_2(\lfloor X/r + t \rfloor) dt,$$

and $H(X; r_2 | r_1) = H(X; r_2) - H(X; r_1)$ for $r_2 < r_1$.

Proposition II.10.4.2 (Uniform smoothing identity). *If U_r is uniform on $[0, r]$ and independent of X , then*

$$H(X; r) = h_2(X + U_r) - \log_2 r.$$

The quantity is invariant under translations and under the simultaneous scaling $(X, r) \mapsto (aX, ar)$, $a > 0$. Moreover

$$\left| H(X; 2^{-n}) - \frac{H(\text{Law}(X), \mathcal{D}_n)}{\log 2} \right| \leq 1.$$

Proof. The density of $X + U_r$ is $g(y) = r^{-1}\mu([y - r, y))$. Writing the translated grid cell masses as $p_j(t) = rg(r(j + 1 - t))$ and integrating in t gives the identity. Translation invariance follows from periodicity in t ; scaling is immediate. For the dyadic comparison, the integer variables $\lfloor X/r \rfloor$ and $\lfloor X/r + t \rfloor$ determine each other up to one binary choice, so their entropies differ by at most one bit. $\square$

Theorem II.10.4.3 (Bounded dyadic entropy deficit). *Let μ be a probability on $[0, 1)$ and let $H_n = H(\mu, \mathcal{D}_n)$. Then*

$$\sup_n (n \log 2 - H_n) < \infty$$

if and only if $\mu = f \, dx$ with

$$\int_0^1 f \log(2 + f) \, dx < \infty.$$

In this case

$$\sup_n (n \log 2 - H_n) = \int_0^1 f \log f \, dx.$$

Proof. Let f_n be the dyadic density, equal to $2^n \mu(I)$ on $I \in \mathcal{D}_n$. Then

$$\int f_n \log f_n = n \log 2 - H_n.$$

If $\mu = f \, dx$, Jensen's inequality for dyadic conditional expectations gives the upper bound. Conversely, a uniform bound on $\int f_n \log f_n$ yields uniform integrability by the explicit tail estimate

$$\int_{\{f_n > M\}} f_n \leq \frac{C + 1/e}{\log M}.$$

Testing dyadic unions and approximating open sets shows that μ vanishes on Lebesgue-null sets. Hence $\mu = f \, dx$. Dyadic conditional expectations converge to f in L^1 ; a subsequence converges almost everywhere, and Fatou applied to $t \log t + 1/e$ gives the reverse entropy inequality. Finally $0 \leq t \log(2 + t) - t \log t \leq 2$ yields the stated $L \log L$ formulation. $\square$

II.10.5 Fourier decay outside a zero-dimensional exceptional set

The Erdős–Kahane mechanism is a parameter-counting argument. Its useful feature here is that the exceptional set depends only on the contraction parameter: after the parameter has been fixed, the same conclusion works simultaneously for every finite digit set with at least two active digits and for every positive weight vector. For $x \in \mathbb{R}$ write

$$\|x\|_{\mathbb{Z}} = \text{dist}(x, \mathbb{Z}).$$

Lemma II.10.5.1 (Erdős–Kahane parameter counting). *For an integer $L \geq 2$, let G_L be the set of $\theta > 1$ such that*

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \min_{1 \leq t \leq \theta} \#\{0 \leq n < N : \|t\theta^n\|_{\mathbb{Z}} \geq L^{-1}\} > L^{-1}.$$

For every compact interval $1 < \Theta_1 < \Theta_2 < \infty$ there is $C = C(\Theta_1, \Theta_2)$ such that

$$\dim_H([\Theta_1, \Theta_2] \setminus G_L) \leq \frac{C \log(CL)}{L}.$$

Proof. Fix $\theta \in [\Theta_1, \Theta_2]$, $t \in [1, \theta]$, and choose $K_n \in \mathbb{Z}$ so that

$$t\theta^n = K_n + \varepsilon_n, \quad |\varepsilon_n| \leq \frac{1}{2}.$$

Call n bad when $|\varepsilon_n| \geq L^{-1}$. From $K_{n+1} + \varepsilon_{n+1} = \theta(K_n + \varepsilon_n)$ and the same relation one step later, elimination of θ gives, once n exceeds a fixed threshold depending only on the compact parameter interval,

$$\left| K_{n+2} - \frac{K_{n+1}^2}{K_n} \right| \leq C_0(|\varepsilon_n| + |\varepsilon_{n+1}| + |\varepsilon_{n+2}|). \quad (\text{II.10.5.1})$$

Without any goodness assumption, K_{n+2} has at most a fixed number $Q = Q(\Theta_1, \Theta_2)$ of possible integer values once (K_n, K_{n+1}) is known. If $L \geq L_0(\Theta_1, \Theta_2)$ and $n, n+1, n+2$ are good, the right side of (II.10.5.1) is $< 1/2$, so K_{n+2} is uniquely determined by (K_n, K_{n+1}) .

Suppose that among $0, \dots, N-1$ there are at most $2N/L$ bad indices. A transition $n \mapsto n+2$ can fail to be deterministic only when one of $n, n+1, n+2$ is bad, so there are at most $6N/L + O(1)$ nondeterministic transitions. The initial pair (K_0, K_1) has only $O_{\Theta_2}(1)$ possibilities. Hence the number of possible integer codes is at most

$$C_1 \sum_{j \leq 6N/L + C_1} \binom{N}{j} Q^j \leq \exp\left(\frac{C_2 N \log(C_2 L)}{L}\right),$$

using $\binom{N}{j} \leq (eN/j)^j$ and enlarging C_2 to absorb finitely many small L .

A late pair determines the parameter to exponentially small accuracy:

$$\frac{K_N}{K_{N-1}} = \theta + O_{\Theta_1, \Theta_2}(\Theta_1^{-N}).$$

Indeed the numerator of the error is $\theta\varepsilon_{N-1} - \varepsilon_N$, while $K_{N-1} \asymp t\theta^{N-1} \geq c\Theta_1^{N-1}$. Thus each code is compatible with an interval of parameters of length at most $C_3\Theta_1^{-N}$.

If $\theta \notin G_L$, then for infinitely many N one can choose $t = t_N \in [1, \theta]$ so that

$$\#\{0 \leq n < N : \|t\theta^n\|_{\mathbb{Z}} \geq L^{-1}\} \leq 2N/L.$$

Therefore $[\Theta_1, \Theta_2] \setminus G_L$ lies in the limsup of covers having at most $\exp(C_2 N \log(C_2 L)/L)$ intervals of length $C_3\Theta_1^{-N}$. If $s > C \log(CL)/L$, the sums of the s -powers of the diameters are summable. Hausdorff–Cantelli gives the asserted dimension bound. For the finitely many $L < L_0$, enlarge C . $\square$

Theorem II.10.5.2 (Common exceptional set for power Fourier decay). *There is a set $E_F \subset (0, 1)$ with $\dim_H E_F = 0$ such that, for every $\lambda \notin E_F$, every finite set of at least two distinct real digits $a_1, \dots, a_m$, and every positive weight vector p , the corresponding homogeneous digit-series measure has power Fourier decay:*

$$|\hat{\mu}(\xi)| \leq C(1 + |\xi|)^{-\gamma}$$

for some $C, \gamma > 0$ depending on the parameter, digits, and weights.

Proof. Define

$$E_F = \{\lambda \in (0, 1) : \lambda^{-1} \notin \bigcup_{L \geq 2} G_L\}.$$

On every compact subinterval of $(1, \infty)$, Lemma II.10.5.1 makes the complement of $\bigcup_L G_L$ have Hausdorff dimension zero. Inversion is locally bi-Lipschitz, and a countable compact exhaustion gives $\dim_H E_F = 0$.

Fix $\lambda \notin E_F$ and put $\theta = \lambda^{-1}$. Choose L with $\theta \in G_L$. After an affine change of the digit set, assume two active digits are 0 and 1, of weights $p_0, p_1 > 0$. Put

$$\Phi(u) = \sum_{j=1}^m p_j e^{-2\pi i a_j u}.$$

Expanding $|\Phi(u)|^2$ and retaining the pair $(0, 1)$ gives

$$|\Phi(u)|^2 \leq 1 - 2p_0 p_1 (1 - \cos 2\pi u) \leq \exp(-c\|u\|_{\mathbb{Z}}^2)$$

for some $c = c(p_0, p_1) > 0$. By Proposition II.10.2.4,

$$\hat{\mu}(\xi) = \prod_{n \geq 0} \Phi(\lambda^n \xi).$$

For $|\xi|$ large choose N so that $\theta^N \leq |\xi| < \theta^{N+1}$ and write $t = |\xi|\theta^{-N} \in [1, \theta)$. Reversing the first N factors gives phases $t\theta^j$, $0 \leq j < N$. The definition of G_L supplies at least $N/(2L)$ indices with $\|t\theta^j\|_{\mathbb{Z}} \geq L^{-1}$ for all sufficiently large N . Hence

$$|\hat{\mu}(\xi)| \leq \exp\left(-\frac{cN}{2L^3}\right) \leq C|\xi|^{-\gamma}$$

for some $\gamma > 0$. Absorb the finitely many remaining frequencies into C . □

II.10.6 The dense factor and Shmerkin's theorem

We now combine the Fourier-decaying sparse factor with a dense factor of full dimension. The dimension statement comes from the analytic-family theorem proved in Chapter II.6; it is restated here in the form needed for the skipped-digit family.

Theorem II.10.6.1 (Analytic-family dimension theorem from Chapter II.6). *Let I be a compact interval and let*

$$f_{i,t}(x) = \lambda_i(t)x + a_i(t), \quad i = 1, \dots, m,$$

be a real-analytic one-parameter family with $0 < |\lambda_i(t)| < 1$. Assume analytic nondegeneracy: for every two distinct infinite symbolic sequences, their coding-point difference is not identically zero on I . Then there is $E \subset I$ of Hausdorff and packing dimension zero, independent of the positive weight vector p , such that for $t \in I \setminus E$ the corresponding self-similar measure is exact dimensional and

$$\dim \mu_{t,p} = \min\{s(t, p), 1\},$$

where $s(t, p)$ is its similarity dimension.

Remark II.10.6.2 (Relation to Chapter II.6). The preceding statement is C-II.6.21, the analytic-family natural-dimension theorem proved in Chapter II.6. Its uniformity in the weight vector and its zero-dimensional exceptional set are both essential in the application below.

Lemma II.10.6.3 (Nondegeneracy of the skipped-digit family). *Fix distinct digits $a_1, \dots, a_m$ and an integer $k \geq 2$. For a block word $u = (i_1, \dots, i_{k-1})$ put*

$$b_u(\lambda) = \sum_{r=1}^{k-1} a_{i_r} \lambda^r, \quad F_{u,\lambda}(x) = \lambda^k x + b_u(\lambda).$$

The real-analytic family $(F_{u,\lambda})_u$, $0 < \lambda < 1$, satisfies the nondegeneracy hypothesis of Theorem II.10.6.1.

Proof. Take two distinct infinite block sequences. Their coding-point difference is a convergent power series in λ . At the first retained digit where the sequences differ, the corresponding coefficient is $a_i - a_j \neq 0$. Hence the difference is not identically zero. This distinguishes identity as an analytic function from an isolated exact overlap at a particular parameter. $\square$

Theorem II.10.6.4 (Shmerkin absolute continuity). *Fix distinct real translations $a_1, \dots, a_m$, $m \geq 2$, and put $H(p) = -\sum_i p_i \log p_i$. There is $E \subset (0, 1)$ with $\dim_H E = 0$ such that for every $\lambda \notin E$ and every positive weight vector p satisfying*

$$\frac{H(p)}{-\log \lambda} > 1,$$

the associated homogeneous self-similar measure is absolutely continuous. The exceptional set is independent of p .

Proof. For each $k \geq 2$, split the digit series into positions divisible by k and their complement. The sparse factor is an affine image of a homogeneous self-similar measure of ratio λ^k , so

outside

$$E_k^F = \{\lambda : \lambda^k \in E_F\}$$

it has positive power Fourier decay by Theorem II.10.5.2. Since $\lambda \mapsto \lambda^k$ is locally bi-Lipschitz away from zero, $\dim_{\text{H}} E_k^F = 0$.

The dense factor is the self-similar measure for the block family in Lemma II.10.6.3, with weights $p_{i_1} \cdots p_{i_{k-1}}$. Its symbolic entropy is $(k-1)H(p)$ and its contraction is λ^k , so its similarity dimension is

$$\left(1 - \frac{1}{k}\right) \frac{H(p)}{-\log \lambda}.$$

Theorem II.10.6.1 gives a zero-dimensional exceptional set E_k^D , independent of p , outside which the dense factor has dimension equal to the minimum of this number and one.

Set

$$E = \bigcup_{k \geq 2} (E_k^F \cup E_k^D).$$

Then $\dim_{\text{H}} E = 0$. Fix $\lambda \notin E$ and p with $H(p)/(-\log \lambda) > 1$. Choose k so large that the displayed dense-factor similarity dimension exceeds one. The dense factor is exact dimensional of value one, while the sparse factor has power Fourier decay. Proposition II.10.3.5 shows that their convolution, which is the original measure, is absolutely continuous. $\square$

II.10.7 Varjú's scale-entropy calculus

Shmerkin's argument used Fourier decay. Varjú's algebraic argument instead controls the entropy lost between nearby scales and shows that these losses are summable. We begin with the scale calculus needed to compare entropy before and after convolution.

For the remainder of the chapter all Shannon entropies and all unsubscripted logarithms inside entropy estimates are base 2. Changing the base in the final algebraic inequality only changes constants.

Definition II.10.7.1 (Separated-cardinality functional). For $A \subset \mathbb{R}$, let $\mathcal{N}_1(A)$ be the maximal cardinality of a 1-separated subset of A .

Definition II.10.7.2 (Homogeneous entropy of a finite measure). Write $\|\sigma\| = \sigma(\mathbb{R})$. If σ is a nonzero finite nonnegative measure with countable support, put

$$H(\sigma) = \|\sigma\| H_2(\sigma/\|\sigma\|).$$

For any nonzero compactly supported finite nonnegative Borel measure on $\mathbb{R}$, put

$$H(\sigma; r_2 \mid r_1) = \|\sigma\| H_2(\sigma/\|\sigma\|; r_2 \mid r_1).$$

For the zero measure both quantities are zero.

Lemma II.10.7.3 (Homogeneous superadditivity on nested grids). *Let $\sigma_1, \dots, \sigma_J$ be finite nonnegative measures. If they have countable support, then*

$$H\left(\sum_{j=1}^J \sigma_j\right) \geq \sum_{j=1}^J H(\sigma_j).$$

If they are compactly supported Borel measures on $\mathbb{R}$, $r > 0$, and $M \in \mathbb{N}$, then

$$H\left(\sum_{j=1}^J \sigma_j; r \mid Mr\right) \geq \sum_{j=1}^J H(\sigma_j; r \mid Mr). \quad (\text{II.10.7.1})$$

Consequently, if σ is a compactly supported finite nonnegative Borel measure and η is a finitely supported finite nonnegative measure, then

$$H(\sigma * \eta; r \mid Mr) \geq \|\eta\| H(\sigma; r \mid Mr). \quad (\text{II.10.7.2})$$

Proof. For a nonnegative summable vector $a = (a_i)_i$, write

$$\Phi(a) = \left(\sum_i a_i\right) H_2\left(\frac{a}{\sum_i a_i}\right), \quad \Phi(0) = 0.$$

If a and b are indexed by the same countable set, concavity of Shannon entropy gives

$$\Phi(a + b) \geq \Phi(a) + \Phi(b).$$

Applying this inequality to the atomic mass vectors proves the first assertion; finite truncation and monotone convergence cover countably many atoms.

We now prove (II.10.7.1). For $t \in [0, M)$ and $k \in \mathbb{Z}$ set

$$p_k^\sigma(t) = \sigma([r(k-t), r(k+1-t))).$$

The cells with indices $M\ell, M\ell + 1, \dots, M\ell + M - 1$ form the single coarse cell

$$[r(M\ell - t), r(M(\ell + 1) - t)).$$

Periodicity of the fine-grid entropy in t , followed by the change of variable $u = t/M$ for the coarse grid, therefore gives

$$H(\sigma; r \mid Mr) = \frac{1}{M} \int_0^M \sum_{\ell \in \mathbb{Z}} \Phi((p_{M\ell+i}^\sigma(t))_{i=0}^{M-1}) dt.$$

For each fixed t and ℓ , the mass vector belonging to $\sum_j \sigma_j$ is the sum of the corresponding mass vectors belonging to the σ_j . The superadditivity of Φ , followed by integration, proves (II.10.7.1).

Finally write $\eta = \sum_j c_j \delta_{y_j}$ and let $T_{y_j}(x) = x + y_j$. Since averaged scale entropy is translation invariant,

$$H(c_j(T_{y_j})_* \sigma; r \mid Mr) = c_j H(\sigma; r \mid Mr).$$

Apply (II.10.7.1) to $\sigma * \eta = \sum_j c_j(T_{y_j})_* \sigma$ and sum over j . □

The integral-ratio hypothesis is the point at which the fine translated grid nests inside the coarse translated grid. We will use homogeneous superadditivity only in this form; it is not asserted for an arbitrary pair of scales.

Lemma II.10.7.4 (Differential-entropy submodularity). *Let X, Y, Z be independent random variables for which the four differential entropies below are finite. Then*

$$h_2(X + Y + Z) + h_2(Y) \leq h_2(X + Y) + h_2(Y + Z).$$

Proof. For densities f_x, g_x , the log-sum inequality gives convexity of relative entropy:

$$D\left(\int f_x d\pi(x) \parallel \int g_x d\pi(x)\right) \leq \int D(f_x \parallel g_x) d\pi(x).$$

Apply this to the Markov kernel that adds the independent variable Z . This yields $I(X; X + Y + Z) \leq I(X; X + Y)$. Independence gives

$$I(X; X + Y) = h_2(X + Y) - h_2(Y), \quad I(X; X + Y + Z) = h_2(X + Y + Z) - h_2(Y + Z),$$

and rearrangement proves the claim. Truncation and Gaussian regularization handle harmless intermediate integrability issues. $\square$

Lemma II.10.7.5 (Integral-scale monotonicity). *Let X, Y be independent bounded random variables and suppose $r_2/r_1 = N \in \mathbb{N}$. Then*

$$H(X + Y; r_1 \mid r_2) \geq H(X; r_1 \mid r_2).$$

The same assertion holds for convolution of compactly supported probabilities.

Proof. Let U be uniform on $[0, r_1]$ and let J be uniform on $\{0, r_1, \dots, (N-1)r_1\}$, independent of everything else. Then $U + J$ is uniform on $[0, r_2]$. By Proposition II.10.4.2, the desired difference is

$$h_2(X + Y + U) - h_2(X + Y + U + J) - h_2(X + U) + h_2(X + U + J).$$

Apply Lemma II.10.7.4 with its first, middle, and third variables equal to $Y, X + U$, and a vanishing Gaussian regularization of J , respectively. This gives the required inequality after rearrangement, without invoking differential entropy of a possibly singular Y . $\square$

Lemma II.10.7.6 (Perturbation of averaged entropy). *Suppose $|X - Y| \leq \delta$ almost surely and $0 < 2\delta \leq r$. Then*

$$|H(X; r) - H(Y; r)| \leq 4 \frac{\delta}{r} \log_2 \frac{2r}{\delta}.$$

Consequently the difference of two conditional entropies over the same pair of scales is bounded by the sum of the corresponding two right-hand sides.

Proof. For fixed translation phase t , the integer variables $\lfloor X/r + t \rfloor$ and $\lfloor Y/r + t \rfloor$ differ only when a grid boundary lies between X and Y . Averaging over $t \in [0, 1]$, the probability of this event is at most δ/r , and on it the integer variables differ by at most one. The elementary comparison $|H(A) - H(B)| \leq H_2(q) + q$ when $\mathbb{P}(A \neq B) \leq q$ and $|A - B| \leq 1$, together with $H_2(q) + q \leq 4q \log_2(2/q)$ for $q \leq 1/2$, proves the result. $\square$

Lemma II.10.7.7 (Scale telescoping and endpoint loss). *For integers $a < b$,*

$$H(\mu; 2^a \mid 2^b) = \sum_{n=a}^{b-1} H(\mu; 2^n \mid 2^{n+1}).$$

If $\sigma_2 < \sigma_1$ are arbitrary real numbers, replacing the endpoints by adjacent integers changes the conditional entropy by at most 2.

Proof. The first identity telescopes. For the second use $0 \leq H(X; r_2) - H(X; r_1) \leq 2 \log_2(r_1/r_2)$ for $r_2 < r_1$, from the smoothing identity and the fact that refining a one-dimensional grid by a factor less than two introduces at most two bits in the averaged partition. $\square$

II.10.8 The high-entropy convolution theorem

The following finite-lattice estimate is the engine behind the high-entropy regime.

Lemma II.10.8.1 (Entropy truncation decomposition). *Let ρ be a probability supported on $[1, N] \cap \mathbb{Z}$ and write $d_N(\rho) = \log_2 N - H_2(\rho)$. Then there are nonnegative functions f, g supported on $[1, N]$ with $\rho = f + g$ such that*

$$\|f - \chi_N\|_2^2 \leq \frac{2d_N(\rho)}{N}, \quad \|f\|_\infty \leq \frac{2}{N}, \quad \|f\|_1 \leq 1, \quad \|g\|_1 \leq 2d_N(\rho),$$

where $\chi_N = N^{-1} \mathbf{1}_{[1, N] \cap \mathbb{Z}}$.

Proof. Define

$$f(n) = \begin{cases} \rho(n), & \rho(n) \leq 2/N, \\ 1/N, & \rho(n) > 2/N, \end{cases} \quad g = \rho - f.$$

For $0 \leq x \leq 2$, convexity gives

$$x \log_2 x \geq \frac{x-1}{\ln 2} + \left(2 - \frac{1}{\ln 2}\right)(x-1)^2,$$

while for $x \geq 2$, $x \log_2 x \geq 2x - 2$. Apply these inequalities with $x = N\rho(n)$ and sum. The linear terms cancel because $\sum_n \rho(n) = 1$, and $2 - 1/\ln 2 \geq 1/2$. $\square$

Lemma II.10.8.2 (Entropy deficit from ℓ^2 distance). *If q is a probability on a set of M points and χ_M is uniform on that set, then*

$$\log_2 M - H_2(q) \leq 2M\|q - \chi_M\|_2^2.$$

Proof. Apply

$$x \log_2 x \leq \frac{x-1}{\ln 2} + \frac{(x-1)^2}{\ln 2}$$

to $x = Mq(j)$ and sum. The linear term vanishes and $1/\ln 2 < 2$. $\square$

Lemma II.10.8.3 (Flat-convolution oscillation and central mass). *Let $f, \tilde{f} : \mathbb{Z} \rightarrow [0, \infty)$ be supported on $[1, N] \cap \mathbb{Z}$ and satisfy $\|f\|_\infty, \|\tilde{f}\|_\infty \leq 2/N$. Then for every $n, m \in \mathbb{Z}$,*

$$|f * \tilde{f}(n) - f * \tilde{f}(n+m)| \leq \frac{3|m|}{N^2} + 2\|f - \chi_N\|_2 \|\tilde{f} - \chi_N\|_2.$$

If in addition

$$\|f - \chi_N\|_2, \|\tilde{f} - \chi_N\|_2 \leq (10\sqrt{N})^{-1}$$

and N is even, then

$$f * \tilde{f}(n) \geq \frac{1}{4N} \quad (N/2 + 1 \leq n \leq 3N/2).$$

Proof. Translate the convolution sum and compare the overlapping summation ranges. The two end strips contribute at most $2|m|/N^2$, and the triangular uniform kernel changes by at most another $|m|/N^2$. On the common range write $f = \chi_N + u$ and $\tilde{f} = \chi_N + v$; Cauchy–Schwarz bounds the nonconstant contribution by $2\|u\|_2\|v\|_2$.

For $N/2 + 1 \leq n \leq 3N/2$, the uniform convolution is at least $1/(2N)$. The two linear error terms are bounded by $N^{-1/2}\|u\|_2$ and $N^{-1/2}\|v\|_2$, and the quadratic term by $\|u\|_2\|v\|_2$. Under the stated hypotheses their sum is $< 1/(4N)$. $\square$

Lemma II.10.8.4 (Central flat–flat entropy estimate). *Let N be even, $M \mid N$, and let $f, \tilde{f}$ satisfy Lemma II.10.8.3, together with*

$$\|f - \chi_N\|_2^2, \|\tilde{f} - \chi_N\|_2^2 \leq \frac{1}{100N}, \quad \|f\|_1, \|\tilde{f}\|_1 \leq 1.$$

*Put $\rho = (f * \tilde{f})|_{[N/2+1, 3N/2]}$. Then*

$$\|\rho\|_1 \log_2 M - H(\rho; 1 \mid M) \leq C \left(N^2 \|f - \chi_N\|_2^2 \|\tilde{f} - \chi_N\|_2^2 + \frac{M \log_2 M}{N} \right).$$

Proof. If $M = 1$, the entropy deficit vanishes, so assume $M \geq 2$. For $a \in \mathbb{Z}$ let $\rho_a =$

$\rho|_{[a+1, a+M]}$ and $p(a) = \|\rho_a\|_1$. Sliding-block averaging gives

$$H(\rho; 1 \mid M) = \frac{1}{M} \sum_a H(\rho_a).$$

For $I = [N/2, 3N/2 - M] \cap \mathbb{Z}$, Lemma II.10.8.3 gives $p(a) \geq M/(4N)$. Its oscillation estimate, the lower bound on $p(a)$, and a sum of M squared coordinate errors show that for $a \in I$ the squared ℓ^2 -distance between $p(a)^{-1}\rho_a$ and the uniform law on its M sites is at most a constant times

$$\frac{M}{N^2} + \frac{N^2}{M} \|f - \chi_N\|_2^2 \|\tilde{f} - \chi_N\|_2^2.$$

Apply Lemma II.10.8.2, multiply by $p(a)$, and sum. The $O(M)$ boundary block positions have total sliding mass $O(M^2/N)$, using $\|\rho\|_\infty \leq 2/N$, and cost at most $CM \log M/N$ after division by M . The identity $\sum_a p(a) = M\|\rho\|_1$ completes the estimate. $\square$

Lemma II.10.8.5 (Central flat-small entropy estimate). *Let N be even and $M \mid N$. Let $f, g : \mathbb{Z} \rightarrow [0, \infty)$ be supported on $[1, N]$, with*

$$\frac{1}{2} \leq \|f\|_1 \leq 1, \quad \|f\|_\infty \leq \frac{2}{N}, \quad \|g\|_1 \leq 1.$$

*Put $\rho = (f * g)|_{[N/2+1, 3N/2]}$. Then*

$$\|\rho\|_1 \log_2 M - H(\rho; 1 \mid M) \leq 8N \|f - \chi_N\|_2^2 \|g\|_1 + 6 \frac{M \log_2 M}{N}.$$

Proof. Normalize f to a probability. Orthogonality to the constant function gives

$$\| \|f\|_1^{-1} f - \chi_N \|_2^2 \leq \|f\|_1^{-2} \|f - \chi_N\|_2^2.$$

Lemma II.10.8.2, first at scale N and then through the sliding M -block identity, gives

$$\|f\|_1 \log_2 M - H(f; 1 \mid M) \leq 8N \|f - \chi_N\|_2^2 + 2M/N.$$

Lemma II.10.7.3, applied to the translates of f in the finite convolution sum, gives $H(f * g; 1 \mid M) \geq \|g\|_1 H(f; 1 \mid M)$. Passing to the central restriction costs only the blocks meeting the boundary strips; their total mass is controlled by $\|f * g\|_1 - \|\rho\|_1 + 4M/N$, using $\|f * g\|_\infty \leq 2\|g\|_1/N$. This proves the stated estimate. $\square$

Proposition II.10.8.6 (Central finite-block estimate). *There is an absolute C such that the following holds. Let N be even, $M \mid N$, and let $\rho, \tilde{\rho}$ be probabilities supported on $[1, N] \cap \mathbb{Z}$. Put $\sigma = (\rho * \tilde{\rho})|_{[N/2+1, 3N/2]}$. Then*

$$\|\sigma\|_1 \log_2 M - H(\sigma; 1 \mid M) \leq C \log_2 M d_N(\rho) d_N(\tilde{\rho}) + C \frac{M \log_2 M}{N}.$$

Proof. If both deficits are below $1/200$, apply Lemma II.10.8.1 to write $\rho = f + g$ and $\tilde{\rho} = \tilde{f} + \tilde{g}$. The central restrictions of $f * \tilde{f}$, $f * \tilde{g}$, and $g * \tilde{f}$ satisfy Lemmas II.10.8.4 and II.10.8.5. Their deficit terms are bounded by $C \log M d_N(\rho) d_N(\tilde{\rho})$. The remaining term $g * \tilde{g}$ has mass at most $4d_N(\rho) d_N(\tilde{\rho})$. Lemma II.10.7.3, applied at the integral scale ratio M , shows that omitting this term costs only its mass times $\log M$.

If exactly one deficit is below $1/200$, decompose its measure. Apply Lemma II.10.8.5. The other deficit is bounded below, so the resulting loss is absorbed according to

$$O(d_N(\rho)) = O(d_N(\rho) d_N(\tilde{\rho})).$$

If both deficits are at least $1/200$, the right side already dominates the trivial upper bound. $\square$

Proposition II.10.8.7 (Integer high-entropy estimate). *There is an absolute C such that, for finitely supported probabilities $v, \tilde{v}$ on $\mathbb{Z}$ and integers N, M with $2 \mid N$ and $M \mid N$,*

$$\log_2 M - H(v * \tilde{v}; 1 \mid M) \leq C \log_2 M \Delta_N(v) \Delta_N(\tilde{v}) + C \frac{M \log_2 M}{N},$$

where $\Delta_N(\eta) = \log_2 N - H(\eta; 1 \mid N)$.

Proof. For $a, b \in \mathbb{Z}$ set $f_a = v|_{[a+1, a+N]}$ and $\tilde{f}_b = \tilde{v}|_{[b+1, b+N]}$, and let $\sigma_{a,b}$ be the restriction of $f_a * \tilde{f}_b$ to $[a+b+N/2+1, a+b+3N/2]$. Sliding-window averaging gives

$$H(v; 1 \mid N) = \frac{1}{N} \sum_a H(f_a), \quad H(\tilde{v}; 1 \mid N) = \frac{1}{N} \sum_b H(\tilde{f}_b).$$

Every point-pair contributing to $v * \tilde{v}$ belongs to N^2 pairs of length- N windows, and the central restriction has a fixed positive fraction of this multiplicity. Apply Proposition II.10.8.6 to the normalized versions of each nonzero pair and multiply back by their masses. Lemma II.10.7.3, again at the integral ratio M , yields the convolution deficit, while the product-deficit sum factors as

$$\frac{1}{N^2} \sum_{a,b} (\|f_a\|_1 \log N - H(f_a)) (\|\tilde{f}_b\|_1 \log N - H(\tilde{f}_b)) = \Delta_N(v) \Delta_N(\tilde{v}).$$

The boundary terms contribute $O(M \log M/N)$ after division by the fixed multiplicity. $\square$

Theorem II.10.8.8 (High-entropy convolution). *There is an absolute C such that if $0 < \alpha < 1/2$ and compactly supported probabilities μ, v satisfy*

$$H(\mu; s \mid 2s), H(v; s \mid 2s) \geq 1 - \alpha$$

for every s with $|\log_2 r - \log_2 s| < 3 \log_2(\alpha^{-1})$, then

$$H(\mu * v; r \mid 2r) \geq 1 - C[\log_2(\alpha^{-1})]^3 \alpha^2.$$

Proof. By scaling reduce to an adjacent dyadic target scale and put $K = \lfloor 3 \log_2 \alpha^{-1} \rfloor$. Choose translation phases for μ and ν and round to integer-valued random variables X_0, Y_0 with errors less than one. The phase-averaging identity from Proposition II.10.4.2 allows the phases to be chosen so that

$$H(X_0; 1 \mid 2^{2K+1}) \geq (2K+1)(1-\alpha), \quad H(Y_0; 1 \mid 2^{2K+1}) \geq (2K+1)(1-\alpha).$$

Apply Proposition II.10.8.7 with $M = 2^{K+1}$ and $N = 2^{2K+1}$. The missing entropy between scales 2^K and 2^{K+1} is at most the missing entropy between 1 and 2^{K+1} , giving

$$1 - H(X_0 + Y_0; 2^K \mid 2^{K+1}) \leq CK^3\alpha^2 + CK2^{-K}.$$

Rounding changes the sum by less than 2, so Lemma II.10.7.6 contributes another $CK2^{-K}$. Since $2^{-K} \leq C\alpha^2$ and $K \leq 3 \log_2 \alpha^{-1}$, scaling back proves the theorem. $\square$

II.10.9 The low-entropy growth theorem

We next isolate the two mechanisms used in the low-entropy regime: a two-point measure creates entropy at a scale where the other factor has a deficit, and a measure with enough entropy contains quantitatively many such two-point pieces.

Proposition II.10.9.1 (Entropy gain from a Bernoulli piece). *Let μ be compactly supported and let η assign mass $1/2$ to two points at distance $t > 0$. Put $d = 1 - H(\mu; t \mid 2t)$. If $d > 0$, $r_2 \leq td/10$, and $r_1 \geq 144td^{-2}$, then*

$$H(\mu * \eta; r_2 \mid r_1) \geq H(\mu; r_2 \mid r_1) + d/3.$$

Proof. Translation and scaling reduce to $\eta = (\delta_0 + \delta_t)/2$. Since $\eta * U_t = U_{2t}$, Proposition II.10.4.2 gives $H(\mu * \eta; t) = H(\mu; 2t) + 1$, so the entropy gain at scale t is exactly d . Put $J(s) = H(\mu * \eta; s) - H(\mu; s)$. Then the desired conditional gain is $J(r_2) - J(r_1)$. The fully proved smoothing estimates (II.9.N28) and (II.9.N27), in the present base-two normalization, give respectively

$$J(r_2) \geq J(t) - \frac{2}{(\log 2)(t/r_2 - 1)}, \quad |J(r_1)| \leq 4\sqrt{t/r_1}.$$

Here $0 < d \leq 1$, so $t/r_2 \geq 10/d$ and

$$\frac{2}{(\log 2)(t/r_2 - 1)} \leq \frac{2d}{9 \log 2} < \frac{d}{3}, \quad 4\sqrt{t/r_1} \leq \frac{d}{3}.$$

Since $J(t) = d$, subtraction proves the claimed gain. $\square$

Lemma II.10.9.2 (Entropy stability under removing small mass). *Let $\rho = (1-\delta)\rho_0 + \delta\rho_1$, $0 < \delta \leq 1/2$. Then for every pair of scales*

$$|H(\rho; r \mid R) - (1-\delta)H(\rho_0; r \mid R) - \delta H(\rho_1; r \mid R)| \leq H_2(\delta).$$

In particular the error is at most $C\delta \log_2(1/\delta)$.

Proof. Adjoin the binary label selecting the component. At each single scale the entropy of the mixture differs from the weighted component entropies by a number between 0 and $H_2(\delta)$. Subtract the two scales. $\square$

Proposition II.10.9.3 (Bernoulli extraction). *Let ρ be finitely supported and $r > 0$. Put $h = H(\rho; r \mid 2r)$. If*

$$H(\rho; r/2 \mid r) \leq \frac{3}{2}h,$$

then

$$\rho = \rho_0 + \eta_1 + \cdots + \eta_N,$$

where ρ_0 is nonnegative, every η_i is an equal-mass two-point measure supported on points at distance in $[r/2, 2r]$, and

$$\sum_i \|\eta_i\| \geq \frac{h}{128(\log_2(h^{-1}) + 1)}.$$

When $h = 0$, the right side is interpreted as 0.

Proof. If $h = 0$, take $N = 0$ and $\rho_0 = \rho$. Henceforth $0 < h \leq 1$. Choose such a decomposition with $\|\rho_0\|$ minimal and put $\delta = \sum_i \|\eta_i\| = 1 - \|\rho_0\|$. Minimality implies no two points of $\text{supp } \rho_0$ have distance in $[r/2, 2r]$. Thus its support is covered by intervals of diameter $< r/2$ separated by gaps $> 2r$, and the sliding-grid formula gives

$$H(\rho_0; r/2 \mid r) = 2H(\rho_0; r \mid 2r). \quad (\text{II.10.9.1})$$

If $\delta \geq 1/2$ there is nothing to prove. Otherwise Lemma II.10.9.2, first between r and $2r$ and then between $r/2$ and r , gives

$$H(\rho; r \mid 2r) \leq H(\rho_0; r \mid 2r) + 3\delta \log_2 \delta^{-1}$$

and, using (II.10.9.1),

$$H(\rho; r/2 \mid r) \geq 2H(\rho; r \mid 2r) - 8\delta \log_2 \delta^{-1}.$$

The hypothesis implies

$$\delta \log_2 \delta^{-1} \geq h/16. \quad (\text{II.10.9.2})$$

Set $L = \log_2 h^{-1} \geq 0$ and $\delta_0 = h/[128(L+1)]$. Since $\delta_0 < 1/2$ and $\log_2(\delta_0^{-1}) = L+7+\log_2(L+1) \leq 7(L+1)$, we have $\delta_0 \log_2(\delta_0^{-1}) \leq 7h/128 < h/16$. The function $x \mapsto x \log_2(x^{-1})$ is increasing on $(0, (e)^{-1}]$, and $\delta_0 \leq 1/128$. Thus $\delta < \delta_0$ contradicts (II.10.9.2). $\square$

Lemma II.10.9.4 (Selection of useful scales). *Fix $0 < \alpha < 1/2$. There is $c_0 = c_0(\alpha) > 0$ such that the following holds. Suppose μ, ν are compactly supported, $\sigma_2 < \sigma_1 < 0$,*

$0 < \beta \leq 1/2$, and

$$\mathcal{N}_1\{\sigma \in [\sigma_2, \sigma_1] : H(\mu; 2^\sigma \mid 2^{\sigma+1}) > 1 - \alpha\} < c_0\beta(\sigma_1 - \sigma_2),$$

while

$$H(v; 2^{\sigma_2} \mid 2^{\sigma_1}) > \beta(\sigma_1 - \sigma_2).$$

Let $K = \lceil \log_2(144/\alpha^2) \rceil + 2$. Then there is a $2K$ -separated set $B \subset \mathbb{Z} \cap [\sigma_2 + K, \sigma_1 - K]$ such that, with $h_n = H(v; 2^n \mid 2^{n+1})$,

$$(i) \ H(v; 2^{n-1} \mid 2^n) < \frac{3}{2}h_n;$$

$$(ii) \ h_n \geq \beta/12;$$

$$(iii) \ H(\mu; t \mid 2t) \leq 1 - \alpha \text{ for } 2^{n-1} \leq t \leq 2^{n+1};$$

$$(iv) \ \sum_{n \in B} h_n \geq c_1\beta(\sigma_1 - \sigma_2) - 1 \text{ for some } c_1 = c_1(\alpha) > 0.$$

Proof. Round the endpoints inward and let $\mathcal{J} = \mathbb{Z} \cap [\sigma_2 + K, \sigma_1 - K]$. If $\mathcal{J}$ is empty, (iv) has a nonpositive right side and $B = \emptyset$ works. Lemma II.10.7.7 and $0 \leq h_n \leq 1$ show that this loses at most $2K + 2$ bits. Put

$$B_1 = \{n \in \mathcal{J} : H(v; 2^{n-1} \mid 2^n) < \frac{3}{2}h_n\}.$$

At every $n \in \mathcal{J} \setminus B_1$ one has $h_n \leq (2/3)h_{n-1}$. On each maximal run of such indices following an index $j \in B_1$, the sum is at most $2h_j$. The initial run contributes at most $3h_{n_0-1} \leq 3$, where n_0 is the first retained integer scale. Writing $\Delta = \sigma_1 - \sigma_2$, the entropy in the interior integer range is at most $3 \sum_{n \in B_1} h_n + 3$. The endpoint loss above therefore gives

$$\sum_{n \in B_1} h_n > \frac{\beta\Delta}{3} - \frac{2K+5}{3}.$$

Deleting indices with $h_n < \beta/12$ removes at most $(\beta/12)(\sigma_1 - \sigma_2 + 1)$ bits. Next delete every n for which (iii) fails. Such an n supplies a witness scale in $[n-1, n+1]$ belonging to the high-entropy set of μ . Color the bad integers by residue modulo 4. Witnesses from one color are 1-separated, so there are at most $4\mathcal{N}_1$ (high-entropy set) bad indices. Since $h_n \leq 1$, choosing $c_0 \leq 1/96$ removes at most $\beta\Delta/24$ further bits. The retained mass is consequently greater than

$$\frac{5\beta\Delta}{24} - \frac{2K+5}{3} - \frac{\beta}{12}.$$

Choose a residue class modulo $2K$ with maximal retained mass. It is $2K$ -separated and carries more than

$$\frac{5\beta\Delta}{48K} - \frac{2K+5}{6K} - \frac{\beta}{24K} \geq \frac{\beta\Delta}{32K} - 1,$$

since $K \geq 12$ and $\beta \leq 1/2$. Thus (iv) holds with $c_1 = 1/(32K)$. $\square$

Theorem II.10.9.5 (Low-entropy growth). *For every $0 < \alpha < 1/2$ there is $c = c(\alpha) > 0$ such that, if $\mu, v, \sigma_2 < \sigma_1 < 0$, and $0 < \beta \leq 1/2$ satisfy the hypotheses of Lemma II.10.9.4,*

then

$$H(\mu * \nu; 2^{\sigma_2} \mid 2^{\sigma_1}) > H(\mu; 2^{\sigma_2} \mid 2^{\sigma_1}) + \frac{c\beta}{\log_2 \beta^{-1} + 1}(\sigma_1 - \sigma_2) - 3.$$

Proof. First suppose ν is finitely supported and take the set B from Lemma II.10.9.4. For $n \in B$, Proposition II.10.9.3 at $r = 2^n$ decomposes ν into Bernoulli pieces of total mass

$$\delta_n \geq c \frac{h_n}{\log_2 \beta^{-1} + 1},$$

because $h_n \geq \beta/12$. Each such piece is supported on a pair at distance $t \in [2^{n-1}, 2^{n+1}]$. Property (iii) gives $1 - H(\mu; t \mid 2t) \geq \alpha$. With $r_2 = 2^{n-K}$ and $r_1 = 2^{n+K}$, Proposition II.10.9.1 applies. Since $r_1/r_2 = 2^{2K}$ is an integer, the convolution consequence of Lemma II.10.7.3 therefore yields

$$H(\mu * \nu; 2^{n-K} \mid 2^{n+K}) \geq H(\mu; 2^{n-K} \mid 2^{n+K}) + c \frac{h_n}{\log_2 \beta^{-1} + 1}.$$

The windows are disjoint because B is $2K$ -separated. Lemma II.10.7.5 lets us insert the remaining convolution factors, and telescoping gives

$$H(\mu * \nu; 2^{\sigma_2} \mid 2^{\sigma_1}) \geq H(\mu; 2^{\sigma_2} \mid 2^{\sigma_1}) + \frac{c}{\log_2 \beta^{-1} + 1} \sum_{n \in B} h_n - 2.$$

Use Lemma II.10.9.4(iv). For general compactly supported ν , approximate almost surely by finitely supported random variables and use Lemma II.10.7.6. $\square$

II.10.10 Algebraic entropy and Mahler measure

The low-entropy theorem becomes effective for algebraic contractions once the entropy carried by a Bernoulli prefix is bounded from below. Mahler measure provides the arithmetic quantity that controls this lower bound.

Definition II.10.10.1. For biased signs $\varepsilon_j \in \{-1, 1\}$ with $\mathbb{P}(\varepsilon_j = 1) = p$, put

$$S_n = \sum_{j=0}^{n-1} \varepsilon_j \lambda^j, \quad h_{\lambda, p} = \lim_{n \rightarrow \infty} \frac{1}{n} H_2(S_n).$$

For algebraic λ , write $M_\lambda = M(\lambda)$ for the Mahler measure of Definition II.8.08.

Proposition II.10.10.2 (Collision dichotomy and entropy-rate existence). *The limit in Definition II.10.10.1 exists and equals $\inf_n H_2(S_n)/n$. If no two finite sign words collide, then $h_{\lambda, p} = H_2(p)$. If a finite collision occurs and $0 < \lambda < 1$ is algebraic, then λ is an algebraic unit.*

Proof. Write $S_{n+m} = S_n + \lambda^n S'_m$, where S'_m is independent with the law of S_m . Since the sum is a deterministic function of the pair, $H_2(S_{n+m}) \leq H_2(S_n) + H_2(S_m)$, and Fekete's lemma gives the limit and infimum formula. In the collision-free case the word-to-sum map is injective.

A collision gives a nonzero polynomial with coefficients in $\{-1, 0, 1\}$ vanishing at λ after division by 2 and removal of an initial power of x . Its leading and constant coefficients are ± 1 . The primitive minimal polynomial of λ divides it in $\mathbb{Z}[x]$, so it is monic with constant term ± 1 . $\square$

Definition II.10.10.3 (Gaussian-smoothed entropy). Let G be a standard Gaussian vector. For an invertible real matrix B and a bounded random vector X , independent of G , put

$$E_B(X) = h_2(X + BG) - h_2(BG), \quad E_{B_1|B_2}(X) = E_{B_1}(X) - E_{B_2}(X).$$

Write $B_1 \preceq B_2$ when $B_2 B_2^T - B_1 B_1^T$ is positive semidefinite.

Lemma II.10.10.4 (Gaussian entropy calculus). *The following hold.*

- (a) *If Z is finite atomic, then $E_B(Z) \leq H_2(Z)$.*
- (b) *If $B_1 \preceq B_2$ and X, Y are independent bounded random vectors, then $E_{B_1|B_2}(X+Y) \geq E_{B_1|B_2}(X)$.*
- (c) $E_I(X) \geq 0$.

Proof. For (a), adjoining the discrete label gives $h_2(Z + BG) \leq H_2(Z) + h_2(BG)$. For (b), choose an independent Gaussian G_M whose covariance is $B_2 B_2^T - B_1 B_1^T$ and apply Lemma II.10.7.4 with its first, middle, and third variables equal to Y , $X + B_1 G$, and G_M , respectively. The Gaussian normalization terms cancel. Part (c) is the nonnegativity of $I(X; X + G)$ by the same relative-entropy/log-sum argument. $\square$

Lemma II.10.10.5 (Matrix Bernoulli lower bound). *Let A be an invertible real matrix with spectral radius < 1 and operator norm at most one, and let*

$$X = \sum_{j \geq 0} \varepsilon_j A^j x.$$

Assume that whenever two sign words give the same scalar prefix S_n , they also give the same vector prefix $Z_n = \sum_{j < n} \varepsilon_j A^j x$. Then

$$h_{\lambda,p} \geq E_{A|I}(\varepsilon_0 x).$$

Proof. The factor assumption gives $H_2(Z_n) \leq H_2(S_n)$. Let X' be an independent copy of X . The identity $X = Z_n + A^n X'$ and Lemma II.10.10.4(a) give $E_{A^n}(X) \leq H_2(Z_n) + E_I(X')$. Adding the independent prefix cannot reduce differential entropy, so $E_I(X) \geq E_I(A^n X')$.

Hence

$$E_{A^n|I}(X) \leq H_2(S_n) + C_{A,x}, \quad (\text{II.10.10.1})$$

with $C_{A,x}$ independent of n .

Telescoping gives $E_{A^n|I}(X) = \sum_{j=1}^n E_{A^j|A^{j-1}}(X)$. Since $\|A\| \leq 1$, we have $A^j(A^j)^T \preceq A^{j-1}(A^{j-1})^T$. Split off the first $j-1$ digits and apply Lemma II.10.10.4(b); after the invertible change by A^{j-1} , $E_{A^j|A^{j-1}}(X) \geq E_{A|I}(X) \geq E_{A|I}(\varepsilon_0 x)$. Combine with (II.10.10.1), divide by n , and use Proposition II.10.10.2. $\square$

Lemma II.10.10.6 (One-defect singular-value model). *Let $0 < \lambda < 1$ be an algebraic unit. There is a real matrix C whose spectrum consists of the Galois conjugates of λ in the open unit disk, such that*

$$\|C\| = 1, \quad \text{the singular values of } C \text{ are } 1, \dots, 1, M_\lambda^{-1}.$$

Proof. Start with a real matrix A_0 whose eigenvalues are precisely those conjugates, using real 2×2 blocks for conjugate pairs. Its spectral radius is < 1 . Since the minimal polynomial is separable, A_0^T has a cyclic vector v . Define

$$q_v(y) = \sum_{n \geq 0} |\langle A_0^n y, v \rangle|^2.$$

The series converges and cyclicity makes q_v positive definite. Choose $B \in GL_d(\mathbb{R})$ with $q_v(y) = \|By\|^2$ and put $C = BA_0B^{-1}$. From $q_v(y) = |\langle y, v \rangle|^2 + q_v(A_0 y)$ one obtains

$$\|Cz\|^2 = \|z\|^2 - |\langle z, w \rangle|^2, \quad w = B^{-T}v.$$

Thus $C^T C = I - ww^T$: all but one singular values are 1, and the remaining one is $|\det C|$. Since λ is a unit, the product of moduli of all conjugates is 1; unit-circle conjugates contribute 1, while those outside the unit disk contribute M_λ . Therefore $|\det C| = M_\lambda^{-1}$. $\square$

Lemma II.10.10.7 (Biased Gaussian increment). *For $p \in (0, 1)$ define*

$$\Phi_p(a) = \sup_{t > 0} \{h_2(ta\varepsilon + G) - h_2(t\varepsilon + G)\}, \quad a > 1,$$

where $\mathbb{P}(\varepsilon = 1) = p$. There is $c_p > 0$ such that

$$\Phi_p(a) \geq c_p \min\{\log_2 a, 1\} \quad (a > 1).$$

Proof. Put $I_p(t) = h_2(t\varepsilon + G) - h_2(G)$. Adding Gaussian noise and the data-processing argument in Lemma II.10.7.4 show that $I_p(t)$ is nondecreasing. Also $\Phi_p(a^2) \leq 2\Phi_p(a)$, since the increment from t to $a^2 t$ is the sum of the increments from t to at and from at to $a^2 t$.

Let $m = \mathbb{E}\varepsilon$, $\eta = \varepsilon - m$, and $\sigma_p^2 = \text{Var } \eta = 4p(1-p) > 0$. Translation invariance reduces the small-signal calculation to $F_p(t) = h_2(t\eta + G)$. The density is a two-Gaussian mixture;

boundedness of η gives integrable domination for the first three derivatives. Differentiating under the integral gives

$$F'_p(0) = 0, \quad F''_p(0) = \sigma_p^2 / \ln 2,$$

so

$$F_p(t) = F_p(0) + \frac{\sigma_p^2}{2 \ln 2} t^2 + O_p(t^3).$$

For $\sqrt{2} \leq a \leq 2$, choose t small and obtain $\Phi_p(a) \geq c_p \log_2 a$. If $1 < a < \sqrt{2}$, iterate the inequality $\Phi_p(a^2) \leq 2\Phi_p(a)$ until $a^{2^k} \in [\sqrt{2}, 2)$. For $a \geq 2$, monotonicity gives a fixed positive lower bound. $\square$

Theorem II.10.10.8 (Biased Mahler entropy bound). *For every $p \in (0, 1)$ there is $c_p > 0$ such that for every algebraic $0 < \lambda < 1$,*

$$h_{\lambda,p} \geq c_p \min\{1, \log_2 M_\lambda\}.$$

Proof. If the prefix map is collision-free, Proposition II.10.10.2 gives $h_{\lambda,p} = H_2(p)$ and the claim follows after decreasing c_p . Otherwise λ is an algebraic unit. Let C be the matrix from Lemma II.10.10.6. If two scalar sign prefixes agree, their difference polynomial is divisible by the minimal polynomial of λ , so it vanishes at every conjugate and gives the same matrix prefix. Lemma II.10.10.5 applies.

Choose a singular-value decomposition $C = UDV^T$ with $D = \text{diag}(1, \dots, 1, M_\lambda^{-1})$ and take $x = tUe_d$. Orthogonal invariance of the Gaussian and the invertible change of variables by C^{-1} give

$$E_{C|I}(\varepsilon x) = h_2(tM_\lambda \varepsilon + G) - h_2(t\varepsilon + G).$$

Therefore $h_{\lambda,p} \geq \Phi_p(M_\lambda)$, and Lemma II.10.10.7 finishes the proof. $\square$

II.10.11 Algebraic block production and absolute continuity

We now pass from the prefix entropy bound to many disjoint coefficient blocks with almost maximal entropy. Repeated high-entropy convolution then makes the total entropy deficit summable, allowing the $L \log L$ criterion to finish the proof.

Fix $0 < \lambda, p < 1$. Let $\mu_{\lambda,p}$ be the law of $\sum_{n \geq 0} \varepsilon_n \lambda^n$. For an interval $I \subset (0, \infty)$ bounded away from zero define the coefficient factor

$$\mu^I = \text{Law} \left(\sum_{n: \lambda^n \in I} \varepsilon_n \lambda^n \right).$$

If $I_1, \dots, I_N$ are disjoint, independence gives

$$\mu_{\lambda,p} = \mu^{I_1} * \dots * \mu^{I_N} * \rho \tag{II.10.11.1}$$

for a remaining probability ρ .

Definition II.10.11.1 (*k*-high entropy). Fix once and for all a sufficiently large absolute integer A . For $r < 2^{-4}$, put $L_3 = \log_2 \log_2 \log_2 r^{-1}$. For integers $0 \leq k \leq \lfloor L_3 \rfloor + 2$, a compactly supported probability ρ satisfies *k*-HE at scale r if

$$H(\rho; t \mid 2t) \geq 1 - 2^{-(2^k + 3k + A)}$$

whenever

$$|\log_2 t - \log_2 r| \leq A(2 + \log_2 \log_2 \log_2 r^{-1} - k) \log_2 \log_2 r^{-1}.$$

Proposition II.10.11.2 (High-entropy amplification). *Let ρ_1, ρ_2 both satisfy *k*-HE at scale r , where $0 \leq k \leq \lfloor L_3 \rfloor + 1$ and L_3 is as in Definition II.10.11.1. For A sufficiently large and r sufficiently small, depending only on A and the absolute constant in Theorem II.10.8.8, the convolution $\rho_1 * \rho_2$ satisfies $(k + 1)$ -HE at scale r .*

Proof. Put $\alpha_k = 2^{-(2^k + 3k + A)}$. Write $L_2 = \log_2 \log_2 r^{-1}$ and $L_3 = \log_2 L_2$. The $(k + 1)$ -HE window has logarithmic radius $A(1 + L_3 - k)L_2$, whereas the *k*-HE hypothesis is available on the larger window of radius $A(2 + L_3 - k)L_2$. Thus every target scale has the extra margin AL_2 . Since $k \leq 1 + L_3$, for sufficiently small r one has $AL_2 \geq 3 \log_2 \alpha_k^{-1}$. Thus Theorem II.10.8.8 applies. Its deficit is at most

$$C(2^k + 3k + A)^3 2^{-2(2^k + 3k + A)}.$$

It is enough that $C(2^k + 3k + A)^3 \leq 2^{3k + A - 3}$. Choose A large so this holds for $k = 0, 1$; the ratio then improves with k . $\square$

Lemma II.10.11.3 (Algebraic separation). *Let $0 < \lambda < 1$ be algebraic and let d be the number of its Galois conjugates on the unit circle. There is $c_\lambda > 0$ such that, for every $\ell \geq 1$ and every $P(x) = \sum_{j=0}^{\ell} b_j x^j$ with $b_j \in \{-1, 0, 1\}$ and $P(\lambda) \neq 0$,*

$$|P(\lambda)| \geq c_\lambda (\ell + 1)^{-d} M_\lambda^{-\ell}.$$

Proof. Let $Q(x) = a \prod_{i=1}^D (x - \lambda_i) \in \mathbb{Z}[x]$ be the primitive minimal polynomial, with $\lambda_1 = \lambda$. The resultant

$$\text{Res}(Q, P) = a^\ell \prod_{i=1}^D P(\lambda_i)$$

is a nonzero integer. For $|\lambda_i| < 1$, a geometric series bounds $|P(\lambda_i)|$ by a constant; for $|\lambda_i| = 1$ it is at most $\ell + 1$; for $|\lambda_i| > 1$ it is at most $C_i |\lambda_i|^\ell$. Solving for $|P(\lambda)|$ yields the claimed Mahler-measure bound. $\square$

Lemma II.10.11.4 (Entropy carried by a coefficient block). *For every $p \in (0, 1)$ there is $c_p > 0$ with the following property. Let $0 < \lambda < 1$ be algebraic, let $0 < \tau < 1$ be*

sufficiently small depending on λ , and put

$$\ell = \left\lfloor \frac{\log_2 \tau^{-1}}{2 \log_2 M_\lambda} \right\rfloor.$$

Then there is $\tau_1 > \tau$ with $\tau_1/\tau \in \mathbb{N}$ such that, for every $\tau^{1/3} < a < 1$,

$$\log_2 \tau_1^{-1} \geq c_p \frac{\log_2 \tau^{-1}}{\log_2(M_\lambda + 1)},$$

and

$$H(\mu^{(a\lambda^\ell, a]}; \tau | \tau_1) \geq c_p \frac{\log_2 \tau^{-1}}{\log_2(M_\lambda + 1)}.$$

Proof. Up to multiplication by a power of λ , the factor is a length- ℓ biased Bernoulli prefix. By Lemma II.10.11.3, distinct atoms are separated by at least $c_\lambda a(\ell+1)^{-d} M_\lambda^{-\ell}$. Since $a > \tau^{1/3}$ and $M_\lambda^{-\ell} \geq \tau^{1/2}$ up to a fixed factor, this exceeds τ for sufficiently small τ . Thus averaging over translated τ -grids does not merge distinct atoms, and

$$H(\mu^{(a\lambda^\ell, a]}; \tau) = H_2(S_\ell) \geq \ell h_{\lambda, p}.$$

Theorem II.10.10.8 gives the required fine-scale lower bound. Every coefficient factor has support in $[-(1-\lambda)^{-1}, (1-\lambda)^{-1}]$, hence $H(\mu^{(a\lambda^\ell, a]}; s) \leq \log_2 s^{-1} + C_\lambda$. Choose $\tau_1/\tau \in \mathbb{N}$ so that $\log_2 \tau_1^{-1}$ is a sufficiently small fixed fraction of the fine-scale lower bound; rounding costs $O(1)$. Subtract the coarse entropy. $\square$

Lemma II.10.11.5 (Many high-entropy scales in one coefficient range). *Fix $0 < \alpha < 1/2$ and $p \in (0, 1)$. Put*

$$K_\lambda = \log_2(M_\lambda + 1) \log_2 \log_2(M_\lambda + 2).$$

There is $c = c(\alpha, p) > 0$ such that, if $\tau > 0$ is sufficiently small and $0 < t < 1$ satisfies

$$1 - \lambda < c \frac{|\log_2 t| \min\{1, \log_2 M_\lambda\}}{|\log_2 \tau| \log_2 \log_2(M_\lambda + 2)}, \quad (\text{II.10.11.2})$$

and $t \geq \tau^{1/6}$, then there are

$$K \geq c K_\lambda^{-1} \log_2 \tau^{-1}$$

scales

$$\tau^{c/\log_2(M_\lambda+1)} > s_1 > \dots > s_K > \tau, \quad s_i > 2s_{i+1},$$

such that

$$H(\mu^{(t^2, t)}; s_i | 2s_i) > 1 - \alpha \quad (1 \leq i \leq K).$$

Proof. Let $\ell = \lfloor \log_2 \tau^{-1} / (2 \log_2 M_\lambda) \rfloor$, and let τ_1 be supplied by Lemma II.10.11.4. Put

$$N = \left\lfloor \frac{\log t}{\ell \log \lambda} \right\rfloor, \quad I_j = (\lambda^{\ell} t, \lambda^{(j-1)\ell} t], \quad 1 \leq j \leq N.$$

Then $I_j \subset (t^2, t)$ and $\lambda^{N\ell} t \geq t^2 \geq \tau^{1/3}$, so every block satisfies Lemma II.10.11.4. With $\beta = c_1 / \log_2(M_\lambda + 1)$,

$$H(\mu^{I_j}; \tau \mid \tau_1) \geq \beta \log_2(\tau_1 / \tau). \quad (\text{II.10.11.3})$$

Set $\rho_i = \mu^{I_1} * \dots * \mu^{I_i}$ and $\rho_0 = \delta_0$. Since conditional entropy is at most $\log_2(\tau_1 / \tau)$, averaging the N increments gives some $0 \leq i < N$ with

$$H(\rho_{i+1}; \tau \mid \tau_1) - H(\rho_i; \tau \mid \tau_1) \leq \frac{1}{N} \log_2(\tau_1 / \tau). \quad (\text{II.10.11.4})$$

The definitions of N, ℓ and $-\log \lambda \leq 2(1 - \lambda)$ near one imply, by (II.10.11.2), $1/N \leq Cc/K_\lambda$ after taking the constant small.

Apply Theorem II.10.9.5 to $\mu = \rho_i, \nu = \mu^{I_{i+1}}, \sigma_2 = \log_2 \tau, \sigma_1 = \log_2 \tau_1$. The entropy hypothesis on ν is (II.10.11.3). If the high-entropy scale set of ρ_i had separated cardinality below the low-entropy threshold, that theorem would give an increment $\geq c_2 K_\lambda^{-1} \log_2(\tau_1 / \tau) - 3$, contradicting (II.10.11.4) for sufficiently small τ . Hence the high-entropy hypothesis fails, yielding at least $c K_\lambda^{-1} \log_2(\tau_1 / \tau) \geq c K_\lambda^{-1} \log_2 \tau^{-1}$ separated logarithmic scales. Ordering them gives $s_i > 2s_{i+1}$. Lemma II.10.11.4 also lets us choose $\tau_1 \leq \tau^{c/\log_2(M_\lambda+1)}$. Since ρ_i is a convolution factor of $\mu^{(t^2, t)}$, Lemma II.10.7.5 transfers the same lower entropy bound to that factor. $\square$

Lemma II.10.11.6 (From one high scale to a 0-HE factor). *Fix the constant A of Definition II.10.11.1. There is $\alpha_0 > 0$ such that the following holds. Let $0 < \lambda, r, s, t < 1$ satisfy*

$$\log_2 t^{-1} > (\log_2 \log_2 r^{-1})^2, \quad r < s < t^4, \quad \lambda > 1 - \alpha_0,$$

and

$$H(\mu^{(t^2, t)}; s \mid 2s) > 1 - \alpha_0.$$

Then $\mu^{(t^3 r/s, r/s)}$ satisfies 0-HE at scale r , once r is sufficiently small in terms of A .

Proof. Take a scale $\tilde{t}$ with

$$|\log_2 \tilde{t} - \log_2 r| \leq \frac{1}{2} (\log_2 \log_2 r^{-1})^2.$$

For small r this contains the entire 0-HE window. Choose an integer q such that

$$|\log_2(s\lambda^q) - \log_2 \tilde{t}| < \log_2 \lambda^{-1}. \quad (\text{II.10.11.5})$$

Then

$$|\log_2 \lambda^q - \log_2(r/s)| < \frac{1}{2} (\log_2 \log_2 r^{-1})^2 + \log_2 \lambda^{-1} < \log_2 t^{-1}$$

for small r . Thus $(t^2 \lambda^q, t \lambda^q) \subset (t^3 r/s, r/s)$. The larger coefficient factor contains a scaled copy of $\mu^{(t^2, t)}$, so Lemma II.10.7.5 and scaling give entropy $> 1 - \alpha_0$ at scale $\lambda^q s$. By the

scale-Lipschitz estimate in Lemma II.10.7.7, (II.10.11.5) changes the conditional entropy by at most $2 \log_2 \lambda^{-1}$. Choose α_0 so small that $\alpha_0 + 2 \log_2(1 - \alpha_0)^{-1} \leq 2^{-(A+1)}$. The required 0-HE threshold follows throughout the window. $\square$

Lemma II.10.11.7 (Disjoint 0-HE interval harvest). *Let α_0 be as in Lemma II.10.11.6, and choose $c_0 > 0$ so that Lemma II.10.11.5 holds with $\alpha = \alpha_0$ and constant c_0 . Let $0 < r, t, \tau < 1$, with r sufficiently small, and suppose*

$$2^{-(\log_2 r^{-1})^{1/2}} > \tau \geq r, \quad (\log_2 \tau^{-1})^{1/2} < \log_2 t^{-1} < \frac{c_0}{4K_\lambda} \log_2 \tau^{-1},$$

and

$$1 - \lambda < c_0 \frac{\log_2 t^{-1} \min\{1, \log_2 M_\lambda\}}{\log_2 \tau^{-1} \log_2 \log_2(M_\lambda + 2)}. \quad (\text{II.10.11.6})$$

Then there are at least

$$\frac{c_0 \log_2 \tau^{-1}}{4K_\lambda \log_2 t^{-1}}$$

pairwise disjoint intervals

$$I \subset \left(\frac{r}{\tau^{c_0/\log_2(M_\lambda+1)}}, \frac{r}{\tau} \right)$$

such that μ^I satisfies 0-HE at scale r .

Proof. The upper bound on $\log_2 t^{-1}$ implies, after reducing c_0 , that $t \geq \tau^{1/6}$. By Lemma II.10.11.5, the number of relevant scales is at least

$$c_0 K_\lambda^{-1} \log_2 \tau^{-1}.$$

They may be enumerated as

$$\tau^{c_0/\log_2(M_\lambda+1)} > s_1 > \dots > s_K > \tau, \quad s_i > 2s_{i+1},$$

with $H(\mu^{(t^2, t)}; s_i \mid 2s_i) > 1 - \alpha_0$. Put $I_i = (t^3 r/s_i, r/s_i)$. The scale assumptions imply the hypotheses of Lemma II.10.11.6, so every μ^{I_i} is 0-HE. The range of s_i gives the stated interval containment after discarding $O(\log_2 t^{-1})$ initial indices.

If $j > i$, then $s_i/s_j > 2^{j-i}$. Hence I_i and I_j are disjoint whenever $j - i \geq 3 \log_2 t^{-1}$. Retain every $\lceil 3 \log_2 t^{-1} \rceil$ -th interval. $\square$

Proposition II.10.11.8 (Production of many 0-HE blocks). *For every $p \in (0, 1)$ there is $c_p > 0$ such that, with*

$$\Psi(M) = \min \left\{ \log_2 M, \frac{1}{\log_2(M+1)[\log_2 \log_2(M+2)]^3} \right\},$$

if $0 < \lambda < 1$ is algebraic and $\lambda > 1 - c_p \Psi(M_\lambda)$, then for every sufficiently small r there are at least $4 \log_2 \log_2 r^{-1}$ pairwise disjoint intervals $I \subset (0, 1]$ such that every factor μ^I satisfies 0-HE at scale r .

Proof. Let c_0 be from Lemma II.10.11.7 and put $a = c_0 / \log_2(M_\lambda + 1)$, with $0 < a < 1/2$. Set

$$N = \left\lfloor \frac{\log_2 \log_2 r^{-1}}{2 \log_2 a^{-1}} \right\rfloor - 1, \quad \tau_n = r^{a^n}, \quad 0 \leq n \leq N.$$

For small r , $r \leq \tau_n < 2^{-(\log_2 r^{-1})^{1/2}}$. Moreover the harvest windows

$$(r/\tau_n^a, r/\tau_n) = (r/\tau_{n+1}, r/\tau_n)$$

are pairwise disjoint up to endpoints.

For each n choose t_n by

$$\frac{\log_2 t_n^{-1}}{\log_2 \tau_n^{-1}} = \frac{c_0}{32K_\lambda \log_2 a^{-1}}. \quad (\text{II.10.11.7})$$

For small r the scale inequalities of Lemma II.10.11.7 hold uniformly in n . Also

$$c_0 \frac{\log_2 t_n^{-1} \min\{1, \log_2 M_\lambda\}}{\log_2 \tau_n^{-1} \log_2 \log_2(M_\lambda + 2)} \geq c\Psi(M_\lambda).$$

Choose c_p small enough. The hypothesis on λ then implies (II.10.11.6) for every n . Applying Lemma II.10.11.7 at the n th scale produces at least

$$\frac{c_0 \log_2 t_n^{-1}}{4K_\lambda \log_2 \tau_n^{-1}} = 8 \log_2 a^{-1}$$

disjoint 0-HE intervals in the n th window. Summing over n gives at least $8(N+1) \log_2 a^{-1} \geq 4 \log_2 \log_2 r^{-1}$ intervals. $\square$

Lemma II.10.11.9 (Comparison of the two algebraic thresholds). *For every $\varepsilon > 0$ there is $c_\varepsilon > 0$ such that, for every $M > 1$,*

$$\Psi(M) \geq c_\varepsilon \min\{\log M, (\log M)^{-1-\varepsilon}\},$$

where the logarithms on the right may be taken in any fixed base.

Proof. Put $x = \log M > 0$. If $0 < x \leq 1$, then $\log(M+1)$ and $\log \log(M+2)$ stay between positive absolute constants, so $\Psi(M) \asymp x$. If $x \geq 1$, then $\log(M+1) \asymp x$ and $\log \log(M+2) \asymp 1 + \log x$, hence

$$\Psi(M) \geq cx^{-1}(1 + \log x)^{-3} \geq c_\varepsilon x^{-1-\varepsilon}.$$

$\square$

Theorem II.10.11.10 (Varjú). *For every $\varepsilon > 0$ and $p \in (0, 1)$ there is $c = c(\varepsilon, p) > 0$ such that, if $0 < \lambda < 1$ is algebraic and*

$$\lambda > 1 - c \min\{\log M_\lambda, (\log M_\lambda)^{-1-\varepsilon}\},$$

then the biased Bernoulli convolution $\mu_{\lambda,p}$ is absolutely continuous and its density belongs to $L \log L$.

Proof. First $M_\lambda > 1$. If $Q(x) = a \prod_i (x - \lambda_i)$ is the primitive minimal polynomial, its nonzero integer constant term gives

$$M_\lambda \prod_{|\lambda_i| < 1} |\lambda_i| \geq 1.$$

Since λ occurs among the factors inside the unit disk, $M_\lambda \geq \lambda^{-1} > 1$.

By Lemma II.10.11.9, after decreasing c the hypothesis implies Proposition II.10.11.8. Fix a sufficiently small r and set

$$k = \lfloor \log_2 \log_2 \log_2 r^{-1} \rfloor + 2.$$

Then $2 \log_2 \log_2 r^{-1} \leq 2^k \leq 4 \log_2 \log_2 r^{-1}$. Proposition II.10.11.8 supplies at least 2^k pairwise disjoint coefficient intervals I_j with every μ^{I_j} 0-HE at scale r . By (II.10.11.1),

$$\mu_{\lambda,p} = \mu^{I_1} * \cdots * \mu^{I_{2^k}} * \rho.$$

Pair the first 2^k factors and apply Proposition II.10.11.2; pair the resulting convolutions and continue. After k generations their convolution, η , is k -HE at scale r . Since $2r/r$ is integral, Lemma II.10.7.5 gives

$$H(\mu_{\lambda,p}; r \mid 2r) \geq H(\eta; r \mid 2r) \geq 1 - 2^{-(2^k + 3k + A)}.$$

Hence

$$H(\mu_{\lambda,p}; r \mid 2r) \geq 1 - (\log_2 r^{-1})^{-2}. \quad (\text{II.10.11.8})$$

Take $r = 2^{-n}$ and sum (II.10.11.8). Since $\sum n^{-2} < \infty$,

$$\sup_N (N - H(\mu_{\lambda,p}; 2^{-N} \mid 1)) < \infty.$$

The fixed term $H(\mu_{\lambda,p}; 1)$ is absorbed into the constant. Affinely rescale the compact support into $[0, 1)$. Proposition II.10.4.2 changes the corresponding dyadic entropy deficits by at most a fixed additive constant. Theorem II.10.4.3 gives a density f with

$$\int f(x) \log(2 + f(x)) dx < \infty.$$

Undo the affine normalization. □

Chapter II.11

L^q Inverse Theorems and Intersection Consequences

II.11.1 Dyadic L^q sums

The basic quantity in this chapter is the decay rate of the q -mass of a measure across dyadic scales.

Definition II.11.1.1 (Dyadic L^q spectrum and dimension). Let $q > 1$. For a finite Borel measure μ on $\mathbb{R}$ and the half-open dyadic partition $\mathcal{D}_m$, set

$$S_m(\mu, q) = \sum_{I \in \mathcal{D}_m} \mu(I)^q, \quad \tau_\mu(q) = \lim_{m \rightarrow \infty} -\frac{1}{m} \log_2 S_m(\mu, q)$$

when the limit exists, and

$$D(\mu, q) = \frac{\tau_\mu(q)}{q-1}.$$

The point of the chapter is to turn near-equality in the L^q convolution inequality into a multiscale branching structure, and then to use that structure to compute L^q dimensions.

Proposition II.11.1.2 (Existence of the dyadic L^q spectrum for self-similar measures). *Let μ be generated by finitely many similarities $x \mapsto \lambda_i x + t_i$ with $0 < |\lambda_i| < 1$ and positive weights. Then $\tau_\mu(q)$ exists for every $q > 1$, without any separation assumption.*

Proof. After a fixed affine normalization assume the attractor is contained in $[0, 1]$. Put $r_* = \min_i |\lambda_i|$ and let

$$\Omega_m = \{u : |\lambda_u| \leq 2^{-m} < |\lambda_{u-}|\}.$$

Then $r_* 2^{-m} < |\lambda_u| \leq 2^{-m}$ and the self-similar identity stopped at this complete antichain is

$$\mu = \sum_{u \in \Omega_m} p_u (f_u)_* \mu.$$

For $I \in \mathcal{D}_m$ let w_I be the total p_u -weight of cylinders $f_u([0, 1])$ meeting I . Such a cylinder is contained in the union I^* of I and its two neighbours, hence $w_I \leq \mu(I^*)$. Bounded overlap of the I^* gives

$$\sum_I w_I^q \leq C_q S_m(\mu, q). \quad (\text{II.11.1.1})$$

For $J \in \mathcal{D}_{m+n}$ contained in I , convexity gives

$$\mu(J)^q \leq w_I^{q-1} \sum_{u: f_u([0,1]) \cap I \neq \emptyset} p_u \mu(f_u^{-1} J)^q.$$

For fixed $u \in \Omega_m$, every $f_u^{-1} J$ has length comparable, with constants depending only on the IFS, to 2^{-n} . Such intervals are covered by $O(1)$ cells of $\mathcal{D}_n$, with bounded overlap as J varies. Hence

$$\sum_{J \subset I} \mu(J)^q \leq C_q w_I^q S_n(\mu, q).$$

Summing over I and using (II.11.1.1) yields

$$S_{m+n}(\mu, q) \leq C_q S_m(\mu, q) S_n(\mu, q).$$

Thus $a_m = -\log_2 S_m(\mu, q)$ satisfies $a_{m+n} \geq a_m + a_n - \log_2 C_q$. The sequence $b_m = a_m - \log_2 C_q$ is superadditive, so Fekete's lemma gives the existence of $\lim b_m/m$, and the additive constant disappears after division by m . $\square$

II.11.2 The additive-combinatorial engine

The inverse theorem has two combinatorial layers. A dense graph with a small restricted sumset first yields large subsets with small doubling. A dyadic regularization then converts that small-doubling structure into a tree whose branching is almost maximal at each active scale. The quantitative imbalanced form is important later, so we prove each step.

Lemma II.11.2.1 (Dense graph gives many alternating paths). *Let $G \subset A \times B$ be a bipartite graph and let $K \geq 1$. If*

$$|G| \geq K^{-1} |A| |B|,$$

then there are $A_0 \subset A$ and $B_0 \subset B$ such that

$$|A_0| \geq \frac{|A|}{4\sqrt{2}K}, \quad |B_0| \geq \frac{|B|}{4K}, \quad (\text{II.11.2.1})$$

and every $(a, b) \in A_0 \times B_0$ is connected by at least

$$\frac{|A||B|}{2^{15}K^6} \quad (\text{II.11.2.2})$$

ordered alternating paths $a - b_1 - a_1 - b$ in G .

Proof. Put

$$A_1 = \left\{ a \in A : \deg_G(a) \geq \frac{|B|}{2K} \right\}.$$

The vertices outside A_1 carry fewer than $|A||B|/(2K)$ edges, hence

$$e(A_1, B) \geq \frac{|A||B|}{2K}. \quad (\text{II.11.2.3})$$

For $a, a' \in A_1$ write

$$d(a, a') = |N(a) \cap N(a')|.$$

Set

$$\eta = \frac{1}{64K^2}, \quad \theta = \frac{|B|}{1024K^4},$$

and call (a, a') *bad* if $d(a, a') < \theta$.

Choose b_0 uniformly from B and put $N = N(b_0) \cap A_1$. By (II.11.2.3),

$$\mathbb{E}|N| \geq \frac{|A|}{2K}, \quad \mathbb{E}|N|^2 \geq \frac{|A|^2}{4K^2}.$$

Let Z be the number of ordered bad pairs in $N \times N$. A fixed bad pair belongs to $N \times N$ with probability less than $\theta/|B| = 1/(1024K^4)$, and therefore

$$\mathbb{E}Z < \frac{|A|^2}{1024K^4} = \frac{\eta|A|^2}{16K^2}.$$

Consequently

$$\mathbb{E} \left(|N|^2 - \frac{2Z}{\eta} \right) \geq \frac{|A|^2}{8K^2}.$$

For some b_0 the corresponding set $A_2 = N(b_0) \cap A_1$ thus satisfies

$$|A_2| \geq \frac{|A|}{2\sqrt{2}K}, \quad Z \leq \frac{\eta}{2}|A_2|^2. \quad (\text{II.11.2.4})$$

For $a \in A_2$, let $z(a)$ be the number of $a' \in A_2$ for which (a, a') is bad, and define

$$A_0 = \{a \in A_2 : z(a) \leq \sqrt{\eta}|A_2|\}.$$

Since $\sum_{a \in A_2} z(a) = Z$, at most $\sqrt{\eta}|A_2|/2$ vertices are deleted. In particular $|A_0| \geq |A_2|/2$, which gives the first inequality in (II.11.2.1).

Every $a \in A_2$ still has degree at least $|B|/(2K)$. Put

$$B_0 = \left\{ b \in B : \deg_{A_2}(b) \geq \frac{|A_2|}{4K} \right\}.$$

The edges from A_2 to $B \setminus B_0$ contribute less than $|A_2||B|/(4K)$. Since $A_2 \subset A_1$, every vertex of A_2 has degree at least $|B|/(2K)$, and hence $e(A_2, B) \geq |A_2||B|/(2K)$. It follows that $e(A_2, B_0) \geq |A_2||B|/(4K)$. As every vertex of B_0 has at most $|A_2|$ neighbors in A_2 , we obtain $|B_0| \geq |B|/(4K)$.

Fix $a \in A_0$ and $b \in B_0$. The vertex b has at least $|A_2|/(4K)$ neighbors $a_1 \in A_2$. At most

$$\sqrt{\eta} |A_2| = \frac{|A_2|}{8K}$$

of them form a bad pair with a . Hence there are at least $|A_2|/(8K)$ choices of a_1 for which $d(a, a_1) \geq \theta$. For each such a_1 there are at least θ choices of $b_1 \in N(a) \cap N(a_1)$. Thus the number of ordered paths $a - b_1 - a_1 - b$ is at least

$$\frac{|A_2|}{8K} \frac{|B|}{1024K^4} \geq \frac{|A||B|}{2^{15}K^6},$$

where the last inequality uses (II.11.2.4). This proves (II.11.2.2). $\square$

Lemma II.11.2.2 (Graph BSG). *Assume $G \subset A \times B$ has density at least K^{-1} and its restricted sumset*

$$S = \{a + b : (a, b) \in G\}$$

satisfies $|S| \leq K' \sqrt{|A||B|}$. Then there are $A' \subset A$ and $B' \subset B$ with

$$|A'| \geq \frac{|A|}{4\sqrt{2}K}, \quad |B'| \geq \frac{|B|}{4K}, \quad (\text{II.11.2.5})$$

and

$$|A' + B'| \leq 2^{15}K^6(K')^3 \sqrt{|A||B|}. \quad (\text{II.11.2.6})$$

Proof. Take A', B' from Lemma II.11.2.1; then (II.11.2.5) holds. For every $x \in A' + B'$ choose one representation $x = a + b$ with $(a, b) \in A' \times B'$. Each of the at least

$$M = \frac{|A||B|}{2^{15}K^6}$$

paths $a - b_1 - a_1 - b$ supplied by Lemma II.11.2.1 gives a triple

$$\Phi(a, b_1, a_1, b) = (a + b_1, a_1 + b_1, a_1 + b) \in S^3.$$

For fixed endpoints (a, b) , the triple determines b_1 and a_1 , so this map is injective on the path set. Moreover

$$x = (a + b_1) - (a_1 + b_1) + (a_1 + b),$$

so a triple cannot arise from chosen endpoint pairs carrying two different sums x . Therefore the path triples belonging to distinct elements of $A' + B'$ are disjoint, and

$$|A' + B'|M \leq |S|^3.$$

Using the bound on $|S|$ and the definition of M gives (II.11.2.6). $\square$

Lemma II.11.2.3 (Petridis–Ruzsa calculus). *If $|A + B| \leq K|A|$, then for all $m, n \geq 0$,*

$$|mB - nB| \leq K^{m+n}|A|.$$

Proof. Choose a nonempty $X \subset A$ minimizing

$$K_0 = \frac{|X + B|}{|X|}.$$

Then $K_0 \leq K$, and every nonempty $Z \subset X$ satisfies

$$|Z + B| \geq K_0|Z|. \quad (\text{II.11.2.7})$$

We first prove Petridis' minimal-growth lemma: for every finite nonempty C ,

$$|X + B + C| \leq K_0|X + C|. \quad (\text{II.11.2.8})$$

We induct on $|C|$. The case $|C| = 1$ is equality. For $|C| > 1$, choose $\gamma \in C$ and put $C' = C \setminus \{\gamma\}$. Define

$$Z = \{x \in X : x + B + \gamma \subset X + B + C'\}.$$

Then

$$|X + B + C| \leq |X + B + C'| + |X + B| - |Z + B| \leq K_0|X + C'| + K_0|X| - K_0|Z|.$$

Also set

$$W = \{x \in X : x + \gamma \in X + C'\}.$$

Since

$$|X + C| = |X + C'| + |X| - |W|$$

and $W \subset Z$ (if $x + \gamma = x' + c'$ then $x + b + \gamma = x' + b + c'$ for every $b \in B$), we get

$$|X + C| \geq |X + C'| + |X| - |Z|.$$

Combining the last two inequalities proves (II.11.2.8). Iterating it with $C = (r - 1)B$ yields

$$|X + rB| \leq K_0^r|X|, \quad r \geq 0. \quad (\text{II.11.2.9})$$

For completeness, the Ruzsa triangle inequality

$$|P - R||Q| \leq |P - Q||Q - R| \quad (\text{II.11.2.10})$$

has the following elementary proof. For each $d \in P - R$ choose one representation $d = p_d - r_d$. The map

$$(d, q) \mapsto (p_d - q, q - r_d)$$

from $(P - R) \times Q$ to $(P - Q) \times (Q - R)$ is injective: the sum of its two coordinates is d , after which the chosen representation determines q .

Apply (II.11.2.10) with $P = mB$, $R = nB$ and $Q = -X$. By (II.11.2.9),

$$|mB - nB||X| \leq |mB + X||X + nB| \leq K_0^{m+n}|X|^2.$$

Cancel $|X|$, use $K_0 \leq K$ and $|X| \leq |A|$, and obtain

$$|mB - nB| \leq K^{m+n}|A|.$$

□

Proposition II.11.2.4 (Imbalanced energy structure). *Fix $\eta > 0$. There are constants $C_\eta, c_\eta > 0$ such that the following holds. Let A, B be finite nonempty subsets of an abelian group, assume $|A| \leq L|B|$ with $L \geq 1$, and suppose*

$$E(A, B) \geq 2\alpha|A||B|^2, \quad 0 < \alpha \leq 1,$$

where

$$E(A, B) = \sum_{b, b' \in B} |A \cap (A + b' - b)|.$$

Then there are finite sets H, X and a translate $H + x_B$ such that

$$|H + H| \leq C_\eta \alpha^{-C_\eta} L^\eta (1 + \log |B|)^{C_\eta} |H|, \quad (\text{II.11.2.11})$$

$$|B \cap (H + x_B)| \geq c_\eta \alpha^{C_\eta} L^{-\eta} (1 + \log |B|)^{-C_\eta} |B|, \quad (\text{II.11.2.12})$$

$$|A \cap (X + H)| \geq c_\eta \alpha^{C_\eta} L^{-\eta} (1 + \log |B|)^{-C_\eta} |A|, \quad (\text{II.11.2.13})$$

$$|X||H| \leq C_\eta \alpha^{-C_\eta} L^\eta (1 + \log |B|)^{C_\eta} |A|. \quad (\text{II.11.2.14})$$

Proof. For $0 < \beta \leq 1$, write

$$\text{Sym}_\beta(A) = \{h : |A \cap (A + h)| \geq \beta|A|\}.$$

The energy hypothesis forces many differences of B into $\text{Sym}_\alpha(A)$. Indeed, if p ordered pairs (b, b') have $b - b' \in \text{Sym}_\alpha(A)$, then

$$E(A, B) \leq p|A| + (|B|^2 - p)\alpha|A|.$$

Since the hypothesis is impossible for $\alpha > 1/2$, we may assume $\alpha \leq 1/2$; if $p < \alpha|B|^2$, the last display is strictly smaller than $2\alpha|A||B|^2$. Thus there is a graph $G_B \subset B \times B$ of density at least α with

$$B -_{G_B} B \subset \text{Sym}_\alpha(A). \quad (\text{II.11.2.15})$$

We use the following elementary symmetry iteration. If $S \subset \text{Sym}_\beta(A)$ is nonempty and

$$A_s = \{a \in A : a + s \in A\},$$

then $|A_s| \geq \beta|A|$ and Cauchy–Schwarz gives

$$\sum_{s, s' \in S} |A_s \cap A_{s'}| = \sum_{a \in A} \#\{s : a \in A_s\}^2 \geq \beta^2 |A| |S|^2.$$

Hence at least $(\beta^2/2)|S|^2$ ordered pairs satisfy

$$|A_s \cap A_{s'}| \geq \frac{\beta^2}{2} |A|,$$

and translation by s' shows that their differences lie in $\text{Sym}_{\beta^2/2}(A)$.

Whenever such a graph occurs, dyadically pigeonhole its positive difference-fiber sizes. If $G' \subset S \times S$ is the retained class and $S^+ = S -_{G'} S$, then for some $R \geq 1$ every nonzero fiber has size in $[R, 2R]$ and

$$\frac{|G'|}{2|S^+|} \leq R \leq \frac{|G'|}{|S^+|}, \quad |S^+| \geq \frac{|G'|}{|S|}. \quad (\text{II.11.2.16})$$

The dyadic pigeonhole costs at most a factor $1 + \log_2 |S|$.

Fix an integer $J \geq 2$, to be chosen only in terms of η . Starting from $S_0 = B$, regularize G_B and put $S_1 = S_0 -_{G_0} S_0$. Iterating the preceding symmetry step, with $\beta_{r+1} = \beta_r^2/2$, produces $S_1, \dots, S_{J+1}$ satisfying

$$|S_r| \geq \alpha^{C_J} (1 + \log |B|)^{-C_J} |B|, \quad 1 \leq r \leq J+1. \quad (\text{II.11.2.17})$$

On the other hand

$$|\text{Sym}_\beta(A)| \leq |A|/\beta,$$

because $\sum_h |A \cap (A + h)| = |A|^2$. Thus

$$\frac{|S_{J+1}|}{|S_1|} \leq C_J \alpha^{-C_J} L (1 + \log |B|)^{C_J}.$$

Multiplying the J successive ratios, some $j \in \{1, \dots, J\}$ obeys

$$\frac{|S_{j+1}|}{|S_j|} \leq C_J \alpha^{-C_J/J} L^{1/J} (1 + \log |B|)^{C_J/J}. \quad (\text{II.11.2.18})$$

Apply Lemma II.11.2.2 to the regularized graph G_j , with the second copy of S_j negated. Its density is bounded below by a fixed power of $\alpha(1 + \log |B|)^{-1}$, while its restricted sumset is S_{j+1} . We obtain $U, V \subset S_j$, each a polynomial fraction of S_j , such that

$$|U - V| \leq M|S_j|, \quad M \leq C_J \alpha^{-C_J} L^{3/J} (1 + \log |B|)^{C_J}.$$

Writing $\theta \geq \alpha^{C_J} (1 + \log |B|)^{-C_J}$ for the common lower density of U and V in S_j , we have

$$|U + (-V)| \leq (M/\theta)|U|.$$

Lemma II.11.2.3, with $A = U$, $B = -V$ and $m = 2, n = 0$, gives

$$|V + V| \leq (M/\theta)^2 |U| \leq (M/\theta)^2 \theta^{-1} |V|.$$

Choose $J = J(\eta)$ so large that the resulting power of L is smaller than $\eta/4$, and put $H = V$. This proves (II.11.2.11) after enlarging the powers of α^{-1} and $1 + \log |B|$.

It remains to transfer H back to large pieces of B and A . At level j , H occupies a polynomial fraction of S_j . Descend through the regularized difference graphs. If

$$|S_{r+1} \cap (H + x_{r+1})| \geq \theta_r |S_{r+1}|,$$

then the uniform lower fiber bound in (II.11.2.16) implies that a polynomial fraction of the edges of G_r have difference in this translate. Pigeonholing the second endpoint gives a

translate x_r for which $H + x_r$ meets a polynomial fraction of S_r . After at most J steps we obtain (II.11.2.12).

Finally $H \subset S_j \subset \text{Sym}_{\beta_j}(A)$, so

$$\sum_{a \in A} |A \cap (H + a)| = \sum_{h \in H} |A \cap (A - h)| \geq \beta_j |A| |H|.$$

Let

$$X_0 = \{a \in A : |A \cap (H + a)| \geq \beta_j |H|/2\}.$$

Then $|X_0| \geq \beta_j |A|/2$. Choose $X \subset X_0$ maximal so that the translates $H + x$, $x \in X$, are pairwise disjoint. Hence

$$|A \cap (X + H)| \geq (\beta_j/2) |X| |H|, \quad X_0 \subset X + (H - H).$$

From (II.11.2.11) and Lemma II.11.2.3, $|H - H|$ is at most a polynomial multiple of $|H|$ with L -exponent at most $2\eta/4$. Therefore maximality gives the lower bound (II.11.2.13), while the first inequality above and $|A \cap (X + H)| \leq |A|$ give (II.11.2.14). All remaining losses are fixed powers of α^{-1} and $1 + \log |B|$ and are absorbed into C_η, c_η . $\square$

Lemma II.11.2.5 (Dyadic tree regularization). *Let $m = m_1 T$ and let A be a nonempty 2^{-m} -set in a bounded interval, represented by one point in each occupied level- m dyadic cell. There are $A_1 \subset A$, a set $S \subset \{0, \dots, m_1 - 1\}$, and, for $s \in S$, integers n_s and dyadic integers R_s such that*

$$sT < n_s < (s+1)T - 4, \quad 2 \leq R_s \leq 2^{n_s - sT},$$

with the following properties.

- (a) *If $s \notin S$, every occupied level- sT parent contains at most one occupied level- $(s+1)T$ child of A_1 .*
- (b) *If $s \in S$, every occupied level- sT parent contains exactly R_s occupied level- n_s cells, each such cell contains exactly one occupied level- $(s+1)T$ child, and two of the occupied level- n_s cells have mutual interval distance in*

$$[2 \cdot 2^{-n_s}, 10 \cdot 2^{-n_s}].$$

- (c) *For an absolute $c > 0$,*

$$|A_1| = \prod_{s \in S} R_s, \quad |A_1| \geq (cT^{-2})^{m_1} |A|.$$

Proof. We begin with one elementary dyadic claim. Suppose $q > p + 8$ and at least 1000 occupied level- q cells lie inside one level- p parent. Then there is an integer r with

$$p < r < q - 4$$

and two occupied level- r cells whose mutual interval distance belongs to $[2 \cdot 2^{-r}, 10 \cdot 2^{-r}]$. Moreover, if r is chosen maximal with this property, every occupied level- r cell contains fewer than 1000 occupied level- q cells.

Indeed, take the leftmost and rightmost occupied level- q cells. Their mutual interval distance D satisfies

$$998 \cdot 2^{-q} \leq D \leq 2^{-p}.$$

Choose r so that $5 \leq D/2^{-r} < 10$. The bounds on D force $p < r < q - 4$, and the level- r cells containing the two endpoints are separated by at least $D - 2 \cdot 2^{-r} \geq 3 \cdot 2^{-r}$ and by less than $10 \cdot 2^{-r}$. If a maximal such level- r cell contained at least 1000 level- q descendants, the same argument inside that cell would produce a qualifying level $r' > r$, a contradiction.

Regularize the blocks from the bottom upward. Suppose the set has already been regularized below level $(s+1)T$. For each occupied level- sT parent I , let $M(I)$ be the number of occupied level- $(s+1)T$ children. If $M(I) < 1000$, retain one child and declare the parent inactive. Otherwise apply the claim with $p = sT$, $q = (s+1)T$, choose the maximal intermediate level $n(I)$, let $M_{n(I)}(I)$ be the number of occupied level- $n(I)$ cells, and choose a dyadic integer $R(I)$ with

$$\frac{M_{n(I)}(I)}{2} < R(I) \leq M_{n(I)}(I).$$

Maximality gives $M(I) \leq 1000M_{n(I)}(I)$. Retaining exactly $R(I)$ intermediate cells and one terminal child in each costs at most an absolute factor 2000 and preserves a separated pair; after changing this absolute constant, the case $R(I) = 1$ is declared inactive.

Because the procedure runs upward, every surviving level- $(s+1)T$ child carries the same number of terminal descendants already selected at lower blocks. Thus the retained point mass is proportional to the number of surviving terminal children. The active type of a parent is the pair $(n(I), \log_2 R(I))$, together with one inactive type. There are at most CT^2 types. Retain, among the level- sT parents, a type carrying maximal retained point mass. The local $1/2000$ loss and this pigeonhole together cost at most cT^{-2} at the block. The selected type is now uniform across all surviving level- sT parents.

Iterating through all m_1 blocks gives the claimed loss $(cT^{-2})^{m_1}$. Deleting only whole already-regularized terminal subtrees preserves every property established at lower levels. At the end, inactive blocks have branching number one and active blocks branching number R_s , whence $|A_1| = \prod_{s \in S} R_s$. Finally, a level- sT parent contains at most $2^{n_s - sT}$ level- n_s cells, so $R_s \leq 2^{n_s - sT}$. $\square$

Proposition II.11.2.6 (Regular tree inside a small-doubling set). *There is an absolute C_0 such that for all sufficiently large block sizes T the following holds. Let $m = m_1T$ and let A be a nonempty 2^{-m} -set in a bounded interval satisfying*

$$|A + A| \leq 2^{2^{-2T-1}m} |A|. \quad (\text{II.11.2.19})$$

Then A contains A_2 and admits block data as in Lemma II.11.2.5 such that every inactive block has one child, while at every active block

$$R_s > 2^{(1-T^{-1/2})(n_s - sT)}. \quad (\text{II.11.2.20})$$

Moreover

$$|A_2| \geq 2^{-C_0(\log_2 T)/\sqrt{T}m} |A|. \quad (\text{II.11.2.21})$$

Proof. Apply Lemma II.11.2.5 and put

$$d_s = n_s - sT, \quad C_s = 2^{d_s}.$$

We first prove the branching-pressure estimate

$$\prod_{s \in S} C_s \leq |2^T A_1|. \quad (\text{II.11.2.22})$$

Inside every active parent, order one of the distinguished separated pairs as left/right. Recursively through active blocks, choose for each binary word $c = (c_s)_{s \in S}$ a point $x_c \in A_1$ that follows the designated left or right descendant at block s . For an integer vector $k = (k_s)_{s \in S}$ with $1 \leq k_s \leq C_s$, define

$$\Phi(k) = \sum_{u=1}^{2^T} x_{c^{(u)}}, \quad c_s^{(u)} = 1 \iff u \leq k_s.$$

If k and ℓ first differ at the active block s , say $k_s < \ell_s$, partition the 2^T summand indices u according to the first active block at which their two threshold bits $c_t^{(u)}(k)$ and $c_t^{(u)}(\ell)$ differ. At block s , exactly $\ell_s - k_s \geq 1$ indices first differ. Each pair of corresponding points follows the designated left and right descendants in the same level- sT parent, so all these displacements have the same sign and each has magnitude at least $2 \cdot 2^{-n_s}$.

For a later active block $t > s$, at most $|k_t - \ell_t| \leq C_t = 2^{n_t - tT}$ indices can first differ there. Each such pair lies in the designated two level- n_t cells of a common parent. Their interval distance is at most $10 \cdot 2^{-n_t}$, and the two cell widths add at most $2 \cdot 2^{-n_t}$; hence its point displacement is at most $12 \cdot 2^{-n_t}$. The total absolute contribution from *all* later first-difference groups is therefore at most

$$12 \sum_{\substack{t > s \\ t \in S}} C_t 2^{-n_t} \leq 12 \sum_{t > s} 2^{-tT} < 15 \cdot 2^{-(s+1)T}$$

for the stipulated sufficiently large T . This grouping is essential: bounding the tail separately for all 2^T summands would not give the needed estimate. Since $n_s < (s+1)T - 4$, the first group contributes more than $32 \cdot 2^{-(s+1)T}$ in one direction. Hence

$$|\Phi(k) - \Phi(\ell)| > 2^{-n_s} > 2^{-m}.$$

Thus Φ is injective at the 2^{-m} covering scale, proving (II.11.2.22).

Set

$$K = 2^{2^{-2T-1}m}.$$

By Lemma II.11.2.3, applied with $B = A$ and $r = 2^T$,

$$|2^T A| \leq K^{2^T} |A| = 2^{2^{-2T-1}m} |A|. \quad (\text{II.11.2.23})$$

Lemma II.11.2.5 gives, for an absolute C_1 and large T ,

$$|A_1| = \prod_{s \in S} R_s \geq 2^{-C_1(\log_2 T)m/T} |A|. \quad (\text{II.11.2.24})$$

Combining (II.11.2.22)–(II.11.2.24),

$$\prod_{s \in S} \frac{C_s}{R_s} \leq 2^{[2^{-T-1} + C_1(\log_2 T)/T]m}. \quad (\text{II.11.2.25})$$

Call an active block good when

$$R_s > C_s^{1-T^{-1/2}}.$$

For every bad active block, $C_s/R_s \geq C_s^{T^{-1/2}}$. Therefore (II.11.2.25) gives

$$\prod_{s \in S \setminus S_{\text{good}}} C_s \leq 2^{[\sqrt{T} 2^{-T-1} + C_1(\log_2 T)/\sqrt{T}]m}.$$

Since $R_s \leq C_s$, the same upper bound holds for the product of the bad R_s . Prune A_1 by keeping one terminal child at each bad active block, identically in every occupied parent. The resulting set A_2 loses exactly the product of those bad branching numbers. Combining this with (II.11.2.24) and enlarging the absolute constant gives (II.11.2.21). Every surviving active block is good and satisfies (II.11.2.20); all other blocks have one child. The separated-pair and one-terminal-child geometry from Lemma II.11.2.5 is unchanged. $\square$

II.11.3 The L^q inverse theorem

The preceding lemmas describe the geometric structure forced by small doubling. We now formulate the inverse theorem that detects this structure from near-equality in the L^q convolution inequality.

Definition II.11.3.1 (Uniform dyadic tree and dyadic measure). Let $D, \ell \geq 1$ be integers and put $m = D\ell$. A 2^{-m} -set A is $(D, \ell, (R_s))$ -uniform if every occupied level- sD dyadic interval has exactly R_s occupied level- $(s+1)D$ descendants. A 2^{-m} -measure is a probability measure supported on $2^{-m}\mathbb{Z} \cap [0, 1)$.

We work first on the circle $\mathbb{R}/\mathbb{Z}$. If the measures are regarded on the real line, their convolution is supported on an interval of length two; splitting that output into its two unit intervals, or equivalently reducing addition modulo one, changes the discrete ℓ^q norm by at most a constant depending only on q . Such a constant is absorbed by $2^{o(m)}$, so the real-line theorem has the same exponential statement.

Theorem II.11.3.2 (Inverse theorem for near-nonflattening). Fix $q \in (1, \infty)$, $\delta > 0$ and $D_0 \in \mathbb{N}$. There are $\varepsilon > 0$ and an integer $D \geq D_0$ with the following property for all

sufficiently large ℓ . Put $m = D\ell$ and let μ, ν be 2^{-m} -measures such that

$$\|\mu * \nu\|_q \geq 2^{-\varepsilon m} \|\mu\|_q. \quad (\text{II.11.3.1})$$

After translating μ and ν by elements of $2^{-m}\mathbb{Z}$, there are sets $A \subset \text{supp } \mu$, $B \subset \text{supp } \nu$ and branching sequences $(R'_s)_{s=0}^{\ell-1}$, $(R''_s)_{s=0}^{\ell-1}$ such that:

- (i) $\|\mu|_A\|_q \geq 2^{-\delta m} \|\mu\|_q$, and $\mu(y) \leq 2\mu(x)$ for all $x, y \in A$;
- (ii) A is $(D, \ell, (R'_s))$ -uniform and every $x \in A$ lies in the middle half of its level- sD dyadic parent for every s ;
- (iii) $\nu(B) \geq 2^{-\delta m}$, and $\nu(y) \leq 2\nu(x)$ for all $x, y \in B$;
- (iv) B is $(D, \ell, (R''_s))$ -uniform and every $y \in B$ lies in the middle half of its level- sD dyadic parent for every s ;
- (v) for every s , either $R''_s = 1$ or

$$R'_s \geq 2^{(1-\delta)D}; \quad (\text{II.11.3.2})$$

(vi) if

$$S = \{s : R'_s \geq 2^{(1-\delta)D}\}, \quad q' = \frac{q}{q-1},$$

then

$$\log_2(\|\nu\|_q^{-q'}) - \delta m \leq D|S| \leq \log_2(\|\mu\|_q^{-q'}) + \delta m. \quad (\text{II.11.3.3})$$

Proof. We give the full reduction because the locations of the losses are important in later flattening arguments. Constants implicit below may depend on q , but not on m . It is enough to treat $0 < \delta < 1$. If $\delta \geq 1$, run the construction below with the retention/regularity parameter $1/2$. In the statement with the original δ , the threshold $2^{(1-\delta)D}$ is at most one, so its set S is automatically all of $\{0, \dots, \ell-1\}$. The branching dichotomy is then tautological, while $\log_2(\|\nu\|_q^{-q'}) \leq m$ and $\log_2(\|\mu\|_q^{-q'}) \geq 0$ make both scale-count inequalities automatic. Hence assume $0 < \delta < 1$.

Step 1: dyadic level sets. For $j \in \mathbb{Z}$ put

$$A_j = \{x : 2^{-j-1} \|\mu\|_q^{q'} < \mu(x) \leq 2^{-j} \|\mu\|_q^{q'}\}, \quad B_j = \{y : 2^{-j-1} 2^{-m} < \nu(y) \leq 2^{-j} 2^{-m}\}.$$

Hölder's inequality gives $\|\mu\|_q^{q'} \geq 2^{-m}$, so no relevant level occurs below $j = -m - O(1)$. Let $L = \lceil 2\varepsilon q' m \rceil$. On the tails $E = \bigcup_{j \geq L} A_j$ and $F = \bigcup_{j \geq L} B_j$,

$$\|\mu|_E\|_q \leq 2^{-L/q'} \|\mu\|_q, \quad \nu(F) \leq 2^{-L}.$$

Young's inequality therefore makes both $\|\mu|_E * \nu\|_q$ and $\|\mu * \nu|_F\|_q$ smaller than a fixed polynomial factor times $2^{-2\varepsilon m} \|\mu\|_q$. For large m , the triangle inequality and (II.11.3.1) leave at least half the convolution norm on pairs of levels with $j, j' < L$. There are only $O_q(m^2)$ such pairs, so dyadic pigeonholing, with the polynomial factor absorbed into $2^{\varepsilon m}$, yields

indices $j, j' \leq 2\epsilon q' m$ and sets $A_1 = A_j, B_1 = B_{j'}$ satisfying

$$\|1_{A_1} * 1_{B_1}\|_q \geq 2^{-2\epsilon m} |A_1|^{1/q} |B_1|, \quad (\text{II.11.3.4})$$

$$\|\mu|_{A_1}\|_q \geq 2^{-2\epsilon m} \|\mu\|_q, \quad (\text{II.11.3.5})$$

$$\nu(B_1) \geq 2^{-2\epsilon m}. \quad (\text{II.11.3.6})$$

In particular, masses are comparable within a factor two on each of A_1, B_1 .

We next convert (II.11.3.4) to energy. Put $f = 1_{A_1} * 1_{B_1}$. If $1 < q < 2$, Hölder gives

$$\sum f^q \leq \left(\sum f\right)^{2-q} \left(\sum f^2\right)^{q-1}.$$

If $q \geq 2$, then $\sum f^q \leq \|f\|_2^2 \|f\|_\infty^{q-2}$ and $\|f\|_\infty \leq |B_1|$. In both cases,

$$E(A_1, B_1) = \|1_{A_1} * 1_{B_1}\|_2^2 \geq 2^{-2 \max(q, q') \epsilon m} |A_1| |B_1|^2. \quad (\text{II.11.3.7})$$

Step 2: an exponential-scale asymmetric BSG model. We record explicitly how Proposition II.11.2.4 is used. Given $\kappa > 0$, choose first $\eta > 0$ so small that the factor L^η in that proposition costs at most $2^{\kappa m/10}$ when the trivial imbalance bound $L \leq 2^m$ is used. Then choose $\tau > 0$ so small that every fixed power of α^{-1} costs at most $2^{\kappa m/10}$ when $\alpha = 2^{-\tau m-1}$. Finally absorb all powers of $1 + \log |B_1|$ into another $2^{\kappa m/10}$ for large m . Thus, whenever

$$E(A_1, B_1) \geq 2^{-\tau m} |A_1| |B_1|^2,$$

Proposition II.11.2.4, followed by translating H and counter-translating X , produces H, X with

$$|H + H| \leq 2^{\kappa m} |H|, \quad (\text{II.11.3.8})$$

$$|A_1 \cap (X + H)| \geq 2^{-\kappa m} |A_1|, \quad (\text{II.11.3.9})$$

$$|B_1 \cap H| \geq 2^{-\kappa m} |B_1|, \quad (\text{II.11.3.10})$$

$$|X||H| \leq 2^{\kappa m} |A_1|. \quad (\text{II.11.3.11})$$

This is the only place where the large imbalance between A_1 and B_1 matters.

We now choose the parameters. Take $D \geq D_0$ large enough that all of

$$D^{-1/8}, \quad \frac{\log_2 D}{D^{1/4}}, \quad \frac{\log_2 D}{D} \quad (\text{II.11.3.12})$$

are smaller than a sufficiently small constant multiple of δ , and also choose D so that $\delta D \notin \mathbb{Z}$. This last requirement is always possible when $0 < \delta < 1$: it is automatic for irrational δ , and for rational $\delta = a/b$ one avoids multiples of b . Put

$$\sigma_D = \delta - \frac{\lfloor \delta D \rfloor}{D} > 0. \quad (\text{II.11.3.13})$$

Set

$$\kappa = 2^{-2D^2-2}$$

and let $\tau = \tau(\kappa)$ be supplied by the preceding paragraph. Choose $\varepsilon > 0$ so small that $2 \max(q, q')\varepsilon < \tau$, all later losses proportional to εm are at most $\delta m/100$, and

$$2qq'\varepsilon < \sigma_D/4. \quad (\text{II.11.3.14})$$

Then (II.11.3.7) implies (II.11.3.8)–(II.11.3.11).

Step 3: uniformization and boundary centering. We use three elementary tree operations, and include their proofs to make the circle-to-line interface explicit.

First, every 2^{-m} -set C contains a (D, ℓ) -uniform subset C_0 with

$$|C_0| \geq (2D)^{-\ell}|C|. \quad (\text{II.11.3.15})$$

Indeed, starting at the bottom of the D -block tree, group the occupied parents according to the dyadic range $[2^r, 2^{r+1})$ containing their number of children, keep the most populated class, and then keep exactly 2^r children in every retained parent. Each block loses at most a factor $2D$.

Second, if C is uniform and $U \subset \{0, \dots, \ell - 1\}$, one may force the branching number to be one at each scale in U by selecting one child in every occupied parent. The resulting subset has size at least

$$\left(\prod_{s \in U} R_s^{-1} \right) |C| \geq 2^{-D|U|} |C|. \quad (\text{II.11.3.16})$$

Third, after translating a 2^{-m} -set by a multiple of 2^{-m} one can retain a subset of size at least $3^{-\ell}|C|$ whose points lie in the middle half of every level- sD parent. To see this, at each block there is a shift in

$$\{-2^{-(sD+2)}, 0, 2^{-(sD+2)}\}$$

that puts at least one third of the currently retained points in the middle half. Start with the finest block and move upward. After the choice at level sD , every shift introduced later in the construction comes from a coarser level $s' < s$ and is a multiple of 2^{-sD} (recall $D \geq 2$), so it does not destroy the centering already obtained at level sD . The total loss is $3^{-\ell}$.

Apply Proposition II.11.2.6 to H with block size $T = D^2$. There is one harmless divisibility point to make explicit. Write $m = m_0 + r$, where $m_0 = D^2 \lfloor m/D^2 \rfloor$ and $0 \leq r < D^2$. Since $m = D\ell$, the remainder r is itself a multiple of D . Let

$$\pi_{m_0}(x) = 2^{-m_0} \lfloor 2^{m_0} x \rfloor, \quad H^{(0)} = \pi_{m_0}(H).$$

Each level- m_0 cell contains at most 2^r points of H , whence $|H^{(0)}| \geq 2^{-r}|H|$. Also

$$|H^{(0)} + H^{(0)}| \leq 3|H + H|. \quad (\text{II.11.3.17})$$

Indeed, for every coarse sum $z = u + v$ choose $x \in H$ in the cell u and $y \in H$ in the cell v . Then $x + y \in [z, z + 2^{1-m_0})$. Since distinct coarse sums are spaced by 2^{-m_0} , any fixed terminal sum can be selected by at most three coarse sums. Consequently

$$\frac{|H^{(0)} + H^{(0)}|}{|H^{(0)}|} \leq 3 \frac{|H + H|}{|H|} = 2^{O_D(1)} \frac{|H + H|}{|H|}.$$

For fixed D and large ℓ this bounded factor is absorbed by the strict exponent room in our choice of κ . Apply Proposition II.11.2.6 to $H^{(0)}$ at depth m_0 . For every retained coarse cell choose one original point of H in that cell. The lifted set has exactly the same branching through level m_0 and a single child at each of the remaining $r/D < D$ fixed- D blocks. Its cardinality is unchanged from the retained coarse set, so relative to H the only additional loss is $2^{-r} = 2^{-O_D(1)}$. Thus all later exponential estimates are unchanged. We may therefore write m below without assuming $D^2 \mid m$.

Since (II.11.3.8) and the definition of κ imply the small-doubling hypothesis of Proposition II.11.2.6, we obtain a set $H_* \subset H$ with active intermediate depths n_t in the D^2 -blocks. Put

$$\Sigma = \sum_t (n_t - tD^2),$$

where the sum is over its active D^2 -blocks. Proposition II.11.2.6 gives

$$\log_2 |H_*| \geq (1 - D^{-1})\Sigma, \quad |H_*| \geq 2^{-C(\log D)D^{-1}m}|H|. \quad (\text{II.11.3.18})$$

If $\Sigma < D^{-1/2}m$, then also $|H_*| \leq 2^\Sigma$ because the branching number of each active intermediate segment is at most $2^{n_t - tD^2}$. Together with (II.11.3.18) this gives $|H| \leq 2^{O(D^{-1/2}m)}$; a singleton already has all required fixed-block branching properties and satisfies (II.11.3.22) below after increasing the absolute constant. We may therefore assume $\Sigma \geq D^{-1/2}m$.

Uniformize H_* at block size D , obtaining H_{**} with branching numbers r_u . By (II.11.3.15),

$$|H_{**}| \geq 2^{-(\log_2(2D)/D)m}|H_*|. \quad (\text{II.11.3.19})$$

Let $U = \{u : r_u > 1\}$. Each active D^2 -block of intermediate length $n_t - tD^2$ meets at most $(n_t - tD^2)/D + 1$ active D -blocks, and hence

$$D|U| \leq \Sigma + \frac{m}{D} \leq (1 + D^{-1/2})\Sigma. \quad (\text{II.11.3.20})$$

Using (II.11.3.18), (II.11.3.19) and $\Sigma \geq D^{-1/2}m$,

$$\log_2 |H_{**}| \geq (1 - C(\log D)D^{-1/2})\Sigma.$$

Together with (II.11.3.20) this yields

$$\frac{1}{|U|} \sum_{u \in U} \frac{\log_2 r_u}{D} \geq 1 - C(\log D)D^{-1/2}. \quad (\text{II.11.3.21})$$

Let

$$U_0 = \{u \in U : r_u < 2^{(1-D^{-1/4})D}\}.$$

Since every summand in (II.11.3.21) is at most one,

$$|U_0| \leq C(\log D)D^{-1/4}|U| \leq C(\log D)D^{-1/4}\frac{m}{D}.$$

Collapse all $u \in U_0$ using (II.11.3.16). Calling the result H_0 and its fixed- D branching numbers R_s , we obtain

$$|H_0| \geq 2^{-c_1(\log_2 D)D^{-1/4}m}|H| \quad (\text{II.11.3.22})$$

and, for every s ,

$$R_s = 1 \quad \text{or} \quad R_s \geq 2^{(1-D^{-1/4})D}. \quad (\text{II.11.3.23})$$

This derives the fixed-block regularization used below from the preceding tree lemmas.

Apply the centering operation and then (II.11.3.15) to $A_1 \cap (X + H)$ and $B_1 \cap H$. This requires independent translations of μ and ν , which are allowed in the theorem. Additive cardinalities are translation invariant, so all estimates involving H remain valid. We obtain uniform sets A and B_2 satisfying the centering clauses and

$$|A| \geq 2^{-c_2(\log_2 D/D)m} |A_1 \cap (X + H)|, \quad (\text{II.11.3.24})$$

$$|B_2| \geq 2^{-c_2(\log_2 D/D)m} |B_1 \cap H|. \quad (\text{II.11.3.25})$$

For D large and ε small, (II.11.3.5), (II.11.3.6), (II.11.3.9), (II.11.3.10) and the factor-two mass comparability already give conclusions (i) and (iii) for A and B_2 , with room to spare for the final pruning of B_2 .

Step 4: the common active scales. We need one tree sumset estimate. If C and H_0 are (D, ℓ) -uniform with branching numbers Q_s and R_s , then

$$|C + H_0| \geq 2^{-\ell} |H_0| \prod_{s: R_s=1} Q_s. \quad (\text{II.11.3.26})$$

To prove it, first use (II.11.3.16) to collapse the branching of C at all scales where H_0 branches. Thus, at each block, at least one of the two trees has a unique child. Starting from the top pair of occupied parents, construct families of parent pairs whose interval sums are pairwise disjoint. If H_0 has one child, retain a nonadjacent subfamily of the occupied children of C ; if C has one child, do the symmetric operation on the children of H_0 . In either case at least half the available children survive and their interval sums are disjoint. Induction through all ℓ blocks therefore produces at least

$$2^{-\ell} |H_0| \prod_{s: R_s=1} Q_s$$

distinct terminal sums, proving (II.11.3.26).

Write

$$S_0 = \{s : R_s = 1\}, \quad S_1 = \{0, \dots, \ell - 1\} \setminus S_0,$$

and let R'_s be the branching numbers of A . From $A \subset X + H$ before the harmless translation, (II.11.3.8), (II.11.3.11), (II.11.3.24), and (II.11.3.22),

$$|A + H_0| \leq 2^{e_D m} |A|, \quad e_D = O\left(\frac{\log D}{D}\right). \quad (\text{II.11.3.27})$$

On the other hand, (II.11.3.26) gives

$$|A + H_0| \geq 2^{-m/D} |H_0| \prod_{s \in S_0} R'_s.$$

Since $|A| = \prod_s R'_s$, $|H_0| = \prod_s R_s$, and (II.11.3.23) holds,

$$\prod_{s \in S_1} R'_s \geq 2^{-(e_D + D^{-1})m} 2^{(1-D^{-1/4})D|S_1|}. \quad (\text{II.11.3.28})$$

If $|S_1| < D^{-1/2}\ell$, put $S_* = \emptyset$ and call all of S_1 exceptional. Otherwise define

$$S_* = \{s \in S_1 : R'_s \geq 2^{(1-D^{-1/8})D}\}, \quad E = S_1 \setminus S_*.$$

Taking logarithms in (II.11.3.28) and using $R'_s \leq 2^D$ shows

$$|E| \leq CD^{-1/8}\ell \quad (\text{II.11.3.29})$$

for all large D . The same estimate is trivially true in the first case. By (II.11.3.12), every $s \in S_*$ satisfies (II.11.3.2).

Step 5: prune the ν -tree. Let $R_s^{(2)}$ be the branching numbers of B_2 . Since $B_2 \subset H$ before translation and $H_0 \subset H$, (II.11.3.8), (II.11.3.25), and (II.11.3.22) give

$$|B_2 + H_0| \leq 2^{f_D m} |H_0|, \quad f_D = O((\log D)D^{-1/4}). \quad (\text{II.11.3.30})$$

Apply (II.11.3.26) to B_2, H_0 to obtain

$$\prod_{s \in S_0} R_s^{(2)} \leq 2^{(f_D + D^{-1})m}. \quad (\text{II.11.3.31})$$

Collapse B_2 at every scale in S_0 using (II.11.3.16); the loss is controlled by (II.11.3.31). Then collapse the scales in E ; by (II.11.3.29) this loses at most $2^{D|E|} = 2^{O(D^{-1/8})m}$. Call the final set B and its branching numbers R_s'' . Then $R_s'' = 1$ for $s \notin S_*$, while on S_* the original branching is retained. For D large enough, the total pruning factor in this paragraph is at least $2^{-\delta m/4}$, so the factor-two mass comparability on B_1 and (II.11.3.6) imply $\nu(B) \geq 2^{-\delta m}$. The centering and mass-comparability properties survive pruning. This proves (iii)–(v), and (i)–(ii) were already established in Step 3.

Step 6: count the active scales. Let

$$S = \{s : R'_s \geq 2^{(1-\delta)D}\}.$$

Since $D^{-1/8} < \delta$, the construction set S_* is contained in S . The set B branches only at scales in $S_* \subset S$, and hence $|B| \leq 2^{D|S|}$. By Hölder's inequality on B and the mass retention $\nu(B) \geq 2^{-\delta_1 m}$,

$$\nu(B) \leq |B|^{1/q'} \|\nu|_B\|_q \leq |B|^{1/q'} \|\nu\|_q,$$

and therefore

$$\|\nu\|_q^{-q'} \leq \nu(B)^{-q'} |B| \leq 2^{q'\delta_1 m + D|S|}.$$

Taking $\delta_1 < \delta/(2q')$ and then m large gives the left inequality in (II.11.3.3).

For the other side, $A \subset A_j$ with $j \leq 2\epsilon q' m$, and therefore

$$\mu(x) \geq \frac{1}{2} 2^{-j} \|\mu\|_q^{q'} \quad (x \in A).$$

Thus

$$\|\mu\|_q^q \geq |A| 2^{-qj-q} \|\mu\|_q^{qq'},$$

which rearranges to

$$|A| \leq 2^{qj+q} \|\mu\|_q^{-q'}. \quad (\text{II.11.3.32})$$

The uniformization procedure in Step 3 kept exactly a power of two of the children at every block. Hence $R'_s = 2^{k_s}$ with $k_s \in \{0, \dots, D\}$. If $s \in S$, then

$$k_s \geq \lceil (1 - \delta)D \rceil = D - \lfloor \delta D \rfloor.$$

Consequently

$$\log_2 |A| = \sum_s k_s \geq (D - \lfloor \delta D \rfloor) |S|,$$

and therefore

$$\begin{aligned} D|S| &\leq \log_2 |A| + \lfloor \delta D \rfloor |S| \\ &\leq \log_2 (\|\mu\|_q^{-q'}) + (2qq'\varepsilon)m + O_q(1) + \frac{\lfloor \delta D \rfloor}{D}m. \end{aligned}$$

By (II.11.3.13)–(II.11.3.14), the last two linear terms are at most $(\delta - \sigma_D/2)m$; the bounded term is absorbed by the remaining $\sigma_D m/2$ for large m . This proves the right inequality in (II.11.3.3) and completes the proof. $\square$

Remark II.11.3.3 (What the dichotomy says). The theorem does *not* assert that B branches almost fully on the active scales. Its precise conclusion is asymmetric: whenever the retained ν -tree branches, the retained μ -tree is almost saturated. The set S in (vi) may contain additional scales on which B has already been collapsed; this only strengthens the lower scale-count inequality. The asymmetry is essential in the flattening argument below.

II.11.4 Pleasant models and a deterministic spectrum

The inverse theorem applies most cleanly when the small-scale rules vary along a uniquely ergodic base. The following abstraction contains both a single homogeneous self-similar measure and the irrational-rotation model used later for convolutions.

Definition II.11.4.1 (Pleasant model). A *pleasant model* consists of a compact metric space X , a continuous uniquely ergodic map $T : X \rightarrow X$ with invariant probability P , a fixed contraction $0 < \lambda < 1$, and a finite atomic probability measure $\Delta(x)$ for each $x \in X$. There is $N_* < \infty$ such that $\# \text{supp } \Delta(x) \leq N_*$ for every x . At P -almost every x the digit law is continuous in the finite-atomic topology: there are a neighborhood U of x , an integer m , and functions $p_j : U \rightarrow (0, 1]$, $a_j : U \rightarrow \mathbb{R}$ continuous at x , with the $a_j(y)$ pairwise distinct and $\Delta(y) = \sum_{j=1}^m p_j(y) \delta_{a_j(y)}$ for $y \in U$. The model also includes a family of non-atomic probabilities $(\mu_x)_{x \in X}$ whose supports lie in a common compact interval, for which $x \mapsto \mu_x$ is weakly continuous at P -almost every x , and which satisfy

$$\mu_x = \Delta(x) * S_\lambda \mu_{Tx}.$$

After one fixed affine change of coordinates we may, and do, assume throughout this section that all μ_x are supported in $[0, 1]$. This changes dyadic levels by only $O(1)$ and hence does not affect any exponential rate below.

For $n \geq 1$ write

$$\mu_{x,n} = \Delta(x) * S_\lambda \Delta(Tx) * \cdots * S_{\lambda^{n-1}} \Delta(T^{n-1}x).$$

Definition II.11.4.2 (Exponential separation for a pleasant model). Let P be the unique invariant probability on X . We say that the pleasant model has *exponential separation* if, for P -almost every x , there is $R > 0$ such that for infinitely many n all symbolic atoms making up $\mu_{x,n}$ are distinct and pairwise λ^{Rn} -separated.

Proposition II.11.4.3 (Hierarchical q -sum cocycle). Let $m(n) = \lceil n \log_2(1/\lambda) \rceil$ and

$$\phi_n^q(x) = \|\mu_x^{(m(n))}\|_q^q.$$

Then

$$\phi_{n+k}^q(x) \leq C \phi_n^q(x) \phi_k^q(T^n x)$$

with C independent of x, n, k .

Proof. Write $m_n = m(n)$ and $m_k = m(k)$. The full measure decomposes as

$$\mu_x = \sum_a \mu_{x,n}(a) (\delta_a * S_{\lambda^n} \mu_{T^n x}).$$

Group the atoms a by their level- m_n dyadic parent I and let w_I be the corresponding total mass. The normalized tail has uniformly bounded support, so a translated tail issued from I meets only $O(1)$ neighbouring level- m_n cells. For a level- m_{n+k} cell J , convexity gives

$$\left(\sum_{a \in I} \mu_{x,n}(a) (S_{\lambda^n} \mu_{T^n x})(J - a) \right)^q \leq w_I^{q-1} \sum_{a \in I} \mu_{x,n}(a) (S_{\lambda^n} \mu_{T^n x})(J - a)^q.$$

After dilation by λ^{-n} , a level- m_{n+k} cell has length comparable to 2^{-m_k} because $|m_{n+k} - m_n - m_k| = O_\lambda(1)$. Summing over J therefore gives at most $C \phi_k^q(T^n x)$ for every fixed a . Summing next over coarse cells gives

$$\phi_{n+k}^q(x) \leq C \phi_k^q(T^n x) \sum_I w_I^q.$$

For completeness, the comparison at level m_n is in the needed direction. A tail translated from a level- m_n parent I meets at most B_λ level- m_n cells, uniformly in x, n, I , because its support diameter is $O(\lambda^n) = O(2^{-m_n})$. Write $v_{I,J}$ for the mass sent from parent I into output cell J . Then

$$w_I^q = \left(\sum_J v_{I,J} \right)^q \leq B_\lambda^{q-1} \sum_J v_{I,J}^q.$$

Each output cell meets at most B'_λ contributing parents, and $\sum_I v_{I,J} = \mu_x(J)$. Hence

$$\sum_I w_I^q \leq B_\lambda^{q-1} B'_\lambda \sum_J \mu_x(J)^q \leq C \phi_n^q(x),$$

proving the asserted cocycle estimate. $\square$

Unique ergodicity turns the subadditive cocycle estimate into a bound that is uniform in the base point. The required principle is the following.

Lemma II.11.4.4 (Uniform upper bound for a regular subadditive cocycle). *Let (X, T) be compact and uniquely ergodic with invariant probability P . Suppose bounded functions F_n satisfy $F_{n+m}(x) \leq F_n(x) + F_m(T^n x)$ and, for every fixed k and every $\eta > 0$, there is a continuous $G \geq F_k$ with $\int (G - F_k) dP < \eta$. Put $L = \inf_k k^{-1} \int F_k dP$. Then for every $\varepsilon > 0$ one has $F_n(x) \leq (L + \varepsilon)n$ for all sufficiently large n , uniformly in x .*

Proof. Fix k . Split an orbit segment into blocks of length k starting at each residue class modulo k . Summing the resulting k subadditive bounds and absorbing the two end blocks gives

$$F_n(x) \leq \frac{1}{k} \sum_{j=0}^{n-1} F_k(T^j x) + C_k.$$

Let $G \geq F_k$ be continuous with $\int (G - F_k) dP < \eta$. Unique ergodicity gives uniform convergence of the Birkhoff averages of G , hence

$$\limsup_{n \rightarrow \infty} \sup_x \frac{F_n(x)}{n} \leq \frac{1}{k} \int F_k dP + \frac{\eta}{k}.$$

Let $\eta \downarrow 0$ and then take the infimum over k . $\square$

For the pleasant models used below the required continuous majorants exist: each finite-prefix q -sum is bounded and has discontinuity set of P -measure zero; taking the local upper envelope at radius δ and then a decreasing Lipschitz majorant gives continuous majorants whose integrals converge to the original integral by dominated convergence. Kingman's subadditive ergodic theorem (proved in the Foundation volume) supplies the almost-sure limiting exponent. We isolate the resulting uniform estimate needed below.

Definition II.11.4.5 (Deterministic L^q spectrum). For $r > 1$, let $m(n) = \lceil n \log_2(1/\lambda) \rceil$ and define

$$T(r) = \lim_{n \rightarrow \infty} -\frac{1}{m(n)} \log_2 \|\mu_x^{(m(n))}\|_r^r \quad (P\text{-a.e. } x).$$

The preceding cocycle estimate and Kingman's theorem show that this limit exists and is independent of x on a set of full P -measure. Set $T(1) = 0$.

Proposition II.11.4.6 (Uniform one-sided spectrum bound). *For every $r > 1$ and every $\zeta > 0$, there is $m_0 = m_0(r, \zeta)$ such that*

$$S_m(\mu_x, r) = \|\mu_x^{(m)}\|_r^r \leq 2^{-(T(r)-\zeta)m} \quad (\text{II.11.4.1})$$

for all $m \geq m_0$ and all $x \in X$. Moreover $T(1) = 0$ and T is concave on $(1, \infty)$.

Proof. Write $a = \log_2(1/\lambda)$ and $m(n) = \lceil an \rceil$. Proposition II.11.4.3 gives $C_r \geq 1$ with $\phi_{n+k}^r(x) \leq C_r \phi_n^r(x) \phi_k^r(T^n x)$. Thus $F_n(x) = \log_2 \phi_n^r(x) + \log_2 C_r$ is subadditive. The bounded support and bounded digit multiplicity make F_n bounded, and the regularity clause above verifies the hypotheses of Lemma II.11.4.4. Kingman's theorem identifies the almost-sure exponent with $T(r)$ from Definition II.11.4.5. The uniform lemma now gives, for every $\zeta > 0$ and all large n , $\phi_n^r(x) \leq 2^{-(T(r)-\zeta/2)m(n)}$ uniformly in x . If $m(n) \leq m < m(n+1)$, refinement decreases the r -sum and the bounded level discrepancy is absorbed in the remaining $\zeta/2$, proving (II.11.4.1). Concavity follows by applying Hölder to the finite-level mass vectors and passing to the limit on a dense set of exponents; $T(1) = 0$ since every 1-sum is one. $\square$

II.11.4.1 Multifractal localization at a differentiability point

Fix for this subsection a point $q > 1$ at which T is differentiable, and put

$$\alpha = T'(q), \quad T^*(a) = \inf_{r \geq 1} \{ar - T(r)\}, \quad \theta = T^*(\alpha) = q\alpha - T(q).$$

To pass from the global spectrum to a local flattening statement, we need to control both the number of cells carrying substantial q -mass and the loss incurred by restricting to a sparse collection. The next four results give those estimates.

Lemma II.11.4.7 (Legendre gap below saturation). *If $T(q) < q - 1$, then*

$$\theta \leq \alpha < 1.$$

Proof. Since $T(1) = 0$ and T is concave,

$$\alpha \leq \frac{T(q) - T(1)}{q - 1} < 1.$$

Also $T^*(\alpha) \leq \alpha - T(1) = \alpha$. At a differentiability point, the tangent-line inequality for the concave function T gives $T^*(\alpha) = q\alpha - T(q)$, as recorded above. $\square$

Lemma II.11.4.8 (Cardinality of a norm-carrying level set). *For every $\rho > 0$ there is $\delta_0 > 0$ such that the following holds for all sufficiently large m , uniformly in x . Suppose $\mathcal{E} \subset \mathcal{D}_m$ and for some $a \geq 0$,*

$$2^{-am} \leq \mu_x(I) \leq 2^{1-am} \quad (I \in \mathcal{E}), \quad (\text{II.11.4.2})$$

and

$$\sum_{I \in \mathcal{E}} \mu_x(I)^q \geq 2^{-(T(q)+\delta_0)m}. \quad (\text{II.11.4.3})$$

Then

$$|\mathcal{E}| \leq 2^{(\theta+\rho)m}.$$

Proof. Put $h = \rho/(3q)$ and choose $\delta_0 > 0$ so small that $q_1 = q - \sqrt{\delta_0} > 1$ and

$$T(q) - T(q_1) \leq \sqrt{\delta_0}(\alpha + h), \quad \sqrt{\delta_0} \leq h/3. \quad (\text{II.11.4.4})$$

By Proposition II.11.4.6, after decreasing δ_0 once more and taking m large,

$$|\mathcal{E}|2^{-aq_1m} \leq S_m(\mu_x, q_1) \leq 2^{-(T(q_1)-\delta_0)m}.$$

On the other hand, (II.11.4.2)–(II.11.4.3) imply

$$|\mathcal{E}|2^{-aqm} \geq 2^{-q}2^{-(T(q)+\delta_0)m}.$$

Eliminating $|\mathcal{E}|$, dividing by $\sqrt{\delta_0 m}$, and using (II.11.4.4) gives, for large m ,

$$a \leq \alpha + 2h.$$

A second use of Proposition II.11.4.6, now at q , gives

$$|\mathcal{E}|2^{-aqm} \leq S_m(\mu_x, q) \leq 2^{-(T(q)-\rho/3)m}.$$

Hence

$$\frac{1}{m} \log_2 |\mathcal{E}| \leq qa - T(q) + \rho/3 \leq q\alpha - T(q) + 2qh + \rho/3 = \theta + \rho,$$

which proves the claim. $\square$

Lemma II.11.4.9 (Heavy cells carry exponentially little q -mass). *For every $\varsigma > 0$ there is $c = c(\varsigma, q) > 0$ such that, for all sufficiently large m and all x ,*

$$\sum_{\substack{I \in \mathcal{D}_m : \\ \mu_x(I) \geq 2^{-m(\alpha-\varsigma)}}} \mu_x(I)^q \leq 2^{-(T(q)+c)m}. \quad (\text{II.11.4.5})$$

Proof. By differentiability choose $h > 0$ so small that

$$T(q+h) \geq T(q) + h\alpha - \frac{1}{4}h\varsigma.$$

For a cell in the sum, $\mu_x(I)^{-h} \leq 2^{h(\alpha-\varsigma)m}$, and therefore Proposition II.11.4.6 at $q+h$, with error $h\varsigma/4$, yields

$$\begin{aligned} \sum_{\mu_x(I) \geq 2^{-m(\alpha-\varsigma)}} \mu_x(I)^q &\leq 2^{h(\alpha-\varsigma)m} S_m(\mu_x, q+h) \\ &\leq 2^{h(\alpha-\varsigma)m} 2^{-[T(q+h)-h\varsigma/4]m} \\ &\leq 2^{-[T(q)+h\varsigma/2]m}. \end{aligned}$$

Thus one may take $c = h\varsigma/2$. $\square$

Lemma II.11.4.10 (Sparse collections have a strict q -mass loss). *For every $\kappa > 0$ there is $\eta = \eta(\kappa, q) > 0$ such that, for all sufficiently large m , all $x \in X$, and every $\mathcal{E} \subset \mathcal{D}_m$ satisfying*

$$|\mathcal{E}| \leq 2^{(\theta-\kappa)m}, \quad (\text{II.11.4.6})$$

one has

$$\sum_{I \in \mathcal{E}} \mu_x(I)^q \leq 2^{-(T(q)+\eta)m}. \quad (\text{II.11.4.7})$$

If $\theta - \kappa < 0$, (II.11.4.6) simply forces $\mathcal{E} = \emptyset$ for all large m .

Proof. Take $\varsigma = \kappa/(2q)$ in Lemma II.11.4.9. The contribution of cells with mass at least $2^{-m(\alpha-\varsigma)}$ is at most $2^{-(T(q)+c)m}$. Every remaining cell has q -th power at most $2^{-q(\alpha-\varsigma)m}$, so by (II.11.4.6) their total contribution is at most

$$2^{(\theta-\kappa)m} 2^{-q(\alpha-\varsigma)m} = 2^{-[T(q)+\kappa-q\varsigma]m} = 2^{-[T(q)+\kappa/2]m}.$$

Absorb the sum of the two exponentials by decreasing the positive gain; for example any $\eta < \min(c, \kappa/2)$ works for all large m . $\square$

For $I \in \mathcal{D}_s$, write $2I$ for the interval with the same center and twice the length, and $\mathcal{D}_{s+m}(I) = \{J \in \mathcal{D}_{s+m} : J \subset I\}$.

Proposition II.11.4.11 (Localized multifractal loss). *Fix $\kappa > 0$. There is $\eta = \eta(\kappa, q) > 0$ such that for all sufficiently large m , uniformly in $x \in X$, $s \geq 0$ and $I \in \mathcal{D}_s$, the following hold.*

(i) If $\mathcal{E} \subset \mathcal{D}_{s+m}(I)$ and

$$|\mathcal{E}| \leq 2^{(\theta-\kappa)m},$$

then

$$\sum_{J \in \mathcal{E}} \mu_x(J)^q \leq 2^{-(T(q)+\eta)m} \mu_x(2I)^q. \quad (\text{II.11.4.8})$$

(ii) For every $\delta > 0$, after increasing the lower threshold on m ,

$$\sum_{J \in \mathcal{D}_{s+m}(I)} \mu_x(J)^q \leq 2^{-(T(q)-\delta)m} \mu_x(2I)^q. \quad (\text{II.11.4.9})$$

Proof. Let n be the smallest integer with $\lambda^n < 2^{-s-2}$ and write the prefix measure as

$$\mu_{x,n} = \sum_j p_j \delta_{y_j}.$$

Retain only those atoms for which the interval $[y_j, y_j + O(\lambda^n)]$ meets I , and denote their total mass by p . The self-similar identity and the minimality of n imply

$$p \leq \mu_x(2I). \quad (\text{II.11.4.10})$$

For $J \subset I$ we may therefore write

$$\mu_x(J) = \sum_j p_j \mu_{T^n x}(\lambda^{-n}(J - y_j)).$$

Convexity in the form $(\sum_j p_j a_j)^q \leq p^{q-1} \sum_j p_j a_j^q$ gives, for any family $\mathcal{E} \subset \mathcal{D}_{s+m}(I)$,

$$\sum_{J \in \mathcal{E}} \mu_x(J)^q \leq p^{q-1} \sum_j p_j \sum_{J \in \mathcal{E}} \mu_{T^n x}(\lambda^{-n}(J - y_j))^q. \quad (\text{II.11.4.11})$$

Because $\lambda^n \asymp_\lambda 2^{-s}$, for each fixed j every interval $\lambda^{-n}(J - y_j)$ has length comparable with 2^{-m} . It can be covered by $O_\lambda(1)$ cells of $\mathcal{D}_m$, and the resulting family has uniformly bounded overlap. Thus, if $|\mathcal{E}| \leq 2^{(\theta-\kappa)m}$, then for large m the covering family has at most $2^{(\theta-\kappa/2)m}$ cells. Lemma II.11.4.10, used with $\kappa/2$, makes the inner sum in (II.11.4.11) at most

$$C_{\lambda,q} 2^{-(T(q)+\eta_0)m}.$$

Absorbing the fixed constant into half of the gain and using (II.11.4.10) proves (i).

For (ii), use Proposition II.11.4.6 in place of Lemma II.11.4.10. The same bounded-covering comparison gives an inner sum at most $C_{\lambda,q} 2^{-(T(q)-\delta/2)m}$, and the fixed constant is absorbed for large m . This proves (II.11.4.9). $\square$

II.11.5 Flattening and the pleasant-model formula

The inverse theorem says that failure of L^q flattening forces many nearly full branches. Multifractal localization says that a norm-carrying part of μ_x cannot support that branching pattern below saturation. Combining the two gives a uniform gain under convolution.

Theorem II.11.5.1 (Flattening). *Let $q > 1$ be a differentiability point of T and assume $T(q) < q - 1$. Put $q' = q/(q - 1)$. Given $\sigma > 0$, there are $\varepsilon = \varepsilon(\sigma, q) > 0$ and m_0 such that, for every $m \geq m_0$, every $x \in X$, and every 2^{-m} -measure ν satisfying*

$$\|\nu\|_q^{q'} \leq 2^{-\sigma m}, \quad (\text{II.11.5.1})$$

one has

$$\|\nu * \mu_x^{(m)}\|_q^q \leq 2^{-(T(q)+\varepsilon)m}. \quad (\text{II.11.5.2})$$

The constants are uniform in the base point x .

Proof. We give the contradiction argument in full because this is where the multifractal information and the inverse theorem meet. All logarithms in the proof are base two. If necessary replace σ by $\min(\sigma, 1)$; proving the theorem with the smaller value only strengthens the conclusion.

Put

$$\alpha = T'(q), \quad \theta = T^*(\alpha) = q\alpha - T(q).$$

By Lemma II.11.4.7, $\theta < 1$. We first dispose of the case $\theta \leq 0$. Choose $\rho > 0$ so small that

$$\theta + \rho < \sigma/4,$$

then choose $\delta > 0$ small compared with σ and with the threshold in Lemma II.11.4.8 for this ρ . Apply Theorem II.11.3.2 with this δ and any fixed large D_0 , obtaining ε_0, D . If both

$$\|\mu_x^{(m)}\|_q^q \geq 2^{-(T(q)+\xi)m} \quad \text{and} \quad \|\nu * \mu_x^{(m)}\|_q^q \geq 2^{-(T(q)+q\varepsilon_0/4)m}$$

hold, where $\xi > 0$ is sufficiently small, the uniform spectrum bound makes the inverse theorem applicable exactly as below. Its scale count gives $m_1 \geq (\sigma - \delta)m$, while its retained q -mass and Lemma II.11.4.8 give $|A| \leq 2^{(\theta+\rho)m}$. Uniformity and the definition of S give at the same time

$$|A| \geq 2^{(1-\delta)m_1} \geq 2^{(1-\delta)(\sigma-\delta)m},$$

a contradiction to $\theta + \rho < \sigma/4$ once δ is small. Thus the flattening conclusion already follows when $\theta \leq 0$. In the remainder assume $\theta > 0$.

Set

$$c_* := \frac{1}{4} \left(1 - \frac{T(q)}{q-1} \right) > 0 \quad (\text{II.11.5.3})$$

and choose

$$0 < \kappa < \min \left\{ \frac{\theta}{4}, \frac{(1-\theta)\sigma}{8} \right\}. \quad (\text{II.11.5.4})$$

Define in advance

$$\gamma := \frac{\kappa c_*}{\theta - \kappa} > 0, \quad (\text{II.11.5.5})$$

and let $\eta > 0$ be supplied by Proposition II.11.4.11 with sparsity parameter $\kappa/2$.

Choose $\delta_0 > 0$ so small that $\delta_0 < \kappa$, and then choose $\delta > 0$ so small that

$$\delta < \min(\delta_0, \sigma/4, c_*), \quad (1 - \theta - \delta_0 - \delta)(\sigma - \delta) \geq 4\kappa, \quad (\text{II.11.5.6})$$

and

$$q\delta < \frac{\eta\gamma}{16}, \quad \delta(1 + T(q) + \eta) < \frac{\eta\gamma}{16}. \quad (\text{II.11.5.7})$$

Take δ also below the threshold supplied by Lemma II.11.4.8 with tolerance δ_0 .

Choose D_0 so large that $D_0 \geq 4$, Proposition II.11.4.11 is valid at every depth $D - 2$ for $D \geq D_0$, and

$$2^{(\theta-\kappa)D} \leq 2^{(\theta-\kappa/2)(D-2)} \quad (D \geq D_0). \quad (\text{II.11.5.8})$$

Apply Theorem II.11.3.2 with q, δ, D_0 , and denote its near-equality parameter by $\varepsilon_0 > 0$ and its block length by $D \geq D_0$. Finally choose $\xi > 0$ so small that

$$\xi < \min \left\{ \frac{\eta\gamma}{16}, \frac{q\varepsilon_0}{4}, (q-1)c_* \right\}. \quad (\text{II.11.5.9})$$

All lower thresholds on m below are then taken uniformly in x .

We first treat $m = D\ell$. If

$$\|\mu_x^{(m)}\|_q^q \leq 2^{-(T(q)+\xi)m}, \quad (\text{II.11.5.10})$$

then Young's inequality immediately gives the desired conclusion, after decreasing the final gain if necessary. Hence suppose

$$\|\mu_x^{(m)}\|_q^q \geq 2^{-(T(q)+\xi)m}. \quad (\text{II.11.5.11})$$

Assume, towards a contradiction, that

$$\|v * \mu_x^{(m)}\|_q^q \geq 2^{-(T(q)+q\varepsilon_0/4)m}. \quad (\text{II.11.5.12})$$

Proposition II.11.4.6, used with error $q\varepsilon_0/4$, yields

$$\|\mu_x^{(m)}\|_q^q \leq 2^{-(T(q)-q\varepsilon_0/4)m}.$$

Taking q -th roots and comparing with (II.11.5.12) gives

$$\|v * \mu_x^{(m)}\|_q \geq 2^{-\varepsilon_0 m/2} \|\mu_x^{(m)}\|_q \geq 2^{-\varepsilon_0 m} \|\mu_x^{(m)}\|_q.$$

The inverse theorem therefore supplies a set $A \subset \text{supp}(\mu_x^{(m)})$ with branching numbers $(R'_s)_{s=0}^{\ell-1}$ and the scale set

$$S = \{s : R'_s \geq 2^{(1-\delta)D}\}.$$

Let

$$m_1 = D|S|, \quad m_2 = m - m_1, \quad S_0 = [\ell] \setminus S.$$

The lower scale-count inequality and (II.11.5.1) give

$$m_1 \geq (\sigma - \delta)m. \quad (\text{II.11.5.13})$$

Indeed, $\|v\|_q^{q'} \leq 2^{-\sigma m}$ is exactly $\log_2(\|v\|_q^{-q'}) \geq \sigma m$. The upper scale-count inequality and (II.11.5.11) give

$$m_1 \leq \left(\frac{T(q) + \xi}{q - 1} + \delta \right) m. \quad (\text{II.11.5.14})$$

By (II.11.5.3) and the choices of δ, ξ , (II.11.5.14) yields the stronger bound

$$m_2 \geq 2c_* m. \quad (\text{II.11.5.15})$$

The inverse theorem also gives factor-two comparability of the masses on A and

$$\sum_{I \in \mathcal{D}_m(A)} \mu_x(I)^q \geq 2^{-q\delta m} \|\mu_x^{(m)}\|_q^q \geq 2^{-(T(q)+q\delta+\xi)m}. \quad (\text{II.11.5.16})$$

Writing the common mass exponent on A as a , Lemma II.11.4.8, with the parameters already chosen, gives

$$|A| \leq 2^{(\theta+\delta_0)m}. \quad (\text{II.11.5.17})$$

Uniformity gives

$$|A| = \prod_{s=0}^{\ell-1} R'_s \geq 2^{(1-\delta)m_1} \prod_{s \in S_0} R'_s.$$

Combining this with (II.11.5.17) yields

$$\log_2 \prod_{s \in S_0} R'_s \leq (\theta + \delta_0)m_2 - (1 - \theta - \delta_0 - \delta)m_1. \quad (\text{II.11.5.18})$$

By (II.11.5.6), (II.11.5.13), and $m_2 \leq m$, the right-hand side is at most

$$(\theta - 2\kappa)m_2. \quad (\text{II.11.5.19})$$

Define the low-branching set

$$S_1 = \{s \in S_0 : \log_2 R'_s \leq (\theta - \kappa)D\}.$$

Since every term in the product is at least one, (II.11.5.19) gives

$$(\theta - \kappa)D|S_0 \setminus S_1| \leq (\theta - 2\kappa)D|S_0|,$$

so

$$|S_1| \geq \frac{\kappa}{\theta - \kappa}|S_0| \geq 2\gamma\ell \geq \gamma\ell, \quad (\text{II.11.5.20})$$

where (II.11.5.15) and the definition (II.11.5.5) were used. This is the promised positive density of low-branching blocks, with all parameters fixed before the inverse theorem was invoked.

It remains to convert (II.11.5.20) into a contradiction in q -mass. Put

$$L_s = -\log_2 \sum_{J \in \mathcal{D}_{sD}(A)} \mu_x(J)^q, \quad 0 \leq s \leq \ell.$$

Because points of A lie in the middle half of every occupied level- sD parent, if $K \in \mathcal{D}_{sD+2}(A)$ then $2K$ is contained in a unique interval of $\mathcal{D}_{sD}(A)$, and each level- sD parent contains at most two such doubled intervals. Proposition II.11.4.11(ii), applied inside every such K with depth $D - 2$, therefore gives

$$L_{s+1} \geq L_s + (T(q) - \delta)(D - 2) - 1. \quad (\text{II.11.5.21})$$

If $s \in S_1$, then the number of level- $(s + 1)D$ cells of A descending from any such K is at most $2R'_s \leq 2^{(\theta - \kappa)D}$. By (II.11.5.8), Proposition II.11.4.11(i), again at depth $D - 2$ but with sparsity parameter $\kappa/2$, improves the recurrence to

$$L_{s+1} \geq L_s + (T(q) + \eta)(D - 2) - 1. \quad (\text{II.11.5.22})$$

Summing (II.11.5.21) and (II.11.5.22), using $|S_1| \geq \gamma\ell$, and taking D_0 still larger if necessary, gives

$$L_\ell \geq (T(q) + \eta\gamma/2)m. \quad (\text{II.11.5.23})$$

Equivalently,

$$\sum_{I \in \mathcal{D}_m(A)} \mu_x(I)^q \leq 2^{-(T(q) + \eta\gamma/2)m}.$$

This contradicts (II.11.5.16) because the final choices ensure $q\delta + \xi < \eta\gamma/2$. Thus (II.11.5.12) is impossible. For multiples of D , the theorem holds with any final gain smaller than $q\epsilon_0/4$.

Finally let m be arbitrary and put $m_0 = D\lfloor m/D \rfloor$. Coarsening a 2^{-m} -measure to scale 2^{-m_0} changes the q' -norm hypothesis by at most a factor $2^{O_q(D)}$, because each coarse atom contains at most 2^D fine atoms. Likewise, dyadic addition and coarsening compare the level- m convolution q -sum with the level- m_0 convolution q -sum up to a factor $2^{O_q(D)}$. Since D is fixed once q, σ are fixed and $m - m_0 < D$, these bounded exponential factors are absorbed by halving both σ and the final gain for all sufficiently large m . This proves (II.11.5.2) for every large m , uniformly in x . $\square$

For the next step, recall the length- n prefix measure $\mu_{x,n}$ defined above, so that

$$\mu_x = \mu_{x,n} * S_{\lambda^n} \mu_{T^n x}. \quad (\text{II.11.5.24})$$

For a measure σ , $\sigma^{(r)}$ denotes its vector of masses on the level- r dyadic cells, and $\|\sigma^{(r)}\|_q^q$ the corresponding q -sum.

Proposition II.11.5.2 (Fine-prefix recursion). *Let $q > 1$ be a differentiability point of the deterministic spectrum T , assume $T(q) < q - 1$, and fix a base point x for which*

$$\lim_{r \rightarrow \infty} \frac{1}{r} \log_2 \|\mu_x^{(r)}\|_q^q = -T(q). \quad (\text{II.11.5.25})$$

Put $a = \log_2(1/\lambda)$ and $m = m(n) = \lceil an \rceil$. Then, for every fixed integer $R \geq 1$,

$$\lim_{n \rightarrow \infty} \frac{\log_2 \|\mu_{x,n}^{(Rm(n))}\|_q^q}{n \log_2 \lambda} = T(q). \quad (\text{II.11.5.26})$$

Proof. At the natural scale m , bounded overlap in (II.11.5.24) gives

$$\|\mu_{x,n}^{(m)}\|_q^q \asymp \|\mu_x^{(m)}\|_q^q, \quad (\text{II.11.5.27})$$

with constants depending only on the model and q . Hence (II.11.5.26) holds for $R = 1$ by (II.11.5.25). We prove the estimate at scale $(R + 1)m$ for a fixed $R \geq 1$.

Decompose the prefix over $I \in \mathcal{D}_m$ as

$$\mu_{x,n} = \sum_I p_I(\mu_{x,n})_I, \quad p_I = \mu_{x,n}(I),$$

where $(\mu_{x,n})_I$ is normalized and zero-mass cells are omitted. Put

$$\rho_{x,I} = S_{\lambda^{-n}}(\mu_{x,n})_I.$$

The component indexed by I in (II.11.5.24) is supported in I enlarged by $O(\lambda^n)$. Since $2^{-m} \leq \lambda^n < 2^{1-m}$, these supports have uniformly bounded overlap. Comparing the pulled-back level- $(R + 1)m$ grid with the level- Rm grid after scaling by λ^{-n} gives

$$\|\mu_x^{((R+1)m)}\|_q^q \leq C \sum_I p_I^q \|\rho_{x,I}^{(Rm)} * \mu_{T^n x}^{(Rm)}\|_q^q. \quad (\text{II.11.5.28})$$

Because the fine dyadic partition refines $\mathcal{D}_m$, the analogous decomposition for the finite prefix has no cross-cell mixing, and the same grid comparison yields

$$\|\mu_{x,n}^{((R+1)m)}\|_q^q \geq c \sum_I p_I^q \|\rho_{x,I}^{(Rm)}\|_q^q. \quad (\text{II.11.5.29})$$

Fix $\sigma > 0$ and let

$$\mathcal{D}' = \{I : p_I > 0, \|\rho_{x,I}^{(Rm)}\|_q^q \leq 2^{-\sigma m}\}.$$

If $q' = q/(q-1)$, then for $I \in \mathcal{D}'$,

$$\|\rho_{x,I}^{(Rm)}\|_q^{q'} \leq 2^{-\sigma m/(q-1)} = 2^{-\sigma_R Rm}, \quad \sigma_R = \frac{\sigma}{R(q-1)}.$$

The flattening theorem, at base point $T^n x$ and scale Rm , therefore gives an $\varepsilon = \varepsilon(\sigma_R, q) > 0$ such that, for large n ,

$$I \in \mathcal{D}' \implies \|\rho_{x,I}^{(Rm)} * \mu_{T^n x}^{(Rm)}\|_q^q \leq 2^{-(T(q)+\varepsilon)Rm}. \quad (\text{II.11.5.30})$$

For $I \notin \mathcal{D}'$, Young's inequality gives the weaker bound

$$\|\rho_{x,I}^{(Rm)} * \mu_{T^n x}^{(Rm)}\|_q^q \leq \|\mu_{T^n x}^{(Rm)}\|_q^q.$$

Insert these estimates into (II.11.5.28). Since $\sum_I p_I^q = \|\mu_{x,n}^{(m)}\|_q^q$, (II.11.5.27) implies

$$\begin{aligned} \|\mu_x^{((R+1)m)}\|_q^q &\leq C 2^{-(\tau+\varepsilon)Rm} \|\mu_x^{(m)}\|_q^q \\ &\quad + C \|\mu_{T^n x}^{(Rm)}\|_q^q \sum_{I \notin \mathcal{D}'} p_I^q. \end{aligned} \quad (\text{II.11.5.31})$$

Here and below $\tau = T(q)$; this avoids conflict with the base dynamics $T : X \rightarrow X$.

By (II.11.5.25), the first term in (II.11.5.31) is exponentially negligible compared with the left side: for all large n it is at most

$$2^{-\varepsilon Rm/2} \|\mu_x^{((R+1)m)}\|_q^q.$$

Consequently

$$\sum_{I \notin \mathcal{D}'} p_I^q \geq c \frac{\|\mu_x^{((R+1)m)}\|_q^q}{\|\mu_{T^n x}^{(Rm)}\|_q^q}. \quad (\text{II.11.5.32})$$

The almost-subadditive construction of the deterministic spectrum gives, for every $\delta > 0$ and all sufficiently large m , uniformly in the base point,

$$\|\mu_y^{(Rm)}\|_q^q \leq 2^{-(\tau-\delta)Rm}. \quad (\text{II.11.5.33})$$

Together with (II.11.5.25), this turns (II.11.5.32) into

$$\sum_{I \notin \mathcal{D}'} p_I^q \geq c 2^{-\tau m - \delta(2R+1)m}.$$

Choose δ with $\delta(2R+1) < \sigma/2$ and absorb the constant c for large n ; then

$$\sum_{I \notin \mathcal{D}'} p_I^q \geq 2^{-(\tau+\sigma)m}. \quad (\text{II.11.5.34})$$

For $I \notin \mathcal{D}'$, $\|\rho_{x,I}^{(Rm)}\|_q^q > 2^{-\sigma m}$. Hence (II.11.5.29) and (II.11.5.34) give, after absorbing a fixed constant,

$$\|\mu_{x,n}^{((R+1)m)}\|_q^q \geq 2^{-(\tau+3\sigma)m}.$$

Conversely, dyadic refinement decreases the q -sum for $q > 1$, so (II.11.5.27) and (II.11.5.25) give

$$\|\mu_{x,n}^{((R+1)m)}\|_q^q \leq 2^{-(\tau-\sigma)m}$$

for large n . Since $\sigma > 0$ is arbitrary,

$$\lim_{n \rightarrow \infty} m^{-1} \log_2 \|\mu_{x,n}^{((R+1)m)}\|_q^q = -T(q).$$

Finally $m/[n \log_2(1/\lambda)] \rightarrow 1$, which is equivalent to (II.11.5.26). $\square$

Theorem II.11.5.3 (Pleasant-model L^q dimension). *Assume the pleasant model has exponential separation. Then for every $q > 1$,*

$$D(\mu_x, q) = \min \left\{ \frac{\int_X \log \|\Delta(x)\|_q^q dP(x)}{(q-1) \log \lambda}, 1 \right\}$$

uniformly in x .

Proof. Write

$$F(q) = \frac{\int_X \log \|\Delta(y)\|_q^q dP(y)}{\log \lambda}.$$

Both numerator and denominator are non-positive, so $F(q) \geq 0$. We first record the universal upper bound on the exponent. Since coarsening can only increase the q -sum,

$$\|\mu_{x,n}^{(m(n))}\|_q^q \geq \|\mu_{x,n}\|_q^q.$$

Merging atoms can only increase the q -sum for $q > 1$, while before merging the symbolic convolution has product q -sum. Hence

$$\|\mu_{x,n}^{(m(n))}\|_q^q \geq \|\mu_{x,n}\|_q^q \geq \prod_{i=0}^{n-1} \|\Delta(T^i x)\|_q^q. \quad (\text{II.11.5.35})$$

The function $y \mapsto \log \|\Delta(y)\|_q^q$ is bounded and continuous P -almost everywhere: uniform boundedness of the atom count gives $-(q-1) \log_2 N_* \leq \log_2 \|\Delta(y)\|_q^q \leq 0$. The uniquely ergodic upper-envelope argument used in Proposition II.11.4.6, now for this additive cocycle, gives

$$\frac{1}{n} \sum_{i=0}^{n-1} \log_2 \|\Delta(T^i x)\|_q^q \longrightarrow \int_X \log_2 \|\Delta(y)\|_q^q dP(y) \quad (\text{II.11.5.36})$$

uniformly in x . Since $m(n)/n \rightarrow \log_2(1/\lambda)$, (II.11.5.35), the bounded-overlap comparison between $\mu_x^{(m(n))}$ and $\mu_{x,n}^{(m(n))}$, and the elementary bound $S_m(\rho, q) \geq 2^{-(q-1)m}$ for a probability measure yield

$$\limsup_{m \rightarrow \infty} -\frac{1}{m} \log_2 S_m(\mu_x, q) \leq \min\{F(q), q-1\} \quad (\text{II.11.5.37})$$

uniformly in x .

It remains to identify the deterministic spectrum T . Because T is concave, it is differentiable except at at most countably many points. Fix a differentiability point $q > 1$ with $T(q) < q - 1$. By the almost-subadditive construction of T , there is a full- P -measure set of base points for which

$$\lim_{m \rightarrow \infty} m^{-1} \log_2 S_m(\mu_x, q) = -T(q). \quad (\text{II.11.5.38})$$

By exponential separation, after intersecting with another full-measure set we may choose one such x and an integer $R \geq 1$ for which, for infinitely many n , all atoms of the prefix $\mu_{x,n}$ are distinct and are at pairwise distance at least λ^{Rn} .

For this x , Proposition II.11.5.2 gives

$$\lim_{n \rightarrow \infty} \frac{\log_2 \|\mu_{x,n}^{(Rm(n))}\|_q^q}{n \log_2 \lambda} = T(q). \quad (\text{II.11.5.39})$$

Along the separated subsequence, $2^{-Rm(n)} \leq \lambda^{Rn}$, so every level- $Rm(n)$ dyadic cell contains at most one prefix atom. Since there are no exact coincidences, every atom corresponds to a unique symbolic word, and therefore

$$\|\mu_{x,n}^{(Rm(n))}\|_q^q = \|\mu_{x,n}\|_q^q = \prod_{i=0}^{n-1} \|\Delta(T^i x)\|_q^q. \quad (\text{II.11.5.40})$$

Take logarithms, divide by $n \log_2 \lambda$, and use (II.11.5.36) and (II.11.5.39). This proves

$$T(q) = F(q) \quad (\text{II.11.5.41})$$

at every differentiability point with $T(q) < q - 1$.

At a differentiability point with $T(q) = q - 1$, the universal inequality $T(q) \leq \min\{F(q), q - 1\}$ from Proposition II.11.4.6 and (II.11.5.37) forces $\min\{F(q), q - 1\} = q - 1 = T(q)$. Hence

$$T(q) = \min\{F(q), q - 1\}$$

at every differentiability point of T . The left side is continuous by concavity. The map F is continuous on $(1, \infty)$ by dominated convergence, using the uniform atom-count bound above. Since differentiability points are dense, the same identity holds for every $q > 1$.

Finally Proposition II.11.4.6 gives

$$\liminf_{m \rightarrow \infty} -m^{-1} \log_2 S_m(\mu_x, q) \geq T(q)$$

uniformly in x , while (II.11.5.37) gives the matching uniform limsup. Thus the limit exists for every x , uniformly in x , and division by $q - 1$ gives the stated formula. $\square$

II.11.6 General self-similar measures

We now remove the homogeneity assumption. The point is not to disguise the variable contraction lengths as a homogeneous system, but to stop every symbolic branch when it first reaches the requested geometric scale. At a fixed stopping scale there are only polynomially many possible contraction ratios. Consequently, separating the stopping

measure by contraction ratio does not change any exponential rate from the fixed-ratio argument above.

After a fixed affine normalization we assume that the self-similar attractor is contained in $[0, 1]$. Write

$$f_i(x) = \lambda_i x + t_i, \quad 0 < \lambda_i < 1,$$

and let $p_i > 0$, $\sum_i p_i = 1$. For a finite word $u = u_1 \cdots u_k$ put

$$f_u = f_{u_1} \circ \cdots \circ f_{u_k}, \quad \lambda_u = \prod_{j=1}^k \lambda_{u_j}, \quad p_u = \prod_{j=1}^k p_{u_j}, \quad t_u = f_u(0),$$

and let u^- denote the word obtained by deleting the last symbol. Put $\lambda_* := \min_i \lambda_i$ and, for $m \in \mathbb{N}$, define the first-crossing antichain

$$\Omega_m = \{u : \lambda_u \leq 2^{-m} < \lambda_{u^-}\}. \quad (\text{II.11.6.1})$$

Thus

$$\lambda_* 2^{-m} < \lambda_u \leq 2^{-m} \quad (u \in \Omega_m). \quad (\text{II.11.6.2})$$

For affine maps $g_j(x) = a_j x + b_j$ define

$$d(g_1, g_2) = \begin{cases} |b_1 - b_2|, & a_1 = a_2, \\ 1, & a_1 \neq a_2. \end{cases}$$

This is the stopping-scale separation notion needed below.

Definition II.11.6.1 (Stopping-word exponential separation). The IFS has *stopping-word exponential separation* if there are $R > 0$ and integers $m_j \rightarrow \infty$ such that

$$d(f_u, f_v) \geq 2^{-Rm_j} \quad \text{for all distinct } u, v \in \Omega_{m_j}.$$

Lemma II.11.6.2 (Stopping pressure identity). Fix $q > 1$, and let $t = t(q)$ be the unique non-negative solution of

$$\sum_i p_i^q \lambda_i^{-t} = 1. \quad (\text{II.11.6.3})$$

Then, for every m ,

$$\sum_{u \in \Omega_m} p_u^q \lambda_u^{-t} = 1. \quad (\text{II.11.6.4})$$

Consequently, with

$$Z_m(q) := \sum_{u \in \Omega_m} p_u^q,$$

one has

$$\lambda_*^t 2^{-tm} \leq Z_m(q) \leq 2^{-tm}, \quad (\text{II.11.6.5})$$

and therefore

$$\lim_{m \rightarrow \infty} -\frac{1}{m} \log_2 Z_m(q) = t(q). \quad (\text{II.11.6.6})$$

Proof. For fixed q , the left side of (II.11.6.3) is continuous and strictly increasing in t . At $t = 0$ it is at most one, with equality only in the one-map degenerate case, and it tends to infinity as $t \rightarrow \infty$ in the non-degenerate case. Hence the root is unique and non-negative.

Put $a_i = p_i^q \lambda_i^{-t}$. Equation (II.11.6.3) says that (a_i) is a probability vector. Under the Bernoulli probability on the symbolic space with digit weights a_i , the cylinders $[u]$, $u \in \Omega_m$, form a measurable partition: along every infinite word the products of contractions tend to zero, so there is a unique first prefix crossing the threshold 2^{-m} . Hence

$$1 = \sum_{u \in \Omega_m} \prod_{j=1}^{|u|} a_{u_j} = \sum_{u \in \Omega_m} p_u^q \lambda_u^{-t},$$

which is (II.11.6.4). By (II.11.6.2),

$$2^{tm} \leq \lambda_u^{-t} < \lambda_*^{-t} 2^{tm}.$$

Insert these two bounds into (II.11.6.4) to obtain (II.11.6.5); (II.11.6.6) follows immediately. $\square$

Define the stopping approximation and its equal-contraction pieces by

$$\mu_m = \sum_{u \in \Omega_m} p_u \delta_{t_u}, \quad \Lambda_m = \{\lambda_u : u \in \Omega_m\}, \quad \mu_{m,\lambda} = \sum_{\substack{u \in \Omega_m \\ \lambda_u = \lambda}} p_u \delta_{t_u}. \quad (\text{II.11.6.7})$$

Thus $\mu_m = \sum_{\lambda \in \Lambda_m} \mu_{m,\lambda}$.

Lemma II.11.6.3 (Natural-scale stopping approximation). *Let N be the number of maps in the IFS. There is a constant $C \geq 1$, depending only on the IFS and q , such that for every m :*

(i)

$$|\Lambda_m| \leq C(m+1)^N; \quad (\text{II.11.6.8})$$

(ii) *the self-similar identity can be regrouped as*

$$\mu = \sum_{u \in \Omega_m} p_u f_u \mu = \sum_{\lambda \in \Lambda_m} \mu_{m,\lambda} * S_\lambda \mu; \quad (\text{II.11.6.9})$$

(iii) *at the natural dyadic scale,*

$$C^{-1} S_m(\mu_m, q) \leq S_m(\mu, q) \leq C S_m(\mu_m, q). \quad (\text{II.11.6.10})$$

In particular $|\Lambda_m| = 2^{o(m)}$.

Proof. If $u \in \Omega_m$ and $n_i(u)$ is the number of occurrences of symbol i in u , then

$$\lambda_u = \prod_{i=1}^N \lambda_i^{n_i(u)}.$$

Since $\lambda_u > \lambda_* 2^{-m}$, the word length is $O(m)$, and hence each $n_i(u)$ lies in $[0, C_0 m]$ for a fixed C_0 . There are at most $C(m+1)^N$ possible exponent vectors, proving (i).

The stopping cylinders form a complete antichain, so iterating the self-similar identity until the first crossing gives

$$\mu = \sum_{u \in \Omega_m} p_u (\delta_{t_u} * S_{\lambda_u} \mu),$$

and grouping words with equal λ_u proves (ii).

For (iii), every tail $S_{\lambda_u} \mu$ is supported in an interval of length at most 2^{-m} , while by (II.11.6.2) its diameter is also bounded below relative to the stopping scale by a fixed IFS constant when the original support is non-trivial. More directly, the mass vector of μ on $\mathcal{D}_m$ is obtained from the weighted atoms of μ_m by spreading each weight among only $O(1)$ neighboring level- m cells. Conversely the weight of every atom of μ_m is controlled by the μ -mass of a fixed $O(1)$ -cell enlargement around that atom. Applying $(a_1 + \dots + a_L)^q \leq L^{q-1} \sum a_j^q$ to these two bounded-overlap relations gives (II.11.6.10). $\square$

The next proposition is the fixed-measure analogue of the multifractal and flattening package already proved for pleasant models. We include the transfer details because this is the only point where the variable stopping lengths enter the argument.

Proposition II.11.6.4 (Fixed self-similar localization and flattening). *Let $q > 1$ be a differentiability point of $\tau(r) = \tau_\mu(r)$, and assume $\tau(q) < q - 1$. Put*

$$\alpha = \tau'(q), \quad \theta = q\alpha - \tau(q).$$

Then the following hold.

- (i) *For every $\rho > 0$ there is $\delta_0 > 0$ such that, for all large m , if $\mathcal{E} \subset \mathcal{D}_m$ has masses comparable within a factor two,*

$$2^{-am} \leq \mu(I) \leq 2^{1-am} \quad (I \in \mathcal{E}),$$

and

$$\sum_{I \in \mathcal{E}} \mu(I)^q \geq 2^{-(\tau(q) + \delta_0)m},$$

then

$$|\mathcal{E}| \leq 2^{(\theta + \rho)m}. \quad (\text{II.11.6.11})$$

- (ii) *For every $\kappa > 0$ there is $\eta > 0$ such that, for all large m ,*

$$|\mathcal{E}| \leq 2^{(\theta - \kappa)m} \implies \sum_{I \in \mathcal{E}} \mu(I)^q \leq 2^{-(\tau(q) + \eta)m}. \quad (\text{II.11.6.12})$$

Moreover, uniformly in $s \geq 0$ and $I \in \mathcal{D}_s$, if $\mathcal{E} \subset \mathcal{D}_{s+m}(I)$ has the same cardinality bound, then

$$\sum_{J \in \mathcal{E}} \mu(J)^q \leq 2^{-(\tau(q) + \eta)m} \mu(2I)^q, \quad (\text{II.11.6.13})$$

and, for every $\delta > 0$ and all sufficiently large m ,

$$\sum_{J \in \mathcal{D}_{s+m}(I)} \mu(J)^q \leq 2^{-(\tau(q) - \delta)m} \mu(2I)^q. \quad (\text{II.11.6.14})$$

(iii) Given $\sigma > 0$, a fixed integer $R \geq 1$, and $B < \infty$, there are $\varepsilon > 0$, m_0 , and $C \geq 1$, depending only on σ, R, q, B and the fixed self-similar system, such that for $m \geq m_0$ every probability ρ supported in $[-B, B]$ satisfying

$$S_{Rm}(\rho, q) \leq 2^{-\sigma m} \quad (\text{II.11.6.15})$$

also satisfies

$$S_{Rm}(\rho * \mu, q) \leq C 2^{-\varepsilon m} S_{Rm}(\mu, q). \quad (\text{II.11.6.16})$$

In particular the same exponential statement holds after dyadic discretization or after a fixed affine rescaling of ρ .

Proof. Because Proposition II.11.1.2 gives the limit defining $\tau(r)$ for every $r > 1$, for every fixed $r > 1$ and $\zeta > 0$,

$$S_m(\mu, r) \leq 2^{-(\tau(r)-\zeta)m} \quad (\text{II.11.6.17})$$

for all sufficiently large m . Concavity of τ follows from Hölder's inequality applied to the level- m mass vectors, and $\tau(1) = 0$. Consequently the elementary Legendre argument of Lemma II.11.4.7 applies with T replaced by τ :

$$\theta \leq \alpha < 1. \quad (\text{II.11.6.18})$$

We first prove (i). Put $h = \rho/(3q)$ and choose $\delta_0 > 0$ so small that $q_1 = q - \sqrt{\delta_0} > 1$ and

$$\tau(q) - \tau(q_1) \leq \sqrt{\delta_0}(\alpha + h), \quad \sqrt{\delta_0} \leq h/3.$$

Using (II.11.6.17) at q_1 and the lower q -mass hypothesis exactly as in Lemma II.11.4.8 gives $a \leq \alpha + 2h$. A second use of (II.11.6.17), now at q , yields

$$\frac{1}{m} \log_2 |\mathcal{E}| \leq qa - \tau(q) + \rho/3 \leq qa - \tau(q) + \rho = \theta + \rho,$$

which is (II.11.6.11).

For the global part of (ii), fix $\kappa > 0$ and set $\varsigma = \kappa/(2q)$. Differentiability gives a small $h > 0$ such that

$$\tau(q+h) \geq \tau(q) + h\alpha - \frac{1}{4}h\varsigma.$$

Hence (II.11.6.17) at $q+h$ shows that the cells with $\mu(I) \geq 2^{-m(\alpha-\varsigma)}$ contribute at most $2^{-(\tau(q)+c)m}$ to the q -sum, for some $c > 0$. On the remaining cells, a family of size at most $2^{(\theta-\kappa)m}$ contributes at most

$$2^{(\theta-\kappa)m} 2^{-q(\alpha-\varsigma)m} = 2^{-[\tau(q)+\kappa/2]m}.$$

After decreasing the positive gain this proves (II.11.6.12).

It remains to localize these estimates. Fix $I \in \mathcal{D}_s$ and stop at Ω_{s+2} . Retain only words u for which the interval $f_u([0, 1])$ meets I , and let p be the sum of their weights. Since $\lambda_u \leq 2^{-s-2}$, every retained $f_u([0, 1])$ lies inside $2I$; hence

$$p \leq \mu(2I). \quad (\text{II.11.6.19})$$

For $J \subset I$ the stopping identity gives

$$\mu(J) = \sum_u p_u \mu(\lambda_u^{-1}(J - t_u)),$$

where only retained words contribute. Therefore

$$\sum_{J \in \mathcal{E}} \mu(J)^q \leq p^{q-1} \sum_u p_u \sum_{J \in \mathcal{E}} \mu(\lambda_u^{-1}(J - t_u))^q. \quad (\text{II.11.6.20})$$

By (II.11.6.2), now at level $s + 2$, the intervals $\lambda_u^{-1}(J - t_u)$ have lengths between fixed multiples of 2^{-m} . For each u they are covered by $O(1)$ cells of $\mathcal{D}_m$, with uniformly bounded overlap. If $|\mathcal{E}| \leq 2^{(\theta-\kappa)m}$, then for large m the covering family has at most $2^{(\theta-\kappa/2)m}$ cells, so the global sparse estimate, with $\kappa/2$, bounds the inner sum in (II.11.6.20) by $C2^{-(\tau(q)+\eta_0)m}$. The fixed constant is absorbed into half the gain, and (II.11.6.19) gives (II.11.6.13). Replacing the sparse estimate by (II.11.6.17) proves (II.11.6.14).

We prove (iii) directly. Put $n = Rm$, $q' = q/(q-1)$ and

$$\sigma_0 = \frac{\sigma}{R(q-1)}.$$

Discretize ρ at level n . Since ρ is supported in a fixed bounded interval, this changes every q -sum appearing below by a multiplicative constant depending only on B and q . The hypothesis then gives

$$\|\rho^{(n)}\|_q^{q'} = S_n(\rho, q)^{1/(q-1)} \leq 2^{-\sigma_0 n}.$$

We claim that there is $c > 0$ such that, for all large n ,

$$S_n(\rho * \mu, q) \leq 2^{-(\tau(q)+c)n}. \quad (\text{II.11.6.21})$$

First suppose $\theta \leq 0$. Fix $0 < \rho_0 < \sigma_0/4$, and let δ_{card} be the threshold in part (i) corresponding to the tolerance ρ_0 . Choose the inverse-theorem regularization parameter $\delta > 0$ so small that

$$\delta < \min\{\sigma_0/4, \delta_{\text{card}}/(2q)\}.$$

Apply Theorem II.11.3.2; write $\varepsilon_* > 0$ for its near-equality constant and D for its block length, and choose $0 < \xi < \min\{\delta_{\text{card}}/2, q\varepsilon_*/4\}$. For the moment take n divisible by D . If $S_n(\mu, q) \leq 2^{-(\tau(q)+\xi)n}$, Young's inequality already gives (II.11.6.21). Otherwise, convergence of the spectrum gives also

$$2^{-(\tau(q)+\xi)n} \leq S_n(\mu, q) \leq 2^{-(\tau(q)-q\varepsilon_*/4)n}$$

for large n . If the convolution had no exponential gain, the inverse theorem would provide a regular set A retaining at least $2^{-q\delta n} S_n(\mu, q)$ of the q -mass. Its scale-count lower bound, together with $\|\rho^{(n)}\|_q^{q'} \leq 2^{-\sigma_0 n}$, gives at least $(\sigma_0 - \delta)n$ near-full-branching levels, hence

$$|A| \geq 2^{(1-\delta)(\sigma_0-\delta)n}.$$

On the other hand the retained q -mass is at least $2^{-(\tau(q)+q\delta+\xi)n}$, so part (i) applies and yields $|A| \leq 2^{(\theta+\rho_0)n} \leq 2^{\rho_0 n}$. Our choices make the two exponents incompatible. This proves (II.11.6.21) when $\theta \leq 0$.

Assume now $\theta > 0$. Set

$$c_* := \frac{1}{4} \left(1 - \frac{\tau(q)}{q-1} \right) > 0$$

and choose

$$0 < \kappa < \min \left\{ \frac{\theta}{4}, \frac{(1-\theta)\sigma_0}{8} \right\}.$$

Let $\eta > 0$ be the strict gain in the local sparse estimate (II.11.6.13), with sparsity parameter $\kappa/2$, and put $\gamma = \kappa c_*/(\theta - \kappa)$. Choose $\delta_0 < \kappa$ and then $\delta > 0$ so small that

$$(1 - \theta - \delta_0 - \delta)(\sigma_0 - \delta) \geq 4\kappa, \quad q\delta < \eta\gamma/16.$$

Take δ below the threshold in part (i), choose the inverse-theorem block length D sufficiently large that the depth- $D-2$ versions of (II.11.6.13)–(II.11.6.14) hold and

$$2^{(\theta-\kappa)D} \leq 2^{(\theta-\kappa/2)(D-2)},$$

and finally choose $\xi > 0$ smaller than $\min\{\eta\gamma/16, q\varepsilon_*/4, (q-1)c_*\}$, where ε_* is the near-equality constant furnished by the inverse theorem for these parameters.

Again take first $n = D\ell$. We may suppose $S_n(\mu, q) \geq 2^{-(\tau(q)+\xi)n}$, since otherwise Young's inequality finishes the proof. If (II.11.6.21) failed with gain $q\varepsilon_*/4$, the inverse theorem would furnish a regular set A and branching numbers R'_s . Let

$$S = \{s : R'_s \geq 2^{(1-\delta)D}\}, \quad n_1 = D|S|, \quad n_2 = n - n_1.$$

Its scale-count inequalities give

$$(\sigma_0 - \delta)n \leq n_1 \leq \left(\frac{\tau(q) + \xi}{q-1} + \delta \right) n, \quad n_2 \geq 2c_*n.$$

The inverse theorem also retains q -mass at least $2^{-(\tau(q)+q\delta+\xi)n}$, while part (i) gives $|A| \leq 2^{(\theta+\delta_0)n}$. Since $|A| \geq 2^{(1-\delta)n_1} \prod_{s \notin S} R'_s$, these inequalities imply

$$\log_2 \prod_{s \notin S} R'_s \leq (\theta - 2\kappa)n_2.$$

Consequently the set

$$S_1 = \{s \notin S : \log_2 R'_s \leq (\theta - \kappa)D\}$$

has cardinality at least $\gamma\ell$.

Define

$$L_s = -\log_2 \sum_{J \in \mathcal{D}_{sD}(A)} \mu(J)^q.$$

The middle-half property supplied by the inverse theorem permits application of the localized estimates at depth $D-2$. Part (ii) gives, for every s ,

$$L_{s+1} \geq L_s + (\tau(q) - \delta)(D-2) - 1,$$

and for $s \in S_1$ it improves this to

$$L_{s+1} \geq L_s + (\tau(q) + \eta)(D-2) - 1.$$

Summing and taking D large gives

$$L_\ell \geq (\tau(q) + \eta\gamma/2)n.$$

This contradicts the retained lower q -mass because $q\delta + \xi < \eta\gamma/2$. Thus (II.11.6.21) holds for multiples of D .

For arbitrary n , replace it by $D\lfloor n/D \rfloor$. Coarsening changes all relevant q -sums by fixed factors, which are absorbed by halving the positive gain. Finally the defining limit for $\tau(q)$ gives $S_n(\mu, q) \geq 2^{-(\tau(q)+c/2)n}$ for all large n , so

$$S_n(\rho * \mu, q) \leq 2^{-cn/2} S_n(\mu, q).$$

Putting $n = Rm$ yields (II.11.6.16); translation and fixed affine rescaling again alter q -sums only by fixed factors. $\square$

Proposition II.11.6.5 (Fine stopping-prefix recursion). *Let $q > 1$ be a differentiability point of $\tau = \tau_\mu$ and assume $\tau(q) < q - 1$. For every fixed integer $R \geq 1$,*

$$\lim_{m \rightarrow \infty} -\frac{1}{m} \log_2 S_{Rm}(\mu_m, q) = \tau(q). \quad (\text{II.11.6.22})$$

Proof. For $R = 1$ this is immediate from Lemma II.11.6.3 and the definition of $\tau(q)$. Fix now $R \geq 1$ and prove the assertion with $R + 1$ in place of R . Throughout the proof

$$L_m := |\Lambda_m| = 2^{o(m)}. \quad (\text{II.11.6.23})$$

For $\lambda \in \Lambda_m$ and $I \in \mathcal{D}_m$, let $a_{\lambda,I} = \mu_{m,\lambda}(I)$, and, when this is positive, let $\mu_{m,\lambda,I}$ be the normalized restriction of $\mu_{m,\lambda}$ to I . By (II.11.6.9), Hölder's inequality in the λ -sum, and bounded overlap of the descendants of different level- m parents,

$$S_{(R+1)m}(\mu, q) \leq CL_m^{q-1} \sum_{\lambda \in \Lambda_m} \sum_{I \in \mathcal{D}_m} a_{\lambda,I}^q S_{(R+1)m}(\mu_{m,\lambda,I} * S_\lambda \mu, q). \quad (\text{II.11.6.24})$$

For fixed λ and I , translate the normalized restriction to the origin and dilate it by λ^{-1} ; call the resulting probability $\rho_{m,\lambda,I}$. Since $\lambda \asymp 2^{-m}$ by (II.11.6.2), changing from level $(R + 1)m$ before this dilation to level Rm afterwards costs only a fixed factor. Thus

$$S_{(R+1)m}(\mu_{m,\lambda,I} * S_\lambda \mu, q) \leq CS_{Rm}(\rho_{m,\lambda,I} * \mu, q). \quad (\text{II.11.6.25})$$

Taking the largest λ -contribution in (II.11.6.24), there is $\lambda_* = \lambda_*(m) \in \Lambda_m$ such that

$$S_{(R+1)m}(\mu, q) \leq CL_m^q \sum_{I \in \mathcal{D}_m} a_I^q S_{Rm}(\rho_I * \mu, q), \quad (\text{II.11.6.26})$$

where $a_I = a_{\lambda_*,I}$ and $\rho_I = \rho_{m,\lambda_*,I}$.

Fix $\sigma > 0$ and split

$$\mathcal{D}_0 = \{I \in \mathcal{D}_m : S_{Rm}(\rho_I, q) \leq 2^{-\sigma m}\}.$$

Proposition II.11.6.4(iii) gives $\varepsilon > 0$ such that, for $I \in \mathcal{D}_0$,

$$S_{Rm}(\rho_I * \mu, q) \leq 2^{-\varepsilon m} S_{Rm}(\mu, q).$$

For every I the same expression is at most $CS_{Rm}(\mu, q)$ by the dyadic form of Young's inequality. Hence

$$S_{(R+1)m}(\mu, q) \leq CL_m^q S_{Rm}(\mu, q) \left(2^{-\varepsilon m} \sum_{I \in \mathcal{D}_0} a_I^q + \sum_{I \notin \mathcal{D}_0} a_I^q \right). \quad (\text{II.11.6.27})$$

Now

$$\sum_I a_I^q \leq S_m(\mu_m, q) = 2^{-\tau(q)m+o(m)}$$

by the natural-scale comparison, while for fixed R ,

$$S_{Rm}(\mu, q) = 2^{-R\tau(q)m+o(m)}, \quad S_{(R+1)m}(\mu, q) = 2^{-(R+1)\tau(q)m+o(m)}.$$

Together with (II.11.6.23), this shows that the first term on the right of (II.11.6.27) is at most half of the left side for all large m . Consequently

$$\sum_{I \notin \mathcal{D}_0} a_I^q \geq 2^{-(\tau(q)+\sigma)m} \quad (\text{II.11.6.28})$$

after increasing the lower threshold on m .

For $I \notin \mathcal{D}_0$, scale comparability gives, for the unnormalized restriction $a_I \mu_{m, \lambda_*, I}$,

$$S_{(R+1)m}(a_I \mu_{m, \lambda_*, I}, q) \geq C^{-1} a_I^q S_{Rm}(\rho_I, q) > C^{-1} a_I^q 2^{-\sigma m}.$$

The level- $(R+1)m$ cells refine the level- m cells, so these unnormalized restriction contributions sum over distinct I . Also $\mu_m \geq \mu_{m, \lambda_*}$ as atomic measures. Using (II.11.6.28),

$$S_{(R+1)m}(\mu_m, q) \geq 2^{-(\tau(q)+2\sigma)m} \quad (\text{II.11.6.29})$$

for all large m . In the other direction, refinement can only decrease the q -sum, and hence

$$S_{(R+1)m}(\mu_m, q) \leq S_m(\mu_m, q) = 2^{-\tau(q)m+o(m)}. \quad (\text{II.11.6.30})$$

Since $\sigma > 0$ was arbitrary, (II.11.6.29)–(II.11.6.30) prove (II.11.6.22) for $R+1$. Together with the base case this proves the proposition for every fixed R . $\square$

Theorem II.11.6.6 (Positive contractions). *Let μ be generated by $x \mapsto \lambda_i x + t_i$, $0 < \lambda_i < 1$, with probabilities p_i . Assume stopping-word exponential separation in the sense of Definition II.11.6.1. For $q > 1$, let $t(q)$ be the unique solution of*

$$\sum_i p_i^q \lambda_i^{-t(q)} = 1.$$

Then

$$D(\mu, q) = \min \left\{ \frac{t(q)}{q-1}, 1 \right\}.$$

Proof. Write $\tau(q) = \tau_\mu(q)$. We first obtain the universal upper bound on $\tau(q)$. By Lemma II.11.6.3, $S_m(\mu, q) \asymp S_m(\mu_m, q)$. Merging atoms can only increase the q -sum, so

$$S_m(\mu_m, q) \geq \sum_{u \in \Omega_m} p_u^q = Z_m(q).$$

Lemma II.11.6.2 therefore gives

$$\tau(q) \leq t(q). \quad (\text{II.11.6.31})$$

Since a probability supported in a fixed bounded interval occupies only $O(2^m)$ cells at level m , convexity also gives $S_m(\mu, q) \geq C_q 2^{-(q-1)m}$ and hence

$$\tau(q) \leq q - 1. \quad (\text{II.11.6.32})$$

Thus

$$\tau(q) \leq \min\{t(q), q - 1\} \quad (\text{II.11.6.33})$$

for every $q > 1$.

Now let q be a differentiability point of τ with $\tau(q) < q - 1$. By exponential separation there are $R_0 > 0$ and $m_j \rightarrow \infty$ such that words in Ω_{m_j} having the same contraction ratio have translations separated by at least $2^{-R_0 m_j}$. Choose an integer $R > R_0$. Then every level- Rm_j dyadic cell contains at most one atom of each $\mu_{m_j, \lambda}$, and hence

$$S_{Rm_j}(\mu_{m_j, \lambda}, q) = \sum_{\substack{u \in \Omega_{m_j} \\ \lambda_u = \lambda}} p_u^q. \quad (\text{II.11.6.34})$$

Using $\mu_{m_j} = \sum_{\lambda} \mu_{m_j, \lambda}$ and Hölder's inequality,

$$\begin{aligned} S_{Rm_j}(\mu_{m_j}, q) &\leq |\Lambda_{m_j}|^{q-1} \sum_{\lambda \in \Lambda_{m_j}} S_{Rm_j}(\mu_{m_j, \lambda}, q) \\ &= |\Lambda_{m_j}|^{q-1} Z_{m_j}(q). \end{aligned} \quad (\text{II.11.6.35})$$

Proposition II.11.6.5 says that the left side has exponent $\tau(q)$ when divided by m_j . The contraction-class count is subexponential by (II.11.6.8), while Lemma II.11.6.2 says that $Z_{m_j}(q)$ has exponent $t(q)$. Taking $-m_j^{-1} \log_2$ in (II.11.6.35) yields

$$\tau(q) \geq t(q).$$

Combined with (II.11.6.31), this proves

$$\tau(q) = t(q) \quad (\text{II.11.6.36})$$

at every differentiability point for which $\tau(q) < q - 1$.

At a differentiability point where $\tau(q) = q - 1$, (II.11.6.33) forces $\min\{t(q), q - 1\} = q - 1 = \tau(q)$. We have therefore proved

$$\tau(q) = \min\{t(q), q - 1\}$$

at every differentiability point of the concave function τ . Such points are dense. The function τ is continuous on $(1, \infty)$ by concavity, and $t(q)$ is continuous because it is the unique root of the continuous, strictly increasing function $t \mapsto \sum_i p_i^q \lambda_i^{-t}$. Passing to the limit from differentiability points proves the identity for every $q > 1$. Division by $q - 1$ gives the stated formula for $D(\mu, q)$. $\square$

Corollary II.11.6.7 (Signed contractions). *The same formula remains valid for non-zero contractions $-1 < \lambda_i < 1$ if the stopping antichain is defined using $|\lambda_u|$, the pressure equation uses $|\lambda_i|$, and exponential separation is imposed on the affine maps f_u as in Definition II.11.6.1.*

Proof. Put $r_u = |\lambda_u|$ and $\varepsilon_u = \text{sgn}(\lambda_u)$. The pressure identity and first-crossing estimates use only r_u and are unchanged. In the contraction-class decomposition, split additionally by the two orientations (r_u, ε_u) ; this only doubles the polynomial class count. A negative tail is a reflected copy of $S_{r_u}\mu$. Reflection, translation, and a bounded change of dyadic scale alter all q -sum comparisons by fixed constants only, so the localization, flattening, and fine-prefix arguments above are unchanged. The separation hypothesis gives the needed fine-scale separation inside each fixed signed contraction class. Therefore the preceding proof applies with $|\lambda_i|$ throughout. $\square$

II.11.7 Irrational resonance and convolution

When two homogeneous systems contract at incommensurable logarithmic rates, their relative scale evolves by an irrational rotation. Encoding that rotation as a pleasant model reduces the convolution formula to the dimension theorem just proved.

Let Δ be a finitely supported probability measure and $0 < \lambda < 1$. We write

$$\eta(\Delta, \lambda) = \bigstar_{n=0}^{\infty} S_{\lambda^n} \Delta.$$

For a finite set $E \subset \mathbb{R}$ and $n \geq 1$, let $\mathcal{P}_{E,n}$ be the family of all non-zero polynomials

$$P(X) = \sum_{k=0}^{n-1} c_k X^k, \quad c_k \in E - E.$$

If E is a singleton, $\mathcal{P}_{E,n}$ is empty and any minimum over this family below is interpreted as $+\infty$.

Theorem II.11.7.1 (Homogeneous convolution formula). *Let*

$$\eta_j = \eta(\Delta_j, \lambda_j), \quad j = 1, 2,$$

where the Δ_j are finitely supported probability measures and $0 < \lambda_j < 1$. Assume

$$\frac{\log \lambda_2}{\log \lambda_1} \notin \mathbb{Q}.$$

After relabeling suppose $\lambda_2 < \lambda_1$. Assume further that there are $R > 0$ and infinitely many integers n for which, simultaneously for $j = 1, 2$,

$$\min_{P \in \mathcal{P}_{\text{supp}(\Delta_j), n}} |P(\lambda_j)| \geq \lambda_1^{Rn}. \quad (\text{II.11.7.1})$$

Then, for every $q > 1$,

$$D(\eta_1 * \eta_2, q) = \min\{D(\eta_1, q) + D(\eta_2, q), 1\}.$$

Proof. Put

$$a_j = \log(1/\lambda_j), \quad 0 < a_1 < a_2,$$

and let $X = \mathbb{R}/a_2\mathbb{Z}$, represented by $[0, a_2)$. Define the irrational rotation

$$T(x) = x + a_1 \pmod{a_2}.$$

It is uniquely ergodic, with invariant probability $P = a_2^{-1} dx$. We build a pleasant model with fixed contraction λ_1 and digit law

$$\Delta(x) = \begin{cases} \Delta_1 * S_{e^x} \Delta_2, & 0 \leq x < a_1, \\ \Delta_1, & a_1 \leq x < a_2. \end{cases} \quad (\text{II.11.7.2})$$

Outside the finitely many phase boundaries and the finitely many solutions of a digit-collision equation, $x \mapsto \Delta(x)$ is continuous in the finite-atomic topology of Definition II.11.4.1; the number of its atoms is uniformly bounded.

We first identify the corresponding infinite convolutions. Let

$$W_n(x) = \left\lfloor \frac{x + na_1}{a_2} \right\rfloor, \quad N_n(x) = \#\{0 \leq i < n : T^i x \in [0, a_1)\}.$$

Here $W_n(x)$ counts wraps, whereas $N_n(x)$ counts inserted Δ_2 digits. If $T^n x$ is represented in $[0, a_2)$, then

$$T^n x = x + na_1 - W_n(x)a_2. \quad (\text{II.11.7.3})$$

Whenever $T^i x \in [0, a_1)$, the Δ_2 digit inserted at time i appears in the global convolution at scale

$$\lambda_1^i e^{T^i x} = e^x \lambda_2^{k_i},$$

where $k_i = W_i(x)$. For $i \geq 1$, membership $T^i x \in [0, a_1)$ is equivalent to a wrap at the preceding step: $W_i(x) = W_{i-1}(x) + 1$. Since $0 < a_1 < a_2$, at most one wrap occurs per step and every wrap is followed by exactly one selected index. The selected wrap counts thus

run through $0, 1, 2, \dots$ if $x < a_1$ (the initial index is selected), and through $1, 2, 3, \dots$ if $x \geq a_1$. Passing to the infinite convolution gives

$$\mu_x = \begin{cases} \eta_1 * S_{e^x} \eta_2, & 0 \leq x < a_1, \\ \eta_1 * S_{e^{x-a_2}} \eta_2, & a_1 \leq x < a_2. \end{cases} \quad (\text{II.11.7.4})$$

In particular

$$\mu_0 = \eta_1 * \eta_2. \quad (\text{II.11.7.5})$$

If both digit laws are supported at one point, the theorem is immediate. In all other cases the measures μ_x are non-atomic. Formula (II.11.7.4) also shows that $x \mapsto \mu_x$ is weakly continuous away from the finitely many phase boundaries. Thus this is a pleasant model.

We next verify Definition II.11.4.2. Fix one of the infinitely many n for which (II.11.7.1) holds. Every difference of two symbolic atoms in the length- n prefix $\mu_{x,n}$ has one of the forms

$$P_1(\lambda_1), \quad c(x)P_2(\lambda_2), \quad P_1(\lambda_1) + c(x)P_2(\lambda_2), \quad (\text{II.11.7.6})$$

where $P_j \in \mathcal{P}_{\text{supp}(\Delta_j),n}$ when non-zero and

$$c(x) = e^x \quad (0 \leq x < a_1), \quad c(x) = e^{x-a_2} \quad (a_1 \leq x < a_2).$$

Indeed, the prefix uses n digits from the first system and at most n digits from the second, so degree $< n$ suffices for both difference polynomials.

The first two types in (II.11.7.6) are at least a fixed multiple of λ_1^{Rn} . Consider the mixed type with both polynomials non-zero. On either of the two arcs on which $c(x)$ is a fixed multiple of e^x , the function

$$F(x) = P_1(\lambda_1) + c(x)P_2(\lambda_2)$$

is monotone and

$$|F'(x)| = |c(x)P_2(\lambda_2)| \geq C^{-1}\lambda_1^{Rn}.$$

Therefore, for $R_0 > R$, the set on that arc where $|F(x)| \leq \lambda_1^{R_0 n}$ has length at most

$$C\lambda_1^{(R_0-R)n}. \quad (\text{II.11.7.7})$$

The number of possible pairs (P_1, P_2) is at most C_0^n for a constant C_0 depending only on the two finite digit supports. Choose R_0 so large that

$$C_0\lambda_1^{R_0-R} < 1.$$

Summing (II.11.7.7) over all pairs and over the common good values of n gives a convergent series. Borel–Cantelli implies that for P -almost every x , all sufficiently large common good n satisfy

$$\text{all distinct atoms of } \mu_{x,n} \text{ are } \lambda_1^{R_1 n} \text{-separated}$$

for some fixed $R_1 > R_0$ (the harmless fixed constants are absorbed by increasing the exponent once). Thus the pleasant model has exponential separation.

Theorem II.11.5.3 is therefore applicable. It remains only to compute the digit average in its formula. Except for finitely many phases x , all sums $a + c(x)b$, with $a \in \text{supp}(\Delta_1)$ and $b \in \text{supp}(\Delta_2)$, are distinct. Hence on $[0, a_1)$, outside this finite exceptional set,

$$\|\Delta(x)\|_q^q = \|\Delta_1\|_q^q \|\Delta_2\|_q^q, \quad (\text{II.11.7.8})$$

while on $[a_1, a_2)$ one has $\|\Delta(x)\|_q^q = \|\Delta_1\|_q^q$. Thus

$$\int_X \log \|\Delta(x)\|_q^q dP(x) = \log \|\Delta_1\|_q^q + \frac{a_1}{a_2} \log \|\Delta_2\|_q^q. \quad (\text{II.11.7.9})$$

Since $a_1/a_2 = \log \lambda_1 / \log \lambda_2$, Theorem II.11.5.3 and (II.11.7.5) give

$$D(\eta_1 * \eta_2, q) = \min \left\{ \frac{\log \|\Delta_1\|_q^q}{(q-1) \log \lambda_1} + \frac{\log \|\Delta_2\|_q^q}{(q-1) \log \lambda_2}, 1 \right\}. \quad (\text{II.11.7.10})$$

The separation hypothesis also applies to each homogeneous system separately, so Theorem II.11.6.6 (or the one-point special case of Theorem II.11.5.3) identifies

$$D(\eta_j, q) = \min \left\{ \frac{\log \|\Delta_j\|_q^q}{(q-1) \log \lambda_j}, 1 \right\}.$$

The two raw exponents in braces are non-negative. If either is at least one, both sides of the desired convolution identity equal one; if both are below one, they equal the corresponding $D(\eta_j, q)$. Thus (II.11.7.10) is exactly the claimed formula. $\square$

II.11.8 A Furstenberg-type intersection consequence

We isolate the elementary packing step used to turn projection mass decay into a bound for a single fiber.

Lemma II.11.8.1 (Homogeneous open-set-condition consequences). *Let $A \subset \mathbb{R}$ be generated by $N \geq 2$ similarities $f_i(x) = \lambda x + t_i$ with a common $0 < |\lambda| < 1$, and suppose the open set condition holds. Put*

$$s = \frac{\log N}{\log(1/|\lambda|)}$$

and let η be the equal-weight self-similar probability. Then there are $c, C, r_0 > 0$ such that

$$cr^s \leq \eta(B(x, r)) \leq Cr^s \quad (x \in A, 0 < r < r_0).$$

Moreover there is $c_0 > 0$ such that for all n and all distinct words u, v of length n ,

$$|f_u(0) - f_v(0)| \geq c_0 |\lambda|^n.$$

In particular the polynomial separation hypothesis used in Theorem II.11.7.1 holds at every level, after a fixed affine rescaling of the digit set.

Proof. Choose a bounded open interval J whose closure lies in a feasible open set for the OSC. The intervals $f_u(J)$, $|u| = n$, are pairwise disjoint and all have length $|J||\lambda|^n$. Since the linear part of every level- n map is the same, the difference of the centers of $f_u(J)$ and $f_v(J)$ is exactly $f_u(0) - f_v(0)$. Two disjoint open intervals of equal length have centers separated by at least their length, giving the separation estimate with $c_0 = |J|$.

Let $D = \text{diam } A$. Given $0 < r < r_0$, choose n so that $D|\lambda|^n \leq r < D|\lambda|^{n-1}$. If $x \in A$, one level- n cylinder $A_u = f_u(A)$ contains x , has diameter at most r , and carries mass N^{-n} . Thus

$$\eta(B(x, r)) \geq N^{-n} \geq cr^s.$$

For the upper bound, if a level- n cylinder A_u meets $B(x, r)$, then the interval $f_u(J)$ lies in a fixed $O(r)$ -enlargement of that ball, because J and A are bounded and both are scaled by $|\lambda|^n$. The intervals $f_u(J)$ are disjoint and each has length comparable to r , so only a bounded number of them can occur. Hence $B(x, r)$ meets at most C_0 level- n cylinders and

$$\eta(B(x, r)) \leq C_0 N^{-n} \leq Cr^s.$$

Finally, differences of level- n cylinder locations are precisely the non-zero digit-difference polynomials evaluated at λ ; the separation estimate therefore gives the common exponential separation required in (II.11.7.1). $\square$

Lemma II.11.8.2 (L^q dimension gives a Frostman upper bound). *Let ν be a compactly supported probability measure on $\mathbb{R}$, let $q > 1$, and suppose $D(\nu, q) > d$. Then there are $C, r_0 > 0$ such that*

$$\nu(B(y, r)) \leq Cr^{(1-1/q)d} \quad (y \in \mathbb{R}, 0 < r < r_0). \quad (\text{II.11.8.1})$$

Proof. Choose $d < d_1 < D(\nu, q)$. For all sufficiently large m ,

$$S_m(\nu, q) \leq 2^{-(q-1)d_1 m}.$$

If $2^{-m-1} < r \leq 2^{-m}$, the interval $B(y, r)$ meets only a fixed number of level- m dyadic cells. Hölder's inequality gives

$$\nu(B(y, r)) \leq C_q S_m(\nu, q)^{1/q} \leq C_q 2^{-m(1-1/q)d_1} \leq Cr^{(1-1/q)d},$$

after absorbing the difference between d and d_1 into the constant for large m . $\square$

Lemma II.11.8.3 (Fiber packing from projection decay). *Let $K \subset \mathbb{R}^d$ be compact and let μ be a probability supported on K . Suppose that for some $s > 0$ and all sufficiently small r ,*

$$\mu(B(z, r)) \geq cr^s \quad (z \in K). \quad (\text{II.11.8.2})$$

Let $\pi : \mathbb{R}^d \rightarrow \mathbb{R}$ be Lipschitz and put $\nu = \pi_ \mu$. If, for some $0 \leq \alpha \leq s$,*

$$\nu(B(y, r)) \leq Cr^\alpha \quad (\text{II.11.8.3})$$

for all y and all sufficiently small r , then every fiber satisfies

$$\overline{\dim}_B(K \cap \pi^{-1}\{y\}) \leq \max\{s - \alpha, 0\}. \quad (\text{II.11.8.4})$$

Proof. Fix y and let $z_1, \dots, z_N$ be a maximal r -separated subset of $K \cap \pi^{-1}\{y\}$. The balls $B(z_i, r/3)$ are disjoint, and if L is a Lipschitz constant for π , their union is contained in $\pi^{-1}(B(y, Lr/3))$. Hence

$$Nc(r/3)^s \leq \sum_{i=1}^N \mu(B(z_i, r/3)) \leq \nu(B(y, Lr/3)) \leq C'(r)^\alpha.$$

Thus $N \leq C''r^{-(s-\alpha)}$. Since a maximal r -separated set is also an r -net, this proves (II.11.8.4). $\square$

Theorem II.11.8.4 (Affine intersections). *Let A_j be attractors of finite homogeneous real self-similar systems satisfying the open set condition, with contraction ratios λ_j such that $\log \lambda_1 / \log \lambda_2$ is irrational. Then for every invertible affine map g ,*

$$\overline{\dim}_B(A_1 \cap g(A_2)) \leq \max\{\dim_H A_1 + \dim_H A_2 - 1, 0\}.$$

Proof. Write $s_j = \dim_H A_j$ and $s = s_1 + s_2$. Let η_j be the uniform self-similar probability on A_j . If either defining system has one map, its attractor is a singleton. Then $A_1 \cap g(A_2)$ is empty or a singleton and its upper box dimension is 0, so the assertion is immediate. Henceforth both systems have at least two maps. Lemma II.11.8.1 gives Ahlfors regularity,

$$cr^{s_j} \leq \eta_j(B(x, r)) \leq Cr^{s_j} \quad (x \in A_j, 0 < r < r_0). \quad (\text{II.11.8.5})$$

In particular $\mu = \eta_1 \times \eta_2$ satisfies the lower bound (II.11.8.2) with exponent s .

Write $g(y) = ay + b$, where $a \neq 0$, and define

$$\pi_a(x, y) = x - ay, \quad \nu_a = (\pi_a)_* \mu = \eta_1 * S_{-a} \eta_2.$$

Rescaling and reflecting the second homogeneous system changes neither its contraction ratio nor its L^q dimension, and Lemma II.11.8.1 gives the separation hypothesis (II.11.7.1) at every level. Therefore Theorem II.11.7.1 gives, for every $q > 1$,

$$D(\nu_a, q) = \min\{s, 1\}. \quad (\text{II.11.8.6})$$

Here we used that the uniform homogeneous measures have $D(\eta_j, q) = s_j$ for all $q > 1$, which follows directly from their equal cylinder masses and the open set condition.

Fix $\alpha < \min\{s, 1\}$. Choose q sufficiently large and then $d < \min\{s, 1\}$ sufficiently close to $\min\{s, 1\}$ so that

$$(1 - 1/q)d > \alpha.$$

Lemma II.11.8.2 and (II.11.8.6) give

$$\nu_a(B(y, r)) \leq C_\alpha r^\alpha.$$

Applying Lemma II.11.8.3 to π_a yields

$$\overline{\dim}_B((A_1 \times A_2) \cap \pi_a^{-1}\{b\}) \leq s - \alpha.$$

The map

$$x \mapsto \left(x, \frac{x-b}{a}\right)$$

is bi-Lipschitz between $A_1 \cap g(A_2)$ and this fiber. Letting $\alpha \uparrow \min\{s, 1\}$ gives

$$\overline{\dim}_B(A_1 \cap g(A_2)) \leq s - \min\{s, 1\} = \max\{s - 1, 0\},$$

as required. □

Chapter II.12

Exact Overlaps and Current Analytic Extensions

II.12.1 From exact overlap to geometric rigidity

A central theme of this volume is that a loss of dimension forces algebraic or geometric structure. In self-similar systems that structure often takes the sharpest possible form: two distinct symbolic words determine the same map. We first prove the unrestricted complete-map entropy-rate formula, including signed unequal contractions and exact collisions. We then prove Rapaport's dimension formula for systems with algebraic contraction ratios and arbitrary translations. Its main device is to pass temporarily to higher-dimensional translation parameters, where entropy growth and algebraic lower bounds can be combined with a backward induction on the ambient dimension.

The same principle then appears in less rigid settings. For analytic maps on the line, Taylor approximation turns separated maps into polynomial actions, so an entropy inverse theorem can again rule out a dimension drop. For three-map homogeneous systems and for systems with algebraic translations, the obstruction is encoded by low-entropy algebraic approximants and Mahler measure. The chapter next treats Varjú's transcendental Bernoulli theorem and Wu's countability results for exceptional parameters and projection directions. It ends with the planar inverse theorem needed for holomorphic self-conformal systems and with the Feng–Rapaport dimension formula.

Each result imported from the literature is stated where it first enters the argument, with its exact hypotheses and citation. Conditional theorems retain their additional hypotheses explicitly; in particular, open questions are never used as unconditional assumptions.

II.12.2 Exact overlaps and the entropy-rate formula

Definition II.12.2.1. Let

$$\Phi = \{\phi_j(x) = \lambda_j x + t_j\}_{j=0}^m, \quad 0 < |\lambda_j| < 1.$$

For a word u , write ϕ_u for the corresponding composition. The IFS has an *exact overlap*

if $\phi_u = \phi_v$ for two distinct words of the same length. For positive weights p_j , put

$$H(p) = - \sum_j p_j \log_2 p_j, \quad \chi = - \sum_j p_j \log_2 |\lambda_j|.$$

II.12.2.1 The unrestricted entropy-rate dimension formula

Let Λ be a finite nonempty alphabet, and let

$$\Phi = (\varphi_i)_{i \in \Lambda}, \quad \varphi_i(x) = r_i x + t_i, \quad t_i \in \mathbb{R}, \quad 0 < |r_i| < 1.$$

The family is indexed: different symbols may specify the same map. Given a probability vector $p = (p_i)_{i \in \Lambda}$ with $p_i > 0$, its self-similar measure is the unique Borel probability measure satisfying

$$\mu = \sum_{i \in \Lambda} p_i (\varphi_i)_* \mu.$$

Equivalently, μ is the law of $\lim_{n \rightarrow \infty} \varphi_{I_1} \circ \cdots \circ \varphi_{I_n}(0)$, where the symbols I_j are independent with law p . The limit exists uniformly in the address: if $r_{\max} = \max_i |r_i|$ and $T_0 = \max_i |t_i|$, all coding limits have absolute value at most $T_0/(1 - r_{\max})$.

Definition II.12.2.2 (Complete-map entropy rate). For a word $w = i_1 \cdots i_n$, write $\varphi_w = \varphi_{i_1} \circ \cdots \circ \varphi_{i_n}$ and $p_w = p_{i_1} \cdots p_{i_n}$. An *exact overlap* is an equality $\varphi_u = \varphi_v$ for distinct words of the same positive length. Equality here means equality of the complete affine maps, including both the signed slope and the translation. We allow all such overlaps. The relevant entropy counts maps rather than addresses. Set

$$G_n = \varphi_{I_1} \circ \cdots \circ \varphi_{I_n}, \quad \mathbb{P}(G_n = g) = \sum_{w \in \Lambda^n: \varphi_w = g} p_w.$$

All logarithms below have base two. For a finite-valued random variable Z , its Shannon entropy is $H(Z) = - \sum_z \mathbb{P}(Z = z) \log \mathbb{P}(Z = z)$, with $0 \log 0 = 0$. Define the random-walk entropy rate and the Lyapunov exponent by

$$h = h_{\text{RW}}(\Phi, p) := \lim_{n \rightarrow \infty} \frac{H(G_n)}{n} = \inf_{n \geq 1} \frac{H(G_n)}{n}, \quad \chi = - \sum_i p_i \log |r_i| > 0. \quad (\text{II.12.2.1})$$

Indeed, a length- $(n + m)$ map is a function of two independent maps with laws G_n, G_m , so $H(G_{n+m}) \leq H(G_n) + H(G_m)$; subadditivity gives the displayed limit and infimum. Also $0 \leq h \leq H(p) := - \sum_i p_i \log p_i$. The ratio h/χ is independent of the common logarithm base.

We use the lower Hausdorff dimension of a measure:

$$\dim_{\text{H}} \nu = \inf \{ \dim_{\text{H}} E : E \subset \mathbb{R} \text{ Borel, } \nu(E) > 0 \}.$$

The Feng–Hu conformal formula proved in Volume I, Theorem T-I.9.08, applies to finite self-similar systems without separation. Indeed, the maps preserve a sufficiently large

compact interval, extend to contracting smooth diffeomorphisms, and have derivative norm and least singular value both equal to $|r_i|$. The system is therefore conformal, and its Bernoulli coding law is ergodic. These observations also cover negative ratios and repeated indexed maps. Thus, for the measure μ above, there is a constant d such that

$$\lim_{s \downarrow 0} \frac{\log \mu(B(x, s))}{\log s} = d \quad \text{for } \mu\text{-almost every } x. \quad (\text{II.12.2.2})$$

For an exact-dimensional measure this constant equals both $\dim_{\text{H}} \mu$ and $\inf\{\dim_{\text{H}} E : \mu(E) = 1\}$. To see the equivalence, restrict to countably many sets on which the local bounds $s^{d+\varepsilon} \leq \mu(B(x, s)) \leq s^{d-\varepsilon}$ hold uniformly for all sufficiently small s . The lower mass bound gives a full-measure union of sets of dimension at most $d + \varepsilon$ by a covering argument; the upper mass bound gives dimension at least $d - \varepsilon$ for every positive-mass set. Let $\varepsilon \downarrow 0$. Consequently either Hausdorff measure-dimension convention gives the same statement below.

Theorem II.12.2.3 (Entropy-rate dimension formula, OpenAI). *For every finite nonempty indexed family $\varphi_i(x) = r_i x + t_i$ on $\mathbb{R}$ with $0 < |r_i| < 1$, and every strictly positive probability vector p , its self-similar measure satisfies*

$$\dim_{\text{H}} \mu_{\Phi, p} = \min \left\{ 1, \frac{h_{\text{RW}}(\Phi, p)}{\chi(\Phi, p)} \right\}.$$

No separation assumption is required; exact overlaps and repeated generators are allowed.

The proof is adapted from [Ope26b]. It proves the entropy-rate dimension conjecture, and in particular the exact-overlap conjecture without an arithmetic assumption. When there are no exact overlaps, $H(G_n) = nH(p)$, and the theorem gives the usual entropy-to-Lyapunov formula. Section II.12.2.6 records this case, the attractor formula, and the homogeneous all-weight consequence, including the three-map family $\{\lambda x, \lambda x + 1, \lambda x + t\}$. The theorem includes critical and supercritical entropy rates. It asserts dimension, without an assertion of absolute continuity.

II.12.2.1.1 How the proof avoids a smallest-separation estimate

The obstacle is not merely the existence of close cylinders. There may be information below every prescribed scale without a useful bound on the smallest positive distance. The proof finds this information in fair pairs, conditions on block types to keep signed contractions fixed, and arranges successive finite bands of gains in disjoint windows. Each new block length is chosen only after the preceding band's endpoint is known. Consequently no estimate on the growth of those endpoints is needed.

II.12.2.1.2 The entropy-rate upper bound

We prove the upper bound before turning to the contradiction argument. Fix a positive integer n . Use the finite set of distinct maps in the support of G_n as a new alphabet, giving each map g its exact probability $q_g = \mathbb{P}(G_n = g) > 0$. Grouping independent coding

symbols into n -blocks shows that this new system has the same measure μ . Its one-symbol entropy is $H(G_n)$ and its mean contraction depth is $n\chi$, since coincident maps have the same absolute slope.

For $0 < \varepsilon < n\chi$, the strong law shows that almost every address in this block system eventually has, at every length j , a prefix of probability at least $2^{-j(H(G_n)+\varepsilon)}$ and absolute contraction at most $2^{-j(n\chi-\varepsilon)}$. There are at most $2^{j(H(G_n)+\varepsilon)}$ such prefixes. For each fixed eventual starting length J , let E_J be the closed set of addresses satisfying both conditions for every $j \geq J$. Its compact coding image is covered by intervals of diameter at most $C2^{-j(n\chi-\varepsilon)}$, where C bounds the attractor diameter. The total a th power of these diameters tends to zero when $a > (H(G_n) + \varepsilon)/(n\chi - \varepsilon)$. The countable union over eventual starting lengths is a full-measure Borel set with this dimension bound. The case of zero attractor diameter is immediate. Letting $\varepsilon \downarrow 0$, then taking the infimum over n , gives

$$d \leq \min\{1, h/\chi\}. \quad (\text{II.12.2.3})$$

In particular, the conclusion of Theorem II.12.2.3 already holds when $h = 0$, including a single-map system. To prove equality in all other cases, it suffices to rule out

$$d < 1, \quad d\chi < h. \quad (\text{II.12.2.4})$$

These inequalities are the contradiction hypothesis in Sections II.12.2.4 and II.12.2.5.

II.12.2.2 Entropy between two scales

We first establish the entropy estimates that turn a separated pair of translations into a definite gain. The decisive property of the self-similar measure is uniform nonsaturation: if its dimension is less than one, every one-bit scale interval leaves a positive entropy deficit. The intermediate entropy identities apply to arbitrary compactly supported probability measures on the line.

Definition II.12.2.4 (Shift-averaged entropy and scale windows). For a finite-valued variable Z , write $H(Z) = -\sum_z \mathbb{P}(Z = z) \log \mathbb{P}(Z = z)$, with $0 \log 0 = 0$. Conditional entropy is the average entropy of the conditional law, and $I(Z; Y) = H(Z) - H(Z | Y)$; conditional mutual information is defined similarly. We use the entropy chain rule and nonnegativity of conditional mutual information. For a bounded real random variable X and $s > 0$, define

$$H_s^0(X) = H(\lfloor X/s \rfloor), \quad \mathcal{H}_s(X) = \int_0^1 H(\lfloor X/s + u \rfloor) du.$$

We use the same notation for the law of X . For $s, t > 0$, put

$$G_{s,t} = \mathcal{H}_s - \mathcal{H}_t, \quad \Delta_s = 1 - G_{s,2s}.$$

As throughout this proof, all logarithms and entropies are in base 2. Write $h(f) = -\int f \log f$ for differential entropy and V_s for an independent uniform random variable on $[0, s]$. A fair bit is a random variable J with $\mathbb{P}(J = 0) = \mathbb{P}(J = 1) = 1/2$. A fair pair at $x \neq y$ is the probability $(\delta_x + \delta_y)/2$, where δ_x denotes unit point mass at x .

The smoothing and nested-grid identities are the averaged-entropy calculus of [Var19a, Section 2.2, Lemmas 5, 6, and 10]; we include their proofs.

Lemma II.12.2.5 (Entropy calculus). *For every compactly supported real law,*

$$\mathcal{H}_s(X) = h(X + V_s) - \log s, \quad \mathcal{H}_s(aX + b) = \mathcal{H}_{s/|a|}(X) \quad (a \neq 0). \quad (\text{II.12.2.5})$$

If $t/s = N$ is a positive integer, then $G_{s,t}$ is concave in the law,

$$0 \leq G_{s,t} \leq \log N, \quad G_{s,t}(X + Y) \geq G_{s,t}(X) \quad (\text{II.12.2.6})$$

for independent bounded X, Y . For arbitrary $z \geq s > 0$,

$$\mathcal{H}_s - \log(z/s + 2) \leq \mathcal{H}_z \leq \mathcal{H}_s + 1. \quad (\text{II.12.2.7})$$

Furthermore,

$$|\mathcal{H}_s(X) - H_s^0(X)| \leq 1, \quad |H_s^0(X + b) - H_s^0(X)| \leq 1. \quad (\text{II.12.2.8})$$

Proof. The density of $X + V_s$ is $f_s(z) = s^{-1}\mathbb{P}(X \in [z - s, z])$ almost everywhere. It is bounded by $1/s$ and has bounded support, so its differential entropy is finite, even when the law of X has atoms. Set $p_j(u) = \mathbb{P}((j - u)s \leq X < (j + 1 - u)s)$. At $z = (j + 1 - u)s$ we have $p_j(u) = sf_s(z)$ almost everywhere. Changing variables in the sum of the integrals $-p_j(u) \log p_j(u)$ therefore gives

$$\mathcal{H}_s(X) = - \int_{\mathbb{R}} f_s(z) \log(sf_s(z)) dz.$$

This proves the first identity. Translation rotates the grid shift modulo 1, and positive scaling changes the cell length. Reflection also preserves averaged entropy: $-X + V_s$ is a translate of $-(X + V'_s)$, where $V'_s = s - V_s$ is independent uniform. These observations prove the signed affine identity.

Suppose $t = Ns$. Let A be independent uniform on $[0, t)$ and define

$$F = \left\lfloor \frac{X + A}{s} \right\rfloor, \quad C = \left\lfloor \frac{X + A}{t} \right\rfloor.$$

These are jointly translated nested grids, with $C = \lfloor F/N \rfloor$. Averaging their entropies gives

$$G_{s,t}(X) = H(F | C, A). \quad (\text{II.12.2.9})$$

Each coarse cell contains N fine cells, proving the two bounds. For a mixture of laws, conditioning additionally on its mixing variable can only reduce the right-hand side; this proves concavity, including for general probability mixtures. A convolution is a mixture of translates, so translation invariance proves the convolution inequality.

Finally, for any two translated grids of lengths $s \leq z$, a z -cell intersects at most $\lfloor z/s \rfloor + 2$ fine cells, and an s -cell intersects at most two coarse cells. The chain rule bounds the two conditional entropies by $\log(z/s + 2)$ and 1. Averaging gives (II.12.2.7). The same argument for two grids of equal length proves (II.12.2.8). $\square$

The integer-ratio hypothesis is essential to our use of concavity. In the argument below it is applied only with ratios 2, a chosen integer K , or the square of a chosen integer M .

Lemma II.12.2.6 (Entropy and dimension). *Let ν be a compactly supported exact-dimensional probability measure on $\mathbb{R}$ of dimension d . Then*

$$\limsup_{s \downarrow 0} \frac{\max\{\mathcal{H}_s(\nu), H_s^0(\nu)\}}{\log(1/s)} \leq d. \quad (\text{II.12.2.10})$$

Proof. Fix $\eta > 0$. Exact dimensionality gives a measurable set E of mass at least $1 - \eta$ and a common threshold $r_0 > 0$ such that

$$\nu(B(x, r)) \geq r^{d+\eta} \quad (x \in E, 0 < r < r_0).$$

Indeed, the sets where this eventual inequality holds below a fixed reciprocal-integer threshold exhaust a set of full measure; rational radii and monotonicity suffice to make this construction measurable.

For any translated grid of length $s < 2r_0$, choose one point of E in each cell meeting E . Within each of the three residue classes of cell indices modulo 3, the chosen points have disjoint balls of radius $s/2$. Each ball has mass at least $(s/2)^{d+\eta}$. Thus at most $3(2/s)^{d+\eta}$ cells meet E . Compact support bounds the total number of occupied cells by C/s , uniformly over grid translations and small s . Conditioning the cell label on membership in E yields

$$H(\text{cell label}) \leq 1 + \nu(E) \log(3(2/s)^{d+\eta}) + \nu(E^c) \log(C/s) \leq (d + 2\eta) \log(1/s) + O_{\eta, \nu}(1).$$

This controls both the fixed grid and its average. Divide by $\log(1/s)$ and let $\eta \downarrow 0$. $\square$

For Bernoulli convolutions, the following nonsaturation property appears in Breuillard–Varjú [BV19, Lemma 13]. The stopping-time argument below extends it to unequal contractions of either sign.

Lemma II.12.2.7 (Uniform nonsaturation). *Let μ be a self-similar probability measure for a finite affine system $x \mapsto r_i x + t_i$ with $0 < |r_i| < 1$. If μ is exact dimensional with dimension $d < 1$, then*

$$\delta := \inf_{\nu > 0} \Delta_\nu(\mu) > 0. \quad (\text{II.12.2.11})$$

No separation or orientation assumption is needed.

Proof. By Lemma II.12.2.5, $0 \leq \Delta_\nu \leq 1$. We show that a vanishing one-bit deficit would force almost maximal entropy in a fixed wider scale interval. Stopped self-similarity then transfers that estimate to every smaller scale, contradicting the entropy bound supplied by exact dimensionality. Suppose there are $\nu_n > 0$ with $\Delta_{\nu_n}(\mu) \rightarrow 0$. For $X \sim \mu$, let λ_ν be the law of $X + V_\nu$. An independent fair bit J gives

$$\Delta_\nu(\mu) = h(X + V_{2\nu}) - h(X + V_\nu) = I(J; X + V_\nu + \nu J). \quad (\text{II.12.2.12})$$

Thus the equal mixture of λ_ν and its translate by ν carries little information about its input label.

Here is a quantitative form of that observation. For two laws P, Q , write $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Observing whether the mixture sample lies in B bounds the information in (II.12.2.12) below by

$$h_2((P(B) + Q(B))/2) - \frac{1}{2}h_2(P(B)) - \frac{1}{2}h_2(Q(B)),$$

where h_2 is binary entropy. Since $h_2''(u) = -1/(\ln(2)u(1-u)) \leq -4/\ln 2$, this expression is at least $(P(B) - Q(B))^2/(2 \ln 2)$. Consequently, for $e_v = \|\lambda_v - T_v \lambda_v\|_{\text{TV}}$, where T_v denotes translation by v ,

$$e_v \leq \sqrt{2 \ln(2) \Delta_v(\mu)}. \quad (\text{II.12.2.13})$$

Fix an integer $K \geq 2$. The laws $P_i = T_{iv} \lambda_v$, $0 \leq i < K$, have densities f_i and satisfy $\|P_i - P_0\|_{\text{TV}} \leq ie_v$. Their common density $f_* := \min_{0 \leq i < K} f_i$ has mass m_v with

$$1 - m_v \leq \sum_{i=1}^{K-1} \int (f_0 - f_i)_+ \leq \frac{K(K-1)}{2} e_v. \quad (\text{II.12.2.14})$$

For each input label, decompose its law into this same common submeasure and a residual submeasure of mass $1 - m_v$. The common/residual flag can be sampled independently of the input label. On the common branch the output gives no information; on the residual branch it gives at most $\log K$. The mutual information of the uniform mixture is therefore at most $(1 - m_v) \log K$. That mixture is $\mu * \text{Law}(V_{Kv})$, and all its conditional differential entropies equal $h(X + V_v)$. For this fixed K ,

$$G_{v_n, Kv_n}(\mu) \longrightarrow \log K. \quad (\text{II.12.2.15})$$

We next transfer this near-saturation to every smaller scale. Put $a_* := \min_i |r_i| > 0$. For $0 < s < v$, stop the independent coding at the first prefix whose absolute contraction is at most s/v . The stopping time is bounded because $\max_i |r_i| < 1$. The stopped contractions have absolute values in $(a_* s/v, s/v]$, and the remaining coding is independent with law μ . Thus μ is a finite mixture of affine images of itself with these contractions. Signed scaling and integer-ratio concavity give

$$\begin{aligned} G_{s, Ks}(\mu) &\geq \inf_{z \in [v, v/a_*]} G_{z, Kz}(\mu) \\ &\geq G_{v, Kv}(\mu) - C_*, \quad C_* := \log(1/a_* + 2) + 1. \end{aligned} \quad (\text{II.12.2.16})$$

For the second inequality, apply (II.12.2.7) at (v, z) and (Kv, Kz) . The loss C_* is independent of K and the scales.

Choose an integer K with $(1 - d) \log K > C_* + 2$. Then choose one $v = v_n$ such that $G_{v, Kv}(\mu) > \log K - 1$, using (II.12.2.15). For every $0 < s < v$,

$$G_{s, Ks}(\mu) > B := \log K - C_* - 1 > d \log K.$$

At $s_m = vK^{-m}$, telescoping gives

$$\mathcal{H}_{s_m}(\mu) - \mathcal{H}_v(\mu) = \sum_{j=1}^m G_{vK^{-j}, vK^{-(j-1)}}(\mu) > mB.$$

Its entropy ratio has lower limit at least $B/\log K > d$, contradicting Lemma II.12.2.6. This proves the lemma. $\square$

The next estimate explains how the deficit produces gain. Unlike the concavity statements, it does not require an integer ratio of scales. The fair-pair comparison follows the method of [Var19a, Proposition 20 and Lemma 22].

Lemma II.12.2.8 (Two-point gain). *Let X be bounded and J an independent fair bit. For $0 < s < v < t$,*

$$G_{s,t}(X + vJ) - G_{s,t}(X) \geq \Delta_v(X) - s/v - v/t. \quad (\text{II.12.2.17})$$

The same estimate holds for convolution with any fair two-point law whose two points have distance v .

Proof. Put $Y = X + vJ$. For each $a > 0$, let U_a be independent uniform on $[0, 1)$ and write $Z_a = \lfloor Y/a + U_a \rfloor$. Translation invariance of the averaged entropy gives

$$D_a := \mathcal{H}_a(Y) - \mathcal{H}_a(X) = I(J; Z_a | U_a).$$

Uniform smoothing also gives $D_v = h(X + V_{2v}) - h(X + V_v) = \Delta_v(X)$.

To compare s and v , use independent shifts U_s, U_v and the same sample Y . The information chain rule gives

$$I(J; Z_v | U_s, U_v) \leq I(J; Z_s | U_s, U_v) + H(Z_v | Z_s, U_s, U_v).$$

The first two terms are D_v and D_s because the unused shift is independent. For each fixed fine grid and each of its cells of positive probability, the conditional sample law is independent of U_v . That cell, of length $s < v$, contains a boundary of the v -grid on a proportion s/v of its shifts. Otherwise the coarse label is fixed; when a boundary occurs there are at most two coarse labels. Averaging therefore gives $H(Z_v | Z_s, U_s, U_v) \leq s/v$, whence $D_s \geq D_v - s/v$.

For the coarse-scale gain, adjoining X to the observation yields

$$D_t \leq I(J; Z_t, X | U_t) = I(J; Z_t | X, U_t).$$

For fixed $X = x$, the two labels at x and $x + v$ differ on a proportion v/t of shifts. Their conditional information is then one bit and is otherwise zero. Thus $D_t \leq v/t$. Subtracting proves (II.12.2.17). The proof allows atoms, since boundary coincidences occur on null sets of shifts. Finally, any fair pair is a translate of $\{0, v\}$ after ordering its two points; translation invariance proves the last assertion, including for reflected pairs. $\square$

II.12.2.3 Pair mass below a fixed scale

We next consider a finite probability law without any separation assumption. Its exact entropy may greatly exceed its entropy at a prescribed scale. The following lemma detects that excess through the mass that can be placed in fair pairs at smaller scales. An upper bound for a sumset will make the lemma useful for the block laws in the next section. Varjú's scale-local decomposition into fair two-point submeasures [Var19a, Proposition 21] is a predecessor of this approach. Here the sumset term controls the total capacity over all finer bands; we prove the needed estimate by randomized nested partitions.

Definition II.12.2.9 (Pair capacity). Let ν be a probability law with finite support $F \subset \mathbb{R}$. For $\ell \in \mathbb{Z}$, let $\mathcal{E}_\ell(F)$ be the set of unordered pairs $\{x, y\} \subset F$ satisfying $2^{-\ell} \leq |x - y| < 2^{1-\ell}$. Define $c_\ell(\nu)$ as the maximum of

$$\sum_{e \in \mathcal{E}_\ell(F)} w_e \quad \text{subject to} \quad w_e \geq 0, \quad \frac{1}{2} \sum_{e \ni x} w_e \leq \nu(\{x\}) \quad (x \in F).$$

When $\mathcal{E}_\ell(F)$ is empty, this maximum is zero.

Thus $c_\ell(\nu)$ is the largest total weight of fair two-point laws in the indicated distance band whose weighted sum is a submeasure of ν . Summing the constraints gives $\sum_e w_e \leq 1$; the feasible set is compact, so the maximum exists and $0 \leq c_\ell(\nu) \leq 1$. For example, if $\nu = p\delta_x + (1-p)\delta_y$ with $x \neq y$, then the capacity in the band containing $|x - y|$ is $2 \min\{p, 1-p\}$, and all its other capacities vanish.

Lemma II.12.2.10 (Finite-law pair bound). *For every integer $k \geq 1$ there is a finite constant C_k such that every finitely supported probability law ν on $\mathbb{R}$ and every $\rho > 0$ satisfy*

$$k(H(\nu) - H_\rho^0(\nu)) \leq \log |kF| + C_k \sum_{\ell: 2^{-\ell} \leq 2\rho} c_\ell(\nu), \quad F = \text{supp } \nu, \quad (\text{II.12.2.18})$$

where $kF = \{x_1 + \dots + x_k : x_i \in F\}$. The constant is independent of the smallest positive distance in F .

Proof. Let $X_1, \dots, X_k$ be independent with law ν , write $\mathbf{X} = (X_1, \dots, X_k)$, and put $Y = X_1 + \dots + X_k$. Knowing the initial length- ρ cells leaves $k(H(\nu) - H_\rho^0(\nu))$ bits of uncertainty about $\mathbf{X}$. Revealing Y costs at most $\log |kF|$ bits. We bound the remaining uncertainty by successively refining those cells. The refinement cuts are random, independent of the samples, to prevent atoms close to a fixed boundary from being charged too often.

The partitions. Set $\rho_h = \rho 4^{-h}$ for integers $h \geq 0$, and start with the half-open grid intervals of length ρ . Suppose a parent at level h has length $L \in [\rho_h/2, 2\rho_h]$. Let N be a nearest integer to L/ρ_{h+1} , with ties resolved by a fixed rule. Then $2 \leq N \leq 8$. Divide the parent into N equal pieces, and perturb its internal cuts independently and uniformly by amounts in $[-\rho_{h+1}/10, \rho_{h+1}/10]$. The unperturbed child lengths belong to $[0.75\rho_{h+1}, 1.25\rho_{h+1}]$; after perturbation they belong to $[0.55\rho_{h+1}, 1.45\rho_{h+1}]$. Thus the cuts remain ordered and every child has length in $[\rho_{h+1}/2, 2\rho_{h+1}]$. Conditional on the previous partitions, every new cut has density at most

$$\frac{5}{\rho_{h+1}} = \frac{20}{\rho_h}. \quad (\text{II.12.2.19})$$

All cuts are sampled independently of $\mathbf{X}$; a parent that contains several coordinates uses the same cuts for each coordinate.

If F is a singleton, the lemma is immediate. Otherwise choose an integer J such that

$$2\rho_J < \min\{|x - y| : x, y \in F, x \neq y\}.$$

Every level- J cell then contains at most one point of F , for every realization of the partitions. No bound on J will be needed.

For a fixed partition scheme, denote by $\mathbf{C}_h$ the tuple of level- h cells containing the samples. Nesting and separation at level J give

$$\begin{aligned} k(H(\nu) - H_\rho^0(\nu)) &= H(\mathbf{X} \mid \mathbf{C}_0) \\ &\leq \log |kF| + H(\mathbf{X} \mid \mathbf{C}_0, Y) \\ &= \log |kF| + \sum_{h=0}^{J-1} H(\mathbf{C}_{h+1} \mid \mathbf{C}_h, Y). \end{aligned} \quad (\text{II.12.2.20})$$

The whole partition scheme is sampled independently of $\mathbf{X}$. We now average this finite chain over the schemes; equivalently, the conditional entropies on its right also condition on that scheme. For the summand at level h , the cell labels involved use only cuts through level $h + 1$. Conditional on those cuts, all later cuts are independent of $\mathbf{X}$ and hence of Y and these labels, so they may be discarded from the conditioning.

One refinement after observing the sum. Fix the partitions through level h , and condition first on a tuple of current cells $A_1, \dots, A_k$ of positive probability. The coordinates remain independent, with laws $\nu(\cdot \mid A_i)$. Write $m_i = \mathbb{E}X_i$ in this conditional law. Each mean belongs to its half-open parent, since it is an average of finitely many points of that parent. Fix the new cuts and let

$$a = \min_{\substack{1 \leq i \leq k \\ c \text{ a new internal cut of } A_i}} |m_i - c|.$$

This minimum is positive almost surely in the new cuts. Call coordinate i exceptional when $|X_i - m_i| \geq a/(4k)$, let p_i be its probability, and put $t = \sum_i p_i$. Let E be the event that at least two coordinates are exceptional, and write $q = \mathbb{P}(E)$. Independence, before conditioning on Y , gives

$$q \leq \sum_{i < j} p_i p_j \leq t^2, \quad q \leq t. \quad (\text{II.12.2.21})$$

Also put

$$B = \left\{ \left| Y - \sum_i m_i \right| \geq a/2 \right\}.$$

This second event is known once Y is known, and $\mathbb{P}(B) \leq t$: if every coordinate is nonexceptional, the sum deviation is less than $a/4$.

On $E^c \cap B^c$, every child label equals the child label of its mean. Indeed, a changed label forces $|X_i - m_i| \geq a$ for some i , including a change at a half-open endpoint. There is then at most one exceptional coordinate, so

$$\left| \sum_j (X_j - m_j) \right| \geq a - \frac{k-1}{4k}a > a/2,$$

a contradiction. Each coordinate has at most eight child labels. In the next display, conditioning on the fixed partitions and new cuts is implicit. By revealing the flag 1_E in addition to Y , we obtain

$$\begin{aligned} H(\mathbf{C}_{h+1} \mid Y, \mathbf{C}_h = (A_i)) &\leq h_2(q) + 3k(q + P(B)) \\ &\leq (3 + 6k)t \leq 9k \sum_i p_i. \end{aligned} \quad (\text{II.12.2.22})$$

Here $h_2(q) \leq 3\sqrt{q}$ and (II.12.2.21) were used. The elementary binary entropy bound follows from $-(1-q)\ln(1-q) \leq q$ and $-\sqrt{q}\ln q \leq 2/e$. There is no binary-entropy cost for B , since it is determined by the conditioned observation Y . This distinction is what makes (II.12.2.22) linear in the exceptional probabilities.

Averaging the cuts and converting to pairs. For a fixed value of X_i , set $u = |X_i - m_i|$. The event $a \leq 4ku$ requires one of at most $7k$ cuts to be within $4ku$ of its corresponding mean. The density bound (II.12.2.19) and a union bound give

$$\mathbb{P}_{\text{cuts}}(a \leq 4ku) \leq 7k \frac{20}{\rho_h}(8ku) = 1120k^2 \frac{u}{\rho_h}.$$

Consequently

$$\mathbb{E}_{\text{cuts}} p_i \leq \frac{1120k^2}{\rho_h} \mathbb{E}|X_i - m_i| \leq \frac{1120k^2}{\rho_h} \mathbb{E}|X_i - X'_i|,$$

where X'_i is an independent copy in the same current cell. The last inequality is Jensen's inequality. These expectations still use the current-cell conditional laws.

For a fixed level- h partition, integrating the cells out gives the symmetric coupling

$$\pi_h = \sum_{A: \nu(A) > 0} \nu(A) (\nu(\cdot \mid A) \otimes \nu(\cdot \mid A)).$$

Both its marginals equal ν , and its pairs have distance at most $2\rho_h$. Each of the k coordinate contributions becomes the same integral against π_h . Thus (II.12.2.22), averaged over the new cuts and current cells, is at most

$$\frac{10080k^4}{\rho_h} \int |x - y| d\pi_h(x, y). \quad (\text{II.12.2.23})$$

For each band $2^{-\ell} \leq |x - y| < 2^{1-\ell}$, the corresponding portion of π_h supplies a feasible mixture in Definition II.12.2.9. Explicitly, the weight assigned to the unordered pair $\{x, y\}$ is $\pi_h(x, y) + \pi_h(y, x) = 2\pi_h(x, y)$. The resulting submeasure has mass $\sum_{y: 2^{-\ell} \leq |x-y| < 2^{1-\ell}} \pi_h(x, y)$ at x , which is at most $\nu(\{x\})$. Its total weight equals the mass of that band under π_h . Therefore (II.12.2.23) is at most

$$20160k^4 \sum_{\ell: 2^{-\ell} \leq 2\rho_h} \frac{2^{-\ell}}{\rho_h} c_\ell(\nu). \quad (\text{II.12.2.24})$$

This estimate also holds after averaging the previous partitions; its right side refers only to the original law ν .

Summing all finer scales. For a fixed ℓ , the factors $2^{-\ell}/\rho_h$ grow by four at each step and are summed only while they are at most two. Hence

$$\sum_{h \geq 0: 2^{-\ell} \leq 2\rho_h} \frac{2^{-\ell}}{\rho_h} \leq \frac{8}{3}.$$

Summing (II.12.2.24) in (II.12.2.20) proves (II.12.2.18); one may take $C_k = 60000k^4$. The separating depth J disappears from the bound. In particular, the estimate includes separations smaller than any prescribed exponential scale. $\square$

For a fixed finite law only finitely many pair capacities are nonzero. Later we will apply Lemma II.12.2.10 to a finite family of finite laws for each fixed block count. Their nonzero capacities therefore lie in a common finite range, although the upper endpoint need not have any effective bound.

II.12.2.4 Conditioning on block types

Let $\Phi = (\varphi_i)_{i \in \Lambda}$ be a finite affine iterated function system on $\mathbb{R}$, where $\varphi_i(x) = r_i x + t_i$ and $0 < |r_i| < 1$. Let $p = (p_i)_{i \in \Lambda}$ be a strictly positive probability vector, and let μ be its self-similar measure. Write G_n for the random complete affine map obtained from n independent symbols with law p , and let $h = \lim_{n \rightarrow \infty} H(G_n)/n$ be its entropy rate. Throughout this section assume

$$d := \dim_{\text{H}} \mu < 1, \quad d\chi < h, \quad \chi = - \sum_{i \in \Lambda} p_i \log |r_i|. \quad (\text{II.12.2.25})$$

All logarithms have base 2. We will find finite conditional coding laws whose exact entropy exceeds their entropy at a prescribed scale by a positive multiple of the number of blocks. Lemma II.12.2.10 will then turn this deficit into pair mass across finite bands of arbitrarily fine scales.

We condition on the number of occurrences of each symbol in a short block. These counts fix its signed contraction. Once all block contractions are fixed, the translation determines the complete composed map. The entropy cost of recording the counts is small, so the conditional translations retain almost all the map entropy. This argument allows distinct words to define the same map. The block-type disintegration is due to Galicer–Saglietti–Shmerkin–Yavicoli [Gal+16, Section 6.4, Lemma 6.6] and is developed further by Saglietti–Shmerkin–Solomyak [SSS18, Lemma 6.2]; Käenmäki–Orponen explicitly allow repeated indexed maps [KO23, Section 2.2, Proposition 2.8]. We retain the signed contraction and estimate the entropy of the complete map after collisions have been combined.

II.12.2.4.1 Parameters and conditional coding laws

Write $m = |\Lambda|$. The second inequality in (II.12.2.25) implies $h > 0$, hence $m \geq 2$. Choose $\alpha > \chi$ such that $d\alpha < h$. For $d > 0$ we may take $\chi < \alpha < h/d$; for $d = 0$ we may take $\alpha = \chi + 1$. Choose an integer $b \geq 1$ sufficiently large that

$$m \log(b+1) < b(h - d\alpha),$$

and set

$$A = b\alpha, \quad A' = \frac{b(\chi + \alpha)}{2}, \quad h_b = bh - m \log(b+1). \quad (\text{II.12.2.26})$$

Then

$$b\chi < A' < A, \quad \gamma := h_b - dA > 0. \quad (\text{II.12.2.27})$$

These parameters remain fixed throughout the rest of the proof.

Group the iid symbols with law p into words $W_0, W_1, \dots \in \Lambda^b$. For $w \in \Lambda^b$ let $\text{type}(w) = (\kappa_i)_{i \in \Lambda}$, where κ_i counts the occurrences of i in w . The set $\mathcal{T}_b$ of types consists of the nonnegative integer vectors with $\sum_i \kappa_i = b$, and has cardinality at most $(b+1)^m$. For $\kappa \in \mathcal{T}_b$, define

$$N_\kappa = \frac{b!}{\prod_i \kappa_i!}, \quad q_\kappa = N_\kappa \prod_i p_i^{\kappa_i}, \quad Q_\kappa = \text{Law}(W_0 \mid \text{type}(W_0) = \kappa).$$

Every word of type κ has probability $\prod_i p_i^{\kappa_i}$, so Q_κ is uniform on its N_κ words. Equivalently, we can generate the original coding by first choosing independent types $\omega_0, \omega_1, \dots$ with distribution q , then choosing the words independently with respective conditional laws $Q_{\omega_0}, Q_{\omega_1}, \dots$. This follows by multiplying the probabilities of any finite sequence of blocks. In particular it preserves the original, possibly unequal, symbol weights p_i .

Write $\omega = (\omega_h)_{h \geq 0}$ for the type sequence and $\theta\omega = (\omega_{h+1})_{h \geq 0}$ for its shift. For $j \geq 0$, let $\mathcal{F}_j$ be the sigma-algebra generated by $\omega_0, \dots, \omega_{j-1}$; $\mathcal{F}_0$ is trivial. For a type κ and an environment ω , define

$$r(\kappa) = \prod_{i \in \Lambda} r_i^{\kappa_i}, \quad R_j = R_j(\omega) = \prod_{h < j} r(\omega_h), \quad \sigma_j = -\log |R_j|, \quad (\text{II.12.2.28})$$

with $R_0 = 1$ and $\sigma_0 = 0$. Thus R_j is $\mathcal{F}_j$ -measurable, including its sign. Let

$$D = \max_{\kappa \in \mathcal{T}_b} (-\log |r(\kappa)|).$$

The increments of σ_j are iid bounded positive random variables of mean $b\chi$, and $0 \leq \sigma_j \leq Dj$.

Given ω , use the independent conditional block draws above to define

$$\nu_{\omega,n} = \text{Law}_\omega \left(\sum_{h=0}^{n-1} R_h \varphi_{W_h}(0) \right), \quad \mu_\omega = \text{Law}_\omega \left(\sum_{h=0}^{\infty} R_h \varphi_{W_h}(0) \right). \quad (\text{II.12.2.29})$$

Here Law_ω denotes the law under those conditional word draws. The infinite sum converges uniformly, since the block translations form a finite set and $\max_i |r_i| < 1$. The law $\nu_{\omega,n}$ depends only on $\mathcal{F}_n$, whereas μ_ω can depend on the entire environment. Averaging over the types gives $\mu = \mathbb{E}\mu_\omega$. For any real $u \neq 0$ and any law λ , write $D_u \lambda = \text{Law}(uX)$ when X has law λ ; this notation retains the sign of u .

For $n \geq 1$, let $\omega_n = (\omega_0, \dots, \omega_{n-1})$ be the tuple of the first n types, and put

$$\rho_n = 2^{-\lceil An \rceil}, \quad E_n = \{\omega : \sigma_n \leq A'n\}, \quad a_n = \lceil An \rceil - 1. \quad (\text{II.12.2.30})$$

Lemma II.12.2.11 (Entropy retained by types). *For a system satisfying (II.12.2.25), with the parameters (II.12.2.26) and conditional laws (II.12.2.29), for every $n \geq 1$ and $s > 0$ we have*

$$\mathbb{E}H(\nu_{\omega,n}) = H(G_{bn} \mid \Omega_n) \geq nh_b, \quad (\text{II.12.2.31})$$

$$\mathbb{E}H_s^0(\nu_{\omega,n}) \leq H_s^0(\mu) + 1. \quad (\text{II.12.2.32})$$

Moreover, there are $\varepsilon > 0$ and an integer n_0 such that for every $n \geq n_0$,

$$\mathbb{E}\left[\mathbf{1}_{E_n}(H(\nu_{\omega,n}) - H_{\rho_n}^0(\nu_{\omega,n}))\right] \geq \varepsilon n. \quad (\text{II.12.2.33})$$

Proof. Fix Ω_n . The composition corresponding to a tuple $(w_0, \dots, w_{n-1})$ of allowed blocks is

$$\varphi_{w_0 \dots w_{n-1}}(x) = R_n x + \sum_{h < n} R_h \varphi_{w_h}(0).$$

Its signed slope R_n is fixed by Ω_n , and its translation has law $\nu_{\omega,n}$. Thus the complete map and its translation determine one another under this conditioning, even when several words give the same map. It follows that

$$\mathbb{E}H(\nu_{\omega,n}) = H(G_{bn} \mid \Omega_n).$$

There are at most $(b+1)^m$ types, and the block types are independent, so $H(\Omega_n) \leq nm \log(b+1)$. The entropy-rate definition gives $H(G_{bn}) \geq bnh$. Therefore

$$\begin{aligned} H(G_{bn} \mid \Omega_n) &\geq H(G_{bn}) - H(\Omega_n) \\ &\geq bnh - nm \log(b+1) = nh_b. \end{aligned}$$

This conditioning inequality does not require Ω_n to be a function of G_{bn} ; equal maps may arise from different type tuples. This proves (II.12.2.31).

We next compare the finite sum with the original measure at a fixed grid scale. Translation changes fixed-grid entropy by at most one bit: for any bounded random variable X and $t \in \mathbb{R}$,

$$H_s^0(X+t) \geq H_s^0(X) - 1. \quad (\text{II.12.2.34})$$

This is the fixed-grid bound in Lemma II.12.2.5, Equation (II.12.2.8).

Condition now only on $\mathcal{F}_n$. The original symbols after the first bn positions are still independent with law p , and are independent of the prefix. Consequently the full coding law conditional on $\mathcal{F}_n$ is

$$\lambda_\omega = \nu_{\omega,n} * D_{R_n} \mu, \quad \mu = \mathbb{E} \lambda_\omega.$$

For each fixed type prefix this convolution is a mixture of translations of $\nu_{\omega,n}$. Applying concavity of fixed-grid entropy first to these translations and then to the type mixture, and using (II.12.2.34), gives

$$H_s^0(\mu) \geq \mathbb{E}H_s^0(\lambda_\omega) \geq \mathbb{E}H_s^0(\nu_{\omega,n}) - 1.$$

All entropies are finite because the laws have bounded support. The sign of R_n affects only the random translation in this argument. This proves (II.12.2.32).

Set $V_n = H(v_{\omega,n}) - H_{\rho_n}^0(v_{\omega,n})$. Quantization is a function of the exact finite value, so

$$0 \leq V_n \leq H(v_{\omega,n}) \leq nb \log m. \quad (\text{II.12.2.35})$$

By Lemma II.12.2.6, $\limsup_{s \downarrow 0} H_s^0(\mu) / \log(1/s) \leq d$. Choose $\eta = \gamma/(4A) > 0$. For all sufficiently large n , the preceding estimates give

$$\begin{aligned} \mathbb{E}V_n &\geq nh_b - (d + \eta)[An] - 1 \\ &\geq \frac{3}{4}\gamma n - (d + \eta) - 1 \geq \frac{1}{2}\gamma n. \end{aligned}$$

The law of large numbers and $b\chi < A'$ imply $\mathbb{P}(E_n^c) \rightarrow 0$. By (II.12.2.35), after increasing the threshold on n ,

$$\mathbb{E}[\mathbf{1}_{E_n^c} V_n] \leq nb \log m \mathbb{P}(E_n^c) \leq \frac{1}{4}\gamma n.$$

Thus (II.12.2.33) holds with $\varepsilon = \gamma/4$. □

II.12.2.4.2 From the deficit to bands of pair mass

The preceding lemma produces entropy that is invisible at scale ρ_n . To apply the finite-law estimate, we must also control the number of possible sums of several independent copies of a conditional prefix. Fixing the types makes that count uniform in the environment.

Proposition II.12.2.12 (Pair mass in finite scale bands). *Under the assumptions and notation of Lemma II.12.2.11, there exist $c > 0$ and an integer n_0 such that, for each $n \geq n_0$, there is a deterministic finite integer $B_n \geq a_n$ satisfying*

$$\mathbb{E} \left[\mathbf{1}_{E_n} \sum_{\ell=a_n}^{B_n} c_\ell(v_{\omega,n}) \right] \geq cn. \quad (\text{II.12.2.36})$$

Here $c_\ell(v)$ is the maximum total weight of fair two-point laws with separations in $[2^{-\ell}, 2^{1-\ell})$ whose weighted sum is a submeasure of v , as in Lemma II.12.2.10. The constants b, A, A', c, n_0 are independent of n and ω ; no growth bound on B_n is asserted.

Proof. Let $\mathcal{D}_b = \{\varphi_w(0) : w \in \Lambda^b\}$. It has at most m^b elements. For fixed types, write $S = \text{supp } v_{\omega,n}$. Since every R_h is fixed by the types,

$$S \subseteq \sum_{h < n} R_h \mathcal{D}_b, \quad kS \subseteq \sum_{h < n} R_h (k\mathcal{D}_b) \quad (k \geq 1).$$

A sum of k elements of $\mathcal{D}_b$ is specified by the number of occurrences of each digit. Each count belongs to $\{0, \dots, k\}$, so $|k\mathcal{D}_b| \leq (k+1)^{m^b}$ and hence

$$\log |kS| \leq nm^b \log(k+1). \quad (\text{II.12.2.37})$$

This is an upper bound regardless of any collisions among the represented sums and regardless of the signs of the R_h .

With b and ε already fixed, choose an integer k so large that

$$m^b \log(k+1) < \frac{k\varepsilon}{2}.$$

Apply Lemma II.12.2.10 to each $v_{\omega,n}$ at scale ρ_n . Its conclusion, with a positive constant C_k depending only on k , is

$$k(H(v_{\omega,n}) - H_{\rho_n}^0(v_{\omega,n})) \leq \log |kS| + C_k \sum_{\ell \geq a_n} c_\ell(v_{\omega,n});$$

the index condition follows from $2^{-\ell} \leq 2\rho_n$ if and only if $\ell \geq a_n$. Multiply by $\mathbf{1}_{E_n}$ and take expectations. Lemma II.12.2.11 and (II.12.2.37) give

$$C_k \mathbb{E} \left[\mathbf{1}_{E_n} \sum_{\ell \geq a_n} c_\ell(v_{\omega,n}) \right] \geq k\varepsilon n - nm^b \log(k+1) \geq \frac{k\varepsilon n}{2}.$$

For fixed n there are finitely many type prefixes and thus finitely many laws $v_{\omega,n}$, all with finite support. Collect the dyadic band indices of all distinct pairs of support points in all these laws into a finite set $J_n \subset \mathbb{Z}$, and set

$$B_n = \max(\{a_n\} \cup J_n).$$

For every environment, $c_\ell(v_{\omega,n}) = 0$ when $\ell > B_n$. Thus the infinite sum in the last inequality equals the sum from a_n to B_n , and (II.12.2.36) follows with $c = k\varepsilon/(2C_k)$. $\square$

The bands in Proposition II.12.2.12 can extend far beyond the scale 2^{-An} . Their finiteness is enough: the next step will select successive block lengths after the earlier upper endpoints B_n are known. No quantitative separation of distinct cylinder translations has been used.

II.12.2.5 Disjoint windows of entropy gain

Proposition II.12.2.12 supplies a linear amount of fair-pair weight in a finite interval of logarithmic depths for every sufficiently long block segment. We now turn this weight into entropy gain. The difficulty is that the depth interval may extend arbitrarily far. We choose several segments whose depth intervals are separated, and show that their gains occupy disjoint windows of block indices. At each physical scale the total gain is bounded, whereas every chosen interval contributes the same positive amount after averaging over scales.

Throughout this section, retain the environment and constants from Section II.12.2.4. Thus $\sigma_j = -\log |R_j|$ is the contraction depth after j blocks, D bounds each block's depth, and $\mathcal{F}_j = \sigma(\omega_0, \dots, \omega_{j-1})$ records the first j types. In particular, $\sigma_j \leq Dj$. The logarithmic-depth bands are $[a_n, B_n]$, where $a_n = \lceil An \rceil - 1$, and their expected total weight on $E_n = \{\sigma_n \leq A'n\}$ is at least cn .

Let θ denote the left shift of the type sequence. Define the suffix laws, including their contraction from the initial scale, by

$$\tau_j = D_{R_j} \mu_{\theta^j \omega}, \quad j \geq 0.$$

The independent conditional block draws give the pathwise identity

$$\tau_j = (D_{R_j} v_{\theta^j \omega, n}) * \tau_{j+n}, \quad n \geq 1. \quad (\text{II.12.2.38})$$

All these laws have compact support. The signs of R_j are retained in the laws; only scale changes use $|R_j|$.

By Lemma II.12.2.7, choose $\delta > 0$ such that

$$\Delta_v(\mu) := 1 - G_{v, 2v}(\mu) \geq \delta \quad (v > 0).$$

Fix an integer $M > 4$ with $5/M < \delta/2$. For an integer target depth $L \geq 1$, write $r = 2^{-L}$ and define

$$g_j = G_{r/M, Mr}(\tau_j). \quad (\text{II.12.2.39})$$

Here and below g_j depends on L and ω . The ratio of the two window scales is the integer M^2 . Lemma II.12.2.5 and (II.12.2.38) therefore imply, for every environment,

$$0 \leq g_j \leq 2 \log M, \quad g_j \geq g_{j+1}. \quad (\text{II.12.2.40})$$

II.12.2.5.1 A gain after averaging the unobserved future

An individual conditional tail need not satisfy the nonsaturation bound for μ . To use that bound, we condition only on the types through the end of the finite block segment producing a fair pair. The unobserved future types then average its tail law back to a scaled copy of μ .

Lemma II.12.2.13 (Conditional pair gain). *For all integers $j \geq 0$, $n \geq 1$, and $L \geq 1$, with g_j as in (II.12.2.39),*

$$\mathbb{E}[g_j - g_{j+n} \mid \mathcal{F}_{j+n}] \geq \frac{\delta}{2} c_{L - \lfloor \sigma_j \rfloor}(v_{\theta^j \omega, n}). \quad (\text{II.12.2.41})$$

Proof. Condition on $\mathcal{F}_{j+n}$ and put $\ell = L - \lfloor \sigma_j \rfloor$. The finite law $v_{\theta^j \omega, n}$, the signed products R_j, R_{j+n} , and ℓ are now fixed. Choose a fair-pair decomposition attaining $c_\ell(v_{\theta^j \omega, n})$. Such a choice is measurable in $\mathcal{F}_{j+n}$: for fixed j, n, L there are only finitely many initial type sequences, so one may fix a maximizer for each of them.

A selected pair of distance $u \in [2^{-\ell}, 2^{1-\ell})$ has distance $v = |R_j|u$ after scaling. Since $r = 2^{-L}$,

$$\frac{v}{r} \in [2^{\lfloor \sigma_j \rfloor - \sigma_j}, 2^{1 + \lfloor \sigma_j \rfloor - \sigma_j}) \subset [1, 4).$$

Thus $r/M < v < Mr$ and

$$\frac{r/M}{v} + \frac{v}{Mr} \leq \frac{5}{M}.$$

Even if R_j is negative, the scaled pair, after ordering its endpoints, is a translate of the fair law on $\{0, v\}$. Translation invariance and Lemma II.12.2.8 show that this pair increases the window entropy of τ_{j+n} by at least

$$\Delta_v(\tau_{j+n}) - \frac{5}{M}.$$

The types beginning at index $j + n$ are independent of $\mathcal{F}_{j+n}$. Since averaging μ_ω over the environment gives μ , their conditional mean is the identity of measures

$$\mathbb{E}[\tau_{j+n} \mid \mathcal{F}_{j+n}] = D_{R_{j+n}}\mu.$$

The functional $\Delta_v = 1 - G_{v,2v}$ is convex by Lemma II.12.2.5. Each chosen distance v is fixed under the present conditioning. Hence Jensen's inequality and signed scaling give

$$\begin{aligned} \mathbb{E}[\Delta_v(\tau_{j+n}) \mid \mathcal{F}_{j+n}] &\geq \Delta_v(D_{R_{j+n}}\mu) \\ &= \Delta_{v/|R_{j+n}|}(\mu) \geq \delta. \end{aligned}$$

Apply concavity of $G_{r/M,Mr}$ to the selected pairs and the residual positive measure in the prefix law. If the residual mass is nonzero, normalize it to a probability law; convolution with that law contributes a nonnegative gain by Lemma II.12.2.5, since the scale ratio is M^2 . The same statement is vacuous when the residual mass is zero. Summing the conditional lower bounds for all the selected pairs therefore gives

$$\mathbb{E}[g_j - g_{j+n} \mid \mathcal{F}_{j+n}] \geq \left(\delta - \frac{5}{M}\right) c_\ell(v_{\theta^j\omega, n}) \geq \frac{\delta}{2} c_\ell(v_{\theta^j\omega, n}),$$

as required. $\square$

II.12.2.5.2 Choosing disjoint block windows

Fix any positive integer q . Choose sufficiently large integers $n_1 < \dots < n_q$ successively so that Proposition II.12.2.12 applies, $a_{n_i} \geq 1$, and

$$a_{n_i} - A'n_i - 1 > B_{n_h} \quad (h < i). \quad (\text{II.12.2.42})$$

This is possible because the previously chosen B_{n_h} are finite and

$$a_n - A'n - 1 = [An] - A'n - 2 \geq (A - A')n - 2 \longrightarrow \infty.$$

In particular, the choice uses no upper bound on how fast B_n grows. Call $[a_{n_i}, B_{n_i}]$ band i . Larger i means finer physical scales, since (II.12.2.42) gives $a_{n_i} > B_{n_h}$ for $h < i$.

Choose an even integer T with

$$\frac{T}{2} > \max_{1 \leq i \leq q} B_{n_i}. \quad (\text{II.12.2.43})$$

For band i , use the candidate starts

$$J_i = \left\{ kn_i : k \in \mathbb{Z}, 0 \leq k \leq \left\lfloor \frac{T}{2n_i} \right\rfloor \right\}. \quad (\text{II.12.2.44})$$

For each target $L \in \{1, \dots, T\}$, retain the window $[j, j + n_i)$, $j \in J_i$, exactly on the event

$$\mathcal{A}_{i,j,L} = \{\sigma_{j+n_i} - \sigma_j \leq A'n_i, \quad a_{n_i} \leq L - [\sigma_j] \leq B_{n_i}\}. \quad (\text{II.12.2.45})$$

This event belongs to $\mathcal{F}_{j+n_i}$, the same sigma-algebra as the conditioning in Lemma II.12.2.13.

The reason for the separation condition is simple. As a block start j moves forward, the adjusted target depth $L - [\sigma_j]$ is nonincreasing. A retained window in a finer band cannot decrease this depth far enough to reach any coarser band. The next lemma makes this order precise for every environment.

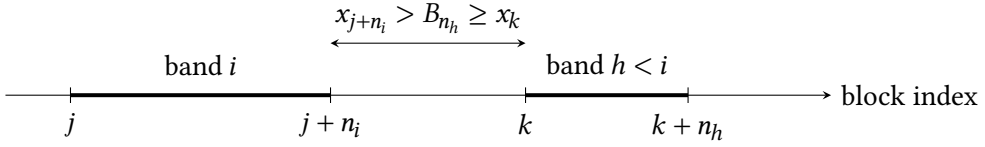


Figure II.12.1: At a fixed target depth, a retained window from a finer band ends before any retained window from a coarser band begins. Here $x_j = L - \lceil \sigma_j \rceil$ is nonincreasing in the block index. The schematic is not to scale.

Lemma II.12.2.14 (Disjoint retained windows). *For each fixed environment and target $L \in \{1, \dots, T\}$, all the retained windows in (II.12.2.45) are pairwise disjoint. Consequently,*

$$\sum_{i=1}^q \sum_{j \in J_i} \mathbf{1}_{\mathcal{A}_{i,j,L}}(g_j - g_{j+n_i}) \leq 2 \log M. \quad (\text{II.12.2.46})$$

Proof. Set $x_j = L - \lceil \sigma_j \rceil$; this is nonincreasing in j . If the band- i window $[j, j + n_i)$ is retained, then

$$\lceil \sigma_{j+n_i} \rceil - \lceil \sigma_j \rceil \leq \sigma_{j+n_i} - \sigma_j + 1 \leq A' n_i + 1,$$

so

$$x_{j+n_i} \geq a_{n_i} - A' n_i - 1. \quad (\text{II.12.2.47})$$

If $[k, k + n_h)$ is a retained window in a coarser band $h < i$, then $x_k \leq B_{n_h}$. By (II.12.2.42) and (II.12.2.47),

$$x_{j+n_i} > B_{n_h} \geq x_k.$$

Monotonicity of x_j forces $k > j + n_i$. Thus the window from the finer band ends before the window from the coarser band starts. Within a single band, distinct starts are distinct multiples of its length n_i , so its half-open windows also do not overlap.

List the retained windows in increasing order as $[u_1, v_1), \dots, [u_N, v_N)$, with $v_t \leq u_{t+1}$. By (II.12.2.40),

$$\sum_{t=1}^N (g_{u_t} - g_{v_t}) \leq g_{u_1} - g_{v_N} \leq 2 \log M.$$

The empty sum is zero. This proves (II.12.2.46). $\square$

Figure II.12.1 shows the order of retained windows from two different bands in this proof.

II.12.2.5.3 Summing over target depths

The disjointness lemma bounds the total gain at one physical scale. Summing over targets recovers every pair scale, in each band. Indeed, for a candidate $j \in J_i$,

$$0 \leq \sigma_j \leq Dj \leq T/2.$$

Because T is even, this also gives $0 \leq \lceil \sigma_j \rceil \leq T/2$. Each integer $\ell \in [a_{n_i}, B_{n_i}]$ therefore occurs exactly once as $L - \lceil \sigma_j \rceil$ for $1 \leq L \leq T$: take $L = \ell + \lceil \sigma_j \rceil$. Its lower bound is 1, and its upper bound is at most $B_{n_i} + T/2 < T$.

Each band has at least $T/(2Dn_i)$ candidate starts, and each contributes expected pair mass at least cn_i after summing the targets. The block length cancels, so every band supplies a fixed multiple of T . Take expectations in (II.12.2.46). Since the retention indicator is $\mathcal{F}_{j+n_i}$ -measurable, the tower property and Lemma II.12.2.13 apply to each summand. Sum the result over $L = 1, \dots, T$ and use the preceding exact correspondence to obtain

$$\begin{aligned} 2T \log M &\geq \frac{\delta}{2} \sum_{i=1}^q \sum_{j \in J_i} \mathbb{E} \left[\mathbf{1}_{E_{n_i}}(\theta^j \omega) \sum_{\ell=a_{n_i}}^{B_{n_i}} c_\ell(v_{\theta^j \omega, n_i}) \right] \\ &\geq \frac{\delta c}{2} \sum_{i=1}^q |J_i| n_i \geq \frac{\delta c q T}{4D}. \end{aligned} \quad (\text{II.12.2.48})$$

The second inequality uses Proposition II.12.2.12 and the identical distribution of the type environment under θ^j ; independence between different candidate windows is not needed. The last inequality follows from the exact count

$$|J_i| = \left\lfloor \frac{T}{2Dn_i} \right\rfloor + 1 \geq \frac{T}{2Dn_i}.$$

After canceling T , (II.12.2.48) bounds the arbitrary integer q by $8D \log M/(\delta c)$. All the constants in this bound were fixed before q was chosen, so this is impossible.

Completion of the proof of Theorem II.12.2.3. The standard upper bound gives $\dim_{\text{H}} \mu \leq \min\{1, h/\chi\}$. If the inequality were strict, then $d = \dim_{\text{H}} \mu$ would satisfy $d < 1$ and $d\chi < h$. Section II.12.2.4 would supply the pair bands used above, and (II.12.2.48) would give a contradiction. Therefore $\dim_{\text{H}} \mu = \min\{1, h/\chi\}$ for the prescribed positive vector p . Since p was arbitrary, the conclusion holds for every such vector, including the cases $h = \chi$ and $h > \chi$. $\square$

II.12.2.6 Overlap conventions and dimension of the attractor

We first reconcile the two usual word-length conventions for an exact overlap. The following observation also appears in [Hoc14, footnote 3]. The nonzero contraction hypothesis is useful here as well as in the proof of the measure theorem.

Lemma II.12.2.15. *For a finite family of maps $x \mapsto r_i x + t_i$ with $0 < |r_i| < 1$, an equality of maps for two distinct nonempty words of arbitrary lengths implies such an equality for two distinct words of the same length.*

Proof. Suppose $\varphi_u = \varphi_v$ and $u \neq v$. If u were a proper prefix of v , writing $v = uw$ and cancelling the invertible map φ_u would give $\varphi_w = \text{id}$. This is impossible because a nonempty word has absolute contraction less than one. The same applies with the words interchanged. Now $\varphi_{uv} = \varphi_u \circ \varphi_v = \varphi_v \circ \varphi_u = \varphi_{vu}$. The words uv and vu have the same length and are distinct: if they were equal, their first $\min\{|u|, |v|\}$ symbols would show that either $u = v$ or the shorter word is a proper prefix of the longer. $\square$

Corollary II.12.2.16 (No exact overlaps). *Let Φ be a finite nonempty indexed family of real similarities with $0 < |r_i| < 1$ and no exact overlaps. For every strictly positive probability vector p ,*

$$\dim_{\mathrm{H}} \mu_{\Phi, p} = \min\{1, H(p)/\chi(\Phi, p)\}.$$

Proof. At every length n , distinct words give distinct complete maps. Hence G_n is an injective function of its symbol word, so $H(G_n) = nH(p)$ and $h_{\mathrm{RW}} = H(p)$. Apply Theorem II.12.2.3. $\square$

Let K_{Φ} be the compact attractor, equivalently the set of all coding limits. Its similarity dimension s_* is the unique number $s_* \geq 0$ satisfying

$$\sum_{i \in \Lambda} |r_i|^{s_*} = 1.$$

For at least two symbols, existence and uniqueness follow because this sum is continuous and strictly decreasing from $|\Lambda| > 1$ to zero. For one symbol its unique solution is zero.

Corollary II.12.2.17. *For a finite nonempty family of similarities of $\mathbb{R}$ with arbitrary real translations and $0 < |r_i| < 1$, absence of exact overlaps implies*

$$\dim_{\mathrm{H}} K_{\Phi} = \min\{1, s_*\}.$$

Proof. For one map the attractor is its fixed point and $s_* = 0$. For at least two maps choose the strictly positive weights $p_i = |r_i|^{s_*}$. Then $H(p) = s_* \chi(\Phi, p)$. Corollary II.12.2.16 gives a measure supported on K_{Φ} of dimension $\min\{1, s_*\}$, which is a lower bound for its set dimension. For the upper bound, if $a > s_*$, the level- n cylinder sets cover K_{Φ} , their diameters tend uniformly to zero, and

$$\sum_{u \in \Lambda^n} (\mathrm{diam} \varphi_u(K_{\Phi}))^a = (\mathrm{diam} K_{\Phi})^a \left(\sum_i |r_i|^a \right)^n \rightarrow 0.$$

The zero-diameter case is immediate. Thus $\dim_{\mathrm{H}} K_{\Phi} \leq s_*$, and the ambient upper bound is one. $\square$

Corollary II.12.2.18. *Let $m \geq 2$, $0 < \lambda < 1$, and $t_1, \dots, t_m \in \mathbb{R}$. If the maps $x \mapsto \lambda x + t_i$ have no exact overlaps, then for every strictly positive probability vector p their self-similar measure has*

$$\dim_{\mathrm{H}} \mu = \min \left\{ 1, \frac{H(p)}{\log(1/\lambda)} \right\}.$$

Proof. Apply Corollary II.12.2.16 with $r_i = \lambda$ for every i . Since the weights sum to one, $\chi = \log(1/\lambda)$. $\square$

Corollary II.12.2.19 (Three maps). *Let $0 < \lambda < 1$ and $t \in \mathbb{R}$. If the indexed maps*

$$x \mapsto \lambda x, \quad x \mapsto \lambda x + 1, \quad x \mapsto \lambda x + t$$

have no exact overlaps, then for every $p_0, p_1, p_2 > 0$ with $p_0 + p_1 + p_2 = 1$, their self-similar measure satisfies

$$\dim_{\text{H}} \mu = \min \left\{ 1, \frac{-\sum_{i=0}^2 p_i \log p_i}{\log(1/\lambda)} \right\}.$$

Proof. Apply Corollary II.12.2.18 with translations $0, 1, t$. □

No claim of absolute continuity, effective separation, or a quantitative rate of convergence follows from these dimension statements. In particular, the finite upper endpoints of the scale ranges used in the proof have no asserted effective bound.

II.12.3 Algebraic contractions: Rapaport's mechanism

The unrestricted theorem above now gives Rapaport's one-dimensional conclusion without an arithmetic assumption. We retain the original mechanism because its higher-dimensional reconstruction, entropy growth, and algebraic lower bounds are useful beyond this corollary.

Rapaport's theorem on the line is obtained from a higher-dimensional translation problem. We begin with that result because it contains the entropy and algebraic mechanisms of the proof; the one-dimensional theorem will then follow by a short reduction.

Theorem II.12.3.1 (Rapaport's higher-dimensional lower bound). *Let $m \geq 2$, let $\lambda_0, \dots, \lambda_m \in (-1, 1) \setminus \{0\}$ be algebraic, and let $p_0, \dots, p_m > 0$. Fix $1 \leq d < m$ and vectors $t_1, \dots, t_m \in \mathbb{R}^d$ spanning $\mathbb{R}^d$. Consider*

$$\phi_0(x) = \lambda_0 x, \quad \phi_j(x) = \lambda_j x + t_j \quad (1 \leq j \leq m).$$

If this IFS has no exact overlaps, then its self-similar measure satisfies

$$\dim \mu \geq \min \left\{ 1, \frac{H(p)}{\chi} \right\}.$$

This is the higher-dimensional input proved in [Rap22, Theorem 9]. The proof below combines its algebraic reconstruction with the Euclidean inverse theorem proved in Chapter II.5.

Proof strategy. The proof has two distinct parts. First, entropy growth shows that a genuine dimension drop would create a nonzero persistently saturated subspace. The averaged quotient-entropy identity rules out that subspace when the dimension is below one. Second, algebraic lower bounds for cylinder differences turn arbitrarily close coincidences

into exact linear relations. A backward induction on d then produces an exact overlap. The lemmas below develop these ingredients in that order before the final induction.

Lemma II.12.3.2 (Full-dimensional fibers force ambient dimension). *Let $V \subset \mathbb{R}^d$ be a k -dimensional subspace, let $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^d/V$ be the quotient map, and disintegrate a probability μ as*

$$\mu = \int \mu_y d\bar{\mu}(y), \quad \bar{\mu} = \pi_*\mu.$$

If $\dim \mu_y = k$ for $\bar{\mu}$ -almost every y , then $\dim \mu \geq k$.

Proof. Let $E \subset \mathbb{R}^d$ be Borel with $\mu(E) > 0$. By disintegration,

$$\mu(E) = \int \mu_y(E) d\bar{\mu}(y),$$

so the set of y for which $\mu_y(E) > 0$ has positive $\bar{\mu}$ -measure. For almost every such y the conditional measure μ_y has dimension k . By the definition of the lower Hausdorff dimension of a measure, every Borel set of positive μ_y -mass has Hausdorff dimension at least k . Hence

$$\dim_H E \geq \dim_H(E \cap \pi^{-1}\{y\}) \geq k$$

for at least one such y . Thus every positive- μ Borel set has Hausdorff dimension at least k , which is precisely $\dim \mu \geq k$. $\square$

II.12.3.1 The higher-dimensional mechanism

We now prove Theorem II.12.3.1. The entropy step is deduced from the full higher-dimensional inverse theorem in Chapter II.5 and the persistent-saturation machinery of Chapter II.7; the algebraic and backward-induction steps are local below.

For this subsection fix $m \geq 2$, algebraic $\lambda = (\lambda_0, \dots, \lambda_m) \in ((-1, 1) \setminus \{0\})^{m+1}$, positive weights $p = (p_j)_{j=0}^m$, and put

$$\Lambda = \{0, \dots, m\}, \quad \beta = \min\{1, H(p)/\chi\}.$$

For $t = (t_1, \dots, t_m) \in (\mathbb{R}^d)^m$ set $t_0 = 0$ and write $\Phi_t = \{\phi_{t,j}(x) = \lambda_j x + t_j\}_{j \in \Lambda}$ and μ_t for the corresponding self-similar measure.

Definition II.12.3.3 (Affine word law and translation/scaling partitions). Let G_d be the group of similarities $g(x) = \lambda_g x + a_g$ with $\lambda_g \neq 0$. For $n \geq 1$ define

$$v_t^{(n)} = \sum_{w \in \Lambda^n} p_w \delta_{\phi_{t,w}}.$$

Let $\mathcal{E}_r$ be the pullback to G_d of the level- $[r]$ dyadic partition of the translation coordinate $a_g \in \mathbb{R}^d$, and let $\mathcal{F}$ be the partition according to the exact scaling coordinate λ_g . Thus $\mathcal{E}_r \vee \mathcal{F}$ records the translation to precision 2^{-r} together with the exact contraction product.

The convolution identity

$$v_t^{(n+k)} = v_t^{(n)} * v_t^{(k)} \quad (\text{II.12.3.1})$$

is immediate from concatenation of independent words.

Lemma II.12.3.4 (Natural-scale entropy formula). *For every $d \geq 1$ and $t \in (\mathbb{R}^d)^m$,*

$$\dim \mu_t = \lim_{n \rightarrow \infty} \frac{1}{\chi n} H(v_t^{(n)}, \mathcal{E}_{\chi n}).$$

Consequently

$$\dim \mu_t \leq \frac{1}{\chi n} H(v_t^{(n)}) \quad (n \geq 1). \quad (\text{II.12.3.2})$$

Proof. Let $\xi = p^{\mathbb{N}}$ be the Bernoulli measure on $\Lambda^{\mathbb{N}}$, let $\Pi\omega$ be the coding point and let $\Pi_n\omega = \phi_{t, \omega_0 \dots \omega_{n-1}}(0)$. Then $\Pi\xi = \mu_t$ and

$$H(v_t^{(n)}, \mathcal{E}_{\chi n}) = H(\Pi_n\xi, \mathcal{D}_{\chi n}^{(d)}).$$

Fix $\varepsilon > 0$. By the law of large numbers there is a set $E_n \subset \Lambda^{\mathbb{N}}$ of measure $1 - o_\varepsilon(1)$ on which

$$2^{-n(\chi+\varepsilon)} \leq |\lambda_{\omega_0 \dots \omega_{n-1}}| \leq 2^{-n(\chi-\varepsilon)}.$$

The tails of the coding series are therefore $O_t(2^{-n(\chi-\varepsilon)})$ on E_n . Restricting to E_n , replacing Π by Π_n , and then removing the restriction changes the level- χn entropy by $O_t(\varepsilon n) + o(n)$: the first change uses bounded overlap of dyadic cells after a displacement $O(2^{-n(\chi-\varepsilon)})$, and the second is the standard entropy bound for discarding a set of probability $o_\varepsilon(1)$. Hence

$$H(\mu_t, \mathcal{D}_{\chi n}^{(d)}) = H(v_t^{(n)}, \mathcal{E}_{\chi n}) + O_t(\varepsilon n) + o(n).$$

Self-similar measures are exact dimensional, so their normalized dyadic entropy tends to their dimension. This is the standard exact-dimensionality input also used in [Rap22, Lemma 6]. Let first $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$.

For the last assertion, entropy is subadditive under convolution of finitely supported random variables, so (II.12.3.1) gives $H(v_t^{(n+k)}) \leq H(v_t^{(n)}) + H(v_t^{(k)})$. Fekete's lemma and $H(v_t^{(N)}, \mathcal{E}_{\chi N}) \leq H(v_t^{(N)})$ imply

$$\dim \mu_t \leq \inf_{N \geq 1} \frac{H(v_t^{(N)})}{\chi N},$$

which yields (II.12.3.2). □

Lemma II.12.3.5 (Affine irreducibility from spanning translations). *If $\text{span}\{t_1, \dots, t_m\} = \mathbb{R}^d$, then Φ_t is affinely irreducible.*

Proof. Choose d linearly independent translations. The fixed point of $x \mapsto \lambda_j x + t_j$ is a nonzero scalar multiple of t_j , while 0 is a fixed point of the map $x \mapsto \lambda_0 x$. Thus the affine span of the attractor contains 0 and d linearly independent points, so it is all of $\mathbb{R}^d$. A common invariant proper affine subspace would contain the attractor, a contradiction. □

Theorem II.12.3.6 (Restricted multidimensional entropy alternative). *Assume Φ_t is affinely irreducible and $\dim \mu_t < \beta$. Then for every $q \geq \chi$,*

$$\lim_{n \rightarrow \infty} \frac{1}{\chi n} H(v_t^{(n)}, \mathcal{E}_{qn} \vee \mathcal{F}) = \dim \mu_t. \quad (\text{II.12.3.3})$$

This is the specialization used in [Rap22, Theorem 9].

Proof. Write $\alpha = \dim \mu_t$. We first prove fine-scale flattening. If it failed, the [chart equivalence for similarity coordinates](#) and the [localization theorem](#) would produce bounded isometry laws with positive normalized entropy and asymptotically zero action gain. The [persistent-subspace theorem](#), based on the [Euclidean inverse theorem](#), would then give a nonzero maximal persistently saturated subspace $V_{\max}(\mu_t)$. The [invariance proposition](#) makes this subspace invariant under the linear parts of the IFS.

We recall the local entropy calculation which rules this out and, in particular, does not use the Falconer–Jin conditional-fiber theorem. Put $V = V_{\max}(\mu_t)$, $r = \dim V \geq 1$, and $\bar{\mu} = (\pi_{V^\perp})_* \mu_t$. Let $\alpha_\pi = \dim \bar{\mu} \geq 0$. For a fixed component window m , write $B_{m,N}$ for the average ambient component entropy and $A_{m,N}$ for the corresponding average quotient component entropy, in the notation of the [quotient entropy lemma](#). Exact dimensionality and the dyadic chain rule give

$$B_{m,N} \longrightarrow \alpha \quad (N \rightarrow \infty),$$

while that lemma gives

$$\lim_{m \rightarrow \infty} \lim_{N \rightarrow \infty} A_{m,N} = \alpha_\pi.$$

Persistent V -saturation says that, for every $\varepsilon > 0$, all large m and then all sufficiently late component levels have expected saturation defect at most $(d+1)\varepsilon + C_d/m$. The exact averaged defect identity (II.7.N16c) is

$$\frac{1}{N} \sum_{n=0}^{N-1} e_{V,m,n}(\mu_t) = r - \frac{1}{N} \sum_{n=0}^{N-1} \int h_m(\mu_t^{x,n}) d\mu_t(x) + \frac{1}{N} \sum_{n=0}^{N-1} A_{V,m,n}(\mu_t),$$

where $A_{V,m,n}$ is the normalized quotient-component entropy from the same lemma. Taking first $N \rightarrow \infty$, then $m \rightarrow \infty$, and finally $\varepsilon \downarrow 0$, gives

$$\alpha - \alpha_\pi \geq r. \quad (\text{II.12.3.4})$$

Thus $\alpha \geq r \geq 1$, contradicting $\alpha < \beta \leq 1$. Fine-scale flattening must therefore hold.

For $q > \chi$, apply that flattening with ratio $q/\chi > 1$. In the present homothetic system the orthogonal part is only a sign and the contraction product has at most $(n+1)^{m+1}$ possible values. Hence replacing the logarithmic similarity grid by the translation grid $\mathcal{E}_{qn}$ and adjoining the exact scaling partition $\mathcal{F}$ changes entropy by only $O_m(\log(n+1)) = o(n)$. We obtain

$$H(v_t^{(n)}, \mathcal{E}_{qn} \vee \mathcal{F}) - H(v_t^{(n)}, \mathcal{E}_{\chi n}) = o(n). \quad (\text{II.12.3.5})$$

At $q = \chi$ the same conclusion follows directly after the $o(n)$ scaling label cost. Lemma II.12.3.4 now gives the asserted limit.

Finally, the contraction product of a word of length n is determined by the multiplicity vector of its $m + 1$ symbols, so there are at most $(n + 1)^{m+1}$ possible values. Therefore

$$0 \leq H(v_t^{(n)}, \mathcal{E}_{qn} \vee \mathcal{F}) - H(v_t^{(n)}, \mathcal{E}_{qn}) \leq (m + 1) \log_2(n + 1) = o(n),$$

and (II.12.3.3) follows. $\square$

Remark II.12.3.7 (Where the entropy argument enters). The proof uses the covariance, Gaussian, subspace-stabilization, and Euclidean inverse theorems developed in Chapter II.5, followed by the localization and persistent-subspace results of Chapter II.7. Inequality (II.12.3.4) comes directly from the averaged entropy defect identity, before any exact-fiber statement is needed. Thus this argument does not require the Falconer–Jin conditional-fiber theorem.

Lemma II.12.3.8 (Lower semicontinuity in the translation parameter). *For fixed contractions λ and probabilities p , the map $t \mapsto \dim \mu_t$ on $(\mathbb{R}^d)^m$ is lower semicontinuous. This is the fixed-linear-part specialization recorded as [Rap22, Theorem 10]; the short proof needed here follows from the earlier projection-entropy formula.*

Proof. Let $\Pi_t : \Lambda^{\mathbb{N}} \rightarrow \mathbb{R}^d$ be the coding map and put $\mathcal{A}_t = \Pi_t^{-1} \mathcal{B}(\mathbb{R}^d)$. The similarities have fixed linear parts and the Bernoulli shift is ergodic. Thus the conformal dimension formula from Volume I, together with the definition of projection entropy, gives

$$\dim \mu_t = \frac{H(p) - \Delta(t)}{\chi}, \quad \Delta(t) = H_\beta(\mathcal{E}_1 | \mathcal{A}_t). \quad (\text{II.12.3.6})$$

Indeed $\sigma^{-1} \mathcal{A}_t$ is tail-measurable, hence independent of the first coordinate, so the first term in the definition of projection entropy is $H(p)$. It remains to prove that Δ is upper semicontinuous.

Fix t_0 . Choose $a \in [0, 1)^d$ so that μ_{t_0} gives zero mass to every boundary hyperplane of every shifted dyadic partition

$$\mathcal{Q}_{r,a} = \{a + 2^{-r}([0, 1)^d + z) : z \in \mathbb{Z}^d\}, \quad r \geq 0.$$

Such an a exists: in each coordinate the pushforward of μ_{t_0} has at most countably many atoms, so one avoids a countable union of forbidden translations. These partitions are nested and generate $\mathcal{B}(\mathbb{R}^d)$ modulo μ_{t_0} . Martingale convergence for finite conditional entropy therefore gives

$$H_\beta(\mathcal{E}_1 | \Pi_{t_0}^{-1} \mathcal{Q}_{r,a}) \downarrow \Delta(t_0). \quad (\text{II.12.3.7})$$

For fixed r , only finitely many cells of $\mathcal{Q}_{r,a}$ meet the attractors for t in a neighborhood of t_0 . Moreover $\Pi_t \rightarrow \Pi_{t_0}$ uniformly as $t \rightarrow t_0$, directly from the uniformly convergent coding series. Since every relevant cell boundary has μ_{t_0} -mass zero, for every $i \in \Lambda$ and every relevant cell Q ,

$$\beta([i] \cap \Pi_t^{-1} Q) \rightarrow \beta([i] \cap \Pi_{t_0}^{-1} Q).$$

The conditional entropy of the two finite partitions is a continuous function of these finitely many joint probabilities. Consequently

$$H_\beta(\mathcal{C}_1 \mid \Pi_t^{-1}\mathcal{Q}_{r,a}) \longrightarrow H_\beta(\mathcal{C}_1 \mid \Pi_{t_0}^{-1}\mathcal{Q}_{r,a}). \quad (\text{II.12.3.8})$$

Let $t_k \rightarrow t_0$ and fix $\varepsilon > 0$. Choose r in (II.12.3.7) so that the left side is below $\Delta(t_0) + \varepsilon$. Since conditioning on the full coding sigma-algebra can only decrease entropy, (II.12.3.8) gives

$$\limsup_{k \rightarrow \infty} \Delta(t_k) \leq H_\beta(\mathcal{C}_1 \mid \Pi_{t_0}^{-1}\mathcal{Q}_{r,a}) < \Delta(t_0) + \varepsilon.$$

Let $\varepsilon \downarrow 0$. Thus Δ is upper semicontinuous, and (II.12.3.6) yields $\dim \mu_{t_0} \leq \liminf_k \dim \mu_{t_k}$. $\square$

For the algebraic part, write $X = (X_0, \dots, X_m)$ and, for $l, n \geq 1$, let $\mathcal{P}(l, n)$ be the integer polynomials in X of total degree $< n$ and coefficient absolute value at most l .

Lemma II.12.3.9 (Exponential algebraic lower bound). *There is $M = M(m, \lambda) > 1$ such that for every $l, n \geq 1$ and $P \in \mathcal{P}(l, n)$,*

$$P(\lambda) = 0 \quad \text{or} \quad |P(\lambda)| \geq l^{-M} M^{-n}.$$

Moreover, a product of at most $k \leq m + 1$ members of $\mathcal{P}(l, n)$ belongs to $\mathcal{P}(l^k n^{C_m k}, kn)$ for a constant C_m depending only on m .

Proof. Put $s = m + 1$. Each polynomial in $\mathcal{P}(l, n)$ has at most n^s monomials, since each of its s exponents is $< n$. Its coefficient ℓ^1 norm is therefore at most ln^s . The coefficient ℓ^1 norm is submultiplicative, so a product of at most k such polynomials has coefficients bounded by $l^k n^{sk}$ and total degree $< k(n - 1) < kn$. Thus the product assertion holds with $C_m = s$, including $n = 1$.

For the lower bound set $K = \mathbb{Q}(\lambda_0, \dots, \lambda_m)$ and $D = [K : \mathbb{Q}]$. Choose an integer $A \geq 1$ such that $A\lambda_j$ is an algebraic integer for every j . Indeed, if $f(X) = aX^e + b_{e-1}X^{e-1} + \dots + b_0$ is the primitive integral minimal polynomial of λ_j , then $a\lambda_j$ is a root of the monic integral polynomial $a^{e-1}f(X/a)$; take a common multiple of the finitely many leading coefficients. Let $t = n - 1$, and put $Q = A^t P(\lambda)$. For each monomial λ^α in P , its contribution to Q is

$$c_\alpha A^{t-|\alpha|} \prod_{j=0}^m (A\lambda_j)^{\alpha_j},$$

which is an algebraic integer. Hence Q is an algebraic integer. If $P(\lambda) \neq 0$, then $Q \neq 0$, and its field norm

$$N_{K/\mathbb{Q}}(Q) = \prod_{\sigma: K \hookrightarrow \mathbb{C}} \sigma(Q)$$

is a nonzero rational algebraic integer, hence a nonzero integer. Consequently its absolute value is at least one.

Let

$$B = \max\{1, |\sigma(\lambda_j)| : \sigma : K \hookrightarrow \mathbb{C}, 0 \leq j \leq m\}.$$

For every embedding σ ,

$$|\sigma(Q)| \leq A^t \ln^s B^t.$$

Using the norm bound and reserving the identity embedding gives

$$|P(\lambda)| = A^{-t}|Q| \geq A^{-Dt}(\ln^s)^{-(D-1)}B^{-t(D-1)}.$$

Because $n \leq 2^n$ for $n \geq 1$, one can choose

$$M > \max\{D-1, A^D B^{D-1} 2^{s(D-1)}\}$$

so that the last quantity is at least $l^{-M}M^{-n}$ for every $l, n \geq 1$. This proves the stated exponential algebraic lower bound without importing a height or product-formula inequality. $\square$

For words $u, v \in \Lambda^n$ with $u \neq v$ and $\lambda_u = \lambda_v$ define

$$L_{u,v}(x_1, \dots, x_m) = \phi_{x,u}(0) - \phi_{x,v}(0),$$

where the same formula is read in one real coordinate. Let $\mathcal{L}_n$ be the family of all these nonzero-word-pair functionals.

Lemma II.12.3.10 (Polynomial coefficients of cylinder differences). *For every $L \in \mathcal{L}_n$ there are $P^1, \dots, P^m \in \mathcal{P}(1, n)$ such that*

$$L(x) = \sum_{j=1}^m P^j(\lambda)x_j.$$

Proof. For a word $w = q_0 \cdots q_{n-1}$,

$$\phi_{x,w}(0) = \sum_{r=0}^{n-1} \lambda_{q_0} \cdots \lambda_{q_{r-1}} x_{q_r}, \quad x_0 = 0.$$

Collecting the terms with $q_r = j$ expresses the coefficient of x_j as a sum of distinct monomials in $\lambda_0, \dots, \lambda_m$ with coefficients 0 or 1. Subtracting the corresponding expression for v gives a polynomial of degree $< n$ with coefficients in $\{-1, 0, 1\}$. $\square$

Proof of Theorem II.12.3.1. We prove the contrapositive by backward induction on d . Fix $1 \leq d < m$, assume the theorem already holds in every dimension $d < d' < m$, and suppose

$$\text{span}\{t_1, \dots, t_m\} = \mathbb{R}^d, \quad \dim \mu_t < \beta.$$

We show that Φ_t has exact overlaps.

After an invertible linear change of coordinates we may suppose that $e_1, \dots, e_d$ occur among $t_1, \dots, t_m$. Write $t_j = (t_j^1, \dots, t_j^d)$ and $t^\ell = (t_1^\ell, \dots, t_m^\ell) \in \mathbb{R}^m$. Choose

$$0 < \varepsilon < \frac{1}{2}(\beta - \dim \mu_t)$$

and then choose $0 < \delta < 1$ sufficiently small in terms of $m, \lambda, t, \varepsilon$. In particular $-\log_2 \delta > \chi$ and all comparisons below between δ^n and the exponential algebraic lower bound of Lemma II.12.3.9 hold for every large n . By Lemma II.12.3.5 and Theorem II.12.3.6, for all sufficiently large n ,

$$\frac{1}{\chi n} H(v_t^{(n)}, \mathcal{E}_{\lceil -\log_2 \delta \rceil n} \vee \mathcal{F}) < \dim \mu_t + \varepsilon < \beta - \varepsilon. \quad (\text{II.12.3.9})$$

Let $\mathcal{A}_n \subset \mathcal{L}_n$ consist of the functionals satisfying

$$|L(t^\ell)| \leq \delta^n \quad (1 \leq \ell \leq d). \quad (\text{II.12.3.10})$$

The entropy in (II.12.3.9) is strictly smaller than $nH(p)$, the entropy of the word distribution. Hence two distinct words $u, v \in \Lambda^n$ of the same contraction product lie in one atom of the partition. Their translation coordinates lie in the same dyadic cube, so each coordinate difference is at most δ^n ; hence $L_{u,v} \in \mathcal{A}_n$. Thus $\mathcal{A}_n \neq \emptyset$ for every large n .

Enumerate $\mathcal{A}_n = \{L_{n,i} : 1 \leq i \leq q_n\}$ and write, using Lemma II.12.3.10,

$$L_{n,i}(x) = \langle a_{n,i}, x \rangle, \quad a_{n,i} = (P_{n,i}^1(\lambda), \dots, P_{n,i}^m(\lambda)).$$

Put $r_n = \dim \text{span}\{a_{n,i} : 1 \leq i \leq q_n\}$. If some $a_{n,i} = 0$, then the corresponding $L_{n,i}$ vanishes identically, and its word pair gives an exact overlap immediately. Hence assume $r_n \geq 1$.

Step 1: the rank bound $r_n \leq m - d$. Take independent $a_1, \dots, a_r$ from the family and put $V = \text{span}\{a_1, \dots, a_r\}$, $r = r_n$. Let $G(a_1, \dots, a_r)$ be their Gram determinant. Every entry of the Gram matrix is a value at λ of a polynomial whose degree is $O(n)$ and whose coefficients grow only polynomially in n ; the product estimate and Lemma II.12.3.9 therefore give

$$G(a_1, \dots, a_r) \geq c_0^n \quad (\text{II.12.3.11})$$

for some $c_0 = c_0(m, \lambda) > 0$, whenever the determinant is nonzero. On the other hand (II.12.3.10) says $|\langle a_i, t^\ell \rangle| \leq \delta^n$. If B is the Gram matrix of the a_i , then the squared norm of the V -component of t^ℓ is $b^T B^{-1} b$, where $b_i = \langle a_i, t^\ell \rangle$. The entries of B are uniformly bounded, while (II.12.3.11) bounds $\det B$ from below; Cramer's rule for B^{-1} therefore gives

$$\text{dist}(t^\ell, V^\perp) = \|\pi_V t^\ell\| \leq C c_0^{-n/2} \delta^n = o(1).$$

Our choice $\delta \ll c_0$ makes the estimate uniform in ℓ . Since $t^1, \dots, t^d$ are linearly independent, their projections to $V^\perp$ remain linearly independent for large n . Thus $d \leq \dim V^\perp = m - r_n$ and hence

$$r_n \leq m - d. \quad (\text{II.12.3.12})$$

We split according to the eventual lower rank.

Case 1: $\liminf r_n = m - d$. By (II.12.3.12), after increasing the starting scale we have $r_n = m - d$ for every n . Relabel coordinates so that $t_j = e_j$ for $1 \leq j \leq d$, put $J = \{d+1, \dots, m\}$ and choose independent rows $a_{n,1}, \dots, a_{n,m-d}$. For every $(m-d)$ -set of columns J_0 , let $P_n^{J_0}(X)$ be the corresponding determinant of the polynomial coefficient matrix $(P_{n,i}^j(X))$. We first claim

$$P_n^J(\lambda) \neq 0. \quad (\text{II.12.3.13})$$

Indeed, if the J -minor had smaller rank, choose a nonzero maximal minor $P_n^F(\lambda)$ containing as many columns of J as possible. Replacing one of its remaining columns by the error vector $(L_{n,i}(t^\ell))_i$ and applying Cramer's rule, the coordinate identity $t_i^\ell = 1$ would force $|P_n^F(\lambda)| = O(\delta^n)$. But P_n^F has degree $O(n)$ and polynomially bounded integer coefficients, contradicting Lemma II.12.3.9 for our choice of δ .

For $k \in J$ and $1 \leq \ell \leq d$, let $J_{k,\ell} = (J \setminus \{k\}) \cup \{\ell\}$. Solving the equations $L_{n,i}(t^\ell) = O(\delta^n)$ in the unknowns $(t_j^\ell)_{j \in J}$ gives

$$\left| t_k^\ell + \frac{P_n^{J_{k,\ell}}(\lambda)}{P_n^J(\lambda)} \right| \leq C\delta^{n/2}. \quad (\text{II.12.3.14})$$

The denominator again has an exponential lower bound. Subtract the approximations at n and $n+1$. After clearing denominators we obtain a polynomial $Q_n \in \mathbb{Z}[X_0, \dots, X_m]$, of degree $O(n)$ and polynomially bounded coefficients, with $|Q_n(\lambda)| = O(\delta^{n/2})$. The algebraic lower bound forces $Q_n(\lambda) = 0$. Hence the ratios in (II.12.3.14) are independent of n , and letting $n \rightarrow \infty$ yields

$$t_k^\ell = -\frac{P_N^{J_{k,\ell}}(\lambda)}{P_N^J(\lambda)}.$$

Substituting these exact identities back into the linear equations and using Cramer's rule gives $L_{N,i}(t^\ell) = 0$ for all i and all ℓ . Since $L_{N,1} = L_{u,v}$ for distinct words of equal contraction product, their translations agree in every coordinate. Thus $\phi_{t,u} = \phi_{t,v}$, an exact overlap.

Case 2: $r := \liminf r_n < m - d$. Choose $n_k \rightarrow \infty$ with $r_{n_k} = r$ and independent rows $a_{n_k,1}, \dots, a_{n_k,r}$. Put $d' = m - r$, so $d < d' < m$. After passing to a subsequence, choose an r -set of columns $J = \{d' + 1, \dots, m\}$ whose determinant has maximal absolute value among all r -minors for every k . In particular it is nonzero. For $j \in J$ and $1 \leq \ell \leq d'$, define $J_{j,\ell} = (J \setminus \{j\}) \cup \{\ell\}$. The maximal-minor choice gives

$$\left| \frac{P_k^{J_{j,\ell}}(\lambda)}{P_k^J(\lambda)} \right| \leq 1.$$

Pass to a further subsequence so all these ratios converge, and define $s_0 = (s_{0,j})_{j=1}^m \in (\mathbb{R}^{d'})^m$ by taking $s_{0,1}, \dots, s_{0,d'}$ to be the standard basis and, for $j \in J$,

$$s_{0,j}^\ell = -\lim_{k \rightarrow \infty} \frac{P_k^{J_{j,\ell}}(\lambda)}{P_k^J(\lambda)}.$$

Let s_k use the same standard-basis columns but the finite ratios at level n_k . Cramer's rule gives

$$L(s_k^\ell) = 0 \quad \text{for every } L \in \mathcal{A}_{n_k}, \quad 1 \leq \ell \leq d'. \quad (\text{II.12.3.15})$$

Indeed it holds first for the chosen independent rows and hence for their entire span, which is the span of $\mathcal{A}_{n_k}$.

If two words $u, v \in \Lambda^{n_k}$ fall in the same atom of $\mathcal{E}_{[-\log_2 \delta]n_k} \vee \mathcal{F}$ for the system t , then $L_{u,v} \in \mathcal{A}_{n_k}$ and (II.12.3.15) shows $\phi_{s_k,u} = \phi_{s_k,v}$. Therefore grouping the word law by atoms gives

$$H(v_{s_k}^{(n_k)}) \leq H(v_t^{(n_k)}, \mathcal{E}_{[-\log_2 \delta]n_k} \vee \mathcal{F}).$$

Using (II.12.3.9) and Lemma II.12.3.4,

$$\dim \mu_{s_k} \leq \frac{H(v_{s_k}^{(n_k)})}{\chi n_k} < \beta - \varepsilon.$$

By construction $s_k \rightarrow s_0$, so Lemma II.12.3.8 gives $\dim \mu_{s_0} \leq \beta - \varepsilon$. Since the first d' translations of s_0 form the standard basis, they span $\mathbb{R}^{d'}$. The induction hypothesis therefore implies that Φ_{s_0} has exact overlaps.

It remains to pull such an overlap back to t . We claim

$$t_j = \sum_{\ell=1}^{d'} s_{0,j}^\ell t_\ell \quad (1 \leq j \leq m). \quad (\text{II.12.3.16})$$

This is immediate for $j \leq d'$. For $j \in J$, solve the equations $L_{n_k,i}(t^a) = O(\delta^{n_k})$ by Cramer's rule using the maximal J -minor. The denominator has the exponential lower bound from Lemma II.12.3.9, so the error term tends to zero; the remaining ratios converge to the coefficients defining s_0 . This proves (II.12.3.16) coordinate by coordinate.

Choose distinct words u, v witnessing an exact overlap for Φ_{s_0} and write $L = L_{u,v} = \sum_{j=1}^m c_j x_j$. Then $L(s_0^\ell) = 0$ for $1 \leq \ell \leq d'$. Using (II.12.3.16),

$$\sum_{j=1}^m c_j t_j = \sum_{\ell=1}^{d'} L(s_0^\ell) t_\ell = 0.$$

Thus $L(t^a) = 0$ in every coordinate $a = 1, \dots, d$; since $\lambda_u = \lambda_v$, we get $\phi_{t,u} = \phi_{t,v}$. Hence Φ_t has an exact overlap also in the second case. The backward induction is complete. $\square$

Remark II.12.3.11 (How the Rapaport argument closes). Theorem II.12.3.1 is proved through the entropy, algebraic/rank, Cramer-rule, determinant-stabilization, and backward-induction steps. The multidimensional inverse theorem from Chapter II.5 supplies the entropy rigidity, while the persistent-subspace contradiction above avoids the independent Falconer–Jin fiber theorem.

Theorem II.12.3.12 (Rapaport: algebraic contractions). *Assume every contraction λ_j is algebraic, while the translations t_j are arbitrary real numbers. If there are no exact overlaps, then the self-similar measure satisfies*

$$\dim \mu = \min \left\{ 1, \frac{H(p)}{\chi} \right\}.$$

Signed contractions are allowed. This is the main theorem of [Rap22].

Proof. For the upper bound, fix $\varepsilon > 0$. By the law of large numbers, for all large n there is a set of words carrying probability at least $1 - \varepsilon$ on which both $-n^{-1} \log p_w$ and $-n^{-1} \log |\lambda_w|$ are within ε of $H(p)$ and χ . Their cylinder images have diameter at most $2^{-n(\chi-\varepsilon)}$ and there are at most $2^{n(H(p)+\varepsilon)}$ such words. The remaining probability can be discarded in the upper Hausdorff-dimension calculation. Letting $\varepsilon \downarrow 0$ gives $\dim \mu \leq \min\{1, H(p)/\chi\}$.

If the IFS has one map, the assertion is trivial. If it has two maps, an invertible affine conjugacy puts it in the form $\lambda_0 x, \lambda_1 x + 1$; all parameters are then algebraic and the Chapter II.6 algebraic-parameter theorem gives the claimed formula. Suppose there are at least three maps. Conjugate by a translation so that $t_0 = 0$. Since there are no exact overlaps, not all remaining translations vanish; hence they span $\mathbb{R}$. Theorem II.12.3.1 with $d = 1$ gives

$$\dim \mu \geq \min \left\{ 1, \frac{H(p)}{\chi} \right\}.$$

Together with the upper bound this is equality. □

Corollary II.12.3.13. *If K is the attractor and s solves $\sum_j |\lambda_j|^s = 1$, then under the hypotheses of Theorem II.12.3.12,*

$$\dim_H K = \min\{1, s\}.$$

This is the set consequence recorded in [Rap22].

Proof. Take $p_j = |\lambda_j|^s$. Then $H(p) = s\chi$, so Theorem II.12.3.12 gives a measure supported on K of dimension $\min\{1, s\}$. The reverse inequality is the standard cylinder-cover estimate. □

II.12.4 Analytic self-conformal systems on the line

Let $I = [0, 1]$ and let $\Phi = \{\phi_i\}_{i \in \Lambda}$ be a finite real-analytic IFS with $0 < |\phi'_i| < 1$ on I . Fix a positive probability vector $p = (p_i)_{i \in \Lambda}$ and let μ be the unique stationary probability

$$\mu = \sum_{i \in \Lambda} p_i (\phi_i)_* \mu.$$

Throughout this section logarithms have base 2, and

$$H(p) = - \sum_{i \in \Lambda} p_i \log_2 p_i, \quad \chi = \chi(\Phi, p) = - \sum_{i \in \Lambda} p_i \int_I \log_2 |\phi'_i(x)| d\mu(x) > 0. \quad (\text{II.12.4.1})$$

Definition II.12.4.1 (Exponential separation for an analytic IFS). The system Φ is *exponentially separated* if there are $c > 0$ and infinitely many integers $n \geq 1$ such that

$$\|\phi_u - \phi_v\|_I \geq c^n \quad (u, v \in \Lambda^n, u \neq v),$$

where $\|f\|_I = \sup_{x \in I} |f(x)|$ and $\phi_{i_1 \dots i_n} = \phi_{i_1} \circ \dots \circ \phi_{i_n}$. Notice that this is a separation condition for maps, not merely for the images of one base point.

Lemma II.12.4.2 (Uniform complex distortion). *There is a common complex neighbor-*

hood of I and a constant $C > 1$ such that for every finite word u ,

$$|\phi'_u(x)| \asymp_C |\phi'_u(y)|, \quad |\phi_u^{(k)}(x)| \leq k! C^k |\phi'_u(0)|.$$

Proof. Choose $\eta > 0$ so that every generator extends holomorphically to the complex η -neighborhood $I^{(\eta)}$ and, after decreasing η ,

$$\eta \leq |\phi'_i(z)| \leq 1 - \eta, \quad |\phi''_i(z)| \leq \eta^{-1} \quad (z \in I^{(\eta)}, i \in \Lambda).$$

The mean-value estimate along the segment from z to a nearest point of I shows that $\phi_i(I^{(\eta)}) \subset I^{(\eta)}$. Applying the chain rule to $\log |\phi'_u|$ and summing the resulting geometric series along the contracting orbit gives a constant C_0 , independent of u , such that

$$|\phi'_u(z)| \leq C_0 |\phi'_u(w)| \quad (z, w \in I^{(\eta)}).$$

This proves the first assertion on I . Put $f_u(z) = \phi_u(z) - \phi_u(0)$. Integration along a path in the tube and the last estimate give

$$\sup_{I^{(\eta)}} |f_u| \leq C_1 |\phi'_u(0)|.$$

For $x \in I$, apply Cauchy's estimate on a disc of fixed radius $\eta/2$ contained in a slightly smaller common tube (using a one-sided disc at an endpoint after extending across it). Thus

$$|\phi_u^{(k)}(x)| = |f_u^{(k)}(x)| \leq k! (2/\eta)^k C_1 |\phi'_u(0)|.$$

Increasing C absorbs C_1 and proves the second assertion. $\square$

We next introduce the entropy notation needed to pass from analytic maps to polynomial actions. For $m \geq 0$, let $\mathcal{D}_m$ be the level- m dyadic partition of $\mathbb{R}$. If θ is a probability measure and $D \in \mathcal{D}_m$ has positive mass, write $\theta_D = \theta|_D / \theta(D)$. For $k \geq 1$, let $\mathcal{P}_k$ be the space of real polynomials of degree at most k , identified with $\mathbb{R}^{k+1}$ through its coefficient vector. If ν is a compactly supported probability on $\mathcal{P}_k$ and θ is a probability on I , then $\nu.\theta$ is the image of $\nu \times \theta$ under evaluation $(P, x) \mapsto P(x)$.

The following argument uses the exact-dimensionality theorem from Volume I. The relevant Feng–Hu projection-entropy formula is stated and derived as Theorem II.12.4.18 below. At this point we need only the entropy consequence

$$\lim_{m \rightarrow \infty} \frac{1}{m} H(\mu, \mathcal{D}_m) = \dim \mu. \quad (\text{II.12.4.2})$$

Importantly, this limit does not assume the dimension formula that we are trying to prove.

Lemma II.12.4.3 (Atomlessness and stopping cylinders). *If the maps in Φ have no common fixed point, then μ is nonatomic. Moreover, with $\phi_\emptyset = \text{id}$ and $p_\emptyset = 1$, put $\Psi_0 = \{\emptyset\}$ and, for $s \geq 1$, let*

$$\Psi_s = \{u : |\phi'_u(0)| \leq 2^{-s} < |\phi'_{u^-}(0)|\},$$

where u^- is the word obtained by deleting the last letter, then Ψ_s is a finite prefix cutset and

$$\mu = \sum_{u \in \Psi_s} p_u(\phi_u)_* \mu, \quad C^{-1}2^{-s} \leq |\phi'_u(x)| \leq C2^{-s} \quad (u \in \Psi_s, x \in I), \quad (\text{II.12.4.3})$$

for a constant C independent of s .

Proof. Suppose that $a = \sup_x \mu\{x\} > 0$. The set $A = \{x : \mu\{x\} = a\}$ is nonempty and finite. Since each ϕ_i is injective,

$$a = \mu\{y\} = \sum_i p_i \mu\{\phi_i^{-1}(y)\} \leq a \quad (y \in A).$$

Equality forces $\phi_i^{-1}(y) \in A$ for every i and $y \in A$. Thus every ϕ_i permutes A . A strict contraction cannot permute a finite set with more than one point, so A is a singleton fixed by every generator, a contradiction.

Every infinite word crosses the derivative threshold because the generators are uniformly contracting. The first-crossing words are prefix-incomparable and finite, hence form a cutset. Iterating stationarity to that cutset gives the first identity in (II.12.4.3). The upper derivative bound follows from distortion. For the lower bound, use the strict inequality for u^- and the positive minimum of $|\phi'_i|$ over $i \in \Lambda$ and I , then absorb this minimum and the distortion constant into C . $\square$

Lemma II.12.4.4 (Dyadic entropy bookkeeping). *Let σ be compactly supported on $\mathbb{R}$.*

(i) *If f is L -bi-Lipschitz on $\text{supp } \sigma$, then*

$$|H(f_*\sigma, \mathcal{D}_m) - H(\sigma, \mathcal{D}_m)| \leq C(1 + \log_2 L).$$

(ii) *Put $H_j = H(\sigma, \mathcal{D}_j)$. If $n \geq m$, then*

$$\frac{1}{n} \sum_{i=1}^n \int \frac{1}{m} H(\sigma_{\mathcal{D}_i(x)}, \mathcal{D}_{i+m}) d\sigma(x) = \frac{H_n}{n} + O\left(\frac{m}{n}\right).$$

The implicit constant is absolute when $\text{diam}(\text{supp } \sigma)$ is bounded by a fixed constant.

(iii) *If $A(x) = ax + b$ with $a \neq 0$ and $r = \lfloor \log_2 |a| \rfloor$, then, whenever $m + r \geq 0$,*

$$|H(A_*\sigma, \mathcal{D}_m) - H(\sigma, \mathcal{D}_{m+r})| \leq C.$$

Here C is an absolute constant, independent of a, b, m , and σ .

Proof. For (i), the image of a level- m dyadic interval meets at most $O(1+L)$ level- m intervals, and the same holds for f^{-1} . The entropy chain rule then bounds each of the two conditional entropies by $\log_2 O(1+L)$.

For (ii), the chain rule gives

$$\int H(\sigma_{\mathcal{D}_i(x)}, \mathcal{D}_{i+m}) d\sigma(x) = H_{i+m} - H_i.$$

After summing in i , all middle terms cancel:

$$\sum_{i=1}^n (H_{i+m} - H_i) = \sum_{j=n+1}^{n+m} H_j - \sum_{j=1}^m H_j.$$

Since refinement by one dyadic level increases entropy by at most one bit, $|H_{n+r} - H_n| \leq r$, while compact support gives $H_j = O(j+1)$. Division by mn proves the formula.

For (iii), the pullback $A^{-1}\mathcal{D}_m$ is a partition into intervals of length $|a|^{-1}2^{-m}$. Since $2^{-m-r-1} < |a|^{-1}2^{-m} \leq 2^{-m-r}$, every atom of $A^{-1}\mathcal{D}_m$ meets only a bounded number of atoms of $\mathcal{D}_{m+r}$, and conversely. The entropy chain rule in the two directions gives the asserted $O(1)$ comparison. The translation b only changes the offset of the interval grid and does not affect this bounded-overlap argument. $\square$

Lemma II.12.4.5 (Uniform decay across scales). *For every $0 < \alpha < 1$ there is $0 < \delta < 1$ such that, for every C^1 map $\psi : I \rightarrow \mathbb{R}$ satisfying $\alpha \leq |\psi'| \leq \alpha^{-1}$,*

$$(\psi_*\mu)(B(x, \delta r)) \leq \alpha(\psi_*\mu)(B(x, r)) \quad (x \in \mathbb{R}, 0 < r < 1).$$

Consequently, after decreasing δ if necessary,

$$\psi_*\mu\left(\bigcup_{D \in \mathcal{D}_s} (\partial D)^{(\delta 2^{-s})}\right) < \alpha \quad (s \geq 0), \quad (\text{II.12.4.4})$$

uniformly over the same class of maps.

Proof. By Lemma II.12.4.3, μ is nonatomic. Hence there is $\eta > 0$ such that every interval of length at most η has μ -mass less than α . Choose an integer q so large that $2C\alpha^{-1}2^{-q} < 1/2$, and then choose $\delta > 0$ so small that $4C\alpha^{-1}2^q\delta < \eta$.

Fix r and take s with $2^{-s-1} < r \leq 2^{-s}$. For $u \in \Psi_{s+q}$, (II.12.4.3) shows that $\psi \circ \phi_u$ has derivative between constant multiples of 2^{-s-q} . Therefore the inverse image of $B(x, \delta r)$ is an interval of length at most $4C\alpha^{-1}2^q\delta < \eta$, and so

$$(\psi \circ \phi_u)_*\mu(B(x, \delta r)) < \alpha.$$

If this quantity is positive, the diameter bound and the choice of q imply $\psi \circ \phi_u(I) \subset B(x, r)$; hence the same component assigns mass one to $B(x, r)$. Summing over the stopping decomposition proves the first claim.

For the consequence, apply the first claim with a smaller constant, with $r = 2^{-s-2}$, to balls centered at the grid $2^{-s}\mathbb{Z}$. The comparison balls have uniformly bounded overlap, while the boundary neighborhood is covered by the corresponding smaller balls. Summing the inequalities yields (II.12.4.4). $\square$

Proposition II.12.4.6 (Uniform lower component entropy). *For every $0 < \alpha < 1$ there is $M_0(\alpha)$ such that, for every $m \geq M_0(\alpha)$, every $s \geq 0$, and every C^1 map ψ with $\alpha \leq |\psi'| \leq \alpha^{-1}$, setting $\theta = \psi_*\mu$ gives*

$$\theta \left\{ x : \frac{1}{m} H(\theta_{\mathcal{D}_s(x)}, \mathcal{D}_{s+m}) > \dim \mu - \alpha \right\} > 1 - \alpha.$$

Proof. Fix an internal parameter $0 < \tau < \alpha/3$. Choose the boundary width δ from Lemma II.12.4.5 so that the set in (II.12.4.4) has θ -mass less than τ^2 . Next choose q so large that every $\psi \circ \phi_u(I)$, $u \in \Psi_{s+q}$, has diameter less than $\delta 2^{-s}$.

Call $u \in \Psi_{s+q}$ good if $\psi \circ \phi_u(I)$ lies in one member of $\mathcal{D}_s$. Every bad image lies in the boundary neighborhood, so the total weight of bad words is less than τ^2 . Markov's inequality now gives a collection $\mathcal{E} \subset \mathcal{D}_s$ with $\theta(\bigcup \mathcal{E}) > 1 - \tau$ such that, conditionally on each $D \in \mathcal{E}$, the good components lying wholly in D have total weight greater than $1 - \tau$.

For such a component, rescaling D to unit size replaces $(\psi \circ \phi_u)_*\mu$ by the image of μ under a bi-Lipschitz map whose constant depends only on α, q and the distortion constant, not on s, u , or D . Lemma II.12.4.4(i) and (II.12.4.2) therefore imply, uniformly in all these choices, that for all sufficiently large m its entropy at level m is at least $m(\dim \mu - \tau)$. Concavity of entropy and the conditional weight bound give

$$\frac{1}{m} H(\theta_D, \mathcal{D}_{s+m}) \geq (1 - \tau)(\dim \mu - \tau) > \dim \mu - 3\tau.$$

Choosing $3\tau < \alpha$ proves the proposition. $\square$

Proposition II.12.4.7 (Uniform entropy dimension of smooth images). *For every $0 < \varepsilon < 1$ there is $M(\varepsilon)$ such that, whenever $m \geq M(\varepsilon)$, there is $N(\varepsilon, m)$ with the following property. If $n \geq N(\varepsilon, m)$ and $\psi \in C^1(I)$ satisfies*

$$\varepsilon \leq |\psi'(x)| \leq \varepsilon^{-1} \quad (x \in I),$$

then, for $\theta = \psi_\mu$, the set of pairs (i, x) , sampled by uniform $i \in \{1, \dots, n\}$ and $x \sim \theta$, for which*

$$\left| m^{-1} H(\theta_{\mathcal{D}_i(x)}, \mathcal{D}_{i+m}) - \dim \mu \right| < \varepsilon$$

has probability greater than $1 - \varepsilon$.

Proof. Fix ε and choose $0 < \alpha < \min\{\varepsilon/2, \varepsilon^2/16\}$. Proposition II.12.4.6 shows, uniformly for every $1 \leq i \leq n$, that

$$\mathbb{P}_{x \sim \theta} \left\{ \frac{1}{m} H(\theta_{\mathcal{D}_i(x)}, \mathcal{D}_{i+m}) > \dim \mu - \alpha \right\} > 1 - \alpha$$

once $m \geq M_0(\alpha)$. Averaging in i gives the same lower-tail estimate for the joint sampling of (i, x) .

On the other hand, Lemma II.12.4.4(ii), followed by part (i) and (II.12.4.2), gives, uniformly over ψ ,

$$\mathbb{E}_{1 \leq i \leq n, x \sim \theta} \frac{1}{m} H(\theta_{\mathcal{D}_i(x)}, \mathcal{D}_{i+m}) \leq \dim \mu + 2\alpha$$

for all sufficiently large $n = N(\alpha, m)$. Each normalized component entropy lies in $[0, 1]$. If q is the probability that it is at least $\dim \mu + \varepsilon$, the preceding lower-tail estimate yields

$$\dim \mu + 2\alpha \geq q(\dim \mu + \varepsilon) + (1 - q - \alpha)(\dim \mu - \alpha).$$

Thus $q(\varepsilon + \alpha) \leq 4\alpha$. Taking $4\alpha/(\varepsilon + \alpha) < \varepsilon/2$ and also $\alpha < \varepsilon/2$ shows that the probability of either tail is less than ε , as required. $\square$

Theorem II.12.4.8 (One-dimensional Shannon entropy inverse theorem). *For every $k_1, k_2 \in \mathbb{N}$ and every $\varepsilon > 0$ there is $\delta > 0$ such that, for all sufficiently large n , if probabilities σ, τ on $[0, 1]$ satisfy*

$$H(\sigma * \tau, \mathcal{D}_n) \leq H(\sigma, \mathcal{D}_n) + \delta n,$$

then there are sets $I, J \subset \{1, \dots, n\}$ with

$$|I \cup J| \geq (1 - \varepsilon)n$$

such that, for every $i \in I$,

$$\sigma(x : H(\sigma^{\mathcal{D}_i(x)}, \mathcal{D}_{k_1}) \geq (1 - \varepsilon)k_1) > 1 - \varepsilon,$$

and, for every $j \in J$,

$$\tau(x : \tau^{\mathcal{D}_j(x)}(Q) \geq 1 - \varepsilon \text{ for some } Q \in \mathcal{D}_{k_2}) > 1 - \varepsilon.$$

Proof. This is the [one-dimensional Shannon entropy inverse theorem](#) proved in Chapter II.4, stated here with the exact quantifiers needed below. The independent L^q inverse theorem of Chapter II.11 is not needed. $\square$

Lemma II.12.4.9 (Nested entropy porosity of smooth-image components). *Assume $d := \dim \mu < 1$. Given $0 < \alpha, \eta < 1$, there is $L = L(\alpha, \eta)$ such that for every $\ell \geq L$ there is $M = M(\alpha, \eta, \ell)$ with the following property. For all sufficiently large n , if $m \geq M$, $m \leq n/4$, and $\psi \in C^1(I, I)$ satisfies*

$$\eta \leq |\psi'(x)| \leq \eta^{-1} \quad (x \in I),$$

then, writing $\theta = \psi_ \mu$, the set of pairs (i, x) with $1 \leq i \leq n - m$ and $x \sim \theta$ for which*

$$\mathbb{P}_{\substack{i \leq j \leq i+m-\ell \\ y \sim \theta_{\mathcal{D}_i(x)}}} \left\{ \frac{1}{\ell} H((\theta_{\mathcal{D}_i(x)})_{\mathcal{D}_j(y)}, \mathcal{D}_{j+\ell}) < d + \alpha \right\} > 1 - \eta \quad (\text{II.12.4.5})$$

has probability greater than $1 - \eta$.

Proof. Put $\varepsilon_* = \min\{\alpha/2, \eta/2, \zeta\}$, where $0 < \zeta < \eta^2/10$ will be fixed at the end. Apply Proposition II.12.4.7 with tolerance ε_* . Increase L so that, for every $\ell \geq L$ and all sufficiently

long scale ranges,

$$\mathbb{P}_{1 \leq r \leq n+m} \left\{ \frac{1}{\ell} H(\theta_{\mathcal{D}_r(y)}, \mathcal{D}_{r+\ell}) \geq d + \alpha \right\} < \zeta. \quad (\text{II.12.4.6})$$

Because $\varepsilon_* \leq \eta$, the derivative hypothesis in the present lemma implies the derivative hypothesis in Proposition II.12.4.7.

For fixed $i \leq j$ and any nonnegative measurable function F of the level- j component, dyadic nesting gives

$$\iint F((\theta_{\mathcal{D}_i(x)})_{\mathcal{D}_j(y)}) d\theta_{\mathcal{D}_i(x)}(y) d\theta(x) = \int F(\theta_{\mathcal{D}_j(y)}) d\theta(y).$$

Indeed, averaging the normalized restrictions over all level- i cells reconstructs θ , and every level- j cell is contained in a unique level- i cell. Average the indicator of the bad event in (II.12.4.5) first in (x, y) and then over the sliding pairs (i, j) . Each global scale r occurs at most m times, while the $O(m + \ell)$ endpoint scales contribute at most $O(m/n + \ell/m)$. By (II.12.4.6), the resulting mean bad probability is at most

$$2\zeta + O(m/n + \ell/m).$$

Choose first $\zeta < \eta^2/10$, then m large compared with ℓ/η^2 , and finally n large compared with m/η^2 . Markov's inequality shows that the inner bad probability is at most η for more than a $1 - \eta$ fraction of (i, x) , proving the lemma. $\square$

Proposition II.12.4.10 (Localized entropy growth under convolution). *For every $0 < a < 1$ there is $\eta_0 = \eta_0(a) > 0$ with the following property. Given an integer $\ell \geq 1$ and $0 < \eta \leq \eta_0$, there are $\rho = \rho(a, \ell, \eta) > 0$ and $M_0 = M_0(a, \ell, \eta)$ such that, whenever $M \geq M_0$, $i \geq 0$, and probabilities ξ, ϑ on $\mathbb{R}$ satisfy*

$$\text{diam}(\text{supp } \xi), \text{diam}(\text{supp } \vartheta) \leq a^{-1} 2^{-i}, \quad (\text{II.12.4.7})$$

$$\frac{1}{M} H(\xi, \mathcal{D}_{i+M}) \geq a, \quad (\text{II.12.4.8})$$

and

$$\mathbb{P}_{i \leq j \leq i+M-\ell} \left\{ \frac{1}{\ell} H(\vartheta_{\mathcal{D}_j(y)}, \mathcal{D}_{j+\ell}) < 1 - a \right\} > 1 - \eta, \quad (\text{II.12.4.9})$$

then

$$\frac{1}{M} H(\xi * \vartheta, \mathcal{D}_{i+M}) \geq \frac{1}{M} H(\vartheta, \mathcal{D}_{i+M}) + \rho.$$

Proof. Choose $0 < \tau < a/100$. Then choose an integer K so large that

$$\frac{h_2(\tau)}{K} + 2\tau < \frac{a}{20},$$

where h_2 is binary entropy. Apply Theorem II.12.4.8 with $k_1 = \ell$, $k_2 = K$ and error τ , and let $\delta_{\text{inv}} > 0$ be the resulting entropy-gap constant. Decrease $\eta_0(a)$ so that

$$\tau + \eta_0/\tau < \frac{a}{20}.$$

We show that the conclusion holds with any fixed $0 < \rho < \delta_{\text{inv}}/2$ once M is large enough.

Suppose otherwise. From (II.12.4.7), choose $q = q(a)$ so that each support is contained in some level- $(i - q)$ dyadic cell. Translate the two measures by the respective left endpoints of those cells and dilate both by 2^{i-q} . The two transformed measures $\tilde{\xi}, \tilde{\vartheta}$ are supported on $[0, 1]$. Since the translations are by level- $(i - q)$ dyadic grid points, the entropy identities are exact:

$$\begin{aligned} H(\tilde{\xi}, \mathcal{D}_{M+q}) &= H(\xi, \mathcal{D}_{i+M}), \\ H(\tilde{\vartheta}, \mathcal{D}_{M+q}) &= H(\vartheta, \mathcal{D}_{i+M}), \\ H(\tilde{\xi} * \tilde{\vartheta}, \mathcal{D}_{M+q}) &= H(\xi * \vartheta, \mathcal{D}_{i+M}). \end{aligned}$$

Put $N = M + q$. Failure of the asserted gain and the choice of ρ imply

$$H(\tilde{\xi} * \tilde{\vartheta}, \mathcal{D}_N) \leq H(\tilde{\vartheta}, \mathcal{D}_N) + \delta_{\text{inv}} N$$

for large M . Apply the inverse theorem with $\sigma = \tilde{\vartheta}$ and $\tau = \tilde{\xi}$, obtaining sets $I, J \subset \{1, \dots, N\}$ with $|I \cup J| \geq (1 - \tau)N$.

Condition (II.12.4.9) is preserved by the dyadic affine normalization, apart from the first and last $O(q + \ell)$ scales. Call a remaining scale r porous if the probability of the low-entropy event is larger than $1 - \tau$. Markov's inequality and (II.12.4.9) show that all but at most $(\eta/\tau)M + O(q + \ell)$ scales are porous. A porous scale cannot belong to I : the inverse-theorem event has probability $> 1 - \tau$ and entropy at least $(1 - \tau)\ell$, whereas the porous event has probability $> 1 - \tau$ and entropy less than $(1 - a)\ell$, and $\tau < a$. Hence

$$|I| \leq \frac{\eta}{\tau} M + O(q + \ell), \quad |J| \geq \left(1 - \tau - \frac{\eta}{\tau} - o(1)\right) N.$$

For every $r \in J$, with probability $> 1 - \tau$ a normalized $\tilde{\xi}$ -component has a level- K cell of mass at least $1 - \tau$. On that event its level- K entropy is at most $h_2(\tau) + \tau K$; on the complementary event it is at most K . Therefore

$$\int H(\tilde{\xi}^{\mathcal{D}_r(x)}, \mathcal{D}_K) d\tilde{\xi}(x) \leq h_2(\tau) + 2\tau K \quad (r \in J).$$

Use the sliding local-entropy identity (the proof of Lemma II.12.4.4(ii), after rescaling components) with window K . The preceding estimate on J , the trivial bound K on its complement, and the lower bound for $|J|$ give

$$\frac{1}{N} H(\tilde{\xi}, \mathcal{D}_N) \leq \frac{h_2(\tau)}{K} + 3\tau + \frac{\eta}{\tau} + o(1) < \frac{a}{4}$$

for all sufficiently large M , after decreasing η_0 once more if needed. On the other hand, (II.12.4.8) gives

$$\frac{1}{N} H(\tilde{\xi}, \mathcal{D}_N) \geq a \frac{M}{M + q} > \frac{a}{2}$$

for large M , a contradiction. This proves the proposition. $\square$

Lemma II.12.4.11 (Convolution does not lose dyadic entropy). *For probabilities ξ, ϑ on $\mathbb{R}$ and every $s \geq 0$,*

$$H(\xi * \vartheta, \mathcal{D}_s) \geq H(\vartheta, \mathcal{D}_s) - 1.$$

Proof. For every $t \in \mathbb{R}$, the partitions $\mathcal{D}_s$ and $\mathcal{D}_s - t$ refine each other with multiplicity at most two, so $|H((T_t)_* \vartheta, \mathcal{D}_s) - H(\vartheta, \mathcal{D}_s)| \leq 1$. Since $\xi * \vartheta$ is the mixture of the translates $(T_t)_* \vartheta$ with $t \sim \xi$, concavity of entropy yields the claim. $\square$

Lemma II.12.4.12 (Local entropy average for polynomial evaluation). *Fix $k \geq 1$ and $R > 1$. Let ν be a probability on $\mathcal{A}_k$ supported on $\{P : \|P\|_2 \leq R\}$, and let θ be a probability on I . If $n \geq m \geq 1$, then*

$$\begin{aligned} \frac{1}{n} H(\nu, \theta, \mathcal{D}_n) &\geq \mathbb{E}_{\substack{1 \leq i \leq n \\ P \sim \nu, x \sim \theta}} \frac{1}{m} H(\nu_{\mathcal{D}_i^k(P)} \cdot \theta_{\mathcal{D}_i(x)}, \mathcal{D}_{i+m}) \\ &\quad - O_{k,R} \left(\frac{1}{m} + \frac{m}{n} \right). \end{aligned}$$

Here zero-mass atoms are ignored in the expectation.

Proof. Put $\zeta = \nu \times \theta$ and $F(P, x) = P(x)$. Let $\mathcal{A}_i$ be the product of the level- i coefficient partition and $\mathcal{D}_i$, and let $\mathcal{B}_j = F^{-1} \mathcal{D}_j$. On the bounded set in question, F is $C_{k,R}$ -Lipschitz. Consequently the image of every atom of $\mathcal{A}_i$ meets at most $C'_{k,R}$ atoms of $\mathcal{D}_i$, and hence

$$H_\zeta(\mathcal{B}_i | \mathcal{A}_i) \leq C_{k,R}.$$

The average, over $z = (P, x) \sim \zeta$, of the entropy of the image of the $\mathcal{A}_i(z)$ -component at level $i + m$ equals $H_\zeta(\mathcal{B}_{i+m} | \mathcal{A}_i)$. The chain rule and the fact that conditioning reduces entropy now give

$$\begin{aligned} H_\zeta(\mathcal{B}_{i+m} | \mathcal{A}_i) &\leq H_\zeta(\mathcal{B}_{i+m} | \mathcal{B}_i) + H_\zeta(\mathcal{B}_i | \mathcal{A}_i) \\ &\leq H(\nu, \theta, \mathcal{D}_{i+m}) - H(\nu, \theta, \mathcal{D}_i) + C_{k,R}. \end{aligned}$$

The left side is precisely the average of the entropy appearing in the statement. Sum over i . The same sliding-window cancellation as in Lemma II.12.4.4(ii), together with the fixed bound $|P(x)| \leq \sqrt{k+1}R$, bounds the sum of the entropy increments by $mH(\nu, \theta, \mathcal{D}_n) + O_{k,R}(m^2)$. Division by mn and rearrangement prove the claim. $\square$

Lemma II.12.4.13 (Polynomial action linearization). *For every $0 < \eta < 1$, $k \geq 1$, and $R > 1$, there is $M = M(\eta, k, R)$ such that the following holds for $m \geq M$. There is $\delta_0 = \delta_0(\eta, k, R, m) > 0$ so that, whenever $0 < \delta < \delta_0$, $P \in \mathcal{A}_k$, $x \in I$, ν is a probability on $\mathcal{A}_k$, θ is a probability on I , and*

$$\|P\|_2 \leq R, \quad \text{supp } \nu \subset B(P, \delta), \quad \text{supp } \theta \subset B(x, \delta),$$

one has, with $\ell = m + \lceil \log_2 \delta^{-1} \rceil$,

$$\left| \frac{1}{m} H(v.\theta, \mathcal{D}_\ell) - \frac{1}{m} H((v.x) * (S_{P'(x)}\theta), \mathcal{D}_\ell) \right| < \eta.$$

Here $v.x$ is evaluation at x and $S_a(y) = ay$.

Proof. For $Q \in \text{supp } v$ and $y \in \text{supp } \theta$, Taylor's formula on I gives

$$|Q(y) - Q(x) - P'(x)(y - x)| \leq C_{k,R} \delta^2.$$

Indeed the Taylor remainder for Q is $O_{k,R}(\delta^2)$, and $|Q'(x) - P'(x)| |y - x| = O_k(\delta^2)$. Choose δ_0 so that $C_{k,R} \delta^2 \leq 2^{-\ell-2}$. Random variables which are pointwise within $2^{-\ell-2}$ have level- ℓ entropies differing by $O(1)$: couple them, observe that their dyadic cell indices differ by at most one, and use the entropy chain rule in both directions. Thus $v.\theta$ has, up to $O(1)$ entropy, the law of $Q(x) + P'(x)(y - x)$. Translation by the fixed number $P'(x)x$ changes dyadic entropy by another $O(1)$. The translated law is $(v.x) * (S_{P'(x)}\theta)$. Taking M so that the total $O(1)/m$ is less than η proves the result. $\square$

Lemma II.12.4.14 (Entropy survives at a polynomial evaluation). *For every $0 < \varepsilon < 1$ and $k \geq 1$ there is $c = c(\varepsilon, k) > 0$ with the following property. If m is sufficiently large and then n is sufficiently large, v is supported on $\{P : \|P\|_2 \leq \varepsilon^{-1}\}$ and*

$$\frac{1}{n} H(v, \mathcal{D}_n^{\mathcal{A}_k}) \geq \varepsilon,$$

while $\psi \in C^1(I, I)$ satisfies $\varepsilon \leq |\psi'| \leq \varepsilon^{-1}$ and $\theta = \psi_* \mu$, then

$$\mathbb{P}_{\substack{1 \leq i \leq n \\ P \sim v, x \sim \theta}} \left\{ \frac{1}{m} H((v_{\mathcal{D}_i^{\mathcal{A}_k(P)}}).x, \mathcal{D}_{i+m}) > c \right\} > c.$$

Proof. Write $d = k + 1$. The coefficient-space version of the sliding entropy identity in Lemma II.12.4.4(ii) gives

$$\mathbb{E}_{i,P} \frac{1}{m} H(v_{\mathcal{D}_i^{\mathcal{A}_k(P)}}, \mathcal{D}_{i+m}^{\mathcal{A}_k}) \geq \varepsilon/2. \quad (\text{II.12.4.10})$$

once m and then n are large enough. (The proof is identical, with one-level entropy increment at most d .) Since the displayed random variable is at most $d + O_k(1/m)$, the set G of pairs (i, P) on which it is at least $\varepsilon/4$ has probability at least a constant $a = a(\varepsilon, k) > 0$.

Lemma II.12.4.5, applied with the present derivative bounds, supplies $r = r(\varepsilon, k) > 0$ such that

$$\sup_y \theta(B(y, r)) < \frac{1}{2d}.$$

Choose $X_0, \dots, X_k$ independently with law θ . Successively excluding the r -balls about the points already chosen shows that the event E on which the X_j are pairwise r -separated has probability at least 2^{-k} . On E , the evaluation map

$$L_X : \mathcal{A}_k \longrightarrow \mathbb{R}^d, \quad Q \longmapsto (Q(X_0), \dots, Q(X_k)),$$

is uniformly bi-Lipschitz: its matrix is Vandermonde, and the determinant $\prod_{a < b} (X_b - X_a)$ is bounded away from zero. The proof of Lemma II.12.4.4(i), word for word in fixed dimension, therefore gives for every component σ occurring above

$$\sum_{j=0}^k H(\sigma.X_j, \mathcal{D}_{i+m}) \geq H(\sigma, \mathcal{D}_{i+m}^{\mathcal{A}_k}) - C_{\varepsilon,k}, \quad (\text{II.12.4.11})$$

where subadditivity bounds the entropy of the joint evaluation vector by the sum of its coordinate entropies.

Integrate (II.12.4.11) first over E and then over $(i, P) \in G$. Because all summands are nonnegative, independence and $\mathbb{P}(E) \geq 2^{-k}$ yield

$$\mathbb{E}_{i,P,x} \frac{1}{m} H((v_{\mathcal{D}_i^{\mathcal{A}_k(P)}})_x, \mathcal{D}_{i+m}) \geq b(\varepsilon, k) > 0. \quad (\text{II.12.4.12})$$

for sufficiently large m . Finally, a level- i coefficient cube has evaluation image of diameter at most $d2^{-i}$ on I , so the normalized entropy in (II.12.4.12) is at most 2 after increasing m by a constant depending on k . The elementary implication $0 \leq Y \leq 2$ and $\mathbb{E}Y \geq b \Rightarrow \mathbb{P}\{Y \geq b/2\} \geq b/4$ proves the statement with $c = \min\{b/4, b/2\}$. $\square$

Theorem II.12.4.15 (Entropy gain for polynomial actions). *Assume $\dim \mu < 1$. For every $k \geq 1$ and $0 < \varepsilon < 1$ there are $\rho = \rho(k, \varepsilon) > 0$ and $N = N(k, \varepsilon)$ such that the following holds for $n \geq N$. Let ν be a compactly supported probability on $\mathcal{A}_k$ such that*

$$\|P\|_2 \leq \varepsilon^{-1}, \quad |P'(x)| \geq \varepsilon \quad (P \in \text{supp } \nu, x \in I), \quad n^{-1}H(\nu, \mathcal{D}_n^{\mathcal{A}_k}) \geq \varepsilon.$$

If $\psi \in C^1(I, I)$ and $\varepsilon \leq |\psi'| \leq \varepsilon^{-1}$, then

$$n^{-1}H(\nu.(\psi_*\mu), \mathcal{D}_n) \geq \dim \mu + \rho.$$

Proof. Write $d = \dim \mu < 1$ and $\theta = \psi_*\mu$. Let $c_0 = c_0(k, \varepsilon) > 0$ be the constant from Lemma II.12.4.14. Choose

$$0 < a < \min \left\{ \frac{c_0}{8}, \frac{1-d}{8}, \frac{\varepsilon}{8(k+1)}, \frac{1}{8} \right\}. \quad (\text{II.12.4.13})$$

Let $\eta_0(a)$ be supplied by Proposition II.12.4.10. Choose $0 < \eta < \min\{\eta_0(a), c_0/16, (1-d)/16, \varepsilon/2\}$. Apply Lemma II.12.4.9 with $\alpha = (1-d)/4$ and this η , obtaining a window length ℓ . For that ℓ , let $\rho_0 > 0$ and M_0 be the constants in Proposition II.12.4.10. Fix

$$0 < \delta < \min \left\{ \frac{c_0}{32}, \frac{\eta}{2}, \frac{\varepsilon}{2}, \frac{c_0\rho_0}{24} \right\}.$$

Now choose the action window m sufficiently large, in this order, so that: (i) $m \geq M_0$; (ii) Proposition II.12.4.6 applies with parameter δ and Lemma II.12.4.9 applies with the already fixed (α, η, ℓ) ; (iii) Lemma II.12.4.14 applies and its conclusion remains valid after a bounded dilation with threshold $c_0/2$; and (iv) all $O_{k,\varepsilon}(1)/m$ entropy errors below are smaller than δ . Finally take $n \gg m$ so that every sliding-average and endpoint loss is smaller than $\min\{\delta, c_0/32\}$.

For $1 \leq i \leq n$, $P \sim \nu$, and $x \sim \theta$, abbreviate

$$\nu_{i,P} = \nu_{\mathcal{D}_i^{\mathcal{A}_k(P)}}, \quad \theta_{i,x} = \theta_{\mathcal{D}_i(x)}, \quad \xi_{i,P,x} = S_{P'(x)^{-1}}(\nu_{i,P}.x).$$

The derivative assumptions and the coefficient bound give

$$\varepsilon \leq |P'(x)| \leq C_{k,\varepsilon}.$$

Moreover, for Q_1, Q_2 in the same level- i coefficient cube,

$$|Q_1(x) - Q_2(x)| \leq (k+1)2^{-i},$$

so

$$\text{diam}(\text{supp } \xi_{i,P,x}) \leq \varepsilon^{-1}(k+1)2^{-i}, \quad \text{diam}(\text{supp } \theta_{i,x}) \leq 2^{-i}. \quad (\text{II.12.4.14})$$

By (II.12.4.13), these satisfy the diameter hypothesis of Proposition II.12.4.10.

We first convert the global action entropy into the average of these local convolutions. Lemma II.12.4.12 gives the corresponding local polynomial actions. For all but $O_{k,\varepsilon,m}(1)$ initial scales, the two component supports lie in balls of radius $C_k 2^{-i}$, which is smaller than the δ_0 in Lemma II.12.4.13. Apply that lemma with $\delta = C_k 2^{-i}$. Its target level differs from $i+m$ by only $O_k(1)$, so monotonicity of dyadic entropy across a bounded number of levels changes the normalized entropy by $O_k(1/m)$. Finally dilate by $P'(x)^{-1}$; Lemma II.12.4.4(i) changes the entropy by $O_{k,\varepsilon}(1)$. Discarding the finitely many initial scales and using $n \gg m$, we obtain

$$\frac{1}{n}H(\nu.\theta, \mathcal{D}_n) \geq \mathbb{E}_{i,P,x} \frac{1}{m}H(\xi_{i,P,x} * \theta_{i,x}, \mathcal{D}_{i+m}) - 3\delta. \quad (\text{II.12.4.15})$$

Let E_1 be the event

$$\frac{1}{m}H(\theta_{i,x}, \mathcal{D}_{i+m}) > d - \delta.$$

By Proposition II.12.4.6, after increasing m , $\mathbb{P}(E_1) > 1 - \delta$. Let E_{por} be the event that the nested component estimate in Lemma II.12.4.9 holds for $\theta_{i,x}$ with inner exceptional probability at most η . After discarding the m terminal values of i (an $o(1)$ fraction because $n \gg m$), we have $\mathbb{P}(E_{\text{por}}) > 1 - 2\eta$.

Let E_{ev} be the evaluation-entropy event from Lemma II.12.4.14. It has probability $> c_0$ and, by the bounded dilation estimate and the choice of m ,

$$\frac{1}{m}H(\xi_{i,P,x}, \mathcal{D}_{i+m}) > \frac{c_0}{2} > a \quad \text{on } E_{\text{ev}}. \quad (\text{II.12.4.16})$$

With the already fixed δ , and after increasing n if necessary,

$$E_2 := E_1 \cap E_{\text{por}} \cap E_{\text{ev}}$$

has probability at least $c_0/2$ for all sufficiently large n . On E_{por} , every low-entropy component in the nested statement has normalized entropy at most

$$d + \frac{1-d}{4} < 1-a,$$

by (II.12.4.13). Thus (II.12.4.14), (II.12.4.16), and Proposition II.12.4.10 give, on E_2 ,

$$\frac{1}{m}H(\xi_{i,P,x} * \theta_{i,x}, \mathcal{D}_{i+m}) \geq \frac{1}{m}H(\theta_{i,x}, \mathcal{D}_{i+m}) + \rho_0 > d - \delta + \rho_0. \quad (\text{II.12.4.17})$$

On all of E_1 , Lemma II.12.4.11 gives

$$\frac{1}{m}H(\xi_{i,P,x} * \theta_{i,x}, \mathcal{D}_{i+m}) > d - 2\delta. \quad (\text{II.12.4.18})$$

The integrand in (II.12.4.15) is nonnegative off E_1 . On E_1 use the baseline $d - 2\delta$, and on E_2 add the extra gain ρ_0 . Since $\mathbb{P}(E_1) > 1 - \delta$ and $\mathbb{P}(E_2) \geq c_0/2$,

$$\mathbb{E}_{i,P,x} \frac{1}{m}H(\xi_{i,P,x} * \theta_{i,x}, \mathcal{D}_{i+m}) \geq (1 - \delta)(d - 2\delta) + \frac{c_0\rho_0}{2} \geq d - 3\delta + \frac{c_0\rho_0}{2}.$$

Together with (II.12.4.15) and the choice $6\delta \leq c_0\rho_0/4$, this gives

$$\frac{1}{n}H(v.\theta, \mathcal{D}_n) \geq d + \frac{c_0\rho_0}{4}.$$

This proves the theorem with $\rho = c_0\rho_0/4$. $\square$

Remark II.12.4.16 (Structure of the entropy-gain proof). The one-dimensional Shannon inverse theorem, Theorem II.12.4.8, supplies the rigidity step. The preceding argument establishes the remaining ingredients—localized convolution growth, nested porosity, the non-loss estimate, and the assembly over parameter scales—in the analytic setting.

Let $\beta = p^{\mathbb{N}}$ on $\Lambda^{\mathbb{N}}$, let Π be the coding map, and let $\{\beta_\omega\}$ be a disintegration of β over $\Pi^{-1}\mathcal{B}(I)$. For $n \geq 1$, let $\mathcal{C}_n$ be the partition of $\Lambda^{\mathbb{N}}$ into level- n cylinders. Define $\Pi_n(\eta) = \phi_{\eta|_n}$ and, for a probability ξ on analytic maps, let $\xi.\mu$ be the image of $\xi \times \mu$ under evaluation.

Lemma II.12.4.17 (Bernoulli entropy on coding fibers). *Let X be a compact metric space, let $(S_i)_{i \in \Lambda}$ be injective contractions of X , and let $\pi : \Lambda^{\mathbb{N}} \rightarrow X$ be their coding map. Let $\beta = p^{\mathbb{N}}$ be a positive Bernoulli measure, put $\mathcal{A} = \pi^{-1}\mathcal{B}(X)$, and let $\{\beta_\omega\}$ be a disintegration of β over $\mathcal{A}$. If $\mathcal{C}_n$ is the level- n cylinder partition and*

$$\Delta = H_\beta(\mathcal{C}_1 \mid \mathcal{A}),$$

then, for β -almost every ω ,

$$\lim_{n \rightarrow \infty} \frac{1}{n}H(\beta_\omega, \mathcal{C}_n) = \Delta. \quad (\text{II.12.4.19})$$

The logarithm may be taken in any fixed base, provided the same base is used on both sides.

Proof. Write σ for the left shift and let $\mathcal{B}_j = \sigma(\eta_1, \dots, \eta_j)$, with $\mathcal{B}_0$ trivial. Put

$$q_i = \mathbb{E}_\beta(\mathbf{1}_{[i]} \mid \mathcal{A}), \quad g = - \sum_{i \in \Lambda} \mathbf{1}_{[i]} \log_2 q_i.$$

Thus $g \in L^1(\beta)$ and $\int g d\beta = \Delta$. On every cylinder $[u]$, $|u| = j$, the coding identity is

$$\pi(\eta) = S_u(\pi(\sigma^j \eta)).$$

Since S_u is injective and is a Borel isomorphism onto its image, knowing u and $\pi(\eta)$ is equivalent to knowing u and $\pi(\sigma^j \eta)$. Consequently

$$\mathcal{B}_j \vee \mathcal{A} = \mathcal{B}_j \vee \sigma^{-j} \mathcal{A} \quad \text{modulo } \beta. \quad (\text{II.12.4.20})$$

The prefix sigma-algebra $\mathcal{B}_j$ is independent of the tail $\sigma^{-j} \mathcal{B}(\Lambda^{\mathbb{N}})$. Therefore (II.12.4.20) and the defining property of conditional expectation give, for every $i \in \Lambda$,

$$\mathbb{E}_\beta(\mathbf{1}_{\{\eta_{j+1}=i\}} \mid \mathcal{B}_j \vee \mathcal{A}) = q_i \circ \sigma^j. \quad (\text{II.12.4.21})$$

Indeed, after replacing $\mathcal{A}$ by $\sigma^{-j} \mathcal{A}$ using (II.12.4.20), both the indicator and the conditioning information other than $\mathcal{B}_j$ are tail-measurable; testing against a product of a prefix cylinder and a tail event proves (II.12.4.21) first on a generating pi-system and then on the whole conditioning sigma-algebra.

Apply the conditional-information chain rule to the successive coordinates and use (II.12.4.21). For every $n \geq 1$,

$$I_\beta(\mathcal{C}_n \mid \mathcal{A}) = \sum_{j=0}^{n-1} g \circ \sigma^j \quad \text{almost everywhere.} \quad (\text{II.12.4.22})$$

The Bernoulli shift is ergodic, so Birkhoff's theorem yields

$$\lim_{n \rightarrow \infty} \frac{1}{n} I_\beta(\mathcal{C}_n \mid \mathcal{A})(\eta) = \Delta \quad \text{for } \beta\text{-almost every } \eta. \quad (\text{II.12.4.23})$$

For a simultaneous version of the disintegration,

$$I_\beta(\mathcal{C}_n \mid \mathcal{A})(\eta) = -\log_2 \beta_\eta(\mathcal{C}_n(\eta)) \quad (n \geq 1)$$

outside one null set. Disintegrating that null set in (II.12.4.23) shows that, for almost every ω ,

$$-\frac{1}{n} \log_2 \beta_\omega(\mathcal{C}_n(\eta)) \longrightarrow \Delta \quad \text{for } \beta_\omega\text{-almost every } \eta. \quad (\text{II.12.4.24})$$

It remains only to pass from pointwise information to entropy; we include the elementary argument because this passage is sometimes hidden inside a conditional Shannon–McMillan–Breiman citation. Let ν be any probability on $\Lambda^{\mathbb{N}}$ for which the left side of (II.12.4.24) tends to a constant a almost everywhere. Fatou's lemma gives

$$\liminf_n \frac{1}{n} H(\nu, \mathcal{C}_n) \geq a.$$

For $\varepsilon > 0$, let G_n be the union of those level- n cylinders C with $\nu(C) \geq 2^{-n(a+\varepsilon)}$. Pointwise convergence implies $\nu(G_n) \rightarrow 1$, and G_n contains at most $2^{n(a+\varepsilon)}$ cylinders. Writing $b_n = \nu(G_n)$ and splitting the entropy between G_n and its complement gives

$$H(\nu, \mathcal{C}_n) \leq 1 + b_n n(a + \varepsilon) + (1 - b_n) n \log_2 |\Lambda|.$$

Hence the entropy limsup is at most $a + \varepsilon$; let $\varepsilon \downarrow 0$. Apply this conclusion to $\nu = \beta_\omega$ and (II.12.4.24) to obtain (II.12.4.19). $\square$

Theorem II.12.4.18 (Feng–Hu projection entropy). *Put*

$$\Delta_F = H_\beta(\mathcal{C}_1 \mid \Pi^{-1}\mathcal{B}(I)).$$

Then μ is exact dimensional,

$$\dim \mu = \frac{H(p) - \Delta_F}{\chi}, \quad (\text{II.12.4.25})$$

and for β -almost every ω ,

$$\lim_{n \rightarrow \infty} \frac{1}{n} H(\beta_\omega, \mathcal{C}_n) = \Delta_F. \quad (\text{II.12.4.26})$$

These are the projection-entropy and conditional Shannon–McMillan–Breiman conclusions of Feng–Hu [FH09]. They follow from the conformal theorem of Volume I and Lemma II.12.4.17.

Proof. Regarded as a system on $\mathbb{R}$, Φ is a finite C^1 conformal IFS in the sense of the definition in Volume I: $0 < |\phi'_i| < 1$ gives local diffeomorphisms, injectivity, and a common contraction constant after shrinking a fixed neighborhood of I . The Bernoulli shift is ergodic and $\Pi_*\beta = \mu$. Hence the Feng–Hu conformal formula from Volume I gives

$$\dim \mu = \frac{h_\Pi(\sigma, \beta)}{\chi}. \quad (\text{II.12.4.27})$$

Let $\mathcal{A} = \Pi^{-1}\mathcal{B}(I)$. By the definition of projection entropy,

$$h_\Pi(\sigma, \beta) = H_\beta(\mathcal{C}_1 \mid \sigma^{-1}\mathcal{A}) - H_\beta(\mathcal{C}_1 \mid \mathcal{A}).$$

The sigma-algebra $\sigma^{-1}\mathcal{A}$ depends only on coordinates $2, 3, \dots$, and is therefore independent of the first coordinate under the Bernoulli law. Its first conditional entropy is consequently $H(p)$, while the second is Δ_F . Substitution into (II.12.4.27) proves (II.12.4.25). Lemma II.12.4.17, applied to (ϕ_i) , proves (II.12.4.26). $\square$

Lemma II.12.4.19 (Vanishing entropy of a conditional symbolic action). *For β -almost every ω ,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} H((\Pi_n)_*\beta_{\omega \cdot \mu}, \mathcal{D}_{[\chi n]}) = 0.$$

Proof. For almost every ω , disintegration gives $\beta_\omega(\Pi^{-1}\{\Pi\omega\}) = 1$, while the ergodic theorem and bounded distortion give

$$\lim_{n \rightarrow \infty} n^{-1} \log_2 |\phi'_{\eta_n}(0)| = -\chi \quad \text{for } \beta_\omega\text{-almost every } \eta.$$

Fix such an ω and $\delta > 0$. For all sufficiently large n , the set U_n of words $u \in \Lambda^n$ with $|\phi'_u(0)| \leq 2^{n(\delta-\chi)}$ has conditional mass at least $1 - \delta$. Whenever $\beta_\omega([u]) > 0$, the fiber property supplies $\eta \in [u]$ with $\Pi\eta = \Pi\omega$. Hence bounded distortion gives

$$\text{supp}(\phi_u\mu) \subset B(\Pi\omega, C2^{n(\delta-\chi)}) \quad (u \in U_n).$$

At dyadic level $\lfloor \chi n \rfloor$, the mixture over U_n therefore meets at most $O(C2^{\delta n})$ cells and has entropy at most $\delta n + O(1)$. The complementary mixture has weight at most δ and is supported on I , so its weighted entropy is at most $\delta \chi n + O(1)$. The entropy-of-mixtures inequality adds only the binary entropy of the two weights. Consequently

$$\limsup_{n \rightarrow \infty} \frac{1}{n} H((\Pi_n)_* \beta_{\omega, \mu}, \mathcal{D}_{\lfloor \chi n \rfloor}) \leq (1 + \chi) \delta.$$

Letting $\delta \downarrow 0$ proves the claim. $\square$

Theorem II.12.4.20 (Analytic self-conformal dimension theorem). *Assume the analytic IFS has no common fixed point and is exponentially separated. Then for every positive probability vector p , the stationary self-conformal measure is exact dimensional and*

$$\dim \mu = \min \left\{ 1, \frac{H(p)}{\chi(\Phi, p)} \right\}.$$

This is the analytic self-conformal theorem of [Rap24]. The proof combines the projection-entropy formula above with the Shannon inverse theorem from Chapter II.4 and the Taylor approximation developed in this section.

Proof. Write $d = \dim \mu$ and $L = \log_2 |\Lambda|$. Theorem II.12.4.18 already gives exact dimensionality and $d = (H(p) - \Delta_F)/\chi$, with $\Delta_F \geq 0$. Hence $d \leq H(p)/\chi$ and, of course, $d \leq 1$. It remains to rule out

$$d < \min\{1, H(p)/\chi\}. \quad (\text{II.12.4.28})$$

Under this assumption $d < 1$ and $\Delta_F > 0$. Put

$$\Delta = \frac{1}{2} \min\{\Delta_F, 1\}, \quad \tau = \frac{\Delta}{2L}.$$

The no-common-fixed-point hypothesis implies $|\Lambda| \geq 2$, so $L \geq 1$ and $0 < \tau \leq 1/4$.

Choice of parameters and separated block lengths. Let $0 < c < 1$ be an exponential-separation constant from Definition II.12.4.1, and enlarge the distortion constant C from Lemma II.12.4.2 whenever harmless. Choose an integer M so large that

$$M > 4(1 + \tau)|\log_2 c| + 10. \quad (\text{II.12.4.29})$$

Next choose k so large that

$$k\chi\tau > 3M + 3\chi(1 + \tau). \quad (\text{II.12.4.30})$$

Fix these M, k . Constants below may depend on them and on Φ, p but not on the final separation level.

We shall use separation only at levels at which it is actually available. Let $N \rightarrow \infty$ run through the infinite sequence supplied by exponential separation and put

$$n = \left\lfloor \frac{N}{1 + \tau} \right\rfloor, \quad n' = N - n, \quad q = \lfloor \chi N \rfloor, \quad r = Mn. \quad (\text{II.12.4.31})$$

Then

$$\frac{n'}{n} = \tau + o(1), \quad \frac{n'}{n}L = \frac{\Delta}{2} + o(1), \quad (\text{II.12.4.32})$$

and, for every distinct $w_1, w_2 \in \Lambda^N$,

$$\|\phi_{w_1} - \phi_{w_2}\|_I \geq c^N. \quad (\text{II.12.4.33})$$

This choice avoids any need to prescribe a separation level in advance.

Choose a small number $\delta > 0$ after M, k have been fixed. It will be sent below a fixed positive entropy gain at the end. We assume from now on that $\delta < \chi/4$ and that it is sufficiently small for all estimates explicitly indicated below. Finally take N sufficiently large in the separation sequence.

The global entropy upper bound. Since dyadic partitions are nested, (II.12.4.2) gives

$$\frac{1}{r}H(\mu, \mathcal{D}_{q+r} \mid \mathcal{D}_q) \leq d + \delta \quad (\text{II.12.4.34})$$

for all sufficiently large chosen N . Stationarity at level N and disintegration give

$$\mu = (\Pi_N)_* \beta \cdot \mu = \int (\Pi_N)_* \beta_\omega \cdot \mu \, d\beta(\omega).$$

Conditional entropy is concave in the underlying probability measure, so (II.12.4.34) implies

$$\int \frac{1}{r}H((\Pi_N)_* \beta_\omega \cdot \mu, \mathcal{D}_{q+r} \mid \mathcal{D}_q) \, d\beta(\omega) \leq d + \delta. \quad (\text{II.12.4.35})$$

Lemma II.12.4.19, dominated convergence, and $N/n = O(1)$ give, after increasing N ,

$$\int \frac{1}{r}H((\Pi_N)_* \beta_\omega \cdot \mu, \mathcal{D}_q) \, d\beta(\omega) < \delta.$$

Using the entropy chain rule in (II.12.4.35), put

$$g(\omega) = \frac{1}{r}H((\Pi_N)_* \beta_\omega \cdot \mu, \mathcal{D}_{q+r}). \quad (\text{II.12.4.36})$$

Then

$$\int g(\omega) \, d\beta(\omega) \leq d + 2\delta. \quad (\text{II.12.4.37})$$

Typical derivative blocks and entropy-rich fibers. For $\ell \geq 1$ let $U_\ell \subset \Lambda^\ell$ consist of the words u with

$$2^{-\ell(\chi+\delta)} \leq |\phi'_u(0)| \leq 2^{-\ell(\chi-\delta)}. \quad (\text{II.12.4.38})$$

The ergodic theorem and bounded distortion imply $\beta([U_\ell]) \rightarrow 1$. Together with (II.12.4.26), disintegration, and Markov's inequality, this shows that, for large chosen N , there is a set $E \subset \Lambda^\mathbb{N}$ with $\beta(E) > 1 - \delta$ such that every $\omega \in E$ satisfies

$$\text{supp } \beta_\omega \subset \Pi^{-1}\{\Pi\omega\}, \quad (\text{II.12.4.39})$$

$$\frac{1}{n}H(\beta_\omega, \mathcal{E}_n) > \Delta, \quad (\text{II.12.4.40})$$

$$\beta_\omega([U_n]) > 1 - \delta, \quad \beta_\omega(\sigma^{-n}[U_{n'}]) > 1 - \delta. \quad (\text{II.12.4.41})$$

Indeed, the averages of the last two conditional masses are respectively $\beta([U_n])$ and $\beta([U_{n'}])$.

Fix $\omega \in E$. Partition U_n into classes $U_{n,j}$, $j \in J$, by unit intervals for $-\log_2 |\phi'_v(0)|$. For large n ,

$$|J| \leq 3\delta n, \quad |\phi'_{v_1}(0)| \leq 2|\phi'_{v_2}(0)| \quad (v_1, v_2 \in U_{n,j}). \quad (\text{II.12.4.42})$$

For $j \in J$ and $u \in U_{n'}$ set

$$F_{j,u} = [U_{n,j}] \cap \sigma^{-n}[u], \quad w_{j,u} = \beta_\omega(F_{j,u}),$$

and, whenever $w_{j,u} > 0$, put

$$g(\omega, j, u) = \frac{1}{r} H((\Pi_N)_*(\beta_\omega)_{F_{j,u}}, \mu, \mathcal{D}_{q+r}). \quad (\text{II.12.4.43})$$

The retained sets have total mass greater than $1 - 2\delta$. Hence there are $0 \leq \delta' < 2\delta$ and a probability ϑ such that

$$\beta_\omega = \sum_{j,u} w_{j,u} (\beta_\omega)_{F_{j,u}} + \delta' \vartheta.$$

Concavity of entropy therefore yields

$$g(\omega) \geq \sum_{j,u} w_{j,u} g(\omega, j, u). \quad (\text{II.12.4.44})$$

The entropy-of-mixtures upper bound, applied now to the symbolic partition $\mathcal{C}_n$, gives

$$\begin{aligned} H(\beta_\omega, \mathcal{C}_n) &\leq \sum_{j,u} w_{j,u} H((\beta_\omega)_{F_{j,u}}, \mathcal{C}_n) + \delta' nL \\ &\quad + \log_2(3\delta n + 1) + n'L + O(1). \end{aligned}$$

Use (II.12.4.40), (II.12.4.32), and then take δ small and N large. We obtain

$$\sum_{j,u} w_{j,u} \frac{1}{n} H((\beta_\omega)_{F_{j,u}}, \mathcal{C}_n) > \frac{\Delta}{3}. \quad (\text{II.12.4.45})$$

Let $Q = Q(\omega)$ be the pairs (j, u) for which

$$\frac{1}{n} H((\beta_\omega)_{F_{j,u}}, \mathcal{C}_n) \geq \frac{\Delta}{6}. \quad (\text{II.12.4.46})$$

Since every normalized cylinder entropy is at most L , (II.12.4.45) implies

$$\sum_{(j,u) \in Q} w_{j,u} > \frac{\Delta}{6L}. \quad (\text{II.12.4.47})$$

A baseline lower bound for every retained component. If $v \in U_n$ and $u \in U_{n'}$, then bounded distortion, the chain rule, and (II.12.4.38) give

$$C^{-1} 2^{-\delta N} \leq 2^q |\phi'_{vu}(x)| \leq C 2^{\delta N} \quad (x \in I),$$

after enlarging C to absorb the integer part in q . Thus $S_{2^q} \circ \phi_{vu}$ is $C 2^{\delta N}$ -bi-Lipschitz. By Lemma II.12.4.4(i),(iii), (II.12.4.2), and concavity,

$$g(\omega, j, u) \geq d - O(\delta) \quad (\text{II.12.4.48})$$

for all retained pairs (j, u) and all sufficiently large N . Here and below $O(\delta)$ has a constant depending only on the already fixed data Φ, p, M, k .

Taylor normalization on an entropy-rich component. Fix $(j, u) \in Q$ and put $a = \phi_u(0)$. For an analytic map f , denote its order- k Taylor polynomial at a by

$$J_{k,a}f(X) = \sum_{\ell=0}^k \frac{f^{(\ell)}(a)}{\ell!} (X - a)^\ell.$$

For $v \in U_{n,j}$ and $x \in I$, Lemma II.12.4.2 and Taylor's formula give

$$\begin{aligned} |(\phi_v - J_{k,a}\phi_v)(\phi_u(x))| &\leq C^{2k+2} |\phi'_u(0)|^{k+1} |\phi'_v(0)| \\ &\leq C^{2k+2} 2^{-(k+1)(\chi-\delta)n'} 2^{-(\chi-\delta)n}. \end{aligned} \quad (\text{II.12.4.49})$$

By (II.12.4.30), after decreasing δ and then increasing N , the right-hand side is at most

$$2^{-q-r-4}. \quad (\text{II.12.4.50})$$

Coupling the exact and Taylorized actions with the same symbolic word and the same point of μ therefore changes their level- $(q+r)$ entropies by only $O(1)$: two coupled values at distance at most 2^{-q-r-4} have dyadic cell indices differing by a bounded amount. Consequently

$$g(\omega, j, u) \geq \frac{1}{r} H((J_{k,a}\Pi_n)_*(\beta_\omega)_{F_{j,u}}, ((\phi_u)_*\mu), \mathcal{D}_{q+r}) - o(1). \quad (\text{II.12.4.51})$$

Let ψ be the affine map with $\psi(a) = 0$ and $\psi(\phi_u(1)) = 1$, and write $\psi(x) = r_\psi x + t_\psi$. Since ϕ_u is monotone, $\psi \circ \phi_u$ maps I onto I . The mean-value theorem and bounded distortion give

$$C^{-1} |\phi'_u(0)|^{-1} \leq |r_\psi| \leq C |\phi'_u(0)|^{-1}. \quad (\text{II.12.4.52})$$

By (II.12.4.42), choose $\alpha > 0$ so that

$$\frac{1}{2} \leq \alpha |\phi'_v(0)| \leq 2 \quad (v \in U_{n,j}). \quad (\text{II.12.4.53})$$

For every v with $\beta_\omega([vu]) > 0$ define

$$P_v(X) = \alpha |r_\psi| [J_{k,a}\phi_v(\psi^{-1}(X)) - \Pi\omega], \quad (\text{II.12.4.54})$$

and let ν be the pushforward of $(\beta_\omega)_{F_{j,u}}$ under $\eta \mapsto P_{\eta|_n}$. Finally set

$$\theta = (\psi \circ \phi_u)_*\mu.$$

By construction, $\nu.\theta$ is the affine rescaling by $\alpha|r_\psi|$, followed by translation by $-\alpha|r_\psi|\Pi\omega$, of the Taylorized action in (II.12.4.51). Moreover, (II.12.4.38), (II.12.4.52), and (II.12.4.53) imply

$$|\log_2(\alpha|r_\psi|) - \chi N| \leq \delta N + O(1). \quad (\text{II.12.4.55})$$

Lemma II.12.4.4(iii), followed by changing at most $\delta N + O(1)$ dyadic levels, now gives

$$g(\omega, j, u) \geq \frac{1}{r} H(\nu.\theta, \mathcal{D}_r) - O(\delta) - o(1). \quad (\text{II.12.4.56})$$

We next verify the hypotheses of Theorem II.12.4.15. First, if $\beta_\omega([vu]) > 0$, the fiber property (II.12.4.39) supplies a point of the cylinder $[vu]$ coding $\Pi\omega$. Hence

$$|\phi_v(a) - \Pi\omega| \leq C|\phi'_v(0)| |\phi'_u(0)|.$$

Using this estimate for the constant coefficient of P_v , and Lemma II.12.4.2 for the other coefficients, we get a constant $R = R(k, C)$ such that

$$\|P_v\|_2 \leq R \quad (v \in \text{supp } \nu). \quad (\text{II.12.4.57})$$

Indeed, for $1 \leq \ell \leq k$ the absolute value of the X^ℓ coefficient is at most

$$2C^\ell |r_\psi|^{1-\ell},$$

which is bounded uniformly once N is large.

For the derivative, the linear term in (II.12.4.54) has absolute value at least $(2C)^{-1}$, whereas the sum of the derivatives of all terms of degree at least two is $O_{k,C}(|r_\psi|^{-1})$. Since $n' \rightarrow \infty$, the latter tends to zero. Thus, for large N ,

$$|P'_v(x)| \geq \frac{1}{4C} \quad (x \in I, v \in \text{supp } \nu). \quad (\text{II.12.4.58})$$

Also, by (II.12.4.52) and bounded distortion,

$$C^{-2} \leq |(\psi \circ \phi_u)'(x)| \leq C^2 \quad (x \in I). \quad (\text{II.12.4.59})$$

It remains to prove that ν has positive coefficient entropy at scale r . Suppose that $\nu_1 \neq \nu_2$ in the support give polynomials in the same atom of $\mathcal{D}_r^{\mathcal{A}_k}$. Then, for every $x \in I$,

$$|P_{\nu_1}(x) - P_{\nu_2}(x)| \leq (k+1)2^{-r}. \quad (\text{II.12.4.60})$$

On the other hand, (II.12.4.33) applied to the distinct words $\nu_1 u, \nu_2 u$ gives an $x_0 \in I$ for which

$$|\phi_{\nu_1 u}(x_0) - \phi_{\nu_2 u}(x_0)| \geq c^N.$$

Set $y_0 = \psi(\phi_u(x_0)) \in I$. By (II.12.4.50), the two Taylor values at $\psi^{-1}(y_0) = \phi_u(x_0)$ differ by at least $c^N/2$ for large N . The factor $\alpha|r_\psi|$ is eventually larger than 1, so

$$|P_{\nu_1}(y_0) - P_{\nu_2}(y_0)| \geq c^N/2.$$

But (II.12.4.29) and $r = Mn$ imply $c^N/2 > (k+1)2^{-r}$ for large N , contradicting (II.12.4.60). Thus distinct n -prefixes occupy distinct level- r coefficient cells, and

$$\frac{1}{r} H(\nu, \mathcal{D}_r^{\mathcal{A}_k}) = \frac{1}{Mn} H((\beta_\omega)_{F_{j,u}}, \mathcal{C}_n) \geq \frac{\Delta}{6M}. \quad (\text{II.12.4.61})$$

Choose once and for all

$$0 < \varepsilon_0 < \min \left\{ R^{-1}, \frac{1}{4C}, C^{-2}, \frac{\Delta}{6M}, \frac{1}{2} \right\}. \quad (\text{II.12.4.62})$$

Theorem II.12.4.15, with the fixed k and ε_0 , supplies $\rho > 0$. Increasing N makes r larger than its threshold. Equations (II.12.4.57)–(II.12.4.61) and (II.12.4.59) then give

$$\frac{1}{r}H(v, \theta, \mathcal{D}_r) \geq d + \rho.$$

Return to (II.12.4.56) and decrease δ if necessary. For every $(j, u) \in Q$ and all sufficiently large chosen N ,

$$g(\omega, j, u) \geq d + \rho - O(\delta). \quad (\text{II.12.4.63})$$

Averaging the gain and contradiction. Combine (II.12.4.44), the baseline (II.12.4.48), the gain (II.12.4.63), the retained mass $> 1 - 2\delta$, and (II.12.4.47). Uniformly for $\omega \in E$,

$$g(\omega) \geq d + \frac{\rho\Delta}{6L} - O(\delta). \quad (\text{II.12.4.64})$$

Since $g \geq 0$ and $\beta(E) > 1 - \delta$, integration gives

$$\int g d\beta \geq d + \frac{\rho\Delta}{6L} - O(\delta).$$

Choose δ so small that the final error plus 2δ is less than $\rho\Delta/(12L)$. This contradicts (II.12.4.37). Therefore (II.12.4.28) is impossible. Together with the upper bounds following from Theorem II.12.4.18, this proves

$$d = \min \left\{ 1, \frac{H(p)}{\chi} \right\},$$

while exact dimensionality is the first assertion of Theorem II.12.4.18. □

Remark II.12.4.21 (The two entropy ingredients). The Taylor and separation argument above rests on two earlier results: the Feng–Hu projection-entropy formula, Theorem II.12.4.18, and the one-dimensional Shannon inverse theorem used in Theorem II.12.4.15. All other estimates in the argument are established in this section.

II.12.5 Three-map homogeneous systems

Consider the homogeneous IFS with translations $0, 1, \tau$ and common contraction λ . Rapaport and Varjú’s analysis connects a dimension drop to algebraic approximation in the two parameters and then repels those approximants from low-entropy analytic curves. We develop that argument below, including its zero-dimensional exceptional-set consequence and its unconditional range. The root-separation and localized power-sum estimates are proved where they enter; algebraic-value lower bounds are obtained directly by field norms.

II.12.5.1 The three-map entropy model

Throughout this subsection logarithms and Shannon entropies use the natural base, and

$$f_1(x) = \lambda x, \quad f_2(x) = \lambda x + 1, \quad f_3(x) = \lambda x + \tau, \quad 0 < \lambda < 1,$$

and the three symbols have probability $1/3$. Let $\xi_0, \xi_1, \dots$ be independent and uniform on $\{1, 2, 3\}$, put $T_1(Y_1, Y_2) = 0$, $T_2(Y_1, Y_2) = Y_1$, $T_3(Y_1, Y_2) = Y_2$, and define

$$A_{\lambda, \tau}^{(n)} = \sum_{j=0}^{n-1} T_{\xi_j}(1, \tau) \lambda^j, \quad h(\lambda, \tau) = \lim_{n \rightarrow \infty} \frac{1}{n} H(A_{\lambda, \tau}^{(n)}). \quad (\text{II.12.5.1})$$

The limit exists because, after conditioning on the first n digits, the last m digits form a translated and dilated copy of an independent $A_{\lambda, \tau}^{(m)}$; hence $H(A_{\lambda, \tau}^{(n+m)}) \leq H(A_{\lambda, \tau}^{(n)}) + H(A_{\lambda, \tau}^{(m)})$, and Fekete's lemma applies.

We shall also use a bi-infinite independent extension $(\xi_j)_{j \in \mathbb{Z}}$ of the digit sequence. If $I \subset (0, \infty)$ is an interval bounded above, define the *scale-window law*

$$\mu_{\lambda, \tau}^I = \text{Law} \left(\sum_{j \in \mathbb{Z}: \lambda^j \in I} T_{\xi_j}(1, \tau) \lambda^j \right). \quad (\text{II.12.5.2})$$

Only finitely many negative indices occur and the positive tail is geometric, so the series converges absolutely. We write $\mu_{\lambda, \tau} = \mu_{\lambda, \tau}^{(0,1]}$ and $\mu_{\lambda, \tau}^{(\lambda^n, 1]} = \mu_{\lambda, \tau}^{(\lambda^n, 1]}$; the latter is the law of $A_{\lambda, \tau}^{(n)}$. Independence and a change of index give, for disjoint intervals and every $k \in \mathbb{Z}$,

$$\mu_{\lambda, \tau}^{I_1 \dot{\cup} I_2} = \mu_{\lambda, \tau}^{I_1} * \mu_{\lambda, \tau}^{I_2}, \quad (\text{II.12.5.3})$$

$$\mathcal{H}(\mu_{\lambda, \tau}^I; r \mid Rr) = \mathcal{H}(\mu_{\lambda, \tau}^{\lambda^k I}; \lambda^k r \mid \lambda^k Rr). \quad (\text{II.12.5.4})$$

The same notation will be used for an arbitrary finite digit D by replacing $T_{\xi_j}(1, \tau)$ with D_j .

Lemma II.12.5.1 (Uniform one-dyadic-digit deficit). *Let ζ be the law of a homogeneous random series $\sum_{j \geq 0} D_j \lambda^j$, where D is a bounded finite random variable. If $\dim \zeta < 1$, then there is $\alpha \in (0, 1/2)$ such that*

$$\mathcal{H}(\zeta^I; r \mid 2r) \leq 1 - \alpha \quad (\text{II.12.5.5})$$

for every $r > 0$ and every interval $I \subset (0, \infty)$ bounded above. Here $\mathcal{H} = \mathcal{H} / \log 2$ is the normalized averaged entropy from Chapter II.9.

Proof. Put $s = \dim \zeta < 1$. The exact-dimensionality theorem and the [entropy-dimension identity](#) give

$$\mathcal{H}(\zeta; 2^{-m}) < m(1 - \alpha_0) \quad (m \geq m_0) \quad (\text{II.12.5.6})$$

for some $\alpha_0 > 0$. The doubling calculation (II.9.N13), which is valid for every probability measure, says

$$1 - \mathcal{H}(\rho; 2u \mid 4u) \leq 4(1 - \mathcal{H}(\rho; u \mid 2u)). \quad (\text{II.12.5.7})$$

Indeed, if χ_u is uniform on $[0, u]$ and $\theta_u = (\delta_0 + \delta_u)/2$, then $\chi_{2u} = \chi_u * \theta_u$; differential-entropy submodularity and the identity $\theta_u * \theta_u = (\theta_{2u} + \delta_u)/2$ give (II.12.5.7) exactly as in the complete derivation of the [dyadic deficit lemma](#).

Suppose that for some $r > 0$, $\mathcal{H}(\zeta; r \mid 2r) \geq 1 - \alpha$. Iterating (II.12.5.7) and adding adjacent scale intervals gives

$$\mathcal{H}(\zeta; r \mid 2^K r) \geq K - 4^K \alpha. \quad (\text{II.12.5.8})$$

Put $c_\lambda = \lceil \log_2 \lambda^{-1} \rceil$. Choose K so large that

$$\frac{c_\lambda}{K + c_\lambda} < \frac{\alpha_0}{3},$$

and then choose $\alpha > 0$ so small that $4^K \alpha / (K + c_\lambda) < \alpha_0/3$. If m is the least positive integer with $\lambda^m \leq 2^{-K}$, then

$$2^{-K-c_\lambda} \leq \lambda^m \leq 2^{-K}.$$

For every $j \geq 0$, the scaled tail $(S_{\lambda^{jm}})_* \zeta$ is a convolution factor of ζ . Integral-ratio convolution monotonicity (II.9.N04), scale covariance, and (II.12.5.8) therefore put the same lower bound on every block $[\lambda^{jm}r, 2^K \lambda^{jm}r]$. These blocks are ordered and leave at most c_λ dyadic digits between consecutive endpoints. Choose $M_J \in \mathbb{N}$ so that 2^{-M_J} is within a factor two of $\lambda^{(J-1)m}r$. The J blocks are disjoint on the logarithmic scale axis, so additivity of conditional entropy and monotonicity give

$$\mathcal{H}(\zeta; 2^{-M_J}) \geq J(K - 4^K \alpha) - O_r(1), \quad M_J \leq J(K + c_\lambda) + O_r(1). \quad (\text{II.12.5.9})$$

Consequently

$$\liminf_{J \rightarrow \infty} \frac{\mathcal{H}(\zeta; 2^{-M_J})}{M_J} \geq \frac{K - 4^K \alpha}{K + c_\lambda} > 1 - \alpha_0,$$

contradicting (II.12.5.6). Hence $\mathcal{H}(\zeta; r \mid 2r) \leq 1 - \alpha$ for all r .

For a bounded-above interval I , choose $q \in \mathbb{Z}$ so that $\lambda^q I \subset (0, 1]$. The law $\zeta^{\lambda^q I}$ is a convolution factor of ζ . Thus (II.9.N04) and scale covariance give

$$\mathcal{H}(\zeta^I; r \mid 2r) = \mathcal{H}(\zeta^{\lambda^q I}; \lambda^q r \mid 2\lambda^q r) \leq \mathcal{H}(\zeta; \lambda^q r \mid 2\lambda^q r) \leq 1 - \alpha,$$

which proves the assertion. □

Proposition II.12.5.2 (Scale-and-convolve bootstrap). *Fix $0 < \lambda < 1$ and $\alpha \in (0, 1/2)$. There is $C = C(\lambda, \alpha) > 1$ with the following property. Let ζ be a homogeneous finite-digit series law satisfying (II.12.5.5), and let $n_1 < \dots < n_N$ and $K_1, \dots, K_N \geq 10$ satisfy*

$$\lambda^{-n_1} \geq \max\{2, \lambda^{-2}\}, \quad (\text{II.12.5.10})$$

$$n_{j+1} \geq K_j n_j \quad (j < N), \quad (\text{II.12.5.11})$$

$$\mathcal{H}(\zeta^{(n_j)}; \lambda^{K_j n_j} \mid \lambda^{10n_j}) \geq \alpha n_j \quad (j \leq N), \quad (\text{II.12.5.12})$$

$$n_j \geq C(\log K_j)^2 \quad (j \leq N). \quad (\text{II.12.5.13})$$

Then

$$\sum_{j=1}^N \frac{1}{\log K_j \log \log K_j} \leq C \left(1 + \frac{1}{n_1} \sum_{j=1}^N \log K_j \right). \quad (\text{II.12.5.14})$$

Proof. We give the window argument using (II.12.5.3), (II.12.5.4), and the arbitrary-measure entropy increment theorem.

We first record the window-transfer estimate. Suppose $0 \leq a_i < b_i \leq 1$ and $0 < r_i < s_i$ satisfy

$$\lambda^{-1}s_1/s_2 \leq a_1/a_2, \quad \lambda r_1/r_2 \geq b_1/b_2, \quad \max\{2, \lambda^{-2}\} \leq s_2/r_2 \leq s_1/r_1. \quad (\text{II.12.5.15})$$

If

$$\mathcal{H}(\zeta^{(a_1, b_1]}; r_1 \mid s_1) \geq \gamma \log_2(s_1/r_1), \quad (\text{II.12.5.16})$$

then

$$\mathcal{H}(\zeta^{(a_2, b_2]}; r_2 \mid s_2) \geq \frac{\gamma}{6} \log_2(s_2/r_2). \quad (\text{II.12.5.17})$$

To prove this, cover $[r_1, s_1]$ minimally by intervals $[\lambda^k r_2, \lambda^k s_2]$. Minimality and the last inequality in (II.12.5.15) bound their number by $3 \log(s_1/r_1) / \log(s_2/r_2)$. Additivity over nested scale intervals therefore gives one member carrying at least the corresponding average. The first two inequalities in (II.12.5.15) imply $(a_1, b_1] \subset \lambda^k(a_2, b_2]$. Apply (II.12.5.4), convolution monotonicity (II.9.N04), and the factor-two endpoint comparison (II.9.N03); this proves (II.12.5.17).

Fix a pair $(n, K) = (n_j, K_j)$ and set

$$J = \max \left\{ 1, \left\lceil \frac{\log(K/10)}{\log(11/10)} \right\rceil \right\}, \quad \mathcal{H}(\zeta^{(n)}; \lambda^{11(11/10)^j n} \mid \lambda^{10(11/10)^j n}) = b_i n.$$

By (II.12.5.12), $\sum_{i < J} b_i \geq \alpha$. Discarding indices with $b_i < \alpha/(2J)$ leaves total mass at least $\alpha/2$. For a retained i , put $T = \lfloor (11/10)^j \rfloor$. The law $\zeta^{(Tn)}$ is the convolution of the T independent scaled copies supported on the disjoint digit windows $(\lambda^{(u+1)n}, \lambda^{un}]$, $0 \leq u < T$. The u th scaled copy has entropy $b_i n$ between

$$\lambda^{(11(11/10)^j + u)n} \quad \text{and} \quad \lambda^{(10(11/10)^j + u)n}.$$

These scale intervals lie inside $[\lambda^{12(11/10)^j n}, \lambda^{10(11/10)^j n}]$, because $T \leq (11/10)^j$. Their dyadic logarithmic length is $(11/10)^j n \log_2 \lambda^{-1}$. Add these T blocks in the order $u = 0, \dots, T-1$, starting from δ_0 . After the u th addition the partial convolution is the contiguous scale-window law $\zeta^{(\lambda^{(u+1)n}, 1]}$. Thus the entropy increment theorem applies at every step with

$$\beta_i = \min \left\{ \frac{1}{4}, \frac{b_i}{4(11/10)^j \log_2 \lambda^{-1}} \right\}.$$

Lemma II.12.5.1 makes the exceptional-scale count in that theorem equal to zero. Moreover $b_i \geq \alpha/(2J) \gg_\alpha 1/\log K$, $(11/10)^j \leq K/10$, and hence $\log(1/\beta_i) \leq C_{\lambda, \alpha} \log K$. The entropy gain from one factor is therefore at least

$$\frac{c_{\lambda, \alpha} b_i n}{\log K} - 3.$$

On the remainder of the common scale interval convolution monotonicity gives no negative gain. Summing over the $T \asymp (11/10)^j$ factors, and using $n \geq C(\log K)^2$ to absorb the total error $3T$, gives

$$\mathcal{H}(\zeta^{(Tn)}; \lambda^{12(11/10)^j n} \mid \lambda^{10(11/10)^j n}) \geq \frac{c_{\lambda} b_i}{\log K} \log_2(\lambda^{-2(11/10)^j n}). \quad (\text{II.12.5.18})$$

Apply (II.12.5.17) to (II.12.5.18) and put $Q_i = \lambda^{-9(11/10)^i n}$. For every $r \leq \lambda^{2Kn}$ and $\max\{2, \lambda^{-2}\} \leq A \leq \lambda^{-n}$ one obtains

$$\mathcal{H}(\zeta^{(Q_i r, Q_i^2 r]}; r \mid Ar) \geq \frac{c_\alpha b_i}{\log K} \log_2 A. \quad (\text{II.12.5.19})$$

Indeed, the substitution in the window-transfer lemma is

$$(a_1, b_1, r_1, s_1) = (\lambda^{Tn}, 1, \lambda^{12(11/10)^i n}, \lambda^{10(11/10)^i n}), \quad (a_2, b_2, r_2, s_2) = (Q_i r, Q_i^2 r, r, Ar).$$

The three hypotheses in (II.12.5.15) follow from $T \leq (11/10)^i < T + 1$, $A \leq \lambda^{-n}$, and $r \leq \lambda^{2Kn}$; the inclusion $Q_i^2 r \leq 1$ uses $(11/10)^i < K/10$ for $i < J$. Choose one residue class modulo 10 carrying maximal retained b_i -mass. For its members put $a_i = \min\{b_i, 1\}$ and relabel them by i . If some $b_i \geq 1$, then $\sum_i a_i \geq 1$; otherwise capping changes no term. Thus, after decreasing $c_\alpha > 0$, these members satisfy

$$\lambda^{-9n} \leq R_i \leq \lambda^{-Kn}, \quad R_{i+1} \geq R_i^2, \quad a_i \geq c_\alpha / \log K, \quad \sum_i a_i \geq c_\alpha, \quad (\text{II.12.5.20})$$

and (II.12.5.19) holds with R_i, a_i , because $a_i \leq b_i$.

Perform this construction for every (n_j, K_j) and use the common scales

$$A = \lambda^{-n_1}, \quad r = \lambda^{2K_N n_N}.$$

Equations (II.12.5.11) and (II.12.5.20) make all windows $(R_{j,i}r, R_{j,i}^2r]$ pairwise disjoint. Hence their laws are independent convolution factors of ζ . Apply the [entropy increment theorem](#) successively with $\beta_{j,i} = c_\alpha a_{j,i} / \log K_j$, decreasing c_α so that $\beta_{j,i} \leq 1/2$. Start with the law of the complementary digit set and add the selected window laws one at a time. At every stage the partial convolution is a convolution factor of ζ ; hence (II.9.N04) and (II.12.5.5) give its one-digit deficit on every scale. Since $a_{j,i} \geq c_\alpha / \log K_j$, one has $\log(1/\beta_{j,i}) \leq C_\alpha \log \log K_j$, and therefore

$$\mathcal{H}(\zeta; r \mid Ar) \geq \sum_{j,i} \left(\frac{c_\alpha a_{j,i}}{\log K_j \log \log K_j} \log_2 A - C_\alpha \right). \quad (\text{II.12.5.21})$$

There are $O_\alpha(\log K_j)$ windows for the j th pair and $\sum_i a_{j,i} \geq c_\alpha$. The scale comparison (II.9.N03) bounds the left side by $2 \log_2 A$. Rearranging (II.12.5.21) gives

$$\sum_{j=1}^N \frac{1}{\log K_j \log \log K_j} \leq C_\alpha \left(1 + \frac{\sum_{j=1}^N \log K_j}{\log A} \right).$$

Because $\log A = n_1 \log(1/\lambda)$, this is (II.12.5.14) after absorbing the fixed factor into $C(\lambda, \alpha)$. $\square$

For $l, n \geq 1$, let $\mathcal{P}_l^{(n)}$ be the integer polynomials of degree $< n$ whose coefficients have modulus at most l . Let

$$\mathcal{Q}^{(n)} = \left\{ P_1(X)Y_1 + P_2(X)Y_2 : P_i \in \mathcal{P}_1^{(n)}, P_1Y_1 + P_2Y_2 = \sum_{j < n} (T_{u_j} - T_{v_j})X^j \right\}, \quad (\text{II.12.5.22})$$

where $u, v \in \{1, 2, 3\}^n$. Thus a labelled exact overlap occurs precisely when a nonzero $Q \in \mathcal{Q}^{(n)}$ satisfies $Q(\lambda, 1, \tau) = 0$.

For $l \geq 1$, let $\mathcal{A} \subset \mathbb{Z}[[X]]$ be the coefficientwise compact set of formal series whose coefficients have modulus at most l , and let

$$\mathcal{R}_l = \{P/Q : P, Q \in \mathcal{A}, Q \neq 0\}.$$

The quotient is first an element of $\mathbb{Q}((X))$ and, because the coefficients are bounded, also a meromorphic function on the open unit disk. We write $\mathcal{R} = \mathcal{R}_1$. Convergence in $\mathbb{Q}((X))$, the field of formal Laurent series, means convergence for the non-Archimedean norm

$$\left| \sum_{j \geq j_0} a_j X^j \right|_X = 2^{-j_0} \quad (a_{j_0} \neq 0), \quad |0|_X = 0;$$

on the bounded sets used here this is exactly coefficientwise convergence. An *admissible curve* is either

$$\gamma_R = \{(x, R(x)) : 0 < x < 1, R(x) \in \mathbb{R}\}, \quad R \in \mathcal{R},$$

or a vertical line $\gamma_{x_0} = \{x_0\} \times \mathbb{R}$. Its entropy rate is

$$h(\gamma) = \lim_{n \rightarrow \infty} n^{-1} H(A_Y^{(n)}), \quad (\text{II.12.5.23})$$

where $A_Y^{(n)}$ is the random function obtained by restricting $\sum_{j < n} T_{\xi_j}(1, y)x^j$ to γ . Two words are identified exactly when their difference polynomial vanishes identically on γ . Subadditivity proves existence of the limit.

II.12.5.2 Four elementary arithmetic estimates

For a polynomial $P = a_d \prod_{j=1}^d (X - z_j)$ write $M(P) = |a_d| \prod_j \max\{1, |z_j|\}$, and for an algebraic number z write $M(z)$ for the Mahler measure of its primitive minimal polynomial. When a statement below permits a transcendental real parameter z , set $M(z) = +\infty$; this is only a convention, not a finite-height assertion.

Lemma II.12.5.3 (Root separation). *If distinct algebraic numbers z, z' are each zeros of a nonzero member of $\mathcal{P}_l^{(n)}$, then*

$$|z - z'| \geq 2^{-n-1} n^{-5n} l^{-4n}.$$

Proof. If z, z' are conjugate, take their square-free minimal polynomial; if not, take the square-free part of the product of their minimal polynomials. The resulting integer polynomial P has degree $d \leq 2n$, contains both roots, and

$$M(P) \leq M(E)M(E') \leq \|E\|_1 \|E'\|_1 \leq n^2 l^2.$$

We prove the required discriminant estimate directly. Write the roots of P as $\alpha_1, \dots, \alpha_d$, let a be its leading coefficient, and put $A_i = \max\{1, |\alpha_i|\}$ and $M = |a| \prod_i A_i = M(P)$. The Vandermonde determinant $V = \det(\alpha_i^{k-1})_{1 \leq k, i \leq d}$ satisfies

$$|a|^{2d-2} |V|^2 = |\text{disc}(P)| \geq 1,$$

because P is square-free and its discriminant is a nonzero integer. Fix two roots, relabelling them so that $A_1 \leq A_2$, and subtract the first Vandermonde column from the second. The first column has Euclidean norm at most $\sqrt{d} A_1^{d-1}$, each remaining ordinary column has norm at most $\sqrt{d} A_i^{d-1}$, and the difference column has norm at most

$$|\alpha_2 - \alpha_1| d^{3/2} A_2^{d-2},$$

since $|\alpha_2^k - \alpha_1^k| \leq |\alpha_2 - \alpha_1| k A_2^{k-1}$. Hadamard's determinant inequality consequently gives

$$1 \leq |a|^{d-1} |V| \leq |\alpha_2 - \alpha_1| d^{(d+2)/2} M^{d-1}.$$

Hence every distinct pair satisfies

$$|\alpha_i - \alpha_j| \geq d^{-(d+2)/2} M(P)^{-(d-1)}.$$

Now $d \leq 2n$ and $M(P) \leq n^2 l^2$, so the last bound is at least $2^{-n-1} n^{-5n} l^{-4n}$, as asserted. $\square$

II.12.5.3 Rational translations

Let $a_1, \dots, a_s$ be distinct integers, let $p_i > 0$, and let D be the random digit with $\mathbb{P}(D = a_i) = p_i$. Denote by μ_λ^D the law of $\sum_{j \geq 0} D_j \lambda^j$, and by $h_D(\lambda)$ the entropy rate of its prefix sums.

Lemma II.12.5.4 (Mahler measure controls arbitrary finite-digit entropy). *There is a nondecreasing function $\Phi_D : [1, \infty) \rightarrow [0, H(D)]$ with $\Phi_D(u) \rightarrow H(D)$ as $u \rightarrow \infty$ such that every algebraic $0 < \eta < 1$ satisfies*

$$h_D(\eta) \geq \Phi_D(M(\eta)/L_D), \quad L_D = \max_{i,j} |a_i - a_j|. \quad (\text{II.12.5.24})$$

Proof. The proof of the [entropy–Mahler theorem](#) works with the binary sign replaced by the nonconstant finite random variable D . We include the two points at which this replacement matters. Let $\eta_1, \dots, \eta_q$ be the conjugates of η in the open unit disk and let A_0 be the real block-diagonal matrix with those eigenvalues. The quadratic-form construction in that theorem conjugates A_0 to a contraction A whose singular values are

$$1, \dots, 1, \tilde{M}(\eta)^{-1}, \quad \tilde{M}(\eta) = \prod_{|\eta_j| < 1} |\eta_j|^{-1}.$$

If no nonzero polynomial with coefficients in the difference set $\{a_i - a_j\}$ vanishes at η , all prefixes are distinct and $h_D(\eta) = H(D)$. Otherwise the primitive minimal polynomial divides such a polynomial after its initial and terminal zero coefficients are removed. Its

nonzero constant coefficient consequently has modulus at most L_D , and the product formula gives

$$\tilde{M}(\eta) = M(\eta)/|a_0| \geq M(\eta)/L_D. \quad (\text{II.12.5.25})$$

For a cyclic vector x , equality of two scalar prefix sums at η implies equality of their vector evaluations at A . The Gaussian entropy comparison from Chapter II.9, exactly as in (II.9.N70)–(II.9.N72), yields

$$h_D(\eta) \geq h_{\text{diff}}(t\tilde{M}(\eta)D + G) - h_{\text{diff}}(tD + G)$$

for every $t > 0$. Define the supremum of the right side with $\tilde{M}$ replaced by u to be $\Phi_D(u)$. Covariance monotonicity from the same lemma makes it nondecreasing. Taking $t = u^{-1/2}$, the first Gaussian mixture has signal size $u^{1/2}$ and the second has signal size $u^{-1/2}$. More precisely, for every $c > 0$,

$$h_{\text{diff}}(cD + G) = h_{\text{diff}}(G) + I(D; cD + G).$$

Dominated convergence for the finite Gaussian-mixture densities gives $I(D; cD + G) \rightarrow 0$ as $c \downarrow 0$. Since the support values of D are distinct, nearest-centre decoding has error tending to zero as $c \rightarrow \infty$; Fano's inequality gives $I(D; cD + G) \rightarrow H(D)$. Hence the value at $t = u^{-1/2}$ tends to $H(D)$. Conversely it is at most $H(D)$ because revealing D can increase differential entropy by at most $H(D)$. Therefore $\Phi_D(u) \rightarrow H(D)$. Combine this with (II.12.5.25). $\square$

Lemma II.12.5.5 (Finite-digit algebraic approximation). *If*

$$\dim \mu_\lambda^D < \min\{1, H(D)/\log \lambda^{-1}\},$$

then for every $\varepsilon > 0$ and N there are $n \geq N$ and an algebraic $\eta \in (0, 1)$ such that

$$|\lambda - \eta| < e^{-n^2}, \quad \deg \eta < n, \quad \dim \mu_\eta^D < \dim \mu_\lambda^D + \varepsilon, \quad (\text{II.12.5.26})$$

and η is a zero of a degree- $< n$ polynomial with coefficients in $\{a_i - a_j\}$.

Proof. Put

$$d = \dim \mu_\lambda^D, \quad \chi = \log \lambda^{-1}, \quad s = \min\{1, H(D)/\chi\},$$

and choose $0 < \delta < \varepsilon/4$ with

$$3\delta < \min\{1, \chi\}(s - d). \quad (\text{II.12.5.27})$$

Write $v_\theta^{(n)}$ for the n -digit law at parameter θ . We first establish the two algebraic facts used in the recursion.

Low entropy gives a common nearby root. Fix a compact interval $J \subset (0, 1)$ containing λ . There is $C_{\text{low}} > 1$ such that, for all large n , all $\theta \in J$, and $0 < r < n^{-C_{\text{low}}n}$,

$$\mathcal{H}(v_\theta^{(n)}; r) < nH(D) \quad (\text{II.12.5.28})$$

implies the existence of a real algebraic $\eta \in (0, 1)$ satisfying

$$|\theta - \eta| < r^{1/C_{\text{low}}}, \quad H(v_\eta^{(n)}) \leq \mathcal{H}(v_\theta^{(n)}; r), \quad (\text{II.12.5.29})$$

where η is a zero of a nonzero degree- $< n$ polynomial with coefficients in $\{a_i - a_j\}$.

Indeed, choose a grid translation t whose quantized entropy is at most the averaged entropy, and let $\mathcal{A}$ be the nonzero difference polynomials P of colliding word pairs. Equation (II.12.5.28) makes $\mathcal{A}$ nonempty and gives $|P(\theta)| \leq r$ for every $P \in \mathcal{A}$. Remove any initial power of X from P . Jensen's formula on a fixed disk containing J shows that P has at most $C_J \log(L_D + 2)$ zeros in a slightly larger disk, while all remaining zeros stay a fixed distance from θ . Factoring P therefore gives a zero η_P with $|\theta - \eta_P| \leq r^{c_{J,D}}$. Choose C_{low} so large that $c_{J,D} > 1/C_{\text{low}}$ and $c_{J,D}C_{\text{low}} > 10 + 4 \log(L_D + 2)$. Then this bound may be weakened to $r^{1/C_{\text{low}}}$, while for $P, Q \in \mathcal{A}$ the stronger bound gives

$$|\eta_P - \eta_Q| \leq 2r^{c_{J,D}} < 2n^{-c_{J,D}C_{\text{low}}n}.$$

The explicit lower bound in Lemma II.12.5.3 forces all η_P to be one common root η . Since $\bar{\eta}$ is another common root at the same distance from the real number θ , the same separation makes η real; for large n it belongs to $(0, 1)$. Every collision of the chosen quantized variable is now an exact collision at η , proving the entropy inequality in (II.12.5.29).

A nearby algebraic point forces maximal fine entropy. There is $C_{\text{near}} > 1$ such that, for large n , if η is a root of a nonzero difference polynomial of degree $< n$ and $|\lambda - \eta| < n^{-C_{\text{near}}n}$, then

$$\mathcal{H}(v_\lambda^{(n)}; r) = nH(D) \quad (0 < r < |\lambda - \eta|^{C_{\text{near}}}). \quad (\text{II.12.5.30})$$

For if strict inequality held, the preceding paragraph would give another root η' with $|\lambda - \eta'| < r^{1/C_{\text{low}}} < |\lambda - \eta|$, after taking $C_{\text{near}} > C_{\text{low}}$. Both roots lie within $2n^{-C_{\text{near}}n}$ of each other, so Lemma II.12.5.3 gives $\eta' = \eta$, contradicting the strict distance inequality.

For later use, subadditivity of prefix entropy gives the elementary bound

$$\dim \mu_\eta^D \leq \frac{H(v_\eta^{(n)})}{n \log \eta^{-1}}. \quad (\text{II.12.5.31})$$

Indeed, $H(v_\eta^{(kn)}) \leq kH(v_\eta^{(n)})$, while the tail comparison at scale η^{kn} and exact dimensionality identify the limit of the left side with the entropy dimension.

We now run the multiscale contradiction. Suppose there is n_0 such that no $m \geq n_0$ admits an η satisfying (II.12.5.26) with ε replaced by 3δ . Lemma II.12.5.1 supplies a deficit $\alpha > 0$. Put

$$\beta = \frac{\delta \min\{1, \chi\}}{\log 2}, \quad \alpha_* = \min\{\alpha, \beta, 1/4\},$$

and let $C_0 = C(\lambda, \alpha_*)$ be the constant in Proposition II.12.5.2. Choose $B, L > 1$ so that

$$B \geq 2C_{\text{near}}, \quad L \geq 2B \max\{C_{\text{low}}, C_{\text{near}}\},$$

and take n_0 sufficiently large for every preceding estimate.

Starting with n_0 , define

$$q_j = \left\lceil \frac{Ln_j \log n_j}{\chi} \right\rceil.$$

If

$$\mathcal{H}(v_\lambda^{(q_j)}; q_j^{-Lq_j}) \geq q_j(d + 2\delta)\chi, \quad (\text{II.12.5.32})$$

set $n_{j+1} = q_j$. By the [natural-scale approximation](#) and [homogeneous flattening theorem](#),

$$\mathcal{H}(v_\lambda^{(m)}; \lambda^{10m}) = dm\chi + o(m), \quad (\text{II.12.5.33})$$

so (II.12.5.32) gives a reservoir of at least $\delta\chi n_{j+1}$ between $n_{j+1}^{-Ln_{j+1}}$ and $\lambda^{10n_{j+1}}$.

If (II.12.5.32) fails, then (II.12.5.27) makes its left side smaller than $q_j H(D)$, and (II.12.5.29) gives a root η_j with

$$\rho_j := |\lambda - \eta_j| < q_j^{-Bq_j}, \quad H(v_{\eta_j}^{(q_j)}) < q_j(d + 2\delta)\chi. \quad (\text{II.12.5.34})$$

For large q_j , (II.12.5.31) and $\log \eta_j^{-1} = \chi + o(1)$ imply $\dim \mu_{\eta_j}^D < d + 3\delta$. The contrary assumption therefore gives

$$\rho_j > e^{-q_j^2}. \quad (\text{II.12.5.35})$$

Let n_{j+1} be maximal with $\rho_j < n^{-Bn}$. Then

$$(n_{j+1} + 1)^{-B(n_{j+1}+1)} \leq \rho_j < n_{j+1}^{-Bn_{j+1}}, \quad n_{j+1} \geq q_j.$$

Pad the polynomial vanishing at η_j with zero coefficients to degree $< n_{j+1}$. By (II.12.5.30) and the choice of L ,

$$\mathcal{H}(v_\lambda^{(n_{j+1})}; (n_{j+1} + 1)^{-L(n_{j+1}+1)}) = n_{j+1}H(D).$$

Changing the fine scale to $n_{j+1}^{-Ln_{j+1}}$ costs $O_L(\log n)$ by (II.9.N03). Subtracting (II.12.5.33) and using (II.12.5.27) gives a reservoir of at least δn_{j+1} for large n_0 .

Thus, with

$$K_j = \frac{L \log n_j}{\chi}$$

the sequences satisfy

$$n_{j+1} \geq K_j n_j, \quad \mathcal{H}(v_\lambda^{(n_j)}; \lambda^{K_j n_j} \mid \lambda^{10n_j}) \geq \beta n_j.$$

Use Proposition II.12.5.2 with α_* . Its remaining size hypotheses hold for large n_0 , and hence

$$\sum_{j=1}^J \frac{1}{\log K_j \log \log K_j} \leq C_0 \left(1 + \frac{1}{n_1} \sum_{j=1}^J \log K_j \right). \quad (\text{II.12.5.36})$$

In the low case, maximality and (II.12.5.35) imply $Bn_{j+1} \log n_{j+1} < q_j^2$; in the high case $n_{j+1} = q_j$. Thus $n_{j+1} < n_j^3$ for large n_0 , and $n_j < n_0^{3^j}$. If $j_0 = \lceil \log \log n_0 \rceil$, then

$$\log K_j \leq C(j + j_0), \quad \log \log K_j \leq 2 \log(j + j_0).$$

Consequently the left side of (II.12.5.36) is at least

$$c(\log^{(2)}(J + j_0) - \log^{(2)}(j_0 + 1)).$$

Choose J to be a sufficiently large fixed power of $j_0 + 1$. The last display then exceeds $2C_0$, while $J \leq 2 \log n_0$ and $n_1^{-1} \sum_{j \leq J} \log K_j < 1$ for large n_0 . This contradicts (II.12.5.36).

We have therefore obtained arbitrarily large n and algebraic η with $|\lambda - \eta| \leq e^{-n^2}$ and $\dim \mu_\eta^D < d + 3\delta < d + \varepsilon$. Its defining polynomial is a word-difference polynomial, completing the proof. $\square$

Theorem II.12.5.6 (Rational-translation dimension formula). *For distinct rational translations $a_1, \dots, a_s$ and positive weights p_i ,*

$$\dim \mu_\lambda^D = \min \left\{ 1, \frac{h_D(\lambda)}{\log \lambda^{-1}} \right\}. \quad (\text{II.12.5.37})$$

In particular, absence of exact overlaps implies the expected-dimension formula.

Proof. After a common rational dilation the translations are distinct integers; this bi-Lipschitz change does not affect dimension or prefix entropy.

Suppose first that λ is algebraic and put $\alpha = \dim \mu_\lambda^D$ and $\ell_n = \lfloor n \log \lambda^{-1} / \log 2 \rfloor$. The exact-dimensionality theorem, the [comparison with entropy dimension](#), and the prefix–tail decomposition give

$$\alpha \leq \min \left\{ 1, \frac{h_D(\lambda)}{\log \lambda^{-1}} \right\}. \quad (\text{II.12.5.38})$$

Indeed, at scale λ^n the infinite law is the convolution of $v_\lambda^{(n)}$ with a uniformly bounded dilate of μ_λ^D , so $\mathcal{H}(\mu_\lambda^D; \lambda^n) \leq H(v_\lambda^{(n)}) + O(1)$; divide by $n \log \lambda^{-1}$ and use subadditivity of prefix entropy.

If $\alpha = 1$, (II.12.5.38) is already equality. Suppose $\alpha < 1$. The [natural-scale approximation](#) and [homogeneous flattening theorem](#), applied to the present finite digit law with its given weights, say for every fixed $q > 1$ that

$$H_{\ell_n}(v_\lambda^{(n)}) = \alpha n \log \lambda^{-1} + o(n), \quad (\text{II.12.5.39})$$

$$H_{\lfloor q\ell_n \rfloor}(v_\lambda^{(n)}) - H_{\ell_n}(v_\lambda^{(n)}) = o(n). \quad (\text{II.12.5.40})$$

The difference of two distinct prefix values is a nonzero integer polynomial in λ , of degree $< n$ and coefficient ℓ^1 -norm at most nL_D . The [exponential algebraic separation lemma](#) therefore supplies $c_0 \in (0, 1)$, independent of n , such that any two distinct prefix values are at distance at least c_0^n . Choose q with $q \log \lambda^{-1} > \log c_0^{-1}$. For all large n , every dyadic cell at level $\lfloor q\ell_n \rfloor$ contains at most one distinct prefix value, and hence

$$H_{\lfloor q\ell_n \rfloor}(v_\lambda^{(n)}) = H(v_\lambda^{(n)}).$$

Equations (II.12.5.39)–(II.12.5.40) now yield $h_D(\lambda) = \alpha \log \lambda^{-1}$. This proves (II.12.5.37) for algebraic λ , without any restriction on the number of digits or on the weights.

Now suppose λ is transcendental and, toward a contradiction, the expected formula fails. Lemma II.12.5.5 gives arbitrarily large q and algebraic η with

$$|\lambda - \eta| < e^{-q^2}, \quad \dim \mu_\eta^D < \min\{1, H(D)/\log \eta^{-1}\} - \varepsilon.$$

The already proved algebraic case and Lemma II.12.5.4 imply a uniform bound $M(\eta) \leq M$.

Choose n by $(2M)^{-n-1} \leq |\lambda - \eta| < (2M)^{-n}$; then $n \gg q^2$. The [super-exponential cylinder concentration theorem](#), applied after the assumed dimension drop, supplies for every fixed $\theta > 0$ and all large n a nonzero $P \in \mathcal{P}_{L_D}^{(n)}$ with $|P(\lambda)| < \theta^n$. Take $\theta = (2M)^{-3K}$, where K is a large fixed integer. Lemma II.12.5.10, with $a = 2$, shows that P vanishes at η to order at least K : its degree hypothesis follows from $n \gg q \log q$, its upper-distance hypothesis from the definition of n , and its lower-distance hypothesis from

$$|\lambda - \eta| \geq (2M)^{-n-1} \geq (2M)^n |P(\lambda)|^{1/K}.$$

On the other hand, the Jensen calculation in Lemma II.12.5.8 bounds, uniformly in n , the number of zeros of a polynomial in $\mathcal{P}_{L_D}^{(n)}$ in a fixed disk $|z| \leq (1 + \lambda)/2$ by $C_{D,\lambda}$. For large q the number η lies in that disk. Choosing $K > C_{D,\lambda}$ is a contradiction. Thus no transcendental counterexample exists, and the theorem follows. $\square$

For the next theorem, let $\mathbb{F}$ be a countable characteristic-zero field and give $\mathbb{F}((X))$ the non-Archimedean norm $|\sum_{j \geq j_0} a_j X^j|_X = 2^{-j_0}$. For $R \in \mathbb{F}[[X]]$ put

$$A_R = \sum_{j \geq 0} T_{\xi_j}(1, R(X)) X^j, \quad A_R^{(n)} = \sum_{j < n} T_{\xi_j}(1, R(X)) X^j.$$

If $Z = \sum a_j X^j$, set $H_X(Z; n) = H(a_0, \dots, a_{n-1})$ and $\dim_X Z = \lim_n n^{-1} H_X(Z; n)$ whenever the limit exists.

Theorem II.12.5.7 (Function-field exact entropy formula). *For every $R \in \mathbb{F}[[X]]$,*

$$\dim_X A_R = h(R) := \lim_{n \rightarrow \infty} n^{-1} H(A_R^{(n)}). \quad (\text{II.12.5.41})$$

Consequently $R \mapsto h(R)$ is lower semicontinuous for coefficientwise convergence.

Proof. We first prove the prefix-separation assertion needed below. If no nonzero $Q \in \bigcup_n \mathcal{Q}^{(n)}$ satisfies $Q(X, 1, R(X)) = 0$, then for arbitrarily large n ,

$$H_X(A_R^{(n)}; 4n) = n \log 3. \quad (\text{II.12.5.42})$$

Otherwise, if $0 \neq Q \in \mathcal{Q}^{(N)}$ vanishes after substitution, then for every $n \geq N$,

$$H_X(A_R^{(n)}; 4n) \geq nh(R). \quad (\text{II.12.5.43})$$

For (II.12.5.42), suppose instead that a collision modulo X^{4n} occurs for every $n \geq N$. Write the corresponding $Q_n = P_{n,1}Y_1 + P_{n,2}Y_2$. Since $Q_n \neq 0$ and $|Q_n(X, 1, R)|_X \leq 2^{-4n}$, necessarily $P_{n,2} \neq 0$ and $|P_{n,2}|_X \geq 2^{-n}$. Hence

$$|P_{n,1}/P_{n,2} + R|_X \leq 2^{-3n}.$$

Two consecutive rational functions therefore satisfy

$$|P_{n,1}P_{n+1,2} - P_{n+1,1}P_{n,2}|_X \leq 2^{-3n}.$$

The polynomial on the left has degree $< 2n + 1$, so it is zero. All the quotients are equal and converge coefficientwise to $-R$, whence $R = -P_{N,1}/P_{N,2}$ and $Q_N(X, 1, R) = 0$, a contradiction.

For (II.12.5.43), let $Q = P_1Y_1 + P_2Y_2 \in \mathcal{Q}^{(N)}$ vanish and suppose $\tilde{Q} = \tilde{P}_1Y_1 + \tilde{P}_2Y_2 \in \mathcal{Q}^{(n)}$ satisfies $|\tilde{Q}(X, 1, R)|_X \leq 2^{-4n}$. As above,

$$|P_1\tilde{P}_2 - \tilde{P}_1P_2|_X \leq 2^{-3n}.$$

Its degree is $< n + N \leq 2n$, so it vanishes. Thus every collision modulo X^{4n} is an actual collision, and $H_X(A_R^{(n)}; 4n) = H(A_R^{(n)}) \geq nh(R)$.

Now put $d = \dim_X A_R$ and suppose $d < h(R)$; set $\varepsilon = h(R) - d$. By the preceding dichotomy there are arbitrarily large n for which

$$H_X(A_R^{(n)}; 4n) \geq nh(R). \quad (\text{II.12.5.44})$$

For every $k, n \geq 1$ one has

$$H_X(A_R^{(n)}; k | k-1) \leq \log 3. \quad (\text{II.12.5.45})$$

Indeed, for $l \geq \max\{n, k\}$, conditional-entropy concavity applied first to $A_R^{(l-k)} + X^{l-k}\tilde{A}_R^{(k)}$ and then to $A_R^{(n)} + X^n\tilde{A}_R^{(l-n)}$ shows

$$H_X(A_R; l | l-1) \geq H_X(A_R^{(n)}; k | k-1).$$

Averaging in l and using $d \leq \log 3$ proves (II.12.5.45).

Choose a selected n so large that $H_X(A_R^{(n)}; n) < n(d + \varepsilon/2)$. Then

$$\frac{1}{3n} \sum_{k=n}^{4n-1} H_X(A_R^{(n)}; k+1 | k) > \varepsilon/6.$$

Let $\mathcal{N}$ consist of those k for which the summand exceeds $\varepsilon/20$. The upper bound (II.12.5.45) gives

$$|\mathcal{N}|/(3n) > \varepsilon/(10 \log 3). \quad (\text{II.12.5.46})$$

Use the independent decomposition $A_R^{(4n)} = A_R^{(n)} + X^n\tilde{A}_R^{(3n)}$. On $k \in \mathcal{N}$, apply Lemma II.12.5.13 with parameter $\min\{\varepsilon/20, 1/\log 3\}$; on the other digits retain the entropy of the second summand by concavity. If $\delta > 0$ is the resulting gain, then

$$\frac{1}{3n} H_X(A_R^{(4n)}; 4n | n) \geq \frac{1}{3n} H_X(A_R^{(3n)}; 3n) + \frac{\varepsilon\delta}{10 \log 3}.$$

The first term tends to d , whereas direct telescoping and Lemma II.12.5.11 give

$$\lim_{n \rightarrow \infty} \frac{1}{3n} H_X(A_R^{(4n)}; 4n | n) = d.$$

This contradiction proves (II.12.5.41). Lower semicontinuity follows from Lemma II.12.5.11. $\square$

Lemma II.12.5.8 (A small value has a nearby zero). *For every $0 < \varepsilon < 1$ there is $c_\varepsilon > 0$ such that, if $l \geq 3$, $0 \neq P \in \mathcal{P}_l^{(n)}$, $\varepsilon \leq |z| \leq 1 - \varepsilon$, and $0 < r < \varepsilon^n 2^{-n}$ satisfies $|P(z)| \leq r$, then P has a zero η with*

$$|z - \eta| \leq (2^n \varepsilon^{-n} r)^{c_\varepsilon / \log l}.$$

Proof. After removing an initial power of X , assume $|P(0)| \geq 1$. Put $\rho = k/(k+1)$ and $a_k = \rho(k+1)^{-1/k}$. Jensen's formula and $|P(\rho e^{it})| \leq l/(1-\rho) = l(k+1)$ show that the number of zeros in $|w| \leq a_k$, counted with multiplicity, is at most

$$k \left(1 + \frac{\log l}{\log(k+1)} \right).$$

Choose $k = k(\varepsilon)$ so that $a_k > 1 - \varepsilon/2$. Hence at most $C_\varepsilon \log l$ zeros lie in that disk. All other zeros are at distance at least $\varepsilon/2$ from z . Factoring P and using that its leading coefficient is a nonzero integer gives

$$r \geq |P(z)| \geq (\varepsilon/2)^n \prod_{|\eta_j| \leq 1-\varepsilon/2} |z - \eta_j|.$$

One of the at most $C_\varepsilon \log l$ factors is at most the claimed quantity, after decreasing c_ε . $\square$

Lemma II.12.5.9 (Values at an algebraic point). *Let z be algebraic and $P \in \mathcal{P}_l^{(n)}$. Either $P(z) = 0$ or*

$$|P(z)| \geq (ln)^{-\deg z} M(z)^{-n}. \quad (\text{II.12.5.47})$$

If, additionally, z is a zero of a nonzero member of $\mathcal{P}_l^{(n)}$, then either $P(z) = 0$ or

$$|P(z)| \geq (ln)^{-2n}. \quad (\text{II.12.5.48})$$

Proof. We give a direct resultant proof, so no height inequality is an unproved premise here. Put $d = \deg z$, and write the primitive integral minimal polynomial of z as

$$f(X) = a \prod_{i=1}^d (X - z_i), \quad a \geq 1, \quad z_1 = z.$$

Let $t = \deg P < n$. Since $P(z) \neq 0$, the Sylvester resultant

$$R = \text{Res}(f, P) = a^t \prod_{i=1}^d P(z_i)$$

is a nonzero integer: its defining determinant has integral entries, and factoring f gives the displayed product identity. Thus $|R| \geq 1$. For every i ,

$$|P(z_i)| \leq \|P\|_1 \max\{1, |z_i|\}^t \leq ln \max\{1, |z_i|\}^t.$$

By the definition $M(z) = a \prod_i \max\{1, |z_i|\}$,

$$|P(z)| \geq (ln)^{-(d-1)} \left(a \prod_{i=2}^d \max\{1, |z_i|\} \right)^{-t} \geq (ln)^{-d} M(z)^{-n},$$

which is (II.12.5.47). The case of a nonzero constant P is immediate and satisfies the same bound.

If a nonzero $E \in \mathcal{P}_l^{(n)}$ vanishes at z , then $d \leq \deg E < n$. Gauss's lemma gives the primitive minimal polynomial f as an integral factor of E , up to a nonzero integer content. Mahler measure is multiplicative on polynomial products, every nonzero integral polynomial has Mahler measure at least one, and Jensen's formula gives

$$M(z) = M(f) \leq M(E) \leq \sup_{|w|=1} |E(w)| \leq \|E\|_1 \leq ln.$$

Substitution in the first bound gives $|P(z)| \geq (ln)^{-d-n} \geq (ln)^{-2n}$, as required. $\square$

Lemma II.12.5.10 (High-multiplicity criterion). *Let $\lambda, \eta \in [0, 1]$, let $n, n', l, k \geq 1$, and let $0 \neq P \in \mathcal{P}_l^{(n')}$. Suppose $\eta \neq \lambda$ is algebraic of degree at most n and*

$$\log a > \frac{(n(k+1) + k + 2) \log n' + (n+1) \log l + \log 2}{n'}.$$

If

$$(aM(\eta))^{n'/k} |P(\lambda)|^{1/k} \leq |\lambda - \eta| \leq (aM(\eta))^{-n'},$$

then P vanishes at η to order at least k .

Proof. Let o be the vanishing order and suppose $o < k$. For $\tilde{P} = P^{(o)}/o!$, Taylor's theorem gives $P(\lambda) = \tilde{P}(\xi)(\lambda - \eta)^o$ for a point ξ between η and λ . Since $\tilde{P} \in \mathcal{P}_{(n')^{o+2}l}^{(n')}$ and $|\tilde{P}'| \leq (n')^{o+2}l$ on $[0, 1]$,

$$|\tilde{P}(\eta)| \leq (n')^{o+2}l|\lambda - \eta| + \frac{|P(\lambda)|}{|\lambda - \eta|^o}.$$

The left side is nonzero. Lemma II.12.5.9 and $o < k$ make it at least $((n')^k ln')^{-n} M(\eta)^{-n'}$. The displayed lower bound on a turns this into

$$2(aM(\eta))^{-n'} < |\lambda - \eta| + \frac{|P(\lambda)|}{|\lambda - \eta|^k},$$

contradicting the two assumed bounds on $|\lambda - \eta|$. $\square$

II.12.5.4 Auxiliary entropy lemmas in the formal power-series field

We prove the three entropy lemmas invoked in Theorem II.12.5.7. All notation is as defined immediately before that theorem.

Lemma II.12.5.11 (Formal entropy monotonicity and semicontinuity). *For every $R \in \mathbb{F}[[X]]$,*

$$\frac{1}{n} H_X(A_R; n) \leq \frac{1}{n+1} H_X(A_R; n+1).$$

Consequently $\dim_X A_R$ exists, and $R \mapsto \dim_X A_R$ is lower semicontinuous for coefficient-wise convergence.

Proof. Let $\tilde{A}_R^{(1)}$ be an independent first digit. Conditional entropy is concave, so

$$H_X(A_R; k+1 | k) = H_X(\tilde{A}_R^{(1)} + XA_R; k+1 | k) \geq H_X(XA_R; k+1 | k) = H_X(A_R; k | k-1).$$

Thus the successive digit entropies are nondecreasing, and their first n averages are nondecreasing. For fixed n , $H_X(A_R; n)$ depends only on the first n coefficients of R , hence is continuous. The dimension is the supremum of these continuous averages and is therefore lower semicontinuous. $\square$

Lemma II.12.5.12 (Entropy growth in a torsion-free group). *For every $\varepsilon > 0$ there is $\delta > 0$ such that, for independent random variables U, V in a countable torsion-free abelian group,*

$$H(U) < \varepsilon^{-1}, \quad H(V) > \varepsilon \implies H(U + V) > H(U) + \delta.$$

Proof. We first record the Kaimanovich–Vershik inequality

$$H(U + V_1 + \cdots + V_N) - H(U) \leq N(H(U + V) - H(U)), \quad (\text{II.12.5.49})$$

where the V_i are independent copies of V . Indeed

$$H(S + V) - H(S) = I(V; S + V)$$

when S and V are independent. Replacing $S = U$ by $S = U + V_1 + \cdots + V_j$ adds independent noise to the observation $U + V_{j+1}$, so the data-processing inequality makes each later increment no larger than the first. Summing proves (II.12.5.49).

If W is uniform on two distinct group elements, torsion-freeness and the binomial bound imply $\max_x \mathbb{P}(W_1 + \cdots + W_N = x) = O(N^{-1/2})$; hence its entropy tends to infinity. Choose $N = N(\varepsilon)$ so that $H(W^{*N}) > \varepsilon^{-1} + 1$. Then (II.12.5.49) gives, uniformly for $H(U) < \varepsilon^{-1}$,

$$H(U + W) > H(U) + N^{-1}. \quad (\text{II.12.5.50})$$

Choose successively $0 < \rho \ll \varepsilon$ and $0 < \eta \ll \rho$. There are three cases. If V has two atoms of mass at least η , split its law as $2\eta W + (1 - 2\eta)\zeta$ and use entropy concavity with (II.12.5.50), obtaining gain $2\eta/N$. If no atom except possibly one has mass $> \eta$, while the largest atom is $< 1 - \rho$, condition off that atom; the remaining law has maximal atom at most η/ρ , hence entropy at least $\log(\rho/\eta)$. Concavity gives gain at least ρ once $\log(\rho/\eta) > \varepsilon^{-1} + 1$. Finally, if an atom has mass at least $1 - \rho$, write the law as the corresponding point mass plus its conditional complement. The binary entropy of this split is $o_{\rho \downarrow 0}(1)$, so $H(V) > \varepsilon$ forces the weighted entropy of the complement to exceed $\varepsilon/2$; concavity, and $(1 - \rho)H(U) \leq \rho/\varepsilon$, give gain at least $\varepsilon/4$. The minimum of these three positive gains is the required δ . $\square$

Lemma II.12.5.13 (One formal digit gains entropy). *For every $\varepsilon > 0$ there is $\delta > 0$ such that independent $\mathbb{F}[[X]]$ -valued U, V satisfy*

$$H_X(U; n | n-1) < \varepsilon^{-1}, \quad H_X(V; n | n-1) > \varepsilon$$

only if

$$H_X(U + V; n \mid n - 1) > H_X(U; n \mid n - 1) + \delta.$$

Proof. Condition both laws on their first $n - 1$ coefficients. On each pair of fibers, the last coefficients are independent $\mathbb{F}$ -valued random variables. Choose $0 < \eta \ll \varepsilon$. If fibers of V whose last coefficient entropy exceeds η have total mass $> \eta$, Markov's inequality makes fibers of U with entropy at most η^{-2} have mass $> 1 - \eta^2/\varepsilon$. Lemma II.12.5.12 on the product of these good fibers, followed by conditional-entropy concavity, gives a uniform gain. Otherwise the exceptional V -fibers have mass at most η but carry more than $\varepsilon - \eta$ total last-coefficient entropy. On them retain the entropy of V ; off them retain that of U . Concavity gives

$$H_X(U + V; n \mid n - 1) \geq (1 - \eta)H_X(U; n \mid n - 1) + \varepsilon - \eta,$$

which exceeds the U -entropy by a fixed positive amount when $\eta \ll \varepsilon$. Taking the smaller gain proves the lemma. $\square$

II.12.5.5 Low-entropy curves are repelled

For $0 < |z| < 1$ and $w \in \mathbb{C}$, let $\mu_{z,w}$ be the planar self-similar measure obtained from the three similarities $u \mapsto zu$, $u \mapsto zu + 1$, $u \mapsto zu + w$, and write $h(z, w)$ for its prefix entropy rate.

Lemma II.12.5.14 (Lower semicontinuity in complex parameters). *The map $(z, w) \mapsto \dim \mu_{z,w}$ is lower semicontinuous on $\{0 < |z| < 1\} \times \mathbb{C}$.*

Proof. Regard multiplication by z as a real planar similarity. The Feng–Hu formula from Volume I gives

$$\dim \mu_{z,w} = \frac{\log 3 - \Delta(z, w)}{\log |z|^{-1}}, \quad \Delta(z, w) = H_\beta(\mathcal{C}_1 \mid \Pi_{z,w}^{-1} \mathcal{B}(\mathbb{C})).$$

Fix (z_0, w_0) and translate all dyadic partitions of $\mathbb{C}$ so that every boundary has zero μ_{z_0, w_0} -mass. Coding maps converge uniformly as $(z, w) \rightarrow (z_0, w_0)$; hence each finite conditional entropy is continuous there. Martingale convergence expresses Δ as the decreasing infimum of those finite conditional entropies, so Δ is upper semicontinuous. The displayed quotient is therefore lower semicontinuous. $\square$

Lemma II.12.5.15 (Compactness away from poles). *Let $R_n \in \mathcal{R} \cap \mathbb{Q}[[X]]$ and suppose the graphs γ_{R_n} approach $(\lambda_0, \tau_0) \in (0, 1) \times \mathbb{R}$. If one complex neighborhood of λ_0 contains no pole of any R_n , then a subsequence converges coefficientwise and uniformly on a smaller neighborhood to $R \in \mathcal{R} \cap \mathbb{Q}[[X]]$, and $R(\lambda_0) = \tau_0$.*

Proof. Write $R_n = P_n/Q_n$ with $P_n, Q_n \in \mathcal{P}_1$ and $|Q_n|_X = 1$. Coefficientwise compactness gives a subsequence $P_n \rightarrow P$, $Q_n \rightarrow Q$. Formal division is continuous when $|Q_n|_X = 1$, so

$R_n \rightarrow P/Q$ coefficientwise. Choose a closed disk B about λ_0 on which no R_n has a pole and whose boundary contains no zero of Q . The bounded coefficients make P_n, Q_n converge uniformly on B . On ∂B , Q_n is eventually bounded away from zero, so $R_n \rightarrow P/Q$ uniformly there and, by the maximum principle, on B . Points of the graphs tending to (λ_0, τ_0) now force $(P/Q)(\lambda_0) = \tau_0$. $\square$

Lemma II.12.5.16 (A singular sequence produces an interval). *Suppose $R_n \in \mathcal{R}$ and the graphs γ_{R_n} approach (λ_0, τ_0) . If every complex neighborhood of λ_0 contains a pole of infinitely many R_n , then some nondegenerate real interval I satisfies*

$$\dim \mu_{\lambda_0, t} \leq \limsup_{n \rightarrow \infty} \frac{h(\gamma_{R_n})}{\log \lambda_0^{-1}} \quad (t \in I). \quad (\text{II.12.5.51})$$

Proof. Pass to a subsequence and choose real $\lambda_n \rightarrow \lambda_0$ with $R_n(\lambda_n) \rightarrow \tau_0$. Around λ_n choose shrinking closed disks B_n whose images $E_n = R_n(B_n)$ meet both circles of radii $|\tau_0| + 1$ and $|\tau_0| + 2$, truncating at the first radius $|\tau_0| + 2$ if necessary. The compact connected sets E_n have, in the Hausdorff metric, a compact connected limit E meeting both circles.

For $w \in E$ choose $z_n \in B_n$ with $R_n(z_n) \rightarrow w$. Then $z_n \rightarrow \lambda_0$, and the word-entropy bound gives

$$\dim \mu_{z_n, R_n(z_n)} \leq \frac{h(z_n, R_n(z_n))}{\log |z_n|^{-1}} \leq \frac{h(\gamma_{R_n})}{\log |z_n|^{-1}}.$$

Lemma II.12.5.14 proves the corresponding upper bound for $\dim \mu_{\lambda_0, w}$. Choose a real-linear map $S : \mathbb{C} \rightarrow \mathbb{R}$ which is the identity on $\mathbb{R}$ and is nonconstant on E . Then $I = S(E)$ is a nondegenerate interval, $S_* \mu_{\lambda_0, w} = \mu_{\lambda_0, S(w)}$, and Lipschitz maps do not increase dimension. This proves (II.12.5.51). $\square$

Lemma II.12.5.17 (Vertical entropy from an interval of small dimensions). *Let $0 < \lambda_0 < 1$ and $0 < h_0 < \log \lambda_0^{-1}$. If a nondegenerate interval I satisfies*

$$\dim \mu_{\lambda_0, t} \leq h_0 / \log \lambda_0^{-1} \quad (t \in I),$$

then the vertical curve γ_{λ_0} has $h(\gamma_{\lambda_0}) \leq h_0$.

Proof. If λ_0 is algebraic, choose algebraic $t \in I \setminus \mathbb{Q}(\lambda_0)$. A relation $P_1(\lambda_0) + tP_2(\lambda_0) = 0$ then forces both coefficients to vanish, so $h(\lambda_0, t) = h(\gamma_{\lambda_0})$. The algebraic part of the proof of Theorem II.12.5.6 applies verbatim to the fixed algebraic tuple (λ_0, t) : the [algebraic separation lemma](#) exponentially separates distinct prefix values, and the [natural-scale approximation](#) together with [homogeneous flattening](#) identifies dimension with prefix entropy rate. It gives

$$\dim \mu_{\lambda_0, t} = \min\{1, h(\gamma_{\lambda_0}) / \log \lambda_0^{-1}\}.$$

Since $h_0 < \log \lambda_0^{-1}$, the assumed dimension bound yields $h(\gamma_{\lambda_0}) \leq h_0$.

If λ_0 is transcendental, choose rational $t \in I \setminus \{0, 1\}$. After a rational dilation the translations $0, 1, t$ are distinct integers, so Theorem II.12.5.6 gives

$$\dim \mu_{\lambda_0, t} = \min\{1, \log 3 / \log \lambda_0^{-1}\}.$$

The dimension hypothesis and $h_0 < \log \lambda_0^{-1}$ imply $\log 3 \leq h_0$. Since every curve entropy is at most $\log 3$, the conclusion follows. $\square$

Proposition II.12.5.18 (Repulsion of low-entropy curves). *Let (λ_0, τ_0) have no exact overlaps. For every*

$$h_0 < \min\{\log 3, \log \lambda_0^{-1}\}$$

there is a neighborhood U of (λ_0, τ_0) which no admissible curve γ with $h(\gamma) \leq h_0$ meets.

Proof. Suppose otherwise. First assume vertical curves γ_{λ_n} of entropy at most h_0 approach the parameter. For every real t , the word-entropy bound gives $\dim \mu_{\lambda_n, t} \leq h_0 / \log \lambda_n^{-1}$. Lower semicontinuity and Lemma II.12.5.17 yield $h(\gamma_{\lambda_0}) \leq h_0 < \log 3$. Hence a nonzero word-difference polynomial vanishes identically on that vertical line, producing an exact overlap at (λ_0, τ_0) , a contradiction.

We may therefore take nonvertical curves γ_{R_n} . Absence of exact overlaps implies $\tau_0 \notin \{0, 1\}$, so in particular $\tau_0 \neq 0$. Replacing R_n by R_n^{-1} and interchanging the two translation coordinates when necessary, we may assume $R_n \in \mathbb{Q}[[X]]$; this operation preserves curve entropy. If the functions are eventually pole-free in a common neighborhood, Lemma II.12.5.15 gives a coefficientwise limit R with $R(\lambda_0) = \tau_0$. Theorem II.12.5.7 gives $h(R) \leq \liminf h(R_n) \leq h_0 < \log 3$. Thus a nonzero difference polynomial vanishes on γ_R , again giving an exact overlap at the base point.

In the remaining case Lemma II.12.5.16 and the bound $h(\gamma_{R_n}) \leq h_0$ give an interval to which Lemma II.12.5.17 applies. The resulting low-entropy vertical curve again produces an exact overlap at the base point. All cases contradict the hypothesis. $\square$

II.12.5.6 Algebraic approximation in the two-parameter family

For $n \geq 1$, let $\mathcal{X}^{(n)}$ be the set of pairs $(\eta, \sigma) \in (0, 1) \times \mathbb{R}$ for which there are

$$Q = P_1 Y_1 + P_2 Y_2, \quad \tilde{Q} = \tilde{P}_1 Y_1 + \tilde{P}_2 Y_2 \quad \text{in } \mathcal{Q}^{(n)}$$

such that both vanish at $(\eta, 1, \sigma)$, $P_2(\eta) \neq 0$, and $P_1 \tilde{P}_2 - P_2 \tilde{P}_1 \neq 0$. Then η is a zero of a nonzero member of $\mathcal{P}_{2n}^{(2n)}$ and $\sigma = -P_1(\eta)/P_2(\eta)$.

Proposition II.12.5.19 (Common-zero trichotomy). *For every $0 < \varepsilon < 1/2$ there is $C > 1$ such that the following holds for all large n . Let $Q_i = P_{i,1} Y_1 + P_{i,2} Y_2 \in \mathcal{Q}^{(n)} \setminus \{0\}$, $1 \leq i \leq q$, let $(\lambda, \tau) \in [\varepsilon, 1 - \varepsilon] \times [-\varepsilon^{-1}, \varepsilon^{-1}]$, and let $0 < r < n^{-Cn \log n}$. If $|Q_i(\lambda, 1, \tau)| \leq r$ for every i , then one of the following holds:*

1. *there is $(\eta, \sigma) \in \mathcal{X}^{(n)}$ with $\|(\lambda, \tau) - (\eta, \sigma)\|_\infty \leq r^{1/(C \log n)}$ and every $Q_i(\eta, 1, \sigma) = 0$;*
2. *there is $\eta \in (0, 1)$ with $|\lambda - \eta| \leq r^{1/(C \log n)}$ and $P_{i,j}(\eta) = 0$ for all i, j ;*
3. *all rows have rank one over $\mathbb{Q}(X)$, $P_{i,1} P_{j,2} = P_{i,2} P_{j,1}$, and $|P_{i,2}(\lambda)| \geq r^{1/2}$ for at least one i .*

Proof. Let d be the rank over $\mathbb{Q}(X)$ of the $q \times 2$ matrix $(P_{i,j})$. Since no row is zero, $d \in \{1, 2\}$. If $d = 1$ and one second entry has modulus at least $r^{1/2}$, alternative 3 holds.

In every remaining case we produce a nonzero $E \in \mathcal{P}_{2n}^{(2n)}$ with $|E(\lambda)| \ll r^{1/2}$. If $d = 2$, take a nonzero 2×2 minor. The matrix at λ has bounded largest singular value and sends $(1, \tau)^T$ to a vector of norm $O(r)$, so its determinant is $O(r)$. If every $|P_{i,2}(\lambda)| < r^{1/2}$, take a nonzero second entry; if none exists, take the nonzero first entry in one row, whose value is then $O(r)$.

Lemma II.12.5.8, with coefficient bound $2n$, gives a zero η of E satisfying

$$|\lambda - \eta| \leq r^{C^{-1/2}/\log n}. \quad (\text{II.12.5.52})$$

The conjugate $\bar{\eta}$ obeys the same estimate, and Lemma II.12.5.3 makes η real; for large n it belongs to $(0, 1)$. For every $P \in \mathcal{P}_{2n}^{(2n)}$ with $|P(\lambda)| \ll r^{1/2}$, the derivative bound $|P'| = O_\varepsilon(n^3)$ and (II.12.5.52) give

$$|P(\eta)| \leq 2n^{-C^{1/2}n/3}.$$

Since η is itself a zero of a member of $\mathcal{P}_{2n}^{(2n)}$, the lower bound (II.12.5.48) forces $P(\eta) = 0$.

If all second entries were small, then $|P_{i,1}(\lambda)| \leq r + |\tau|r^{1/2}$, so every entry vanishes at η and alternative 2 holds. Otherwise $d = 2$. If every entry vanishes at η , alternative 2 again holds; choose a row, say the first, which is nonzero at η . We claim that

$$|P_{1,2}(\eta)| > n^{-C^{1/2}n/8}. \quad (\text{II.12.5.53})$$

Otherwise (II.12.5.48), with degree and coefficient bounds $2n$, forces $P_{1,2}(\eta) = 0$. The derivative bound used in (II.12.5.52) and $Q_1(\lambda, 1, \tau) = O(r)$ then give

$$|P_{1,1}(\eta)| \leq |P_{1,1}(\eta) - P_{1,1}(\lambda)| + |Q_1(\lambda, 1, \tau)| + |\tau| |P_{1,2}(\lambda) - P_{1,2}(\eta)| \leq n^{-C^{1/2}n/3}.$$

Another application of (II.12.5.48) forces $P_{1,1}(\eta) = 0$, contradicting the choice of the row. This proves (II.12.5.53). Put $\sigma = -P_{1,1}(\eta)/P_{1,2}(\eta)$. Every minor has value $O(r)$ at λ , hence vanishes at η by the preceding argument; therefore every $Q_i(\eta, 1, \sigma) = 0$, and $d = 2$ makes $(\eta, \sigma) \in \mathcal{X}^{(n)}$. Finally

$$\left| \tau + \frac{P_{1,1}(\lambda)}{P_{1,2}(\lambda)} \right| \leq r^{1/2},$$

where the same derivative estimate and (II.12.5.53) ensure $|P_{1,2}(\lambda)| \geq \frac{1}{2}n^{-C^{1/2}n/8} \geq r^{1/2}$. The quotient derivative bound, the lower bound on the two denominators, and (II.12.5.52) give $|\tau - \sigma| \leq r^{1/(C \log n)}$. Enlarging C also weakens the λ estimate to this exponent, proving alternative 1. $\square$

Proposition II.12.5.20 (Low scale entropy yields a point or a curve). *On the same compact parameter rectangle, for all sufficiently large n and $0 < r < n^{-Cn \log n}$, the inequality*

$$\mathcal{H}(\mu_{\lambda, \tau}^{(n)}; r) < n \log 3$$

implies one of the following:

1. some $(\eta, \sigma) \in \mathcal{X}^{(n)}$ lies within $r^{1/(C \log n)}$ of (λ, τ) and

$$H(A_{\eta, \sigma}^{(n)}) \leq \mathcal{H}(\mu_{\lambda, \tau}^{(n)}; r);$$

2. an admissible curve γ lies within the same distance and

$$H(A_\gamma^{(n)}) \leq \mathcal{H}(\mu_{\lambda, \tau}^{(n)}; r).$$

Proof. Choose a grid translation $t \in [0, 1]$ for which the entropy of $\lfloor r^{-1} A_{\lambda, \tau}^{(n)} + t \rfloor$ does not exceed the averaged entropy. Let $\mathcal{A}$ be the nonzero word-difference polynomials whose value at $(\lambda, 1, \tau)$ has modulus at most r . The strict entropy drop makes $\mathcal{A}$ nonempty. Apply Proposition II.12.5.19 to all of $\mathcal{A}$. In alternative 1, words colliding in the quantized variable also collide at (η, σ) , which gives conclusion 1. In alternative 2 they collide identically on the vertical curve $\{\eta\} \times \mathbb{R}$. In alternative 3 choose an index with large denominator; all collision polynomials vanish on $y = -P_{i,1}(x)/P_{i,2}(x)$, whose graph is within $r^{1/2}$ of the base point. These last two cases give conclusion 2, with the asserted entropy inequality because the curve partition is coarser than the chosen quantized partition. $\square$

Proposition II.12.5.21 (A closest algebraic point forces fine entropy). *Fix a no-exact-overlap parameter (λ, τ) in a compact rectangle. For every $\varepsilon > 0$ there is $C > 1$ such that, for all large n , if $(\eta, \sigma) \in \mathcal{X}^{(n)}$ and*

$$\|(\lambda, \tau) - (\eta, \sigma)\|_\infty \leq n^{-Cn},$$

then

$$\frac{\mathcal{H}(\mu_{\lambda, \tau}^{(n)}; r)}{n \log \lambda^{-1}} > \min\{1, \log 3 / \log \lambda^{-1}\} - \varepsilon$$

whenever $0 < r < \|(\lambda, \tau) - (\eta, \sigma)\|_\infty^{C \log n}$.

Proof. Put

$$s = \min\{1, \log 3 / \log \lambda^{-1}\}, \quad h_0 = (s - \varepsilon/2) \log \lambda^{-1}.$$

Choose the constant C in the statement larger than twice the constant C_0 supplied by Proposition II.12.5.20, and also large enough for the separation estimates below. Suppose the conclusion failed at a scale r . Then

$$\mathcal{H}(\mu_{\lambda, \tau}^{(n)}; r) \leq n(s - \varepsilon) \log \lambda^{-1} < n \log 3,$$

so the low-entropy proposition applies. If it produced a curve γ , subadditivity of its prefix entropies would give

$$h(\gamma) \leq n^{-1} H(A_\gamma^{(n)}) \leq (s - \varepsilon) \log \lambda^{-1} < h_0.$$

The curve is within $r^{1/(C_0 \log n)}$ of the base point. For large n this contradicts the fixed neighborhood furnished by Proposition II.12.5.18.

We must therefore be in the point alternative. It produces $(\tilde{\eta}, \tilde{\sigma}) \in \mathcal{X}^{(n)}$ with

$$\|(\lambda, \tau) - (\tilde{\eta}, \tilde{\sigma})\|_{\infty} \leq r^{1/(C_0 \log n)} < \|(\lambda, \tau) - (\eta, \sigma)\|_{\infty}^{C/C_0} < \|(\lambda, \tau) - (\eta, \sigma)\|_{\infty}.$$

Both first coordinates are zeros of members of $\mathcal{P}_{2n}^{(2n)}$ and are within $2n^{-Cn}$, so Lemma II.12.5.3 gives $\tilde{\eta} = \eta$. If the second coordinates differed, write each as a quotient of two members of $\mathcal{P}_1^{(n)}$. Their difference has a nonzero numerator $B \in \mathcal{P}_{2n}^{(2n)}$. Since $|\eta| \leq 1 - \varepsilon_1$ for some fixed $\varepsilon_1 > 0$, both denominators have modulus $O_{\varepsilon_1}(1)$ at η , and hence

$$0 < |B(\eta)| \ll n^{-Cn}.$$

But η is a zero of a member of $\mathcal{P}_{2n}^{(2n)}$, so (II.12.5.48), applied with degree bound $2n$ and coefficient bound $2n$, gives $|B(\eta)| \geq (4n^2)^{-4n}$, a contradiction for large C . Thus the two points coincide, contradicting strict closeness. $\square$

Theorem II.12.5.22 (Rapaport–Varjú algebraic approximation). *Suppose (λ, τ) has no exact overlaps and*

$$d := \dim \mu_{\lambda, \tau} < \min\{1, \log 3 / \log \lambda^{-1}\}.$$

For every $\varepsilon > 0$ and N there are $n \geq N$ and $(\eta, \sigma) \in \mathcal{X}^{(n)}$ such that

$$|\lambda - \eta|, |\tau - \sigma| \leq e^{-n^{1/\varepsilon}}, \quad (\text{II.12.5.54})$$

$$\frac{H(A_{\eta, \sigma}^{(n)})}{n \log \eta^{-1}} \leq d + \varepsilon, \quad (\text{II.12.5.55})$$

$$h(\gamma) \geq \min\{\log 3, \log \lambda^{-1}\} - \varepsilon \quad \text{for every admissible } \gamma \ni (\eta, \sigma). \quad (\text{II.12.5.56})$$

In particular, η is a zero of a nonzero member of $\mathcal{P}_{2n}^{(2n)}$ and $\sigma = P_1(\eta)/P_2(\eta)$ with $P_i \in \mathcal{P}_1^{(n)}$ and $P_2(\eta) \neq 0$.

Proof. Write

$$\chi = \log \lambda^{-1}, \quad s = \min\{1, \log 3 / \chi\}.$$

It is enough to prove the assertion with a smaller positive tolerance. We may therefore assume $\varepsilon < \frac{1}{2} \min\{1, \chi, s\chi\}$. Choose $0 < \delta < \varepsilon/4$ so that

$$3\delta < \min\{1, \chi\}(s - d). \quad (\text{II.12.5.57})$$

In particular

$$d + 2\delta < s - \delta/\chi. \quad (\text{II.12.5.58})$$

For $m \geq 1$ set

$$\mathcal{F}_m = \left\{ (\eta, \sigma) \in \mathcal{X}^{(m)} : \frac{H(A_{\eta, \sigma}^{(m)})}{m \log \eta^{-1}} \leq d + 3\delta \right\}.$$

Suppose, contrary to the desired conclusion, that for some m_0 ,

$$\|(\lambda, \tau) - (\eta, \sigma)\|_{\infty} > \exp(-m^{1/\delta}) \quad \text{for all } m \geq m_0, \quad (\eta, \sigma) \in \mathcal{F}_m. \quad (\text{II.12.5.59})$$

Lemma II.12.5.1 supplies $\alpha \in (0, 1/2)$ for $\mu_{\lambda, \tau}$. Put

$$c_\delta = \delta \min\{1, \chi\} / \log 2, \quad \alpha_* = \min\{\alpha, c_\delta, 1/4\},$$

and let $C_{\text{boot}} = C(\lambda, \alpha_*)$ be the constant in Proposition II.12.5.2. Let C_{low} be the constant in Proposition II.12.5.20, and let C_{near} be the constant in Proposition II.12.5.21 used with tolerance $\delta/(2\chi)$. Choose constants $B, L > 1$, in this order, so that

$$B \geq 2C_{\text{near}}, \quad L \geq 2B \max\{C_{\text{low}}, C_{\text{near}}\}, \quad (\text{II.12.5.60})$$

and enlarge them whenever one of the preceding propositions requires it. Finally choose n_0 so large that all those results apply, that $n_0 \geq m_0$, and that every $o(n)$ estimate below has its displayed bound.

Inductively construct $n_0 < n_1 < \dots$. Given n_j , put

$$q_j = \left\lceil \frac{Ln_j(\log n_j)^2}{\chi} \right\rceil. \quad (\text{II.12.5.61})$$

There are two cases. If

$$\mathcal{H}(\mu_{\lambda, \tau}^{(q_j)}; q_j^{-Lq_j \log q_j}) \geq q_j(d + 2\delta)\chi, \quad (\text{II.12.5.62})$$

set $n_{j+1} = q_j$. The [natural-scale approximation](#) and [fixed-ratio flattening theorem](#) give

$$\mathcal{H}(\mu_{\lambda, \tau}^{(m)}; \lambda^{10m}) = dm\chi + o(m). \quad (\text{II.12.5.63})$$

Subtracting (II.12.5.63) from (II.12.5.62) yields, for large n_0 ,

$$\mathcal{H}(\mu_{\lambda, \tau}^{(n_{j+1})}; n_{j+1}^{-Ln_{j+1} \log n_{j+1}} | \lambda^{10n_{j+1}}) \geq \delta\chi n_{j+1}. \quad (\text{II.12.5.64})$$

Suppose instead that (II.12.5.62) fails. By (II.12.5.58), its left side is strictly smaller than $q_j \log 3$, so Proposition II.12.5.20 applies at $r = q_j^{-Lq_j \log q_j}$. Its curve alternative would give an admissible curve within $q_j^{-(L/C_{\text{low}})q_j}$ and with

$$h(\gamma) \leq q_j^{-1} H(A_\gamma^{(q_j)}) < (d + 2\delta)\chi < s\chi - \delta.$$

For large n_0 this contradicts Proposition II.12.5.18, used with $h_0 = s\chi - \delta/2$. Hence the point alternative holds and gives $(\eta_j, \sigma_j) \in \mathcal{X}^{(q_j)}$ such that

$$\rho_j := \|(\lambda, \tau) - (\eta_j, \sigma_j)\|_\infty < q_j^{-Bq_j}, \quad H(A_{\eta_j, \sigma_j}^{(q_j)}) < q_j(d + 2\delta)\chi. \quad (\text{II.12.5.65})$$

Since $\eta_j \rightarrow \lambda$, the second inequality and $\log \eta_j^{-1} = \chi + o(1)$ imply $(\eta_j, \sigma_j) \in \mathcal{F}_{q_j}$. In particular $\rho_j > 0$ by the no-exact-overlap assumption, and (II.12.5.59) gives

$$\rho_j > \exp(-q_j^{1/\delta}). \quad (\text{II.12.5.66})$$

Let n_{j+1} be the largest integer m for which $\rho_j < m^{-Bm}$. Equation (II.12.5.65) gives $n_{j+1} \geq q_j$, while maximality gives

$$(n_{j+1} + 1)^{-B(n_{j+1}+1)} \leq \rho_j < n_{j+1}^{-Bn_{j+1}}. \quad (\text{II.12.5.67})$$

Padding the two defining word pairs with a common tail shows $\mathcal{X}^{(q_j)} \subset \mathcal{X}^{(n_{j+1})}$. Thus Proposition II.12.5.21 applies at n_{j+1} . By (II.12.5.67) and (II.12.5.60), the scale

$$r_* = (n_{j+1} + 1)^{-L(n_{j+1}+1) \log n_{j+1}}$$

is smaller than $\rho_j^{C_{\text{near}} \log n_{j+1}}$. Therefore

$$\mathcal{H}(\mu_{\lambda, \tau}^{(n_{j+1})}; r_*) > n_{j+1}(s\chi - \delta/2). \quad (\text{II.12.5.68})$$

The scale comparison (II.9.N03) changes r_* to $n_{j+1}^{-L n_{j+1} \log n_{j+1}}$ at a cost $O_L((\log n_{j+1})^2) = o(n_{j+1})$. Subtracting (II.12.5.63) and using (II.12.5.57) gives

$$\mathcal{H}(\mu_{\lambda, \tau}^{(n_{j+1})}; n_{j+1}^{-L n_{j+1} \log n_{j+1}} \mid \lambda^{10n_{j+1}}) \geq \delta n_{j+1}. \quad (\text{II.12.5.69})$$

Thus, in both cases, with

$$K_j = \frac{L(\log n_j)^2}{\chi}$$

we have, for $j \geq 1$,

$$n_{j+1} \geq K_j n_j, \quad \mathcal{H}(\mu_{\lambda, \tau}^{(n_j)}; \lambda^{K_j n_j} \mid \lambda^{10n_j}) \geq c_\delta n_j. \quad (\text{II.12.5.70})$$

Proposition II.12.5.2, with parameter α_* , has all its hypotheses once n_0 is large: (II.12.5.70) gives the growth and reservoir, $K_j \geq 10$, and $n_j \geq C_{\text{boot}}(\log K_j)^2$. Consequently, for every $J \geq 1$,

$$\sum_{j=1}^J \frac{1}{\log K_j \log \log K_j} \leq C_0 \left(1 + \frac{1}{n_1} \sum_{j=1}^J \log K_j \right), \quad (\text{II.12.5.71})$$

where C_0 depends only on the fixed parameter and δ .

It remains to contradict this estimate. In the high-entropy case, $n_{j+1} = q_j < n_j^{2/\delta}$ for large n_j . In the low-entropy case, (II.12.5.67) and (II.12.5.66) imply

$$B n_{j+1} \log n_{j+1} < -\log \rho_j < q_j^{1/\delta},$$

and hence again $n_{j+1} < q_j^{1/\delta} < n_j^{2/\delta}$ for large n_j . Inductively,

$$n_j < n_0^{(2/\delta)^j}. \quad (\text{II.12.5.72})$$

Put $j_0 = \lceil \log \log n_0 \rceil$. From (II.12.5.72) and the definition of K_j ,

$$\log K_j \leq C_\delta(j + j_0), \quad \log \log K_j \leq 2 \log(j + j_0) \quad (\text{II.12.5.73})$$

for large n_0 . Integral comparison therefore gives

$$\sum_{j=1}^J \frac{1}{\log K_j \log \log K_j} \geq c'_\delta (\log^{(2)}(J + j_0) - \log^{(2)}(j_0 + 1)). \quad (\text{II.12.5.74})$$

Choose

$$J = \left\lceil \exp^{(2)}(\log^{(2)}(j_0 + 1) + A) \right\rceil$$

with A so large that the right side of (II.12.5.74) exceeds $2C_0$. This J is a fixed power of $j_0 + 1$; hence $J \leq 2 \log n_0$ for large n_0 . Equation (II.12.5.73) then gives

$$\frac{1}{n_1} \sum_{j=1}^J \log K_j \leq \frac{C_\delta J(J + j_0)}{n_0} < 1.$$

The right side of (II.12.5.71) is therefore at most $2C_0$, contradicting (II.12.5.74). This proves the existence of arbitrarily large m and $(\eta, \sigma) \in \mathcal{F}_m$ with

$$\|(\lambda, \tau) - (\eta, \sigma)\|_\infty \leq \exp(-m^{1/\delta}).$$

Since $\delta < \varepsilon$ and $3\delta < \varepsilon$, this implies (II.12.5.54) and (II.12.5.55).

Finally apply Proposition II.12.5.18 with $h_0 = s\chi - \varepsilon$. All sufficiently large approximating points lie in the resulting neighborhood of (λ, τ) , so every admissible curve through one of them has entropy greater than h_0 . This proves (II.12.5.56). The description of $\mathcal{X}^{(m)}$ gives the algebraicity assertion; replacing P_1 by $-P_1$ removes the sign in the displayed quotient. $\square$

Corollary II.12.5.23 (Zero-dimensional exceptional set). *The set of no-exact-overlap parameters at which the expected-dimension formula fails has Hausdorff dimension zero.*

Proof. Let F_n be the set of all algebraic pairs with the description in the last sentence of Theorem II.12.5.22. Counting the choice of the root polynomial, the root, and the two quotient polynomials gives

$$|F_n| \leq 2n(4n + 1)^{2n} 3^{2n} \leq (5n)^{5n}.$$

For every N , Theorem II.12.5.22 with $\varepsilon = 1/2$ covers the exceptional set by the balls $\overline{B}(x, e^{-n^2})$, $n \geq N$, $x \in F_n$. For every $s > 0$ its s -dimensional Hausdorff content is at most

$$\sum_{n \geq N} (5n)^{5n} (2e^{-n^2})^s \longrightarrow 0 \quad (N \rightarrow \infty).$$

Thus every positive-dimensional Hausdorff content vanishes. $\square$

II.12.5.7 Jet entropy and the final contradiction

For $R \in \mathcal{R} \cap \mathbb{Q}[[X]]$, a point $x \in (0, 1)$ which is not a pole of R , and $K \geq 1$, define

$$B_{R,x,K}^{(n)} = (A_R^{(n)}(x), (A_R^{(n)})'(x), \dots, (A_R^{(n)})^{(K-1)}(x))$$

and

$$h(R, x, K) = \lim_{n \rightarrow \infty} n^{-1} H(B_{R,x,K}^{(n)}). \quad (\text{II.12.5.75})$$

The limit exists by subadditivity. The limiting law is the stationary measure of the affine maps

$$y \mapsto \Theta(x, K)y + v_i(R, x, K), \quad \Theta(x, K) = \begin{pmatrix} x & 0 & \cdots & 0 \\ 1 & x & \cdots & 0 \\ 0 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & K-1 & x \end{pmatrix}, \quad (\text{II.12.5.76})$$

where the coordinates of v_i are $T_i(1, R(x)), T_i(0, R'(x)), \dots, T_i(0, R^{(K-1)}(x))$.

Lemma II.12.5.24 (Equal-Lyapunov projection-entropy formula). *Let A be an invertible contraction of $\mathbb{R}^K$ satisfying*

$$\log \|A^n\| = -\chi n + o(n), \quad \log s_{\min}(A^n) = -\chi n + o(n).$$

For a finite affine IFS $Ay + t_i$ with Bernoulli weights, its stationary measure is exact dimensional and

$$\dim \nu = \frac{H(p) - H_\beta(\mathcal{C}_1 \mid \Pi^{-1}\mathcal{B}(\mathbb{R}^K))}{\chi}. \quad (\text{II.12.5.77})$$

In particular the dimension is lower semicontinuous when A, t_i vary inside this equal-Lyapunov class with χ varying continuously.

Proof. Let $\Omega = \Lambda^{\mathbb{N}}$, let $\beta = p^{\mathbb{N}}$, let $\mathcal{C}_n$ be the length- n cylinder partition, and let Π be the coding map. The conditional Shannon–McMillan theorem used in the proof of the Feng–Hu formula gives, for β -almost every ω ,

$$-\frac{1}{n} \log \frac{\beta([\omega]_n \cap \Pi^{-1}B(\Pi\omega, r))}{\beta(\Pi^{-1}B(\Pi\omega, r))} \rightarrow H(p) - H_\beta(\mathcal{C}_1 \mid \Pi^{-1}\mathcal{B}(\mathbb{R}^K)) \quad (\text{II.12.5.78})$$

after first letting $n \rightarrow \infty$ and then $r \downarrow 0$. This part of that proof is purely symbolic and is unchanged here.

We spell out the only geometric replacement. For each $\delta > 0$ and all large n ,

$$B(0, e^{-(\chi+\delta)n}) \subset A^n B(0, 1) \subset B(0, e^{-(\chi-\delta)n}).$$

Consequently the image of every level- n cylinder lies in a ball of radius $Ce^{-(\chi-\delta)n}$, and the inverse image under its affine map of a ball of radius $e^{-(\chi+\delta)n}$ has uniformly bounded diameter. The two ball–cylinder comparisons in the proof of that theorem, combined with (II.12.5.78), therefore give for $\nu = \Pi_*\beta$ and ν -almost every x

$$\frac{h_\pi}{\chi + \delta} \leq \underline{d}_\nu(x) \leq \bar{d}_\nu(x) \leq \frac{h_\pi}{\chi - \delta}, \quad h_\pi = H(p) - H_\beta(\mathcal{C}_1 \mid \Pi^{-1}\mathcal{B}).$$

Letting $\delta \downarrow 0$ proves exact dimensionality and (II.12.5.77).

For lower semicontinuity, translate the dyadic grids so their boundaries have zero mass at the base parameter. The coding maps converge uniformly, so conditional entropy

given each finite pulled-back grid is continuous. Their decreasing limit is the conditional entropy in (II.12.5.77), hence is upper semicontinuous. The quotient in that formula is lower semicontinuous. $\square$

Lemma II.12.5.25 (Jet dimension and entropy). *For the stationary jet law $\nu_{R,x,K}$,*

$$\dim \nu_{R,x,K} \leq h(R, x, K) / \log x^{-1}.$$

Proof. The Jordan matrix in (II.12.5.76) satisfies $\|\Theta^n\| = x^n e^{o(n)}$ and $s_{\min}(\Theta^n) = x^n e^{o(n)}$. Hence Lemma II.12.5.24 gives exact dimensionality. Let $\mathcal{D}_r$ be the partition into half-open K -cubes of side r . The stationary decomposition is

$$\nu_{R,x,K} = \nu_{R,x,K}^{(n)} * \Theta(x, K)^n \nu_{R,x,K}.$$

The second factor is supported in a set of diameter $O(\|\Theta^n\|) = x^n e^{o(n)}$ and hence meets only $e^{o(n)}$ cells of $\mathcal{D}_{x^n}$. Entropy of a discrete mixture and translation of a grid by at most one neighboring cell in each coordinate give

$$H(\nu_{R,x,K}; \mathcal{D}_{x^n}) \leq H(B_{R,x,K}^{(n)}) + o(n).$$

For an exact-dimensional measure, cube entropy divided by $\log r^{-1}$ tends to its dimension along any geometric scale by the [entropy-dimension identity](#). Therefore

$$\dim \nu_{R,x,K} \leq \liminf_n \frac{H(B_{R,x,K}^{(n)}) + o(n)}{n \log x^{-1}},$$

which is the claim. $\square$

Lemma II.12.5.26 (Finite multiplicity detects formal coefficients). *For every $0 < \varepsilon < 1/2$ and integers $N, l \geq 1$ there is K such that the following holds. If $x \in [\varepsilon, 1 - \varepsilon]$, $R \in \mathcal{R}_l \cap \mathcal{Q}[[X]]$ has no pole at x , and $P_1, P_2 \in \mathcal{A}$ satisfy*

$$(P_1 + P_2 R)^{(j)}(x) = 0 \quad (0 \leq j < K),$$

then $|P_1 + P_2 R|_X \leq 2^{-N}$.

Proof. Write $R = U/V$ with $U, V \in \mathcal{A}$ and $|V|_X = 1$. If the first nonzero coefficient of $P_1 + P_2 R$ occurred before N , then

$$F = X^{-m}(P_1 V + P_2 U) \in \mathbb{Z}[[X]], \quad m < N,$$

would have first nonzero coefficient of modulus at least one and coefficients bounded by $2Nl^2(j+1)$. Jensen's formula on a fixed circle inside the unit disk bounds, in terms of ε, N, l , the order of any zero of such an F in $[\varepsilon, 1 - \varepsilon]$. Since $P_1 + P_2 R = X^m F/V$, choosing K above that bound contradicts the assumed multiplicity. $\square$

Proposition II.12.5.27 (Stable jets retain formal entropy). *For $R_0 \in \mathcal{R} \cap \mathbb{Q}[[X]]$ and $\delta > 0$ there is K such that*

$$|R - R_0|_X \leq 2^{-K}, \quad x \in [\delta, 1 - \delta] \implies h(R, x, K) \geq h(R_0) - \delta,$$

whenever $R \in \mathcal{R} \cap \mathbb{Q}[[X]]$ has no pole at x .

Proof. Choose N so that $N^{-1}H_X(A_{R_0}; N) > h(R_0) - \delta$ and choose $K \geq N$ from Lemma II.12.5.26, with $\varepsilon = \delta$, coefficient bound $l = 2$, and this value of N . We give the symbolic conditional-entropy argument because the relation used in it is not a partition.

Let $\Omega = \{1, 2, 3\}^{\mathbb{N}}$ with Bernoulli law β , and let $\mathcal{D}_n$ be its length- n cylinder partition. Let $\mathcal{E}_n$ identify two sequences when their n -prefixes have the same first K derivatives at x ; then

$$H_\beta(\mathcal{E}_n) = H(B_{R,x,K}^{(n)}). \quad (\text{II.12.5.79})$$

For $\omega, \omega' \in \Omega$, write $\omega \sim_n \omega'$ if some continuations of their first n symbols have equal K -jets at a common finite level $m \geq n$, and put $S_n(\omega) = \{\omega' : \omega \sim_n \omega'\}$. This is a union of level- n cylinders, $S_{n+N}(\omega) \subset S_n(\omega)$, and every $\mathcal{E}_n$ -atom is contained in $S_n(\omega)$. Hence

$$H_\beta(\mathcal{E}_n) \geq \int -\log \beta(S_n(\omega)) d\beta(\omega). \quad (\text{II.12.5.80})$$

For each $D \in \mathcal{D}_n$, partition D by the formal coefficient block in positions $n, \dots, n + N - 1$ of $\sum_j T_{\omega_j}(1, R(X))X^j$; call this partition $\mathcal{F}_D$. Independence and the shift by X^n give

$$H_{\beta_D}(\mathcal{F}_D) = H_X(A_R; N). \quad (\text{II.12.5.81})$$

If $D, D' \in \mathcal{D}_n$ contain a related pair, their $\mathcal{F}$ -atoms have a canonical matching: related pairs at level $n + N$ must lie in matched atoms. Indeed, subtract two jet equalities realizing the relations. Their first n terms cancel, and after division by X^n the difference has the form $P_1 + P_2R$, with $P_i \in \mathcal{P}_2$, whose first K derivatives vanish at x . Lemma II.12.5.26 makes its first N formal coefficients zero. Thus the formal block on one side determines the block on the other, and symmetry gives a bijection.

Fix D and $F \in \mathcal{F}_D$. Let $E(F)$ be the union, over all level- n cylinders D' contained in $S_n(D)$, of the atom of $\mathcal{F}_{D'}$ matched with F . Then $S_{n+N}(\omega) \subset E(F)$ for $\omega \in F$, while the matched families partition $S_n(D)$; consequently

$$\sum_{F \in \mathcal{F}_D} \beta(E(F)) = \beta(S_n(D)).$$

The log-sum inequality, applied with $y_F = \beta(F)$ and $z_F = \beta(E(F))$, now gives

$$\begin{aligned} & \int_D -\log \frac{\beta(S_{n+N}(\omega))}{\beta(S_n(\omega))} d\beta(\omega) \\ & \geq \sum_{F \in \mathcal{F}_D} \beta(F) \log \frac{\beta(S_n(D))}{\beta(E(F))} \geq \beta(D) H_{\beta_D}(\mathcal{F}_D). \end{aligned} \quad (\text{II.12.5.82})$$

Sum (II.12.5.82) over D and then over $n = 0, N, \dots, (m-1)N$. Equations (II.12.5.80) and (II.12.5.81) yield

$$H(B_{R,x,K}^{(mN)}) \geq mH_X(A_R; N).$$

Finally $|R - R_0|_X \leq 2^{-K}$ and $K \geq N$ make the first N coefficients identical, so $H_X(A_R; N) = H_X(A_{R_0}; N)$. Divide by mN and let $m \rightarrow \infty$. $\square$

Theorem II.12.5.28 (Turán's localized power-sum estimate). *Let $n \geq 1$, $m \geq 0$, and let $z_1, \dots, z_n$ and $b_1, \dots, b_n$ be complex numbers with $z_1 \neq 0$. Order the points so that*

$$|z_1 - z_2| \leq \dots \leq |z_1 - z_n|.$$

Fix $0 < \delta_2 < \delta_1 < n/(m+n+1)$, and let h be the largest index for which $|z_1 - z_h| < \delta_2|z_1|$. If either $h = n$ or $|z_1 - z_{h+1}| > \delta_1|z_1|$, then some $j \in \{m+1, \dots, m+n\}$ satisfies

$$\left| \sum_{i=1}^n b_i z_i^j \right| \geq 2 \left(\frac{\delta_1 - \delta_2}{12e} \right)^n \left| \sum_{i=1}^h b_i \right| |z_1|^j.$$

This is the gap case of [Tur84, Theorem 10.2]; the following proof gives the precise constant needed here without invoking the general two-sided theorem.

Proof. Divide every z_i by z_1 and write $w_i = z_i/z_1$, so that $w_1 = 1$. Put $\Delta = \delta_1 - \delta_2$ and $\rho = (\delta_1 + \delta_2)/2$. We first suppose the w_i distinct. The gap assumption says that $|w_i - 1| < \delta_2$ for $i \leq h$, whereas $|w_i - 1| > \delta_1$ for $i > h$.

For a polynomial $F(z) = \sum_k f_k z^k$ write $\|F\|_1 = \sum_k |f_k|$. Define

$$P(z) = \prod_{i=h+1}^n \frac{z - w_i}{1 - w_i} = \prod_{i=h+1}^n \left(1 + \frac{z-1}{1-w_i} \right).$$

The empty product is one. Since $\delta_1 < 1$,

$$\|P\|_1 \leq \prod_{i>h} \frac{1 + |w_i|}{|1 - w_i|} \leq (3/\delta_1)^{n-h}. \quad (\text{II.12.5.83})$$

On $|z - 1| \leq \rho$, the function $g(z) = z^{-(m+1)}/P(z)$ is holomorphic. Moreover

$$\max_{|z-1|=\rho} |g(z)| \leq (1-\rho)^{-(m+1)} (2\delta_1/\Delta)^{n-h} \leq e^n (2\delta_1/\Delta)^{n-h} =: M. \quad (\text{II.12.5.84})$$

Indeed, $|1 + (z-1)/(1-w_i)| \geq 1 - \rho/\delta_1 = \Delta/(2\delta_1)$; also $\rho < \delta_1 < n/(m+n+1)$, so, with $x = m+1$,

$$(1-\rho)^{-x} < (1+n/x)^x \leq e^n.$$

Let R be the Newton interpolation polynomial of degree at most $h-1$ for g at $w_1, \dots, w_h$:

$$R(z) = \sum_{k=0}^{h-1} c_k \prod_{i=1}^k (z - w_i).$$

By the residue theorem, the Newton divided-difference coefficient is the following contour integral (its residues at the w_i give the usual interpolation formula):

$$c_k = \frac{1}{2\pi i} \int_{|\zeta-1|=\rho} \frac{g(\zeta)}{\prod_{i=1}^{k+1} (\zeta - w_i)} d\zeta, \quad |c_k| \leq \frac{M\rho}{(\Delta/2)^{k+1}}.$$

Because $|w_i| \leq 1 + \delta_2$ for $i \leq h$, the polynomial factor in the k th term has coefficient norm at most $(2 + \delta_2)^k \leq 3^k$. With $a = 6/\Delta \geq 6$,

$$\|R\|_1 \leq \frac{M\rho}{3} \sum_{r=1}^h a^r \leq \frac{M\rho}{3} \frac{a}{a-1} a^h \leq \frac{2M}{5} a^h. \quad (\text{II.12.5.85})$$

Thus $Q = PR$ has degree at most $n-1$, vanishes at every w_i with $i > h$, and satisfies $w_i^{m+1}Q(w_i) = 1$ for $i \leq h$. By (II.12.5.83)–(II.12.5.85),

$$\|Q\|_1 \leq \frac{2}{5} e^n (2\delta_1/\Delta)^{n-h} (3/\delta_1)^{n-h} (6/\Delta)^h = \frac{2}{5} (6e/\Delta)^n.$$

Writing $Q(z) = \sum_{k=0}^{n-1} q_k z^k$ now yields the exact identity

$$\sum_{i=1}^h b_i = \sum_{i=1}^n b_i w_i^{m+1} Q(w_i) = \sum_{k=0}^{n-1} q_k \sum_{i=1}^n b_i w_i^{m+1+k}.$$

Consequently at least one $j \in \{m+1, \dots, m+n\}$ obeys

$$\left| \sum_i b_i w_i^j \right| \geq \frac{5}{2} \left(\frac{\Delta}{6e} \right)^n \left| \sum_{i \leq h} b_i \right| \geq 2 \left(\frac{\Delta}{12e} \right)^n \left| \sum_{i \leq h} b_i \right|.$$

Multiplication by $|z_1|^j$ gives the assertion.

For coincident w_i , perturb $w_2, \dots, w_n$ to distinct points while keeping $w_1 = 1$. The strict inner and outer inequalities preserve the same cluster size and gap for sufficiently small perturbations. The finite maximum in the conclusion is continuous in the points, so passage to the limit completes the proof. $\square$

Lemma II.12.5.29 (Finite power-sum stability). *For every integer $M \geq 1$ and every $\varepsilon > 0$ there is $\delta > 0$ with the following property. If U, W are finite complex multisets, $|U|, |W| \leq M$, every member has modulus at least one, and*

$$\left| \sum_{u \in U} u^j - \sum_{w \in W} w^j \right| \leq \delta \quad (1 \leq j \leq 2M),$$

then $|U| = |W|$ and

$$\left| 1 - \frac{\prod_{u \in U} u}{\prod_{w \in W} w} \right| < \varepsilon.$$

Proof. Put

$$\eta = \min \left\{ \frac{1}{8M}, \frac{\varepsilon}{8Me} \right\}, \quad \rho_* = \frac{\eta}{2^{2M+1}}, \quad c_* = 2 \left(\frac{\rho_*}{24e} \right)^{2M},$$

and take $0 < \delta < c_*$. For each $w \in W$, one of the radii

$$\rho_w \in \{\eta/2^r : 1 \leq r \leq 2M+1\}$$

has the property

$$(B(w, 2\rho_w|w|) \setminus B(w, \rho_w|w|)) \cap (U \cup W) = \emptyset. \quad (\text{II.12.5.86})$$

Indeed, the displayed annuli have disjoint interiors and there are $2M+1$ of them, whereas $U \cup W$ has at most $2M$ members, counted with multiplicity.

Choose $W' \subset W$ greedily in decreasing order of $\rho_w|w|$, retaining w precisely when it has not already been covered by a retained ball $B(w', \rho_{w'}|w'|)$. The retained balls cover W . They are pairwise disjoint: if two met, and the first had the larger radius, the centre of the second would lie in the doubled first ball; by (II.12.5.86) it would then lie in the first ball, contrary to its retention.

Fix $w \in W'$. Apply Theorem II.12.5.28 with $m = 0$ to the at most $2M$ labelled members of $U \cup W$: attach coefficient $+1$ to a member of U , coefficient -1 to a member of W , take $z_1 = w$, and order the other labelled points by distance from w . Use

$$\delta_2 = \rho_w, \quad \delta_1 = \frac{3}{2}\rho_w.$$

Because $\eta \leq 1/(8M)$, the required inequality $\delta_1 < n/(n+1)$ holds. Condition (II.12.5.86) supplies the required gap after the cluster $B(w, \rho_w|w|)$. If the signed number of labelled points in this ball were nonzero, Turán's estimate and $|w| \geq 1$ would give, for some $1 \leq j \leq 2M$,

$$\left| \sum_{u \in U} u^j - \sum_{v \in W} v^j \right| \geq 2 \left(\frac{\rho_w}{24e} \right)^n \geq 2 \left(\frac{\rho_*}{24e} \right)^{2M} = c_*.$$

This contradicts the choice of δ . Consequently every retained ball contains equally many members of U and W .

The retained balls are disjoint and cover W , so $|W| \leq |U|$. Interchanging U and W gives the reverse inequality. Hence $|U| = |W|$, all members of both multisets lie in the retained balls, and we may choose a bijection $f : W \rightarrow U$ which stays inside each retained ball. If w and $f(w)$ lie in the ball centred at $w' \in W'$, then

$$|w - f(w)| < 2\rho_{w'}|w'| \leq \frac{2\eta}{1-\eta}|w| < 4\eta|w|.$$

Writing $f(w)/w = 1 + a_w$, we have $|a_w| < 4\eta$ and therefore

$$\left| 1 - \frac{\prod_{u \in U} u}{\prod_{w \in W} w} \right| \leq \prod_{w \in W} (1 + |a_w|) - 1 \leq e^{4M\eta} - 1 \leq 4Me^{1/2}\eta < \varepsilon.$$

This proves the assertion. □

Lemma II.12.5.30 (Compactness or derivative blow-up). *Fix $(\lambda_0, \tau_0) \in (0, 1) \times (\mathbb{R} \setminus \{0\})$. There is K_0 such that whenever $R_n \in \mathcal{R} \cap \mathcal{Q}[[X]]$, $x_n \rightarrow \lambda_0$, and $R_n(x_n) \rightarrow \tau_0$, either a subsequence converges coefficientwise to $R_0 \in \mathcal{R} \cap \mathcal{Q}[[X]]$ with $R_0(\lambda_0) = \tau_0$, or*

$$\limsup_n |R_n^{(j)}(x_n)| = \infty \quad \text{for some } 1 \leq j \leq K_0.$$

Proof. Write $R_n = P_n/Q_n$ with $P_n, Q_n \in \mathcal{P}_1$ and $|Q_n|_X = 1$, and pass to coefficientwise limits P, Q . The normalization gives $Q(0) \neq 0$. We first assume $P \neq 0$. Choose a disk $B(\lambda_0, \kappa)$ whose closure lies in the unit disk and which contains no zero of P or Q other than possibly λ_0 . Coefficientwise convergence is uniform on compact subdisks. Rouché's theorem therefore makes the numbers of zeros of P_n, Q_n in this disk eventually constant, say k_1, k_2 . Jensen's formula, applied on a slightly larger fixed disk, gives $k_1, k_2 \leq M = M(\lambda_0)$. Denote these zero multisets by U_n, W_n and factor

$$P_n(z) = F_{1,n}(z) \prod_{u \in U_n} (z - u), \quad Q_n(z) = F_{2,n}(z) \prod_{w \in W_n} (z - w).$$

The zero-free factors converge uniformly on the smaller disk to $P(z)/(z - \lambda_0)^{k_1}$ and $Q(z)/(z - \lambda_0)^{k_2}$, respectively, and $F_{2,n}$ is uniformly bounded away from zero there.

Assume that $R_n^{(j)}(x_n)$ is bounded for $0 \leq j \leq 2M$. Since $R_n(x_n) \rightarrow \tau_0 \neq 0$, induction in the identities for derivatives of $\log R_n$ shows that $(\log R_n)^{(j)}(x_n)$ is bounded for $1 \leq j \leq 2M$. The zero-free factors and all their derivatives are uniformly bounded, so logarithmic differentiation of the preceding factorization gives

$$\left| \sum_{u \in U_n} (x_n - u)^{-j} - \sum_{w \in W_n} (x_n - w)^{-j} \right| = O_j(1) \quad (1 \leq j \leq 2M). \quad (\text{II.12.5.87})$$

Every u, w tends to λ_0 . Choose $\kappa_n \downarrow 0$ slowly enough that

$$\left| \frac{\kappa_n}{x_n - u} \right|, \left| \frac{\kappa_n}{x_n - w} \right| \geq 1$$

for all local zeros and that multiplying the j th bound in (II.12.5.87) by κ_n^j makes it tend to zero. Lemma II.12.5.29, applied with error tending to zero, yields $k_1 = k_2$ and

$$\frac{\prod_{w \in W_n} (x_n - w)}{\prod_{u \in U_n} (x_n - u)} \rightarrow 1. \quad (\text{II.12.5.88})$$

It follows that

$$\tau_0 = \lim_n R_n(x_n) = \lim_n \frac{F_{1,n}(x_n)}{F_{2,n}(x_n)} = \frac{P(z)/(z - \lambda_0)^{k_1}}{Q(z)/(z - \lambda_0)^{k_1}} \Big|_{z=\lambda_0}.$$

Thus the coefficientwise limit $R_0 = P/Q$ has a removable singularity at λ_0 with value τ_0 , giving the first alternative.

If $P = 0$, choose $q \in \{-1, 1\}$ with $q + \tau_0 \neq 0$ and apply the preceding argument to $R_n + q = (P_n + qQ_n)/Q_n$, now with coefficient bound two. Its derivatives of positive order are those of R_n . The compact alternative would have coefficientwise limit equal to the constant q , but value $q + \tau_0$ at λ_0 , impossible. Hence a derivative of order at most a constant $K_0 = 2M$, enlarged to cover this coefficient bound, must be unbounded. This proves the dichotomy. $\square$

Lemma II.12.5.31 (A singular jet spreads to every real parameter). *Suppose $R_n \in \mathcal{R} \cap \mathcal{Q}[[X]]$, $x_n \rightarrow \lambda_0$, $R_n(x_n) \rightarrow \tau_0$, and $|R_n^{(K-1)}(x_n)| \rightarrow \infty$, while all lower derivatives remain bounded. Then*

$$\dim \mu_{\lambda_0, t} \leq \liminf_n \frac{h(R_n, x_n, K)}{\log \lambda_0^{-1}} \quad (t \in \mathbb{R}).$$

Proof. After a subsequence the top derivative is nonzero and has fixed sign. Choose invertible diagonal matrices Δ_n whose first diagonal entry is one, whose last is $R_n^{(K-1)}(x_n)$, and for which the absolute values of successive diagonal ratios tend to infinity. Conjugating the jet IFS by Δ_n^{-1} gives linear parts $\Delta_n^{-1}\Theta(x_n, K)\Delta_n \rightarrow \lambda_0 I$ and translation vectors converging to

$$(T_i(1, \tau_0), 0, \dots, 0, T_i(0, 1)).$$

For all large n these conjugated linear maps are contractions and remain in the equal-Lyapunov class. Their stationary measures are the bi-Lipschitz images $\Delta_n^{-1}\nu_{R_n, x_n, K}$, so their dimensions are unchanged. Lemmas II.12.5.24 and II.12.5.25 give, for the limiting self-similar measure ν ,

$$\dim \nu \leq \liminf_n \dim(\Delta_n^{-1}\nu_{R_n, x_n, K}) \leq \liminf_n \frac{h(R_n, x_n, K)}{\log x_n^{-1}} = \liminf_n \frac{h(R_n, x_n, K)}{\log \lambda_0^{-1}}.$$

For every t , the map $\pi_t(y) = y_1 + (t - \tau_0)y_K$ sends ν to $\mu_{\lambda_0, t}$. Lipschitz projection cannot increase dimension. $\square$

Proposition II.12.5.32 (Uniform lower bound for nearby jet entropies). *Let (λ_0, τ_0) have no exact overlaps. If*

$$h_0 < \min\{h(\lambda_0, \tau_0), \log \lambda_0^{-1}\},$$

then some K and some neighborhood U of (λ_0, τ_0) satisfy

$$h(R, x, K) > h_0$$

whenever $(x, R(x)) \in U$ and $R \in \mathcal{R}$ has no pole at x .

Proof. Otherwise choose $K_n \rightarrow \infty$ and counterexamples R_n, x_n tending to the base parameter. Since $\tau_0 \neq 0$, invert R_n and interchange the two translation coordinates if necessary so that $R_n \in \mathcal{Q}[[X]]$. Apply Lemma II.12.5.30. In the compact branch, fix $0 < \delta < h(R_0) - h_0$.

Proposition II.12.5.27 supplies a fixed jet order K_* . For large n , coefficientwise convergence gives $|R_n - R_0|_X \leq 2^{-K_*}$ and $K_n \geq K_*$. Since adjoining jet coordinates cannot decrease discrete entropy,

$$h(R_n, x_n, K_n) \geq h(R_n, x_n, K_*) \geq h(R_0) - \delta > h_0.$$

Here $h(\lambda_0, \tau_0) \leq h(R_0)$ because evaluation at the base point is a factor of the random function on the limit curve. This is a contradiction.

In the derivative-blow-up branch, pass to a subsequence and let j be the first positive derivative order which diverges. Put $K = j + 1$; all lower derivatives are bounded, and $K_n \geq K$ eventually. Monotonicity in the number of jet coordinates gives $h(R_n, x_n, K) \leq h(R_n, x_n, K_n) \leq h_0$. Lemma II.12.5.31 therefore gives $\dim \mu_{\lambda_0, t} \leq h_0 / \log \lambda_0^{-1}$ for every t . Lemma II.12.5.17 then makes the vertical curve entropy at most h_0 , whereas the base-point entropy is no larger than that curve entropy. This contradicts $h_0 < h(\lambda_0, \tau_0)$. $\square$

Lemma II.12.5.33 (Low three-symbol entropy bounds Mahler measure). *For every $\varepsilon > 0$ there is M_ε such that*

$$h(\eta, \sigma) < \frac{2}{3} \log 2 - \varepsilon \implies M(\eta) \leq M_\varepsilon$$

for algebraic $0 < \eta < 1$ and arbitrary σ .

Proof. Let $\tilde{\xi}_j$ be fair on the first two translations 0, 1. For $A \subset \{0, \dots, n-1\}$, condition the three-symbol word on the set of positions at which its symbol belongs to $\{1, 2\}$ and put symbol 3 elsewhere. If $Y_n = \sum_{j \in A} \tilde{\xi}_j \eta^j$, then the [entropy–Mahler theorem](#) and subadditivity give $H(Y_n) \geq n(\log 2 - \delta)$. Revealing the $n - |A|$ omitted fair digits costs at most $(n - |A|) \log 2$, and therefore

$$H\left(\sum_{j \in A} \tilde{\xi}_j \eta^j\right) \geq n(\log 2 - \delta) - (n - |A|) \log 2.$$

Entropy concavity now gives, whenever $\Phi(M(\eta)) \geq \log 2 - \delta$,

$$H(A_{\eta, \sigma}^{(n)}) \geq \sum_A (2/3)^{|A|} (1/3)^{n-|A|} ((\log 2 - \delta)n - \log 2(n - |A|)) = n \left(\frac{2}{3} \log 2 - \delta \right).$$

Let $n \rightarrow \infty$, take $\delta < \varepsilon$, and use $\Phi(u) \rightarrow \log 2$ to choose M_ε . $\square$

Conditional theorem II.12.5.34 (Global theorem conditional on the Mahler question). *The additional hypothesis below is precisely [RV24, Question 1.5]. Assume the following statement: for every $\varepsilon > 0$ there is $M < \infty$ such that, whenever*

$$(\lambda, \tau) \in (\varepsilon, 1 - \varepsilon) \times \mathbb{R}, \quad h(\lambda, \tau) \leq \min\{\log 3, \log \lambda^{-1}\} - \varepsilon$$

and every admissible curve through (λ, τ) has entropy at least $\min\{\log 3, \log \lambda^{-1}\} - M^{-1}$, one has $M(\lambda) \leq M$. Then every no-exact-overlap three-map system has the expected dimension.

Theorem II.12.5.35 (Unconditional Rapaport–Varjú range). *If $2^{-2/3} < \lambda < 1$ and the three-map system has no exact overlaps, then*

$$\dim \mu_{\lambda, \tau} = \min\{1, \log 3 / \log \lambda^{-1}\}.$$

The endpoint is not included.

Proof. We prove Theorem II.12.5.35, and Theorem II.12.5.34 under its stated hypothesis. Suppose the expected formula fails at (λ_0, τ_0) . The fully proved algebraic-contraction theorem II.12.3.12 shows that λ_0 is transcendental; also $\tau_0 \neq 0$ because otherwise two defining maps coincide. Choose

$$\dim \mu_{\lambda_0, \tau_0} \log \lambda_0^{-1} < h_0 < \min\{\log 3, \log \lambda_0^{-1}\}$$

and take K and a neighborhood U from Proposition II.12.5.32. Choose $r_U > 0$ such that the Euclidean ball $B((\lambda_0, \tau_0), r_U)$ lies in U . Increase K , if necessary, until $K^{-1} < r_U/2$. The proposition remains valid with the same U because adding jet coordinates cannot decrease their entropy rate.

Apply Theorem II.12.5.22 with tolerances tending to zero and with arbitrarily large lower cutoff. For the resulting $(\eta, \sigma) \in \mathcal{X}^{(n)}$,

$$|\lambda_0 - \eta| < e^{-n^2}, \quad h(\eta, \sigma) \leq \frac{1}{n} H(A_{\eta, \sigma}^{(n)}) \leq (d + o(1)) \log \eta^{-1} < h_0.$$

The first inequality is subadditivity of prefix entropy; the last uses $\eta \rightarrow \lambda_0$ and the strict choice of h_0 .

In the conditional theorem, first fix a positive entropy gap $\rho < \min\{\lambda_0, 1 - \lambda_0\}/3$ with $h_0 + 3\rho < \min\{\log 3, \log \lambda_0^{-1}\}$ and invoke the assumed Mahler statement, obtaining $M = M(\rho)$. Then choose the approximation tolerance smaller than both ρ and $(2M)^{-1}$. For large n , its curve clause gives every admissible curve through (η, σ) entropy at least $\min\{\log 3, \log \eta^{-1}\} - M^{-1}$, while the preceding display gives $h(\eta, \sigma) \leq \min\{\log 3, \log \eta^{-1}\} - \rho$. The assumed statement therefore gives $M(\eta) \leq M$. In the unconditional range, $\log \lambda_0^{-1} < \frac{2}{3} \log 2$; choose h_0 below $\frac{2}{3} \log 2$ and apply Lemma II.12.5.33 with the fixed gap $\frac{2}{3} \log 2 - h_0$ to obtain the same uniform bound. Increase it so that $M \geq 4$. Thus in either case

$$\deg \eta \leq 2n, \quad M(\eta) \leq M. \quad (\text{II.12.5.89})$$

Choose n' so that $M^{-4n'} < |\lambda_0 - \eta| < M^{-3n'}$; then $n' \gg_M n^2$. Choose once and for all

$$0 < c < \min\{\lambda_0, (2M)^{-6K}\}. \quad (\text{II.12.5.90})$$

The [natural-scale approximation](#) and the [fixed-ratio flattening theorem](#) show, for all large n' , that the averaged entropy of the n' -prefix at scale $c^{n'}$ is at most $h_0 n'$. Choose a grid translate attaining this average, and let $\mathcal{A}$ be the partition of words according to the corresponding interval of length $c^{n'}$ containing $\sum_{j < n'} T_{\omega_j}(1, \tau_0) \lambda_0^j$ satisfies

$$H_\beta(\mathcal{A}) \leq h_0 n'. \quad (\text{II.12.5.91})$$

Let $\tilde{\mathcal{Q}}$ contain all word-difference polynomials $Q = P_1Y_1 + P_2Y_2$ arising from pairs in one atom of $\mathcal{A}$.

Some such Q has $P_2(\eta) \neq 0$. Otherwise $|P_1(\lambda_0)| \leq c^{n'} + |\tau_0 P_2(\lambda_0)|$ and the mean-value theorem give

$$|P_1(\eta)| \leq c^{n'} + (1 + |\tau_0|)n'^2|\eta - \lambda_0|.$$

The choice (II.12.5.90), (II.12.5.89), and Lemma II.12.5.9 force $P_1(\eta) = 0$: a nonzero value is at least $(n')^{-2n}M^{-n'}$, whereas the preceding upper bound is $o((n')^{-2n}M^{-n'})$ because $n' \gg n^2$. Every collision polynomial would then vanish on the vertical curve through η , making that curve's entropy at most h_0 , contrary to the curve clause of Theorem II.12.5.22.

Fix $Q = P_1Y_1 + P_2Y_2$ with $P_2(\eta) \neq 0$ and put $R = -P_1/P_2$. The same value lower bound gives

$$|R(\eta) - \tau_0| \leq (n')^{4n+2}(2 + |\tau_0|)M^{-2n'} < K^{-1}. \quad (\text{II.12.5.92})$$

For any $\tilde{Q} = \tilde{P}_1Y_1 + \tilde{P}_2Y_2$ in $\tilde{\mathcal{Q}}$,

$$|(P_1\tilde{P}_2 - P_2\tilde{P}_1)(\lambda_0)| \leq n'c^{n'}.$$

If this cross-product is nonzero, it lies in $\mathcal{P}_{2n'}^{(2n')}$. Apply Lemma II.12.5.10 with $a = 2$, algebraic-degree bound $2n$, polynomial length $2n'$, coefficient bound $2n'$, and multiplicity K . The logarithmic hypothesis follows from $n' \gg n^2$; the upper distance bound follows from $M \geq 4$ and $|\lambda_0 - \eta| < M^{-3n'}$; and (II.12.5.90) gives, for large n' , the lower bound

$$(2M)^{2n'/K}(n'c^{n'})^{1/K} < M^{-4n'} < |\lambda_0 - \eta|.$$

Hence η is a zero of order at least K of every nonzero cross-product; a zero cross-product has this property trivially. Dividing by P_2 shows that $(\tilde{P}_1 + R\tilde{P}_2)^{(j)}(\eta) = 0$ for $0 \leq j < K$.

Thus any two words in one $\mathcal{A}$ -atom have identical K -jets at (R, η) , so (II.12.5.91) yields

$$H(B_{R,\eta,K}^{(n')}) \leq h_0n', \quad h(R, \eta, K) \leq h_0.$$

But (II.12.5.92) and $|\eta - \lambda_0| < e^{-n^2} < r_U/2$ put $(\eta, R(\eta))$ in the neighborhood from Proposition II.12.5.32, which asserts the strict opposite inequality. This contradiction proves both the unconditional theorem and the conditional implication. $\square$

II.12.6 Algebraic translations with arbitrary contraction

Here the translations are algebraic, while the common contraction is arbitrary: it may be negative or transcendental. The Feng–Feng argument combines height bounds, separation of collision roots, and an entropy–Mahler estimate. These ingredients and the final contradiction are proved below. The one research-scale result retained from the source is the multiscale algebraic-approximation theorem, which is stated with its exact hypotheses before use.

II.12.6.1 Parameters, collision polynomials, and height

Definition II.12.6.1 (Feng–Feng collision data and Garsia entropy). All logarithms and entropies in this subsection use the natural base. Fix $m \geq 2$, distinct real algebraic numbers $t_1, \dots, t_m$, and positive weights $p_1, \dots, p_m$ with sum one. Repeated translations, if originally present, are first combined by adding their weights. Put

$$\nu = \sum_{j=1}^m p_j \delta_{t_j}, \quad H(p) = - \sum_{j=1}^m p_j \log p_j, \quad \mathcal{D} = \{t_i - t_j : 1 \leq i, j \leq m\}.$$

For $n \geq 1$, let

$$\mathcal{P}_{\mathcal{D}}^{(n)} = \left\{ \sum_{r=0}^{n-1} a_r X^r : a_r \in \mathcal{D} \right\}.$$

If $\xi_0, \xi_1, \dots$ are independent with law ν , define

$$S_n(\eta) = \sum_{r=0}^{n-1} \xi_r \eta^r, \quad h_{\eta, \nu} = \lim_{n \rightarrow \infty} \frac{1}{n} H(S_n(\eta)).$$

The limit exists by subadditivity. A collision between two length- n words is exactly an identity $P(\eta) = 0$ for the corresponding $P \in \mathcal{P}_{\mathcal{D}}^{(n)}$. Hence all prefixes are distinct if no nonzero polynomial in $\bigcup_n \mathcal{P}_{\mathcal{D}}^{(n)}$ vanishes at η .

The case $m = 1$ is degenerate: the invariant measure is a point mass, $H(p) = 0$, and every dimension assertion below is immediate. Thus the standing assumption $m \geq 2$ loses no content.

For an algebraic number a , write $\deg a = [\mathbb{Q}(a) : \mathbb{Q}]$ and

$$H(a) = M(a)^{1/\deg a}$$

for its absolute multiplicative height. Here $M(a)$ is the Mahler measure of the primitive integral minimal polynomial, as in the definition in Chapter II.8. For a polynomial $F = \sum c_\alpha X^\alpha$ put $\|F\|_1 = \sum |c_\alpha|$.

Theorem II.12.6.2 (Height evaluation and lower bound). *Let $0 \neq F \in \mathbb{Z}[X_1, \dots, X_s]$ have partial degree at most L_i in X_i , and let $a_1, \dots, a_s$ be algebraic. Then*

$$H(F(a_1, \dots, a_s)) \leq \|F\|_1 \prod_{i=1}^s H(a_i)^{L_i}. \quad (\text{II.12.6.1})$$

If $0 \neq a$ belongs to a number field E , then

$$|a| \geq H(a)^{-[E:\mathbb{Q}]}. \quad (\text{II.12.6.2})$$

These inequalities are also recorded in [Mas16, Propositions 14.7 and 14.13].

Proof. We give the local-height calculation, isolating the sole local-field construction it uses. For each rational prime ℓ , define $|q|_\ell = \ell^{-v_\ell(q)}$ on $\mathbb{Q}^\times$ and let $\mathbb{Q}_\ell$ be its completion. Equip

a fixed algebraic closure $\overline{\mathbb{Q}}_\ell$ with the extension of this non-Archimedean absolute value. The existence and uniqueness of that extension over a complete field is the elementary local-field extension theorem; it is the only local-field input in this proof. For a number field E of degree D and $b \in E$, put

$$Q_E(b) = \prod_{\sigma: E \hookrightarrow \mathbb{C}} \max\{1, |\sigma b|\} \prod_{\ell \text{ prime}} \prod_{\tau: E \hookrightarrow \overline{\mathbb{Q}}_\ell} \max\{1, |\tau b|_\ell\}.$$

There are exactly D embeddings in each inner family, counted separately even when their restrictions to a subfield agree. Only finitely many prime factors differ from 1.

We claim that

$$Q_E(b) = H(b)^D. \quad (\text{II.12.6.3})$$

Let $d = [Q(b) : Q]$ and write the primitive minimal polynomial as $f(X) = c_d \prod_{j=1}^d (X - b_j)$ with $c_d > 0$. Each conjugate b_j occurs D/d times among the restrictions of the embeddings of E , both over $\mathbb{C}$ and over $\overline{\mathbb{Q}}_\ell$. The non-Archimedean Gauss norm $\|P\|_{\ell, G} = \max_k |X^k| P|_\ell$ is multiplicative. Indeed, after scaling two polynomials to Gauss norm 1, reduce their coefficients modulo the maximal ideal of the valuation ring: the two nonzero residue polynomials have nonzero product, so the product still has norm 1. Thus

$$1 = \|f\|_{\ell, G} = |c_d|_\ell \prod_{j=1}^d \max\{1, |b_j|_\ell\},$$

where the first equality uses primitiveness of f . The rational product formula, an immediate consequence of unique prime factorization, gives $\prod_\ell |c_d|_\ell^{-1} = c_d$. Multiplication over primes and then over the D/d repetitions of each conjugate proves (II.12.6.3), including $b = 0$.

Now take $E = Q(a_1, \dots, a_s)$ and put $b = F(a_1, \dots, a_s)$. At each complex embedding σ ,

$$\max\{1, |\sigma b|\} \leq \|F\|_1 \prod_i \max\{1, |\sigma a_i|\}^{L_i}.$$

At each ℓ -adic embedding τ , the ultrametric inequality and $|c_\alpha|_\ell \leq 1$ for integral coefficients give the same bound without $\|F\|_1$. Multiply all these inequalities and use (II.12.6.3); since there are D complex embeddings, the result is

$$H(b)^D \leq \|F\|_1^D \prod_i H(a_i)^{DL_i}.$$

Taking D -th roots proves (II.12.6.1).

Finally, for $0 \neq a \in E$, the field-norm identity and the rational product formula give

$$1 = \prod_{\sigma: E \hookrightarrow \mathbb{C}} |\sigma a| \prod_{\ell} \prod_{\tau: E \hookrightarrow \overline{\mathbb{Q}}_\ell} |\tau a|_\ell \leq |a| Q_E(a) = |a| H(a)^D,$$

where the distinguished complex embedding is the given inclusion $E \subset \mathbb{C}$. Rearrangement proves (II.12.6.2). $\square$

The algebraic-integer generator needed in the norm-polynomial construction is elementary, but its precise form is worth proving here rather than hiding it inside the height citation.

Lemma II.12.6.3 (Integral primitive element). *If $E/\mathbb{Q}$ is a finite extension, there is an algebraic integer $\theta \in E$ with $E = \mathbb{Q}(\theta)$. In particular, for every finite set $\mathcal{D} \subset E$, there is an integer $L \geq 1$ such that $L\mathcal{D} \subset \mathbb{Z}[\theta]$.*

Proof. First, every algebraic number β has a positive integer multiple which is an algebraic integer. Indeed, write its primitive integral minimal polynomial as

$$f(X) = aX^e + b_{e-1}X^{e-1} + \cdots + b_0, \quad a > 0.$$

Then $a\beta$ is a root of the monic integral polynomial

$$a^{e-1}f(X/a) = X^e + b_{e-1}X^{e-1} + ab_{e-2}X^{e-2} + \cdots + a^{e-1}b_0.$$

It remains to show that a field generated by finitely many algebraic integers has one integral generator. Suppose $E = \mathbb{Q}(\alpha, \beta)$ with both α, β algebraic integers, and put $D = [E : \mathbb{Q}]$. There are exactly D embeddings $\sigma : E \hookrightarrow \mathbb{C}$, since every finite extension of $\mathbb{Q}$ is separable. For two distinct embeddings σ, τ , the equality

$$\sigma(\alpha + c\beta) = \tau(\alpha + c\beta)$$

holds for at most one $c \in \mathbb{Q}$ if $\sigma(\beta) \neq \tau(\beta)$, and for no c otherwise. Choose an integer c outside the finite union of these exceptional sets. The D images of $\theta = \alpha + c\beta$ are then distinct. Hence $[\mathbb{Q}(\theta) : \mathbb{Q}] \geq D$, and equality must hold because $\theta \in E$. Thus $E = \mathbb{Q}(\theta)$; also θ is an algebraic integer because sums and products of algebraic integers are algebraic integers. Induction handles any finite list of generators.

Finally, $1, \theta, \dots, \theta^{D-1}$ is a $\mathbb{Q}$ -basis of E . Express the finitely many elements of $\mathcal{D}$ in this basis and clear all their rational denominators with one positive integer L . Then $L\mathcal{D} \subset \mathbb{Z}[\theta]$. $\square$

Lemma II.12.6.4 (Uniform number of roots in a smaller disk). *For every $\kappa > 0$ there is $N_\kappa < \infty$, depending only on κ and $\mathcal{D}$, such that every nonzero $P \in \bigcup_n \mathcal{P}_{\mathcal{D}}^{(n)}$ has at most N_κ nonzero roots in $|z| \leq 1 - \kappa$, counted with multiplicity.*

Proof. Let

$$d_0 = \min\{|a| : a \in \mathcal{D} \setminus \{0\}\}, \quad D_0 = \max_{a \in \mathcal{D}} |a|, \quad \rho = D_0/d_0.$$

Choose k so large that

$$a_k := \frac{k}{k+1} [\rho(k+1)]^{-1/k} > 1 - \kappa$$

and set $r = k/(k+1)$. Remove the initial zero coefficients of P and divide by its first nonzero coefficient. The resulting polynomial Q has $|Q(0)| = 1$ and every coefficient has modulus at most ρ . If $z_1, \dots, z_N$ are the nonzero roots of P in $|z| \leq 1 - \kappa$, Jensen's formula gives

$$\sum_{j=1}^N \log \frac{r}{|z_j|} \leq \frac{1}{2\pi} \int_0^{2\pi} \log |Q(re^{i\theta})| d\theta \leq \log \frac{\rho}{1-r} = \log(\rho(k+1)).$$

Since $|z_j| < a_k$ and $\log(r/a_k) = k^{-1} \log(\rho(k+1))$, the left side is at least $Nk^{-1} \log(\rho(k+1))$. Thus $N \leq k$. Take $N_\kappa = k$. $\square$

Lemma II.12.6.5 (Integral norm polynomial). *Put $K = \mathbb{Q}(\mathcal{D})$ and $d = [K : \mathbb{Q}]$. There is $C_{\mathcal{D}} \geq 1$ such that, if $0 \neq P \in \mathcal{P}_{\mathcal{D}}^{(n)}$ and $P(\eta) = 0$, then some nonzero $F \in \mathbb{Z}[X]$ satisfies*

$$F(\eta) = 0, \quad \deg F \leq dn, \quad \|F\|_{\infty} \leq C_{\mathcal{D}}(n+1)^d.$$

Proof. Choose θ and L as supplied by Lemma II.12.6.3 for $K = \mathbb{Q}(\mathcal{D})$. Let $\sigma_1, \dots, \sigma_d$ be the embeddings of K into $\mathbb{C}$. For $P(X) = \sum_{r=0}^{n-1} a_r X^r$, set

$$F(X) = \prod_{j=1}^d \left(\sum_{r=0}^{n-1} \sigma_j(La_r) X^r \right).$$

Every coefficient of F is an algebraic integer. The Galois group of the normal closure of K permutes the embeddings $K \hookrightarrow \overline{\mathbb{Q}}$ and hence permutes the factors in the displayed product. It therefore fixes every coefficient, which is consequently rational. Thus every coefficient is a rational algebraic integer, and therefore an integer. The identity embedding is one factor, so $F(\eta) = 0$. Finally

$$\|F\|_{\infty} \leq \|F\|_1 \leq \prod_{j=1}^d \sum_{r < n} |\sigma_j(La_r)| \leq C_{\mathcal{D}}(n+1)^d.$$

The degree bound is immediate. □

Proposition II.12.6.6 (Separation of algebraic collision roots). *There is $C > 0$, depending only on $\mathcal{D}$, such that for all large n the following holds. If distinct algebraic numbers η_1, η_2 are each roots of nonzero members of $\mathcal{P}_{\mathcal{D}}^{(n)}$, then*

$$|\eta_1 - \eta_2| > n^{-Cn}.$$

Proof. Lemma II.12.6.5 places each η_i among the roots of an integral polynomial F_i of degree at most dn and coefficient bound $C_{\mathcal{D}}(n+1)^d$. In particular

$$M(F_i) \leq \|F_i\|_1 \leq (dn+1)C_{\mathcal{D}}(n+1)^d = n^{O_{\mathcal{D}}(1)}.$$

Take the square-free product of the irreducible factors of $F_1 F_2$ containing η_1, η_2 . It has degree $O_{\mathcal{D}}(n)$ and Mahler measure $n^{O_{\mathcal{D}}(1)}$. The discriminant separation estimate proved in Lemma II.12.5.3 therefore gives

$$\log |\eta_1 - \eta_2| \geq -C_1 n \log n$$

for a constant depending only on $\mathcal{D}$. Increasing C proves the assertion. □

Lemma II.12.6.7 (Derivative-value estimate). *Let $P \in \mathcal{P}_{\mathcal{D}}^{(N)}$, let $\lambda, \eta \in (-1, 1)$ be*

distinct, and let q be the order of vanishing of P at η . Then, with $D_0 = \max_{a \in \mathcal{D}} |a|$,

$$\left| \frac{P^{(q)}(\eta)}{q!} \right| \leq N^{q+2} D_0 |\lambda - \eta| + \frac{|P(\lambda)|}{|\lambda - \eta|^q}. \quad (\text{II.12.6.4})$$

Proof. Put $Q = P^{(q)}/q!$. Taylor's theorem, using the vanishing of the first $q - 1$ derivatives at η , gives a point ζ between η and λ such that $P(\lambda) = Q(\zeta)(\lambda - \eta)^q$. The coefficients of Q have modulus at most $N^q D_0$, whence $|Q'(x)| \leq N^{q+2} D_0$ for $|x| < 1$. The mean-value theorem between η and ζ now gives (II.12.6.4). $\square$

Lemma II.12.6.8 (Direct algebraic-value bound for field coefficients). *Let $K/\mathbb{Q}$ be a number field of degree d , let $0 \neq Q \in K[X]$ have degree $< N$, and suppose an integer $L \geq 1$ makes every coefficient of LQ an algebraic integer. Put*

$$B = \max \left\{ 2, \max_{\sigma: K \hookrightarrow \mathbb{C}} \|\sigma(LQ)\|_1 \right\}.$$

If η is algebraic with $|\eta| \leq 1$, $e = \deg \eta$, and $Q(\eta) \neq 0$, then

$$|Q(\eta)| \geq L^{-1} (2B)^{-de} M(\eta)^{-dN}. \quad (\text{II.12.6.5})$$

Proof. Form the auxiliary polynomial

$$R(X, T) = \prod_{\sigma: K \hookrightarrow \mathbb{C}} (T + \sigma(LQ)(X)) = \sum_{j=0}^d R_j(X) T^j.$$

Every coefficient of R is an algebraic integer, because the coefficients of each factor are algebraic integers. The Galois group of a normal closure of K permutes all the factors, so these coefficients are also rational. Thus $R \in \mathbb{Z}[X, T]$. Each R_j has degree $< dN$ and $\|R_j\|_1 \leq \binom{d}{j} B^{d-j} \leq (2B)^d$.

Let h be the number of factors $\sigma(LQ)(\eta)$ equal to zero. Since the identity factor is nonzero, $h < d$. The coefficient of T^h in the specialization $R(\eta, T)$ is

$$R_h(\eta) = \prod_{\substack{\sigma: K \hookrightarrow \mathbb{C} \\ \sigma(LQ)(\eta) \neq 0}} \sigma(LQ)(\eta) \neq 0.$$

For any nonzero $F \in \mathbb{Z}[X]$ of degree t with $F(\eta) \neq 0$, the integral resultant with the primitive minimal polynomial $f_\eta(X) = a \prod_{i=1}^e (X - \eta_i)$ gives

$$1 \leq |\text{Res}(f_\eta, F)| = |a|^t \prod_{i=1}^e |F(\eta_i)|.$$

Since $|F(\eta_i)| \leq \|F\|_1 \max(1, |\eta_i|)^t$, it follows that

$$|F(\eta)| \geq \|F\|_1^{-(e-1)} M(\eta)^{-t}.$$

Apply this to $F = R_h$. Then

$$|R_h(\eta)| \geq (2B)^{-d(e-1)} M(\eta)^{-dN}.$$

All the other nonzero factors in the displayed product for $R_h(\eta)$ have modulus at most B , because $|\eta| \leq 1$. Dividing by their product and using $d - h - 1 \leq d - 1$ yields

$$|LQ(\eta)| \geq (2B)^{-d(e-1)} B^{-(d-h-1)} M(\eta)^{-dN} \geq (2B)^{-de} M(\eta)^{-dN}.$$

Division by L proves (II.12.6.5). □

Proposition II.12.6.9 (High multiplicity from a narrow algebraic window). *Let $d = [\mathbb{Q}(\mathcal{D}) : \mathbb{Q}]$. For every $\alpha > 0$ and integer $k \geq 1$ there is n_0 such that the following holds. Suppose $n \geq n_0$, $\eta \in (-1, 1) \setminus \{0\}$ is algebraic with $\deg \eta \leq n$, $0 < |\lambda - \eta| \leq 1$, and $N \geq (k+2)n^{1+\alpha}$. If $0 \neq P \in \mathcal{P}_{\mathcal{D}}^{(N)}$ and*

$$(2M(\eta))^{dN/k} |P(\lambda)|^{1/k} \leq |\lambda - \eta| \leq (2M(\eta))^{-dN}, \quad (\text{II.12.6.6})$$

then P vanishes at η to order at least k .

Proof. Let q be the vanishing order and suppose $q < k$. Put $Q = P^{(q)}/q!$, so $Q(\eta) \neq 0$. Write the finitely many nonzero members of $\mathcal{D}$ as $s_1, \dots, s_r$. Put $K = \mathbb{Q}(\mathcal{D})$, so $[K : \mathbb{Q}] = d$. Lemma II.12.6.3 gives an integer $L \geq 1$, depending only on $\mathcal{D}$, such that $L\mathcal{D}$ consists of algebraic integers. Every coefficient of Q has the form $\binom{j+q}{q} a_{j+q}$ with $a_{j+q} \in \mathcal{D}$; therefore, for a constant $C_{\mathcal{D}} \geq 2$,

$$\max_{\sigma: K \hookrightarrow \mathbb{C}} \|\sigma(LQ)\|_1 \leq C_{\mathcal{D}} N^{q+1}.$$

Apply Lemma II.12.6.8 with $e = \deg \eta \leq n$. After enlarging a fixed constant $C_* \geq 2$ to absorb L and the factor 2,

$$|Q(\eta)| \geq (C_* N^k)^{-dn} M(\eta)^{-dN}. \quad (\text{II.12.6.7})$$

On the other hand, put $D_0 = \max_{a \in \mathcal{D}} |a|$. Lemma II.12.6.7, $q < k$, and (II.12.6.6) imply

$$|Q(\eta)| \leq (N^{k+2} D_0 + 1) (2M(\eta))^{-dN}.$$

Combining this with (II.12.6.7) and cancelling $M(\eta)^{-dN}$ yields

$$2^{dN} \leq (N^{k+2} D_0 + 1) (C_* N^k)^{dn}.$$

The logarithm of the right side is $O_{\mathcal{D},k}(n \log N)$, whereas the logarithm of the left side is $dN \log 2$. Since $N \geq (k+2)n^{1+\alpha}$, this is impossible for all sufficiently large n . Hence $q \geq k$. □

II.12.6.2 A field-relative entropy–Mahler estimate

Definition II.12.6.10 (Relative inward Mahler factor and Gaussian gain). Let $E \subset \mathbb{C}$ be a number field and let η be algebraic over E . Define the inward relative Mahler factor

$$\tilde{M}_E(\eta) = \prod_{\eta'} \frac{1}{\min\{1, |\eta'| \}}, \quad (\text{II.12.6.8})$$

where the product is over all conjugates of η over E , counted once. We identify $\mathbb{C}$ with $\mathbb{R}^2$, write h_{diff} for two-dimensional differential entropy, and call $G = G_1 + iG_2$ a standard complex Gaussian when G_1, G_2 are independent standard real Gaussians. For a finitely supported complex random variable Z , independent of G , put

$$\Phi_Z(a) = \sup_{t>0} (h_{\text{diff}}(taZ + G) - h_{\text{diff}}(tZ + G)), \quad a \geq 1. \quad (\text{II.12.6.9})$$

Lemma II.12.6.11 (Field-relative Gaussian entropy bound). Let $u_1, \dots, u_m \in E$ be distinct, let Z take the value u_j with probability p_j , and suppose $0 < |\eta| < 1$. Then

$$h_{\eta, \text{Law}(Z)} \geq \Phi_Z(\tilde{M}_E(\eta)). \quad (\text{II.12.6.10})$$

Moreover Φ_Z is nondecreasing and $\Phi_Z(a) \rightarrow H(p)$ as $a \rightarrow \infty$.

Proof. Let $\eta_1, \dots, \eta_q$ be the conjugates of η over E that lie in the open unit disk, and put $A_0 = \text{diag}(\eta_1, \dots, \eta_q)$ on $\mathbb{C}^q$. Choose v so that $(A_0^*)^r v$, $r \geq 0$, spans $\mathbb{C}^q$, and set

$$q(x) = \sum_{r \geq 0} |\langle A_0^r x, v \rangle|^2.$$

This is a positive-definite Hermitian form. If $q(x) = \|Bx\|^2$, then $A = BA_0B^{-1}$ satisfies

$$\|Ay\|^2 = \|y\|^2 - |\langle B^{-1}y, v \rangle|^2.$$

Consequently A is a contraction, $q-1$ of its complex singular values are one, and the remaining one is

$$|\det A| = \prod_{j=1}^q |\eta_j| = \tilde{M}_E(\eta)^{-1}.$$

For $x \in \mathbb{C}^q$, define

$$X_{A,x}^{(n)} = \sum_{r=0}^{n-1} Z_r A^r x, \quad X_{A,x} = \sum_{r=0}^{\infty} Z_r A^r x,$$

where the Z_r are independent copies of Z . This series converges absolutely: A is similar to A_0 and $\rho(A) = \max_j |\eta_j| < 1$, so $\sum_r \|A^r\| < \infty$. If two scalar prefixes agree at η , their difference is a polynomial in $E[X]$ divisible by the minimal polynomial of η over E ; it therefore vanishes at every η_j , and hence at A . Thus

$$H(S_n(\eta)) \geq H(X_{A,x}^{(n)}). \quad (\text{II.12.6.11})$$

Realify $\mathbb{C}^q$ as $\mathbb{R}^{2q}$ and use the same letters for the realifications. We now give the covariance-smoothed entropy calculation. Let $X = X_{A,x}$ and let X' be an independent copy. The independent decomposition $X = X_{A,x}^{(n)} + A^n X'$ and the [Gaussian entropy inequalities](#) give

$$\mathbf{H}(X; A^n \mid I) \leq \mathbf{H}(X_{A,x}^{(n)}; A^n), \quad (\text{II.12.6.12})$$

$$\mathbf{H}(X_{A,x}^{(n)}; A^n) \leq H(X_{A,x}^{(n)}). \quad (\text{II.12.6.13})$$

Indeed, in the first line the second summand contributes $\mathbf{H}(A^n X'; A^n) = \mathbf{H}(X'; I) = \mathbf{H}(X; I)$, which cancels the terminal term in the between-covariance entropy; the second line is the discrete-mixture bound used in (II.9.N71).

Since $\|A\| \leq 1$, one has $A^{r+1} \preceq A^r$. Scaling and the addition monotonicity in the same Gaussian entropy lemma therefore imply, for $0 \leq r < n$,

$$\begin{aligned} \mathbf{H}(X; A^{r+1} \mid A^r) &\geq \mathbf{H}(A^r X'; A^{r+1} \mid A^r) \\ &= \mathbf{H}(X'; A \mid I) \\ &\geq \mathbf{H}(Zx; A \mid I). \end{aligned}$$

For the first inequality use $X = X_{A,x}^{(r)} + A^r X'$ and add the independent prefix; the equality is the scaling identity. For the last inequality use $X' = Zx + AX''$ and add the independent tail. Telescoping and (II.12.6.12)–(II.12.6.13) yield the finite- n inequality

$$H(X_{A,x}^{(n)}) \geq n \mathbf{H}(Zx; A \mid I). \quad (\text{II.12.6.14})$$

Take a complex singular-value decomposition $A = UDV^*$ and choose $x = tUe_q$, where $D = \text{diag}(1, \dots, 1, \tilde{M}_E(\eta)^{-1})$. Scalar multiplication by every u_j commutes with U . In the term smoothed by A , first replace V^*G by a standard Gaussian and then apply U^* ; in the term smoothed by I , apply U^* . The first $q-1$ complex Gaussian coordinates cancel, and scaling the last coordinate gives

$$\mathbf{H}(Zx; A \mid I) = h_{\text{diff}}(t\tilde{M}_E(\eta)Z + G) - h_{\text{diff}}(tZ + G).$$

Combine this identity with (II.12.6.11) and (II.12.6.14), divide by n , let $n \rightarrow \infty$, and take the supremum over t to obtain (II.12.6.10).

For monotonicity, the scaling rule for differential entropy gives

$$h_{\text{diff}}(taZ + G) - h_{\text{diff}}(tZ + G) = \mathbf{H}(tZ; a^{-1}I \mid I).$$

If $1 \leq a \leq b$, covariance monotonicity from that lemma gives $\mathbf{H}(tZ; b^{-1}I \mid I) \geq \mathbf{H}(tZ; a^{-1}I \mid I)$. Taking suprema proves that Φ_Z is nondecreasing.

For the limit, use $t = a^{-1/2}$. For every $c > 0$,

$$h_{\text{diff}}(cZ + G) = h_{\text{diff}}(G) + I(Z; cZ + G).$$

As $c \downarrow 0$, dominated convergence for the finite Gaussian-mixture densities gives $I(Z; cZ + G) \rightarrow 0$. As $c \rightarrow \infty$, nearest-centre decoding recovers Z with error tending to zero, so Fano's inequality gives $I(Z; cZ + G) \rightarrow H(Z) = H(p)$. Consequently

$$h_{\text{diff}}(a^{1/2}Z + G) - h_{\text{diff}}(a^{-1/2}Z + G) \rightarrow H(p).$$

The opposite inequality $\Phi_Z(a) \leq H(p)$ follows because revealing the discrete variable Z can increase entropy by at most $H(Z)$. Hence $\Phi_Z(a) \rightarrow H(p)$. $\square$

Theorem II.12.6.12 (Feng–Feng entropy versus Mahler measure). *For every $h < H(p)$ there is $M_0 < \infty$, depending only on h , the translations, and the weights, such that every algebraic $\eta \in \mathbb{C}$ with $0 < |\eta| < 1$ and $M(\eta) \geq M_0$ satisfies*

$$h_{\eta, \nu} \geq h.$$

Proof. Translation of all digits by $-t_1$ does not change prefix collisions, so assume $t_1 = 0$ and hence $t_j \in K := \mathbb{Q}(\mathcal{D})$. If no nonzero collision polynomial vanishes at η , all word values are distinct and $h_{\eta, \nu} = H(p)$. We may therefore fix $0 \neq P \in \mathcal{P}_{\mathcal{D}}^{(n)}$ with $P(\eta) = 0$.

Choose an algebraic integer θ with $K = \mathbb{Q}(\theta)$ and $L \geq 1$ with $L\mathcal{D} \subset \mathbb{Z}[\theta]$. If $\sigma_1, \dots, \sigma_d$ are the embeddings of K , the polynomial

$$F(X) = \prod_{j=1}^d \sigma_j(LP)(X)$$

belongs to $\mathbb{Z}[X]$ by the norm argument in Lemma II.12.6.5. If r is the first index for which the coefficient a_r of P is nonzero, the first nonzero coefficient of F is $\prod_j \sigma_j(La_r)$. Since a_r belongs to the fixed finite set $\mathcal{D}$, its modulus is at most a constant C_1 independent of P and n .

Let $f = b_s X^s + \dots + b_0$ be the primitive integral minimal polynomial of η . Gauss's lemma shows that f divides F after the initial power of X is removed. Therefore $|b_0| \leq C_1$. Vieta's formula gives

$$\tilde{M}_{\mathbb{Q}}(\eta) = \frac{M(\eta)}{|b_0|} \geq C_1^{-1} M(\eta). \quad (\text{II.12.6.15})$$

Let $g \in K[X]$ be an irreducible polynomial of η , scaled so that its coefficients are algebraic integers. For every embedding $\sigma_j : K \rightarrow \mathbb{C}$, let $g_j = \sigma_j(g)$. The product $G_0 = \prod_j g_j$ is, up to a nonzero rational scalar, an integral polynomial divisible by f . The inward factor of a polynomial means the product of $1/\min\{1, |z|\}$ over its nonzero roots. It is multiplicative under products and increases on taking multiples. Consequently

$$\prod_{j=1}^d \tilde{M}(g_j) \geq \tilde{M}_{\mathbb{Q}}(\eta).$$

For some j_0 we have

$$\tilde{M}(g_{j_0}) \geq C_1^{-1/d} M(\eta)^{1/d}. \quad (\text{II.12.6.16})$$

We may increase the eventual threshold M_0 so that the right side is larger than one. Since g_{j_0} is irreducible over $\sigma_{j_0}(K)$, the conjugates over that field of any one of its roots are exactly all its roots. The strict inequality $\tilde{M}(g_{j_0}) > 1$ ensures that at least one root η_{j_0} lies in the open unit disk, and for this root

$$\tilde{M}_{\sigma_{j_0}(K)}(\eta_{j_0}) = \tilde{M}(g_{j_0}).$$

Put $\tilde{t}_i = \sigma_{j_0}(t_i)$ and $\tilde{\nu} = \sum_i p_i \delta_{\tilde{t}_i}$. For every difference polynomial $Q \in K[X]$,

$$Q(\eta) = 0 \iff \sigma_{j_0}(Q)(\eta_{j_0}) = 0,$$

because these conditions are respectively divisibility by g and by g_{j_0} . Thus the collision partitions of every prefix coincide, and

$$h_{\eta,v} = h_{\eta_{j_0},\tilde{v}}. \quad (\text{II.12.6.17})$$

Lemma II.12.6.11, (II.12.6.16), and (II.12.6.17) yield

$$h_{\eta,v} \geq \Phi_{\tilde{Z}} \left(C_1^{-1/d} M(\eta)^{1/d} \right),$$

where $\tilde{Z}$ has law $\tilde{v}$. There are only d possible embedded digit laws, and each corresponding Φ tends to $H(p)$. Choosing one common threshold for this finite family proves [FF25, Theorem 1.4]. $\square$

II.12.6.3 Algebraic approximation and the final contradiction

Definition II.12.6.13 (Algebraic collision approximants). For $0 \neq \eta \in (-1, 1)$ let μ_η be the self-similar measure for $\{x \mapsto \eta x + t_j\}_{j=1}^m$ and the weights p . For $a > 0$ and $n \geq 1$, define

$$E_a^{(n)} = \left\{ \eta \in (-1, 1) \setminus \{0\} : \dim \mu_\eta < a \text{ and } P(\eta) = 0 \text{ for some } 0 \neq P \in \mathcal{P}_{\mathcal{D}}^{(n)} \right\}.$$

Lemma II.12.6.14 (Entropy dimension at an algebraic contraction). *If $0 \neq \eta \in (-1, 1)$ is algebraic, then*

$$\dim \mu_\eta = \min \left\{ 1, \frac{h_{\eta,v}}{\log |\eta|^{-1}} \right\}. \quad (\text{II.12.6.18})$$

Proof. Put $\alpha = \dim \mu_\eta$ and $\ell_n = \lfloor n \log |\eta|^{-1} / \log 2 \rfloor$. The exact-dimensionality theorem, the entropy-dimension identity, and the prefix-tail decomposition give

$$\alpha \leq \min \left\{ 1, \frac{h_{\eta,v}}{\log |\eta|^{-1}} \right\}.$$

If $\alpha = 1$, this is already equality. Suppose $\alpha < 1$. The natural-scale approximation and homogeneous flattening theorem give, for every fixed $q > 1$,

$$\begin{aligned} H_{\ell_n}(S_n(\eta)) &= \alpha n \log |\eta|^{-1} + o(n), \\ H_{\lfloor q\ell_n \rfloor}(S_n(\eta)) - H_{\ell_n}(S_n(\eta)) &= o(n). \end{aligned}$$

Distinct prefix values differ by $P(\eta)$ for a nonzero $P \in \mathcal{P}_{\mathcal{D}}^{(n)}$. Put $K = Q(\mathcal{D})$, $d = [K : \mathbb{Q}]$, and $e = \deg \eta$. Choose the fixed integer L of Lemma II.12.6.3. The coefficient bound for $\mathcal{D}$ gives $\max_{\sigma} \|\sigma(LP)\|_1 \leq C_{\mathcal{D}} n$. Lemma II.12.6.8 therefore yields, whenever $P(\eta) \neq 0$,

$$|P(\eta)| \geq L^{-1} (2C_{\mathcal{D}} n)^{-de} M(\eta)^{-dn}.$$

Since $n \leq 2^n$, one may fix $c_\eta \in (0, 1)$ such that $|P(\eta)| \geq c_\eta^n$ for all n and all these nonzero values. Choose q so that $q \log |\eta|^{-1} > \log c_\eta^{-1}$. At dyadic level $\lfloor q\ell_n \rfloor$ the mesh is, for all large

n , smaller than half the minimum separation between distinct atoms. Hence every cell of every translated dyadic grid contains at most one distinct atom, and therefore

$$H_{\lfloor q\ell_n \rfloor}(S_n(\eta)) = H(S_n(\eta)).$$

The preceding two asymptotics now imply $h_{\eta,v} = \alpha \log |\eta|^{-1}$, proving (II.12.6.18). $\square$

The next step explains how a very fine entropy collision produces one algebraic parameter shared by *all* word pairs in the same grid cells. The bounded-root and separation estimates above make this selection uniform even though the digit differences are algebraic rather than integral.

Lemma II.12.6.15 (Common collision root at a fine scale). *Let J be a compact subset of $(-1, 0) \cup (0, 1)$. There are constants $C, L > 1$ and n_J such that, whenever $n \geq n_J$, $\theta \in J$, and $0 < r < n^{-Ln}$ satisfy*

$$\mathcal{H}(\text{Law } S_n(\theta); r) < nH(p),$$

there is a real root $\eta \in (-1, 0) \cup (0, 1)$ of a nonzero member of $\mathcal{P}_{\mathcal{D}}^{(n)}$ for which

$$|\theta - \eta| < r^{1/C}, \quad H(S_n(\eta)) \leq \mathcal{H}(\text{Law } S_n(\theta); r).$$

Proof. Choose $0 < \kappa < R_0 < R_1 < 1$ such that $\kappa \leq |\theta| \leq R_0$ for all $\theta \in J$, and put $c = 1 - R_0/R_1 > 0$. Lemma II.12.6.4 supplies an integer $N \geq 1$ such that every nonzero difference polynomial has at most N nonzero roots in $|z| \leq R_1$, counted with multiplicity. Also put $d_0 = \min\{|a| : a \in \mathcal{D} \setminus \{0\}\} > 0$.

We first show that a nonzero $P \in \mathcal{P}_{\mathcal{D}}^{(n)}$ with $|P(\theta)| \leq r$ has a zero very close to θ . Write $P(X) = X^k Q(X)$ with $Q(0) \neq 0$. Since $k < n$, $|Q(\theta)| \leq \kappa^{-n}r$. If every zero ζ of Q obeyed $|\theta - \zeta| \geq a$, where $0 < a < R_1$, the product factorization about its constant term would give

$$|Q(\theta)| = |Q(0)| \prod_{Q(\zeta)=0} |1 - \theta/\zeta| \geq d_0 c^n (a/R_1)^N.$$

Here roots outside $|z| \leq R_1$ contribute at least c , and each of the at most N inner roots contributes at least a/R_1 ; replacing their actual number by N only weakens the bound. Thus there is a constant $C_1 = C_1(J, \mathcal{D}) > 1$ such that $|P(\theta)| \leq r$ forces a zero η_P with

$$|\theta - \eta_P| \leq C_1^n r^{1/N}.$$

For $r < n^{-Ln}$ and sufficiently large n this is less than $r^{1/(2N)}$. Choose L so large, relative to the exponent in Proposition II.12.6.6, that

$$2r^{1/(2N)} < n^{-C_{\text{sep}} n}.$$

All the zeros η_P selected from polynomials with $|P(\theta)| \leq r$ must then coincide; denote the common zero by η . Since the polynomials have real coefficients, $\bar{\eta}$ is also a zero, and root separation applied to $\eta, \bar{\eta}$ makes η real. It lies in the same component of $(-1, 0) \cup (0, 1)$ as θ once n is large.

Choose a translation of the length- r grid for which the entropy of the quantized $S_n(\theta)$ does not exceed its averaged entropy. The strict entropy deficit means that two distinct words occupy one cell, and their difference is a nonzero polynomial P with $|P(\theta)| < r$. Thus the preceding common root exists. Every pair of words in one cell has difference polynomial either zero or of this same small-value type, hence the pair has exactly the same value at η . The partition into grid cells therefore refines the partition into values of $S_n(\eta)$, giving the asserted entropy inequality. Take $C = 2N$, enlarging it slightly if needed to make the distance inequality strict. $\square$

Lemma II.12.6.16 (A nearby root forces maximal finer entropy). *Fix $\lambda \in (-1, 0) \cup (0, 1)$. There is $B_* > 1$ such that, for all sufficiently large n , if η is a root of a nonzero $P \in \mathcal{P}_{\mathcal{D}}^{(n)}$ and $0 < \rho := |\lambda - \eta| < n^{-B_* n}$, then*

$$\mathcal{H}(\text{Law } S_n(\lambda); r) = nH(p) \quad (0 < r < \rho^{B_*}).$$

Proof. Apply Lemma II.12.6.15 on a compact neighborhood J of λ , and let C, L be its constants. Let C_{sep} be the exponent in Proposition II.12.6.6. Choose B_* so large that $B_* > C$, $B_*^2 > L$, and $B_* > C_{\text{sep}} + 2$. If the displayed averaged entropy were smaller than $nH(p)$, then $r < \rho^{B_*} < n^{-Ln}$ and the common-root lemma would give another collision root η' with

$$|\lambda - \eta'| < r^{1/C} < \rho^{B_*/C} < \rho.$$

Thus $\eta' \neq \eta$, but $|\eta' - \eta| < 2\rho < n^{-C_{\text{sep}} n}$ for large n , contrary to root separation. The entropy cannot exceed $nH(p)$, so equality holds. $\square$

Corollary II.12.6.17 (Signed finite-digit scale-and-convolve bound). *Let $0 < |\lambda| < 1$, let D be a bounded finite real random variable, put $\sigma = |\lambda|$, and let ζ be the law of $\sum_{j \geq 0} D_j \lambda^j$ for independent copies D_j of D . Using a bi-infinite independent extension of those digits, let ζ^I be the law of $\sum_{j \in \mathbb{Z}: \sigma^j \in I} D_j \lambda^j$ for any interval $I \subset (0, \infty)$ bounded above, and put $\zeta^{(n)} = \zeta^{(\sigma^n, 1]}$. If $\dim \zeta < 1$, there is $\alpha_0 > 0$ such that every scale-window law ζ^I satisfies*

$$\mathcal{H}(\zeta^I; r \mid 2r) \leq 1 - \alpha_0 \quad (r > 0).$$

For each $0 < \alpha \leq \alpha_0$ there is $C > 1$ such that, if $n_1 < \dots < n_N$ and $K_1, \dots, K_N \geq 10$ satisfy

$$\begin{aligned} \sigma^{-n_1} &\geq \max\{2, \sigma^{-2}\}, \\ n_{j+1} &\geq K_j n_j \quad (1 \leq j < N), \\ n_j &\geq C(\log K_j)^2 \quad (1 \leq j \leq N), \end{aligned}$$

and

$$\mathcal{H}(\zeta^{(n_j)}; \sigma^{K_j n_j} \mid \sigma^{10n_j}) \geq \alpha n_j \quad (1 \leq j \leq N),$$

then

$$\sum_{j=1}^N \frac{1}{\log K_j \log \log K_j} \leq C \left(1 + \frac{1}{n_1} \sum_{j=1}^N \log K_j \right).$$

Proof. For $\lambda > 0$ these are Lemma II.12.5.1 and Proposition II.12.5.2. Suppose $\lambda < 0$. Disjoint windows convolve, and independence plus an index shift give

$$\zeta^{\sigma^k I} = (S_{\lambda^k})_* \zeta^I.$$

For every $a \neq 0$, reflection of the uniform smoothing interval is a translate of that interval. The differential-entropy representation (II.9.02) and translation invariance therefore give the exact identity

$$\mathcal{H}((S_a)_* \xi; |a|r \mid |a|Rr) = \mathcal{H}(\xi; r \mid Rr).$$

These are precisely the two structural identities used in (II.12.5.3)–(II.12.5.4). In the proof of Lemma II.12.5.1, every scaled tail is consequently a reflected or unreflected convolution factor with the same entropy, so its deficit argument applies with σ . In the proof of Proposition II.12.5.2, the window-transfer step uses only those two identities and convolution monotonicity; the amplified-block step applies the arbitrary-measure entropy increment theorem to the independent window factors; and the final separated-window step uses only their one-digit deficit. None of these steps depends on the sign of λ . Replacing each positive scale λ^j there by σ^j therefore proves the displayed conclusion with the same constants. $\square$

Theorem II.12.6.18 (Feng–Feng algebraic approximation). *Let $0 \neq \lambda \in (-1, 1)$ and suppose*

$$\dim \mu_\lambda < \min \left\{ 1, \frac{H(p)}{\log |\lambda|^{-1}} \right\}.$$

For every $\varepsilon > 0$ and $n_0 \geq 1$, there are $n \geq n_0$ and

$$\eta \in E_{\dim \mu_\lambda + \varepsilon}^{(n)} \quad \text{such that} \quad |\lambda - \eta| \leq \exp(-n^{1/\varepsilon}).$$

This is [FF25, Theorem 1.3]; we give the multiscale proof below.

Proof. Write

$$\sigma = |\lambda|, \quad \chi = \log \sigma^{-1}, \quad d = \dim \mu_\lambda, \quad s = \min\{1, H(p)/\chi\}, \quad \nu_\theta^{(n)} = \text{Law } S_n(\theta).$$

If λ is itself a collision root, padding its polynomial with zero coefficients proves the assertion for arbitrarily large n . We may therefore assume it is not such a root. Choose $\delta > 0$ so small that

$$4\delta < \varepsilon, \quad d + 4\delta < s, \quad (d + 4\delta)\chi < H(p), \quad (\text{II.12.6.19})$$

and fix $A > \max\{2, \varepsilon^{-1}\}$.

Corollary II.12.6.17 supplies a one-digit deficit $\alpha > 0$ for every window law and the scale-and-convolve bound with σ , including when $\lambda < 0$.

For use below, the entropy bound at any nonzero contraction θ is

$$\dim \mu_\theta \leq \frac{H(S_n(\theta))}{n \log |\theta|^{-1}} \quad (n \geq 1). \quad (\text{II.12.6.20})$$

Indeed, prefix subadditivity gives $H(S_{kn}(\theta)) \leq kH(S_n(\theta))$; comparing the prefix with the infinite series at scale $|\theta|^{kn}$ costs $O(1)$ entropy, and the entropy–dimension identity gives (II.12.6.20) after $k \rightarrow \infty$. The natural-scale approximation and homogeneous flattening results from Chapter II.6 give

$$\mathcal{H}(v_\lambda^{(n)}; \sigma^{10n}) = dn\chi + o(n). \quad (\text{II.12.6.21})$$

Suppose for a contradiction that, for some n_0 , no $n \geq n_0$ and no collision root η of degree- $< n$ type satisfy both

$$|\lambda - \eta| \leq e^{-n^A}, \quad \dim \mu_\eta < d + 3\delta. \quad (\text{II.12.6.22})$$

Enlarge n_0 whenever a fixed preceding estimate requires it. Apply Lemma II.12.6.15 in a compact neighborhood of λ , and denote its constants by $C_{\text{low}}, L_{\text{low}}$. Let B_* be supplied by Lemma II.12.6.16. Choose $B, L > 1$, in that order, so large that

$$B > 2B_*, \quad L > 2B \max\{C_{\text{low}}, B_*, L_{\text{low}}\}.$$

Set

$$\beta = \min\left\{\alpha, \frac{\delta\chi}{2\log 2}, \frac{H(p) - d\chi}{4\log 2}, \frac{1}{4}\right\} > 0,$$

and let C_0 be the constant in Corollary II.12.6.17 for β .

Starting from an arbitrarily large n_0 , construct $n_0 < n_1 < n_2 < \dots$ recursively. Given n_j , put

$$q_j = \left\lfloor \frac{Ln_j \log n_j}{\chi} \right\rfloor.$$

If

$$\mathcal{H}(v_\lambda^{(q_j)}; q_j^{-Lq_j}) \geq q_j(d + 2\delta)\chi, \quad (\text{II.12.6.23})$$

set $n_{j+1} = q_j$. Equation (II.12.6.21) shows that the entropy between $q_j^{-Lq_j}$ and σ^{10q_j} is at least $\delta\chi n_{j+1}$ for sufficiently large n_0 .

Otherwise the strict reverse of (II.12.6.23) and (II.12.6.19) put the fine-scale entropy below $q_j H(p)$. Lemma II.12.6.15 gives a real collision root η_j with

$$\rho_j := |\lambda - \eta_j| < q_j^{-Bq_j}, \quad H(S_{q_j}(\eta_j)) < q_j(d + 2\delta)\chi. \quad (\text{II.12.6.24})$$

By (II.12.6.20) and $\log |\eta_j|^{-1} = \chi + o(1)$, its dimension is less than $d + 3\delta$ for large n_0 . The contrary assumption (II.12.6.22) therefore forces

$$\rho_j > e^{-q_j^A}. \quad (\text{II.12.6.25})$$

Let n_{j+1} be the largest integer n for which $\rho_j < n^{-Bn}$. This integer exists, is at least q_j , and satisfies

$$(n_{j+1} + 1)^{-B(n_{j+1}+1)} \leq \rho_j < n_{j+1}^{-Bn_{j+1}}.$$

The polynomial defining η_j has degree $< q_j \leq n_{j+1}$, so Lemma II.12.6.16 applies at level n_{j+1} . Since $L > BB_*$, its fine-scale conclusion gives

$$\mathcal{H}\left(v_\lambda^{(n_{j+1})}; (n_{j+1} + 1)^{-L(n_{j+1}+1)}\right) = n_{j+1}H(p).$$

The factor-two scale comparison (II.9.N03) changes the displayed fine scale to $n_{j+1}^{-Ln_{j+1}}$ at a cost $O_L(\log n_{j+1})$. Subtract (II.12.6.21); by the choice of β , the resulting entropy reservoir is at least $\beta n_{j+1} \log 2$ for large n_0 .

Each stage has produced the reservoir at the *new* index n_{j+1} . Thus reindexing the stage $j - 1$ as j and putting $K_j = L \log n_j / \chi$ yields

$$n_{j+1} \geq K_j n_j, \quad \mathcal{H}(v_\lambda^{(n_j)}; \sigma^{K_j n_j} \mid \sigma^{10n_j}) \geq \beta n_j \quad (j \geq 1).$$

For large n_0 the two size conditions of Corollary II.12.6.17 also hold, and hence

$$\sum_{j=1}^J \frac{1}{\log K_j \log \log K_j} \leq C_0 \left(1 + \frac{1}{n_1} \sum_{j=1}^J \log K_j\right) \quad (J \geq 1). \quad (\text{II.12.6.26})$$

The growth of this sequence is too slow for (II.12.6.26). In the high case $n_{j+1} = q_j < n_j^2$ for large n_j . In the low case, (II.12.6.25) and $\rho_j < n_{j+1}^{-Bn_{j+1}}$ give $Bn_{j+1} \log n_{j+1} < q_j^A$, and thus $n_{j+1} < n_j^{A+2}$ for large n_j . Put $T = A + 2$ and $j_0 = \max\{3, \lceil \log \log n_0 \rceil\}$. Induction gives $\log n_j \leq T^j \log n_0$, so

$$\log K_j \leq C_A(j + j_0), \quad \log \log K_j \leq C_A \log(j + j_0).$$

The left side of (II.12.6.26) is therefore at least

$$c_A \sum_{j=1}^J \frac{1}{(j + j_0) \log(j + j_0)} \geq c_A \log \frac{\log(J + j_0 + 1)}{\log(j_0 + 1)}.$$

Choose a fixed exponent p so large that, for $J = \lceil (j_0 + 1)^p \rceil$, this lower bound exceeds $3C_0$. On the other hand

$$\frac{1}{n_1} \sum_{j=1}^J \log K_j \leq \frac{C_A J(J + j_0)}{n_1} \rightarrow 0 \quad (n_0 \rightarrow \infty),$$

contradicting (II.12.6.26).

Thus arbitrarily large n admit a collision root η with $|\lambda - \eta| \leq e^{-n^A}$ and $\dim \mu_\eta < d + 3\delta < d + \varepsilon$. Because $A > \varepsilon^{-1}$, this is stronger than the stated distance bound. The root belongs to $E_{d+\varepsilon}^{(n)}$ by definition. $\square$

Theorem II.12.6.19 (Feng–Feng full-dimension theorem). *Let*

$$\Phi = \{x \mapsto \lambda x + t_j\}_{j=1}^m, \quad 0 \neq \lambda \in (-1, 1),$$

have algebraic translations and positive weights p_j . If Φ has no exact overlaps, then

$$\dim \mu_\lambda = \min \left\{ 1, \frac{H(p)}{\log |\lambda|^{-1}} \right\}.$$

In particular, signed contractions are included. This is [FF25, Theorem 1.2]. The proof below derives it from the approximation theorem, Theorem II.12.6.18, and the arithmetic and entropy estimates proved above.

Proof. The standard entropy–Lyapunov upper bound gives “ $\leq$ ”. If λ is algebraic, Theorem II.12.3.12 gives the reverse inequality. We may therefore suppose that λ is transcendental and, toward a contradiction, that

$$s := \dim \mu_\lambda < \min \left\{ 1, \frac{H(p)}{L} \right\}, \quad L = \log |\lambda|^{-1}.$$

Choose $\varepsilon > 0$ so small that $1/\varepsilon > 4$ and

$$s + 4\varepsilon < \min \left\{ 1, \frac{H(p)}{L} \right\}. \quad (\text{II.12.6.27})$$

By continuity of $x \mapsto \log |x|^{-1}$ away from zero, there are a neighborhood U of λ and a number $h_* < H(p)$ such that

$$(s + \varepsilon) \log |x|^{-1} \leq h_* \quad (x \in U). \quad (\text{II.12.6.28})$$

Choose h_1 with $h_* < h_1 < H(p)$. Theorem II.12.6.12 supplies $B \geq 2$ such that

$$M(\eta) \geq B \implies h_{\eta,v} \geq h_1 > h_*. \quad (\text{II.12.6.29})$$

Let $d = [Q(\mathcal{D}) : \mathbb{Q}]$, choose $\kappa > 0$ so that $|x| \leq 1 - \kappa$ for $x \in U$, and let N_κ be given by Lemma II.12.6.4. Fix

$$k = N_\kappa + 1.$$

Apply Theorem II.12.6.18 with n_0 arbitrarily large. It gives $n \geq n_0$ and an algebraic $\eta \in U$ such that

$$\dim \mu_\eta < s + \varepsilon, \quad 0 < |\lambda - \eta| \leq e^{-n^{1/\varepsilon}}, \quad Q(\eta) = 0 \quad (\text{II.12.6.30})$$

for some $0 \neq Q \in \mathcal{P}_{\mathcal{D}}^{(n)}$. The distance is nonzero because λ is transcendental. The last relation gives

$$\deg \eta \leq dn. \quad (\text{II.12.6.31})$$

By Lemma II.12.6.14, (II.12.6.30), and (II.12.6.28), the inequality $\dim \mu_\eta < s + \varepsilon < 1$ places us in the non-saturated branch of the algebraic dimension formula, so

$$h_{\eta,v} = \dim \mu_\eta \log |\eta|^{-1} < h_*.$$

Thus (II.12.6.29) forces

$$M(\eta) < B. \quad (\text{II.12.6.32})$$

Let N be the unique nonnegative integer satisfying

$$(2B)^{-d(N+1)} \leq |\lambda - \eta| < (2B)^{-dN}. \quad (\text{II.12.6.33})$$

Equations (II.12.6.30) and (II.12.6.33) imply $N \gg n^{1/\varepsilon}$; in particular, for large n_0 ,

$$N \geq (k+2)(dn)^2. \quad (\text{II.12.6.34})$$

The strict dimension drop at λ and the [super-exponential concentration theorem](#) imply that, for all large N , two distinct length- N words have translation difference at most $(2B)^{-3kdN}$. Hence there is $0 \neq P \in \mathcal{P}_{\mathcal{D}}^{(N)}$ with

$$|P(\lambda)| \leq (2B)^{-3kdN}. \quad (\text{II.12.6.35})$$

It is nonzero because a polynomial with algebraic coefficients cannot vanish at the transcendental number λ .

From (II.12.6.32), (II.12.6.33), and (II.12.6.35),

$$\begin{aligned} (2M(\eta))^{dN/k} |P(\lambda)|^{1/k} &\leq (2B)^{-2dN} \leq |\lambda - \eta|, \\ |\lambda - \eta| &< (2B)^{-dN} \leq (2M(\eta))^{-dN}. \end{aligned}$$

Apply Proposition II.12.6.9 with the degree bound dn in place of n , with $\alpha = 1$, and with (II.12.6.34). It says that P vanishes at η to order at least $k = N_\kappa + 1$. But $\eta \in U$ has $0 < |\eta| \leq 1 - \kappa$, contradicting Lemma II.12.6.4. This contradiction proves the theorem, including the case $\lambda < 0$. $\square$

II.12.7 Transcendental Bernoulli parameters: Varjú's theorem

For $\lambda \in (0, 1)$ let ν_λ be the Bernoulli convolution

$$\nu_\lambda = \text{Law} \left(\sum_{n=0}^{\infty} X_n \lambda^n \right), \quad \mathbb{P}(X_n = 1) = \mathbb{P}(X_n = -1) = \frac{1}{2},$$

with independent signs. Varjú proved the theorem below in [Var19b]. This subsection records the arithmetic counterpart of the exceptional-parameter results below. It is deliberately placed here, rather than in the absolute-continuity chapter, because the conclusion is a *dimension theorem*: full dimension does not by itself imply absolute continuity (compare Section 10.1).

The proof uses three results from earlier chapters of Volume II. We restate the precise forms needed here so that the argument can be read without interrupting its arithmetic thread.

Proposition II.12.7.1 (Algebraic approximation). *The Breuillard–Varjú approximation theorem [BV19] gives the following. If $\dim \nu_\lambda < 1$ for some $\lambda \in [1/2, 1)$, then there are*

arbitrarily large integers d and algebraic numbers $\xi = \xi(d) \in (1/2, 1)$ such that

$$\deg \xi \leq d, \quad \dim v_\xi < 1, \quad |\lambda - \xi| \leq e^{-d^2}.$$

Proof. This is the [Breuillard–Varjú approximation theorem](#) proved in Chapter II.9, with its parameter renamed d . $\square$

Proposition II.12.7.2 (Small polynomials from dimension drop). *Suppose $\dim v_\lambda < 1$ with $\lambda \in [1/2, 1)$. For every $0 < \theta < 1$ and all sufficiently large n there is a nonzero polynomial P of degree at most n , with coefficients in $\{-1, 0, 1\}$, such that*

$$|P(\lambda)| < \theta^n.$$

If

$$\tilde{v}_\lambda = \text{Law} \left(\sum_{\substack{n \geq 0 \\ n \not\equiv 2 \pmod{3}}} X_n \lambda^n \right)$$

has dimension < 1 , then the same conclusion holds with P in the restricted class

$$\mathcal{Q} = \left\{ \sum a_j x^j : a_j \in \{-1, 0, 1\}, a_j = 0 \text{ whenever } j \equiv 2 \pmod{3} \right\}.$$

Proof. The unrestricted assertion is the [small-polynomial consequence of entropy drop](#); the restricted-digit assertion is the [corresponding residue-class lemma](#). Both were proved earlier in this volume. $\square$

Proposition II.12.7.3 (Dimension formula at algebraic parameters). *For algebraic $\xi \in [1/2, 1)$,*

$$\dim v_\xi = \min \left\{ 1, \frac{h_\xi}{\log_2 \xi^{-1}} \right\}, \quad h_\xi = \lim_{N \rightarrow \infty} \frac{1}{N} H_2 \left(\sum_{n=0}^{N-1} X_n \xi^n \right).$$

Proof. This is the [algebraic entropy–dimension formula](#) proved in Chapter II.9. $\square$

Together these propositions isolate the earlier entropy theory needed for the final arithmetic argument below.

The next lemma is the high-Mahler consequence needed to keep the algebraic approximants in a fixed arithmetic complexity class.

Lemma II.12.7.4 (Large Mahler measure forces almost maximal Garsia entropy). *For every $0 < h < 1$ there is $C(h) > 1$ such that every algebraic $\xi \in (0, 1)$ with $M(\xi) \geq C(h)$ satisfies $h_\xi \geq h$. Consequently, if $\lambda \in (1/2, 1)$ is fixed and $\xi_j \rightarrow \lambda$ are algebraic with $\dim v_{\xi_j} < 1$, then $M(\xi_j)$ is bounded uniformly in j .*

Proof. The proof of Theorem II.10.10.8, specialized to the fair sign law, actually gives the stronger estimate

$$h_\xi \geq \Phi(M(\xi)), \quad \Phi(a) = \sup_{t>0} (h_2(taX + G) - h_2(tX + G)),$$

where X is a fair sign and G is an independent standard Gaussian. Put $t = a^{-1/2}$. Then $tX + G \Rightarrow G$ with convergence of differential entropy, while $taX + G = a^{1/2}X + G$ is a mixture of two Gaussians whose overlap tends to zero. Hence

$$h_2(tX + G) \longrightarrow h_2(G), \quad h_2(taX + G) \longrightarrow 1 + h_2(G),$$

so $\Phi(a) \rightarrow 1$. This proves the first assertion.

For the consequence, choose h with $\log_2 \lambda^{-1} < h < 1$. For all large j , $\log_2 \xi_j^{-1} < h$. If $M(\xi_j) \geq C(h)$, then Proposition II.12.7.3 gives

$$\dim v_{\xi_j} \geq \min \left\{ 1, \frac{h}{\log_2 \xi_j^{-1}} \right\} = 1,$$

a contradiction. The finitely many initial terms can be absorbed into the bound. □

We now prove the arithmetic separation statement that is local to Varjú's short 2019 argument. The classical transversality input used here is the one already developed in the Bernoulli-convolution background chapter from Solomyak and Peres–Solomyak [Sol95; PS96]: there is an absolute $\delta > 0$ such that $[0, 2^{-2/3}]$ is an interval of δ -transversality for $1 + \sum_{n \geq 1} a_n x^n$, $a_n \in \{-1, 0, 1\}$, and $[0, 2^{-1/2}]$ is such an interval for the two residue-restricted classes obtained by forbidding one of the two nonzero classes modulo 3. These are the [classical transversality estimate](#) and its [residue-restricted form](#) from Chapter II.8.

Lemma II.12.7.5 (Uniform value bound at an algebraic point). *Let ξ be algebraic of degree d and let P be a nonzero polynomial of degree at most n with coefficients in $\{-1, 0, 1\}$. If $P(\xi) \neq 0$, then*

$$|P(\xi)| \geq M(\xi)^{-n} (n+1)^{1-d}.$$

Proof. Let $Q(x) = a_d \prod_{j=1}^d (x - \xi_j)$ be the primitive irreducible polynomial of $\xi = \xi_1$. The algebraic norm calculation underlying Lemma II.10.11.3 gives

$$a_d^n \prod_{j=1}^d P(\xi_j) \in \mathbb{Z} \setminus \{0\}.$$

Therefore its absolute value is at least one. For each conjugate,

$$|P(\xi_j)| \leq (n+1) \max\{1, |\xi_j|\}^n.$$

Solving for the $j = 1$ factor and using the definition of Mahler measure gives the claimed inequality. □

Proposition II.12.7.6 (Varjú's annular polynomial separation). *There is an absolute d_0 with the following property. Let*

$$\xi \in (1/2, 2^{-1/2})$$

be algebraic of degree at most d , with $d \geq d_0$ and $M(\xi) \leq M$. Let $n > 10d \log d$, and let $P \neq 0$ belong to $\mathcal{Q}$ with $\deg P \leq n$. Then

$$|P(\lambda)| > (20M)^{-n}$$

whenever $\lambda \in (1/2, 2^{-1/2})$ and

$$(5M)^{-n} \leq |\lambda - \xi| \leq (5M)^{-n+1}.$$

If $\xi \leq 2^{-2/3}$, the same conclusion holds for every nonzero P of degree at most n with coefficients in $\{-1, 0, 1\}$.

Proof. Assume to the contrary that $|P(\lambda)| \leq (20M)^{-n}$. Divide by the lowest nonzero monomial and change sign if necessary so that the new polynomial has constant term 1. Since $\lambda > 1/2$, this costs at most a factor 2^n , hence the normalized polynomial still satisfies

$$|P(\lambda)| \leq (10M)^{-n}.$$

It lies in the unrestricted transversality class when $\xi \leq 2^{-2/3}$ and, in the restricted case, in one of the two residue classes covered by the $[0, 2^{-1/2}]$ transversality theorem.

We first claim that $P(\xi) \neq 0$. Otherwise both $P(\xi)$ and $P(\lambda)$ are smaller than the transversality threshold for all sufficiently large n . Along the interval joining ξ and λ , transversality gives a derivative of fixed sign and magnitude at least δ . Hence

$$|P(\xi) - P(\lambda)| \geq \delta|\xi - \lambda| \geq \delta(5M)^{-n},$$

contradicting $P(\xi) = 0$ and $|P(\lambda)| \leq (10M)^{-n}$ once n is large.

Lemma II.12.7.5 now gives

$$|P(\xi)| \geq (n+1)^{1-d} M^{-n} \geq (2M)^{-n},$$

where the last inequality follows from $n > 10d \log d$ after increasing d_0 . On the other hand $|P'(t)| \leq n^2$ for $0 \leq t \leq 1$, so

$$|P(\xi) - P(\lambda)| \leq n^2|\xi - \lambda| \leq n^2(5M)^{-n+1}.$$

For large d (and hence large n) this is smaller than $(2M)^{-n} - (10M)^{-n}$, contradicting the preceding two inequalities. $\square$

Lemma II.12.7.7 (Passing to a power cannot increase dimension). *For every $\lambda \in (0, 1)$ and $k \geq 1$,*

$$\dim v_{\lambda^k} \leq \dim v_{\lambda}.$$

Proof. Split the random series according to indices divisible by k :

$$v_\lambda = v_{\lambda^k} * \rho$$

for a probability ρ . If E is Borel with $v_\lambda(E) > 0$, then by Fubini $v_{\lambda^k}(E - t) > 0$ for t in a set of positive ρ -measure. Since Bernoulli convolutions are exact dimensional, the dimension equals the infimum of the Hausdorff dimensions of positive-measure Borel sets. Thus $\dim_{\text{H}} E = \dim_{\text{H}}(E - t) \geq \dim v_{\lambda^k}$. Taking the infimum over E proves the claim. $\square$

Theorem II.12.7.8 (Varjú: all transcendental Bernoulli parameters have full dimension). *If $\lambda \in (1/2, 1)$ is transcendental, then*

$$\dim v_\lambda = 1.$$

Proof. Assume for contradiction that $\dim v_\lambda < 1$. By Lemma II.12.7.7, replacing λ by a suitable positive power can only decrease the dimension. We may therefore assume

$$\lambda \in (1/2, 2^{-1/2}).$$

The replacement preserves transcendence.

Proposition II.12.7.1 supplies arbitrarily large d and algebraic ξ such that

$$\deg \xi \leq d, \quad \dim v_\xi < 1, \quad |\lambda - \xi| \leq e^{-d^2}.$$

By Lemma II.12.7.4, after restricting to sufficiently large d we have a uniform bound $M(\xi) \leq M$, where M depends on λ but not on d . Choose the integer n determined by

$$(5M)^{-n} \leq |\lambda - \xi| \leq (5M)^{-n+1}.$$

Because $|\lambda - \xi| \leq e^{-d^2}$, we have $n > 10d \log d$ for all sufficiently large d .

If $\xi \leq 2^{-2/3}$, apply Proposition II.12.7.2 with $\theta = (20M)^{-1}$. Since its conclusion is available at every sufficiently large scale, it yields a nonzero $\{-1, 0, 1\}$ polynomial of degree at most n with

$$|P(\lambda)| < (20M)^{-n},$$

contradicting Proposition II.12.7.6.

If $\xi > 2^{-2/3}$, the full Bernoulli convolution is the convolution of $\tilde{v}_\lambda$ with the omitted residue-class factor. The proof of Lemma II.12.7.7 therefore also gives

$$\dim \tilde{v}_\lambda \leq \dim v_\lambda < 1.$$

The restricted part of Proposition II.12.7.2 yields $P \in \mathcal{Q} \setminus \{0\}$ of degree at most n with the same upper bound $|P(\lambda)| < (20M)^{-n}$, again contradicting Proposition II.12.7.6. Both cases are impossible. $\square$

Corollary II.12.7.9 (Countability of the Bernoulli dimension-drop set). *The set*

$$\{\lambda \in (1/2, 1) : \dim v_\lambda < 1\}$$

is contained in the algebraic numbers and hence is at most countable.

Proof. Theorem II.12.7.8 excludes every transcendental parameter. □

Remark II.12.7.10. Theorem II.12.7.8 is a full-dimension theorem, not an absolute-continuity theorem. The distinction is essential: Chapter II.10 begins with a singular probability of dimension one precisely to prevent these two conclusions from being conflated.

II.12.8 Countable exceptional parameters in transverse affine families

We now consider how parameter transversality can force a deficient dimension to be a strict local minimum. The condition must compare *distinct* symbolic addresses; imposing a positive derivative bound on the diagonal would be impossible.

Definition II.12.8.1 (Off-diagonal β -transversality). Let $J \subset \mathbb{R}$ be an open interval, let Λ be a finite alphabet, and let

$$\Phi_t = \{f_{i,t}(x) = r_{i,t}x + s_{i,t} : i \in \Lambda\}, \quad 0 < |r_{i,t}| < 1 \quad (t \in J),$$

be a family of affine IFSs. Assume $r_{i,t}$ and $s_{i,t}$ are C^2 in t and that, on each compact $J' \subset J$, the first two t -derivatives of the coding map

$$\pi_t(\omega) = \lim_{n \rightarrow \infty} f_{\omega_1,t} \circ \cdots \circ f_{\omega_n,t}(0), \quad \omega \in \Lambda^{\mathbb{N}},$$

are bounded uniformly over $\omega \in \Lambda^{\mathbb{N}}$ and $t \in J'$. For distinct ω, ω' let $\ell(\omega, \omega')$ be the length of their longest common prefix, put $d_{\Sigma}(\omega, \omega') = 2^{-\ell(\omega, \omega')}$, and set $\Delta_{\omega, \omega'}(t) = \pi_t(\omega) - \pi_t(\omega')$.

The family is *off-diagonally β -transverse* on J if there are $\beta \geq 0$ and $c_{\beta} > 0$ such that, for every $t \in J$ and every distinct ω, ω' ,

$$|\Delta_{\omega, \omega'}(t)| \leq c_{\beta} d_{\Sigma}(\omega, \omega')^{\beta} \implies |\partial_t \Delta_{\omega, \omega'}(t)| \geq c_{\beta} d_{\Sigma}(\omega, \omega')^{\beta}.$$

For positive weights p_i with $\sum_i p_i = 1$, write

$$s_{\text{sim}}(t) = \frac{-\sum_i p_i \log p_i}{-\sum_i p_i \log |r_{i,t}|}.$$

A parameter t has *deficient dimension* when $\dim \mu_t < \min\{1, s_{\text{sim}}(t)\}$, where μ_t is the self-similar measure of (Φ_t, p) .

This is the source condition of [Wu25, Definition 1.9]; β need not be at most one.

Theorem II.12.8.2 (Strict local improvement). *Let $(\Phi_t, p)_{t \in J}$ satisfy Definition II.12.8.1. If $t_0 \in J$ has deficient dimension, then there are $\delta, \varepsilon > 0$ such that*

$$\dim \mu_t \geq \dim \mu_{t_0} + \delta$$

for every $t \in J$ with $0 < |t - t_0| < \varepsilon$. This is [Wu25, Theorem 1.4]. Its inverse-entropy input is the fully proved Proposition P-II.6.06; the remaining parameter argument is proved next.

Proof. We use base-two entropy throughout. Write $H_k(\theta) = H(\theta, \mathcal{D}_k)$ and

$$e_\ell(\theta^{x,j}) = \ell^{-1} H(\theta^{\mathcal{D}_\ell(x)}, \mathcal{D}_\ell).$$

Fix t_0 , abbreviate $\mu_0 = \mu_{t_0}$, and put

$$d = \dim \mu_0, \quad h = - \sum_i p_i \log_2 p_i, \quad \chi = - \sum_i p_i \log_2 |r_{i,t_0}|, \quad s = h/\chi.$$

Thus $0 \leq d < \min(1, s)$. Work on a compact parameter interval about t_0 , so that all contraction ratios are bounded away from 0 and 1, all attractors lie in a common bounded interval, and the first two parameter derivatives of every coding point are uniformly bounded. Constants below may depend on this interval but never on the moving parameter or the scale.

Choose a letter $a \in \Lambda$. Conjugate every system by the parameter-dependent translation $x \mapsto x - \pi_t(a^\infty)$. The map $f_{a,t}$ now fixes 0 for every t . This does not change dimensions, contraction ratios, coding-point differences, or the transversality condition. The new coding maps retain uniform C^2 parameter bounds. Thus

$$z_I(t) := f_{I,t}(0) = \pi_t(Ia^\infty) \tag{W0}$$

is simultaneously a finite-word anchor and a genuine symbolic coding point. The choice of a moving origin is needed in the transversality step.

A finite-scale self-similar entropy inequality. Uniformly for all nearby u and all integers $k, l \geq 1$,

$$H_{k+l}(\mu_u) \geq H_k(\mu_u) + H_l(\mu_u) - C, \quad H_k(\mu_u) \leq k \dim \mu_u + C. \tag{II.12.8.1}$$

To prove the first inequality, stop a Bernoulli word at the first prefix I for which $|r_{I,u}| \leq 2^{-k}$. Its ratio lies in $[r_* 2^{-k}, 2^{-k}]$ for one fixed $r_* > 0$. If $X \sim \mu_u$ is an independent tail, set $Y = f_{I,u}(X)$ and let Q be the level- k cell of $f_{I,u}(0)$. Since the support of $(f_{I,u})_* \mu_u$ has diameter $O(2^{-k})$, the two coarse variables Q and $\mathcal{D}_k(Y)$ determine each other up to $O(1)$ possibilities. Also $H(Q \mid \mathcal{D}_{k+l}(Y)) = O(1)$. Hence the chain rule, concavity of conditional entropy, and Lemma L-II.6.01 give

$$\begin{aligned} H_{k+l}(Y) &\geq H(Q) + H(\mathcal{D}_{k+l}(Y) \mid Q) - C \\ &\geq H_k(Y) + \sum_I p_I H_{k+l}((f_{I,u})_* \mu_u) - C \\ &\geq H_k(\mu_u) + H_l(\mu_u) - C. \end{aligned}$$

Here $Y \sim \mu_u$, and the signed-affine comparison in the last line is uniform over the stopping ratios. Iterating with $l = k$, dividing by the total scale, and invoking the exact dimensional entropy limit [Lemma L-II.6.02](#) proves the second inequality of [\(II.12.8.1\)](#).

We will use the following quantitative component stability fact. Let X, Y be coupled bounded real random variables with $|X - Y| \leq A2^{-mn}$, where A is fixed. If $d > 0$, fix $0 < a < d$ with $ma > 2$. Exact dimensionality of μ_0 implies that for every $v > 0$ there are F_v and C_v such that $\mu_0(F_v) > 1 - v$ and

$$\mu_0(B(x, r)) \leq C_v r^a \quad (x \in F_v, 0 < r < 1). \quad (\text{W1})$$

When X is any bounded-slope affine image of a μ_0 -point, at most $O(2^{j+\ell})$ level- $(j + \ell)$ grid boundaries meet its support. The inverse image of a boundary strip of width $A2^{-mn}$ has length $O(2^{-mn})$. Therefore, uniformly in the affine translation and for $0 \leq j \leq n$,

$$\mathbb{P}\{\mathcal{D}_{j+\ell}(X) \neq \mathcal{D}_{j+\ell}(Y)\} \leq v + C_v t^{j+\ell} 2^{-man} \leq v + o_n(1). \quad (\text{II.12.8.2})$$

The first term discards F_v^c ; the second follows by covering each boundary strip meeting F_v by a ball centered in F_v and applying [\(W1\)](#). This is why m must depend on the positive dimension d .

For completeness, if two joint laws of a level- j parent cell and a level- $(j + \ell)$ descendant differ in total variation by at most v_1 , denote their parent masses by p_Q, q_Q and their conditional descendant laws by P_Q, Q_Q . Direct subtraction of the joint probabilities gives

$$\sum_Q \min(p_Q, q_Q) \|P_Q - Q_Q\|_{\text{TV}} \leq 2v_1. \quad (\text{W2})$$

Outside parent cells of mass $O(\sqrt{v_1})$, conditional total variation is $O(\sqrt{v_1})$. Entropy continuity on their 2^ℓ children bounds the normalized entropy difference by

$$O\left(\sqrt{v_1} + \frac{h_2(\sqrt{v_1})}{\ell}\right). \quad (\text{W3})$$

Equations [\(II.12.8.2\)](#)–[\(W3\)](#) are the precise stability statements used below, rather than an appeal to weak convergence.

Here is the resulting lower-profile estimate. For each fixed integer $B \geq 1$, compact nonzero slope interval $c_- \leq |c| \leq c_+$, and $\zeta > 0$, one can fix a block length ℓ and then n_0 so that, whenever $|u - t_0| \leq 2^{-mn}$ and $n \geq n_0$, every $\theta = (A_{c,b})_* \mu_u$ in that slope range satisfies

$$\frac{1}{Bn} \sum_{j=0}^{Bn-1} \theta\{x : e_\ell(\theta^{x,j}) < d - \zeta\} < \zeta. \quad (\text{II.12.8.3})$$

Indeed, couple μ_u and μ_0 by the same infinite word; the coding derivative bound gives $|X_u - X_0| \leq C|u - t_0|$. Choose the tolerance in the affine-uniform component theorem [T-II.6.05](#) much smaller than ζ^4 , then choose its ℓ , choose $v \ll \zeta^4$ in [\(II.12.8.2\)](#), and finally enlarge n_0 . Equations [\(W2\)](#)–[\(W3\)](#) transfer the one-block average component estimate of μ_0 to every bounded-slope affine image of μ_u , uniformly over $0 \leq j < n$.

For $q = 0, \dots, B - 1$, decompose θ at the first-cylinder stopping scale 2^{-qn} of μ_u . After magnification by the exact dyadic factor 2^{qn} , level- $(qn + j)$ components become level- j

components; each summand is an affine image of μ_u with slope in another fixed compact nonzero range. The one-block estimate therefore applies uniformly to its levels $qn, \dots, (q+1)n-1$. On any level- j parent cell, the entropy of the whole mixture component is at least the weighted mean of the summand-component entropies. If the total cylinder-point mass of summands below $d - \zeta/2$ is less than $\zeta^2/4$, Markov's inequality shows that on all but $\zeta/2$ of the mixture mass their conditional weight is less than $\zeta/2$; concavity then gives component entropy at least $d - \zeta$. This proves (II.12.8.3). The finitely many q -blocks use the same ℓ , chosen before n .

We also need an upper-profile consequence under a hypothetical small dimension. Put $g = 1 - d > 0$, $\gamma = g/2$, and take $B = 3m$ in (II.12.8.3). If $\dim \mu_u \leq d + \delta$, then (II.12.8.1), Lemma L-II.6.01, and its local entropy-average identity give, for $N = (3m - 1)n$,

$$\frac{1}{N} \sum_{j=0}^{N-1} \int e_{\ell}(\theta^{x,j}) d\theta(x) \leq d + \delta + O(\ell/N) \quad (\theta = (A_{c,b})_* \mu_u).$$

The lower-profile bound and the elementary comparison of the mean with the mass above $1 - \gamma = d + g/2$ yield

$$\mathcal{U}_{N,\ell}(\theta; 1 - \gamma) \leq \frac{2(\delta + 2\zeta + O(\ell/N))}{g}. \quad (\text{II.12.8.4})$$

If $d = 0$, nonnegativity of every component entropy gives the same conclusion with $\gamma = 1/2$ directly from the preceding mean bound; no Frostman set or lower-profile estimate is used.

Stopping words and symbolic entropy lost at t_0 . At t_0 define the prefix-free stopping partition

$$\mathcal{W}_n = \{I : |r_{I,t_0}| \leq 2^{-n} < |r_{I^-,t_0}|\}.$$

For $I \in \mathcal{W}_n$, $|r_{I,t_0}| \asymp 2^{-n}$ and $|I| \leq C_0 n$. The stopped Bernoulli word has law p_I and entropy

$$H(I) = \sum_{I \in \mathcal{W}_n} p_I \log_2(1/p_I) = sn + O(1). \quad (\text{II.12.8.5})$$

In fact $-\log_2 |r_{I,t_0}| \in [n, n + C]$. The elementary stopping identity, applied first to $-\log_2 |r_{i,t_0}|$ and then to $-\log_2 p_i$, says $\mathbb{E}|I| = (n + O(1))/\chi$ and $\mathbb{E}[-\log_2 p_I] = h\mathbb{E}|I|$, proving the display. Also $|\mathcal{W}_n| \leq |\Lambda|^{C_0 n + 1}$.

Let $Q(I)$ be the level- n dyadic cell containing the anchor $z_I(t_0)$, let $A_Q = \{I \in \mathcal{W}_n : Q(I) = Q\}$, and put $a_Q = \sum_{I \in A_Q} p_I$. For $a_Q > 0$, define

$$\eta_Q^u = \sum_{I \in A_Q} \frac{p_I}{a_Q} \delta_{z_I(u)}, \quad \nu_Q^u = \sum_{I \in A_Q} \frac{p_I}{a_Q} (f_{I,u})_* \mu_u.$$

The same-word coding coupling gives $|\pi_u(\omega) - \pi_{t_0}(\omega)| \leq C|u - t_0|$. When $|u - t_0| \leq 2^{-mn}$ with $m > 1$, a point in the stopped cylinder $f_{I,u}(K_u)$ lies within $C2^{-n}$ of $z_I(t_0)$. Thus the label $Q(I)$ and the level- n cell of that point determine each other up to boundedly many possibilities. The same is true when the point is recorded at any finer level $K \geq n$. The chain rule therefore gives

$$H_K(\mu_u) = H((a_Q)_Q) + \sum_Q a_Q H_K(\nu_Q^u) + O(1), \quad H((a_Q)_Q) = H_n(\mu_0) + O(1). \quad (\text{II.12.8.6})$$

The error is uniform in K, n, u ; the identity is simply the entropy of the joint pair consisting of $Q(I)$ and the fine location, followed by the bounded conditional entropy of $Q(I)$ given that location.

Since $H_n(\mu_0) = dn + o(n)$, the entropy chain rule for $I \mapsto Q(I)$ and (II.12.8.5) give

$$\sum_Q a_Q H((p_I/a_Q)_{I \in A_Q}) = (s-d)n + o(n).$$

The conditional entropy is at most $C_1 n$ in each group. As $s-d > 0$, there are fixed $a_0, a_1 > 0$ and a family $\mathcal{B}_1$ of coarse cells for which

$$\sum_{Q \in \mathcal{B}_1} a_Q \geq a_0, \quad H((p_I/a_Q)_{I \in A_Q}) \geq a_1 n \quad (Q \in \mathcal{B}_1). \quad (\text{II.12.8.7})$$

For example one may take $a_1 < (s-d)/4$ and then choose $a_0 > 0$ using the upper bound $C_1 n$.

Fix the transversality constants β, c_β from Definition II.12.8.1. Choose an integer m large enough that

$$m > \beta C_0 + 4, \quad m > C_1 + 4, \quad ma > 2 \text{ if } d > 0,$$

where $a \in (0, d)$ was fixed in the stability argument. Given $u \neq t_0$ sufficiently near t_0 , choose n so that

$$2^{-m(n+1)} \leq |u - t_0| < 2^{-mn}, \quad L = 3mn, \quad N = L - n = (3m-1)n. \quad (\text{II.12.8.8})$$

Then $n \rightarrow \infty$ as $u \rightarrow t_0$. Uniform C^1 dependence of the finitely many contraction ratios and $|I| \leq C_0 n$ show that, for all large n ,

$$(r_*/2)2^{-n} \leq |r_{I,u}| \leq 2 \cdot 2^{-n} \quad (I \in \mathcal{W}_n). \quad (\text{W4})$$

For either $u = t_0$ or a parameter satisfying (II.12.8.8), the number of distinct signed ratios $r_{I,u}$, $I \in \mathcal{W}_n$, is at most $(C_0 n + 2)^{|\Lambda|}$: each product is determined by the symbol-count vector. The entropy of the ratio label is consequently $O(\log n)$.

Most coarse groups do not grow at t_0 . Concavity and the signed-affine comparison show, uniformly in Q , that

$$H_L(v_Q^{t_0}) \geq H_N(\mu_0) - C = dN - o(n).$$

On the other hand, (II.12.8.6) at t_0 and $H_L(\mu_0) = dL + o(n)$ imply $\sum_Q a_Q H_L(v_Q^{t_0}) = dN + o(n)$. Consequently, for any fixed small $\varepsilon > 0$, all sufficiently large n admit a family $\mathcal{B}_2$ such that

$$\sum_{Q \in \mathcal{B}_2} a_Q > 1 - \varepsilon, \quad H_L(v_Q^{t_0}) \leq dN + \varepsilon n \quad (Q \in \mathcal{B}_2). \quad (\text{II.12.8.9})$$

Indeed the uniform lower error is $o(n)$, so the positive excess has mean $o(n)$; apply Markov with threshold εn .

We next prove that the conditional anchor law at t_0 has little fine-scale entropy on these groups:

$$H_L(\eta_Q^{t_0}) < \frac{1}{4} a_1 n \quad (Q \in \mathcal{B}_2), \quad (\text{II.12.8.10})$$

provided ε was fixed sufficiently small. Split A_Q into its distinct signed-ratio classes r , write $w_{Q,r}$ for their conditional weights, and denote the corresponding anchor and cylinder laws

by $\eta_{Q,r}^{t_0}$ and $v_{Q,r}^{t_0}$. The mixture entropy upper bound and the $O(\log n)$ ratio-label entropy show that, if (II.12.8.10) fails, then

$$\sum_r w_{Q,r} H_L(\eta_{Q,r}^{t_0}) \geq \frac{1}{5} a_1 n$$

for all large n . Each such entropy is at most $N + O(1)$, since the anchors lie in one level- n cell. Hence a fixed positive ratio-class mass $b_0 > 0$ has $H_L(\eta_{Q,r}^{t_0}) \geq a_1 n/10$. Within one ratio class,

$$v_{Q,r}^{t_0} = \eta_{Q,r}^{t_0} * (S_r)_* \mu_0. \quad (\text{W5})$$

Magnify both factors by 2^n . By (W4), the tail becomes an affine image of μ_0 with slope in a fixed compact nonzero range. After separate integer translations both factors lie in one fixed interval $[-R, R]$; their convolution is then translated by an integer, which changes no dyadic entropy or component profile. By Theorem T-II.6.05, the tail's upper component profile above $(1+d)/2$ is arbitrarily small for a fixed sufficiently large block length and then all large N . The anchor factor has normalized level- N entropy at least $a_1/[20(3m-1)]$. Apply the fully proved Proposition P-II.6.06 with a porosity threshold strictly below that number. It gives a uniform $c_0 > 0$ such that each of the selected ratio classes satisfies

$$H_L(v_{Q,r}^{t_0}) \geq H_L((S_r)_* \mu_0) + c_0 N - O(1) = dN + c_0 N - o(n).$$

For all other classes, conditioning on the translation and using the bounded same-level entropy effect of translation gives $H_L(v_{Q,r}^{t_0}) \geq dN - o(n)$. Concavity over classes would therefore give

$$H_L(v_Q^{t_0}) \geq dN + b_0 c_0 N - o(n),$$

contradicting (II.12.8.9) once $\varepsilon < b_0 c_0 (3m-1)/2$. This proves (II.12.8.10).

The moving anchors separate at the perturbed parameter. Suppose $I \neq J$ belong to $\mathcal{W}_n$ and $z_I(t_0), z_J(t_0)$ lie in the same level- L dyadic cell. The addresses Ia^∞, Ja^∞ are distinct because $\mathcal{W}_n$ is prefix-free. Their common-prefix length is at most $C_0 n$. Thus their difference $\Delta(t) = z_I(t) - z_J(t)$ satisfies

$$|\Delta(t_0)| \leq 2^{-L} \leq c_\beta 2^{-\beta C_0 n}$$

for all large n , and transversality gives $|\Delta'(t_0)| \geq c_\beta 2^{-\beta C_0 n}$. The uniform second-derivative bound and $|u - t_0| < 2^{-mn}$, with $m > \beta C_0 + 4$, ensure that Δ' keeps its sign and at least half its magnitude on the interval between t_0 and u . It follows from (II.12.8.8) that

$$|\Delta(u)| \geq \frac{1}{2} c_\beta |u - t_0| 2^{-\beta C_0 n} - 2^{-L} > 2^{-L}$$

for all sufficiently large n . In particular $z_I(u)$ and $z_J(u)$ occupy different level- L cells. Conditional on the level- L cell of $z_I(t_0)$, the level- L cell of $z_I(u)$ therefore determines I . For $Q \in \mathcal{B}_1 \cap \mathcal{B}_2$, the ordinary entropy chain rule yields

$$H_L(\eta_Q^u) \geq H(I \mid \mathcal{D}_L(z_I(t_0)), I \in A_Q) = H((p_I/a_Q)_{I \in A_Q}) - H_L(\eta_Q^{t_0}) \geq \frac{3}{4} a_1 n. \quad (\text{II.12.8.11})$$

Entropy growth at u . We show that a sufficiently small dimension $\dim \mu_u - d$ is incompatible with (II.12.8.11). For $Q \in \mathcal{B}_1 \cap \mathcal{B}_2$, split A_Q by its signed ratios $r_{I,u}$. The ratio-label entropy is $O(\log n)$, while every conditional anchor law has level- L entropy

at most $N + O(1)$. Hence (II.12.8.11) implies that a fixed positive ratio-class mass $b_1 > 0$ consists of classes with

$$H_L(\eta_{Q,r}^u) \geq \frac{1}{4}a_1n. \quad (\text{II.12.8.12})$$

The number b_1 depends only on a_1, m , not on Q, u, n . As in (W5), $v_{Q,r}^u = \eta_{Q,r}^u * (S_r)_* \mu_u$.

Fix

$$0 < \rho < \frac{a_1}{64(3m-1)} < \frac{a_1}{16(3m-1)} =: \beta_1.$$

Choose $\zeta > 0$, and then $\delta > 0$, sufficiently small that (II.12.8.4) is at most ρ for all large n whenever $\dim \mu_u \leq d + \delta$. In the positive-dimensional case choose the fixed block length ℓ supplied by (II.12.8.3); when $d = 0$, any fixed $\ell \geq 1$ works. After magnification by 2^n and integer centering, all tail factors $(S_r)_* \mu_u$ become probabilities on one fixed $[-R, R]$, and their upper component profiles obey the same bound ρ . By (II.12.8.12), their anchor factors have normalized level- N entropy at least β_1 for all large n . Thus Proposition P-II.6.06 with the fixed parameters $R, \gamma, \ell, \rho, \beta_1$ gives a uniform $c_1 > 0$ such that

$$H_L(v_{Q,r}^u) \geq H_L((S_r)_* \mu_u) + c_1N - O(1) \quad (\text{II.12.8.13})$$

for the ratio classes in (II.12.8.12). For every other class the same estimate holds without the c_1N term, by conditioning on its translation.

The common-symbol coupling gives $|\pi_u(\omega) - \pi_{t_0}(\omega)| \leq C2^{-mn}$. Since $m > 1$, bounded cell-neighbor multiplicity implies

$$H_n(\mu_u) = H_n(\mu_0) + O(1) = dn + o(n).$$

Iterate (II.12.8.1) $3m - 1$ times and apply signed-affine entropy comparison with (W4). Uniformly over all stopped ratios $r = r_{I,u}$,

$$H_L((S_r)_* \mu_u) \geq H_N(\mu_u) - O(1) \geq dN - o(n). \quad (\text{II.12.8.14})$$

Concavity over the ratio classes, followed by (II.12.8.13), now gives

$$\begin{aligned} H_L(v_Q^u) &\geq dN + b_1c_1N - o(n) \quad (Q \in \mathcal{B}_1 \cap \mathcal{B}_2), \\ H_L(v_Q^u) &\geq dN - o(n) \quad (a_Q > 0). \end{aligned}$$

The $o(n)$ is uniform in Q and the allowed parameter u . Choose the fixed error ε in (II.12.8.9) small enough for both the earlier argument and $\varepsilon < a_0/2$. Then $\sum_{Q \in \mathcal{B}_1 \cap \mathcal{B}_2} a_Q \geq a_0/2$. Substitute these estimates and $H((a_Q)) = dn + o(n)$ in (II.12.8.6). We obtain

$$H_L(\mu_u) \geq dL + \frac{1}{2}a_0b_1c_1N - o(n) \geq (d + 2\delta_*)L$$

for one fixed $\delta_* > 0$ and all sufficiently large n . The second part of (II.12.8.1) yields $\dim \mu_u \geq d + \delta_*$, after decreasing δ_* once to absorb C/L . At the earlier choice of δ , require in addition $\delta < \delta_*$. This contradicts $\dim \mu_u \leq d + \delta$. The choices depended only on t_0 and the family, so one common neighborhood and one positive δ work for every $u \neq t_0$ in that neighborhood. $\square$

Lemma II.12.8.3 (Strict local minima are countable). *For an arbitrary function f on an open real interval, the set of points t for which $f(t) < f(t')$ for every sufficiently close $t' \neq t$ is at most countable.*

Proof. For rational $a < b$, let $E_{a,b}$ consist of points $t \in (a, b)$ with $f(x) > f(t)$ for all $x \in (a, b) \setminus \{t\}$. Each $E_{a,b}$ has at most one point, and every strict local minimum belongs to some $E_{a,b}$. There are only countably many rational intervals. $\square$

Corollary II.12.8.4 (Countable exceptional set). *For every family satisfying Definition II.12.8.1, the deficient-dimension parameters form an at most countable set. The source states this as [Wu25, Theorem 1.2]; the deduction below is included for completeness.*

Proof. Apply Lemma II.12.8.3 to $t \mapsto \dim \mu_t$. $\square$

II.12.9 A combinatorial entropy-growth theorem and CP projections

We now isolate the finite combinatorial input used in Wu's countable-exceptional-direction argument. Throughout this section, logarithms are base two. For $x \in \mathbb{R}$, $\mathcal{D}_k(x)$ denotes the level- k dyadic interval containing x , and for a probability σ and a dyadic cell Q of positive mass, σ^Q denotes the normalized restriction to Q , rescaled affinely to $[0, 1)$.

The first lemma is the asymmetric Balog–Szemerédi–Gowers form needed below. It is not imported as a black box: it is an exponential-scale corollary of Proposition II.11.2.4 from Chapter II.11.

Lemma II.12.9.1 (Exponential-scale asymmetric BSG). *For every $\varepsilon > 0$ there is $\delta > 0$ such that, for all sufficiently large n , the following holds. Let $X, Y \subset [-2, 2] \cap 2^{-n}\mathbb{Z}$ be nonempty finite sets and let $G \subset X \times Y$. Suppose*

$$|X| \leq 2^n |Y|, \quad |G| \geq 2^{-\delta n} |X| |Y|,$$

and the restricted sumset

$$S_G = \{x + y : (x, y) \in G\}$$

satisfies

$$|S_G| \leq 2^{\delta n} |X|.$$

Then there are $X' \subset X$ and $Y' \subset Y$ such that

$$|X'| \geq 2^{-\varepsilon n} |X|, \quad |Y'| \geq 2^{-\varepsilon n} |Y|, \quad |X' + Y'| \leq 2^{\varepsilon n} |X'|.$$

Proof. Let $r_G(s) = |\{(x, y) \in G : x + y = s\}|$. Cauchy–Schwarz gives

$$E(X, Y) \geq \sum_s r_G(s)^2 \geq \frac{|G|^2}{|S_G|} \geq 2^{-3\delta n} |X| |Y|^2.$$

Apply Proposition II.11.2.4 with $L = 2^n$, with

$$\alpha = 2^{-3\delta n-1},$$

and with its small exponent parameter $\eta > 0$ to be chosen momentarily. Enlarging one harmless constant, the proposition supplies finite sets H, Z and a translate $H + y_0$ for which

$$\begin{aligned} |H + H| &\leq \Lambda |H|, \\ |Y \cap (H + y_0)| &\geq \Lambda^{-1} |Y|, \\ |X \cap (Z + H)| &\geq \Lambda^{-1} |X|, \\ |Z| |H| &\leq \Lambda |X|, \end{aligned}$$

where

$$\Lambda \leq C_\eta 2^{(3C_\eta \delta + \eta)n} (1 + n)^{C_\eta}.$$

Here we used $|Y| \leq O(2^n)$, so the logarithmic factor in Proposition II.11.2.4 is polynomial in n .

Put

$$X' = X \cap (Z + H), \quad Y' = Y \cap (H + y_0).$$

Then

$$X' + Y' \subset Z + H + H + y_0,$$

so

$$|X' + Y'| \leq |Z| |H + H| \leq \Lambda^2 |X| \leq \Lambda^3 |X'|.$$

Choose first $\eta > 0$ so small that $4\eta < \varepsilon/2$, then choose $\delta > 0$ so small that $12C_\eta \delta < \varepsilon/2$. Since $(1 + n)^{C_\eta} = 2^{o(n)}$, for all sufficiently large n we have $\Lambda^3 \leq 2^{\varepsilon n}$ and $\Lambda \leq 2^{\varepsilon n}$. The three asserted conclusions follow. $\square$

We reuse the one-dimensional Shannon entropy inverse theorem already recorded as Theorem II.12.4.8. The next elementary identity converts its first alternative into a bound on the number of such scales.

Lemma II.12.9.2 (Sliding local-entropy average). *For every probability σ on a bounded interval and integers $n, T \geq 1$,*

$$H(\sigma, \mathcal{D}_n) = \frac{1}{T} \sum_{k=0}^{n-1} \int H(\sigma^{\mathcal{D}_k(x)}, \mathcal{D}_T) d\sigma(x) + O(T),$$

where the implicit constant is absolute after enlarging the ambient interval by a fixed amount.

Proof. Let

$$h_r = H(\sigma, \mathcal{D}_{r+1} \mid \mathcal{D}_r), \quad r \geq 0.$$

The dyadic chain rule gives

$$H(\sigma, \mathcal{D}_n) = \sum_{r=0}^{n-1} h_r.$$

For fixed k ,

$$\int H(\sigma^{\mathcal{D}_k(x)}, \mathcal{D}_T) d\sigma(x) = \sum_{r=k}^{k+T-1} h_r,$$

up to an $O(1)$ boundary term if the bounded ambient interval is not itself dyadic. Summing over $0 \leq k < n$, every h_r with $T \leq r \leq n-1$ is counted exactly T times, whereas the first and last T levels contribute at most $O(T^2)$ because $0 \leq h_r \leq 1 + O(1/T)$ in one dimension. Divide by T . $\square$

We next need a quantitative porosity statement. The formulation separates the density of scales on which the original measure loses mass from the density of scales on which the selected set has only one descendant. This avoids an unnecessary equality of the two parameters.

Lemma II.12.9.3 (Mass loss against a one-branch subtree). *Fix $0 < \tau < 1$, an integer $\ell \geq 1$, and numbers $0 \leq a < b \leq 1$. There is $\rho = \rho(\tau, \ell, b-a) > 0$ such that the following holds for all sufficiently large n .*

Let η be a probability on $[0, 1)$. Suppose $A \subset [0, 1)$ has the property that every $x \in A$ satisfies

$$|\{1 \leq k \leq n - \ell : \eta(\mathcal{D}_{k+\ell}(x)) \leq \tau \eta(\mathcal{D}_k(x))\}| \geq (1-a)n.$$

If $D \subset A$ and every $x \in D$ satisfies

$$|\{1 \leq k \leq n - \ell : \#\{Q \in \mathcal{D}_{k+\ell} : Q \subset \mathcal{D}_k(x), Q \cap D \neq \emptyset\} = 1\}| \geq bn,$$

then

$$\eta(D) \leq 2^{-\rho n}.$$

Proof. For a dyadic cell $Q \in \mathcal{D}_k$ meeting D , define

$$a_Q = \frac{\sum \{\eta(Q') : Q' \in \mathcal{D}_{k+1}, Q' \subset Q, Q' \cap D \neq \emptyset\}}{\eta(Q)}.$$

Starting from mass 1 at $[0, 1)$, distribute mass down the dyadic tree by keeping only children meeting D and, inside each parent Q , renormalizing the original η -weights by the factor a_Q . This constructs a probability ν on level- n cells meeting D and, for every $x \in D$,

$$\nu(\mathcal{D}_n(x)) = \eta(\mathcal{D}_n(x)) \prod_{k=0}^{n-1} a_{\mathcal{D}_k(x)}^{-1}. \quad (\text{II.12.9.1})$$

Let G_x be the set of scales at which both displayed hypotheses hold. By inclusion-exclusion,

$$|G_x| \geq (b-a)n - O(\ell).$$

If $k \in G_x$, there is only one level- $(k+\ell)$ descendant of $\mathcal{D}_k(x)$ meeting D . Therefore

$$\prod_{j=0}^{\ell-1} a_{\mathcal{D}_{k+j}(x)} = \frac{\eta(\mathcal{D}_{k+\ell}(x))}{\eta(\mathcal{D}_k(x))} \leq \tau.$$

Multiply these inequalities over $k \in G_x$. Each factor $a_{\mathcal{D}_r(x)}$ appears in at most ℓ of the products and lies in $(0, 1]$. Hence

$$\left(\prod_{r=0}^{n-1} a_{\mathcal{D}_r(x)} \right)^\ell \leq \tau^{|G_x|},$$

so, for all sufficiently large n ,

$$\prod_{r=0}^{n-1} a_{\mathcal{D}_r(x)}^{-1} \geq \tau^{-\frac{b-a}{2\ell}n}.$$

Combining this with (II.12.9.1) gives

$$v(\mathcal{D}_n(x)) \geq \eta(\mathcal{D}_n(x)) \tau^{-\frac{b-a}{2\ell}n}.$$

Summing over the level- n cells meeting D yields

$$\eta(D) \leq \tau^{\frac{b-a}{2\ell}n}.$$

Thus one may take any

$$0 < \rho < \frac{b-a}{2\ell} \log_2(1/\tau).$$

□

The tree regularization already proved in Chapter II.11 gives a particularly convenient substitute for the measure-regularization step in Wu's original proof.

Lemma II.12.9.4 (Uniform block tree extracted from a finite set). *For every sufficiently large block length T there is $\kappa_T > 0$, with $\kappa_T \rightarrow 0$ as $T \rightarrow \infty$, such that the following holds. For all sufficiently large n and every nonempty*

$$E \subset [0, 1] \cap 2^{-n}\mathbb{Z},$$

there is $E_ \subset E$ satisfying*

$$|E_*| \geq 2^{-\kappa_T n - O(T)} |E|$$

with the following property. If $m = \lfloor n/T \rfloor T$ and λ_ is the uniform probability on E_* , then for every block boundary $rT \leq m$, all level- rT dyadic cells meeting E_* have the same λ_* -mass.*

Proof. Choose one point of E in each occupied level- m cell. This loses at most the factor $2^{n-m} \leq 2^T$. Apply Lemma II.11.2.5 to the resulting 2^{-m} -set. Its retention estimate gives

$$|E_*| \geq (cT^{-2})^{m/T} 2^{-T} |E| \geq 2^{-C(\log T)n/T - O(T)} |E|.$$

Set $\kappa_T = C(\log T)/T$.

At every block, Lemma II.11.2.5 gives the same branching number for every occupied parent. Hence every occupied level- rT cell has exactly

$$\prod_{s=r}^{m/T-1} R_s$$

terminal descendants, where inactive blocks have $R_s = 1$. The uniform measure on E_* therefore assigns equal mass to all occupied level- rT cells. $\square$

Theorem II.12.9.5 (Wu's finite combinatorial growth theorem). *Fix $\gamma \in (0, 1)$ and an integer $\ell \geq 1$. There is $\delta > 0$ such that, for all sufficiently large n , the following holds. Let*

$$A \subset [0, 1] \cap 2^{-n}\mathbb{Z}, \quad \eta \in \mathcal{P}([0, 1] \cap 2^{-n}\mathbb{Z}),$$

with $\eta(A) \geq 1/2$, and suppose that every $a \in A$ satisfies

$$|\{1 \leq k \leq n : \eta(\mathcal{D}_k(a)) \geq 2\eta(\mathcal{D}_{k+\ell}(a))\}| \geq (1 - \gamma/2)n. \quad (\text{II.12.9.2})$$

Let $B \subset [0, 1] \cap 2^{-n}\mathbb{Z}$, and for each $a \in A$ let $B_a \subset B$. If

$$|B| \leq 2^{(1-\gamma)n}, \quad |B_a| \geq |B|^{1-\delta} \quad (a \in A),$$

then

$$\left| \bigcup_{a \in A} (a + B_a) \right| \geq |B|^{1+\delta}.$$

Proof. We give the parameter order first. Apply Lemma II.12.9.3 with

$$\tau = \frac{1}{2}, \quad a = \frac{\gamma}{2}, \quad b = \frac{3\gamma}{4},$$

and denote the resulting exponent by $\rho > 0$. Choose $0 < \tau_0 \ll \gamma$ so small that

$$\frac{1-\gamma}{(1-\tau_0)^2} + 2\tau_0 < 1 - \frac{7\gamma}{8}. \quad (\text{II.12.9.3})$$

Choose T so large that

$$\kappa_T + \frac{1}{T} < \frac{\rho}{8}, \quad \frac{\ell}{T} < \frac{1}{20}, \quad (\text{II.12.9.4})$$

where κ_T is from Lemma II.12.9.4. Apply Theorem II.12.4.8 with $k_1 = T$, $k_2 = 2T$ and error τ_0 ; let $\delta_{\text{ent}} > 0$ be the entropy-gap tolerance. Finally choose

$$0 < \zeta < \min \left\{ \frac{\delta_{\text{ent}}}{2}, \frac{\rho}{8}, \frac{\gamma}{100} \right\},$$

and let $\delta_{\text{BSG}} > 0$ be supplied by Lemma II.12.9.1 for output exponent ζ . The theorem will follow once $\delta > 0$ is chosen sufficiently small compared with δ_{BSG} and ζ .

Suppose, to the contrary, that for arbitrarily large n there are data satisfying the hypotheses and

$$\left| \bigcup_{a \in A} (a + B_a) \right| < |B|^{1+\delta}. \quad (\text{II.12.9.5})$$

We may assume $|B| \geq 2$, since the conclusion is trivial otherwise.

Step 1: a terminal-mass class. For $j \geq 0$ put

$$A(j) = \{a \in A : 2^{-n(j+1)/T} < \eta(\{a\}) \leq 2^{-nj/T}\}.$$

The classes with $j \geq T + 1$ carry total mass at most $2^{-n/T}$, because there are at most $2^n + 1$ atoms. Hence some $j_0 \in \{0, \dots, T\}$ gives a class

$$A_0 = A(j_0)$$

with

$$\eta(A_0) \geq \frac{1/2 - 2^{-n/T}}{T + 1}, \quad (\text{II.12.9.6})$$

and for $a, a' \in A_0$,

$$2^{-n/T} \leq \frac{\eta(\{a\})}{\eta(\{a'\})} \leq 2^{n/T}. \quad (\text{II.12.9.7})$$

This is the only place where the assumption $\eta(A) \geq 1/2$ is used quantitatively.

Step 2: the asymmetric BSG reduction. Let

$$G = \{(b, a) \in B \times A_0 : b \in B_a\}.$$

We intentionally place B on the first side. This is the correct orientation for the asymmetric BSG estimate, because the restricted sumset in (II.12.9.5) is naturally measured relative to $|B|$. We have

$$|B| \leq 2^n |A_0|,$$

and

$$|G| \geq |A_0| |B|^{1-\delta} \geq 2^{-\delta n} |A_0| |B|.$$

Moreover

$$|S_G| \leq |B|^{1+\delta} \leq 2^{\delta n} |B|.$$

For $\delta \leq \delta_{\text{BSG}}$, Lemma II.12.9.1 therefore gives sets

$$B' \subset B, \quad A' \subset A_0$$

satisfying

$$|B'| \geq 2^{-\zeta n} |B|, \quad |A'| \geq 2^{-\zeta n} |A_0|, \quad |A' + B'| \leq 2^{\zeta n} |B'|. \quad (\text{II.12.9.8})$$

Step 3: regularize the A' -tree. Apply Lemma II.12.9.4 to A' and obtain

$$A'' \subset A', \quad |A''| \geq 2^{-\kappa_T n - O(T)} |A'|.$$

Let ν_A and ν_B be the uniform probabilities on A'' and B' . Since distinct 2^{-n} -grid points occupy distinct level- n cells,

$$H(\nu_B, \mathcal{D}_n) = \log |B'|.$$

Also, by (II.12.9.8),

$$H(v_B * v_A, \mathcal{D}_n) \leq \log |A'' + B'| \leq \log |B'| + \zeta n.$$

Our choice $\zeta < \delta_{\text{ent}}$ makes Theorem II.12.4.8 applicable. We obtain $I, J \subset \{1, \dots, n\}$ with

$$|I \cup J| \geq (1 - \tau_0)n, \quad (\text{II.12.9.9})$$

and the following alternatives:

$$v_B \left(x : H(v_B^{\mathcal{D}_i(x)}, \mathcal{D}_T) \geq (1 - \tau_0)T \right) > 1 - \tau_0, \quad i \in I, \quad (\text{II.12.9.10})$$

$$v_A \left(x : v_A^{\mathcal{D}_j(x)}(Q) \geq 1 - \tau_0 \text{ for some } Q \in \mathcal{D}_{2T} \right) > 1 - \tau_0, \quad j \in J. \quad (\text{II.12.9.11})$$

The entropy of v_B is bounded by

$$H(v_B, \mathcal{D}_n) = \log |B'| \leq \log |B| \leq (1 - \gamma)n.$$

On the other hand, (II.12.9.10) and Lemma II.12.9.2 give

$$(1 - \gamma)n \geq |I|(1 - \tau_0)^2 - O(T).$$

Hence, for n large and using (II.12.9.3),

$$|I| \leq \left(1 - \frac{7\gamma}{8} + \tau_0 \right) n.$$

Together with (II.12.9.9), this yields

$$|J| \geq \frac{4\gamma}{5}n \quad (\text{II.12.9.12})$$

for our choice of τ_0 and all large n .

Step 4: atomic components force one-branch blocks. Discard from J the $O(T)$ indices within distance $2T$ of the terminal block boundary $m = \lfloor n/T \rfloor T$, and call the remaining set J_0 . For $j \in J_0$, let F_j be the set of x for which the event in (II.12.9.11) holds. Then

$$v_A(F_j) > 1 - \tau_0.$$

Fix $j \in J_0$ and $x \in F_j$. Choose the unique integer k such that

$$j < kT \leq j + T.$$

Then $(k + 1)T \leq j + 2T \leq m$. There is a level- $(j + 2T)$ cell $Q_0 \subset \mathcal{D}_j(x)$ with

$$v_A(Q_0) > (1 - \tau_0)v_A(\mathcal{D}_j(x)).$$

Let $R_0 \in \mathcal{D}_{(k+1)T}$ contain Q_0 . By Lemma II.12.9.4, every occupied level- $(k + 1)T$ cell has the same v_A -mass. If two such cells inside $\mathcal{D}_j(x)$ met A'' , then each would have mass at most half of $v_A(\mathcal{D}_j(x))$, contradicting $\tau_0 < 1/2$ and the preceding display. Thus there is exactly one occupied level- $(k + 1)T$ cell inside $\mathcal{D}_j(x)$.

Consequently, for every

$$kT \leq r \leq (k+1)T - \ell,$$

there is exactly one cell of $\mathcal{D}_{r+\ell}$ contained in $\mathcal{D}_r(x)$ that meets A'' .

By Fubini and Markov, there is $F \subset A''$ with

$$v_A(F) > 1 - \sqrt{\tau_0} \quad (\text{II.12.9.13})$$

and such that every $x \in F$ belongs to at least $(1 - \sqrt{\tau_0})|J_0|$ of the sets F_j . Each block of length T contains at most T indices j , so (II.12.9.12) and our choice of T imply that, for every $x \in F$,

$$|\{1 \leq r \leq n - \ell : \#\{Q \in \mathcal{D}_{r+\ell} : Q \subset \mathcal{D}_r(x), Q \cap F \neq \emptyset\} = 1\}| \geq \frac{3\gamma}{4}n \quad (\text{II.12.9.14})$$

for all sufficiently large n . Indeed, every block hit by one of the relevant j 's contributes at least $T - \ell$ singleton scales, and the block intervals are disjoint.

Step 5: contradiction with the original branching. Because $F \subset A_0 \subset A$, hypothesis (II.12.9.2) and (II.12.9.14) allow us to apply Lemma II.12.9.3; hence

$$\eta(F) \leq 2^{-\rho n}. \quad (\text{II.12.9.15})$$

We now obtain the opposite lower bound. From (II.12.9.13),

$$|F| \geq (1 - \sqrt{\tau_0})|A''|.$$

Combining (II.12.9.8), Lemma II.12.9.4, and (II.12.9.7),

$$\begin{aligned} \eta(F) &\geq \eta(A_0) 2^{-n/T} \frac{|F|}{|A_0|} \\ &\geq c_T (1 - \sqrt{\tau_0}) 2^{-n/T} 2^{-\zeta n} 2^{-\kappa_T n - O(T)}, \end{aligned}$$

where $c_T > 0$ is independent of n by (II.12.9.6). The choices (II.12.9.4) and $\zeta < \rho/8$ make the exponent on the last line strictly smaller than ρn for all sufficiently large n . Thus

$$\eta(F) > 2^{-\rho n},$$

contradicting (II.12.9.15). The assumed small union (II.12.9.5) is therefore impossible, and the theorem follows. $\square$

Remark II.12.9.6 (Two bookkeeping points in the source proof). The proof above follows Wu's revised Section 4.2 but makes two quantitative points explicit. First, the asymmetric BSG lemma is applied to the ordered pair (B, A_0) , not (A_0, B) ; this makes the small restricted sumset automatically small relative to the first side. Second, the terminal-mass pigeonhole uses $\eta(A) \geq 1/2$, giving the factor $1/2$ in (II.12.9.6). Neither change affects the theorem, but both remove hidden exponent bookkeeping from the compressed argument.

II.12.9.1 CP distributions and countable exceptional directions

For the remainder of the discussion let $\mathcal{D}_n^{(2)}$ denote the half-open dyadic squares of side 2^{-n} in $[0, 1)^2$. If $D \in \mathcal{D}_n^{(2)}$, let S_D be the orientation-preserving homothety from D onto $[0, 1)^2$. For $x \in D$ and a probability μ with $\mu(D) > 0$, write

$$\mu^D = (S_D)_* \frac{\mu|_D}{\mu(D)}.$$

Definition II.12.9.7 (Dyadic CP distribution). The magnification map is

$$M(\mu, x) = (\mu^{\mathcal{D}_1^{(2)}(x)}, S_{\mathcal{D}_1^{(2)}(x)}x).$$

A probability distribution Q on $\mathcal{P}([0, 1)^2) \times [0, 1)^2$ is *adapted* if, conditionally on its first coordinate being μ , its second coordinate has law μ . It is a *CP distribution* if it is adapted and M -invariant, and it is *ergodic* if the resulting measure-preserving system is ergodic. We identify Q with its measure marginal whenever the point coordinate is not displayed.

For an ergodic CP distribution define

$$\dim Q = \int H(\mu, \mathcal{D}_1^{(2)}) dQ(\mu).$$

CP invariance gives the equivalent formula $\dim Q = \ell^{-1} \int H(\mu, \mathcal{D}_\ell^{(2)}) dQ(\mu)$ for every $\ell \geq 1$. For a fixed orthogonal projection π , the notation $\dim(\pi Q)$ denotes the common exact dimension of $\pi\mu$ for Q -almost every μ ; existence and constancy are part of the cited fixed-projection theorem stated below.

We next state the two-dimensional, base-two centering theorem needed for Wu's argument. The accompanying derivation follows Hochman's CP/scenery equivalence and conditional-fiber mechanism [Hoc10]; the measurable quasi-Palm reconstruction is supplied by the cited theorem.

Definition II.12.9.8 (Centered sceneries, quasi-Palm laws, and EFDs). Let $B = B(0, 1) \subset \mathbb{R}^d$, and let $\mathcal{M}^*$ be the space of Radon measures ν such that $0 \in \text{supp } \nu$ and $\nu(B) = 1$, equipped with the weak topology on bounded sets. Write

$$T_x \nu(A) = \nu(A + x), \quad S_t \nu(A) = \nu(e^{-t} A),$$

and, whenever the denominator is positive, put

$$\rho^* = \frac{\rho}{\rho(B)}, \quad T_x^* \nu = (T_x \nu)^*, \quad S_t^* \nu = (S_t \nu)^*.$$

For $L \in GL_d(\mathbb{R})$, $L^* \nu = (L_* \nu)^*$ whenever it is defined, and $L^* P$ denotes the push-forward of P by this normalized action. A Borel probability P on $\mathcal{M}^*$ is *quasi-Palm* if for every Borel class $\mathcal{A} \subset \mathcal{M}^*$,

$$P(\mathcal{A}) = 0 \iff \nu\{x \in B : T_x^* \nu \in \mathcal{A}\} = 0 \quad \text{for } P\text{-almost every } \nu.$$

A *fractal distribution* is a quasi-Palm probability invariant under every S_t^* , $t \geq 0$. It is an *ergodic fractal distribution* or *EFD* if this scaling flow is ergodic.

A Radon measure η *uniformly scales* to an EFD P if, for η -almost every y ,

$$\frac{1}{T} \int_0^T \delta_{S_t^* T_y^* \eta} dt \longrightarrow P \quad (\text{US})$$

weakly as $T \rightarrow \infty$.

Lemma II.12.9.9 (Continuous centering package). *Let R be an ergodic dyadic CP distribution in $\mathbb{R}^d$. There is an EFD $P = \text{cent}(R)$ with the following properties.*

(i) *If $\mathcal{A}$ is a Borel class of locally finite measures invariant under translation, positive homothety, restriction to a positive-mass bounded neighborhood and scalar normalization, then*

$$R\{\theta : \theta \in \mathcal{A}\} = P(\mathcal{A}). \quad (\text{CT})$$

(ii) *P -almost every measure uniformly scales to P .*

(iii) *If $L \in GL_d(\mathbb{R})$, then the normalized image distribution L^*P is an EFD. Moreover, in the coordinate system with basis $Le_1, \dots, Le_d$, there is an ergodic base-two CP distribution R_L whose continuous centering is L^*P .*

This is the specialized content of [Hoc10, Theorem 1.14 and Proposition 1.28]. The argument below is a detailed guide to the construction but does not reproduce the full measurable scenery formalism.

Proof. We give the construction because it is exactly what removes the apparent coordinate dependence of a CP process.

Step 1: the extended CP law and continuous centering. Take the two-sided natural extension of the CP system. A typical extended state consists of a locally finite measure $\tilde{\theta}$, a marked point $z \in \text{supp } \tilde{\theta}$, and all its dyadic ancestors, with the property that one forward magnification deletes the oldest ancestor. The extension is obtained by the ordinary natural-extension construction for the measure-preserving system (R, M) ; consistency of the measures on nested ancestors follows from the definition of dyadic magnification. Denote the extended law by R^e .

For $0 \leq t < \log 2$, center the state at its marked point and zoom by the continuous phase t :

$$C_t(\tilde{\theta}, z) = S_t^* T_z^* \tilde{\theta}.$$

Define

$$P = \frac{1}{\log 2} \int_0^{\log 2} (C_t)_* R^e dt. \quad (\text{C})$$

Advancing the scenery flow by s merely translates the phase t modulo $\log 2$; every wrap through $\log 2$ is one application of M . Thus P is scaling-flow invariant. The constant-roof suspension over an ergodic transformation is ergodic as an $\mathbb{R}$ -flow, and (C) is a factor of that suspension. Hence the scaling flow on P is ergodic.

Adaptedness gives the quasi-Palm property. Indeed, if $\mathcal{N}$ is a P -null Borel class, apply (C), then Fubini, to the indicator of $\mathcal{N}$. For almost every extended measure the marked point is sampled from that measure, so after replacing the marked point z by a second sampled point $z + y$ we obtain

$$\int \tilde{\theta}\{z + y : S_t^* T_{z+y}^* \tilde{\theta} \in \mathcal{N}\} dR^e(\tilde{\theta}, z) = 0$$

after restricting to any fixed bounded window. Translating back says that for P -almost every ν , for ν -almost every y in that window, $T_y^* \nu \notin \mathcal{N}$. An increasing union of windows gives the forward quasi-Palm implication. For the reverse implication, fix a bounded neighborhood $B \ni 0$ and a positive-mass bounded set C . The same two-point Fubini calculation, now first restricting the recentered measure to B , gives the following kernel domination: the law obtained by sampling a point from $\nu|_C$, translating it to the origin, restricting to the unit window, and normalizing is absolutely continuous with respect to the average of the analogous laws obtained after one further recentering inside B . To verify this, choose a ball $B(0, \rho) \subset B$; whenever the first sampled point lies in C , a second point in its ρ -ball contributes to the latter kernel with positive weight, and Fubini exchanges the order of the two samples. Consequently a class which is null after almost every admissible recentering is already null before recentering. Together with the forward implication this is the usual quasi-Palm equivalence of null classes. Therefore P is an EFD.

The same formula proves (CT). Membership in $\mathcal{A}$ is unchanged by every operation appearing in C_t , and the restriction from the extended state to its distinguished unit cell is one more positive-mass restriction and normalization. Hence the indicator of $\mathcal{A}$ has the same value before and after centering, almost surely.

Step 2: uniform scaling for an EFD. Apply the flow ergodic theorem to a countable convergence-determining family of bounded continuous functions on the compact restricted-measure space. It gives, for P -almost every centered measure ν , convergence of the scenery averages at the origin to P . The quasi-Palm equivalence proved in Step 1 transfers the same full-measure statement from the origin to ν -almost every spatial point. Hence P -almost every measure uniformly scales to P , as in (US).

Step 3: changing coordinates and returning to a CP law. The map $\nu \mapsto L^* \nu$ commutes with the scaling flow. Translation by y before applying L is translation by Ly afterwards, so the quasi-Palm null-set condition is preserved as well. Hence $L^* P$ is an EFD.

It remains to recover a dyadic CP law in the new coordinates. Choose a $L^* P$ -typical measure ν and a ν -typical point x . For a translated dyadic grid whose translation is generic, form the empirical laws of the normalized level- n dyadic components containing x , together with the normalized location of x inside the component. Compactness gives subsequential limits. Adaptedness of every limit follows from the martingale-difference identity

$$\frac{1}{N} \sum_{n < N} \left[f(\nu_n, x_n) - \int f(\nu_n, y) d\nu_n(y) \right] \longrightarrow 0$$

a.s. for bounded continuous f . Magnification invariance follows by shifting the empirical sum by one term. Finally compare a dyadic component with the continuous scenery during the corresponding interval of logarithmic scales. If the marked point is at relative distance at least 2^{-m} from the component boundary, the two normalized measures agree on a fixed

central window after at most $m + O(1)$ extra magnifications. Averaging first in the random grid translation gives bad-level density $O(2^{-m})$; then $m \rightarrow \infty$. Since ν generates L^*P , every resulting CP limit has continuous centering L^*P . Decomposing such a limit into ergodic CP components and using ergodicity of L^*P , one component still has centering L^*P . This is the required R_L . $\square$

Lemma II.12.9.10 (Exact dimension of an ergodic dyadic CP law). *Let R be an ergodic dyadic CP distribution on $[0, 1]^d$. Then for R -almost every measure θ , the measure θ is exact dimensional with one common dimension*

$$\alpha_R = \int -\log_2 \theta(D_1(x)) dR(\theta, x), \quad (\text{ED})$$

where $D_1(x)$ is the first-level dyadic cell containing the marked point. The same statement holds for every ergodic coordinate factor of R .

Proof. Put $I(\theta, x) = -\log_2 \theta(D_1(x))$. Adaptedness gives

$$\int I dR = \int H(\theta, \mathcal{D}_1^{(d)}) dR(\theta) \leq d,$$

so $I \in L^1(R)$. If $(\theta_k, x_k) = M^k(\theta, x)$, the dyadic magnification chain rule telescopes:

$$-\log_2 \theta(D_n(x)) = \sum_{k=0}^{n-1} I(\theta_k, x_k).$$

Birkhoff and ergodicity therefore give

$$-\frac{1}{n} \log_2 \theta(D_n(x)) \longrightarrow \alpha_R$$

for R -almost every pointed state. Disintegrating the adapted law shows that this holds for R -almost every θ and θ -almost every x . The dyadic differentiation lemma identifies these dyadic local exponents with Euclidean lower and upper local dimensions, proving exact dimensionality. A coordinate projection of an ergodic CP law is an ergodic CP factor, so the same proof applies to every coordinate factor. $\square$

Lemma II.12.9.11 (Coordinate dimension conservation for an ergodic CP law). *Let R be an ergodic dyadic CP distribution on $[0, 1]^{m_1} \times [0, 1]^{m_2}$, and let $p(u, v) = u$ be the coordinate projection. For R -almost every measure θ , write*

$$\theta = \int \delta_u \times \theta_u d\bar{\theta}(u), \quad \bar{\theta} = p\theta.$$

Then θ and $\bar{\theta}$ are exact dimensional and, for $\bar{\theta}$ -almost every u , θ_u is exact dimensional with one common dimension δ , satisfying

$$\dim \theta = \dim \bar{\theta} + \delta. \quad (\text{DC})$$

Proof. The coordinate projection of the CP state is itself an ergodic dyadic CP factor. Lemma II.12.9.10, applied to the source and this factor, gives exact dimensionality of θ and $\bar{\theta}$; write their dimensions as a and b .

We prove the fiber identity. Let $I'_n(u)$ and $I''_n(v)$ be the level- n dyadic cells in the first and second coordinate blocks. For a state $(\theta; u, v)$ define

$$J_n(\theta; u, v) = \frac{\theta(I'_n(u) \times I''_1(v))}{\theta(I'_n(u) \times [0, 1]^{m_2})}, \quad K_n = -\log_2 J_n.$$

By martingale convergence in the base coordinate,

$$J_n \longrightarrow J_\infty(\theta; u, v) = \theta_u(I''_1(v)) \quad \text{for almost every state.} \quad (\text{MC})$$

The family K_n has an integrable maximal function. To see this, fix one first-level fiber atom A . For the increasing base sigma-algebras $\mathcal{F}_n$, select, in increasing n , the first atoms on which $\mathbb{P}(A \mid \mathcal{F}_n) < 2^{-N}$, discarding descendants of already selected atoms. These selected atoms are disjoint, and hence

$$\mathbb{P}\left(A \cap \left\{ \sup_n [-\log_2 \mathbb{P}(A \mid \mathcal{F}_n)] > N \right\}\right) \leq 2^{-N}.$$

Summing over the finitely many first-level fiber atoms and then over N proves integrability of $\sup_n K_n$.

We use the elementary triangular ergodic lemma: if $f_n \rightarrow f$ a.e. and $\sup_n |f_n| \in L^1$, then

$$\frac{1}{N} \sum_{k=0}^{N-1} f_{N-k}(T^k z) - \frac{1}{N} \sum_{k=0}^{N-1} f(T^k z) \longrightarrow 0 \quad (\text{TE})$$

for almost every z . Indeed, for fixed q all but the last q terms are bounded by the Birkhoff average of $\sup_{n \geq q} |f_n - f|$; let first $N \rightarrow \infty$, then $q \rightarrow \infty$, and use dominated convergence. The last q terms are handled by truncation of the common integrable maximal function.

Apply (TE) to K_n . Birkhoff and ergodicity show that the average of K_∞ converges to a constant δ . Because magnification restricts a conditional fiber to its selected dyadic child, the product of the successive conditional probabilities telescopes:

$$\sum_{k=0}^{N-1} K_\infty(M^k(\theta; u, v)) = -\log_2 \theta_u(I''_N(v)). \quad (\text{F})$$

Hence θ_u has dyadic local dimension δ at almost every v . The standard dyadic differentiation lemma (dyadic cells and balls have the same lower and upper local dimensions almost everywhere) upgrades this to Euclidean exact dimensionality.

There is a second telescoping identity. At time k , retaining $N - k$ future base digits gives

$$\sum_{k=0}^{N-1} K_{N-k}(M^k(\theta; u, v)) = -\log_2 \theta(I'_N(u) \times I''_N(v)) + \log_2 \bar{\theta}(I'_N(u)). \quad (\text{T})$$

Divide by N . The left side tends to δ by (TE); exact dimensionality of θ and $\bar{\theta}$, together with dyadic differentiation, makes the right side tend to $a - b$. Thus $\delta = a - b$, proving (DC). $\square$

Lemma II.12.9.12 (Conditional fibers of an EFD form an EFD factor). *Let P be an EFD on $\mathbb{R}^2$ and let $\pi \in G(2, 1)$. For P -almost every ν , the conditional measure on the fiber through the origin is defined up to normalization. After identifying $\ker \pi$ with $\mathbb{R}$, set*

$$\mathcal{F}_\pi(\nu) = (\nu_{\pi^{-1}(0)})^*.$$

*Then $P_\pi = (\mathcal{F}_\pi)_*P$ is an EFD, $\mathcal{F}_\pi$ is a scaling-flow factor, and for P -almost every ν and ν -almost every x , the translated conditional fiber through x uniformly scales to P_π . This is the conditional-fiber factor mechanism of [Hoc10, Theorem 1.30]; the proof below records the null-set transfer used later.*

Proof. Disintegrate ν over π . For a fixed ν , the conditional fiber is defined for ν -almost every base point. The quasi-Palm property of P transfers this full-measure statement to the fiber through the origin for P -almost every ν . The origin belongs to the support of that conditional measure, again by disintegration and quasi-Palm, so the normalization in $\mathcal{F}_\pi$ is legitimate.

Uniqueness of disintegration gives the equivariance identity

$$\mathcal{F}_\pi(S_t^* \nu) = S_t^* \mathcal{F}_\pi(\nu) \quad (\text{FE})$$

whenever both sides are defined. Therefore P_π is scaling invariant, and it is ergodic because it is a factor of the ergodic scaling flow on P .

It remains only to check the quasi-Palm property. Let $\mathcal{N}$ be a P_π -null Borel class. Its inverse image under $\mathcal{F}_\pi$ is P -null. Quasi-Palm for P says that, for P -almost every ν and ν -almost every x , the recentered measure $T_x^* \nu$ avoids this inverse image. Disintegrating the x -integral over the fibers and using uniqueness of conditional measures shows that, for P_π -almost every fiber η and η -almost every y , the recentered fiber $T_y^* \eta$ avoids $\mathcal{N}$. This is the forward quasi-Palm null-set implication for P_π . The reverse implication follows from the same two-point kernel domination used in Step 1 of Lemma II.12.9.9, now carried out inside the one-dimensional conditional fiber: disintegration first selects the fiber, and Fubini then exchanges the two points sampled from that fiber. Hence the null classes are equivalent under fiber recentering. Therefore P_π is an EFD. Lemma II.12.9.9(ii), applied to P_π , gives the uniform-scaling assertion. $\square$

Theorem II.12.9.13 (CP centering and conditional fibers). *Let Q be an ergodic dyadic CP distribution on $\mathbb{R}^2$, let $\pi \in G(2, 1)$, and put*

$$\alpha = \dim Q, \quad \beta_\pi = \dim(\pi Q).$$

Then for Q -almost every μ , both μ and $\pi\mu$ are exact dimensional, with dimensions α and β_π , respectively. If $\kappa = \alpha - \beta_\pi > 0$, then for Q -almost every μ and μ -almost every x , the conditional measure $\mu_{\pi^{-1}(\pi x)}$ is exact dimensional of dimension κ and, after translating the fiber to $\mathbb{R}$, it uniformly scales to an ergodic non-atomic fractal distribution. The cited scenery inputs are [Hoc10, Theorems 1.14, 1.22 and 1.30, Proposition 1.28].

Proof. Let $P = \text{cent}(Q)$. By Lemma II.12.9.9, P is an EFD. Exact dimensionality of a measure, and the value of its dimension, are invariant under translation, homothety, positive-mass restriction and normalization. Lemma II.12.9.10 gives a constant α such that Q -almost every μ is exact dimensional of dimension α . This is $\dim Q$.

Fix the direction π . Choose an orthogonal map L for which $p \circ L = \pi$, with $p(u, v) = u$. The image L^*P is an EFD. By Lemma II.12.9.9(iii), in these coordinates it is the centering of an ergodic dyadic CP distribution R . The coordinate projection of R is a one-dimensional ergodic CP factor; hence Lemma II.12.9.10 shows that $p\theta$ is exact dimensional with one common dimension for R -almost every θ . The centering-transfer identity (CT) moves this statement to L^*P , and then back through L and from P to Q . Denote the resulting common projection dimension by β_π . This proves the first assertion. The quantifiers are important: the argument is for each fixed π ; no simultaneous conull set over all directions is asserted.

The exact dimension class in Lemma II.12.9.10 is preserved by continuous centering and by the orthogonal map L . Hence $\dim R = \alpha$ and $\dim(pR) = \beta_\pi$. Applying Lemma II.12.9.11 to R therefore gives the common conditional-fiber dimension

$$\delta = \dim R - \dim(pR) = \alpha - \beta_\pi = \kappa.$$

Dimension conservation is unchanged by the translations, homotheties and normalizations in continuous centering, so the same value holds for the conditional fibers of P , hence, by (CT), for the conditional fibers of a Q -typical μ .

Finally Lemma II.12.9.12 applied to P gives an EFD P_π to which the translated conditional fibers uniformly scale. The preceding paragraph shows that P_π -typical measures have exact dimension κ . If $\kappa > 0$, such measures have no atoms, so P_π is non-atomic. This proves the final assertion. $\square$

Remark II.12.9.14 (What scenery theory supplies). Wu's Proposition 4.4 uses exact dimensionality of a fixed projection, while Proposition 4.6 uses continuous centering and the conditional-fiber factor. The corresponding general results are Hochman's Theorems 1.14, 1.22, and 1.30 and Proposition 1.28 in [Hoc10]. The argument above specializes their centering, coordinate-change, fiber-factor, and dimension-conservation conclusions. The measurable quasi-Palm reconstruction and CP reconstruction are invoked from those cited results.

Lemma II.12.9.15 (Uniform component entropy from CP invariance). *Under the hypotheses of Theorem II.12.9.13, fix $\epsilon > 0$. For all sufficiently large block lengths r there is a set $\mathcal{A} = \mathcal{A}(\pi, r, \epsilon)$ of measure coordinates with $Q(\mathcal{A}) > 1 - \epsilon$ such that for every $\mu \in \mathcal{A}$,*

$$\mu \left\{ x : \left| \frac{1}{r} H(\mu^{\mathcal{D}_r^{(2)}(x)}, \mathcal{D}_r^{(2)}) - \alpha \right| < \epsilon \right\} > 1 - \epsilon, \quad (\text{II.12.9.16})$$

$$\mu \left\{ x : \left| \frac{1}{r} H(\pi(\mu^{\mathcal{D}_r^{(2)}(x)}), \mathcal{D}_r) - \beta_\pi \right| < \epsilon \right\} > 1 - \epsilon. \quad (\text{II.12.9.17})$$

Moreover, after decreasing $\mathcal{A}$, for every $\mu \in \mathcal{A}$ there is D_μ with $\mu(D_\mu) > 1 - \epsilon$ and a

scale r_0 such that for $x \in D_\mu$ and $0 < \rho < r_0$,

$$\rho^{\alpha+\epsilon} \leq \mu(B(x, \rho)) \leq \rho^{\alpha-\epsilon}, \quad (\text{II.12.9.18})$$

$$\rho^{\beta_\pi+\epsilon} \leq \pi\mu(B(\pi x, \rho)) \leq \rho^{\beta_\pi-\epsilon}. \quad (\text{II.12.9.19})$$

This follows from the preceding scenery theorem; compare [Wu25, Proposition 4.4].

Proof. The uniform ball estimates are the Egorov form of exact dimensionality in Theorem II.12.9.13. We prove the component statements, since this is the part in which CP invariance is actually used.

Let $F_{r,\epsilon}$ be the set of measure coordinates θ for which $r^{-1}H(\theta, \mathcal{D}_r^{(2)}) = \alpha \pm \epsilon$. Exact dimensionality and the entropy-dimension theorem give $Q(F_{r,\epsilon}) \rightarrow 1$. Adaptedness and M -invariance give

$$Q(M^{-r}(F_{r,\epsilon} \times [0, 1)^2)) = \int \mu\{x : \mu^{\mathcal{D}_r^{(2)}(x)} \in F_{r,\epsilon}\} dQ(\mu) = Q(F_{r,\epsilon}).$$

Markov's inequality therefore gives (II.12.9.16) outside a set of measure coordinates of Q -mass $o_r(1)$.

For the projection, let $F_{r,\epsilon}^\pi$ be the set of θ for which $r^{-1}H(\pi\theta, \mathcal{D}_r) = \beta_\pi \pm \epsilon$. Exact dimensionality of $\pi\theta$ for Q -almost every θ implies $Q(F_{r,\epsilon}^\pi) \rightarrow 1$. The same adaptedness/invariance identity, with $F_{r,\epsilon}^\pi$ in place of $F_{r,\epsilon}$, proves (II.12.9.17). $\square$

Definition II.12.9.16 (Dyadic spreading). A probability η on $\mathbb{R}$ is (n, r, ϵ) -dyadically spreading if some translate $\tilde{\eta}$ satisfies

$$\tilde{\eta}\left\{x : \frac{1}{n} |\{1 \leq k \leq n : \tilde{\eta}(\mathcal{D}_k(x)) \leq 2\tilde{\eta}(\mathcal{D}_{k+r}(x))\}| < \epsilon\right\} > 1 - \epsilon.$$

Thus at all but an ϵ -fraction of scales, along all but ϵ of the mass, the r -step dyadic descendant carries less than half the mass of its ancestor.

Lemma II.12.9.17 (Positive-dimensional CP fibers spread). Suppose $\alpha - \beta_\pi > 0$. For every $\epsilon > 0$ there are integers r_*, n_* and a set of measure coordinates $\mathcal{A}$ with $Q(\mathcal{A}) > 1 - \epsilon$ such that for every $\mu \in \mathcal{A}$ and $n \geq n_*$ there is a set D with $\mu(D) > 1 - \epsilon$ for which, for every $x \in D$, the fiber measure $\mu_{\pi^{-1}(\pi x)}$, viewed on $\mathbb{R}$, is (n, r_*, ϵ) -dyadically spreading. This uses the conditional-fiber EFD input from [Hoc10, Theorem 1.30] in the form employed by [Wu25, Proposition 4.6].

Proof. By Theorem II.12.9.13, a typical conditional fiber uniformly scales to an ergodic non-atomic fractal distribution P_π . Fix $\epsilon > 0$. Non-atomicity and compactness imply that for some $\rho > 0$, the set of measures θ satisfying

$$\sup_{I: |I| \leq \rho} \theta(I) < \epsilon$$

has P_π -mass larger than $1 - \epsilon$. Hence, for a typical fiber η and η -almost every point y , the scaling orbit spends more than $1 - 2\epsilon$ of its time in this set. Equivalently, for large T ,

$$\frac{1}{T} \int_0^T \sup_{|I| \leq \rho} \eta^{y,t}(I) dt < 3\epsilon. \quad (*)$$

Translate η by a generic h . For Lebesgue almost every h , $\eta + h$ -almost every y is normal for the doubling map. Choose r so large that a level- k dyadic interval contains $B(y, 2^{-k-r})$ for at least $1 - \epsilon$ of the integers k . Choose r still larger, depending on ρ , so that $(*)$ implies

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{\eta(B(y, 2^{-k-r}))}{\eta(B(y, 2^{-k}))} < 4\epsilon.$$

Markov's inequality shows that, outside at most $8\epsilon n$ scales,

$$\eta(B(y, 2^{-k-r})) > 2\eta(B(y, 2^{-k-2r})).$$

Combining this with the dyadic-containment scales and $\mathcal{D}_{k+2r+1}(y) \subset B(y, 2^{-k-2r})$, we obtain

$$\limsup_{n \rightarrow \infty} \frac{1}{n} |\{1 \leq k \leq n : \eta(\mathcal{D}_k(y)) \leq 2\eta(\mathcal{D}_{k+2r+1}(y))\}| < 10\epsilon.$$

Renaming 10ϵ as ϵ , and applying Egorov first on the fiber and then on the CP distribution, yields uniform integers $r_* = 2r + 1$ and n_* and proves the lemma. $\square$

Lemma II.12.9.18 (Entropy survives a high-mass restriction). *Let σ be a probability, let E satisfy $\sigma(E) > 1 - \epsilon$, and let $\mathcal{P}$ be a finite partition with N atoms. If $\sigma_E = \sigma|_E / \sigma(E)$, then*

$$H(\sigma_E, \mathcal{P}) \geq H(\sigma, \mathcal{P}) - C\sqrt{\epsilon} \log_2 N$$

for $0 < \epsilon < 1/4$, with an absolute constant C .

Proof. Write $\delta = 1 - \sigma(E) < \epsilon$ and $\sigma = (1 - \delta)\sigma_E + \delta\sigma_{E^c}$. The entropy of a mixture is at most the mixture of the entropies plus the binary entropy, hence

$$H(\sigma, \mathcal{P}) \leq (1 - \delta)H(\sigma_E, \mathcal{P}) + \delta \log_2 N + h_2(\delta).$$

Rearranging and using $h_2(\delta) \leq C\sqrt{\delta} \log_2 N$ after enlarging C (the cases $N = 1, 2$ are harmless) proves the claim. $\square$

Lemma II.12.9.19 (At most one zero-dimensional projection). *If Q is ergodic and $\dim Q > 0$, there is at most one $\pi \in G(2, 1)$ with $\dim(\pi Q) = 0$. This local deduction depends on the fixed-projection exact-dimensionality input [Hoc10, Theorem 1.22].*

Proof. Suppose $\pi_1 \neq \pi_2$ both have zero projection dimension. By Theorem II.12.9.13, for Q -almost every μ , $\pi_j \mu$ is exact dimensional of dimension zero. Thus for every $s > 0$ there are Borel sets $E_j \subset \mathbb{R}$ of full $\pi_j \mu$ -measure with packing dimension at most s . The linear map $T = (\pi_1, \pi_2) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is invertible, and $T\mu(E_1 \times E_2) = 1$. Hence

$$\dim \mu = \dim T\mu \leq \dim_P(E_1 \times E_2) \leq \dim_P E_1 + \dim_P E_2 \leq 2s.$$

Letting $s \downarrow 0$ contradicts $\dim \mu = \dim Q > 0$. $\square$

Theorem II.12.9.20 (Countable exceptional directions for an ergodic CP distribution).

Let Q be an ergodic dyadic CP distribution. The set of directions π for which

$$\dim(\pi Q) < \min\{1, \dim Q\}$$

is at most countable. This is [Wu25, Theorem 1.5]; the combinatorial and two-direction parts are proved below, while the scenery inputs remain cited as stated above.

Proof. Put $\alpha = \dim Q$. The assertion is trivial if $\alpha = 0$, so assume $\alpha > 0$. Suppose for contradiction that the exceptional set is uncountable. By a countable pigeonhole there are $\kappa > 0$ and an uncountable Borel set E such that

$$0 < \beta_\pi := \dim(\pi Q) \leq \min\{1, \alpha\} - 4\kappa \quad (\pi \in E), \quad (\text{II.12.9.20})$$

and, after another countable pigeonhole, $\beta_\pi \geq b > 0$ on E . Here Lemma II.12.9.19 allows us to discard the possible single direction with $\beta_\pi = 0$. Choose a non-atomic probability ν supported on a compact uncountable subset of E .

Fix errors in the order

$$0 < \epsilon_0 \ll \epsilon_1 \ll \epsilon_2 \ll \kappa b.$$

For each $\pi \in E$, Lemma II.12.9.17 gives a spreading block and a starting scale. Since these are integer-valued data, a countable pigeonhole yields a positive- ν -measure Borel subset on which the same integers r_*, n_* work. By inner regularity replace it by a compact positive-measure subset E_1 . A further countable pigeonhole in the values of β_π , followed again by compact restriction, gives a positive-measure compact set $E_2 \subset E_1$ such that

$$|\beta_\pi - \beta_{\pi'}| < \epsilon_2 \quad (\pi, \pi' \in E_2). \quad (\text{II.12.9.21})$$

Fix $\pi_0 \in E_2$ and write $\beta_0 = \beta_{\pi_0}$.

Because $\nu|_{E_2}$ is non-atomic, choose distinct $\pi_1, \pi_2 \in E_2$ so close that, with

$$L = \lfloor -\log_2 \text{dist}(\pi_1, \pi_2) \rfloor,$$

we have $L \geq n_*$ and L is larger than every scale threshold needed below. Thus

$$\text{dist}(\pi_1, \pi_2) = 2^{-L+O(1)}. \quad (\text{II.12.9.22})$$

Only now, with the block length L fixed, apply Lemma II.12.9.15 separately to π_1 and π_2 . Each resulting set of good measure coordinates has Q -mass $1 - o_L(1)$, while the spreading lemma has a good set of Q -mass at least $1 - \epsilon_0$. Hence their intersection has positive Q -mass. Choose μ in this intersection and also in the full-measure set on which the two exact-dimensionality conclusions of Theorem II.12.9.13 hold.

Apply Egorov to the source and projected local-dimension limits for this fixed μ , and Fubini–Markov to the conditional π_1 -fibers and to the level- L source components. We obtain a set $\tilde{D}$ with $\mu(\tilde{D}) > 1 - \epsilon_3$, where $\epsilon_3 = o_{\epsilon_0, \epsilon_1}(1)$, and a scale threshold r_1 with $L \geq r_1$, such that for every $x \in \tilde{D}$:

- (i) the π_1 -fiber through x gives $\tilde{D}$ conditional mass at least $1 - \epsilon_3$ and is (L, r_*, ϵ_0) -dyadically spreading;
- (ii) $\tilde{D}$ has relative μ -mass at least $1 - \epsilon_3$ inside the level- L source square containing x ;
- (iii) the source and projected ball estimates hold for π_1, π_2 , and the level- L component entropy estimates (II.12.9.16)–(II.12.9.17) hold at x for both directions.

By (II.12.9.19) and (II.12.9.21), for $j = 1, 2$,

$$N_\rho(\pi_j \tilde{D}) \leq C \rho^{-\beta_0 - 2\epsilon_1 - 2\epsilon_2} \quad (0 < \rho \leq 2^{-r_1}). \quad (\text{II.12.9.23})$$

Likewise the upper mass bound in (II.12.9.19) gives

$$N_\rho(\pi_1 \tilde{D}) \geq c \rho^{-\beta_0 + 2\epsilon_1 + 2\epsilon_2}. \quad (\text{II.12.9.24})$$

Taking $\rho = 2^{-L}$, choose

$$z_1, \dots, z_M \in \pi_1 \tilde{D}, \quad M \geq c 2^{L(\beta_0 - 3\epsilon_1 - 3\epsilon_2)},$$

with mutual separation at least $8 \cdot 2^{-L}$.

Fix one $z = z_i$. Let

$$\mathcal{A}_z = \{A \in \mathcal{D}_L^{(2)} : A \cap \tilde{D} \cap \pi_1^{-1}(z) \neq \emptyset\}.$$

For each $A \in \mathcal{A}_z$, choose $x_A \in A \cap \tilde{D} \cap \pi_1^{-1}(z)$ and put

$$B'_A = A \cap \tilde{D}, \quad B_A = \pi_1(B'_A), \quad B = \bigcup_{A \in \mathcal{A}_z} B_A.$$

The set B has diameter $O(2^{-L})$. At resolution 2^{-2L} , (II.12.9.23) therefore gives

$$N_{2^{-2L}}(B) \leq C 2^{L(\beta_0 + 4\epsilon_1 + 4\epsilon_2)}. \quad (\text{II.12.9.25})$$

On the other hand, by the projected component entropy estimate at x_A , property (ii), and Lemma II.12.9.18,

$$H(\pi_1((\mu|_{\tilde{D}})^A), \mathcal{D}_L) \geq L(\beta_0 - 3\epsilon_1 - 2\epsilon_2 - o_{\epsilon_3}(1)).$$

Since a probability supported on N atoms has entropy at most $\log_2 N$, this implies

$$N_{2^{-2L}}(B_A) \geq c 2^{L(\beta_0 - 4\epsilon_1 - 3\epsilon_2 - o_{\epsilon_3}(1))} \quad (A \in \mathcal{A}_z). \quad (\text{II.12.9.26})$$

We now put the preceding geometric data into the literal finite model of Theorem II.12.9.5. Parametrize the line $\pi_1^{-1}(z)$ by arclength, translate a fixed bounded interval containing the relevant fiber to $[0, 1]$, and atomize it by level- L dyadic intervals. Give each occupied atom the mass of the conditional π_1 -fiber restricted to $\tilde{D}$ and renormalized. The resulting grid probability η_L has mass one, and the set of atoms meeting the good fiber set has η_L -mass $> 1 - \epsilon_3 > 1/2$. Up to shifting the scale index by an absolute constant (caused by the chosen affine parametrization), property (i) says precisely that these atoms obey the block

branching hypothesis of Theorem II.12.9.5 with block r_* . Enlarging r_* by that absolute constant does not affect its uniformity.

Likewise, translate and dilate the π_1 -coordinate by 2^L and replace every level- $2L$ interval meeting $2^L B$ by its grid index in $\{0, \dots, 2^L - 1\}$; call the resulting set $\mathcal{B}$. For each fiber atom represented by A , let $\mathcal{B}_A \subset \mathcal{B}$ be the indices meeting $2^L B_A$. Bounded-endpoint multiplicities give

$$|\mathcal{B}| \asymp N_{2^{-2L}}(B), \quad |\mathcal{B}_A| \asymp N_{2^{-2L}}(B_A).$$

Finally discretize $2^L(\pi_2(x_A) - \pi_1(x_A))$ at mesh 2^{-L} . By (II.12.9.22), this displacement coordinate is a bounded bi-Lipschitz image of the arclength coordinate on the π_1 -fiber, so the same atomized probability satisfies the branching hypothesis after another absolute endpoint adjustment.

Now (II.12.9.20) and (II.12.9.25) imply, for small enough errors,

$$|\mathcal{B}| \leq 2^{L(1-\kappa)},$$

while (II.12.9.26), (II.12.9.25), and $\beta_0 \geq b$ imply

$$|\mathcal{B}_A| \geq |\mathcal{B}|^{1-\delta_*}$$

with δ_* smaller than the constant required by Theorem II.12.9.5. Also, for every occupied fiber atom, (II.12.9.26) gives

$$|\mathcal{B}| \geq |\mathcal{B}_A| \geq 2^{L(\beta_0 - o_{\epsilon_1, \epsilon_2, \epsilon_3}(1))}.$$

Applying Theorem II.12.9.5 to the atomized fiber probability and the family $\{\mathcal{B}_A\}$ therefore gives $|\bigcup_A (a_A + \mathcal{B}_A)| \geq |\mathcal{B}|^{1+\delta}$. Since $\beta_0 \geq b$, after choosing the errors much smaller than δb , this is at least $2^{L(\beta_0 + c_*)}$ for some $c_* = c_*(b, \kappa, r_*) > 0$. Undoing the bounded endpoint discretizations yields

$$N_{2^{-2L}}\left(\bigcup_{A \in \mathcal{A}_z} (B_A + \pi_2(x_A) - z)\right) \geq 2^{L(\beta_0 + c_*)}. \quad (\text{II.12.9.27})$$

For $y \in B'_A$, both y and x_A lie in the same level- L square. Hence

$$|\pi_2(y) - (\pi_1(y) + \pi_2(x_A) - \pi_1(x_A))| \leq C \text{dist}(\pi_1, \pi_2) 2^{-L} \leq C 2^{-2L}.$$

Thus (II.12.9.27) yields

$$N_{2^{-2L}}\left(\pi_2\left(\tilde{D} \cap \pi_1^{-1}(B(z, 2^{-L+1}))\right)\right) \geq c 2^{L(\beta_0 + c_*)}. \quad (\text{II.12.9.28})$$

The tubes corresponding to the separated z_i 's have uniformly bounded overlap after projection by π_2 : indeed every point moves by $O(2^{-L})$ when π_1 is replaced by π_2 , while the centers are separated by a fixed multiple of 2^{-L} . Hence summing (II.12.9.28) over the M choices of z_i , and losing only an absolute multiplicity constant, gives

$$N_{2^{-2L}}(\pi_2 \tilde{D}) \geq c 2^{2L\beta_0 + L(c_* - 3\epsilon_1 - 3\epsilon_2)}.$$

For $\epsilon_1, \epsilon_2 \ll c_*$, this contradicts the upper bound (II.12.9.23) at $\rho = 2^{-2L}$. The exceptional set is therefore at most countable. $\square$

Remark II.12.9.21 (From the CP theorem to its applications). The proof of Theorem II.12.9.20 establishes the two-direction argument from restriction entropy, uniform projection covering estimates, fiber discretization, Theorem II.12.9.5, and the transverse $O(2^{-2L})$ comparison. Continuous centering, fixed-projection exact dimensionality, and the fiber factor come from [Hoc10]. The packing- and Assouad-dimension consequences below additionally use the cited scenery-generation and microset-realization theorems.

II.12.9.2 Packing- and Assouad-dimension applications

For a bounded set E in a metric space, let $N_r(E)$ be the least number of sets of diameter at most r needed to cover E . For a nonempty set $F \subset \mathbb{R}^d$, its *Assouad dimension* is

$$\dim_A F = \inf \left\{ s \geq 0 : \begin{array}{l} \text{there is } C \geq 1 \text{ such that, for all } x \in F \text{ and } 0 < r < R, \\ N_r(F \cap B(x, R)) \leq C(R/r)^s \end{array} \right\}.$$

For a Borel probability η , its packing dimension in this subsection is the lower packing dimension

$$\dim_p \eta = \inf \{ \dim_p E : \eta(E) > 0 \}.$$

Let $\mathcal{D}_k^{(d)}(x)$ be the level- k dyadic cube containing x and let $\eta^{x,k}$ be the normalized restriction to that cube, affinely rescaled to $[0, 1)^d$. The *upper uniform entropy dimension* of $\eta \in \mathcal{P}(\mathbb{R}^d)$ is the supremum of the numbers $\alpha \geq 0$ for which

$$\liminf_{\varepsilon \downarrow 0} \limsup_{\ell \rightarrow \infty} \limsup_{n \rightarrow \infty} \eta \left\{ x : \# \{ 1 \leq k \leq n : |H(\eta^{x,k}, \mathcal{D}_\ell^{(d)}) - \ell \alpha| \leq \ell \varepsilon \} \geq (1 - \varepsilon)n \right\} = 1. \quad (\text{II.12.9.29})$$

The order of the three limits in (II.12.9.29) is part of the definition.

Theorem II.12.9.22 (Wu: packing dimension of projections). *Let η be a Borel probability on $\mathbb{R}^2$ with upper uniform entropy dimension α . Then*

$$\dim_p(\pi_* \eta) \geq \min\{1, \alpha\}$$

for every $\pi \in G(2, 1)$ outside an at most countable set. This is [Wu25, Theorem 1.13].

Theorem II.12.9.23 (Wu: Assouad dimension of projections). *Let $F \subset \mathbb{R}^2$ be nonempty. Then*

$$\{ \pi \in G(2, 1) : \dim_A(\pi F) < \min\{1, \dim_A F\} \}$$

is at most countable. This is [Wu25, Theorem 1.15].

Research boundary — Wu's projection applications

Theorems II.12.9.22 and II.12.9.23 are Theorems 1.13 and 1.15 of [Wu25]. We cite their deductions from CP distributions, including the scenery-generation and microset-realization arguments required to pass from typical measures to packing and Assouad dimension.

II.12.10 Holomorphic self-conformal systems

The last part concerns holomorphic systems on a planar domain. We use Bernoulli weights and exponential separation, together with the geometric nondegeneracy conditions of [FR26]. The following definition gives the precise geometric and analytic conventions.

Definition II.12.10.1 (Holomorphic IFS, adapted domain, and separation). Let $\Omega \subset \mathbb{C}$ be a bounded domain and let Λ be a finite nonempty set. A *holomorphic IFS on Ω* is a family $\Phi = \{\varphi_i : i \in \Lambda\}$ for which a common domain $\Omega' \supset \overline{\Omega}$ exists, every $\varphi_i : \Omega' \rightarrow \mathbb{C}$ is holomorphic and injective, $\varphi_i(\overline{\Omega}) \subset \Omega$, and $0 < |\varphi'_i(z)| < 1$ for $z \in \overline{\Omega}$. For a word $u = i_1 \cdots i_n$ put $\varphi_u = \varphi_{i_1} \circ \cdots \circ \varphi_{i_n}$. Its attractor K_Φ is the compact invariant set $K_\Phi = \bigcup_i \varphi_i(K_\Phi)$; for a positive probability vector p , the stationary law μ is defined by $\mu = \sum_i p_i(\varphi_i)_* \mu$, with $K_\Phi = \text{supp } \mu$.

The IFS is *exponentially separated* when there are $c > 0$ and infinitely many n such that

$$\|\varphi_u - \varphi_v\|_\Omega \geq c^n \quad (u, v \in \Lambda^n, u \neq v), \quad \|f\|_\Omega = \sup_{z \in \Omega} |f(z)|.$$

Put

$$H(p) = - \sum_i p_i \log_2 p_i, \quad \chi = - \sum_i p_i \int \log_2 |\varphi'_i(z)| d\mu(z).$$

A *regular real-analytic curve* $\Gamma \subset \Omega$ is an embedded, possibly disconnected, real-analytic one-manifold without boundary: at each $z \in \Gamma$ there are a neighborhood $U \subset \Omega$, an open interval $J \subset \mathbb{R}$, and a real-analytic homeomorphism $\gamma : J \rightarrow \Gamma \cap U$ with $\gamma'(t) \neq 0$ throughout J . It is *closed in Ω* if it is relatively closed there, and is Φ -*invariant* if $\varphi_i(\Gamma) \subset \Gamma$ for every i . A map on a planar domain is *homothetic* if it has the form $z \mapsto rz + a$ with $r \in \mathbb{R} \setminus \{0\}$ and $a \in \mathbb{C}$. The IFS is *holomorphically conjugate to a homothetic IFS* if an injective holomorphic $\psi : \Omega \rightarrow \mathbb{C}$ makes every $\psi\varphi_i\psi^{-1}$ homothetic on $\psi(\Omega)$.

An *adapted domain* for Φ is a bounded domain Ω_0 with $\overline{\Omega} \subset \Omega_0 \subset \Omega'$ such that every generator maps Ω_0 into Ω , $\varphi_i(\overline{\Omega}) \subset \Omega$, and there are $0 < m < q < 1$ for which

$$m \leq |\varphi'_i(z)| \leq q \quad (i \in \Lambda, z \in \Omega_0).$$

These are the source hypotheses of Feng–Rapaport; the adapted domain used in the normalization arguments is supplied by the following lemma.

Lemma II.12.10.2 (Existence of an adapted domain). *Every holomorphic IFS in Definition II.12.10.1 has an adapted domain Ω_0 .*

Proof. Since $\overline{\Omega}$ is compact in Ω' and Λ is finite, there is $\delta_1 > 0$ such that the closed $2\delta_1$ -neighborhood of $\overline{\Omega}$ lies in Ω' . For each i , the compact set $\varphi_i(\overline{\Omega})$ lies in the open set Ω . Uniform continuity on a neighborhood of $\overline{\Omega}$ therefore gives $\delta_2 > 0$ such that $\varphi_i((\overline{\Omega})^{(\delta_2)}) \subset \Omega$ simultaneously for all i . Compactness and the strict derivative bounds on $\overline{\Omega}$ give $0 < m_0 \leq |\varphi'_i| \leq q_0 < 1$ there; continuity yields the same bounds with $m_0/2$ and $(1 + q_0)/2$ on a sufficiently small neighborhood of $\overline{\Omega}$. Choose $\delta > 0$ smaller than all three

neighborhood thresholds and let $\Omega_0 = \{z : \text{dist}(z, \Omega) < \delta\}$. This is a bounded domain containing $\overline{\Omega}$, its closure lies in Ω' , every $\varphi_i(\Omega_0)$ lies in $\Omega \subset \Omega_0$, and the displayed derivative bounds hold on Ω_0 . These are exactly the adapted-domain conditions. $\square$

The entropy-increase argument used in that setting needs the planar form of Hochman's convolution inverse theorem [Hoc20]. Since this is the only place in the present volume where the multidimensional convolution theorem is used, we give a self-contained two-dimensional proof in the precise bounded- support form consumed below.

Definition II.12.10.3 (Planar dyadic components, concentration and saturation). Let $\mathcal{D}_n^{(2)}$ be the half-open dyadic squares of side 2^{-n} in $\mathbb{R}^2$. For a compactly supported probability σ put

$$H_n(\sigma) = \frac{1}{n} H(\sigma, \mathcal{D}_n^{(2)}), \quad n \geq 1.$$

If $D = \mathcal{D}_i^{(2)}(x)$ has positive σ -mass, let $\sigma^{x,i}$ be the normalized restriction of σ to D , followed by the orientation- preserving affine rescaling of D onto $[0, 1]^2$.

Let $V \leq \mathbb{R}^2$ be a linear subspace. A probability θ on a fixed bounded set is (V, η) -concentrated if some affine translate of V has η -neighbourhood of θ -mass at least $1 - \eta$. It is (V, η, m) -saturated if

$$H_m(\theta) \geq H_m((\pi_{V^\perp})_* \theta) + \dim V - \eta. \quad (\text{II.12.10.1})$$

Here a fixed orthonormal identification of $V^\perp$ with $\mathbb{R}^{2-\dim V}$ is used to define its dyadic entropy. Changing that identification changes normalized entropy by $O(1/m)$ only.

The definitions are insensitive to the harmless normalizations that occur in the proof. In particular, a bounded translation changes dyadic entropy by $O(1)$ bits, and replacing a line by another line whose angle is at most 2^{-m} changes the relevant scale- m entropies by $O(1)$ bits. Thus all such changes cost $O(1/m)$ after normalization. Also, if $U \leq V$ and a measure is (V, η, m) -saturated, then it is $(U, \eta + O(1/m), m)$ -saturated: quotient first by $V^\perp$ and then by the remaining $\dim V - \dim U$ directions, whose entropy is at most their dimension up to $O(1/m)$.

Lemma II.12.10.4 (Local entropy comparison for convolutions). *Fix a support bound R . For compactly supported probabilities α, β on $[-R, R]^2$, integers $1 \leq m \leq n$, and independent $x \sim \alpha, y \sim \beta$,*

$$\begin{aligned} H_n(\alpha * \beta) - H_n(\alpha) &\geq \mathbb{E}_{0 \leq i < n} \mathbb{E}_{x,y} [H_m(\alpha^{x,i} * \beta^{y,i}) - H_m(\alpha^{x,i})] \\ &\quad - O_R\left(\frac{1}{m} + \frac{m}{n}\right). \end{aligned} \quad (\text{II.12.10.2})$$

For every fixed k one also has the dyadic Kaimanovich–Vershik estimate

$$H_n(\alpha * \beta^{*k}) - H_n(\alpha) \leq k(H_n(\alpha * \beta) - H_n(\alpha)) + O_{R,k}(1/n). \quad (\text{II.12.10.3})$$

Proof. Let X, Y be independent with laws α, β . Conditional on the level- i dyadic cells of

X and Y , the level- i cell of $X + Y$ is determined up to one of a bounded number of carry vectors. After rescaling the two conditioning cells to unit size, the conditional law of the fine part of $X + Y$ is therefore a translate of $\alpha^{x,i} * \beta^{y,i}$, again up to a bounded carry choice. The entropy chain rule over the window $[i, i + m]$ gives an $O(1)$ -bit discrepancy. Summing in i makes the middle conditional entropies telescope; the first and last m scales contribute $O_R(m^2)$ bits. Division by mn gives (II.12.10.2).

For (II.12.10.3), quantize $X, Y_1, \dots, Y_k$ to the level- n lattice. The discrete Kaimanovich–Vershik inequality says that the entropy increments produced by successively adding independent copies of the same variable are non-increasing, hence

$$H(X + Y_1 + \dots + Y_k) - H(X) \leq k(H(X + Y_1) - H(X)).$$

Passing between the quantized sums and the level- n dyadic cells introduces only a bounded number of carry states, hence $O_{R,k}(1)$ bits. Divide by n . $\square$

Lemma II.12.10.5 (Saturation transfer). *There is an absolute constant C such that for probabilities α, β on a fixed bounded subset of $\mathbb{R}^2$,*

$$H_m(\alpha * \beta) \geq H_m(\alpha) - C/m. \quad (\text{II.12.10.4})$$

If β is (V, η, m) -saturated while α is not $(V, 2\eta, m)$ -saturated, then

$$H_m(\alpha * \beta) \geq H_m(\alpha) + \eta - C/m. \quad (\text{II.12.10.5})$$

Proof. For fixed b , translation by b changes the entropy of a dyadic partition by at most $O(1)$ bits, because every translated dyadic square meets only a bounded number of unshifted squares and conversely. Since $\alpha * \beta = \int (T_b)_* \alpha d\beta(b)$, entropy concavity gives (II.12.10.4).

Let $W = V^\perp$. The quantity $H_m(\theta) - H_m((\pi_W)_* \theta)$ is, up to an $O(1/m)$ dyadic carry term, the normalized conditional entropy in the V -directions over the W -projection. Conditional entropy is concave in the joint law, and a translation changes it by only $O(1/m)$. Since $\alpha * \beta = \int (T_a)_* \beta d\alpha(a)$, saturation of β is therefore preserved by convolution by α , with an additional $O(1/m)$ loss. Consequently

$$H_m(\alpha * \beta) \geq H_m((\pi_W)_*(\alpha * \beta)) + \dim V - \eta - O(1/m).$$

Projection commutes with convolution, so (II.12.10.4) applied in W gives

$$H_m((\pi_W)_*(\alpha * \beta)) \geq H_m((\pi_W)_* \alpha) - O(1/m).$$

Failure of $(V, 2\eta, m)$ -saturation means $H_m(\alpha) < H_m((\pi_W)_* \alpha) + \dim V - 2\eta$. Combining the last three inequalities proves (II.12.10.5). $\square$

Lemma II.12.10.6 (Covariance gap gives concentration). *Let Θ be a random probability on $[0, 1]^2$, let $A = \mathbb{E} \Sigma(\Theta)$, and suppose that for some $V \leq \mathbb{R}^2$ of dimension r ,*

$$V = \text{span}\{\text{eigenvectors of } A \text{ with eigenvalue at least } s\}, \quad \lambda_{r+1}(A) \leq t.$$

Then

$$\mathbb{P}(\Theta \text{ is } (V, Ct^{1/8})\text{-concentrated}) \geq 1 - Ct^{1/4},$$

with an absolute C .

Proof. Choose an orthonormal basis of $V^\perp$. For a unit vector u in that basis,

$$\mathbb{E} \text{Var}_\Theta \langle u, X \rangle = u^T A u \leq t.$$

Markov's inequality shows that, outside probability $O(t^{1/4})$, every such transverse variance is at most $t^{3/4}$. Conditional Chebyshev, centered at the mean of the component, then puts all but $O(t^{1/2})$ of its mass in a tube of width $O(t^{1/8})$ around the affine translate of V through that mean. There are at most two transverse basis vectors, so the stated bounds follow after changing C . $\square$

Lemma II.12.10.7 (Finite-scale Gaussian saturation). *Fix $0 < \eta < 1$, an integer m sufficiently large in terms of η , and $s > 0$. There are integers $k = k(s, \eta, m)$ and $p = p(s, \eta, m)$ and a number $t_0 = t_0(s, \eta, m, k) > 0$ with the following property. They may be chosen so that $\ell = \lfloor \log_2 \sqrt{k} \rfloor - p \geq 1$.*

Let σ be a probability on a fixed bounded set, fix a scale i , and put $A_i = \mathbb{E}\Sigma(\sigma^{x,i})$. Order its eigenvalues decreasingly and use the conventions $\lambda_0(A_i) = +\infty$ and $\lambda_3(A_i) = 0$. Assume that, for some $r \in \{0, 1, 2\}$,

$$\lambda_r(A_i) \geq s, \quad \lambda_{r+1}(A_i) \leq t_0,$$

*and write W_i for the active eigenspace. Set $\tau = \sigma^{*k}$ and*

$$\ell = \lfloor \log_2 \sqrt{k} \rfloor - p.$$

Then, after decreasing t_0 if necessary,

$$\mathbb{P}(\tau^{z,i-\ell} \text{ is } (W_i, \eta, m)\text{-saturated}) > 1 - \eta. \quad (\text{II.12.10.6})$$

For $r = 0$ the same conclusion is automatic with $W_i = \{0\}$. The result is the planar specialization of the [Gaussian smoothing theorem of Chapter II.5](#).

Proof. For $r = 0$, saturation along $W_i = \{0\}$ is automatic. Suppose $r \geq 1$. Apply the [Gaussian smoothing theorem](#), part (2), in dimension two with error η , window m , spectral lower bound $s/2$, and the random level- i rescaled component of σ . Its mean covariance is exactly A_i . The theorem first supplies p and k_1 and then permits every $k \geq k_1$. Choose

$$k \geq \max\{k_1, 2^{2(p+1)}\}$$

before its transverse threshold t_0 . Thus $\ell = \lfloor \log_2 \sqrt{k} \rfloor - p \geq 1$. Whenever $\lambda_r(A_i) \geq s$ and $\lambda_{r+1}(A_i) \leq t_0$, the components of $\tau = \sigma^{*k}$ at level

$$i - \lfloor \log_2 \sqrt{k} \rfloor + p = i - \ell$$

are (W_i, η, m) -saturated outside probability η . This is precisely (II.12.10.6). The theorem chooses k, p before t_0 , which is the parameter order required by the recursive spectral-gap selection below. $\square$

Lemma II.12.10.8 (Planar persistence of concentration subspaces). *Fix $0 < \eta < 1/10$, an integer shift $\ell \geq 0$, and $m \geq 1$. There is $a = a(\eta, \ell, m) > 0$ such that the following holds for all sufficiently large n . Suppose σ is supported on $[0, 1]^2$ and subspaces $W_0, \dots, W_n \leq \mathbb{R}^2$ satisfy*

$$\mathbb{P}_{i=q}(\sigma^{x,i} \text{ is } (W_q, a)\text{-concentrated}) > 1 - a \quad (0 \leq q \leq n). \quad (\text{II.12.10.7})$$

Then there are $V_q \leq W_q$ such that

$$\mathbb{P}_{i=q}(\sigma^{x,i} \text{ is } (V_q, \eta)\text{-concentrated}) > 1 - \eta \quad (\text{II.12.10.8})$$

for every q , and for at least $(1 - \eta)n$ indices $q \in \{\ell, \dots, n\}$ the spaces $V_q, V_{q-\ell}$ have the same dimension and, when this dimension is one, their angle is less than $2^{-m}\eta$.

Proof. We use a finite-block version of the persistence argument. This avoids any compactness shortcut and makes explicit why only finitely many direction changes can occur. Put

$$\vartheta = 2^{-m}\eta/20$$

and choose an integer $M > 20/\eta$. We shall choose a so small, in terms of η, ℓ, m, M , that every error generated in the next M steps is below $\min\{\eta/20, \vartheta^2/20\}$.

We first record the elementary one-step mechanism. Suppose that at a scale q typical components are (U, ρ) -concentrated, and put $j = q + \ell$. Conditional Markov applied inside the level- q cells shows that, outside a set of component pairs of probability $O(\sqrt{\rho})$, the level- j component is concentrated near the inherited translate of U : after the level- j rescaling the tube width is at most $2^\ell \rho$, and the mass lost is at most $O(\sqrt{\rho})$. Hence it is $(U, C_\ell \sqrt{\rho})$ -concentrated. Intersect this good event with (II.12.10.7) at level j . The component is then simultaneously concentrated near translates of U and W_j .

There is a deterministic subspace $E \leq W_j$ with $\dim E \leq \dim U$ for which these simultaneous concentrations imply $(E, F(\rho))$ -concentration, where

$$F(\rho) = C_{\ell, \vartheta} \sqrt{\rho}$$

provided $a \leq \rho$. Indeed, if U and W_j are lines making angle less than ϑ , take $E = W_j$; if their angle is at least ϑ , the intersection of the two tubes inside a fixed bounded square lies in a ball of radius $O_\vartheta(C_\ell \sqrt{\rho})$, so take $E = \{0\}$. If one of the spaces is $\{0\}$ or $\mathbb{R}^2$, take respectively the smaller compatible space. Thus either the dimension drops, or the dimension is unchanged and, in the one-dimensional case, the new line makes angle less than ϑ with U . This is the planar tube-intersection step behind subspace stabilization.

If $\ell = 0$, take $V_q = W_q$ for every q ; then the alignment statement is tautological and (II.12.10.8) follows from $a < \eta$. We may therefore assume $\ell \geq 1$. Split each residue class modulo ℓ into consecutive blocks of M indices. At the first index q of a block set $V_q = W_q$. Move through the block in steps of ℓ and apply the preceding one-step mechanism, setting

the resulting space E equal to the next V_j . The dimensions are non-increasing inside the block and therefore can drop at most twice. The concentration error is bounded by an iterate of F starting from a . Since M is fixed, $F^M(a) \rightarrow 0$ as $a \downarrow 0$; choose a so small that all these iterates, including their probability errors and the factors 2^ℓ , are below η . This proves (II.12.10.8) throughout every block.

For a pair $q - \ell, q$ lying inside the same block, either a dimension drop occurs or the two spaces have the same dimension; in the latter case their one-dimensional angle is less than $\vartheta < 2^{-m}\eta$. There are at most two drops and one block boundary per M transitions in each residue class. Hence the exceptional proportion is at most $3/M + O(\ell/n) < \eta$ for all sufficiently large n . This proves the lemma. $\square$

Proposition II.12.10.9 (Repeated-convolution structure in the plane). *For every $0 < \eta < 1/20$ and every sufficiently large integer $m = m(\eta)$ there are an integer $k = k(\eta, m)$ and $n_0 = n_0(\eta, m)$ such that, for every $n \geq n_0$ and every probability σ on a fixed bounded subset of $\mathbb{R}^2$, there are subspaces $V_0, \dots, V_n \leq \mathbb{R}^2$ for which, with $\tau = \sigma^{*k}$,*

$$\mathbb{P}_{0 \leq i \leq n}(\sigma^{x,i} \text{ is } (V_i, \eta)\text{-concentrated}) > 1 - \eta, \quad (\text{II.12.10.9})$$

$$\mathbb{P}_{0 \leq i \leq n}(\tau^{z,i} \text{ is } (V_i, \eta, m)\text{-saturated}) > 1 - \eta. \quad (\text{II.12.10.10})$$

The parameters are uniform over the probability σ . This is the planar repeated-convolution structure underlying [Hoc20, Theorem 2.8]. The proof below derives it from the covariance, Gaussian smoothing, and subspace-persistence lemmas developed above.

Proof. A fixed affine normalization reduces to σ supported on $[0, 1)^2$. For each level q set

$$A_q = \mathbb{E}_{i=q} \Sigma(\sigma^{x,i}).$$

Its eigenvalues lie in a fixed compact interval. Starting from a fixed $s_0 > 2$, choose recursively

$$s_{r+1} = \rho(s_r),$$

where $\rho(s)$ is chosen in the following noncircular order. After s and an internal error much smaller than η are fixed, Lemma II.12.10.7 fixes $p(s)$ and then $k(s)$ with $\ell(s) = \lfloor \log_2 \sqrt{k(s)} \rfloor - p(s) \geq 1$. Lemma II.12.10.8 now supplies its concentration threshold $a(\eta/20, \ell(s), m) > 0$. Only then choose $\rho(s) > 0$ below the Gaussian transverse threshold and so small that Lemma II.12.10.6 has concentration probability and tube error below this a . These choices depend on s, η, m , not on σ or the scale q .

Take $L > 16/\eta$. For $1 \leq r \leq L$ let $N_r(A_q)$ be the number of eigenvalues of A_q lying in $(s_r, s_{r-1}]$. Since A_q has only two eigenvalues,

$$\sum_{r=1}^L N_r(A_q) \leq 2.$$

Averaging first in q and then in r yields an index r for which $N_r(A_q) = 0$ for more than a $(1 - \eta/8)$ -proportion of q . Fix this r , write $s = s_{r-1}, t = s_r$, choose the corresponding k, p, ℓ ,

and for each good q let W_q be the span of the eigenvectors of A_q with eigenvalue at least s . On bad q put $W_q = \mathbb{R}^2$.

For good q there is a genuine spectral gap: the active eigenvalues are at least s and the first inactive one is at most t . Lemma II.12.10.6, with the recursive choice of t , therefore gives concentration of typical level- q components of σ near W_q . Lemma II.12.10.7 gives saturation of typical level- $(q - \ell)$ components of τ along W_q whenever $q \geq \ell$. The bad scales, their ℓ -shifts, and the boundary $q < \ell$ cost only $\eta/4 + O(\ell/n)$ in the final averaged probabilities. On bad q the choice $W_q = \mathbb{R}^2$ makes the concentration hypothesis of the persistence lemma automatic. For good q the recursive choice of t was made small enough that the concentration error is below the number $a(\eta/20, \ell, m)$ required by that lemma.

Apply Lemma II.12.10.8 to the family (W_q) . For all but an $O(\eta)$ -proportion of q , it gives $V_q \leq W_q$ and V_q is within angle $2^{-m}O(\eta)$ of $V_{q+\ell}$. Concentration of $\sigma^{x,q}$ transfers from W_q to the chosen V_q by the construction of the persistence lemma. Saturation of $\tau^{z,q}$ initially holds along $W_{q+\ell}$; it passes to the smaller space $V_{q+\ell}$ and then, by the angle bound and dyadic stability, to V_q . Enlarging the internal error constants to η proves (II.12.10.9)–(II.12.10.10). $\square$

Theorem II.12.10.10 (Planar inverse entropy theorem, restricted form). *For every support bound $R > 0$ and $0 < \varepsilon < 1/20$ there is $m_0 = m_0(\varepsilon)$ such that, for every $m \geq m_0$, there are $\delta = \delta(R, \varepsilon, m) > 0$ and $n_0 = n_0(R, \varepsilon, m)$ with the following property. If $n \geq n_0$ and compactly supported probabilities μ, ν on $[-R, R]^2$ satisfy*

$$H_n(\mu * \nu) \leq H_n(\mu) + \delta,$$

then there are subspaces $V_0, \dots, V_n \leq \mathbb{R}^2$ such that

$$\begin{aligned} \mathbb{P}_{0 \leq i \leq n}(\mu^{x,i} \text{ is } (V_i, \varepsilon, m)\text{-saturated}) &> 1 - \varepsilon, \\ \mathbb{P}_{0 \leq i \leq n}(\nu^{y,i} \text{ is } (V_i, \varepsilon)\text{-concentrated}) &> 1 - \varepsilon. \end{aligned}$$

This large- m form is the version used in the Feng–Rapaport entropy-increase argument. It is the bounded-support planar specialization of [Hoc20, Theorem 2.8]; the proof below uses the Gaussian smoothing theorem from Chapter II.5.

Proof. Fix R, ε and take m large enough that every $O(1/m)$ error below is at most $\varepsilon/100$. Apply Proposition II.12.10.9 to ν with $\eta = \varepsilon/20$. It gives an integer k , a measure $\tau = \nu^{*k}$, and subspaces V_i such that, outside total probability $O(\eta)$, the ν -component is (V_i, η) -concentrated and the τ -component is (V_i, η, m) -saturated.

Suppose that the desired saturation conclusion for μ fails. After slightly tightening the internal tolerances, there is then a set of scale/component pairs of probability at least 3η on which $\mu^{x,i}$ is not $(V_i, 2\eta, m)$ -saturated. Intersecting with the good event for τ leaves probability at least 2η . Lemma II.12.10.5 gives on this event

$$H_m(\mu^{x,i} * \tau^{z,i}) - H_m(\mu^{x,i}) \geq \eta - O(1/m),$$

while the same difference is always at least $-O(1/m)$. Hence the averaged local increment

is at least $c\eta^2$ for an absolute $c > 0$. By Lemma II.12.10.4, after taking n sufficiently large,

$$H_n(\mu * \tau) - H_n(\mu) \geq \frac{c}{2}\eta^2. \quad (\text{II.12.10.11})$$

On the other hand, the Kaimanovich–Vershik estimate (II.12.10.3) and the hypothesis of the theorem give

$$H_n(\mu * \tau) - H_n(\mu) \leq k\delta + O_{R,k}(1/n).$$

Choose first $\delta < c\eta^2/(8k)$ and then n so large that the last error is $< c\eta^2/8$. This contradicts (II.12.10.11). Therefore μ is saturated along V_i at all but $O(\eta) < \varepsilon$ of the scale/component pairs. The concentration conclusion for ν already comes from (II.12.10.9); enlarging η to ε finishes the proof. $\square$

Remark II.12.10.11 (Relation to the source inverse theorem). The argument above is the $d = 2$, bounded-support specialization of Hochman’s multidimensional convolution inverse theorem [Hoc20, Theorem 2.8]. The scale-shifted common-subspace stabilization is isolated as Lemma II.12.10.8, and the transfer from a saturated perturbation to entropy growth of the target is Lemma II.12.10.5. The proof also displays the covariance-gap selection, Gaussian smoothing of repeated convolutions, and Kaimanovich–Vershik amplification; Gaussian saturation follows from the [Gaussian theorem in Chapter II.5](#).

Definition II.12.10.12 (Complex polynomial actions and coefficient partitions). For $k \geq 1$, let $\mathcal{P}_k(\mathbb{C})$ be the complex vector space of polynomials

$$q(X) = a_0 + a_1X + \cdots + a_kX^k, \quad a_j \in \mathbb{C},$$

identified with $\mathbb{C}^{k+1} \cong \mathbb{R}^{2k+2}$ through the coefficient vector. We write

$$\|q\|_2 = \left(\sum_{j=0}^k |a_j|^2 \right)^{1/2},$$

and let $\mathcal{D}_n^{\mathcal{P}_k(\mathbb{C})}$ be the pullback of the standard level- n dyadic partition of $\mathbb{R}^{2k+2}$. If $\nu \in \mathcal{M}(\mathcal{P}_k(\mathbb{C}))$ and $\theta \in \mathcal{M}(\mathbb{C})$, then $\nu.\theta$ denotes the push-forward of $\nu \times \theta$ under $(q, z) \mapsto q(z)$. We also write $\nu.z$ for the push-forward of ν under evaluation $q \mapsto q(z)$.

Lemma II.12.10.13 (Planar sliding-window entropy identity). *Let σ be supported on a fixed bounded subset of $\mathbb{C}$, and let $1 \leq L \leq n$. Then*

$$\mathbb{E}_{1 \leq i \leq n} \int \frac{1}{L} H(\sigma^{z,i}, \mathcal{D}_L^{(2)}) d\sigma(z) = H_n(\sigma) + O\left(\frac{L}{n}\right),$$

where $H_n(\sigma) = n^{-1}H(\sigma, \mathcal{D}_n^{(2)})$. The same statement holds in every fixed finite dimension, with the implicit constant depending only on the dimension and the support bound.

Proof. If $E_j = H(\sigma, \mathcal{D}_j^{(2)})$, then the chain rule gives

$$\int H(\sigma^{z,i}, \mathcal{D}_L^{(2)}) d\sigma(z) = E_{i+L} - E_i.$$

Summing in i makes the middle terms telescope. Since one dyadic refinement in the plane increases entropy by at most two bits and the support is bounded, the two boundary blocks contribute $O(L^2)$. Division by Ln proves the claim. The fixed-dimensional version is identical. $\square$

Lemma II.12.10.14 (Entropy cost of point concentration). *There is an absolute constant C with the following property. Let $0 < \kappa < 1/8$, put $L = \lceil \log_2 \kappa^{-1} \rceil$, and let σ be a probability on $[0, 1]^2$ which is $(\{0\}, \kappa)$ -concentrated in the sense of Definition II.12.10.3. Then*

$$\frac{1}{L} H(\sigma, \mathcal{D}_L^{(2)}) \leq C \left(\kappa + \frac{1 + h_2(\kappa)}{L} \right),$$

where $h_2(t) = -t \log_2 t - (1-t) \log_2 (1-t)$.

Proof. There is a ball B of radius κ with $\sigma(B) \geq 1 - \kappa$. Since $2^{-L} \asymp \kappa$, the ball B meets at most an absolute number C_0 of level- L dyadic squares. Split $\sigma = (1-t)\sigma_0 + t\sigma_1$, where $t \leq \kappa$, σ_0 is supported on those C_0 squares, and σ_1 is arbitrary. The entropy of a mixture is bounded above by the entropy of the selector plus the average component entropies, hence

$$H(\sigma, \mathcal{D}_L^{(2)}) \leq h_2(t) + (1-t) \log_2 C_0 + t(2L + C_1).$$

Use $t \leq \kappa$ and divide by L . $\square$

Proposition II.12.10.15 (Planar convolution growth from the inverse theorem). *For every $0 < a < 1/10$ and support bound $R > 1$ there are $\eta_0 = \eta_0(a) > 0$ and, for every sufficiently large block length m , constants $\delta = \delta(a, R, m) > 0$ and $n_0 = n_0(a, R, m)$ such that the following holds for $n \geq n_0$.*

Let ξ, θ be probabilities on $[-R, R]^2$. Assume

$$H_n(\xi) > a, \tag{II.12.10.12}$$

and that, for some $0 < \eta < \eta_0$, with probability greater than $1 - \eta$ over $0 \leq i \leq n$ and $z \sim \theta$,

$$H_m(\theta^{z,i}) < 2 - a, \tag{II.12.10.13}$$

$$\inf_{W \in \mathbb{RP}^1} H_m((\pi_W)_* \theta^{z,i}) > H_m(\theta^{z,i}) - 1 + a. \tag{II.12.10.14}$$

Then

$$H_n(\xi * \theta) \geq H_n(\theta) + \delta. \tag{II.12.10.15}$$

This is the planar growth consequence of Hochman's inverse theorem [Hoc20, Theorem 2.8] used in [FR26].

Proof. Choose first $0 < \kappa \ll a$ so small that

$$2\kappa + C \left(\kappa + \frac{1 + h_2(\kappa)}{\lceil \log_2 \kappa^{-1} \rceil} \right) < a/4, \quad (\text{II.12.10.16})$$

with C larger than the constants in Lemma II.12.10.14 and in the planar component average. Next choose $m \geq m_0(\kappa)$ from Theorem II.12.10.10, and decrease η_0 so that $\eta_0 + 3\kappa < a/20$.

Suppose, contrary to (II.12.10.15), that $H_n(\xi * \theta) \leq H_n(\theta) + \delta$, where δ is the inverse-theorem threshold for the tolerance κ . Theorem II.12.10.10, applied with target θ and perturbation ξ , produces subspaces $V_i \leq \mathbb{R}^2$ such that, outside probability $O(\kappa)$,

$$\theta^{z,i} \text{ is } (V_i, \kappa, m)\text{-saturated,} \quad \xi^{y,i} \text{ is } (V_i, \kappa)\text{-concentrated.} \quad (\text{II.12.10.17})$$

Intersect with the event (II.12.10.13)–(II.12.10.14).

If $\dim V_i = 2$, saturation gives $H_m(\theta^{z,i}) \geq 2 - \kappa$, contradicting (II.12.10.13). If $\dim V_i = 1$, saturation gives, with $W = V_i^\perp$,

$$H_m((\pi_W)_* \theta^{z,i}) \leq H_m(\theta^{z,i}) - 1 + \kappa,$$

contradicting (II.12.10.14). Thus on the common good event we must have $V_i = \{0\}$, and hence $\xi^{y,i}$ is $(\{0\}, \kappa)$ -concentrated.

It follows that such point concentration holds for more than a $1 - (\eta + 2\kappa)$ proportion of scale/component pairs (i, y) . Put $L = \lceil \log_2 \kappa^{-1} \rceil$. Lemmas II.12.10.14 and II.12.10.13, using the trivial bound $L^{-1}H(\xi^{y,i}, \mathcal{D}_L^{(2)}) \leq 2 + O_R(1/L)$ on the exceptional pairs, give for large n

$$H_n(\xi) \leq C \left(\kappa + \frac{1 + h_2(\kappa)}{L} \right) + 2(\eta + 2\kappa) + o(1) < a,$$

by (II.12.10.16) and the choice of η_0 . This contradicts (II.12.10.12). Hence the desired positive increment holds after choosing δ below the inverse-theorem threshold and then n sufficiently large. $\square$

Lemma II.12.10.16 (Complex polynomial action: local entropy average). *Fix $k \geq 1$ and $R > 1$. Let ν be a probability on $\mathcal{A}_k(\mathbb{C})$ supported on $\{q : \|q\|_2 \leq R\}$, and let θ be a probability on the unit disk. If $n \geq m \geq 1$, then*

$$\begin{aligned} \frac{1}{n}H(\nu, \theta, \mathcal{D}_n^{(2)}) &\geq \mathbb{E}_{\substack{1 \leq i \leq n \\ q \sim \nu, z \sim \theta}} \frac{1}{m}H\left(\nu_{\mathcal{D}_i^{\mathcal{A}_k(\mathbb{C})}(q)}, \theta_{\mathcal{D}_i^{(2)}(z)}, \mathcal{D}_{i+m}^{(2)}\right) \\ &\quad - O_{k,R}\left(\frac{1}{m} + \frac{m}{n}\right). \end{aligned} \quad (\text{II.12.10.18})$$

Proof. Apply the entropy chain rule to the evaluation map $F(q, z) = q(z)$ on the bounded coefficient-disk set. This map is uniformly Lipschitz there. Hence the image of a level- i product atom meets only $O_{k,R}(1)$ level- i dyadic squares. Exactly as in Lemma II.12.4.12, the conditional entropy of the level- $(i + m)$ image over the level- i product atom is bounded above by the level- $[i, i + m]$ entropy increment of ν, θ , up to $O_{k,R}(1)$ bits. Sum over i and use sliding-window cancellation. $\square$

Lemma II.12.10.17 (Complex polynomial action: Taylor linearization). *For every $0 < \zeta < 1$, $k \geq 1$, $R > 1$ and sufficiently large m , there is $\delta_0 = \delta_0(\zeta, k, R, m) > 0$ such that the following holds. Suppose $q \in \mathcal{A}_k(\mathbb{C})$, $z \in \mathbb{D}$, and probabilities ν, θ satisfy*

$$\|q\|_2 \leq R, \quad \text{supp } \nu \subset B(q, \delta), \quad \text{supp } \theta \subset B(z, \delta), \quad 0 < \delta < \delta_0.$$

With $L = m + \lceil \log_2 \delta^{-1} \rceil$,

$$\left| \frac{1}{m} H(\nu, \theta, \mathcal{D}_L^{(2)}) - \frac{1}{m} H((\nu \cdot z) * (S_{q'(z)} \theta), \mathcal{D}_L^{(2)}) \right| < \zeta. \quad (\text{II.12.10.19})$$

Proof. For $r \in \text{supp } \nu$ and $w \in \text{supp } \theta$, Taylor's formula gives

$$r(w) = r(z) + q'(z)(w - z) + O_{k,R}(\delta^2).$$

Indeed, the quadratic remainder is $O_{k,R}(\delta^2)$ and $|r'(z) - q'(z)| |w - z| = O_k(\delta^2)$. Choose δ_0 so that the error is at most 2^{-L-3} . Coupled random variables which differ by that amount can occupy only a bounded number of neighboring level- L dyadic squares, so their entropies differ by $O(1)$ bits. A fixed translation by $q'(z)z$ changes entropy by another $O(1)$ bits. Divide by m . $\square$

Lemma II.12.10.18 (Entropy survives at a complex polynomial evaluation). *Fix $0 < \epsilon < 1$ and $k \geq 1$. There is $c = c(\epsilon, k) > 0$ such that the following holds for all sufficiently large m and then all sufficiently large n . Let ν be supported on $\{q \in \mathcal{A}_k(\mathbb{C}) : \|q\|_2 \leq \epsilon^{-1}\}$ and suppose*

$$\frac{1}{n} H(\nu, \mathcal{D}_n^{\mathcal{A}_k(\mathbb{C})}) \geq \epsilon. \quad (\text{II.12.10.20})$$

Let θ be a probability on the unit disk for which there exists $r = r(\epsilon, k) > 0$ such that

$$\sup_{w \in \mathbb{C}} \theta(B(w, r)) < \frac{1}{2(k+1)}. \quad (\text{II.12.10.21})$$

Then

$$\mathbb{P}_{\substack{1 \leq i \leq n \\ q \sim \nu, z \sim \theta}} \left\{ \frac{1}{m} H\left((\nu_{\mathcal{D}_i^{\mathcal{A}_k(\mathbb{C})}(q)}) \cdot z, \mathcal{D}_{i+m}^{(2)}\right) > c \right\} > c. \quad (\text{II.12.10.22})$$

Proof. The coefficient space has real dimension $2(k+1)$. The fixed-dimensional version of Lemma II.12.10.13, applied there, shows that for large m and then large n ,

$$\mathbb{E}_{i,q} \frac{1}{m} H(\nu_{\mathcal{D}_i^{\mathcal{A}_k(\mathbb{C})}(q)}, \mathcal{D}_{i+m}^{\mathcal{A}_k(\mathbb{C})}) \geq \epsilon/2.$$

Thus a set G of pairs (i, q) of probability $a = a(\epsilon, k) > 0$ has coefficient-component entropy at least $\epsilon/4$.

Choose $Z_0, \dots, Z_k$ independently with law θ . By (II.12.10.21), successively excluding the r -balls around points already chosen shows that the event E on which the Z_j are pairwise

r -separated has probability bounded below by a positive constant depending only on k . On E , the complex Vandermonde map

$$L_Z : \mathcal{A}_k(\mathbb{C}) \longrightarrow \mathbb{C}^{k+1}, \quad q \longmapsto (q(Z_0), \dots, q(Z_k)),$$

is uniformly bi-Lipschitz on coefficient space, because $|\det L_Z| = \prod_{a < b} |Z_b - Z_a|$ is bounded below. Therefore, for every coefficient component σ occurring above,

$$\sum_{j=0}^k H(\sigma, Z_j, \mathcal{D}_{i+m}^{(2)}) \geq H(\sigma, \mathcal{D}_{i+m}^{\mathcal{A}_k(\mathbb{C})}) - C_{\epsilon, k}.$$

Integrating over E and $(i, q) \in G$ gives

$$E_{i,q,z} \frac{1}{m} H((\nu_{\mathcal{D}_i^{\mathcal{A}_k(\mathbb{C})}(q)}) \cdot z, \mathcal{D}_{i+m}^{(2)}) \geq b(\epsilon, k) > 0.$$

The displayed normalized entropy is bounded by an absolute constant depending only on k , because a level- i coefficient cube has evaluation image of diameter $O_k(2^{-i})$ on the unit disk. A final averaging/Markov argument yields (II.12.10.22). $\square$

Lemma II.12.10.19 (Bounded distortion on the adapted domain). *Fix the adapted domain Ω_0 of Lemma II.12.10.2. There is $C \geq 1$, depending only on the IFS and these domains, such that for every word u and $z, w \in \Omega_0$,*

$$C^{-1} \leq \frac{|\varphi'_u(z)|}{|\varphi'_u(w)|} \leq C.$$

If $\psi : \Omega_0 \rightarrow \mathbb{C}$ is holomorphic and injective with $\epsilon \leq |\psi'| \leq \epsilon^{-1}$ on Ω_0 , and $z_ \in \Omega$, then $|\varphi'_u(z_*)|^{-1} \psi \circ \varphi_u$ is $C\epsilon^{-1}$ -bi-Lipschitz on Ω .*

Proof. Let d_0 be the intrinsic path distance in Ω_0 . Compact containment $\overline{\Omega} \subset \Omega_0$ and compactness imply that for some C_0 ,

$$d_0(z, w) \leq C_0 |z - w| \quad (z, w \in \Omega).$$

Indeed, a short segment lies in Ω_0 when $|z - w|$ is below half the distance from $\overline{\Omega}$ to $\mathbb{C} \setminus \Omega_0$; for more distant points, a finite disk cover of $\overline{\Omega}$ and paths joining its centers give a uniform path-length bound. Since $\sup_{\Omega_0} |\varphi'_i| \leq q < 1$, images of paths show

$$d_0(\varphi_v z, \varphi_v w) \leq q^{|v|} d_0(z, w).$$

Because Ω_0 is a fixed positive neighborhood of the compact connected set $\overline{\Omega}$, a finite disk-chain cover gives it finite intrinsic diameter L_0 . Write the logarithm of $|\varphi'_u|$ as the sum of the logarithms of the derivative factors, indexed by suffix length k . For the finitely many $k < k_0$, each difference is at most $\log(q/m)$. Choose k_0 so large that $q^{k_0} L_0$ is less than half the distance from $\overline{\Omega}$ to $\mathbb{C} \setminus \Omega_0$. For $k \geq k_0$, both suffix images lie in Ω and their joining segment lies in Ω_0 ; on the relevant compact neighborhood of $\overline{\Omega}$, $|\varphi''_i|/|\varphi'_i|$ is uniformly bounded. The

k -th difference is consequently at most $C_1 q^k d_0(z, w) \leq C_1 q^k L_0$. Summing the geometric series proves the derivative-ratio assertion.

The same estimate compares $|\varphi'_u(\xi)|$ to $|\varphi'_u(z_*)|$ uniformly along paths in Ω_0 of bounded length joining points of Ω : after one nonempty suffix the images lie in Ω , and the zero-length suffix costs only $\log(q/m)$. Along such a path the derivative of $|\varphi'_u(z_*)|^{-1}\psi \circ \varphi_u$ is therefore bounded above by $C\epsilon^{-1}$. The path-length bound gives the global upper Lipschitz estimate.

For the lower bound, the normalized composition is injective on Ω_0 and has derivative at least $C^{-1}\epsilon$ on Ω . Every $z \in \Omega$ has a disk of the same fixed radius inside Ω_0 . The Koebe distortion and quarter theorems, applied on this disk, give

$$|g(z) - g(w)| \geq c\epsilon \min\{|z - w|, r_0\} \geq c_{\Omega_2}\epsilon|z - w| \quad (z, w \in \Omega),$$

where $g = |\varphi'_u(z_*)|^{-1}\psi \circ \varphi_u$. Enlarging C proves the claimed bi-Lipschitz bound. $\square$

Definition II.12.10.20 (Holomorphic normalization and stopping words). In this definition and the holomorphic proofs that follow, fix the adapted domain Ω_0 supplied by Lemma II.12.10.2. Translating the coordinate once, choose $0 \in \Omega$; this changes none of the hypotheses or conclusions. For the remainder of the holomorphic argument write $\Sigma = \Lambda^{\mathbb{N}}$, let $\beta = p^{\mathbb{N}}$, let $\sigma : \Sigma \rightarrow \Sigma$ be the shift, and let $\Pi : \Sigma \rightarrow K_{\Phi}$ be the coding map, so that $\Pi_*\beta = \mu$. For $0 < \epsilon < 1$, let $\mathcal{F}_{\epsilon}$ be the class of holomorphic injections $\psi : \Omega_0 \rightarrow \mathbb{C}$ satisfying

$$\epsilon \leq |\psi'(z)| \leq \epsilon^{-1} \quad (z \in \Omega_0).$$

For a finite word u , put

$$r_u = |\varphi'_u(0)|, \quad S_u(z) = r_u^{-1}z.$$

For integers $l, s \geq 1$ and $0 \leq j < l$, let $\Psi(j, l; s)$ be the minimal cutset consisting of words $u = u_0 u_1 \cdots u_q$, with $|u_0| = j$ and $|u_t| = l$ for $t \geq 1$, for which $r_u < 2^{-s}$ but every preceding block-prefix has derivative scale at least 2^{-s} . Write $\Psi_s = \Psi(0, 1; s)$. If $\mathbf{U}_n$ is the Bernoulli word of deterministic length n , and $\mathbf{I}(j, l; s)$ the Bernoulli word chosen from $\Psi(j, l; s)$, then

$$\mu = \sum_{v \in \Psi(j, l; s)} p_v (\varphi_v)_* \mu.$$

We regard $\mathbb{R}P^1$ as the compact multiplicative group $\mathbb{C}^*/\mathbb{R}^*$, writing $z\mathbb{R}$ for the line generated by $z \neq 0$.

Lemma II.12.10.21 (Uniform holomorphic normalization). *There is a distortion constant $C \geq 1$ such that, for every $0 < \epsilon < 1$, $\psi \in \mathcal{F}_{\epsilon}$, and finite word u , $S_u \circ \psi \circ \varphi_u$ is $C\epsilon^{-1}$ -bi-Lipschitz on Ω . Moreover, for each fixed l there is $C_l \geq 1$ such that*

$$C_l^{-1}2^{-s} \leq r_v < 2^{-s} \quad (v \in \Psi(j, l; s)). \quad (\text{II.12.10.23})$$

Consequently, if $d = \dim \mu$, then for each fixed $0 < \epsilon < 1$ and angular gap $\kappa > 0$,

uniformly in $u, \psi \in \mathcal{F}_\epsilon$, and lines V, W whose angle is at least κ ,

$$\frac{1}{m} H(S_u(\psi\varphi_u)_*\mu, \pi_V^{-1}\mathcal{D}_m \vee \pi_W^{-1}\mathcal{D}_m) = d + o_{m \rightarrow \infty; \epsilon, \kappa}(1). \quad (\text{II.12.10.24})$$

Proof. Lemma II.12.10.19, with $z_* = 0$, proves both $|\varphi'_u(z)| \asymp r_u$ and the global bi-Lipschitz assertion; in particular the lower Lipschitz estimate uses injectivity and Koebe's theorem, not merely a lower derivative bound. At the first block at which the derivative scale crosses 2^{-s} , bounded distortion and the fixed block length l give (II.12.10.23).

The two-projection map $z \mapsto (\pi_V z, \pi_W z)$ is bi-Lipschitz with constant depending only on κ . Its product dyadic partition at scale m is therefore commensurable, with bounded multiplicity, with the planar dyadic partition. Exact dimensionality of μ gives $m^{-1} H(\mu, \mathcal{D}_m^{(2)}) \rightarrow d$; a family with fixed bi-Lipschitz constant changes this entropy by only $O_{\epsilon, \kappa}(1)$ bits. This proves (II.12.10.24). $\square$

Remark II.12.10.22 (Geometric ingredients). Lemma II.12.10.21 uses exact dimensionality and the entropy–dimension identity from Volume I. The next lemmas establish the curve-nullity, boundary-avoidance, direction-cocycle, projection-improvement, stopping-scale, and component-transfer properties needed for the holomorphic argument. Throughout that package write $d = \dim \mu$. The curve-nullity and boundary-avoidance statements require no common fixed point and no invariant regular real-analytic curve; the direction-spreading and improved-projection statements additionally require nonconjugacy to a homothetic IFS. Each dependent result states its own hypotheses below.

Lemma II.12.10.23 (No atoms and amplification of positive mass). *Under the hypothesis that the generators have no common fixed point, μ is nonatomic. If $E \subset \Omega$ is Borel with $\mu(E) > 0$, then for every $\eta > 0$ there is a finite word u such that*

$$(\varphi_u)_*\mu(E) > 1 - \eta, \quad \text{diam}(\varphi_u(\Omega)) < \eta. \quad (\text{II.12.10.25})$$

Proof. If x were an atom of maximal mass, stationarity $\mu = \sum_i p_i(\varphi_i)_*\mu$ and positivity of the weights would force every inverse branch through x to carry an atom of the same maximal mass. The finite set of maximal atoms would therefore be invariant under every φ_i . Since sufficiently long compositions are strict contractions, such a finite invariant set must be a singleton fixed by every generator, contrary to the hypothesis.

For the second assertion choose a μ -density point x of E , then a radius r with $\mu(\partial B(x, r)) = 0$ and $\mu(E \cap B(x, r)) > (1 - \eta/4)\mu(B(x, r))$. Choose a much finer stopping cutset Ψ_s , so every cylinder image has diameter much smaller than a thin annulus around $\partial B(x, r)$. The total weight of cylinders meeting $B(x, r)$ but not contained in it is bounded by the mass of that annulus and can be made $< (\eta/4)\mu(B(x, r))$. Decomposing μ over Ψ_s and averaging now produces a cylinder contained in $B(x, r)$ whose normalized copy gives E mass $> 1 - \eta$. Taking s still larger gives the diameter condition in (II.12.10.25). $\square$

Proposition II.12.10.24 (Real-analytic curves are null). *Assume that the generators have no common fixed point and that there is no regular real-analytic curve in Ω , closed in Ω , which is invariant under every map in Φ . Then*

$$\mu(\Gamma) = 0 \quad (\text{II.12.10.26})$$

for every regular real-analytic curve $\Gamma \subset \Omega$.

Proof. Suppose that a regular real-analytic curve has positive mass. In one analytic chart there are a compact interval J , a complex neighbourhood $V \supset J$, and a holomorphic injection $f : V \rightarrow \Omega$ with $\mu(f(J)) > 0$. On $f(V)$ put $h = \text{Im}(f^{-1})$. After shrinking to a relatively compact domain $W \subset\subset f(V)$ containing $f(J)$, the harmonic function h has nonvanishing gradient on $\overline{W}$.

By Lemma II.12.10.23 choose words u_n with $(\varphi_{u_n})_*\mu(f(J)) > 1 - 1/n$ and $\varphi_{u_n}(\Omega) \subset W$. Put $r_n = r_{u_n}$ and

$$h_n(z) = r_n^{-1}h(\varphi_{u_n}(z)).$$

The bounded-distortion estimate and the chain rule make h_n uniformly Lipschitz and give a uniform positive lower bound for $|\nabla h_n|$. Since each zero set has positive mass, choose $z_n \in K_\Phi$ with $h_n(z_n) = 0$; boundedness of K_Φ then makes h_n uniformly bounded. Harmonic compactness gives a subsequence converging, together with its gradients on compact sets, to a harmonic h_0 with $|\nabla h_0| > 0$.

For every $\rho > 0$, uniform convergence on K_Φ and $\mu(h_n^{-1}\{0\}) > 1 - 1/n$ give $\mu\{|h_0| < \rho\} > 1 - \rho$ for all sufficiently large n . Hence $\mu(h_0^{-1}\{0\}) = 1$, and support considerations give $K_\Phi \subset h_0^{-1}\{0\}$. By the real-analytic implicit function theorem the latter is a regular real-analytic curve. Let Γ_0 be the union of its connected components meeting K_Φ . It is closed in Ω .

It remains to see invariance. Fix a component $C \subset \Gamma_0$ and a point of $C \cap K_\Phi$. A sufficiently small cylinder copy $\varphi_u(K_\Phi)$ lies inside C . For every generator i , the harmonic function $h_0 \circ \varphi_i$ vanishes on the uncountable set $\varphi_u(K_\Phi) \subset C$, hence on all of C by real-analytic unique continuation along the curve. Thus $\varphi_i(C)$ is contained in a component of $h_0^{-1}\{0\}$ meeting K_Φ . Therefore $\varphi_i(\Gamma_0) \subset \Gamma_0$ for every i , contradicting the geometric hypothesis. $\square$

Proposition II.12.10.25 (Uniform line and dyadic-boundary avoidance). *Assume that the generators have no common fixed point and that there is no invariant regular real-analytic curve closed in Ω . For every $0 < \epsilon < 1$ there is $\delta = \delta(\epsilon) > 0$ such that for all $s \geq 1$, $u \in \Psi_s$, $\psi \in \mathcal{F}_\epsilon$, affine real lines $a + V$,*

$$(\psi\varphi_u)_*\mu((a + V)^{(\delta 2^{-s})}) < \epsilon. \quad (\text{II.12.10.27})$$

Consequently, after decreasing δ ,

$$\psi_*\mu\left(\bigcup_{D \in \mathcal{D}_s^{(2)}} (\partial D)^{(\delta 2^{-s})}\right) < \epsilon \quad (s \geq 1, \psi \in \mathcal{F}_\epsilon). \quad (\text{II.12.10.28})$$

Proof. If (II.12.10.27) failed, there would be $s_k, u_k, \psi_k, a_k, V_k$ with tube widths $o(2^{-s_k})$ and mass at least ϵ . Normalize $F_k = r_{u_k}^{-1}(\psi_k \varphi_{u_k} - \psi_k \varphi_{u_k}(0))$. Lemma II.12.10.21 makes F_k a normal family of injective holomorphic maps with derivative bounded above and away from zero. After a subsequence, $F_k \rightarrow F$ locally uniformly, $V_k \rightarrow V$, and the normal components of the translated lines converge. The limit F is injective and nonconstant, and weak convergence yields $(F_*\mu)(a + V) \geq \epsilon$ for a limiting affine line. Its inverse image is a regular real-analytic curve of positive μ -mass, contradicting Proposition II.12.10.24.

For (II.12.10.28), decompose μ over Ψ_s . The support of each $(\psi \varphi_u)_*\mu$ has diameter $O_\epsilon(2^{-s})$, so it meets only $O_\epsilon(1)$ vertical and horizontal level- s grid lines. Apply (II.12.10.27) to these finitely many lines, with a smaller internal error, and average over the stopping decomposition. $\square$

Lemma II.12.10.26 (Trivial projection lower bound). *For every $0 < \epsilon < 1$, all sufficiently large m , every finite word u , every $\psi \in \mathcal{F}_\epsilon$, and every line $V \in \mathbb{RP}^1$,*

$$\frac{1}{m} H(S_u(\pi_V \psi \varphi_u)_*\mu, \mathcal{D}_m) \geq d - 1 - \epsilon. \quad (\text{II.12.10.29})$$

Moreover, for every word u there is a line V_u such that, for every line W outside the ϵ -ball about V_u ,

$$\frac{1}{m} H(S_u(\pi_W \varphi_u)_*\mu, \mathcal{D}_m) > \frac{d}{2} - \epsilon. \quad (\text{II.12.10.30})$$

Proof. For the first claim use the perpendicular line $V^\perp$ in (II.12.10.24). The entropy of the $V^\perp$ projection is at most $m + O(1)$, and the conditional-entropy inequality bounds the joint entropy by the sum of the two marginal entropies. This gives (II.12.10.29).

For the second claim set $H(W) = m^{-1} H(S_u(\pi_W \varphi_u)_*\mu, \mathcal{D}_m)$ and choose V_u so that $H(V_u) \leq \inf_W H(W) + \epsilon/4$; attainment of the infimum is not needed. If $d(V_u, W) \geq \epsilon$, the two-projection form (II.12.10.24) and the conditional-entropy inequality give

$$d - \epsilon/4 < H(V_u) + H(W) \leq 2H(W) + \epsilon/4,$$

and hence $H(W) > d/2 - \epsilon/4$, which implies (II.12.10.30). $\square$

Lemma II.12.10.27 (Harmonic level sets). *If h is a nonconstant harmonic function on a planar domain and $c \in \mathbb{R}$, then $h^{-1}\{c\}$ is the union of a countable set and a possibly empty regular real-analytic curve.*

Proof. Locally write $h = \text{Re } F$ with F holomorphic. The critical points of h are the zeros of F' , hence are discrete and therefore countable. Away from them, the real-analytic implicit function theorem gives a regular real-analytic one-manifold. A second-countable domain has only countably many local charts, so these regular pieces form one regular real-analytic curve, possibly disconnected. $\square$

Lemma II.12.10.28 (Real derivative direction forces homothety). *Assume that the generators have no common fixed point and that there is no invariant regular real-*

analytic curve closed in Ω . Let $\psi : \Omega \rightarrow \mathbb{C}$ be holomorphic and injective, fix $j \in \Lambda$, and set $g = \psi \varphi_j \psi^{-1}$. If

$$g'(\psi(z))\mathbb{R} = \mathbb{R} \quad (z \in K_\Phi), \quad (\text{II.12.10.31})$$

then g is homothetic.

Proof. Choose a logarithm branch on $U = \mathbb{C} \setminus i\mathbb{R}_{\geq 0}$ and set $\Omega_1 = \{z : g'(\psi(z)) \in U\}$. By (II.12.10.31), $\mu(\Omega_1) = 1$. On Ω_1 ,

$$h(z) = \text{Im} \log_U(g'(\psi(z)))$$

is harmonic. Some connected component G of Ω_1 has positive μ -mass. On $K_\Phi \cap G$, h takes only the two values 0 and $-\pi$. If h were nonconstant on G , Lemma II.12.10.27, nonatomicity, and Proposition II.12.10.24 would force $\mu(G \cap h^{-1}\{0, -\pi\}) = 0$, a contradiction. Thus h is constant on G . The holomorphic function $g' \circ \psi$ therefore maps the open set G into one real ray. By the open mapping theorem it is constant there, and hence everywhere by analytic continuation. The constant has real direction by (II.12.10.31), so $g(z) = rz + a$ with $0 \neq r \in \mathbb{R}$. $\square$

Proposition II.12.10.29 (The derivative-direction cocycle is not a coboundary). *Assume that the generators have no common fixed point, that there is no invariant regular real-analytic curve closed in Ω , and that Φ is not holomorphically conjugate to a homothetic IFS. Define*

$$\alpha(\omega) = \varphi'_{\omega_0}(\Pi(\sigma\omega))\mathbb{R}, \quad \alpha_n(\omega) = \prod_{t=0}^{n-1} \alpha(\sigma^t \omega). \quad (\text{II.12.10.32})$$

Then

$$\alpha_n(\omega) = \varphi'_{\omega|_n}(\Pi(\sigma^n \omega))\mathbb{R}, \quad (\text{II.12.10.33})$$

and there is no Borel $f : \Sigma \rightarrow \mathbb{RP}^1$ satisfying

$$\alpha(\omega) = f(\omega)^{-1} f(\sigma\omega) \quad \text{for } \beta\text{-a.e. } \omega. \quad (\text{II.12.10.34})$$

Proof. The chain rule gives (II.12.10.33). Suppose (II.12.10.34) holds. Iteration and insertion of an arbitrary finite prefix u give

$$f(\omega) = \varphi'_u(\Pi\omega)\mathbb{R} f(u\omega) \quad \text{for every } u \text{ and a.e. } \omega. \quad (\text{II.12.10.35})$$

By Lusin's theorem choose a positive-measure compact set on which f is continuous. The martingale density theorem provides η in this set and a sequence $n_k \rightarrow \infty$ such that, after a subsequence,

$$f(\eta|_{n_k} \omega) \longrightarrow f(\eta) \quad \text{for a.e. } \omega. \quad (\text{II.12.10.36})$$

Normalize

$$\psi_k(z) = \frac{\varphi_{\eta|_{n_k}}(z) - \varphi_{\eta|_{n_k}}(0)}{|\varphi'_{\eta|_{n_k}}(0)|}.$$

Bounded distortion makes $\{\psi_k\}$ locally bounded and keeps $|\psi'_k|$ bounded away from zero. Montel's theorem and the compactness lemma for injective holomorphic functions give, after a subsequence, an injective holomorphic limit ψ with nonvanishing derivative. In particular,

$$\varphi'_{\eta|_{n_k}}(z)\mathbb{R} \longrightarrow \psi'(z)\mathbb{R} \quad (z \in \Omega).$$

Using (II.12.10.35) with $u = \eta|_{n_k}$ and (II.12.10.36) yields

$$f(\omega) = \psi'(\Pi\omega)\mathbb{R} f(\eta) \quad \text{for a.e. } \omega. \quad (\text{II.12.10.37})$$

Fix $j \in \Lambda$ and let $g = \psi\varphi_j\psi^{-1}$. Combining the one-step identity (II.12.10.35) with (II.12.10.37) gives $g'(\psi\Pi\omega)\mathbb{R} = \mathbb{R}$ for β -a.e. ω . Since $\Pi_*\beta = \mu$ and $K_\Phi = \text{supp } \mu$, continuity extends this to every $z \in K_\Phi$. Lemma II.12.10.28 shows that every conjugated generator $\psi\varphi_j\psi^{-1}$ is homothetic. This contradicts the hypothesis that Φ is not holomorphically conjugate to a homothetic IFS. $\square$

Proposition II.12.10.30 (Uniform spreading of cocycle directions). *Assume that the generators have no common fixed point, that there is no invariant regular real-analytic curve closed in Ω , and that Φ is not holomorphically conjugate to a homothetic IFS. There is $\delta_0 > 0$ such that for every continuous $h : \Sigma \rightarrow \mathbb{RP}^1$ and for β -a.e. ω , all sufficiently large N satisfy*

$$\frac{1}{N} \# \{1 \leq i \leq N : d(\alpha_i(\omega)h(\sigma^i\omega), V) > \delta_0\} > \delta_0 \quad \text{for every } V \in \mathbb{RP}^1. \quad (\text{II.12.10.38})$$

Proof. Consider the skew product $T(\omega, V) = (\sigma\omega, \alpha(\omega)V)$ on $\Sigma \times \mathbb{RP}^1$. The product of β with Haar measure is T -invariant; choose an ergodic component λ . Its projection to Σ is the ergodic measure β . Disintegrate $\lambda = \int \delta_\omega \times \xi_\omega d\beta(\omega)$. Invariance and the Bernoulli conditional law of the first symbol give the transfer identity

$$\xi_\omega = \sum_{i \in \Lambda} p_i \alpha(i\omega)_* \xi_{i\omega} \quad \text{for } \beta\text{-a.e. } \omega. \quad (\text{II.12.10.39})$$

If ξ_ω were a point mass for a.e. ω , its atom would give a measurable graph f . Since every $p_i > 0$, a convex combination of point masses can equal $\delta_{f(\omega)}$ only when each summand equals it. Equation (II.12.10.39) would therefore imply $f(\sigma\eta) = \alpha(\eta)f(\eta)$ for β -a.e. η , precisely the forbidden coboundary relation. Hence there are $\delta > 0$ and a set E of positive β -measure on which ξ_ω is not δ -concentrated. Decrease δ so that $\beta(E) > 4\delta$.

For continuous h , push λ forward under $(\omega, V) \mapsto Vh(\omega)$, obtaining a probability ϑ on $\mathbb{RP}^1$. If ϑ placed mass $> 1 - \delta^2$ in a δ -ball, Fubini and Markov would imply that outside a set of base measure $< \delta$, the rotated fiber $h(\omega)_*\xi_\omega$ places mass $> 1 - \delta$ in that ball. This contradicts the definition of E . Thus ϑ is uniformly nonconcentrated.

For λ -a.e. (ω, V_0) , Birkhoff's theorem makes the orbit generic for λ . Its image under the preceding continuous map is $\alpha_i(\omega)V_0h(\sigma^i\omega)$; multiplying by the fixed line V_0^{-1} shows that $\alpha_i(\omega)h(\sigma^i\omega)$ is equidistributed for a rotation of ϑ . Since the projection of the generic set to Σ has full β -measure, the empirical measures converge to a rotation of ϑ for almost every base point. Cover $\mathbb{RP}^1$ by a finite $\delta/4$ -net. Weak convergence for the finitely many

corresponding balls, followed by the triangle inequality, makes the lower density estimate uniform in the centre line. Replacing δ by a smaller universal δ_0 gives (II.12.10.38). $\square$

Lemma II.12.10.31 (Holomorphic tail linearization). *For every $0 < \epsilon < 1$, all sufficiently large m , and then all sufficiently large k , the following holds. Let u be a finite word, $\psi \in \mathcal{F}_\epsilon$, $i \geq 1$, $\omega \in \Sigma$, and $Z \in \mathbb{RP}^1$. Put $v = (\sigma^i \omega)|_k$ and*

$$L(i, \omega) = \frac{Z}{(\psi \varphi_u)'(\Pi \omega)} \mathbb{R} \alpha_i(\omega)^{-1}. \quad (\text{II.12.10.40})$$

Then

$$\begin{aligned} & \frac{1}{m} H(S_{u\omega|_{i+k}}(\pi_Z \psi \varphi_{u\omega|_{i+k}})_* \mu, \mathcal{D}_m) \\ & \geq \frac{1}{m} H(S_v(\pi_{L(i, \omega)} \varphi_v)_* \mu, \mathcal{D}_m) - \epsilon. \end{aligned} \quad (\text{II.12.10.41})$$

Proof. Set $a = \Pi(\sigma^i \omega)$ and $F = \psi \circ \varphi_{u\omega|_i}$. On $\varphi_v(K_\Phi)$, Taylor's formula at a , Cauchy estimates on a fixed complex neighbourhood, and bounded distortion give

$$|F(w) - F(a) - F'(a)(w - a)| \leq C_\epsilon r_{u\omega|_i} r_v^2.$$

Since $r_v \leq C r_{\max}^k$, choosing $k \gg m, \epsilon^{-1}$ makes this error smaller than 2^{-m} times the scale $r_{u\omega|_{i+k}}$. Replacing F by its affine part therefore costs only $O(1)$ bits at the normalized scale m .

Write $b = F'(a)/|F'(a)|$. Rotation by b^{-1} costs $O(1)$ entropy bits and turns $\pi_Z(F'(a)w)$ into $|F'(a)|\pi_{b^{-1}Z}(w)$. The chain rule gives

$$b\mathbb{R} = \frac{(\psi \varphi_u)'(\Pi \omega)}{|(\psi \varphi_u)'(\Pi \omega)|} \mathbb{R} \alpha_i(\omega),$$

so $b^{-1}Z\mathbb{R} = L(i, \omega)$. Finally bounded distortion compares $|F'(a)|r_v$ with $r_{u\omega|_{i+k}}$, changing entropy by another $O_\epsilon(1)$ bits. Dividing by m proves the lemma. $\square$

Proposition II.12.10.32 (Positive-density $d/2$ projection improvement). *Assume that the generators have no common fixed point, that there is no invariant regular real-analytic curve closed in Ω , and that Φ is not holomorphically conjugate to a homothetic IFS. There exists $\delta_1 > 0$ such that for every $0 < \epsilon < 1$, all sufficiently large m , then all sufficiently large N , every line Z , finite word u , and $\psi \in \mathcal{F}_\epsilon$,*

$$\mathbb{P}_{1 \leq i \leq N} \left\{ \frac{1}{m} H(S_{uU_i}(\pi_Z \psi \varphi_{uU_i})_* \mu, \mathcal{D}_m) > \frac{d}{2} - \epsilon \right\} > \delta_1. \quad (\text{II.12.10.42})$$

Proof. Let δ_0 be supplied by Proposition II.12.10.30. Put $e = \min\{\epsilon/4, \delta_0/4\}$. Since $\mathcal{F}_\epsilon \subset \mathcal{F}_e$, choose m large and then choose $k \gg m$ so that Lemma II.12.10.31 applies with error e . For each word $v \in \Lambda^k$, Lemma II.12.10.26 supplies a bad line B_v such that every line outside its e -neighbourhood has normalized projection entropy $> d/2 - e$. The cylinder function $h(\omega) = B_{\omega|_k}$ is continuous.

For fixed ω , Lemma II.12.10.31 says that the prefix $\omega|_{i+k}$ is good whenever $L(i, \omega)$ lies outside the ϵ -ball about $h(\sigma^i \omega)$. By (II.12.10.40) this is equivalent to saying that $\alpha_i(\omega)h(\sigma^i \omega)$ stays away from the line $Z((\psi\varphi_u)'(\Pi\omega))^{-1}\mathbb{R}$. The spreading proposition holds simultaneously for every centre line, so for a.e. ω a proportion at least δ_0 of large i 's are good, because $\epsilon \leq \delta_0/4$. For each such index the two entropy errors total at most $2e \leq \epsilon/2$, proving the strict $d/2 - \epsilon$ bound. Integrate in ω , discard the first k indices, and use that $\omega|_i$ has the law of $\mathbf{U}_i$. One may take $\delta_1 = \delta_0/2$. $\square$

Lemma II.12.10.33 (Deterministic lengths versus stopping scales). *For every sufficiently large block length l there is $M = M(l)$ such that, for $0 \leq j < l$ and $N \geq 1$,*

$$\mathbb{E}_{1 \leq t \leq N} \delta_{\mathbf{U}_{j+lt}} \ll_M \mathbb{E}_{1 \leq s \leq MN} \delta_{\mathbf{I}(j,l;s)}. \quad (\text{II.12.10.43})$$

In particular, an event having probability ρ under the left-hand mixture has probability at least ρ/M under the right-hand mixture.

Proof. Let C be the distortion constant, and put $r_{\max} = \max_{i,z} |\varphi'_i(z)| < 1$ and $r_{\min} = \min_{i,z} |\varphi'_i(z)| > 0$. Choose l with $Cr_{\max}^l < 1/2$, and then M with $r_{\min}^{2l} \geq 2^{-M}$. If $u = u_0 u_1 \cdots u_t$ with $|u_0| = j$ and $|u_q| = l$ for $q \geq 1$, then each added l -block reduces r_u by a factor below $1/2$. For $t \leq N$, if s is the integer with $2^{-s-1} \leq r_u < 2^{-s}$, then $s \leq MN$, and the preceding block prefixes all have derivative at least 2^{-s} . Thus $u \in \Psi(j, l; s)$ for some $s \leq MN$. The Bernoulli weight of each word is the same on both sides, and summing proves (II.12.10.43) with Radon–Nikodym derivative bounded by M . $\square$

Lemma II.12.10.34 (Stopping-scale local entropy average). *Fix l . For every $0 < \epsilon < 1$, all sufficiently large m , and then all sufficiently large n , uniformly in $0 \leq j < l$, finite words u , lines Z , and $\psi \in \mathcal{F}_\epsilon$,*

$$\begin{aligned} & \frac{1}{n} H(S_u(\pi_Z \psi \varphi_u)_* \mu, \mathcal{D}_n) \\ & \geq \mathbb{E}_{1 \leq s \leq n} \mathbb{E} \left[\frac{1}{m} H(S_{u\mathbf{I}(j,l;s)}(\pi_Z \psi \varphi_{u\mathbf{I}(j,l;s)})_* \mu, \mathcal{D}_m) \right] - \epsilon. \end{aligned} \quad (\text{II.12.10.44})$$

Proof. Apply the chain rule in sliding windows of length m to the measure $S_u(\pi_Z \psi \varphi_u)_* \mu$. Apart from the first and last m scales this gives the average of $m^{-1} H(\cdot, \mathcal{D}_{s+m} \mid \mathcal{D}_s)$, with total error $O(m/n)$. For each s , decompose $\varphi_u \mu = \sum_{v \in \Psi(j,l;s)} p_v \varphi_{uv} \mu$. Conditional entropy is concave, so the level- s window entropy is bounded below by the Bernoulli average of the corresponding cylinder window entropies. By (II.12.10.23) and bounded distortion, each such cylinder has diameter $O_{\epsilon,l}(2^{-s})$ after the outer normalization. Thus conditioning on the level- s cell, removing that conditioning, and replacing the scale factor 2^s by r_v^{-1} each cost only $O_{\epsilon,l}(1)$ bits. Choose m so these errors are $< \epsilon m/2$, and then $n \gg m/\epsilon$. $\square$

Proposition II.12.10.35 (Uniform projection gap for normalized cylinders). *Assume that the generators have no common fixed point, that there is no invariant regular real-*

analytic curve closed in Ω , and that Φ is not holomorphically conjugate to a homothetic IFS. If $d < 2$, there is $\gamma_0 > 0$ such that for every $0 < \epsilon < 1$, all sufficiently large n , every line Z , finite word u , and $\psi \in \mathcal{F}_\epsilon$,

$$\frac{1}{n}H(S_u(\pi_Z\psi\varphi_u)_*\mu, \mathcal{D}_n) \geq d - 1 + \gamma_0. \quad (\text{II.12.10.45})$$

Proof. Let δ_1 be given by Proposition II.12.10.32. Choose a large block length l , let $M = M(l)$ be from Lemma II.12.10.33, and then choose a small internal error $\eta > 0$. For a large auxiliary projection block m , let $\mathcal{V}$ be the set of words v for which

$$\frac{1}{m}H(S_{uv}(\pi_Z\psi\varphi_{uv})_*\mu, \mathcal{D}_m) > \frac{d}{2} - \eta.$$

Put $n' = \lfloor n/M \rfloor$. Apply Proposition II.12.10.32 over deterministic lengths up to $ln' + l - 1$. For large n , endpoint losses are negligible, so one residue class $j \pmod{l}$ satisfies

$$\mathbb{P}_{1 \leq t \leq n'} \{ \mathbf{U}_{j+lt} \in \mathcal{V} \} > \delta_1/2.$$

Lemma II.12.10.33, used with $N = n'$, transfers this to stopping scales $1 \leq s \leq Mn'$. Since $Mn' \leq n$ and $n/(Mn') \rightarrow 1$, the stopping words $\mathbf{I}(j, l; s)$ belong to $\mathcal{V}$ on a proportion

$$\rho := \mathbb{P}_{1 \leq s \leq n} \{ \mathbf{I}(j, l; s) \in \mathcal{V} \} \geq \frac{\delta_1}{4M} \quad (\text{II.12.10.46})$$

for all sufficiently large n .

Use Lemma II.12.10.34. On every stopping word, Lemma II.12.10.26 gives the baseline $d - 1 - \eta$; on the fraction ρ in $\mathcal{V}$ use $d/2 - \eta$. Therefore

$$\frac{1}{n}H(S_u(\pi_Z\psi\varphi_u)_*\mu, \mathcal{D}_n) \geq d - 1 + \rho \left(1 - \frac{d}{2} \right) - 4\eta.$$

Since $d < 2$, choose η smaller than one eighth of the positive middle term. The resulting positive constant can be fixed as γ_0 , independently of Z, u, ψ . $\square$

Theorem II.12.10.36 (Holomorphic line nonsaturation). *Assume the holomorphic IFS satisfies the Feng–Rapaport geometric hypotheses: its generators have no common fixed point, there is no invariant regular real-analytic curve closed in Ω , and the IFS is not holomorphically conjugate to a homothetic IFS. Let μ be a Bernoulli self-conformal measure with $d = \dim \mu < 2$. Then there exists $\gamma > 0$ such that for every $0 < \epsilon < 1$, all sufficiently large m , every outer scale $n \geq 1$, and every $\psi \in \mathcal{F}_\epsilon$, if $\theta = \psi_*\mu$, then*

$$\mathbb{P}_{z \sim \theta} \left\{ \inf_{W \in \mathbb{RP}^1} \frac{1}{m} H((\pi_W)_*\theta^{z,n}, \mathcal{D}_m) > d - 1 + \gamma \right\} > 1 - \epsilon. \quad (\text{II.12.10.47})$$

Proof. Let γ_0 be from Proposition II.12.10.35; eventually take $\gamma = \gamma_0/8$. Choose an internal boundary error $\eta \ll \epsilon\gamma_0$ and let δ be supplied by Proposition II.12.10.25 with error parameter η^2 (note that $\psi \in \mathcal{F}_\epsilon \subset \mathcal{F}_{\eta^2}$ after making η smaller). Choose k so large that every $\psi\varphi_u(\Omega)$,

$u \in \Psi_{n+k}$, has diameter below $\delta 2^{-n}$. If such a cylinder is not contained in one level- n dyadic square, its image lies in the $\delta 2^{-n}$ -neighbourhood of the grid boundary. Hence the total Bernoulli weight of these bad words is $< \eta^2$.

By Markov's inequality there is a collection $\mathcal{E} \subset \mathcal{D}_n^{(2)}$ with $\theta(\bigcup \mathcal{E}) > 1 - \eta$ such that, conditionally on every $D \in \mathcal{E}$, words $u \in \Psi_{n+k}$ whose full image $\psi\varphi_u(\Omega)$ lies in D carry conditional weight $> 1 - \eta$. For such a word and every line W , Proposition II.12.10.35 applied at length $m - k$, together with (II.12.10.23), gives

$$\frac{1}{m} H((\pi_W)_*(\psi\varphi_u)_*\mu, \mathcal{D}_{n+m}) > d - 1 + \frac{\gamma_0}{2}, \quad (\text{II.12.10.48})$$

provided $k \ll m$; the difference between the natural normalized scale $m - k + O(1)$ and m costs at most $O(k/m)$.

The conditional measure θ_D is a convex combination of the cylinder pieces meeting D , and the pieces wholly contained in D have total weight $> 1 - \eta$. Entropy concavity and (II.12.10.48) therefore imply, uniformly in W ,

$$\frac{1}{m} H((\pi_W)_*\theta_D, \mathcal{D}_{n+m}) > d - 1 + \frac{\gamma_0}{4}.$$

Rescaling D to the unit square changes the projected entropy by only $O(1)$ bits, so for large m the normalized component $\theta^{z,n}$ satisfies the same bound with $\gamma_0/8$ whenever $D = \mathcal{D}_n^{(2)}(z) \in \mathcal{E}$. Since these cells carry mass $> 1 - \eta > 1 - \epsilon$, (II.12.10.47) follows. $\square$

Proposition II.12.10.37 (Uniform lower entropy of holomorphic components). *Let $d = \dim \mu$. For every $0 < \epsilon < 1$ and $\tau > 0$ there is $M = M(\epsilon, \tau)$ such that, whenever $n \geq 1$, $m \geq M$, $\Psi \in \mathcal{F}_\epsilon$, and $\theta = \Psi_*\mu$,*

$$\mathbb{P}_{z \sim \theta} \left\{ \frac{1}{m} H(\theta^{z,n}, \mathcal{D}_m^{(2)}) > d - \tau \right\} > 1 - \tau. \quad (\text{II.12.10.49})$$

Proof. The assertion is immediate when $d = 0$, so suppose $d > 0$. Fix small auxiliary numbers $t, b, \delta > 0$, to be chosen in that order in terms of ϵ, τ, d . Exact dimensionality gives, for all sufficiently large m , a Borel set $G_m \subset K_\Phi$ with $\mu(G_m) > 1 - \delta$ such that

$$\mu(B(x, C_\epsilon 2^{-m})) \leq 2^{-m(d-t)} \quad (x \in G_m). \quad (\text{II.12.10.50})$$

Here C_ϵ is a fixed constant large enough for all the inverse Lipschitz estimates below; its presence does not change the limit defining exact dimension.

Use the stopping decomposition

$$\theta = \sum_{u \in \Psi_n} p_u \theta_u, \quad \theta_u = (\Psi\varphi_u)_*\mu.$$

By Lemma II.12.10.21 and (II.12.10.23), every θ_u meets at most J_ϵ level- n dyadic squares, and the inverse image under $\Psi\varphi_u$ of a level- $(n+m)$ square containing $\Psi\varphi_u(x)$ lies in $B(x, C_\epsilon 2^{-m})$. For a level- n square D , put $\alpha_{u,D} = \theta_u(D)$, and when this is positive let $\theta_{u,D}$ be the normalized restriction of θ_u to D . If $\alpha_{u,D} \geq b$, then for every $x \in G_m$ with $\Psi\varphi_u(x) \in D$, the level- $(n+m)$

atom of $\theta_{u,D}$ containing $\Psi\varphi_u(x)$ has mass at most $b^{-1}2^{-m(d-t)}$. Since the conditional mass of such points is at least $1 - \delta/b$, the entropy-as-mean-information identity gives

$$\frac{1}{m}H(\theta_{u,D}, \mathcal{D}_{n+m}^{(2)}) \geq (1 - \delta/b) \left(d - t - \frac{\log_2(1/b)}{m} \right). \quad (\text{II.12.10.51})$$

The total mass of pairs (u, D) with $\alpha_{u,D} < b$ is at most $J_\epsilon b$, because each θ_u meets at most J_ϵ squares. Choose $t \ll \tau$, then b so small that $\sqrt{J_\epsilon b} \ll \tau$, then $\delta \ll b\tau$, and finally m large. Equation (II.12.10.51) is then $> d - \tau/2$ for every pair with $\alpha_{u,D} \geq b$.

For a square D of positive θ -mass, θ_D is the mixture of the measures $\theta_{u,D}$ with weights $p_u \alpha_{u,D} / \theta(D)$. Let b_D be the total weight of pairs with $\alpha_{u,D} < b$. Averaging over D gives $\sum_D \theta(D) b_D \leq J_\epsilon b$. Markov's inequality shows that outside θ -mass $\sqrt{J_\epsilon b}$, one has $b_D \leq \sqrt{J_\epsilon b}$. Concavity of entropy and (II.12.10.51) then give

$$\frac{1}{m}H(\theta_D, \mathcal{D}_{n+m}^{(2)}) \geq (1 - \sqrt{J_\epsilon b})(d - \tau/2) > d - \tau.$$

Rescaling D to the unit square identifies this entropy with $H(\theta^{z,n}, \mathcal{D}_m^{(2)})$ for $z \in D$, proving (II.12.10.49). $\square$

Proposition II.12.10.38 (Feng–Rapaport target component package). *Under the hypotheses of Theorem II.12.10.36, let $d = \dim \mu < 2$. Fix $0 < \epsilon < 1$. There exist $a = a(\mu) > 0$ and, for every sufficiently small $\eta_{\text{in}} > 0$, arbitrarily large inner block lengths ℓ such that the following holds. For every outer accuracy $\zeta > 0$, every sufficiently large outer block m , every sufficiently large averaging range n , and every $\theta = \Psi_*\mu$ with $\Psi \in \mathcal{F}_\epsilon$ and $\Psi(\Omega) \subset \mathbb{D}$, let i be uniform on $\{1, \dots, n\}$ and $z \sim \theta$. With probability $> 1 - \zeta$ over (i, z) one has*

$$\left| \frac{1}{m}H(\theta^{z,i}, \mathcal{D}_m^{(2)}) - d \right| < \zeta. \quad (\text{II.12.10.52})$$

Independently of this baseline event, with probability $> 1 - \eta_{\text{in}}$ over (i, z) the following holds with conditional probability $> 1 - \eta_{\text{in}}$ over the nested scale/component pair (j, w) inside the next m levels; here j is uniform on $\{0, \dots, m-1\}$ and $w \sim \theta^{z,i}$:

$$\frac{1}{\ell}H((\theta^{z,i})^{w,j}, \mathcal{D}_\ell^{(2)}) < 2 - a, \quad (\text{II.12.10.53})$$

$$\inf_{W \in \mathbb{RP}^1} \frac{1}{\ell}H((\pi_W)_*(\theta^{z,i})^{w,j}, \mathcal{D}_\ell) > \frac{1}{\ell}H((\theta^{z,i})^{w,j}, \mathcal{D}_\ell^{(2)}) - 1 + a. \quad (\text{II.12.10.54})$$

Moreover the family of normalized images satisfies the uniform small-ball condition (II.12.10.21) for every fixed k and ϵ after choosing $r = r(k, \epsilon) > 0$.

Proof. First derive the two-sided typical entropy assertion used below. For $\theta = \Psi_*\mu$, the uniform bi-Lipschitz bound in Lemma II.12.10.21 with the empty word implies

$$\sup_{\Psi \in \mathcal{F}_\epsilon} \left| \frac{1}{N}H(\Psi_*\mu, \mathcal{D}_N^{(2)}) - d \right| \longrightarrow 0. \quad (\text{II.12.10.55})$$

Indeed, a level- N square and its image under any of these maps meet at most $O_\epsilon(1)$ level- N squares, in either direction; the two entropies thus differ by $O_\epsilon(1)$ bits, while $N^{-1}H(\mu, \mathcal{D}_N^{(2)}) \rightarrow d$. If $h_s = H(\theta, \mathcal{D}_{s+1}^{(2)} \mid \mathcal{D}_s^{(2)})$, the dyadic chain rule gives exactly

$$\mathbb{E}_{z \sim \theta} H(\theta^{z,i}, \mathcal{D}_m^{(2)}) = \sum_{s=i}^{i+m-1} h_s.$$

Since $0 \leq h_s \leq 2$, averaging over $1 \leq i \leq n$ and using (II.12.10.55) gives

$$\mathbb{E}_{i,z} \frac{1}{m} H(\theta^{z,i}, \mathcal{D}_m^{(2)}) = d + o_{n \rightarrow \infty; \epsilon, m}(1) + O(m/n), \quad (\text{II.12.10.56})$$

uniformly in Ψ . For a prescribed $\zeta > 0$, apply Proposition II.12.10.37 with $\tau \ll \zeta^2$. It bounds the probability of values below $d - \tau$ by τ , at every outer scale i . Equation (II.12.10.56) then bounds the probability of values above $d + \zeta$ by $O((\tau + o(1))/\zeta)$. Taking m and then n large, with τ chosen sufficiently small, proves (II.12.10.52) with probability $> 1 - \zeta$. Notice that ℓ has not entered this choice.

Let $\gamma > 0$ be supplied by Theorem II.12.10.36; choose

$$0 < a < \min\{(2-d)/8, \gamma/8\} \quad \text{and} \quad 0 < q < \min\{\eta_{\text{in}}^2/16, \epsilon/2, a/4\}.$$

Apply the just-proved two-sided assertion at block length ℓ with accuracy q , and apply Theorem II.12.10.36 at the same length with error q . Both apply for every sufficiently large ℓ , so ℓ may be chosen arbitrarily large. Since $\mathcal{F}_\epsilon \subset \mathcal{F}_q$, the line theorem applies uniformly to the present maps. For a global component $\theta^{x,s}$, each conclusion fails on probability at most q when s is averaged over a sufficiently long range; the line conclusion in fact has this bound at every s . On their intersection,

$$\ell^{-1}H(\theta^{x,s}, \mathcal{D}_\ell^{(2)}) < d + q < 2 - a$$

and

$$\inf_W \ell^{-1}H((\pi_W)_* \theta^{x,s}, \mathcal{D}_\ell) > d - 1 + \gamma > \ell^{-1}H(\theta^{x,s}, \mathcal{D}_\ell^{(2)}) - 1 + a.$$

Conditionally on an outer cell $\mathcal{D}_i^{(2)}(z)$, a nested component at level j is precisely the global component at level $s = i + j$, after the common dyadic rescaling. The marginal law of the sampled point remains θ . The distribution of $s = i + j$, with i uniform on $1, \dots, n$ and j uniform on $0, \dots, m-1$, differs from the uniform distribution on $1, \dots, n$ by total variation $O(m/n)$. Thus the average conditional proportion of nested failures is at most $2q + O(m/n)$. Take $n \gg m/q$ and apply Markov's inequality to the outer pair (i, z) . Because $q < \eta_{\text{in}}^2/16$, outside a set of outer probability $< \eta_{\text{in}}$ the conditional failure probability is $< \eta_{\text{in}}$. This proves the nested assertion independently of ζ .

Finally, μ is nonatomic by Lemma II.12.10.23, so compactness of K_Φ gives

$$\sup_x \mu(B(x, r)) \longrightarrow 0 \quad (r \downarrow 0).$$

The same uniform inverse Lipschitz bound yields

$$\sup_{\Psi \in \mathcal{F}_\epsilon, w \in \mathbb{C}} (\Psi_* \mu)(B(w, r)) \leq \sup_x \mu(B(x, C_\epsilon r)).$$

Choose r so that the right side is below $1/[2(k+1)]$, proving (II.12.10.21). $\square$

Theorem II.12.10.39 (Polynomial-action entropy growth). *Assume the hypotheses of Theorem II.12.10.36 and $d = \dim \mu < 2$. For every $k \geq 1$ and $0 < \epsilon < 1$ there are $\rho = \rho(k, \epsilon) > 0$ and $N = N(k, \epsilon)$ such that the following holds for $n \geq N$. Let ν be a compactly supported probability on $\mathcal{A}_k(\mathbb{C})$ satisfying*

$$\|q\|_2 \leq \epsilon^{-1}, \quad |q'(z)| \geq \epsilon \quad (q \in \text{supp } \nu, z \in \mathbb{D}), \quad \frac{1}{n} H(\nu, \mathcal{D}_n^{\mathcal{A}_k(\mathbb{C})}) \geq \epsilon. \quad (\text{II.12.10.57})$$

If $\theta = \Psi_ \mu$ for some $\Psi \in \mathcal{F}_\epsilon$ with $\Psi(\Omega) \subset \mathbb{D}$, then*

$$\frac{1}{n} H(\nu, \theta, \mathcal{D}_n^{(2)}) \geq d + \rho. \quad (\text{II.12.10.58})$$

This deduction follows [FR26]. The proof combines the planar inverse theorem with the target-component and evaluation-entropy estimates established above.

Proof. Let $c_0 = c_0(k, \epsilon) > 0$ be the constant from Lemma II.12.10.18. The coefficient bound and $|q'(z)| \geq \epsilon$ give the diameter constant $C_{k,\epsilon}$ in (II.12.10.59) below; fix once and for all a support bound $R > 10 + C_{k,\epsilon}$, independent of i, m, n, q, z . Apply Proposition II.12.10.38 and choose $a > 0$ so small that

$$a < \min\{c_0/8, (2-d)/16, \epsilon/16\}.$$

Decreasing the gap parameter supplied by the target package preserves its two inequalities. Let $\eta_0(a) > 0$ be the tolerance from Proposition II.12.10.15 and choose

$$0 < \eta_{\text{in}} < \min\{\eta_0(a)/4, c_0/16, a/100\}.$$

Now choose an arbitrarily large inner block length ℓ supplied by Proposition II.12.10.38, large enough that Proposition II.12.10.15 applies with source threshold a , target gaps a , and this fixed support bound R ; let $\rho_0 > 0$ be its entropy increment. Only after ρ_0 is fixed choose an outer accuracy

$$0 < \zeta < \min\{c_0/16, a/100, c_0\rho_0/128\}.$$

Choose the outer action block $m \gg \ell$ so that Proposition II.12.10.38, Lemma II.12.10.18, and Lemma II.12.10.17 all apply, and so that $1/(m+1) < \eta_0(a)/4$; finally take $n \gg m$ so that every endpoint and bounded-grid error below is at most ζ .

For $1 \leq i \leq n$, $q \sim \nu$ and $z \sim \theta$, write

$$v_{i,q} = \nu_{\mathcal{D}_i^{\mathcal{A}_k(\mathbb{C})}(q)}, \quad \theta_{i,z} = \theta_{\mathcal{D}_i^{(2)}(z)}, \quad \xi_{i,q,z} = S_{q'(z)}^{-1}(v_{i,q} \cdot z).$$

The coefficient-cube diameter and (II.12.10.57) imply

$$\text{diam}(\text{supp } \xi_{i,q,z}) \leq C_{k,\epsilon} 2^{-i}, \quad \text{diam}(\text{supp } \theta_{i,z}) \leq C 2^{-i}. \quad (\text{II.12.10.59})$$

The local action average, followed by Taylor linearization and the bounded entropy cost of the dilation $q'(z)^{-1}$, yields

$$\frac{1}{n} H(\nu, \theta, \mathcal{D}_n^{(2)}) \geq \mathbb{E}_{i,q,z} \frac{1}{m} H(\xi_{i,q,z} * \theta_{i,z}, \mathcal{D}_{i+m}^{(2)}) - 4\zeta. \quad (\text{II.12.10.60})$$

Let E_1 be the event that the baseline estimate (II.12.10.52) holds for $\theta_{i,z}$. Then $\mathbb{P}(E_1) > 1 - \zeta$. Let E_{tar} be the event that the nested full-plane and line nonsaturation conclusions (II.12.10.53)–(II.12.10.54) hold for $\theta_{i,z}$ with conditional probability $> 1 - \eta_{\text{in}}$; the target package gives $\mathbb{P}(E_{\text{tar}}) > 1 - \eta_{\text{in}}$. Finally, by Lemma II.12.10.18, the event

$$E_{\text{ev}} = \left\{ \frac{1}{m} H(\xi_{i,q,z}, \mathcal{D}_{i+m}^{(2)}) > c_0/2 \right\}$$

has probability $> c_0/2$ once m is large; bounded multiplication by $q'(z)^{-1}$ changes normalized entropy by $o(1)$. Hence

$$E_2 = E_1 \cap E_{\text{tar}} \cap E_{\text{ev}}$$

has probability at least $c_0/2 - \zeta - \eta_{\text{in}} > c_0/4$.

Fix a triple in E_2 and rescale the two measures in (II.12.10.59) by 2^i . For the target subtract the integer lower-left corner of $2^i \mathcal{D}_i^{(2)}(z)$; the resulting measure is exactly $\theta^{z,i}$. For the source subtract the integer parts of any one support point. Both translated measures lie in the fixed square $[-R, R]^2$. Their level- m dyadic entropies equal the original level- $(i+m)$ entropies exactly: multiplication by 2^i and integer translations preserve the dyadic grid, and the convolution is translated by the sum of the two integer vectors. The target is literally the normalized measure used in the target package, so its nested component and projection hypotheses apply without any translation-invariance claim for one-dimensional dyadic entropy. The source has normalized entropy $> a$, while the target satisfies the hypotheses of Proposition II.12.10.15 on the nested ℓ -blocks. That proposition averages over $0 \leq j \leq m$, whereas the target package uses $0 \leq j < m$; the extra endpoint costs at most $1/(m+1)$, so its failure probability is still below $\eta_0(a)$. Consequently

$$\frac{1}{m} H(\xi_{i,q,z} * \theta_{i,z}, \mathcal{D}_{i+m}^{(2)}) \geq \frac{1}{m} H(\theta_{i,z}, \mathcal{D}_{i+m}^{(2)}) + \rho_0 \geq d - \zeta + \rho_0. \quad (\text{II.12.10.61})$$

On all of E_1 , Lemma II.12.10.5, specifically the non-loss inequality (II.12.10.4), gives the baseline

$$\frac{1}{m} H(\xi_{i,q,z} * \theta_{i,z}, \mathcal{D}_{i+m}^{(2)}) \geq d - 2\zeta. \quad (\text{II.12.10.62})$$

The integrand in (II.12.10.60) is nonnegative up to $o(1)$ outside E_1 . Therefore

$$\mathbb{E}_{i,q,z} \frac{1}{m} H(\xi_{i,q,z} * \theta_{i,z}, \mathcal{D}_{i+m}^{(2)}) \geq d - 4\zeta + \frac{c_0 \rho_0}{4}.$$

Our choice $\zeta < c_0 \rho_0 / 128$ and (II.12.10.60) prove (II.12.10.58), for example with $\rho = c_0 \rho_0 / 16$. $\square$

Remark II.12.10.40 (Comparison with Feng–Rapaport’s formulation). Theorem II.12.10.39 uses the normalization of this volume for Feng–Rapaport’s entropy increase theorem. Their statement uses a measure on complex degree- k polynomials, a uniform coefficient bound, a uniform lower bound for $|q'|$ on the disk, and positive coefficient-space entropy; the conclusion is a definite gain above $\dim \mu$. The proof above separates the survival of entropy at an evaluation point from the application of the planar inverse theorem. Its remaining geometric ingredient is Theorem II.12.10.36.

For $n \geq 1$ define

$$\Pi_n : \Sigma \rightarrow \mathcal{O}(\Omega), \quad \Pi_n(\eta) = \varphi_{\eta|_n},$$

and, for a probability ξ on holomorphic maps, let $\xi \times \mu$ denote the image of $\xi \times \mu$ under evaluation. Let $\mathcal{C}_n$ be the partition of Σ into level- n cylinders and let $\{\beta_\omega\}_{\omega \in \Sigma}$ be a disintegration of β over $\Pi^{-1}\mathcal{B}(\mathbb{C})$.

Theorem II.12.10.41 (Feng–Hu entropy formula for holomorphic coding). *Put*

$$\Delta_F^{\mathbb{C}} = H_\beta(\mathcal{C}_1 \mid \Pi^{-1}\mathcal{B}(\mathbb{C})).$$

Then μ is exact dimensional,

$$\dim \mu = \frac{H(p) - \Delta_F^{\mathbb{C}}}{\chi}, \quad (\text{II.12.10.63})$$

and for β -almost every ω ,

$$\lim_{n \rightarrow \infty} \frac{1}{n} H(\beta_\omega, \mathcal{C}_n) = \Delta_F^{\mathbb{C}}. \quad (\text{II.12.10.64})$$

This is the conformal projection-entropy theorem of Feng–Hu [FH09]. It follows here from the conformal theorem of Volume I and Lemma II.12.4.17.

Proof. Identify $\mathbb{C}$ with $\mathbb{R}^2$. The real derivative of an injective holomorphic generator is multiplication by the nonzero complex number $\varphi'_i(z)$, hence is conformal. The standing uniform-contraction and compact-containment hypotheses give the remaining conditions in the definition of a conformal IFS, and the Bernoulli shift is ergodic. The conformal dimension theorem therefore gives

$$\dim \mu = \frac{h_\Pi(\sigma, \beta)}{\chi}. \quad (\text{II.12.10.65})$$

With $\mathcal{A} = \Pi^{-1}\mathcal{B}(\mathbb{C})$, the projection-entropy definition gives

$$h_\Pi(\sigma, \beta) = H_\beta(\mathcal{C}_1 \mid \sigma^{-1}\mathcal{A}) - H_\beta(\mathcal{C}_1 \mid \mathcal{A}).$$

The first coordinate is independent of $\sigma^{-1}\mathcal{A}$, since the latter is measurable with respect to coordinates 2, 3, Thus the first term is $H(p)$ and the second is $\Delta_F^{\mathbb{C}}$. Substitution into (II.12.10.65) proves (II.12.10.63). Lemma II.12.4.17, now applied to the holomorphic coding, proves (II.12.10.64). $\square$

Lemma II.12.10.42 (Vanishing entropy of a conditional holomorphic action). *For β -almost every ω ,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} H((\Pi_n)_* \beta_\omega \cdot \mu, \mathcal{D}_{[\chi n]}^{(2)}) = 0. \quad (\text{II.12.10.66})$$

Proof. For almost every ω , disintegration gives $\beta_\omega(\Pi^{-1}\{\Pi\omega\}) = 1$, and the ergodic theorem together with bounded distortion gives

$$\lim_{n \rightarrow \infty} n^{-1} \log_2 |\varphi'_{\eta|_n}(0)| = -\chi \quad \text{for } \beta_\omega\text{-almost every } \eta.$$

Fix such an ω and $\delta > 0$. For all sufficiently large n , the set U_n of words $u \in \Lambda^n$ satisfying $|\varphi'_u(0)| \leq 2^{n(\delta-\chi)}$ has conditional mass at least $1 - \delta$. If $\beta_\omega([u]) > 0$, then some point of $[u]$ codes $\Pi\omega$. The uniform bi-Lipschitz normalization in Lemma II.12.10.21 therefore gives

$$\text{supp}((\varphi_u)_*\mu) \subset B(\Pi\omega, C2^{n(\delta-\chi)}) \quad (u \in U_n).$$

At planar dyadic level $\lfloor \chi n \rfloor$ the good mixture consequently meets $O(C^2 2^{2\delta n})$ squares, so its entropy is at most $2\delta n + O(1)$. The complementary mixture has weight at most δ and is supported in a fixed bounded set, hence contributes at most $2\chi\delta n + O(1)$ after the entropy-of-mixtures inequality. Thus

$$\limsup_{n \rightarrow \infty} \frac{1}{n} H((\Pi_n)_* \beta_{\omega \cdot \mu}, \mathcal{D}_{\lfloor \chi n \rfloor}^{(2)}) \leq 2(1 + \chi)\delta.$$

Let $\delta \downarrow 0$. □

Lemma II.12.10.43 (Comparison of block homothetic conjugacies). *Fix $m \geq 1$. Let $\psi_1, \psi_2 : \Omega \rightarrow \mathbb{C}$ be holomorphic injections and assume that, for $j = 1, 2$ and every $u \in \Lambda^m$, $\psi_j \varphi_u \psi_j^{-1}$ is homothetic. Then there is $w \in \mathbb{C}^*$ such that*

$$\psi'_2(z)\mathbb{R} = \psi'_1(z)w\mathbb{R} \quad (z \in K_\Phi). \quad (\text{II.12.10.67})$$

Proof. Fix $z = \Pi\omega \in K_\Phi$ and put $u_t = (\sigma^{tm}\omega)|_m$. If $h_{j,u} = \psi_j \varphi_u \psi_j^{-1}$, then

$$\varphi_{\omega|_{rm}} = \psi_j^{-1} h_{j,u_0} \cdots h_{j,u_{r-1}} \psi_j.$$

The derivative of the middle composition has real direction because every $h_{j,u}$ is homothetic. Evaluating at 0 gives

$$\varphi'_{\omega|_{rm}}(0)\mathbb{R} = \psi'_j(\varphi_{\omega|_{rm}}(0))^{-1} \psi'_j(0)\mathbb{R}.$$

The left side is independent of j . Letting $r \rightarrow \infty$ and using $\varphi_{\omega|_{rm}}(0) \rightarrow z$ yields

$$\psi'_1(z)^{-1} \psi'_1(0)\mathbb{R} = \psi'_2(z)^{-1} \psi'_2(0)\mathbb{R},$$

which is (II.12.10.67) with $w = \psi'_2(0)/\psi'_1(0)$. □

Lemma II.12.10.44 (Arbitrarily large admissible block systems). *Assume (M1)–(M4) of Theorem II.12.10.45. For arbitrarily large m , the block IFS*

$$\Phi_m = \{\varphi_u : u \in \Lambda^m\}$$

also satisfies (M1)–(M4).

Proof. Conditions (M1), (M2), and (M4) are inherited by every block system. Indeed, if all m -block maps fixed x , then $\varphi_i^m(x) = x$ for every i ; uniqueness of the attracting fixed point of each generator would make x a common fixed point of the original IFS. If a regular real-analytic curve, closed in Ω , were Φ_m -invariant, then it would contain the block attractor K_Φ , contradicting Proposition II.12.10.24 because $\mu(K_\Phi) = 1$.

For exponential separation, let $N_j \rightarrow \infty$ be separating levels for Φ . After passing to a subsequence, $N_j \equiv r \pmod{m}$ for a fixed $0 \leq r < m$, say $N_j = mk_j + r$. Fix a word $a \in \Lambda^r$ if $r > 0$. For distinct $v, w \in \Lambda^{mk_j}$, separation of va and wa gives

$$\|\varphi_v - \varphi_w\|_\Omega \geq \|\varphi_{va} - \varphi_{wa}\|_\Omega \geq c^{mk_j+r} \geq (c^{m+r})^{k_j}.$$

Thus Φ_m is exponentially separated.

It remains to obtain arbitrarily large m for which (M3) holds. Suppose instead that for some m both Φ_m and Φ_{m+1} were holomorphically conjugate to homothetic IFSs, by ψ_1 and ψ_2 . Both conjugacies then make every block of length $m(m+1)$ homothetic, so Lemma II.12.10.43 gives $\psi_2'(z)\mathbb{R} = \psi_1'(z)w\mathbb{R}$ on K_Φ for some $w \neq 0$. Fix $i \in \Lambda$ and set $g = \psi_2\varphi_i\psi_2^{-1}$. For every $u \in \Lambda^m$, the map $\psi_2\varphi_{ui}\psi_2^{-1}$ is homothetic. Hence, for $z \in K_\Phi$,

$$\begin{aligned} \mathbb{R} &= \psi_2'(\varphi_{ui}z) \varphi_u'(\varphi_i z) (\psi_2^{-1})'(\psi_2\varphi_i z) g'(\psi_2 z)\mathbb{R} \\ &= \psi_1'(\varphi_{ui}z) \varphi_u'(\varphi_i z) (\psi_1^{-1})'(\psi_1\varphi_i z) g'(\psi_2 z)\mathbb{R} \\ &= (\psi_1\varphi_u\psi_1^{-1})'(\psi_1\varphi_i z) g'(\psi_2 z)\mathbb{R} = g'(\psi_2 z)\mathbb{R}. \end{aligned}$$

In the second line the factors w and w^{-1} from (II.12.10.67) cancel; in the last line the first factor has real direction because the m -block conjugate is homothetic. Lemma II.12.10.28 therefore makes g homothetic. Since this holds for every generator i , the original IFS is holomorphically conjugate to a homothetic one, contrary to (M3). Hence for every m at least one of Φ_m, Φ_{m+1} satisfies (M3), proving the claim. $\square$

For $t \geq 0$ define the conformal pressure and similarity dimension by

$$P_\Phi(t) = \lim_{m \rightarrow \infty} \frac{1}{m} \log_2 \left(\sum_{u \in \Lambda^m} \|\varphi_u'\|_\Omega^t \right), \quad P_\Phi(s(\Phi)) = 0. \quad (\text{II.12.10.68})$$

Submultiplicativity gives existence of the limit, and uniform contraction makes P_Φ continuous and strictly decreasing from $P_\Phi(0) = \log_2 |\Lambda|$ to $-\infty$.

Theorem II.12.10.45 (Feng–Rapaport holomorphic dimension theorem). *Let Φ be a finite holomorphic IFS on a bounded planar domain Ω in the sense of Definition II.12.10.1. Assume:*

- (M1) *the maps in Φ do not have a common fixed point;*
- (M2) *no regular real-analytic curve in Ω , closed in Ω , is invariant under all maps in Φ ;*
- (M3) *Φ is not holomorphically conjugate to a homothetic IFS;*
- (M4) *Φ is exponentially separated.*

Then every Bernoulli self-conformal measure with positive probability vector p satisfies

$$\dim \mu = \min \left\{ 2, \frac{H(p)}{\chi} \right\}. \quad (\text{II.12.10.69})$$

Moreover the attractor satisfies

$$\dim_H K_\Phi = \min\{2, s(\Phi)\}. \quad (\text{II.12.10.70})$$

The measure conclusion is [FR26, Theorem 1.5]; the attractor conclusion is [FR26, Corollary 1.6].

Proof. Fix the adapted domain from Lemma II.12.10.2. Write $d = \dim \mu$ and $L = \log_2 |\Lambda|$. By Theorem II.12.10.41, $d = (H(p) - \Delta_F^{\mathbb{C}})/\chi$, so $d \leq H(p)/\chi$, while $d \leq 2$. Assume for contradiction that

$$d < \min\{2, H(p)/\chi\}. \quad (\text{II.12.10.71})$$

Then $d < 2$ and $\Delta_F^{\mathbb{C}} > 0$. Put

$$\Delta = \frac{1}{2} \min\{\Delta_F^{\mathbb{C}}, 1\}, \quad \tau = \frac{\Delta}{2L}. \quad (\text{II.12.10.72})$$

The no-common-fixed-point hypothesis gives $|\Lambda| \geq 2$.

Separated block lengths and the global upper bound. Let $0 < c < 1$ be an exponential-separation constant. Enlarge the uniform bounded-distortion constant C when harmless. Choose

$$M > 4(1 + \tau)|\log_2 c| + 10, \quad (\text{II.12.10.73})$$

and then choose k so large that

$$k\chi\tau > 3M + 3\chi(1 + \tau). \quad (\text{II.12.10.74})$$

Let $N \rightarrow \infty$ run only through actual separating levels and set

$$n = \left\lfloor \frac{N}{1 + \tau} \right\rfloor, \quad n' = N - n, \quad q = \lfloor \chi N \rfloor, \quad r = Mn. \quad (\text{II.12.10.75})$$

Then $n'/n = \tau + o(1)$ and $(n'/n)L = \Delta/2 + o(1)$, while

$$\|\varphi_{w_1} - \varphi_{w_2}\|_{\Omega} \geq c^N \quad (w_1 \neq w_2 \in \Lambda^N). \quad (\text{II.12.10.76})$$

Fix $0 < \delta < \chi/4$, to be made sufficiently small after all fixed parameters, and then take N large.

Exact dimensionality implies entropy dimension d . Since $q/r = O(1)$, the entropy chain rule therefore gives, after increasing N ,

$$\frac{1}{r} H(\mu, \mathcal{D}_{q+r}^{(2)} \mid \mathcal{D}_q^{(2)}) \leq d + \delta. \quad (\text{II.12.10.77})$$

Stationarity and disintegration yield

$$\mu = (\Pi_N)_* \beta \cdot \mu = \int (\Pi_N)_* \beta_{\omega} \cdot \mu \, d\beta(\omega).$$

Conditional entropy is concave in the measure, hence (II.12.10.77) bounds the average conditional entropy of these fiber actions by $d + \delta$. Lemma II.12.10.42, dominated convergence, and $N/n = O(1)$ imply

$$\int \frac{1}{r} H((\Pi_N)_* \beta_{\omega} \cdot \mu, \mathcal{D}_q^{(2)}) \, d\beta(\omega) < \delta.$$

Consequently, with

$$g(\omega) = \frac{1}{r} H((\Pi_N)_* \beta_{\omega} \cdot \mu, \mathcal{D}_{q+r}^{(2)}), \quad (\text{II.12.10.78})$$

we have

$$\int g d\beta \leq d + 2\delta. \quad (\text{II.12.10.79})$$

Entropy-rich derivative packets. For $\ell \geq 1$ let $U_\ell \subset \Lambda^\ell$ consist of the words u with

$$2^{-\ell(\chi+\delta)} \leq |\varphi'_u(0)| \leq 2^{-\ell(\chi-\delta)}. \quad (\text{II.12.10.80})$$

The ergodic theorem and bounded distortion give $\beta([U_\ell]) \rightarrow 1$. Using (II.12.10.64), disintegration and Markov's inequality, there is, for all large chosen N , a set $E \subset \Sigma$ with $\beta(E) > 1 - \delta$ such that every $\omega \in E$ satisfies

$$\text{supp } \beta_\omega \subset \Pi^{-1}\{\Pi\omega\}, \quad (\text{II.12.10.81})$$

$$\frac{1}{n} H(\beta_\omega, \mathcal{C}_n) > \Delta, \quad (\text{II.12.10.82})$$

$$\beta_\omega([U_n]) > 1 - \delta, \quad \beta_\omega(\sigma^{-n}[U_{n'}]) > 1 - \delta. \quad (\text{II.12.10.83})$$

Fix $\omega \in E$. Partition U_n into classes $U_{n,j}$ by unit intervals for $-\log_2 |\varphi'_v(0)|$. There is an index set J with

$$|J| \leq 3\delta n, \quad |\varphi'_{v_1}(0)| \leq 2|\varphi'_{v_2}(0)| \quad (v_1, v_2 \in U_{n,j}). \quad (\text{II.12.10.84})$$

For $j \in J$ and $u \in U_{n'}$ put

$$F_{j,u} = [U_{n,j}] \cap \sigma^{-n}[u], \quad w_{j,u} = \beta_\omega(F_{j,u}),$$

and, when $w_{j,u} > 0$, set

$$g(\omega, j, u) = \frac{1}{r} H((\Pi_N)_* (\beta_\omega)_{F_{j,u}} \cdot \mu, \mathcal{D}_{q+r}^{(2)}). \quad (\text{II.12.10.85})$$

The retained packets have total mass $> 1 - 2\delta$, so entropy concavity gives

$$g(\omega) \geq \sum_{j,u} w_{j,u} g(\omega, j, u). \quad (\text{II.12.10.86})$$

The entropy-of-mixtures upper bound for $\mathcal{C}_n$, together with (II.12.10.82), gives

$$\sum_{j,u} w_{j,u} \frac{1}{n} H((\beta_\omega)_{F_{j,u}}, \mathcal{C}_n) > \frac{\Delta}{3} \quad (\text{II.12.10.87})$$

for small δ and large N ; here the bookkeeping loss is at most $2\delta nL + \log_2(3\delta n + 1) + n'L + O(1)$, and $n'L/n = \Delta/2 + o(1)$. Let $Q = Q(\omega)$ consist of those pairs with

$$\frac{1}{n} H((\beta_\omega)_{F_{j,u}}, \mathcal{C}_n) \geq \frac{\Delta}{6}. \quad (\text{II.12.10.88})$$

Since every normalized cylinder entropy is at most L ,

$$\sum_{(j,u) \in Q} w_{j,u} > \frac{\Delta}{6L}. \quad (\text{II.12.10.89})$$

Every retained $v \in U_n$, $u \in U_{n'}$ satisfies $2^q |\varphi'_{vu}(z)| \in [C^{-1}2^{-\delta N}, C2^{\delta N}]$ uniformly in z . Thus $2^q \varphi_{vu}$ is bi-Lipschitz with distortion scale $2^{O(\delta N)}$. After the dilation $z \mapsto 2^q z$, level $q+r$ becomes level r up to $O(1)$ grid multiplicity. Comparing this image with μ therefore shifts the fine dyadic level by at most $O(\delta N)$; since $r = Mn$ and $N/n = 1 + \tau + o(1)$, exact dimensionality gives

$$\frac{1}{r} H((\varphi_{vu})_* \mu, \mathcal{D}_{q+r}^{(2)}) \geq d - O(\delta) - o(1).$$

Concavity over the retained prefix mixture absorbs the $o(1)$ for large N and gives the uniform baseline

$$g(\omega, j, u) \geq d - O(\delta) \quad (\text{II.12.10.90})$$

for all retained pairs.

Taylor normalization and coefficient separation. Fix $(j, u) \in Q$, put $a = \varphi_u(0)$, and let

$$P_{k,a} f(X) = \sum_{\ell=0}^k \frac{f^{(\ell)}(a)}{\ell!} (X - a)^\ell.$$

Cauchy estimates on the fixed complex neighbourhood, bounded distortion, and (II.12.10.80) give, for $v \in U_{n,j}$ and $z \in \Omega$,

$$\begin{aligned} |(\varphi_v - P_{k,a} \varphi_v)(\varphi_u(z))| &\leq C^{2k+2} |\varphi'_u(0)|^{k+1} |\varphi'_v(0)| \\ &\leq C^{2k+2} 2^{-(k+1)(\chi-\delta)n'} 2^{-(\chi-\delta)n}. \end{aligned} \quad (\text{II.12.10.91})$$

By (II.12.10.74), after decreasing δ and increasing N , this is at most 2^{-q-r-4} . Coupling exact and Taylorized actions therefore changes level- $(q+r)$ planar entropy by only $O(1)$ bits.

Let $\psi(z) = r_\psi(z - a)$, $r_\psi > 0$, be chosen so that

$$\sup_{z \in \Omega} |\psi(\varphi_u(z))| = 1. \quad (\text{II.12.10.92})$$

The bi-Lipschitz part of Lemma II.12.10.21 implies

$$C^{-1} \leq r_\psi |\varphi'_u(0)| \leq C. \quad (\text{II.12.10.93})$$

Choose $\alpha > 0$ from the derivative bin so that

$$\frac{1}{2} \leq \alpha |\varphi'_v(0)| \leq 2 \quad (v \in U_{n,j}). \quad (\text{II.12.10.94})$$

For every v occurring in the packet define

$$q_v(X) = \alpha r_\psi [P_{k,a} \varphi_v(\psi^{-1}(X)) - \Pi \omega], \quad (\text{II.12.10.95})$$

and let ν be the pushforward of $(\beta_\omega)_{F_{j,u}}$ under $\eta \mapsto q_{\eta|_n}$. Put $\theta = (\psi \circ \varphi_u)_* \mu$. Then $\text{supp } \theta \subset \overline{\mathbb{D}}$ and

$$|\log_2(\alpha r_\psi) - \chi N| \leq \delta N + O(1). \quad (\text{II.12.10.96})$$

The planar version of the bounded-overlap dyadic scaling estimate therefore gives

$$g(\omega, j, u) \geq \frac{1}{r} H(v, \theta, \mathcal{D}_r^{(2)}) - O(\delta) - o(1). \quad (\text{II.12.10.97})$$

We verify the hypotheses of Theorem II.12.10.39. If $\beta_\omega([vu]) > 0$, the fiber property (II.12.10.81) supplies a point of $[vu]$ coding $\Pi\omega$, whence

$$|\varphi_v(a) - \Pi\omega| \leq C|\varphi'_v(0)| |\varphi'_u(0)|.$$

Together with Cauchy estimates, (II.12.10.93), and (II.12.10.94), this gives a constant $R = R(k, C)$ with

$$\|q_v\|_2 \leq R. \quad (\text{II.12.10.98})$$

The linear part of q'_v has modulus at least $(2C)^{-1}$, whereas all higher-order derivative terms are $O_{k,C}(r_\psi^{-1})$. Since $n' \rightarrow \infty$, for large N ,

$$|q'_v(z)| \geq \frac{1}{4C} \quad (z \in \mathbb{D}). \quad (\text{II.12.10.99})$$

Also (II.12.10.93) and Lemma II.12.10.19, on the full adapted domain Ω_0 , give

$$C^{-2} \leq |(\psi \circ \varphi_u)'(z)| \leq C^2 \quad (z \in \Omega_0). \quad (\text{II.12.10.100})$$

The map $\Psi = \psi \circ \varphi_u$ is holomorphic and injective on Ω_0 . Moreover, (II.12.10.92) says $\sup_{z \in \Omega} |\Psi(z)| = 1$; since Ψ is nonconstant, the maximum-modulus principle gives $\Psi(\Omega) \subset \mathbb{D}$. Hence $\Psi \in \mathcal{F}_{\epsilon_0}$ for the choice of ϵ_0 below, and $\theta = \Psi_*\mu$ meets the target hypothesis of Theorem II.12.10.39.

Finally suppose distinct prefixes v_1, v_2 produced polynomials in the same atom of $\mathcal{D}_r^{\mathcal{R}(\mathbb{C})}$. Then

$$|q_{v_1}(z) - q_{v_2}(z)| \leq C_k 2^{-r} \quad (z \in \mathbb{D}). \quad (\text{II.12.10.101})$$

By (II.12.10.76) there is $x_0 \in \Omega$ with $|\varphi_{v_1 u}(x_0) - \varphi_{v_2 u}(x_0)| \geq c^N$. Set $z_0 = \psi(\varphi_u(x_0)) \in \overline{\mathbb{D}}$. The Taylor-error bound shows that the corresponding two Taylor values differ by at least $c^N/2$. Multiplication by αr_ψ can only increase this lower bound for all large N . Hence $|q_{v_1}(z_0) - q_{v_2}(z_0)| \geq c^N/2$, contradicting (II.12.10.101) by (II.12.10.73). Thus distinct prefixes occupy distinct coefficient cells, and

$$\frac{1}{r} H(v, \mathcal{D}_r^{\mathcal{R}(\mathbb{C})}) = \frac{1}{Mn} H((\beta_\omega)_{F_{j,u}}, \mathcal{C}_n) \geq \frac{\Delta}{6M}. \quad (\text{II.12.10.102})$$

Choose

$$0 < \epsilon_0 < \min \left\{ R^{-1}, \frac{1}{4C}, C^{-2}, \frac{\Delta}{6M}, \frac{1}{2} \right\}. \quad (\text{II.12.10.103})$$

Theorem II.12.10.39 supplies $\rho > 0$. For sufficiently large N , the polynomial-norm and coefficient-entropy estimates above give

$$\frac{1}{r} H(v, \theta, \mathcal{D}_r^{(2)}) \geq d + \rho.$$

Returning to (II.12.10.97),

$$g(\omega, j, u) \geq d + \rho - O(\delta) \quad ((j, u) \in Q). \quad (\text{II.12.10.104})$$

Combining (II.12.10.86), (II.12.10.90), (II.12.10.89), and (II.12.10.104) gives, uniformly on E ,

$$g(\omega) \geq d + \frac{\rho\Delta}{6L} - O(\delta).$$

Since $\beta(E) > 1 - \delta$ and $g \geq 0$, integration contradicts (II.12.10.79) once δ is sufficiently small. Therefore (II.12.10.71) is impossible, proving (II.12.10.69).

The attractor. The usual cylinder cover and bounded distortion give $\dim_{\text{H}} K_{\Phi} \leq \min\{2, s(\Phi)\}$. We prove the reverse inequality without using an equilibrium-state black box. For $m \geq 1$ put

$$Z_m = \sum_{u \in \Lambda^m} \|\varphi'_u\|_{\Omega}^{s(\Phi)}, \quad p_u^{(m)} = \frac{\|\varphi'_u\|_{\Omega}^{s(\Phi)}}{Z_m}.$$

By (II.12.10.68), $m^{-1} \log_2 Z_m \rightarrow 0$. Let

$$A_m = \sum_{u \in \Lambda^m} p_u^{(m)} (-\log_2 \|\varphi'_u\|_{\Omega}).$$

Uniform contraction gives $A_m \geq c_0 m$ for some $c_0 > 0$, while uniform bounded distortion gives, for the block Bernoulli measure associated to $(\Phi_m, p^{(m)})$,

$$H(p^{(m)}) = s(\Phi) A_m + \log_2 Z_m, \quad \chi(\Phi_m, p^{(m)}) = A_m + O(1). \quad (\text{II.12.10.105})$$

Hence

$$\frac{H(p^{(m)})}{\chi(\Phi_m, p^{(m)})} \rightarrow s(\Phi). \quad (\text{II.12.10.106})$$

By Lemma II.12.10.44, there are arbitrarily large m for which Φ_m satisfies (M1)–(M4). Applying the already proved measure statement to $(\Phi_m, p^{(m)})$, whose measure is supported on the same attractor K_{Φ} , gives

$$\dim_{\text{H}} K_{\Phi} \geq \min \left\{ 2, \frac{H(p^{(m)})}{\chi(\Phi_m, p^{(m)})} \right\}.$$

Letting such $m \rightarrow \infty$ and using (II.12.10.106) proves (II.12.10.70). $\square$

Remark II.12.10.46 (From measures to the attractor). The proof of Theorem II.12.10.45 first obtains the measure formula from conditional symbolic entropy and Taylor separation. For the set formula, Lemmas II.12.10.43 and II.12.10.44 provide suitable block systems, while the probabilities $p_u^{(m)} \propto \|\varphi'_u\|_{\Omega}^{s(\Phi)}$ approximate the pressure root directly. The Feng–Hu theorem, Theorem II.12.10.41, connects symbolic entropy to dimension, and Lemma II.12.10.7 begins the planar inverse argument that supplies the required entropy increase.

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